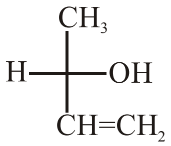
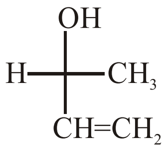
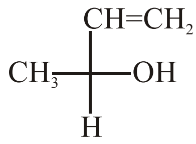
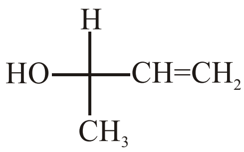
22. Which of the following combinations amongst the four Fischer projections represents the same absolute configurations ?


(I)

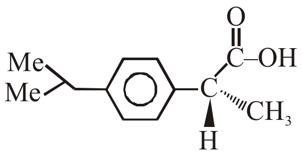

(II)

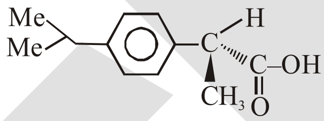

(III)

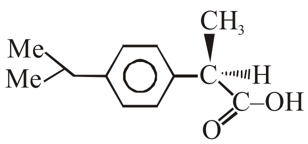

(IV)

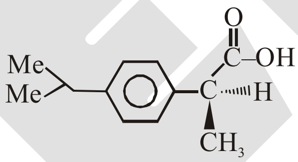
(A) (II) and (III) (B) (I) and (IV) (C) (II) and (IV) (D) (III) and (IV)

23. The S-ibuprofen is responsible for its pain relveing property. Which one of the structure shown is S-ibuprofen :

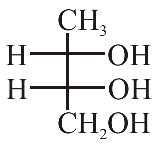
(A) 

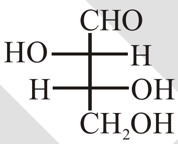
(B) 

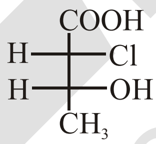
(C) 

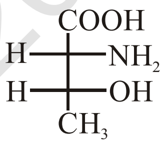
(D) 

24. Which of the following is a 'threo' isomer :

(A) 

(B) 

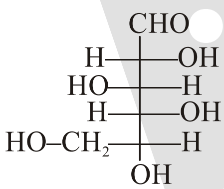
(C) 

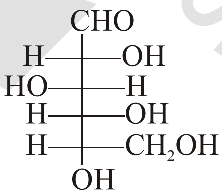
(D) 

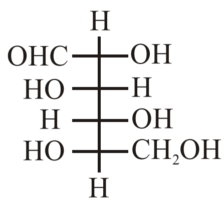
25. Number of possible stereoisomers of glucose are :-

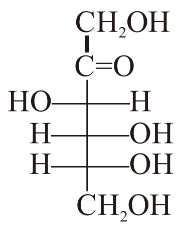
(A) 10 (B) 8 (C) 16 (D) 20

26. Which of the following is not D sugar :

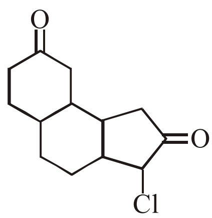
(A) 

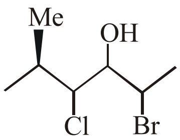
(B) 

(C) 

(D) 

27. Number of chiral centres in [X] & [Y] is a & b respectively. The value of (a–b) is :

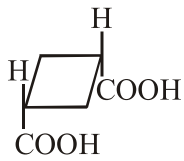
[X] = 

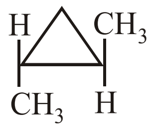
[Y] = 

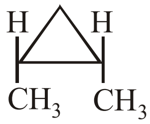
(A) 1 (B) 2 (C) 3 (D) 4

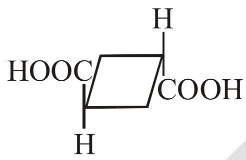
Session 2019-20
node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Stereo Isomerism\Eng.p65

28. Which one of the following is resolvable :

(A) 

(B) 

(C) 

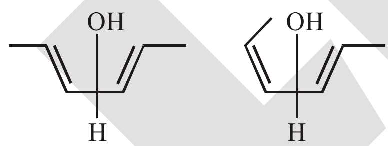
(D) 

29. How many stereoisomers can exist for the following acid.

H–C(OH).CO_2H
|
H–C.CO_2H
|
H–C(OH).CO_2H

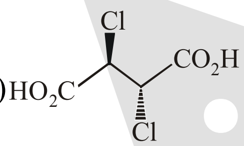
(A) Two (B) Four (C) Eight (D) Six

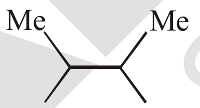
30. Incorrect relationship between given compounds are

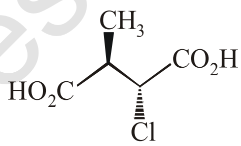


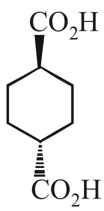
(A) Both are geometrical isomers
(B) Both are stereo isomers
(C) Both are enantiomers
(D) Both are diastereomers

31. Identify meso compound.

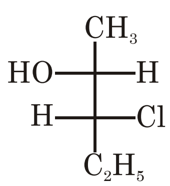
(A) 

(B) 

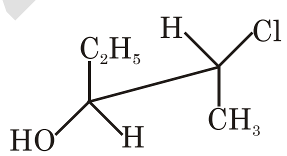
(C) 

(D) 

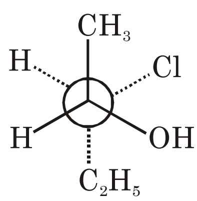
32. The two projection formulae that represent a pair of enantiomers are :-



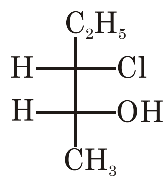
(I)



(II)



(III)



(IV)

(A) I and II
(B) III and IV
(C) I and III
(D) II and IV

ALLEN Session 2019-20
node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Stereo Isomerism\Eng.p65

33. A pure sample of 2-chlorobutane shows rotation of PPL by 30° in standard conditions. When above sample is made impure by mixing its opposite form, so that the composition of the mixture becomes 87.5% d-form and 12.5% ℓ-form, then what will be the observed rotation for mixture.

(A) – 22.5° (B) + 22.5° (C) + 7.5° (D) – 7.5°

34. When an optically active compound is placed in a 10 dm tube is present 20 gm in a 200 ml solution rotates the PPL by 30°. Calculate the angle of rotation & specific angle of rotation if above solution is diluted to 1 Litre.

(A) 16° & 36° (B) 6° & 30° (C) 3° & 30° (D) 6° & 36°

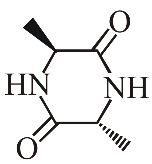
35. Identify % optical purity if 6 gm (+)-2-butanol is mixed with 2 gm (–)-2-butanol.

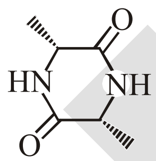
(A) 50 % (B) 66.6 % (C) 33.3 % (D) 75 %

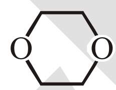
36. A mixture of d and ℓ, 2-bromobutane contain 75% d-2-bromobutane. Calculate enantiomeric excess.

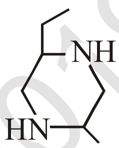
(A) 75% (B) 25% (C) 50% (D) 100%

37. Which of the following is example of meso compound?

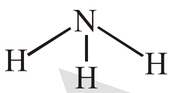
(A) 

(B) 

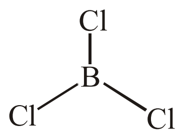
(C) 

(D) 

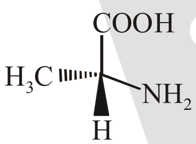
38. Which of the following has C_2 & C_3 axis of symmetry ?

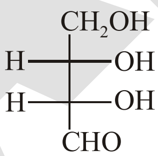
(A) 

(B) H_2CCl_2 (C) $H_3C–Cl$

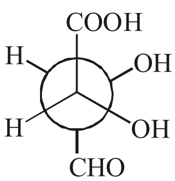
(D) 

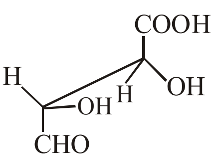
39. Configuration of I & II respectively will be :





(A) D, D (B) L , D (C) D , L (D) L, L

40. 



, compound related as :

(A) Enantiomers (B) Conformation (C) Identicle (D) Diastereomers

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Stereo Isomerism\Eng.p65

EXERCISE # II (JEE-ADVANCE ORIENTED LEVEL-I)

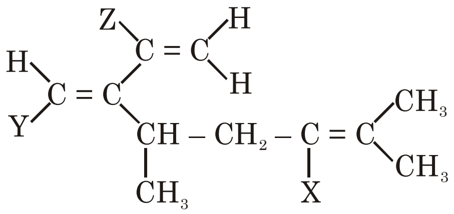
Single correct Option Type :

1. Molecular formula $C_5H_{10}O$ can have :

(A) 6-Aldehyde, 4-Ketone (B) 5-Aldehyde, 3-Ketone

(C) 4-Aldehyde, 3-Ketone (D) 5-Aldehyde, 2-Ketone

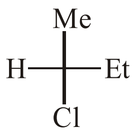
2. In the given halogenoalkene M, atoms X, Y and Z represents hydrogen or bromine or chlorine. To show cis-trans isomerism, what could be the identities of atoms X, Y and Z?

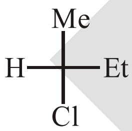


(M)

	X	Y	Z
1	Cl	H	Br
2	H	Br	Cl
3	Cl	Br	H

(A) 1, 2 and 3 (B) 1 and 2 only (C) 2 and 3 only (D) 1 and 3 only

3. **Statement–1 :**  is a chiral resolvable molecule.

Statement–2 :  is non-superimposable on its mirror image.

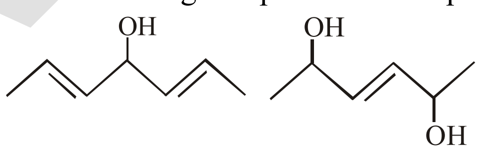
(A) Statement–1 is true, Statement–2 is true; Statement–2 is not the correct explanation of Statement–1

(B) Statement–1 is true, Statement–2 is true ; Statement–2 is the correct explanation of Statement–1

(C) Statement–1 is true, Statement–2 is false

(D) Statement–1 is false, Statement–2 is true

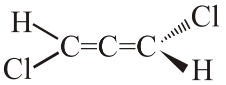
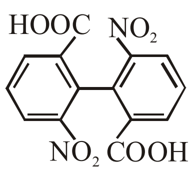
4. Total number of stereoisomer of following compounds are respectively :-



(A) 4, 6 (B) 8 (C) 6,6 (D) 8, 8

5. Which of the following compounds are optically active ?

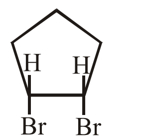
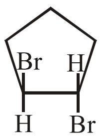
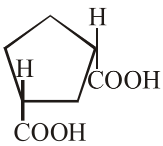
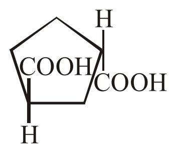
(A) $CH_3.CHOH.CH_2.CH_3$ (B) $H_2C=CH.CH_2.CH=CH_2$

(C)  (D) 

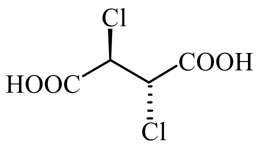
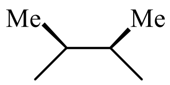
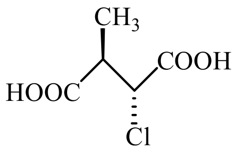
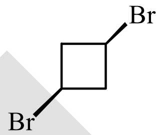
ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Stereo Isomerism\Eng.p65

6. Which out the following are Non-resolvable :

(A)  (B)  (C)  (D) 

7. Identify compound(s) which is/are not meso :

(A)  (B)  (C)  (D) 

8. Which of the following statements for a meso compound is/are correct :

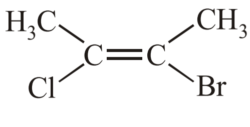
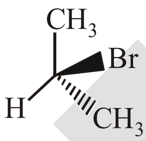
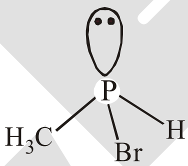
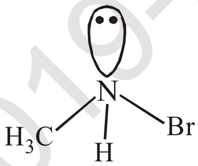
(A) The meso compound has either a plane or centre of symmetry

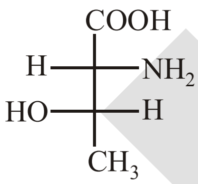
(B) The meso compound is optically inactive due to internal compensation.

(C) The meso compound is achiral

(D) The meso compound is formed when equal amounts of two enantiomers are mixed

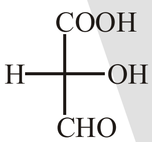
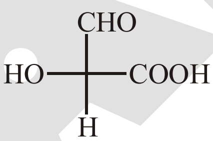
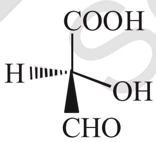
9. Among the following the non- resolvable compound is/are :

(A)  (B)  (C)  (D) 

10. Compound  is/are ?

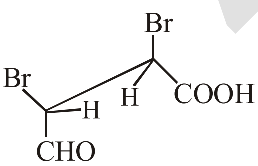
(A) (2R, 3S), L (B) L, Erythreo (C) Threo , D (D) (2R, 3S), D

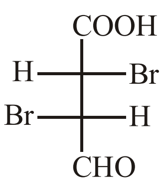
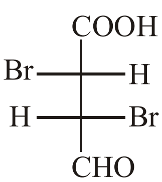
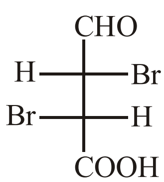
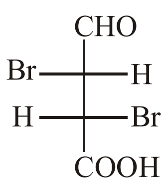
11. Relation between compounds are :

(A) I & II = Enantiomers (B) II & III = Enantiomers

(C) I & II = Identical (D) II & III = Identical

12.  has fisher diagram ?

(A)  (B)  (C)  (D) 

ALLEN Session 2019-20

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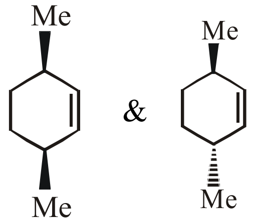
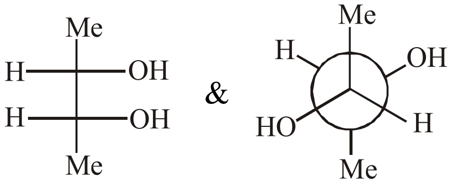
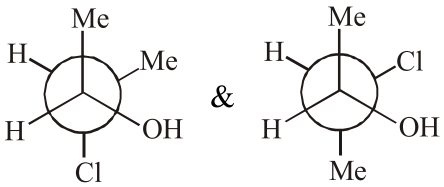
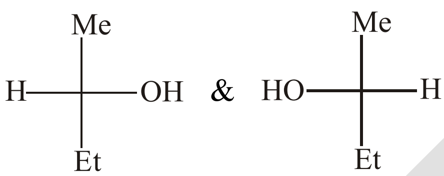
13. Identify correct relation between pair of compounds ?

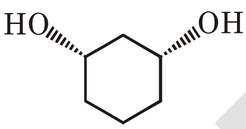
(A)  Diastereomers

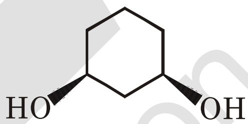
(B)  Identical

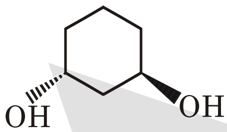
(C)  Diastereomers

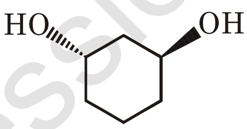
(D)  Enantiomers

14. Which two of the following compounds represents a pair of enantiomers?

(I) 

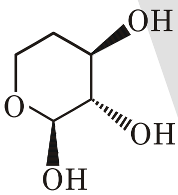
(II) 

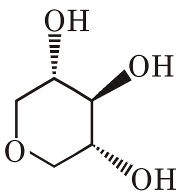
(III) 

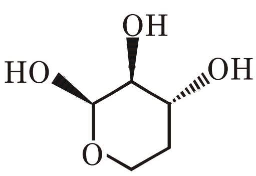
(IV) 

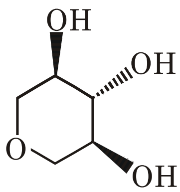
(A) I & II (B) II & III (C) III & IV (D) II & IV

15. Which two of the following compounds are diastereomers?

(I) 

(II) 

(III) 

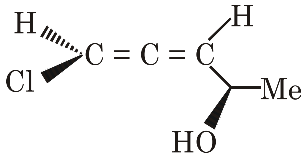
(IV) 

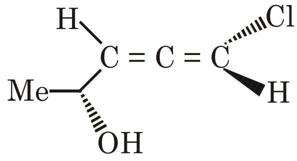
(A) I & II (B) II & IV (C) III & IV (D) I & III

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Stereo Isomerism\Eng.p65

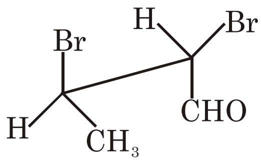
16. The correct relation between the following compounds is :



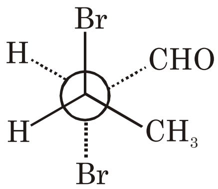


(A) Enantiomers
(B) Diastereomers
(C) Homomers (Identical)
(D) Constitutional isomers

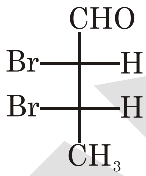
17. Identify the correct statement regarding following molecules?



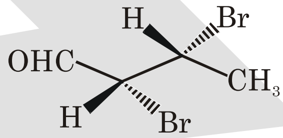
(M)



(N)



(O)

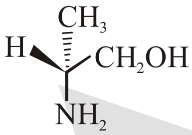
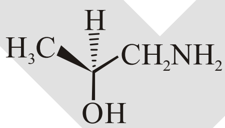


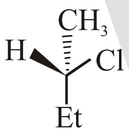
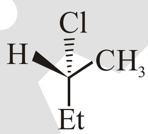
(P)

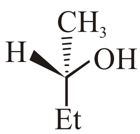
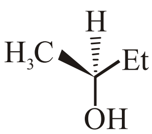
(A) **M** and **O** are diastereomers
(B) **N** and **P** are enantiomers
(C) **M** and **N** are identical
(D) **O** and **P** are diastereomers

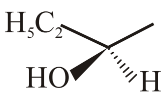
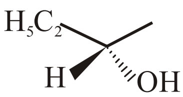
Matrix Match Type :

18. **Column I** | **Column II**

(A)  & 

(B)  & 

(C)  & 

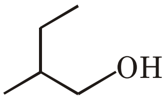
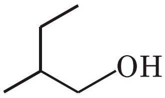
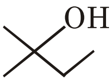
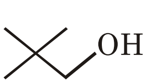
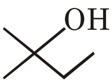
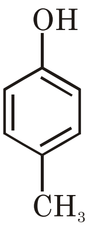
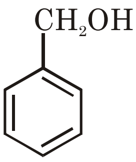
(D)  & 

(P) Structural
(Q) Identical
(R) Enantiomers
(S) Diastereomers

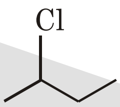
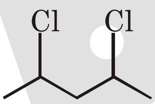
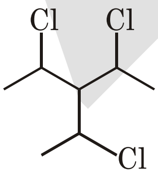
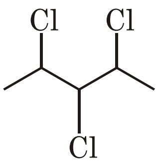
ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Stereo Isomerism\Eng.p65

19. Match the column :-

	Column-I			Column-II
(1)	 & 		**(P)**	Position isomers
(2)	 & 		**(Q)**	Chain isomers
(3)	 & 		**(R)**	Homologues
(4)	 & 		**(S)**	Functional isomers

20. Match the column :-

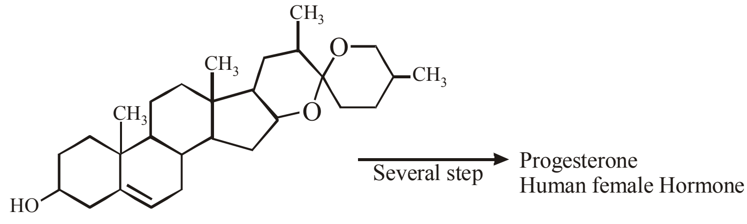
	Column-I (Compounds)		Column-II (Total number of stereoisomers)
(1)		**(P)**	8
(2)		**(Q)**	4
(3)		**(R)**	3
(4)		**(S)**	2

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Stereo Isomerism\Eng.p65

Subjective Type :

21.

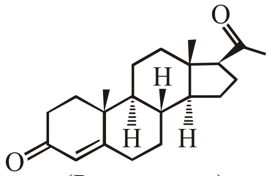


Diosgenin

What is number of chiral centres present in Diosgenen is :

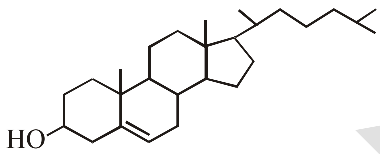
22. Calculate the total number of chrial carbon atoms in.

(i)



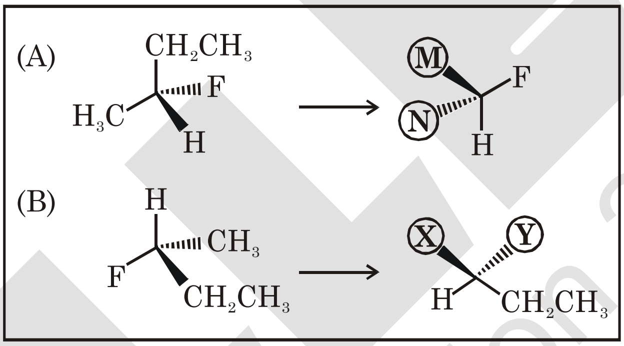
(Progesterone)

(ii)



23. Total number of isomeric (including stereo) bromochlorofluoroiodo propadiene.

24. Re-orient the molecule at the left to match the partially drawn perspective at the right. Find the two missing substituents at their correct positions.



(A) **M** $=CH_3CH_2-$
N $=CH_3-$

(B) **X** $=CH_3-$
Y $=F-$

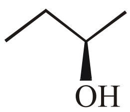
(C) **M** $=CH_3-$
N $=CH_3CH_2-$

(D) **X** $=F-$
Y $=CH_3-$

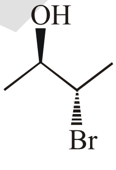
25. Find out the total number of cyclic isomers of C_6H_{12} which are optically active ?

26. How many of the given compounds are chiral :

(i)

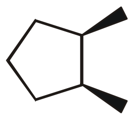


(ii)

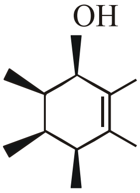


(iii)

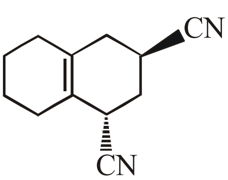
(iv)



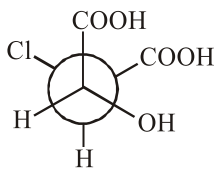
(v)



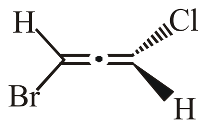
(vi)



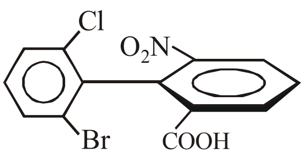
(vii)



(viii)

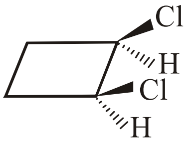
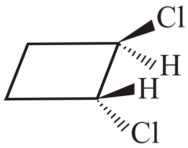


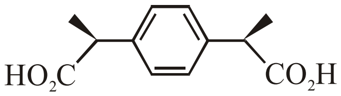
(ix)

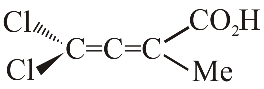
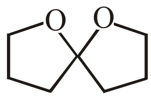


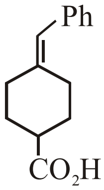
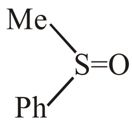
Session 2019-20

27. With reasons, state whether each of the following compounds I to VIII is chiral

(I)    (II) 

(III) 

(IV)    (V) 

(VI)    (VII) 

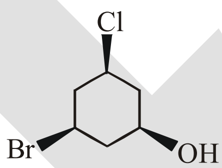
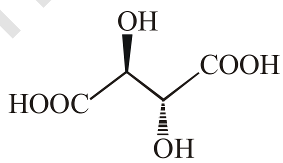
28. How many cyclopentane structures (including stereo) are possible for C_7H_{14}.

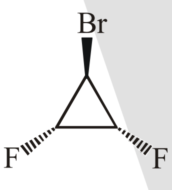
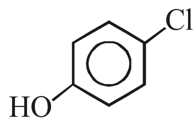
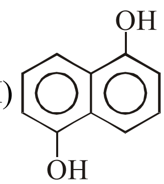
29. The number of diastereoisomers (excluding enantiomers) for 1-bromo-2-chloro-3-iodocyclopropane.

30. Identify total number of stereoisomers for the following compound :

$$\underset{}{CH_2}\text{-}\underset{|}{\underset{OH}{CH}}\text{-}\underset{|}{\underset{Cl}{CH}}\text{-}CH{=}CH\text{-}CH_3$$

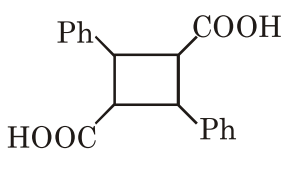
31. How many of the given molecule / species are chiral :

(I)    (II)    (III)    (IV) 

(V)    (VI)    (VII)    (VIII) 

(IX) CH_4

32. Total geometrical isomers possible for :



(A) 3   (B) 4   (C) 5   (D) 8

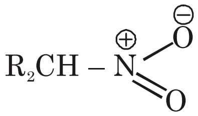
ALLEN Session 2019-20

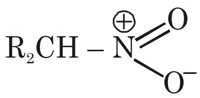
node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Stereo Isomerism\Eng.p65

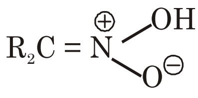
EXERCISE # III (JEE-ADVANCE ORIENTED LEVEL # II)

Single Correct Type :

1. The correct statements describing the relationship between :

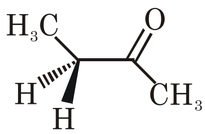

(X)

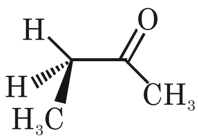

(Y)


(Z)

(A) **X** and **Y** are resonance structures and **Z** is a tautomer
(B) **X** and **Y** are tautomers and **Z** is resonance structure
(C) **X**, **Y** and **Z** are all resonance structures
(D) **X**, **Y** and **Z** all are tautomers

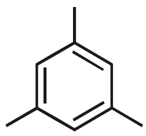
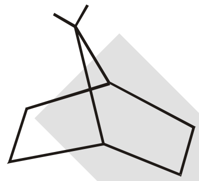
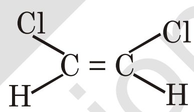
2. The correct statements about conformations X and Y of 2-butanone are :

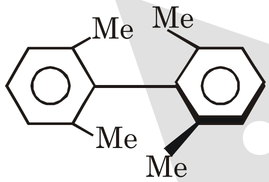
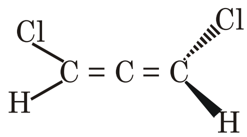

(X)


(Y)

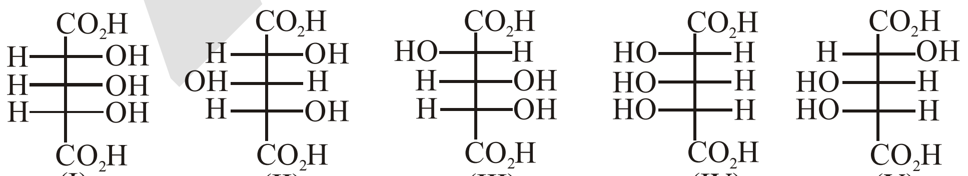
(a) **X** is more stable than **Y** (b) **Y** is more stable than **X**
(c) Methyl groups in **X** are anti (d) Methyl groups in **Y** are gauche
(A) a and d (B) a and c (C) b and c (D) a, c and d

3. Among the following, the number of molecules that possess C_2–axis of symmetry is :

, , CH_4, , BF_3,

, $CHCl_3$, 2,5-dimethylthiophene, 

4. Observe the given compounds and answer the following questions.


(I) (II) (III) (IV) (V)

(i) Which of the above formulae represent identical compounds ?
(A) I and II (B) I and IV (C) II and IV (D) III and IV

(ii) Which of the above compounds are enantiomers ?
(A) II and III (B) III and IV (C) III and V (D) I and V

ALLEN Session 2019-20
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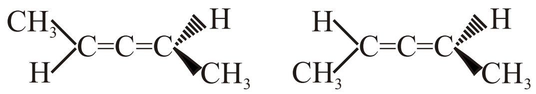
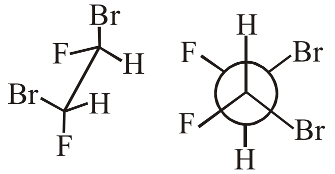
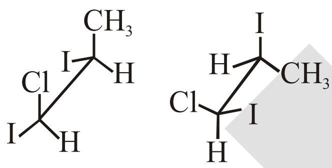
5. Which of the following option is correct regarding the given compounds :

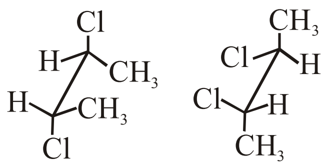


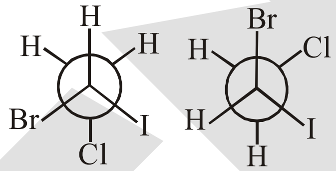
(A) Both are identical
(B) Both are optically inactive
(C) Both are enantiomers
(D) Geometrical isomer

6. Which of the following pairs of compound is/are identical ?

(A) 

(B) 

(C) 

(D) 

Multiple Correct Type :

7. Which of the following statements is/are not correct for D-(+) glyceraldehyde :

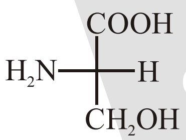
(A) The symbol D indicates the dextrorotatory nature of the compound

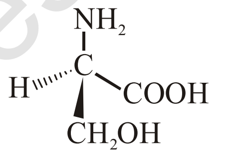
(B) The sign(+) indicates the dextrorotatory nature of the compound

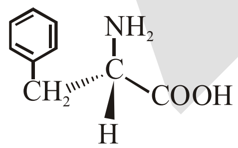
(C) The symbol D indicates that (–OH) group lies left to the chiral centre in the conventionally correct Fischer projection diagram

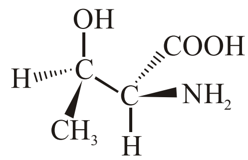
(D) The symbol D indicates that (–OH) group lies right to the chiral centre in the conventionally correct Fischer projection diagram

8. Which of the following are correct representation of L-amino acids :

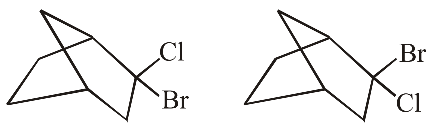
(A) 

(B) 

(C) 

(D) 

9. Identify relation between these two compounds :



(A) Homomers
(B) Enantiomers
(C) Diastereomers
(D) Positional Isomers

ALLEN Session 2019-20
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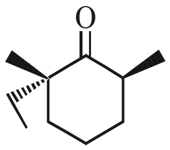
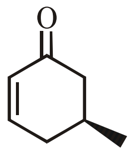
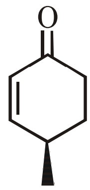
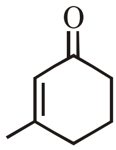
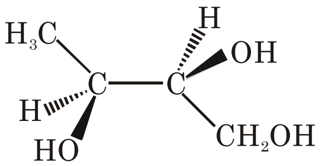
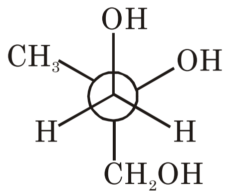
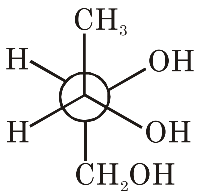
10. Which of the following undergoes racemisation in alkaline medium?

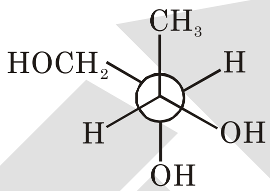
(A)  (B)  (C)  (D) 

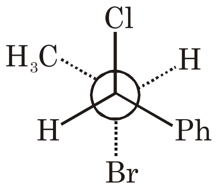
11. Which compound is different from the others?

(A) 

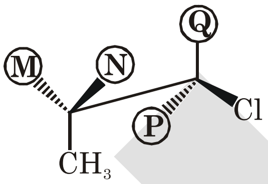
(B) 

(C) 

(D) 

12. 

(X)



(Y)

What would be the correct match to get (**Y**) as a diastereomers of (**X**)?

(A) **M** = –H **N** = –Br **Q** = –Ph **P** = –H

(B) **M** = –H **N** = –Br **Q** = –H **P** = –Ph

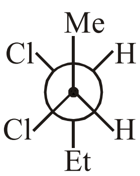
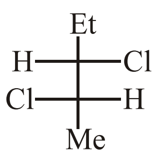
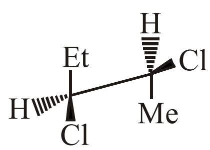
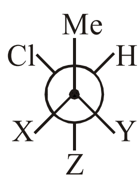
(C) **M** = –Br **N** = –H **Q** = –Ph **P** = –H

(D) **M** = –Br **N** = –H **Q** = –H **P** = –Ph

Comprehension Type :

Paragraph for Question 13 to 14

Four compounds are given below 'S' is a stereoisomer of P.

(P)  (Q)  (R)  (S) 

13. P & Q are related as :

(A) Identical (B) Enantiomer (C) Diastereomer (D)Positional isomerism

14. Which of the above structures represented is Sawhorse projection :-

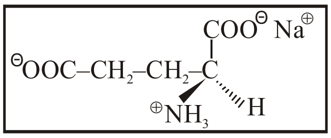
(A) P (B) Q (C) R (D) S

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Stereo Isomerism\Eng.p65

Paragraph for Question 15 to 17

S(+) Mono sodium Glutamate (MSG) is a flavour enhancer used in many foods. Fast foods often contain substantial amount of MSG and is widely used in Chinese food. If one mole of above MSG was placed in 845 ml solution and passed through 200 mm tube, the observed rotation was found to be + 9.6°.



15. Find out the specific rotation of (–) MSG :

(A) + 24° (B) +56.8º (C) –48° (D) None of these

16. Find out the approximate percentage composition of (–) MSG in a mixture containing (+) MSG and (–) MSG whose specific optical rotation is –20° :

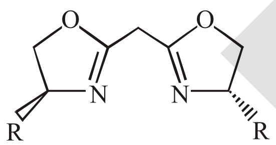
(A) 83.3% (B) 16.7% (C) 91.6% (D) 74%

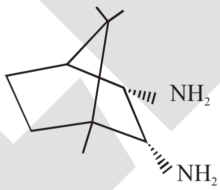
17. If 33.8 g of (+) MSG was put in 338 ml solution and was mixed with 16.9g of (–) MSG put in 169 ml solution and the final solution was passed through 400 mm tube. Find out observed rotation of the final solution :

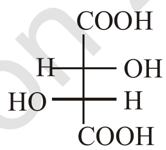
(A) +1.6° (B) +4.8° (C) +3.2° (D) None of these

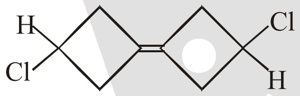
Paragraph for Q.18 to Q.19

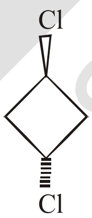
Among the following structures ?

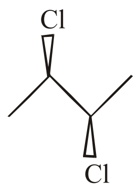
(I) 

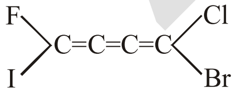
(II) 

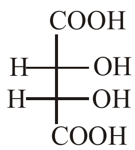
(III) 

(IV) 

(V) 

(VI) 

(VII) 

(VIII) 

18. Optically active compound is -

(A) III (B) IV (C) V (D) VII

19. Which of the following will not show optical isomerism -

(A) I (B) II (C) V (D) VIII

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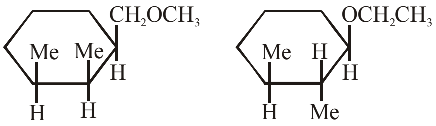
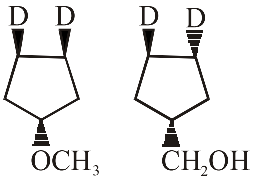
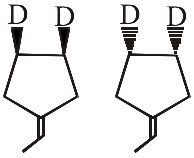
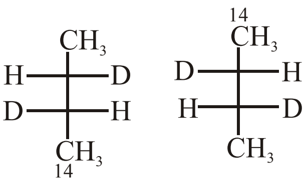
Matrix Match Type :

20. **Column-I (Compounds)** — **Column-II (Relation)**

(A) 

(B) 

(C) 

(D) 

(P) Metamers

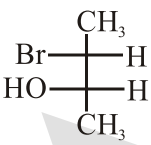
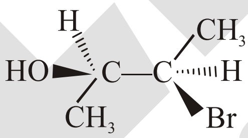
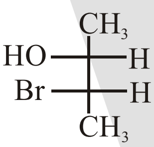
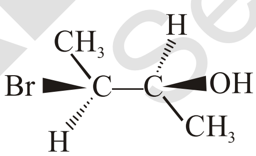
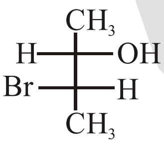
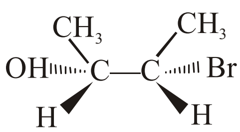
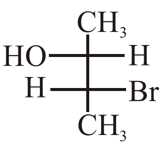
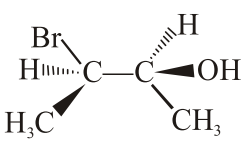
(Q) Functional Isomer

(R) Geometrical isomer

(S) Enantiomer

(T) Diastereomer

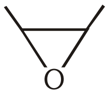
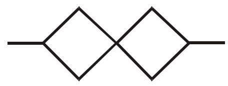
21. Match List-I, II, III with each other :

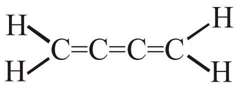
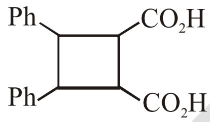
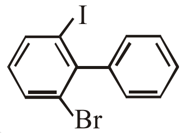
List - I	List - II	List - III
(A) 	(1) 	(i) (2R, 3R)
(B) 	(2) 	(ii) (2S,3S)
(C) 	(3) 	(iii) (2S,3R)
(D) 	(4) 	(iv) (2R,3S)

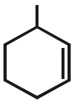
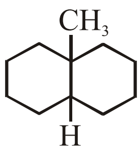
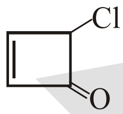
ALLEN Session 2019-20
node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Stereo Isomerism\Eng.p65

Subjective Type :

22. In what stereoisomeric forms would you expect the following compounds to exist ?

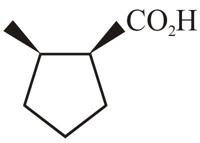
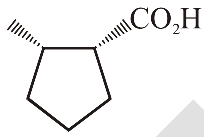
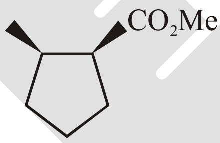
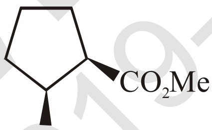
(a) EtCH(CO_2H)Me (b) MeCH(CO_2Et)CO_2H (c)  (d) 

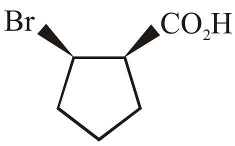
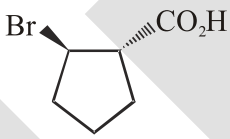
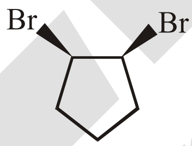
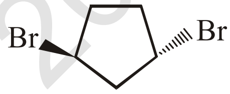
(e)  (f) Et(Me)C=C=C(Me)Et (g)  (h) 

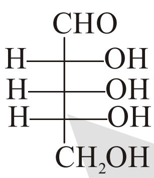
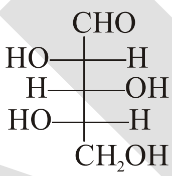
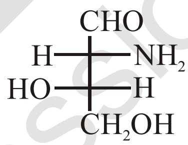
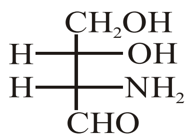
(I)  (J)  (k) 

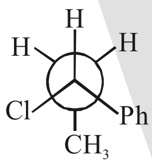
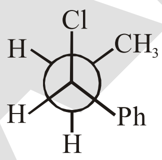
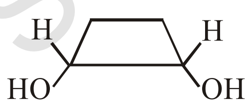
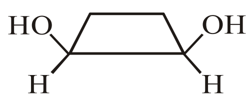
23. Calculate the number of Benzenoid isomers possible for C_6H_3ClBrI.

24. What are the relationships between the following pairs of isomers ?

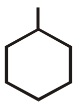
(a)  and  (b)  and 

(c)  and  (d)  and 

(e)  and  (f)  and 

(g)  and  (h)  and 

(i)  and 

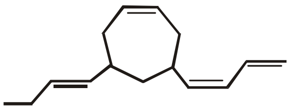
25. X $\xrightarrow[\text{pt}]{2H_2}$ 

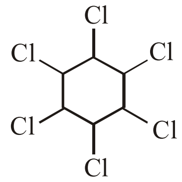
Find out total number of structures of X.

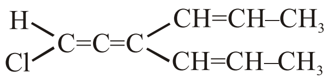
26. Calculate the number of chiral center in the molecule Ethyl 2,2-dibromo-4-ethyl-6-methoxy cyclohexane carboxylate.

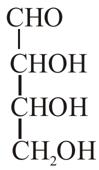
node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Stereo Isomerism\Eng.p65

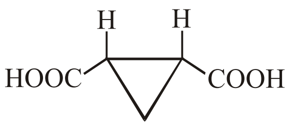
27. Calculate the total number of stereoisomers possible for

(i) 

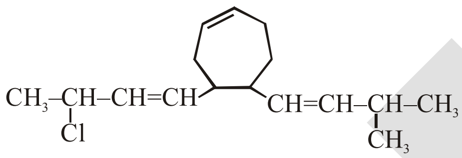
(ii) 

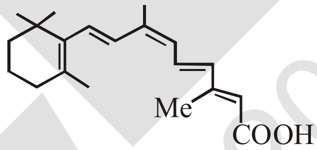
(iii) 

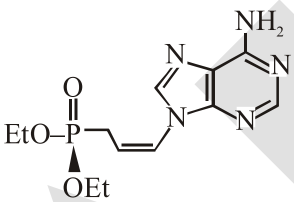
(iv) 

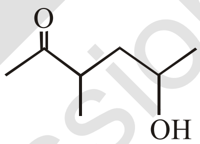
(v) 

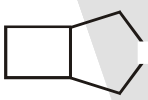
(vi) 

(vii) 

(viii) 

(ix) 

(x) 

(xi) 

28. How many different chloroethanes are there from the formula $C_2H_{6-n}Cl_n$ (where n can be any integer from 1 to 6)?

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Stereo Isomerism\Eng.p65

EXERCISE # IV (A) (J-MAINS)

1. Recemic mixture is formed by mixing two : **[AIEEE-2002]**

(1) Isomeric compounds (2) Chiral compounds

(3) Meso compounds (4) Enantiomers with chiral carbon

2. Following types of compounds I and II **[AIEEE-2002]**

(I) $CH_3CH{=}CHCH_3$

(II) $CH_3—\underset{\large CH_2CH_3}{\underset{|}{C}H}—OH$, are studied in terms of isomerism in-

(1) Chain isomerism (2) Position isomerism

(3) Conformers (4) Stereo isomerism

3. Among the following four structures I to IV **[AIEEE-2003]**

$C_2H_5—\underset{\large CH_3}{\underset{|}{C}H}—C_3H_7$ (I) $C_2H_5—\underset{\large O}{\underset{||}{C}}—\underset{\large CH_3}{\underset{|}{C}H}—C_2H_5$ (II) $H—\overset{\large H}{\overset{|}{\underset{\large H}{\underset{|}{C}}}}{}^{\oplus}$ (III) $C_2H_5—\underset{\large CH_3}{\underset{|}{C}H}—C_2H_5$ (IV)

It is true that-

(1) All four are chiral compounds

(2) Only I and II are chiral compounds

(3) Only III is a chiral compound

(4) Only II and IV are chiral compounds

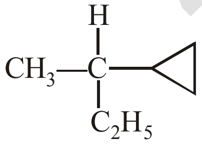
4. Which of the following will have a meso-isomer also- **[AIEEE-2004]**

(1) 2–chlorobutane (2) 2,3–dichlorobutane

(3) 2,3–dichloropentene (4) 2–hydroxy propanoic acid

5. Amongst the following compounds, the optically active alkane having lowest molecular mass is **[AIEEE-2004]**

(1) $CH_3—CH_2—C\equiv CH$ (2) $CH_3—CH_2—\overset{\large CH_3}{\overset{|}{C}H}—CH_3$

(3) 

(4) $CH_3—CH_2—CH_2—CH_3$

6. Which of following compounds is not chiral **[AIEEE-2005]**

(1) 1–chloropentane (2) 2–chloropentane

(3) 1–chloro–2–methyl pentane (4) 3–chloro–2–methyl pentane

ALLEN Session 2019-20
node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Stereo Isomerism\Eng.p65

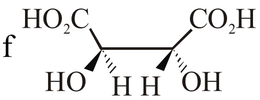
7. Of the five isomeric hexanes, the isomer which can give two monochlorinated compounds is :

(1) 2-methyl pentane (2) 2,2–dimethyl butane **[AIEEE-2005]**

(3) 2,3–dimethyl butane (4) n-hexane

8. Which types of isomerism is shown by 2,3–dichloro butane- **[AIEEE-2005]**

(1) Structural (2) Geometric (3) Optical (4) Diastereo

9. The absolute configuration of 

is : **[AIEEE-2008]**

(1) S, S (2) R, R (3) R, S (4) S, R

10. The number of stereoisomers possible for a compound of the molecular formula CH_3–CH=CH–CH(OH)–Me is :- **[AIEEE-2009]**

(1) 4 (2) 6 (3) 3 (4) 2

11. Out of the following, the alkene that exhibits optical isomerism is :- **[AIEEE-2010]**

(1) 2-methyl-2-pentene

(2) 3-methyl-2-pentene

(3) 4-methyl-1-pentene

(4) 3-methyl-1-pentene

12. The optically inactive compound from the following is :- **[JEE-MAIN-2015]**

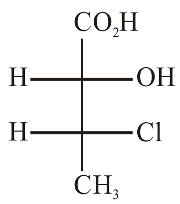
(1) 2-chloropropanal

(2) 2-chlorobutane

(3) 2-chloro-2-methylbutane

(4) 2-chloropentane

13. The absolute configuration of : **[JEE-MAIN-2016]**

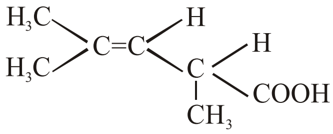


(1) (2R, 3R) (2) (2R, 3S) (3) (2S, 3R) (4) (2S, 3S)

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Stereo Isomerism\Eng.p65

EXERCISE # IV (B) (J-ADVANCE OBJECTIVE)

1. The  shows : **[IIT-1995]**

(A) Geometrical isomerism
(B) Optical isomerism
(C) Geometrical & optical isomerism
(D) tautomerism

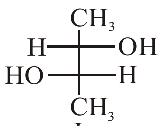
2. How many optically active stereoisomers are possible for butane –2,3–diol : **[IIT-1997]**

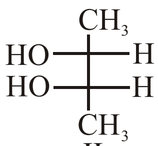
(A) 1 (B) 2 (C) 3 (D) 4

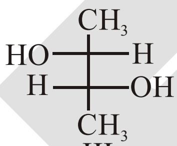
3. The number of possible enantiomeric pairs that can be produced during monochlorination of 2–methyl butane is : **[IIT-1997]**

(A) 2 (B) 3 (C) 4 (D) 1

4. Identify the pairs of enantiomers and diastereomers from the following compounds I, II and III


I


II


III

[IIT-2000]

5. Which of the following compounds exhibits stereoisomerism– **[IIT-2002]**

(A) 2–Methylbutene–1
(B) 3–Methylbutyne–1
(C) 3–Methylbutanoic acid
(D) 2–Methylbutanoic acid

6. On monochlorination of 2-methylbutane, the total number of chiral compounds formed is :

(A) 2 (B) 4 (C) 6 (D) 8 **[IIT-2004]**

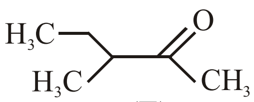
7. **Statement-I :** Molecules that are not superimposable on their mirror images are chiral

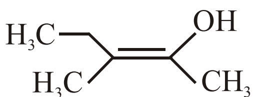
Because

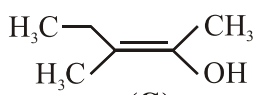
Statement-II : All chiral molecules have chiral centres. **[IIT-2007]**

(A) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1
(B) Statement-1 is True, Statement-2 is True ; Statement-2 is NOT a correct explanation for Statement-1
(C) Statement-1 is True, Statement-2 is False.
(D) Statement-1 is False, Statement-2 is True.

8. The correct statement(s) concerning the structures E, F and G is (are) **[IIT-2008]**


(E)

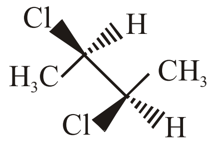

(F)


(G)

(A) **E, F** and **G** are resonance structures
(B) **E, F** and **E, G** are tautomers
(C) **F** and **G** are geometrical isomers
(D) **F** and **G** are diastereomers

ALLEN Session 2019-20
node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Stereo Isomerism\Eng.p65

9. The correct statement(s) about the compound given below is (are) : **[IIT-2008]**



(A) The compound is optically active

(B) The compound possesses centre of symmetry

(C) The compound possesses plane of symmetry

(D) The compound possesses axis of symmetry

10. The correct statement(s) about the compound $H_3C(HO)HC - CH = CH - CH(OH)CH_3$ **(X)** is (are) :

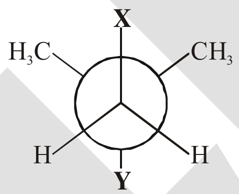
(A) The total number of stereoisomers possible for X is 6 **[IIT-2009]**

(B) The total number of diastereomers possible for X is 3

(C) If the stereochemistry about the double bond in X is trans, the number of enantiomers possible for X is 4

(D) If the stereochemistry about the double bond in X is cis, the number of enantiomers possible for X is 2

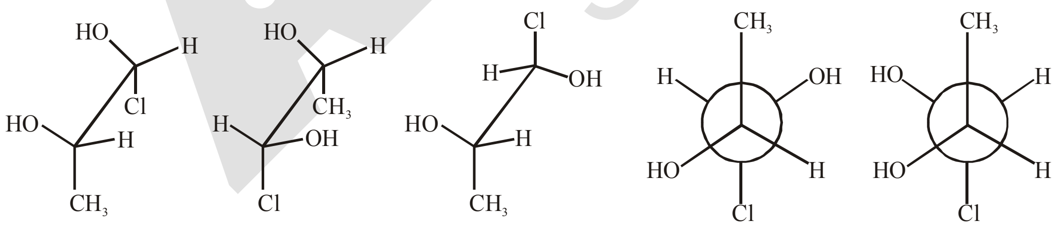
11. In the Newman projection for 2,2–dimethylbutane **[IIT-2010]**



X and **Y** can respectively be –

(A) H and H (B) H and C_2H_5 (C) C_2H_5 and H (D) CH_3 and CH_3

12. Which of the given statement(s) about N,O,P and Q with respect to M is (are) correct ? **[JEE-2012]**



(A) M and N are non-mirror image stereoisomers

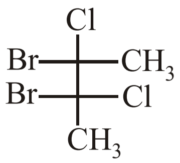
(B) M and O are identical

(C) M and P are enantiomers

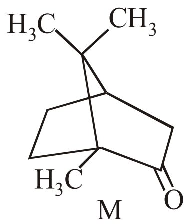
(D) M and Q are identical

Session 2019-20

13. The total number(s) of **<u>stable</u>** conformers with **non-zero** dipole moment for the following compound is (are) **[JEE-2014]**

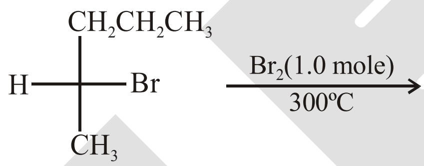


14. The total number of stereoisomers that can exist for M is : **[JEE-2015]**



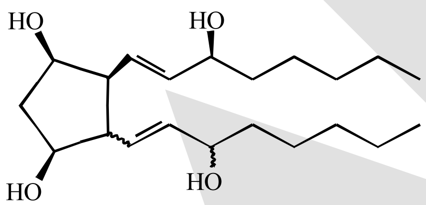
M

15. In the following monobromination reaction, the number of possible chiral products is : **[JEE-2016]**



(1.0 mole)
(enatiomerically pure)

16. For the given compound X, the total number of optically active stereoisomers is____. **[IIT-JEE 2018]**



X

This type of bond indicates that the configuration at the specific carbon and the geometry of the double bond is fixed

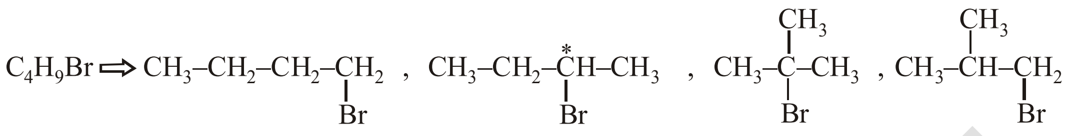
This type of bond indicates that the configuration at the specific carbon and the geometry of the double bond is NOT fixed

ALLEN Session 2019-20

ANSWER-KEY

EXERCISE # I (MAINS ORIENTED)

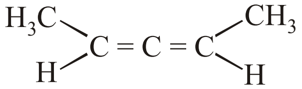
1. Ans. (C) 2. Ans. (C) 3. Ans. (D) 4. Ans. (D)
5. Ans. (A) 6. Ans. (A) 7. Ans. (A) 8. Ans. (D)
9. Ans. (B)



(Optically active)

Optically active isomers $\Rightarrow$ 2

10. Ans. (B) 11. Ans. (D) 12. Ans. (A)
13. Ans. (C)



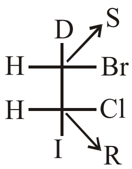
14. Ans. (B) 15. Ans. (C) 16. Ans. (D)
17. Ans. (C)
18. Ans. (A)

Cl–CH=C=C=CH–Cl is a planar structure

19. Ans. (A)
20. Ans. (C)

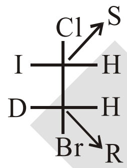
Meso tartaric acid and d-fartaric acid and not mirror image of each other so they are diastereomers.

21. Ans. (A)



(I)

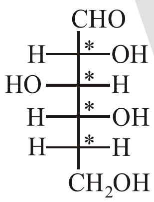
and



(II)

Enantiomers (I & II are mirror image of each other)

22. Ans. (C) 23. Ans. (D) 24. Ans. (B)
25. Ans. (C)

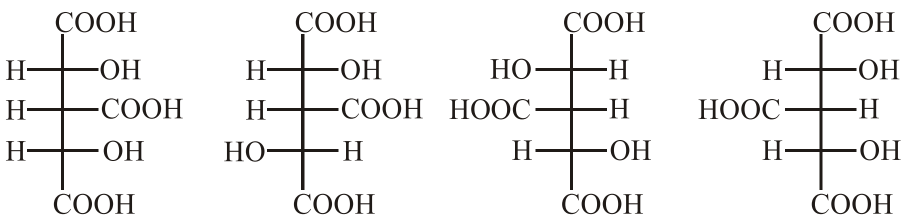


$2^n \Rightarrow 2^4 \Rightarrow 16$

26. Ans. (B) 27. Ans. (B)
28. Ans. (B)

Optically active compounds are resolvable.

29. Ans. (B)



ALLEN Session 2019-20

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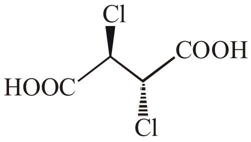
30. Ans. (C)

31. Ans. (A)

POS is present in 

32. Ans. (C)

33. Ans. (B)

e.e = 87.5 – 12.5 $\Rightarrow$ 75 % of

$$ee = \frac{\text{Rotation by mixture}}{\text{Rotation by pure isomer}} \times 100$$

$$75 = \frac{X}{30} \times 100$$

$$X = \frac{75 \times 30}{100} = +22.5^o$$

34. Ans. (B)

ℓ = 10 dm

c = 20 gm/200 ml

α = 30º

$$[\alpha]_{specific} = \frac{\alpha_{obs}}{c.l} = \frac{30}{\frac{20}{200} \times 10} \Rightarrow 30^o$$

$\alpha_{obs.}$ after dilution

$\alpha_{obs} = \alpha_{sp}$ c.p.

$$\alpha_{obs} = 30 \times \frac{20}{1000} \times 10 \Rightarrow 6^o$$

35. Ans. (A)

$$\text{\% optical purity} = \frac{|d - \ell|}{d + \ell} \times 100 = \frac{4}{8} \times 100 = 50\%$$

36. Ans. (C) **37. Ans. (A)** **38. Ans. (D)** **39. Ans. (C)** **40. Ans. (A)**

EXERCISE-II (JEE-ADVANCE ORIENTED LEVEL-I)

Single correct Option Type :

1. Ans. (B)

$C_5H_{10}O$:

(i) $CH_3–CH_2–CH_2–CH_2–\underset{\underset{O}{||}}{C}–H$

(ii) $CH_3–CH_2–\overset{CH_3}{\underset{*}{CH}}–\underset{\underset{O}{||}}{C}–H$ Optical active (Two)

(iii) $CH_3–\underset{CH_3}{CH}–CH_2–\underset{\underset{O}{||}}{C}–H$

(iv) $CH_3–\overset{CH_3}{\underset{CH_3}{C}}–\underset{\underset{O}{||}}{C}–H$

Total $\Rightarrow$ 5 aldehyde

Ketones

(i) $CH_3–\overset{\overset{O}{||}}{C}–CH_2–CH_2–CH_3$

(ii) $CH_3–CH_2–\overset{\overset{O}{||}}{C}–CH_2–CH_3$

(iii) $CH_3–\overset{\overset{O}{||}}{C}–\underset{CH_3}{CH}–CH_3$

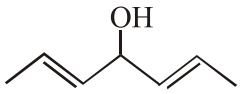
Total $\Rightarrow$ 3 ketones

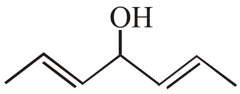
ALLEN Session 2019-20

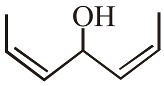
2. **Ans. (C)**

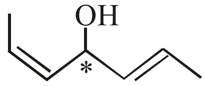
3. **Ans. (B)**

4. **Ans. (A)**

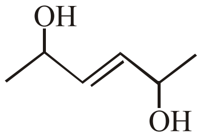


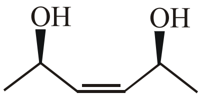
⇒ (i)  (Trans, Trans)

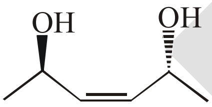
(ii)  (Cis , Cis)

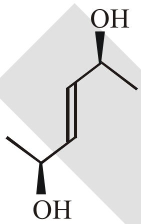
(iii)  (Cis , Trans) Two ⇒ R + S

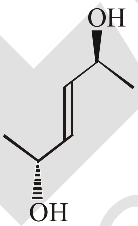
Total is isomers = 4



⇒ (i) 

(ii)  ($\pm$) Two

(iii)  ($\pm$) Two

(iv) 

Total = 6 isomer

5. **Ans. (A,C,D)**

6. **Ans. (A,C)**

Optically active compounds are resolvable and A & C are optically inactive

7. **Ans. (B,C,D)** 8. **Ans. (A,B,C)** 9. **Ans. (A,B,D)**

10. **Ans. (C,D)** 11. **Ans. (A,D)** 12. **Ans. (A,C)**

13. **Ans. (A,B,C,D)** 14. **Ans. (C)** 15. **Ans. (D)**

16. **Ans. (C)** 17. **Ans. (D)**

18. **Ans. (A)→P ; (B)→R ; (C)→Q ; (D)→R**

19 **Ans. (1–R, 2–P, 3–Q, 4–S)**

20. **Ans. (1-S, 2–R, 3–Q, 4–Q)**

ALLEN Session 2019-20

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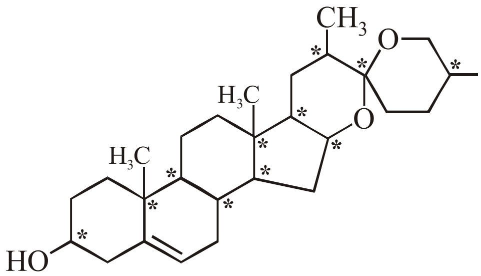
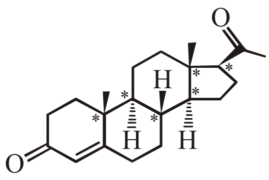
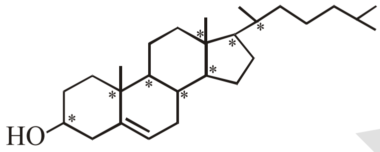
21. Ans. (11)

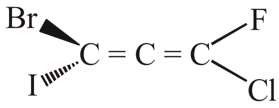


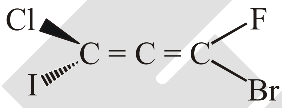
22. Ans. (i) 6, (ii) 8

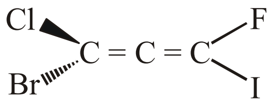
Explanation

(i)  (ii) 

23. Ans. (6)

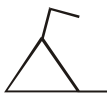
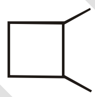
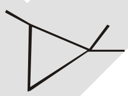
 + Enantiomer

 + Enantiomer

 + Enantiomer

24. Ans. (A,B)

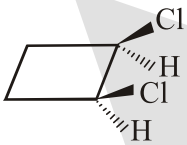
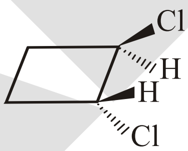
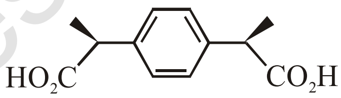
25. Ans. (8)

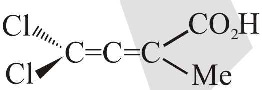
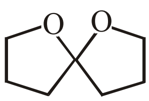
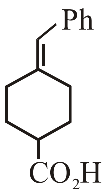
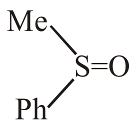
  

4 Optically Active Isomers 2 Optically Active 2 Optically Active

2 + 2 + 4 = 8

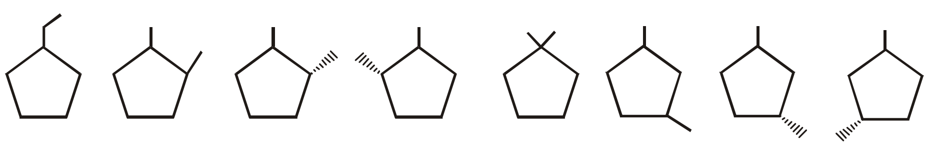
26. Ans. (6)

(I)  (II)  (III) 

(IV)  (V)  (VI)  (VII) 

27. Ans. Achiral : I, III , IV ; chiral : II,V, VI, VII

28. Ans. (8)



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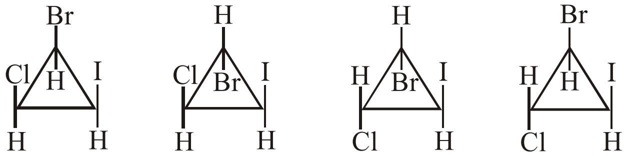
29. Ans. (4)



30. Ans. (8)

$$CH_3-\underset{OH}{\underset{|}{\overset{*}{C}H}}-\underset{OH}{\underset{|}{\overset{*}{C}H}}-\boxed{CH=CH}-CH_3$$

Stereogenic centre = 3

total number of stereoisomer = $2^3 = 8$

31. Ans. (3)

32. Ans.(C)

EXERCISE#III (JEE-ADVANCE ORIENTED LEVEL # II)

Single Correct Type :

1. Ans. (A)
2. Ans. (D)
3. Ans. (8)
4. Ans. (i) (B) ; (ii) (C)
5. Ans. (C)
6. Ans. (C)

Multiple Correct Type :

7. Ans. (A,C)
8. Ans. (A,C,D)
9. Ans. (C)
10. Ans. (C)
11. Ans. (B)
12. Ans. (A,D)

Comprehension Type :

13. Ans. (B)
14. Ans. (C)

15. Ans. (D)

M.W. of MSG = 169

$$C = \frac{169 \text{ gm}}{845 \text{ ml}}$$

$\ell = 200 \text{ mm} = 2 \text{ dm}$

$\alpha_{obs} = +9.6^o$

$$[\alpha]_{sp} = \frac{\alpha_{obs}}{C.\ell.} = \frac{9.6}{\frac{169}{845}\times 2} = -24^o$$

16. Ans. (C)

$$ee = \frac{[\alpha]_{mixture}}{[\alpha]_{pure}}\times 100 = \frac{-20^o}{-24^o}\times 100 = 83.3^o$$

$$\therefore \quad RM = 100 - 83.3 \Rightarrow 16.7\ \% \begin{cases} \rightarrow d = 8.35 \\ \rightarrow l = 8.35 \end{cases}$$

Total (–) MSG = 83.3 + 8.35

= 91.6 %

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17. Ans. (C)

$$C = \frac{33.8 - 16.9\ g}{338 + 169\ ml} = \frac{16.9\ g}{507\ ml}$$

ℓ = 400 mm = 4 dm

$$\alpha_{obs} = [\alpha]_{sp.}\ c.\ell = 24 \times \frac{16.9}{507} \times 4$$

$= + 3.2^{o}$

18. Ans. (A) **19. Ans. (C)**

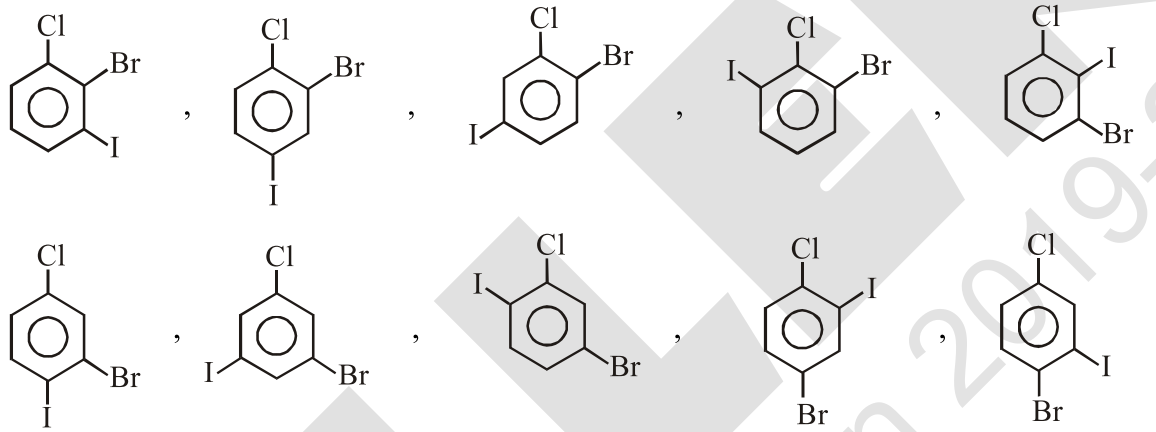
Matrix Match Type :

20. Ans. (A)$\rightarrow$P ; (B)$\rightarrow$Q ; (C)$\rightarrow$R,S ; (D)$\rightarrow$S **21. Ans. (A)-4-iii ; (B)-3-iv ; (C)-2-ii ; (D)-1-i**

Subjective Type :

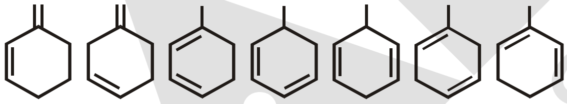
22. Ans. Optical : a, b, c, d, f, g, i, j, k ; Geometrical isomer : c, g, j ; None : e, h

23. Ans. (10)



24. Ans. (a) Enantiomers, (b) Enantiomers, (c) Geometrical isomers & Diastereomers, (d) Positional, (e) Optical, (Diastereomers), (f) Diastereomers (g) Enantiomers, (h) Identical, (i) Geometrical isomers (Diastereomers)

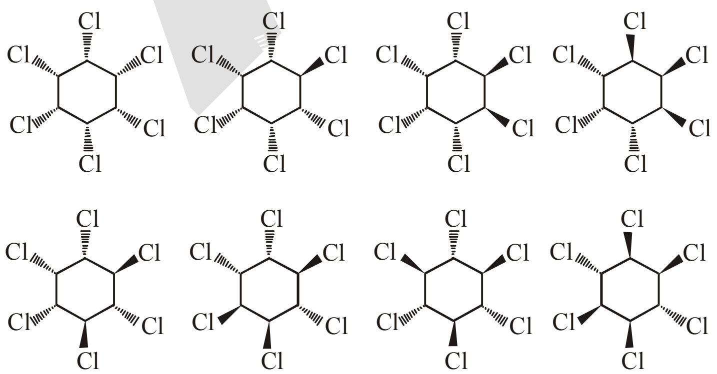
25. Ans. (7)



26. Ans. (3)

27. Ans. (i) 2^4 (ii) 9 (iii) 4 (iv) 4 (v) 3, (vi) 4 (vii) 2^5, (viii) 2^4, (ix) 2, (x) 4, (xi) 3

(ii)



28. Ans. (9)

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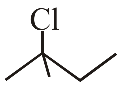
EXERCISE # IV (A) (J-MAINS)

1. **Ans. (4)** **2.** **Ans. (4)** **3.** **Ans. (2)** **4.** **Ans. (2)**

5. **Ans. (3)** **6.** **Ans. (1)** **7.** **Ans.(3)** **8.** **Ans. (3)**

9. **Ans. (2)** **10.** **Ans. (1)** **11.** **Ans. (4)**

12. **Ans. (3)**

Sol. 

It achiral \ optically inactive

13. **Ans. (3)**

EXERCISE # IV (B) (J-ADVANCE OBJECTIVE)

1. **Ans. (B)** **2.** **Ans. (B)** **3.** **Ans. (A)**

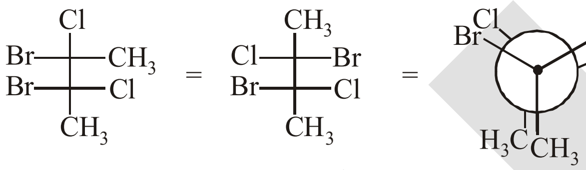
4. **Ans.** Enantiomers - I and III ; Diastereomers - I & II and II & III

5. **Ans. (D)** **6.** **Ans. (B)** **7.** **Ans. (C)**

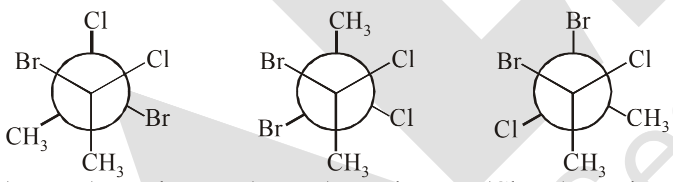
8. **Ans. (B,C,D)** **9.** **Ans. (A,D)** **10.** **Ans. (A,D)**

11. **Ans. (B,D)** **12.** **Ans. (A,B,C)**

13. **Ans. (3)**



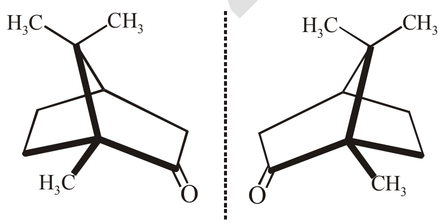
Stable conformer (with $\mu \neq 0$)



(Me-Me) gauche (Br-Me) gauche (Cl-Me) gauche

14. **Ans. (2)**

Sol. M is a organic compound known as camphor. M contains two **rigid** chiral centre so it can exist only in **two** enantiomeric forms.



15. **Ans. (5)** **16.** **Ans.(7)**

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Stereo Isomerism\Eng.p65

SOLID STATE

ALLEN
Session 2019-20

SOLID STATE

1. THE SOLID STATE :

The solid are characterised by incompressibility, rigidity and mechanical strength. The molecules, atoms or ions in solids are closely packed i.e. they are held together by strong forces and can not move about at random. Thus solids have definite volume, shape, slow diffusion, low vapour pressure and possesses the unique property of being rigid.

2. AMORPHOUS AND CRYSTALLINE SOLIDS

Solids can be classified as *crystalline* or *amorphous* on the basis of the nature of order present in the arrangement of their constituent particles. A crystalline solid usually consists of a large number of small crystals, each of them having a definite characteristic geometrical shape. In a crystal, the arrangement of constituent particles (atoms, molecules or ions) is ordered. It has ***long range order*** which means that there is a regular pattern of arrangement of particles which repeats itself periodically over the entire crystal. Sodium chloride and quartz are typical examples of crystalline solids. An amorphous solid (Greek *amorphos* = no form) consists of particles of irregular shape. The arrangement of constituent particles (atoms, molecules or ions) in such a solid has only ***short range order***. In such an arrangement, a regular and periodically repeating pattern is observed over short distances only.

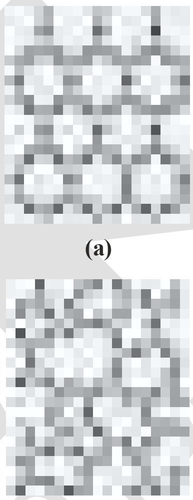

(a)
(b)

Fig. 1.1 : Two dimensional structure of (a) quartz and (b) quartz glass

Such portions are scattered and in between, the arrangement is disordered. The structures of quartz (crystalline) and quartz glass (amorphous) are shown in Fig. 1.1 (a) and (b) respectively. While the two structures are almost identical, yet in the case of amorphous quartz glass there is no *long range order*. The structure of amorphous solids is similar to that of liquids. Glass, rubber and plastics are typical examples of amorphous solids. Due to the differences in the arrangement of the constituent particles, the two types of solids differ in their properties.

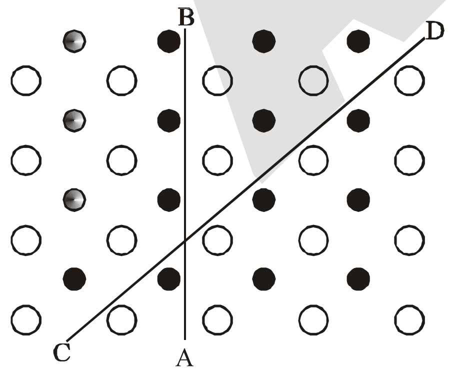


Fig. 1.2 : Anisotropy in crystals is due to different arrangement of particles along different directions.

Crystalline solids have a sharp melting point. On the other hand, amorphous solids soften over a range of temperature and can be moulded and blown into various shapes. On heating they may become crystalline at some temperature. Some glass objects from ancient civilisations are found to become milky in appearance because of some crystallisation. Like liquids, amorphous solids have a tendency to flow, though very slowly. Therefore, sometimes these are called ***pseudo solids*** or ***super cooled liquids***. Glass panes fixed to windows or doors of old buildings are invariably found to be slightly thicker at the bottom than at the top. This is because the glass flows down very slowly and makes the bottom portion slightly thicker.

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Crystalline solids are ***anisotropic*** in nature, that is, some of their physical properties like electrical resistance or refractive index show different values when measured along different directions in the same crystals. This arises from different arrangement of particles in different directions. This is illustrated in Fig. 1.2. Since the arrangement of particles is different along different directions, the value of same physical property is found to be different along each direction. Amorphous solids on the other hand are ***isotropic*** in nature. It is because there is no *long range* order in them and arrangement is irregular along all the directions. Therefore, value of any physical property would be same along any direction. These differences are summarised in Table below :

Distinction between Crystalline and Amorphous Solids

Property	Crystalline Solids	Amorphous Solids
Shape	Definite characteristic geometrical shape	Irregular shape
Melting point	Melt at a sharp and characteristic temperature	Gradually soften over a range of temperature
Cleavage property	When cut with a sharp edged tool, they split into two pieces and the newly generated surfaces are plane and smooth	When cut with a sharp edged tool, they cut into two pieces with irregular surfaces
Heat of fusion	They have a definite and characteristic heat of fusion	They do not have definite heat of fusion
Anisotropy	Anisotropic in nature	Isotropic in nature
Nature	True solids	Pseudo solids or super cooled liquids
Order in arrangement of constituent particles	Long range order	Only short range order.

Ex.1 ***Classify the following as amorphous or crystalline solids :***

(a) Polyurethane (b) Napthalene

(c) Benzoic acid (d) Teflon

(e) Potassium nitrate (f) Cellophane

(g) Polyvinyl chloride (h) Fibre glass

(i) Copper

Sol. Crystalline : (b) , (c) , (e) , (i)

Amorphous : (a) , (d) , (f) , (g), (h)

Note : Polymeric substances are generally amorphous.

Session 2019-20

3. TYPES OF THE CRYSTALLINE SOLID

Types of Solid	Constituent Particles	Bonding/ Attractive forces	Examples	Physical Nature	Electrical Conductivity	Melting Point
(1) Molecular Solids (i) Non polar	Molecules	Dispersion or London forces	Ar, CCl_4, H_2, I_2 CO_2	Soft	Insulator	Very low
(ii) Polar		Dipole-dipole	HCl, SO_2	Soft	Insulator	Low
(iii) Hydrogen bonded		Hydrogen bonding	H_2O (ice)	Hard	Insulator	Low
(2) Ionic Solids	Ions	Coulombic or electrostatic	NaCl, MgO, ZnS, CaF_2	Hard but brittle	Insulator in solid state but conductors in molten state and in aqueous solutions	High
(3) Metallic Solids	Positive ions in a sea of delocalised electrons	Metallic bonding	Fe, Cu, Ag, Mg	Hard but malleable and ductile	Conductors in solid state as well as in molten state	Fairly high
(4) Covalent or network Solids	Atoms	Covalent bonding	SiO_2 (quartz) SiC, C (diomond) AIN,	Hard	Insulators	Very high
			C(graphite)	Soft	Conductor	

Ex.2 ***Classify the following solids in different categories based on the nature of intermolecular forces operating in them :***

(a) Potassium sulphate (K_2SO_4) **(b)** Tin (Sn)

(c) Benzene (C_6H_6) **(d)** Urea (NH_2CONH_2)

(e) Ammonia (NH_3) **(f)** Water (H_2O)

(g) Zinc sulphide (ZnS) **(h)** Graphite (C)

(i) Rubidium (Rb) **(j)** Argon (Ar)

(k) Silicon carbide (SiC) **(*l*)** Bronze

Sol. Ionic solids : (a), (g)

Metallic solids : (b), (i), (*l*)

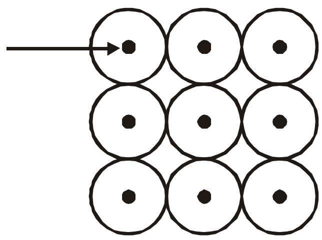
Molecular solids : (c), (d), (e), (f), (j)

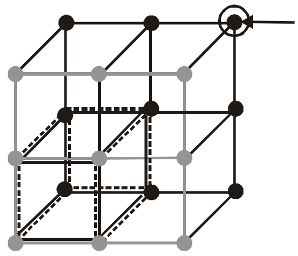
Covalent network solids.: (h), (k)

Session 2019-20

4.0 SOME BASIC DEFINITION :

4.1 SPACE LATTICE (CRYSTAL LATTICE :





The main characteristic of crystalline solids is a regular and repeating pattern of constituent particles. If the three dimensional arrangement of constituent particles in a crystal is represented diagrammatically, in which each particle is depicted as a point, the arrangement is called crystal lattice. Thus, ***a regular three dimensional arrangement of points in space is called a crystal lattice***. A portion of a crystal lattice is shown in Fig. The following are the characteristics of a crystal lattice:

(a) Each point in a lattice is called ***lattice point*** or ***lattice site***.

(b) Each point in a crystal lattice represents one constituent particle which may be an atom, a molecule (group of atoms) or an ion.

(c) Lattice points are joined by straight lines to bring out the geometry of the lattice.

4.2. UNIT CELL :

Unit cell is the smallest portion of a crystal lattice which , when repeated in different directions, generates the entire lattice.

A unit cell is characterized by the edge lengths a, b and c along the three edges of the unit cell and the angles α, β and γ between the pair of edges : bc, ca and ab, respectively.

TYPE OF UNIT CELLS -

4.2.1 Primitive and Centred Unit cells

Unit cells can be broadly divided into two categories, primitive and centred unit cells.

(a) Primitive Unit Cells (P)

When constituent particles are present only on the corner positions of a unit cell, it is called as **primitive unit cell.**

(b) Centred Unit Cells

When a unit cell contains one or more constituent particles present at positions other than corners in addition to those at corners, it is called a centred unit cell. **Centred unit cells** are of three types:

(i) **Body-Centred Unit Cells (I) :** Such a unit cell contains one constituent particle (atom, molecule or ion) at its body-centre besides the ones that are at its corners.

(ii) **Face-Centred Unit Cells (F) :** Such a unit cell contains one constituent particle present at the centre of each face, besides the ones that are at its corners.

(iii) **End-Centred Unit Cells (E) :** In such a unit cell, one constituent particle is present at the centre of any two opposite faces besides the ones present at its corners.

ALLEN Session 2019-20
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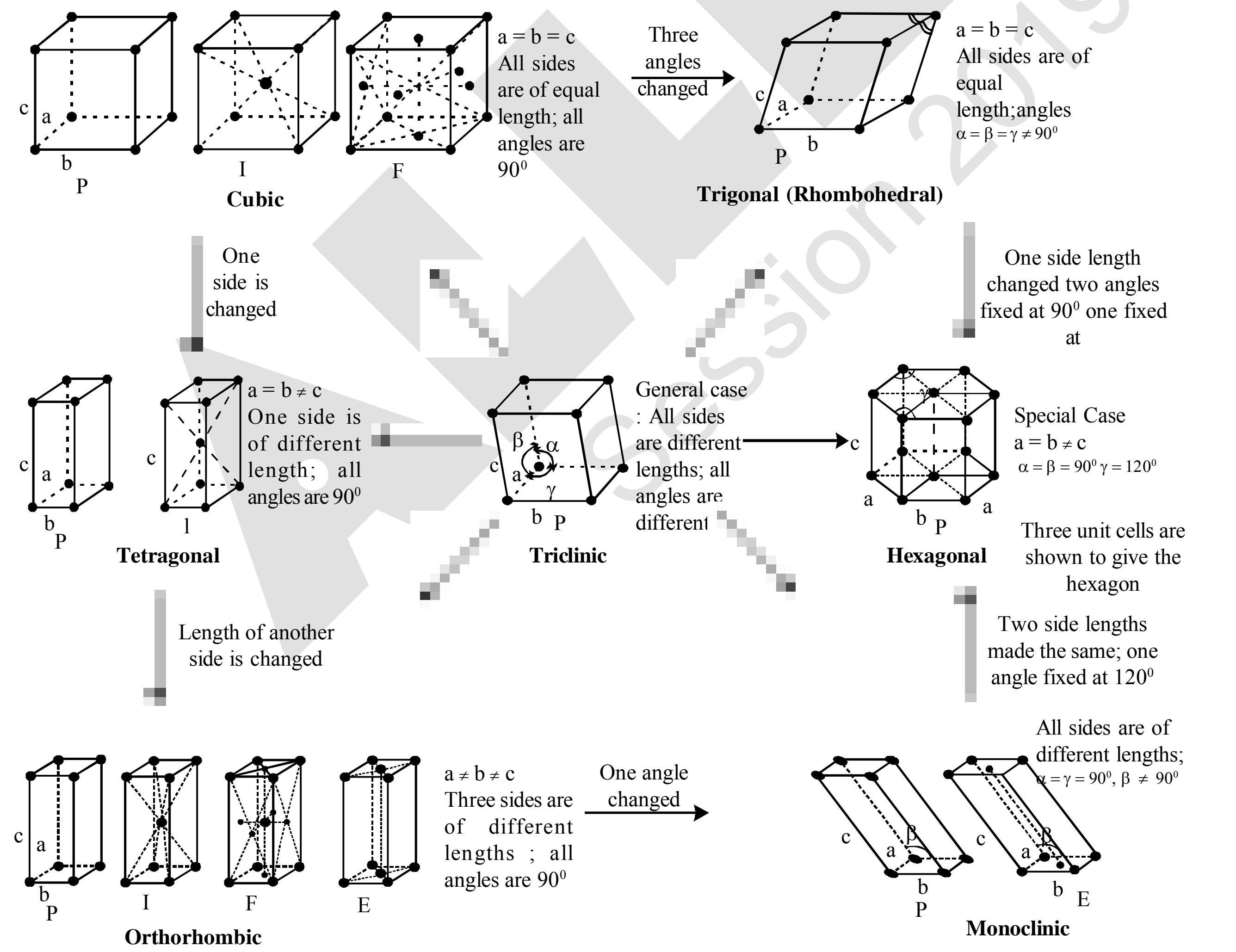
4.3. THE SEVEN CRYSTAL SYSTEMS

On the basis of the classification of symmetry, the lattice have been divided into seven systems. These can be grouped into 7 crystal systems. These seven systems with the characteristics of their axes (angles and intercepts) along with some examples of each are given in the following table :

Seven Primitive Unit cells and their Possible Variations as Centred Unit Cells

Crystal system	Possible Variations	Axial distance or edge lengths	Axial angles	Examples
Cubic	Primitive, body-centred, Face centre	a = b = c	$\alpha = \beta = \gamma = 90°$	NaCl, Zinc blende, Cu
Tetragonal	Primitive, Body-centred	$a = b \neq c$	$\alpha = \beta = \gamma = 90°$	White tin, SnO_2, TiO_2, $CaSO_4$
Orthorhombic	Primitive, Body-centred, Face-centred, End-centred	$a \neq b \neq c$	$\alpha = \beta = \gamma = 90°$	Rhombic sulphur, KNO_3, $BaSO_4$
Hexagonal	Primitive	$a = b \neq c$	$\alpha = \beta = 90°$, $\gamma = 120°$	Graphite, ZnO, CdS,
Rhombohedral or Trigonal	Primitive	a = b = c	$\alpha = \beta = \gamma \neq 90°$	Calcite ($CaCO_3$), HgS (Cinnabar)
Monoclinic	Primitive, End-centred	$a \neq b \neq c$	$\alpha = \gamma = 90°$, $\beta \neq 90°$	Monoclinic sulphur, $Na_2SO_4 \cdot 10H_2O$
Triclinic	Primitive	$a \neq b \neq c$	$\alpha \neq \beta \neq \gamma \neq 90°$	$K_2Cr_2O_7$, $CuSO_4.5H_2O$, H_3BO_3

4.4. BRAVAIS LATTICES : There are 14 Bravais lattices :



Session 2019-20
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4.5. CO-ORDINATION NUMBER :

The number of nearest particles around a specific particle in a given crystalline substance is called ***co-ordination*** number.

4.6. PACKING EFFICIENCY OR PACKING DENSITY (P.E.) :

Packing efficiency is defined as the ratio of volume occupied by the constituent particles to the total volume of the crystalline substance.

$$\text{P.E.} = \frac{\text{Volume occupied by particles present in a crystal}}{\text{Volume of crystal}}$$

$$\text{P.E.} = \frac{\text{Volume occupied by particles present in unit cell}}{\text{Volume of Unit Cell}}$$

$\text{P.E.} = \dfrac{Z \times (4/3)\pi r^3}{V}$, where Z = number of atoms present in unit cell

4.7. *DENSITY OF THE CRYSTAL :*

Density of crystal = Density of an unit cell = $\dfrac{\text{Mass of unit cell}}{\text{Volume of unit cell}}$

Mass of the unit cell = Number of particles present in a unit cell × Mass of one particles = Z × m

But mass of one particles (m) = $\dfrac{\text{Particle mass}}{\text{Avogadro Number}} = \dfrac{M}{N_A}$

Mass of an unit cell = $Z \times \dfrac{M}{N_A}$

Density of an unit cell = $\dfrac{Z \times \dfrac{M}{N_A}}{V}$

∴ **Density of Crystal, d = Density of an unit cell = $\dfrac{Z \times M}{V \times N_A}$ g cm^{-3}**

5.0 ANALYSIS OF CUBIC CRYSTAL :

5.1. GEOMETRY OF A CUBE

Number of corners = 8

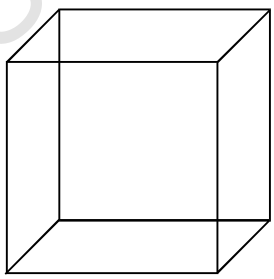
Number of faces = 6

Number of edges = 12

Number of cube centre = 1

Number of cube diagonals = 4

Number of face diagonals = 12

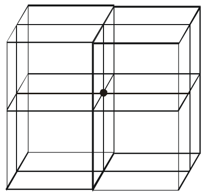


ALLEN Session 2019-20

5.2 CONTRIBUTION OF A CONSTITUENT PARTICLE AT DIFFERENT SITES OF CUBE :

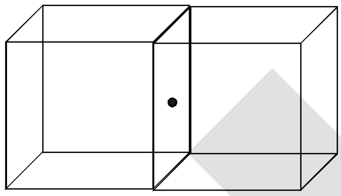
5.2.1 A corner of a cube is common in 8 cubes.

So — th part of a particle is present at this corner of cube.



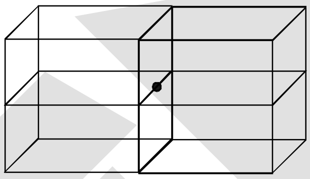
5.2.2 A face of a cube is common is 2 cubes.

So —th part of a particle is present at the face of a cube.



5.2.3 An edge of a cube is common in four cubes,

so —th part of particle is present at the edge of a cube

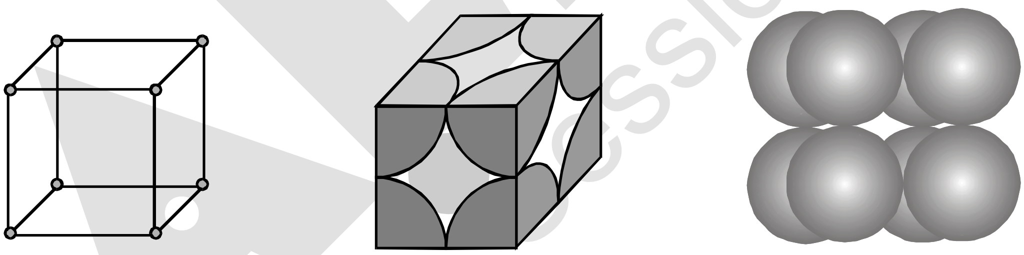


5.3 TYPE OF CUBIC UNIT CELL :

5.3.1 Simple/Primitive/Basic Unit cell (Simple cubic, SC) ;

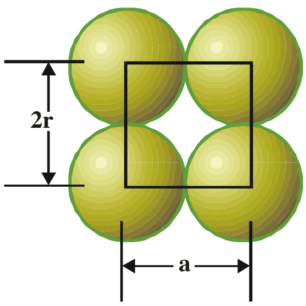
A unit cell having lattice point only at corners called as primitive or simple unit cell. In this case there is one particle at each of the eight corners of the unit cell.

Considering a particles at one corner as the centre, it will be found that this atom is surrounded by six equidistant neighbours (particles) and thus the co-ordination number will be six. If 'a' is the side of the unit cell, then the distance between the nearest neighbours shall be equal to 'a'.



(a) Relationship between Edge length 'a' and Particle radius 'r' :-

i.e. , $\boxed{\quad -\quad}$



(One face of SC)

ALLEN Session 2019-20

(b) Number of particles present in unit cell (Z) : In this case one particle lies at each corner.

Hence simple cubic unit cell contains a total of $\frac{1}{8}\times 8 = 1$ particle /unit cell.

(c) Packing efficiency (P. E.) :

$$\text{P.E.} = \frac{\text{Volume occupied by particles present in unit cell}}{\text{Volume of unit cell}} = \frac{Z\times\frac{4}{3}\pi r^3}{V} \qquad \left[\text{Volume of atom} = \frac{4}{3}\pi r^3\right]$$

$$\text{P.E.} = \frac{1\times\frac{4}{3}\times\pi\times\left(\frac{a}{2}\right)^3}{a^3} = \frac{\pi}{6} = 0.52 \quad \text{or} \quad 52\% \qquad \left[r = \frac{a}{2} \text{ and } V = a^3, Z = 1\right]$$

In SC, 52% of total volume is occupied by particles.

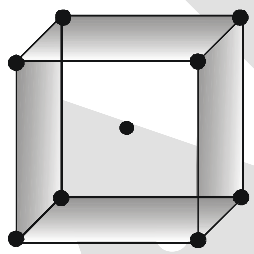
$\therefore$ Void space $\approx$ (100 – 52)= 48 %

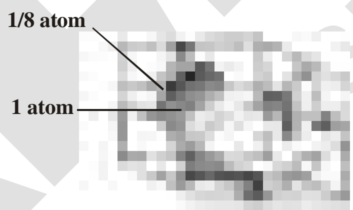
(d) Coordination number (CN)

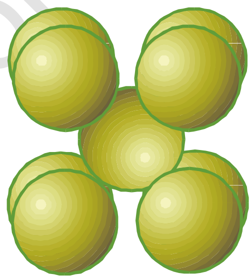
Nearest neighbour	Distance	Number
1	a	6
2	$\sqrt{2}\,a$	12
3	$\sqrt{3}\,a$	8

5.3.2. Body Centred Cubic unit cell (BCC):

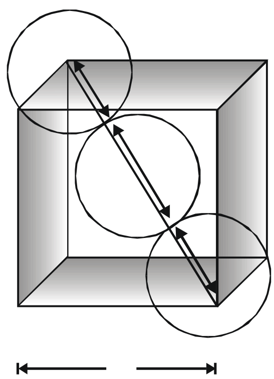
A unit cell having lattice point at the body centre in addition to the lattice points at every corner is called as body centered unit cell. Here the central particle is surrounded by eight equidistant particles and hence the co-ordination number is eight. The nearest distance between two particles will be $\frac{a\sqrt{3}}{2}$







(a) Relationship between edge length 'a' and particle radius 'r' :



In BCC, along cube diagonal all particles touches each other and the length of cube diagonal is $\sqrt{3}a$.

So, $\sqrt{3}a = 4r$

i.e. $r = \frac{\sqrt{3}a}{4}$

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(b) Number of particle present in unit cell (Z):

$$Z = \underbrace{\left(\frac{1}{8}\times 8\right)}_{\text{(Corner)}} + \underbrace{(1 \times 1)}_{\text{(Body centre)}} = 1 + 1 = 2 \text{ particles/unit cell.}$$

In this case one particle lies at the each corner of the cube. Thus contribution of the 8 corners is $\left(\frac{1}{8}\times 8\right) = 1$, while that of the body centred is 1 in the unit cell. Hence total number of particles per unit cell is 1 + 1 = 2

(c) Packing efficiency :

$$\text{P.E.} = \frac{Z\times\frac{4}{3}\pi r^3}{V} = \frac{2\times\frac{4}{3}\times\pi\left(\frac{\sqrt{3}a}{4}\right)^3}{a^3} = \frac{\sqrt{3}\pi}{8} = 0.68 \text{ or } 68\% \qquad \left[\, Z = 2,\ r = \frac{\sqrt{3}a}{4},\ V = a^3\right]$$

In BCC, 68% of total volume is occupied by particles.

$\therefore$ Void space = 100 – 68 = 32 %

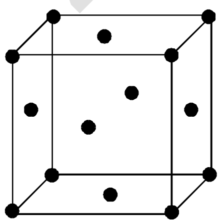
(d) Coordination number (CN)

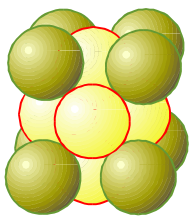
Nearest neighbour	Distance	Number
1	$\sqrt{3}\frac{a}{2}$	8
2	a	6
3	$\sqrt{2}\,a$	12

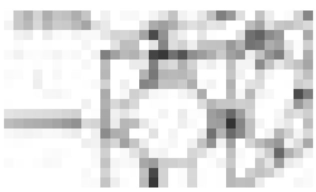
5.3.3 Face Centred Cubic unit cell (FCC):

A unit cell having lattice point at every face centre in addition to the lattice point at every corner called as face centred unit cell.

In this case there are eight particles at the eight corners of the unit cell and six particles at the centre of six faces. Considering a particle at the face centre as origin, it will be found that this face is common to two cubes and there are twelve points surrounding it situated at a distance which is equal to half the face diagonal of the unit cell. Thus the co-ordination number will be twelve and the distance between the two nearest particles will be $\frac{}{\sqrt{}}$.







ALLEN Session 2019-20

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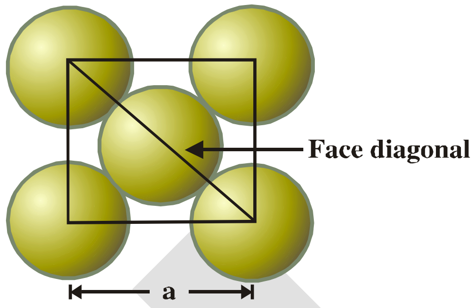
(a) Relationship between edge length 'a' and atomic radius 'r' :

In FCC, along the face diagonal all atoms touches each other and the length of face diagonal is $\sqrt{2}a$.

So $\quad 4r = \sqrt{2}a$

i.e. $\quad r = \dfrac{a}{2\sqrt{2}}$



(b) Number of particles per unit cell : (Z)

$$Z = \underset{\text{Corner}}{\left(\frac{1}{8}\times 8\right)} + \underset{\text{faces}}{\left(6\times\frac{1}{2}\right)} = 1 + 3 = 4 \text{ particles/unit cell}$$

In this case, one particle lies at the each corner of the cube and one particle lies at the centre of each face of the cube. It may noted that only 1/2 of each face sphere lie within the unit cell and there are six such faces. The total contribution of 8 corners is $\left(\frac{1}{8}\times 8\right) = 1$, while that of 6 face centred particles is $\left(\frac{1}{2}\times 6\right) = 3$ in the unit cell.

Hence, total number of particles per unit cell is 1 + 3 = 4

(c) Packing efficiency :

$$\text{P.E.} = \frac{Z\times\frac{4}{3}\pi r^3}{V} \qquad \left[Z = 4,\ r = \frac{a}{2\sqrt{2}},\ V = a^3\right]$$

$$= \frac{4\times\frac{4}{3}\pi\times\left(\frac{a}{2\sqrt{2}}\right)^3}{a^3} = \frac{\pi}{3\sqrt{2}} = 0.74 \text{ or } 74\%$$

In FCC, 74% of total volume is occupied by particles. This is maximum for crystals having identical particles.

$\therefore$ Void space = 100 – 74 = 26 %

(d) Coordination number (CN)

Nearest neighbour	Distance	Number
1	$\frac{a}{\sqrt{2}}$	12
2	a	6
3	$\sqrt{\frac{3}{2}}a$	24

5.4 SUMMARY OF CUBIC CRYSTAL :

Unit cell	No. of particles per unit cell (Z)	2r =	CN	Volume occupied by particles (%)
Simple cube (SC)	1	a	6	52
Body centred cube (BCC)	2	$\frac{a\sqrt{3}}{2}$	8	68
Face centred cube (FCC)	4	$\frac{a}{\sqrt{2}}$	12	74

Ex.3. *An element (atomic mass = 60) having face centred cubic crystal has a density of 6.23 g cm^{-3}.* What is the edge length of the unit cell ? (Avogadro constant, N_A = *6.02 × 10^{23} mol^{-1}*)

Sol. *Since element has fcc structure hence there are 4 atoms in a unit cell (Z = 4), Atomic mass is 60 (M = 60), N_A = 6.02 × 10^{23} and d = 6.23 g cm^{-3}.*

$$\because \quad \frac{\times}{\times}$$

or $$\frac{\times}{\times} = \frac{(\qquad)}{(\qquad)(\quad\times\quad^{-})}$$

$$= 64 \times 10^{-24}\ cm^3$$

Let ℓ be the length of the edge of the unit cell.

$\therefore \quad \ell^3 = V = 64 \times 10^{-24}\ cm^3 \qquad or \qquad \ell = 4.0 \times 10^{-8}\ cm$

Ex.4. The density of Al is 5.4 g/cm^3. If it crystallise in fcc lattice, determine its atomic radius.

Sol. $d = \frac{ZM}{N_A.V} \Rightarrow 5.4 = \frac{4\times 27}{N_A \times (2\sqrt{2}r)^3}$

$\therefore$ r = 1.136 × 10^{-8} cm = 1.136 Å

Ex.5. A solid crystallises in cubic crystal in which 'X' atoms occupy all the corners and body centres and 'Y' atoms occupy all the face-centres. What is the simplest formula of solid?

Sol. $Z_X = 8\times\frac{1}{8}+1 = 2$

$Z_Y = 6\times\frac{1}{2} = 3$

$\therefore$ *Simplest formula of solid = X_2Y_3*

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ALLEN Session 2019-20

6. CLOSE PACKING OF IDENTICAL SOLID SPHERES

The solids which have non-directional bonding, their structures are determined on the basis of geometrical consideration. For such solids, it is found that the lowest energy structure is that in which each particle is surrounded by the greatest possible number of neighbours. In order to understand the structure of such solids, let us consider the particles as hard sphere of equal size in three directions. Although there are many ways to arrange the hard spheres but the one in which maximum available space is occupied will be economical which is known as **closed packing**.

To clearly understand the packing of these spheres, the packing can be categorised as :

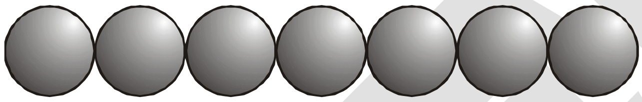
(i) Close packing in one dimension.

(ii) Close packing in two dimension.

(iii) Close packing in three dimension.

6.1 CLOSE PACKING IN ONE DIMENSION :

In one dimension , only one arrangement of spheres is possible as shown in fig.



Each sphere is touched by other two spheres, hence coordination numbers of packing is two.

6.2 CLOSE PACKING IN TWO DIMENSION :

Two possible types of two dimensional packing are

(i) Square close packing in two dimension.

(ii) Hexagonal close packing in two dimension.

(i) Square close packing in two dimension :

When two rows are placed in such a manner, that spheres of one row are placed immediately below of the other, the resulting packing is called two dimensional square close packing.

(i) Since all the rows are identical, the packing is called AAA .... type packing.

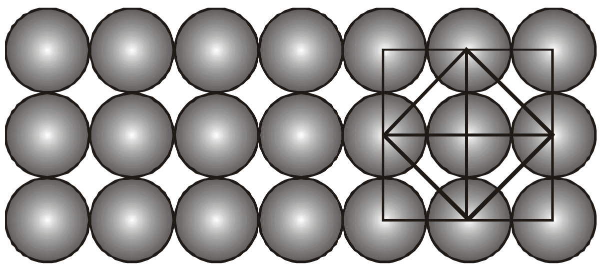
(ii) Each sphere is touched by four other, hence coordination number is four.

(iii) If centres of spheres are connected, square cells are formed , hence it also called two dimensional square packing.

(iv) This type of packing is not very effective in terms of utilisation of space.

(v) Packing efficiency in 2-D $= \frac{\times \pi}{} = \frac{\times \pi}{} = \frac{\pi}{} =$ or 78%

(vi) Packing efficiency in 3-D $= \frac{\times - \pi \left(- \right)}{} =$ or 52% [In 3-D, its unit cell is simple cubic]



Session 2019-20

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(ii) Hexagonal close packing in two dimension :

If various one dimensional close pack rows are placed in such a way that spheres of top row fits in depression of bottom row spheres, the resulting packing is called two dimensional hexagonal close packing.

(i) Every third row sphere comes exactly at top of first row sphere, hence the packing is called ABABAB ...... packing.

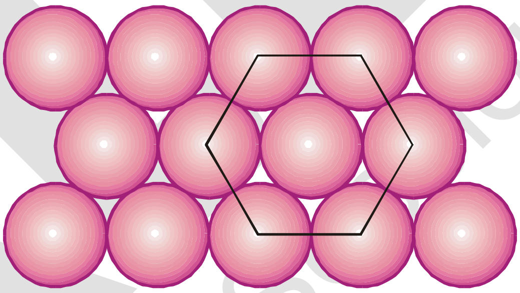
(ii) If centres are joined, hexagonal unit cells are formed. Hence this is called two dimensional hexagonal close packing.

(iii) This packing is most efficient in utilising space in two dimensional arrangement.

(iv) Each sphere is touched by six other, hence coordination number is six.

(v) Packing efficiency in 2-D = $\dfrac{\times \pi \left(- \right)}{\dfrac{\sqrt{\ }}{\ } \times} = \dfrac{\pi}{\sqrt{\ }} =$ or 90%

(vi) Packing efficiency in 3-D = $\dfrac{\times - \pi \left(- \right)}{\dfrac{\sqrt{\ }}{\ } \times \ \times} = \dfrac{\pi}{\sqrt{\ }} =$ or 60%



6.3 CLOSE PACKING IN THREE DIMENSIONS :

When two dimensional packing structure are arranged one above the other, depending upon type of two dimensional arrangement in a layer, and the relative positions of spheres in above or below layer, various types of three dimensional packing results. To define 3-D lattice, six lattice parameters are required - 3 edge lengths & 3 angles.

(i) Simple cubic packing (A A A A ......)

(ii) Hexagonal close packing (AB AB AB ......)

(iii) Cubic close packing or face centred cubic (ABC ABC...)

Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Solid State\Eng\01_Theory

6.3.1 Three dimensional close packing from square two dimensional packing (Simple cubic packing in three dimension)

The two dimensional square close packed layer are placed, in such a manner that spheres in each layer comes immediately on top of below layer, simple cubic packing results. Important points :

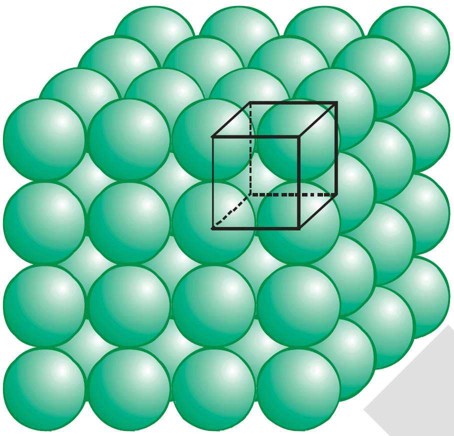
(i) Spheres all aligned vertically and horizontally in all directions.

(ii) The unit cell for this packing is simple cubic unit cell.

(iii) In this packing, only 52% of available space is occupied by spheres.

(iv) Each sphere is in contact will six spheres and hence coordination number is 6.

(v) Packing efficiency = $\dfrac{1\times\frac{4}{3}\pi r^3}{(2r)^3} = \dfrac{\pi}{6} \approx 0.52$



Simple cubic lattice formed by A A A ... arrangement

Coordination number = 6

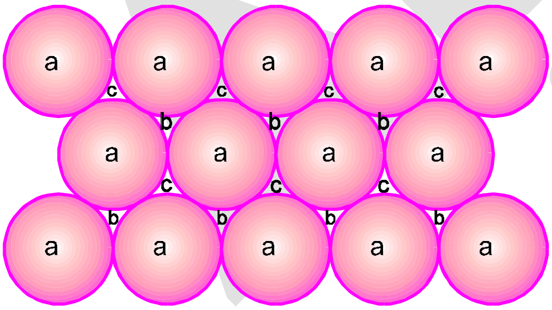
First neighbour = 6 at a distance = a

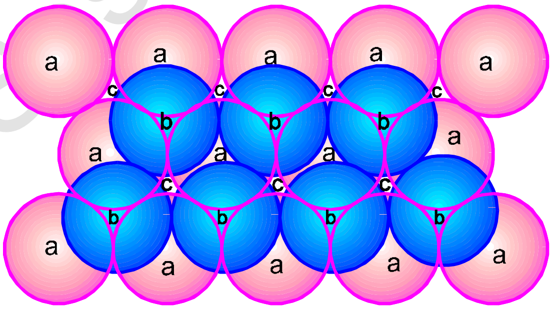
Second neighbour = 12 at ($\sqrt{2}$a) distance

Third neighbour = 8 at ($\sqrt{3}$a) distance

6.3.2 Three dimensional close packing from hexagonal two dimensional packing :

In hexagonal close packing, there are two types of the voids (open space or space left) which are divided into two sets 'b' and 'c' for convenience. The spaces marked 'c' are curved triangular spaces with tips pointing upwards whereas spaces marked 'b' are curved triangular spaces with tips pointing downwards.





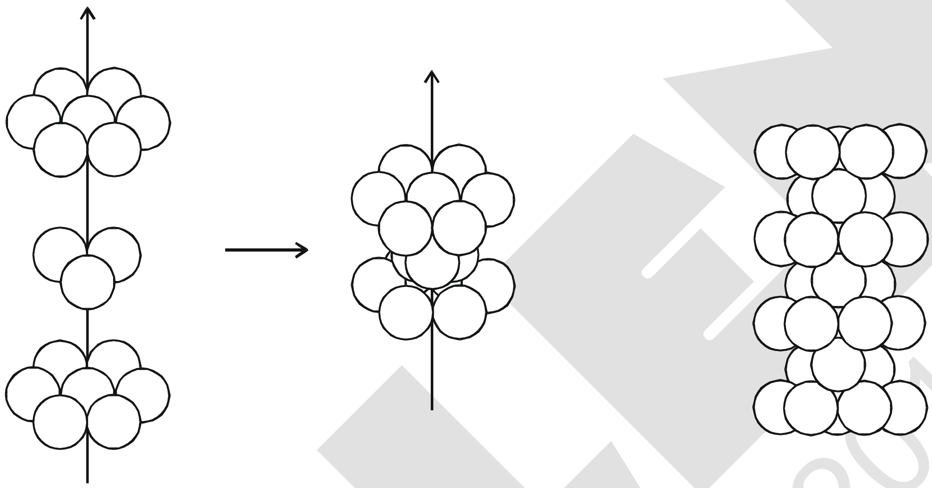
Now we extend the arrangement of spheres in three dimensions by placing second close packed layer (hexagonal close packing) (B) on the first layer (A). The spheres of second layer may placed either on space denoted by 'b' or 'c'. It may be noted that it is not possible to place spheres on both types of voids (i.e. b and c). Thus half of the voids remain unoccupied by the second layer. The second layer also have voids of the types 'b' and in order to build up the third layer, there are following ways :

(I) Hexagonal close packing (HCP)

(i) In one way, the spheres of the third layer lie on the spaces of second layer (B) in such a way that they lie directly above those in the first layer(A). In other words we can say that the third layer becomes indentical to the first layer. Such arrangement is called AB AB AB ....type packing or hexagonal close packing (hcp).

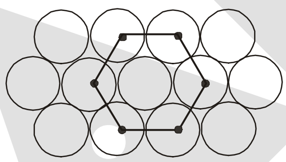
(ii) Maximum possible space is occupied by spheres.

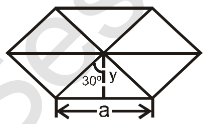
(iii) Each sphere is touched by 12 other spheres in 3D (6 is one layer, 3 in top layer and 3 in bottom) and hence the coordination number is 12.



(iv) Packing efficiency of HCP units
Relation between a, b, c and R :

a = b = 2R

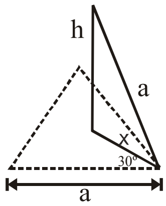




$\tan 30^\circ = \frac{\quad}{\times}$ So $y = \frac{\times\sqrt{\quad}}{\times} = \frac{\sqrt{\quad}}{\quad}\,a.$

Base Area $= 6\left[\frac{\quad}{\quad} \times \frac{\sqrt{\quad}}{\quad}\right] = \frac{\sqrt{\quad}}{\quad}$

Calculation of c :



$\cos 30^\circ = \frac{\quad}{\times}$ $x = \frac{\quad}{\times\sqrt{\quad}} = \frac{\quad}{\sqrt{\quad}}$

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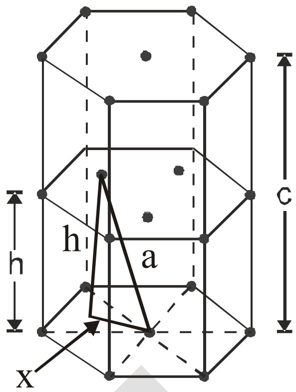


Applying pythogoras theorem : $x^2 + h^2 = a^2$

So $h^2 = a^2 - x^2 = a^2 - \frac{\ }{\ } = \frac{\ }{\ } a^2$

$h = \sqrt{\frac{\ }{\ }}\, a$ so $c = 2h = 2\sqrt{\frac{\ }{\ }}\, a$

So volume of hexagon = area of base × height

$= \frac{\sqrt{\ }}{\ } \times a^2 \times 2\sqrt{\frac{\ }{\ }}\, a = \frac{\sqrt{\ }}{\ } \times (2R)^2 + 2\sqrt{\frac{\ }{\ }} \times (2R) = 24\sqrt{\ }\, R^3$

(v) Effective no. of particles (Z)

$Z = 3 + 2 \times \frac{\ }{\ } + 12 \times \frac{\ }{\ } = 3 + 1 + 2 = 6.$

It must be noted that all three spheres of 'B' layer are not exactly inside the unit cell. But the contribution of three spheres are taken because the same volume of other spheres in that layer is also inside the unit cell.

(vi) Packing efficiency $= \frac{\times \frac{\ }{\ } \pi}{\sqrt{\ }} = \frac{\pi}{\sqrt{\ }} = 0.74$ or 74%.

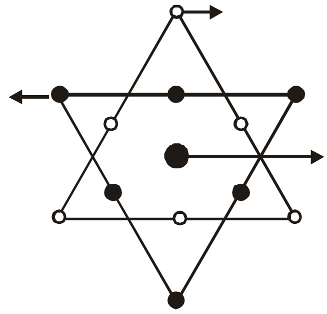
(vii) Density (d) $= \frac{\ }{\ } = \left[\frac{\times}{\times}\right]$

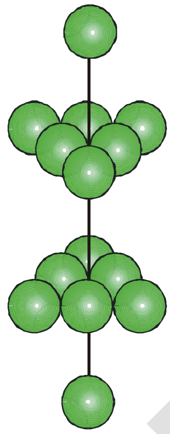
(II) Cubic close packing (CCP) or face centered cubic (FCC)

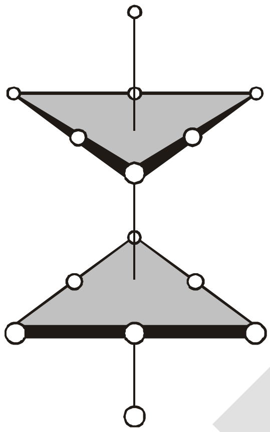
In second way, the spheres of the third layer (C) lie on the second layer (B) in such a way that they lie over the unoccupied spaces 'C' of the first layer(A). If this arrangement is continued in the same order every fourth layer becomes identical to the first. Such arrangement of particle is called ABC ABC ABC.... or cubic close packing (ccp) or face centered cubic (fcc).

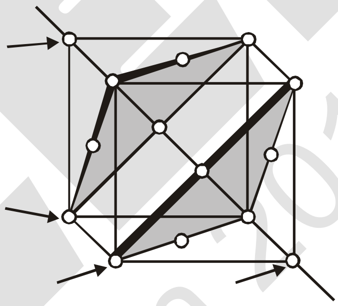
It may be noted that in ccp (or fcc) structures, each sphere is surrounded by 12 spheres hence the coordination number of each sphere is 12. The spheres occupy 74% of the total volume and 26% of is the empty space in both (hcp and ccp) structures.

ALLEN Session 2019-20
node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Solid State\Eng\01_Theory









ALLEN Session 2019-20

(i) Relation between 'a' and 'R' :

$a \neq 2R$

$\sqrt{\ } \ =$ (sphere are touching along the face diagonal)

(ii) Effective no. of particles per unit cell (Z)

$Z = -\times \ + -\times \ =$

(iii) Packing fraction :

$$P.F. = \frac{\times - \pi}{\times \ \times} \times \sqrt{\ } \times \ = \frac{\pi}{\sqrt{\ }} = 0.74 \text{ or } 74\%$$

(iv) Density (d) $= \dfrac{\times}{\times}$

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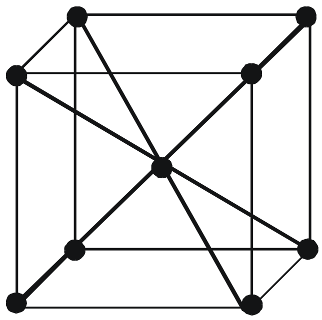
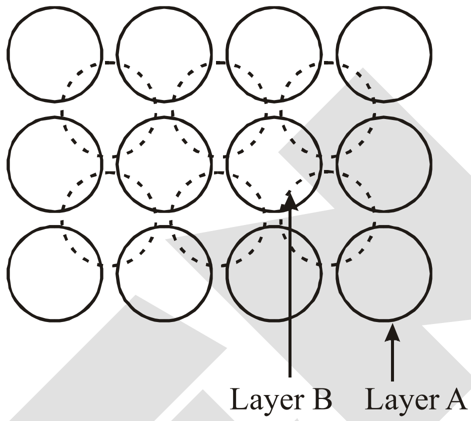
6.3.3 Body centred cubic (bcc):

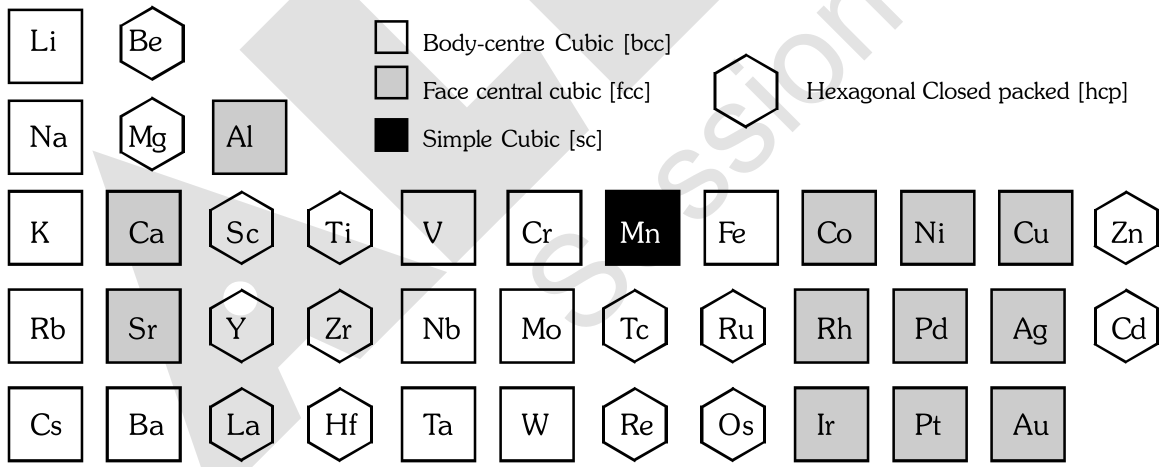
There is another possible arrangement of packing of spheres known as body centred cubic (bcc) arrangement. This arrangement is observed in square close packing in which there is suitable space between the spheres in each layer. In bcc arrangement, the spheres of the second layer lie at the space (hollows or voids) in the first layer.





Thus each sphere of the second layer touches four spheres of the first layer. Now spheres of the third layer are placed exactly about the spheres of first layer. In this way each sphere of the second layer touches eight spheres (four of 1st layer and four of IIIrd layer). Therefore coordination number of each sphere is 8 in bcc sturcture. The spheres occupy 68% of the total volume 32% of the volume is the empty space.

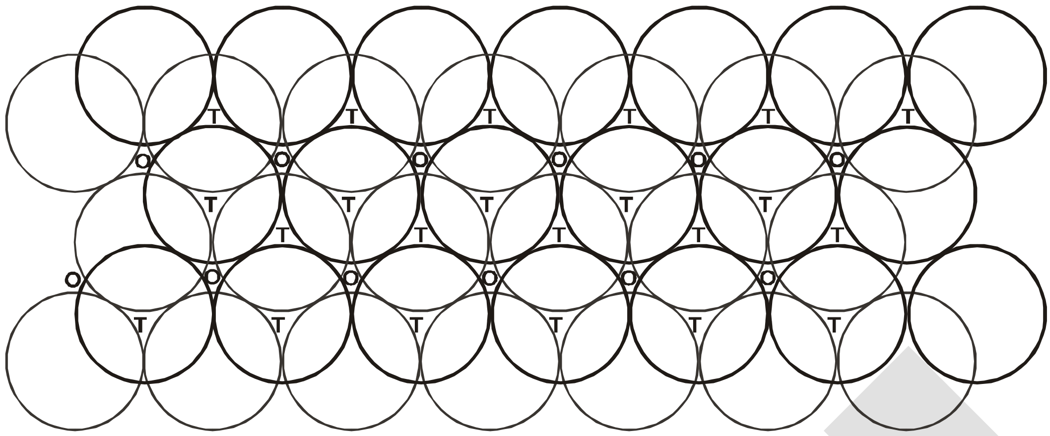
7. STRUCTURES OF VARIOUS ELEMENTS



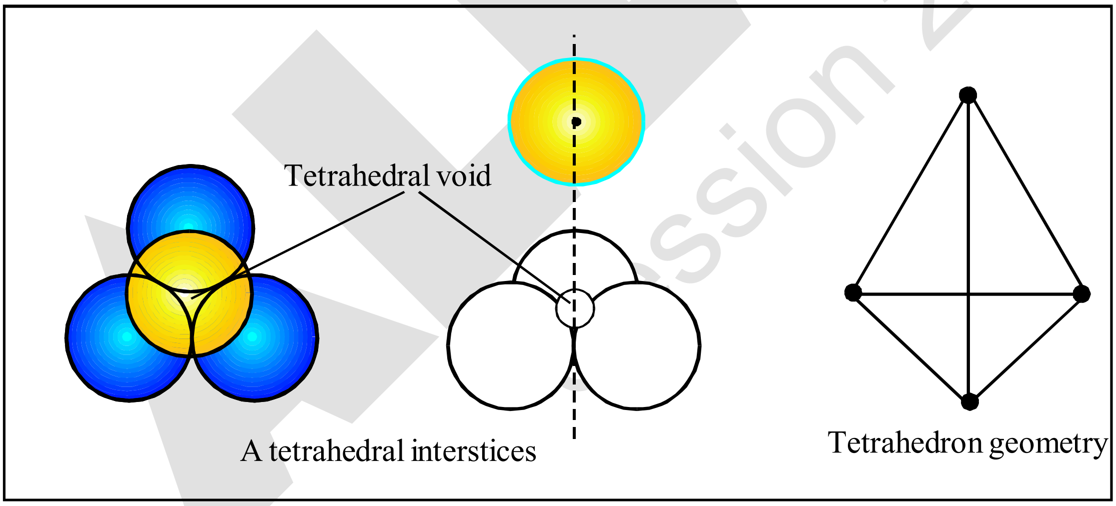
8. INTERSTICES OR VOIDS OR HOLES IN CRYSTALS

It has been shown that the particles are closely packed in the crystals even than there is some empty space left in between the spheres. This is known as interstices (or interstitial site or hole or empty space or voids). In three dimentional close packing (CCP & HCP) the interstices are of two types : (i) tetrahedral voids and (ii) octahedral voids.

Session 2019-20
node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Solid State\Eng\01_Theory



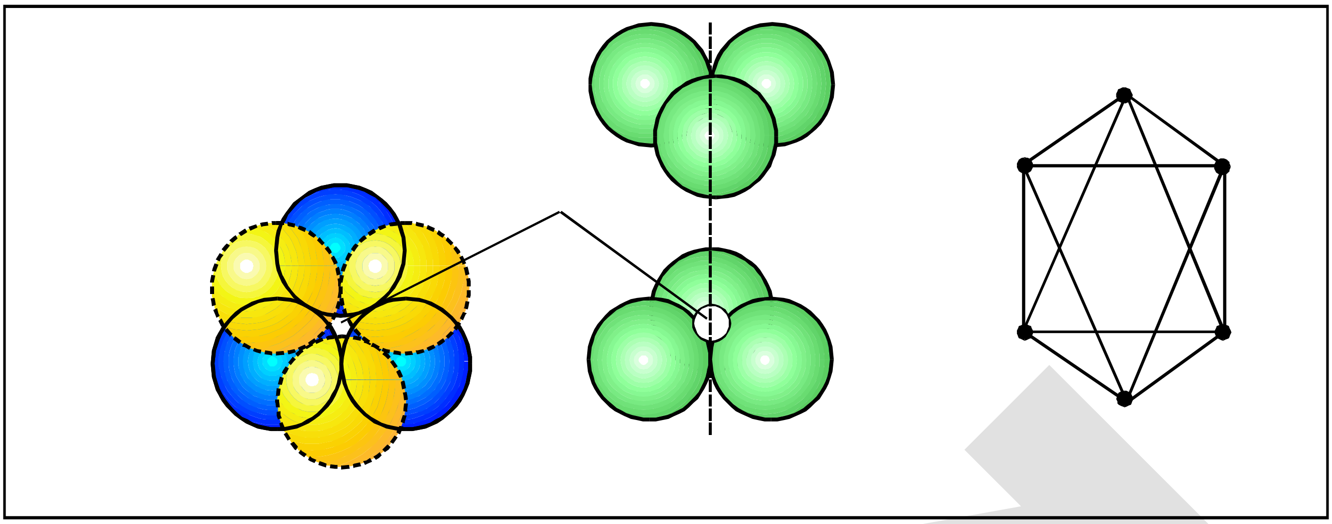
(i) Tetrahedral voids : We have seen that in hexagonal close packing (hcp) and cubic close packing (ccp), each sphere of second layer touches with three spheres of first layer. Thus, they leave a small space in between which is known as **tetrahedral site or interstices**. In another words, the vacant space between 4 touching spheres is called as tetrahedral void. Since a sphere touches three spheres in the below layer and three spheres in the above layer hence there are two tetrahedral sites associated with one sphere. It may by noted that a tetrahedral site does not mean that the site is tetrahedral in geometry but it means that this site is surrounded by four spheres and the centres of these four spheres lie at the apices of a regular tetrahedron.



(ii) Octahedral voids : Hexagonal close packing (hcp) and cubic close packing (ccp) also form another type of interstices (or site), which is called octahedral site (or interstices). In another words, the vacant space between 6 touching spheres is called as octahedral void.

In the figure two layers of close packed spheres are shown. The spheres of first layer are shown by full circles while that of second layer by dotted circles. Two triangles are drawn by joining the centres of three touching spheres of both the layers.

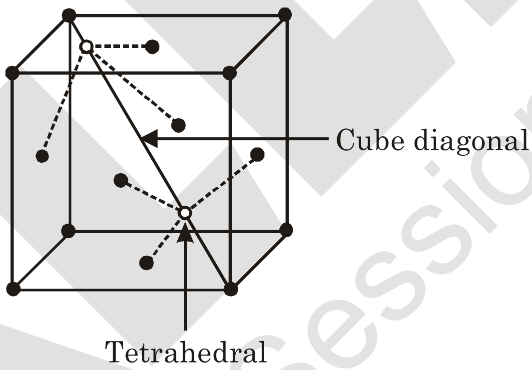
Session 2019-20
node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Solid State\Eng\01_Theory



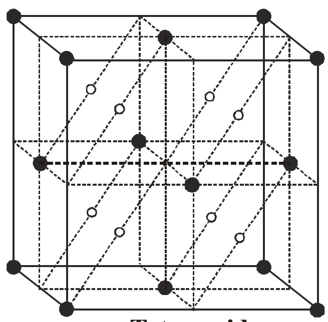
The apices of these triangles point are in opposite directions. On super imposing these triangles on one another, an octahedral site is created. It may be noted that an octahedral site does not mean that the hole is octahedral in shape but it means that this site is surrounded by six nearest neighbour lattice points arranged octahedrally.

8.1 POSITIONS OF TETRAHEDRAL VOIDS IN AN FCC UNIT CELL :

In FCC, one corner and its three face centred atom of faces meeting at that corner form a tetrahedral void. In FCC, two tetrahedral voids are obtained along one cube diagonal. So in FCC 8 tetrahedral voids are present.



Alternatively, the centre of tetrahedral void is located on the centre of body diagonal of each small cube of volume $\left(-\right)$.



Total number of particles per unit cell $= - \times \quad + \quad \times - =$

Total number of tetrahedral void = 8

$\therefore$ Effective number of tetrahedral void per particle = 2.

ALLEN Session 2019-20

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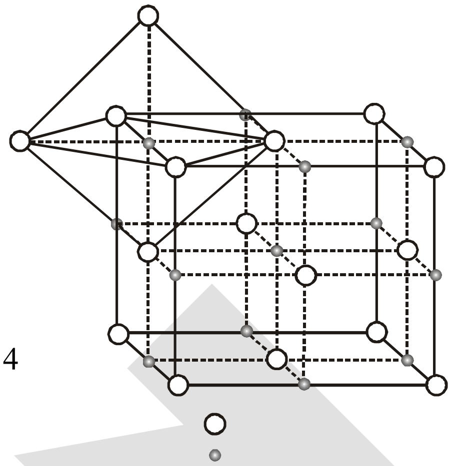
8.2 POSITION OF OCTAHEDRAL VOID IN FCC UNIT CELL :



Position of octahedral void is at mid-point of each edge (total 12 edges in a cube) and at the centre of cube.

Each octahedral void located at mid point of edge contributes 1/4 to the unit cell. The octahedral void situated at the centre contributes 1.

In FCC, total number of octahedral voids are

$$\underset{\text{(Cube centre)}}{(1 \times 1)} + \underset{\text{(edge)}}{(12 \times -)} = 1 + 3 = 4$$

In FCC, number of particles = 4

∴ Effective number of octahedral voids per particle = 1

8.3 POSITION OF TETRAHEDRAL AND OCTAHEDRAL VOID IN HCP:

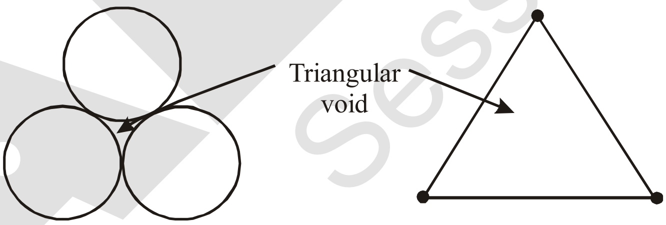
(i) Above & below each sphere, there is a tetrahedral void, hence number of tetrahedral void per unit cell = 12

(ii) Other than TV, all voids are OV & number of OV per unit cell = 2 × 3 = 6

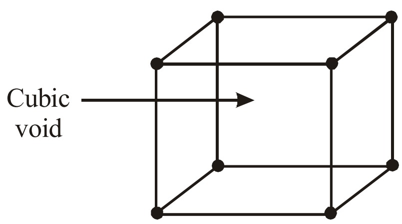
Note : In any closed packing of maximum efficiency, there are two TV & one OV per particle

8.4 Triangular and cubic voids :

(i) Triangular void : It is formed by three spheres in a plane.



(ii) Cubic void : It is formed in simple cubic unit cell by eight spheres at the corners of cube.

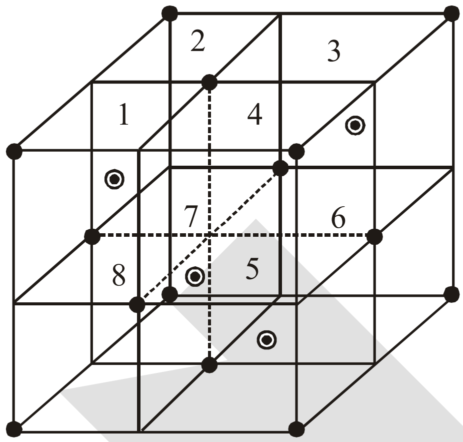


Session 2019-20
node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Solid State\Eng\01_Theory

9. CRYSTAL OF DIAMOND

The crystal is FCC for C atom & alternate tetrahedral voids are also occupied by carbon atoms.

* $Z = \frac{1}{8} \times 8 + 6 \times \frac{1}{2} + 4 = 8$
* CN = 4
* $\frac{\sqrt{3}a}{4} = 2r = d_{C-C}$
* Number of C–C bonds/unit cell = 4 × 4 =16
* Number of C–C bonds/C-atom = $\frac{16}{8} = 2$
* $PE = \frac{8 \times \frac{4}{3}\pi r^2}{\left(\frac{8r}{\sqrt{3}}\right)^3} = \frac{\sqrt{3}\pi}{16} \simeq 0.34 \text{ or } 34\%$



CLASS ILLUSTRATION

Ex.6 A solid crystallises in close packing for 'P' atoms and 25% of tetrahedral voids are occupied by 'Q' atoms. What is the simplest formula of solid ?

Sol. $Z_P = 1 \text{(say)}$

$$Z_Q = \frac{25}{100} \times 2 = \frac{1}{2}$$

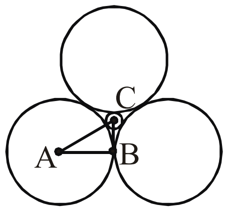
∴ Simplest formula of solid = $PQ_{1/2} = P_2Q$

Ex.7 Calculate the radius (r) of largest sphere which may be fitted in the (i) triangular voids (ii) tetrahedral voids (iii) octahedral voids (iv) cubic voids made by identical spheres of radius, 'R', without disturbing the crystal.

Sol. (i) AB = R

AC = R + r

∠BAC = 30°

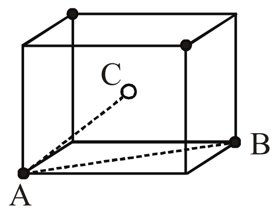


Now, $\cos 30° = \frac{AB}{AC} \Rightarrow \frac{\sqrt{3}}{2} = \frac{R}{R+r}$

$$r = \left(\frac{2}{\sqrt{3}} - 1\right)R = 0.155R$$

(ii) $AB = \sqrt{2}a = 2R$

$$AC = \frac{\sqrt{3}\,a}{2} = R + r$$



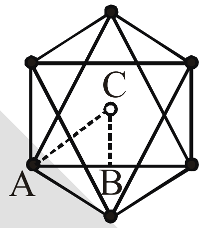
ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Solid State\Eng\01_Theory

Now, $\frac{R+r}{R}=\frac{\sqrt{3}}{\sqrt{2}}$

$\therefore \quad r=\left(\sqrt{\frac{3}{2}}-1\right)R=0.225R$

(iii) AB = R



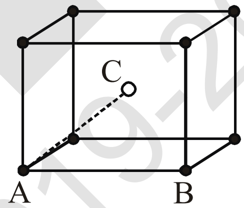
AC = R + r

$\angle BAC = 45°$

Now, $\cos 45°=\frac{AB}{AC}\Rightarrow\frac{1}{\sqrt{2}}=\frac{R}{R+r}$

$\therefore \quad r=(\sqrt{2}-1)R=0.414R$

(iv) AB = a = 2R



$AC=\frac{\sqrt{3}\,a}{2}=R+r$

Now, $\frac{R+r}{R}=\sqrt{3}$

$\therefore \quad r=(\sqrt{3}-1)R=0.732R$

Ex.8 An element (molar mass = 60 gm/mole) crystallises in CCP lattice. If density is 6.25 gm/cm³ and the distance between next nearest neighbour is dÅ, then the value of 'd' is (N_A = 6 × 10²³).

Sol. **Ans.(4)**

$d=\frac{ZM}{N_A.a^3}$

$6.25=\frac{4\times60}{6\times10^{23}\times a^3}\Rightarrow a=4\times10^{-8}\text{cm}=4\text{Å}.$

10 PACKING IN IONIC SOLID

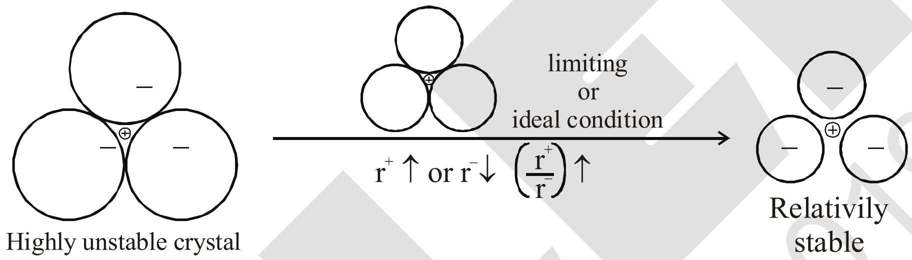
(i) Normally anions are bigger than cations, hence ionic solid are considered as the packing of anions and cations are supposed to occupy the voids.

(ii) Oppositively charged particles should lie closer and smilarly charged particles should lie away from each other.

(iii) Each ion tend to maximise its coordination number (Number of oppositively charged ions around it).

Point (ii) and (iii) contradict each other because on increase the CN, the repulsion between like charges will also increases. Hence, in all the ionic solids, there must be a balance with the help of relative size of ions to ensure maximum CN and minimum repulsion between like charges.

10.1 LIMITING OR IDEAL RADIUS RATION $\left(\frac{r^+}{r^-}\right)$:



The minimum $\frac{r^+}{r^-}$ values for the existence of a cation in a particular void is called **limiting radius ratio** for that void.

With the increase in radius ratio, space between anion will increase & hence the cation may tend for higher coordination number.

Voids	CN	Limiting r^+/r^-	Range of r^+/r^-
Triangular	3	0.155	0.155 – 0.225
Tetrahedral	4	0.225	0.225 – 0.414
Octahedral	6	0.414	0.414 – 0.732
Cubic	8	0.732	0.732 – 1.000

* For values lesser than 0.155, crystal will not exist.

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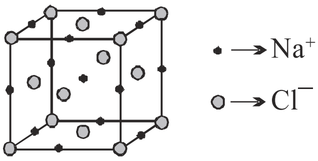
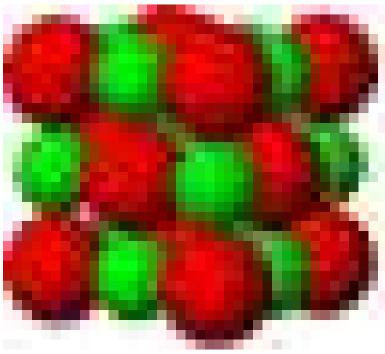
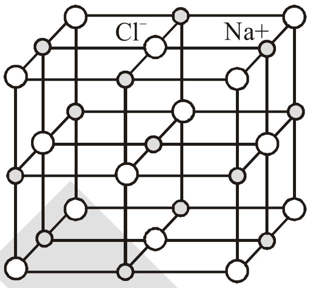
10.2 *TYPES OF IONIC STRUCTURES:*

10.2.1 **Rock salt structure** (NaCl) :



Rock salt structure





The bigger Cl^- forms cubic close packing and small Na^+ occupy positions of all octahedral voids. The radius ratio $\frac{+}{-}$ lie in the range 0.414 – 0.732.

(i) Each Na^+ is surrounded by six Cl^- and each Cl^- is surrounded by six Na^+ ion. [6:6 coordination]

(ii) No. of Na^+ and Cl^- in each unit cell is 4.

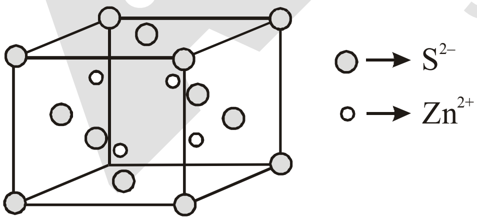
(iii) Number of formula units of NaCl per unit cell is equal to 4.

(iv) The density of NaCl crystal is given by $d = \left(\frac{\times}{\times}\right)$

(v) The edge length of NaCl unit cell is given by $(2r^+ + 2r^-) \Rightarrow \boxed{- = \ ^{+} + \ ^{-}}$

10.2.2 **Zinc blende (sphalerite) structure** (ZnS) :

Larger ion (S^{2-}) forming ccp arrangement and smaller ion (Zn^{2+}) filling half or alternate tetrahedral voids



Zinc blende structure

(i) C.N. of $Zn^{2+} = 4$; C.N. of $S^{2-} = 4$ [4 : 4 coordination]

(ii) Formula units of ZnS per unit cell = 4. **(iii)** $d_{ZnS} = \frac{\times}{\times}$ **(iv)** $_{+} + \quad = \frac{\sqrt{\ }}{\ }$

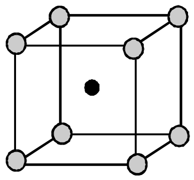
10.2.3 Cesium chloride structure (CsCl) **:**

Cl^- at the corners of cube and Cs^+ in the center (cubic void).

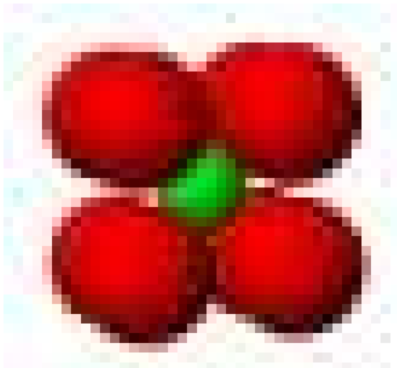
(i) C.N. of $Cs^+ = 8$; C.N. of $Cl^- = 8$ [8 : 8 coordination]

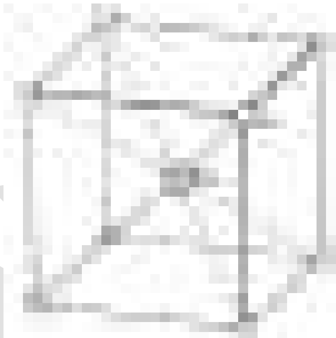
(ii) Formula units of CsCl per unit cell = 1

(iii) $d_{CsCl} = \frac{}{\times}$ **(iv)** $r_{Cs^+} + r_{Cl^-} = \frac{\sqrt{}}{} \Rightarrow \boxed{{}^{+} + {}^{-} = \frac{\sqrt{}}{}}$



Cesium chloride structure





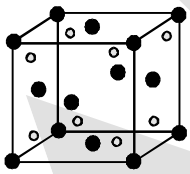
10.2.4 Fluorite structure (CaF_2) **:**

Ca^{2+} forming ccp arrangement and F^- filling all tetrahedral voids.

(i) C.N. of $F^- = 4$; C.N. of $Cs^+ = 8$ [8 : 4 coordination]

(ii) Formula units of CaF_2 per unit cell = 4

(iii) $d_{CaF_2} = \frac{\times}{\times}$ **(iv)** $r_{Ca^{2+}} + r_{F^-} = \frac{\sqrt{}}{}$



Fluorite structure

10.2.5 Antifluorite structure (Li_2O) **:**

O^{2-} ion forming ccp and Li^+ taking all tetrahedral voids.

(i) C.N. of $Li^+ = 4$

C.N. of $O^{2-} = 8$

(ii) Formula units of Li_2O ; per unit cell = 4

(iii) $= \frac{\times}{\times}$ **(iv)** $r_{Li^+} + r_{O^{2-}} = \frac{\sqrt{}}{}$

10.2.6 Corundum Structure (Al_2O_3) **:**

O^{2-} forming hcp and Al^{3+} filling 2/3 octahedral voids.

Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Solid State\Eng\01_Theory

10.2.7 Rutile structure (TiO_2) **:**

O^{2-} forming hcp while Ti^{4+} ions occupy half of the octahedral voids.

10.2.8 Pervoskite structure ($CaTiO_3$) **:**

Ca^{2+} in the corner of cube, O^{2-} at the face center and Ti^{4+} at the centre of cube.

10.2.9 Spinel and inverse spinel structure ($MgAl_2O_4$) **:**

O^{2-} forming fcc, Mg^{2+} filling 1/8 of tetrahedral voids and Al^{3+} taking half of octahedral voids.

In an inverse spinel structure, O^{2-} ion form FCC lattice, A^{2+} ions occupy 1/8 of the tetrahedral voids and trivalent cation occupies 1/8 of the tetrahedral voids and 1/4 of the octahedral voids.

Ex.9 A solid A^+B^- has NaCl type close packed structure. If the anion has a radius of 250 pm, what should be the ideal radius for the cation ? Can a cation C^+ having a radius of 180 pm be slipped into the tetrahedral site of the crystal A^+B^- ? Give reason for your answer.

Sol. *In Na^+Cl^- crystal each Na^+ ion is surrounded by 6 Cl^+ ions and vice versa. Thus Na^+ ion is placed in octahedral hole.*

The limiting radius ratio for octahedral site = 0.414

or $\quad \frac{{}^{+}}{{}_{-}} = \frac{}{} =$

Given that radius of anion (B^-) R = 250 pm

i.e. radius of cation (A^+) $\quad r = 0.414\ R = 0.414 \times 250$ *pm*

or $\quad r = 103.5$ *pm*

Thus ideal radius for cation (A^+) is r = 103.5 pm.

We know that (r/R) for tetrahedral hole is 0.225.

$\therefore \quad \frac{}{} =$

or $\quad r = 0.225 \quad R = 0.225 \times 250 = 56.25$ *pm*

Thus ideal radius for cation is 56.25 pm for tetrahedral hole. But the radius of C^+ is 180 pm. It is much larger than ideal radius i.e. 56.25 pm. Therefore we can not slip cation C^+ into the tetrahedral site.

Ex.10 A compound forms hexagonal close-packed structure. What is the total number of voids in 0.5 mole of it? How many of these are tetrahedral voids?

Sol. *Since, for every atom forming hcp structure there are two tetrahedral voids and one octahedral void. Total voids = 3 × 0.5 = 1.5 mol and total tetrahedral void = 2 × 0.5 = 1 mol.*

Ex.11 A compound is formed by two elements M and N. The element N forms ccp and atoms of M occupy 1/3rd of tetrahedral voids. What is the formula of the compound?

Sol. *In one unit cell of ccp, no. of N = 4 ; total no. of tetrahedral void = 8 ; occupied tetrahedral void by M = 8/3 ; Empirical formula :* $N_4M_{8/3} = N_3M_2$.

ALLEN Session 2019-20
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11. IMPERFECTIONS IN SOLIDS

Although crystalline solids have short range as well as long range order in the arrangement of their constitutent particles, yet crystals are not perfect. Usually a solid consists of an aggregate of large number of small crystals. These small crystals have defects in them. This happens when crystallisation process occurs at fast or moderate rate. Single crystals are formed when the process of crystallisation occurs at extremely slow rate. Even these crystals are not free of defects.

The defects are basically irregularities in the arrangement of constituent particles. Broadly speaking, the defects are of two types, namely, **point defects** and **line defects**. Point defects are the irrgularties or deviations from ideal arrangement around a point or an atom in a crystalline substance, whereas the line defects are the irregularities or deviations from ideal arrangement in entire rows of lattice points. These irregularities are called crystal defects. We shall confine our discussion to point defects only.

11.1 Types of Point Defects

Point defect can be classified into three types:

(i) Stoichiometric defects

(ii) Non-stoichiometric defects

(iii) Impurity added defect

(i) Stoichiometric Defect

These are the point defect that do not disturb the stoichiometry of the solid. They are also called ***intrinsic*** or **thermodynamic defects**. Basically these are of two types; vacancy defects and interstitial defect.

(a) Vacancy Defect :

When some of the lattice sites are vacant, the crystal is said to have vacancy defect. This results in decrease in density of the substance. This defect can also develop when a substance is heated.

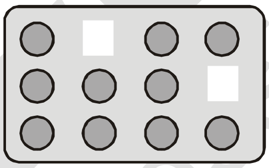


Fig. : Vacancy defects

(b) Interstitial Defect : When some constituent particles (atoms or molecules) occupy an interstitial site. the crystal is said to have interstitial defect. This defect increases the density of the substance.

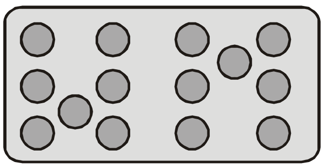
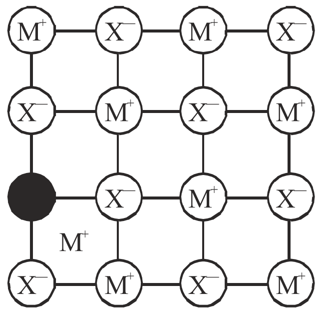


Fig. : Interstitial defects

Vacancy and interstitial defects as explained above can be shown by non-ionic solids. Ionic solids must always maintain electrical neutrality. Rather than simple vacancy or interstitial defects, they show these defects as **Frenkel and Schottky defects**.

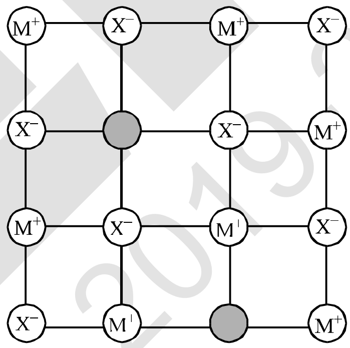
(c) Frenkel Defect : This defect is shown by ionic solids.
The smaller ion (usually cation) is delocalised from its normal site to an interstitial site . It creates a vacancy defect at its original site and an interstitial defect at its new location.
Frenkel defect is also called **dislocation defect**. It does not change the density of the solid.
Frenkel defect is shown by ionic substance in which there is a large difference in the size of ions, for example, ZnS, AgCl, AgBr and AgI due to small size of Zn^{2+} and Ag^+ ions.



Frenkel Defect

Influences : Makes solid crystals good conductor. In Frenkel defect , ions in interstitial sites increases the dielectric constant.

(d) Schottky Defect : It is basically a vacancy defect in ionic solids. In order to maintain electrical neutrality, the number of missing cations and anions are equal (in stoichiometric ratio)
Like simple vacancy defect, Schottky defect also decreases the density of the substance. Number of such defects in ionic solids is quite significant. For example, in NaCl there are approximately,
10^6 Schottky pairs per cm^3 at room temperature. In 1 cm^3 there are about 10^{22} ions. Thus, there is one Schottky defect per 10^{16} ions. Schottky defect is shown by ionic substances in which the cation and anion are of almost similar sizes. For example, NaCl, KCl, CsCl and AgBr. It may be noted that AgBr shows both, Frenkel as well as Schottky defects.



Schottky Defect

Influence : The presence of large number of schottky defects in crystal results in significant decrease in its density.

(ii) Non-Stoichiometric Defects

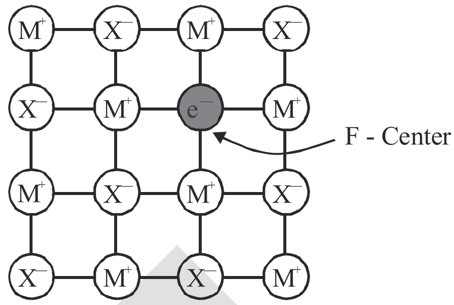
The defects discussed so far do not disturbs the stoichiometry of the crystalline substance. However, a large number of non-stoichiometric inorganic solids are known which contain the constituent elements in non-stoichiometric ratio due to defects in their crystal structures. These defects are of two types:
(a) Metal excess defect.
(b) Metal deficiency defect.

ALLEN Session 2019-20

(a) Metal Excess Defect

(I) Metal excess defect due to anionic vacancies :

Alkali halides like NaCl and KCl show his type of defect. When crystals of NaCl are heated in an atmosphere of sodium vapour, the sodium atoms are deposited on the surface of the crystal. The Cl^- ions diffuse to the surface of the crystal and combine with Na atoms to give NaCl. This happens by loss of electron by sodium atoms to form Na^+ ions.



Metal excess defects due to anion vacancies

The released electrons diffuse into the crystal and occupy anionic sites . As a result the crystal now has an excess of sodium. The anionic sites occupied by unpaired electrons are called ***F-centres*** (from the German word ***Farbenzenter*** for colour centre).

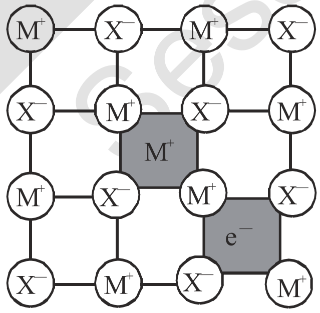
They impart yellow colour to the crystals of NaCl. The colour results by excitation of these electrons when they absorb energy from the visible light falling on the crystals. Similarly, excess of lithium makes LiCl crystals pink and excess of potassium makes KCl crystals violet (or lilac).

(II) *Metal excess defect due to the presence of extra cations at interstitial sites :*

Zinc oxide is white in colour at room temperature. On heating it loses oxygen and turns yellow.

$$ZnO \xrightarrow{\text{heating}} Zn^{2+} + \frac{1}{2}O_2 + 2e^-$$

Now there is excess of zinc in the crystal and its formula becomes $Zn_{1+x}O$. The excess Zn^{2+} ions move to interstitial sites and the electrons to neighbouring interstitual sites.


Metal excess defects due to interstitial cation

(b) Metal Deficiency Defect

There are many solids which are difficult to prepare in the stoichiometric composition and contain less amount of the metal as compared to the stoichiometric proportion. A typical example of this type is FeO which is mostly found with a composition of $Fe_{0.95}O$. It may actually range from $FeO_{0.93}O$ to $Fe_{0.96}O$. In crystals of FeO, some Fe^{2+} cations are missing and the loss of positive charge is made up by the presence of required number of Fe^{3+} ions.

ALLEN Session 2019-20
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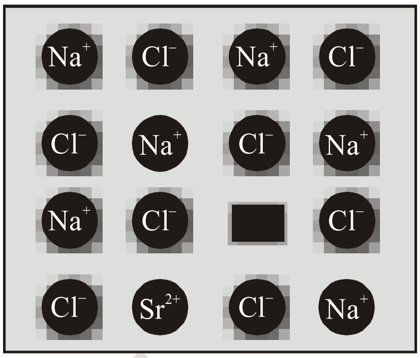
(iii) Impurity Defects

If molten NaCl containing a little amount of $SrCl_2$ is crystallised, some of the sites of Na^+ ions are occupied by Sr^{2+}. Each Sr^{2+} replaces two Na^+ Ions. It occupies the site of one ion and the other site remains vacant. The cationic vacancies thus produced are equal in number to that of Sr^{2+} ions. Another similar example is the solid solution of $CdCl_2$ and AgCl.



Introduction of cation vacancy in NaCl by substitution of Na^+ by Sr^{2+}

Note: (i) As temperature increases, no. of defects increases exponentially.

(ii) For defect formation : $\Delta H > 0$, $\Delta S > 0$

$\therefore$ More spontaneous at higher temperatures.

(iii) No matter how many imperfections are present in a crystal, it is always electrically neutral (no net charge).

12. ELECTRICAL PROPERTIES

Solids exhibit an amazing range of electrical conductivities, extending over 27 orders of magnitude ranging from 10^{-20} to 10^{7} ohm^{-1} m^{-1}. Solids can be classified into three types on the basis of their conductivities.

(i) Conductors : The solids with conductivities ranging between 10^4 to 10^7 $ohm^{-1}m^{-1}$ are called conductors. Metals having conductivities in the order of 10^7 $ohm^{-1}m^{-1}$ are good conductors.

(ii) Insulators : These are the solids with very low conductivities ranging between 10^{-20} to 10^{-10} $ohm^{-1}m^{-1}$.

(iii) Semiconductors : These are the solids with conductivities in the intermediate range from 10^{-6} to 10^4 $Ohm^{-1}m^{-1}$.

12.1 : Conduction of Electricity in Metals

A conductor may conduct electricity through movement of electrons or ions. Metallic conductors belong to the former category and electrolytes to the latter.

Metals conduct electricity in solid as well as molten state. The conductivity of metals depend upon the number of valence electrons available per atom. The atomic orbitals of metal atoms from molecular orbitals which are so close in energy to each other as to form a band. If this band is partially filled or it overlaps with a higher energy unoccupied conduction band, then electrons can flow easily under an applied electric field and the metal shows conductivity.

ALLEN Session 2019-20

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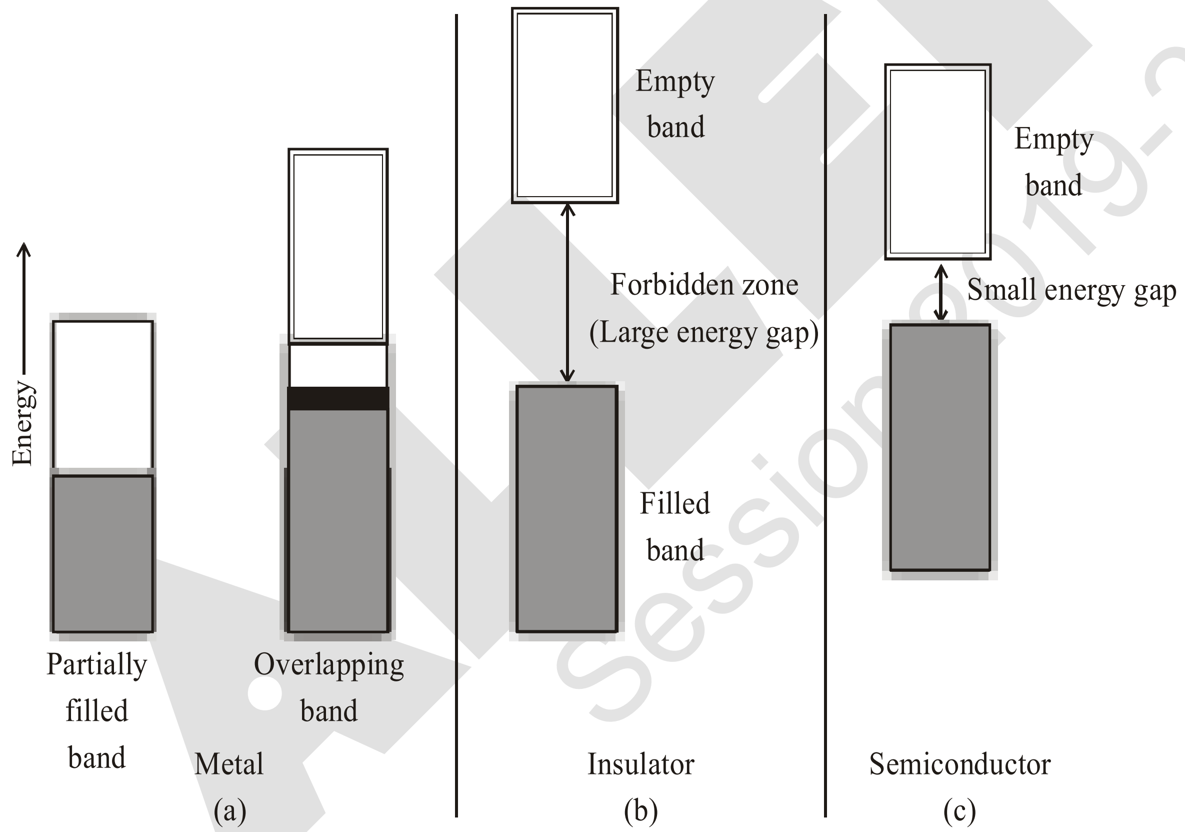
12.2 : Insulator :

If the gap between filled valence band and the next higher unoccupied band (conduction band) is large, electron cannot jump to it and such a substance has very small conductivity and it behaves as an insulator.

12.3 : Conduction of Electricity in Semi-conductor

In case of semiconductors, the gap between the valence band and conduction band is small. Therefore, some electrons may jump to conduction band and show some conductivity. Electrical conductivity of semconductors increases with rise in temperature, since more electrons can jump to the conduction band. Substances like silicon and germanium show this type of behaviour and are called **intrinsic semiconductors**.



Distinction among (a) metals (b) insulators and (c) semiconductors.
In each case, an unshaded area represents a conduction band.

The conducitivity of these interinsic semiconductors is too low to be of practical use. Their conductivity is increased by adding an appropriate amount of suitable impurity. This process is called **doping**. Doping can be done with an impurity which is electron rich or electron deficient as compared to the intrinsic semiconductor silicon or germanium. Such impurities introduce ***electrical defect*** in them.

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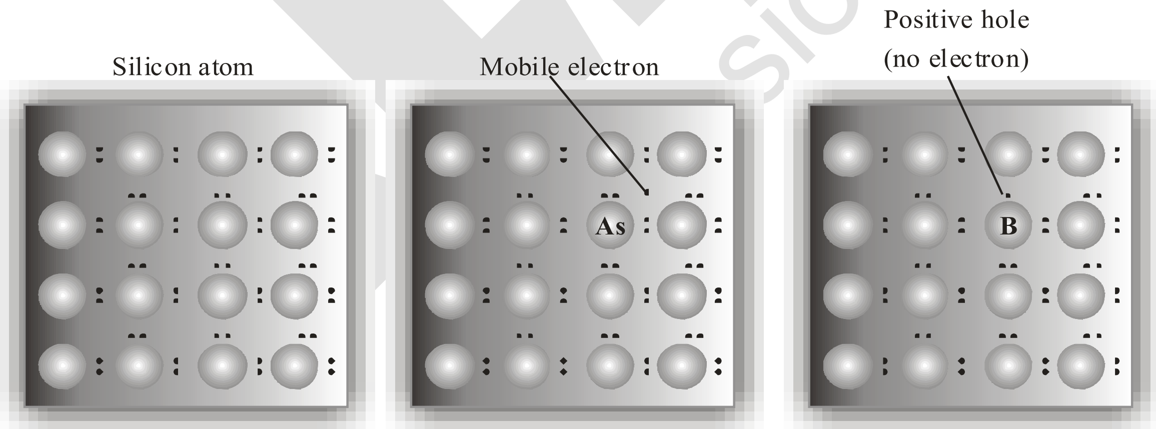
(a) Electron – rich impurities

Silicon and germanium belong to group 14 of the periodic table and have four valence electrons each. In their crystals each atom forms four covalent bonds with its neighbours . When doped with a group 15 element like P or As, which contains five valence electrons, they occupy some of the lattice sites in silicon or germanium crystal. Four out of five electrons are used in the formation of four covalent bonds with the four neighbouring silicon atoms. The fifth electron is extra and becomes delocalised. These delocalised electrons increase the conductivity of doped silicon (or germanium). Here the increase in conductivity is due to the negatively charged electron, hence silicon doped with electron-rich impurity is called ***n-type semiconductor***.

(b) Electron – deficit impurities

Silicon or germanium can also be doped with a group 13 element like B, Al or Ga which contains only three valence electrons. The place where the fourth valence electron is missing is called ***electron hole or electron vacancy*** . An electron from a neighbouring atom can come and fill the electron hole, but in doing so it would leave an electron hole at its original position. If it happens, it would appear as if the electron hole has moved in the direction opposite to that of the electron that filled it. Under the influence of electric field, electrons would move towards the positively charged plate through electronic holes, but it would appear as if electron holes are positively charged and are moving towards negatively charged plate. This type of semi conductors are called ***p-type semiconductors***.



Perfect crystal (a) | *n*-type (b) | *p*-type (b)

Creation of n-type and p-type semi conductors by doping groups 13 and 15 elements

(c) Applications of n-type and p-type semiconductors

Various combinations of *n*-type and *p*-type semiconductors are used for making electronic components. Diode is a combination of *n*-type and *p*-type semiconductor and is used as a rectifier. Transistors are made by sandwiching a layer of one type of semiconductor between two layers of the other type of semiconductor. *npn* and *pnp* type of transistors are used to detect or amplify radio or audio signals. The solar cell is an efficient photo-diode used for conversion of light energy into electrical energy.

Germanium and Silicon are group 14 elements and therefore, have a characteristic valency of four and form four bonds as in diamond. A large variety of solid state materials have been prepared by combination of group 13 and 15 or 12 and 16 to simulate average valence of four as in Ge or Si. Typical compounds of groups 13-15 are InSb, AIP and GaAs. Gallium arsenide (GaAs) semiconductor have very fast response and have revolutionised the design of semiconductor devices. ZnS, CdS, CdSe and HgTe are examples of groups 12-16 compounds. In these compounds, the bonds are not perfectly covalent and the ionic character depends on the electronegativities of the two elements.

It is interesting to learn that transition metal oxides show marked differences in electrical properties. TiO, CrO_2 and ReO_3 behave like metals. Rhenium oxide, ReO_3 is like metallic copper in its conductivity and appearance. Certain other oxides like VO, VO_2, VO_3 and TiO_3 show metallic or insulating properties depending on temperature.

13. MAGNETIC PROPERTIES :

Every substance has some magnetic properties associated with it. The origin of these properties lies in the electrons. Each electron in an atom behaves like a tiny magnet. Its magnetic moment originates from two types of motions (i) its orbital motion around the nucleus and (ii) its spin around its own axis . Electron being a charged particle and undergoing these motions can be considered as a small loop of current which possesses a magnetic moment. Thus, each electron has a permanent spin and an orbital magnetic moment associated with it. Magnitude of this magnetic moment is very small and is measured in the unit called Bohr magneton, μ_B. It is equal to 9.27×10^{-24} A m^2.

On the basis of their magnetic properties, substances can be classified into five categories: (i) paramagnetic (ii) diamagnetic (iii) ferromagnetic (iv) antiferromagnetic and (v) ferrimagnetic.

(i) **Paramagnetism :**

Paramagnetic substances are weakly attracted by a magnetic field. They are magnetised in a magnetic field in the same direction. They lose their magnetism in the absence of magnetic field. Paramagnetism is due to presence of one or more unpaired electrons which are attracted by the magnetic field. O_2, Cu^{2+}, Fe^{3+}, Cr^{3+} are some examples of such substances.

(ii) **Diamagnetism :**

Diamagnetic substances are weakly repelled by a magnetic field. H_2O, NaCl and C_6H_6 are some examples of such substances. They are weakly magnetised in a magnetic field in opposite direction. Diamagnetism is shown by those substances in which all the electrons are paired and there are no unpaired electrons. Pairing of electrons cancels their magnetic moments and they lose their magnetic character.

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Solid State\Eng\01_Theory

ALLEN Session 2019-20

(iii) **Ferromagnetism :**

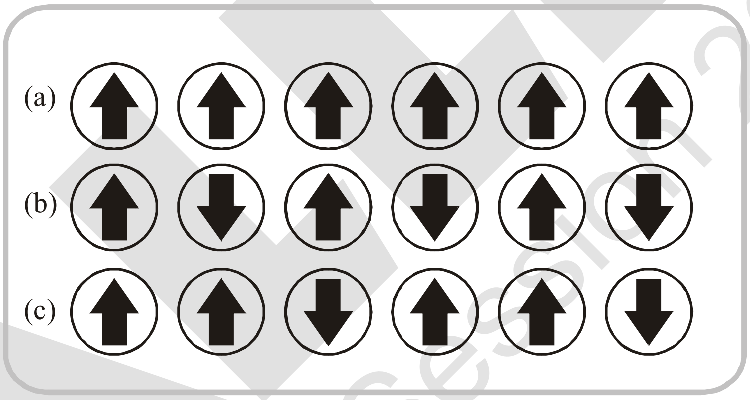
A few substances like iron, cobalt, nickel, gadolinium and CrO_2 are attracted very strongly by a magnetic field. Such substances are called ferromagnetic substances. Besides strong attractions, these substances can be permanently magnetised. In solid state, the metal ions of ferromagnetic substances are grouped together into small regions called ***domains***. Thus, each domain acts as a tiny magnet. In an unmagnetised piece of a ferromagnetic substance the domains are randomly oriented and their magnetic moments get cancelled. When the substance is placed in a magnetic field all the domains get oriented in the direction of the magnetic field and a strong magnetic effect is produced. This ordering of domains persist even when the magnetic field is removed and the ferromagnetic substance becomes a permanent magnet.

(iv) **Antiferromagnetism :**

Substances like MnO showing antiferromagnetism have domain structure similar to ferromagnetic substance, but their domains are oppositely oriented and cancel out each other's magnetic moment

(v) **Ferrimagnetism :**

Ferrimagnetism is observed when the magnetic moments of the domains in the substance are aligned in parallel and anti-parallel directions in unequal numbers. They are weakly attracted by magnetic field as compared to ferromagnetic substances. Fe_3O_4 (magnetite) and ferrites like $MgFe_2O_4$ and $ZnFe_2O_4$ are examples of such substances. These substances also lose ferrimagnetism on heating and become paramagnetic.



Schematic alignment of magnetic moments in (a) ferromagnetic (b) antiferromagnetic and (c) ferrimagnetic

EXERCISE # S-I

METALLIC CRYSTALS

Cubic crystals

1. A closed packed structure of uniform spheres has the edge length of 600 pm. Calculate the radius of sphere, if it exist in $(\sqrt{2} = 1.4, \sqrt{3} = 1.7)$

(a) simple cubic lattice (b) BCC lattice (c) FCC lattice

2. Xenon crystallises in the face-centred cubic lattice and the edge length of the unit cell is $438\sqrt{2}$ pm. What is the nearest neighbour distance and what is the radius of xenon atom?

3. The effective radius of the iron atom is $\sqrt{2}$ Å. It has FCC structure. Calculate its density (Fe = 56 amu , $N_A = 6 \times 10^{23}$)

4. Calculate the ratio of densities if same element undergoes fcc as well as simple cubic packing. Assume same atomic radius in both crystals.

5. Potassium has body-centred cubic structure with the nearest neighbour distance $260\sqrt{3}\,\text{pm}$. Its density would be ($\frac{1}{(5.2)^2} = 0.036$, $N_A = 6 \times 10^{23}$, K = 39)

6. A cubic solid is made up of two elements 'A' and 'B'. Atoms 'B' are at the corners of the cube and 'A' at the body centre. What is the simplest formula of compound ?

7. An element 'X' crystallizes in bcc. Find volume of unit cell in $(\text{Å})^3$, if atomic radius is $\sqrt{3}$ Å.

8. A lattice has simple cube unit cell then number of faces which meets at a corner of a cube in this lattice is.

9. How many next nearest neighbours does potassium have in bcc lattice ?

10. Number of crystal systems having, only 2 types of Bravais lattices = x,
Number of crystal systems having, atleast 2 interfacial angles equal = y,
all the three interfacial angles and all the three axes lenghts equal = z
Then find y – (x + z).

PACKING IN SOLIDS

11. A cubic solid is made by atoms 'A' forming close pack arrangement, 'B' occupying one-fourth of tetrahedral void and C occupying half of the octahedral voids. What is the formula of compound ?

12. If number of nearest neighbours, next nearest (2^{nd} nearest) neighbour and next to next nearest (3^{rd} nearest) neighbours are x, y and z respectively for body centred cubic unit cell, then calculate value of $\frac{xy}{z}$ is.

13. In FCC unit cell, what fraction of edge is not covered by atoms ?

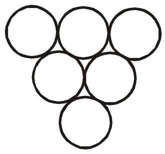
14. Element A crystallizes in hcp. Calculate total number of tetrahedral voids in 24 micrograms of A ($N_A = 6 \times 10^{23}$, Atomic mass of A = 48)

15. For ABC ABC ABC .... packing, distance between two succesive tetrahedral void is X and distance between two successive octahedral void is y in an unit cell, then $\frac{y\sqrt{2}}{X}$ is

ALLEN Session 2019-20

16. An element 'M' crystallizes in ABAB....type packing. If adjacent layer A & B are $10\frac{\sqrt{2}}{\sqrt{3}}$ pm apart, then calculate radius of largest sphere which can be fitted in the void without disturbing the lattice arrangement (Given : $\sqrt{2} = 1.414$)

17. A 3d unit cell is such that one of its planes has the following arrangement of atoms.



What will be the number of next nearest neighbour in such unit cell?

18. The density of solid Argon is 1.6 gm/ml at –233°C. If the atomic volume of Argon is assumed to be $\frac{5}{3}\times10^{-23}$ cm^3 then what % of solid Argon is apparently empty space ? (Ar = 40, $N_A = 6 \times 10^{23}$)

19. An element crystallizes in a structure having FCC unit cell of an edge 200 pm. Calculate the density, if 200 g of this element contains 2.4×10^{24} atoms. ($N_A = 6 \times 10^{23}$)

20. Find packing fraction of three dimensional unit cell of AAAAA.......type hypothetical arrangement in which hexagonal packing is taken in layer.

IONIC CRYSTALS

21. If the radius of Mg^{2+} ion, Cs^{+} ion, O^{2-} ion, S^{2-} ion and Cl^{-} ion are 0.65 Å , 1.69 Å, 1.40 Å, 1.84 Å, and 1.81 Å respectively, calculate the co-ordination numbers of the cations in the crystals of MgS, MgO and CsCl.

22. The two ions A^{+} and B^{-} have radii 88 pm and 200 pm respectively. In the closed packed crystal of compound AB, predict the co-ordination number of A^{+}.

23. Spinel is a important class of oxides consisting of two types of metal ions with the oxide ions arranged in CCP pattern. The normal spinel has one-eight of the tetrahedral holes occupied by one type of metal ion and one half of the octahedral hole occupied by another type of metal ion. Such a spinel is formed by Zn^{2+}, Al^{3+} and O^{2-}, with Zn^{2+} in the tetrahedral holes. Give the formula of spinel.

24. KF crystallizes in the NaCl type structure. If the radius of K^{+} ion is 132 pm and that of F^{-} ion is 135 pm, what is the shortest K–F distance? What is the edge length of the unit cell? What is the closet K^{+}– K^{+} distance?

25. CsCl has bcc unit cell with edge length 400 pm. Calculate the interionic distance in CsCl.

26. The density of KBr is 2.38 g cm^{-3} . The length of the edge of the unit cell is 700 pm. Find the number of fomula unit of KBr present in the single unit cell.

($N_A = 6 \times10^{23}$ mol^{-1} , At. mass : K = 39, Br = 80)

27. A crystal of lead(II) sulphide has NaCl structure. In this crystal the shortest distance between Pb^{+2} ion and S^{2-} ion is 300 pm. What is the length of the edge of the unit cell in lead sulphide? Also calculate the unit cell volume.

ALLEN Session 2019-20

28. If the length of the body diagonal for CsCl which crystallises into a cubic structure with Cl^- ions at the corners and Cs^+ ions at the centre of the unit cell, is 7 Å and the radius of the Cs^+ ion is 1.69 Å, what is the radius of Cl^- ion?

29. Rbl crystallizes in (8 : 8) structure in which each Rb^+ is surrounded by eight iodide ions each of radius 2.17 Å. Find the length of one side of RbI unit cell , assuming anion, anion contact.

30. Solid AB has NaCl type structure. If the radius of A^+ and B^- are 0.8 Å and 1.2Å respectively and formula mass of AB is 48 g/mole, what is the density of AB solid.

Take : Avogadro's number = 6×10^{23}

PROBLEMS RELATED WITH DEFECTS IN SOLID

31. The composition of a sample of wustite is $Fe_{0.93}O_{1.0}$. What percentage of iron is present in the form of Fe(III)?

32. If NaCl is dopped with 10^{-3} mol % $SrCl_2$, what is the number of cation vacancies per mole of NaCl? ($N_A = 6 \times 10^{23}$)

33. AgCl has the same structure as that of NaCl. The edge length of unit cell of AgCl is found to be 523.5 pm and the density of AgCl is 6.0 g cm^{-3}. Find the percentage of sites that are unoccupied.

[Ag = 108 , $(5.235)^3 = 143.5$]

34. A non stoichiometric compound Fe_7S_8 consist of iron in both Fe^{+2} and Fe^{+3} form and sulphur is present as sulphide ions. Calculate cation vacancies as a percentage of Fe^{+2} initially present in the ideal crystal.

35. The density of ZnS crystal (Zinc blende structure) having 10% Frenkel defect is

[$r_{Zn^{2+}} = 40\sqrt{3}\text{pm}, r_{S^{2-}} = 110\sqrt{3}\text{pm}$, Zn = 65.2, S = 32]

ALLEN Session 2019-20

EXERCISE # S-II

1. An element (atomic weight = 125) crystallises in simple cubic structure. Diameter of the largest atom which can be placed without disturbing unit cell is 366 pm. If the density of element 'x' gm/cm^3 , the value of $\left(\frac{6x}{5}\right)$ is. [**Given** : $\sqrt{3}$ = 1.732 , $N_A = 6 \times 10^{23}$]

2. The density of diamond from the fact that it has face centred cubic structure with two atoms per lattice point and unit cell edge length of 3.6 Å, is 'x' gm / cm^3, then the value of (1.458x) is ($N_A = 6 \times 10^{23}$)

3. Iron crystallizes in several modifications. At about 910°C, the body-centred cubic 'α' form undergoes a transition to the face-centred cubic 'γ' form. Assuming that the distance between nearest neighbours is the same in the two forms at the transition temperature, the ratio of the density of γ iron to that of α iron at the transition temperature, x : 1, then the value of $(3\sqrt{6}x)$ is.

4. What is the percent by mass of titanium in rutile, a mineral that contain titanium and oxygen, if structure can be described as a closest packed array of oxide ions, with titanium in one half of the octahedral holes. What is the oxidation number of titanium? (Ti = 48)

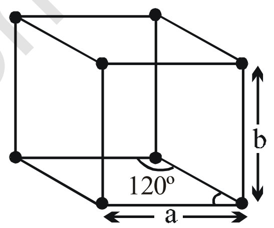
5 . An element crystallizes into a structure which may be described by a cubic type of unit cell having one atom on each corner of the cube and two extra atoms on one of its body diagonal. If the volume of this unit cell is 2.4×10^{-23} cm^3 and density of element is 7.2 g cm^{-3}, calculate the number of atoms present in 288 g of element.

6. What will be packing fraction of solid in which atoms are present at corners and cubic void is occupied. The insertion of the sphere into void does not disturb simple cubic lattice.

7. Ice crystallizes in a hexagonal lattice. At the low temperature at which the structure was determined, the lattice constants where **a** = 5 Å, and **b** = $3.5\sqrt{3}$ Å

How many molecules are contained in the given unit cell?

[density (ice) = $\frac{16}{17.5}$ gm/cm^3)]



8. The mineral hawleyite, one form of CdS, crystallizes in one of the cubic lattices, with edge length 5.87Å. The density of hawleyite is 4.63 g cm^{-3}. (Cd = 112)

(i) In which cubic lattice does hawleyite crystallize?

(ii) Find the Schottky defect in g cm^{-3}.

9. KCl crystallizes in the same type of lattice as does NaCl. Given that $\frac{r_{Na^+}}{r_{Cl^-}}=0.5$ and $\frac{r_{Na^+}}{r_{K^+}}=0.7$

Calculate :

(a) The ratio of the sides of unit cell for KCl to that for NaCl and

(b) The ratio of densities of NaCl to that for KCl.

$$\left(\left(\frac{8}{7}\right)^3 = 1.5, \frac{74.5}{58.5} = 1.25\right)$$

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ALLEN Session 2019-20

10. A cubic unit cell contains manganese ions at the corners and fluoride ions at the center of each edge.

(a) What is the empirical formula of the compound?

(b) What is the co-ordination number of the Mn ion?

(c) Calculate the edge length of the unit cell, if the radius of Mn ion is 0.65 Å and that of F^- ion is 1.36 Å.

11. Potassium crystallizes in a body-centred cubic lattice with edge length, a = 5.2 Å.

(a) What is the distance between nearest neighbours?

(b) What is the distance between next-nearest neighbours?

(c) How many nearest neighbours does each K atom have?

(d) How many next-nearest neighbours does each K atom have?

(e) What is the density of crystalline potassium?

(Given : K = 39, $(5.2)^3 = 140$)

12. An element X (atomic weight = 24 gm/mole) forms a face centred cubic lattice. If the edge length of the lattice is 4×10^{-8} cm and the observed density is 2.40×10^3 kg/m^3 , calculate the percentage occupancy of lattice points by X element. (Given : $N_A = 6 \times 10^{23}$)

13. Calculate the density (in gm/cm^3) of NaCl type ionic solid (MW = 75 gm/mol) if distance between two nearest cations is $250\sqrt{2}$ pm (avogadro number = 6×10^{23})

14. The olivine series of minerals consists of crystals in which Fe and Mg ions may substitute for each other causing substitutional impurity defect without changing the volume of the unit cell. In olivine series of minerals, oxide ion exist as FCC with Si^{4+} occupying 1/4 th of octahedral voids and divalent ions occupying 1/4th of tetrahedral voids. The density of forsterite (magnesium silicate) is 3.0 g/cc and that of fayalite (ferrous silicate) is 4.0 g/cc. Find the formula of forsterite and fayalite minerals and the mass percentage of fayalite in an olivine with a density of 3.5 g/cc.

Session 2019-20

EXERCISE # O-I

SINGLE CORRECT :

1. Which of the following are the correct axial distances and axial angles for rhombohedral system ?
(A) $a = b = c, \alpha = \beta = \gamma \neq 90^0$ (B) $a = b \neq c, \alpha = \beta = \gamma = 90^0$
(C) $a \neq b = c, \alpha = \beta = \gamma = 90^0$ (D) $a \neq b \neq c, \alpha \neq \beta \neq \gamma \neq 90^0$

2. $a \neq b \neq c, \alpha \neq \beta \neq \gamma \neq 90^0$ represents
(A) tetragonal system (B) orthorhombic system
(C) monoclinic system (D) triclinic system

3. Diamond belongs to the crystal system :
(A) Cubic (B) triclinic (C) tetragonal (D) hexagonal

4. A match box exhibits -
(A) Cubic geometry (B) Monoclinic geometry
(C) Tetragonal geometry (D) Orthorhombic geometry

5. Which of the following solids substances will have same refractive index when measured in different directions?
(A) NaCl (B) Monoclinic sulphur (C) Rubber (D) Graphite

6. In the body-centred cubic unit cell & face centred cubic unit cell, the radius of atom in terms of edge lengh(a) of the unit cell is respectively:
(A) $\frac{a}{2}, \frac{a}{2\sqrt{2}}$ (B) $\frac{a}{2\sqrt{2}}, \frac{\sqrt{3}a}{4}$ (C) $\frac{\sqrt{3}a}{4}, \frac{a}{2\sqrt{2}}$ (D) $\frac{\sqrt{3}a}{2}, \frac{a}{2\sqrt{2}}$

7. Crystal system in which maximum number of Bravais lattices are possible is
(A) Cubic (B) Triclinic (C) Orthorhombic (D) Rhombohedral

8. Number of atoms at second nearest position from a given atom in a BCC structure is
(A) 8 (B) 6 (C) 12 (D) 4

9. Percentage area of each face covered by atoms in a FCC unit cell is -
(A) 60.4% (B) 68% (C) 74% (D) 78.5%

10. Correct sequence of the coordination number in SC, FCC & BCC is-
(A) 6, 8, 12 (B) 6, 12, 8 (C) 8, 12, 6 (D) 8, 6, 12

11. If 'Z' is the number of atoms in the unit cell that represents the closest packing sequence ---A B C A B C ---, the number of tetrahedral voids in the unit cell is equal to
(A) Z (B) 2Z (C) Z/2 (D) Z/4

12. The interstitial hole is called tetrahedral because
(A) It is formed by four spheres.
(B) Partly same and partly different.
(C) It is formed by four spheres the centres of which form a regular tetrahedron.
(D) None of the above three.

13. The size of an octahedral void formed in a closest packed lattice as compared to tetrahedral void is
(A) Equal (B) Smaller (C) Larger (D) Not definite

ALLEN Session 2019-20

14. If the anions (A) form hexagonal closest packing and cations (C) occupy only 2/3 octahedral voids in it, then the general formula of the compound is

(A) CA (B) CA_2 (C) C_2A_3 (D) C_3A_2

15. Which one of the following schemes of ordering closest packed sheets of equal sized spheres do not generate close packed lattice.

(A) ABCABC (B) ABACABAC (C) ABBAABBA (D) ABCBCABCBC

16 . Copper metal crystallizes in FCC lattice. Edge length of unit cell is 362 pm. The radius of largest atom that can fit into the voids of copper lattice without disturbing it is

(A) 53 pm (B) 45 pm (C) 93 pm (D) 87 pm

17. Packing fraction in 2-D hexagonal arrangement of identical sphere is

(A) $\frac{\pi}{3\sqrt{2}}$ (B) $\frac{\pi}{3\sqrt{3}}$ (C) $\frac{\pi}{2\sqrt{3}}$ (D) $\pi/6$

18. In fcc unit cell smallest distance between octahedral void & tetrahedral void is -

(a = edge length of unit cell)

(A) $\frac{a}{\sqrt{2}}$ (B) $\frac{\sqrt{3}a}{2}$ (C) a (D) $\frac{\sqrt{3}a}{4}$

19. What is not true regarding hexagonal close packing (hcp)

(A) packing fraction is 0.74

(B) coordination number is 12

(C) ABC ABC.....type packing

(D) Containing both tetrahedral and octahedral voids

20. In which of the following arrangement distance between two nearest neighbours is maximum, considering identical sized atoms in all arrangements ?

(A) Simple cubic (B) bcc (C) fcc (D) equal in all

21. How many unit cell are there in 1 gram cubic crystal of NaCl ?

(A) $\frac{4\times N_A}{58.5}$ (B) $\frac{N_A}{58.5}$ (C) $\frac{N_A}{58.5\times 4}$ (D) $\frac{N_A}{58.5\times 8}$

22. The density of CaF_2 (fluorite structure) is 3.18 g/cm^3. The length of the side of the unit cell is (Ca = 40, F = 19)

(A) 253 pm (B) 344 pm (C) 546 pm (D) 273 pm

23. The coordination number of cation and anion in Fluorite CaF_2 and CsCl are respectively

(A) 8:4 and 6:3 (B) 6:3 and 4:4 (C) 8:4 and 8:8 (D) 4:2 and 2:4

24. A compound XY crystallizes in 8 : 8 lattice with unit cell edge lenght of 480 pm. If the radius of Y^- is 225 pm, then the radius of X^+ is

(A) 127.5 pm (B) 190.68 pm (C) 225 pm (D) 255 pm

25. The mass of a unit cell of CsCl corresponds to

(A) 1 Cs^+ and 1 Cl^- (B) 1 Cs^+ and 6 Cl^- (C) 4 Cs^+ and 4 Cl^- (D) 8 Cs^+ and 1 Cl^-

ALLEN Session 2019-20

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26. An ionic compound AB has ZnS type structure. If the radius A^+ is 22.5 pm, then the ideal radius of B^- would be

(A) 54.35 pm (B) 100 pm (C) 145.16 pm (D) none of these

27. Edge length of M^+X^- (NaCl structure) is 7.2 Å. Assuming $M^+ - X^-$ contact along the cell edge, radius of X^- ion is ($r_{M^+} = 1.6$Å) :

(A) 2.0 Å (B) 5.6 Å (C) 2.8 Å (D) 38 Å

28. NH_4Cl crystallizes in CsCl type lattice with a unit cell edge length of 387 pm. The distance between the oppositively charged ions in the lattice is

(A) 335.1 pm (B) 83.77 pm (C) 274.46 pm (D) 137.23 pm

29. $r_{Na^+} = 95$ pm and $r_{Cl^-} = 181$ pm in NaCl (rock salt) structure. What is the shortest distance between Na^+ ions?

(A) 778.3 pm (B) 276 pm (C) 195.7 pm (D) 390.3 pm

30. AB crystallises itself as NaCl crystal. If $r_+ = \frac{2}{\sqrt{6}}$ and $r_- = \sqrt{6}$, the edge length of cube is

(A) $2\sqrt{3}$ (B) $\frac{4}{\sqrt{3}}$ (C) $\frac{8}{\sqrt{6}}$ (D) $\frac{16}{\sqrt{6}}$

31. Which of the following is the most likely to show schottky defect ?

(A) CaF_2 (B) ZnS (C) AgCl (D) CsCl

32. In the Schottky defect, in AB type ionic solids

(A) cations are missing from the lattice sites and occupy the interstitial sites

(B) equal number of cations and anions are missing

(C) anions are missing and electrons are present in their place

(D) equal number of extra cations and electrons are present in the interstitial sites

33. Choose the correct option.

(A) Two adjacent face centre atom doesn't touch each other in fcc unit cell because they are not nearest atom of face each other in fcc lattice

(B) Number of nearest Na^+ ions of another Na^+ in Na_2O crystal will be 24.

(C) Minimum distance between two cubical voids in simple cube unit cell lattice will be 'a' where 'a' is length of edge of unit cell

(D) By defects in solids, density of solids either remains constant or decreases but it can never increase.

34. The measured density of AgI is 6.94 g/cm^{-3} and the theoretical density is 5.67 g/cm^{-3}. These data indicate that solid AgI has -

(A) Schottky defect (B) Frenkel defect

(C) Interstitial impurities defect (D) Both (1) and (2)

35. Which of the following statement is **CORRECT** ?

(A) A metal can show only non- stoichiometric defects

(B) Schottky defect reduces the density of a solid due to significant increase in volume.

(C) Impurity defect always change the density.

(D) Solids having F-centres may have metal excess defect due to missing anions.

ALLEN Session 2019-20

EXERCISE # O-II

SINGLE CORRECT :

1. The only incorrect statement for the packing of identical spheres in two dimension is :

(A) For square close packing , coordination number is 4.

(B) For hexagonal close packing, coordination number is 6.

(C) There is only one void per atom in both, square and hexagonal close packing.

(D) Hexagonal close packing is more efficiently packed than square close packing.

2. Correct statement for ccp is :

(A) Each octahedral void is surrounded by 6 spheres and each sphere is surrounded by 4 octahedral voids

(B) Each octahedral void is surrounded by 6 spheres and each sphere is surrounded by 6 octahedral voids

(C) Each octahedral void is surrounded by 6 spheres and each sphere is surrounded by 8 octahedral voids

(D) Each octahedral void is surrounded by 6 spheres and each sphere is surrounded by 12 octahedral voids

3. Which of the following statements is correct in the rock-salt structure of an ionic compounds?

(A) coordination number of cation is four whereas that of anion is six.

(B) coordination number of cation is six whereas that of anion is four.

(C) coordination number of each cation and anion is four.

(D) coordination number of each cation and anion is six.

MORE THAN ONE MAY BE CORRECT :

4. Which of the following statements is/are correct :

(A) In an anti-fluorite structure, anions form FCC lattice and cations occupy all tetrahedral voids.

(B) If the radius of cations and anions are 0.2 Å and 0.95 Å , then coordination number of cation in the crystal is 4.

(C) Each sphere is surrounded by six voids in two dimensional hexagonal close packed layer.

(D) 8 Cs^+ ions occupy the second nearest neighbour locations of a Cs^+ ion in CsCl crystals.

5. Select correct statement(s)

(A) Density of crystal always increases due to substitutional impurity defect.

(B) An ion is transferred from a lattice site to an interstitial position in Frenkel defect.

(C) In AgCl, the silver ion is displaced from its lattice position to an interstitial position. Such a defect is called a frenkel defect

(D) None

6. Lead metal has a density of 11.34 g/cm^3 and crystallizes in a face–centred lattice. Choose the correct alternatives (Pb = 208 , $N_A = 6 \times 10^{23}$)

(A) the volume of one unit cell is 1.22×10^{-22} cm^3.

(B) the volume of one unit cell is 1.22×10^{-19} cm^3.

(C) the atomic radius of lead is 175 pm.

(D) the atomic radius of lead is 155.1 pm.

ALLEN Session 2019-20

7. Which of the following statement(s) is/are correct ?

(A) NaCl is a 'AB' crystal lattice that can be interpreted to be made up of two individual fcc unit cells of A^+ and B^- fused together in such a manner that the corner of one unit cell becomes the edge centre of the other.

(B) In a face centred cubic unit cell, the body centre is an octahedral void.

(C) In fcc unit cell, octahedral and tetrahedral voids are equal in number.

(D) Tetrahedral voids = 2 × octahedral voids, is valid for ccp and hcp.

8. Select the correct statement (s) :

(A) CsCl mainly shows Schottky defect
(B) ZnS mainly shows Frenkel defect
(C) NaCl unit cell contain $4Na^+$ and $4Cl^-$
(D) Truncated octahedron have 24 corners.

9. Select the correct statement(s) –

(A) The ionic crystal of AgBr has Schottky defect.

(B) The unit cell having crystal parameters, $a = b \neq c$, $\alpha = \beta = 90°$, $\gamma = 120°$ is hexagonal

(C) Ionic compounds having Frenkel defect has high r^+/r^- ratio.

(D) The co-ordination number of Na^+ ion in NaCl is 6

10. Which of the following is/are true ?

(A) Ratio of nearest neighbours in simple cubic cell to next nearest neighbours in face centred is cubic cell is 1.

(B) Packing efficiency of a unit cell in which atoms are present at each corner and each edge centre is about 26 % in metallic crystal.

(C) Distance between two planes in FCC or HCP arrangement is same for a metal existing in both forms, with same atomic radius.

(D) If number of unit cell along one edge are 'x' , then total number of unit cell in cube = x^3

ASSERTION / REASON :

11. **Statement-1** : In Antifluorite structure (Li_2O), the oxide ions occupy c.c.p. (cubic close packing) and Li^+ ions, 100% tetrahedral voids.

Statement-2 : The distance of the nearest neighbours in antifluorite structure is $\frac{\sqrt{3}a}{4}$, where 'a' is the edge length of the cube

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

12. **Statement-1 :** In FCC unit cell, packing efficiency is more when all tetrahedral voids are filled with spheres of maximum possible size as compared with packing efficiency when all octahedral voids are filled in similar way.

Statement-2 : Tetrahedral voids are more in the number than octahederal voids in FCC.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

Session 2019-20

13. **Statement-1 :** In diamond , carbon atoms occupy alternate tetrahedral voids in the FCC lattice formed by the carbon atoms.

Statement-2 : In diamond, packing fraction is more than 74%.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

14. **Statement-1 :** Due to Frenkel defect, there is no effect on the density of the crystalline solid.

Statement-2 : In Frenkel defect, no cation or anion leaves the crystal.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

15. **Statement-1 :** Conductivity of silicon increased by doping it with group 15 element.

Statement-2 : Doping means introduction of small amount of impurities like P or As into pure silicon crytstal.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

COMPREHENSION :

Paragraph for Q.16 & Q.17

Solid balls of radius 17.32 cm crystallises in bcc pattern. During one such crystallisation, some oxygen gas is trapped. This trapped oxygen at 640K creates pressure of 5 atm.

Assume :

(i) BCC arrangement is not disturbed due to trapping of gas.

(ii) Gas is uniformly distributed inside unit cell

[Take R = 0.08 atm-litre/mole-K , $N_A = 6 \times 10^{23}$, Mass of a solid ball = 64 gms]

16. Number of oxygen molecules present in an unit cell is -

(A) 2.4×10^{24} (B) 1.2×10^{24} (C) 6×10^{23} (D) 3×10^{23}

17. Calculate percentage increase in density due to trapping of gas

(A) 16.67 % (B) 33.33 % (C) 100% (D) 50%

Paragraph for Q.18 to Q.21

Calcium crystallizes in a cubic unit cell with density 3.2 g/cc. Edge-length of the unit cell is 437 pm.

18. The type of unit cell is

(A) Simple cubic (B) BCC (C) FCC (D) Edge-centred

19. The nearest neighbour distance is

(A) 154.5 pm (B) 309 pm (C) 218.5 pm (D) 260 pm

20. The number of nearest neighbours of a Ca atom are

(A) 4 (B) 6 (C) 8 (D) 12

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21. If the metal is melted, density of the molten metal was found to be 3 g/cc. What will be the percentage of empty space in the molten metal?

(A) 31% (B) 36% (C) 28% (D) 49%

***TABLE TYPE* COMPREHENSION :**

Column-I	Column-II	Column-III
(A) NaCl (Rock salt) structure	(i) Cation - FCC Anion - Tetrahedral void	(I) All tetrahedral voids are occupied
(B) CsCl structure	(ii) Anion - FCC Cation - Tetrahedral voids	(II) All octahedral voids are occupied
(C) ZnS (zinc blende) structure	(iii) Anion - SC Cation - Cubic voids	(III) 50 % of tetrahedral voids are occupied
(D) CaF_2 (fluorite) structure	(iv) Anion - FCC Cation - Octahedral void	(IV) All octahedral voids are empty

22. Which of the following is correct match ?

(A) A, i, I (B) A, ii, IV (C) A, iv, II D) A, iv, IV

23. Which of the following is incorrect match ?

(A) B, iii, I (B) C, ii, III (C) D, i, I D) D, i, IV

24. Which of the following is correct match ?

(A) D, ii, I (B) B, iv, IV (C) D, iii, I D) C, ii, IV

MATCH THE COLUMN :

25. Match the column

	Column I		Column II
(A)	Tetragonal and Hexagonal	(P)	are two crystal systems
(B)	Cubic and Rhombohedral	(Q)	$a = b \neq c$
(C)	Monoclinic and Triclinic	(R)	$a \neq b \neq c$
(D)	Cubic and Hexagonal	(S)	$a = b = c$

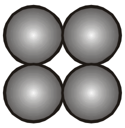
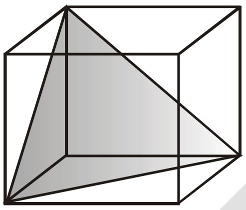
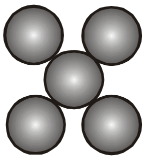
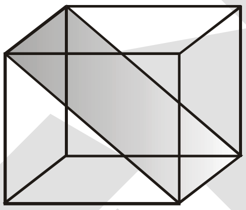
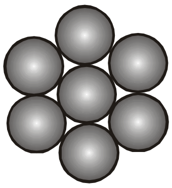
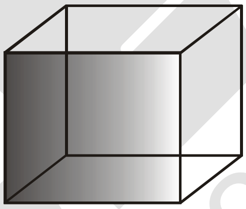
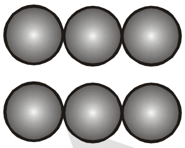
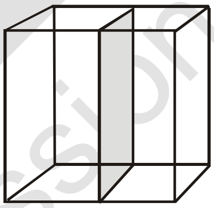
26. Match the column:

	Column I		Column II
(A)	Rock salt structure	(P)	Co-ordination number of cation is 4
(B)	Zinc Blend structure	(Q)	$\frac{\sqrt{3}a}{4} = r_+ + r_-$
(C)	Flourite structure	(R)	Co-ordination number of cation and anion are same
		(S)	Distance between two nearest anion is $\frac{a}{\sqrt{2}}$

ALLEN Session 2019-20

***MATCHING LIST TYPE* :**

27. Match the column

Column I (Arrangement of the atoms/ions)		**Column II** (Planes in fcc lattice)	
(P)		(1)	
(Q)		(2)	
(R)		(3)	
(S)		(4)	

Code :

	P	Q	R	S
(A)	4	3	1	2
(B)	4	3	2	1
(C)	3	2	1	4
(D)	1	2	4	3

Session 2019-20

28. **Column I** **Column II**

[Distance in terms of edge length of cube (a)]

(P)	0.866 a	(1)	Shortest distance between cation & anion in CsCl structure.
(Q)	0.707 a	(2)	Shortest distance between two cation in CaF_2 structure.
(R)	0.433 a	(3)	Shortest distance between carbon atoms in diamond.
(S)	a	(4)	shortest distance between next nearest cations in rock salt structrue.

Code:

	P	Q	R	S
(A)	4	3	1	2
(B)	1	2	3	4
(C)	3	2	1	4
(D)	1	2	4	3

ALLEN
Session 2019-20

EXERCISE # J-MAIN

1. The no. of atoms per unit cell in B.C.C. & F.C.C. is respectively : **[AIEEE-02]**

(1) 8, 10 (2) 2, 4 (3) 1, 2 (4) 1, 3

2. How many unit cells are present in a cube-shaped ideal crystal of NaCl of mass 1.00g ?

(1) 1.28×10^{21} unit cells (2) 1.71×10^{21} unit cells **[AIEEE-03]**

(3) 2.57×10^{21} unit cells (4) 5.14×10^{21} unit cells

3. What type of crystal defect is indicated in the diagram below ? **[AIEEE-04]**

Na^+	Cl^-	Na^+	Cl^-	Na^+	Cl^-
Cl^-		Cl^-	Na^+		Na^+
Na^+	Cl^-		Cl^-	Na^+	Cl^-
Cl^-	Na	$^+Cl^-$	Na^+		Na^+

(1) Frenkel defect (2) Schottky defect

(3) Interstitial defect (4) Frenkel and Schottky defects

4. An ionic compound has a unit cell consisting of A ions at the corners of a cube and B ions on the centres of the faces of the cube. The empirical formula of this compound would be- **[AIEEE-05]**

(1) A_2B (2) AB (3) A_3B (4) AB_3

5. Lattice energy of an ionic compound depends upon - **[AIEEE-05]**

(1) Size of the ion only (2) Charge on the ion only

(3) Charge on the ion and size of the ion (4) Packing of ions only

6. Total volume of atoms present in a face-centred cubic unit cell of a metal is (r is atomic radius) : **[AIEEE-06]**

(1) —π (2) —π (3) —π (4) —π

7. In a compound, atoms of element Y form ccp lattice and those of element X occupy 2/3rd of tetrahedral voids. The formula of the compound will be - **[AIEEE-08]**

(1) X_4Y_3 (2) X_2Y_3 (3) X_2Y (4) X_3Y_4

8. The edge length of a face centred cubic cell of an ionic substance is 508 pm. If the radius of the cation is 110 pm, the radius of the anion is :- **[AIEEE-10]**

(1) 144 pm (2) 288 pm (3) 398 pm (4) 618 pm

9. Percentages of free space in cubic close packed structure and in body centred packed structure are respectively :- **[AIEEE-10]**

(1) 48% and 26% (2) 30% and 26% (3) 26% and 32% (4) 32% and 48%

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10. The radius of a calcium ion is 94 pm and of the oxide ion is 146 pm. The possible crystal structure of calcium oxide will be :- **[Jee-Main (online)-12]**

(1) Octahedral (2) Tetrahedral (3) Pyramidal (4) Trigonal

11. Ammonium chloride crystallizes in a body centred cubic lattice with edge length of unit cell of 390 pm. If the size of chloride ion is180 pm, the size of ammonium ion would be: **[Jee-Main (online)-12]**

(1) 158 pm (2) 174 pm (3) 142 pm (4) 126 pm

12. A solid has 'bcc' structure. If the distance of nearest approach between two atoms is 1.73 Å, the edge length of the cell is :- **[Jee-Main (online)-12]**

(1) 314.20 pm (2) 216 pm (3) 200 pm (4) 1.41 pm

13. Among the following the incorrect statement is :- **[Jee-Main (online)-12]**

(1) Density of crystals remains unaffected due to Frenkel defect

(2) In BCC unit cell the void space is 32%

(3) Electrical conductivity of semiconductors and metals increases with increase in temperature

(4) Density of crystals decreases due to Schottky defect

14. Copper crystallises in fcc with a unit cell edge length of 361pm. What is the radius of copper atom? **[AIEEE-2011]**

(1) 181pm (2) 128pm (3) 157pm (4) 108pm

15 Lithium forms body centred cubic structure. The length of the side of its unit cell is 351 pm. Atomic radius of the lithium will be :- **[Jee-Main (offline)-12]**

(1) 152 pm (2) 75 pm (3) 300 pm (4) 240 pm

16. In a face centred cubic lattice, atoms of A form the corner points and atoms of B form the face centred points. If two atoms of A are missing from the corner points, the formula of the ionic compound is **[Jee-Main (online)-13]**

(1) AB_2 (2) AB_3 (3) AB_4 (4) A_2B_5

17. Which one of the following statements about packing in solids is **incorrect** ? **[Jee-Main (online)-13]**

(1) Void space in ccp mode of packing is 26%

(2) Coordination number in hcp mode of packing is 12

(3) Void space in hcp mode of packing is 32%

(4) Coordination number in bcc mode of packing is 8

18. An element having an atomic radius of 0.14 nm crystallizes in an fcc unit cell. What is the length of a side of the cell ? **[Jee-Main (online)-13]**

(1) 0.96 nm (2) 0.4 nm (3) 0.24 nm (4) 0.56 nm

ALLEN Session 2019-20

19. Experimentally it was found that a metal oxide has formula $M_{0.98}O$. Metal M, is present as M^{2+} and M^{3+} in its oxide. Fraction of the metal which exists as M^{3+} would be :- **[Jee-Main (offline)-13]**

(1) 7.01% (2) 4.08% (3) 6.05% (4) 5.08

20. The total number of octahedral void(s) per atom present in a cubic close packed structure is :- **[Jee-Main (online)-14]**

(1) 1 (2) 2 (3) 3 (4) 4

21. In a monoclinic unit cell, the relation of sides and angles are respectively **[Jee-Main (online)-14]**

(1) $a \neq b \neq c$ and $\alpha \neq \beta \neq \gamma \neq 90^o$ (2) $a \neq b \neq c$ and $\beta = \gamma = 90^o \neq \alpha$

(3) $a = b \neq c$ and $\alpha = \beta = \gamma = 90^o$ (4) $a \neq b \neq c$ and $\alpha = \beta = \gamma = 90^o$

22. The appearance of colour in solid alkali metal halides is generally due to : **[Jee-Main (online)-14]**

(1) Frenkel defect (2) F-centres (3) Schottky defect (4) Interstitial position

23. In a face centred cubic lattice atoms A are at the corner points and atoms B at the face centred points. If atom B is missing from one of the face centred points, the formula of the ionic compound is : **[AIEEE-2011, Jee-Main (online)-14]**

(1) AB_2 (2) A_2B_3 (3) A_5B_2 (4) A_2B_5

24. CsCl crystallises in body centred cubic lattice. if 'a' is its edge length then which of the following expression is correct : **[Jee-Main (offline)-14]**

(1) $r_{Cs^+} + r_{Cl^-} = \frac{\sqrt{3}}{2}a$ (2) $r_{Cs^+} + r_{Cl^-} = \sqrt{3}a$ (3) $r_{Cs^+} + r_{Cl^-} = 3a$ (4) $r_{Cs^+} + r_{Cl^-} = \frac{3a}{2}$

25. Sodium metal crystallizes in a body centred cubic lattice with a unit cell edge of 4.29Å. The radius of sodium atom is approximately :- **[Jee-Main (offline)-15]**

(1) 5.72Å (2) 0.93Å (3) 1.86Å (4) 3.022Å

26. Which of the following compounds is metallic and ferromagnetic ? **[Jee-Main (offline)-16]**

(1) MnO_2 (2) TiO_2 (3) CrO_2 (4) VO_2

27. A metal crystallises in a face centred cubic structure. If the edge length of its unit cell is 'a', the closest approach between two atoms in metallic crystal will be :- **[Jee-Main (offline)-17]**

(1) 2a (2) $2\sqrt{2}\,a$ (3) $\sqrt{2}\,a$ (4) $\frac{a}{\sqrt{2}}$

28. Which type of effect 'defect' has the presence of cations in the interstitial sites -

(1) Vacancy defect (2) Frenkel defect **[Jee-Main (offline)-18]**

(3) Metal deficiency defect (4) Schottky defect

29. All of the following share the same crystal structure except :- **[Jee-Main (online)-18]**

(1) RbCl (2) CsCl (3) LiCl (4) NaCl

30. Which of the following arrangements shows the schematic alignment of magnetic moments of antiferromagnetic substance ? **[Jee-Main (online)-18]**

(1) $\uparrow \downarrow \downarrow \downarrow \downarrow \uparrow$ (2) $\uparrow \uparrow \uparrow \uparrow \uparrow \uparrow$

(3) $\uparrow \uparrow \downarrow \uparrow \uparrow \downarrow$ (4) $\uparrow \downarrow \uparrow \downarrow \uparrow \downarrow$

ALLEN Session 2019-20

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EXERCISE # J-ADVANCED

1. A metal crystallises into two cubic phases, FCC and BCC whose unit cell lengths are 3.5 and 3.0 Å respectively. Calculate the ratio of densities of FCC and BCC. **[JEE-1999]**

2. The coordination number of a metal crystallising in a hcp structure is **[JEE-2000]**

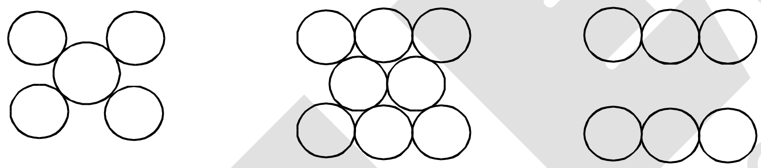
(A) 12 (B) 4 (C) 8 (D) 6

3. In any ionic solid [MX] with schottky defects , the number of positive and negative ions are same. **[T/F]** **[JEE-2000]**

4. In a solid "AB" having NaCl structure "A" atoms occupy the corners of the cubic unit cell. If all the face-centred atoms along one of the axes are removed, then the resultant stoichiometry of the solid is

(A) AB_2 (B) A_2B (C) A_4B_3 (D) A_3B_4 **[JEE-2000]**

5. The figures given below show the location of atoms in three crystallographic planes in FCC lattice. Draw the unit cell for the corresponding structure and identify these planes in your diagram. **[JEE-2000]**



6. A substance A_xB_y crystallises in a FCC lattice in which atoms "A" occupy each corner of the cube and atoms "B" occupy the centres of each face of the cube. Identify the correct composition of the substance A_xB_y.

(A) AB_3

(B) A_4B_3

(C) A_3B

(D) composition cannot be specified **[JEE-2002]**

7. Marbles of diameter 10 mm each are to be arranged on a flat surface so that their centres lie within the area enclosed by four lines of length each 40 mm. Sketch the arrangement that will give the maximum number of marbles per unit area, that can be enclosed in this manner and deduce the expression to calculate it. **[JEE 2003]**

8. (i) AB crystallizes in a rock salt structure with A : B = 1 : 1. The shortest distance between A and B is $Y^{1/3}$ nm. The formula mass of AB is 6.023 Y amu where Y is any arbitrary constant. Find the density in kg m^{-3}.

(ii) If measured density is 20 kg m^{-3}. Identify the type of point defect. **[JEE–2004]**

9. Which of the following FCC structure contains cations in alternate tetrahedral voids?

(A) NaCl (B) ZnS (C) Na_2O (D) CaF_2 **[JEE 2005]**

10. An element crystallises in FCC lattice having edge length 400 pm. Calculate the maximum diameter which can be placed in interstital sites without disturbing the structure. **[JEE 2005]**

11. The edge length of unit cell of a metal having atomic weight 75 g/mol is 5 Å which crystallizes in cubic lattice. If the density is 2 g/cc then find the radius of metal atom. ($N_A = 6 \times 10^{23}$). Give the answer in pm. **[JEE 2006]**

12. Match the crystal system / unit cells mentioned in Column I with their characteristic features mentioned in Column II. Indicate your answer by darkening the appropriate bubbles of the 4 × 4 matrix given in the ORS.

Column I	**Column II**
(A) simple cubic and face-centred cubic	(P) have these cell parameters a = b = c and $\alpha = \beta = \gamma$
(B) cubic and rhombohedral	(Q) are two crystal systems
(C) cubic and tetragonal	(R) have only two crystallographic angles of 90°
(D) hexagonal and monoclinic	(S) belong to same crystal system.

[JEE 2007]

Paragraph for Question No.13 to 15

In hexagonal systems of crystals, a frequently encountered arrangement of atoms is described as a hexagonal prism. Here, the top and bottom of the cell are regular hexagons and three atoms are sandwiched in between them. A space-filling model of this structure, called hexagonal close-packed (HCP), is constituted of a sphere on a flat surface surrounded in the same plane by six identical spheres as closely as possible. Three spheres are then placed over the first layer so that they touch each other and represent the second layer. Each one of these three spheres touches three spheres of the bottom layer. Finally, the second layer is covered with a third layer that is identical to the bottom layer in relative position. Assume radius of every sphere to be 'r'.

13. The number of atoms in this HCP unit cells is **[JEE 2008]**

(A) 4 (B) 6 (C) 12 (D) 17

14. The volume of this HCP unit cell is **[JEE 2008]**

(A) $24\sqrt{2}\,r^3$ (B) $16\sqrt{2}\,r^3$ (C) $12\sqrt{2}\,r^3$ (D) $\frac{64}{3\sqrt{3}}\,r^3$

15. The empty space in this HCP unit cell is **[JEE 2008]**

(A) 74% (B) 47.6 % (C) 32% (D) 26%

16. The correct statement(s) regarding defects in solid is (are) **[JEE 2009]**

(A) Frenkel defect is usually favoured by a very small difference in the sizes of cation and anion.

(B) Frenkel defect is a dislocation defect

(C) Trapping of an electron in the lattice leads to the formation of F-center.

(D) Schottky defects have no effect on the physical properties of solids.

ALLEN Session 2019-20

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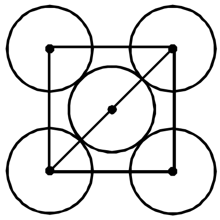
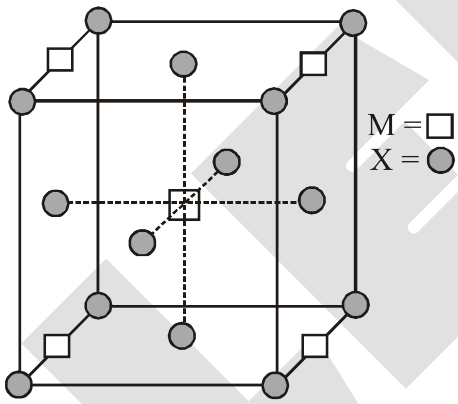
17. The packing effeciency of the two-dimensional square unit cell shown below is **[JEE-2010]**



(A) 39.27% (B) 68.02% (C) 74.05% (D) 78.54%

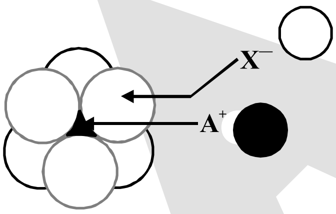
18. The number of hexagonal faces that present in a truncated octahedron is. **[JEE-2011]**

19. A compound M_pX_q has cubic close packing (ccp) arrangement of X. Its unit cell structure is shown below. The empirical formula of the compound is : **[JEE-2012]**



(A) MX (B) MX_2 (C) M_2X (D) M_5X_{14}

20. The arrangement of X^- ions around A^+ ion in solid AX is given in the figure (not drawn to scale). If the radius of X^- is 250 pm, the radius of A^+ is - **[JEE-2013]**



(A) 104 pm (B) 125 pm (C) 183 pm (D) 57 pm

21. If the unit cell of a mineral has cubic close packed (ccp) array of oxygen atoms with m fraction of octahedral holes occupied by aluminium ions and n fraction of tetrahedral holes occupied by magnesium ions m and n respectively, are - **[JEE-2015]**

(A) $\frac{1}{2},\frac{1}{8}$ (B) $1,\frac{1}{4}$ (C) $\frac{1}{2},\frac{1}{2}$ (D) $\frac{1}{4},\frac{1}{8}$

22. The **CORRECT** statement(s) for cubic close packed (ccp) three dimensional structure is (are)

(A) The number of the nearest neighbours of an atom present in the topmost layer is 12

(B) The efficiency of atom packing is 74% **[JEE-2016]**

(C) The number of octahedral and tetrahedral voids per atom are 1 and 2, respectively

(D) The unit cell edge length is $2\sqrt{2}$ times the radius of the atom

23. A crystalline solid of a pure substance has a face-centred cubic structure with a cell edge of 400 pm. If the density of the substance in the crystal is 8g cm^{-3}, then the number of atoms present in 256g of the crystal is $N \times 10^{24}$. The value of N is **[JEE-2017]**

24. Consider an ionic solid MX with NaCl structure. Construct a new structure (Z) whose unit cell is constructed from the unit cell of MX following the sequential instructions given below. Neglect the charge balance. **[JEE-2018]**

(i) Remove all the anions (X) except the central one

(ii) Replace all the face centered cations (M) by anions (X)

(iii) Remove all the corner cations (M)

(iv) Replace the central anion (X) with cation (M)

The value of $\left(\frac{\text{number of anions}}{\text{number of cations}}\right)$ in Z is____.

ALLEN Session 2019-20

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ANSWER KEY

EXERCISE # S-I

1. Ans. (a) 300 pm, (b) 255 pm, (c) 210 pm
2. Ans.438 pm, 219 pm
3. Ans.5.83 g cm^{-3}
4. Ans.($\sqrt{2}:1$)
5. Ans. (0.9 gm/cm^{-3})
6. Ans.AB
7. Ans.(64)
8. Ans.(12)
9. Ans.(6)
10. Ans (2)
11. Ans.A_2BC
12. Ans. (4)
13. Ans.(0.293)
14. Ans. (6×10^{17})
15. Ans. (2) $\dfrac{\frac{a}{\sqrt{2}} \times \sqrt{2}}{a/2} = 2$
16. Ans.(2.07 pm)
17. Ans.(6)
18. Ans. (60%)
19. Ans. 41.67 g cm^{-3}
20. Ans.0.60
21. Ans.4, 6, 8
22. Ans.(6)
23. Ans.$ZnAl_2O_4$
24. Ans.267 pm, 534 pm, 377.6 pm
25. Ans.346.4 pm
26. Ans.4
27. Ans.a =600 pm, V=2.16×10^{-22} cm^3
28. Ans.1.81Å
29. Ans.4.34 Å
30. Ans.(5 gm/cm^3)
31. Ans.15.05 %
32. Ans.6.0×10^{18}
33. Ans.10
34. Ans. 12.5%
35. Ans. 3.0 gm/cm^3

EXERCISE # S-II

1. Ans.(2)
2. Ans.(5)
3. Ans.Ans.8
4. Ans.60%, +4
5. Ans.5×10^{24}
6. Ans.0.72
7. Ans. 4
8 Ans. (i) FCC (ii) 0.116 g/cc
9. Ans. (a) 1.143, (b) 1.2
10. Ans.(a) MnF_3, (b) 6, (c) 4.02Å
11. Ans. (a) 4.5 Å, (b) 5.2 Å, (c) 8, (d) 6, (e) 0.929 g/cm^3
12 Ans. 96%
13. Ans (4)
14. Ans. Mg_2SiO_4, Fe_2SiO_4, 57.14%

ALLEN Session 2019-20

EXERCISE # O-I

1.	Ans.(A)	2.	Ans.(D)	3.	Ans.(A)
4.	Ans.(D)	5.	Ans.(C)	6.	Ans.(C)
7.	Ans. (C)	8.	Ans (B)	9.	Ans (D)
10.	Ans (B)	11.	Ans.(B)	12.	Ans.(C)
13.	Ans.(C)	14.	Ans.(C)	15.	Ans.(C)
16.	Ans.(A)	17.	Ans.(B)	18.	Ans (D)
19.	Ans (C)	20.	Ans (D)	21.	Ans.(C)
22.	Ans.(C)	23.	Ans.(C)	24.	Ans.(B)
25.	Ans.(A)	26.	Ans.(B)	27.	Ans.(A)
28.	Ans.(A)	29.	Ans.(D)	30.	Ans.(D)
31.	Ans.(D)	32.	Ans.(B)	33.	Ans(C)
34.	Ans (C)	35.	Ans.(D)		

EXERCISE # O-II

1.	Ans. (C)	2.	Ans.(B)	3.	Ans.(D)
4.	Ans.(A,C)	5.	Ans.(B,C)	6.	Ans.(A,C)
7.	Ans.(A,B,D)	8.	Ans.(A,B,C,D)	9.	Ans.(A,B,D)
10.	Ans.(A,B,C,D)	11.	Ans.(B)	12.	Ans.(D)
13.	Ans.(C)	14	Ans.(A)	15.	Ans.(A)
16.	Ans.(B)	17.	Ans.(D)	18.	Ans.(C)
19.	Ans.(B)	20.	Ans.(D)	21.	Ans.(A)
22.	Ans.(C)	23.	Ans.(A)	24.	Ans.(D)

25. Ans.(A) $\rightarrow$ P, Q ; (B) $\rightarrow$ P,S ; (C) $\rightarrow$ P,R ; (D) $\rightarrow$ P

26. Ans.(A) $\rightarrow$ R,S ; (B) $\rightarrow$ P,Q,R,S ; (C) $\rightarrow$ Q

27.	Ans.(A)	28.	Ans.(B)

EXERCISE # J-MAIN

1.	Ans.(2)	2.	Ans.(3)	3.	Ans.(2)
4.	Ans.(4)	5.	Ans.(3)	6.	Ans.(3)
7.	Ans.(1)	8.	Ans.(1)	9.	Ans.(3)
10.	Ans.(1)	11.	Ans.(1)	12.	Ans.(3)
13.	Ans.(3)	14.	Ans.(2)	15	Ans.(1)
16.	Ans.(3)	17.	Ans.(3)	18.	Ans.(2)
19.	Ans.(2)	20.	Ans.(1)	21.	Ans.(2)
22.	Ans.(2)	23.	Ans.(4)	24.	Ans.(1)
25.	Ans.(3)	26.	Ans.(3)	27	Ans. (4)
28.	Ans. (2)	29.	Ans. (2)	30.	Ans. (4)

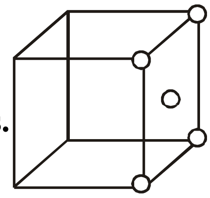
ALLEN Session 2019-20

EXERCISE # J-ADVANCED

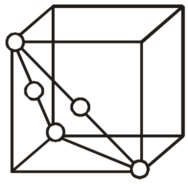
1. **Ans.1.259** **2.** **Ans.(A)** **3.** **Ans. True**

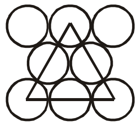
4. **Ans.(D)**

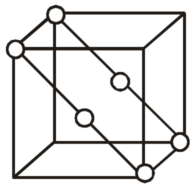
5. **Ans.**



fcc plan







6. **Ans.(A)** **7.** **Ans.18** **8.** **Ans.(i) = 5 kg m^{-3}**

9. **Ans.(B)** **10.** **Ans. 117.1 pm** **11.** **Ans. 216.5 pm**

12. **Ans. (A) P, S ; (B) -P,Q ; (C) - Q ; (D) - Q,R**

13. **Ans. (B)** **14.** **Ans. (A)** **15.** **Ans. (D)**

16. **Ans. (B,C)** **17.** **Ans. (D)** **18.** **Ans. (8)**

19. **Ans.(B)** **20.** **Ans. (A)** **21.** **Ans. (A)**

22. **Ans.(B,C,D)** **23.** **Ans.(2)** **24.** **Ans.(3)**

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IDEAL GAS

1. INTRODUCTION

Matter, as we know, broadly exist in three states - solid, liquid and gas.

There are always two opposite tendencies between particles of matter which determine the state of matter :

- Intermolecular forces.
- The molecular motion / random motion (energy of particles)

Intermolecular forces are the forces of attraction and repulsion between atoms or molecules. Attractive intermolecular forces are known as ***vander Waals forces***. These are dispersion forces, dipole-dipole forces & dipole induced forces. When two molecules are brought very close, they will exert repulsive forces. Magnitude of the repulsion rises very rapidly as the distance separating the molecules decreases. This is the reason that liquid & solids are hard to compress. In these states, molecules are already in close contact, therefore they resist further compression (in that case repulsive interaction will increase)

Thermal energy is the energy of a body arising from motion of its atoms or molecules. It is directly proportional to the temperature of the substance. It is the measure of average kinetic energy of the particles of the matter and is thus responsible for movement of particles. This movement of particles is called ***thermal motion***. Intermolecular forces tend to keep the molecules together but thermal energy of the molecules tends to keep them apart. Three states of matter are the result of balance between intermolecular forces and the thermal energy of the molecules.

2. GENERAL CHARACTERISTICS OF SOLID, LIQUID & GAS

Each physical state of matter possesses characteristics properties of its own. For example,

❖ **Solids :**

All solids show the following characteristics :

(i) Solids are rigid and incompressible.

(ii) Solids have fixed shape and definite volume.

(iii) Solids have their melting and boiling points above room temperature.

(iv) Density of solid is high.

❖ **Liquids :**

All liquids show the following characteristics :

(i) Liquids are almost incompressible but less incompressible than solids.

(ii) Liquids have fixed volume but no fixed shape.

(iii) Liquids have their melting points below room temperature and boiling points above room temperature, under normal conditions.

(iv) Density of liquids is lower than that of solids but much higher than that of gases.

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❖ **Gases :**

All gases show the following characteristics :

(i) Gases are highly compressible, i.e. gases can be compressed easily by applying pressure.

(ii) Gases have no fixed volume and shape. Gases fill the container of any size and shape completely.

(iii) Gases can diffuse into each other rapidly.

(iv) Gases have their melting and boiling points both below room temperature.

(v) Gases generally have low density.

3. MEASURABLE PROPERTIES OF GASES

The characteristics of gases are described fully in terms of four parameters or measurable properties :

(I) Amount of the gas (i.e., mass or number of moles).

(II) Volume (V) of the gas.

(III) Temperature (T)

(IV) Pressure (P)

I. Amount of the gas :

(i) The mass of a gas can be determined by weighing the container in which the gas is enclosed and again weighing the container after removing the gas. The difference between the two masses gives the mass of the gas.

Mass of gas (m) = Mass of filled container – mass of empty container

(ii) The mass of the gas is related to the number of moles of the gas as

Moles of gas (n) = Mass in grams / Molar mass = m/M

(iii) Mass is expressed in gram or kg.

II. Gas volume :

(i) Since gases occupy the entire space available to them, the measurement of volume of a gas only requires a measurement of the container confining the gas.

(ii) Volume is expressed in litres (L), millilitres (mL) or cubic centimeters (cm^3) or cubic meters (m^3).

(iii) $1\ L = 1000\ mL$; $1\ L = 1\ dm^3 = 10^{-3}\ m^3$

$1\ m^3 = 10^3\ dm^3 = 10^6\ cm^3 = 10^6\ mL = 10^3\ L$

$1\ mL$ or $1\ cc = 1\ cm^3$

III. Temperature :

(i) Gases expand on increasing the temperature.

(ii) Temperature is measured in degree centigrade (°C) or Celsius degree with the help of thermometers. Temperature is also measured in degree Fahrenheit (°F).

(iii) S.I. unit of temperature is kelvin (K) or absolute degree

$K = °C + 273$

(iv) Relation between °F and °C is

$°C/5 = (°F - 32) / 9$

Session 2019-20

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IV. Pressure :

Force exerted by the gas per unit area of the walls of the container in all directions. Thus,

Pressure (P) = Force(F) / Area(A)

❖ **Atmospheric pressure :**

The pressure exerted by atmosphere on earth's surface at sea level is called atmospheric pressure. Generally its unit is atm.

Pressure(P) = Force(F) / Area(A)

= Mass(m) × Acceleration(g) / Area(a)

= Volume × density × Acceleration(g) / Area(a)

= Area (a) × height (h) × density (ρ) × Acceleration(g) / Area(a)

Pressure(P) = hρg

where h = Height of mercury column in the barometer.

ρ = Density of mercury.

g = Acceleration due to gravity.

Pressure does not depend on the cross section of tube, but only on the vertical height of the Hg. If area is doubled, volume also gets doubled and mass will also gets doubled. Now it will rest on twice area but pressure exerted remains same.

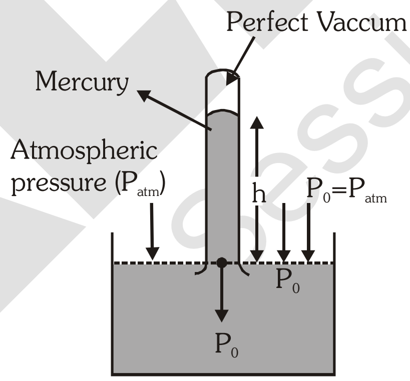
1 atm = 1.013 bar = 1.013×10^5 N/m^2 = 1.013×10^5 Pa

1 atm = 76 cm of Hg = 760 mm of Hg = 760 torr

□ **PRESSURE MEASURING DEVICES**

Generally, the instruments used for the calculation of pressure of a gas are ***barometer*** and ***manometer.***

(i) Barometer : A barometer is an instrument that is used for the measurement of atmospheric pressure. The construction of the barometer is as follows -



A thin narrow calibrated capillary tube is filled up to the brim, with a liquid such as mercury, and is inverted into a trough filled with the same fluid. Now depending on the external atmospheric pressure, the level of the mercury inside the tube will adjust itself, the reading of which can be monitored. When the mercury column inside the capillary comes to rest, then the net forces on the column should be balanced.

⇒ $P_0 \times A = \rho A \times gh$

⇒ $\boxed{P_0 = \rho gh}$; where ρ is the density of the fluid.

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(ii) Manometer :

(a) Open end manometer :

It consists of a U-shaped tube partially filled with mercury. One limb of the tube is shorter than the other. The shorter limb is connected to the vessel containing the gas whereas the longer limb is open as shown in fig. The mercury in the longer tube is subjected to the atmospheric pressure while mercury in the shorter tube is subjected to the pressure of the gas.

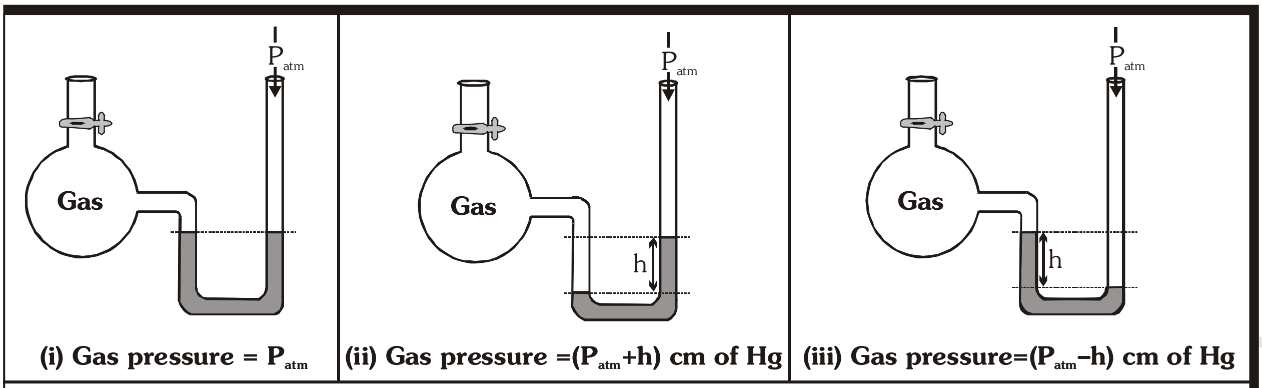


Fig : Measurement of gas pressure

Where P_{atm} = 76 cm of Hg and h = Height in cm of Hg

There are three possibilities as described below :

(i) If the level of Hg in the two limbs is same, then gas pressure = atmospheric pressure (P_{atm}).

(ii) If the level of Hg in the longer limbs is higher, gas pressure

$= P_{atm}$ + (difference between the two levels)

$= P_{atm} + h.$

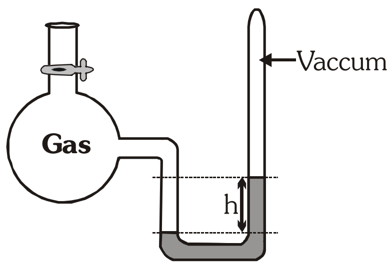
(iii) If the level of Hg in the shorter limb is higher, then gas pressure

$= P_{atm}$ – (difference between the two levels)

$= P_{atm} - h.$

(b) Closed end manometer :

This is generally used to measure low gas pressure. It also consists of U-tube with one limb shorter than the other and partially filled with mercury as shown in fig. The space above mercury on the closed end is completely evacuated. The shorter limb is connected to the vessel containing gas. The gas exerts pressure on the mercury in the shorter limb and forces its level down.



Closed end manometer

Gas pressure = [Difference in the Hg level in two limbs]

Ex.1 Why mercury is used in the barometer tube?

Sol. Mercury, a liquid with very high density, is normally used in the barometer because it does not stick to the surface of the glass tube. Mercury is also non-volatile at room temperature. Therefore, there are hardly any vapours of mercury above the liquid column and their pressure, if any , can be neglected. Due to high density of mercury, height of mercury column will be small and can be easily measured.

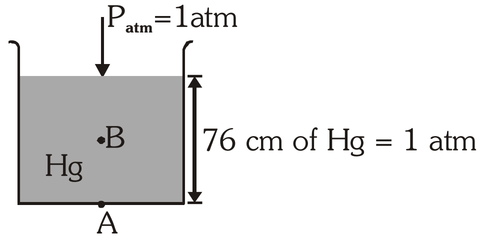
Ex.2 An open tank is filled with Hg upto a height of 76cm.

Find the pressure at the

(a) Bottom (A) of the tank

(b) Middle (B) of the tank.

(If atmospheric pressure is 1 atm)



Sol. **(a)** At bottom,

$P_A = P_{atm} + P_{Hg}$

$= 1 + 1 = 2$ atm

(b) At middle,

$P_B = P_{atm} + P_{Hg}$

$= 1 + \frac{1}{2} = 1.5$ atm

Ex.3 Find the height of water upto which water must be filled to create the same pressure at the bottom, as in above problem.

Given that $d_w = 1$ gm/cm^3, $d_{Hg} = 13.6$ gm/cm^3, $h_{Hg} = 76$ cm

Sol. $P_{water} = P_{Hg}$

$h_w d_w g = h_{Hg} d_{Hg} g$

$h_w d_w = h_{Hg} d_{Hg}$

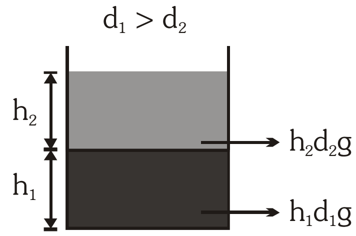
$h_w \times 1\ \text{g/cm}^3 = 76\ \text{cm} \times 13.6\ \text{g/cm}^3$

$h_w = 1033.6$ cm

Ex.4 What will be the pressure if two immiscible fluid is filled according to given diagram.

(a) Find the pressure at the bottom of tank.

(b) Find the pressure at the middle point of bottom layer.

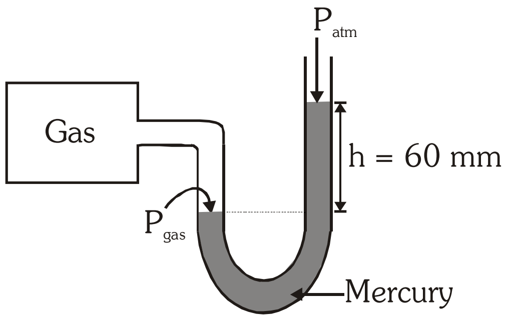


Sol. (a) $P_{atm} + h_2d_2g + h_1d_1g$;

(b) $P_{atm} + h_2d_2g + \frac{h_1}{2} d_1g$

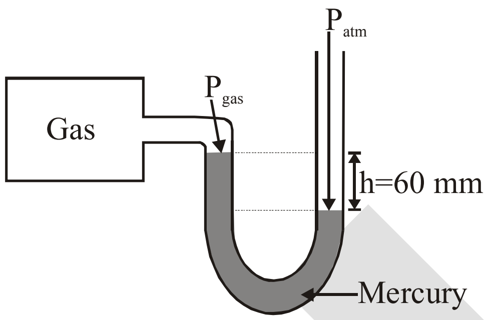
Ex.5 Find the pressure of the gas inside a container if the open manometer attached to the container shows a difference of 60 mm.

Sol. **Case-I :** $P_{atm} < P_{gas}$



$P_{gas} = P_{atm} + 60$ mm
$= 760$ mm $+ 60$ mm
$= 820$ mm of Hg

Case-II : $P_{atm} > P_{gas}$



$P_{atm} = P_{gas} + 60$ mm
$760 \text{ mm} = P_{gas} + 60$
$P_{gas} = 700$ mm

4. GAS LAWS

The behaviour of the gases is governed by same general laws, which were discovered as a result of their experimental studies. These laws are relationships between measurable properties of gases. Some of these properties like pressure, volume, temperature and mass are very important because relationships between these variables describe state of the gas.

The first reliable measurement on properties of gases was made by Anglo-Irish scientist Robert Boyle in 1662. The law which he formulated is known as ***Boyle's Law***. Later on attempts to fly in air with the help of hot air balloons motivated Jaccques Charles and Joseph Lewis Gay Lussac to discover additional gas laws. Contribution from *Avogadro* and others provided lot of information about gaseous state.

4.1 Boyle's Law :

For a fixed amount of gas at constant temperature, the volume occupied by the gas is inversely proportional to the pressure applied on the gas or pressure of the gas.

$\Rightarrow \quad V \propto \dfrac{1}{P}$

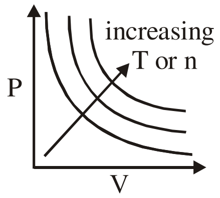
$\Rightarrow \quad$ Hence , PV = const. (K)

$\Rightarrow \quad \boxed{P_1V_1 = P_2V_2}$

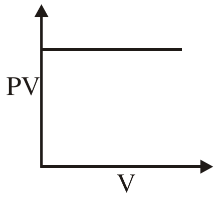
Session 2019-20

❖ **Graphical representation of Boyle's law :**

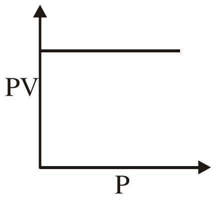
(i) P v/s V :



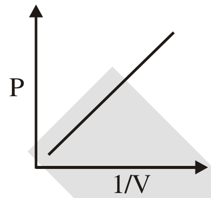
(ii) PV v/s V :



(iii) PV v/s P :



(iv) P v/s 1/V :



4.2. Charle's Law :

For a fixed amount of gas at constant pressure, volume occupied by the gas is directly proportional to temperature of the gas on absolute scale of temperature.

$\Rightarrow$ $V \alpha T$

$\Rightarrow$ $V = KT$

$$\boxed{\frac{V}{T} = \text{constant (K)}}$$

T = Temperature on absolute scale, kelvin scale or ideal gas scale.

$$\boxed{\frac{V_1}{T_1} = \frac{V_2}{T_2}}$$

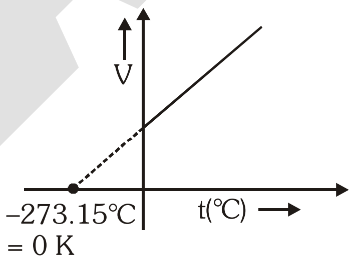
$\boxed{V = V_0 + bt}$ $\boxed{V = b'T}$

t = temperature on centigrade scale. T = absolute temperature (K)

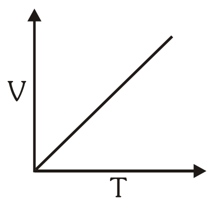
V_0 = volume of gas at 0°C. b, b' = constants

❖ **Graphical representation of Charle's Law :**

(i) V v/s t (°C) :



(ii) V v/s T :



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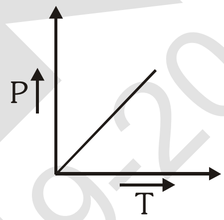
- **Important Points :**
  - Since volume is proportional to absolute temperature, the volume of a gas should be theoretically zero at absolute zero temperature.
  - In fact, no substance exists as gas at a temperature near absolute zero, though the straight line plots can be inter polated to zero volume. Absolute zero can never be attained practically though it can be approached only.
  - By considering –273.15°C as the lowest approachable limit, Kelvin developed temperature scale which is known as absolute scale.

4.3. Gay-lussac's law :

For a fixed amount of gas at constant volume, pressure of the gas is directly proportional to temperature of the gas on absolute scale of temperature.

$\Rightarrow \quad P \propto T$

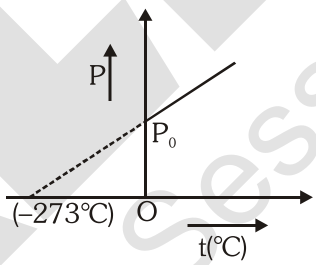
$\frac{P}{T} = \text{constant (K)}$



$\frac{P_1}{T_1} = \frac{P_2}{T_2}$ Temperature on absolute scale, kelvin scale or ideal gas scale.

Note: Originally, the law was developed on the centigrade scale, where it was found that pressure is a linear function of temperature $\Rightarrow \boxed{P = P_0 + bt}$,

where 'b' is a constant and 'P_0' is pressure at zero degree centigrade.



But for kelvin scale : $\boxed{P = b''T}$ where T is in K.

4.4. Avogadro's law :

Equal volumes of all the gases under similar conditions of temperature and pressure contains equal number of molecules or moles of molecules (not atoms).

$V \propto N$ (Temperature and pressure constant)

$V \propto n$ (Temperature and pressure constant)

Where, N = number of molecules, n = number of moles of molecules

$$\boxed{\frac{V_1}{N_1} = \frac{V_2}{N_2} \text{ or } \frac{V_1}{n_1} = \frac{V_2}{n_2}}$$

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\01_Ideal gas\Theory.p65

Since volume of a gas is directly proportional to the number of moles; one mole of each gas at standard temperature and pressure (STP) will have same volume.

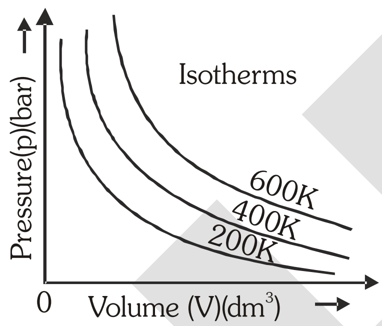
Standard temperature and pressure means 273.15 K (0°C) temperature and 1 bar (i.e., exactly 10^5 pascal) pressure. At STP, molar volume of an ideal gas or a combination of ideal gases is 22.71098 L mol^{-1}.

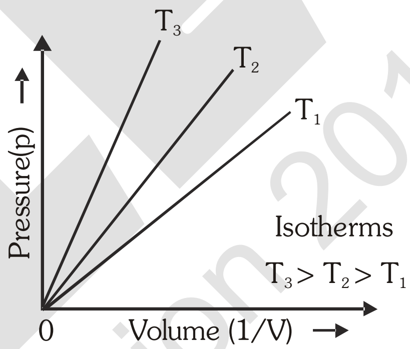
Molar volume in litres per mole of some gases at 273.15 K and 1 bar (STP).

Argon	22.37
Carbon dioxide	22.54
Dinitrogen	22.69
Dioxygen	22.69
Dihydrogen	22.72
Ideal gas	22.71

Ex.6 What is a pressure-volume isotherm ?

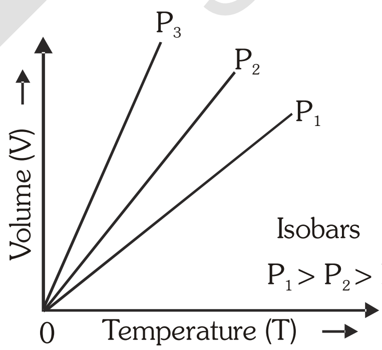
Sol. Graph between P & V at constant temperature is called ***PV-isotherm***.





Ex.7 What is Isobar ?

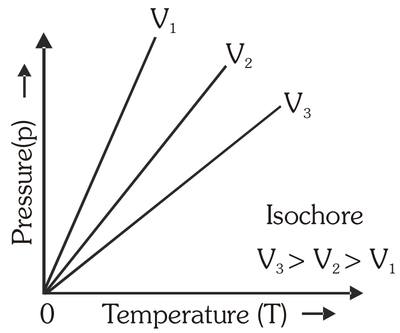
Sol. Graph plotted at constant pressure is called isobar. Graph between V & T at constant pressure is called VT-isobar.



ALLEN Session 2019-20

Ex.8 What is Isochore ?

Sol. Graph plotted at constant volume is called isochore. Graph between P & T at constant volume is called PT-isochore.



5. IDEAL GAS EQUATION

A single equation which is combination of Boyle's, Charle's & Avogadro's Law is known as ***Ideal gas equation***. Or we can say a single equation which describe the simultaneous effects of the change in temperature & pressure on volume of the given amount of the gas is called the ***Ideal gas equation.***

$PV = nRT$

According to Boyle's law, $V \propto \frac{1}{P}$ (at constant T and n)

According to Charle's law, $V \propto T$ (at constant P and n)

According to Avogadro's law, $V \propto n$ (at constant T and P)

According to the three laws, $V \propto \frac{nT}{P}$ or $PV \propto nT$

or $PV = nRT$ [Equation of state or combined gas law]

Where R is a constant called ***Universal gas constant***, which does not depend on variables (P, V, n, T) and nature of gas.

- Molar volume is the volume of 1 mole of gas.

Molar volume $(V_m) = \frac{\text{Volume}}{\text{mole}}$

$PV_m = RT$

Volume of 1 mole of an ideal gas under STP conditions (273.15 K and 1 bar pressure) is 22.7 L mol^{-1}.

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\01_Ideal gas\Theory.p65

- **Dimension of R :**

$$R = \frac{PV}{nT} = \frac{\text{Pressure} \times \text{Volume}}{\text{Mole} \times \text{Temperature}} = \frac{(\text{Force / Area}) \times (\text{Area} \times \text{Length})}{\text{Mole} \times \text{Temperature (K)}}$$

$$= \frac{\text{Force} \times \text{Length}}{\text{Mole} \times \text{Temperature(K)}} = \frac{\text{Work or energy}}{\text{Mole} \times \text{Temperature (K)}}$$

- **Physical significance of R :**

The dimensions of R are energy per mole per kelvin and hence it represents the amount of work (or energy) that can be obtained from one mole of a gas when its temperature is raised by 1 K isobarically.

- **Units of R :**

(i) In lit-atm $R = \frac{1\,\text{atm} \times 22.4\,\text{lit.}}{1\,\text{mol} \times 273\,\text{K}}$ **= 0.0821 lit-atm mol^{-1}K^{-1}**

(ii) In C.G.S system $R = \frac{1 \times 76 \times 13.6 \times 980\ \text{dyne cm}^{-2} \times 22400\,\text{cm}^3}{1\,\text{mol} \times 273\,\text{K}}$

= 8.314 × 10^7 erg mole^{-1} K^{-1}.

(iii) In M.K.S.system R **= 8.314 Joule mole^{-1} K^{-1}.** [10^7 erg = 1 joule]
(SI units)

(iv) In calories, $R = \frac{8.314 \times 10^7\ \text{erg mole}^{-1}\text{K}^{-1}}{4.184 \times 10^7\,\text{erg}}$

= 1.987 ≈ **2 calorie mol^{-1} K^{-1}.**

Ex.9 A sample of gas occupies 100 dm^3 at 1 bar pressure and at T °C. If the volume of the gas is reduced to 5 dm^3 at the same temperature, what additional pressure must be applied ?

Sol. From the given data :

P_1 = 1 bar P_2 = ?

V_1 = 100 dm^3 V_2 = 5 dm^3

Since the temperature is constant, Boyle's law can be applied

$$P_1V_1 = P_2V_2 = P_2 = \frac{P_1V_1}{V_2}$$

$$P_2 = \frac{(1\,\text{bar}) \times (100\,\text{dm}^3)}{(5\,\text{dm}^3)} = 20\,\text{bar}$$

∴ Additional pressure applied = 20 – 1 **= 19 bar**

Ex.10 A human adult breathes in approximately 0.50 dm^3 of air at 1.00 bar with each breath. If an air tank holds 100 dm^3 of air at 200 bar, how many breathes the tank will supply ?

Sol. From the given data :

Volume of air in the tank (V_1) = 100 dm^3

Session 2019-20

Pressure of air in the tank (P_1) = 200 bar

Pressure of air in each breath (P_2) = 1 bar

$\therefore$ Volume of air at 1 bar pressure (V_2) = ?

Since temperature is constant, Boyle's law is applicable

$$P_1V_1 = P_2V_2$$

or $$V_2 = \frac{P_1V_1}{P_2}$$

$$V_2 = \frac{(200\,\text{bar}) \times (100\,\text{dm}^3)}{(1\,\text{bar})} = 20{,}000\,\text{dm}^3$$

Volume of air breathed in each breath = 0.50 dm^3

Total no. of breathes the tank will supply $= \dfrac{\text{total volume of air}}{\text{volume of air in each breath}} = \dfrac{20000}{0.50}$

= 40,000 breaths.

Ex.11 A certain amount of a gas at 27°C and 1 bar pressure occupies a volume of 25 m^3. If the pressure is kept constant and the temperature is raised to 77°C, what will be the volume of the gas ?

Sol. From the available data :

$V_1 = 25\ m^3$ $T_1 = 27 + 273 = 300$ K

$V_2 = ?$ $T_2 = 77 + 273 = 350$ K

Since the pressure of the gas is constant, Charles law is applicable

$$\frac{V_1}{V_2} = \frac{T_1}{T_2}$$

or $$V_2 = \frac{V_1 \times T_2}{T_1}$$

$$V_2 = \frac{(25\,\text{m}^3) \times (350\,\text{K})}{(300\,\text{K})} = \mathbf{29.17\ m^3}$$

Ex.12 A gas at a pressure of 5.0 bar is heated from 0°C to 546°C and simultaneously compressed to one third of its original volume. What will be the final pressure ?

Sol. From the available data :

$V_1 = V\ dm^3$ $V_2 = \dfrac{V}{3}\,dm^3$

$P_1 = 5$ bar $P_2 = ?$

$T_1 = 0 + 273 = 273$ K $T_2 = 546 + 273 = 819$ K

According to Gas equation $\dfrac{P_1V_1}{T_1} = \dfrac{P_2V_2}{T_2}$ or $P_2 = \dfrac{P_1V_1T_2}{T_1V_2}$

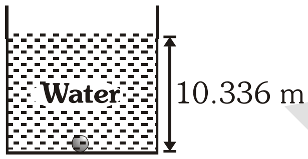
By substituting the values, $P_2 = \dfrac{(5\ \text{bar}) \times (V\,\text{dm}^3) \times (819\,\text{K})}{(273\,\text{K}) \times (V/3\ \text{dm}^3)}$ **= 45 bar**

Ex.13 The density of a gas is found to be 1.56 g dm^{-3} at 0.98 bar pressure and 65°C. Calculate the molar mass of the gas. **Use R = 0.083 dm^3 bar K^{-1} mol^{-1}**

Sol. We know that :

$$M = \frac{dRT}{P} = \frac{(1.56\,g\,dm^{-3}) \times (0.083\,dm^3\,bar\,K^{-1}\,mol^{-1}) \times 338K}{(0.98\,bar)} = \mathbf{44.66\ g\ mol^{-1}}$$

Ex.14 **(a)** Radius of a bubble at the bottom of the tank shown below was found to be 1 cm, then find the radius of the bubble at the surface of water considering the temperature at the surface & bottom being same.



(b) If absolute temperature at the surface is 4 times that at the bottom, then find radius of bubble at the surface.

Sol. **(a)** $\frac{P_bV_b}{n_bT_b} = \frac{P_sV_s}{n_sT_s}$

$$2 \times \frac{4}{3}\pi r_b^3 = 1 \times \frac{4}{3}\pi r_s^3$$

$$2 \times 1^3 = r_s^3$$

$$r_s = 2^{1/3}\,cm$$

(b) Given that $T_s = 4 \times T_b$

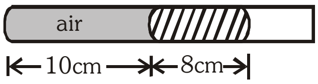
$$\frac{P_bV_b}{n_bT_b} = \frac{P_sV_s}{n_sT_s}$$

$$2 \times \frac{\frac{4}{3}\pi r_b^3}{T_b} = 1 \times \frac{\frac{4}{3}\pi r_s^3}{4 \times T_b}$$

$$r_s^3 = 8$$

r_s **= 2 cm**

Ex.15 A 10 cm length of air is trapped by a column of Hg, 8 cm long in capillary tube horizontally fixed as shown below at 1 atm pressure.



(fig. 1)

Calculate the length of air column when the tube is fixed at same temperature.

(a) Vertically with open end up

(b) Vertically with open end down

(c) At an angle of 45° from horizontal with open end up

ALLEN Session 2019-20
node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\01_Ideal gas\Theory.p65

Sol. **(a)** When the capillary tube is held vertically with open end up (fig.2),

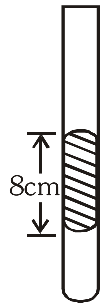
The pressure on the air column = atmospheric pressure + pressure of 8 cm Hg column

= 76 + 8 = 84 cm of Hg.

Let, at this condition the length of the air column = ℓ_2

and the length of air column when capillary is horizontally fixed = ℓ_1 = 10 cm and pressure on air column = 1atm.

Let the cross section area of the capillary = a cm^2



(fig. 2)

$\therefore$ 76 × 10 × a = 84 × ℓ_2 × a or $\ell_2 = \frac{76 \times 10}{84}$ **= 9.04 cm**

(b) When the capillary tube is held vertically with open end down (fig.3), the pressure on the air column

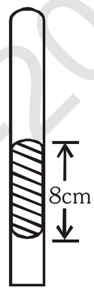
= atmospheric pressure – pressure of 8 cm Hg column

= 76 – 8 = 68 cm of Hg.

Let at this condition the length of air column = ℓ_3.

$\therefore$ 68 × ℓ_3 × a = 76 × 10 × a

or $\ell_3 = \frac{76 \times 10}{68}$ **= 11.17 cm**



(fig. 3)

(c) When the capillary is held at 45° with open end up, the weight of Hg is partially borne by the gas and partially by the Hg. The pressure on the gas due to Hg column

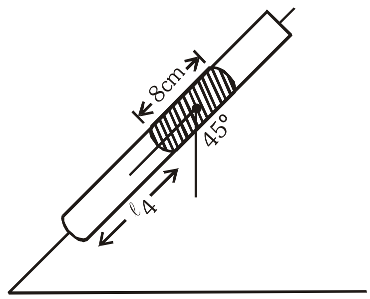
$= 8 \times \sin 45° = 8 \times \frac{1}{\sqrt{2}} = \frac{8}{\sqrt{2}}$ cm of Hg

$\therefore$ total pressure on the gas $= \left(76 + \frac{8}{\sqrt{2}}\right)$ cm of Hg.

Let length of air column at this pressure = ℓ_4.

$\therefore \ell_4 \times a \times \left(76 + \frac{8}{\sqrt{2}}\right) = 10 \times a \times 76$

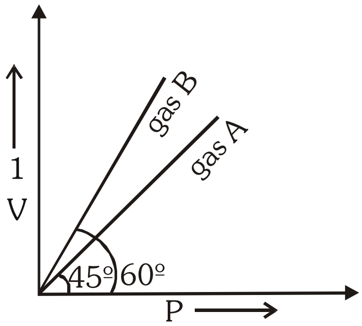
$\therefore \ell_4 = \left(\frac{10 \times 76}{76 + 8/\sqrt{2}}\right)$ **= 9.3 cm**



(fig. 4)

ALLEN Session 2019-20

Ex.16 At constant temperature of 273 K, $\left(\frac{1}{V}\right)$ v/s P are plotted for two ideal gases A and B as shown.



Find out the number of moles of gas A and B.

Sol. $PV = nRT,\ P = \frac{1}{V}nRT \quad \Rightarrow \quad \frac{1}{V} = \frac{1}{nRT}P$

Comparing by equation : y = mx + C

C = 0

$m = \frac{1}{nRT} \quad (\because m = \tan\theta)$

$\tan\theta = \frac{1}{nRT}$

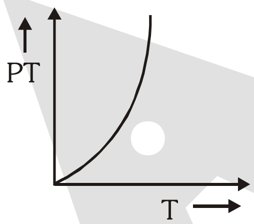
$n_A = \frac{1}{RT\tan\theta} \Rightarrow n_A = \frac{1}{0.0821 \times 273 \times \tan 45^\circ} \Rightarrow n_A = \frac{1}{22.4}$

$n_B = \frac{1}{RT\tan 60^\circ} = \frac{1}{22.4\sqrt{3}}$

Ex.17 (a) Plot the curve between PT vs T at const V & constant no. of moles.

(b) Find the number of moles of gas taken when the volume of the vessel is 82.1 ml and $\frac{d}{dT}[PT]$ at 300 K = 300 for the given curve.

Sol. (a)



(b) From graph, P = KT $\Rightarrow$ $PT = KT^2$

$\frac{d(PT)}{dT} = 2KT$

$300 = 2\,K \times 300$

$K = \frac{1}{2}$

$\frac{nR}{V} = \frac{1}{2} \Rightarrow \frac{n \times 0.0821}{0.0821} = \frac{1}{2} \quad \Rightarrow \quad n = \frac{1}{2}$

ALLEN Session 2019-20

Ex.18 A glass bulb of 2 L capacity is filled by helium gas at 10 atm pressure. Due to a leakage the gas leaks out. What is the volume of gas leaked if the final pressure in container is 1 atm.

Sol. $P_1 \times 2\ L = P_2 \times V$

$\Rightarrow \quad 10 \times 2 = 1 \times V$

$\Rightarrow \quad 20 = V$

$V = V' + 2\ L$

$20 = V' + 2\ L$

$V' = \mathbf{18\ L}$

$\Rightarrow$ total volume of gas leaked

Ex.19 LPG is a mixture of n-butane & iso-butane. What is the volume of oxygen needed to burn 1 kg of LPG at 1 atm, 273 K ?

Sol. During the burning of LPG following reaction takes place -

$$C_4H_{10} + \frac{13}{2}O_2 \longrightarrow 4CO_2 + 5\,H_2O$$

Now, for complete combustion of 1 mole of LPG $\longrightarrow \frac{13}{2}$ moles of O_2 are required.

$\therefore$ for $\frac{1000}{58}$ moles of LPG $\longrightarrow \frac{13}{2} \times \frac{1000}{58}$ moles of O_2 are required.

Thus, the volume of oxygen needed to burn 1 kg of LPG at 1 atm & 273K would be

Vol. of O_2 = Moles of $O_2 \times 22.4$ L $= \frac{13}{2} \times \frac{1000}{58} \times 22.4$ = **2510 L**

Ex.20 3.6 g of an ideal gas was injected into a bulb of internal volume of 8 L at pressure P atm and temp T K. The bulb was then placed in a thermostat maintained at (T + 15)K, 0.6 g of the gas was let off to keep the original pressure. Find P and T if mol weight of gas is 44.

Sol. Here, we are given with 2 conditions.

Initially :

Pressure , P_1 = P atm

Volume, V_1 = 8 L

Moles, $n_1 = \frac{3.6}{44}$

Temperature T_1 = T K

Finally $\Rightarrow$ At (T + 15)K 0.6 g of gas was let off to keep pressure constant.

$\therefore$ Pressure, P_2 = P atm

Volume, V_2 = 8 L (const.)

Moles, $n_2 = \frac{3}{44}$

ALLEN Session 2019-20
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Temp, $T_2 = (T + 15)$ K

Now, applying -

$$\frac{P_1V_1}{n_1T_1} = \frac{P_2V_2}{n_2T_2}$$

$$\Rightarrow \quad \frac{P \times 8}{\frac{3.6}{44} \times T} = \frac{P \times 8}{\left(\frac{3}{44}\right) \times (T + 15)}$$

T = 75 K

Now, putting the value of temp. in the following equation we get :

PV = nRT (putting initial condition)

$$P(8) = \frac{3.6}{44}(0.08)(75)$$

$\Rightarrow$ **P = 0.062 atm**

Ex.21 The best vacuum so far attained in laboratory is 10^{-10} mm of Hg. What is the number of molecules of gas remain per cm^3 at 20°C in this vacuum ?

Sol. Given conditions -

$$P = 10^{-10} \text{ mmHg} = \frac{10^{-10}}{760} \text{atm} \; ; \; V = 1 \text{ cm}^3 = \frac{1}{1000} L \; ; \; T = 20°C = 293 \text{ K}$$

No. of molecules, N = ?

Now, applying

PV = nRT

$\therefore$ number of molecules per cm^3, $N = \frac{N_A \times P \times V}{RT}$

$$\Rightarrow \quad N = \frac{6.023 \times 10^{23} \times (10^{-10}/760) \times (1/1000)}{(0.0821)(293)}$$

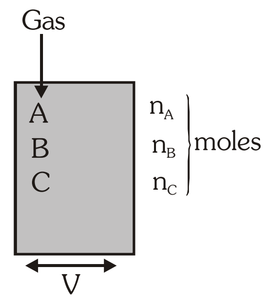
N = 3.29 × 10^6 molecules

6. DALTON'S LAW OF PARTIAL PRESSURES

6.1 Partial pressure :

In a mixture of non-reacting gases, partial pressure of any component gas is defined as the pressure exerted by the individual gas if whole of the volume of mixture had been occupied by this component only.

Partial pressure of component gases are -



$P_A = \frac{n_ART}{V}$ = partial pressure of A

$P_B = \frac{n_BRT}{V}$ = partial pressure of B

$P_C = \frac{n_CRT}{V}$ = partial pressure of C

ALLEN Session 2019-20
node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\01_Ideal gas\Theory.p65

6.2 Dalton's Law :

Dalton's law of partial pressure states "at a given temperature, the total pressure exerted by two or more non-reacting gases occupying a definite volume is equal to the sum of the partial pressures of the component gases."

$$P_{Total} = p_1 + p_2 + p_3 + ... \text{ (At constant V and T)}$$

$$= \left(\frac{n_1}{V} + \frac{n_2}{V} + \frac{n_3}{V} + ...\right) RT = (n_1 + n_2 + n_3 + ...) \frac{RT}{V} = \frac{nRT}{V}$$

Where $n = n_1 + n_2 + n_3 + ... =$ Total moles, V = Total volume

$$P_{Total} = \sum p_i = \frac{RT}{V}\sum n_i$$

- **Dalton's law of partial pressure is applicable only to non-reacting gases.**

If the two non-reacting gases A and B having n_A and n_B number of moles respectively are filled in a vessel of volume V at temperature T, then

$$\frac{p_A}{P} = \frac{n_A RT/V}{(n_A + n_B)RT/V} = \frac{n_A}{n_A + n_B} = x_A \text{ (mole fraction of A)}$$

$p_A = x_A \times P$, Similarly $p_B = x_B \times P$

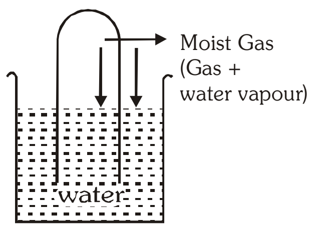
Partial pressure of a component = Mole fraction × total pressure.

6.3 It has been observed that gases are generally collected over water and therefore are moist.

$$\mathbf{P_{dry\ gas} = P_{moist\ gas} - P_{water\ vapour}}$$

or Pressure of dry gas = Pressure of moist gas – aqueous tension

The pressure exerted by water vapour is constant when it is in equilibrium with liquid water at a particular temperature. It is called vapour pressure of water or ***aqueous tension***, which varies with the temperature and becomes 760 mm at 100°C.



6.4 Relative Humidity (RH) $= \dfrac{\text{Partial pressure of water in air}}{\text{Vapour pressure of water (aq.tension)}}$

7. AMAGAT'S LAW OF PARTIAL VOLUME

7.1 Partial Volume :

Partial volume of any component is defined as the volume occupied by that particular component when it is kept at same total pressure and temperature as of the mixture.

7.2 Amagat's law :

According to this law at constant temperature and pressure, the total volume of mixture of non-reacting gases is equal to the sum of partial volumes of each component present in mixture.

$$V_T = V_1 + V_2 + V_3 + ...$$

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\01_Ideal gas\Theory.p65

Let us consider three non-reacting gases A, B and C are present in a container which have no. of moles n_A, n_B and n_C respectively. For each gas partial volume is

$$V_A = n_A\left(\frac{RT}{P}\right) = \text{partial volume of A}$$

$$V_B = n_B\left(\frac{RT}{P}\right) = \text{partial volume of B}$$

$$V_C = n_C\left(\frac{RT}{P}\right) = \text{partial volume of C}$$

- **Total volume** :

$$V_T = V_A + V_B + V_C = (n_A + n_B + n_C)\left(\frac{RT}{P}\right) = n_T\left(\frac{RT}{P}\right)$$

$$\frac{V_A}{V_T} = \frac{n_A}{n_T} = x_A \quad \text{(Mole fraction of gas A)}$$

$$\frac{V_B}{V_T} = \frac{n_B}{n_T} = x_B \quad \text{(Mole fraction of gas B)}$$

$$\frac{V_C}{V_T} = \frac{n_C}{n_T} = x_C \quad \text{(Mole fraction of gas C)}$$

$\therefore$ ***Partial volume of a gas = Mole fraction × Total volume***

Ex.22 (a) Find the total pressure and partial pressure of each component if a container of volume 8.21 lit. contains 2 moles of A and 3 mole of B at 300K.

(b) What will be the final pressure and partial pressure of each component if 5 moles of C is also added to the container at same temperature.

Sol. (a) $P_A = \frac{n_A RT}{V} = \frac{2 \times 0.821 \times 300}{8.21} = 6\,\text{atm}$

$P_B = \frac{n_B RT}{V} = \frac{3 \times 0.821 \times 300}{8.21} = 9\,\text{atm}$

Total pressure $P_T = P_A + P_B = 6 + 9 = 15$ atm

(b) $P_C = \frac{n_C RT}{V} = \frac{5 \times 0.821 \times 300}{8.21} = 15\,\text{atm}$

Note: If we add or remove a non reacting gas partial pressure of other gases remains unchanged.

$P_T = P_A + P_B + P_C = 6 + 9 + 15 = 30$ atm

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\01_ideal gas\Theory.p65

Ex.23 If 2 lit. of gas A at 1.5 atm and 3 lit. of gas B at 2 atm are mixed in a 5 lit. container then find the final pressure, considering all are at same temperature.

Sol. $n_A = \frac{P_A V_A}{RT} = \frac{2\times1.5}{RT} = \frac{3}{RT}$

$n_B = \frac{P_B V_B}{RT} = \frac{3\times2}{RT} = \frac{6}{RT}$

$\because$ $n_T = n_A + n_B$

$\frac{P_T V_T}{RT} = \frac{3}{RT} + \frac{6}{RT}$

$P_T \times 5 = 3 + 6$

$P_T = 1.8$ atm

Ex.24 One mole of N_2 and 3 moles of H_2 are taken in a container of capacity 8.21 lit. at 300 K to produce NH_3. Find the partial pressures of N_2 and H_2 if partial pressure of NH_3 after sufficient time was found to be 3 atm.

Sol.

	N_2	+ $3H_2$	$\longrightarrow$ $2NH_3$
mole at t = 0	1	3	
mole at $t = t_{eq}$	1–x	3–3x	2x

Given that

$P_{NH_3} = 3$

$\frac{n_{NH_3}RT}{V} = 3$

$\frac{2x\times0.0821\times300}{8.21} = 3$

$x = \frac{1}{2}$

$n_{N_2} = 1 - x = \frac{1}{2}$

$n_{H_2} = 3 - 3x = \frac{3}{2}$

$n_{NH_3} = 2x = 1$

$P_{N_2} = \frac{n_{N_2}RT}{V} = \frac{1}{2}\times\frac{0.0821\times300}{8.21} = 1.5\text{atm}$

$P_{H_2} = \frac{n_{H_2}RT}{V} = \frac{3}{2}\times\frac{0.0821\times300}{8.21} = 4.5\text{atm}$

ALLEN Session 2019-20

Ex.25 2 moles of $NH_3(g)$ and 1 mole of $HCl(g)$ are taken in a container of capacity 8.21 lit at 300K to produce $NH_4Cl(s)$. Find the total pressure after the reaction.

Sol. $NH_3(g) + HCl(g) \longrightarrow NH_4Cl(s)$

initial	2	1
after rxn	1	0

In this reaction HCl is L.R. so it will be completely consumed. We don't consider pressure due to solid.

$$\therefore \quad P_T = \frac{nRT}{V} = \frac{1 \times 0.821 \times 300}{8.21} = 3\,\text{atm}$$

Ex.26 A closed container containing O_2 and some liquid water was found to exert 740 mm pressure at 27°C.

(a) Then calculate the pressure exerted by O_2 if aqueous tension at 27°C is 20 mm.

(b) What will be the final pressure if volume is reduced to half.
(consider volume of liquid water negligible)

(c) What will be the final pressure if volume is doubled.

Sol. (a) $P_T = P_{\text{dry gas}} + P_{\text{aq. tension}}$

$$740 = P_{O_2} + 20$$

$$P_{O_2} = 740 - 20 = 720 \text{ mm}$$

(b) $\left(P_{O_2} V_{O_2}\right)_{\text{initial}} = \left(P_{O_2} V_{O_2}\right)_{\text{final}}$ (Boyle's law)

$$720 \times V = P_{O_2} \times \frac{V}{2}$$

$$P_{O_2} = 1440\,\text{mm}$$

$$P_T = P_T = P_{O_2} + P_{\text{aq.}}$$

$$= 1440 + 20 = 1460 \text{ mm}$$

(c) $\left(P_{O_2} V_{O_2}\right)_{\text{initial}} = \left(P_{O_2} V_{O_2}\right)_{\text{final}}$

$$720 \times V = P_{O_2} \times 2V$$

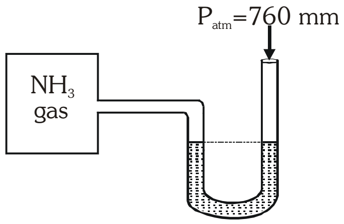
$$P_{O_2} = 360 \text{ mm}$$

$$P_T = P_{O_2} + P_{\text{aq.}}$$

$= 360 + 20 =$ **380 mm**

ALLEN Session 2019-20

Ex.27 A manometer attached to a flask contains NH_3 gas have no difference in mercury level initially as shown in diagram. After the sparking into the flask, it have difference of 19 cm in mercury level in two columns. Calculate % dissociation of ammonia.



Sol. $P_{NH_3} = 76$ cm of Hg initially

$$2NH_3 \longrightarrow N_2 + 3H_2$$

Before sparking   76

After sparking   76 – 2x   x   3x

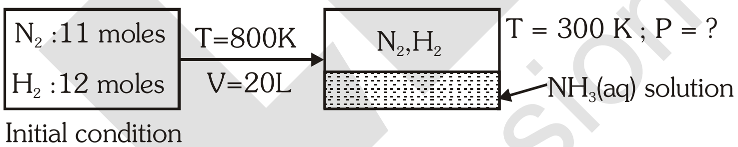
Total pressure after sparking = 76 + 2x

= 76 + 19 = 95

$$x = \frac{19}{2} = 9.75 \text{ cm of Hg}$$

$$\% \text{ dissociation} = \frac{2 \times 9.76}{76} \times 100 \Rightarrow 25\ \%$$

Ex.28 11 moles N_2 and 12 moles of H_2 mixture reacted in 20 litre vessel at 800 K. After equilibrium was reached, 6 mole of H_2 was present. 3.58 litre of liquid water is injected in equilibrium mixture and resultant gaseous mixture suddenly cooled to 300 K. What is the final pressure of gaseous mixture? Assume (i) all NH_3 dissolved in water (ii) no change in volume of liquid (iii) no reaction of N_2 and H_2 at 300 K (iv) Vapour pressure of water to be negligible :



Sol. The reaction will occur as follows -

$$N_2 \quad + \quad 3H_2 \longrightarrow 2NH_3$$

initially at

t = 0   11 moles   12 moles   0

$t = t_{eq.}$ 9 moles   6 moles   4 moles

As 6 moles of H_2 are present at equilibrium it means that 6 moles of H_2 are reacted from the initial condition and along with that 2 moles of N_2 are also reacted to form 4 moles of NH_3.

Now, the final conditions will be -

Volume = (20 – 3.58) = 16.42 L   [As 3.58 L of water is injected in vessel]

Total moles = 9 + 6 = 15 moles   [All NH_3 dissolved in water]

Temperature = 300 K

Now applying, PV = nRT

$$P = \frac{(15) \times (0.0821) \times (300)}{(16.42)}$$

Total pressure, P_T = **22.5 atm**

Ex.29 What is the difference in the density of dry air at 1 atm and 25°C and moist air with 50% relative humidity under the same condition. The vapour pressure of water at 25°C is 23.7 Torr and dry air has 75.5% N_2 and 24.5% O_2.

Sol. $$\text{M (dry air)} = \frac{M_1(O_2)X_1(\%\text{ of } O_2) + M_2(N_2)X_2(\%\text{ of } N_2)}{X_1 + X_2}$$

$$= \frac{32 \times 24.5 + 28 \times 75.5}{100} = 28.98 \text{ g mol}^{-1}$$

$$\text{d (dry air)} = \frac{PM(\text{air})}{RT} = \frac{1 \times 28.98}{0.0821 \times 298} = 1.184 \text{ g L}^{-1}$$

$$= 1.184 \text{ kg m}^{-3} \left(\because 1 \text{g L}^{-1} = \frac{10^{-3}\text{kg}}{10^{-3}\text{m}^3} = 1 \text{kg m}^{-3} \right)$$

$$\text{Relative humidity (50\%)} = \frac{\text{partial pressure of } H_2O \text{ in air}}{\text{vapour pressure of } H_2O}$$

$$\therefore \quad P(H_2O) = 0.50 \times 23.7 \text{ Torr} = 11.85 \text{ torr} = \frac{11.85}{760} \text{ atm} = 0.0156 \text{ atm}$$

$$\% \text{ of } H_2O \text{ vapour in air} = \frac{0.0156 \times 100}{1} = 1.56\%$$

% of N_2 and O_2 in air = 98.44%

$$\text{M (wet air)} = \frac{28.98 \times 98.44(\text{air}) + 18 \times 1.56 (\text{water vapour})}{100} = 28.81 \text{ g mol}^{-1}$$

$$\text{d (wet air)} = \frac{PM(\text{wet air})}{RT} = \frac{1 \times 28.81}{0.0821 \times 298}$$

$$= 1.177 \text{ g L}^{-1} = 1.177 \text{ kg m}^{-3}$$

difference = 1.184 – 1.177 = 0.007 kg m^{-3}

Ex.30 A gaseous mixture at 760 mm in a cylinder has 65% N_2, 15% O_2 and 20 % CO_2 by volume. Calculate the partial pressure of each gas.

Sol. Partial pressure of gas = mole fraction of gas × total pressure = volume fraction of gas × total pressure

$$p_{N_2} = \frac{65}{100} \times 760 = 494 \text{ mm}$$

$$p_{O_2} = \frac{15}{100} \times 760 = 114 \text{ mm}$$

$$p_{CO_2} = \frac{20}{100} \times 760 = 152 \text{ mm}$$

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\01_Ideal gas\Theory.p65

8. PROBLEM RELATED WITH DIFFERENT TYPE OF CONTAINERS

I. Closed Container :

In this case gas can neither go outside nor it can come inside. So number of moles of gas is always constant.

Closed container can be of following types -

(a) Closed rigid container : In this case number of moles constant, volume constant.

At this condition :

Initial Final

$$\frac{P_1}{T_2} = \frac{P_2}{T_2}$$

Example : Gas cylinder

(b) Closed non rigid container : (fitted with freely movable piston)

In this kind of container inside pressure is always equal to outside pressure, i.e., atmospheric pressure so that n = constant , p = constant

Ex: Balloon, Water bubble

Initial Final

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

Ex.31 A balloon is inflated to $\frac{7}{8}$ of its maximum volume at 27°C then calculate the minimum temperature above which it will burst.

Sol. $\frac{V_1}{T_1} = \frac{V_2}{T_2}$

$$\frac{7V}{8 \times 300} = \frac{V}{T}$$

T = 342.8 K

Ex.32 A gas cylinder containing cooking gas can with stand a pressure of 18 atm. The pressure gauge of cylinder indicates 12 atm at 27°C. Due to sudden fire in building the temperature start rising at what temperature will the cylinder explode.

Sol. $\frac{P_1}{T_2} = \frac{P_2}{T_2}$

$$\frac{12}{300} = \frac{18}{T}$$

T = 450 K

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\01_Ideal gas\Theory.p65

II. Tyre tube type container :

In this case temperature is always constant. Initially on adding gas volume of tube will increase and pressure of tube will remain constant until it will gain maximum volume.

$\because\ V \propto n$

Initial Final

$$\frac{V_1}{n_2} = \frac{V_2}{n_2}$$

after attaining maximum volume on adding gas pressure reach to a maximum possible pressure.

Hence at this condition volume constant or temperature constant.

V = constant T = constant

$P \propto n$

$$\frac{P_1}{n_1} = \frac{P_2}{n_2}$$

Ex.33 A tyre tube of maximum volume 8.21 lit. can withstand a pressure of 10 atm. Initially the tube is empty.

(i) Calculate the number of moles required to inflate completely the tube upto a pressure of 1 atm & 300 K temperature.

(ii) Calculate the minimum number of moles required to burst the tyre tube at 300 K.

Sol. (i) PV = nRT

$1 \times 8.21 = n_1 \times 0.0821 \times 300$

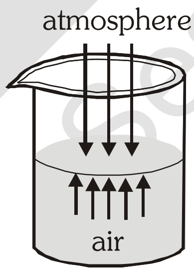
$$n_1 = \frac{1}{3}$$

(ii) PV = nRT

$10 \times 8.21 = n_2 \times 0.0821 \times 300$

$$n_2 = \frac{10}{3}$$

III. Open rigid container : When air is heated in an open vessel , pressure is always atmospheric pressure i.e., constant and volume is also constant.



At this condition

Initial Final

$n_1\,T_1 = n_2T_2$

n_1 = initial number of moles

n_2 = final number of moles

$$\mathbf{n_{initial} = n_{final} + n_{removed.}}$$

ALLEN Session 2019-20

Ex.34 An open flask contains air at 27°C. Calculate the temperature at which it should be heated so that

(a) $1/3^{rd}$ of air measured in the container at 27°C escape out.

(b) $1/3^{rd}$ of air measured in the container at final temperature escape out.

Sol. **(a)** $\mathbf{n_{initial} - n_{final} = n_{expelled}}$

$$\frac{PV}{R\times 300}-\frac{PV}{R\times T}=\frac{1}{3}\times\frac{PV}{R\times 300}$$

$$\frac{1}{300}-\frac{1}{T}=\frac{1}{900}$$

$$\frac{1}{T}=\frac{1}{300}-\frac{1}{900}=\frac{3-1}{900}=\frac{2}{900}$$

$\Rightarrow$ T = 450 K = 177°C

(b) $\mathbf{n_{initial} - n_{final} = n_{expelled}}$

$$\frac{PV}{R\times 300}-\frac{PV}{R\times T}=\frac{1}{3}\times\frac{PV}{R\times T}$$

$$\frac{1}{300}-\frac{1}{T}=\frac{1}{3\times T}$$

$\Rightarrow$ T = 400 K = 127°C

Ex.35 A bulb of unknown volume containing air is heated from 27°C to 227°C at constant pressure. The expelled air is measured at different temperature of determine volume of container. What will be volume of container if -

(a) 200 ml of air measured at 227°C was expelled.

(b) 200 ml of air measured at 27°C was expelled.

(c) 200 ml of air measured at 127°C was expelled.

Sol. **(a)** $\mathbf{n_{initial} - n_{final} = n_{expelled}}$

$$\frac{PV}{R\times 300}-\frac{PV}{R\times 500}=\frac{P\times 200}{R\times 500}$$

$$\frac{V}{3}-\frac{V}{5}=\frac{200}{5}$$

$$\frac{5V-3V}{15}=\frac{200}{5}$$

$\Rightarrow$ $2V=\frac{200\times 15}{5}$ $\Rightarrow$ V = 300 ml

(b) $$\frac{PV}{R\times 300}-\frac{PV}{R\times 500}=\frac{P\times 200}{R\times 300}$$

$$\frac{2V}{15}=\frac{200}{3}\Rightarrow V=\frac{100\times 15}{2\times 3}=500\text{ml}$$

(c) $$\frac{PV}{R\times 300}-\frac{PV}{R\times 500}=\frac{P\times 200}{R\times 400}$$

$$\frac{2V}{15}=\frac{200}{4} \quad\Rightarrow\quad V=\frac{200\times 15}{2\times 4}=375\text{ml}$$

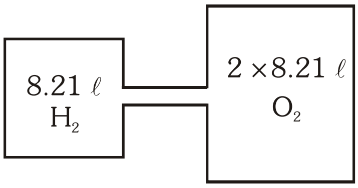
Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\01_Ideal gas\Theory.p65

IV. Connected Container :

If containers are connected for substantial time then gases move from one container to another container till partial pressure of each component of mixtures becomes equal in all connected containers (irrespective whether containers have same or different temperature and volumes)

Ex.36 A container of 8.21 lit. capacity is filled with 1 mole of H_2 at 300 K and it is connected to another container of capacity 2×8.21 lit. containing 4 moles of O_2 at 300 K, then find the final pressure & partial pressure of each gas.



Sol. $P_fV_f = n_fRT$

$P_f (3 \times 8.21) = 5 \times 0.0821 \times 300$

$P_f = 5$ atm

$$P_{H_2} = x_{H_2} P_f = \frac{1}{1+4} \times 5 = 1 \text{atm}$$

$$P_{O_2} = x_{O_2} P_f = \frac{4}{1+4} \times 5 = 4 \text{atm}$$

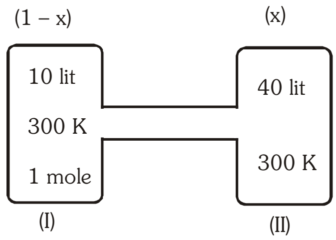
Ex.37 A 10 litre container consist of 1 mole of gas at 300 K. It is connected to another container having volume 40 litre and is initially at 300 K. The nozzle connecting two containers is opened for a long time and once the movement of gas stopped, the larger container was heated to a temperature of 600 K. Calculate

(a) Moles and pressure of gas in both the containers before heating.

(b) Moles and pressure in two containers after heating.

(Assume that initially the larger container is completely evacuated.)

Sol. **(a)** Before heating :



$P_I = P_{II}$

$$\frac{(1-x)R \times 300}{10} = \frac{x \times R \times 300}{40}$$

x = 0.8 moles

$n_I = 1 - x = 0.2$ mole

$n_{II} = x = 0.8$ mole

$$\text{Pressure} = \frac{x \times R \times T}{V} = \frac{0.8 \times R \times 300}{40} = 0.492 \text{ atm}$$

(b) After heating :

$$\frac{(1-x_1)R \times 300}{10} = \frac{x_1 \times R \times 600}{40}$$

$x_1 = 0.67$ moles, Given $T_1 = 600$ K

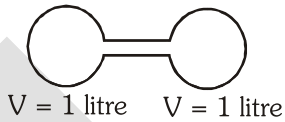
$$\text{Pressure} = \frac{x_1 \times R \times T_1}{V} = \frac{0.67 \times .0821 \times 600}{40} = 0.821 \text{ atm}$$

Ex.38 One litre flask contains air, water vapour and a small amount of liquid water at a pressure of 200 mmHg. If this is connected to another one litre evacuated flask, what will be the final pressure of the gas mixture at equilibrium? Assume that temperature to be 50ºC. Aqueous tension at 50°C = 93 mmHg.

Sol. The aqueous tension remains same in both the flask. Also flask are at same temperature

$\therefore \quad P_1V_1 = P_2V_2$

where $P_1 = 200 - 93 = 107$ mm



$V_1 = 1$ litre

$V_2 = 2$ litre

$\therefore \quad 107 \times 1 = P \times 2$

$P = 53.5$ mm

Since aqueous tension is also present in flask, equivalent to 93 mm.

$\therefore$ pressure of gaseous mixture = 93 + 53.5 = 146.5 mmHg.

9. GRAHAM'S LAW OF DIFFUSION & EFFUSION

9.1 Diffusion :

The ***diffusion*** is the process of gradual mixing of molecules of one gas with molecules of another gas due to their molecular motion (kinetic energy). The diffusion always proceeds from a region of high concentration to a region of lower concentration (or high partial pressure to low partial pressure). For example, when a bottle of perfume is opened at one end of the room, the person sitting at the other end of the room can smell the perfume because of the diffusion process of perfume molecules.

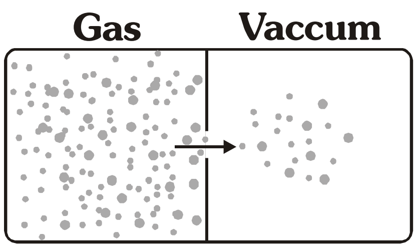
Note : Initially

When stop cock is removed flow will be from both sides, N_2 will try to equalise its partial pressure in both the vessels, and so will O_2.

Finally

9.2 Effusion :

The effusion is the process of forcing a gas through a pin hole or small orifice from one compartment to another empty (vacuum) compartment.



ALLEN Session 2019-20

9.3 Graham's Law :

Under similar condition of pressure (partial pressure) and temperature, the rate of diffusion of different gases is inversely proportional to square root of their density.

$\Rightarrow$ rate of diffusion, $\boxed{r \propto \dfrac{1}{\sqrt{d}}}$

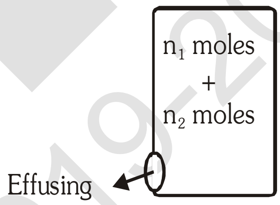
$$\Rightarrow \quad \frac{r_1}{r_2} = \frac{\sqrt{d_2}}{\sqrt{d_1}} = \frac{\sqrt{M_2}}{\sqrt{M_1}} = \frac{\sqrt{V.D_2}}{\sqrt{V.D_1}}$$

where, d = density of gas
V.D = vapour density
M = molar mass of gas

- **Under conditions of same temperature but different pressure, we have -**

$$\boxed{r \propto \frac{P}{\sqrt{M}}}$$

$$\boxed{\frac{r_1}{r_2} = \frac{P_1}{P_2}\sqrt{\frac{M_2}{M_1}}}$$



If both gases are present in the same container at same temperature.

$P \propto n$

$$\Rightarrow \quad \frac{r_1}{r_2} = \frac{n_1}{n_2}\sqrt{\frac{M_2}{M_1}}$$

Rate of diffusion/effusion can be expressed as -

$$r = \frac{\text{volume diffused}}{\text{time taken}} \text{ or } \frac{\text{moles diffused}}{\text{time taken}} \text{ or } \frac{\text{pressure dropped}}{\text{time taken}}$$

or $$\frac{\text{distance travelled in horizontal tube of uniform cross} - \text{section}}{\text{time taken}}$$

Ex.39 32 ml of He effuses through a fine orifice in 1 minute. Then what volume of CH_4 will diffuse in 1 minute under the similar condition.

Sol. $\because$ $r \propto \dfrac{1}{\sqrt{M}}$

$$r = \frac{\text{volume diffused}}{\text{time}}$$

$\because$ time is same so

$$\frac{V_{CH_4}}{V_{He}} = \sqrt{\frac{M_{He}}{M_{CH_4}}}$$

$$\frac{V_{CH_4}}{32} = \sqrt{\frac{4}{16}}$$

$$V_{CH_4} = \frac{1}{2} \times 32 = 16\,\text{mL}$$

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\01_Ideal gas\Theory.p65

Session 2019-20

Ex.40 20 dm^3 of Ne diffuse through a porous partition in 60 seconds. What volume of SO_3 will diffuse under similar conditions in 30 sec. (Atomic wt. of Ne = 20, S = 32)

Sol. $\frac{r_{N_e}}{r_{SO_3}} = \sqrt{\frac{M_{SO_3}}{M_{Ne}}} \Rightarrow \frac{V_{Ne}/t_{Ne}}{V_{SO_3}/t_{SO_3}} = \sqrt{\frac{M_{SO_3}}{M_{Ne}}}$

$\Rightarrow \quad V_{SO_3} = \frac{V_{Ne} \times t_{SO_3}}{t_{Ne}} \times \sqrt{\frac{M_{Ne}}{M_{SO_3}}} = \frac{20 \times 30}{60} \times \sqrt{\frac{20}{80}} = \frac{10}{2} = 5\,dm^3$

Ex.41 A gaseous mixture of O_2 and X containing 20% (mole %) of X, diffused through a small hole in 234 seconds while pure O_2 takes 224 seconds to diffuse through the same hole. Molecular weight of X is :

Sol. $\frac{t_{mix}}{t_{O_2}} = \sqrt{\frac{M_{mix}}{M_{O_2}}}$

$\frac{234}{224} = \sqrt{\frac{M_{mix}}{32}}$

$\therefore \quad M_{mix} = 34.921.$

As the mixture contains 20% (mole %) of X, the molar ratio of O_2 and X may be represented as 0.8n : 0.2n, n being the total no. of moles.

$\therefore \quad M_{mix} = \frac{32 \times 0.8n + M_x \times 0.2n}{n} = 34.921$

$\therefore \quad M_x$ (mol. wt. of X) = 46.6

Ex.42 A mixture of H_2 and O_2 in 2 : 1 mole ratio is allowed to diffuse through a orifice. Calculate the composition of gases coming out initially.

Sol. $\frac{r_1}{r_2} = \frac{n_1}{n_2}\sqrt{\frac{M_2}{M_1}}$

where n_1 and n_2 are the moles of gases taken.

$\frac{n'_1}{n'_2} = \frac{n_1}{n_2}\sqrt{\frac{M_2}{M_1}}$

where n'_1 and n'_2 are the moles of gases diffused.

$\frac{n'_1}{n'_2} = \frac{2}{1}\sqrt{\frac{32}{2}} = \frac{8}{1}$

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\01_Ideal gas\Theory.p65

Ex.43 One mole of nitrogen gas at 0.8 atm takes 38 s to diffuse through a pinhole, whereas one mole of an unknown compound of xenon with fluorine at 1.6 atm takes 57 s to diffuse through the same hole. Determine the molecular formula of the compound.

Sol. or $\frac{r_2}{r_1} = \frac{p_2}{p_1}\left(\frac{M_1}{M_2}\right)^{1/2} \Rightarrow \frac{t_1}{t_2} = \left(\frac{p_2}{p_1}\right)\left(\frac{M_1}{M_2}\right)^{1/2}$

or $M_2 = \left(\frac{p_2 t_2}{p_1 t_1}\right)^2 M_1$

$$M_2 = \left(\frac{1.6}{0.8} \times \frac{57}{38}\right)^2 \times 28 = 252 \text{ g mol}^{-1}$$

Let the molecular formula of the unknown compound be XeF_n.

$M_{xe} + nM_F = 252$ or $131 + 19\,n = 252$

$$n = \frac{252 - 131}{19} = 6.36 \simeq 6$$

Hence, the molecular formula of the gas is XeF_6.

Ex.44 At 1200ºC, mixture of Cl_2 and Cl atoms (both in gaseous state) effuses 1.16 times as fast as krypton effuses under identical conditions. Calculate the fraction of chlorine molecules dissociated into atoms. M (Kr) = 83.8 g mol^{-1}.

Sol. $Cl_2 \rightleftharpoons 2Cl$

$$\frac{r(Cl_2 \text{ and } Cl \text{ mix})}{r(Kr)} = 1.16\sqrt{\frac{M(Kr)}{M_{av}(Cl_2 + Cl)}} = \sqrt{\frac{83.8}{M_{av}}}$$

$\therefore$ $M_{av} = \frac{83.8}{(1.16)^2} = 62.28 \text{ g mol}^{-1}$

	$Cl_2 \rightleftharpoons$	$2Cl$
Initial mole	1	0
After dissociation	(1 – x)	2x

(x = degree of dissociation)

Total moles after dissociation = 1 – x + 2x = (1 + x)

$\therefore$ $\frac{(1-x)M(Cl_2) + 2x\,M(Cl)}{(1+x)} = 62.28 \Rightarrow \frac{(1-x) \times 71 + 2x \times 35.5}{1+x} = 62.28$

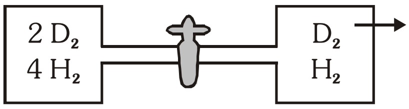
x = 0.14

$\therefore$ % dissociation = 14%

Session 2019-20

Ex.45 A mixture containing 2 moles of D_2 and 4 moles of H_2 is taken inside a container which is connected to another empty container through a nozzle. The nozzle is opened for certain time and then closed. The second bulb was found to contain 4 gm D_2. Then find % by moles of the lighter gases in second container.

Sol. **I** **II**



$w_{D_2} = 4$ gm

$n_{D_2} = \frac{4}{4} = 1$

lighter gas = H_2

$$\frac{r_{H_2}}{r_{D_2}} = \frac{n_{H_2}}{n_{D_2}}\sqrt{\frac{M_{D_2}}{M_{H_2}}} = \frac{n'_{H_2}/t}{n'_{D_2}/t}$$

$$(\%\text{ mole of } H_2)_{II} = \left(\frac{n'_{H_2}}{n'_{H_2} + n'_{D_2}}\right) \times 100$$

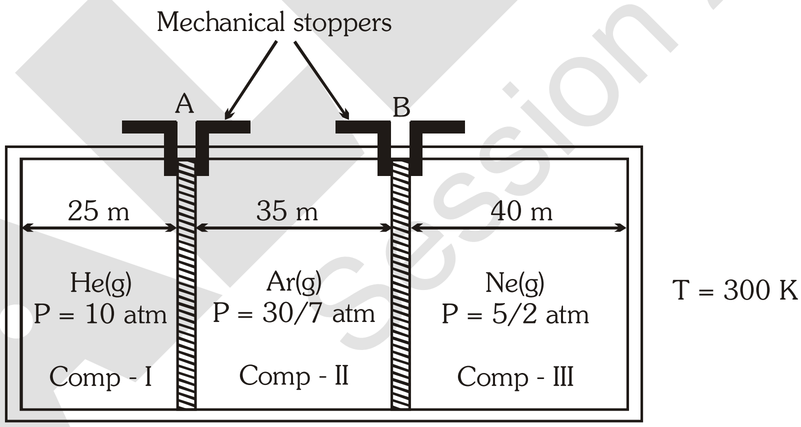
$\therefore \quad \frac{n_{H_2}}{n_{D_2}}\sqrt{\frac{M_{D_2}}{M_{H_2}}} = \frac{n'_{H_2}}{n'_{D_2}} \quad \Rightarrow \quad \frac{4}{2}\sqrt{\frac{4}{2}} = \frac{n'_{H_2}}{1}$

$\Rightarrow \quad n'_{H_2} = 2\sqrt{2} \quad \Rightarrow \quad n'_{D_2} = 1$

$$(\%\text{ mole of } H_2)_{II} = \frac{(2\sqrt{2})}{(2\sqrt{2}+1)} \times 100 = \frac{2 \times 1.44}{[(2 \times 1.44) + 1]} \times 100 = \frac{(2.8)}{(2.8+1)} \times 100 = \frac{(2.8)}{(2.8+1)} \times 100$$

= 73.87%

Ex.46 The figure shows **initial conditions** of a uniform cylinder with frictionless pistons A and B held in shown position by mechanical stoppers.



If the mechanical stoppers holding piston A and B as shown in figure are removed.

[Assume that temperature remains constant.]

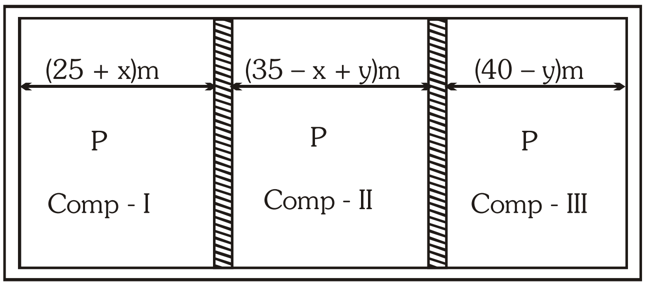
(a) Pressure developed in each compartment in final state.

(b) What will be the final position of piston A (with respect to far left end of container)

(c) What will be the final position of piston B (with respect to far left end of container)

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\01_Ideal gas\Theory.p65
ALLEN Session 2019-20

Sol. Piston will move whenever partial pressure will not equalize in each compartment.



(a) $n_T = n_1 + n_2 + n_3$

$P_fV_f = P_1V_1 + P_2V_2 + P_3V_3$

$P_f\ell_fA = P_1\ell_1A + P_2\ell_2A + P_3\ell_3A$

$P(25 + 35 + 40) = 10 \times 25 + \frac{30}{7} \times 35 + \frac{5}{2} \times 40$

$P = 5$ atm

(b) For comp I

$P_fV_f = P_1V_1$

$P_f\ell_fA = P_1\ell_1A$

$(25 + x)\, 5 = 10 \times 25$

$x = 25$ m

Final position of piston A (with respect to far left end of container)

$= (25 + x) = 25 + 25 = 50$ m

(c) For comp II

$P_fV_f = P_2V_2$

$P_f\ell_fA = P_2\ell_2A$

$(35 - x + y)\, 5 = \frac{30}{7} \times 35$

$(35 - 25 + y)\, 5 = 150$

$y = 20$ m

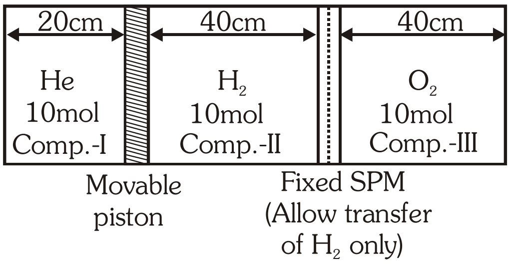
Final position of piston B (with respect to far left end of container)

$= (25 + x) + (35 - x + y) = 80$m

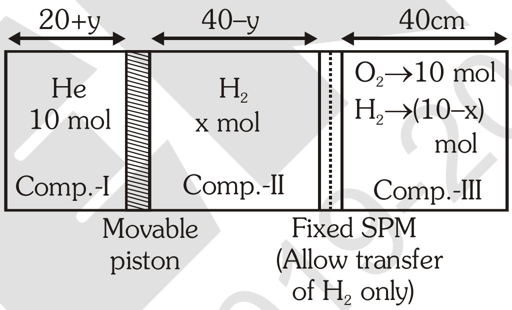
ALLEN Session 2019-20

Ex.47 Few gases are filled in a container as shown in diagram and allowed to attain equilibrium. Then calculate the moles of H_2 in compartment II.

(Assuming temperature remains constant through out the process.)



Sol. Piston move till pressure on both side becomes equal and H_2 will pass in III compartment till pressure of H_2 in II and III compartment becomes equal let the final position of containers as shown



For H_2:

$(PH_2)_{II} = (PH_2)_{III}$

$$\left(\frac{nRT}{V}\right)_{II} = \left(\frac{nRT}{V}\right)_{III}$$

$$\frac{x}{40-y} = \frac{10-x}{40} \qquad \text{............(i)}$$

Finally total pressure of gases are equal

$(P_T)_I = (P_T)_{II} = (P_T)_{III}$

$$\frac{10}{20+y} = \frac{x}{40-y} = \frac{10-x}{40} \qquad \text{..........(ii)}$$

$$\frac{10}{20+y} = \frac{10-x}{40} \quad \Rightarrow \quad x = 10 - \frac{400}{20+y}$$

$$\frac{10}{20+y} = \frac{10 - \dfrac{400}{20+y}}{40-y}$$

$400 - 10y = 10\,(20 + y) - 400$

$400 - 10y = 200 + 10y - 400$

$600 = 20\,y \Rightarrow y = 30$

$x = 2$

Moles of H_2 in compartment II = x = 2 mol

Moles of H_2 in compartment II = 10 – x = 8 mol

Ex.48 Why a heavier gas from a gas mixture effuses at slower rates ?

Sol. In a gas mixture, the average kinetic energy of each gas $\left(\frac{1}{2}mv^2\right)$ is the same $\left(\frac{3}{2}RT\right)$. Hence, heavier gas has smaller speed.

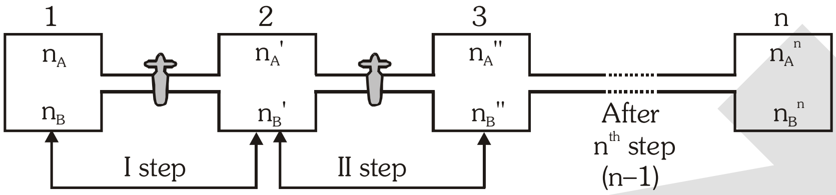
Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\01_Ideal gas\Theory.p65

9.4 APPLICATION OF GRAHAM'S LAW OF DIFFUSION IN ENRICHMENT OF ISOTOPES

❖ **Enrichment factor or Isotopic separation factor :**

When two gases present in a container are allowed to diffuse in another container then gas having lower molecular mass will diffuse more and if this process is continued for large number of steps then we can obtain a mixture which is rich in a gas having lower molecular mass. Hence ultimately in the ultimate (last) container amount of lighter gas is larger as compared to heavier gas.



where, $M_A > M_B$ [similar condition of T and V] & $n_B^n >>>> n_A^n$

Enrichment factor (f) is defined as the ratio of final ratio of moles in the mixture [after n^{th} step] with initial mole ratio of the mixture.

$$f = \frac{\left(\dfrac{n_A^n}{n_B^n}\right)}{\left(\dfrac{n_A}{n_B}\right)} = \left(\sqrt{\frac{M_B}{M_A}}\right)^n = \frac{\text{final molar ratio}}{\text{initial molar ratio}}$$

Ex.49 A mixture of N_2 and H_2 has initially mass ratio of 196 : 1 then find after how many steps we can obtain a mixture containing 1 : 14 mole ratio of N_2 and H_2.

Sol.

	N_2	H_2
	196	1
mole	$\frac{196}{28} = 7$	$\frac{1}{2}$

ratio $\quad \frac{n_{N_2}}{n_{H_2}} = \frac{7}{\frac{1}{2}} = \frac{14}{1}$

$$\frac{n'_{N_2}}{n'_{H_2}} = \frac{n_{N_2}}{n_{H_2}}\left(\sqrt{\frac{M_{H_2}}{M_{N_2}}}\right)^n = \frac{1}{14} = \frac{14}{1}\left(\sqrt{\frac{2}{28}}\right)^n, \; n = 4$$

Ex.50 Find the ratio of moles of SO_2 to CH_4 after fifth diffusion steps if their initially mole ratio is 8 : 1.

Sol. $$\frac{n'_{SO_2}}{n'_{CH_4}} = \frac{n_{SO_2}}{n_{CH_4}}\left(\sqrt{\frac{M_{CH_4}}{M_{SO_2}}}\right)^5$$

$$\frac{n'_{SO_2}}{n'_{CH_4}} = \frac{8}{1}\left(\sqrt{\frac{16}{64}}\right)^5 = \frac{n'_{SO_2}}{n'_{CH_4}} = \frac{1}{4}$$

ALLEN Session 2019-20
node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\01_Ideal gas\Theory.p65

10. KINETIC THEORY OF GASES

This is a theoretical model for ideal gas which can correlate the experimental facts (like Boyle's law, Charle's law& Avogadro's law etc.). It was presented by ***Bernoulli in 1738*** and developed in 1860 by ***Clausius, Maxwell, Kroning*** *and* ***Boltzmann***. Postulates of kinetic theory of gases are :

(i) All the gases consist of very small molecules or atoms whose volume is negligible compared to volume of container.

(ii) There are no attractive or repulsive forces between the molecules.

(iii) The gaseous molecules are under a continuous state of motion which is unaffected by gravity (the random straight line motion is known as brownian motion)

(iv) Due to the continuous motion, collision between gaseous molecules and with the wall of container occurs. The collision with the wall of container are responsible for pressure exerted by the gas on the wall of container.

(v) The molecule moves with different speed, however the speed of each molecule keep on changing as the collision occur.

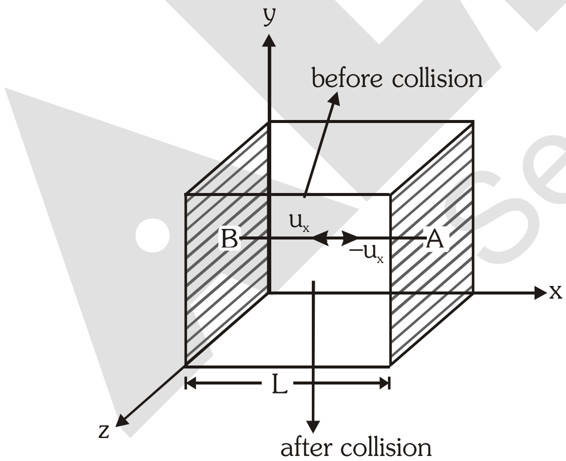
(vi) Collision among gas particles molecules is perfectly elastic, i.e., there is no loss in kinetic energy and moment during such collision.

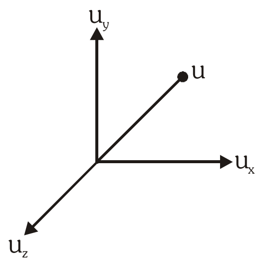
(vii) The average kinetic energy of gas particles will depends on absolute temperature only.

10.1 Kinetic gas equation :

Let us consider a cube of side L, that has N molecules each of mass m moving with velocity u in all direction and thus colliding with one each other and against sides of the container. Velocity u can be resolved into three components u_x, u_y and u_z along there axis such that

$$u^2 = u_x^2 + u_y^2 + u_z^2$$





For a simplest case we consider motion of a molecule along x-axis only in which it moves towards face B with velocity u_x. After collision against face B it moves towards face A with velocity $(-u_x)$ collision being elastic (which results in change in direction but not speed)

$\therefore$ Momentum before collision on face B = mu_x

Momentum after collision on face B = $-mu_x$

Change in momentum due to one collision on face B = $mu_x - (-mu_x) = 2mu_x$

To strike face B again distance travelled = 2L

Time taken to strike face B again = $\frac{2L}{u_x}$ seconds

$\therefore$ Number of collisions per second on face B along x-axis = $\frac{u_x}{2L}$

$\therefore$ Rate of change in momentum due to $\frac{u_x}{2L}$ collisions per second on face B along x-axis.

$= 2mu_x \cdot \frac{u_x}{2L} = \frac{mu_x^2}{L}$

Similarly for y-axis change in momentum per second = $\frac{mu_y^2}{L}$ and for z-axis = $\frac{mu_z^2}{L}$

Net force by N molecules on a wall , $F_x = \frac{mu_{x_1}^2}{L} + \frac{mu_{x_2}^2}{L} .... + \frac{mu_{x_N}^2}{L} = \frac{M}{L}.\Sigma u_x^2$

Now pressure $= \frac{\text{Force}}{\text{Area of six faces}} = \frac{\frac{m}{L}.\Sigma u_x^2}{L^2} = \frac{m.\Sigma u_x^2}{L^3} = \frac{m.\Sigma u^2}{V}$ [L^3 = volume V]

$\therefore$ $PV = m.\Sigma u_x^{\ 2}$

As $\Sigma u_x^2 = \Sigma u_y^2 = \Sigma u_z^2$ and $\Sigma u_x^2 + \Sigma u_y^2 + \Sigma u_z^2 = \Sigma u^2$

$\therefore$ $PV = m.\Sigma u_x^{\ 2} = \frac{1}{3} m.\Sigma u^2$

$$PV = \frac{1}{3} m \left(\frac{u_1^2 + u_1^2 + ...u_N^2}{N} \right).N$$

$\therefore$ $PV = \frac{1}{3} mN\, u_{rms}^2$

This equation is called kinetic gas equation.

10.2 Kinetic energy of gas molecules :

Total translational K.E. of molecules

$= \frac{1}{2} mu_2^2 + \frac{1}{2} mu_2^2 + ... + \frac{1}{2} m.u_N^{\ 2} = \frac{1}{2} m.\Sigma u^2 = \frac{1}{2} mN.u_{rms}^2 = \frac{3}{2} PV = \frac{3}{2} nRT$

$\therefore$ Average translational K.E. per mole $= \frac{3}{2} RT$

and $(K.E.)_{\text{per molecule}} = \frac{3}{2}\left(\frac{R}{N_A}\right)T = \frac{3}{2} kT$

Where k = Boltzman constant $= \frac{R}{N_A} = \frac{8.314 J/molK}{6.02 \times 10^{23}} = 1.3806 \times 10^{-23}$ J K^{-1}

- $(K.E.)_{\text{per molecule}}$ and $(K.E.)_{\text{per mol}}$ is only depend on absolute temperature. It is does not depend on the nature of gas. This conclusion is known as ***"Maxwell's Generalisation"***.

ALLEN Session 2019-20

Ex.51 Calculate the kinetic energy of 8 gram methane (CH_4) at 27°C temperature.

Sol. $n = \frac{8}{16} = \frac{1}{2}$, T = (27° + 273) = 300 K, R = 8.314 J mol^{-1} K^{-1}

$$(K.E.)_{n\ mol} = n \times \frac{3}{2}RT = \frac{1}{2} \times \frac{3}{2} \times 8.314 \times 300 = 1870.65\ J$$

Ex.52 Calculate the pressure exerted by 10^{23} gas molecules, each of mass 10^{-25} kg, in a container of volume 1×10^{-3} m^3 and having root mean square velocity of 10^3 ms^{-1}. Also calculate total kinetic energy and Temperature of the gas.

Sol. By kinetic theory

$$P = \frac{1}{3}\frac{mNu^2}{V} = \frac{1 \times 10^{-25} \times 10^{23} \times (10^3)^2}{3 \times 10^{-3}} = 3.33 \times 10^6\ N\ m^{-2}$$

$$\text{Total KE} = \left(\frac{1}{2}mu_{rms}^2\right) \times N = \frac{1}{2} \times 10^{-25} \times (10^3)^2 \times 10^{23} = \frac{1}{2} \times 10^4 = 0.5 \times 10^4\ J$$

$$\text{Also total KE} = \frac{3}{2}nRT, \text{ where } n\ (\text{mole}) = \frac{10^{23}}{N_A} = \frac{10^{23}}{6.023 \times 10^{23}}$$

$$0.5 \times 10^4 = \frac{3}{2} \times \frac{10^{23}}{6.023 \times 10^{23}} \times 8.314 \times T$$

$$\therefore \quad T = \frac{0.5 \times 10^4 \times 2 \times 6.023}{3 \times 8.314} = 2415\ K$$

10.3 Root Mean Square Velocity (u_{rms}) by kinetic gas equation :

$$PV = \frac{1}{3}mN\,u_{rms}^2$$

$$\therefore \quad u_{rms} = \sqrt{\frac{3PV}{mN}} = \sqrt{\frac{3P}{d}} = \sqrt{\frac{3RT}{M}}$$

10.4 Diffrent kind of speed of molecules :

(i) Average or mean speed, $u_{av} = \frac{u_1 + u_2 + \ldots\ldots.. + u_N}{N} = \sqrt{\frac{8RT}{\pi M}}$

(ii) Root mean square speed, $u_{rms} = \sqrt{\frac{u_1^2 + u_2^2 + \ldots\ldots.. + u_N^2}{N}} = \sqrt{\frac{3RT}{M}}$

(iii) Most probable speed, $u_{mp} = \sqrt{\frac{2RT}{M}}$

It is the speed at which maximum fraction of molecules are travelling

ALLEN Session 2019-20
node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\01_Ideal gas\Theory.p65

❖ **Ratio of speeds :**

$$U_{rms} : U_{avg} : U_{mps} = \sqrt{\frac{3RT}{M}} : \sqrt{\frac{8RT}{\pi M}} : \sqrt{\frac{2RT}{M}}$$

$$= \sqrt{3} : \sqrt{\frac{8}{\pi}} : \sqrt{2} = 1.22 : 1.13 : 1 : 00 = 1.00 : 0.92 : 0.816$$

11. MAXWELL'S DISTRIBUTION OF SPEEDS

It has already been pointed out that a gas is a collection of tiny particles separated from one another by large empty spaces and moving rapidly at random in all directions. In the course of their motion, they collide with one another and also with the walls of the container. Due to frequent collisions, the speeds and directions of motion of the molecules keep on changing. Thus, all the molecules in a sample of gas do not have same speed. Although it is not possible to find out the speeds of individual molecule , yet from probability considerations it has become possible to work out the distribution of molecules in different speed intervals. This distribution is referred to as the Maxwell-Boltzmann distribution in honour of the scientists who developed it. It may be noted that the distribution of speeds remains constant at a particular temperature although individual speeds of molecules may change.

$$dN = 4\pi N \left[\frac{M}{2\pi RT}\right]^{\frac{3}{2}} e^{-\frac{Mu^2}{2RT}} u^2 du = 4\pi N \left[\frac{m}{2\pi kT}\right]^{3/2} e^{\frac{-mu^2}{2KT}} .u^2.du$$

Here, dN = Number of molecules having speeds between u and u + du.

N = Total number of molecules.

M = Molar mass of gas (kg/mol)

u = Root mean square velocity

du = Velocity interval

$$\frac{dN}{N} = 4\pi\left[\frac{M}{2\pi RT}\right]^{\frac{3}{2}} e^{-\frac{Mu^2}{2RT}} u^2 du$$

Here, dN / N = fraction of molecules having speeds between u and u + du.

and $\frac{1}{N}\left(\frac{dN}{du}\right)$ = fraction of molecules having speed between u to u + du per unit interval of speed.

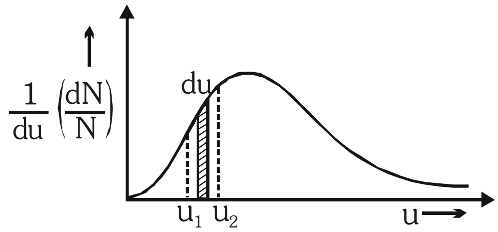
= Maxwell distribution function.

According to this expression, the fraction $\frac{dN}{N}$ of molecules depends only on temperature having speeds between u and u + du for a gas of molar mass M. Thus for a given temperature, this fraction has a constant value.

ALLEN Session 2019-20

11.1 Properties of Maxwell's graph :

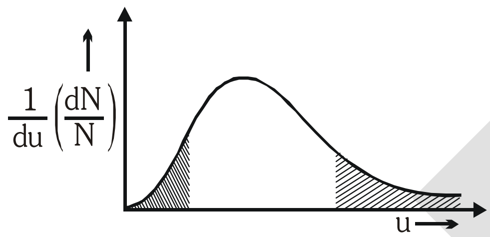
I.



Area between u_1 and u_2 = $\int_{u_1}^{u_2} \frac{1}{du}\left(\frac{dN}{N}\right)du = \int_{u_1}^{u_2}\left(\frac{dN}{N}\right)$

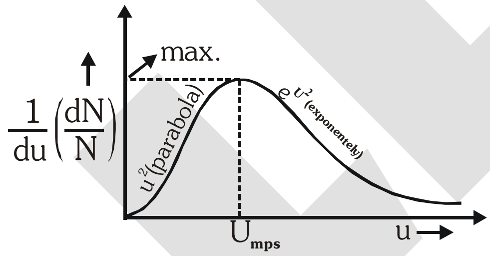
Hence, total fraction of particles with speed between u_1 and u_2
= Area under the curve represents fraction of molecules.

II.



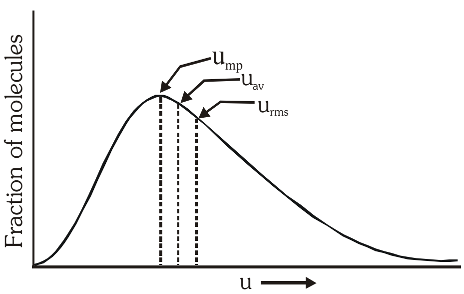
It can be seen from the above figure, that the fraction of molecules having either very low speeds or very high speeds are small in numbers.

III.



The curve at any temperature is parabolic near the origin, since the factor u^2 is dominant in this region, the exponential function being approximately equal to unity. At high values of u, however, the exponential factor dominates the behaviour of the function, causing it to decrease rapidly in value. As a consequence of the contrasting behaviour of two factors, the product function passes through a maximum at a speed known as the most probable speed (u_{mps}). Thus, the most probable speed is the speed possessed by the maximum fraction of the molecules.

IV Graph between fraction of molecules vs molecular speeds :



ALLEN Session 2019-20
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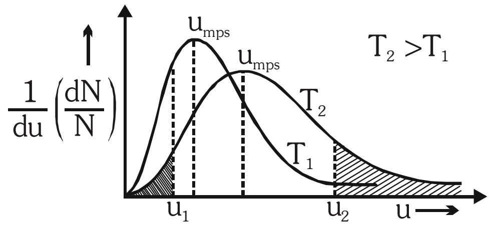
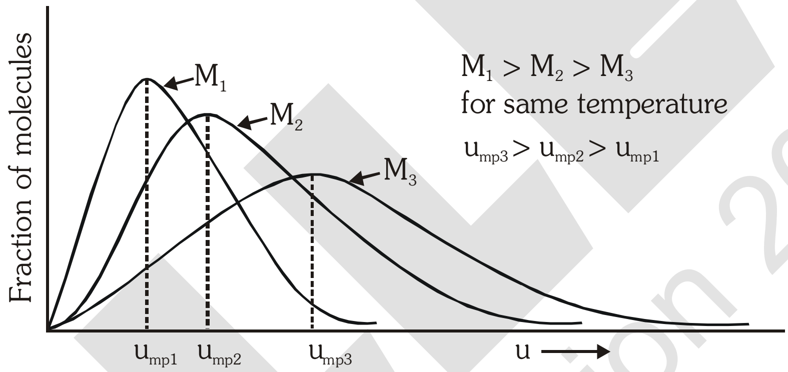
V.



Total area under the curve will be constant and will be unity at all temperatures. The above figure illustrates the distribution of speeds at two temperatures T_1 and T_2. Since the total no. of molecules is same at both temperatures, increase in the K.E. of the molecules results decrease in fraction of molecules having lower speed range and increase in fraction of molecules having higher speed range on increasing the temperature.

On increasing temperature the value of u_{mps}(most probable speed) will increase. Also the curve at the higher temperature T_2 has its u_{mps} shifted to a higher value compared with that for T_1, whereas corresponding fraction of molecules has decreased. But at the same time, the curve near u_{mps} has become broader at the higher temperature indicating the more molecules possess speeds near to most probable speed.

VI.

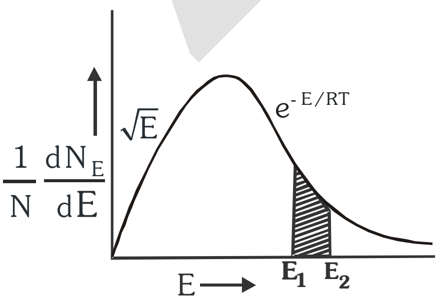


At a given temperature u_{mps} will be more for lighter gas (M_3) but fraction of molecules moving with u_{mps} will be more for heavier gas (M_1).

Note : Effect of M and T are opposite.

11.2 Maxwell Distribution of kinetic energy :

By Maxwell equation : $dN = 4\pi N\left(\frac{M}{2\pi RT}\right)^{3/2} . e^{\frac{-Mu^2}{2RT}} u^2.du = 2\pi\left(\frac{1}{\pi RT}\right)^{3/2} \sqrt{E}.e^{-E/RT}.dE$



The shaded area of this graph indicate the fraction of particles having energy between E_1 and E_2

Ex.53 Four particles have speed 2, 3, 4 and 5 cm/s respectively. Find their avg. & rms speed :

Sol. $U_{avg.} = \frac{U_1 + U_2 + U_3 + ...U_N}{N}$

$U_{avg.} = \frac{2+3+4+5}{4} = 3.5\,cm/s$

$U_{r.m.s.} = \sqrt{\frac{U_1^2 + U_2^2 + U_3^2 + ...}{N}}$

$u_{rms} = \sqrt{\frac{2^2 + 3^2 + 4^2 + 5^2}{4}} = \frac{\sqrt{54}}{2}\,cm/s$

Ex.54 At what temperature do the average speed of $CH_{4(g)}$ molecule equal the average speed of O_2 molecule at 300 K ?

Sol. $(U_{avg})_{CH_4} = (U_{avg})_{O_2}$

$\sqrt{\frac{8RT}{\pi \times 16}} = \sqrt{\frac{8 \times R \times 300}{\pi \times 32}}$

T = 150 K

Ex.55 At 27°C find the ratio of root mean square speeds of ozone to oxygen :-

Sol. $\frac{U_{rms}(O_3)}{U_{rms}(O_2)} = \sqrt{\frac{\frac{3RT}{M.W_{O_3}}}{\frac{3RT}{M.W_{O_2}}}} = \sqrt{\frac{M.W_{O_2}}{M.W_{O_3}}} = \sqrt{\frac{32}{48}} = \sqrt{\frac{2}{3}}$

Ex.56 The temperature at which U_{rms} of He becomes equal to U_{mp} of CH_4 at 500 K.

Sol. $(U_{rms})_{He} = (U_{mp})_{CH_4}$

$\sqrt{\frac{3RT_{He}}{M_{He}}} = \sqrt{\frac{2RT_{CH_4}}{M_{CH_4}}}$

$\frac{3T}{4} = \frac{2 \times 500}{16}$

$T = \frac{250}{3}K$

Ex.57 Calculate the root mean square speed of H_2 molecules under following condition.

(a) 2 mole of H_2 at 27°C.

(b) 3 mole of H_2 in a 5 lit container at 10^5 Pa.

(c) 4 mole of H_2 at the density of 1 gm/ml at 10^5 Pa.

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\01_Ideal gas\Theory.p65

Sol. (a) $U_{r.m.s.} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3\times 8.314\,J/mol\,K\times 300\,K}{2\times 10^{-3}\,kg}} = 1934.24$ m/sec.

(b) $U_{r.m.s.} = \sqrt{\frac{3PV}{nM}} = \sqrt{\frac{3\times 10^{5}\,Pa\times 5\times 10^{-3}\,m^{3}}{3\,mol\times 2\times 10^{-3}\,kg}} = 500$ m/sec.

(c) $U_{r.m.s.} = \sqrt{\frac{3P}{d}} = \sqrt{\frac{3\times 10^{5}\,Pa}{10^{3}\,kg/m^{3}}} = 17.32$ m/sec.

Ex.58 Calculate the fraction of N_2 molecules at 1 atm and 27°C whose speeds are in the range of $(U_{mp} - 0.005\,U_{mp})$ to $(U_{mp} + 0.005\,U_{mp})$?

Sol. Most probable speed,

$$U_{mp} = \sqrt{\frac{2RT}{M}} = \sqrt{\frac{2\times 8.314\times 300}{28\times 10^{-3}}} = 422.09 \text{ m/sec}$$

$du = (U_{mp} + 0.005\,U_{mp}) - (U_{mp} - 0.005\,U_{mp})$

$= 2 \times 0.005 \times U_{mp} = 4.22$ m/sec

From Maxwell's equation :

$$\frac{dN}{N} = 4\pi\left[\frac{M}{2\pi RT}\right]^{\frac{3}{2}} e^{-\frac{MU_{mp}^2}{2RT}}\, U_{mp}^{2} du$$

$$\frac{dN}{N} = 4\pi\left(\frac{M}{2\pi RT}\right)^{\frac{3}{2}} .e^{-\frac{M}{2RT}\times\frac{2RT}{M}} .U_{mp}^2.du$$

$$\frac{dN}{N} = 4 \times 3.14 \times \left(\frac{28\times 10^{-3}}{2\times 3.14\times 8.314\times 300}\right)^{\frac{3}{2}} e^{-1}\times (422.09)^2 \times 4.22$$

$$\frac{dN}{N} = 8.3\times 10^{-3}$$

Ex.59 Calculate the fraction of N_2 molecules whose speeds are in the range of U_{mp} to $(U_{mp} + f \times U_{mp})$?

Sol. $$\frac{dN}{N} = 4\pi\left[\frac{M}{2\pi RT}\right]^{\frac{3}{2}} e^{-\frac{MU^2}{2RT}}\, U^2 du$$

Here, $U = U_{mp} = \sqrt{\frac{2RT}{M}}$, $du = f \times U_{mp}$

$$\Rightarrow \frac{dN}{N} = 4\pi\left[\frac{M}{2\pi RT}\right]^{\frac{3}{2}} e^{-\frac{M}{2RT}\times\frac{2RT}{M}} \times \frac{2RT}{M}\times f\times\left(\frac{2RT}{M}\right)^{\frac{1}{2}}$$

$$\frac{dN}{N} = 4\pi \times \frac{1}{\pi\sqrt{\pi}} e^{-1}\times f = \frac{4f}{e\sqrt{\pi}}$$

ALLEN Session 2019-20

This fraction does not depend upon U_{mp}, or Temperature and depends only on 'f' factor.

(i) Therefore, we can conclude that at two different temperatures the fraction of molecules between U_{mp_1} to $(U_{mp_1} + f \times U_{mp_1})$ and between U_{mp_2}, to $(U_{mp_2} + f \times U_{mp_2})$ are same.

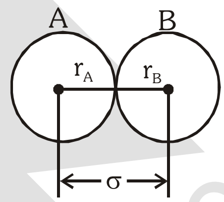
(ii) Even for two different gases say N_2 and O_2, the fraction of molecules between U_{mp} to $(U_{mp} + f \times U_{mp})$ are same.

12. COLLISION PARAMETERS :

Assumption :

All the particles (molecules or atoms) have rigid , similar shape and size and are spherical in nature that will not change after collision.

12.1 Collision diameter : It is the closest distance between the centres of two molecules taking part in collision.



Collision diameter $(\sigma) = r_A + r_B$

12.2 Collision Frequency :

It is the total number of molecular collisions taking place per second per unit volume of the gas. The no. of collisions made by a single molecule with other molecules per unit time **(collision number)** are given by

$$Z_1 = \sqrt{2}\,\pi\sigma^2\, U_{avg} N^*$$

The total number of bimolecular collision per unit time is given as Z_{11} **(collision frequency)**

$$Z_{11} = \frac{1}{2}(Z_1 N^*) = \frac{1}{2} \times N^* \times \sqrt{2}\,\pi\sigma^2\, U_{avg} N^*$$

$$Z_{11} = \frac{1}{\sqrt{2}}\pi\sigma^2\, U_{avg} N^{*2}$$

(f) If the collisions involve two unlike molecules, the no. of bimolecular collision is given as Z_{12}.

$$Z_{12} = \pi\sigma_{12}^2 \left(\frac{8kT}{\pi\mu}\right)^{\frac{1}{2}} N_1^* N_2^*$$

Where N_1^* and N_2^* are the no. of molecules per unit volume of the two types of gases, σ_{12} is the average diameter of two molecules, that is $\sigma_{12} = \frac{\sigma_1 + \sigma_2}{2}$ & μ is the reduced mass,

that is $\mu = \frac{m_1 m_2}{(m_1 + m_2)}$, m_1 & m_2 are the mass of single molecule respectively 1 and 2.

12.3 Mean free path :

The mean free path is the average distance travelled by a molecule between two successive collisions. We can express it as follows :

$$\lambda = \frac{\text{Average distance travelled per unit time}}{\text{No. of collsions made by single molecule per unit time}}$$

$$= \frac{U_{avg}}{Z_1} = \frac{U_{avg}}{\sqrt{2}\pi\sigma^2 U_{avg} N^*} = \frac{1}{\sqrt{2}\pi\sigma^2 N^*}$$

ALLEN Session 2019-20

12.4 Wall collision :

It represents the total number of molecules colliding at the wall per unit area per unit time.

$$Z_w = \frac{1}{4}.N^*.u_{av} = \frac{P.N_A}{\sqrt{2\pi MRT}}$$

12.5 Rate of effusion :

If the corss-section area of orifice in the vessel is 'A', then the number of molecules effusing out per unit time is

$$r_{eff.} = Z_w.A = \frac{P.N_A.A}{\sqrt{2\pi MRT}}$$

Ex.60. Calculate λ, Z_1 and Z_{11} for oxygen at 298 K and 10^{-3} mm Hg. Given $\sigma = 3.61 \times 10^{-8}$ cm.

Sol. $N^* = \frac{P}{kT} = \frac{10^{-3} \times 101325}{760 \times 1.38 \times 10^{-23} \times 298} = 0.324 \times 10^{20}$

$U_{avg} = \sqrt{\frac{8}{\pi}\frac{RT}{M}} = \sqrt{\frac{8}{3.14} \times \frac{8.314 \times 298}{32 \times 10^{-3}}} = 444.138$ m/sec

$Z_1 = \sqrt{2}\,\pi\,\sigma^2\,U_{avg}\,N^* = \sqrt{2} \times 3.14 \times 3.61 \times 10^{-10}\text{ m} \times 444.138\text{ m} \times 0.324 \times 10^{20}$

$= 8.326 \times 10^3\ \text{sec}^{-1}$

$Z_{11} = \frac{1}{2} Z_1 N^* = \frac{1}{2} \times 8.326 \times 10^3\ \text{sec}^{-1} \times 0.324 \times 10^{20} = 13.488 \times 10^{22}\ \text{m}^{-3}\ \text{sec}^{-1}$;

$\lambda = \frac{U_{avg.}}{Z_1} = \frac{444.138\text{ m/sec}}{8.326 \times 10^3\ \text{sec}^{-1}} = 5.334 \times 10^{-2}$ m

Ex.61 Two flask A and B have equal volume. A is maintained at 300 K and B at 600 K while A contains H_2 gas , B has an equal mass of CH_4 gas. Assuming ideal behaviour for both the gases, find the following.

(a) Flask containing greater number of moles

(b) Flask in which pressure is greater

(c) Flask in which U_{avg} of the molecules are greater

(d) Flask with greater mean free path of molecules (Collision diameters of H_2 & CH_4 may be taken same)

(e) Flask with greater molar kinetic energy.

(f) Flask in which the total kinetic energy is greater

(g) Flask in which Z_1 and Z_{11} are greater

Sol.

A	B
H_2 (300K)	CH_4 (600K)

(a) $N_{H_2} = \frac{m}{2}N_A$; $N_{CH_4} = \frac{m}{16}N_A$

$\therefore$ molecules of H_2 in flask A > molecules of CH_4 in flask B

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\01_Ideal gas\Theory.p65

(b) $P_AV = n_ART_A$; $P_BV = n_BRT_B$

$$\Rightarrow \quad \frac{P_A}{P_B} = \frac{n_AT_A}{n_BT_B} = \frac{m/2}{m/16} \times \frac{300}{600} = 4$$

$\therefore$ pressure of H_2 in flask A > pressure of CH_4 in flask B

(c) $(U_{avg})_A = \sqrt{\frac{8}{\pi}\frac{RT_A}{M_A}}$; $(U_{avg})_B = \sqrt{\frac{8}{\pi}\frac{RT_B}{M_B}}$

$$\frac{(U_{avg})_A}{(U_{avg})_B} = \sqrt{\frac{T_A}{T_B} \times \frac{M_B}{M_A}} = \sqrt{\frac{300}{600} \times \frac{16}{2}} = 2$$

$\therefore$ U_{avg} of H_2 in flask A > U_{avg} of CH_4 in flask B

(d) $\lambda = \frac{1}{\sqrt{2}\pi\sigma^2 N^*} = \frac{1}{\sqrt{2}\pi\sigma^2 \times P} \times kT$ [where N* = P/kT]

$$\frac{\lambda_A}{\lambda_B} = \frac{T_A}{T_B} \times \frac{P_B}{P_A} = \frac{300}{600} \times \frac{1}{4} = \frac{1}{8}$$

$\therefore$ $\lambda_B > \lambda_A$

(e) molar K.E. = $\frac{3}{2}RT$

$\therefore$ $T_B > T_A$

$\therefore$ KE of CH_4 in flask B > KE of H_2 in flask A

(f) $(KE)_{total} = \frac{3}{2}RT \times n$

$$\frac{(KE)_{T,A}}{(KE)_{T,B}} = \frac{(300 \times \frac{m}{2})}{(600 \times \frac{m}{16})} = \frac{1}{2} \times 8 = 4$$

$\therefore$ $(KE)_{T,A} > (KE)_{T,B}$

(g) (i) $Z_1 = \sqrt{2}\pi\sigma^2 U_{avg} N^*$

$$\frac{(Z_1)_A}{(Z_1)_B} = \frac{(U_{avg})_A}{(U_{avg})_B} \times \frac{N_A{}^*}{N_B{}^*} = 2 \times \frac{N_A}{N_B} = 2 \times 8 = 16$$

$\therefore$ $(Z_1)_A > (Z_1)_B$ $(Z_1)_A > (Z_1)_B$

(ii) $Z_{11} = \frac{1}{\sqrt{2}}\pi\sigma^2 U_{avg} N^{*2}$

$$\frac{(Z_{11})_A}{(Z_{11})_B} = \frac{(U_{avg})_A}{(U_{avg})_B} \times \frac{(N_A{}^*)^2}{(N_B{}^*)^2} = 2 \times \frac{N_A}{N_B} = 2 \times 8^2 = 128$$

$\therefore$ $(Z_{11})_A > (Z_{11})_B$

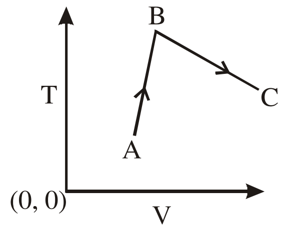
ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\01_Ideal gas\Theory.p65

EXERCISE # O-I

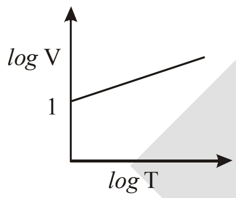
Single Correct :

1. In the given isobaric process shown by graph between T & V.



(A) Moles decreases throughout (B) Moles first increases then decreases
(C) Moles first decreases then increases (D) Moles cannot be predicted form given data

2. At constant pressure of 0.821 atm ; log V vs log T is ploted as shown in-
Then number of moles present in experiment -



(A) 1 (B) 10 (C) 100 (D) 0.1

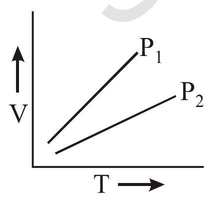
3. A gas at a pressure of 5.0 atm is heated from 0°C to 546°C and simultaneously compressed to one-third of its orginal volume. Hence final pressure is
(A) 10 atm (B) 30 atm (C) 45 atm (D) 5 atm

4. A flask containing air (open to the atmosphere) is heated from 300 K to 500 K. Then % of air escaped to the atmosphere is -
(A) 20.0 (B) 40 (C) 60 (D) 80

5. 10 g of a gas at 1 atm, 273 K occupies 5 litres. The temp. at which the volume becomes double for the same mass of gas at the same pressure is ?
(A) 273K (B) -273^0C (C) 273^0C (D) 546^0C

6. A gas is found to have a formula $[CO]_x$. If its vapour density is 70 the value of x is
(A) 2.5 (B) 3.0 (C) 5.0 (D) 6.0

7. V versus T curves at constant pressure P_1 and P_2 for an ideal gas are shown in Fig. Which is correct



(A) $P_1 > P_2$ (B) $P_1 < P_2$ (C) $P_1 = P_2$ (D) All

8. A container when is empty weighs 50 gm. After certain liquid of density 25 gm/dm^3 is filled its mass becomes equal to 100 gm. The volume of the container will be :
(A) 0.25 dm^3 (B) 0.5 dm^3 (C) 1 dm^3 (D) 2 dm^3

9. A vessel contains mono atomic 'He' at 1 bar and 300 K, determine its number density -
(A) $2.4 \times 10^{25}\ m^{-3}$ (B) $6.8 \times 10^{23}\ m^{-3}$ (C) $4.8 \times 10^{26}\ m^{-3}$ (D) $9.2 \times 10^{27}\ m^{-3}$

ALLEN Session 2019-20
node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\01_Ideal gas\Exercises.p65

10. A rigid container containing 10 gm gas at some pressure and temperature. The gas has been allowed to escape from the container due to which pressure of the gas becomes half of its initial pressure and temperature become $(2/3)^{rd}$ of its initial. The mass of gas (in gms) escaped is

(A) 7.5 (B) 1.5 (C) 2.5 (D) 3.5

11. The density of gas A is twice that of B at the same temperature the molecular weight of gas B is thrice that of A. The ratio of pressure of gas A and gas B will be

(A) 1 : 6 (B) 7 : 8 (C) 6 : 1 (D) 1 : 4

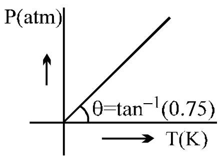
12. In a rigid container NH_3 is kept at certain temperature, if on doubling the temperature it is completely dissociated into N_2 and H_2. Find final pressure to initial pressure ratio :

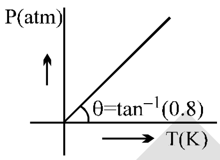
(A) 4 (B) 2 (C) $\frac{1}{2}$ (D) $\frac{1}{4}$

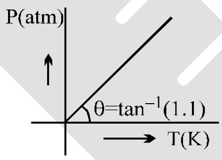
13. Gas A (1 mol) dissociates in a closed rigid container of volume 0.16 lit. as per following reaction.

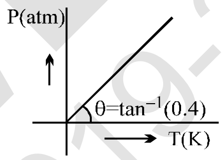
$2A\,(g) \longrightarrow 3B\,(g) + 2C\,(g)$

If degree of dissociation of A is 0.4 and remains constant in entire range of temperature, then the correct P vs T graph is [Given R = 0.08 lit-atm/mol/K]

(A) 

(B) 

(C) 

(D) 

14. A gaseous reaction,

$3A \longrightarrow 2B$

is carried out in a 0.0821 litre closed container initially containing 1 mole of gas A. After sufficient time a curve of P (atm) vs T (K) is plotted and the angle with x-axis was found to be 42.95°. The degree of association of gas A is [Given : tan 42.95 = 0.8]

(A) 0.4 (B) 0.6 (C) 0.5 (D) 0.8

15. 4.0 g of argon has pressure P and temperature T K in a vessel. On keeping the vessel at 50° higher temperature, 0.8 g of argon was given out to maintain the pressure P. The original temperature was :

(A) 73 K (B) 100 K (C) 200 K (D) 510 K

16. The total pressure exerted by a number nonreacting gases is equal to the sum of partial pressure of the gases under the same conditions is known as :

(A) Boyle's law (B) Charle's law (C) Avogadro's law (D) Dalton's law

17. Dalton's law cannot be applied for which gaseous mixture at normal temperatures:

(A) O_2 and N_2 (B) NH_3 and HCl (C) He and N_2 (D) CO_2 and O_2

18. A closed vessel contains helium and ozone at a pressure of P atm. The ratio of He and oxygen atoms is 1 : 1. If helium is removed from the vessel, the pressure of the system will reduce to :

(A) 0.5 P atm (B) 0.75 P atm (C) 0.25 P atm (D) 0.33 P atm

19. At constant temperature 200 cm^3 of N_2 at 720 mm and 400 cm^3 of O_2 at 750 mm pressure are put together in a one litre flask. The final pressure of mixture is

(A) 111 mm (B) 222 mm (C) 333 mm (D) 444 mm

20. If saturated vapours are compressed slowly (temperature remaining constant) to half the initial volume, the vapour pressure will :

(A) Become four times (B) become doubled (C) Remain unchanged (D) Become half

ALLEN Session 2019-20

21. A box of 1L capacity is divided into two equal compartments by a thin partiion which are filled with 2g H_2 and 16g CH_4 respectively. The pressure in each compartment is recorded as P atm. The total pressure when partition is removed will be :

(A) P (B) 2P (C) P/2 (D) P/4

22. A vessel has N_2 gas and water vapours at a total pressure of 1 atm. The vapour pressure of water is 0.3 atm. The contents of this vessel are transferred to another vessel having one third of the capacity of original volume, completely at the same temperature the total pressure of this system in the new vessel is -

(A) 3.0 atm (B) 1 atm (C) 3.33 atm (D) 2.4 atm

23. Which gas effuses fastest under identical conditions -

(A) N_2 (B) O_2 (C) Cl_2 (D) CH_4

24. The rate of diffusion of methane at a given temperature is twice that of a gas X. The molecular weight of X is :

(A) 64 (B) 32 (C) 4.0 (D) 8.0

25. The rate of diffusion of two gases A and B is in the ratio of 1 : 4 and that of B and C in the ratio of 1 : 3 the rate of diffusion of C with respect to A is -

(A) $\frac{1}{12}$ (B) 12 (C) 6 (D) 4

26. A gas X diffuses three times faster than another gas Y the ratio of their densities i.e., $D_x : D_y$ is

(A) 1/3 (B) 1/9 (C) 1/6 (D)1/12

27. Rate of diffusion of a gas is :

(A) directly proportional to its density

(B) directly proportional to its molecular weight

(C) directly proportional to the square of its molecular weight

(D) inversely proportional to the square root of its molecular weight

28. Since the atomic weights of carbon nitrogen and oxygen are 12, 14 and 16 respectively, among the following pairs of gases, the pair that will diffuse at the same rate is :

(A) CO_2 and N_2O (B) CO_2 and N_2O_3

(C) CO_2 and CO (D) CO_2 and NO

29. If the four tubes of a car are filled to the same pressure with N_2, O_2, H_2 and He separately then which one will be filled first :

(A) N_2 (B) O_2 (C) H_2 (D) He

30. The increasing order of effusion among the gases, H_2, O_2, NH_3 and CO_2 is –

(A) H_2, CO_2, NH_3, O_2 (B) H_2, NH_3, O_2, CO_2

(C) H_2, O_2, NH_3, CO_2 (D) CO_2, O_2, NH_3, H_2

31. The rate of diffusion of hydrogen is about–

(A) $\frac{1}{2}$ that of Helium (B) 1.4 times that of Helium

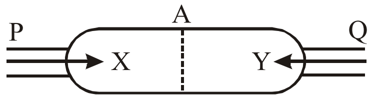
(C) twice that of H (D) Four times that of Helium

32. Under identical conditions of pressure and temperature. 4 L of gaseous mixture (H_2 and CH_4) effuses through a hole in 5min whereas 4 L of a gas X of molecular mass 36 takes to 10 min to effuse through the same hole. The mole ratio of $H_2 : CH_4$, in the mixture is -

(A) 1 : 2 (B) 2 : 1 (C) 2 : 3 (D) 1 : 1

ALLEN Session 2019-20

33. 3 mole of gas "X" and 2 moles of gas "Y enters from end "P" and "Q" of the cylinder respectively. The cylinder has the area of cross-section A, shown as under -



The length of the cylinder is 150 cm. The gas "X" intermixes with gas "Y" at the point A. If the molecular weight of the gases X and Y is 20 and 80 respectively, then what will be the distance of point A from Q?

(A) 75cm (B) 50cm (C) 37.5 cm (D) 90 cm

34. Under identical experiment conditions which of the following pairs of gases will be most easy to separate by using effusion process -

(A) H_2 and T_2 (B) SO_2 and SO_3 (C) NH_3 and CH_4 (D) $U^{235}O_2$ and $U^{238}O_2$

35. Certain volume of He gas takes 10 sec for its diffusion, how much time will be taken by CH_4 gas to disffuse it's same volume under identical conditions -

(A) 5 sec (B) 10 sec (C) 20 sec (D) 40 sec

36. A football bladder contains equimolar proportions of H_2 and O_2. The composition by mass of the mixture effusing out of punctured football is in the ratio ($H_2 : O_2$)

(A) 1 : 4 (B) $2\sqrt{2} : 1$ (C) $1 : 2\sqrt{2}$ (D) 4 : 1

37. Consider the following pairs of gases A and B.

	A	B
(a)	CO	N_2
(b)	O_2	O_3
(c)	$^{235}UF_6$	$^{238}UF_6$

Relative rates of effusion of gases A to B under similar condition is in the order:

(A) a < b < c (B) a < c < b (C) a > b > c (D) a > c > b

38. For the reaction

$$2NH_3(g) \longrightarrow N_2(g) + 3H_2(g),$$

what is the % of NH_3 converted if the mixture diffuses twice as fast as that of SO_2 under similar conditions.

(A) 3.125 % (B) 31.25 % (C) 6.25 % (D) 62.5 %

39. A 4 : 1 molar mixture of He & CH_4 kept in a vessel at 20 bar pressure. Due to a hole in the vessel, gas mixture leaks out. What is the composition of mixture effusing out initially -

(A) 8 : 1 (B) 4 : 1 (C) 1 : 4 (D) 4 : 3

40. Calculate the ratio of rate of effusion of O_2 and H_2 from a container containing 16gm O_2 and 2gm H_2

(A) 1 : 8 (B) 8 : 1 (C) 1 : 4 (D) 4 : 1

41. The number of effusion steps required to convert a mixture of H_2 and O_2 from 240 : 1600 (by mass) to 3072 : 20 (by mass) is

(A) 2 (B) 4 (C) 5 (D) 6

42. The density of the gaseous mixture in a vessel (CH_4 and He) at 2 atmosphere pressure and 300 K is 0.9756 g/lit. If a small pin-hole is made on the wall of the vessel, through which gases effuse, then which of the followings is the correct composition (by volume) of the gases CH_4 and He effusing out initially ?

(A) 1 : 1 (B) 2 : 1 (C) 3 : 1 (D) 1 : 4

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\01_Ideal gas\Exercises.p65

43. At STP, the order of root mean square speed of molecules H_2, N_2, O_2 and HBr is :
(A) $H_2 > N_2 > O_2 > HBr$ (B) $HBr > O_2 > N_2 > H_2$
(C) $HBr > H_2 > O_2 > N_2$ (D) $N_2 > O_2 > H_2 > HBr$

44. Four particles have speed 2, 3, 4 and 5 cm/s respectively. Their rms speed is :
(A) 3.5 cm/s (B) $\left(\frac{27}{2}\right)$ cm/s (C) $\sqrt{54}$ cm/s (D) $\left(\frac{\sqrt{54}}{2}\right)$ cm/s

45. A flask has 10 molecules out of which four molecules are moving at 7ms^{-1} & the remaining are moving at same speed of X ms^{-1}. If Urms of the gas is 5 ms^{-1}. The value of 'X' will be
(A) 5 (B) 3 (C) 9 (D) 16

46. If the average velocity of N_2 molecules is 0.3 m/sec. at 27^{0}C, then the velocity of 0.6 m/sec will take place at :
(A) 273 K (B) 927 K (C) 1000 K (D) 1200 K

47. Which of the following expression does not give root mean square velocity –
(A) $\left(\frac{3RT}{M_w}\right)^{\frac{1}{2}}$ (B) $\left(\frac{3P}{DM_w}\right)^{\frac{1}{2}}$ (C) $\left(\frac{3P}{D}\right)^{\frac{1}{2}}$ (D) $\left(\frac{3PV}{nM_w}\right)^{\frac{1}{2}}$

48. Which one of the following gases would have the highest R.M.S. velocity at 25^{0}C ?
(A) Oxygen (B) Carbon dioxide (C) Sulphur dioxide (D) Carbon monoxide

49. At what temperature would the rms speed of a gas molecule have twice its value at 100°C ?
(A) 4192 K (B) 1492 K (C) 9142 K (D) 2491 K

50. Two flasks X and Y have capacity 1L and 2L respectively and each of them contains 1 mole of a gas. The temperature of the flask are so adjusted that average speed of molecules in X is twice as those in Y. The pressure in flask X would be
(A) same as that in Y (B) half of that in Y (C) twice of that in Y (D) 8 times of that in Y

51 Temperature at which most probable speed of O_2 becomes equal to root mean square speed of N_2 is [Given : N_2 at 427°C]
(A) 732 K (B) 1200 K (C) 927 K (D) 800 K

52. The density ratio of O_2 and H_2 is 16 : 1. The ratio of their U_{rms} is :-
(A) 4 : 1 (B) 16 : 1 (C) 1 : 4 (D) 1 : 16

53. Which of the gas have highest fraction of molecules at 27ºC in most probable speed region -
(A) H_2 (B) N_2 (C) O_2 (D) CO_2

54. The av. K.E./mole of an ideal monoatomic gas at 27^{0}C is
(A) 900 cal (B) 1800 cal (C) 300 cal (D) None

55. The average kinetic energy of an ideal gas per molecule in SI units at 25^{0}C will be :
(A) 6.17×10^{-21} kJ (B) 6.17×10^{-21} J (C) 6.17×10^{-20} J (D) 7.16×10^{-20} J

ALLEN Session 2019-20

56. At what temperature will be total kinetic energy (KE) of 0.30 mole of He be the same as the total KE of 0.40 mole of Ar at 400 K :
(A) 400 K (B) 373 K (C) 533 K (D) 300 K

57. Average K.E. of CO_2 at 27^0C is E. The average kinetic energy of N_2 at the same temperature will be
(A) E(B) 22E (C) E/22 (D) $E/\sqrt{2}$

58. If a gas expands at constant temperature then :
(A) No. of gaseous molecule decreases (B) kinetic energy of molecule decreases
(C) K.E. remain same (D) K.E. increases

59. Total translational kinetic energy possessed by 8 gm methane at 273°C
(R = 2 Cal/mol-K)
(A) 819 calorie (B) 409.5 calorie (C) 1638 calorie (D) None of these

60. Ideal gas equation in terms of K.E. per unit volume E, is-
(A) $P = \frac{3}{2}RT$ (B) $P = \frac{2}{3}E$ (C) $P = \frac{2}{3}RT$ (D) $P = \frac{3}{2}RT$

61. Three gases A, B and C are at same temperature. If their r.m.s. speed are in the ratio $1 : \frac{1}{\sqrt{2}} : \frac{1}{\sqrt{3}}$ then their molar masses will be in the ratio :
(A) 1 : 2 : 3 (B) 3 : 2 : 1 (C) $1 : \sqrt{2} : \sqrt{3}$ (D) $\sqrt{3} : \sqrt{2} : 1$

62. The following gases are present under similar condition of T, P & V. The longest mean free path stands for
(A) H_2 (B) N_2 (C) O_2 (D) Cl_2

63. At constant volume, Z_{11} is directly proportional to -
(A) $\sqrt{P}$ (B) P (C) T^2 (D) T

64. At constant pressure, Z_{11} is directly proportional to -
(A) $\sqrt{T}$ (B) $\frac{1}{T}$ (C) $\frac{1}{\sqrt{T}}$ (D) $\frac{1}{T^{3/2}}$

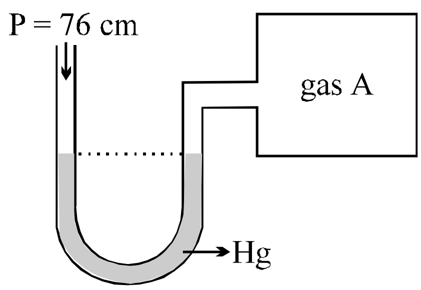
65. On increasing the temperature of an ideal gas at constant volume, the mean free path -
(A) Increases
(B) Decreases
(C) Remains unchanged
(D) May or may not change depending on the size of gas molecules

ALLEN Session 2019-20

EXERCISE # O-II

One or more may be correct :

1. An open ended mercury manometer is used to measure the pressure exerted by a trapped gas as shown in the figure. Initially manometer shows no difference in mercury level in both columns as shown in diagram.



After sparking 'A' dissociates according to following reaction

$$2A(g) \longrightarrow 3B(g) + 2C(g)$$

If pressure of Gas "A" decreases to 0.8 atm. Then (Assume temperature to be constant and is 300 K)

(A) total pressure increased by 1.3 atm

(B) total pressure increased by 0.3 atm

(C) total pressure increased by 22.3 cm of Hg

(D) difference in mercury level is 228 mm.

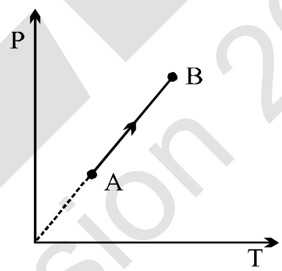
2. Select the correct option for an ideal gas undergoing a process as shown in diagram.

(A) If 'n' is changing , 'V' must also be changing.

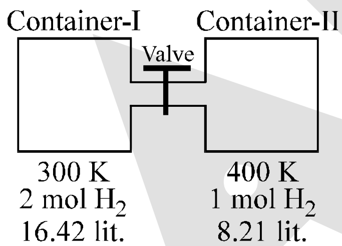
(B) If 'n' is constant , 'V' must be constant.

(C) If 'n' is constant , 'V' must be changing.

(D) If 'n' is changing , 'V' must be constant.



3. Select the correct option(s):



(A) Pressure in container-I is 3 atm before opening the valve.

(B) Pressure after opening the valve is 3.57 atm.

(C) Moles in each compartment are same after opening the valve.

(D) Pressure in each compartment are same after opening the valve.

4. When an equimolar mixture of two gases A and B [$M_A > M_B$] is allowed to effuse through a Pin hole select incorrect statement -

(A) B comes out at a faster rate

(B) Relative rate of effusion of A increases with time

(C) Rate of effusion of B will always be greater

(D) Initially, with equal molar ratio rate of effusion of B is greater than rate of effusion of A.

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\01_Ideal gas\Exercises.p65

5. Select the correct option(s) for an ideal gas

(A) Most probable speed increases with increase in temperature

(B) Fraction of particles moving with most probable speed increases with increase in temperature

(C) Fraction of particles moving with most probable speed are more for Cl_2 than H_2 under similar condition of T, P & V.

(D) Most probable speed is more for Cl_2 than H_2 at same temperature

6. A closed vessel at temperature T contain a mixture of two diatomic gases A and B. Molar mass of A is 16 times that of B and mass of gas A contained in the vessel is 2 times that of B. Which of the following statements are correct-

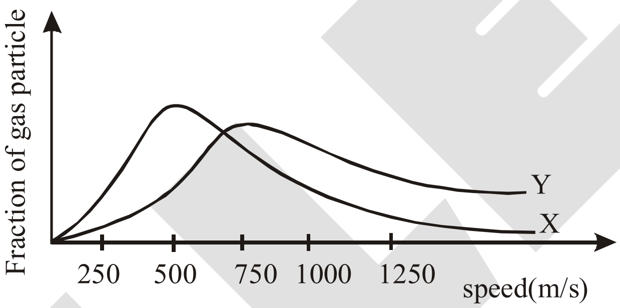
(A) Average kinetic energy per molecule of A is equal to that of B.

(B) Root mean square velocity of B is four times that of A

(C) Pressure exerted by B is eight time of that exerted by A

(D) Number of molecules of B, in the cylinder, is eight time that of A

7. The graph below shows the distribution of molecular speed of two ideal gases X and Y at 200K. on the basis of the below graph identify the correct statements -



(A) If gas X is methane, then gas Y can be CO_2

(B) Fraction of molecules of X must be greater than Y in a particular range of speed at 200K

(C) Under identical conditions rate of effusion of Y is greater than that of X

(D) The molar kinetic energy of gas X at 200K is equal to the molar kinetic energy of Y at 200K

8. Identify the correct statements when a fixed amount of ideal gas is heated in a container fitted with a movable piston always operating at constant pressure.

(A) Average distance travelled between successive collisions will decreases.

(B) Collisions frequency increases since speed of the molecules increases with increase in temperature.

(C) Average relative speed of approach remains unaffected.

(D) Average angle of approach remains unaffected.

9. Choose the correct statement(s) among the following

(A) Average molecular speed of gases increases with decrease in fraction of molecules moving slowly

(B) Rate of effusion of gases increases with increase in collision frequency at constant volume.

(C) Rate of effusion is inversily proportional to molecular weight of gas

(D) Mean free path does not change with change in temperature at constant pressure.

10. Which of the following quantities is the same for all ideal gases at the same temperature :

(A) The kinetic energy of 1 mol

(B) The kinetic energy of 1 g

(C) The number of molecules in 1 mol

(D) The number of molecules in 1 g

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module#Ideal gas, Solid state & Chemical Eq\Eng\01_Ideal gas\Exercises.p65

11. Choose the correct statement(s) among the following -

(A) The mean free path (λ) of gaseous molecules is directly proportional to temperature of gas at constant volume

(B) The mean free path (λ) of gaseous molecules is inversely proportional to pressure of gas at constant volume

(C) The mean free path (λ) of gaseous molecules is directly proportional to volume of gas at constant T

(D) The mean free path (λ) of gaseous molecules is directly proportional to volume of gas at constant P

12. Which statement is/are correct for postulates of kinetics theory of gases -

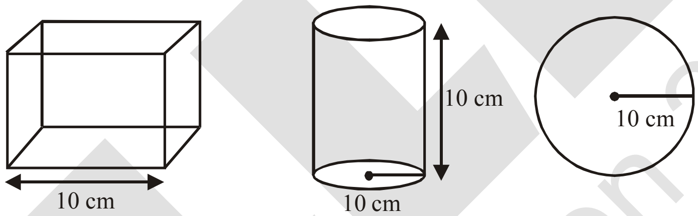
(A) Gases are composed of molecules whose size is negligible compared with the average distance between them

(B) Molecules moves randomly in straight lines in all directions and at various speeds.

(C) When molecules collide with one another the collisions are elastic. In an elastic collision the loss of kinetic energy takes place

(D) The average kinetic energy of a molecule is proportional to the absolute temperature.

13. Three closed containers are filled with equal amount of same ideal gas.



Container-I (Cube) **Container-II (Cylinder)** **Container-III (Sphere)**

If pressure is same in all container then -

(A) $(U_{rms})_3 > (U_{rms})_2 > (U_{rms})_1$ (B) $(U_{rms})_3 = (U_{rms})_2 = (U_{rms})_1$

(C) $\lambda_1 < \lambda_2 < \lambda_3$ ($\lambda \rightarrow$ Mean free path) (D) $(z_1)_1 < (z_1)_2 < (z_1)_3$

14. Which of the following statements is (are) true -

(A) The ratio of the average speed to the rms speed is independent of the temperature

(B) The square of the mean squared speed of the molecule is equal to the mean square speed at a certain temperature

(C) Mean kinetic energy of the gas molecules at any given temperature is independent of the mean speed

(D) The difference between rms speed and average speed at any temperature for different gases diminished as larger molar masses are considered

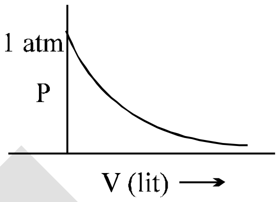
ALLEN Session 2019-20

Paragraph for Q.15 & 17

Question No. 18 to 20 are based on the following Passage. Read it carefully & answer the questions that follow

On the recently discovered 10^{th} planet it has been found that the gases follow the relationship $Pe^{V/2} = nCT$ where C is constant other notation are as usual (V in lit., P in atm and T in Kelvin). A curve is plotted

between P and V at 500 K & 2 moles of gas as shown in figure



15. The value of constant C is

(A) 0.01 (B) 0.001 (C) 0.005 (D) 0.002

16. Find the slope of the curve plotted between P Vs T for closed container of volume 2 lit. having same moles of gas

(A) $\frac{e}{2000}$ (B) 2000 e (C) 500 e (D) $\frac{2}{1000e}$

17. If a closed container of volume 200 lit. of O_2 gas (ideal gas) at 1 atm & 200 K is taken to planet. Find the pressure of oxygen gas at the planet at 821 K in same container

(A) $\frac{10}{e^{100}}$ (B) $\frac{20}{e^{50}}$ (C) 1 atm (D) 2 atm

Paragraph for Question 18 to 20

The constant motion and high velocities of gas particles lead to some important practical consequences. One such consequence is that as minimum rapidly when they come in contact. The mixing of different gases by random molecular motion and with frequent collisions is called diffusion A similar process in which gas molecules escape through a tiny hole into a vaccume is called effusion.

18. Helium gas at 1 atm and SO_2 at 2 atm pressure, temperature being the same, are released separately at the same moment into 1 m long evacuated tubes of equal diameters. If helium reaches the other end of the tube in t sec, what distance SO_2 would traverse in the same time interval in the other tube ?

(A) 25 cm (B) 50 cm (C) 60 cm (D) 75 cm

19. 4 g of H_2 effused through a pinhole in 10 sec at constant temperature and pressure. The amount of oxygen effused in the same time interval and at the same conditions of temperature and pressure would be :

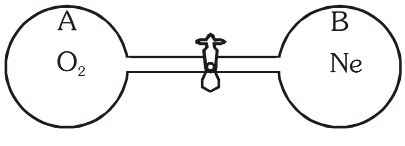
(A) 4 g (B) 8 g (C) 16 g (D) 32 g

20. For 10 min. each at 27°C, from two identical bulbs helium and an unknown gas X at equal pressures are leaked into a common vessel of 3 L capacity. The resulting pressure is 4.1 atm and the mixture contains 0.4 mol of helium. The molar mass of gas X is :

(A) 16 (B) 32 (C) 64 (D) None of these

Paragraph for Question 21 to 22

Initially, flask A contained oxygen gas at 27°C and 950 mm of Hg, and flask B contained neon gas at 27°C and 900 mm. Finally, the two flasks were joined by means of a narrow tube of negligible volume equipped with a stopcork and gases were allowed to mixup freely. The final pressure in the combined system was found to be 910 mm of Hg.



21. What is the correct relationship between volumes of the two flasks ?

(A) $V_B = 3V_A$ (B) $V_B = 4V_A$ (C) $V_B = 5V_A$ (D) $V_B = 4.5V_A$

22. How many moles of gas are present in flask A in the final condition, if volume of flask B is 304 litre ? ($R = 0.08$ atm L mol^{-1} K^{-1})

(A) 7.58 (B) 3.79 (C) 15.16 (D) None of these

TABLE TYPE QUESTION :

Column-I (Gases at different) conditions)	Column-II (Value of speed (m/s)	Column-III
(A) CH_4 at 27°C	(P) U_{rms} = 342.5	(I) Molar K.E. = 3750 Joule
(B) SO_2 at 27°C	(Q) U_{mp} = 1100	(II) Molar K.E. = 5000 Joule
(C) 4 gm He at 1 atm and 24.6 litre	(R) U_{rms} = 685	(III) Average K.E. per gram = 937.5 Joule
(D) 32 gm O_2 at 127°C	(S) U_{rms} = 550	(IV) Average K.E. per gram = 234 Joule.

(Given : $R = \frac{25}{3}\frac{J}{mol \times K}$, $\sqrt{30} = 5.48$, $\sqrt{\frac{2}{3}} = 0.8$, $\sqrt{5} = 2.2$**)**

23. Which of the following is correct

(A) A ; R ; 4 (B) A ; R ; 2 (C) A ; Q ; 4 (D) B ; R ; 1

24. Which of the following is correct

(A) B ; Q ; 1 (B) B ; Q ; 2 (C) B ; P ; 1 (D) C ; S ; 3

25. Which of the following is correct

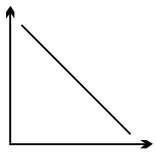
(A) C ; Q ; 3 (B) D ; R ; 2 (C) D ; S ; 1 (D) D ; S ; 4

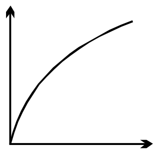
ALLEN Session 2019-20

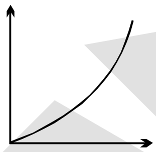
Match the column :

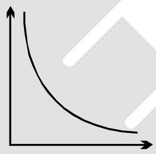
26. Match the entries in column I with entries in Column II and then pick out correct options.

Column I **Column II**

(A) $\frac{1}{V^2}$ vs P for ideal gas at constant T and n. (P) 

(B) V vs $\frac{1}{T}$ for ideal gas at constant P and n (Q) 

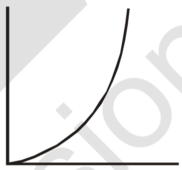
(C) log P vs log V for ideal gas at constant T and n. (R) 

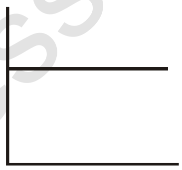
(D) V vs $\frac{1}{P^2}$ for ideal gas at constant T and n. (S) 

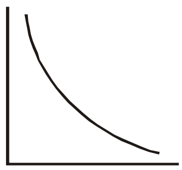
27. Match the column-

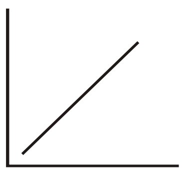
Column-I **Column-II**

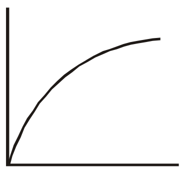
(For an ideal gas at constant pressure) **(correct graph)**

(A) V v/s T (P) 

(B) TV v/s $\left(\frac{1}{T^2}\right)$ (Q) 

(C) $\frac{V}{T}$ v/s V (R) 

(D) TV v/s T (S) 

(T) 

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # ideal gas, Solid state & Chemical Eq\Eng\01_Ideal gas\Exercises.p65

JEE-MAINS

1. According to the kinetic theory of gases, in an ideal gas, between two successive collisions a gas molecule travels - **[AIEEE-2003]**

(1) In a straight line path(2) with an accelerated velocity
(3) In a circular path (4) In a wavy path

2. What volume of hydrogen gas, at 273K and 1 atm, pressure will be consumed in obtaining 21.6g of elemental boron (atomic mass = 10.8) from the reduction of boron trichloride by hydrogen ? **[AIEEE-2003]**

(1) 44.8 L (2) 22.4 L (3) 89.6 L (4) 67.2 L

3. As the temperature is raised from 20°C to 40°C, the average kinetic energy of neon atoms changes by factor of which of the following ? **[AIEEE-2004]**

(1) 1/2 (2) $\sqrt{(313/293)}$ (3) 313/293 (4) 2

4. Equal masses of methane and oxygen are mixed in an empty container at 25°C. The fraction of the total pressure exerted by oxygen is - **[AIEEE-2008]**

(1) 2/3 (2) $\frac{1}{3}\times\frac{273}{298}$ (3) $\frac{1}{3}$ (4) $\frac{1}{2}$

5. The molecular velocity of any gas is :- **[AIEEE-2011]**

(1) inversely proportional to the square root of temperature
(2) inversely proportional to absolute temperature
(3) directly proportional to square of temperature
(4) directly proportional to square root of temperature

6. α, v and u represent most probable velocity, average velocity and root mean square velocity respectively of a gas at a particular temperature. The correct order among the following is -**[JEE(Main)-2012]**

(1) $\alpha > u > v$ (2) $v > u > \alpha$ (3) $u > v > \alpha$ (4) $u > \alpha > v$

7. An open vessel at 300 K is heated till $\frac{2}{5}$ th of the air in it is expelled. Assuming that the volume of the vessel remains constant, the temperature to which the vessel is heated is :- **[JEE(Main-online)-2012]**

(1) 750 K (2) 400 K (3) 500 K (4) 1500K

8. For 1 mol of an ideal gas at constant temperature T, the plot of (log P) against(log V) is a (P : Pressure, V : Volume) :- **[JEE(Main-online)-2012]**

(1) Straight line parallel to x-axis (2) Curve starting at origin
(3) Straight line with a negative slope (4) Straight line passing through origin

9. The relationship among most probable velocity, average velocity and root mean square velocity is respectively :- **[JEE(Main-online)-2012]**

(1) $\sqrt{2}:\sqrt{8/\pi}:\sqrt{3}$ (2) $\sqrt{2}:\sqrt{3}:\sqrt{8/\pi}$ (3) $\sqrt{3}:\sqrt{8/\pi}:\sqrt{2}$ (4) $\sqrt{8/\pi}:\sqrt{3}:\sqrt{2}$

10. Which one of the following is the wrong assumption of kinetic theory of gases? **[JEE(Main-online)-2013]**

(1) All the molecules move in straight line between collision and with same velocity.
(2) Molecules are separated by great distances compared to their sizes.
(3) Pressure is the result of elastic collision of molecules with the container's wall.
(4) Momentum and energy always remain conserved.

Session 2019-20

11. By how many folds the temperature of a gas would increase when the root mean square velocity of the gas molecules in a container of fixed volume is increased from 5×10^4 cm/s to 10×10^4 cm/s ?

[JEE(Main-online)-2013]

(1) Four (2) three (3) Two (4) Six

12. For gaseous state, if most probable speed is denoted by C, average speed by $\overline{C}$ and mean square speed by C, then for a large number of molecules the ratios of these speeds are :-

[JEE(Main-offline)-2013]

(1) C : $\overline{C}$: C = 1.225 : 1.128 : 1 (2) C : $\overline{C}$: C = 1.128 : 1.225 : 1

(3) C : $\overline{C}$: C = 1 : 1.128 : 1.225 (4) C : $\overline{C}$: C = 1 : 1.225 : 1.128

13. A gaseous compound of nitrogen and hydrogen contains 12.5%(by mass) of hydrogen. The density of the compound relative to hydrogen is 16. The molecular formula of the compound is :

[JEE(Main-online)-2014]

(1) NH_2 (2) NH_3 (3) N_3H (4) N_2H_4

14. The initial volume of a gas cylinder is 750.0 mL. If the pressure of gas inside the cylinder changes from 840.0 mm Hg to 360.0 mm Hg, the final volume the gas will be **[JEE(Main-online)-2014]**

(1)1.750 L (2) 7.50 L (3) 3.60 L (4) 4.032 L

15. The temperature at which oxygen molecules have the same root mean square speed as helium atoms have at 300 K is : **[JEE(Main-online)-2014]**

(Atomic masses : He = 4 u, O = 16 u)

(1) 1200 K (2) 600 K (3) 300 K (4) 2400 K

16. Which of the following is not an assumption of the kinetic theory of gases ?

(1) Gas particles have negligible volume. **[JEE-Mains (online)-2015]**

(2) A gas consists of many identical particles which are in continual motion.

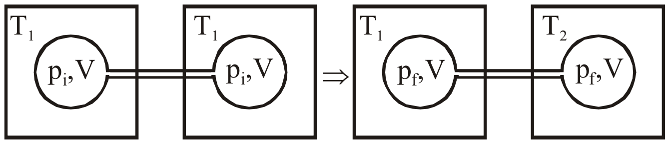
(3) At high pressure, gas particles are difficult to compress.

(4) Collisions of gas particles are perfectly elastic.

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\01_Ideal gas\Exercises.p65

17. Two closed bulbs of equal volume(V) containing an ideal gas initially at pressure p_i and temperature T_1 are connected through a narrow tube of negligible volume as shown in the figure below. The temperature of one of the bulbs is then raised to T_2. The final pressure p_f is :- **[JEE-Mains-2016]**



(1) $2p_i\left(\frac{T_1T_2}{T_1+T_2}\right)$ (2) $p_i\left(\frac{T_1T_2}{T_1+T_2}\right)$ (3) $2p_i\left(\frac{T_1}{T_1+T_2}\right)$ (4) $2p_i\left(\frac{T_2}{T_1+T_2}\right)$

18. At 300 K, the density of a certain gaseous molecule at 2 bar is double to that of dinitrogen (N_2) at 4 bar. The molar mass of gaseous molecule is:- **[JEE-Mains-2017(ONLINE)]**

(1) 28 g mol^{-1} (2) 56 g mol^{-1} (3) 224 g mol^{-1} (4) 112 g mol^{-1}

19. Assuming ideal gas behaviour, the ratio of density of ammonia to that of hydrogen chloride at same temperature and pressure is : **[JEE-Mains-2018(ONLINE)]**

(Atomic wt. of Cl = 35.5 u)

(1) 0.64 (2) 1.64 (3) 1.46 (4) 0.46

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\01_Ideal gas\Exercises.p65

JEE-ADVANCED

1. Calculate the total pressure in a 10 litre cylinder which contains 0.4 g He, 1.6 g oxygen and 1.4 g of nitrogen at 27ºC. Also calculate the partial pressure of He gas in the cylinder. Assume ideal behavious for gases. **[JEE 1997]**

2. According to Graham's law , at a given temperature the ratio of the rates of diffusion $\frac{r_A}{r_B}$ of gases A and B is given by : **[JEE 1998]**

   (A) $\frac{P_A}{P_B}\left(\frac{M_A}{M_B}\right)^{1/2}$ (B) $\left(\frac{M_A}{M_B}\right)\left(\frac{P_A}{P_B}\right)^{1/2}$ (C) $\frac{P_A}{P_B}\left(\frac{M_B}{M_A}\right)^{1/2}$ (D) $\frac{M_A}{M_B}\left(\frac{P_B}{P_A}\right)^{1/2}$

3. An evacuated glass vessel weighs 50.0 g when empty, 148.0 gm when filled with a liquid of density 0.98 g /mL and 50.5 g when filled with an ideal gas at 760 mm Hg at 300 k . Determine the molecular weight of the gas . **[JEE 1998]**

4. The pressure exerted by 12 g of an ideal gas at temperature t ºC in a vessel of volume V is one atmp . When the temperature is increased by 10 degrees at the same volume, the pressure increases by 10 %. Calculate the temperature 't' and volume 'V'. [molecular weight of gas = 120] **[JEE 1999]**

5. One mole of N_2 gas at 0.8 atmp takes 38 sec to diffuse through a pin hole, whereas one mole of an unknown compound of Xenon with F at 1.6 atmp takes 57 sec to diffuse through the same hole . Calculate the molecular formula of the compound.(At. wt. Xe = 138, F = 19) **[JEE 1999]**

6. The r.m.s. velocity of hydrogen is $\sqrt{7}$ times the r.m.s. velocity of nitrogen. If T is the temperature of the gas : **[JEE 2000]**

   (A) $T(H_2) = T(N_2)$ (B) $T(H_2) > T(N_2)$ (C) $T(H_2) < T(N_2)$ (D) $T(H_2) = \sqrt{7}\ T(N_2)$

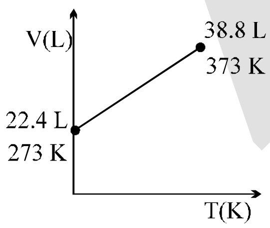
7. **Statement-1 :** The pressure of a fixed amount of an ideal gas is proportional to its temperature.

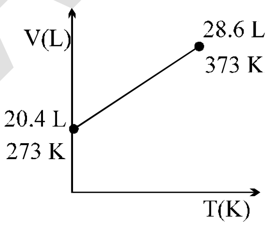
   Statement-2 : Frequency of collision and their impact both increase in proportion to the square root of temperature. True/False. **[JEE 2000]**

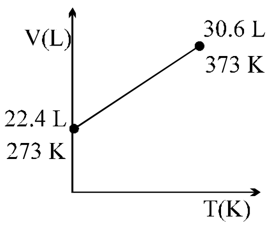
8. The root mean square velocity of an ideal gas at constant pressure varies with density as

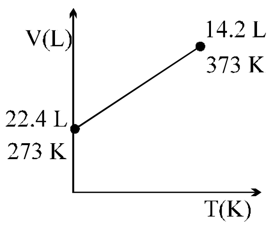
   (A) d^2 (B) d (C) $d^{1/2}$ (D) $1/d^{1/2}$ **[JEE 2001]**

9. Which one of the following V, T plots represents the behaviour of one mole of an ideal gas at one atmp? **[JEE 2002]**

   (A) 

   (B) 

   (C) 

   (D) 

10. The average velocity of gas molecules is 400 m/sec. Calculate its (rms) velocity at the same temperature. **[JEE 2003]**

11. The ratio of the rate of diffusion of helium and methane under identical condition of pressure and temperature will be

    (A) 4 (B) 2 (C) 1 (D) 0.5 **[JEE 2005]**

ALLEN Session 2019-20

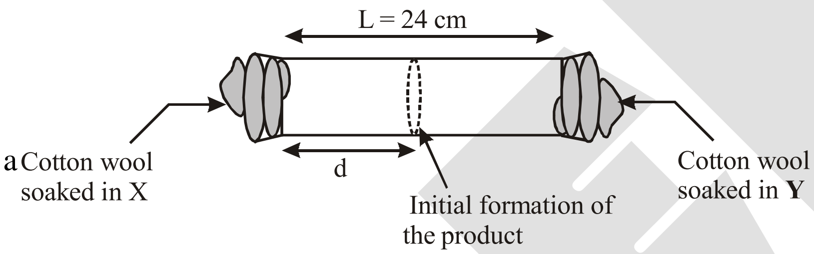
node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\01_Ideal gas\Exercises.p65

12. At 400 K, the root mean square (rms) speed of a gas X (molecular weight = 40) is equal to the most probable speed of gas Y at 60 K. The molecular weight of the gas Y is **[JEE 2009]**

13. To an evacuated vessel with movable piston under external pressure of 1 atm., 0.1 mol of He and 1.0 mol of an unknown compound (vapour pressure 0.68 atm. at 0°C) are introduced. Considering the ideal gas behaviour, the total volume (in litre) of the gases at 0°C is close to **[JEE 2011]**

Paragraph for Question 14 & 15

X and Y are two volatile liquids with molar weights of 10g mol^{-1} and 40g mol^{-1} respectively. Two cotton plugs, one soaked in X and the other soaked in Y, are simultaneously placed at the ends of a tube of length L = 24 cm, as shown in the figure. The tube is filled with an inert gas at 1 atmosphere pressure and a temperature of 300K. Vapours of X and Y react to form a product which is first observed at a distance d cm from the plug soaked in X. Take X and Y to have equal molecular diameters and assume ideal behaviour for the inert gas and the two vapours. **[JEE 2014]**



14 The value of d in cm (shown in the figure), as estimated from Graham's law, is -

(A) 8 (B) 12 (C) 16 (D) 20

15. The experimetnal value of d is found to be smaller than the estimate obtained using Graham's law. This is due to -

(A) Larger mean free path for X as compared to that of Y

(B) Larger mean free path for Y as compared to that of X

(C) Increased collision frequency of Y with the inert gas as compared to that of X with the inert gas

(D) Increased collision frequency of X with the inert gas as compared to that of Y with the inert gas

16. A closed tank has two compartments A and B, both filled with oxygen (assumed to be ideal gas). The partition separating the two compartments is fixed and is a perfect heat insulator (Figure 1). If the old partition is replaced by a new partition which can slide and conduct heat but does NOT allow the gas to leak across (Figure 2), the volume (in m^3) of the compartment A after the system attains equilibrium is____. **[JEE 2018]**

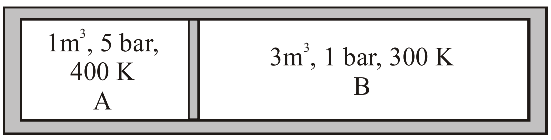


Figure 1

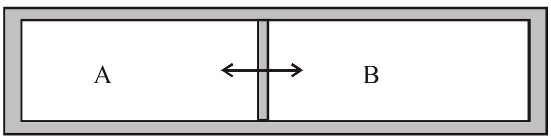


Figure 2

ANSWER KEY

EXERCISE - O-I

1.	Ans (C)	2.	Ans (C)	3.	Ans. (C)	4.	Ans. (B)
5.	Ans.(C)	6.	Ans.(C)	7.	Ans.(B)	8.	Ans. (D)
9.	Ans.(A)	10.	Ans.(C)	11.	Ans.(C)	12.	Ans. (A)
13.	Ans.(B)	14.	Ans.(B)	15.	Ans. (C)	16.	Ans.(D)
17.	Ans.(B)	18.	Ans. (C)	19.	Ans.(D)	20.	Ans.(C)
21.	Ans.(A)	22.	Ans(D)	23	Ans.(D)	24.	Ans.(A)
25.	Ans(B)	26.	Ans.(B)	27.	Ans.(D)	28.	Ans.(A)
29.	Ans.(C)	30.	Ans.(D)	31.	Ans.(B)	32.	Ans(D)
33.	Ans(C)	34.	Ans (A)	35.	Ans.(C)	36.	Ans.(A)
37.	Ans.(B)	38.	Ans.(C)	39.	Ans.(A)	40.	Ans.(A)
41.	Ans.(C)	42.	Ans.(A)	43.	Ans.(A)	44.	Ans.(D)
45.	Ans.(B)	46.	Ans.(D)	47.	Ans.(B)	48.	Ans.(D)
49.	Ans. (B)	50.	Ans.(D)	51.	Ans.(B)	52.	Ans.(C)
53.	Ans.(D)	54.	Ans.(A)	55.	Ans.(B)	56.	Ans.(C)
57.	Ans.(A)	58.	Ans.(C)	59.	Ans.(A)	60	Ans(B)
61.	Ans (A)	62.	Ans. (A)	63.	Ans.(A)	64.	Ans.(D)
65.	Ans.(C)						

EXERCISE O-II

1.	Ans.(B,D)	2.	(A,B)	3.	Ans.(A,D)	4.	Ans. (C)
5.	Ans.(A,C)	6.	Ans. (A,B,C,D)	7.	Ans. (C,D)	8.	Ans. (D)
9.	Ans. (A, B)	10.	Ans.(A,C)	11.	Ans.(C,D)	12.	Ans.(A, B, D)
13.	Ans (A, C)	14.	Ans.(A,C,D)				
15.	Ans.(B)	16	Ans.(D)	17.	Ans.(A)	18	Ans.(B)
19	Ans.(C)	20	Ans.(C)	21	Ans. (B)	22.	Ans. (B)
23.	Ans.(A)	24.	Ans.(C)				
25.	Ans.(A)	26.	Ans.(A) R, (B) S, (C) P, (D) Q				
27.	(A)-S; (B)- R ; (C) - Q; (D) – P						

ALLEN Session 2019-20

JEE-MAINS

1.	Ans (1)	2.	Ans (4)	3.	Ans(3)	4.	Ans(3)
5.	Ans(4)	6.	Ans(3)	7.	Ans.(3)		
8.	Ans.(3)	9.	Ans.(1)	10.	Ans.(1)	11.	Ans.(1)
12.	Ans (3)	13.	Ans.(4)	14.	Ans.(1)	15.	Ans.(4)
16.	Ans. (3)	17.	Ans. (4)	18.	Ans.(4)	19.	Ans.(4)

JEE-ADVANCED

1.	Ans.0.492 atmp ; 0.246 atmp	2.	Ans.(C)
3.	Ans.123	4.	–173°C , 0.82 L
5.	Ans.XeF_6	6.	Ans.(C)
7.	Ans.Both statements are correct	8.	Ans.(D)
9.	Ans.(C)	10.	Ans.434.17 m/sec
11.	Ans.(B)	12	Ans.(4)
13.	Ans.(7)	14	Ans.(C)
15	Ans.(D)	16	Ans.(2.22)

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\01_Ideal gas\Exercises.p65

SOLID STATE

1. THE SOLID STATE :

The solid are characterised by incompressibility, rigidity and mechanical strength. The molecules, atoms or ions in solids are closely packed i.e. they are held together by strong forces and can not move about at random. Thus solids have definite volume, shape, slow diffusion, low vapour pressure and possesses the unique property of being rigid.

2. AMORPHOUS AND CRYSTALLINE SOLIDS

Solids can be classified as *crystalline* or *amorphous* on the basis of the nature of order present in the arrangement of their constituent particles. A crystalline solid usually consists of a large number of small crystals, each of them having a definite characteristic geometrical shape. In a crystal, the arrangement of constituent particles (atoms, molecules or ions) is ordered. It has ***long range order*** which means that there is a regular pattern of arrangement of particles which repeats itself periodically over the entire crystal. Sodium chloride and quartz are typical examples of crystalline solids. An amorphous solid (Greek *amorphos* = no form) consists of particles of irregular shape. The arrangement of constituent particles (atoms, molecules or ions) in such a solid has only ***short range order***. In such an arrangement, a regular and periodically repeating pattern is observed over short distances only.

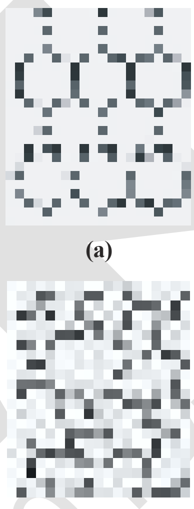


Fig. 1.1 : Two dimensional structure of (a) quartz and (b) quartz glass

Such portions are scattered and in between, the arrangement is disordered. The structures of quartz (crystalline) and quartz glass (amorphous) are shown in Fig. 1.1 (a) and (b) respectively. While the two structures are almost identical, yet in the case of amorphous quartz glass there is no *long range order*. The structure of amorphous solids is similar to that of liquids. Glass, rubber and plastics are typical examples of amorphous solids. Due to the differences in the arrangement of the constituent particles, the two types of solids differ in their properties.

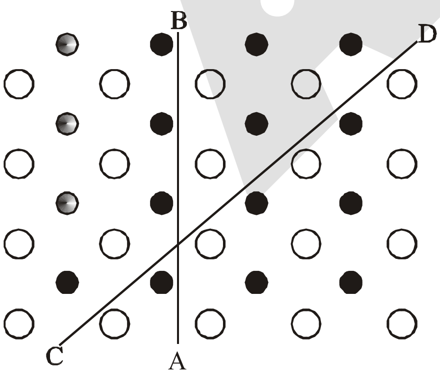


Fig. 1.2 : Anisotropy in crystals is due to different arrangement of particles along different directions.

Crystalline solids have a sharp melting point. On the other hand, amorphous solids soften over a range of temperature and can be moulded and blown into various shapes. On heating they may become crystalline at some temperature. Some glass objects from ancient civilisations are found to become milky in appearance because of some crystallisation. Like liquids, amorphous solids have a tendency to flow, though very slowly. Therefore, sometimes these are called ***pseudo solids*** or ***super cooled liquids***. Glass panes fixed to windows or doors of old buildings are invariably found to be slightly thicker at the bottom than at the top. This is because the glass flows down very slowly and makes the bottom portion slightly thicker.

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\02_Solid State\Theory.p65

Crystalline solids are ***anisotropic*** in nature, that is, some of their physical properties like electrical resistance or refractive index show different values when measured along different directions in the same crystals. This arises from different arrangement of particles in different directions. This is illustrated in Fig. 1.2. Since the arrangement of particles is different along different directions, the value of same physical property is found to be different along each direction. Amorphous solids on the other hand are ***isotropic*** in nature. It is because there is no *long range* order in them and arrangement is irregular along all the directions. Therefore, value of any physical property would be same along any direction. These differences are summarised in Table below :

Distinction between Crystalline and Amorphous Solids

Property	Crystalline Solids	Amorphous Solids
Shape	Definite characteristic geometrical shape	Irregular shape
Melting point	Melt at a sharp and characteristic temperature	Gradually soften over a range of temperature
Cleavage property	When cut with a sharp edged tool, they split into two pieces and the newly generated surfaces are plane and smooth	When cut with a sharp edged tool, they cut into two pieces with irregular surfaces
Heat of fusion	They have a definite and characteristic heat of fusion	They do not have definite heat of fusion
Anisotropy	Anisotropic in nature	Isotropic in nature
Nature	True solids	Pseudo solids or super cooled liquids
Order in arrangement of constituent particles	Long range order	Only short range order.

Ex.1 Classify the following as amorphous or crystalline solids :

(a) Polyurethane (b) Napthalene

(c) Benzoic acid (d) Teflon

(e) Potassium nitrate (f) Cellophane

(g) Polyvinyl chloride (h) Fibre glass

(i) Copper

Sol. Crystalline : (b) , (c) , (e) , (i)

Amorphous : (a) , (d) , (f) , (g), (h)

Note : Polymeric substances are generally amorphous.

Session 2019-20

3. TYPES OF THE CRYSTALLINE SOLID

Types of Solid	Constituent Particles	Bonding/ Attractive forces	Examples	Physical Nature	Electrical Conductivity	Melting Point
(1) Molecular Solids (i) Non polar	Molecules	Dispersion or London forces	Ar, CCl_4, H_2, I_2 CO_2	Soft	Insulator	Very low
(ii) Polar		Dipole-dipole	HCl, SO_2	Soft	Insulator	Low
(iii) Hydrogen bonded		Hydrogen bonding	H_2O (ice)	Hard	Insulator	Low
(2) Ionic Solids	Ions	Coulombic or electrostatic	NaCl, MgO, ZnS, CaF_2	Hard but brittle	Insulator in solid state but conductors in molten state and in aqueous solutions	High
(3) Metallic Solids	Positive ions in a sea of delocalised electrons	Metallic bonding	Fe, Cu, Ag, Mg	Hard but malleable and ductile	Conductors in solid state as well as in molten state	Fairly high
(4) Covalent or network Solids	Atoms	Covalent bonding	SiO_2 (quartz) SiC, C (diomond) AlN,	Hard	Insulators	Very high
			C(graphite)	Soft	Conductor	

Ex.2 Classify the following solids in different categories based on the nature of intermolecular forces operating in them :

(a) Potassium sulphate (K_2SO_4) **(b)** Tin (Sn)

(c) Benzene (C_6H_6) **(d)** Urea (NH_2CONH_2)

(e) Ammonia (NH_3) **(f)** Water (H_2O)

(g) Zinc sulphide (ZnS) **(h)** Graphite (C)

(i) Rubidium (Rb) **(j)** Argon (Ar)

(k) Silicon carbide (SiC) **(*l*)** Bronze

Sol. Ionic solids : (a), (g)

Metallic solids : (b), (i), (*l*)

Molecular solids : (c), (d), (e), (f), (j)

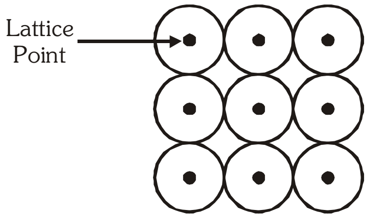
Covalent network solids.: (h), (k)

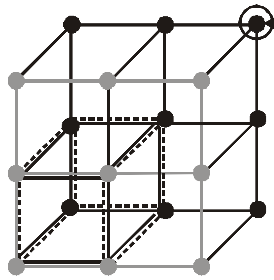
Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\02_Solid State\Theory.p65

4.0 SOME BASIC DEFINITION :

4.1 SPACE LATTICE (CRYSTAL LATTICE :





The main characteristic of crystalline solids is a regular and repeating pattern of constituent particles. If the three dimensional arrangement of constituent particles in a crystal is represented diagrammatically, in which each particle is depicted as a point, the arrangement is called crystal lattice. Thus, ***a regular three dimensional arrangement of points in space is called a crystal lattice***. A portion of a crystal lattice is shown in Fig. The following are the characteristics of a crystal lattice:

(a) Each point in a lattice is called ***lattice point*** or ***lattice site***.

(b) Each point in a crystal lattice represents one constituent particle which may be an atom, a molecule (group of atoms) or an ion.

(c) Lattice points are joined by straight lines to bring out the geometry of the lattice.

4.2. UNIT CELL :

Unit cell is the smallest portion of a crystal lattice which , when repeated in different directions, generates the entire lattice.

A unit cell is characterized by the edge lengths a, b and c along the three edges of the unit cell and the angles α, β and γ between the pair of edges : bc, ca and ab, respectively.

TYPE OF UNIT CELLS -

4.2.1 Primitive and Centred Unit cells

Unit cells can be broadly divided into two categories, primitive and centred unit cells.

(a) Primitive Unit Cells (P)

When constituent particles are present only on the corner positions of a unit cell, it is called as **primitive unit cell.**

(b) Centred Unit Cells

When a unit cell contains one or more constituent particles present at positions other than corners in addition to those at corners, it is called a centred unit cell. **Centred unit cells** are of three types:

(i) **Body-Centred Unit Cells (I) :** Such a unit cell contains one constituent particle (atom, molecule or ion) at its body-centre besides the ones that are at its corners.

(ii) **Face-Centred Unit Cells (F) :** Such a unit cell contains one constituent particle present at the centre of each face, besides the ones that are at its corners.

(iii) **End-Centred Unit Cells (E) :** In such a unit cell, one constituent particle is present at the centre of any two opposite faces besides the ones present at its corners.

ALLEN Session 2019-20

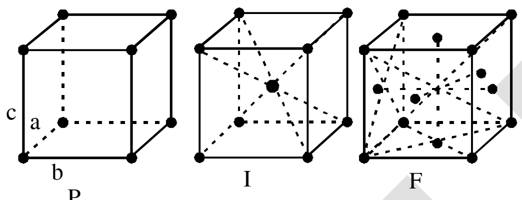
4.3. THE SEVEN CRYSTAL SYSTEMS

On the basis of the classification of symmetry, the lattice have been divided into seven systems. These can be grouped into 7 crystal systems. These seven systems with the characteristics of their axes (angles and intercepts) along with some examples of each are given in the following table :

Seven Primitive Unit cells and their Possible Variations as Centred Unit Cells

Crystal system	Possible Variations	Axial distance or edge lengths	Axial angles	Examples
Cubic	Primitive, body-centred, Face centre	$a = b = c$	$\alpha = \beta = \gamma = 90°$	NaCl, Zinc blende, Cu
Tetragonal	Primitive, Body-centred	$a = b \neq c$	$\alpha = \beta = \gamma = 90°$	White tin, SnO_2, TiO_2, $CaSO_4$
Orthorhombic	Primitive, Body-centred, Face-centred, End-centred	$a \neq b \neq c$	$\alpha = \beta = \gamma = 90°$	Rhombic sulphur, KNO_3, $BaSO_4$
Hexagonal	Primitive	$a = b \neq c$	$\alpha = \beta = 90°$, $\gamma = 120°$	Graphite, ZnO, CdS,
Rhombohedral or Trigonal	Primitive	$a = b = c$	$\alpha = \beta = \gamma \neq 90°$	Calcite ($CaCO_3$), HgS (Cinnabar)
Monoclinic	Primitive, End-centred	$a \neq b \neq c$	$\alpha = \gamma = 90°$, $\beta \neq 90°$	Monoclinic sulphur, $Na_2SO_4 \cdot 10H_2O$
Triclinic	Primitive	$a \neq b \neq c$	$\alpha \neq \beta \neq \gamma \neq 90°$	$K_2Cr_2O_7$, $CuSO_4.5H_2O$, H_3BO_3

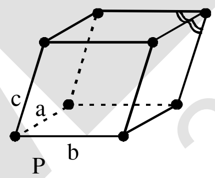
4.4. BRAVAIS LATTICES : There are 14 Bravais lattices :



Cubic

$a = b = c$ All sides are of equal length; all angles are 90^0

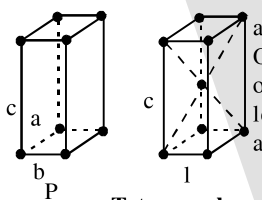
Three angles changed



$a = b = c$ All sides are of equal length;angles $\alpha = \beta = \gamma \neq 90^0$

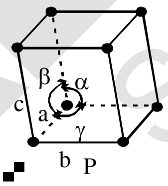
Trigonal (Rhombohedral)

One side is changed



$a = b \neq c$ One side is of different length; all angles are 90^0

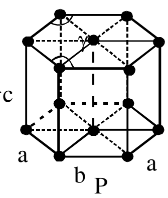
Tetragonal



General case : All sides are different lengths; all angles are different

Triclinic

One side length changed two angles fixed at 90^0 one fixed at



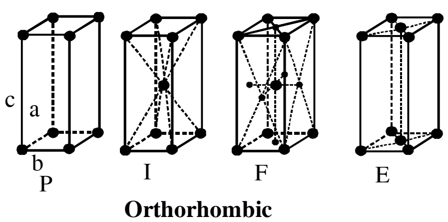
Special Case $a = b \neq c$ $\alpha = \beta = 90^0$ $\gamma = 120^0$

Hexagonal

Three unit cells are shown to give the hexagon

Length of another side is changed

Two side lengths made the same; one angle fixed at 120^0

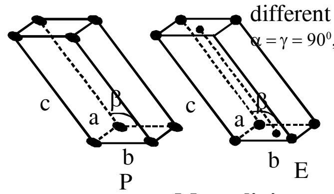


$a \neq b \neq c$ Three sides are of different lengths ; all angles are 90^0

Orthorhombic

One angle changed

All sides are of different lengths; $\alpha = \gamma = 90^0$, $\beta \neq 90^0$



Monoclinic

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module#Ideal gas, Solid state & Chemical Eq\Eng\02_Solid State\Theory.p65

4.5. CO-ORDINATION NUMBER :

The number of nearest particles around a specific particle in a given crystalline substance is called ***co-ordination*** number.

4.6. PACKING EFFICIENCY OR PACKING DENSITY (P.E.) :

Packing efficiency is defined as the ratio of volume occupied by the constituent particles to the total volume of the crystalline substance.

$$\text{P.E.} = \frac{\text{Volume occupied by particles present in a crystal}}{\text{Volume of crystal}}$$

$$\text{P.E.} = \frac{\text{Volume occupied by particles present in unit cell}}{\text{Volume of Unit Cell}}$$

$\text{P.E.} = \dfrac{Z \times (4/3)\pi r^3}{V}$, where Z = number of atoms present in unit cell

4.7. *DENSITY OF THE CRYSTAL :*

Density of crystal = Density of an unit cell = $\dfrac{\text{Mass of unit cell}}{\text{Volume of unit cell}}$

Mass of the unit cell = Number of particles present in a unit cell $\times$ Mass of one particles = $Z \times m$

But mass of one particles (m) = $\dfrac{\text{Particle mass}}{\text{Avogadro Number}} = \dfrac{M}{N_A}$

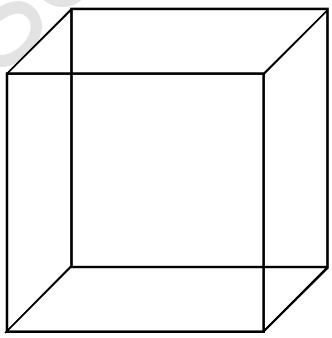
Mass of an unit cell = $Z \times \dfrac{M}{N_A}$

Density of an unit cell = $\dfrac{Z \times \dfrac{M}{N_A}}{V}$

$\therefore$ **Density of Crystal, d = Density of an unit cell =** $\mathbf{\dfrac{Z \times M}{V \times N_A}}$ **g cm^{-3}**

5.0 ANALYSIS OF CUBIC CRYSTAL :

5.1. GEOMETRY OF A CUBE



Number of corners = 8

Number of faces = 6

Number of edges = 12

Number of cube centre = 1

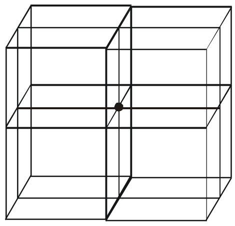
Number of cube diagonals = 4

Number of face diagonals = 12

5.2 CONTRIBUTION OF A CONSTITUENT PARTICLE AT DIFFERENT SITES OF CUBE :

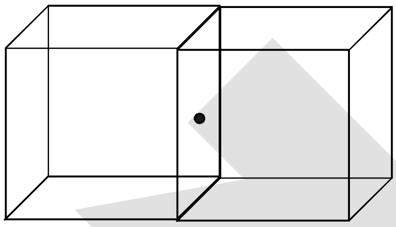
5.2.1 A corner of a cube is common in 8 cubes.

So $\frac{1}{8}$ th part of a particle is present at this corner of cube.



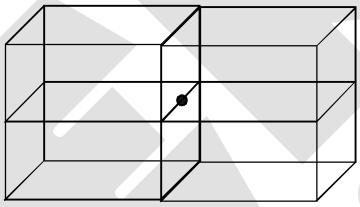
5.2.2 A face of a cube is common is 2 cubes.

So $\frac{1}{2}$ th part of a particle is present at the face of a cube.



5.2.3 An edge of a cube is common in four cubes,

so $\frac{1}{4}$ th part of particle is present at the edge of a cube

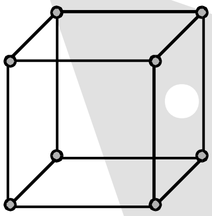


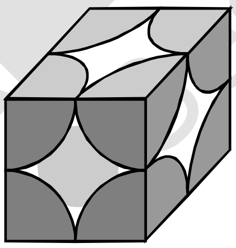
5.3 TYPE OF CUBIC UNIT CELL :

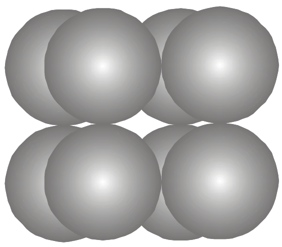
5.3.1 Simple/Primitive/Basic Unit cell (Simple cubic, SC) ;

A unit cell having lattice point only at corners called as primitive or simple unit cell. In this case there is one particle at each of the eight corners of the unit cell.

Considering a particles at one corner as the centre, it will be found that this atom is surrounded by six equidistant neighbours (particles) and thus the co-ordination number will be six. If 'a' is the side of the unit cell, then the distance between the nearest neighbours shall be equal to 'a'.



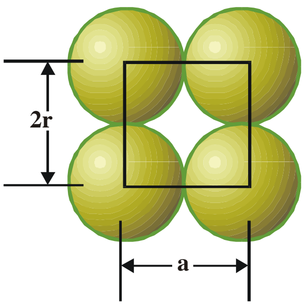




(a) Relationship between Edge length 'a' and Particle radius 'r' :-

$a = 2r$

i.e., $\boxed{r = \frac{a}{2}}$



(One face of SC)

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\02_Solid State\Theory.p65

(b) Number of particles present in unit cell (Z) : In this case one particle lies at each corner. Hence simple cubic unit cell contains a total of $\frac{1}{8}\times 8 = 1$ particle /unit cell.

(c) Packing efficiency (P. E.) :

$$\text{P.E.} = \frac{\text{Volume occupied by particles present in unit cell}}{\text{Volume of unit cell}} = \frac{Z\times\frac{4}{3}\pi r^3}{V} \qquad \left[\text{Volume of atom} = \frac{4}{3}\pi r^3\right]$$

$$\text{P.E.} = \frac{1\times\frac{4}{3}\times\pi\times\left(\frac{a}{2}\right)^3}{a^3} = \frac{\pi}{6} = 0.52 \quad \text{or} \quad 52\% \qquad \left[r = \frac{a}{2} \text{ and } V = a^3, Z = 1\right]$$

In SC, 52% of total volume is occupied by particles.

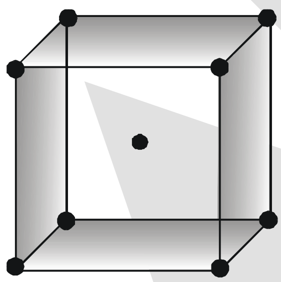
$\therefore$ Void space $\approx$ (100 – 52)= 48 %

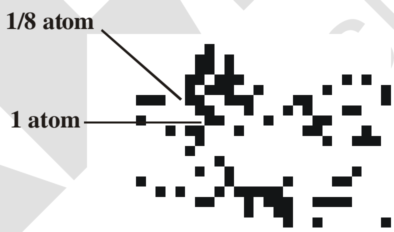
(d) Coordination number (CN)

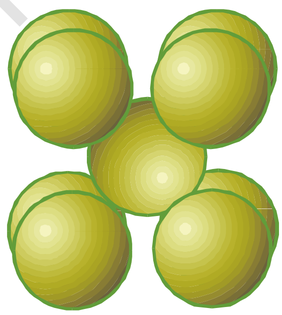
Nearest neighbour	Distance	Number
1	a	6
2	$\sqrt{2}\,a$	12
3	$\sqrt{3}\,a$	8

5.3.2. Body Centred Cubic unit cell (BCC):

A unit cell having lattice point at the body centre in addition to the lattice points at every corner is called as body centered unit cell. Here the central particle is surrounded by eight equidistant particles and hence the co-ordination number is eight. The nearest distance between two particles will be $\frac{a\sqrt{3}}{2}$





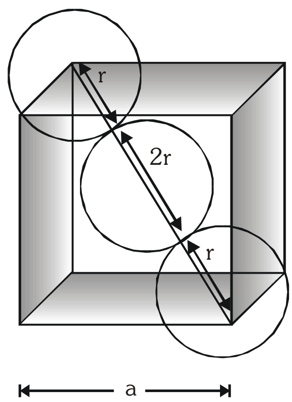


(a) Relationship between edge length 'a' and particle radius 'r' :

In BCC, along cube diagonal all particles touches each other and the length of cube diagonal is $\sqrt{3}a$.

So, $\sqrt{3}a = 4r$

i.e. $r = \frac{\sqrt{3}a}{4}$



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(b) Number of particle present in unit cell (Z):

$$Z = \underset{\text{(Corner)}}{\left(\frac{1}{8}\times 8\right)} \quad + \quad \underset{\text{(Body centre)}}{(1\times 1)} = 1 + 1 = 2 \text{ particles/unit cell.}$$

In this case one particle lies at the each corner of the cube. Thus contribution of the 8 corners is $\left(\frac{1}{8}\times 8\right) = 1$, while that of the body centred is 1 in the unit cell. Hence total number of particles per unit cell is 1 + 1 = 2

(c) Packing efficiency :

$$\text{P.E.} = \frac{Z\times\frac{4}{3}\pi r^3}{V} = \frac{2\times\frac{4}{3}\times\pi\left(\frac{\sqrt{3}a}{4}\right)^3}{a^3} = \frac{\sqrt{3}\pi}{8} = 0.68 \text{ or } 68\% \qquad \left[Z = 2,\ r = \frac{\sqrt{3}a}{4},\ V = a^3\right]$$

In BCC, 68% of total volume is occupied by particles.

$\therefore$ Void space = 100 – 68 = 32 %

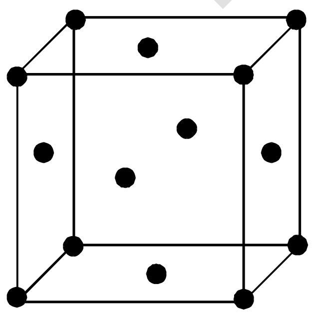
(d) Coordination number (CN)

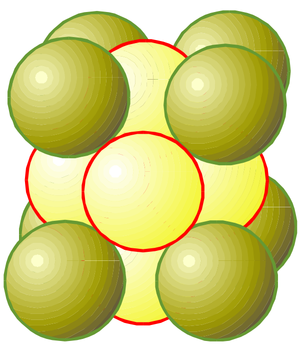
Nearest neighbour	Distance	Number
1	$\sqrt{3}\,\frac{a}{2}$	8
2	a	6
3	$\sqrt{2}\,a$	12

5.3.3 Face Centred Cubic unit cell (FCC):

A unit cell having lattice point at every face centre in addition to the lattice point at every corner called as face centred unit cell.

In this case there are eight particles at the eight corners of the unit cell and six particles at the centre of six faces. Considering a particle at the face centre as origin, it will be found that this face is common to two cubes and there are twelve points surrounding it situated at a distance which is equal to half the face diagonal of the unit cell. Thus the co-ordination number will be twelve and the distance between the two nearest particles will be $\frac{a}{\sqrt{2}}$.





ALLEN Session 2019-20

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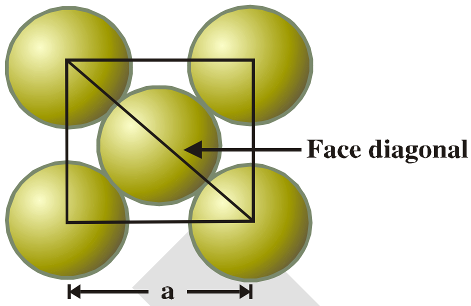
(a) Relationship between edge length 'a' and atomic radius 'r' :

In FCC, along the face diagonal all atoms touches each other and the length of face diagonal is $\sqrt{2}a$.

So $4r = \sqrt{2}a$

i.e. $r = \dfrac{a}{2\sqrt{2}}$



(b) Number of particles per unit cell : (Z)

$$Z = \left(\frac{1}{8}\times 8\right) + \left(6\times\frac{1}{2}\right) = 1 + 3 = 4 \text{ particles/unit cell}$$

Corner faces

In this case, one particle lies at the each corner of the cube and one particle lies at the centre of each face of the cube. It may noted that only 1/2 of each face sphere lie within the unit cell and there are six such faces. The total contribution of 8 corners is $\left(\frac{1}{8}\times 8\right) = 1$, while that of 6 face centred particles is $\left(\frac{1}{2}\times 6\right) = 3$ in the unit cell.

Hence, total number of particles per unit cell is $1 + 3 = 4$

(c) Packing efficiency :

$$\text{P.E.} = \frac{Z\times\frac{4}{3}\pi r^3}{V} \qquad \left[Z = 4,\ r = \frac{a}{2\sqrt{2}},\ V = a^3\right]$$

$$= \frac{4\times\frac{4}{3}\pi\times\left(\frac{a}{2\sqrt{2}}\right)^3}{a^3} = \frac{\pi}{3\sqrt{2}} = 0.74 \text{ or } 74\%$$

In FCC, 74% of total volume is occupied by particles. This is maximum for crystals having identical particles.

$\therefore$ Void space = $100 - 74 = 26\ \%$

(d) Coordination number (CN)

Nearest neighbour	Distance	Number
1	$\frac{a}{\sqrt{2}}$	12
2	a	6
3	$\sqrt{\frac{3}{2}}a$	24

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Session 2019-20

5.4 SUMMARY OF CUBIC CRYSTAL :

Unit cell	No. of particles per unit cell (Z)	2r =	CN	Volume occupied by particles (%)
Simple cube (SC)	1	a	6	52
Body centred cube (BCC)	2	$\frac{a\sqrt{3}}{2}$	8	68
Face centred cube (FCC)	4	$\frac{a}{\sqrt{2}}$	12	74

Ex.3. An element (atomic mass = 60) having face centred cubic crystal has a density of 6.23 g cm^{-3}. What is the edge length of the unit cell ? (Avogadro constant, N_A = 6.02 × 10^{23} mol^{-1})

Sol. *Since element has fcc structure hence there are 4 atoms in a unit cell (Z = 4), Atomic mass is 60 (M = 60), N_A = 6.02 × 10^{23} and d = 6.23 g cm^{-3}.*

$$\because \quad d = \frac{Z \times M}{V \times N_A}$$

$$or \quad V = \frac{Z \times M}{d \times N_A} = \frac{(4)\left(60 \text{ g mol}^{-1}\right)}{\left(6.23 \text{ g cm}^{-3}\right)\left(6.02 \times 10^{23} \text{mol}^{-1}\right)}$$

$$= 64 \times 10^{-24} \text{ cm}^3$$

Let ℓ be the length of the edge of the unit cell.

$$\therefore \quad \ell^3 = V = 64 \times 10^{-24} \text{ cm}^3 \quad or \quad \ell = 4.0 \times 10^{-8} \text{ cm}$$

Ex.4. The density of Al is 5.4 g/cm^3. If it crystallise in fcc lattice, determine its atomic radius.

Sol. $d = \frac{ZM}{N_A.V} \Rightarrow 5.4 = \frac{4 \times 27}{N_A \times (2\sqrt{2}r)^3}$

$\therefore$ r = 1.136 × 10^{-8} cm = 1.136 Å

Ex.5. A solid crystallises in cubic crystal in which 'X' atoms occupy all the corners and body centres and 'Y' atoms occupy all the face-centres. What is the simplest formula of solid?

Sol. $Z_X = 8 \times \frac{1}{8} + 1 = 2$

$Z_Y = 6 \times \frac{1}{2} = 3$

$\therefore$ *Simplest formula of solid = X_2Y_3*

ALLEN Session 2019-20

6. CLOSE PACKING OF IDENTICAL SOLID SPHERES

The solids which have non-directional bonding, their structures are determined on the basis of geometrical consideration. For such solids, it is found that the lowest energy structure is that in which each particle is surrounded by the greatest possible number of neighbours. In order to understand the structure of such solids, let us consider the particles as hard sphere of equal size in three directions. Although there are many ways to arrange the hard spheres but the one in which maximum available space is occupied will be economical which is known as **closed packing**.

To clearly understand the packing of these spheres, the packing can be categorised as :

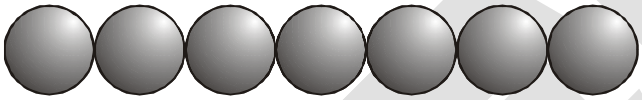
(i) Close packing in one dimension.

(ii) Close packing in two dimension.

(iii) Close packing in three dimension.

6.1 CLOSE PACKING IN ONE DIMENSION :

In one dimension , only one arrangement of spheres is possible as shown in fig.



Close packing of spheres in one dimension

Each sphere is touched by other two spheres, hence coordination numbers of packing is two.

6.2 CLOSE PACKING IN TWO DIMENSION :

Two possible types of two dimensional packing are

(i) Square close packing in two dimension.

(ii) Hexagonal close packing in two dimension.

(i) Square close packing in two dimension :

When two rows are placed in such a manner, that spheres of one row are placed immediately below of the other, the resulting packing is called two dimensional square close packing.

(i) Since all the rows are identical, the packing is called AAA .... type packing.

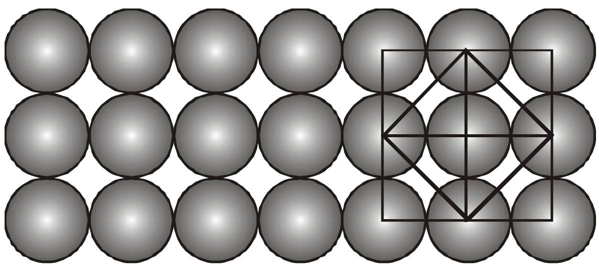
(ii) Each sphere is touched by four other, hence coordination number is four.

(iii) If centres of spheres are connected, square cells are formed , hence it also called two dimensional square packing.

(iv) This type of packing is not very effective in terms of utilisation of space.

(v) Packing efficiency in 2-D = $\frac{1\times\pi r^2}{a^2} = \frac{1\times\pi(a/2)^2}{a^2} = \frac{\pi}{4} = 0.78$ or 78%

(vi) Packing efficiency in 3-D = $\frac{1\times\frac{4}{3}\pi\left(\frac{a}{2}\right)^3}{a^3} = 0.52$ or 52% [In 3-D, its unit cell is simple cubic]



ALLEN Session 2019-20

(ii) Hexagonal close packing in two dimension :

If various one dimensional close pack rows are placed in such a way that spheres of top row fits in depression of bottom row spheres, the resulting packing is called two dimensional hexagonal close packing.

(i) Every third row sphere comes exactly at top of first row sphere, hence the packing is called ABABAB ...... packing.

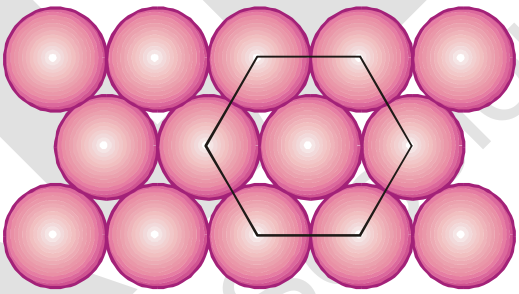
(ii) If centres are joined, hexagonal unit cells are formed. Hence this is called two dimensional hexagonal close packing.

(iii) This packing is most efficient in utilising space in two dimensional arrangement.

(iv) Each sphere is touched by six other, hence coordination number is six.

(v) Packing efficiency in 2-D = $\dfrac{3\times\pi\left(\dfrac{a}{2}\right)^2}{\dfrac{a^2\sqrt{3}}{4}\times 6} = \dfrac{\pi}{2\sqrt{3}} = 0.90 \text{ or } 90\%$

(vi) Packing efficiency in 3-D = $\dfrac{3\times\dfrac{4}{3}\pi\left(\dfrac{a}{2}\right)^3}{\dfrac{a^2\sqrt{3}}{4}\times 6\times a} = \dfrac{\pi}{3\sqrt{3}} = 0.60 \text{ or } 60\%$



6.3 CLOSE PACKING IN THREE DIMENSIONS :

When two dimensional packing structure are arranged one above the other, depending upon type of two dimensional arrangement in a layer, and the relative positions of spheres in above or below layer, various types of three dimensional packing results. To define 3-D lattice, six lattice parameters are required - 3 edge lengths & 3 angles.

(i) Simple cubic packing (A A A A ......)

(ii) Hexagonal close packing (AB AB AB ......)

(iii) Cubic close packing or face centred cubic (ABC ABC...)

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6.3.1 Three dimensional close packing from square two dimensional packing (Simple cubic packing in three dimension)

The two dimensional square close packed layer are placed, in such a manner that spheres in each layer comes immediately on top of below layer, simple cubic packing results. Important points :

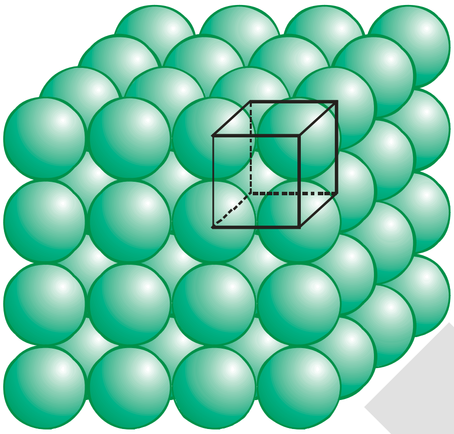
(i) Spheres all aligned vertically and horizontally in all directions.

(ii) The unit cell for this packing is simple cubic unit cell.

(iii) In this packing, only 52% of available space is occupied by spheres.

(iv) Each sphere is in contact will six spheres and hence coordination number is 6.

(v) Packing efficiency = $\dfrac{1\times\frac{4}{3}\pi r^3}{(2r)^3} = \dfrac{\pi}{6} \approx 0.52$



Simple cubic lattice formed by A A A ... arrangement

Coordination number = 6

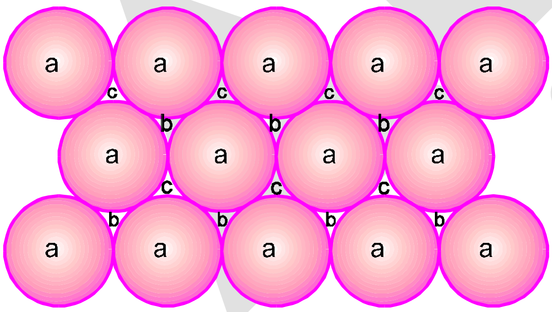
First neighbour = 6 at a distance = a

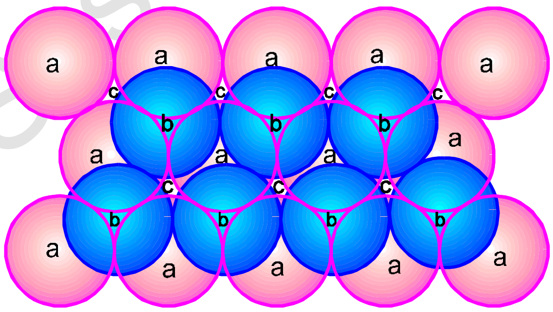
Second neighbour = 12 at ($\sqrt{2}$a) distance

Third neighbour = 8 at ($\sqrt{3}$a) distance

6.3.2 Three dimensional close packing from hexagonal two dimensional packing :

In hexagonal close packing, there are two types of the voids (open space or space left) which are divided into two sets 'b' and 'c' for convenience. The spaces marked 'c' are curved triangular spaces with tips pointing upwards whereas spaces marked 'b' are curved triangular spaces with tips pointing downwards.





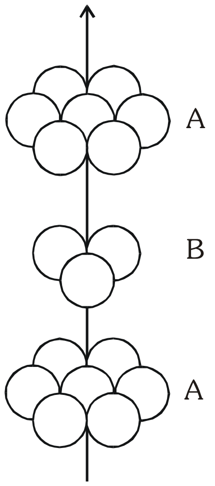
Now we extend the arrangement of spheres in three dimensions by placing second close packed layer (hexagonal close packing) (B) on the first layer (A). The spheres of second layer may placed either on space denoted by 'b' or 'c'. It may be noted that it is not possible to place spheres on both types of voids (i.e. b and c). Thus half of the voids remain unoccupied by the second layer. The second layer also have voids of the types 'b' and in order to build up the third layer, there are following ways :

(I) Hexagonal close packing (HCP)

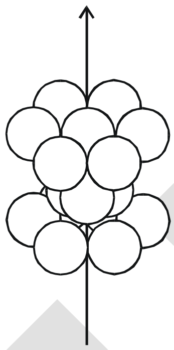
(i) In one way, the spheres of the third layer lie on the spaces of second layer (B) in such a way that they lie directly above those in the first layer(A). In other words we can say that the third layer becomes indentical to the first layer. Such arrangement is called AB AB AB ....type packing or hexagonal close packing (hcp).

(ii) Maximum possible space is occupied by spheres.

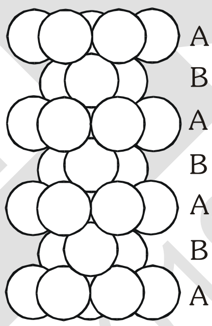
(iii) Each sphere is touched by 12 other spheres in 3D (6 is one layer, 3 in top layer and 3 in bottom) and hence the coordination number is 12.



Exploded view

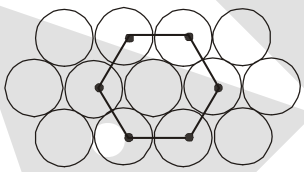


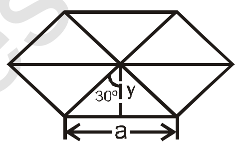
Hexagonal closest-packed



(iv) Packing efficiency of HCP units
Relation between a, b, c and R :

$a = b = 2R$

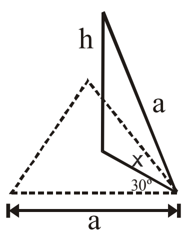




$\tan 30^\circ = \dfrac{a}{2 \times y}$ So $y = \dfrac{a \times \sqrt{3}}{2 \times 1} = \dfrac{\sqrt{3}}{2}a.$

$$\text{Base Area} = 6\left[\frac{1}{2}\frac{a}{} \times \frac{\sqrt{3}}{2}\frac{a}{}\right] = \frac{6\sqrt{3}a^2}{4}$$

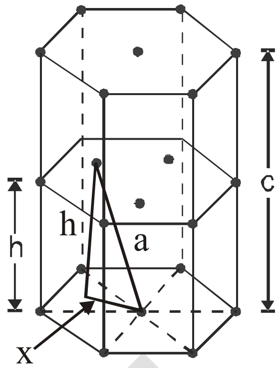
Calculation of c :



$\cos 30^\circ = \dfrac{a}{2 \times x}$ $x = \dfrac{2a}{2 \times \sqrt{3}} = \dfrac{a}{\sqrt{3}}$

ALLEN Session 2019-20

Applying pythogoras theorem : $x^2 + h^2 = a^2$



So $h^2 = a^2 - x^2 = a^2 - \frac{a^2}{3} = \frac{2}{3}a^2$

$h = \sqrt{\frac{2}{3}}\,a$ so $c = 2h = 2\sqrt{\frac{2}{3}}\,a$

So volume of hexagon = area of base × height

$$= \frac{6.\sqrt{3}}{4} \times a^2 \times 2\sqrt{\frac{2}{3}}\,a = \frac{6.\sqrt{3}}{4} \times (2R)^2 + 2\sqrt{\frac{2}{3}} \times (2R) = 24\sqrt{2}\,R^3$$

(v) Effective no. of particles (Z)

$$Z = 3 + 2 \times \frac{1}{2} + 12 \times \frac{1}{6} = 3 + 1 + 2 = 6.$$

It must be noted that all three spheres of 'B' layer are not exactly inside the unit cell. But the contribution of three spheres are taken because the same volume of other spheres in that layer is also inside the unit cell.

(vi) Packing efficiency $= \dfrac{6 \times \frac{4}{3}\pi \;\; R^3}{24\sqrt{2} \;\; R^3} = \dfrac{\pi}{3\sqrt{2}} = 0.74$ or 74%.

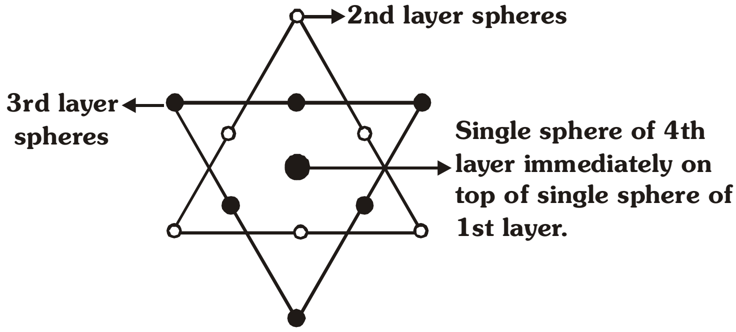
(vii) Density (d) $= \dfrac{\text{mass}}{\text{volume}} = \left[\dfrac{Z \times M}{N_A \times \text{volume}}\right]$

(II) Cubic close packing (CCP) or face centered cubic (FCC)

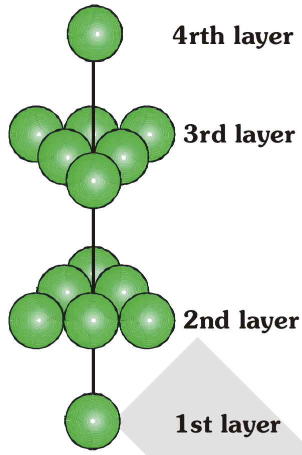
In second way, the spheres of the third layer (C) lie on the second layer (B) in such a way that they lie over the unoccupied spaces 'C' of the first layer(A). If this arrangement is continued in the same order every fourth layer becomes identical to the first. Such arrangement of particle is called ABC ABC ABC.... or cubic close packing (ccp) or face centered cubic (fcc).

It may be noted that in ccp (or fcc) structures, each sphere is surrounded by 12 spheres hence the coordination number of each sphere is 12. The spheres occupy 74% of the total volume and 26% of is the empty space in both (hcp and ccp) structures.

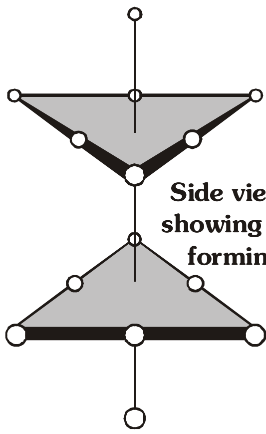
Session 2019-20



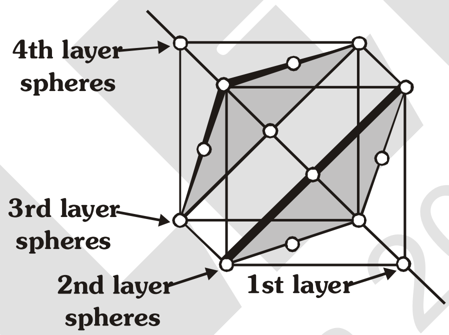
Open structure show top view of 14 spheres forming ccp packing.



Side view exploded space filling diagram showing 14 selected atom forming ccp packing



Side view open structure showing 14 selected atoms forming ccp packing



fcc unit cell showing 14 spheres

(i) Relation between 'a' and 'R' :

$a \neq 2R$

$\sqrt{2}a = 4R$ (sphere are touching along the face diagonal)

(ii) Effective no. of particles per unit cell (Z)

$$Z = \frac{1}{8} \times 8 + \frac{1}{2} \times 6 = 4$$

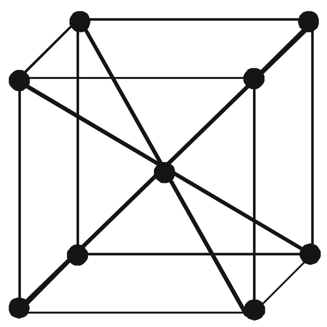
(iii) Packing fraction :

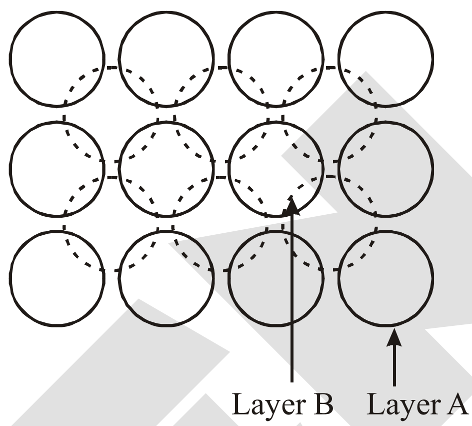
$$\text{P.F.} = \frac{4 \times \frac{4}{3}\pi \ R^3}{4 \times 4 \times 4 \ \ R^3} \times \sqrt{2} \times 2 = \frac{\pi}{3\sqrt{2}} = 0.74 \text{ or } 74\%$$

(iv) Density (d) $= \dfrac{Z \times M}{N_A \times a^3}$

6.3.3 Body centred cubic (bcc):

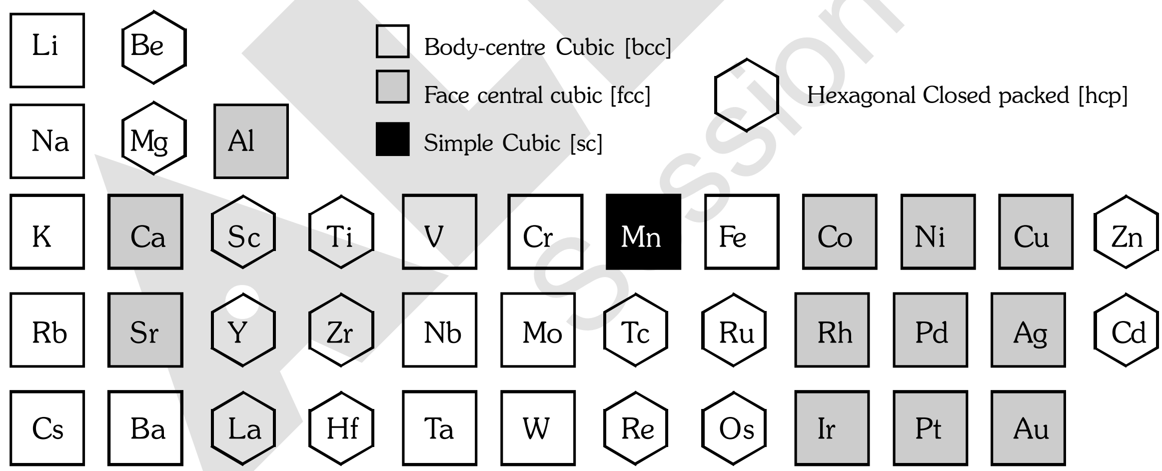
There is another possible arrangement of packing of spheres known as body centred cubic (bcc) arrangement. This arrangement is observed in square close packing in which there is suitable space between the spheres in each layer. In bcc arrangement, the spheres of the second layer lie at the space (hollows or voids) in the first layer.





Thus each sphere of the second layer touches four spheres of the first layer. Now spheres of the third layer are placed exactly about the spheres of first layer. In this way each sphere of the second layer touches eight spheres (four of 1st layer and four of IIIrd layer). Therefore coordination number of each sphere is 8 in bcc sturcture. The spheres occupy 68% of the total volume 32% of the volume is the empty space.

7. STRUCTURES OF VARIOUS ELEMENTS



8. INTERSTICES OR VOIDS OR HOLES IN CRYSTALS

It has been shown that the particles are closely packed in the crystals even than there is some empty space left in between the spheres. This is known as interstices (or interstitial site or hole or empty space or voids). In three dimentional close packing (CCP & HCP) the interstices are of two types : (i) tetrahedral voids and (ii) octahedral voids.

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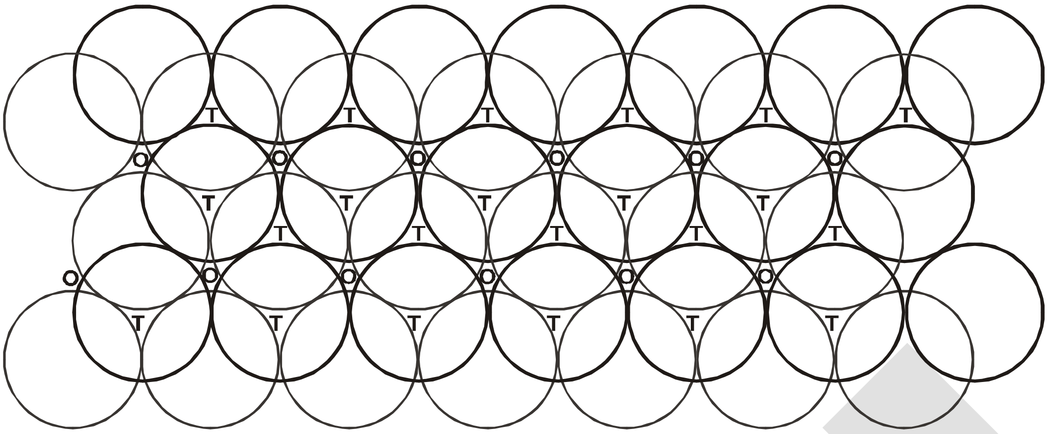
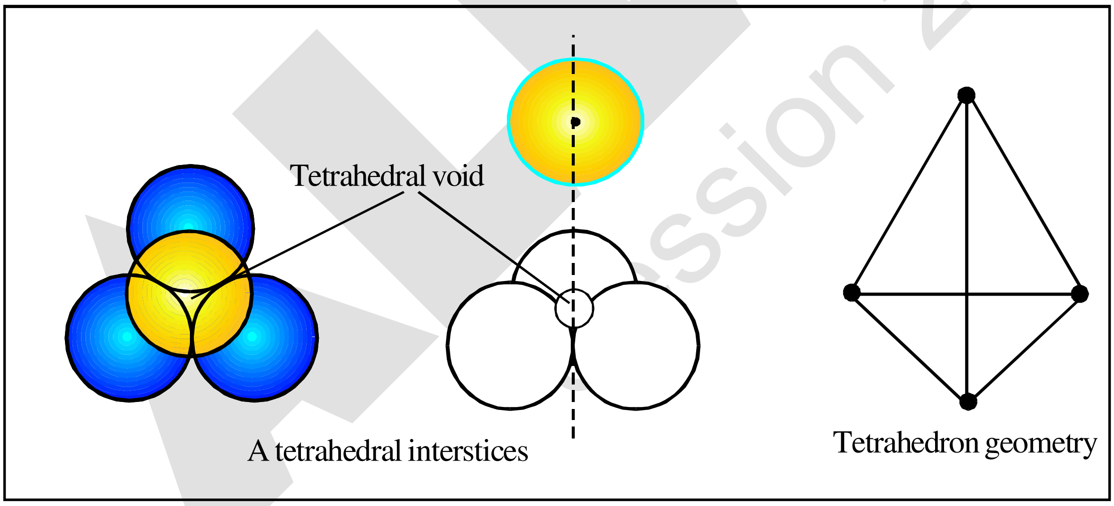


A stack of two layers of close packed spheres and voids generated in them. T = Tetrahedral void; O = Octahedral void

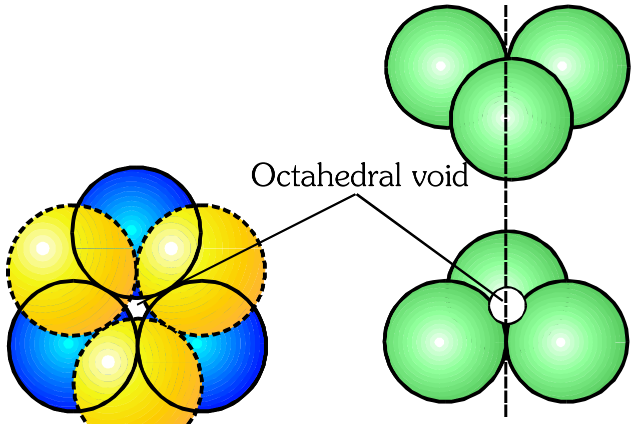
(i) Tetrahedral voids : We have seen that in hexagonal close packing (hcp) and cubic close packing (ccp), each sphere of second layer touches with three spheres of first layer. Thus, they leave a small space in between which is known as **tetrahedral site or interstices**. In another words, the vacant space between 4 touching spheres is called as tetrahedral void. Since a sphere touches three spheres in the below layer and three spheres in the above layer hence there are two tetrahedral sites associated with one sphere. It may by noted that a tetrahedral site does not mean that the site is tetrahedral in geometry but it means that this site is surrounded by four spheres and the centres of these four spheres lie at the apices of a regular tetrahedron.



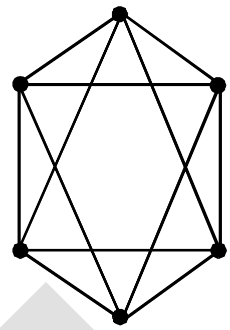
(ii) Octahedral voids : Hexagonal close packing (hcp) and cubic close packing (ccp) also form another type of interstices (or site), which is called octahedral site (or interstices). In another words, the vacant space between 6 touching spheres is called as octahedral void.

In the figure two layers of close packed spheres are shown. The spheres of first layer are shown by full circles while that of second layer by dotted circles. Two triangles are drawn by joining the centres of three touching spheres of both the layers.

ALLEN Session 2019-20
node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\02_Solid State\Theory.p65



Octahedral interstices

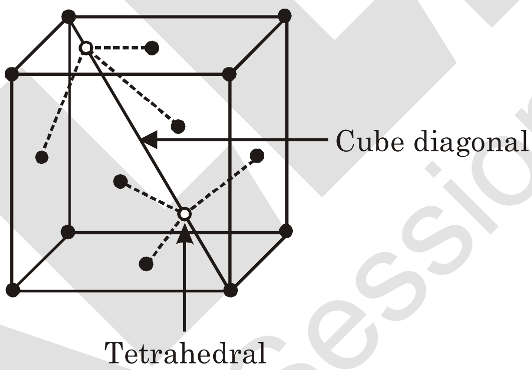


Octahedral geometry

The apices of these triangles point are in opposite directions. On super imposing these triangles on one another, an octahedral site is created. It may be noted that an octahedral site does not mean that the hole is octahedral in shape but it means that this site is surrounded by six nearest neighbour lattice points arranged octahedrally.

8.1 POSITIONS OF TETRAHEDRAL VOIDS IN AN FCC UNIT CELL :

In FCC, one corner and its three face centred atom of faces meeting at that corner form a tetrahedral void. In FCC, two tetrahedral voids are obtained along one cube diagonal. So in FCC 8 tetrahedral voids are present.

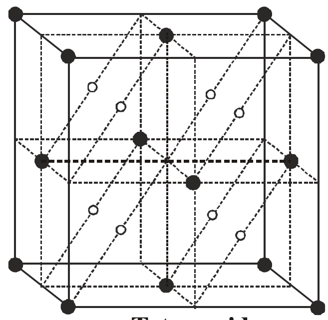


Alternatively, the centre of tetrahedral void is located on the centre of body diagonal of each small cube of volume $\left(\frac{a^3}{8}\right)$.

Total number of particles per unit cell $= \frac{1}{2}\times 6 + 8\times\frac{1}{8} = 4$

Total number of tetrahedral void = 8

$\therefore$ Effective number of tetrahedral void per particle = 2.



○ Tetra void
● Atom forming fcc packing

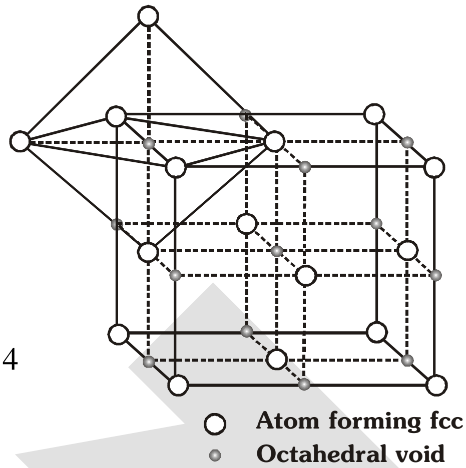
Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\02_Solid State\Theory.p65

8.2 POSITION OF OCTAHEDRAL VOID IN FCC UNIT CELL :

Position of octahedral void is at mid-point of each edge (total 12 edges in a cube) and at the centre of cube.

Each octahedral void located at mid point of edge contributes 1/4 to the unit cell. The octahedral void situated at the centre contributes 1.



In FCC, total number of octahedral voids are

$$\underset{\text{(Cube centre)}}{(1 \times 1)} + \underset{\text{(edge)}}{\left(12 \times \frac{1}{4}\right)} = 1 + 3 = 4$$

In FCC, number of particles = 4

$\therefore$ Effective number of octahedral voids per particle = 1

8.3 POSITION OF TETRAHEDRAL AND OCTAHEDRAL VOID IN HCP:

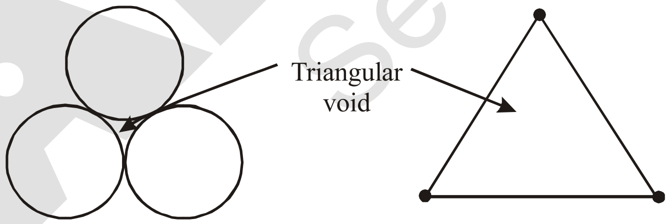
(i) Above & below each sphere, there is a tetrahedral void, hence number of tetrahedral void per unit cell = 12

(ii) Other than TV, all voids are OV & number of OV per unit cell = $2 \times 3 = 6$

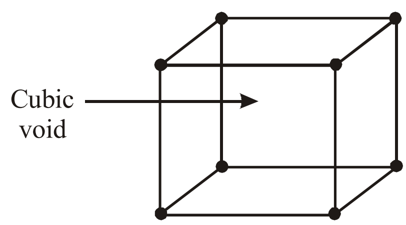
Note : In any closed packing of maximum efficiency, there are two TV & one OV per particle

8.4 Triangular and cubic voids :

(i) Triangular void : It is formed by three spheres in a plane.



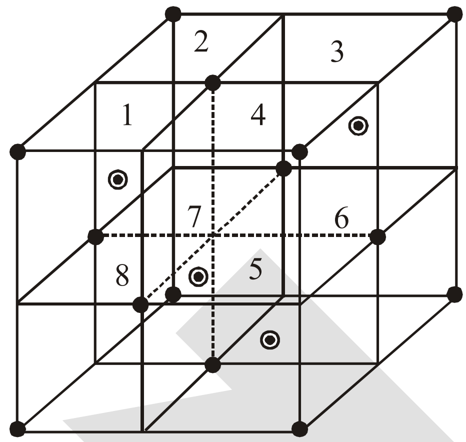
(ii) Cubic void : It is formed in simple cubic unit cell by eight spheres at the corners of cube.



Session 2019-20

9. CRYSTAL OF DIAMOND

The crystal is FCC for C atom & alternate tetrahedral voids are also occupied by carbon atoms.



* $Z = \frac{1}{8}\times 8 + 6\times\frac{1}{2} + 4 = 8$
* CN = 4
* $\frac{\sqrt{3}a}{4} = 2r = d_{C-C}$
* Number of C–C bonds/unit cell = 4 × 4 =16
* Number of C–C bonds/C-atom = $\frac{16}{8} = 2$
* $PE = \frac{8\times\frac{4}{3}\pi r^2}{\left(\frac{8r}{\sqrt{3}}\right)^3} = \frac{\sqrt{3}\pi}{16} \simeq 0.34 \text{ or } 34\%$

CLASS ILLUSTRATION

Ex.6 A solid crystallises in close packing for 'P' atoms and 25% of tetrahedral voids are occupied by 'Q' atoms. What is the simplest formula of solid ?

Sol. $Z_P = 1(\text{say})$

$Z_Q = \frac{25}{100}\times 2 = \frac{1}{2}$

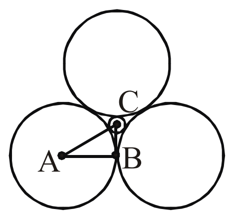
$\therefore$ Simplest formula of solid = $PQ_{1/2} = P_2Q$

Ex.7 Calculate the radius (r) of largest sphere which may be fitted in the (i) triangular voids (ii) tetrahedral voids (iii) octahedral voids (iv) cubic voids made by identical spheres of radius, 'R', without disturbing the crystal.

Sol. (i) AB = R

AC = R + r

∠BAC = 30°

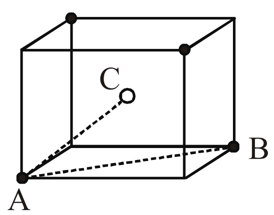


Now, $\cos 30° = \frac{AB}{AC} \Rightarrow \frac{\sqrt{3}}{2} = \frac{R}{R+r}$

$r = \left(\frac{2}{\sqrt{3}} - 1\right)R = 0.155R$

(ii) $AB = \sqrt{2}a = 2R$

$AC = \frac{\sqrt{3}\,a}{2} = R + r$



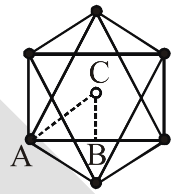
node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\02_Solid State\Theory.p65

ALLEN Session 2019-20

Now, $\dfrac{R+r}{R}=\dfrac{\sqrt{3}}{\sqrt{2}}$

$\therefore \quad r=\left(\sqrt{\dfrac{3}{2}}-1\right)R=0.225R$

(iii) $AB=R$



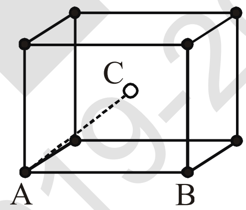
$AC=R+r$

$\angle BAC=45°$

Now, $\cos 45°=\dfrac{AB}{AC}\Rightarrow\dfrac{1}{\sqrt{2}}=\dfrac{R}{R+r}$

$\therefore \quad r=\left(\sqrt{2}-1\right)R=0.414R$

(iv) $AB=a=2R$



$AC=\dfrac{\sqrt{3}\,a}{2}=R+r$

Now, $\dfrac{R+r}{R}=\sqrt{3}$

$\therefore \quad r=\left(\sqrt{3}-1\right)R=0.732R$

Ex.8 An element (molar mass = 60 gm/mole) crystallises in CCP lattice. If density is 6.25 gm/cm³ and the distance between next nearest neighbour is dÅ, then the value of 'd' is ($N_A = 6 \times 10^{23}$).

Ans. (4)

$d=\dfrac{ZM}{N_A.a^3}$

$6.25=\dfrac{4\times 60}{6\times 10^{23}\times a^3}\Rightarrow a=4\times 10^{-8}\text{cm}=4\text{Å}.$

Session 2019-20

10 PACKING IN IONIC SOLID

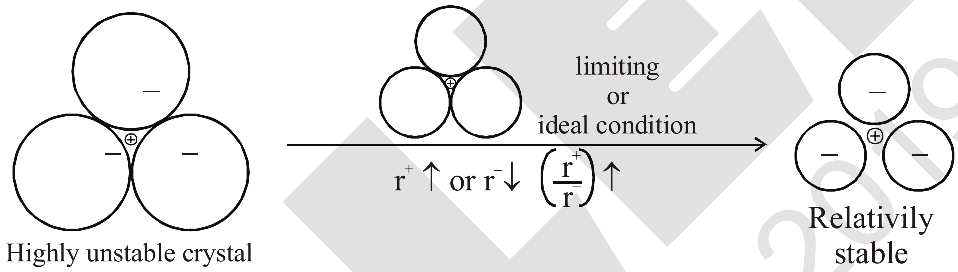
(i) Normally anions are bigger than cations, hence ionic solid are considered as the packing of anions and cations are supposed to occupy the voids.

(ii) Oppositively charged particles should lie closer and smilarly charged particles should lie away from each other.

(iii) Each ion tend to maximise its coordination number (Number of oppositively charged ions around it).

Point (ii) and (iii) contradict each other because on increase the CN, the repulsion between like charges will also increases. Hence, in all the ionic solids, there must be a balance with the help of relative size of ions to ensure maximum CN and minimum repulsion between like charges.

10.1 LIMITING OR IDEAL RADIUS RATION $\left(\frac{r^+}{r^-}\right)$:



The minimum $\frac{r^+}{r^-}$ values for the existence of a cation in a particular void is called **limiting radius ratio** for that void.

With the increase in radius ratio, space between anion will increase & hence the cation may tend for higher coordination number.

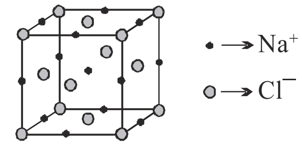
Voids	CN	Limiting r^+/r^-	Range of r^+/r^-
Triangular	3	0.155	0.155 – 0.225
Tetrahedral	4	0.225	0.225 – 0.414
Octahedral	6	0.414	0.414 – 0.732
Cubic	8	0.732	0.732 – 1.000

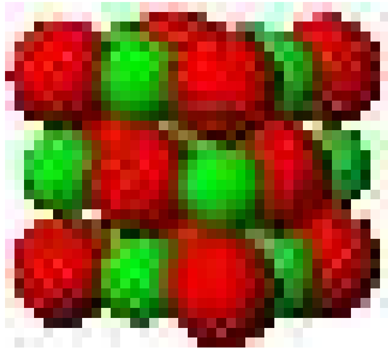
* For values lesser than 0.155, crystal will not exist.

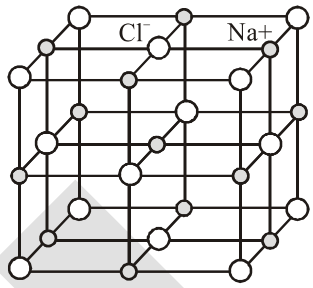
ALLEN Session 2019-20
node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\02_Solid State\Theory.p65

10.2 *TYPES OF IONIC STRUCTURES:*

10.2.1 Rock salt structure (NaCl) :







The bigger Cl^- forms cubic close packing and small Na^+ occupy positions of all octahedral voids. The radius ratio $\frac{r^+}{r^-}$ lie in the range 0.414 – 0.732.

(i) Each Na^+ is surrounded by six Cl^- and each Cl^- is surrounded by six Na^+ ion. [6:6 coordination]

(ii) No. of Na^+ and Cl^- in each unit cell is 4.

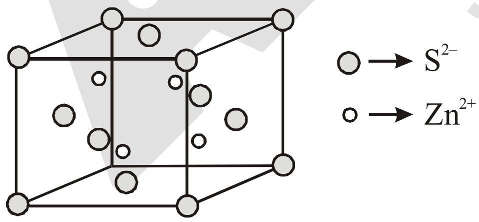
(iii) Number of formula units of NaCl per unit cell is equal to 4.

(iv) The density of NaCl crystal is given by $d = \left(\frac{4 \times M_{NaCl}}{N_A \times a^3}\right)$

(v) The edge length of NaCl unit cell is given by $(2r^+ + 2r^-) \Rightarrow \boxed{\frac{\mathbf{a}}{2} = \mathbf{r}^+ + \mathbf{r}^-}$

10.2.2 Zinc blende (sphalerite) structure (ZnS) :

Larger ion (S^{2-}) forming ccp arrangement and smaller ion (Zn^{2+}) filling half or alternate tetrahedral voids



Zinc blende structure

(i) C.N. of $Zn^{2+} = 4$; C.N. of $S^{2-} = 4$ [4 : 4 coordination]

(ii) Formula units of ZnS per unit cell = 4. **(iii)** $d_{ZnS} = \frac{4 \times M_{ZnS}}{N_A \times a^3}$ **(iv)** $r_{Zn^{2+}} + r_{S^{2-}} = \frac{a\sqrt{3}}{4}$

Session 2019-20

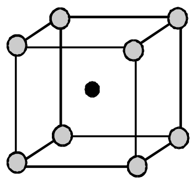
10.2.3 Cesium chloride structure (CsCl) :

Cl^- at the corners of cube and Cs^+ in the center (cubic void).

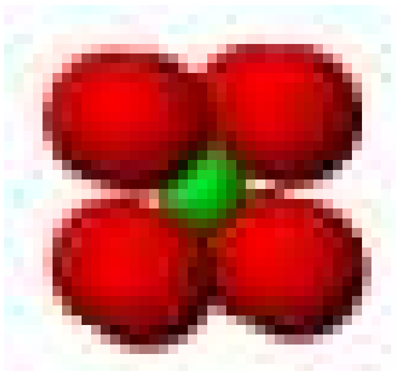
(i) C.N. of $Cs^+ = 8$; C.N. of $Cl^- = 8$ [8 : 8 coordination]

(ii) Formula units of CsCl per unit cell = 1

(iii) $d_{CsCl} = \dfrac{M_{CsCl}}{N_A \times a^3}$ **(iv)** $r_{Cs^+} + r_{Cl^-} = \dfrac{a\sqrt{3}}{2} \Rightarrow \boxed{\mathbf{r^+ + r^- = \dfrac{a\sqrt{3}}{2}}}$



Cesium chloride structure



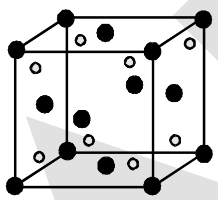
10.2.4 Fluorite structure (CaF_2) :

Ca^{2+} forming ccp arrangement and F^- filling all tetrahedral voids.

(i) C.N. of $F^- = 4$; C.N. of $Cs^+ = 8$ [8 : 4 coordination]

(ii) Formula units of CaF_2 per unit cell = 4

(iii) $d_{CaF_2} = \dfrac{4 \times M_{CaF_2}}{N_A \times a^3}$ **(iv)** $r_{Ca^{2+}} + r_{F^-} = \dfrac{a\sqrt{3}}{4}$



Fluorite structure

10.2.5 Antifluorite structure (Li_2O) :

O^{2-} ion forming ccp and Li^+ taking all tetrahedral voids.

(i) C.N. of $Li^+ = 4$

C.N. of $O^{2-} = 8$

(ii) Formula units of Li_2O ; per unit cell = 4

(iii) $d_{Li_2O} = \dfrac{4 \times M_{Li_2O}}{N_A \times a^3}$ **(iv)** $r_{Li^+} + r_{O^{2-}} = \dfrac{a\sqrt{3}}{4}$

10.2.6 Corundum Structure (Al_2O_3) :

O^{2-} forming hcp and Al^{3+} filling 2/3 octahedral voids.

Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\02_Solid State\Theory.p65

10.2.7 Rutile structure (TiO_2) **:**

O^{2-} forming hcp while Ti^{4+} ions occupy half of the octahedral voids.

10.2.8 Pervoskite structure ($CaTiO_3$) **:**

Ca^{2+} in the corner of cube, O^{2-} at the face center and Ti^{4+} at the centre of cube.

10.2.9 Spinel and inverse spinel structure ($MgAl_2O_4$) **:**

O^{2-} forming fcc, Mg^{2+} filling 1/8 of tetrahedral voids and Al^{3+} taking half of octahedral voids.

In an inverse spinel structure, O^{2-} ion form FCC lattice, A^{2+} ions occupy 1/8 of the tetrahedral voids and trivalent cation occupies 1/8 of the tetrahedral voids and 1/4 of the octahedral voids.

Ex.9 A solid A^+B^- has NaCl type close packed structure. If the anion has a radius of 250 pm, what should be the ideal radius for the cation ? Can a cation C^+ having a radius of 180 pm be slipped into the tetrahedral site of the crystal A^+B^- ? Give reason for your answer.

Sol. *In Na^+Cl^- crystal each Na^+ ion is surrounded by 6 Cl^+ ions and vice versa. Thus Na^+ ion is placed in octahedral hole.*

The limiting radius ratio for octahedral site = 0.414

or $\frac{A^+}{B^-} = \frac{r}{R} = 0.414$

Given that radius of anion (B^-) R = 250 pm

i.e. radius of cation (A^+) $r = 0.414\,R = 0.414 \times 250$ pm

or $r = 103.5$ pm

Thus ideal radius for cation (A^+) is r = 103.5 pm.

We know that (r/R) for tetrahedral hole is 0.225.

$\therefore \quad \frac{r}{R} = 0.225$

or $r = 0.225\,R = 0.225 \times 250 = 56.25$ pm

Thus ideal radius for cation is 56.25 pm for tetrahedral hole. But the radius of C^+ is 180 pm. It is much larger than ideal radius i.e. 56.25 pm. Therefore we can not slip cation C^+ into the tetrahedral site.

Ex.10 A compound forms hexagonal close-packed structure. What is the total number of voids in 0.5 mole of it? How many of these are tetrahedral voids?

Sol. *Since, for every atom forming hcp structure there are two tetrahedral voids and one octahedral void. Total voids = 3 × 0.5 = 1.5 mol and total tetrahedral void = 2 × 0.5 = 1 mol.*

Ex.11 A compound is formed by two elements M and N. The element N forms ccp and atoms of M occupy 1/3rd of tetrahedral voids. What is the formula of the compound?

Sol. *In one unit cell of ccp, no. of N = 4 ; total no. of tetrahedral void = 8 ; occupied tetrahedral void by M = 8/3 ; Empirical formula :* $N_4M_{8/3} = N_3M_2$.

ALLEN Session 2019-20

11. IMPERFECTIONS IN SOLIDS

Although crystalline solids have short range as well as long range order in the arrangement of their constitutent particles, yet crystals are not perfect. Usually a solid consists of an aggregate of large number of small crystals. These small crystals have defects in them. This happens when crystallisation process occurs at fast or moderate rate. Single crystals are formed when the process of crystallisation occurs at extremely slow rate. Even these crystals are not free of defects.

The defects are basically irregularities in the arrangement of constituent particles. Broadly speaking, the defects are of two types, namely, **point defects** and **line defects**. Point defects are the irrgularties or deviations from ideal arrangement around a point or an atom in a crystalline substance, whereas the line defects are the irregularities or deviations from ideal arrangement in entire rows of lattice points. These irregularities are called crystal defects. We shall confine our discussion to point defects only.

11.1 Types of Point Defects

Point defect can be classified into three types:

(i) Stoichiometric defects

(ii) Non-stoichiometric defects

(iii) Impurity added defect

(i) Stoichiometric Defect

These are the point defect that do not disturb the stoichiometry of the solid. They are also called ***intrinsic*** or **thermodynamic defects**. Basically these are of two types; vacancy defects and interstitial defect.

(a) Vacancy Defect :

When some of the lattice sites are vacant, the crystal is said to have vacancy defect. This results in decrease in density of the substance. This defect can also develop when a substance is heated.

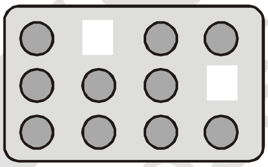


Fig. : Vacancy defects

(b) Interstitial Defect : When some constituent particles (atoms or molecules) occupy an interstitial site. the crystal is said to have interstitial defect. This defect increases the density of the substance.

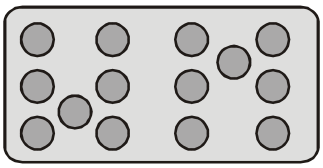


Fig. : Interstitial defects

Vacancy and interstitial defects as explained above can be shown by non-ionic solids. Ionic solids must always maintain electrical neutrality. Rather than simple vacancy or interstitial defects, they show these defects as **Frenkel and Schottky defects**.

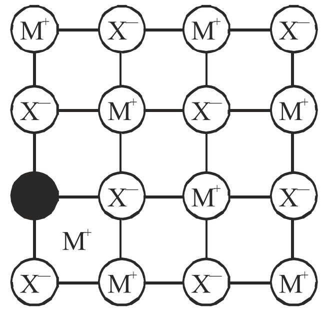
node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\02_Solid State\Theory.p65

(c) Frenkel Defect : This defect is shown by ionic solids.

The smaller ion (usually cation) is delocalised from its normal site to an interstitial site . It creates a vacancy defect at its original site and an interstitial defect at its new location.

Frenkel defect is also called **dislocation defect**. It does not change the density of the solid.

Frenkel defect is shown by ionic substance in which there is a large difference in the size of ions, for example, ZnS, AgCl, AgBr and AgI due to small size of Zn^{2+} and Ag^+ ions.



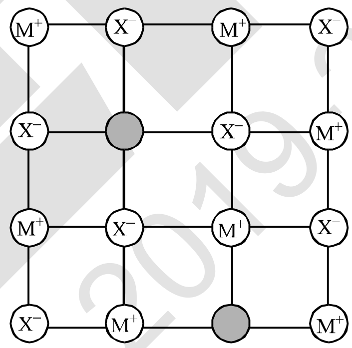
Frenkel Defect

Influences : Makes solid crystals good conductor. In Frenkel defect , ions in interstitial sites increases the dielectric constant.

(d) Schottky Defect : It is basically a vacancy defect in ionic solids. In order to maintain electrical neutrality, the number of missing cations and anions are equal (in stoichiometric ratio)

Like simple vacancy defect, Schottky defect also decreases the density of the substance. Number of such defects in ionic solids is quite significant. For example, in NaCl there are approximately,

10^6 Schottky pairs per cm^3 at room temperature. In 1 cm^3 there are about 10^{22} ions. Thus, there is one Schottky defect per 10^{16} ions. Schottky defect is shown by ionic substances in which the cation and anion are of almost similar sizes. For example, NaCl, KCl, CsCl and AgBr. It may be noted that AgBr shows both, Frenkel as well as Schottky defects.



Schottky Defect

Influence : The presence of large number of schottky defects in crystal results in significant decrease in its density.

(ii) Non-Stoichiometric Defects

The defects discussed so far do not disturbs the stoichiometry of the crystalline substance. However, a large number of non-stoichiometric inorganic solids are known which contain the constituent elements in non-stoichiometric ratio due to defects in their crystal structures. These defects are of two types:

(a) Metal excess defect.

(b) Metal deficiency defect.

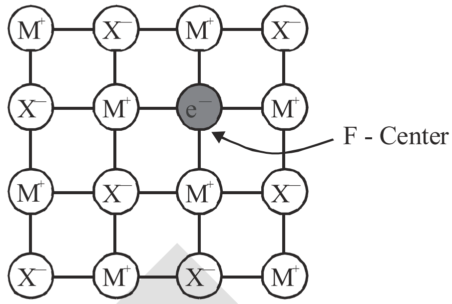
ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\02_Solid State\Theory.p65

(a) Metal Excess Defect

(I) Metal excess defect due to anionic vacancies :

Alkali halides like NaCl and KCl show his type of defect. When crystals of NaCl are heated in an atmosphere of sodium vapour, the sodium atoms are deposited on the surface of the crystal. The Cl^- ions diffuse to the surface of the crystal and combine with Na atoms to give NaCl. This happens by loss of electron by sodium atoms to form Na^+ ions.



Metal excess defects due to anion vacancies

The released electrons diffuse into the crystal and occupy anionic sites . As a result the crystal now has an excess of sodium. The anionic sites occupied by unpaired electrons are called ***F-centres*** (from the German word ***Farbenzenter*** for colour centre).

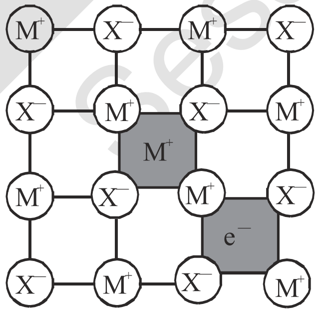
They impart yellow colour to the crystals of NaCl. The colour results by excitation of these electrons when they absorb energy from the visible light falling on the crystals. Similarly, excess of lithium makes LiCl crystals pink and excess of potassium makes KCl crystals violet (or lilac).

(II) ***Metal excess defect due to the presence of extra cations at interstitial sites :***

Zinc oxide is white in colour at room temperature. On heating it loses oxygen and turns yellow.

$$ZnO \xrightarrow{\text{heating}} Zn^{2+} + \frac{1}{2}O_2 + 2e^-$$

Now there is excess of zinc in the crystal and its formula becomes $Zn_{1+x}O$. The excess Zn^{2+} ions move to interstitial sites and the electrons to neighbouring interstitual sites.



Metal excess defects due to interstitial cation

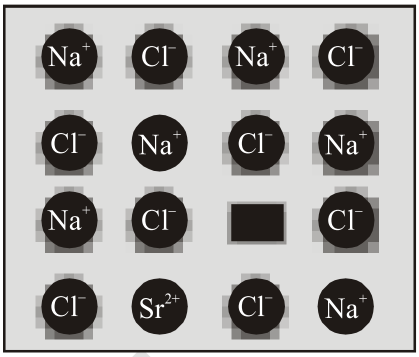
(b) Metal Deficiency Defect

There are many solids which are difficult to prepare in the stoichiometric composition and contain less amount of the metal as compared to the stoichiometric proportion. A typical example of this type is FeO which is mostly found with a composition of $Fe_{0.95}O$. It may actually range from $FeO_{0.93}O$ to $Fe_{0.96}O$. In crystals of FeO, some Fe^{2+} cations are missing and the loss of positive charge is made up by the presence of required number of Fe^{3+} ions.

ALLEN Session 2019-20
node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\02_Solid State\Theory.p65

(iii) Impurity Defects

If molten NaCl containing a little amount of $SrCl_2$ is crystallised, some of the sites of Na^+ ions are occupied by Sr^{2+}. Each Sr^{2+} replaces two Na^+ Ions. It occupies the site of one ion and the other site remains vacant. The cationic vacancies thus produced are equal in number to that of Sr^{2+} ions. Another similar example is the solid solution of $CdCl_2$ and AgCl.



Introduction of cation vacancy in NaCl by substitution of Na^+ by Sr^{2+}

Note: (i) As temperature increases, no. of defects increases exponentially.

(ii) For defect formation : $\Delta H > 0$, $\Delta S > 0$

$\therefore$ More spontaneous at higher temperatures.

(iii) No matter how many imperfections are present in a crystal, it is always electrically neutral (no net charge).

12. ELECTRICAL PROPERTIES

Solids exhibit an amazing range of electrical conductivities, extending over 27 orders of magnitude ranging from 10^{-20} to 10^{7} ohm^{-1} m^{-1}. Solids can be classified into three types on the basis of their conductivities.

(i) Conductors : The solids with conductivities ranging between 10^4 to 10^7 $ohm^{-1}m^{-1}$ are called conductors. Metals having conductivities in the order of 10^7 $ohm^{-1}m^{-1}$ are good conductors.

(ii) Insulators : These are the solids with very low conductivities ranging between 10^{-20} to 10^{-10} $ohm^{-1}m^{-1}$.

(iii) Semiconductors : These are the solids with conductivities in the intermediate range from 10^{-6} to 10^4 $Ohm^{-1}m^{-1}$.

12.1 : Conduction of Electricity in Metals

A conductor may conduct electricity through movement of electrons or ions. Metallic conductors belong to the former category and electrolytes to the latter.

Metals conduct electricity in solid as well as molten state. The conductivity of metals depend upon the number of valence electrons available per atom. The atomic orbitals of metal atoms from molecular orbitals which are so close in energy to each other as to form a band. If this band is partially filled or it overlaps with a higher energy unoccupied conduction band, then electrons can flow easily under an applied electric field and the metal shows conductivity.

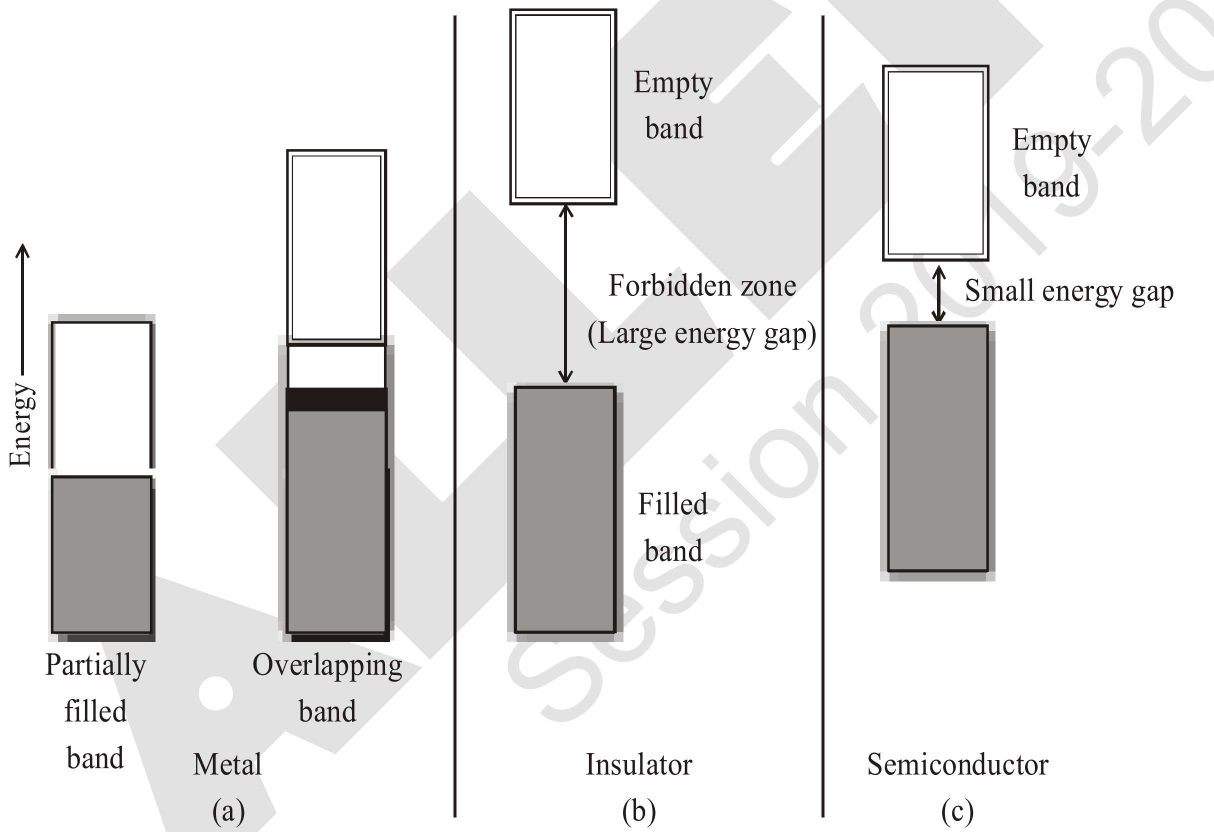
Session 2019-20

12.2 : Insulator :

If the gap between filled valence band and the next higher unoccupied band (conduction band) is large, electron cannot jump to it and such a substance has very small conductivity and it behaves as an insulator.

12.3 : Conduction of Electricity in Semi-conductor

In case of semiconductors, the gap between the valence band and conduction band is small. Therefore, some electrons may jump to conduction band and show some conductivity. Electrical conductivity of semconductors increases with rise in temperature, since more electrons can jump to the conduction band. Substances like silicon and germanium show this type of behaviour and are called **intrinsic semiconductors**.



Distinction among (a) metals (b) insulators and (c) semiconductors. In each case, an unshaded area represents a conduction band.

The conducitivity of these interinsic semiconductors is too low to be of practical use. Their conductivity is increased by adding an appropriate amount of suitable impurity. This process is called **doping**. Doping can be done with an impurity which is electron rich or electron deficient as compared to the intrinsic semiconductor silicon or germanium. Such impurities introduce ***electrical defect*** in them.

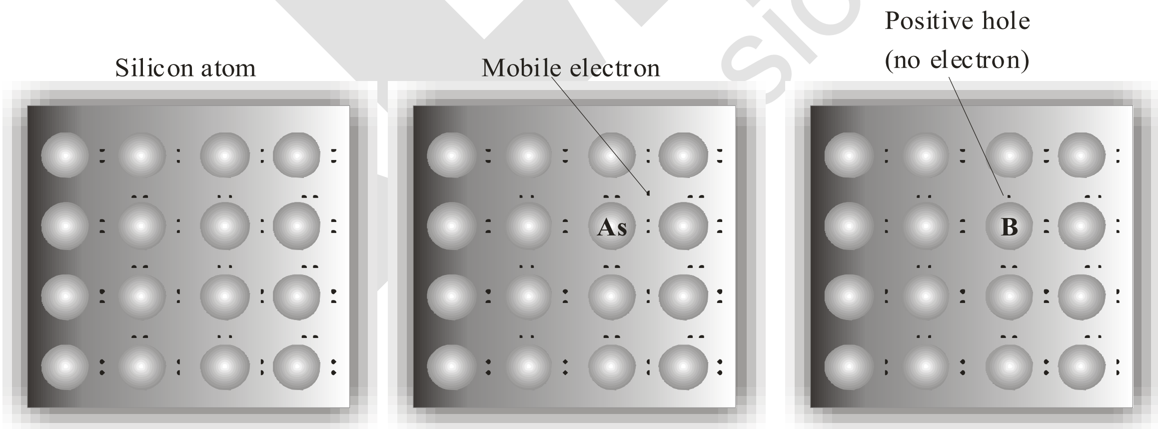
node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\02_Solid State\Theory.p65

(a) Electron – rich impurities

Silicon and germanium belong to group 14 of the periodic table and have four valence electrons each. In their crystals each atom forms four covalent bonds with its neighbours . When doped with a group 15 element like P or As, which contains five valence electrons, they occupy some of the lattice sites in silicon or germanium crystal. Four out of five electrons are used in the formation of four covalent bonds with the four neighbouring silicon atoms. The fifth electron is extra and becomes delocalised. These delocalised electrons increase the conductivity of doped silicon (or germanium). Here the increase in conductivity is due to the negatively charged electron, hence silicon doped with electron-rich impurity is called ***n-type semiconductor***.

(b) Electron – deficit impurities

Silicon or germanium can also be doped with a group 13 element like B, Al or Ga which contains only three valence electrons. The place where the fourth valence electron is missing is called ***electron hole or electron vacancy*** . An electron from a neighbouring atom can come and fill the electron hole, but in doing so it would leave an electron hole at its original position. If it happens, it would appear as if the electron hole has moved in the direction opposite to that of the electron that filled it. Under the influence of electric field, electrons would move towards the positively charged plate through electronic holes, but it would appear as if electron holes are positively charged and are moving towards negatively charged plate. This type of semi conductors are called ***p-type semiconductors***.



Perfect crystal (a) | *n*-type (b) | *p*-type (b)

Creation of n-type and p-type semi conductors by doping groups 13 and 15 elements

(c) Applications of n-type and p-type semiconductors

Various combinations of *n*-type and *p*-type semiconductors are used for making electronic components. Diode is a combination of *n*-type and *p*-type semiconductor and is used as a rectifier. Transistors are made by sandwiching a layer of one type of semiconductor between two layers of the other type of semiconductor. *npn* and *pnp* type of transistors are used to detect or amplify radio or audio signals. The solar cell is an efficient photo-diode used for conversion of light energy into electrical energy.

Germanium and Silicon are group 14 elements and therefore, have a characteristic valency of four and form four bonds as in diamond. A large variety of solid state materials have been prepared by combination of group 13 and 15 or 12 and 16 to simulate average valence of four as in Ge or Si. Typical compounds of groups 13-15 are InSb, AIP and GaAs. Gallium arsenide (GaAs) semiconductor have very fast response and have revolutionised the design of semiconductor devices. ZnS, CdS, CdSe and HgTe are examples of groups 12-16 compounds. In these compounds, the bonds are not perfectly covalent and the ionic character depends on the electronegativities of the two elements.

It is interesting to learn that transition metal oxides show marked differences in electrical properties. TiO, CrO_2 and ReO_3 behave like metals. Rhenium oxide, ReO_3 is like metallic copper in its conductivity and appearance. Certain other oxides like VO, VO_2, VO_3 and TiO_3 show metallic or insulating properties depending on temperature.

13. MAGNETIC PROPERTIES :

Every substance has some magnetic properties associated with it. The origin of these properties lies in the electrons. Each electron in an atom behaves like a tiny magnet. Its magnetic moment originates from two types of motions (i) its orbital motion around the nucleus and (ii) its spin around its own axis . Electron being a charged particle and undergoing these motions can be considered as a small loop of current which possesses a magnetic moment. Thus, each electron has a permanent spin and an orbital magnetic moment associated with it. Magnitude of this magnetic moment is very small and is measured in the unit called Bohr magneton, μ_B. It is equal to 9.27×10^{-24} A m^2.

On the basis of their magnetic properties, substances can be classified into five categories: (i) paramagnetic (ii) diamagnetic (iii) ferromagnetic (iv) antiferromagnetic and (v) ferrimagnetic.

(i) **Paramagnetism :**

Paramagnetic substances are weakly attracted by a magnetic field. They are magnetised in a magnetic field in the same direction. They lose their magnetism in the absence of magnetic field. Paramagnetism is due to presence of one or more unpaired electrons which are attracted by the magnetic field. O_2, Cu^{2+}, Fe^{3+}, Cr^{3+} are some examples of such substances.

(ii) **Diamagnetism :**

Diamagnetic substances are weakly repelled by a magnetic field. H_2O, NaCl and C_6H_6 are some examples of such substances. They are weakly magnetised in a magnetic field in opposite direction. Diamagnetism is shown by those substances in which all the electrons are paired and there are no unpaired electrons. Pairing of electrons cancels their magnetic moments and they lose their magnetic character.

(iii) **Ferromagnetism :**

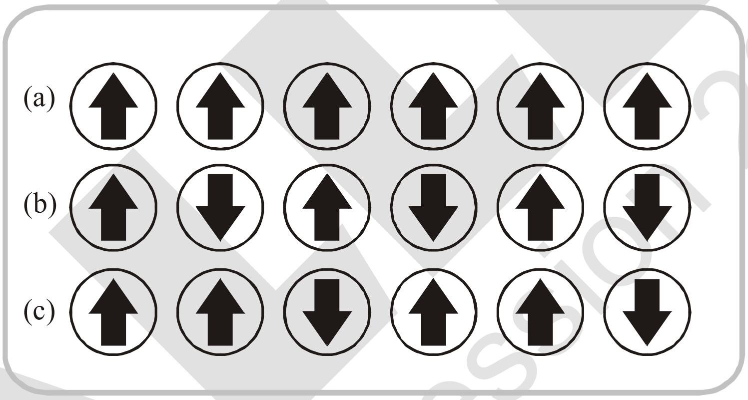
A few substances like iron, cobalt, nickel, gadolinium and CrO_2 are attracted very strongly by a magnetic field. Such substances are called ferromagnetic substances. Besides strong attractions, these substances can be permanently magnetised. In solid state, the metal ions of ferromagnetic substances are grouped together into small regions called ***domains***. Thus, each domain acts as a tiny magnet. In an unmagnetised piece of a ferromagnetic substance the domains are randomly oriented and their magnetic moments get cancelled. When the substance is placed in a magnetic field all the domains get oriented in the direction of the magnetic field and a strong magnetic effect is produced. This ordering of domains persist even when the magnetic field is removed and the ferromagnetic substance becomes a permanent magnet.

(iv) **Antiferromagnetism :**

Substances like MnO showing antiferromagnetism have domain structure similar to ferromagnetic substance, but their domains are oppositely oriented and cancel out each other's magnetic moment

(v) **Ferrimagnetism :**

Ferrimagnetism is observed when the magnetic moments of the domains in the substance are aligned in parallel and anti-parallel directions in unequal numbers. They are weakly attracted by magnetic field as compared to ferromagnetic substances. Fe_3O_4 (magnetite) and ferrites like $MgFe_2O_4$ and $ZnFe_2O_4$ are examples of such substances. These substances also lose ferrimagnetism on heating and become paramagnetic.



Schematic alignment of magnetic moments in (a) ferromagnetic (b) antiferromagnetic and (c) ferrimagnetic

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\02_Solid State\Theory.p65

EXERCISE : S-I

METALLIC CRYSTALS

Cubic crystals

1. A closed packed structure of uniform spheres has the edge length of 600 pm. Calculate the radius of sphere, if it exist in $(\sqrt{2}=1.4, \sqrt{3}=1.7)$

(a) simple cubic lattice (b) BCC lattice (c) FCC lattice

2. Xenon crystallises in the face-centred cubic lattice and the edge length of the unit cell is $438\sqrt{2}$ pm. What is the nearest neighbour distance and what is the radius of xenon atom?

3. The effective radius of the iron atom is $\sqrt{2}$ Å. It has FCC structure. Calculate its density (Fe = 56 amu , $N_A = 6 \times 10^{23}$)

4. Calculate the ratio of densities if same element undergoes fcc as well as simple cubic packing. Assume same atomic radius in both crystals.

5. Potassium has body-centred cubic structure with the nearest neighbour distance $260\sqrt{3}\,\text{pm}$. Its density would be $\left(\frac{1}{(5.2)^2} = 0.036\right.$, $N_A = 6 \times 10^{23}$, K = 39)

6. A cubic solid is made up of two elements 'A' and 'B'. Atoms 'B' are at the corners of the cube and 'A' at the body centre. What is the simplest formula of compound ?

7. An element 'X' crystallizes in bcc. Find volume of unit cell in $(\text{Å})^3$, if atomic radius is $\sqrt{3}$ Å.

8. A lattice has simple cube unit cell then number of faces which meets at a corner of a cube in this lattice is.

9. How many next nearest neighbours does potassium have in bcc lattice ?

10. Number of crystal systems having, only 2 types of Bravais lattices = x,
Number of crystal systems having, atleast 2 interfacial angles equal = y,
all the three interfacial angles and all the three axes lenghts equal = z
Then find y – (x + z).

PACKING IN SOLIDS

11. A cubic solid is made by atoms 'A' forming close pack arrangement, 'B' occupying one-fourth of tetrahedral void and C occupying half of the octahedral voids. What is the formula of compound ?

12. If number of nearest neighbours, next nearest (2nd nearest) neighbour and next to next nearest (3rd nearest) neighbours are x, y and z respectively for body centred cubic unit cell, then calculate value of $\frac{xy}{z}$ is.

13. In FCC unit cell, what fraction of edge is not covered by atoms ?

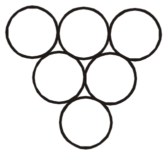
14. Element A crystallizes in hcp. Calculate total number of tetrahedral voids in 24 micrograms of A ($N_A = 6 \times 10^{23}$, Atomic mass of A = 48)

15. For ABC ABC ABC .... packing, distance between two succesive tetrahedral void is X and distance between two successive octahedral void is y in an unit cell, then $\frac{y\sqrt{2}}{X}$ is

ALLEN Session 2019-20

16. An element 'M' crystallizes in ABAB....type packing. If adjacent layer A & B are $10\frac{\sqrt{2}}{\sqrt{3}}$ pm apart, then calculate radius of largest sphere which can be fitted in the void without disturbing the lattice arrangement (Given : $\sqrt{2} = 1.414$)

17. A 3d unit cell is such that one of its planes has the following arrangement of atoms.



What will be the number of next nearest neighbour in such unit cell?

18. The density of solid Argon is 1.6 gm/ml at –233°C. If the atomic volume of Argon is assumed to be $\frac{5}{3}\times10^{-23}$ cm^3 then what % of solid Argon is apparently empty space ? (Ar = 40, $N_A = 6 \times 10^{23}$)

19. An element crystallizes in a structure having FCC unit cell of an edge 200 pm. Calculate the density, if 200 g of this element contains 2.4×10^{24} atoms. ($N_A = 6 \times 10^{23}$)

20. Find packing fraction of three dimensional unit cell of AAAAA.......type hypothetical arrangement in which hexagonal packing is taken in layer.

IONIC CRYSTALS

21. If the radius of Mg^{2+} ion, Cs^+ ion, O^{2-} ion, S^{2-} ion and Cl^- ion are 0.65 Å , 1.69 Å, 1.40 Å, 1.84 Å, and 1.81 Å respectively, calculate the co-ordination numbers of the cations in the crystals of MgS, MgO and CsCl.

22. The two ions A^+ and B^- have radii 88 pm and 200 pm respectively. In the closed packed crystal of compound AB, predict the co-ordination number of A^+.

23. Spinel is a important class of oxides consisting of two types of metal ions with the oxide ions arranged in CCP pattern. The normal spinel has one-eight of the tetrahedral holes occupied by one type of metal ion and one half of the octahedral hole occupied by another type of metal ion. Such a spinel is formed by Zn^{2+}, Al^{3+} and O^{2-}, with Zn^{2+} in the tetrahedral holes. Give the formula of spinel.

24. KF crystallizes in the NaCl type structure. If the radius of K^+ ion is 132 pm and that of F^- ion is 135 pm, what is the shortest K–F distance? What is the edge length of the unit cell? What is the closet K^+– K^+ distance?

25. CsCl has bcc unit cell with edge length 400 pm. Calculate the interionic distance in CsCl.

26. The density of KBr is 2.38 g cm^{-3} . The length of the edge of the unit cell is 700 pm. Find the number of fomula unit of KBr present in the single unit cell.

($N_A = 6 \times10^{23}$ mol^{-1} , At. mass : K = 39, Br = 80)

ALLEN Session 2019-20

27. A crystal of lead(II) sulphide has NaCl structure. In this crystal the shortest distance between Pb^{+2} ion and S^{2-} ion is 300 pm. What is the length of the edge of the unit cell in lead sulphide? Also calculate the unit cell volume.

28. If the length of the body diagonal for CsCl which crystallises into a cubic structure with Cl^- ions at the corners and Cs^+ ions at the centre of the unit cell, is 7 Å and the radius of the Cs^+ ion is 1.69 Å, what is the radius of Cl^- ion?

29. Rbl crystallizes in (8 : 8) structure in which each Rb^+ is surrounded by eight iodide ions each of radius 2.17 Å. Find the length of one side of RbI unit cell , assuming anion, anion contact.

30. Solid AB has NaCl type structure. If the radius of A^+ and B^- are 0.8 Å and 1.2Å respectively and formula mass of AB is 48 g/mole, what is the density of AB solid.

Take : Avogadro's number = 6×10^{23}

PROBLEMS RELATED WITH DEFECTS IN SOLID

31. The composition of a sample of wustite is $Fe_{0.93}O_{1.0}$. What percentage of iron is present in the form of Fe(III)?

32. If NaCl is dopped with 10^{-3} mol % $SrCl_2$, what is the number of cation vacancies per mole of NaCl? ($N_A = 6 \times 10^{23}$)

33. AgCl has the same structure as that of NaCl. The edge length of unit cell of AgCl is found to be 523.5 pm and the density of AgCl is 6.0 g cm^{-3}. Find the percentage of sites that are unoccupied.

[Ag = 108 , $(5.235)^3 = 143.5$]

34. A non stoichiometric compound Fe_7S_8 consist of iron in both Fe^{+2} and Fe^{+3} form and sulphur is present as sulphide ions. Calculate cation vacancies as a percentage of Fe^{+2} initially present in the ideal crystal.

35. The density of ZnS crystal (Zinc blende structure) having 10% Frenkel defect is

[$r_{Zn^{2+}} = 40\sqrt{3}\text{pm}, r_{S^{2-}} = 110\sqrt{3}\text{pm}$, Zn = 65.2, S = 32]

ALLEN Session 2019-20

EXERCISE : S-II

1. An element (atomic weight = 125) crystallises in simple cubic structure. Diameter of the largest atom which can be placed without disturbing unit cell is 366 pm. If the density of element 'x' gm/cm^3 , the value of $\left(\frac{6x}{5}\right)$ is

[**Given** : $\sqrt{3}$ = 1.732 , $N_A = 6 \times 10^{23}$]

2. The density of diamond from the fact that it has face centred cubic structure with two atoms per lattice point and unit cell edge length of 3.6 Å, is 'x' gm / cm^3, then the value of (1.458x) is ($N_A = 6 \times 10^{23}$)

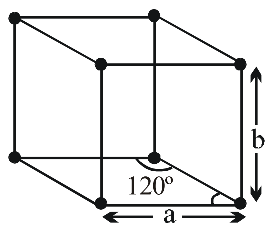
3. Iron crystallizes in several modifications. At about 910°C, the body-centred cubic 'α' form undergoes a transition to the face-centred cubic 'γ' form. Assuming that the distance between nearest neighbours is the same in the two forms at the transition temperature, the ratio of the density of γ iron to that of α iron at the transition temperature, x : 1, then the value of $(3\sqrt{6}x)$ is.

4. What is the percent by mass of titanium in rutile, a mineral that contain titanium and oxygen, if structure can be described as a closest packed array of oxide ions, with titanium in one half of the octahedral holes. What is the oxidation number of titanium? (Ti = 48)

5 . An element crystallizes into a structure which may be described by a cubic type of unit cell having one atom on each corner of the cube and two extra atoms on one of its body diagonal. If the volume of this unit cell is 2.4×10^{-23} cm^3 and density of element is 7.2 g cm^{-3}, calculate the number of atoms present in 288 g of element.

6. What will be packing fraction of solid in which atoms are present at corners and cubic void is occupied. The insertion of the sphere into void does not disturb simple cubic lattice.

7. Ice crystallizes in a hexagonal lattice. At the low temperature at which the structure was determined, the lattice constants where **a** = 5 Å, and **b** = $3.5\sqrt{3}$ Å How many molecules are contained in the given unit cell?



[density (ice) = $\frac{16}{17.5}$ gm/cm^3)]

8. The mineral hawleyite, one form of CdS, crystallizes in one of the cubic lattices, with edge length 5.87Å. The density of hawleyite is 4.63 g cm^{-3}. (Cd = 112)

(i) In which cubic lattice does hawleyite crystallize?

(ii) Find the Schottky defect in g cm^{-3}.

9. KCl crystallizes in the same type of lattice as does NaCl. Given that $\frac{r_{Na^+}}{r_{Cl^-}}=0.5$ and $\frac{r_{Na^+}}{r_{K^+}}=0.7$ Calculate:

(a) The ratio of the sides of unit cell for KCl to that for NaCl and

ALLEN
Session 2019-20
node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\02_Solid State\Exercise.p65

(b) The ratio of densities of NaCl to that for KCl.

$\left(\left(\frac{8}{7}\right)^3 = 1.5, \frac{74.5}{58.5} = 1.25\right)$

10. A cubic unit cell contains manganese ions at the corners and fluoride ions at the center of each edge.

(a) What is the empirical formula of the compound?

(b) What is the co-ordination number of the Mn ion?

(c) Calculate the edge length of the unit cell, if the radius of Mn ion is 0.65 Å and that of F^- ion is 1.36 Å.

11. Potassium crystallizes in a body-centred cubic lattice with edge length, a = 5.2 Å.

(a) What is the distance between nearest neighbours?

(b) What is the distance between next-nearest neighbours?

(c) How many nearest neighbours does each K atom have?

(d) How many next-nearest neighbours does each K atom have?

(e) What is the density of crystalline potassium?

(Given : K = 39, $(5.2)^3 = 140$)

12 An element X (atomic weight = 24 gm/mole) forms a face centred cubic lattice. If the edge length of the lattice is 4×10^{-8} cm and the observed density is 2.40×10^3 kg/m^3 , calculate the percentage occupancy of lattice points by X element. (Given : $N_A = 6 \times 10^{23}$)

13 Calculate the density (in gm/cm^3) of NaCl type ionic solid (MW = 75 gm/mol) if distance between two nearest cations is $250\sqrt{2}$ pm (avogadro number = 6×10^{23})

14. The olivine series of minerals consists of crystals in which Fe and Mg ions may substitute for each other causing substitutional impurity defect without changing the volume of the unit cell. In olivine series of minerals, oxide ion exist as FCC with Si^{4+} occupying 1/4 th of octahedral voids and divalent ions occupying 1/4th of tetrahedral voids. The density of forsterite (magnesium silicate) is 3.0 g/cc and that of fayalite (ferrous silicate) is 4.0 g/cc. Find the formula of forsterite and fayalite minerals and the mass percentage of fayalite in an olivine with a density of 3.5 g/cc.

ALLEN Session 2019-20

EXERCISE : O-I

SINGLE CORRECT :

1. Which of the following are the correct axial distances and axial angles for rhombohedral system ?

(A) $a = b = c, \alpha = \beta = \gamma \neq 90^0$ (B) $a = b \neq c, \alpha = \beta = \gamma = 90^0$

(C) $a \neq b = c, \alpha = \beta = \gamma = 90^0$ (D) $a \neq b \neq c, \alpha \neq \beta \neq \gamma \neq 90^0$

2. $a \neq b \neq c, \alpha \neq \beta \neq \gamma \neq 90^0$ represents

(A) tetragonal system (B) orthorhombic system

(C) monoclinic system (D) triclinic system

3. Diamond belongs to the crystal system :

(A) Cubic (B) triclinic (C) tetragonal (D) hexagonal

4. A match box exhibits -

(A) Cubic geometry (B) Monoclinic geometry

(C) Tetragonal geometry (D) Orthorhombic geometry

5. Which of the following solids substances will have same refractive index when measured in different directions?

(A) NaCl (B) Monoclinic sulphur (C) Rubber (D) Graphite

6. In the body-centred cubic unit cell & face centred cubic unit cell, the radius of atom in terms of edge lengh(a) of the unit cell is respectively:

(A) $\frac{a}{2}, \frac{a}{2\sqrt{2}}$ (B) $\frac{a}{2\sqrt{2}}, \frac{\sqrt{3}a}{4}$ (C) $\frac{\sqrt{3}a}{4}, \frac{a}{2\sqrt{2}}$ (D) $\frac{\sqrt{3}a}{2}, \frac{a}{2\sqrt{2}}$

7. Crystal system in which maximum number of Bravais lattices are possible is

(A) Cubic (B) Triclinic (C) Orthorhombic (D) Rhombohedral

8. Number of atoms at second nearest position from a given atom in a BCC structure is

(A) 8 (B) 6 (C) 12 (D) 4

9. Percentage area of each face covered by atoms in a FCC unit cell is -

(A) 60.4% (B) 68% (C) 74% (D) 78.5%

10. Correct sequence of the coordination number in SC, FCC & BCC is-

(A) 6, 8, 12 (B) 6, 12, 8 (C) 8, 12, 6 (D) 8, 6, 12

11. If 'Z' is the number of atoms in the unit cell that represents the closest packing sequence ---A B C A B C ---, the number of tetrahedral voids in the unit cell is equal to

(A) Z (B) 2Z (C) Z/2 (D) Z/4

12. The interstitial hole is called tetrahedral because

(A) It is formed by four spheres.

(B) Partly same and partly different.

(C) It is formed by four spheres the centres of which form a regular tetrahedron.

(D) None of the above three.

13. The size of an octahedral void formed in a closest packed lattice as compared to tetrahedral void is

(A) Equal (B) Smaller (C) Larger (D) Not definite

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\02_Solid State\Exercise.p65

14. If the anions (A) form hexagonal closest packing and cations (C) occupy only 2/3 octahedral voids in it, then the general formula of the compound is

(A) CA (B) CA_2 (C) C_2A_3 (D) C_3A_2

15. Which one of the following schemes of ordering closest packed sheets of equal sized spheres do not generate close packed lattice.

(A) ABCABC (B) ABACABAC (C) ABBAABBA (D) ABCBCABCBC

16 . Copper metal crystallizes in FCC lattice. Edge length of unit cell is 362 pm. The radius of largest atom that can fit into the voids of copper lattice without disturbing it is

(A) 53 pm (B) 45 pm (C) 93 pm (D) 87 pm

17. Packing fraction in 2-D hexagonal arrangement of identical sphere is

(A) $\frac{\pi}{3\sqrt{2}}$ (B) $\frac{\pi}{3\sqrt{3}}$ (C) $\frac{\pi}{2\sqrt{3}}$ (D) $\pi/6$

18. In fcc unit cell smallest distance between octahedral void & tetrahedral void is -

(a = edge length of unit cell)

(A) $\frac{a}{\sqrt{2}}$ (B) $\frac{\sqrt{3}\,a}{2}$ (C) a (D) $\frac{\sqrt{3}\,a}{4}$

19. What is not true regarding hexagonal close packing (hcp)

(A) packing fraction is 0.74

(B) coordination number is 12

(C) ABC ABC.....type packing

(D) Containing both tetrahedral and octahedral voids

20. In which of the following arrangement distance between two nearest neighbours is maximum, considering identical sized atoms in all arrangements ?

(A) Simple cubic (B) bcc (C) fcc (D) equal in all

21. How many unit cell are there in 1 gram cubic crystal of NaCl ?

(A) $\frac{4\times N_A}{58.5}$ (B) $\frac{N_A}{58.5}$ (C) $\frac{N_A}{58.5\times 4}$ (D) $\frac{N_A}{58.5\times 8}$

22. The density of CaF_2 (fluorite structure) is 3.18 g/cm^3. The length of the side of the unit cell is (Ca = 40, F = 19)

(A) 253 pm (B) 344 pm (C) 546 pm (D) 273 pm

23. The coordination number of cation and anion in Fluorite CaF_2 and CsCl are respectively

(A) 8:4 and 6:3 (B) 6:3 and 4:4 (C) 8:4 and 8:8 (D) 4:2 and 2:4

24. A compound XY crystallizes in 8 : 8 lattice with unit cell edge lenght of 480 pm. If the radius of Y^- is 225 pm, then the radius of X^+ is

(A) 127.5 pm (B) 190.68 pm (C) 225 pm (D) 255 pm

25. The mass of a unit cell of CsCl corresponds to

(A) 1 Cs^+ and 1 Cl^- (B) 1 Cs^+ and 6 Cl^- (C) 4 Cs^+ and 4 Cl^- (D) 8 Cs^+ and 1 Cl^-

ALLEN Session 2019-20

26. An ionic compound AB has ZnS type structure. If the radius A^+ is 22.5 pm, then the ideal radius of B^- would be

(A) 54.35 pm (B) 100 pm (C) 145.16 pm (D) none of these

27. Edge length of M^+X^- (NaCl structure) is 7.2 Å. Assuming $M^+ - X^-$ contact along the cell edge, radius of X^- ion is ($r_{M^+} = 1.6$Å) :

(A) 2.0 Å (B) 5.6 Å (C) 2.8 Å (D) 38 Å

28. NH_4Cl crystallizes in CsCl type lattice with a unit cell edge length of 387 pm. The distance between the oppositively charged ions in the lattice is

(A) 335.1 pm (B) 83.77 pm (C) 274.46 pm (D) 137.23 pm

29. r_{Na^+} = 95 pm and r_{Cl^-} = 181 pm in NaCl (rock salt) structure. What is the shortest distance between Na^+ ions?

(A) 778.3 pm (B) 276 pm (C) 195.7 pm (D) 390.3 pm

30. AB crystallises itself as NaCl crystal. If $r_+ = \frac{2}{\sqrt{6}}$ and $r_- = \sqrt{6}$, the edge length of cube is

(A) $2\sqrt{3}$ (B) $\frac{4}{\sqrt{3}}$ (C) $\frac{8}{\sqrt{6}}$ (D) $\frac{16}{\sqrt{6}}$

31. Which of the following is the most likely to show schottky defect ?

(A) CaF_2 (B) ZnS (C) AgCl (D) CsCl

32. In the Schottky defect, in AB type ionic solids

(A) cations are missing from the lattice sites and occupy the interstitial sites

(B) equal number of cations and anions are missing

(C) anions are missing and electrons are present in their place

(D) equal number of extra cations and electrons are present in the interstitial sites

33. Choose the correct option.

(A) Two adjacent face centre atom doesn't touch each other in fcc unit cell because they are not nearest atom of face each other in fcc lattice

(B) Number of nearest Na^+ ions of another Na^+ in Na_2O crystal will be 24.

(C) Minimum distance between two cubical voids in simple cube unit cell lattice will be 'a' where 'a' is length of edge of unit cell

(D) By defects in solids, density of solids either remains constant or decreases but it can never increase.

34. The measured density of AgI is 6.94 g/cm^{-3} and the theoretical density is 5.67 g/cm^{-3}. These data indicate that solid AgI has -

(A) Schottky defect (B) Frenkel defect

(C) Interstitial impurities defect (D) Both (1) and (2)

35. Which of the following statement is **CORRECT** ?

(A) A metal can show only non- stoichiometric defects

(B) Schottky defect reduces the density of a solid due to significant increase in volume.

(C) Impurity defect always change the density.

(D) Solids having F-centres may have metal excess defect due to missing anions.

ALLEN Session 2019-20

EXERCISE : O-II

SINGLE CORRECT :

1. The only incorrect statement for the packing of identical spheres in two dimension is :

(A) For square close packing , coordination number is 4.

(B) For hexagonal close packing, coordination number is 6.

(C) There is only one void per atom in both, square and hexagonal close packing.

(D) Hexagonal close packing is more efficiently packed than square close packing.

2. Correct statement for ccp is :

(A) Each octahedral void is surrounded by 6 spheres and each sphere is surrounded by 4 octahedral voids

(B) Each octahedral void is surrounded by 6 spheres and each sphere is surrounded by 6 octahedral voids

(C) Each octahedral void is surrounded by 6 spheres and each sphere is surrounded by 8 octahedral voids

(D) Each octahedral void is surrounded by 6 spheres and each sphere is surrounded by 12 octahedral voids

3. Which of the following statements is correct in the rock-salt structure of an ionic compounds?

(A) coordination number of cation is four whereas that of anion is six.

(B) coordination number of cation is six whereas that of anion is four.

(C) coordination number of each cation and anion is four.

(D) coordination number of each cation and anion is six.

MORE THAN ONE MAY BE CORRECT :

4. Which of the following statements is/are correct :

(A) In an anti-fluorite structure, anions form FCC lattice and cations occupy all tetrahedral voids.

(B) If the radius of cations and anions are 0.2 Å and 0.95 Å , then coordination number of cation in the crystal is 4.

(C) Each sphere is surrounded by six voids in two dimensional hexagonal close packed layer.

(D) 8 Cs^+ ions occupy the second nearest neighbour locations of a Cs^+ ion in CsCl crystals.

5. Select correct statement(s)

(A) Density of crystal always increases due to substitutional impurity defect.

(B) An ion is transferred from a lattice site to an interstitial position in Frenkel defect.

(C) In AgCl, the silver ion is displaced from its lattice position to an interstitial position. Such a defect is called a frenkel defect

(D) None

6. Lead metal has a density of 11.34 g/cm^3 and crystallizes in a face–centred lattice. Choose the correct alternatives (Pb = 208 , $N_A = 6 \times 10^{23}$)

(A) the volume of one unit cell is 1.22×10^{-22} cm^3.

(B) the volume of one unit cell is 1.22×10^{-19} cm^3.

(C) the atomic radius of lead is 175 pm.

(D) the atomic radius of lead is 155.1 pm.

7. Which of the following statement(s) is/are correct ?

(A) NaCl is a 'AB' crystal lattice that can be interpreted to be made up of two individual fcc unit cells of A^+ and B^- fused together in such a manner that the corner of one unit cell becomes the edge centre of the other.

(B) In a face centred cubic unit cell, the body centre is an octahedral void.

(C) In fcc unit cell, octahedral and tetrahedral voids are equal in number.

(D) Tetrahedral voids = 2 × octahedral voids, is valid for ccp and hcp.

8. Select the correct statement (s) :

(A) CsCl mainly shows Schottky defect
(B) ZnS mainly shows Frenkel defect
(C) NaCl unit cell contain $4Na^+$ and $4Cl^-$
(D) Truncated octahedron have 24 corners.

9. Select the correct statement(s) –

(A) The ionic crystal of AgBr has Schottky defect.

(B) The unit cell having crystal parameters, $a = b \neq c$, $\alpha = \beta = 90°$, $\gamma = 120°$ is hexagonal

(C) Ionic compounds having Frenkel defect has high r^+/r^- ratio.

(D) The co-ordination number of Na^+ ion in NaCl is 6

10. Which of the following is/are true ?

(A) Ratio of nearest neighbours in simple cubic cell to next nearest neighbours in face centred is cubic cell is 1.

(B) Packing efficiency of a unit cell in which atoms are present at each corner and each edge centre is about 26 % in metallic crystal.

(C) Distance between two planes in FCC or HCP arrangement is same for a metal existing in both forms, with same atomic radius.

(D) If number of unit cell along one edge are 'x' , then total number of unit cell in cube = x^3

ASSERTION / REASON :

11. **Statement-1** : In Antifluorite structure (Li_2O), the oxide ions occupy c.c.p. (cubic close packing) and Li^+ ions, 100% tetrahedral voids.

Statement-2 : The distance of the nearest neighbours in antifluorite structure is $\frac{\sqrt{3}a}{4}$, where 'a' is the edge length of the cube

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

12. **Statement-1 :** In FCC unit cell, packing efficiency is more when all tetrahedral voids are filled with spheres of maximum possible size as compared with packing efficiency when all octahedral voids are filled in similar way.

Statement-2 : Tetrahedral voids are more in the number than octahederal voids in FCC.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\02_Solid State\Exercise.p65

13. **Statement-1**: In diamond , carbon atoms occupy alternate tetrahedral voids in the FCC lattice formed by the carbon atoms.

कथन-2: In diamond, packing fraction is more than 74%.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

14. **Statement-1 :** Due to Frenkel defect, there is no effect on the density of the crystalline solid.

Statement-2 : In Frenkel defect, no cation or anion leaves the crystal.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

15. **Statement-1 :** Conductivity of silicon increased by doping it with group 15 element.

Statement-2 : Doping means introduction of small amount of impurities like P or As into pure silicon crytstal.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

COMPREHENSION :

Paragraph for Q.16 & Q.17

Solid balls of radius 17.32 cm crystallises in bcc pattern. During one such crystallisation, some oxygen gas is trapped. This trapped oxygen at 640K creates pressure of 5 atm.

Assume :

(i) BCC arrangement is not disturbed due to trapping of gas.

(ii) Gas is uniformly distributed inside unit cell

[Take R = 0.08 atm-litre/mole-K , $N_A = 6 \times 10^{23}$, Mass of a solid ball = 64 gms]

16. Number of oxygen molecules present in an unit cell is -

(A) 2.4×10^{24} (B) 1.2×10^{24} (C) 6×10^{23} (D) 3×10^{23}

17. Calculate percentage increase in density due to trapping of gas

(A) 16.67 % (B) 33.33 % (C) 100% (D) 50%

Paragraph for (Que. 18 to 21)

Calcium crystallizes in a cubic unit cell with density 3.2 g/cc. Edge-length of the unit cell is 437 pm.

18. The type of unit cell is

(A) Simple cubic (B) BCC (C) FCC (D) Edge-centred

19. The nearest neighbour distance is

(A) 154.5 pm (B) 309 pm (C) 218.5 pm (D) 260 pm

20. The number of nearest neighbours of a Ca atom are

(A) 4 (B) 6 (C) 8 (D) 12

21. If the metal is melted, density of the molten metal was found to be 3 g/cc. What will be the percentage of empty space in the molten metal?

(A) 31% (B) 36% (C) 28% (D) 49%

***TABLE TYPE* COMPREHENSION :**

	Column-I		Column-II		Column-III
(A)	NaCl (Rock salt) structure	(i)	Cation - FCC Anion - Tetrahedral void	(I)	All tetrahedral voids are occupied
(B)	CsCl structure	(ii)	Anion - FCC Cation - Tetrahedral voids	(II)	All octahedral voids are occupied
(C)	ZnS (zinc blende) structure	(iii)	Anion - SC Cation - Cubic voids	(III)	50 % of tetrahedral voids are occupied
(D)	CaF_2 (fluorite) structure	(iv)	Anion - FCC Cation - Octahedral void	(IV)	All octahedral voids are empty

22. Which of the following is correct match ?

(A) A, i, I (B) A, ii, IV (C) A, iv, II D) A, iv, IV

23. Which of the following is incorrect match ?

(A) B, iii, I (B) C, ii, III (C) D, i, I D) D, i, IV

24. Which of the following is correct match ?

(A) D, ii, I (B) B, iv, IV (C) D, iii, I D) C, ii, IV

MATCH THE COLUMN :

25. Match the column

	Column I		Column II
(A)	Tetragonal and Hexagonal	(P)	are two crystal systems
(B)	Cubic and Rhombohedral	(Q)	$a = b \neq c$
(C)	Monoclinic and Triclinic	(R)	$a \neq b \neq c$
(D)	Cubic and Hexagonal	(S)	$a = b = c$

26. Match the column:

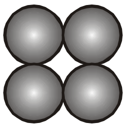
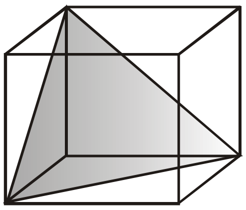
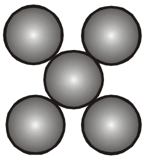
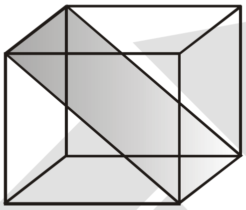
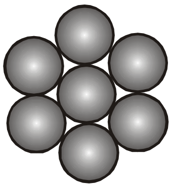
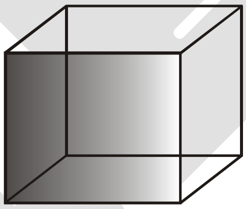
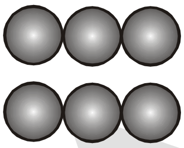
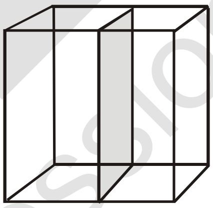
	Column I		Column II
(A)	Rock salt structure	(P)	Co-ordination number of cation is 4
(B)	Zinc Blend structure	(Q)	$\frac{\sqrt{3}a}{4} = r_+ + r_-$
(C)	Flourite structure	(R)	Co-ordination number of cation and anion are same
		(S)	Distance between two nearest anion is $\frac{a}{\sqrt{2}}$

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\02_Solid State\Exercise.p65

MATCHING LIST TYPE :

27. Match the column

Column I **(Arrangement of the atoms/ions)**	**Column II** **(Planes in fcc lattice)**
(P) 	(1) 
(Q) 	(2) 
(R) 	(3) 
(S) 	(4) 

Code :

	P	Q	R	S
(A)	4	3	1	2
(B)	4	3	2	1
(C)	3	2	1	4
(D)	1	2	4	3

ALLEN Session 2019-20

28. **Column I** **Column II**

[Distance in terms of edge length of cube (a)]

	Column I		Column II
(P)	0.866 a	(1)	Shortest distance between cation & anion in CsCl structure.
(Q)	0.707 a	(2)	Shortest distance between two cation in CaF_2 structure.
(R)	0.433 a	(3)	Shortest distance between carbon atoms in diamond.
(S)	a	(4)	shortest distance between next nearest cations in rock salt structrue.

Code :

	P	Q	R	S
(A)	4	3	1	2
(B)	1	2	3	4
(C)	3	2	1	4
(D)	1	2	4	3

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\02_Solid State\Exercise.p65

EXERCISE : J-MAIN

1. The no. of atoms per unit cell in B.C.C. & F.C.C. is respectively : **[AIEEE-02]**

(1) 8, 10 (2) 2, 4 (3) 1, 2 (4) 1, 3

2. How many unit cells are present in a cube-shaped ideal crystal of NaCl of mass 1.00g ? **[AIEEE-03]**

(1) 1.28×10^{21} unit cells (2) 1.71×10^{21} unit cells

(3) 2.57×10^{21} unit cells (4) 5.14×10^{21} unit cells

3. What type of crystal defect is indicated in the diagram below ? **[AIEEE-04]**

Na^+	Cl^-	Na^+	Cl^-	Na^+	Cl^-
Cl^-		Cl^-	Na^+		Na^+
Na^+	Cl^-		Cl^-	Na^+	Cl^-
Cl^-	Na	$^+Cl^-$	Na^+		Na^+

(1) Frenkel defect (2) Schottky defect

(3) Interstitial defect (4) Frenkel and Schottky defects

4. An ionic compound has a unit cell consisting of A ions at the corners of a cube and B ions on the centres of the faces of the cube. The empirical formula of this compound would be- **[AIEEE-05]**

(1) A_2B (2) AB (3) A_3B (4) AB_3

5. Lattice energy of an ionic compound depends upon - **[AIEEE-05]**

(1) Size of the ion only (2) Charge on the ion only

(3) Charge on the ion and size of the ion (4) Packing of ions only

6. Total volume of atoms present in a face-centred cubic unit cell of a metal is (r is atomic radius) : **[AIEEE-06]**

(1) $\frac{24}{3}\pi r^3$ (2) $\frac{12}{3}\pi r^3$ (3) $\frac{16}{3}\pi r^3$ (4) $\frac{20}{3}\pi r^3$

7. In a compound, atoms of element Y form ccp lattice and those of element X occupy $2/3^{rd}$ of tetrahedral voids. The formula of the compound will be - **[AIEEE-08]**

(1) X_4Y_3 (2) X_2Y_3 (3) X_2Y (4) X_3Y_4

8. The edge length of a face centred cubic cell of an ionic substance is 508 pm. If the radius of the cation is 110 pm, the radius of the anion is :- **[AIEEE-10]**

(1) 144 pm (2) 288 pm (3) 398 pm (4) 618 pm

9. Percentages of free space in cubic close packed structure and in body centred packed structure are respectively :- **[AIEEE-10]**

(1) 48% and 26% (2) 30% and 26% (3) 26% and 32% (4) 32% and 48%

ALLEN Session 2019-20
node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\02_Solid State\Exercise.p65

10. The radius of a calcium ion is 94 pm and of the oxide ion is 146 pm. The possible crystal structure of calcium oxide will be :- **[Jee-Main (online)-12]**

(1) Octahedral (2) Tetrahedral (3) Pyramidal (4) Trigonal

11. Ammonium chloride crystallizes in a body centred cubic lattice with edge length of unit cell of 390 pm. If the size of chloride ion is180 pm, the size of ammonium ion would be:

[Jee-Main (online)-12]

(1) 158 pm (2) 174 pm (3) 142 pm (4) 126 pm

12. A solid has 'bcc' structure. If the distance of nearest approach between two atoms is 1.73 Å, the edge length of the cell is :- **[Jee-Main (online)-12]**

(1) 314.20 pm (2) 216 pm (3) 200 pm (4) 1.41 pm

13. Among the following the incorrect statement is :- **[Jee-Main (online)-12]**

(1) Density of crystals remains unaffected due to Frenkel defect

(2) In BCC unit cell the void space is 32%

(3) Electrical conductivity of semiconductors and metals increases with increase in temperature

(4) Density of crystals decreases due to Schottky defect

14. Copper crystallises in fcc with a unit cell edge length of 361pm. What is the radius of copper atom?

[AIEEE-2011]

(1) 181pm (2) 128pm (3) 157pm (4) 108pm

15 Lithium forms body centred cubic structure. The length of the side of its unit cell is 351 pm. Atomic radius of the lithium will be :- **[Jee-Main (offline)-12]**

(1) 152 pm (2) 75 pm (3) 300 pm (4) 240 pm

16. In a face centred cubic lattice, atoms of A form the corner points and atoms of B form the face centred points. If two atoms of A are missing from the corner points, the formula of the ionic compound is

[Jee-Main (online)-13]

(1) AB_2 (2) AB_3 (3) AB_4 (4) A_2B_5

17. Which one of the following statements about packing in solids is **incorrect** ?

(1) Void space in ccp mode of packing is 26% **[Jee-Main (online)-13]**

(2) Coordination number in hcp mode of packing is 12

(3) Void space in hcp mode of packing is 32%

(4) Coordination number in bcc mode of packing is 8

18. An element having an atomic radius of 0.14 nm crystallizes in an fcc unit cell. What is the length of a side of the cell ? **[Jee-Main (online)-13]**

(1) 0.96 nm (2) 0.4 nm (3) 0.24 nm (4) 0.56 nm

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\02_Solid State\Exercise.p65

19. Experimentally it was found that a metal oxide has formula $M_{0.98}O$. Metal M, is present as M^{2+} and M^{3+} in its oxide. Fraction of the metal which exists as M^{3+} would be :- **[Jee-Main (offline)-13]**

(1) 7.01% (2) 4.08% (3) 6.05% (4) 5.08

20. The total number of octahedral void(s) per atom present in a cubic close packed structure is :- **[Jee-Main (online)-14]**

(1) 1 (2) 2 (3) 3 (4) 4

21. In a monoclinic unit cell, the relation of sides and angles are respectively **[Jee-Main (online)-14]**

(1) $a \neq b \neq c$ and $\alpha \neq \beta \neq \gamma \neq 90^\circ$ (2) $a \neq b \neq c$ and $\beta = \gamma = 90^\circ \neq \alpha$

(3) $a = b \neq c$ and $\alpha = \beta = \gamma = 90^\circ$ (4) $a \neq b \neq c$ and $\alpha = \beta = \gamma = 90^\circ$

22. The appearance of colour in solid alkali metal halides is generally due to : **[Jee-Main (online)-14]**

(1) Frenkel defect (2) F-centres (3) Schottky defect (4) Interstitial position

23. In a face centred cubic lattice atoms A are at the corner points and atoms B at the face centred points. If atom B is missing from one of the face centred points, the formula of the ionic compound is : **[AIEEE-2011, Jee-Main (online)-14]**

(1) AB_2 (2) A_2B_3 (3) A_5B_2 (4) A_2B_5

24. CsCl crystallises in body centred cubic lattice. if 'a' is its edge length then which of the following expression is correct : **[Jee-Main (offline)-14]**

(1) $r_{Cs^+} + r_{Cl^-} = \frac{\sqrt{3}}{2}a$ (2) $r_{Cs^+} + r_{Cl^-} = \sqrt{3}a$ (3) $r_{Cs^+} + r_{Cl^-} = 3a$ (4) $r_{Cs^+} + r_{Cl^-} = \frac{3a}{2}$

25. Sodium metal crystallizes in a body centred cubic lattice with a unit cell edge of 4.29Å. The radius of sodium atom is approximately :- **[Jee-Main (offline)-15]**

(1) 5.72Å (2) 0.93Å (3) 1.86Å (4) 3.022Å

26. Which of the following compounds is metallic and ferromagnetic ? **[Jee-Main (offline)-16]**

(1) MnO_2 (2) TiO_2 (3) CrO_2 (4) VO_2

27. A metal crystallises in a face centred cubic structure. If the edge length of its unit cell is 'a', the closest approach between two atoms in metallic crystal will be :- **[Jee-Main (offline)-17]**

(1) 2a (2) $2\sqrt{2}\,a$ (3) $\sqrt{2}\,a$ (4) $\frac{a}{\sqrt{2}}$

28. Which type of effect 'defect' has the presence of cations in the interstitial sites - **[Jee-Main (offline)-18]**

(1) Vacancy defect (2) Frenkel defect

(3) Metal deficiency defect (4) Schottky defect

29. All of the following share the same crystal structure except :- **[Jee-Main (online)-18]**

(1) RbCl (2) CsCl (3) LiCl (4) NaCl

30. Which of the following arrangements shows the schematic alignment of magnetic moments of antiferromagnetic substance ? **[Jee-Main (online)-18]**

(1) $\uparrow\ \downarrow\ \downarrow\ \downarrow\ \downarrow\ \uparrow$ (2) $\uparrow\ \uparrow\ \uparrow\ \uparrow\ \uparrow\ \uparrow$

(3) $\uparrow\ \uparrow\ \downarrow\ \uparrow\ \uparrow\ \downarrow$ (4) $\uparrow\ \downarrow\ \uparrow\ \downarrow\ \uparrow\ \downarrow$

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\02_Solid State\Exercise.p65

EXERCISE : J-ADVANCED

1. A metal crystallises into two cubic phases, FCC and BCC whose unit cell lengths are 3.5 and 3.0 Å respectively. Calculate the ratio of densities of FCC and BCC. **[JEE-1999]**

2. The coordination number of a metal crystallising in a hcp structure is **[JEE-2000]**

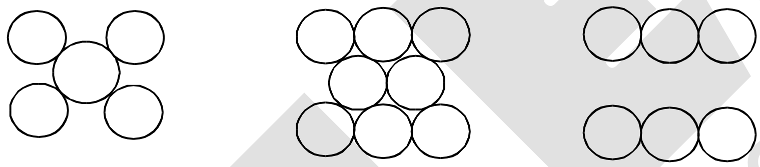
(A) 12 (B) 4 (C) 8 (D) 6

3. In any ionic solid [MX] with schottky defects , the number of positive and negative ions are same. **[T/F]** **[JEE-2000]**

4. In a solid "AB" having NaCl structure "A" atoms occupy the corners of the cubic unit cell. If all the face-centred atoms along one of the axes are removed, then the resultant stoichiometry of the solid is

(A) AB_2 (B) A_2B (C) A_4B_3 (D) A_3B_4 **[JEE-2000]**

5. The figures given below show the location of atoms in three crystallographic planes in FCC lattice. Draw the unit cell for the corresponding structure and identify these planes in your diagram. **[JEE-2000]**



6. A substance A_xB_y crystallises in a FCC lattice in which atoms "A" occupy each corner of the cube and atoms "B" occupy the centres of each face of the cube. Identify the correct composition of the substance A_xB_y.

(A) AB_3 (B) A_4B_3

(C) A_3B (D) composition cannot be specified **[JEE-2002]**

7. Marbles of diameter 10 mm each are to be arranged on a flat surface so that their centres lie within the area enclosed by four lines of length each 40 mm. Sketch the arrangement that will give the maximum number of marbles per unit area, that can be enclosed in this manner and deduce the expression to calculate it. **[JEE 2003]**

8. (i) AB crystallizes in a rock salt structure with A : B = 1 : 1. The shortest distance between A and B is $Y^{1/3}$ nm. The formula mass of AB is 6.023 Y amu where Y is any arbitrary constant. Find the density in kg m^{-3}.

(ii) If measured density is 20 kg m^{-3}. Identify the type of point defect. **[JEE–2004]**

9. Which of the following FCC structure contains cations in alternate tetrahedral voids?

(A) NaCl (B) ZnS (C) Na_2O (D) CaF_2 **[JEE 2005]**

10. An element crystallises in FCC lattice having edge length 400 pm. Calculate the maximum diameter which can be placed in interstital sites without disturbing the structure. **[JEE 2005]**

ALLEN Session 2019-20

11. The edge length of unit cell of a metal having atomic weight 75 g/mol is 5 Å which crystallizes in cubic lattice. If the density is 2 g/cc then find the radius of metal atom. ($N_A = 6 \times 10^{23}$). Give the answer in pm.
[JEE 2006]

12. Match the crystal system / unit cells mentioned in Column I with their characteristic features mentioned in Column II. Indicate your answer by darkening the appropriate bubbles of the 4×4 matrix given in the ORS.

Column I	**Column II**
(A) simple cubic and face-centred cubic	(P) have these cell parameters $a = b = c$ and $\alpha = \beta = \gamma$
(B) cubic and rhombohedral	(Q) are two crystal systems
(C) cubic and tetragonal	(R) have only two crystallographic angles of 90°
(D) hexagonal and monoclinic	(S) belong to same crystal system.

[JEE 2007]

Paragraph for Question No.13 to 15

In hexagonal systems of crystals, a frequently encountered arrangement of atoms is described as a hexagonal prism. Here, the top and bottom of the cell are regular hexagons and three atoms are sandwiched in between them. A space-filling model of this structure, called hexagonal close-packed (HCP), is constituted of a sphere on a flat surface surrounded in the same plane by six identical spheres as closely as possible. Three spheres are then placed over the first layer so that they touch each other and represent the second layer. Each one of these three spheres touches three spheres of the bottom layer. Finally, the second layer is covered with a third layer that is identical to the bottom layer in relative position. Assume radius of every sphere to be 'r'.

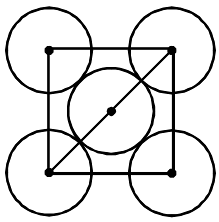
13. The number of atoms in this HCP unit cells is **[JEE 2008]**
(A) 4 (B) 6 (C) 12 (D) 17

14. The volume of this HCP unit cell is **[JEE 2008]**
(A) $24\sqrt{2}\,r^3$ (B) $16\sqrt{2}\,r^3$ (C) $12\sqrt{2}\,r^3$ (D) $\frac{64}{3\sqrt{3}}\,r^3$

15. The empty space in this HCP unit cell is **[JEE 2008]**
(A) 74% (B) 47.6 % (C) 32% (D) 26%

16. The correct statement(s) regarding defects in solid is (are) **[JEE 2009]**
(A) Frenkel defect is usually favoured by a very small difference in the sizes of cation and anion.
(B) Frenkel defect is a dislocation defect
(C) Trapping of an electron in the lattice leads to the formation of F-center.
(D) Schottky defects have no effect on the physical properties of solids.

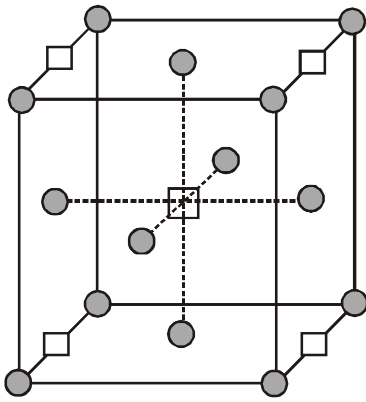
17. The packing effeciency of the two-dimensional square unit cell shown below is **[JEE-2010]**



(A) 39.27% (B) 68.02% (C) 74.05% (D) 78.54%

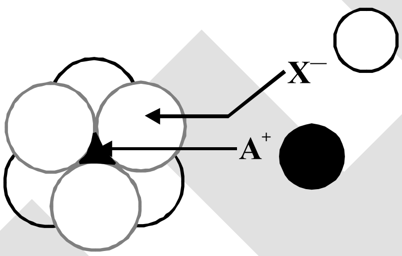
18. The number of hexagonal faces that present in a truncated octahedron is. **[JEE-2011]**

19. A compound M_pX_q has cubic close packing (ccp) arrangement of X. Its unit cell structure is shown below. The empirical formula of the compound is : **[JEE-2012]**



(A) MX (B) MX_2 (C) M_2X (D) M_5X_{14}

20. The arrangement of X^- ions around A^+ ion in solid AX is given in the figure (not drawn to scale). If the radius of X^- is 250 pm, the radius of A^+ is - **[JEE-2013]**



(A) 104 pm (B) 125 pm (C) 183 pm (D) 57 pm

21. If the unit cell of a mineral has cubic close packed (ccp) array of oxygen atoms with m fraction of octahedral holes occupied by aluminium ions and n fraction of tetrahedral holes occupied by magnesium ions m and n respectively, are - **[JEE-2015]**

(A) $\frac{1}{2},\frac{1}{8}$ (B) $1,\frac{1}{4}$ (C) $\frac{1}{2},\frac{1}{2}$ (D) $\frac{1}{4},\frac{1}{8}$

22. The **CORRECT** statement(s) for cubic close packed (ccp) three dimensional structure is (are)

(A) The number of the nearest neighbours of an atom present in the topmost layer is 12

(B) The efficiency of atom packing is 74% **[JEE-2016]**

(C) The number of octahedral and tetrahedral voids per atom are 1 and 2, respectively

(D) The unit cell edge length is $2\sqrt{2}$ times the radius of the atom

ALLEN Session 2019-20

23. A crystalline solid of a pure substance has a face-centred cubic structure with a cell edge of 400 pm. If the density of the substance in the crystal is $8 g\ cm^{-3}$, then the number of atoms present in 256g of the crystal is $N \times 10^{24}$. The value of N is **[JEE-2017]**

24. Consider an ionic solid MX with NaCl structure. Construct a new structure (Z) whose unit cell is constructed from the unit cell of MX following the sequential instructions given below. Neglect the charge balance. **[JEE-2018]**

(i) Remove all the anions (X) except the central one

(ii) Replace all the face centered cations (M) by anions (X)

(iii) Remove all the corner cations (M)

(iv) Replace the central anion (X) with cation (M)

The value of $\left(\dfrac{\text{number of anions}}{\text{number of cations}}\right)$ in Z is____.

ALLEN Session 2019-20

ANSWER KEY

S-I

1. Ans. (a) 300 pm, (b) 255 pm, (c) 210 pm
2. Ans.438 pm, 219 pm
3. Ans.5.83 g cm^{-3}
4. Ans.($\sqrt{2}:1$)
5. Ans. (0.9 gm/cm^{-3})
6. Ans. A,B
7. Ans.(64)
8. Ans.(12)
9. Ans.(6)
10. Ans (2)
11. Ans.A_2BC
12. Ans. (4)
13. Ans.(0.293)
14. Ans. (6×10^{17})
15. Ans. (2) $\dfrac{\frac{a}{\sqrt{2}} \times \sqrt{2}}{a/2} = 2$
16. Ans.(2.07 pm)
17. Ans.(6)
18. Ans. (60%)
19. Ans. 41.67 g cm^{-3}
20. Ans.0.60
21. Ans.4, 6, 8
22. Ans.(6)
23. Ans.$ZnAl_2O_4$
24. Ans.267 pm, 534 pm, 377.6 pm
25. Ans.346.4 pm
26. Ans.4
27. Ans.a =600 pm, V=2.16×10^{-22} cm^3
28. Ans.1.81Å
29. Ans.4.34 Å
30. Ans.(5 gm/cm^3)
31. Ans.15.05 %
32. Ans.6.0×10^{18}
33. Ans.10
34. Ans. 12.5%
35. Ans. 3.0 gm/cm^3

S-II

1. Ans.(2)
2. Ans.(5)
3. Ans.Ans.8
4. Ans.60%, +4
5 . Ans.5×10^{24}
6. Ans.0.72
7. Ans. 4
8 Ans. (i) FCC (ii) 0.116 g/cc
9. Ans. (a) 1.143, (b) 1.2
10. Ans.(a) MnF_3, (b) 6, (c) 4.02Å
11. Ans. (a) 4.5 Å, (b) 5.2 Å, (c) 8, (d) 6, (e) 0.929 g/cm^3
12 Ans. 96%
13. Ans (4)
14. Ans. Mg_2SiO_4, Fe_2SiO_4, 57.14%

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\02_Solid State\Exercise.p65

O-I

1.	Ans.(A)	2.	Ans.(D)	3.	Ans.(A)
4.	Ans.(D)	5.	Ans.(C)	6.	Ans.(C)
7.	Ans. (C)	8.	Ans (B)	9.	Ans (D)
10.	Ans (B)	11.	Ans.(B)	12.	Ans.(C)
13.	Ans.(C)	14.	Ans.(C)	15.	Ans.(C)
16.	Ans.(A)	17.	Ans.(B)	18.	Ans (D)
19.	Ans (C)	20.	Ans (D)	21.	Ans.(C)
22.	Ans.(C)	23.	Ans.(C)	24.	Ans.(B)
25.	Ans.(A)	26.	Ans.(B)	27.	Ans.(A)
28.	Ans.(A)	29.	Ans.(D)	30.	Ans.(D)
31.	Ans.(D)	32.	Ans.(B)	33.	Ans(C)
34.	Ans (C)	35.	Ans.(D)		

O-II

1.	Ans. (C)	2.	Ans.(B)	3.	Ans.(D)
4.	Ans.(A,C)	5.	Ans.(B,C)	6.	Ans.(A,C)
7.	Ans.(A,B,D)	8.	Ans.(A,B,C,D)	9.	Ans.(A,B,D)
10.	Ans.(A,B,C,D)	11.	Ans.(B)	12.	Ans.(D)
13.	Ans.(C)	14	Ans.(A)	15.	Ans.(A)
16.	Ans.(B)	17.	Ans.(D)	18.	Ans.(C)
19.	Ans.(B)	20.	Ans.(D)	21.	Ans.(A)
22.	Ans.(C)	23.	Ans.(A)	24.	Ans.(D)

25. Ans.(A) $\rightarrow$ P, Q ; (B) $\rightarrow$ P,S ; (C) $\rightarrow$ P,R ; (D) $\rightarrow$ P

26. Ans.(A) $\rightarrow$ R,S ; (B) $\rightarrow$ P,Q,R,S ; (C) $\rightarrow$ Q

27.	Ans.Ans.(A)	28.	Ans.(B)

J-MAIN

1.	Ans.(2)	2.	Ans.(3)	3.	Ans.(2)
4.	Ans.(4)	5.	Ans.(3)	6.	Ans.(3)
7.	Ans.(1)	8.	Ans.(1)	9.	Ans.(3)
10.	Ans.(1)	11.	Ans.(1)	12.	Ans.(3)
13.	Ans.(3)	14.	Ans.(2)	15	Ans.(1)
16.	Ans.(3)	17.	Ans.(3)	18.	Ans.(2)
19.	Ans.(2)	20.	Ans.(1)	21.	Ans.(2)
22.	Ans.(2)	23.	Ans.(4)	24.	Ans.(1)
25.	Ans.(3)	26.	Ans..(3)	27	Ans. (4)
28.	Ans. (2)	29.	Ans. (2)	30.	Ans. (4)

ALLEN Session 2019-20

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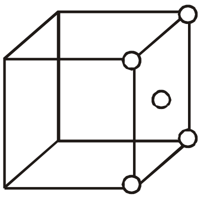
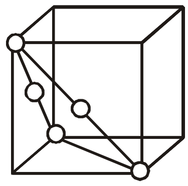
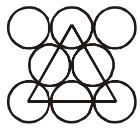
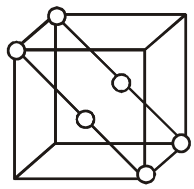
1. Ans.1.259
2. Ans.(A)
3. Ans.True
4. Ans.(D)
5. Ans.



fcc plan







6. Ans.(A)
7. Ans.18
8. Ans.(i) = 5 kg m^{-3}
9. Ans.(B)
10. Ans. 117.1 pm
11. Ans. 216.5 pm
12. Ans. (A) P, S ; (B) -P,Q ; (C) - Q ; (D) - Q,R
13. Ans. (B)
14. Ans. (A)
15. Ans. (D)
16. Ans. (B,C)
17. Ans. (D)
18. Ans. (8)
19. Ans.(B)
20. Ans. (A)
21. Ans. (A)
22. Ans.(B,C,D)
23. Ans.(2)
24. Ans.(3)

ALLEN Session 2019-20

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CHEMICAL EQUILIBRIUM

KEY CONCEPTS

TYPES OF REACTION:

(i) Reversible (ii) Irreversible

STATE OF EQUILIBRIUM

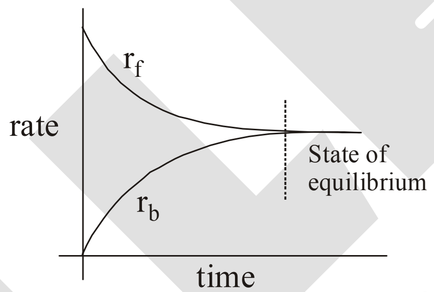
Most of the chemical reaction do not go to completion in a closed system and attain a state of equilibrium.

There are two approaches to understand nature of equilibrium. The One stems from thermodynaics. Equilibrium criteria is explained on the basis of thermodynamic function like ΔH (change in enthalpy), ΔS (change in entropy) and ΔG (change in Gibb's function). At equilibrium macroscopic properties of the system like concentration, pressure ect. become constant at constant temperature.

Other approach comes from kinetics as developed by Guldberg and Waage (1863). Equilibrium is said to have reached in a physical or chemical system when rate of forward and reverse processes are equal.

At equilibrium

Rate of forward reaction = Rate of backward reaction.



Example:

$N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)$

Starting with pure H_2 and N_2 as reaction proceeds in forward direction. Ammonia is formed. At initially conc. of H_2 and N_2 drops and attain a steady value at equilibrium. On the others hand conc. of NH_3 increases and at equilibrium attains a constant value.

Concentration time graphs for

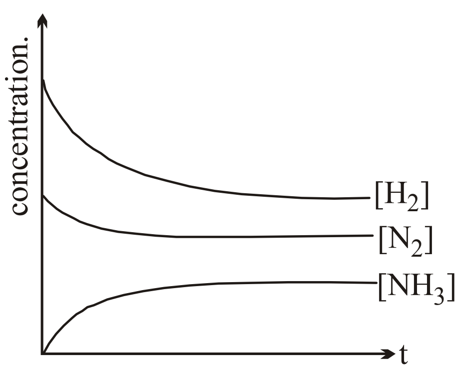


fig (1)

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IMPORTANT CHARACTERISTIC OF EQUILIBRIUM

(i) Equilibrium is possible only in closed system.

(ii) The rate of forward process at equilibrium is equal to rate of backward process.

(iii) All measurable properties of system remain constant over time.State of chemical equilibrium is characterised by equilibrium constant. Equilibrium constant have constant value at a given temperature.

(iv) At equilibrium entropy of universe is maxmized.

(v) Both, *Kinetic* and *Thermodynamics* theories can be invoked to understand the extent to which a reaction proceed to forward direction. *e.g.* If extent of reaction is too large for forward direction (equilibrium is tilted heavily to forward direction) than

(a) Specific rate of forward reaction >>> specific rate of backward reaction

(b) Gibb's function of product is vary small as compared to Gibb's function of reactant.

LAW OF CHEMICAL EQUILIBRIUM OR MASS ACTION **(By Guldberg and Waage)**

A general equation for a reversible reaction may be written

$$mA + nB \rightleftharpoons xC + yD$$

rate of reaction in inforward direction$(r_f) \propto [A]^m [B]^m$; $r_f = k_f [A]^m [B]^m$

rate of reaction in inforward direction$(r_b) \propto [C]^x [D]^y$; $r_b = k_b [C]^x [D]^y$

At equilibrium, $r_f = r_b$

$$K_{eq} \text{ or } K_c = \frac{[C]^x[D]^y}{[A]^m[B]^m} \qquad \text{where } K_c = k_f / k_b$$

where we use [ ] to indicate "molar concentration."

***Note* :** We should calculate the value of K from the activities of the reactants and products rather than from their concentrations. However, the activity of a dilute solute is usefully approximated by its molar concentration, so we will use molar concentrations. However, for gases we can use molar concentrations of gases and partial pressure in our equilibrium calculations, The activity of a pure solid or pure liquid is constant , and the activity of a solvent in a dilute solution is also constant.Thus these species (solids, liquids, and solvents) are omitted from reactions quotients and equilibrium calculations.

UNIT OF EQUILIBRIUM CONSTANT $(K_C, K_C^o, K_P \& K_P^o)$

We have already noted that the value of an equilibrium constant has meaning only when we give the corresponding balanced chemical equation. Its value changes for the new equation obtained by multiplying or dividing the original equation by a number. The value for equilibrium constant, K_C is calculate substituting the concentration in mol/L and for K_P by substituting partial pressure in Pa, kPa, etc. in atm. Thus, units of equilibrium constant will turn out to be units based on molarity or pressure, unless the sum of the exponents in the numerator is equal to the sum of the exponents in the denominator. Thus for the reaction:

$H_2(g) + I_2(g) \rightleftharpoons 2HI$, K_C and K_P do not have any unit

$N_2(g) + 3H_2(g) \rightleftharpoons 2\,NH_3$, K_C has unit $(mol/L)^{-2}$ and K_P has unit bar^{-2} or

$N_2O_4\,(g) \rightleftharpoons 2NO_2$, K_C has unit mol / L and K_P has unit bar

However, these days we express equilibrium constants in dimensionless quantities by deviding concentration by 1 M and partial pressure by 1 bar.

* **Relationship between K_P & K_C**

$K_P = K_C\,(RT)^{\Delta ng}$

ALLEN Session 2019-20

TYPE OF EQUILIBRIUM

(a) *Homogeneous chemical equilibria*

A homogeneous equilibrium is equilibrium with in a single phase i.e. when physical state of all the reactants and product are same.

(ii) *Liquid phase homogeneous equilibrium*

Example : (i) $I_2\,(aq) + I^-\,(aq) \rightleftharpoons I_3^-\,(aq)$

Eq. constants is $K = \dfrac{[I_3^-(aq)]}{[I_2(aq)][I^-(aq)]}$

(iii) *Homogeneous equilibria in gases*

Example : (i) $C_2H_6(g) \rightleftharpoons C_2H_4(g) + H_2(g)$

Eq. constants is

$K_C = \dfrac{[C_2H_4O)][H_2]}{[C_2H_6(g)]}$ $\qquad K_P = \dfrac{[P_{C_2H_4}][P_{H_2}]}{[P_{C_2H_6}]}$

[ ] represents concentration in mol/ litre at equilibrium $\qquad P_{C_2H_4}$ & other are partial pressure at equilibrium

(ii) $3O_2(g) \rightleftharpoons 2O_3(g)$

Eq. constants is

$K_C = \dfrac{[O_3]^2}{[O_2]^3}$ $\qquad K_P = \dfrac{P_{O_3}^2}{P_{O2}^3}$

(b) *Hetrogeneous equilibria*

If reactants and product are found in two or more phases, the equilibria describing them is called hetrogeneous equilibrium.

Example: $CaO(s) + CO_2(g) \rightleftharpoons CaCO_3(s)$

Eq. constants is $K_P = \dfrac{1}{P_{CO_2}}$ $\qquad K_C = \dfrac{1}{[CO_2(g)]}$

CHARACTERSTICS OF EQUILIBRIUM CONSTANT

(i) The expression for equilibrium constant, K is applicable only when concentrations of the reactants and products have attained their equilibrium values and do not change with time.

(ii) The value of equilibrium constant is independent of initial concentration of the reactants and product.

(iii) Equilibrium constant has one unique value for a particular reaction represented by a balanced equation at a given temperature.

(iv) The equilibrium constant for the reverse reaction is equal to the inverse of the equilibrium constant for the forward reaction.

$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

$K_p = \dfrac{P_{HI^2}}{P_{H_2}.P_{I_2}}$; $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$

$K_P' = \dfrac{1}{K_P}$

(v) The equilibrium constant K, for a reaction is related to the equilibrium constant of the corresponding reaction whose equation is obtained by multiplying or dividing the equation.

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g) \quad ; \qquad \frac{1}{2}H_2(g) + \frac{1}{2}I_2(g) \rightleftharpoons HI(g)$$

$$K_P'' = \frac{P_{HI}}{P_{H_2}^{1/2}.P_{I_2}^{1/2}} = \sqrt{K_p}$$

(vi) If reaction is performed is steps

$A \rightleftharpoons B$; overall reaction

Step-1 : $A(g) \rightleftharpoons C(g) + D(g)$ $\quad K_{p_1}$

Step-2 : $C(g) \rightleftharpoons E(g)$ $\quad K_{p_2}$

Step-3 : $D(g) + E(g) \rightleftharpoons B(g)$ $\quad K_{p_3}$

$A(g) \rightleftharpoons B(g)$ $\quad K_p$ then $K_p = K_{p_1} . K_{p_2} . K_{p_3}$

APPLICATION OF EQULIBRIUM CONSTANT.

Now we will consider some applications of equilibrium constant and use it to answer question like:

(i) predicting the extent of a reaction on the basis of its magnitude.

(ii) predicting the direction of the reaction, and

(iii) calculating equilibrium concentration.

(i) Predicting the extent of a reaction

The magnitude of equilibrium constant is very useful especially in reactions of industrial importance. An equilibrium constant tells us whether we can expect a reaction mixture to contain a high or low concentration of product(s) at equilibrium. (It is important to note that an equilibrium constant tells us nothing about the rate at which equilibrium is reached). In the expression of K_C or K_P, product of the concentrations of products is written in numerator and the product of the concentrations of reactants is written in denominator. High value of equilibrium constant indicates that product(s) concentration is high and its low value indicates that concentration of the product(s) in equilibrium mixture is low.

For reaction, $H_2(g) + Br_2(g) \rightleftharpoons 2HBr(g)$, the value of

$$K_P = \frac{(P_{HBr})^2}{(P_{H_2})(P_{Br_2})} = 5.4 \times 10^{18}$$

The large value of equilibrium constant indicates that concentration of the product, HBr is very high and reaction goes nearly to completion.

Similarly, equilibrium constant for the reaction $H_2(g) + Cl_2(g) \rightleftharpoons 2HCl(g)$ aty 300 K is very high and reaction goes virtually to completion.

$$K_C = \frac{[HCl]^2}{[H_2][Cl_2]} = 4.0 \times 10^{31}$$

Thus, large value of K_P or K_C (larger than about 10^3), favour the products strongly. For intermedicate values of K (approximately in the range of 10^{-3} to 10^3), the concentrations of reactants and products are comparable. Small values of equilibrium constant (smaller than 10^{-3}), favour the reactants strongly.

At 298 K for reaction, $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$

$$K_C = \frac{[NO]^2}{[N_2][O_2]} = 4.8 \times 10^{-31}$$

The very small value of K_C implies that reactants N_2 and O_2 will be the predominant species in the reaction mixture at equilibrium.

ALLEN Session 2019-20
node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module#Ideal gas, Solid state & Chemical Eq\Eng\03_Chemical Equilibrium\Theory & Ex..p65

(ii) Predicting the direction of the reaction.

The equilibrium constant is also used to find in which direction an rabidity reaction mixture of reactants and products will proceed. For this purpose, we calculate the reaction quotient, Q. The reaction quotient is defined in the same way as the equilibrium constant (with molar concentrations to give Q_C, or with partial pressure to give Q_P) at any stage of reaction. For a general reaction:

$$aA + bB \rightleftharpoons cC + dD$$

$$Q_C = \frac{[C]^c[D]^d}{[A]^a[B]^b}$$

Then, if $Q_C > K_c$, the reaction will proceed in the direction of reactants (reverse reaction).

if $Q_C < K_c$, the reaction will move in the direction of the products

if $Q_C = K_c$, the reaction mixture is already at equilibrium.

In the reaction, $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$, if the molar concentrations of H_2 , I_2 and HI are 0.1 mol L^{-1} respectively at 783 K, then reaction quotient at this stage of the reaction is

$$Q_C = \frac{[HI]^2}{[H_2][I_2]} = \frac{(0.4)^2}{(0.1)(0.2)} = 8$$

K_C for this reaction at 783 K is 46 and we find that $Q_C < K_C$. The reaction, therefore, will move to right i.e. more $H_2(g)$ and $I_2(g)$ will react to form more HI (g) and their concentration will decrease till $Q_C = K_C$.

(iii) Calculating equilibrium concentration.

(a)	A	+	B	$\longrightarrow$	C
at time t = 0	a		b		
at equlb.	a – x		b – x		x

$$K_c = \frac{(x/V)}{\left(\frac{a-x}{V}\right)\left(\frac{b-x}{V}\right)} \quad ; \quad K_p = \frac{\left(\frac{x}{a+b-x}P_T\right)}{\left(\frac{a-x}{a+b-x}P_T\right)\left(\frac{b-x}{a+b-x}P_T\right)}$$

(b) Equilibrium constant expressions in term of 'α'

	$PCl_5(g) \longrightarrow$	$PCl_3(g)$ +	$Cl_2(g)$
Initial partial pr.	1	–	–
At. Eq.	1–α	α	α

$$K_P = \frac{[p_{PCl_3(g)}][p_{Cl_2(g)}]}{p_{[PCl_5(g)]}} \quad ; \quad K_C = \frac{\alpha^2}{1-\alpha}\left(\frac{1}{V}\right)$$

$$K_P = \frac{\alpha.\alpha}{1-\alpha} \times \frac{P_T}{(1+\alpha)}$$

$$K_P = \frac{\alpha^2}{1-\alpha^2}P_T$$

ALLEN Session 2019-20
node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module#Ideal gas, Solid state & Chemical Eq\Eng\03_Chemical Equilibrium\Theory&Ex.p65

(IV) Degree of dissociation in terms of molar mass and vapour density

(a) $A(g) \rightleftharpoons nB(g)$

$$\alpha = \frac{M_T - M_O}{M_O(n-1)} \quad \text{or} \quad \alpha = \frac{D_T - D_O}{D_O(n-1)}$$

M_T = Theoretical molar mass of reactant
M_O = Observed molar mass of mixture

$$D_T = \frac{M_T}{2} \quad \text{and} \quad D_O = \frac{M_O}{2}$$

(b) $nA\ (g) \rightleftharpoons A_n\ (g)$

Initial con^n	a	0
at eq^m	$1(1-\alpha)$	$\frac{a\alpha}{n}$

$$\alpha = \frac{M_T - M_O}{M_O\left(\frac{1}{n}-1\right)}$$

FACTOR'S AFFECTING EQUILIBRIA (Le-chatelier's principle)

(i) Effect of change in concentration on equilibrium

A chemical system at equilibrium can be shifted out of equilibrium by adding or removing one more of reactants or products. Shifting out of equilibrium doesn't mean that value of equilibrium constant change. Any alteration of concentration of reactant or product will disturb the equilibrium and concentration of reactant and product one readjust to one again attain equilibrium concentration.

In other word, as we add or remove reactant (or product) the ratio of equilibrium concentration become 'Q' (reaction quotient) and depending upon.

$Q < K$: equilibrium will shift in forward direction.

$Q > K$: equilibrium will shift in backward direction.

Example : Fe^{3+} (a) + SCN^- (aq) $\rightleftharpoons$ Fe $(SCN)^{2+}$(aq)

(i) adding Fe^3 or SCN^- will more $\frac{[Fe(SCN)^{2+}]}{[Fe^{3+}][SCN^-]}$ = Q less then K_C and equilibria will shift in forward direction.

(ii) Removing $Fe(SCN)^{2+}$ will have same effect

(iii) Adding $Fe(SCN)^{2+}$ from outside source in equilibrium mixture will have effect of increasing 'Q' hence reaction shift in backward direction.

(ii) Effect of change in pressure

Sometimes we can change the position of equilibrium by changing the pressure on a system. However, changes in pressure have a measurable effect only in system where gases are involved – and then only when the chemical reaction produces a change in the total number of gas molecules in the system.

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\03_Chemical Equilibrium\Theory & Ex..p65

As we increase the pressure of a gaseous system at equilibrium, either by decreasing the volume of the system or by adding more of the equilibrium mixture, we introduce a stress by increasing the number of molecules per unit of volume. In accordance with Le Chatelier's principle, a chemical reaction that reduces the total number of molecules per unit of volume will be favored because this relieves the stress. The reverse reaction would be favoured by a decrease in pressure.

Consider what happens when we increase the pressure on a system in which NO, O_2 and NO_2 are in equilibrium.

Example : $2NO(g) + O_2\,(g) \rightleftharpoons 2NO_2(g)$

The formation of additional amounts of NO_2 decreases the total number of molecules in the system, because each time two molecules of NO_2 form, a total of three molecules of NO and O_2 react. This reduces the total pressure exerted by the system and reduces, but does not completely relieve, the stress of the increased pressure. On the other hand, a decrease in the pressure on the system favors decomposition of NO_2 into NO and O_2 which tends to restore the pressure.

Let us now consider the reaction

$$N_2(g) + O_2\,(g) \rightleftharpoons 2NO(g)$$

Because there is no change in the total number of molecules in the system during reaction, a change in pressure does not favor either formation or decomposition of gaseous nitric oxide.

(iii) ***Effect of change in temperature on equilibrium***

Changing concentration or pressure upsets an equilibrium because the reaction quotient is shifted away from the equilibrium value. Changing the temperature of a system at equilibrium has a different effect: A change in temperature changes the value of the equilibrium constant. However, we can predict the effect of the temperature change by treating it as a stress on the system and applying Le Chatelier's principle. When hydrogen reacts with gaseous iodine, energy is released as heat is evolved.

$$H_2(g) + I_2(g) \rightleftharpoons 2I(g) \quad \Delta H = -9.4 \text{ kJ (exothermic)}$$

Because this reaction is exothermic, we can write it with heat as a product.

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g) + 9.4 \text{ kJ}$$

Increasing the temperature of the reaction increases the amount of energy present. Thus, increasing the temperature has the effect of increasing the amount of one of the products of this reaction. The reaction shifts to the left to relieve the stress, and there is an increase in the concentration of H_2 and I_2 and a reduction in the concentration of HI. When we change the temperature of a system at equilibrium, the equilibrium constant for the reaction changes. Lowering the temperature in the HI system increases the equilibrium constant from 50.0 at 400°C to 67.5 at 357°C. At equilibrium at the lower temperature, the concentration of HI has increased and the concentrations of H_2 and I_2 have decreased. Raising the temperature decreases the value of the equilibrium constant from 67.5 at 357°C to 50.0 at 400°C.

van't Hoff equation

(a) $$\frac{d(\ell nK)}{dT} = \frac{\Delta H^\circ}{RT^2}$$

$$ln\left(\frac{K_2}{K_1}\right) = -\frac{\Delta H^\circ}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

ALLEN Session 2019-20

(iv) Effect of catalyst on equilibrium

A catalyst has no effect on the value of an equilibrium constant or on equilibrium concentrations. The catalyst merely increase the rates of both the forward and the reverse reactions to the same extent so that equilibrium is reached more rapidly.

All of these effects change in concentration or pressure, change in temperature, and the effect of a catalyst on a chemical equilibrium play a role in the industrial synthesis of ammonia from nitrogen and hydrogen according to the equation.

$$N_2 + 3H_2 \rightleftharpoons 2NH_3$$

One way to increase the yield of ammonia is to increase the pressure on the system in which N_2, H_2 and NH_3 are in equilibrium or are coming to equilibrium.

$$N_2\,(g)\ 3H_2(g) \rightleftharpoons 2NH_3(g)$$

The formation of additional amounts of ammonia reduces the total pressure exerted by the system and somewhat reduces the stress of the increased pressure.

Although increasing the pressure of a mixture of N_2 , H_2 and NH_3 increase the yield ammonia, at low temperatures the rate of formation of ammonia is slow. At room temperature, for example, the reaction is so slow that if we prepared a mixture of N_2 and H_2, no detectable amount of ammonia would form during our lifetime. Attempts to increase the rate of the reaction by increasing the temperature are counterproductive. The formation of ammonia from hydrogen and nitrogen is an exothermic process:

$$N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g) \qquad \Delta H = -92.2 \text{ kJ}$$

Thus increasing the temperature to increase the rate lowers the yield. If we lower the temperature to shift the equilibrium to the right to favour the formation of more ammonia, equilibrium is reached more slowly because of the large decrease of reaction rate with decreasing temperature.

Part of the rate of formation lost by operating at lower temperatures can be recovered by using a catalyst to increase the reaction rate. Iron powder is one catalyst used. However, as we have seen, a catalyst serves equally well to increase the rate of a reverse reaction in this case, the decomposition of ammonia into its constituent elements. Thus the net effect of the iron catalyst on the reaction is to cause equilibrium to be reached more rapidly.

A THERMODYNAMIC RELATIONSHIP :

$$\Delta G = \Delta G^\circ + RT\ln Q \qquad [Q = \text{Reaction quotient}]$$

$$\Delta G^\circ = -RT\ln K_{eq} \qquad (\text{At eq}^m.\ (\Delta G)_{T,P} = 0)$$

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

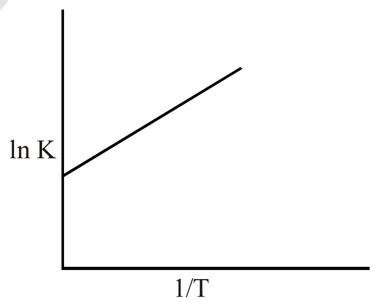
$$\ln K_{eq} = \frac{-\Delta_r H^\circ}{RT} + \frac{\Delta_r S^\circ}{R}$$

$$\ln K_1 = \frac{\Delta_r S^\circ}{R} - \frac{\Delta_r H^\circ}{RT_1}$$

$$\ln K_2 = \frac{\Delta_r S^\circ}{R} - \frac{\Delta_r H^\circ}{KT_2}$$

$$\ln\left(\frac{K_2}{K_1}\right) = \frac{\Delta H^\circ}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

For Exothermic reaction



EQUILIBRIUM CONSTANT AS PER KINETICS :

$$A(g) \underset{K_b}{\overset{K_f}{\rightleftharpoons}} B\,(g) + C(g)$$

$$\frac{-d(A)}{dt} = K_f\,[A] - K_b\,[B]\,[C]$$

ALLEN Session 2019-20
node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module#Ideal gas, Solid state & Chemical Eq\Eng\03_Chemical Equilibrium\Theory & Ex..p65

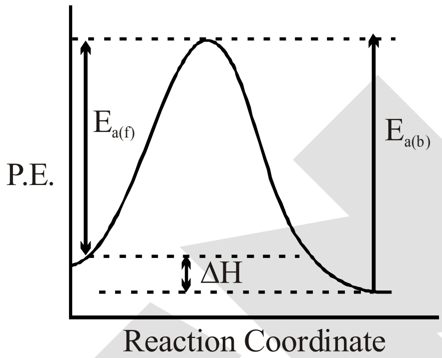
At eq^m $\frac{-d[A]}{dt} = 0$ $\qquad \frac{K_f}{K_b} = \frac{[B][C]}{[A]} = K_c$

where $k = A \cdot e^{-E_a/RT}$; A : pre-expotential factor

$k_f = A_f \cdot e^{-E_{a(f)}/RT}$ $\qquad E_a$: activation energy

$k_b = A_b \cdot e^{-E_{a(b)}/RT}$

$$K_{eq} = \frac{k_f}{k_b} = \frac{A_f.e^{-E_{a(f)}/RT}}{A_b.e^{-E_{a(b)}/RT}}$$



$k = A \cdot e^{-\Delta H/RT}$

where $\Delta H = E_{a(f)} - E_{a(b)}$

$$\ln K_1 = \ln A - \frac{\Delta H}{RT_1}$$

$$\ln K_2 = \ln A - \frac{\Delta H}{RT_2}$$

$$\ln \frac{K_2}{K_1} = \frac{\Delta H}{R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$$

SIMULTANEOUS EQUILIBRIA :

$X(s) \rightleftharpoons A(g) + B(g)$ $\qquad K_{P_1} = (x+y)x$

$\qquad (x+y) \quad x$

$Y(s) \rightleftharpoons A(g) + C(g)$ $\qquad K_{P_2} = (x+y)y$

$\qquad (y+x) \quad y$

$\frac{K_{P_1}}{K_{P_2}} = \frac{x}{y}$ $\qquad P_{total} = 2x + 2y$

SEQUENTIAL EQUILIBRIUM :

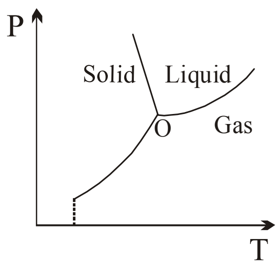
$A(s) \longrightarrow B(g) + C(g)$ $\qquad K_{C_1} = (x)(x-y)$

a–x $\quad$ x $\quad$ x–y

$C(g) \longrightarrow D(g)$ $\qquad K_{C_2} = \frac{y}{x-y}$

x–y $\quad$ y

PHASE DIAGRAM FOR WATER:



ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module#Ideal gas, Solid state & Chemical Eq\Eng\03_Chemical Equilibrium\Theory&Ex.p65

EXERCISE # O-I

Only one is correct

1. x $\rightleftharpoons$ y reaction is said to be in equilibrium, when :-
(A) Only 10% conversion of x to y takes place
(B) Complete conversion of x to y has taken place
(C) Conversion of x to y is only 50% complete
(D) The rate of change of x to y is just equal to the rate of change of y to x in the system

2. The equilibrium concentration of B $[(B)_e]$ for the reversible reaction A $\rightleftharpoons$ B can be evaluated by the expression:-
(A) $K_C[A]_e^{-1}$ (B) $\frac{k_f}{k_b}[A]_e^{-1}$ (C) $k_f k_b^{-1}[A]_e$ (D) $k_f\, k_b\, [A]^{-1}$

3. In a chemical equilibrium, the rate constant for the backward reaction is 2×10^{-4} and the equilibrium constant is 1.5. The rate constant for the forward reaction is:-
(A) 2×10^{-3} (B) 5×10^{-4} (C) 3×10^{-4} (D) 9.0×10^{-4}

4. For which reaction is $K_p = K_c$:-
(A) $2NOCl(g) \rightleftharpoons 2NO(g) + Cl_2(g)$
(B) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$
(C) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$
(D) $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

5. For the reaction
$CuSO_4.5H_2O_{(s)} \rightleftharpoons CuSO_4.3H_2O_{(s)} + 2H_2O_{(g)}$
Which one is correct representation :–
(A) $K_p = (P_{(H_2O)})^2$ (B) $K_c = [H_2O]^2$ (C) $K_p = K_c(RT)^2$ (D) All

6. $\log \frac{K_p}{K_c} + \log RT = 0$ is true relationship for the following reaction:-
(A) $PCl_5 \rightleftharpoons PCl_3 + Cl_2$
(B) $2SO_2 + O_2 \rightleftharpoons 2SO_3$
(C) $N_2 + 3H_2 \rightleftharpoons 2NH_3$
(D) (B) and (C) both

7. For a reaction $N_2 + 3H_2 \rightleftharpoons 2NH_3$, the value of K_C does not depends upon :-
(a) Initial concentration of the reactants
(b) Pressure
(c) Temperature
(d) Catalyst
(A) Only c (B) a, b, c (C) a, b, d (D) a, b, c, d

8. For any reversible reaction if concentration of reactants increases then effect on equilibrium constant:-
(A) Depends on amount of concentration
(B) Unchange
(C) Decrease
(D) Increase

9. If some He gas is introduced into the equilibrium $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$ at constant pressure and temperature then equilibrium constant of reaction:
(A) Increase (B) Decrease (C) Unchange (D) Nothing can be said

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\03_Chemical Equilibrium\Theory & Ex..p65

10. The equilibrium constant for the reaction ;
$N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$ at temperature
T is 4×10^{-4}. The value of K_c for the reaction.

$NO(g) \rightleftharpoons \frac{1}{2} N_2(g) + \frac{1}{2} O_2(g)$ at the same temperature is :

(A) 0.02 (B) 50 (C) 4×10^{-4} (D) 2.5×10^{-2}

11. The equilibrium constant for the given reaction :

$$SO_3(g) \rightleftharpoons SO_2(g) + \frac{1}{2} O_2(g) \; ; K_c = 5 \times 10^{-2}$$

The value of K_c for the reaction :
$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$, will be
(A) 400 (B) 2.40×10^{-3} (C) 9.8×10^{-2} (D) 4.9×10^{-2}

12. For the following three reactions 1, 2 and 3, equilibrium constants are given :
(1) $CO(g) + H_2O(g) \rightleftharpoons CO_2(g) + H_2(g)$; K_1
(2) $CH_4(g) + H_2O(g) \rightleftharpoons CO(g) + 3H_2(g)$; K_2
(3) $CH_4(g) + 2H_2O(g) \rightleftharpoons CO_2(g) + 4H_2(g)$; K_3
Which of the following relations is correct ?

(A) $K_1 \sqrt{K_2} = K_3$ (B) $K_2K_3 = K_1$ (C) $K_3 = K_1K_2$ (D) $K_3 \cdot K_2^3K_1^2$

13. Sulfide ion in alkaline solution reacts with solid sulfur to form polysulfide ions having formulas S_2^{2-}, S_3^{2-}, S_4^{2-} and so on. The equilibrium constant for the formation of S_2^{2-} is 12 (K_1) & for the formation of S_3^{2-} is 132 (K_2), both from S and S^{2-}.What is the equilibrium constant for the formation of S_3^{2-} from S_2^{2-} and S?
(A) 11 (B) 12 (C) 132 (D) None of these

14. What should be the value of K_c for the reaction $2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)}$, if the amount are SO_3 = 48 g, SO_2= 12.8 g and O_2 = 9.6 g at equilibrium and the volume of the container is one litre?
(A) 64 (B) 30 (C) 42 (D) 8.5

15. In the reaction, $PCl_5 \rightleftharpoons PCl_3 + Cl_2$, the amount of each PCl_5, PCl_3 and Cl_2 is 2 mole at equilibrium and total pressure is 3 atmosphere. The value of K_p will be
(A) 1.0 atm. (B) 3.0 atm. (C) 2.9 atm. (D) 6.0 atm.

16. For the reaction : $P \rightleftharpoons Q + R$. Initially 2 moles of P was taken. Up to equilibrium 0.5 moles of P was dissociated. What would be the degree of dissociation :-
(A) 0.5 (B) 1 (C) 0.25 (D) 4.2

17. 4 moles of PCl_5 are heated at constant temperature in closed container. If degree of dissociation for PCl_5 is 0.5 calculate total number of moles at equilibrium :-
(A) 4.5 (B) 6 (C) 3 (D) 4

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\03_Chemical Equilibrium\Theory & Ex..p65

ALLEN Session 2019-20

18. In the reaction 2P(g) + Q(g) [redacted] 3R(g) + S(g). If 2 moles each of P and Q taken initially in a 1 litre flask. At equilibrium which is true:-

(A) [P] < [Q] (B) [P] = [Q] (C) [Q] = [R] (D) None of these

19. For the reaction $A + 2B \rightleftharpoons 2C + D$, initial concentration of A is a and that of B is 1.5 times that of A. Concentration of A and D are same at equilibrium. What should be the concentration of B at equilibrium ?

(A) $\frac{a}{4}$ (B) $\frac{a}{2}$ (C) $\frac{3a}{4}$ (D) All of the above.

20. For the reaction $3\,A(g) + B(g) \rightleftharpoons 2\,C(g)$ at a given temperature, $K_c = 9.0$. What must be the volume of the flask, if a mixture of 2.0 mol each of A, B and C exist in equilibrium?

(A) 6L (B) 9L (C) 36 L (D) None of these

21. For the following gases equilibrium. $N_2O_4\,(g) \rightleftharpoons 2NO_2\,(g)$

K_p is found to be equal to K_c. This is attained when temperature is

(A) 0°C (B) 273 K (C) 1 K (D) 12.19 K

22. The degree of dissociation of SO_3 is α at equilibrium pressure p^0.

K_p for $2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$

(A) $\frac{p^0\alpha^3}{2(1-\alpha)^3}$ (B) $\frac{p^0\alpha^3}{(2+\alpha)(1-\alpha)^2}$ (C) $\frac{p^0\alpha^2}{2(1-\alpha)^2}$ (D) None of these

23. For the reaction : $2HI\,(g) \rightleftharpoons H_2(g) + I_2(g)$, the degree of dissociated (α) of HI(g) is related to equilibrium constant K_P by the expression

(A) $\frac{1+2\sqrt{K_p}}{2}$ (B) $\sqrt{\frac{1+2K_p}{2}}$ (C) $\sqrt{\frac{2K_p}{1+2K_p}}$ (D) $\frac{2\sqrt{K_p}}{1+2\sqrt{K_p}}$

24. The equilibrium constant for the reaction

$A(g) + 2B(g) \rightleftharpoons C(g)$

is 0.25 dm^6mol^{-2}. In a volume of 5 dm^3, what amount of A must be mixed with 4 mol of B to yield 1 mol of C at equilibrium.

(A) 3 moles (B) 24 moles (C) 26 moles (D) None of these

25. The equilibrium constant K_C for the reaction,

$A(g) + 2B(g) \rightleftharpoons 3C(g)$ is 2×10^{-3}

What would be the equilibrium partial pressure of gas C if initial pressure of gas A & B are 1 & 2 atm respectively.

(A) 0.0625 atm (B) 0.1875 atm (C) 0.21 atm (D) None of these

26. A 20.0 litre vessel initially contains 0.50 mole each of H_2 and I_2 gases. These substances react and finally reach an equilibrium condition. Calculate the equilibrium concentration of HI if $K_{eq} = 49$ for the reaction $H_2 + I_2 \rightleftharpoons 2HI$.

(A) 0.78 M (B) 0.039 M (C) 0.033 M (D) 0.021 M

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module#Ideal gas, Solid state & Chemical Eq\Eng\03_Chemical Equilibrium\Theory & Ex..p65

27. At 675 K, H_2(g) and CO_2 (g) react to form CO(g) and H_2O (g), K_p for the reaction is 0.16. If a mixture of 0.25 mole of H_2(g) and 0.25 mol of CO_2 is heated at 675 K, mole % of CO(g) in equilibrium mixture is :

(A) 7.14 (B) 14.28 (C) 28.57 (D) 33.33

28. The vapour density of N_2O_4 at a certain temperature is 30. What is the % dissociation of N_2O_4 at this temperature?

(A) 53.3% (B) 106.6% (C) 26.7% (D) None

29. The equilibrium constant K_P (in atm) for the reaction is 9 at 7 atm and 300 K.

$A_2 (g) \rightleftharpoons B_2(g) + C_2 (g)$

Calculate the average molar mass (in gm/mol) of an equilibrium mixture.

Given : Molar mass of A_2, B_2 and C_2 are 70, 49 & 21 gm/mol respectively.

(A) 50 (B) 45 (C) 40 (D) 37.5

30. Vapour density of the equilibrium mixture of the reaction

$2NH_3 (g) \rightleftharpoons N_2 (g) + 3H_2 (g)$ is 6.0

Percent dissociation of ammonia gas is:

(A) 13.88 (B) 58.82 (C) 41.66 (D) None of these

31. The equilibrium $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ shifts forward if :-

(A) A catalyst is used.

(B) An adsorbent is used to remove SO_3 as soon as it is formed.

(C) Small amounts of reactants are removed.

(D) None of these

32. In manufacture of NO, the reaction $N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)}$, ΔH +ve is favourable if :-

(A) Pressure is increased (B) Pressure is decreased

(C) Temperature is increased (D) Temperature is decreased

33. For the reaction $PCl_5 \rightleftharpoons PCl_3 + Cl_2$, the degree of dissociation varies inversely as the square root of pressure of the system. Supposing at constant temperature If the volume is increased 16 times the intial volume, the degree of dissociation for this reaction will becomes :-

(A) 4 times (B) $\frac{1}{4}$ times (C) 2 times (D) $\frac{1}{2}$ times

34. In which of the following reactions, increase in the pressure at constant temperature does not affect the moles at equliibrium :

(A) $2NH_3(g) \rightleftharpoons N_2(g) + 3H_2(g)$ (B) $C(g) + \frac{1}{2}O_2(g) \rightleftharpoons CO(g)$

(C) $H_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons H_2O(g)$ (D) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

35. Change in volume of the system does not alter the number of moles in which of the following equilibrium

(A) $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$ (B) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

(C) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ (D) $SO_2Cl_2(g) \rightleftharpoons SO_2(g) + Cl_2(g)$

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module#Ideal gas, Solid state & Chemical Eq\Eng\03_Chemical Equilibrium\Theory&Ex..p65

ALLEN Session 2019-20

36. The conditions favourable for the reaction :

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$; $\Delta H° = -198$ kJ

are :

(A) low temperature, high pressure
(B) any value of T and P
(C) low temperature and low pressure
(D) high temperature and high pressure

37. Densities of diamond and graphite are 3.5 and 2.3 gm/mL.

C (diamond) $\rightleftharpoons$ C (graphite) $\Delta_r H = -1.9$ kJ/mole

favourable conditions for formation of diamond are

(A) high pressure and low temperature
(B) low pressure and high temperature
(C) high pressure and high temperature
(D) low pressure and low temperature

38. The equilibrium $SO_2Cl_2(g) \rightleftharpoons SO_2(g) + Cl_2(g)$ is attained at 25°C in a closed rigid container and an inert gas, helium is introduced. Which of the following statements is/are correct.

(A) concentrations of SO_2, Cl_2 and SO_2Cl_2 do not change
(B) more chlorine is formed
(C) concentration of SO_2 is reduced
(D) more SO_2Cl_2 is formed

39. The yield of product in the reaction

$$2A(g) + B(g) \rightleftharpoons 2C(g) + Q \text{ kJ}$$

would be lower at :

(A) low temperature and low pressure
(B) high temperature & high pressure
(C) low temperature and to high pressure
(D) high temperature & low pressure

40. What is the effect of the reduction of the volume of the system for the equilibrium

$2C(s) + O_2(g) \rightleftharpoons 2CO(g)$?

(A) The equilibrium will be shifted to the left by the increased pressure caused by the reduction in volume.
(B) The equilibrium will be shifted to the right by the decreased pressure caused by the reduction in volume.
(C) The equilibrium will be shifted to the left by the increased pressure caused by the increase in volume.
(D) The equilibrium will be shifted to the right by the increased pressure caused by the reduction in volume.

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module#Ideal gas, Solid state & Chemical Eq\Eng\03_Chemical Equilibrium\Theory&Ex..p65

J-MAIN

1. The equilibrium constants at 298 K for a reaction $A + B \rightleftharpoons C + D$ is 100. If the initial concentration of all the four species were 1 M each, then equilibrium concentration of D (in mol L^{-1}) will be : **[JEE-MAINS-16]**

(1) 1.182 (2) 0.182 (3) 0.818 (4) 1.818

2. For the reaction $SO_{2(g)} + \frac{1}{2}O_{2(g)} \rightleftharpoons SO_{3(g)}$, if $K_p = K_C (RT)^x$ where the symbols have usual meaning then the value of x is : (assuming ideality) **[JEE-MAINS-14]**

(1) $\frac{1}{2}$ (2) 1 (3) –1 (4) $-\frac{1}{2}$

3. For the decomposition of the compound, represented as **[JEE-MAINS(online)-14]**

$NH_2COONH_4(s) \rightleftharpoons 2NH_3(g) + CO_2(g)$

the $K_P = 2.9 \times 10^{-5}$ atm^3.

If the reaction is started with 1 mol of the compound, the total pressure at equilibrium would be

(1) 38.8×10^{-2} atm (2) 1.94×10^{-2} atm

(3) 5.82×10^{-2} atm (4) 7.66×10^{-2} atm

4. In reaction $A + 2B \rightleftharpoons 2C + D$, initial concentration of B was 1.5 times of $|A|$, but at equilibrium the concentrations of A and B became equal. The equilibrium constant for the reaction is : **[JEE-MAINS(online)-13]**

(1) 4 (2) 6 (3) 12 (4) 8

5. $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g), K_1$ (1) **[JEE-MAINS(online)-13]**

$N_2(g) + O_2(g) \rightleftharpoons 2NO(g), K_2$ (2)

$H_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons H_2O(g), K_3$ (3)

The equation for the equilibrium constant of the reaction

$2\ NH_3(g) + \frac{5}{2}O_2(g) \rightleftharpoons 2NO(g) + 3H_2O(g), (K_4)$

in terms of K_1, K_2 and K_3 is :

(1) $\frac{K_1K_3^2}{K_2}$ (2) $\frac{K_2K_3^3}{K_1}$ (3) $\frac{K_1K_2}{K_3}$ (4) $K_1\ K_2\ K_3$

6. One mole of $O_2(g)$ and two moles of $SO_2(g)$ were heated in a closed vessel of one litre capacity at 1098 K. At equilibrium 1.6 moles of $SO_3(g)$ were found. The equilibrium constant K_C of the reaction would be :- **[JEE-MAINS(online)-12]**

(1) 60 (2) 80 (3) 30 (4) 40

ALLEN Session 2019-20

7. K_1, K_2 and K_3 are the equilibrium constants of the following reactions (I), (II) and (III), respectively

(I) $N_2 + 2O_2 \rightleftharpoons 2NO_2$ (II) $2NO_2 \rightleftharpoons N_2 + 2O_2$ **[JEE-MAINS(online)-12]**

(III) $NO_2 \rightleftharpoons \frac{1}{2}N_2 + O_2$

The correct relation from the following is :

(1) $K_1 = \sqrt{K_2} = K_3$ (2) $K_1 = \frac{1}{K_2} = \frac{1}{K_3}$ (3) $K_1 = \frac{1}{K_2} = K_3$ (4) $K_1 = \frac{1}{K_2} = \frac{1}{(K_3)^2}$

8. The value of Kp for the equilibrium reaction $N_2O_4(g) \rightleftharpoons 2NO_2(g)$ is 2. The percentage dissociation of $N_2O_4(g)$ at a pressure of 0.5 atm is **[JEE-MAINS(online)-12]**

(1) 71 (2) 50 (3) 88 (4) 25

9. 8 mol of $AB_3(g)$ are introduced into a 1.0 dm^3 vessel. If it dissociates as $2AB_3(g) \rightleftharpoons A_2(g) + 3B_2(g)$

At equilibrium, 2mol of A_2 are found to be present. The equilibrium constant of this reaction is :- **[JEE-MAINS(online)-12]**

(1) 36 (2) 3 (3) 27 (4) 2

10. The equilibrium constant (K_C) for the reaction $N_2(g) + O_2(g) \longrightarrow 2NO(g)$ at temperature T is 4×10^{-4}. The value of K_C for the reaction. $NO(g) \longrightarrow \frac{1}{2}N_2(g) + \frac{1}{2}O_2(g)$ at the same temperature is :- **[AIEEE-2012]**

(1) 50.0 (2) 0.02 (3) 2.5×10^2 (4) 4×10^{-4}

11. A vessel at 1000 K contains CO_2 with a pressure of 0.5 atm. Some of the CO_2 is converted into CO on the addition of graphite. If the total pressure at equilibrium is 0.8 atm, the value of K is :- **[AIEEE-2011]**

(1) 0.3 atm (2) 0.18 atm (3) 1.8 atm (4) 3 atm

12. The equlibrium constants K_{p_1} and K_{p_2} for the reaction $X \rightleftharpoons 2Y$ and $Z \rightleftharpoons P + Q$, respectively are in the ratio of 1 : 9. If the degree of dissociation of X and Z be equal then the ratio of total pressure at these equilibria is **[AIEEE-2008]**

(1) 1 : 36 (2) 1 : 1 (3) 1 : 3 (4) 1 : 9

13. Phosphorus pentachloride dissociates as follows, in a closed reaction vessel, **[AIEEE-2006]**

$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

If total pressure at equilibrium of the reaction mixture is P and degree of dissociation of PCl_5 is x, the partial pressure of PCl_3 will be-

(1) $\left(\frac{2x}{1-x}\right)P$ (2) $\left(\frac{x}{x-1}\right)P$ (3) $\left(\frac{x}{1-x}\right)P$ (4) $\left(\frac{x}{x+1}\right)P$

14. The equilibrium constant for the reaction $SO_3(g) \rightleftharpoons SO_2(g) + \frac{1}{2}O_2(g)$ is $K_C = 4.9\times10^{-2}$. The value of K_C for the reaction **[AIEEE-2006]**

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ will be

(1) 2.40×10^{-3} (2) 9.8×10^{-2} (3) 4.9×10^{-2} (4) 416

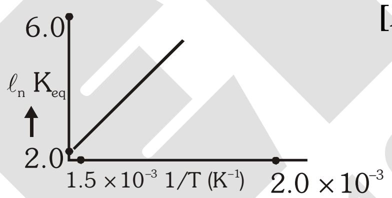
ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\03_Chemical Equilibrium\Theory & Ex..p65

15. For the reaction $2NO_{2(g)} \rightleftharpoons 2NO_{(g)} + O_{2(g)}$, ($K_c = 1.8 \times 10^{-6}$ at 184° C) (R = 0.831 kJ(mol.K))
When K_p and k_c are compared at 184°C it is found that **[AIEEE-2005]**
(1) K_p is less than K_c
(2) K_p is greater than K_c
(3) Whether K_p is greater than, less than or equal to K_c depends upon the total gas pressure
(4) $K_p = K_c$

16. The exothermic formation of ClF_3 is represented by the equation **[AIEEE-2005]**
$Cl_{2(g)} + 3F_{2(g)} \rightleftharpoons 2ClF_{3(g)}$; $\Delta H_r = -329$ kJ
Which of the following will increase the quantity of ClF_3 in an equilibrium mixture of Cl_2, F_2 and ClF_3?
(1) Removing Cl_2
(2) Increasing the temperature
(3) Adding F_2
(4) Increasing the volume of the container

17. A schematic plot of $\ell n\ K_{eq}$ verus inverse of temperature for a reaction is shown below. The reaction must be **[AIEEE-2005]**



(1) endothermic
(2) exothermic
(3) highly spontaneous at ordinary temperature
(4) one with negligible enthalpy change

18. What is the equilibrium expression for the reaction $P_{4(s)} + 5O_{2(g)} \rightleftharpoons P_4O_{10(s)}$? **[AIEEE-2004]**
(1) $K_C = [P_4O_{10}] / [P_4] [O_2]^5$
(2) $K_C = [P_4O_{10}] / 5 [P_4] [O_2]$
(3) $K_C = [O_2]^5$
(4) $K_C = 1 / [O_2]^5$

19. For the reaction $CO_{(g)} + Cl_{2(g)} \rightleftharpoons COCl_{2(g)}$ the $\frac{K_P}{K_C}$ is equal to - **[AIEEE-2004]**
(1) $\frac{1}{RT}$
(2) RT
(3) $\sqrt{RT}$
(4) 1.0

20. The equilibrium constant for the reaction
$N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)}$ at temperature T is 4×10^{-4}. The value of K_C for the reaction
$NO_{(g)} \rightleftharpoons \frac{1}{2}N_{2(g)} + \frac{1}{2}O_{2(g)}$ **[AIEEE-2004]**
(1) 2.5×10^2
(2) 50
(3) 4×10^{-4}
(4) 0.02

21. For the reaction equilibrium,
N_2O_4 (g) $\rightleftharpoons$ $2NO_2$(g) the concentration of N_2O_4 and NO_2 at equilibrium are 4.8×10^{-2} and 1.2×10^{-2} mol L^{-1} respectively. The value of K_C for the reaction is- **[AIEEE-2003]**
(1) 3×10^{-3} mol L^{-1}
(2) 3×10^3 mol L^{-1}
(3) 3.3×10^2 mol L^{-1}
(4) 3×10^{-1} mol L^{-1}

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module#Ideal gas, Solid state & Chemical Eq\Eng\03_Chemical Equilibrium\Theory&Ex.p65

22. Consider the reaction equilibrium **[AIEEE-2003]**

$2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2\ SO_{3(g)}$; $\Delta H° = -198$ kJ

On the basis of Le-Chatelier's principle, the condition favourable for the forward reaction is -

(1) Lowering the temperature and increasing the pressure

(2) Any value of temperature as well as pressure

(3) Lowering of temperature as well as pressure

(4) Increasing temperature as well as pressure

23. Reaction $CO_{(g)} + \frac{1}{2} O_{2(g)} \rightleftharpoons CO_{2(g)}$. The value of $\frac{K_P}{K_c}$ is - **[AIEEE-2002]**

(1) $\frac{1}{RT}$ (2) $\sqrt{RT}$ (3) $\frac{1}{\sqrt{RT}}$ (4) RT

24. One of the following equilibrium is not affected by change in volume of the flask - **[AIEEE-2002]**

(1) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ (2) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

(3) $N_2(g) + O_2 \rightleftharpoons 2NO(g)$ (4) $SO_2Cl_2(g) \rightleftharpoons SO_2(g) + Cl_2(g)$

ALLEN Session 2019-20

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J-ADVANCE

1. The thermal dissociation equilibrium of $CaCO_3(s)$ is studied under different conditions.
$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$ **[JEE 2013]**
For this equilibrium, the correct statement(s) is(are)
(A) ΔH is dependent on T
(B) K is independent of the initial amount of $CaCO_3$
(C) K is dependent on the pressure of CO_2 at a given T
(D) ΔH is independent of the catalyst, if any

2. If $Ag^+ + NH_3 \rightleftharpoons [Ag(NH_3)]^+$; $K_1 = 1.6 \times 10^3$ and **[JEE 2006]**
$[Ag(NH_3)]^+ + NH_3 \rightleftharpoons [Ag(NH_3)_2]^+$; $K_2 = 6.8 \times 10^3$. The formation constant of $[Ag(NH_3)_2]^+$ is :
(A) 6.08×10^{-6} (B) 6.8×10^{-6} (C) 1.6×10^3 (D) 1.088×10^7

3. Consider the following equilibrium in a closed container : $N_2O_4(g) \rightleftharpoons 2NO_2(g)$.
At a fixed temperature, the volume of the reaction container is halved. For this change, which of the following statements holds true regarding the equilibrium constant (K_P) and degree of dissociation (α) : **[JEE 2002]**
(A) Neither K_P nor α changes (B) Both K_P and α change
(C) K_P changes, but α does not change (D) K_P does not change, but α changes

4. At constant temperature, the equilibrium constant (K_P) for the decomposition reaction. $N_2O_4 \rightleftharpoons 2NO_2$ is expressed by $K_P = 4x^2P/(1 - x^2)$ where P is pressure, x is extent of decomposition. Which of the following statement is true ? **[JEE 2001]**
(A) K_P increases with increase of P (B) K_P increases with increase of x
(C) K_P increases with decrease of x (D) K_P remains constant with change in P or x

5. When 3.06g of solid NH_4HS is introduced into a two litre evacuated flask at 27°C, 30% of the solid decomposes into gaseous ammonia and hydrogen sulphide. **[JEE 2000]**
(i) Calculate K_C & K_P for the reaction at 27°C.
(ii) What would happen to the equilibrium when more solid NH_4HS is introduced into the flask?

6. When two reactants A and B are mixed to give products C and D, the reaction quotient Q, at the initial stages of the reaction : **[JEE 2000]**
(A) is zero (B) decrease with time
(C) independent of time (D) increases with time

7. For the reversible reaction : **[JEE 2000]**
$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ at 500°C. The value of K_p is 1.44×10^{-5}, when partial pressure is measured in atmospheres. The corresponding value of K_c with concentration in mol L^{-1} is :
(A) $1.44 \times 10^{-5} /(0.082 \times 500)^2$ (B) $1.44 \times 10^{-5} /(8.314 \times 773)^2$
(C) $1.44 \times 10^{-5} /(0.082 \times 500)^2$ (D) $1.44 \times 10^{-5} /(0.082 \times 773)^{-2}$

8. The degree of dissociation is 0.4 at 400K & 1.0 atm for the gaseous reaction $PCl_5 \rightleftharpoons PCl_3 + Cl_2(g)$. Assuming ideal behaviour of all gases. Calculate the density of equilibrium mixture at 400K & 1.0 atm pressure. **[JEE 1999]**

ALLEN Session 2019-20

ANSWER KEY

EXERCISE # O-I

1.	(D)	2.	(C)	3.	(C)	4.	(C)
5.	(D)	6.	(B)	7.	(C)	8.	(B)
9.	(C)	10.	(B)	11.	(A)	12.	(C)
13.	(A)	14	(B)	15.	(A)	16.	(C)
17.	(B)	18.	(A)	19.	(B)	20.	(A)
21.	(D)	22.	(B)	23.	(D)	24.	(C)
25.	(B)	26.	(B)	27.	(B)	28.	(A)
29.	(C)	30.	(C)	31.	(B)	32.	(C)
33.	(A)	34.	(D)	35.	(A)	36.	(A)
37.	(C)	38.	(A)	39.	(D)	40.	(A)

J-MAIN

1.	Ans.(4)	2.	Ans.(4)	3.	Ans(3)	4.	Ans(1)
5.	Ans(2)	6.	Ans(2)	7.	Ans(4)	8.	Ans(1)
9.	Ans(3)	10.	Ans.(1)	11.	Ans.(3)	12.	Ans.(1)
13.	Ans.(4)	14.	Ans.(4)	15.	Ans.(2)	16.	Ans.(3)
17.	Ans.(2).	18.	Ans.(4)	19.	Ans.(1)	20.	Ans.(2)
21.	Ans.(1)	22.	Ans.(1)	23.	Ans.(3)	24.	Ans.(3)

J-ADVANCE

1.	Ans.(A,B,D)	2.	Ans.(D)	3.	Ans.(D)	4.	Ans.(D)

5. (i) $K_c = 8.1 \times 10^{-5}$ mol^2 L^2 ; $K_p = 4.91 \times 10^{-2}$ atm^2 (ii) No effect;

6. Ans.(D)

7. Ans.(D)

8. Ans. 4.54 g dm^{-3}

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Module # Ideal gas, Solid state & Chemical Eq\Eng\03_Chemical Equilibrium\Theory & Ex..p65

NURTURE COURSE

REAL GAS

REAL GAS

1. Introduction :

An ideal gas is a hypothetical gas whose pressure, volume and temperature behaviour is completely described by the ideal gas equation. Actually no gas is ideal or perfect in nature. All gases are real gases.

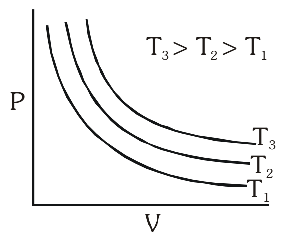
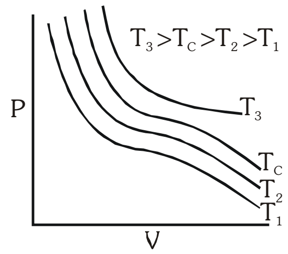
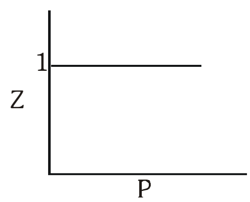
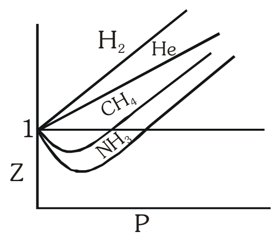
Real gases do not obey the ideal gas laws exactly under all conditions of temperature and pressure.

Real gases deviates from ideal behaviour because of mainly two assumptions of "Kinetic theory of gases".

(i) The volume of gas particle is negligible compared to volume of container (while the real gas particle may have some significant volume).

(ii) There is no interaction between gaseous particles (while attraction forces exist between real gas particles).

1.1 Comparision between Real and Ideal gas :

IDEAL GAS	REAL GAS
(i) PV = nRT	**(i)** PV $\neq$ nRT $\Rightarrow$ If PV > nRT (Gas is less compressible than ideal gas). If PV < nRT (Gas is more compressible than ideal gas).
(ii) No liquifaction is possible.	**(ii)** Liquifaction is possible below a certain temperature.
	 $\Rightarrow$ Follow critical phenomena and can not liquefy above T_C.
(iii) 	**(iii)** 
(iv) No interaction force is present between gas particles.	**(iv)** Interaction force exist between gas particles which vary depending upon conditions.
(v) Volume of gas particles is negligible w.r.t. volume of container.	**(v)** Volume of gas particles has significant value and can not be neglected normally w.r.t. volume of container.

node06\B0AH-AI\Kota\JEE(Advanced)\Nurture\Chem\Sheet\REAL GAS\Eng\01_Theory.p65

1.2 VANDER WAAL EQUATION OF REAL GASES

The ideal gas equation does not consider the effect of attractive forces and molecular volume.

Van der Waal corrected the ideal gas equation by taking the effect of

(a) Molecular volume

(b) Molecular attraction

(A) Volume correction :

In the ideal gas equation, $P_iV_i = nRT$, V_i represents the ideal volume where the molecules can move freely. In real gases, a part of the total volume is occupied by the gas molecules. Hence the free volume V_i is the total volume V minus the volume occupied by the gas molecules.

Real volume of gas = Actual volume of container – volume occupied by molecules in motion.

$$V_i = V - nb \text{ for n mole of gas}$$

Where b is termed the 'excluded volume' or 'co-volume' per mole.

It is constant and characteristic for each gas.

$$b = 4 \times \text{volumes of one molecules} \times N_A$$

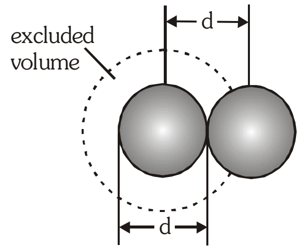
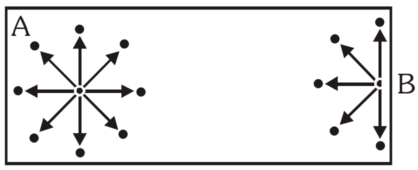


Illustration of excluded volume

(B) Pressure correction :

In order to take account the effect of intermolecular forces of attraction, let us consider a molecule A in the midst of the vessel.

This molecule is surrounded by other molecules in a symmetrical manner and is being attracted uniformly on all sides by the neighbouring molecules with the result that this molecule on the whole experiences no net force of attraction.



Now, consider a molecule B near the side of the vessel, which is about to strike one of its sides, thus contributing towards the total pressure of the gas. There are molecules only in one side of the vessel, i.e. towards its centre, with the result of that, this molecule experiences a net force of attraction towards the centre of the vessel. This results in decreasing the velocity of the molecule, and hence its momentum. Thus, the molecule does not contribute as much force as it would have, had there been no force of attraction. Thus, the pressure of a real gas would be smaller than the corresponding pressure of an ideal gas.

Van der Waals noted that the total force of attraction on any molecule about to hit a wall is proportional to the concentration of neighbouring molecules, n/V. However, the number of molecules about to hit the wall per unit wall area is also proportional to the concentration n/V. Therefore, the force per unit wall area, or pressure, is reduced from that assumed in the ideal gas wall by a factor proportional to n^2/V^2. Letting a be the proportionality constant, we can write

$$P\text{ (actual)} = P\text{(ideal)} - \frac{an^2}{V^2} \quad \text{or} \quad P\text{ (ideal)} = P\text{(actual)} + \frac{an^2}{V^2}$$

'a' is a constant which depends upon the nature of the gas,

Combining the two corrections,

for 1 mole of gas

$$\left(P + \frac{a}{V^2}\right)(V - b) = RT$$

and for n mole of gas $\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$

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The constants 'a' and 'b' :

Van der Waals constant for attraction 'a' and volume 'b' are characteristic constants for a given gas.

(i) The 'a' values for a given gas are measure of intermolecular forces of attraction. More are the intermolecular forces of attraction, more will be the value of a.

(ii) The gas having higher value of 'a' can be liquefied easily and therefore H_2 and He are not liquefied easily.

(iii) Unit of 'a' is atm lit^2 mole^{-2} or dyne cm^4 mole^{-2} or Nm^{4}mol^{-2}

(iv) Unit of 'b' is lit mole^{-1} or cm^3 mole^{-1} or m^3 mol^{-1}

❑ **The van der Walls constants for some common gases**

Gas	a (atmL^{2}mol^{-2})	b (L mol^{-1})
Ammonia	4.17	0.0371
Argon	1.35	0.0322
Carbon dioxide	3.59	0.0427
Carbon monoxide	1.49	0.0399
Chlorine	6.49	0.0562
Ethane	5.49	0.0638
Ethanol	2.56	0.087
Ethylene	4.47	0.0571
Helium	0.034	0.0237
Hydrogen	0.024	0.0266
Hydrogen chloride	3.67	0.0408
Hydrogen bromide	4.45	0.0433
Methane	2.25	0.0428
Neon	0.21	0.0171
Nitric oxide	1.34	0.0279
Nitrogen	1.39	0.0319
Oxygen	1.36	0.0318
Sulphur dioxide	3.71	0.0564
Water	5.44	0.0305

1.3 COMPRESSIBILITY FACTOR (Z) :

The extent to which a real gas departs from the ideal behaviour may be expressed in terms of compressibility factor (Z),

$$Z = \frac{(PV)_{real}}{(PV)_{ideal}} = \frac{V_m}{V_{m(ideal)}} = \frac{PV_m}{RT} \quad [V_m = \text{molar volume}]$$

1.4 Plots of compressibility factor vs pressure :

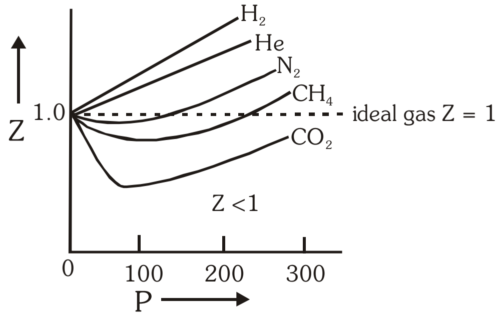
(i) For an ideal gas Z = 1 and is independent of temperature and pressure.

(ii) Exceptional behaviour of H_2 and He :
For these gases Z > 1 at 0ºC.

(iii) Effect of pressure :
At very low P, $PV_m \approx RT$ i.e. $Z \approx 1$
At low P, $PV_m < RT$ i.e. $Z < 1 \Rightarrow$ attractive forces dominant
At high P, $PV_m > RT$ i.e. $Z > 1 \Rightarrow$ repulsive forces dominant

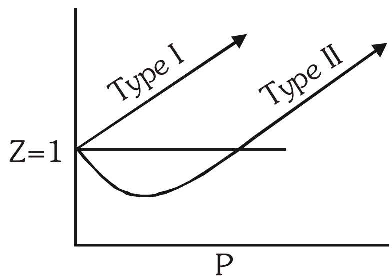
(iv) For the gases which are easily liquefied (e.g. CO_2) Z dips sharply below the ideal line in the low pressure region.

(v) Effect of temperature : An increase in temperature shows a decrease in deviation from ideal behaviours, Z approaches unity with increase in temperature.



1.5 Verification of compressibility factor using Van Der Waal's equation :

Variation of Z with P for real gas at any temperature is given by following graph.



Van der waal equation :

$$\left(P + \frac{a}{V_m^2}\right)(V_m - b) = RT$$

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(i) At low pressure and constant temperature

At low pressure V_m will be high hence b can be neglected in comparision to V_m. But $\frac{a}{V_m^2}$ can't be neglected as pressure is low. Thus equation would be

$$\left(P+\frac{a}{V_m^2}\right)V_m = RT$$

$$PV_m + \frac{a}{V_m} = RT$$

$$\frac{PV_m}{RT} + \frac{a}{V_mRT} = 1$$

$$\mathbf{Z = 1 - \frac{a}{V_mRT}} \quad \Rightarrow \quad Z < 1$$

Substituting $V_m = \frac{RT}{P}$ in above equation ; $\mathbf{Z = 1 - \frac{aP}{R^2T^2}}$

At low pressure, real gas is easily compressible as compared to an ideal gas.

(ii) At high pressure and constant temperature

At high pressure the V_m will be low. So b can't be neglected in comparision to V_m but $\frac{a}{V_m^2}$ can be neglected as compared to much higher values of P.

Then van der Waals' equation will be

$P(V_m - b) = RT$

$PV_m - Pb = RT$

$$\frac{PV_m}{RT} = \frac{Pb}{RT} + 1$$

$$\mathbf{Z = \frac{Pb}{RT} + 1} \quad \Rightarrow \quad (Z > 1)$$

At high pressure, gas is more difficult to compress as compared to an ideal gas.

(iii) At low pressure and very high temperature.

V_m will be very large, hence 'b' can be neglected and $\frac{a}{V_m^2}$ can also be neglected as V_m is very large.

$PV_m = RT$ (ideal gas condition)

(iv) For H_2 or He $a \simeq 0$ because molecules are smaller in size or vander Waal's forces will be very weak.

$P(V_m - b) = RT$

So $\mathbf{Z = 1 + \frac{Pb}{RT}}$

This explains type I plot.

Ex.1. Calculate the pressure exerted by 5 mole of CO_2 in one litre vessel at 47°C using van der waal's equation. Also report the pressure of gas if it behaves ideal in nature.

Given that a = 3.592 atm $lt^2\ mol^{-2}$, b = 0.0427 L/mol. Also, if the volume occupied by CO_2 molecules is negligible, then calculate the pressure exerted by one mole of CO_2 gas at 273 K.

Sol. Vander waal's equation

$$\left[p+\frac{n^2a}{V^2}\right][V - nb] = nRT$$

$n_{CO_2} = 5$, V = 1 litre, T = 320 K, a = 3.592, b = 0.0427

$$\therefore \quad \left[P+25\times\frac{3.592}{1}\right][1 - 5 \times 0.0427] = 5 \times 0.0821 \times 320$$

$$\therefore \quad P = 77.218 \text{ atm}$$

For ideal behaviour of gas, PV = nRT

$$\therefore \quad P \times 1 = 5 \times 0.0821 \times 320$$

$$\therefore \quad P = 131.36 \text{ atm}$$

For one mole $\left[P+\frac{a}{V^2}\right][V - b] = RT$

$$\therefore \quad P = \frac{RT}{V} - \frac{a}{V^2}$$

$$\therefore \quad P = \frac{0.0821\times 273}{22.4} - \frac{3.592}{(22.4)^2}$$

$$\therefore \quad P = 0.9922 \text{ atm}$$

The volume occupied by 1 mole at 273 K is 22.4 litre if b is negligible.

Ex.2 One mole of CCl_4 vapours at 77°C occupies a volume of 35.0 L. If vander waal's constants are a = 20.39 L^2 atm mol^{-2} and b = 0.1383 L mol^{-1}, calculate compressibility factor Z under,

(a) low pressure region. (b) high pressure region.

Sol. (a) Under low pressure region, V is high

$$\therefore \quad (V - b) \approx V$$

$$\left(P+\frac{a}{V^2}\right)V = RT$$

$$PV + \frac{a}{V} = RT$$

$$\frac{PV}{RT}+\frac{a}{RTV} = 1$$

$$Z = \frac{PV}{RT} = \left(1-\frac{a}{RTV}\right) = 1 - \frac{20.39}{0.0821\times 350\times 35} = 0.98$$

(b) Under high pressure region, P is high,

$$\left(P+\frac{a}{V^2}\right) \approx P$$

$$\therefore \quad P(V - b) = RT$$

$$PV - Pb = RT$$

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$$\frac{PV}{RT}-\frac{Pb}{RT} = 1$$

$\therefore \quad Z = \frac{PV}{RT} = 1 + \frac{Pb}{RT} \qquad \left(\because \frac{PV}{RT}=1, \text{ or } \frac{P}{RT}=\frac{1}{V}\right)$

$$Z = 1 + \frac{b}{V} = 1 + \frac{0.1383}{35} = 1 + 0.004 = 1.004$$

Ex.3 One way of writing the equation of state for real gases is $PV = RT\left[1+\frac{B}{V}+...\right]$ where B is a constant. Derive an approximate expression for B in terms of van der Waal's constants a and b.

Sol. According to van der Waal's equation

$$\left[P+\frac{a}{V^2}\right][V - b] = RT \text{ or } \quad P = \frac{RT}{(V-b)}-\frac{a}{V^2}$$

Multiply by V, then

$$PV = \frac{RTV}{(V-b)}-\frac{a}{V} \quad \text{or} \quad PV = RT\left[\frac{V}{V-b}-\frac{a}{VRT}\right]$$

or $\quad PV = RT\left[\left(1-\frac{b}{V}\right)^{-1}-\frac{a}{VRT}\right]$

$\because \quad \left[1-\frac{b}{V}\right]^{-1} = 1 + \frac{b}{V} + \left(\frac{b}{V}\right)^2 + ...$

$\therefore \quad PV = RT\left[1+\frac{b}{V}+...-\frac{a}{VRT}\right]$

$$PV = RT\left[1+\left(b-\frac{a}{RT}\right).\frac{1}{V}+...\right]$$

$\therefore \quad B = b - \frac{a}{RT}$

2. BOYLE TEMPERATURE :

(i) It is temperature at which a real gas behave ideally in a wide range of pressure.

(ii) (a) Tamperature < Boyle temperature

Z < 1, low pressure range

Z > 1, high pressure range

(b) Temperature = Boyle temperature

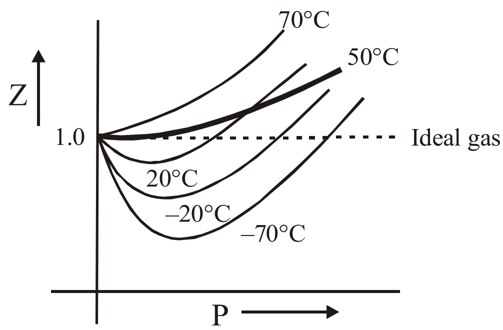
Z = 1, low pressure range

Z > 1, high pressure range

(c) At temperature > Boyle temperature

Z > 1, at all pressure

(d) $T \rightarrow \infty, Z \rightarrow 1$



On increasing the temperature, the thermal energy increases and simultaneously the attractive forces decreases. Hence a temperature comes at which the thermal energy become too high that it balances the effect of attraction and gas molecules becomes independent.

If at Boyle temperature, pressure is increased, molecules come more closer. Due to repulsive force, Z becomes greater that 1.

2.1 CALCULATION OF T_B :

(i) $PV_m = RT = \left(1 + \frac{B}{V_m} + \underbrace{\frac{C}{V_m^2} + ...}_{\text{negligible}}\right)$ at low pressure.

At T = T_B, the second initial coefficient should be 0

B = 0

or $b - \frac{a}{RT_B} = 0$

$\therefore \boxed{T_B = \frac{a}{Rb}}$

(ii) Calculus method :

At Boyle temperature, $\left(\frac{\partial Z}{\partial P}\right)_T = 0$ at low pressure.

Ex.4 Derive the expresion for compresibility factor of a vanderwaal gas at Boyle temperature.

Solution : $Z = \frac{V_m}{V_m - b} - \frac{a}{V_m - RT}$

At Boyle temperature,

$$Z = \frac{V_m}{V_m - b} - \frac{a}{V_m R \times \frac{a}{Rb}}$$

$$Z = \frac{V_m}{V_m - b} - \frac{b}{V_m}$$

$$Z = \frac{V_m^2 - V_m b + b^2}{V_m (V_m - b)}$$

$$Z = 1 + \frac{b^2}{V_m (V_m - b)}$$

Ex.5 Calculate the volume occupied by 2 moles of a vanderwall gas at 5 atm 800 K.

Given : a = 4.0 atm ℓ^2 mol^{-2} , b = 0.0625 ℓ mol^{-1} , R = 0.08 ℓ-atm/K-mol

Solution : $T_B = \frac{a}{Rb} = 800K$

Gas behave ideally at given condition.

PV = nRT

5 × V = 2 × 0.08 × 800

V = 25.6 litre

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3. LIQUEFACTION OF GASES AND CRITICAL POINTS

The phenomenon of converting a gas into liquid is known as liquefaction. The liquefaction of a gas takes place when the intermolecular forces of attraction become so high that they exist in liquid state. A gas can be liquefied by :

(a) **Increasing pressure :** An increase in pressure results decrease in intermolecular distance.

(b) **Decreasing temperature** : A decrease in temperature results decrease in kinetic energy of molecules.

Note : Due to absence of intermolecular forces, ideal gases can never be liquified.

3.1 Andrews Isotherms :

The essential conditions for liquefaction of gases were discovered by Andrews (1869) as a result of his study of P–V–T relationship for CO_2.The types of isotherms are shown in figure.

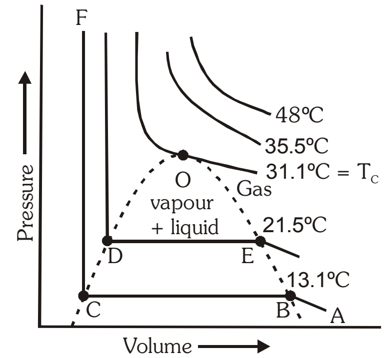


Fig : Isotherms for carbon dioxide

Observations from figure :

(a) **At low temperatures :** For the curve ABCF, as the pressure increases, volume of the curve decreases (curve A to B). At point B, at constant pressure, liquefaction commences and the volume decreases rapidly (because gas is converted to liquid with higher density). At point C, liquefaction is complete. The line CF represents the variation of V with P of the liquid state. The steepness of the line CF indicates that the liquid cannot be easily compressed. Thus AB, represent gaseous state, BC represent liquid and vapour in equilibrium and CF represent liquid state.

The pressure corresponding to the line BC is vapour pressure of the liquid at that temperature.

(b) **At lower temperatures** : Similar type of curve as in case (A) is obtained but the width of the horizontal portion is reduced. The pressure corresponding to this portion is higher than at lower temperatures.

(c) **At high temperatures** : (say 48°C), the isotherms are like those of ideal gas. Gas does not liquify, even at very high pressure.

(d) **At temperature (31.1°C)** : The horizontal portion is reduced to a point.

The isotherm at T_C is called **critical isotherm**.

At point O, $\frac{dP}{dV} = 0$.

The point O is called the **point of inflection**.

3.2 Critical parameters or critical constants :

Critical temperature (T_C) : The temperature above which a system can never be liquefied by the application of pressure alone i.e. the temperature above which a liquid cannot exit is called the critical temperature T_C.

Critical pressure (P_C) :The minimum pressure required to liquefy the system at the temperature T_C is called the critical pressure P_C.

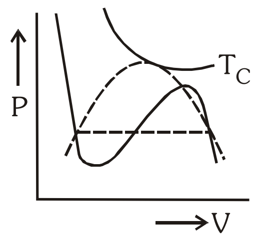
Critical volume (V_C) :The volume occupied by one mole of the system at critical temperature, T_C and critical pressure, P_C is called the critical volume (V_C) of the gas.

3.3 Determination of value of P_C, V_C and T_C :

The Vander waal's equation is

$$\left(P+\frac{a}{V_m^2}\right)(V_m - b) = RT$$

or $$V_m^3 - \left(b+\frac{RT}{P}\right)V_m^2 + \frac{a}{P}V_m - \frac{ab}{P} = 0 \qquad \text{.... (1)}$$



This equation has three roots in V_m for given values of a, b, P and T. It is found that either all the three roots are real or one is real and the other two are imaginary.

At temperature lower than T_C , the isotherm exhibits a maximum and a minimum for certain values of pressures, the equation gives three roots of volume e.g., V_1, V_2 and V_3 at pressure P_1. On increasing the temperature, the three roots become closer to each other and ultimately at critical temperature, they become identical. Thus, the cubic equation V_m can be written as

$$(V_m - V')(V_m - V'')(V_m - V''') = 0$$

At the critical point $V' = V'' = V''' = V_C$

$\therefore$ the equation becomes,

$$(V_m - V_C)^3 = 0$$

or $$V_m^3 - V_C^3 - 3V_C V_m^2 + 3V_C^2 V_m = 0 \qquad \text{.....(2)}$$

By comparing the coefficients in eq.(1) and eq(2)

$$3V_C = b + \frac{RT_C}{P_C}, \quad 3V_C^2 = \frac{a}{P_C}, \quad V_C^3 = \frac{ab}{P_C}$$

By solving, $\mathbf{V_C = 3b}$, $\mathbf{P_C = \frac{a}{27b^2}}$ **and** $\mathbf{T_C = \frac{8a}{27Rb}}$

- The value of critical compressibility factor in terms of vander wall's constants is given by

$$Z = \frac{P_C V_C}{RT_C} = \frac{\frac{a}{27b^2}\times 3b}{R\times\frac{8b}{27Rb}} = \frac{3}{8} = 0.375$$

If we compare the value of $\frac{P_C V_C}{RT_C} = 0.375$, with the experimental values, it has been found that the agreement is very poor.

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Critical constants of gases

Gas	P_C (atm)	$V_{m,c}$ (cm^3mol^{-1})	T_C (K)
He	2.26	57.9	5.2
Ne	26.9	41.7	44.4
Ar	48.1	75.2	150.7
Xe	58.0	119.0	289.7
H_2	12.8	65.5	33.3
O_2	50.1	78.2	154.8
N_2	33.5	90.1	126.2
CO_2	72.8	94.0	304.2
H_2O	218.0	55.6	647.3
NH_3	111.5	72.5	405.0
CH_4	45.6	98.7	190.6
C_2H_6	48.2	148.0	305.4

Ex.6 The critical temperature and pressure of CO_2 gas are 304.2 K and 72.9 atm respectively. What is the radius of CO_2 molecule assuming it to behave as vander Waal's gas ?

Sol. T_C = 304.2 K P_C = 72.9 atm

$$T_C = \frac{8a}{27Rb} \qquad P_C = \frac{a}{27b^2}$$

$$\therefore \quad \frac{T_C}{P_C} = \frac{\frac{8a}{27Rb}}{\frac{a}{27b^2}} = \frac{8a}{27Rb} \times \frac{27b^2}{a} = \frac{8b}{R}$$

or $b = \dfrac{RT_C}{8P_C} = \dfrac{1}{8} \times \dfrac{0.082 \times 304.2}{72.9} = 0.04277 \text{ lit mol}^{-1}$

$$b = 4\,N_A \times \frac{4}{3}\pi r^3 = 42.77 \text{ cm}^3$$

or $r = (4.24)^{1/3} \times 10^{-8}$ cm $= 1.62 \times 10^{-8}$ cm

$\therefore$ radius of CO_2 molecule = 1.62 Å

4. THE LIQUID STATE

Liquid state is intermediate between gaseous and solid states. The liquids posses fluidity like gases but incompressibility like solids.

The behaviour of liquids explained above gives some characteristic properties to the liquids such as definite shape, incompressibilty, diffusion, fluidity (or viscosity), evaporation (or vapour pressure), surface tension, etc.

The following general characteristics are exhibited by liquids :

(i) Shape :

Liquids have no shape of their own but assume the shape of the container in which they are kept. No doubt, liquids are mobile but they do not expand like gases as to fill up all the space offered to them but remain confined to the lower part of the container.

(ii) Volume :

Liquids have definite volume as the molecules of a liquid are closely packed and the cohesive forces are strong.

(iii) Density :

As the molecules in liquids are closely packed, the densities of liquids are much higher than in gaseous state. For example, density of water at 100° C and 1 atmospheric pressure is 0.958 g mL^{-1} while that of water vapour under similar conditions as calculated from ideal gas law $\left(d = \frac{MP}{RT}\right)$ is 0.000588 g mL^{-1}.

(iv) Compressibility :

The molecules in a liquid are held in such close contact by their mutual attractive forces (cohesive forces) that the volume of any liquid decreases very little on increasing pressure. Thus, liquids are relatively incompressible compared to gases.

(v) Diffusion :

When two miscible liquids are put together, there is slow mixing as the molecules of one liquid move into the other liquid. As the space available for movement of molecules in liquids is much less and their velocities are small. Liquids diffuse slowly in comparison to gases.

(vi) Evaporation :

The process of changes of liquid into vapour state on standing is termed **evaporation**. Evaporation may be explained in terms of motion of molecules. At any given temperature all the motion of molecules do not possess the same kinetic energy (average kinetic energy is, however same). Some molecules move slowly, some at intermediate rates and some move very fast. A rapidly moving molecule near the surface of the liquid may possess sufficient kinetic energy to overcome the attraction of its neighbours and escape. Evaporation depends on the following factor.

(a) Nature of the liquid : The evaporation depends on the strength of intermolecular forces (cohesive forces). The liquids having low intermolecular forces evaporate faster in comparison to the liquids having high intermolecular forces. For example, ether evaporates more quickly than alcohol, and alcohol evaporates more quickly than water, as the intermolecular forces in these liquids are in the order :

Ether < Alcohol < Water

$$\underset{\text{Increasing extent of hydrogen bonding}}{\xrightarrow{\text{Alcohol < glycol < glycerol}}}$$

(b) Surface area : Evaporation is a surface phenomenon. Larger the surface area, greater is the oportunity of the molecules to escape. Thus, rate of evaporation increases with increase of surface area.

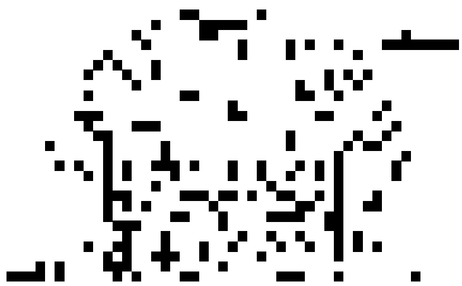
(c) Temperature : Rate of evaporation increases with the increase of temperature as the kinetic energy of the molecules increases with the rise of temperature.

(d) Flow of air current over the surface : Flow of air helps the molecules to go away from the surface of liquid and, therefore, increases the evaporation of liquid in open vessel.

(vii) Heat of vaporisation :

The quantity of heat required to evaporate a unit mass of a given liquid at constant temperature is known as heat of vaporisation. The heat of vaporisation depends on the strength of the intermolecular forces with in the liquid. The value of heat of vaporisation generally decreases with increase in temperature. It becomes zero at the critical temperature. When the vapour is condensed into a liquid, heat is evolved. This is called **heat of condensation**. It is numerically equal to the heat of vaporisation at the same temperature.

(viii) Vapour pressure :



When the space above the liquids is closed, the molecules cannot escape into open but strike the walls of the container, rebound and may strike the surface of the liquid, where they may be trapped. The return of the molecules from the vapour state of the liquid state is known as **condensation**. As evaporation proceeds, the number of molecules in the vapour state increases and, in turn, the rate of condensation increases.

The rate of condensation soon becomes equal to the rate of the evaporation, i.e., the vapour in the closed container is in equilibrium with the liquid.

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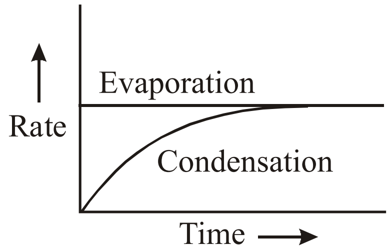

At equilibrium the concentration of molecules in the vapour phase remains unchanged. The pressure exerted by the vapour in equilibrium with liquid, at a given temperature, is called the **vapour pressure**. Mathematically, it may be given by ideal gas equation, assuming ideal behaviour.

$$P = \frac{n}{V}RT = CRT$$

where C is the concentration of vapour, in mol/litre.

Since the rate of evaporation increases and rate of condensation decreases with increasing temperature, vapour pressure of liquids always increases as temperature increases. At any given temperature, the vapour pressures of different liquids are different because their cohesive forces are different. Easily vaporisable liquids are called **volatile liquids** and they have relatively high vapour pressure. Vapour pressure values (in mm of Hg) for water, alcohol and ether at different temperatures are given in the following table :

Substance	Temperatures				
	0° C	20° C	40° C	80° C	100° C
Water	4.6	17.5	55.0	355.5	760.3
Ethyl alcohol	12.2	43.9	812.6	1693.3	
Diethyl ether	185.3	442.2	921.1	2993.6	4859.4

The vapour pressure of a given liquid at two differnt temperatures can be compared with the help of **Clausius-Clapeyron equation.**

$$\log\times\frac{P_2}{P_1} = \frac{\Delta H}{2.303R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

Where ΔH is the latent heat of vaporisation and R is the molar gas constant.

(ix) Boiling point :

The temperature at which the vapour pressure of the liquid becomes equal to the atmospheric pressure is called the **boiling point** of the liquid. When a liquid is heated under a given applied pressure, bubbles of vapour begin to form below the surface of the liquid. They rise to the surface and burst releasing the vapour into the air. This process is called **boiling**. The normal boiling point is the temperature at which the vapour pressure of a liquid is equal to exactly one atmospheric pressure (760 mm of Hg). Figure shows that normal boiling points of di-ethyl ether, ethyl alcohol and water are 34.6° C, 78.5° C and 100° C respectively.

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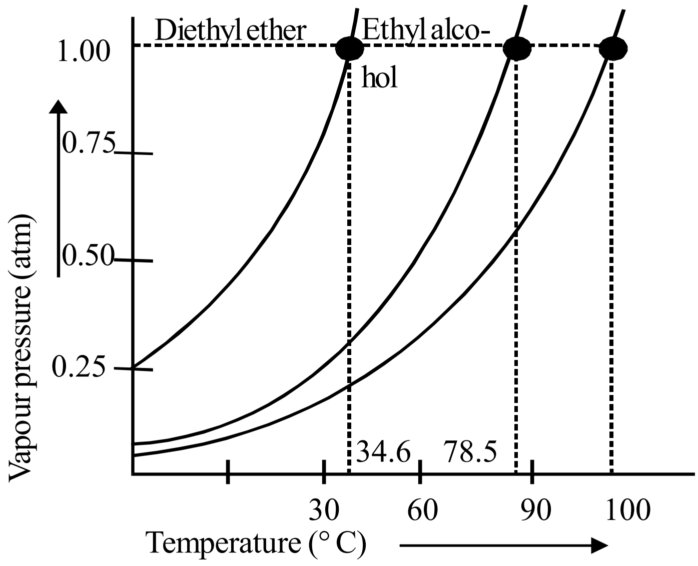


The temperature of the liquid remains constant until all the liquid has been vaporised. Heat must be added to the boiling liquid to maintian the temperature because in the boiling process, the high energy molecules are lost by the liquid.

The boiling point of a liquid changes with the change in external pressure. A liquid may boil at temperatures higher than normal under external pressures greater than one atmoshpere; conversely, the boiling point of a liquid may be lowered than normal below one atmosphere. Thus, at high altitudes where the atmospheric pressure is less than 760 mm, water boils at temperatures below its normal boiling water.

Boiling and evaporation are similar processes (conversion of liquid into vapour) but differ in following respects :

(a) Evaporation takes place spontaneously at all temperatures but boiling occurs at a particular temperature at which the vapour prssure is equal to the atmospheric pressure.

(b) Evaporation is surface phenomenon. It occurs only at the surface of the liquid whereas boiling involves formation of bubbles below the surface of the liquid.

Note. : Boiling does not occur when liquid is heated in a closed vessel.

(x) Freezing point :

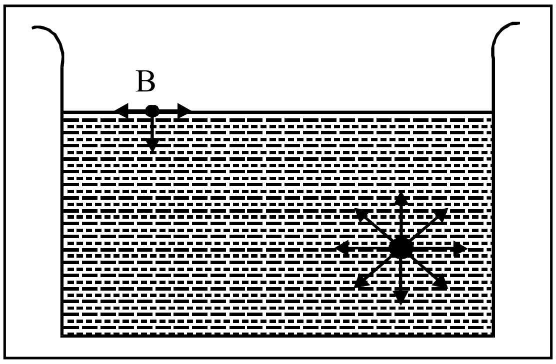
When a liquid is cooled, the kinetic energy of the molecules goes on decreasing. A stage comes when the intermolecular forces become stronger and the rotational motion is siezed. At this stage, the formation of solid begins and the liquid is seen to freeze out. The point (temperature) at which the vapour pressure of solid and liquid forms of a substance become equal is termed as **freezing point.**

Normal freezing point of a liquid is the temperature at which is liquid and solid forms are in equilibrium with each other under a pressure of one atmosphere. The freezing point of a liquid is the same as the melting point of its solid. The amount of heat that must be removed to freeze a unit mass of the liquid at the freezing point, which is called the **heat of fusion.**

The freezing point of a liquid is affected by the change of external pressure. With increased external pressure, the freezing point of some liquids rises while of others falls. But the effect of pressures is very small because solid as well as liquid are almost incompressible.

(xi) Surface tension :

It is the property of liquids caused by the intermolecular attractive forces. A molecule within the bulk of the liquid is attracted equally in all the directions by the neighbouring molecules. The resultant force on any one molecule in the centre of the liquid is, therefore, zero. However, the molecules on the surface of the liquid are attracted only inward and sideways. This unbalanced molecular attraction pulls some of the molecules into the bulk of the liquid, i.e., are pulled inward and the surface area is minimized.



Surface tension is a measure of this inward force on the surface of the liquid. It acts downwards perpendicular to the plane of the surface. The unit of surface tension is dyne cm^{-1}. Surface tension is, thus, defined **as the force acting on the surface at right angles to any line of unit length.**

As the intermolecular forces of attraction decreases with the rise of temperature, the surface tension of a liquid, thus, decreases with increase in temperature. Similarly, addition of chemicals to a liquid may reduce its surface tension. For example, addition of chemicals like soaps, detergents, alcohol, camphor, etc., lowers the surface tension of water.

Many common phenomenon can be explained with the help of surface tension. Some are described here :

(a) Small droplets are spherical in shape : The surface tension acting on the surface of the liquid tries to minimise the surface area of a given mass of a liquid. It is known that for a given volume, a sphere has the minimum surface area. On account of this, drops of liquids acquire a spherical shape.

(b) Insects can walk on the surface of water : Many insects can walk on the surface of water without drowing. This is due to the existence of surface tension. The surface tension makes the water surface to behave like an elastic membrane and prevents the insects from drowning.

(c) Cleaing action of soap and detergents : Soap and detergent solutions due to their lower surface tensions penetrate into the fibre and surround the greasy substances and wash them away.

(d) Capillary action : The tendency of a liquid to rise into narrow pores and tiny openings is called capillary action. The liquids rise in the capillary tubes due to the surface tension.

(e) Surface Energy : The work required to be done to increase or extend surface area by unit area is called surface energy. The units of surface energy are, therfore, erg per sq. cm (or joule per sq. metre, i.e. J m^{-2} in S.I. system)

(xii) Viscosity :

It is one of the characteristic properties of liquids. Viscosity is a measure of resistance to flow which arises due to the internal friction between layers of fluid as they slip past one another while liquid flows. Strong intermolecular forces between molecules hold them together and resist movement of layers past one another.

When a liquid flows over a fixed surface, the layer of molecules in the immediate contact of surface is stationary. The velocity of upper layers increase as the distance of layers from the fixed layer increases. This type of flow in which there is a regular gradation of velocity in passing from one layer to the next is called **laminar flow**. If we choose any layer in the flowing liquid , the layer above it accelerates its flow and the layer below this retards its flow.

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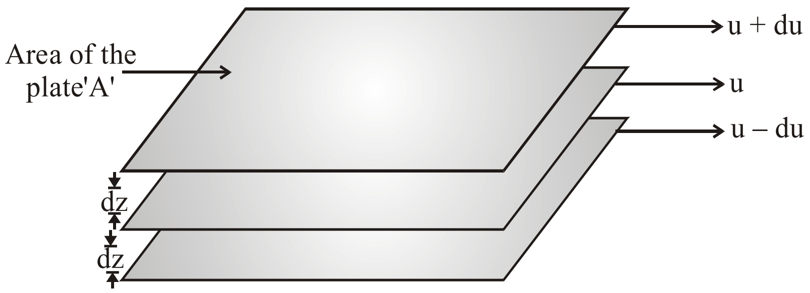

Gradation of velocity in the laminar flow

If the velocity of the layer at a distance dz is changed by a value du then velocity gradient is given by the amount $\frac{du}{dz}$. A force is required to maintain the flow of layers. This force is proportional to the area of contact of layers and velocity gradient i.e.

$F \propto A$ (A is the area of contact)

$F \propto A \frac{du}{dz}$ (where, $\frac{du}{dz}$ is velocity gradient; the change in velocity with distance)

$$F \propto A.\frac{du}{dz} \Rightarrow F = \eta A \frac{du}{dz}$$

'η' is proportionality constant and is called **coefficient of viscosity**. Viscosity coefficient is the force when velocity gradient is unity and the area of contact is unit area. Thus 'h' is measure of viscosity. SI unit of viscosity coefficient is 1 newton second per square metre ($N\ s\ m^{-2}$) = pascal second ($Pa\ s = 1 kg\ m^{-1}s^{-1}$). In cgs system the unit of coefficient of viscosity is poise (named after great scientist Jean Louise Poiseuille).

1 poise = $1\ g\ cm^{-1}s^{-1} = 10^{-1} kg\ m^{-1}s^{-1}$

Greater the viscosity, the more slowly the liquid flows. Hydrogen bonding and van der Waals forces are strong enough to cause high viscosity. Glass is an extremely viscous liquid. It is so viscous that many of its properties resemble solids. However, property of flow of glass can be experienced by measuring the thickness of windowpanes of old buildings. These become thicker at the bottom than at the top.

Viscosity of liquids decreases as the temperature rises because at high temperature molecules have high kinetic energy and can overcome the intermolecular forces to slip past one another between the layers.

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EXERCISE # S-I

1. Calculate the pressure exerted by 22 g of carbon dioxide in 0.5 dm^3 at 300 K using:

(a) the ideal gas law and

(b) Van der Waal's equation respectively.

Given : [a = 3.6 atm litre^{2}mol^{-2} , b = 0.04 litre mol^{-1}, R = 0.08 L-atm/K-mol]

2. Calculate from the Van der Waals equation, the temperature at which 192 g of SO_2 would occupy a volume of 6 dm^3 at 15 atm pressure.[a = 5.68 atm L^2 mol^{-2}, b = 0.06 L mol^{-1}]

3. The density of water vapour at 328.4 atm and 800 K is 135.0 g/dm^3. Determine the molar volume, V_m and the compression factor of water vapour .

4. At 300 K and under a pressure of 10.1325 MPa, the compressibility factor of O_2 is 0.9. Calculate the mass of O_2 necessary to fill a gas cylinder of 45 dm^3 capacity under the given conditions.
[R = 0.08L-atm/K-mol]

5. 1 mole of CCl_4 vapours at 27°C occupies a volume of 40 L. If Van der Waals constants are 24.6 L^2 atm mol^{-2} and 0.125 Lmol^{-1}, then, calculate compressibility factor in

(a) Low pressure region (b) High Pressure region [R = 0.082 L -atm/K-mol]

6. If at 200 K & 500 atm, density of CH_4 is 0.246 gm/ml then its compressibility factor (Z) is approx 2.0×10^x. 'x' is:

7. Certain mass of a gas occupy 500 ml at 2 atm and 27ºC. Calculate the volume occupied by same mass of the gas at 0.3 atm and 227ºC. The compressibility factors of gas at the given condition are 0.8 and 0.9, respectively.

BOYLE TEMPERATURE

8. The vander waal's constant for a gas are a = 1.92 atm L^2 mol^{-2}, b = 0.06L mol^{-1}. If R = 0.08 L atm $K^{-1}mol^{-1}$, what is the Boyle's temperatrue of this gas.

9. The Van der Waals constant for O_2 are a = 1.642 atm L^2 mol^{-2} and b = 0.04 L mol^{-1}. Calculate the temperature at which O_2 gas behaves ideally for longer range of pressure.

LIQUIFICATION OF GASES, CRITICAL PHENOMENON

10. The Van der Waals constants for gases A, B and C are as follows

Gas	**a[/atm L^2 mol^{-2}]**	**b[/L mol^{-1}]**
A	8.21	0.050
B	4.105	0.030
C	1.682	0.040

Which gas has (i) the highest critical temperature, (ii) the largest molecular volume, and (iii) most ideal behaviour around 500 K?

11. For a real gas, if at critical conditions molar volume of gas is 8.21 litre at 3 atm, then critical temperature (in K) will be :

12. An unknown gas behaves ideally at 540K in low pressure region, then calculate the temperature (in K) below which it can be liquified by applying pressure

EXERCISE # S-II

1. The density of mercury is 13.6 g/cm^3. Estimate the value of **'b'** (in cm^3/ mole).
2. The molarity of O_2 gas at 72 atm and 300K is 6M. Calculate the value of Z for O_2.
(Use : R = 0.08 atm-litre/K-mole).
3. Calculate the amount of He (in gm) present in the 10 litre container at 240 atm and 300K. Given value of "b" for He is 0.08 dm^3 mol^{-1} ; R = 0.08 atm lit mol^{-1} K^{-1}.
4. For a real gas (mol. mass = 30) if density at critical point is 0.40 g/cm^3 and its $T_c = \frac{2\times10^5}{821}$ K, then calculate Van der Waals constant a (in atm L^{2}mol^{-2}).
5. Calculate the volume occupied by 0.2 mole of a Vander waal gas at 27ºC and 0.0821 atm.
[a = 4.105 L^2 atm mol^{-2}, b = $\frac{1}{6}$Lmol^{-1}]
6. At what pressure and 127ºC, the density of O_2 gas becomes 1.6 g/L ?
[a = 4.0 atm L^2 mol^{-2} , 0.4 Lmol^{-1}, R = 0.08 L - atm/ K-mol]

EXERCISE # O-1

1. The correct expression for the Van der Waals equation of state is :

(A) $\left(P+\frac{a}{n^2V^2}\right)(V-nb)=nRT$ (B) $\left(P+\frac{an^2}{V^2}\right)(V-nb)=\Delta nRT$

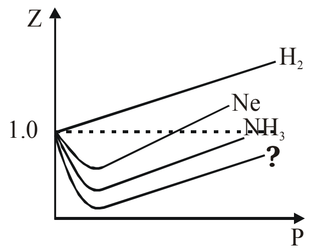
(C) $\left(P+\frac{an^2}{V^2}\right)(V-b)=nRT$ (D) $\left(P+\frac{an^2}{V^2}\right)(V-nb)=nRT$

2. At relatively high pressure, Van der Waals equation reduces to :

(A) $PV_m=RT$ (B) $PV_m=RT+\frac{a}{V_m}$

(C) $PV_m = RT + Pb$ (D) $PV_m = RT-\frac{a}{V_m^2}$

3. Observe the following Z vs P graph.



The missing gas in the above graph can be :

(A) He (B) Ar (C) C_5H_{12} (D) All are correct

4. Correct option regarding a container containing 1 mol of a gas in 22.4 litre container at 273 K is

(A) If compressibility factor (Z) > 1 then 'P' will be less than 1 atm.

(B) If compressibility factor (Z) > 1 then 'P' will be greater than 1 atm.

(C) If 'b' dominates, pressure will be less than 1 atm.

(D) If 'a' dominates, pressure will be greater than 1 atm.

5. If 'V' is actual volume of 1 molecule of gas then, excluded volume (b) of 1 mole of gaseous molecule is -

(A) $4\ N_AV$ (B) N_AV (C) V/N_A (D) V

6. Consider the equation $Z=\frac{PV}{RT}$, Which of the following statements is correct :

(A) When Z > 1 real gases are easier to compress

(B) When Z = 1 real gases are easier to compress

(C) When Z > 1 real gases are difficult to compress

(D) When Z < 1 real gases are difficult to compress

7. Compressibility factor of ideal gas is :-

(A) z > 1 (B) z > 1 (C) z = 1 (D) $z=\infty$

8. The density of a gaseous substance at 1 atm pressure and 750 K is 0.30 g/lt. If the molecular weight of the substance is 27, the dominant forces existing among gas molecules is -

(A) Attractive (B) Repulsive

(C) Both (A) and (B) (D) None of these

9. The third virial coefficient of a He gas is 4×10^{-2} $(\text{lit/mol})^2$, then what will be volume of 2 mole He gas at 1 atm 273K (273K > T_B)

(A) 22.0 lit (B) 44.0 lit (C) 44.8 lit (D) 45.3 lit

10. At low pressure the vander waals equation is reduced to -

(A) $Z = \dfrac{pV_m}{RT} = 1 - \dfrac{a}{RTV_m}$ (B) $Z = \dfrac{pV_m}{RT} = 1 + \dfrac{a}{RT}p$

(C) $pV_m = RT$ (D) $Z = \dfrac{pV_m}{RT} = 1 - \dfrac{a}{RT}$

11. The values of Van der Waals constant 'a' for the gases O_2, N_2, NH_3 and CH_4 are 1.360, 1.390, 4.170 and 2.253 L atm mol^{-2} respectively. The gas which can most easily be liquefied is :

(A) O_2 (B) N_2 (C) NH_3 (D) CH_4

12. The values of critical temperatures of few gases are given gases :

Gases :	H_2	He	O_2	N_2
T_C(K)	33.2	5.2	154.3	126

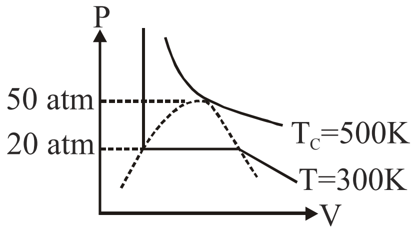
From the above data arrange the given gases in the increasing order of ease of their liquification.

(A) O_2, N_2, H_2, He (B) He, N_2, O_2, H_2

(C) He, H_2, N_2, O_2 (D) H_2, N_2, O_2, He

EXERCISE # O-II

1. For real gas the P–V curve was experiementally plotted and it had the following appearance. With respect to liquification, choose the incorrect statement :



(A) At T = 500 K, P = 40 atm, the state will be liquid
(B) At T = 300 K, P = 50 atm, the state will be gas
(C) At T < 300 K, P = 20 atm, the state will be gas
(D) At 300 K < T < 500 K, P > 50 atm, the state will be liquid

2. Select the incorrect statement (s)

(A) The critical constant for a Vander Waal's gas is $V_C = 3b$, $P_C = \frac{a}{27b^2}$ and $T_C = \frac{a}{27Rb}$
(B) At 56 K a gas may be liquified if its critical temperature is –156°C.
(C) U_{avg} of gas in a rigid container can be doubled when the pressure is quadrupled by pumping in more gas at constant temperature
(D) At extremely low pressure, all real gases behave ideally.

3. A 1 litre vessel contains 2 moles of a vanderwaal's gas.

Given data : $a = 2.5$ atm-L^2 mole^{-2} T = 240 K
$b = 0.4$ L-mole^{-1} RT = 20 L-atm mole^{-1}

Identify the correct options about the gas sample :
(A) Pressure of gas = 190 atm
(B) Compressibility factor = 4.75
(C) Attraction forces are dominant in the gaseous sample
(D) T_B (Boyle temperature) = 75 K

4. Choose the correct statement(s) among the following -
(A) A gas having higher value T_C is easy to liquify
(B) The radius of molecules of gas having same value of T_C/P_C is same
(C) Hydrogen gas can be liquified at its boyle temperature by application of pressure.
(D) Real gas show negative deviation from ideal behaviour at low pressure condition.

5. Select the INCORRECT statement(s):
(A) At Boyle's temperature a real gas behaves like an ideal gas irrespective of pressure.
(B) At critical condition, a real gas behaves like an ideal gas.
(C) On increasing the temperature four times,collision frequency (Z_1) becomes double at constant volume.
(D) At high pressure Van der Waals constant 'b' dominates over 'a'.

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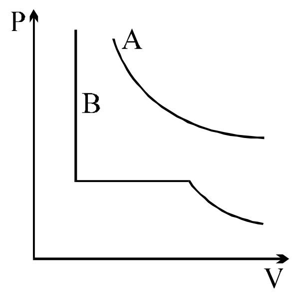
Question No. 6 & 7 (2 questions)

For two gases A and B, P v/s V isotherms are shown at same temperature, T K . T_A & T_B are critical temperatures of A & B respectively



6. Which of the following is true?

(A) $T_A < T < T_B$ (B) $T_A > T > T_B$

(C) $T_A > T_B > T$ (D) none of above

7. The correct statement(s) is/are

(I) Pressure correction term will be more negligible for gas B at T K.

(II) The curve for gas 'B' will be of same shape as for gas A if $T > T_B$

(III) Gas 'A' will show same P v/s V curve as of gas 'B' if $T > T_A$

(A) III only (B) II and III (C) II only (D) All

Match the column :

8. Match the column :

	Column-I		**Column-II**
(A)	Boyle's temperature	(P)	Depends on 'a' and 'b'
(B)	Compressibility factor	(Q)	Depends on identity of real gas
(C)	Real gas with very large molar volume	(R)	The temperature at which $\frac{dZ}{dP} = 0$ at low pressure region.
(D)	Critical temperature	(S)	PV = nRT
		(T)	$\frac{8a}{27R.b.}$

EXERCISE # JEE-MAINS

1. In Van der Waals equation of state of the gas law, the constant 'b' is a measure of : **[AIEEE-04]**
(1) intermolecular repulsions
(2) intermolecular attractions
(3) volume occupied by the molecules
(4) intermolecular collisions per unit volume

2. 'a' and 'b' are Van der Waals constants for gases. Chlorine is more easily liquefied than ethane because :- **[AIEEE-2011]**
(1) a for Cl_2 < a for C_2H_6 but b for Cl_2 > b for C_2H_6
(2) a for Cl_2 > a for C_2H_6 but b for Cl_2 < b for C_2H_6
(3) a and b for Cl_2 > a and b for C_2H_6
(4) a and b for Cl_2 < a and b for C_2H_6

3. When does a gas deviate the most from it's ideal behaviour ? **[JEE-MAINS(ONLINE)-2015]**
(1) At high pressure and low temperature
(2) At high pressure and high temperature
(3) At low pressure and low temperature
(4) At low pressure and high temperature

4. If Z is the compressibility factor, Van der Waals equation **at low** pressure can be written as : **[JEE-MAINS-2014]**

(1) $Z = 1 - \dfrac{Pb}{RT}$ (2) $Z = 1 + \dfrac{Pb}{RT}$ (3) $Z = 1 + \dfrac{RT}{Pb}$ (4) $Z = 1 - \dfrac{a}{V_mRT}$

5. Among the following, the incorrect statement is : **[JEE-Mains-2017(ONLINE)]**
(1) At low pressure, real gases show ideal behaviour
(2) At very large volume, real gases show ideal behaviour
(3) At Boyle's temperature, real gases show ideal behaviour
(4) At very low temperature, real gases show ideal behaviour

EXERCISE # JEE-ADVANCED

1. One way of writing the equation for state for real gases is, **[JEE 1997]**

$$P\overline{V} = RT\left[1 + \frac{B}{\overline{V}} + ...\right] \quad \text{where B is a constant.}$$

Derive an approximate expression for 'B' in terms of Van der Waals constants **'a'** & **'b'**.

2. Using Vander Waals equation, calculate the constant "a" when 2 moles of a gas confined in a 4 litre flask exerts a pressure of 11.0 atm at a temperature of 300 K. The value of "b" is 0.05 litre mol^{-1}.

3. A gas will approach ideal behaviour at : **[JEE 1999]**

(A) low temperature and low pressure

(B) low temperature and high pressure

(C) low pressure and high temperature

(D) high temperature and high pressure .

4. The compression factor (compressibility factor) for one mole of a Van der Waals gas at 0° C and 100 atmosphere pressure is found to be 0.5. Assuming that the volume of a gas molecule is negligible, calculate the Van der Waals constant 'a'. **[JEE 2001]**

5. The density of the vapour of a substance at 1 atm pressure and 500 K is 0.36 Kg m^{-3}. The vapour effuses through a small hole at a rate of 1.33 times faster than oxygen under the same condition.

Determine **[JEE 2002]**

(i) mol. wt.; (ii) molar volume ; (iii) compression factor (z) of the vapour and

(iv) which forces among the gas molecules are dominating, the attractive or the repulsive

6. Positive deviation from ideal behaviour takes place because of **[JEE 2003]**

(A) molecular attraction between atoms and $\frac{PV}{nRT} > 1$

(B) molecular attraction between atoms and $\frac{PV}{nRT} < 1$

(C) finite size of atoms and $\frac{PV}{nRT} > 1$

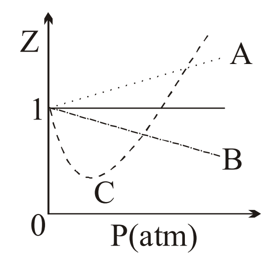
(D) finite size of atoms and $\frac{PV}{nRT} < 1$

7. For a real gas obeying Van der Waals equation a graph is plotted between PV_m (y-axis) and P(x-axis) where V_m is molar volume. Find y-intercept of the graph. **[JEE 2004]**

8. The given graph represents the variation of Z

(compressibility factor = $\frac{PV}{nRT}$) versus P, for three real

gases A, B and C. Identify the only **INCORRECT** statement.



(A) for the gas A, a = 0 and its dependence on P is linear at all pressure

(B) for the gas B, b = 0 and its dependence on P is linear at all pressure

(C) for the gas C, which is typical real gas for which neither a nor b = 0. By knowing the minima and the point of intersection, with Z = 1, a and b can be calculated.

(D) At high pressure, the slope is positive for all real gases A, B and C. **[JEE 2006]**

9. Match gases under specific conditions listed in Column I with their properties / laws in Column II. Indicate your answer by darkening the appropriate bubbles of the 4 × 4 matrix given in the ORS.

	Column I		**Column II**
(A)	Hydrogen gas (P = 200 atm, T = 273 K)	(P)	Compressibility factor $\neq 1$
(B)	Hydrogen gas (P ~ 0, T = 273 K)	(Q)	Attractive forces are dominant
(C)	CO_2 (P = 1 atm, T = 273 K)	(R)	PV = nRT
(D)	Real gas with very large molar volume	(S)	P (V– nb) = nRT

[JEE 2007]

10. A gas described by Van der Waals equation **[JEE 2008]**

(A) behaves similar to an ideal gas in the limit of large molar volumes

(B) behaves similar to an ideal gas in the limit of large pressures

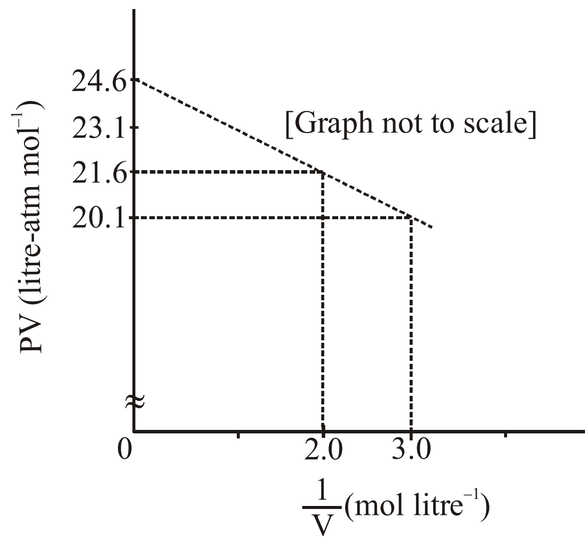
(C) is characterised by Van der Waals coefficients that are dependent on the identity of the gas but are independent of the temperature

(D) has the pressure that is lower than the pressure exerted by the same gas behaving ideally

11. The term that corrects for the attractive forces present in a real gas in the Vander Waals' equation is

(A) nb (B) $\frac{an^2}{V^2}$ (C) $-\frac{an^2}{V^2}$ (D) –nb **[JEE 2009]**

12. For one mole of a Van der Waals gas when b = 0 and T = 300 K, the PV vs. 1/V plot is shown below. The value of the Van der Waals constant a (atm. litre^2 mol^{-2}) is **[JEE 2012]**



(A) 1.0 (B) 4.5 (C) 1.5 (D) 3.0

ANSWER KEY

EXERCISE # S-I

1. (a) 24.0 atm, (b) 21.4 atm
2. 388K
3. Molar vol = 0.1333 L/mol; Z = 0.667
4. 6.67 kg
5. (a) 0.975; (b) 1.003
6. Ans. (0)
7. 6.25 L
8. Ans. (400K)
9. 500 K
10. (i) A, (ii) A, (iii) C
11. Ans. (800)
12. Ans. (160K)

EXERCISE # S-II

1. 58.82
2. Ans.(0.5)
3. Ans. (222.22)
4. 1.6875
5. 6L
6. 1.622 atm.

EXERCISE # O-I

1.	Ans. (D)	2.	Ans.(C)	3.	Ans.(C)	4.	Ans.(B)
5.	Ans. (A)	6.	Ans.(C)	7.	Ans.(C)	8.	Ans.(B)
9.	Ans. (D)	10.	Ans. (A)	11.	Ans. (D)	12.	Ans.(C)

EXERCISE # O-II

1.	Ans.(A,B,C)	2.	Ans.(A,C)	3.	Ans.(A,B,D)	4.	Ans.(A,B,D)
5	Ans.(A,B)	6.	Ans.(A)	7.	Ans.(C)		

8. Ans.(A) - P,Q,R,S ; (B) - P, Q ; (C) - S ; (D) - P, Q, T

EXERCISE # JEE-MAINS

1.	Ans.(3)	2.	Ans. (2)	3.	Ans. (1)	4.	Ans. (4)
5.	Ans. (4)						

EXERCISE # JEE-ADVANCED

1. $\left[B = b - \frac{a}{RT}\right]$
2. 6.52 atm L^2mol^{-2}
3. Ans. (C)
4. Ans. 1.256 atm L^2mol^{-2}
5. Ans.(i) 18 g/mol , (ii) 50 L mol^{-1} , (iii) 1.218 , (iv) repulsive
6. Ans.(C)
7. Ans.RT
8. Ans.(D)
9. Ans.(A) P, S; (B) R; (C) P, Q; (D) R
10. Ans.(A,C,D)
11. Ans.(B)
12. Ans.(C)

LIQUID SOLUTION

1. SOLUTION

It is the homogeneous mixture of two or more components.

The substance which dissolve other substance is **solvent** & the substance which is dissolved is **solute**, independent of their quantity. If both are soluble in each other then the substance present in larger amount by mole is solvent.

* A solution may exist in any physical state.

Types of Solution :

	Solvent	Solute	Examples
1.	Gas	Gas	Mixture of gases, eg. air.
2.	Gas	Liquid	$CHCl_3(\ell) + N_2(g)$
3.	Gas	Solid	Camphor (s) + N_2(g).
4.	Liquid	Gas	CO_2 gas dissolve in water (aerated drink), soda water.
5.	Liquid	Liquid	Mixture of miscible liquids e.g. alcohol in water.
6.	Liquid	Solid	Salt in water, sugar in water.
7.	Solid	Gas	hydrogen over palladium.
8.	Solid	Liquid	Mercury in zinc, mercury in gold, i.e. all amalgams.
9.	Solid	Solid	Alloys e.g. copper in gold, zinc in copper.

2. SOLUBILITY

Maximum amount of solute which can be dissolved in a specified amount of solvent at constant temperature is to solubility. Solubility is affected by

1. nature of solute and solvent
2. temperature and
3. pressure

3. SOLUBILITY OF SOLID IN A LIQUID

Polar solutes are soluble in polar solvent and non polar solutes are soluble in non polar solvent due to similar intermolecular forces.

When solid solutes are dissolved in solvent, then following equilibrium exits.

Solute + Solvent $\rightleftharpoons$ Solution. ; ΔH may be positive or negative

Such a solution in which no more solute can be dissolved at the same temperature and pressure is called a **saturated solution**. An unsaturated solution is one in which more solute can be dissolved at the same temperature. The solution which is in dynamic equilibrium with undissolved solute is the saturated solution and contains the maximum amount of solute dissolved in a given amount of solvent.

3.1 Effect of temperature :

The solubility of a solid in a liquid is significantly affected by temperature changes, obeying ***Le Chateliers Principle***. In general, if in a nearly saturated solution, the dissolution process is endothermic ($\Delta_{sol}H > 0$), the solubility should increase with rise in temperature and if it is exothermic ($\Delta_{sol}H < 0$) the solubility should decrease. These trends are also observed experimentally.

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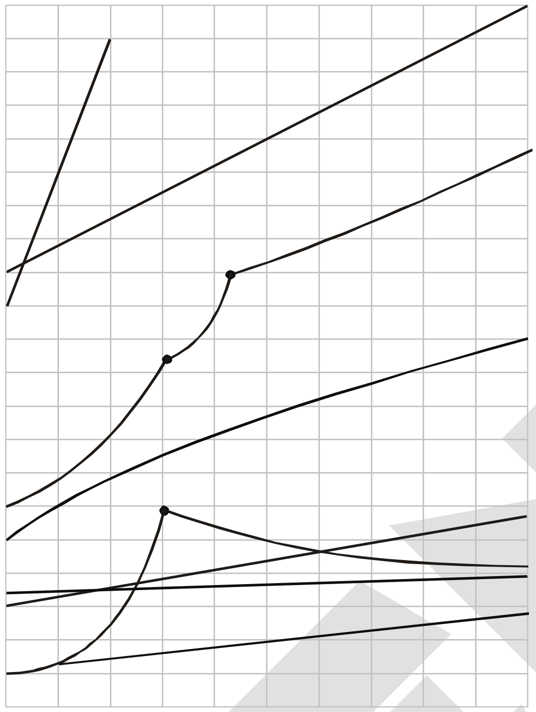


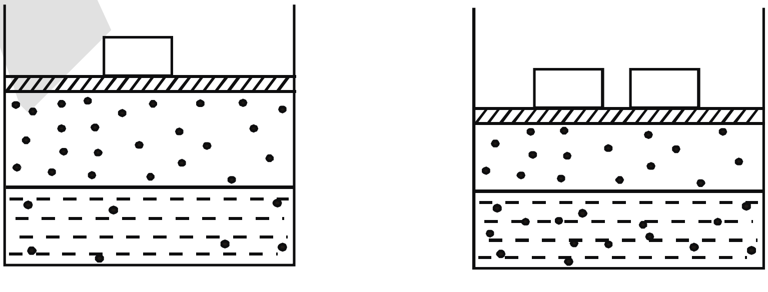
3.2 Effect of pressure :

Pressure does not have any significant effect on solubility of solids in liquids. It is so because solids and liquids are highly incompressible and practically remain unaffected by changes in pressure.

4. SOLUBILITY OF GASES IN LIQUID

Certain gases are highly soluble in water like NH_3, HCl, etc, and certain gases are less soluble in water like O_2, N_2, He, etc. Solubility of gases is greatly effected by pressure and temperature. Increasing pressure increases solubility and increase in temperature decreases solubility since dissolution of any gas in any liquid is exothermic in nature.

4.1 Henry' Law :



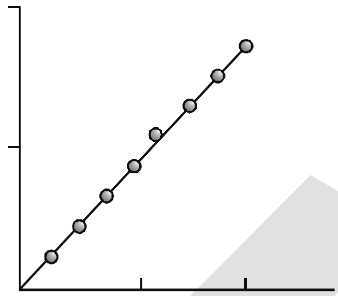
Effect of pressure on the solubility of a gas. The concentration of dissolved gas is proportional to the pressure on the gas above the solution.

The partial pressure of the gas in vapour phase (p) is proportional to the mole fraction of the gas (x) in the solution, at constant temperature

$$P = K_H X$$

K_H = Henry's Law Constant

* Henry's Law Constant depends on nature of gas and liquid as well as temperature.
* K_H increases with increases in temperature.
* Greater K_H means low solubility.



Values of Henry's Law Constant for Some Selected Gases in Water

Gas	Temperature/K	K_H /kbar
He	293	144.97
H_2	293	69.16
N_2	293	76.48
N_2	303	88.84
O_2	293	34.86
O_2	303	46.82
Argon	298	40.3
CO_2	298	1.67
Formaldehyde	298	1.83×10^{-5}
Methane	298	0.413
Vinyl chloride	298	0.611

4.2 Limitations of Henry' Law :

(1) It is valid only for ideal behaviour of gas. As none of the gas is ideal, this law may be applied at low pressure and high temperature.

(2) It gives better result when the solubility of gas in the liquid is low.

(3) The gas should neither dissociate nor associate in the liquid.

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4.3 Henry' Law application :

(1) To increase the solubility of CO_2 in soft drinks and soda water, the bottle is sealed under high pressure.

(2) Scuba divers must cope with high concentrations of dissolved gases while breathing air at high pressure underwater. Increased pressure increases the solubility of atmospheric gases in blood. When the divers come towards surface, the pressure gradually decreases. This releases the dissolved gases and leads to the formation of bubbles of nitrogen in the blood. This blocks capillaries and creates a medical condition known as bends, which are painful and dangerous to life. To avoid bends, as well as, the toxic effects of high concentrations of nitrogen in the blood, the tanks used by scuba divers are filled with air diluted with helium (11.7% helium, 56.2% nitrogen and 32.1% oxygen).

(3) At high altitudes the partial pressure of oxygen is less than that at the ground level. This leads to low concentrations of oxygen in the blood and tissues of people living at high altitudes or climbers. Low blood oxygen causes climbers to become weak and unable to think clearly, symptoms of a condition known as anoxia.

4.4 Effect of Temperature :

Solubility of any gas in any liquid decreases with rise in temperature as dissolution is an exothermic process.

At constant pressure, $\ell n\dfrac{C_2}{C_1} = \dfrac{\Delta H_{sol}}{R}\left(\dfrac{1}{T_1} - \dfrac{1}{T_2}\right)$

where C = molar concentration of gas in solution

Ex.1. If N_2 gas is bubbled through water at 293 K, how many millimoles of N_2 gas would dissolve in 1 litre of water? Assume that N_2 exerts a partial pressure of 0.987 bar. Given that Henry's law constant for N_2 at 293 K is 76.48 kbar.

Sol: *x (Nitrogen) =* ———— = ———— = × −

As 1 litre of water contains 55.5 mol of it, therefore if n represents number of moles of N_2 in solution,

x (Nitrogen) = ———— = ——— = × −
+

(n in denominator is neglected as it is < < 55.5)

Thus n = $1.29 \times 10^{-5} \times 55.5$ mol = 7.16×10^{-4} mol = 0.716 m mol

Ex.2 The air is a mixture of a number of gases. The major components are oxygen and nitrogen with approximate proportion of 20% is to 79% by volume at 298 K. The water is in equilibrium with air at a pressure of 10 atm. At 298 K, if the Henry's law constants for oxygen and nitrogen are 3.30×10^7 mm and 6.51×10^7 mm respectively, calculate the composition of these gases in water.

Sol: *Percentage of oxygen (O_2) in air = 20%*

Percentage of nitrogen (N_2) in air = 79%

Also, it is given that water is in equilibrium with air at a total pressure of 10 atm, that is, (10 × 760) mm Hg = 7600 mm Hg

Partial pressure of oxygen, = —— × *= 1520 mm Hg*

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Partial pressure of nitrogen, $= \frac{}{} \times$ $= 6004$ *mm of Hg*

Now, according to Henry's law :

$p = K_H x$

For oxygen :

$=$

$\Rightarrow$ $= \frac{}{} = \frac{}{\times}$ *(Given* $K_H = 3.30 \times 10^7$ *mm of Hg* $= 4.61 \times 10^{-5}$

For nitrogen,

$=$

$\Rightarrow$ $= \frac{}{} = \frac{}{\times} = 9.22 \times 10^{-5}$

Hence, the mole fractions of oxygen and nitrogen in water are 4.61×10^{-5} *and* 9.22×10^{-5} *respectively.*

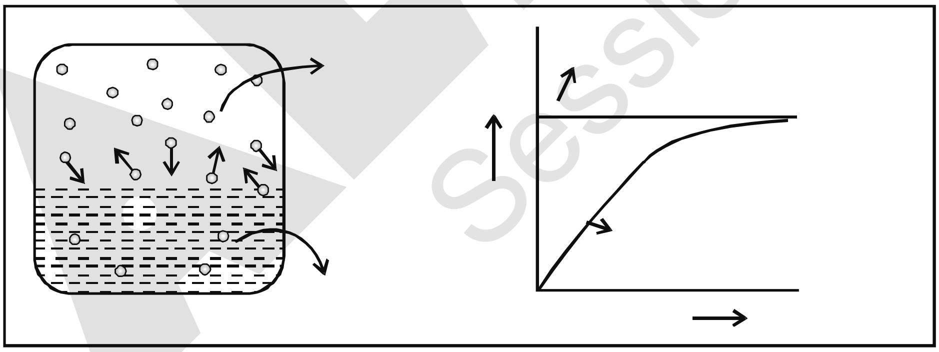
5. *Vapour pressure :*

The Evaporation of a Liquid in a Closed Container

When a liquid is taken in a closed vessel at constant temperature, then there are two process which takes place.

(i) evaporation (ii) condensation

In the constant evaporation from the surface particles continue to break away from the surface of the liquid As the gaseous particles bounce around, some of them will hit the surface of the liquid again, and will be trapped there. This is called condensation. The rate of condensation increases with time, but rate of evaporation remain constant. There will rapidaly be an equilibrium set up in which the number of particles leaving the surface is exactly balanced by the number rejoining it.



This pressure at equilibrium is called the **vapour pressure** (also known as **saturation vapour pressure**) **of the liquid at that temperature.**

5.1 Effect of Temperature on vapour pressure

When the space above the liquid is saturated with vapour particles, you have this equilibrium occuring on the surface of the liquid :

$$\rightleftharpoons \qquad \Delta$$

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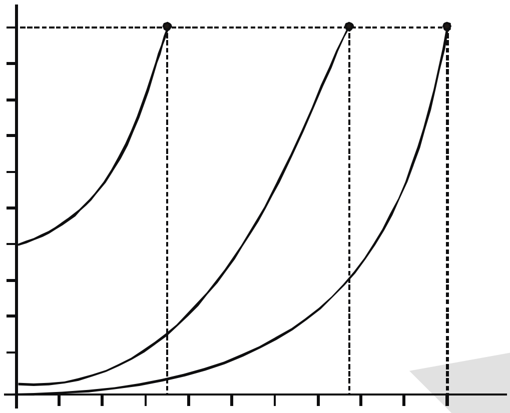
The forward change (liquid to vapour) is endothermic. It needs heat to convert the liquid into the vapour. According to Le Chatelier, increasing the temperature of a system in a dynamic equilibrium favours the endothermic change. That means increasing the temperature increase the amount of vapour present, and so increases the vapour pressure.



Effect of nature : V.pr α ______________________

The dependence of vapour pressure of a liquid on temperature is given by clausius-clapeyron equation : $\ln \frac{P_2}{P_1} = \frac{\Delta H_{vap.}}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$

5.2 Nature of liquid : Weaker the intermolecular attraction , higher will be the vapour pressure.

6. VAPOUR PRESSURE OF LIQUID SOLUTION

6.1 Raoult's law :

The partial pressure of any volatile constituents of a solution at a constant temperature is equal to the vapour pressure of pure constituents multiplied by the mole fraction of that constituent in the solution, at equilibrium.

6.2 Vapour pressure of liquid – liquid solution :

Let P_A and P_B be the partial vapour pressures of two constituents A and B in solution and P_A^0 and P_B^0 the vapour pressures in pure state respectivity.

Thus, according Raoult's law

$$= \frac{\quad}{\quad + \quad} = \qquad ....(1)$$

and $$= \frac{\quad}{\quad + \quad} = \qquad ....(2)$$

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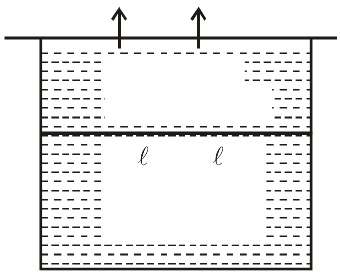
If total pressure be P_S, then

$$P_S = P_A + P_B = X_A P_A^0 + X_B P_B^0$$

$$P_S = X_A P_A^0 + (1 - X_A) P_B^0 \qquad [\because X_A + X_B = 1]$$

$$P_S = X_A P_A^0 - X_A P_B^0 + P_B^0$$

$$P_S = X_A [P_A^0 - P_B^0] + P_B^0$$



6.2.1 Relation between Dalton's Law and Raoult's Law :

The composition of the vapour in equilibrium with the solution can be calculated applying Dalton's law of partial pressures. Let the mole fraction of A and B in vapours be Y_A and Y_B respectively.

$$p_A = Y_A P_S = X_A P_A^0 \qquad ....(1)$$

$$p_B = Y_B P_S = X_B P_B^0 \qquad ....(2)$$

Now, $\qquad X_A = \dfrac{Y_A \cdot P_S}{P_A^0} \qquad$ and $\qquad X_B = \dfrac{Y_B \cdot P_S}{P_B^0}$

As, $\qquad X_A + X_B = 1, \quad \dfrac{Y_A \cdot P_S}{P_A^0} + \dfrac{Y_B \cdot P_S}{P_B^0} = 1$

$$\therefore \quad \frac{1}{P_S} = \frac{Y_A}{P_A^0} = \frac{Y_B}{P_B^0}$$

Hence, the total vapour pressure of solution may be calculated from liquid composition at equilibrium as

$$P_S = X_A \cdot P_A^0 + X_B \cdot P_B^0$$

and from vapour composition at equilibrium at

$$\frac{1}{P_S} = \frac{Y_A}{P_A^0} + \frac{Y_B}{P_B^0}$$

6.2.2 Comparision between liquid and vapour composition :

$$\frac{Y_A}{Y_B} = \frac{P_A / P_S}{P_B / P_S} = \frac{P_A}{P_B} = \frac{X_A \cdot P_A^0}{X_B \cdot P_B^0}$$

If A is more volatile than B $\left(P_A^0 > P_B^0\right)$, then $\dfrac{Y_A}{Y_B} > \dfrac{X_A}{X_B}$

It means that the mole-fraction of A is vapour form is relatively greater than that in liquid form, relative to B. Hence, in any ideal solution, vapour is always more richer in the more volatile component.

6.2.3 Raoult's law as a special case of Henry's law :

From Raoult's law, the vapour pressure of volatile component in the solution is $P = P^0.X$.

In the solution of gas in liquid, one component is so volatile that it exist as gas and its pressure is given by Henry's law as $P = K_H.X$

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In both laws, the partial pressure of volatile component is directly proportional to its mole-fraction is solution. Only the proportionality constant K_H differs from P^0. Hence, Raoult's law becomes a special case of Henry's law in which K_H becomes P^0.

6.3 Vapour pressure of solution of solids in liquid :

Let us assume A = non volatile solid & B = volatile liquid

According to Raoult's law –

$\because \quad P_s = X_A P_A^{\,0} + X_B P_B^{\,0}$

for A, $P_A^{\,0} = 0$

$\therefore \quad P_s = X_B P_B^{\,0}$ ....(5)

Let $P_B^{\,0} = P^0$ = Vapour pressure of pure state of solvent.

here X_B is mole fraction of solvent

$= \frac{}{+}$ ....(6)

$\propto \frac{}{+}$ i.e. vapour pressure of solution $\propto$ mole fraction of solvent

$\Rightarrow \quad P_S = X_B P_B^{\,0} \Rightarrow P_S = (1 - X_A)\, P_B^{\,0}$

$\Rightarrow \quad P_S = P_B^{\,0} - X_A\, P_B^{\,0}$

$\Rightarrow \quad \frac{-}{} =$

or $\frac{-}{} =$ ....(7)

$\boxed{\frac{-}{} = \frac{}{+}}$ ....(8)

or $\frac{}{-} = \frac{+}{}$

or $\frac{}{-} = + \frac{}{}$ or $\frac{}{-} - \; = \frac{}{}$

$\frac{}{-} = \frac{}{}$

$\boxed{\frac{-}{} = \frac{}{} = \frac{}{}}$ ....(9)

ALLEN Session 2019-20

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6.4 Ideal and Non ideal solutions :

The solutions which obey Raoult's law over the entire range of concentration are known as ideal solutions.

Table : Comparison between Ideal and Non-ideal solutions

Ideal solutions	Non-ideal solutions	
	+ve deviation from Raoult's law	**– deviation from Raoult's law**
1. Obeys Raoult's law at every concentrations.	Do not obey Raoult's law.	Do not obey Raoult's law.
2. $\Delta H_{mix} = 0$; Neither heat is evolved nor absorbed during dissolution.	$\Delta H_{mix} > 0$. Endothermic dissolution; heat is absorbed.	$\Delta H_{mix} < 0$; exothermic dissolution heat is evolved.
3. $\Delta V_{mix} = 0$; total volume of solution is equal to sum of volumes of the components.	$\Delta V_{mix} > 0$. Volume is increased after dissolution.	$\Delta V_{mix} < 0$; volume is decreased during dissolution.
4. = + = + i.e., = =	> > ∴ + > +	< < ∴ + < +
5. A — A, A — B, B — B interactions should be same, i.e., 'A' and 'B' are identical in shape, size and character.	A — B, attraction force should be weaker than A — A and B — B attractive forces. 'A' and 'B' have different shape, size and character.	A — B, attraction force should be greater than A — A and B — B attractive forces. 'A' and 'B' have different shape, size and charater.
6. Escaping tendency of 'A' and 'B' should be same in pure liquids and in the solution.	'A' and 'B' escape easily showing high vapour pressure than the expected value.	Escaping tendency of both components A and B is lowered showing lower vapour pressure than expected ideally.
Example :	**Example :**	**Example :**
dilute solutions ;	acetone + ethanol ;	acetone + aniline ;
benzene + toluene ;	acetone + CS_2 ;	acetone + chloroform ;
n-hexane + n-heptane ;	water + methanol ;	CH_3OH + CH_3COOH ;
chlorobenzene + bromobenzene ;	water + ethanol ;	H_2O + HNO_3 ;
ethyl bromide + ethyl iodide ;	CCl_4 + $CHCl_3$;	
n-butyl chloride + n-butyl bromide	CCl_4 + toluene ;	water + HCl ;
CCl_4 + $SiCl_4$;	acetone + benzene	acetic acid + pyridine ;
$C_2H_4Br_2 + C_2H_4Cl_2$	CCl_4 + CH_3OH ;	
$C_2H_5Br + C_2H_5Cl$	cyclohexane + ethanol	$HNO_3 + CHCl_3$

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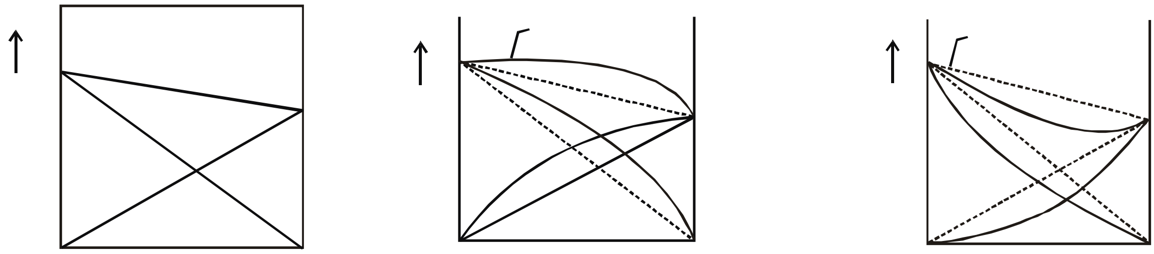

Ex.3. ***1 mole heptane (V.P. = 92 mm of Hg) is mixed with 4 mole Octane (V.P. = 31 mm of Hg), form an ideal solution. Find out the vapour pressure of solution.***

Sol. *Total mole = 1 + 4 = 5*

Mole fraction of heptane = $X_A = 1/5$

Mole fraction of octane = $X_B = 4/5$

$P_S = X_A P_A^0 + X_B P_B^0 = -\times \quad + -\times \quad = 43.2$ *mm of Hg.*

Ex.4. ***At 88°C, benzene has a vapour pressure of 900 torr and toluene has a vapour pressure of 360 torr. What is the mole fraction of benzene in the mixture with toluene that will be boil at 88°C at 1 atm pressure, benzene – toluene form an ideal solution.***

Sol. $P_S = 760$ *torr, because solution boils at 88°C*

$\therefore 760 = 900x + 360(1-x)$

$x = 0.74$ *where 'x' is mole fraction* C_6H_6.

Ex.5. ***The vapour pressure of benzene at 90°C is 1020 torr. A solution of 5 g of a solute in 58.5 g benzene has vapour pressure 990 torr. What is the molecular weight of solute ?***

Sol. $\frac{-}{\quad} = \frac{\times}{\times} \Rightarrow \frac{-}{\quad} = \frac{\times}{\times} \Rightarrow m = 220$

Ex.6. ***Two liquids A and B form an ideal solution. At 300 K, the vapour pressure of a solution containing 1 mole of A and 3 mole of B is 550 mm of Hg. At the same temperature, if one more mole of B is added to this solution, the vapour pressure of the solution increases by 10 mm of Hg. Determine the vapour pressure of A and B in their pure states.***

Sol. *Let the vapour pressure of pure A be = ; and the vapour pressure of pure B be = .*

Total vapour pressure of solution (1 mole A + 3 mole B)

$= \quad + \quad$ *[*X_A *is mole fraction of A and* X_B *is mole fraction of B]*

$= - \quad + -$ *or* $= \quad +$ *.......(i)*

Total vapour pressure of solution (1 mole A + 4 mole B) = $- \quad + -$

$= - \quad + -$

ALLEN Session 2019-20

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= + ...(ii)

Solving eqs. (i) and (ii)

= 600 mm of Hg = vapour pressure of pure B

= 400 mm of Hg = vapour pressure of pure A

Ex.7. ***Liquids 'A' and 'B' form an ideal solution. Calculate the vapour pressure of solution having 40 mole-percent of A in the vapour at equilibrium.*** $\left(P_A^0 = 80\,\text{cm Hg},\ P_B^0 = 30\,\text{cm Hg}\right)$

Sol. $$\frac{1}{P_{\text{total}}} = \frac{Y_A}{P_A^0} + \frac{Y_B}{P_B^0} = \frac{0.4}{80} + \frac{0.6}{30} = \frac{1}{40}$$

P_{total} *= 40 cm Hg*

Ex.8. ***Liquids 'A' and 'B' form an ideal solution. Calculate the molar-fraction of 'A' in vapour form above the liquid solution containing 25 mole-percent of 'A' at equilibrium*** $\left(P_A^0 = 0.2\,\text{atm},\ P_B^0 = 0.5\,\text{atm}\right)$

Sol. $$Y_A = \frac{P_A}{P_{\text{total}}} = \frac{X_A \cdot P_A^0}{X_A \cdot P_A^0 + X_B \cdot P_B^0} = \frac{0.25 \times 0.2}{0.25 \times 0.2 + 0.75 \times 0.5} = \frac{2}{17}$$

Ex.9. ***Liquids 'A' and 'B' form an ideal solution. At 80°C,*** $P_A^0 = 0.4\,\text{bar}$ ***and*** $P_B^0 = 0.8\,\text{bar}$. ***All the vapour above the liquid solution containing equal moles of both the liquids at equilibrium is collected in another empty vessel and condensed. Now, the condesate is heated to 80°C and all the vapours above the liquid solution at equilibrium is again collected in another empty vessel and condensed. What is the mole-fraction of 'B' is new condensate ?***

Sol. *For the first condensate,*

$$\frac{n_B'}{n_A'} = \frac{X_B'}{X_A'} = \frac{Y_B}{Y_A} = \frac{X_B}{X_A} \cdot \frac{P_A^0}{P_B^0} = \frac{n_B}{n_A} \times \frac{P_B^0}{P_A^0}$$

For second condensate,

$$\frac{n_B''}{n_A''} = \frac{X_B''}{X_A''} = \frac{Y_B'}{Y_A'} = \frac{X_B' \cdot P_B^0}{X_A' \cdot P_A^0} = \frac{n_B}{n_A} \times \left(\frac{P_B^0}{P_A^0}\right)^2 = \frac{x}{x} \times \left(\frac{0.8}{0.4}\right) = \frac{4}{1}$$

$\therefore$ *Mole fraction of* $B = \frac{4}{5} = 0.8$

Note : For multi-step condensation at constant temperature,

$$\boxed{\frac{n_B^f}{n_A^f} = \frac{n_B^i}{n_B^i} \cdot \left(\frac{P_B^0}{P_A^0}\right)^n}$$ *n : number of steps*

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Ex.10. ***Liquid A & B form an ideal solution. In a cylinder piston arrangement, 2 moles of vapours of liquid A & 3 moles of vapours of liquid B are taken at 0.3 atm.*** $P_A^0 = 0.4\,\text{atm}$, $P_B^0 = 0.6\,\text{atm}$.

(i) ***Predict whether vapours will condense or not ?***

(ii) ***If the vapours are compressed slowly & isothermly, at what pressure I^{st} drop of liquid will form***

(iii) ***If the initial volume of vapours was 10L, at what volume I^{st} drop of liquid will form ?***

(iv) ***What is the composition of I^{st} drop of liquid formed ?***

(v) ***If the vapours are further compressed slowly & isothermally, at what P almost complete condensation will occur ?***

(vi) ***What is the composition of last traces of vapours remained ?***

(vii) ***What is the composition of system at 0.58 atm ?***

(viii) ***What is the composition of system at 0.51 atm ? Also calculate moles of A & B in liquid & vapour form.***

(ix) ***At what P, half of the total amount of vapours will condense ?***

Sol. *(i)* $\dfrac{1}{P_T} = \dfrac{Y_A}{P_A^0} + \dfrac{Y_B}{P_B^0} = \dfrac{2/5}{0.4} + \dfrac{3/5}{0.6} \quad \Rightarrow \quad P_{total} = 0.5 \text{ atm} > 0.3 \text{ atm}$

$\therefore$ *100% gas, no condensation*

(ii) *0.5 atm.*

(iii) $P_iV_i = P_fV_f$

$$V_f = \frac{0.3\times10}{0.5} = 6\,L$$

(iv) $P_{total} = X_A \cdot P_A^0 + X_B \cdot P_B^0$

or $0.5 = X_A \times 0.4 + (1 - X_A) \times 0.6 \quad \Rightarrow \quad X_A = 0.5$

(v) $P_T = 0.4 \times 0.4 + 0.6 \times 0.6 = 0.52$ *atm*

Note: *For a pure liquid, there is a fixed P(V.P.) below & above which, the system will be 100% gas & 100% liquid, respectively but for a solution, a pressure range exist in which both physical states will be present.*

(vi) $Y_A = \dfrac{X_A \cdot P_A^0}{P_{total}} = \dfrac{0.4\times0.4}{0.52} = \dfrac{4}{13}$

(vii) *100% liquid, A = 2mole and B = 3 mole*

(viii) $P_{total} = X_AP_A^0 + X_BP_B^0$

$0.51 = X_A \times (0.4) + (1 - X_A) \times 0.6 \quad \& \quad Y_A = \dfrac{X_AP_A^0}{P_{total}} = \dfrac{0.45\times0.4}{0.51} = 0.35$

$X_A = 0.45$

ALLEN Session 2019-20

A = 2mol
B = 3 mol

Liquid	Vapour
A = x mol	A = (2 – x)
B = y mol	B = (3 – y)

$$X_A = \frac{9}{20} = \frac{x}{x+y} \qquad \textit{and} \qquad Y_A = \frac{18}{51} = \frac{2-x}{(2-x)+(3-y)}$$

$\therefore$ *x = 1.09 ; y = 1.33*

(ix)

A = 2mole
B = 3 mole

Liquid = 2.5 mole	Vapour = 2.5 mole
A = z mole	A = (2 – z) mole
B = (2.5 – z) mole	B = (0.5 – z) mole

$$P_T = \frac{z}{2.5}\times 0.4 + \frac{2.5-z}{2.5}\times 0.6$$

$$\frac{2-z}{2.5} = \frac{z\times 0.4}{2.5\times P_T}$$

On solving, z = 1.12, P_T = 0.5104 atm

6.5 Graphs for ideal Binary solution of liquid A & liquid B :

(Assume $P_A^0 < P_B^0$)

I. Vapour pressure V/s liquid composition

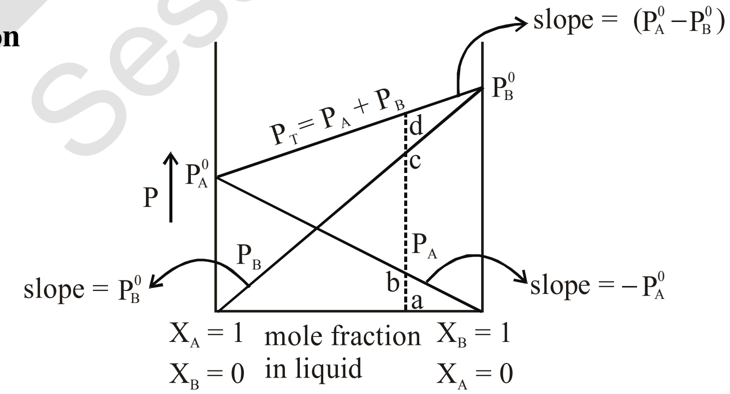
$$P_A = X_A P_A^0 = (1 - X_B)P_A^0$$

$$P_A = P_A^0 - X_B P_A^0$$

$$P_B = X_B P_B^0$$

$$P_T = X_A P_A^0 - X_B P_B^0 = (1 - X_B)P_A^0 + X_B P_B^0$$

$$\therefore \quad P_T = P_A^0 + X_B\left(P_B^0 - P_A^0\right)$$



As $\quad P_T = P_A + P_B, \qquad$ ad = ab + ac

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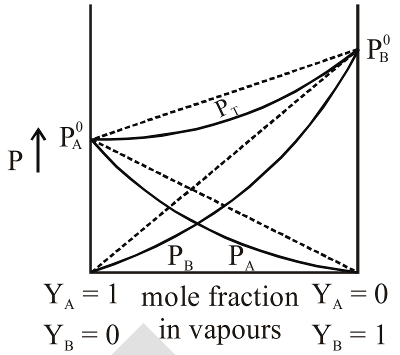
II. Vapour pressure V/s vapour composition

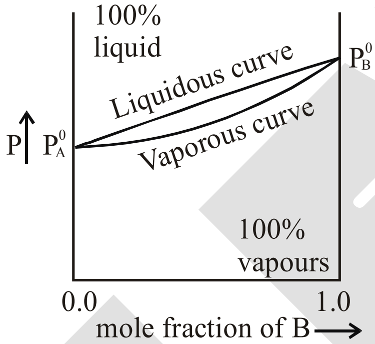
$$\frac{1}{P_T} = \frac{Y_A}{P_A^0} + \frac{Y_B}{P_B^0}$$

$$\frac{1}{P_T} = \frac{1}{P_A^0} + Y_B\left(\frac{1}{P_B^0} - \frac{1}{P_A^0}\right)$$

$$\frac{1}{y} = c + mx$$



III. Vapour pressure V/s composition



(i) The V.P. of ideal solution always lie in between the V.P. of pure components.

(ii) Below vapourous curve, the system will by 100% vapour & above liquidous and curve, 100% liquid. Both the physical states exists only in between the curves.

(iii) At any composition, the physical state of system may be changed by changing the pressure.

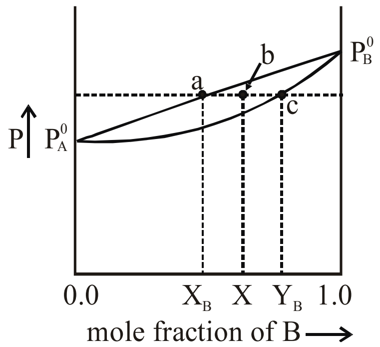
(iv) At any pressure in between P_A^0 and P_B^0, the physical state of system may be changed by changing the composition.

(v) Length ab $\propto$ total moles of vapour

Length cb $\propto$ total moles of liquid

$$\frac{ab}{cb} = \frac{\text{total moles of vapour}}{\text{total moles of liquid}}$$

or $\dfrac{X - X_B}{Y_B - X} = \dfrac{n_{A(g)} + n_{B(g)}}{n_{A(\ell)} + n_{B(\ell)}}$ (Lever's rule)



6.6 Boiling point :

Boiling point of a liquid is the temperature at which the vapour pressure of the liquid become equal to the atmospheric pressure.

Session 2019-20

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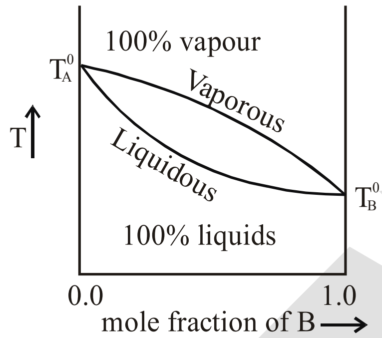
Ex.11 Liquid 'A' and 'B' form an ideal solution. At 27°C, the vapour pressure of pure liquids 'A' and 'B' are 0.6 atm and 1.2 atm, respectively. What the coposition of liquid solution boiling at 27°C?

Sol. $P_T = X_A \cdot P_A^0 + X_B \cdot P_B^0$

or $1 = X_A \times 0.6 + (1 - X_A) \times 1.2 \quad \Rightarrow X_A = \dfrac{1}{3}$

6.7 Boiling point curves for ideal binary solution :



(i) The boiling point of ideal solutioin always lie in between the boiling points of pure liquids.

(ii) Below liquidus curve, the system in 100% liquid and above vapourus curve, the systm is 100% vapour. Both the physical states exists only in between the curves.

(iii) At any composition, the physical state of system may be changed by changing the temperature.

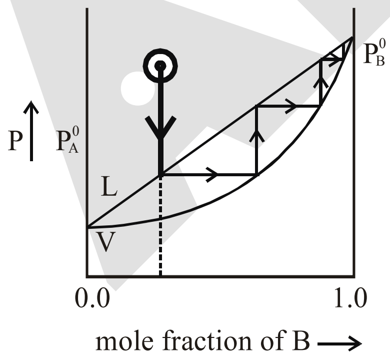
(iv) At any temperature in between T_A^0 and T_B^0, the physical state of system may be changed by changing the composition.

6.8 Distillation

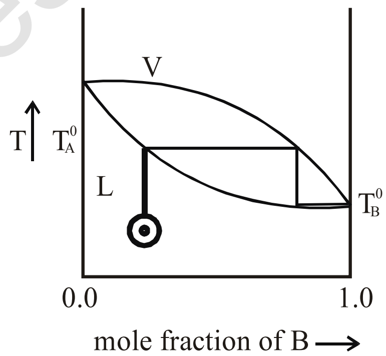
It is the method of seperation of liquids by converting them into vapours (boiling).

Distillation

Isothermal



Isobaric

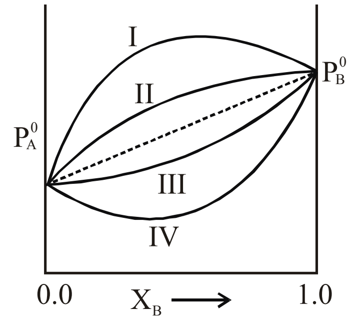


(i) The seperation of liquid by distillation occurs because at any T or P, the composition of distillate or condensate is different than the composition of original liquid.

(ii) With the elemination of vapour above the liquid, the boiling point of residual liquid increases.

(iii) The boiling point of distillate is less than that of original liquid.

6.9 Graphs for Non-ideal solutions and azeotropic mixture



(I) Large positive deviation : $(V.P.)_{solution} > P_B^0$ at some composition

(II) Small positive deviation : $P_A^0 < (V.P.)_{solution} < P_B^0$

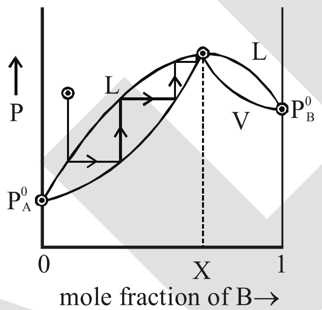
(III) Small negative deviation : $P_A^0 < (V.P.)_{solution} < P_B^0$

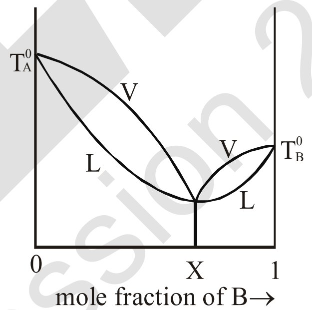
(IV) Large negative deviation : $(V.P.)_{solution} < P_A^0$ at some composition

6.9.1 Konowaloff's rule :

In ideal or non-idal solution the vapour is always more rich in the component, addition of which in liquid, increase the vapour pressure of solution.

6.9.2 Large (+)ve derivation (Minimum boiling azeotrope)





The solution of large positive deviation can not be seperated by distillation becasue at composition 'x', the liquid & vapour composition becomes identical.

Such solution which can not be saperated by distillation are called **azeotropic mixture or constant boiling solution.**

At any composition less than X, traces of pure A may be obtained but not pure B. Similarly at composition greater than X, traces of pure B may be obtained but not pure A.

ALLEN Session 2019-20

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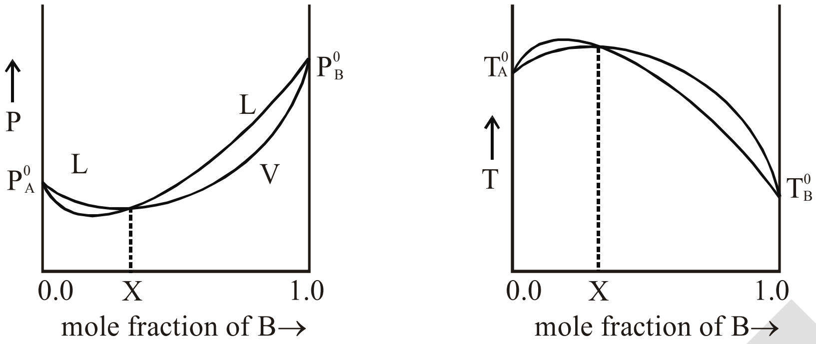
5.9.3 Large (–)ve derivation (Maximum boiling azeotrope)



Note : Azeotrope is not formed in ideal solution or solution of small deviations.

7. COLLIGATIVE PROPERTIES

Properties of a solution which depends on the number of solute particles irrespective of their nature, relative to the total number of particles present in solution are called colligative properties.

The following properties are colligative properties of solution :

(i) Relative lowering of vapour pressure.

(ii) Elevation in boiling point.

(iii) Depression in freezing point.

(iv) Osmotic pressure.

7.1 Lowering of vapour pressure :

When a non-volatile solute 'A' is dissolved in a pure solvent 'B', the vapour pressure of the solvent is lowered i.e. the vapour pressure of a solution is always lower than that of pure solvent, because the escaping tendency of solvent molecules decreases.

If at a certain temperature P° is the vapour pressure of pure solvent, and P_s is the vapour pressure of solution then

Lowering of vapour pressure = $P° - P_s$

Relative lowering of vapour pressure $= \frac{-}{\quad}$

from equation (8)

$$\frac{-}{\quad} = \frac{\Delta}{\quad} = \frac{\quad}{+} =$$

Session 2019-20

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For a very dilute solution $n_A <<< n_B$

so $\dfrac{-}{\qquad} = \dfrac{}{\quad} = \dfrac{}{\quad} \times \dfrac{}{\quad}$

Ex.12. Calculate the vapour pressure lowering caused by addition of 50 g of sucrose (molecular mass = 342) to 500 g of water if the vapour pressure of pure water at 25°C is 23.8 mm Hg.

Sol. According to Raoult's law,

$\dfrac{-}{\qquad} = \dfrac{}{+}$ or $\Delta = \dfrac{}{+}$

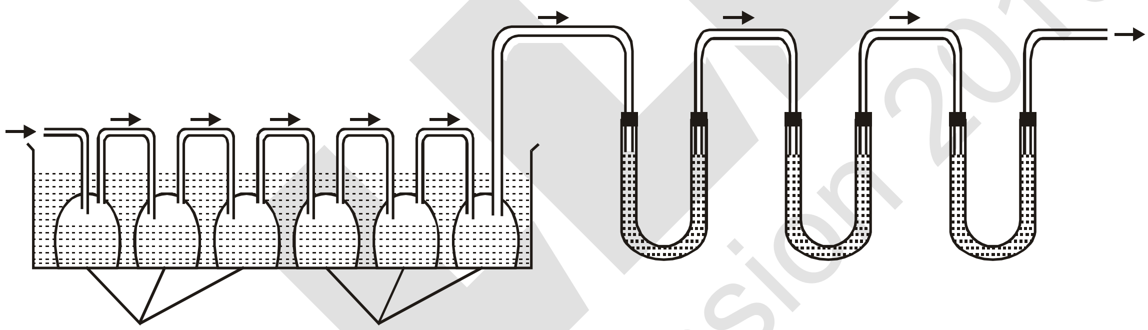
Given : n = $\dfrac{}{\quad}$ = 0.146 ; N = $\dfrac{}{\quad}$ = 27.78 and p_0 = 23.8 mmHg

Substituting the values in the above equation,

$\Delta = \dfrac{}{+} \times \quad = \quad$ mm Hg

7.1.1 Measurement of Lowering in Vapour Pressure by Dynamic Method (Ostwald and Walker Method)

The apparatus used is shown in Fig. It consists of two sets of bulbs. The first set of three bulbs is filled with solution to half of their capcity and second set of another three bulbs is filled with the pure solvent.



Each set is separately weighed accurately. Both sets are connected to each other and then with the accurately weighed set of guard tubes filled with anhydrous calcium chloride or some other dehydrating agents like P_2O_5, conc. H_2SO_4 etc. The bulbs of solution and pure solvent are kept in a thermostat maintained at a constant temperature.

A current of pure dry air is bubbled through the series of bulbs as shown in fig. The air gets saturated with the vapour in each set of bulbs. The air takes up an amount of vapours proportional to the vapour pressure of the solution first and then it takes up more amount of vapours from the solvent which is proportional to the difference in the vapour pressure of the solvent and the vapour pressure of solution, i.e., $p_0 - p_s$.

The two sets of bulbs are weighed again. The guard tubes are also weighed.

Loss in mass in the solution bulbs $\propto p_s$

Session 2019-20

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Loss in mass in the solvent bulbs $\propto (p_0 - p_s)$

Total loss in both sets of bulbs $\propto [p_s + (p_0 - p_s)] \propto p_0$

Total loss in mass of both sets of bulbs is equal to gain in mass of guard tubes.

Thus, $\frac{-}{} = \frac{}{} = \frac{}{}$

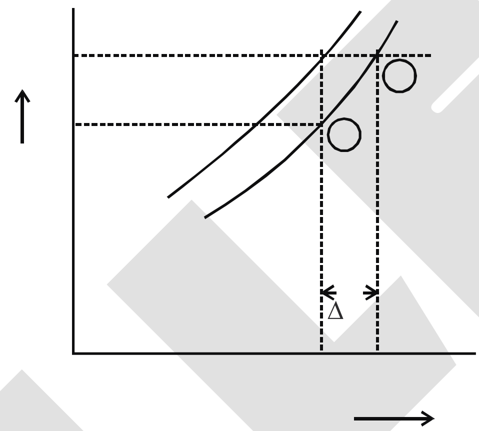
Further, we know from Raoult's law,

$$\frac{-}{} = \frac{}{+}$$

$\therefore$ $$\frac{}{} = \frac{}{+}$$

6.2 Elevation in boiling point (Ebullioscopy) :

solvent $(\ell) \rightleftharpoons$ solvent (g)



Δ

When a non-volatile solute A is dissolved in a pure solvent B, its vapour pressure decreases and hence the boiling point increases. The difference ΔT_b of boiling points of the solution and pure solvent is called elevation in boiling point.

If T_b^0 is the boiling point of pure solvent and T_b is the boiling point of the solution then, $T_b > T_b^0$

and the elevation in boiling point $\quad \Delta T_b = T_b - T_b^0$

Experiments have shown that for dilute solutions, the elevation of boiling point (ΔT_b) is directly proportional to the molal concentration of the solute in a solution. Thus

$$\Delta T_b \propto m$$

or $$\Delta T_b = K_b.m$$

where K_b = boiling point elevation constant or molal elevation constant or ebulioscopic constant.

Molal elevation constant is characteristic of a particular solvent and can be calculated from the thermodynamical relationship.

$$K_b = \frac{RT_b^2.M}{1000\ \Delta H_{vap}}$$

where, R is molar gas constant = 2 cal/mol–K , M = molar mass of solvent

T_b is the boiling point of the pure solvent (in K)

and ΔH_{vap} is the latent heat of vaporisation of pure solvent

For water $K_b = \dfrac{2\times(373)^2}{1000\times540} = 0.515$ K-kg/mol

The molal elevation constant for some common solvents are given in the following table

Solvent	B.P. (°C)	Molal elevation constant
Water	100.0	0.52
Acetone	56.0	1.70
Chloroform	61.2	3.63
Carbon tetrachloride	76.8	5.03
Benzene	80.0	2.53
Ethyl alcohol	78.4	1.20

Ex.13. 0.15 g of a substance dissolved in 15 g of solvent boiled at a temperature higher by 0.216°C than that of the pure solvent. What is the molecular weight of the substance. [K_b for solvent = 2.16 K-kg/mol]

Sol. ***Given : K_b = 2.16°C, w = 0.15 g, ΔT_b = 0.216°C, W = 15 g***

ΔT_b = molality × K_b

$$\Delta T_b = \frac{w}{m\times W}\times1000\times K_b$$

$$0.216 = \frac{0.15}{m\times15}\times1000\times2.16$$

$$m = \frac{0.15\times1000\times2.16}{0.216\times15} = \mathbf{100}$$

Ex.14 The rise in boiling point of a solution containing 1.8 g glucose in 100 g of a solvent is 0.1°C. The molal elevation constant of the liquid is –

Sol. *ΔT_b = 0.1°C, m = 180, W = 100, w = 1.8*

$$K_b = \frac{\Delta T_b\times m\times W}{1000\times w} = \frac{180\times0.1\times100}{1000\times1.8} = 1.0$$

7.3 Depression in freezing point (Cryoscopy) :

The freezing point of a liquid is that temperature at which the liquid and its solid state exist in equilibrium with each other. It may also be defined as the temperature at which the liquid and solid states of a substance have the same vapour pressure.

ALLEN Session 2019-20

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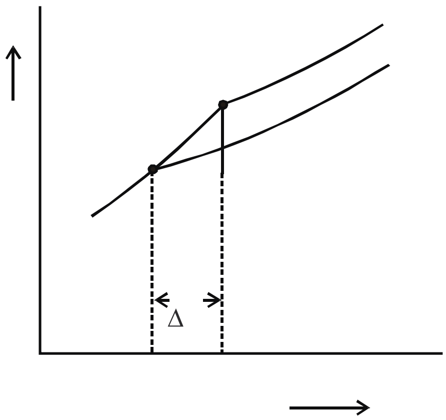

When a non-volatile non-electrolyte is dissolved in a pure solvent, the vapour pressure of the solvent is lowered and it become equal to that of solid solvent at lower temperature.

If T_f^0 is the freezing point of pure solvent and (T_f) is the freezing point of its solution then,

$$T_f < T_f^0$$

The difference in the freezing point of pure solvent and solution is the depression of freezing point (ΔT_f) Thus,

$$T_f^0 - T_f = \Delta T_f$$

Similar to elevation in boiling point, depression in freezing point for dilute solution is directly proportional to its molality,

$$\Delta T_f \propto m$$

or $\Delta T_f = K_f . m$

where K_f is called freezing point depression constant or molal depression constant or cryoscopic constant.

K_f is characteristic of a particular solvent and can be calculated from the thermodynamical relationship

$$K_f = \frac{R \cdot T_f^{0^2} \cdot M}{1000 \cdot \Delta H_{fus}}$$

where, T_f^0 is the freezing point of pure solvent in (Kelvin) and ΔH_{fus} is the latent heat of fusion pure solvent.

For water,

$$= \frac{\times}{\quad} = 1.86 \text{ K-kg/mol}$$

The molal depression constant for some common solvents are given in the following table

Solvent	**F.P.(°C)**	**Molal depression solvents**
Water	0.0	1.86
Ethyl alcohol	– 114.6	1.99
Chloroform	– 63.5	4.79
Carbon tetrachloride	– 22.8	31.8
Benzene	5.5	5.12
Camphor	179.0	39.70

ALLEN Session 2019-20
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Ex.15. If freezing point of a solution prepared from 1.25 g of a non electrolyte and 20 g of water is 271.9 K, the molar mas of the solute will be – (K_f of water = 1.86 K-kg/mol)

Solution : Given $T_f = 271.9$ K

$w = 1.25$ g $\quad W = 20$ g $\quad K_f = 1.86$

$\Delta T_f = T_f^0 - T_f = 273 - 271.9 = 1.1$ K

$\Delta T_f =$ molality $\times K_f$

$$\Rightarrow \quad \Delta \; = \frac{}{\times} \times \qquad \times$$

$$\text{or} \qquad = \frac{\times \quad \times}{\Delta \quad \times} = \frac{\times \quad \times}{\times} = 105.68 \text{ mol/kg}$$

Ex.16. Molal depression constant for water is 1.86. What is freezing point of a 0.05 molal solution of a non electrolyte in water ? (K_f of water = 1.86 K-kg/mol)

Solution : $\Delta T_f =$ molality $\times K_f$

$= 0.05 \times 1.86 = 0.093°C$

$T_f = T_f^0 - 0.093 = 0 - 0.093$

$T_f = -0.093°C$

Ex.17. 2m aqueous urea solution is cooled to –7.44°C. Calculate the mass percent of water present in solution, which will separate as ice, (K_f of water = 1.86 K-kg/mol)

Sol. Let the initial mass of water in the solution = 1kg

$\therefore$ Moles of solute = 2

Now, $\Delta T_f = K_f m = K_f . \dfrac{n_{solute}}{Kg_{solvent}}$ $\quad$ or $7.44 = 1.86 \times \dfrac{2}{Kg_{solvent}}$

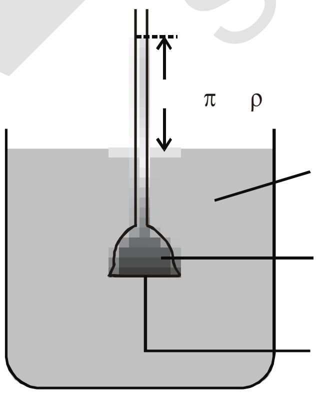
$\therefore$ $Kg_{solvent}$ left in the solution = 0.5

$\therefore$ Mass percent of water separated as ice $= \dfrac{0.5}{1} \times 100 = 50\%$

7.4 Osmosis and osmotic pressure :

7.4.1 Osmosis :

Osmosis is defined as the spontaneous flow of solvent molecules through semipermeable membrane from a pure solvent to solution.



Level of solution rises in the funnel due to osmosis of solvent

ALLEN Session 2019-20
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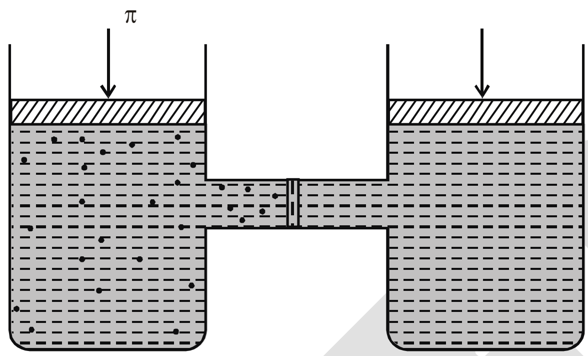
7.4.2 Semi-permeable membrane (SPM) :

There are the membranes (substances) which allow selective movement of particles across them. For a solution of solid solute in a liquid solvent, ideal SPM allow free movement of solvent particles across it, but not solute particles. These membranes contain a network of submicroscopic holes or pores through which small solvent molecules may pass but not the bigger solute particle.

7.4.3 Osmotic presssure (π) : It is the pressure which should be applied on the solution to just prevent osmostis or The hydrostatic pressure built up on the solution which just stops the osmosis.

osmotic pressure = hydrostatic pressure

$\pi = \rho gh$



The excess pressure equal to the osmotic pressure must be applied on the solution side to prevent osmosis

7.4.4 Van't Hoff laws :

(i) The osmotic pressure (π) of a solution is directly proportional to its molar concentration(C), when the temperature is kept constant. (Van't Hoff-Boyle's law)

thus $\pi \propto C$ (when temperature is constant)

(ii) Concentration remaining same, the osmotic pressure of a dilute solution is directly proportional to its absolute temperature (T). (Van't Hoff-Charle's law)

$\pi \propto T$ (when C is constant)

Combining the two laws, i.e., when concentration and temperature both are changing, the osmotic pressure will be given by :

$\pi \propto C.T$ or $\pi = CRT$

Where R = Universal gas constant.

7.4.5 Isotonic or iso-osmotic solution :

Solutions which have the same osmotic pressures at a given temperature are called isotonic or iso-osmotic solutions

When isotonic solutions are separated by semipermeable membrane, no osmosis occurs between them. For example, the osmotic pressure associated with the fluid inside the blood cell is equivalent to that of 0.9% (mass/volume) sodium chloride solution, called normal saline solution and it is safe to inject intravenously.

On the other hand, if we place the cells in a solution containing more than 0.9% (mass/volume) sodium chloride, water will flow out of the cells and they would shrink. Such a solution is called **hypertonic**. If the salt concentration is less than 0.9% (mass/volume), the solution is said to be **hypotonic**. In this case, water will flow into the cells if placed in this solution and they would swell.

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ALLEN Session 2019-20

Ex.18. A cane sugar solution has an osmotic pressure of 2.46 atm at 300 K. What is the strength of the solution.

Sol. $\pi = CRT$

$$or\ C = \frac{\pi}{RT} = \frac{2.46}{300 \times 0.0821} = 0.1\,M$$

Ex.19 A solution containing 8.6 g urea in one litre was found to be isotonic with 0.5% (wt./vol.) solution of an organic, non-volatile solution. The molecular weight of latter is

Sol. *Solutions are isotonic*

$$so\ \pi_1 = \pi_2$$

$$\frac{n_1}{V_1}RT = \frac{n_2}{V_2}RT \qquad \{R\ \&\ T\ are\ constant\}$$

$$so,\ \frac{n_1}{V_1} = \frac{n_2}{V_2}$$

$$or\ \left(\frac{w_1}{m_1 \times v_1}\right)_{urea} = \left(\frac{w_2}{m_2 \times v_2}\right)_{organic}$$

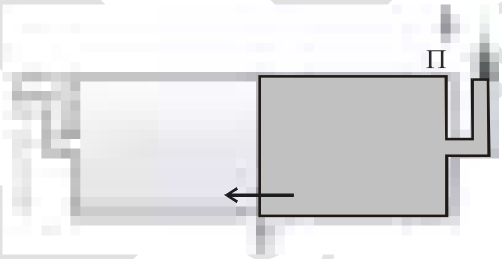
$$or\ \frac{8.6}{60 \times 1000} = \frac{0.5}{m_2 \times 100}$$

$$m_2 = 34.89\ gm/mol$$

7.5 Reverse Osmosis and water purification :

If external pressure greater than osmotic pressure is applied, the flow of solvent molecules can be made to proceed from solution towards pure solvent, i.e., in reverse direction of the ordinary osmosis.

Reverse osmosis is used for the desalination of sea water for getting fresh drinking water.



8 Abnormal colligative properties :

It has been observed that difference in the observed and calculated molecular masses of solute is due to association or dissociation of solute molecules in solution. It results in a change in the number of particles in solution.

Van't Hoff in 1880, introduced a factor, called Van't Hoff factor (i). The factor 'i' is defined as

$$= \frac{\quad}{\quad} = \frac{\quad}{\quad}$$

$$= \frac{\quad}{\quad}$$

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In case of association of solute particles in solution, the observed molecular weight of solute being more than the normal, the value of factor 'i' is less than unity (i.e. $i < 1$), while for dissociation the value of i is greater than unity (i.e. $i > 1$), because the observed molecular weight has lesser value than normal molecular weight.

8.1 Calculation of 'i' in case of dissociation

$$A_xB_y \longrightarrow xA^{y+} + yB^{x-}$$

	A_xB_y	xA^{y+}	yB^{x-}
Initial mol	1	0	0
after dissociation	$1-\alpha$	$x\alpha$	$y\alpha$

Total no. of solute particles $= 1-\alpha + x\alpha + y\alpha = 1-\alpha+\alpha(x+y)$

or $\quad i = 1-\alpha+n\alpha \quad$ [where $x + y = n$ (total ions.)]

or $\quad \mathbf{i = 1+\alpha(n-1)}$

For strong electrolytes : $\alpha = 1$ or 100%, so $i = n$ (total no. of ions)

$NaCl \rightarrow Na^+ + Cl^-$ $\quad (i = 2)$

$K_2SO_4 \rightarrow 2K^+ + SO_4^{2-}$ $\quad (i = 3)$

For complex compound

$K_4[Fe(CN)_6] \rightarrow 4K^+ + [Fe(CN)_6]^{4-} \Rightarrow \quad n = 5$

S. No.	Solute type	Example	Ionisation	No. of particles in the solution from 1 mole solute (n)	van't Hoff factor ('i')	Abnormal molecular mass
1.	Non-electrolyte	Urea, sucrose, glucose, fructose	—	1	1	m_{normal}
2.	Binary electrolyte AB type	NaCl, KCl, HCl CH_3COOH NH_4OH, NaOH etc.	$\rightleftharpoons\ {}^{+} +\ {}^{-}$ $-\alpha \quad \alpha \quad \alpha$	2	$1+\alpha$	$+\alpha$
3.	Ternary electrolyte AB_2 type or A_2B type	$CaCl_2$, $BaCl_2$ H_2SO_4, $K_2[PtCl_6]$	$\rightleftharpoons\ {}^{+} +\ {}^{-}$ $-\alpha \quad \alpha \quad \alpha$ $\rightleftharpoons\ {}^{+} +\ {}^{-}$ $-\alpha \quad \alpha \quad \alpha$	3 3	$1+2\alpha$ $1+2\alpha$	$+\ \alpha$ $+\ \alpha$
4.	Quanternary electrolyte AB_3 or A_3B type	$AlCl_3$, $K_3[Fe(CN)_6]$	$\rightleftharpoons\ {}^{+} +\ {}^{-}$ $-\alpha \quad \alpha \quad \alpha$ $\rightleftharpoons\ {}^{+} +\ {}^{-}$ $-\alpha \quad \alpha \quad \alpha$	4 4	$1+3\alpha$ $1+3\alpha$	$+\ \alpha$ $+\ \alpha$
5.	General electrolyte AB_{n-1}	One mole of solute giving 'n' ions in the solution	${}^{-} \rightleftharpoons\ {}^{+}\ {}^{-}\ +\ -\ {}^{-}$ $-\alpha \quad \alpha \quad \alpha$	n	$1+(n-1)\alpha$	$[\ +\ \ -\ \ \alpha]$

8.2 Calculation of 'i' in case of assocation

	$nA \longrightarrow$	$(A)_n$
Initial mol	1	0
after dissociation	$1-\alpha$	α/n

$$= \frac{-\alpha + \frac{\alpha}{}}{}\ , \quad \alpha = \text{degree of assocation}$$

$$i = 1 + \alpha\left(\frac{1}{n} - 1\right)$$

Ex.20. Determine the amount of $CaCl_2$ (i = 2.47) dissolved in 2.5 litre of water such that its osmotic pressure is 0.75 atm at 27°C.

Sol. *We know that,*

$$\pi = \quad \Rightarrow \pi = $$

$$\Rightarrow \quad = \frac{\pi}{} = \frac{\times \quad \times}{\times \quad \times} = 3.42\ g$$

Hence, the required amount of $CaCl_2$ is 3.42 g.

Ex.21 19.5 g of CH_2FCOOH is dissolved in 500 g of water. The depression in the freezing point of water observed is 1.0°C. Calculate the van't Hoff factor and dissociation constant of fluoroacetic acid.

Sol. *It is given that :*

$w_1 = 500\ g$

$w_2 = 19.5\ g$

$K_f = 1.86\ K\ kg\ mol^{-1}$

$\Delta T_f = 1\ K$

We know that :

$$= \frac{\times \quad \times}{\Delta \quad \times} = \frac{\times \quad \times}{\times} = 72.54\ mol^{-1}$$

Therefore, observed molar mass of CH_2FCOOH, $(M_2)_{obs} = 72.54$ mol

The calculated molar mass of CH_2FCOOH is :

$(M_2)_{cal} = 14 + 19 + 12 + 16 + 16 + 1 = 78\ g\ mol^{-1}$

Therefore, van't Hoff factor, $= \frac{}{} = \frac{}{} = 1.0753$

Let α be the degree of dissociation of CH_2FCOOH

	CH_2FCOOH	$\rightleftharpoons$ CH_2FCOO^-	$+$ H^+	
Initial conc.	C *mol* L^{-1}	0	0	
At equilibrium	$C(1-\alpha)$	$C\alpha$	$C\alpha$	*Total* $= C(1+\alpha)$

$$\therefore \quad = \frac{(\quad + \alpha)}{\quad}$$

$\Rightarrow i = 1 + \alpha \Rightarrow \alpha = i - 1$

$= 1.0753 - 1 = 0.0753$

Now, the value of K_a *is given as :*

$$= \frac{\quad - \quad -}{\quad} = \frac{\alpha \quad \alpha}{(\quad - \alpha)} = \frac{\alpha}{-\alpha}$$

Taking the volume of the solution as 500 mL. We have the concentration : 19.5 M

$$= \frac{\quad}{\quad} \times \qquad = 0.5\ M$$

Therefore, $= \dfrac{\alpha}{-\alpha}$

$$= \frac{\times(\qquad)}{-} = \frac{\times}{\quad} = 0.00307 \ \text{(approximately)} = 3.07 \times 10^{-3}$$

EXERCISE # S-I

(Raoult's law)

1. At 25°C, the vapour pressure of methyl alcohol is 96.0 torr. What is the mole fraction of CH_3OH in a solution in which the (partial) vapor pressure of CH_3OH is 24.0 torr at 25°C?

2. The vapour pressure of ethanol and methanol are 44.0 mm and 88.0 mm Hg respectively. An ideal solution is prepared at the same temperature by mixing 69 g of ethanol with 40 g of methanol. Calculate total vapour pressure of the solution.

3. Liquide 'A' and 'B' form an ideal solution. The vapour pressure of solution containing equal moles of both liquids is 80 cm Hg. At the same temperature, the vapour pressure of solution containing 25 mole percent of liquid 'A' is 70 cm Hg. Calculate P_A^0 and P_B^0.

4. Liquids 'A'' and 'B' form an ideal solution. Calculate the mole-fraction of 'A' in the vapours above the liquid solution containing the liquids 'A' and 'B' in 2 : 3 mole ratio, at equilibrium.

[Given : $P_A^0 = 0.4$ atm, $P_B^0 = 0.8$ atm]

5. Liquids 'P' and 'Q' form an ideal solution. At equilibrium, the vapours contain 40% molecules of 'P'. Calculate the vapour presure of solution.

[Given : $P_P^0 = 0.4$ bar, $P_Q^0 = 0.6$ bar]

6. Liquids 'X' and 'Y' form an ideal solution. The vapour pressure of solution may be expressed as : **P[/cmHg] = (80 – 25x)**, where 'x' is the mole-fraction of liquid 'X' in the liquid solution at equilibrium. Calculate the vapour pressures of pure liquids 'X' and 'Y'.

7. Liquid 'R' and 'S'' form an ideal solution. The mole-fraction of 'R' in liquid and vapour phases at equilibrium are 0.25 and 0.40, respectively. If the vapour pressure of solution is 0.50 bar, calculate P_R^0 and P_S^0

Colligative properties

8. The vapour pressure of pure liquid solvent A is 0.80 atm. When a nonvolatile substance B is added to the solvent its vapour pressure drops to 0.60 atm. What is the mole fraction of component B in the solution?

9. Calculate the relative lowering in vapour pressure if 100 g of a nonvolatile solute (mol.wt.100) are dissolved in 432 g water.

10. The vapour pressure of pure benzene at 30° C is 640 mm of Hg and the vapour pressure of a solution of a solute in C_6H_6 at the same temperature is 624 mm of Hg. Calculate molality of solution.

11. The vapour pressure of pure benzene at a certain temperature is 640 mm of Hg. A nonvolatile nonelectrolyte solid weighing 2.175 g is added to 39.0 of benzene. The vapour pressure of the solution is 600 mm of Hg. What is molecular weight of solid substance?

12. The vapour pressure of water is 17.54 mm Hg at 293 K. Calculate vapour pressure of 0.5 molal solution of a solute in it.

13. When 10.5 g of a nonvolatile substance is dissolved in 742 g of ether, its boiling point is raised 0.25°C. What is the molecular weight of the substance? Molal boiling point constant for ether is 2.12°C·kg/mol.

14. Calculate the molal elevation constant, K_b for water and the boiling point of 0.1 molal urea solution. Latent heat of vaporisation of water is 9.72 kcal mol^{-1} at 373.15 K. [$(373.15)^2 = 258 \times 540$]

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Liquid Solution\Eng\02_Exercise

ALLEN Session 2019-20

15. Pure benzene freeze at 5.45°C. A solution containing 6.72 g of $C_2H_2Cl_4$ in 120 g of benzene was observed to freeze at 3.75°C. What is the molal freezing point constant of benzene?

16. The freezing point of a solution containing 2.40 g of a compound in 60.0 g of benzene is 0.10°C lower than that of pure benzene. What is the molecular weight of the compound? (K_f is 5.12°C/m for benzene)

17. Normal boiling point of diethyl ether is 327° and at 190 mmHg boiling points in 27°C. What is the value of $\Delta H^{\circ}_{vap.}$ in kJ/mole.

(Use : R = 8.3 J/K-mole, ln2 = 0.7)

18. A 6.84% solution (w/v) of cane-sugar (Mol. weight = 342) is isotonic with 0.8%(w/v) solution of non-volatile solute. Find molecular weight of solute.

19. Calculate the osmotic pressure of 12% (w/v) aq. urea solution at 27°C.

20. Calculate the osmotic pressure of a solution containing 18 gm glucose and 17.1 gm canesugar ($C_{12}H_{22}O_{11}$) per litre, at 27°C.

21. A storage battery contains a solution of H_2SO_4 38% by weight. What will be the Van't Hoff factor if the $\Delta T_{f(experiment)}$ is 22.8K. [Given K_f = 1.86 mol^{-1} Kg]

22. A certain mass of a substance, when dissolved in 100 g C_6H_6, lowers the freezing point by 1.28°C. The same mass of solute dissolved in 100g water lowers the freezing point by 1.40°C. If the substance has normal molecular weight in benzene and is completely ionized in water, into how many ions does it dissociate in water? K_f for H_2O and C_6H_6 are 1.86 and 5.12K kg mol^{-1}.

23. 2.0 g of benzoic acid dissolved in 25.0g of benzene shows a depression in freezing point equal to 1.96K. Molal depression constant (K_f) of benzene is 4.9 K.kg.mol^{-1}. What is the percentage association of the acid?

24. A decimolar solution of potassium ferrocyanide is 50% (w/v) dissociated at 300K. Calculate the osmotic pressure of the solution. R=8.314 JK^{-1} mol^{-1})

25. A 1.2% solution (w/v) of NaCl is isotonic with 7.2% solution (w/v) of glucose. Calculate degree of ionization and Van't Hoff factor of NaCl.

Henry's law

26. How many gm O_2 gas will disolve in 100 gm water at 9 bar and 27°C? (K_H = 40 Kbar)

27. The Henry law constant for dissolution of a gas in aqueous medium is 3×10^2 atm. At what partial pressure of the gas (in atm), the molality of gas in aqueous solution will be $\frac{5}{9}$ m.

EXERCISE # S-II

1. At 90°C, the vapour pressure of toluene is 400 torr and that of σ-xylene is 150 torr. What is the composition of the liquid mixture that boils at 90°C, when the pressure is 0.50 atm? What is the composition of vapour produced?

2. The vapour pressure of an aqueous solution of glucose is 750 mm Hg at 373 K. Calculate molality and mole fraction of solute.

3. Calculate the amount of ice that will separate out of cooling a solution containing 50g of ethylene glycol in 200 g water to –9.3°C. (K_f for water = 1.86 K mol^{-1} kg)

4. A solution of crab hemocyanin, a pigmented protein extracted from crabs, was prepared by dissolving 0.750 g in 125 cm^3 of an aqueous medium. At 4°C an osmotic pressure rise of 2.77 mm of the solution was observed. The solution had a density of 1.013 g/cm^3. Determine the molecular weight of the protein. (g = 10 m/s^2)

5. The vapour pressure of an aqueous solution is found to be 750 torr at certain temperature 'T'. If 'T' is the temperature at which pure water boils under atmospheric pressure, calculate the atmospheric pressure. The boiling point of solution is 101.04ºC. (K_b = 0.52 K kg mol^{-1}).

6. How many grams of sucrose (mol.wt. = 342) should be dissolved in 100 gm water in order to produce a solution with 105°C difference between the freezing point & boiling point temperature at 1 atm? **(Unit: K_f = 2 K.kg mol^{-1} ; k_b = 0.5 K.kg mol^{-1})**

7. An aqueous solution containing 288 gm of a non-volatile compound having the stoichiometric composition $C_xH_{2x}O_x$ in 90 gm water boils at 101.36°C at 1.00 atmospheric pressure. What is the molecular formula? $K_b(H_2O)$ = 0.52 K mol^{-1} kg

8. A complex is represented as $CoCl_3 \cdot xNH_3$. It's 0.1 molal solution in a solution shows ΔT_f = 0.558°C. K_f for H_2O is 1.86 K mol^{-1} kg . Assuming 100% ionisation of complex and coordination no. of Co is six, calculate formula of complex.

9. Phenol associates in benzene to a certain extent to form a dimer. A solution containing 18.8×10^{-3} kg phenol in 1 kg of benzene has its freezing point depressed by 0.768 K. Calculate the fraction of phenol that has dimerised. K_f for benzene = 5.12 kg mol^{-1}K.

10. The molar volume of liquid benzene (density = 0.877 g ml^{-1}) increases by a factor of 2750 as it vaporizes at 20°C and that of liquid toluene (density = 0.867gml^{-1}) increases by a factor of 7720 at 20°C. A Solution of benzene & toluene has a vapour pressure of 46.0 torr. Find the mole fraction of benzene in the vapour above the solution.

11. A solution of 0.1 M CH_3COOH is placed between parallel electrodes of cross-section area 4cm^2, separated by 2cm. For this solution resistance measured is 100Ω. If the elevation in boiling point of the 0.1 M CH_3COOH solution is 'x'K, then the value of (160x) is.

K_b = 0.5 K kg/mol; $\wedge_m^\infty (H^+)$ = 300 Scm^2 $mole^{-1}$; $\wedge_m^\infty (CH_3COO^-)$ = 100 Scm^2 $mole^{-1}$

12. Cane sugar underoges the inversion as follow

$$C_{12}H_{22}O_{11} + H_2O \rightarrow C_6H_{12}O_6 + C_6H_{12}O_6$$

If solution of 0.025 moles of sugar in 200 gm of water show depresion in freezing point 0.372ºC, then what % sucrose has inverted. ($K_f(H_2O)$ = 1.86 K kg mol^{-1})

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Liquid Solution\Eng\02_Exercise

13. When 0.1 M $Pb(NO_3)_2$ solution is titrated with 0.1 M KI solution then what will be the osmotic pressure (in atm) of solution when equivalence point is reached at 300 K .

(Take : R = 0.08 atm L/mol-k)

14. Using the following information determine the boiling point of a mixture contains 1560 gm benzene and 1125 gm chlorobenzene, when the external pressure is 1000 torr. Assume the solution is ideal.

Temperature (0°C)	Vapour pressure of benzene(torr)	Vapour pressure of chlorobenzene(torr)
80	750	120
90	1000	200
100	1350	300
110	1800	400
120	2200	540

15. The cryoscopic constant for acetic acid is 3.6 K kg/mol. A solution of 1 g of a hydrocarbon in 100 g of acetic acid freezes at 16.14°C instead of the usual 16.60°C. The hydrocarbon contains 92.3% carbon. What is the molecular formula?

ALLEN Session 2019-20

EXERCISE # O-I

Single correct :

1. The boiling point of C_6H_6, CH_3OH, $C_6H_5NH_2$ and $C_6H_5NO_2$ are 80^0C, 65^0C, 184^0C and 212^0C respectively which will show highest vapour pressure at room temperature :

(A) C_6H_6 (B) CH_3OH (C) $C_6H_5NH_2$ (D) $C_6H_5NO_2$

2. Mole fraction of A vapours above the solution in mixture of A and B ($X_A = 0.4$) will be

[Given : P_A° = 100 mm Hg and P_B° = 200 mm Hg]

(A) 0.4 (B) 0.8 (C) 0.25 (D) none of these

3. At a given temperature, total vapour pressure in Torr of a mixture of volatile components A and B is given by

$$P_{Total} = 120 - 75\ X_B$$

hence, vapour pressure of pure A and B respectively (in Torr) are

(A) 120, 75 (B) 120, 195 (C) 120, 45 (D) 75, 45

4. Two liquids A & B form an ideal solution. What is the vapour pressure of solution containing 2 moles of A and 3 moles of B at 300 K? [Given : At 300 K, Vapour pr. of pure liquid A (P_A^o) = 100 torr, Vapour pr. of pure liquid B (P_B^o) = 300 torr]

(A) 200 torr (B) 140 torr (C) 180 torr (D) None of these

5. If Raoult's law is obeyed, the vapour pressure of the solvent in a solution is directly proportional to

(A) Mole fraction of the solvent

(B) Mole fraction of the solute

(C) Mole fraction of the solvent and solute

(D) The volume of the solution

6. 1 mole of heptane (V. P. = 92 mm of Hg) was mixed with 4 moles of octane (V. P. = 31mm of Hg). The vapour pressure of resulting ideal solution is :

(A) 46.2 mm of Hg (B) 40.0 mm of Hg

(C) 43.2 mm of Hg (D) 38.4 mm of Hg

7. Mole fraction of A vapours above solution in mixture of A and B ($X_A = 0.4$) will be :-

(P_A° = 100mm, P_B° = 200mm)

(A) 0.4 (B) 0.8 (C) 0.25 (D) None

8. The vapour pressure of a pure liquid 'A' is 70 torr at 27^0C. It forms an ideal solution with another liquid B. The mole fraction of B is 0.2 and total vapour pressure of the solution is 84 torr at 27^0C. The vapour pressure of pure liquid B at 27^0C is

(A) 14 (B) 56 (C) 140 (D) 70

9. At 88 °C benzene has a vapour pressure of 900 torr and toluene has a vapour pressure of 360 torr. What is the mole fraction of benzene in the mixture with toluene that will boil at 88 °C at 1 atm. pressure, benzene - toluence form an ideal solution:

(A) 0.416 (B) 0.588 (C) 0.688 (D) 0.740

ALLEN Session 2019-20
node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Liquid Solution\Eng\02_Exercise

10. The exact mathematical expression of Raoult's law is (n = moles of solute ; N = moles of solvent)

(A) $\dfrac{P^0 - P_s}{P^0} = \dfrac{n}{N}$ (B) $\dfrac{P^0 - P_s}{P^0} = \dfrac{N}{n}$ (C) $\dfrac{P^0 - P_s}{P_s} = \dfrac{n}{N}$ (D) $\dfrac{P^0 - P_s}{P^0} = n \times N$

11. The vapour pressure of a solvent decreased by 10 mm of Hg when a non-volatile solute was added to the solvent. The mole fraction of solute in solution is 0.2, what would be mole fraction of the solvent if decrease in vapour pressure is 20 mm of Hg

(A) 0.2 (B) 0.4 (C) 0.6 (D) 0.8

12. The vapour pressure of a solution having solid as solute and liquid as solvent is :

(A) Directly proportional to mole fraction of the solvent

(B) Inversely proportional to mole fraction of the solvent

(C) Directly proportional to mole fraction of the solute

(D) Inversely propartional to mole fraction of the solute

13. One mole of non volatile solute is dissolved in two moles of water. The vapour pressure of the solution relative to that of water is

(A) – (B) – (C) – (D) –

14. The vapour pressure of pure A is 10 torr and at the same temperature when 1 g of B is dissolved in 20 gm of A, its vapour pressure is reduced to 9.0 torr. If the molecular mass of A is 200 amu, then the molecular mass of B is :

(A) 100 amu (B) 90 amu (C) 75 amu (D) 120 amu

15. The vapour pressure of a pure liquid solvent (X) is decreased to 0.60 atm. from 0.80 atm on addition of a non volatile substance (Y). The mole fraction of (Y) in the solution is:-

(A) 0.20 (B) 0.25 (C) 0.5 (D) 0.75

16. Among the following, that does not form an ideal solution is :

(A) C_6H_6 and $C_6H_5CH_3$ (B) C_2H_5Cl and C_6H_5OH

(C) C_6H_5Cl and C_6H_5Br (D) C_2H_5Br and C_2H_5I

17. Colligative properties of the solution depend upon

(A) Nature of the solution (B) Nature of the solvent

(C) Number of solute particles (D) Number of moles of solvent

18. Elevation of boiling point of 1 molar aqueous glucose solution (density = 1.2 g/ml) is

(A) K_b (B) 1.20 K_b (C) 1.02 K_b (D) 0.98 K_b

19. When common salt is dissolved in water

(A) Melting point of the solution increases

(B) Boiling point of the solution increases

(C) Boiling point of the solution decreases

(D) Both Melting point and Boiling point is decreases

20. What should be the freezing point of aqueous solution containing 17 gm of C_2H_5OH in 1000 gm of water (water K_f = 1.86 deg – kg mol^{-1})

(A) – 0.69°C (B) – 0.34°C (C) 0.0°C (D) 0.34°C

Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Liquid Solution\Eng\02_Exercise

21. If mole fraction of the solvent in solution decreases then :
(A) Vapour pressure of solution increases
(B) B. P. decreases
(C) Osmotic pressure increases
(D) All are correct

22. 5% (w/v) solution of sucrose is isotonic with 1% (w/v) solution of a compound 'A' then the molecular weight of compound 'A' is -
(A) 32.4 (B) 68.4 (C) 121.6 (D) 34.2

23. Osmotic pressure of a sugar solution at 24^0C is 2.5 atmosphere. The concentration of the solution in mole per litre is :
(A) 10.25 (B) 1.025 (C) 1025 (D) 0.1025

24. A solution containing 4 g of a non volatile organic solute per 100 ml was found to have an osmotic pressure equal to 500 cm of mercury at 27^0C. The molecular weight of solute is :
(A) 14.97 (B) 149.7 (C) 1697 (D) 1.497

25. If a 6.84% (wt. / vol.) solution of cane-sugar (mol. wt. 342) is isotonic with 1.52% (wt./vol.) solution of thiocarbamide, then the molecular wight of thiocarbamide is :
(A) 152 (B) 76 (C) 60 (D) 180

26. Which of the following aqueous solution will show maximum vapour pressure at 300 K?
(A) 1 M NaCl (B) 1 M $CaCl_2$ (C) 1 M $AlCl_3$ (D) 1 M $C_{12}H_{22}O_{11}$

27. The correct relationship between the boiling points of very dilute solution of $AlCl_3$ (T_1K) and $CaCl_2$ (T_2K) having the same molar concentration is
(A) $T_1 = T_2$ (B) $T_1 > T_2$ (C) $T_2 > T_1$ (D) $T_2 \leq T_1$

28. 1.0 molal aqueous solution of an electrolyte A_2B_3 is 60% ionised. The boiling point of the solution at 1 atm is ($K_{b(H_2O)} = 0.52$ K kg mol^{-1})
(A) 274.76 K (B) 377 K (C) 376.4 K (D) 374.76 K

29. The freezing point depression of a 0.1 M a solution of weak acid (HX) is –0.20°C.
What is the value of equilibrium constant for the reaction?

$$HX\,(aq) \rightleftharpoons H^+(aq) + X^-\,(aq)$$

[Given : K_f for water = 1.8 kg mol^{-1} K. & Molality = Molarity]
(A) 1.46×10^{-4} (B) 1.35×10^{-3} (C) 1.21×10^{-2} (D) 1.35×10^{-4}

30. The Vant Hoff factor (i) for a dilute solution of $K_3[Fe(CN)_6]$ is (Asuming 100% ionsation) :
(A) 10 (B) 4 (C) 5 (D) 0.25

31. The substance A when dissolved in solvent B shows the molecular mass corresponding to A_3. The vant Hoff's factor will be -
(A) 1 (B) 2 (C) 3 (D) 1/3

ALLEN Session 2019-20

32. The value of observed and calcutated molecular wieght of silver nitrate are 92.64 and 170 respectively. The degree of dissociation of silver nitrate is :

(A) 60% (B) 83.5 % (C) 46.7% (D) 60.23%

33. The freezing point of 1 molal NaCl solution assuming NaCl to be 100% dissociated in water is : (K_f = 1.86 K Molality^{-1})

(A) –1.86 °C (B) –3.72 °C (C) +1.86 °C (D) + 3.72 °C

34. What is the freezing point of a solution containing 8.1 gm. of HBr in 100gm. water assuming the acid to be 90% ionised (K_f for water=1.86 K molality^{-1}) :-

(A) 0.85°C (B) –3.53°C (C) 0°C (D) –0.35°C

35. If a ground water contains H_2S at concentration of 2 mg/l, determine the pressure of H_2S in head space of a closed tank containing the ground water at 20ºC. Given that for H_2S, Henry's constant is equal to 6.8×10^3 bar at 20ºC.

(A) 720 Pa (B) 77×10^2 Pa (C) 553 Pa (D) 55×10^2 Pa

36. A pressure cooker reduces cooking time for food because -

(A) The higher pressure inside the cooker crushes the food material

(B) Cooking involves chemical changes helped by a rise in temperature

(C) Heat is more evenly distributed in the cooking space

(D) Boiling point of water involved in cooking is increased

ALLEN Session 2019-20

EXERCISE # O-II

Single correct :

1. An ideal solution was obtained by mixing (MeOH) methanol and (EtOH) ethanol. If the partial vapour pressure of methanol and ethanol are 2.619 K Pa and 4.556 K Pa respectively, the composition of vapour (in terms of mole fraction) will be -

(A) 0.635 MeOH, 0.365 EtOH (B) 0.365 MeOH, 0.635 EtOH

(C) 0.574 MeOH, 0.326 EtOH (D) 0.173 MeOH, 0.827 EtOH

2. Molar volume of liquid A(d = 0.8gm/ml) increase by factor of 2000 when it vapourises at 200K. Vapour pressure of liquid A at 200K is [R = 0.08 L-atm/mol-K]

(Molar mass of A = 80g/mol)

(A) 0.4 atm (B) 8 atm (C) 0.8 atm (D) 0.08 atm

3. Assuming each salt to be 90 % dissociated, which of the following will have highest boiling point?

(A) Decimolar $Al_2(SO_4)_3$

(B) Decimolar $BaCl_2$

(C) Decimolar Na_2SO_4

(D) A solution obtained by mixing equal volumes of (B) and (C)

4. The vapour pressure of a saturated solution of sparingly soluble salt (XCl_3) was 17.20 mm Hg at 27°C. If the vapour pressure of pure H_2O is 17.25 mm Hg at 300 K, what is the solubility of sparingly soluble salt XCl_3 in mole/Litre.

(A) 4.04×10^{-2} (B) 8.08×10^{-2} (C) 2.02×10^{-2} (D) 4.04×10^{-3}

5. A solution has a 1 : 4 mole ratio of pentane to hexane. The vapour pressures of the pure hydrocarbons at 20°C are 440 mmHg for pentane and 120 mmHg for hexane. The mole fraction of pentane in the vapour phase would be :-

(A) 0.200 (B) 0.478 (C) 0.549 (D) 0.786

6. For which of the following vant' Hoff's factor is not correctly matched -

	Salt	**Degree of dissociation** (α)	***i***
(A)	Na_2SO_4	50 %	2
(B)	$K_3[Fe(CN)_6]$	75%	3.25
(C)	$[Ag(NH_3)_2]Cl$	80 %	1.8
(D)	$[Cr(NH_3)_5Cl]SO_4$	90 %	2.8

7. In the depression of freezing point experiment, it is found that

(I) The vapour pressure of the solution is less than that of pure solvent.

(II) The vapour pressure of the solution is more than that of pure solvent.

(III) Only solute molecules solidify at the freezing point.

(IV) Only solvent molecules solidify at the freezing point.

(A) I, II (B) II, III (C) I, IV (D) I, II, III

Session 2019-20

8. **Statement-1** :Additon of ethylene glycol (non-volatile) to water lowers the freezing point of water hence used as antifreeze.

Statement-2 : Addition of any substance to water lowers its freezings point of water.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

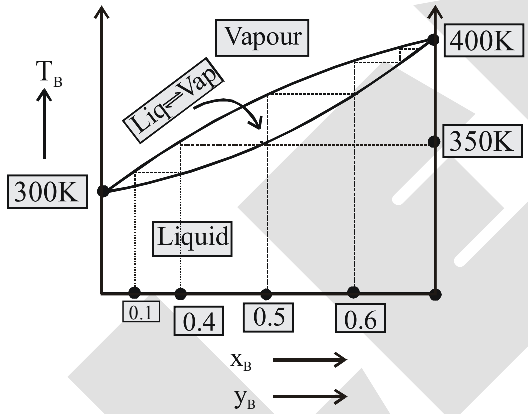
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

More than one may be correct

9. For an ideal solution having two liquid A(P_A^0 and liquid B(P_B^0) boiling point versus composition graph is given below, then select incorrect statement (s) :



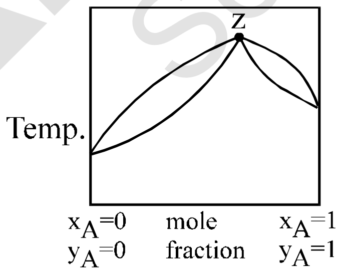
(A) B is less volatile than A

(B) If mole percent of A in liquid phase is 40% then mole percent of A in vapour phase is 50%

(C) If mole percent of B in liquid phase is 10% then mole percent of B in vapour phase is 40%

(D) If the mole percent of A in solution is 50% then its boiling point is 350K

10 A liquid mixture having composition corresponding to point z in the figure shown is subjected to distillation at constant pressure.



Which of the following statement is correct about the process

(A) The composition of distillate differs from the mixture

(B) The boiling point goes on changing

(C) The mixture has lowest vapour pressure than for any other composition.

(D) Composition of an azeotrope alters on changing the external pressure.

11. Which of the following is correct for a non-ideal solution of liquids A and B, showing negative deviation?

(A) $\Delta H_{mix} = -ve$ (B) $\Delta V_{mix} = -ve$

(C) $\Delta S_{mix} = +ve$ (D) $\Delta G_{mix} = -ve$

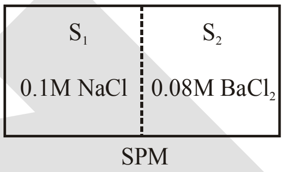
12. Two solutions S_1 and S_2 containing 0.1M NaCl(aq.) and 0.08M $BaCl_2$(aq.) are separated by semipermeable membrane. Which among the following statement(s) is/are correct -

(A) S_1 and S_2 are isotonic

(B) S_1 is hypertonic and S_2 is hypotonic

(C) S_1 is hypotonic and S_2 is hypertonic

(D) Osmosis will take place to from S_1 to S_2



13. For an ideal binary liquid solution with $P_A^\circ > P_B^\circ$, which relation between X_A (mole fraction of A in liquid phase) and Y_A(mole fraction of A in vapour phase) is correct?

(A) $Y_A < Y_B$ (B) $X_A > X_B$ (C) $\frac{Y_A}{Y_B} > \frac{X_A}{X_B}$ (D) $\frac{Y_A}{Y_B} < \frac{X_A}{X_B}$

14. Which of the following plots represents an ideal binary mixture?

(A) Plot of P_{total} v/s $1/X_B$ is linear (X_B = mole fraction of 'B' in liquid phase).

(B) Plot of P_{total} v/s Y_A is linear (Y_B = mole fraction of 'A' in vapour phase)

(C) Plot of $\frac{1}{P_{total}}$ v/s Y_A is linear

(D) Plot of $\frac{1}{P_{total}}$ v/s Y_B is non linear

Paragraph for Q.15 to Q.17

An ideal solution is obtained by mixing a non-volatile solute B with a volatile solvent A (molar mass = 60). If the mass ratio of A : B in solution is 10 : 1 and vapour pressure of pure A is 400 mm and vapour pressure decreases by 4% on forming the above solution at 300K.

15. The mole fraction of solute in the solution is -

(A) 0.96 (B) 0.04 (C) 0.16 (D) 0.84

16. The molality the solution is -

(A) 1 (B) $\frac{36}{25}$ (C) $\frac{25}{36}$ (D) $\frac{36}{25}$

17. The molar mass of B in the solution is -

(A) 1440 (B) 14400 (C) 4 (D) 144

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Liquid Solution\Eng\02_Exercise

Table type question :

	Column-I		Column-II		Column-III
(1)	$C_6H_6 + C_6H_5{-}CH_3$	(a)	$\Delta H = 0$	(P)	$\Delta G = -ve$
(2)	$CHCl_3 + CH_3COCH_3$	(b)	$\Delta H = +ve$	(Q)	Form minimum boiling azerotrope
(3)	$CCl_4 + CH_3COCH_3$	(c)	$\Delta H = -ve$	(R)	Form maximum boiling azerotrope
(4)	$C_2H_5OH + H_2O$	(d)	$\Delta S = +ve$	(S)	No azeotrope

18. Select the correct match -

(A) 1 , a, R (B) 2, b, P (C) 3, b, Q (D) 4, d, R

19. Select the correct match -

(A) 1 , d, S (B) 3, b, S (C) 2, b, S (D) 4, c, P

20. Select the incorrect match -

(A) 1 , d, P (B) 2, c, R (C) 3, d, S (D) 3, b, P

Match the column :

21.

	Column–I (Colligative properties)		Column–II (Aqueous solution) (Assume m = M)
(A)	$\Delta T_f = 0.3 \times K_f$	(P)	$0.1\ m - Ca(NO_3)_2$
(B)	$\Delta T_b = 0.28 \times K_b$	(Q)	$0.14\ m - NaBr$
(C)	$\pi = 0.19 \times RT$	(R)	$0.1\ m - MgCl_2 (\alpha = 0.9)$
(D)	$\dfrac{P^0 - P}{P^0} = \dfrac{\left(\dfrac{\Delta T_f}{K_f}\right)}{\left(\dfrac{1000}{18}\right) + \left(\dfrac{\Delta T_f}{K_f}\right)}$	(S)	$0.28\ m -$ Urea
		(T)	$0.1\ m - HA$ (monobasic acid, $K_a = 0.81$)

ALLEN Session 2019-20

EXERCISE # (J-MAIN)

1. The degree of dissociation (1) of a weak electrolyte, A_xB_y is related to van't Hoff factor (i) by the expression :- **[AIEEE-2011]**

(1) $\alpha=\frac{x+y-1}{i-1}$ (2) $\alpha=\frac{x+y+1}{i-1}$ (3) $\alpha=\frac{i-1}{(x+y-1)}$ (4) $\alpha=\frac{i-1}{x+y+1}$

2. K_f for water is 1.86 K kg mol^{-1}. If your automobile radiator holds 1.0 kg of water, how many grams of ethylene glycol ($C_2H_6O_2$) must you add to get the freezing point of the solution lowered to –2.8°C ? **[AIEEE-2012]**

(1) 27 g (2) 72 g (3) 93 g (4) 39 g

3. A solution containing 0.85 g of $ZnCl_2$ in 125.0 g of water freezes at –0.23ºC. The apparent degree of dissociation of the salt is : **[JEE (MAIN)-2012 ONLINE]**

(k_f for water = 1.86 K kg mol^{-1}, atomic mass ; Zn = 65.3 and Cl = 35.5)

(1) 1.36% (2) 2.47% (3) 73.5% (4) 7.35%

4. Liquids A and B form an ideal solution. At 30ºC, the total vapour pressure of a solution containing 1 mol of A and 2 moles of B is 250 mm Hg. The total vapour pressure becomes 300 mm Hg when 1 more mol of A is added to the first solution. The vapour pressures of pure A and B at the same temperature are **[JEE (MAIN)-2012 ONLINE]**

(1) 450, 150 mm Hg (2) 250, 300 mm Hg (3) 125, 150 mm Hg (4) 150, 450 mm Hg

5. The freezing point of a 1.00 m aqueous solution of HF is found to be –1.91°C. The freezing point constant of water, K_f, is 1.86 K kg mol^{-1}. The percentage dissociation of HF at this concentration is **[JEE (MAIN)-2012 ONLINE]**

(1) 2.7% (2) 30% (3) 10% (4) 5.2%

6. How many grams of methyl alcohol should be added to 10 litre tank of water to prevent its freezing at 268 K ? **[JEE (MAIN)-2013 ONLINE]**

(K_f for water is 1.86 K kg mol^{-1})

(1) 899.04 g (2) 786 g (3) 860 g (4) 880.07 g

7. Vapour pressure of pure benzene is 119 torr and that of toluene is 37.0 torr at the same temperature. Mole fraction of toluene in vapour phase which is in equilibrium with a solution of benzene and toluene having a mole fraction of toluene 0.50, will be : **[JEE (MAIN)-2013 ONLINE]**

(1) 0.137 (2) 0.205 (3) 0.237 (4) 0.435

8. A molecule M associates in a given solvent according to the equation $M \rightleftharpoons (M)_n$. For a certain concentration of M, the van't Hoff factor was found to be 0.9 and the fraction of associated molecules was 0.2. The value of n is : **[JEE (MAIN)-2013 ONLINE]**

(1) 2 (2) 4 (3) 5 (4) 3

9. 12g of a nonvolatile solute dissolved in 108g of water produces the relative lowering of vapour pressure of 0.1. The molecular mass of the solute is : **[JEE (MAIN)-2013 ONLINE]**

(1) 60 (2) 80 (3) 40 (4) 20

10. The molarity of a solution obtained by mixing 750 mL of 0.5(M)HCl with 250 mL of 2(M)HCl will be :- **[JEE (MAIN)-2013]**

(1) 0.875 M (2) 1.00 M (3) 1.75 M (4) 0.975 M

ALLEN Session 2019-20

11. The observed osmotic pressure for a 0.10 M solution of $Fe(NH_4)_2(SO_4)_2$ at 25°C is 10.8 atm. The expected and experimental (observed) values of Van't Hoff factor (i) will be respectively :
(R= 0.082 L atm k^{-} mol^{-1}) **[JEE (MAIN)-2014 ONLINE]**
(1) 3 and 5.42 (2) 5 and 3.42 (3) 4 and 4.00 (4) 5 and 4.42

12. For an ideal Solution of two components A and B, which of the following is true ?
(1) $\Delta H_{mixing} < 0$ (zero) **[JEE(MAIN)-2014 ONLINE]**
(2) A – A, B – B and A – B interactions are identical
(3) A – B interaction is stronger than A – A and B – B interactions
(4) $\Delta H_{mixing} > 0$ (zero)

13. Consider separate solution of 0.500 M $C_2H_5OH(aq)$, 0.100 M $Mg_3(PO_4)_2(aq)$, 0.250 M KBr(aq) and 0.125 M $Na_3PO_4(aq)$ at 25°C. Which statement is true about these solutions, assuming all salts to be strong electrolytes ? **[JEE (MAIN)-2014]**
(1) 0.125 M Na_3PO_4 (aq) has the highest osmotic pressure.
(2) 0.500 M C_2H_5OH (aq) has the highest osmotic pressure.
(3) They all have the same osmotic pressure.
(4) 0.100 M $Mg_3(PO_4)_2$ (aq) has the highest osmotic pressure.

14. Determination of the molar mass of acetic acid in benzene using freezing point depression is affected by :
[JEE (MAIN)-2015 ONLINE]
(1) association (2) dissociation (3) complex formation (4) partial ionization

15. A solution at 20°C is composed of 1.5 mol of benzene and 3.5 mol of toluene. If the vapour pressure of pure benzene and pure toluene at this temperature are 74.7 torr and 22.3 torr, respectively, then the total vapour pressure of the solution and the benzene mole fraction in equilibrium with it will be, respectively :
[JEE (MAIN)-2015 ONLINE]
(1) 38.0 torr and 0.589 (2) 30.5 torr and 0.389
(3) 35.8 torr and 0.280 (4) 35.0 torr and 0.480

16. The vapour pres sure of acetone at 20°C is 185 torr. When 1.2 g of non-volatile substance was dissolved in 100 g of acetone at 20°C, its vapour pressure was 183 torr. The molar mass (g mol^{-1}) of the substance is : **[JEE (MAIN)-2015]**
(1) 128 (2) 488 (3) 32 (4) 64

17. An aqueous solution of a salt MX_2 at certain temperature has a van't Hoff factor of 2. The degree of dissociation for this solution of the salt is : **[JEE (MAIN-2016-ONLINE]**
(1) 0.50 (2) 0.80 (3) 0.67 (4) 0.33

18. The solubility of N_2 in water at 300 K and 500 torr partial pressure is 0.01 g L^{-1}. The solubility (in g L^{-1}) at 750 torr partial pressure is : **[JEE (MAIN-2016-ONLINE]**
(1) 0.02 (2) 0.005 (3) 0.015 (4) 0.0075

19. 18 g glucose ($C_6H_{12}O_6$) is added to 178.2g at 100ºC water. The vapour pressure of water (in torr) for this aqueous solution is : **[JEE (MAIN)-2016]**
(1) 759.0 (2) 7.6 (3) 76.0 (4) 752.4

Session 2019-20

20. The freezing point of benzene decreases by 0.45°C when 0.2 g of acetic acid is added to 20 g of benzene. If acetic acid associates to form a dimer in benzene, percentage association of acetic acid in benzene will be :- **(K_f for benzene = 5.12 K kg mol^{-1})** **[JEE (MAIN)-2017]**

(1) 64.6% (2) 80.4% (3)74.6% (4) 94.6%

21. 5 g of Na_2SO_4 was dissolved in x g of H_2O. The change in freezing point was found to be 3.82°C. If Na_2SO_4 is 81.5% ionised, the value of x **[JEE (MAIN-2017-ONLINE]**

(K_f for water = 1.86°C kg mol^{-1}) is approximately.

(Molar mass of S = 32 g mol^{-1} and that of Na = 23 g mol^{-1})

(1) 45 g (2) 65 g (3) 15 g (4) 25 g

22. A solution is prepared by mixing 8.5 g of CH_2Cl_2 and 11.95 g of $CHCl_3$. If vapour pressure of CH_2Cl_2 and $CHCl_3$ at 298 K are 415 and 200 mmHg respectively, the mole fraction of $CHCl_3$ in vapour form is: (Molar mass of Cl = 35.5 g mol^{-1}) **[JEE (MAIN-2017-ONLINE]**

(1) 0.486 (2) 0.325 (3) 0.162 (4) 0.675

23. For 1 molal aqueous solution of the following compounds, which one will show the highest freezing point ?

(1) $[Co(H_2O)_5Cl]Cl_2.H_2O$ **[JEE (MAIN)-2018]**

(2) $[Co(H_2O)_4Cl_2]Cl.2H_2O$

(3) $[Co(H_2O)_3Cl_3].3H_2O$

(4) $[Co(H_2O)_6]Cl_3$

24. Two 5 molal solutions are prepared by dissolving a non-electrolyte non-volatile solute separately in the solvents X and Y. The molecular weights of the solvents are M_X and M_Y, respectively where $M_X = \frac{3}{4} M_Y$. The relative lowering of vapour pressure of the solution in X is "m" times that of the solution in Y. Given that the number of moles of solute is very small in comparison to that of solvent, the value of "m" is - **[JEE (MAIN-2018-ONLINE]**

(1) $\frac{3}{4}$ (2) $\frac{4}{3}$ (3) $\frac{1}{2}$ (4) $\frac{1}{4}$

25. The mass of a non-volatile, non-electrolyte solute (molar mass = 50 g mol^{-1}) needed to be dissolved in 114 g octane to reduce its vapour pressure by 75%, is :- **[JEE (MAIN-2018-ONLINE]**

(1) 50 g (2) 37.5 g (3) 75 g (4) 150 g

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Liquid Solution\Eng\02_Exercise

EXERCISE # (J-ADVANCED)

1. To 500 cm^3 of water, 3×10^{-3} kg of acetic acid is added. If 23% of acetic acid is dissociated, what will be the depression in freezing point ? K_f and density of water are 1.86 K kg^{-1} mol^{-1} and 0.997 g cm^{-3} respectively. **[JEE 2000]**

2. The vapour pressure of two miscible liquids (A) and (B) are 300 and 500 mm of Hg respectively. In a flask 10 mole of (A) is mixed with 12 mole of (B). However, as soon as (B) is added, (A) starts polymerising into a completely insoluble solid. The polymerisation follows first-order kinetics. After 100 minute, 0.525 mole of a solute is dissolved which arrests the polymerisation completely. The final vapour pressure of the solution is 400 mm of Hg. Estimate the rate constant of the polymerisation reaction. Assume negligible volume change on mixing and polymerisation and ideal behaviour for the final solution. **[JEE 2001]**

3. Match the boiling point with K_b for x, y and z, if molecular weight of x, y and z are same. **[JEE 2003]**

	b.pt.	$\mathbf{K_b}$
x	100	0.68
y	27	0.53
z	253	0.98

4. During depression of freezing point in a solution, the following are in equilibrium **[JEE 2003]**

(A) liquid solvent-solid solvent (B) liquid solvent-solid solute

(C) liquid solute-solid solute (D) liquid solute-solid solvent

5. 1.22 g of benzoic acid is dissolved in (i) 100 g acetone (K_b for acetone = 1.7) and (ii)100 g benzene (K_b for benzene = 2.6). The elevation in boiling points T_b is 0.17°C and 0.13°C respectively.

(a) What are the molecular weights of benzoic acid in both the solutions?

(b) What do you deduce out of it in terms of structure of benzoic acid? **[JEE 2004]**

6. A 0.004 M solution of Na_2SO_4 is isotonic with a 0.010 M solution of glucose at same temperature. The apparent degree of dissociation of Na_2SO_4 is

(A) 25% (B) 50% (C) 75% (D) 85% **[JEE 2004]**

7. The elevation in boiling point, when 13.44 g of freshly prepared $CuCl_2$ are added to one kilogram of water, is [Some useful data, K_b (H_2O) = 0.52 kg K mol^{-1}, mol. wt. of $CuCl_2$ = 134.4 gm]

(A) 0.05 (B) 0.1 (C) 0.16 (D) 0.21 **[JEE 2005]**

8. 72.5 g of phenol is dissolved in 1 kg of a solvent (k_f = 14) which leads to dimerization of phenol and freezing point is lowered by 7 kelvin. What percent of total phenol is present in dimeric form? **[JEE 2006]**

9. When 20 g of naphtholic acid ($C_{11}H_8O_2$) is dissolved in 50 g of benzene (K_f = 1.72 K kg mol^{-1}), a freezing point depression of 2 K is observed. The van't Hoff factor (*i*) is **[JEE 2007]**

(A) 0.5 (B) 1 (C) 2 (D) 3

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Liquid Solution\Eng\02_Exercise

Paragraph for Question No. Q.10 to Q.12

Properties such as boiling point, freezing point and vapour pressure of a pure solvent change when solute molecules are added to get homogeneous solution. These are called colligative properties. Applications of colligative properties are very useful in day-to-day life. One of its examples is the use of ethylene glycol and water mixture as anti-freezing liquid in the radiator of automobiles.

A solution **M** is prepared by mixing ethanol and water. The mole fraction of ethanol in the mixture is 0.9.

Given : Freezing point depression constant of water $\left(K_f^{water}\right)$ = 1.86 K kg mol^{-1}

Freezing point depression constant of ethanol $\left(K_f^{ethanol}\right)$ = 2.0 K kg mol^{-1}

Boiling point elevation constant of water $\left(K_b^{water}\right)$ = 0.52 K kg mol^{-1}

Boiling point elevation constant of ethanol $\left(K_b^{ethanol}\right)$ = 1.2 K kg mol^{-1}

Standard freezing point of water = 273 K

Standard freezing point of ethanol = 155.7 K

Standard boiling point of water = 373 K

Standard boiling point of ethanol = 351.5 K

Vapour pressure of pure water = 32.8 mm Hg

Vapour pressure of pure ethanol = 40 mm Hg

Molecular weight of water = 18 g mol^{-1}

Molecular weight of ethanol = 46 g mol^{-1}

In answering the following questions, consider the solutions to be ideal dilute solutions and solutes to be non-volatile and non-dissociative.

10. The freezing point of the solution **M** is **[JEE 2008]**

(A) 268.7 K (B) 268.5 K (C) 234.2 K (D) 150.9 K

11. The vapour pressure of the solution **M** is **[JEE 2008]**

(A) 39.3 mm Hg (B) 36.0 mm Hg (C) 29.5 mm Hg (D) 28.8 mm Hg

12. Water is added to the solution **M** such that the mole fraction of water in the solution becomes 0.9. The boiling point of this solution is **[JEE 2008]**

(A) 380.4 K (B) 376.2 K (C) 375.5 K (D) 354.7 K

13. The Henry's law constant for the solubility of N_2 gas in water at 298 K is 1.0×10^5 atm. The mole fraction of N_2 in air is 0.8. The number of moles of N_2 from air dissolved in 10 moles of water at 298 K and 5 atm pressure is- **[JEE 2009]**

(A) 4.0×10^{-4} (B) 4.0×10^{-5} (C) 5.0×10^{-4} (D) 4.0×10^{-5}

14. For a dilute solution containing 2.5 g of a non-volatile non-electrolyte solute in 100 g of water, the elevation in boiling point at 1 atm pressure is 2°C. Assuming concentration of solute is much lower than the concentration of solvent, the vapour pressure (mm of Hg) of the solution is-(take K_b=0.76 K kg mol^{-1}) **[JEE 2011]**

(A) 724 (B) 740 (C) 736 (D) 718

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Liquid Solution\Eng\02_Exercise

15. The freezing point (in °C) of a solution containing 0.1 g of $K_3[Fe(CN)_6]$ (Mol. Wt. 329) in 100 g of water (K_f= 1.86 K kg mol^{-1}) is - **[JEE 2011]**

(A) -2.3×10^{-2} (B) -5.7×10^{-2} (C) -5.7×10^{-3} (D) -1.2×10^{-2}

16. For a dilute solution containing 2.5 g of a non-volatile non-electrolyte solute in 100 g of water, the elevation in boiling point at 1 atm pressure is 2°C. Assuming concentration of solute is much lower than the concentration of solvent, the vapour pressure (mm of Hg) of the solution is

(take K_b=0.76 K kg mol^{-1}) **[JEE 2012]**

(A) 724 (B) 740 (C) 736 (D) 718

17. Benzene and naphthalene form an ideal solution at room temperature. For this process, the true statement(s) is(are) **[J-Adv. 2013]**

(A) ΔG is positive (B) ΔS_{system} is positive (C) $\Delta S_{surroundings} = 0$. (D) $\Delta H = 0$

18. If the freezing point of a 0.01 molal aqueous solution of a cobalt (III) chloride-ammonia complex (which behaves as a strong eletrolyte) is – 0.0558°C, the number of chloride (s) in the coordination sphere of the complex is- [K_f of water = 1.86 K kg mol^{-1}] **[JEE-Adv. 2015]**

19. Mixture(s) showing positive deviation from Raoult's law at 35°C is (are) **[JEE-Adv. 2016]**

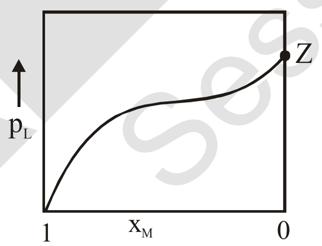
(A) carbon tetrachloride + methanol

(B) carbon disulphide + acetone

(C) benzene + toluene

(D) phenol + aniline

20. For a solution formed by mixing liquids L and M, the vapour pressure of L plotted against the mole fraction of M in solution is shown in the following figure, Here x_L and x_M represent mole fractions of L and M, respectively, in the solution. the correct statement(s) applicable to this system is(are)



(A) Attractive intramolecular interactions between L–L in pure liquid L and M–M in pure liquid M are stronger than those between L–M when mixed in solution **[JEE-Adv. 2017]**

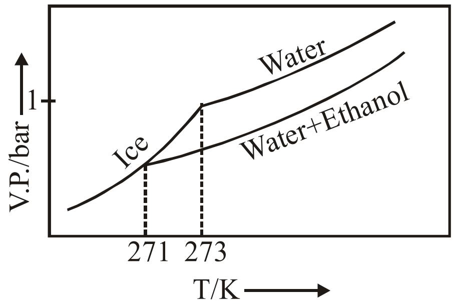
(B) The point Z represents vapour pressure of pure liquid M and Raoult's law is obeyed when $x_L \rightarrow 0$

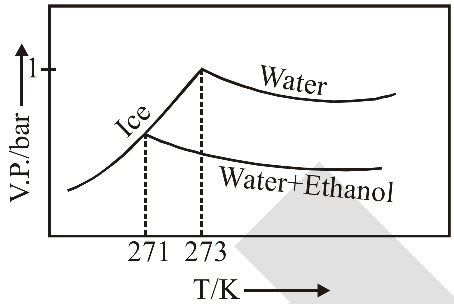
(C) The point Z represents vapour pressure of pure liquid L and Raoult's law is obeyed when $x_L \rightarrow 1$

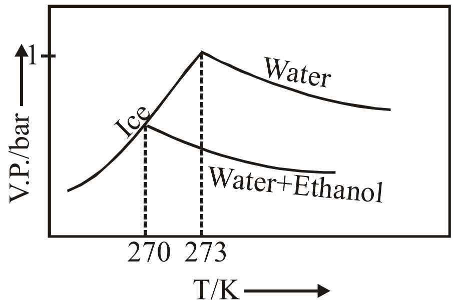
(D) The point Z represents vapour pressure of pure liquid M and Raoult's law is obeyed from $x_L = 0$ to $x_L = 1$

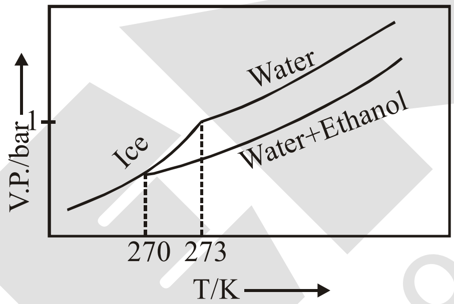
node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Liquid Solution\Eng\02_Exercise

21. Pure water freezes at 273 K and 1 bar. The addition of 34.5 g of ethanol to 500 g of water changes the freezing point of the solution. Use the freezing point depression constant of water as 2 K kg mol^{-1}. The figures shown below represents plots of vapour pressure (V.P.) versus temperature (T). [Molecular weight of ethanol is 46 g mol^{-1}]
Among the following, the option representing change in the freezing point is - **[JEE-Adv. 2017]**

(A) 

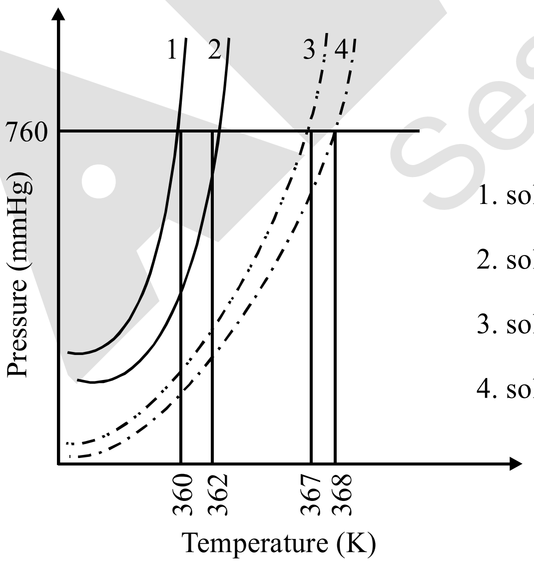
(B) 

(C) 

(D) 

22. Liquids A and B form ideal solution over the entire range of composition. At temperature T, equimolar binary solution of liquids A and B has vapour pressure 45 Torr. At the same temperature, a new solution of A and B having mole fractions x_A and x_B, respectively, has vapour pressure of 22.5 Torr. The value of x_A/x_B in the new solution is____. **[JEE-Adv. 2018]**
(Given that the vapour pressure of pure liquid A is 20 Torr at temperature T)

23. The plot given below shows P–T curves (where P is the pressure and T is the temperature) for two solvents X and Y and isomolal solutions of NaCl in these solvents. NaCl completely dissociates in both the solvents.



1. solvent X
2. solution of NaCl in solvent X
3. solvent Y
4. solution of NaCl in solvent Y

On addition of equal number of moles a non-volatile solute S in equal amount (in kg) of these solvents, the elevation of boiling point of solvent X is three times that of solvent Y. Solute S is known to undergo dimerization in these solvents. If the degree of dimerization is 0.7 in solvent Y, the degree of dimerization in solvent X is ___. **[JEE-Adv. 2018]**

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ANSWER-KEY

EXERCISE # S-I

1.	Ans. 0.25	2.	Ans. 64.0 mm Hg
3.	Ans. P_A^0 = 100 cmHg, P_B^0 = 60 cmHg.	4.	Ans. 0.25
5.	Ans. 0.5 bar	6.	Ans. P_X^0 = 55 cmHg, P_Y^0 = 80 cmHg
7.	Ans. P_R^0 = 0.8 bar, P_S^0 = 0.4 bar	8.	Ans. 0.25
9.	Ans. 0.04	10.	Ans. 0.32 m
11.	Ans. 65.25	12.	Ans. 17.38 mm Hg
13.	Ans. 120 g/mol	14.	Ans. K_b= 0.516kg mol K^{-1}, T_b = 373.20 K
15.	Ans. 5.1 K-kg/mol	16.	Ans. 2048 g/mol
17.	Ans. (6.972)	18.	Ans. 40 g/mol
19.	Ans. 49.26 atm	20.	Ans. 3.69 atm
21.	Ans. 1.96	22.	Ans. 3 ions
23.	Ans. 78 %	24.	Ans. 7.482 ×105 Nm^{-2}
25.	Ans. 0.95; 1.95	26.	Ans. 0.04
27.	Ans. (3)		

EXERCISE # S-II

1.	Ans. 92 mol% toluene;96.8 mol % toluene	2.	Ans. 0.741 m, 0.0136
3.	Ans. 38.71 g	4.	Ans. 4.8×10^4 g/mol
5.	Ans. (777)	6.	Ans. (68.4)
7.	Ans. $C_{40}H_{80}O_{40}$	8.	Ans. $[Co(NH_3)_5Cl]Cl_2$
9.	Ans. a = 0.50	10.	Ans. 0.73
11.	Ans. (9)	12.	Ans (60%)
13.	Ans. (3.2)	14.	Ans. (100°C)
15.	Ans. C_6H_6		

EXERCISE # O-I

1.	Ans. (B)	2.	Ans. (C)	3.	Ans. (C)	4.	Ans. (D)
5.	Ans. (A)	6.	Ans. (C)	7.	Ans. (C)	8.	Ans. (C)
9.	Ans. (D)	10.	Ans. (C)	11.	Ans. (C)	12.	Ans. (A)
13.	Ans. (A)	14.	Ans. (B)	15.	Ans. (B)	16.	Ans. (B)
17.	Ans. (C)	18.	Ans. (D)	19.	Ans. (B)	20.	Ans. (A)
21.	Ans. (C)	22.	Ans. (B)	23.	Ans. (D)	24.	Ans. (B)
25.	Ans. (B)	26.	Ans. (D)	27.	Ans. (B)	28.	Ans. (D)
29.	Ans. (B)	30.	Ans. (B)	31.	Ans. (D)	32.	Ans. (B)
33.	Ans. (B)	34.	Ans. (B)	35.	Ans. (A)	36.	Ans. (D)

ALLEN Session 2019-20

EXERCISE # O-II

1. Ans. (B) **2.** Ans. (D) **3.** Ans. (A) **4.** Ans. (A)

5. Ans. (B) **6.** Ans. (D) **7.** Ans. (C) **8.** Ans. (C)

9. Ans. (C) **10.** Ans. (C, D) **11.** Ans. (A,B,C,D) **12.** Ans. (C,D)

13 Ans. (C) **14.** Ans. (C) **15.** Ans. (B) **16.** Ans. (C)

17. Ans. (D) **18.** Ans. (C) **19.** Ans. (A) **20.** Ans. (C)

21. Ans. (A) $\rightarrow$ P ; (B) $\rightarrow$ Q, R, S ; (C) $\rightarrow$ T ; (D) $\rightarrow$ P, Q, R, S, T

EXERCISE # (J-MAIN)

1. Ans. (3) **2.** Ans. (3) **3.** Ans. (3) **4.** Ans. (1)

5. Ans. (1) **6.** Ans. (3) **7.** Ans. (3) **8.** Ans. (1)

9. Ans. (4) **10.** Ans. (1) **11.** Ans. (4) **12.** Ans. (2)

13. Ans. (3) **14.** Ans. (1) **15.** Ans. (1) **16.** Ans. (4)

17. Ans. (1) **18.** Ans. (3) **19.** Ans. (4) **20.** Ans. (4)

21. Ans. (1) **22.** Ans. (2) **23.** Ans. (3) **24.** Ans. (1)

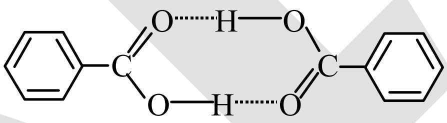
25. Ans. (4)

EXERCISE # (J-ADVANCED)

1. Ans. 0.23 K **2.** Ans. $(1.0 \times 10^{-4})\text{min}^{-1}$

3. Ans. $K_b(x) = 0.68$, $K_b(y) = 0.53$, $K_b(z) = 0.98$ **4.** Ans. (A)

5. Ans. (a)122, (b) It means that benzoic acid remains as it is in acetone while it dimerises in benzene as



6. Ans. (C) **7.** Ans. (C)

8. Ans. 35% (degree of asso = 70%) **9.** Ans. (A)

10. Ans. (D) **11.** Ans. (B) **12.** Ans. (B) **13.** Ans. (A)

14 Ans. (A) **15.** Ans. (A) **16.** Ans. (A) **17.** Ans. (B,C,D)

18. Ans. (1) **19.** Ans. (A,B) **20.** Ans. (A,C) **21.** Ans. (D)

22. Ans. (19) **23.** Ans. (0.05)

CHEMICAL KINETICS

1. INTRODUCTION

Chemical kinetics deals with the rates of chemical processes. Any chemical process may be broken down into a sequence of one or more single-step processes known either as elementary processes, elementary reactions, or elementary steps. Elementary reactions usually involve either a single reactive collision between two molecules, which we refer to as a bimolecular step, or dissociation/isomerisation of a single reactant molecule, which we refer to as a unimolecular step. Very rarely, under conditions of extremely high pressure, a termolecular step may occur, which involves simultaneous collision of three reactant molecules. An important point to recognise is that, many reactions that are written as a single reaction equation, in actual fact, consist of a series of elementary steps. This will become extremely important as we learn more about the theory of chemical reaction rates.

As a general rule, elementary processes involve a transition between two atomic or molecular states separated by a potential barrier. The potential barrier constitutes the activation energy of the process, and determines the rate at which it occurs. When the barrier is low, the thermal energy of the reactants will generally be high enough to surmount the barrier and move over to products, and the reaction will be fast. However, when the barrier is high, only a few reactants will have sufficient energy, and the reaction will be much slower. The presence of a potential barrier to reaction is also the source of the temperature dependence of reaction rates.

The huge variety of chemical species, types of reaction, and the accompanying potential energy surfaces involved means that the time scale over which chemical reactions occur covers many orders of magnitude, from very slow reactions, such as iron rusting, to extremely fast reactions, such as the electron transfer processes involved in many biological systems or the combustion reactions occurring in flames.

A study into the kinetics of a chemical reaction is usually carried out with one or both of two main goals in mind :

(i) Analysis of the sequence of elementary steps giving rise to the overall reaction. i.e. the reaction mechanism.

(ii) Determination of the absolute rate of the reaction and/or its individual elementary steps.

2. CLASSIFICATION OF REACTION

(i) There are certain reactions which are too slow. **Ex.** Rusting of Iron, weathering of rocks.

(ii) Instantaneous reactions i.e. too fast. **Ex.** Detonation of explosives, acid-base neutralization, precipitation of AgCl by NaCl and $AgNO_3$.

(iii) Neither too fast nor too slow. **Ex.** Combination of H_2 and Cl_2 in presence of light, hydrolysis of ethyl acetate catalysed by acid, decomposition of Azomethane.

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3. Types of Rates of chemical reaction :

For a reaction R $\longrightarrow$ P ,

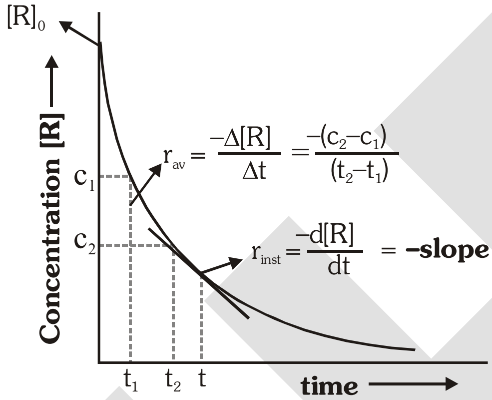
- **Average rate** $= \frac{\text{Total change in concentration}}{\text{Total time taken}} = \frac{|\Delta c|}{\Delta t} = -\frac{\Delta[\text{Reactant}]}{\Delta t} = \frac{\Delta[\text{Product}]}{\Delta t}$

- **Instantaneous rate** : Rate of reaction at a particular instant.

$$\mathbf{R_{instantaneous}} = \lim_{\Delta t \to 0}\left[\frac{|\Delta c|}{\Delta t}\right] = \left|\frac{dc}{dt}\right| = -\frac{d[R]}{dt} = \frac{d[P]}{dt}$$

Instantaneous rate can be determined from slope of a tangent at time t on curve drawn for concentration versus time.

3.1 Initial Rate : Instantaneous rate at 't = 0' is called initial rate [Slope of tangent at t = 0].



❖ **Reaction rates and stoichiometry :**

We have seen that for stoichiometrically simple reactions of the type A $\rightarrow$ B, the rate can be either expressed in terms of the decrease in reactant concentration with time, $-\Delta[A]/\Delta t$ or the increase in product concentration with time, $\Delta[B]/\Delta t$. For more complex reactions, we must be careful in writing the rate expressions. Consider for example, the reaction,

2 A $\longrightarrow$ B

Two moles of A disappear for each mole of B that forms – that is, the rate of disappearance of A is twice as fast as the rate of appearance of B. We write the overall rate of reaction as either

$$\text{Rate} = -\frac{1}{2}\frac{\Delta[A]}{\Delta t} = \frac{\Delta[B]}{\Delta t}$$

In general, for the reaction, **aA + bB $\longrightarrow$ cC + dD**

the rate is given by

$$\text{Rate} = -\frac{1}{a}\frac{\Delta[A]}{\Delta t} = -\frac{1}{b}\frac{\Delta[B]}{\Delta t} = \frac{1}{c}\frac{\Delta[C]}{\Delta t} = \frac{1}{d}\frac{\Delta[D]}{\Delta t}$$

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Ex. : **For the reaction in terms : $N_2 + 3H_2 \longrightarrow 2NH_3$**

Rate of reaction in terms of $N_2 = -\frac{d[N_2]}{dt}$ = rate of disappearance of N_2

Rate of reaction in terms of $H_2 = -\frac{d[H_2]}{dt}$ = rate of disappearance of H_2

Rate of reaction in terms of $NH_3 = \frac{d[NH_3]}{dt}$ = rate of appearance of NH_3

These rates are not all equal. Therefore, by convention, the rate of a reaction is defined as

- **Rate of reaction** $= -\frac{d[N_2]}{dt} = -\frac{1}{3}\frac{d[H_2]}{dt} = \frac{1}{2}\frac{d[NH_3]}{dt}$

Note: The value of rate of reaction is dependent on the stoichiometric coefficients used in the reaction while the rate of increase or decrease in amount of any species will be fixed value under given conditions.

Ex.1 **The following reaction was studied in a closed vessel**

$2N_2O_5(g) \longrightarrow 4NO_2(g) + O_2(g)$

It was found that the concentration of NO_2 increases by 2.0×10^{-2} mol L^{-1} in five seconds. Calculate.

(i) **The rate of reaction**

(ii) **The rate of decrease of concentration of N_2O_5**

Sol. **(i)** Rate of reaction $= \frac{1}{4}\frac{d[NO_2]}{dt}$

But $\frac{d[NO_2]}{dt} = \frac{2.0 \times 10^{-2}\,\text{mol L}^{-1}}{5\text{s}} = 4 \times 10^{-3}$ mol L^{-1} s^{-1}

Rate of reaction $\frac{1}{4} \times 4 \times 10^{-3} = 10^{-3}$ mol L^{-1} s^{-1}

(ii) Rate of decrease of conc. of N_2O_5

$= -\frac{d[N_2O_5]}{dt} = -\frac{1}{2} \times$ Rate of formation of $NO_2 = +\frac{1}{2}\frac{d[NO_2]}{dt}$

$= +\frac{1}{2} \times 4 \times 10^{-3} = 2 \times 10^{-3}$ mol L^{-1} s^{-1}

3.3 FACTORS AFFECTING RATE OF CHEMICAL REACTION

(i) Concentration

(ii) Temperature

(iii) Nature of reactants and products

(iv) Catalyst

(v) pH of the solution

(vi) Dielectric constant of the medium.

(vii) Radiations / light

(viii) Pressure

(ix) Electrical and magnetic field.

The first four factors generally affect rate of almost all reactions while other factors are specific to some reactions only.

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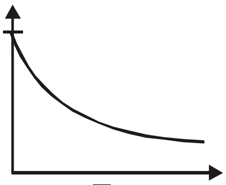
(i) Effect of concentration :

For most of the reactions, rate depends on concentration of reactants. So rate of reaction decreases with passage of time, since concentration of reactants decreases .



(ii) Effect of temperature : Generally rate of reaction increases on increasing temperature.

(iii) Effect of nature of reactants and products :

(a) Physical state of reactants : Gaseous state > Liquid state > Solid state (Decreasing order of rate of reaction)

Because collisions in gaseous systems are more effective than condensed systems (solid & liquid).

(b) Physical size of reactants : In heterogeneous reactions, as we decreases the particle size, rate of reaction increases since surface area increases.

(c) Chemical nature of reactants :

- If more bonds are to be broken, the rate of reaction will be slow.
- Similarly if bond strength in reactants is more, rate of reaction will be slow.

(iv) Effect of Catalyst :

- Presence of positive catalyst lowers down the activation energy hence increases the rate of reaction.
- Presence of negative catalyst increases activation energy hence decreases the rate of reaction.

(v) Effect of pH of solution : Ex. $[Fe(CN)_6]^{4-} \xrightarrow{(Tl^{3+})} [Fe(CN)_6]^{3-}$

This reaction takes place with appreciable rate in acidic medium, but does not take place in basic medium.

(vi) Effect of dielectric constant of the medium : More is the dielectric constant of the medium greater will be the rate of ionic reactions.

(vii) Effect of radiations / light : Radiations are useful for photochemical reactions.

(viii) Effect of pressure : Pressure is important factor for gaseous reactions.

(ix) Effect of electrical & magnetic field : Electric and magnetic fields are rate determining factors if a reaction involves polar species.

4. RATE LAW (DEPENDENCE OF RATE ON CONCENTRATION OF REACTANTS) :

The representation of rate of reaction in terms of the concentration of the reactants is called the rate law. It can only be established by experiments.

Generally rate law expressions are not simple and these may differ for the same reaction on conditions under which the reaction is being carried out. But for large number of reactions starting with pure reactants we can obtain simple rate laws. For these reactions :

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Rate $\propto$ (conc.)$^{\text{order}}$

Rate = K (conc.)$^{\text{order}}$ **This is the differential rate equation or rate expression.**

Where K = Rate constant = specific reaction rate = rate of reaction when concentration is unity

unit of K = (conc)$^{1-\text{order}}$ time^{-1}

Note: Value of K is a constant for a given reaction, depending only on temperature and catalyst use.

4.1 Order of reaction :

Let there be a reaction , $m_1A + m_2B \longrightarrow$ products.
Now, if on the basis of experiment, we find that

$$R \propto [A]^p [B]^q$$

where p may or may not be equal to m_1 and similarily q may or may not be equal to m_2.

p is order of reaction with respect to reactant A and q is order of reaction with respect to reactant B and (p + q) is **overall order of the reaction.**

Note: Order of a reaction can be 'zero', any whole number, fractional number or even be negative with respect to a particular reactant.

❖ **Examples showing different values of order of reactions :**

	Reaction	Rate law	Order
(i)	$2N_2O_5(g) \rightarrow 4NO_2(g) + O_2(g)$	$R = K[N_2O_5]^1$	1
(ii)	$5Br^-(aq) + BrO_3^-(aq) + 6H^+(aq) \rightarrow 3Br_2(\ell) + 3H_2O(\ell)$	$R = K[Br^-][BrO_3^-][H^+]^2$	1 + 1 + 2 = 4
(iii)	H_2 (Para) $\rightarrow H_2$ (ortho)	$R = K[H_{2\,(Para)}]^{3/2}$	3/2
(iv)	$NO_2(g) + CO(g) \rightarrow NO(g) + CO_2(g)$	$R = K[NO_2]^2[CO]^o$	2 + 0 = 2
(v)	$2O_3(g) \rightarrow 3O_2(g)$	$R = K[O_3]^2[O_2]^{-1}$	2 – 1 = 1
(vi)	$H_2 + Cl_2 \xrightarrow{hv} 2\,HCl$	$R = K[H_2]^o[Cl_2]^o$	0 + 0 = 0

The reaction (ii) does not take place in one single step. It is almost impossible for all the 12 molecules of the reactants to be in a state of encounter simultaneously. Such a reaction is called ***Complex reaction*** and takes places in a sequence of a number of ***Elementary reactions***. For an elementary reaction, the sum of stoichiometric coefficients of reactants = order of the reaction. But for complex reactions, order is to be experimentally calculated.

4.2 Molecularity of reaction :

The number of molecules that react in an elementary step is the molecularity of the elementary reaction. Molecularity is defined only for the elementary reactions and not for complex reactions. No elementary reactions involving more than three molecules are known, because of very low probability of simultaneous collision of more than three molecules.

The rate law for the elementary reaction

aA + bB $\longrightarrow$ Products, Rate = k[A]^a[B]^b , where a + b = 1, 2 or 3

For an elementary reaction, the orders in the rate law equal the coefficients of the reactants. It must be noted that the order is defined for complex as well as elementary reactions and is always experimentally calculated from the mechanism of the reaction, usually by the slowest step of the mechanism known as rate determining step (RDS) of the reaction.

Comparision between molecularity and order of reaction			
Molecularity of Reaction		***Order of Reaction***	
1	It is defined as the no. of molecules of reactant taking part in a chemical reaction. $NH_4NO_2 \rightarrow N_2 + 2H_2O$, molecularity = 1	1	It is defined as the sum of the power of concentration terms that appear in rate law. $NH_4NO_2 \rightarrow N_2 + 2H_2O$. Rate = k[$NH_4NO_2$] , order =1
2	It is always a whole number. It can neither be zero nor fractional.	2	It may be zero, fractional or integer.
3	It is derived from RDS in the mechanism of reaction	3	It is derived from rate expression.
4	It is theoretical value.	4	It is experimental value.
5	Reactions with molecularity > 3 are rare.	5	Reactions with order of reaction > 4 are also rare.
6	Molecularity is independent of pressure and temperature.	6	Order of reaction may depend upon pressure and temperature.

5. INTEGRATED RATE LAW :

For a single reactant reaction where the chemical equation has the form

A $\rightarrow$ products

and the rate law is assumed to be of the form

Rate = – d[A]/dt = k [A]^m

Where m is the order of the reaction with respect to substance A. Three important cases can be treated : m = 0, m = 1, and m = 2. These are called ***zeroth order***, ***first order***, and ***second order***, respectively.

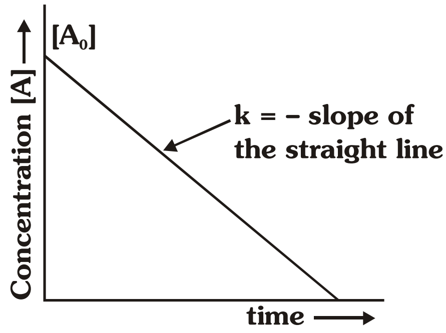
5.1 Zero - order reaction :

For a reaction where the chemical equation has the form

A $\rightarrow$ products

and the rate law is assumed to be of the form

Rate = – d[A]/dt = k [A]^0



$$-\int_{C_0}^{C_t} d[A] = k\int_0^t dt$$

$$k = \frac{C_0 - C_t}{'t'}$$

or **kt = C$_0$ – C$_t$** or $C_t = C_0 - kt$

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- Unit of K is same as that of Rate = mol lit^{-1} sec^{-1}.
- Time for completion $= \dfrac{C_0}{k}$
- $t_{1/2}$ (half life period)

At $\quad t_{1/2}$, $C_t = \dfrac{C_0}{2}$,

So $\quad kt_{1/2} = \dfrac{C_0}{2} \qquad \Rightarrow \qquad t_{1/2} = \dfrac{C_0}{2k}$

$\therefore \quad t_{1/2} \propto C_0$

❖ **Examples of Zero order reactions :**

Generally decomposition of gases on metal surfaces at high concentrations follow zero order kinetics as rate depends on surface area of catalyst.

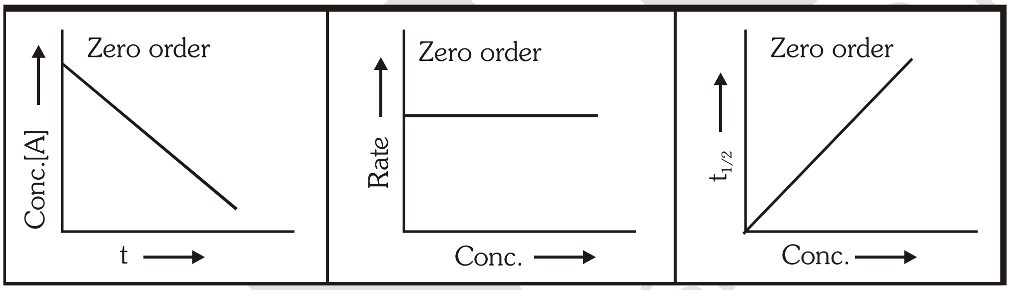
$\boxed{2PH_3\,(g) \xrightarrow{Ni} 2\,P(s) + 3H_2(g)}$ **Rate = K [PH$_3$]^0**

$2HI\,(g) \xrightarrow{Au} H_2(g) + I_2(g)$ **Rate = K [HI]^0**

$2NH_3(g) \xrightarrow{Pt} N_2(g) + 3H_2(g)$ **Rate = K [NH$_3$]^0**

$H_2(g) + Cl_2(g) \xrightarrow{hv} 2\,HCl(g)$ **Rate = K [H$_2$]^0 [Cl$_2$]^0**

❖ **Graphs :**



Ex.2 For the zero order reaction : A → P , K = 10^{-2} (mol/litre) sec^{-1} If initial concentration of A is 0.3M , then find concentration of A left at 10 sec.

(A) 0 M **(B) 0.2 M** **(C) 0.1 M** **(D) 0.15 M**

Sol. (B)

$[A]_t = [A]_0 - Kt = 0.3 - 10^{-2} \times 10 = 0.2$ M

5.2 First order reaction :

Consider a first order reaction with single reactant.

	A	$\longrightarrow$	**Products**
t = 0	a		0
t = t	a–x		

Rate = – d[A]/dt = k [A]^1

$\therefore \quad \dfrac{dx}{dt} = k\,(a-x)^1 \quad$ or $\quad \dfrac{dx}{a-x} = kdt.$

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- **On solving** $k = \frac{2.303}{t} \log \frac{a}{a-x} = \frac{2.303}{t} \log \frac{C_0}{C_t}$

$K = \boxed{\frac{2.303}{t}\log\frac{C_0}{C_t}}$ $\boxed{\text{Wilhemy formula : } C_t = C_0 e^{-kt}}$ $\boxed{\text{Interval formula : } k = \frac{2.303}{(t_2 - t_1)}\log\frac{C_1}{C_2}}$

- Unit of $k = sec^{-1}$, min^{-1}, etc.
- **Half-life time ($t_{1/2}$) :**

$$k = \frac{2.303}{t_{1/2}} \log \frac{2C_0}{C_0} \Rightarrow t_{1/2} = \frac{2.303 \log 2}{k} = \frac{\ln 2}{k} = \frac{0.693}{k}$$

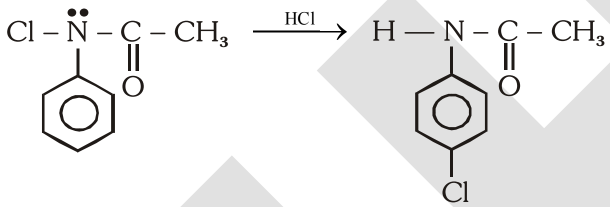
∴ Half-life period for a first order reaction is a constant quantity at given temperature.

- **Examples of first order reactions :**

(i) Decomposition of azoisopropane

$$(CH_3)_2CH-N=N-CH(CH_3)_2(g) \xrightarrow{\Delta} N_2(g) + C_6H_{14}(g)$$

(ii) Conversion of N–chloro acetanilide into p–chloroacetanilide



(iii) $H_2O_2(aq.) \longrightarrow H_2O(l) + \frac{1}{2}O_2(g)$

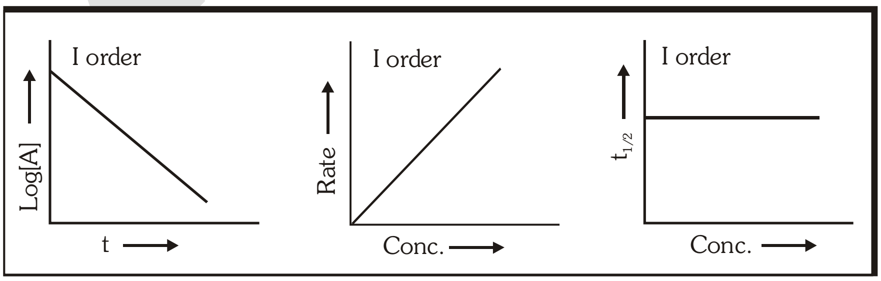
(iv) $NH_4NO_2 \longrightarrow 2H_2O(g) + N_2(g)$

(v) Radioactive decay

All radioactive decays always follow first order kinetics.

$${}^{226}_{88}Ra \longrightarrow {}^{222}_{86}Rn + {}_2He^4$$

- **Graphs :**



ALLEN Session 2019-20

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Ex.3 **Calculate $\frac{t_{0.75}}{t_{0.50}}$ for a first order reaction :**

Sol. $k = \frac{2.303}{t_{3/4}} \log \frac{C_0}{\frac{1}{4}C_0} = \frac{2.303}{t_{1/2}} \log \frac{C_0}{\frac{C_0}{2}} \Rightarrow \frac{t_{3/4}}{t_{1/2}} = \frac{\log\ 4}{\log\ 2} = \frac{2\ \log\ 2}{\log\ 2} = 2$

Ex.4 **90% of a first order reaction was completed in 10 hours. When will 99.9% of the reaction complete ?**

Sol. $K = \frac{2.303}{t}\log\frac{a}{a?\ x}$

$a = 100$, $x = 90$, $t = 10$

So $K = \frac{2.303}{10}\log\frac{100}{10}$

$K = 2.303 \times 10^{-1}$ hour^{-1}

Now for 99.9% completion -

$a = 100$ and $x = 99.9$

$$t = \frac{2.303}{K}\log\frac{100}{0.1} = \frac{2.303}{2.303\times10^{-1}}\times 3 = 30 \text{ hours}$$

Ex.5 **A first order reaction is 90% complete after 40 min. Calculate the half life of reaction.**

Sol. $a = 100,\ \ x = 90$

$K_1 = \frac{2.303}{t}\log\frac{a}{a-x}$

$= \frac{2.303}{40}\log\frac{100}{10}$

$= \frac{2.303}{40} = 5.757 \times 10^{-2}$ min^{-1}

$t_{1/2} = \frac{0.693}{K_1} = \frac{0.693}{5.757\times10^{-2}} = 12.03$ min.

5.3.1 **Second order reaction :**

• **Case : 1**

$$\begin{array}{llll} A & + & A & \longrightarrow \text{Products} \\ a & & a & \\ (a-x) & & (a-x) & \end{array}$$

$\therefore \quad \frac{dx}{dt} = k\,(a-x)^2$

$\Rightarrow \quad \frac{1}{(a-x)} - \frac{1}{a} = kt \quad$ or $\quad \frac{1}{C_t} - \frac{1}{C_0} = kt$

ALLEN Session 2019-20
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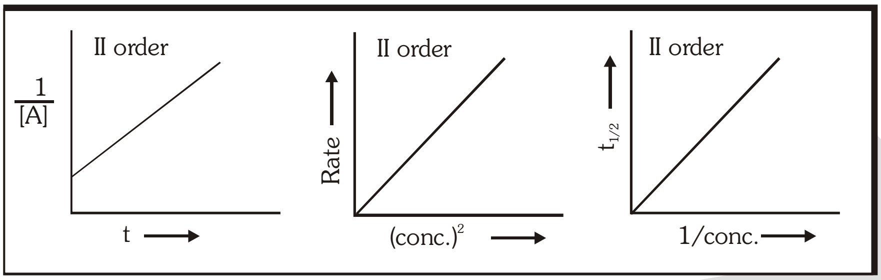
- Unit of $k = Lmol^{-1} sec^{-1}$

- Half life, $t_{½} = \dfrac{1}{k.C_o}$

  - **Graphs :**



5.3.2 Case : 2

A + B $\longrightarrow$ products.

a b 0

a – x b – x

Rate law

$$\frac{dx}{dt} = k\,(a-x)\,(b-x)$$

$$\int_0^x \frac{dx}{(a-x)\quad(b-x)} = \int_0^t kdt$$

$$\boxed{k = \frac{2.303}{t(a-b)}\log\frac{b(a-x)}{a(b-x)}}$$ (no need to remember)

where $a \neq b$

Ex.6 For a second order reaction in which both the reactants have equal initial concentration, the time taken for 60% completion of reaction is 3000 second. What will be the time taken for 20% of the reaction?

Sol. $k_2 = \dfrac{1}{t}\dfrac{x}{a(a-x)}$

Let a = 1,

$$k_2 = \frac{1}{t}\frac{x}{(1-x)}$$

$$= \frac{1}{3000}\left(\frac{0.6}{1-0.6}\right)$$

$$= \frac{1}{3000}\times\frac{0.6}{0.4} = \frac{1}{2000}$$

So time for the 20% completion :

$$t = \frac{1}{k_2}\frac{x}{a(a-x)}$$

$$= 2000\times\frac{0.20}{0.80} = 500 \text{ sec.}$$

Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Chemical Kinetics\Eng\01_Theory

5.4 Pseudo order reaction :

For, $A \quad + \quad B \longrightarrow$ Products

t=0 $\quad a \qquad b$

t=t $\quad a-x \qquad b-x$

$$\textbf{Rate} = K[A]^n[B]^m$$

If $[B] = \text{constant} \Rightarrow [b-x] \approx \text{constant}$

$\text{Rate} = K'[A]^n$

when $K' = K[B]^n$

- **Case 1 :**
  If concentration of B is much greater than A, then $[B] = \text{Constant} \Rightarrow [b-x] \approx \text{Constant}$.
- **Case 2 :**
  If B is a catalyst , then $[B] = \text{Constant} \Rightarrow [b-x] \approx \text{Constant}$.
- **Case 3 :**
  If B is a solvent , then $[B] = \text{Constant} \Rightarrow [b-x] \approx \text{Constant}$.

5.4.1 PSEUDO FIRST ORDER REACTIONS :

A second order (or of higher order) reactions can be converted into a first order reaction if the other reactant is taken in large excess. Such first order reactions are known as **pseudo first order reactions**.

$\therefore$ For $A + B \longrightarrow$ Products $\quad$ **[Rate] = K $[A]^1$ $[B]^1$**

$$K = \frac{2.303}{t\ (a-b)} \log \frac{b(a-x)}{a(b-x)} \qquad \text{(from 5.3.2)}$$

Now if 'B' is taken in large excess b > > a.

$$\therefore \quad K = \frac{2.303}{-bt} \log \frac{(a-x)}{a} \quad \Rightarrow \quad K = \frac{2.303}{bt} \log \frac{a}{a-x}$$

$$\Rightarrow \quad K.b = \frac{2.303}{t} \log \frac{a}{a-x} \quad \Rightarrow \quad K' = \frac{2.303}{t} \log \frac{a}{a-x}$$

K' is pseudo first order rate constant.

K' will have units of first order.

K will have units of second order.

ALLEN Session 2019-20

❖ **Table :** Characteristics of Zero, First, Second and n[th] order reactions of the type **A $\longrightarrow$ Products**

	Zero order	**First order**	**Second order**	**nth order**
Differential Rate law	$-\dfrac{d[A]}{dt} = k[A]^\circ$	$-\dfrac{d[A]}{dt} = k[A]$	$-\dfrac{d[A]}{dt} = k[A]^2$	$-\dfrac{d[A]}{dt} = k[A]^n$
Integrated Rate law	$[A]_t = [A]_0 - kt$	$\ln[A]_t = -kt + \ln[A]_0$	$\dfrac{1}{[A]_t} = kt + \dfrac{1}{[A]_0}$	$\dfrac{1}{[A_t]^{n-1}} - \dfrac{1}{[A_0]^{n-1}} = (n-1)\ kt$
Linear graph	$[A]_t$ v/s t	In [A] v/s t	$\dfrac{1}{[A]}$ v/s t	$\dfrac{1}{[A_t]^{n-1}}$ v/s t
Half-life	$t_{1/2} = \dfrac{[A]_0}{2k}$ (depends on $[A]_0$)	$t_{1/2} = \dfrac{0.693}{k}$ (independent of $[A]_0$)	$t_{1/2} = \dfrac{1}{k[A]_0}$ (depends on $[A]_0$)	$t_{1/2} \propto \dfrac{1}{[A_0]^{n-1}}$

6 Methods to determine order of a reaction :

6.1 Initial rate method :

By comparison of different initial rates of a reaction by varying the concentration of one of the reactants while others are kept constant.

$r = k\,[A]^a\,[B]^b\,[C]^c$ if [B] = constant and [C] = constant

then for two different initial concentrations of A we have

$$r_{0_1} = k\,[A_0]_1^{\,a} \quad r_{0_2} = k\,[A_0]_2^{\,a} \quad \Rightarrow \quad \frac{r_{0_1}}{r_{0_2}} = \left(\frac{[A_0]_1}{[A_0]_2}\right)^a$$

or in log form we have
$$a = \frac{\log\ (r_{0_1}/r_{0_2})}{\log\ ([A_0]_1/[A_0]_2)}$$

6.2 Integrated rate law method :

It is method of hit and trial. By checking if the kinetic data (experimental data) best fits into a particular integrated rate law, we determine the order. It can also be done graphically.

Ex.7 The rate of decomposition of N_2O_5 in CCl_4 solution has been studied at 318 K and the following results have been obtained :

t/min	0	135	342	683	1693
c/M	2.08	1.91	1.67	1.35	0.57

Find the order of the reaction and calculate its rate constant. What is the half-life period?

ALLEN session 2019-20

Sol. It can be shown that these data will not satisfy the integrated rate law of zero order. We now try integrated first order equation i.e., $k = \frac{\ln(c_0/c)}{t}$.

t / min	c / M	$k = \frac{\ln(c_0/c)}{t}\ \text{min}^{-1}$
0	2.08	6.32×10^{-4}
135	1.91	6.30×10^{-4}
339	1.68	6.32×10^{-4}
683	1.35	6.32×10^{-4}
1680	0.72	6.31×10^{-4}

It can be seen that the value of k is almost constant for all the experimental results and hence it is first order reaction with $k = 6.31 \times 10^{-4}\ \text{min}^{-1}$.

$$t_{1/2} = \frac{0.693}{6.31 \times 10^{-4}\ \text{min}^{-1}} = 1.094 \times 10^{3}\ \text{min}^{-1}$$

6.3 Method of half lives :

The half lives of each order is unique so by comparing half lives we can determine order for n^{th} order reaction $t_{1/2} \propto \frac{1}{[A_0]^{n-1}}$ (Remember)

$$\frac{(t_{1/2})_1}{(t'_{1/2})_2} = \frac{[A'_0]_2^{n-1}}{[A_0]_1^{n-1}} \qquad \text{(Remember)}$$

Ex.8 In a decomposition reaction, the reaction was found to be 50% complete in 210 seconds when the initial pressure of the mixture was 200 mm. In a second experiment the time of half reaction was 140 seconds when the initial pressure was 300 mm. Calculate the total order of the reaction.

Sol. For a n^{th} order reaction $t_{1/2} \propto \frac{1}{c_0^{n-1}}$

$$\frac{210}{140} = \left(\frac{300}{200}\right)^{n-1} \Rightarrow n = 2$$

6.4 Ostwald's isolation method :

This method is useful for reactions which involve a large number of reactants. In this method, the concentration of all the reactants are taken in large excess except that of one, so if

$$\textbf{Rate} = k\,[A]^a\,[B]^b\,[C]^c = k_0\,[A]^a$$

Then value of 'a' can be calculated by previous methods and similarly 'b' and 'c' can also be calculated.

Session 2019-20

6.5 Initial rate method :

For reaction : $A + B \longrightarrow$ Products

Initial rate, $r_0 = K.[A_0]^n [B_0]^m$

Now, order with respect to A may be determined by comparing the initial rate of reaction at different initial concentration of A but fixed initial concentration of B.

Ex.9 Consider the following data for the reaction :

$$A + B \longrightarrow \text{Products}$$

Run	Initial concentration	Initial concentration	Initial rate (mol s^{-1})
1	0.10 M	1.0 M	2.1×10^{-3}
2.	0.20 M	1.0 M	8.4×10^{-3}
3.	0.20 M	2.0 M	8.4×10^{-3}

Determine the order of reaction with respect to A and with respect B and the over all order of reaction.

Sol. The rate law may be expressed as :

Rate = $k [A]^p [B]^q$

Comparing experiments 2 and 3

$(\text{Rate})_2 = k[0.2]^p [1.0]^q = 8.4 \times 10^{-3}$ (1)

$(\text{Rate})_3 = k[0.2]^p [2.0]^q = 8.4 \times 10^{-3}$ (2)

Dividing equation (2) by (1)

$$\frac{(\text{Rate})_3}{(\text{Rate})_2} = \frac{k[0.2]^p[2.0]^q}{k[0.2]^p[1.0]^q} = \frac{8.4\times10^{-3}}{8.4\times10^{-3}}$$

$[2]^q = [2]^0$

or $q = 0$

Comparing experiments 1 and 2,

$(\text{Rate})_2 = k[0.20]^p [1.0]^q = 8.4 \times 10^{-3}$ (3)

$(\text{Rate})_1 = k[0.10]^p [1.0]^q = 2.1 \times 10^{-3}$ (4)

Dividing equation (3) by (4)

$$\frac{(\text{Rate})_2}{(\text{Rate})_1} = \frac{k[0.20]^p[1.0]^q}{k[0.10]^p[1.0]^q} = \frac{8.4\times10^{-3}}{2.1\times10^{-3}} = 4$$

$[2]^p = [2]^2$ or $p = 2$

so order with respect to A = 2

order with respect to B = 0

overall order = 2

Session 2019-20

7. APPLICATION OF FIRST ORDER REACTION

(Methods to monitor the progress of the reaction)

7.1 Case : 1

First order gaseous reaction :

Progress of gaseous reactions can be monitored by measuring total pressure at a fixed volume & temperature. This method can be applied for those reactions also in which a gas is produced because of decomposition of a solid or liquid. We can get an idea about the concentration of reacting species at a particular time by measuring pressure.

The pressure data can be given in terms of

(i) Partial pressure of the reactant

(ii) Total pressure of the reaction system

(iii) Pressure at only some points of time.

Ex.10 Find the expression for K in terms of P_0, P_t and n. For the reaction

$$\mathbf{A(g) \longrightarrow nB(g)}$$

Sol.

$$A(g) \longrightarrow nB(g)$$

$$P_0$$

$$P_A = (P_0 - x) \qquad nx$$

$\therefore$ P_t (Total pressure at time 't') $= P_0 - x + nx = P_0 + (n-1)x$

$\therefore$ $x = \dfrac{P_t - P_0}{n-1}$

$\therefore$ $P_A = P_0 - \dfrac{P_t - P_0}{n-1} = \dfrac{P_0 n - P_t}{n-1}$

$\therefore$ $a \propto p_0$ & $(a-x) \propto P_A = \dfrac{nP_0 - P_t}{n-1}$

$\therefore$ $k = \dfrac{2.303}{t} \log \dfrac{P_0(n-1)}{(nP_0 - P_t)}$

Final total pressure after infinite time $= P_f = nP_0$

Do not remember the formula but derive it for each question.

7.2 Case : 2

Volume measurement :

(i) By measuring the volume of product formed we can monitor the progress of reactions.

Ex.11 Study of a reaction whose progress is monitored by measuring the volume of a escaping gas.

$$\mathbf{NH_4NO_2(s) \xrightarrow{\Delta} 2H_2O(\ell) + N_2(g)}$$

Sol. Let, V_t be the volume of N_2 collected at time 't'

ALLEN Session 2019-20

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V_∞ = be the volume of N_2 collected at the end of the reaction.

$a \propto V_\infty$ and $x \propto V_t$

$(a-x) \propto V_\infty - V_t$

$$\therefore \quad k = \frac{2.303}{t} \log \frac{V_\infty}{V_\infty - V_t}$$

(ii) By titration method : By measuring the volume of titrating agent we can monitor amount of reactant remaining or amount of product formed at any time. It is the **titre value** . Here the milliequivalent or millimoles are calculated using valence factors.

Ex.12 From the following data show that the decomposition of hydrogen peroxide in aqueous solution is a first - order reaction. What is the value of the rate constant?

Time in minutes	0	10	20	30	40
Volume V in mL	25.0	20.0	15.7	12.5	9.6

Where V is the volume in mL of standard $KMnO_4$ solution required to react with a definite volume of hydrogen peroxide solution.

Sol. The equation for a first order reaction is,

$$2H_2O_2(\ell) \xrightarrow{\Delta} 2H_2O(\ell) + O_2(g)$$

the volume of $KMnO_4$ used, evidently corresponds to the undecomposed hydrogen peroxide.

Hence the volume of $KMnO_4$ used, at zero time corresponds to the initial concentration a and the volume used after time t, corresponds to (a – x) at that time. Inserting these values in the above equation, we get

When $t = 10$ min. $k_1 = \frac{2.303}{10} \log \frac{25}{20.0} = 0.022318\ \text{min}^{-1} = 0.000372\ \text{s}^{-1}$

When $t = 20$ min. $k_1 = \frac{2.303}{20} \log \frac{25}{15.7} = 0.023265\ \text{min}^{-1} = 0.000387\ \text{s}^{-1}$

When $t = 30$ min. $k_1 = \frac{2.303}{30} \log \frac{25}{12.5} = 0.02311\ \text{min}^{-1} = 0.000385\ \text{s}^{-1}$

When $t = 40$ min. $k_1 = \frac{2.303}{40} \log \frac{25}{9.6} = 0.023932\ \text{min}^{-1} = 0.0003983\ \text{s}^{-1}$

The constancy of k, shows that the decomposition of H_2O_2 in aqueous solution is a **first order** reaction. The average value of the rate constant is $0.0003879\ \text{s}^{-1}$.

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Ex.13 Study of acid hydrolysis of an ester.

$$CH_3COOCH_3 + \underset{(excess)}{H_2O} \xrightarrow[(HCl)]{H^+} CH_3COOH + CH_3OH$$

The progress of this reaction is monitored or determined by titrating the fixed volume of reaction mixture at different time intervals against a standard solution of NaOH using phenolphthalein as indicator. Find out rate constant of the reaction in terms of volume of NaOH consumed at $t = 0$, V_0, at $t = \infty$, V_∞ & at time t, V_t.

Sol. Let, V_0 = Volume of NaOH used at t = 0 [this is exclusively for HCl.]

V_t = Volume of NaOH used at 't'

V_∞ = Volume of NaOH used at $t = \infty$

$a \propto (V_\infty - V_0)$

$(a - x) \propto (V_\infty - V_t)$

$x \propto (V_t - V_0)$

$$k = \frac{2.303}{t} \log \frac{V_\infty - V_0}{V_\infty - V_t}$$

7.3 Case : 3

Optical rotation measurement :

It is used for optically active sample. It is applicable if there is at least one optically active species involved in chemical reaction.

• The optically active species may be present in reactant or product.

It is found that $(r_\infty - r_0) \propto a$ (a = concentration , x = amount consumed)

$(r_\infty - r_t) \propto (a - x)$

Where are r_0, r_t, r_∞ are angle of optical rotation at time t = 0, t = t and $t = \infty$.

$$k = \frac{2.303}{t} \log \frac{r_\infty - r_0}{r_\infty - r_t}$$ (Remember)

Ex.14 Study of hydrolysis of sucrose, progress of this reaction is monitored with the help of polarimeter because a solution of sucrose is dextrorotatory and on hydrolysis, the mixture of glucose and fructose obtained becomes laevorotatory. That's why this reaction is also known as inversion of cane sugar.

$$\underset{}{C_{12}H_{22}O_{11}} + \underset{\text{excess}}{H_2O} \xrightarrow{H^+} \underset{\text{glucose}}{C_6H_{12}O_6} + \underset{\text{fructose}}{C_6H_{12}O_6}$$

Let the readings in the polarimeter be

$t = 0$, θ_0 ; t = 't', θ_t and at $t = \infty$, θ_∞

Then calculate rate constant 'k' in terms of these readings.

Sol. The principle of the experiment is that change in the rotation is directly proportional to the amount of sugar hydrolysed.

$\therefore \quad a \propto (\theta_0 - \theta_\infty) \quad ; \quad (a - x) \propto (\theta_t - \theta_\infty) \; ; \; x \propto (\theta_0 - \theta_t)$

$$k = \frac{2.303}{t} \log\left(\frac{\theta_0 - \theta_\infty}{\theta_t - \theta_\infty}\right)$$

7.4 Case : 4

First order growth reaction :

For bacteria multiplication or virus growth use following concept

Consider a growth reaction,

Time	**Population (or colony)**
0	a
t	(a + x)

$$\frac{dx}{dt} = k\,(a + x) \quad \text{or} \quad \frac{dx}{(a+x)} = kdt$$

On integration

$\log_e (a + x) = kt + C \quad$ at $\quad t = 0 \; ; \; x = 0 \Rightarrow C = \log_e a$

$$kt = -\log_e \frac{a}{(a+x)} = -\frac{2.303}{t} \log_{10}\left(\frac{a}{(a+x)}\right)$$

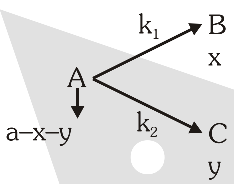
$$\boxed{k = \frac{2.303}{t} \log_{10}\left(\frac{a+x}{a}\right)}$$

❖ **Generation time :**

At t = generation time , $x = a \quad \therefore \quad t = \frac{0.693}{K}$

8. SOME SPECIAL CASES :

8.1 FIRST ORDER PARALLEL OR COMPETING OR SIMULTANEOUS REACTIONS



At $\quad t = 0 \quad [A] = a \quad [B] = [C] = 0$

(i) Differential rate law :

$$\frac{d[B]}{dt} = k_1 [A] \; ; \; \frac{d[C]}{dt} = k_2 [A]$$

and, $$\frac{-d[A]}{dt} = \frac{d[B]}{dt} + \frac{d[C]}{dt} \Rightarrow \frac{-d[A]}{dt} = (k_1 + k_2)[A] = k_{eff}[A]$$

$k_{eff} = k_1 + k_2$ = overall rate constant for the disappearance of 'A'

(ii) Integral rate law :

$$[A]_t = ae^{-k_{eff}t} = ae^{-(k_1+k_2)t}$$

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Chemical Kinetics\Eng\01_Theory

$$\frac{d[B]}{dt} = k_1 [A] \Rightarrow \frac{d[B]}{dt} = k_1 a e^{-(k_1+k_2)t}$$

$$[B] = \left(\frac{k_1 a}{k_1 + k_2}\right)(1 - e^{-(k_1+k_2)t})$$

Similarly, $[C] = \left[\frac{k_2 a}{k_1 + k_2}\right](1 - e^{-(k_1+k_2)t})$

(iii) Composition of product :

$$\frac{[B]}{[C]} = \frac{k_1}{k_2}$$

8.2 FIRST ORDER REVERSIBLE REACTION

$$A \underset{k_b}{\overset{k_f}{\rightleftharpoons}} B$$

	A	B
$t = 0$	a	0
$t = t$	$a - x$	x
$t = t_{eq.}$	$a - x_{eq.}$	$x_{eq.}$

$x_{eq.}$ = eq conc. of product

(i) $\frac{d[A]}{dt} = \frac{d[B]}{dt} = 0$ **($\because$ At equilibrium, conc. will not changed)**

(ii) $\frac{d[A]}{dt} = -k_f[A] + k_b[B] \Rightarrow \frac{d[B]}{dt} = -k_b[B] + k_f[A]$

$$\frac{d(a-x)}{dt} = -k_f(a-x) + k_b x$$

$$-\frac{dx}{dt} = -k_f a + (k_f + k_b)x$$

At eq^m, $\frac{dx}{dt} = 0 = k_f a + (k_f + k_b)x_{eq}$

$k_f a = (k_f + k_b)x_{eq}$

$$-\frac{dx}{dt} = -(k_f + k_b)x_{eq} + (k_f + k_b)x$$

$$\frac{dx}{dt} = (k_f + k_b)(x_{eq} - x)$$

$$\int_0^x \frac{dx}{(x_{eq} - x)} = (k_f + k_b)\int_0^t dt$$

$\mathbf{k_f + k_b = \frac{1}{t}\ \ell n\left(\frac{x_{eq.}}{x_{eq.} - x}\right)}$ (No need to remember this equation)

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8.3 FIRST ORDER SEQUENTIAL REACTION

$$A \xrightarrow{k_1} B \xrightarrow{k_2} C$$

All first order reactions

t = 0	a	0	0
t = t	a – x	y	z

(i) For 'A'

$$\frac{-d[A]}{dt} = r_1 = k_1 [A]$$

$$\frac{-d[A]}{[A]} = k_1\, dt$$

$$\mathbf{[A]_t = [A]_0}\, e^{-k_1 t}$$

$$a - x = a e^{-k_1 t}$$

$$x = a\,(1 - e^{-k_1 t})$$

(ii) For 'B'

$$\frac{dy}{dt} = k_1 a e^{-kt} - k_2 y$$

$$\frac{dy}{dt} + k_2 y = k_1 a e^{-kt}$$

$$dy + k_2 y dt = k_1\, a e^{-kt} dt$$

$$e^{k_2 t} dy + k_2\, y e^{k_2 t} = k_1 a e^{-k_1 t}.e^{k_2 t} \quad dt$$

$$\int_0^t d(k_2 y e^{k_2 t}) = \int_0^t k_1 a e^{(k_2 - k_1)t} dt$$

$$\mathbf{k_2 y}\, e^{k_2 t} \mathbf{=} \left(\frac{\mathbf{k_1 a}}{\mathbf{k_2 - k_1}}\right) \mathbf{e^{(k_2 - k_1)t} + y} \qquad \text{(No need to remember)}$$

(iii) Calculate time at which concentration of B will be maximum.

$$\frac{dy}{dt} = 0$$

$$-k_1 e^{-k_1 t} + k_2 e^{-k_2 t} = 0$$

$$e^{-k_2 t} = \frac{k_1}{k_2} e^{-k_1 t}$$

$$e^{k_1 t} = \frac{k_1}{k_2} e^{k_2 t} \quad \Rightarrow \quad k_1 t = \ell n \frac{k_1}{k_2} + k_2 t$$

$$\mathbf{t_{max.} = \frac{1}{(k_1 - k_2)}\ \ell n \frac{k_1}{k_2}}$$

ALLEN Session 2019-20

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(iv) $[B]_{max} = a \times \left[\frac{k_2}{k_1}\right]^{\frac{k_2}{k_1 - k_2}} = [A_0]\left[\frac{k_2}{k_1}\right]^{\frac{k_2}{k_1 - k_2}}$

8.3.1 Case -I $k_1 >> k_2$

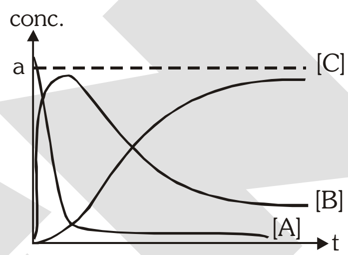
$$A \xrightarrow{k_1} B \xrightarrow{k_2} C$$

$$\begin{array}{ccc} & a & \\ \left[\begin{array}{c}\text{Tends to zero}\\ \text{very fast}\end{array}\right] & a - x & x \end{array}$$

$[A] = ae^{-k_1 t}$

$[B] = ae^{-k_2 t}$

$[C] = a(1 - e^{-k_2 t})$

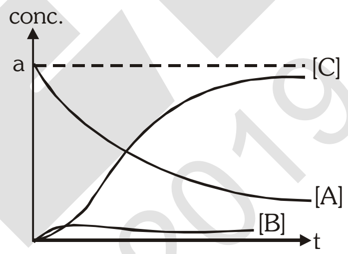


8.3.2 Case -II $k_2 >> k_1$

$[B]_t \rightarrow 0$

$[A] = ae^{-k_1 t}$

$[C] = a(1 - e^{-k_1 t})$



Note : Both Case (I) and Case (II) show that rate of overall reaction depends on rate of slowest step (RDS.)

9. EFFECT OF TEMPERATURE ON RATE OF REACTION :

In early days the effect of temperature on reaction rate was expressed in terms of ***temperature coefficient*** (μ) which was defined as the ratio of rate of reaction at two different temperature differing by 10ºC (usually these temperatures were taken as 25ºC and 35ºC)

$$\text{T.C, } \mu = \frac{K_{T+10}}{K_T} \approx 2 \text{ to } 3 \quad \textbf{(for most of the reactions)}$$

Exception : For some reactions, temperature coefficient is also found to be less than unity.

For example, $2NO + O_2 \rightarrow 2NO_2$ rate of reaction decreases on increasing temperature.

Note : Rate of reaction increases on increasing temperature (Generally ; T.C > 1)

ALLEN Session 2019-20
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Ex.15 For a reaction T.C. = 2, Calculate $\frac{k_{40°C}}{k_{25°C}}$ for this reaction. Assuming T.C remains constant.

Sol. $\frac{k_2}{k_1} = (T.C.)^{\frac{\Delta t}{10}} = (2)^{\frac{15}{10}} = (2)^{\frac{3}{2}} = \sqrt{8}$

But the method of temperature coefficient was not exact to explain the effect of temperature on reaction rate. For that a new theory was evolved.

9.1 Collision theory of reaction rate :

It was developed by ***Max Trautz*** and ***William lewis.*** It gives insight in to the energetics and mechanistic aspects of reactions.

It is based upon kinetic theory of gases.

According the this theory :

(i) A chemical reaction takes place due to the collision among reactant molecules. The number of collisions taking place per second per unit volume of the reaction mixture is known as collision frequency (Z).

(ii) Every collision does not bring a chemical change. The collision that actually produce the products are effective collision. For a collision to be effective the following two barriers are to be cleared.

❖ **Energy barrier :** The minimum amount of absolute energy which the colliding molecules must possess as to make the chemical reaction to occur is known as **threshold energy**.

- Reactant molecules having energy equal or greater than the threshold are called ***active molecules*** and those having energy less than the threshold are called ***passive molecules***.
- At a given temperature there exists a dynamic equilibrium between active and passive molecules. The process of transformation from passive to active molecules being endothermic, increase of temperature increases the number of active molecules and hence the reaction.

  Passive molecules $\rightleftharpoons$ Active molecules, ΔH = +ve
- "The minimum amount of excess energy required by reactant molecules to participate in a reaction is called **activation energy (E_a)".**

CONCEPT OF ENERGY OF ACTIVATION (E_a) :

- The average extra amount of energy which the reactant molecules (having energy less than the threshold) must acquire so that their mutual collision may lead to the breaking of bond(s) and hence the energy is known as energy of activation of the reaction. It is denoted by the symbol E_a. Thus,

  E_a = Threshold energy – Actual average energy of reactants

  E_a is expressed in kcal mole^{-1} or kJ mole^{-1}.

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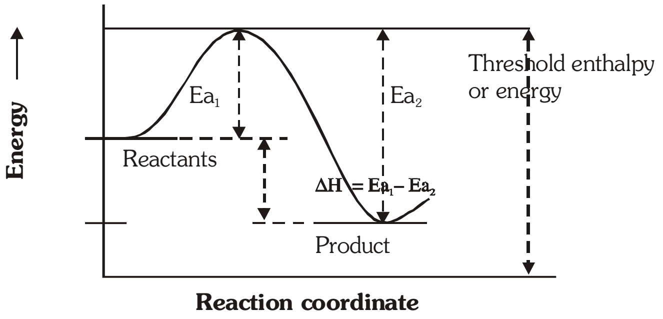
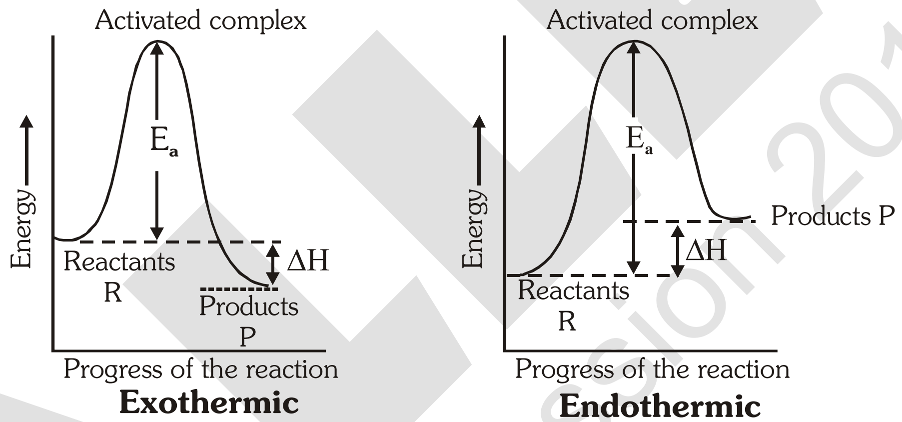
- The essence of Collision Theory of reaction rate is that there exists an energy barrier in the reaction path between reactant(s) and product(s) and for reaction to occur, the reactant molecules must climb over the top of the barrier which they do by collision. The existence of energy barrier and concept of E_a can be understood from the following diagram.



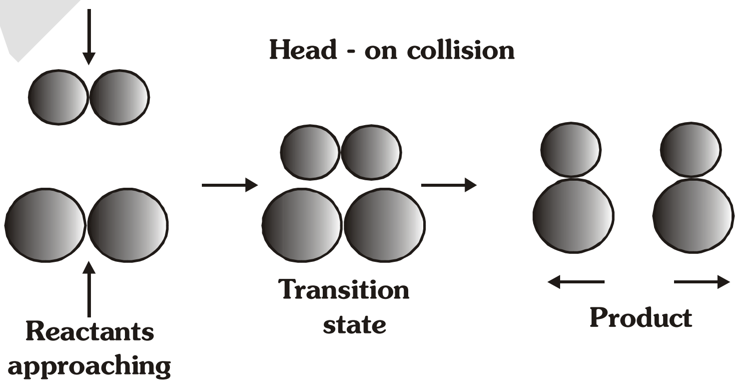
ΔH = Enthalpy change during the reaction ($R \rightarrow P$)
Ea_1 = Energy of activation of the forward reaction
Ea_2 = Energy of activation of the backward reaction

From the figure above it can be concluded that the minimum activation energy of any exothermic reaction will be zero while minimum activation energy for any endothermic reaction will be equal to ΔH.
Greater the height of energy barrier, greater will be the energy of activation and more slower will be the reaction at a given temperature.



❖ **Orientation barrier :** Energy alone does not determine the effectiveness of the collision. The reacting molecules must collide in proper direction to make collision effective. Following diagrams can explain importance of suitable direction for collision.

Consider reaction : $A_2 + B_2 \rightarrow 2AB$



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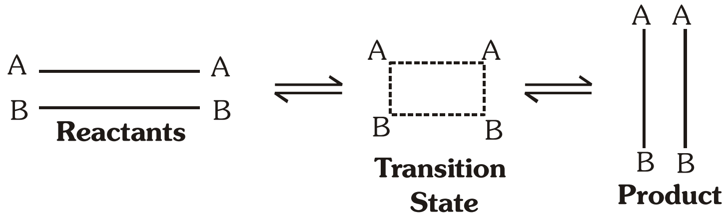
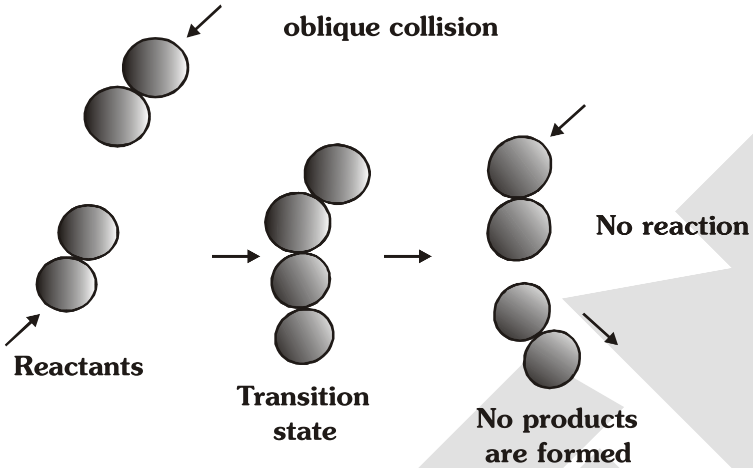
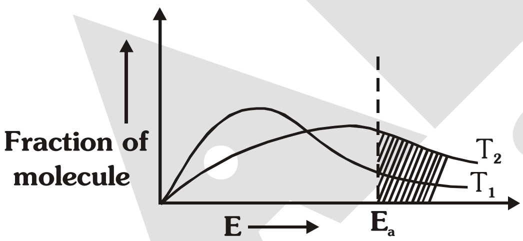




(iii) **Rate of any chemical reaction** = Collision frequency × fraction of the total number of effective collision.

= Collision frequency × fraction of the total number of collision in which K.E. of the colliding molecules equals to E_a or exceeds over it × fraction of collision in proper orientation.

From Maxwellian distribution, it is found that fraction of molecules having excess energy greater than threshold energy lead to the formation of product.



$T_2 > T_1$

$e^{-E_a/RT} \rightarrow$ Represents fraction of molecules having K.E. greater than or equal to E_a.

Rate $\propto e^{-E_a/RT}$

Dependence of rate on temperature is due to dependence of k on temperature.

$k \propto e^{-E_a/RT}$

$\mathbf{k = Ae^{-E_a/RT}}$ **[Arrhenius equation]**

ALLEN Session 2019-20
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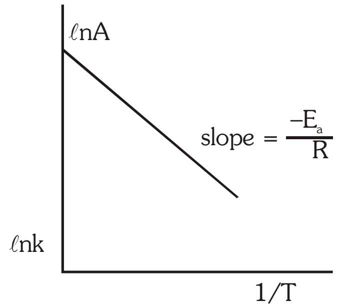
'A' is **pre-exponential** factor or **frequency factor** representing collisions taking place with proper orientation. A and E_a are assumed to be independent of temperature.

$$\ell nk = \ell n\, A - \frac{E_a}{RT}$$

As $T \to \infty$, $K \to A$



At temperature T_1, Rate constant = k_1
At temperature T_2, Rate constant = k_2

$$\ell nk_1 = \ell nA - \frac{E_a}{RT_1} \quad \& \ \ell nk_2 = \ell nA - \frac{E_a}{RT_2} \quad \Rightarrow \quad \ell n\frac{\mathbf{k_2}}{\mathbf{k_1}} = \frac{\mathbf{E_a}}{\mathbf{R}}\left(\frac{1}{\mathbf{T_1}} - \frac{1}{\mathbf{T_2}}\right)$$

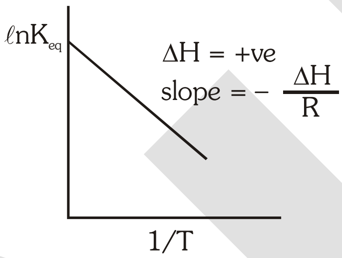
9.2 REVERSIBLE REACTIONS

$$k_f = A_f\, e^{-E_{af}/RT}$$

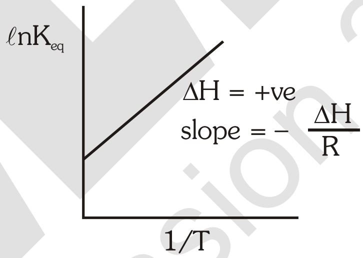
$$k_b = A_b\, e^{-E_{ab}/RT}$$

$$K_{eq} = \frac{k_f}{k_b} = \frac{A_f\ e^{-E_{af}/RT}}{A_b\ e^{-E_{ab}/RT}} = \left(\frac{A_f}{A_b}\right) e^{-(E_{af}-E_{ab})/RT}$$

$$\ell n\, K_{eq} = -\frac{\Delta H}{RT} + \ell n\left(\frac{A_f}{A_b}\right)$$



endothermic



exothermic

Ex.16 For a reaction, temperature coefficient = 2, then calculate the activation energy (in kJ) of the reaction.

Sol. **Given :** Temperature coefficient $= \frac{K_2}{K_1} = 2$

$T_1 = 25 + 273 = 298$ k
$T_2 = 35 + 273 = 308$ k

$$\log\frac{K_2}{K_1} = \frac{E_a}{2.303R}\left(\frac{T_2 - T_1}{T_1T_2}\right)$$

$$\log 2 = \frac{E_a}{2.303 \times 8.314} \times \left(\frac{10}{298 \times 308}\right)$$

$E_a = 52.31$ kJ mol^{-1}

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Ex.17 For first order gaseous reaction, log k when plotted against $\frac{1}{T}$, it give a straight the with a slope of –8000. Calculate the activation energy of the reaction.

Sol. For an arrhenius equation $k = Ae^{-E_a/RT}$

$$\log k = \log A - \frac{E_a}{2.303R} \times \frac{1}{T}$$

when curve is plotted between log k and $\frac{1}{T}$. A straight line is obtained and a slope of this curve

$$= -\frac{E_a}{2.303R}$$

Then, $\frac{E_a}{2.303R} = 8000$

or $E_a = 8000 \times 2.303 \times 2 = 36848$ calories

Ex.18 The slope of the plot of log k vs $\frac{1}{T}$ for a certain reaction was found to be – 5.4 × 10^3. Calculate the energy of activation of the reaction. If the rate constant of the reaction is 1.155 × 10^{-2} sec^{-1} at 373 K , what is its frequency factor ?

Sol. **(a)** Slope $= \frac{-E}{2.303\ \ R} = -5.4 \times 10^3$

$E_a = 5.4 \times 10^3 \times 2.303 \times 1.987$

$= 24.624$ cal mol^{-1}

(b) $K = Ae^{-E/RT}$; $\log 1.155 \times 10^{-2} = \log A - \frac{24.624}{2.303 \times 1.987 \times 373}$

or $A = 1.764 \times 10^3\ sec^{-1}$

10. CATALYST AND CATALYSIS :

A catalyst is a substance, which increases the rate of a reaction without itself being consumed at the end of the reaction, and the phenomenon is called ***catalysis***.

Thermal decomposition of $KClO_3$ is found to be accelerated by the presence of MnO_2. Here MnO_2 acts as a catalyst.

$$\mathbf{2KClO_3 + [MnO_2] \longrightarrow 2KCl + 3O_2 \uparrow + [MnO_2]}$$

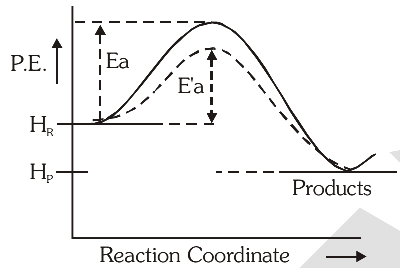
- MnO_2 can be received in the same composition and mass at the end of the reaction.
- The word catalyst should not be used when the added substance reduces the rate of reaction. The substance is then called **inhibitor**.
- Catalyst are generally foreign substances but sometimes one of the product may act as a catalyst and such catalyst is called **"*auto catalyst*"** and the phenomena is called ***auto catalysis***.

In the permanganate titration of oxalic acid initially there is slow discharge of the colour of permanganate solution but afterwards the discharge of the colour become faster. This is due to the formation of $MnSO_4$ during the reaction which acts as a catalyst for the same reaction. Thus, $MnSO_4$ is an **"*auto catalyst*"** for this reaction. This is an example of auto catalyst.

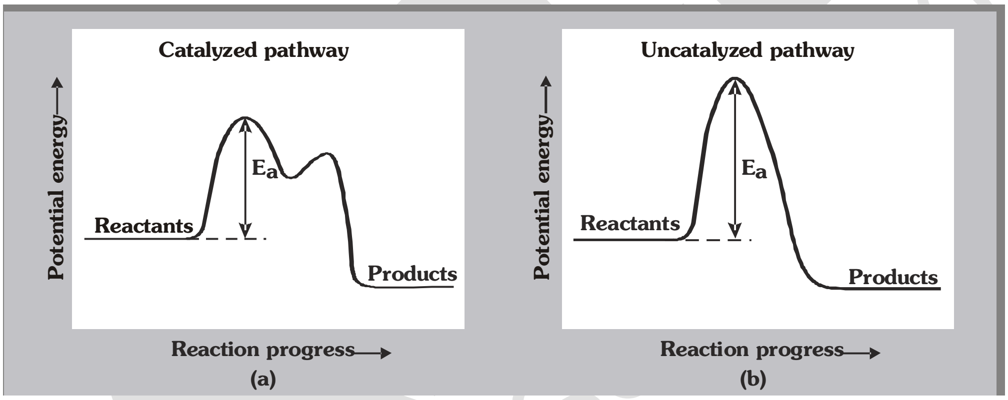
$$\mathbf{2KMnO_4 + 3H_2SO_4 + 5H_2C_2O_2 \longrightarrow K_2SO_4 + 8H_2O + 10CO_2 + MnSO_4}$$

10.1 General characteristics of catalyst :

- A catalyst does not initiate the reaction normally. It simply fastens it.
- Only a small amount of catalyst can catalyse the reaction.
- A catalyst does not alter the position of equilibrium i.e. magnitude of equilibrium constant and hence ΔG^o. It simply lowers the time needed to attain equilibrium. This means if a reversible reaction in absence of catalyst completes to go to the extent of 75% till attainment of equilibrium, and this state of equilibrium is attained in 20 minutes then in presence of a catalyst, the reaction will still go to 75% of completion on the attainment of equilibrium but the time needed for this will be less than 20 minutes.



A catalyst drives the reaction through a low energy path and hence E_a is less. That is, the function of the catalyst is to lower down the activation energy.
E_a = Energy of activation in absence of catalyst.
E_a' = Overall Energy of activation in presence of catalyst.
$E_a - E_a'$ = Lowering of activation energy by catalyst.



10.2 Comparison of rates of reaction in presence and absence of catalyst :

If k and k_{cat} be the rate constant of a reaction at a given temperature T , E_a and E'_a are the activation energies of the reaction in absence and presence of catalyst, respectively, the

$$\frac{k_{cat}}{k} = \frac{Ae^{-E'a/RT}}{Ae^{-Ea/RT}} = Ae^{(E_a - E'_a)/RT}$$

Since $E_a - E'_a$ is positive so $k_{cat} > k$. the ratio $\frac{k_{cat}}{k}$ gives the number of times the rate of reaction will increase by the use of catalyst at a given temperature.
The rate of reaction in the presence of catalyst at any temperature T_1 may be made equal to the rate of reaction in absence of catalyst but for this sake we will have to raise the temperature. Let this temperature be $T_2 = e^{-E'_a/RT_1} = e^{-E_a/RT_2}$ or $\frac{E'_a}{T_1} = \frac{E_a}{T_2}$

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11 CALCULATION OF RATE LAW / ORDER

11.1 When first step is rate determining step.

Ex.19 Calculate order and rate law of reaction -

$$2NO_2 + F_2 \longrightarrow 2NO_2F$$

with help of mechanism

I. $NO_2 + F_2 \xrightarrow{K_1} NO_2F + F$ **(slow)**

II. $NO_2 + F \xrightarrow{K_2} NO_2F$ **(fast)**

Sol. According to RDS

Rate = $k_1\,[NO_2]\,[F_2]$

11.2 Equilibrium approach :

Ex.20 For the reaction , $H^+ + HNO_2 + C_6H_5NH_2 \xrightarrow{Br^-} C_6H_5N_2^+ + 2H_2O$, the mechanism is ,

$H^+ + HNO_2 \underset{k_2}{\overset{k_1}{\rightleftharpoons}} \underset{\text{intermediate}}{H_2NO_2^+}$ **(fast equilibrium step)**

$H_2NO_2^+ + Br^- \xrightarrow{k_3} NOBr + H_2O$ **(slow)**

$NOBr^- + C_6H_5NH_2 \xrightarrow{k_4} C_6H_5N_2^+ + Br^- + H_2O$ **(fast)**

Derive the rate law expression for the reaction

Sol. $r = k_3\,[Br^-]\,[H_2NO_2^+]$

$$K_{eq} = \frac{k_1}{k_2} = \frac{[H_2NO_2^+]}{[H^+][HNO_2]}$$

$$[H_2NO_2^+] = \left[\frac{k_1}{k_2}\right][H^+]\,[HNO_2]$$

$$r = \frac{k_1k_3}{k_2}[H^+]\,[HNO_2]\,[Br^-].$$

Note : Rate law can have reactant product or catalyst concentration terms but not any intermediate terms.

ALLEN Session 2019-20

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11.3 CASE-III : STEADY STATE APPROXIMATION :

Ex.21 For the reaction , $2O_3 \rightarrow 3O_2$, the mechanism is

$$O_3 \xrightarrow{k_1} O_2 + O$$

$$O_2 + O \xrightarrow{k_2} O_3$$

$$O_3 + O \xrightarrow{k_3} 2O_2$$

Derive the rate law expression for the reaction

Sol. rate = $\frac{-1}{2}\frac{d[O_3]}{dt} = \frac{1}{3}\frac{d[O_2]}{dt}$

$$\frac{d[O_3]}{dt} = -k_1[O_3] + k_2[O_2][O] - k_3[O_3][O]$$

$$\frac{d[O_2]}{dt} = k_1[O_3] - k_2[O_2][O] + k_3[O_3][O]$$

At steady state $\frac{d[O]}{dt} = 0$

$$\frac{d[O]}{dt} = k_1[O_3] - k_2[O_2][O] - k_3[O_3][O] = 0$$

$$[O] = \frac{k_1[O_3]}{k_2[O_2]+k_3[O_3]}$$

$$\frac{d[O_3]}{dt} = -k_1[O_3] + \frac{\{k_2[O_2]\ \ k_1[O_3]\}}{k_2[O_2]+k_3[O_3]} - \frac{k_1\ \ k_3[O_3]^2}{k_2[O_2]+k_3[O_3]}$$

$$= -k_1[O_3] + \frac{k_1k_2[O_2]\ \ [O_3] - k_1k_3[O_3]^2}{k_2[O_2]+k_3[O_3]}$$

$$= \frac{k_1k_2[O_2][O_3] - k_1k_3[O_3]^2 + k_1k_2[O_2][O_3] - k_1k_3[O_3]^2}{k_2[O_2]+k_3[O_3]}$$

$$= \frac{-2k_1\ \ k_3[O_3]^2}{k_2[O_2]+k_3[O_3]}$$

$$\left[\frac{-1}{2}\ \frac{d}{dt}[O_3]\right] = \frac{k_1\ \ k_3\ \ [O_3]^2}{k_2[O_2]+k_3[O_3]}$$

$$\text{Rate} = -\frac{1}{2}\frac{d}{dt}[O_3]$$

So, Rate (r) $= \frac{k_1\ \ k_3[O_3]^2}{k_2[O_2]+k_3[O_3]}$

if 3rd step is RDS then $k_1 >> k_3$

$k_2 >> k_3$

Rate (r) = $\frac{k_1k_3}{k_2}[O_3]^2[O_2]^{-1}$ (ie. first order)

Session 2019-20

MISSLENIOUS PREVIOUS YEARS QUESTION

1. Which of the following statement(s) is (are) correct **[JEE 1999]**

(A) A plot of $\log K_p$ versus $1/T$ is linear

(B) A plot of $\log [X]$ versus time is linear for a first order reaction, $X \longrightarrow P$

(C) A plot of $\log P$ versus $1/T$ is linear at constant volume.

(D) A plot of P versus $1/V$ is linear at constant temperature.

Ans. **(A,B,D)**

2. The rate constant for an isomerisation reaction $A \rightarrow B$ is $4.5 \times 10^{-3}\ min^{-1}$. If the initial concentration of A is 1 M. Calculate the rate of the reaction after 1 h. **[JEE 1999]**

Ans. $\mathbf{3.435 \times 10^{-3}}$ **M/min**

Sol. $r = k[A]$

$r = K[A]_0\ e^{-kt}$

$r = (4.5 \times 10^{-3})\ e^{-4.5 \times 10^{-3} \times 60}$ M/min.

3. A hydrogenation reaction is carried out at 500 K. If the same reaction is carried out in the presence of a catalyst at the same rate, the temperature required is 400 K. Calculate the activation energy of the reaction if the catalyst lowers the activation barrier by 20 $kJmol^{-1}$. **[JEE 2000]**

Ans. **100 $\mathbf{kJmol^{-1}}$**

Sol. $\ln\left(\frac{r_2}{r_2}\right) = -\frac{1}{R}\left(\frac{Ea_2}{T_2} - \frac{Ea_1}{T_1}\right)$

$\ln(1) = -\frac{1}{R}\left(\frac{Ea_1 - 20}{40} - \frac{Ea_1}{500}\right)$

$Ea_1 = 100$

4. The rate constant for the reaction **[JEE SCR 2000]**

$2N_2O_5 \longrightarrow 4NO_2 + O_2$

is $3.0 \times 10^{-5}\ sec^{-1}$. if the rate is 2.4×10^{-5} mol $litre^{-1}\ sec^{-1}$, then the concentration of N_2O_5 (in mol $litre^{-1}$) is

(A) 1.4 (B) 1.2 (C) 0.004 (D) 0.8

Ans. **(D)**

Sol. $r = k[N_2O_5]^2$

$2.4 \times 10^{-5} = 3 \times 10^{-5}[N_2O_5]^2$

5. If I is the intensity of absorbed light and C is the concentration of AB for the photochemical process : $AB + h\nu \longrightarrow AB$, the rate of formation of AB is directly proportional to **[JEE SCR 2001]**

(A) C (B) I (C) I^2 (D) CI

Ans. **(B)**

Sol. For phot chemical reaction

$r \propto I$

6. The rate of a first order reaction is 0.04 mole $litre^{-1}\ s^{-1}$ at 10 minutes and 0.03 mol $litre^{-1}\ s^{-1}$ at 20 minutes after initiation. Find the half life of the reaction. **[JEE 2001]**

Ans. $\mathbf{t_{1/2} = 24.14}$ **min**

Sol. $\left(\frac{\ln 2}{t_{1/2}}\right)(20 - 10) = \ln\left(\frac{0.04}{0.03}\right)$

Session 2019-20

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7. Consider the chemical reaction, $N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)$. The rate of this reaction can be expressed in term of time derivative of concentration of $N_2(g)$, $H_2(g)$ or NH_3 (g). Identify the correct relationship amongst the rate expressions. **[JEE SCR 2002]**

(A) Rate = – d[N_2]/dt = – 1/3 d[H_2]/dt = 1/2d[NH_3]/dt

(B) Rate = – d[N_2]/dt = – 3 d[H_2]/dt = 2d[NH_3]/dt

(C) Rate = d[N_2]/dt = 1/3 d[H_2]/dt =1/2d[NH_3]/dt

(D) Rate = – d[N_2]/dt = – d[H_2]/dt = d[NH_3]/dt

Ans. **(A)**

Sol. $\frac{-d[N_2]}{dt} = -\frac{1}{3}\frac{d[H_2]}{dt} = \frac{1}{2}\frac{d[NH_3]}{dt}$

8. In a first order reaction the concentration of reactant decreases from 800 mol/dm^3 to 50 mol/dm^3 in 2×10^4 sec. The rate constant of reaction in sec^{-1} is **[JEE SCR 2003]**

(A) 2×10^4 (B) 3.45×10^{-5}

(C) 1.3486×10^{-4} (D) 2×10^{-4}

Ans. **(C)**

Sol. $K(2 \times 10^4) = \ln\left(\frac{800}{50}\right)$

9. The reaction, X $\longrightarrow$ Product follows first order kinetics. In 40 minutes the concentration of X changes from 0.1 M to 0.025 M. Then the rate of reaction when concentration of X is 0.01 M

(A) 1.73×10^{-4} M min^{-1} (B) 3.47×10^{-5} M min^{-1}

(C) 3.47×10^{-4} M min^{-1} (D) 1.73×10^{-5} M min^{-1} **[JEE SCR 2004]**

Ans. **(C)**

Sol. $K(40) = \ln\left(\frac{0.1}{0.25}\right)$

$K = \frac{\ln 2}{20}$

$r = K[X]$

$= \frac{\ln 2}{20} \times 0.1$

10. $2X(g) \longrightarrow 3Y(g) + 2Z(g)$

Time (in Min)	0	100	200
Partial pressure of X (in mm of Hg)	800	400	200

Assuming ideal gas condition. Calculate

(a) Order of reaction

(b) Rate constant (K_X)

(c) Time taken for 75% completion of reaction

(d) Total pressure when P_X = 700 mm. **[JEE 2005]**

ALLEN Session 2019-20

Ans. **(a) 1, (b) 6.93×10^{-3} min^{-1} , (c) 200, (d) 950 mm**

Sol. (a) As half life is constant so it will be of first order.

(b) $K_x(100) = \ln\left(\frac{800}{400}\right)$

(c) $t = 75\% = 2t_{50\%} = 2\left(\frac{\ln 2}{K_x}\right)$

(d) $2X(g) \longrightarrow 3Y(g) + 2Z(g)$

$800 - x \qquad \frac{3x}{2} \qquad x$

Here x = 100

So $P_T = 800 + \frac{3x}{2} = 950$

11. Which of the following statement is incorrect about order of reaction? **[JEE 2005]**

(A) Order of reaction is determined experimentally

(B) It is the sum of power of concentration terms in the rate law expression

(C) It does not necessarily depend on stoichiometric coefficients

(D) Order of the reaction can not have fractional value.

Ans. **(D)**

12. Consider a reaction aG + bH $\rightarrow$ Products. When concentration of both the reactants G and H is doubled, the rate increases by eight times. However, when concentration of G is doubled keeping the concentration of H fixed, the rate is doubled. The overall order of the reaction is :

[JEE 2006]

(A) 0 (B) 1 (C) 2 (D) 3

Ans. **(D)**

Sol. $\frac{r_2}{r_1}\left(\frac{[G\ ]_2}{[G\ ]_1}\right)^{\alpha}\left(\frac{[H]_2}{[H]_1}\right)^{\alpha}$

$8 = 2^{\alpha} . 2^{\beta}$ ....(1)

$2 = 2^{\alpha}$ ....(2)

$\alpha = 1, \beta = 2$

13. Under the same reaction conditions, initial concentration of 1.386 mol dm^{-3} of a substance becomes half in 40 seconds and 20 seconds through first order and zero order kinetics, respectively. Ratio $\left(\frac{k_1}{k_0}\right)$ of the rate constants for first order (k_1) and zero order (k_0) of the reactions is **[JEE 2008]**

(A) 0.5 mol^{-1} dm^3 (B) 1.0 mol dm^{-3}

(C) 1.5 mol dm^{-3} (D) 2.0 mol^{-1} dm^3

Ans. **(A)**

ALLEN Session 2019-20

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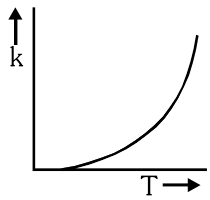
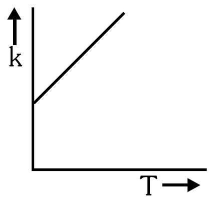
Sol. $K_1 = \dfrac{\ln 2}{40}$

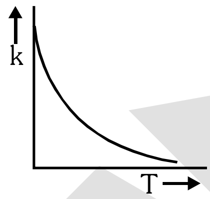
$K_0 = \dfrac{1.386}{2 \times 20}$

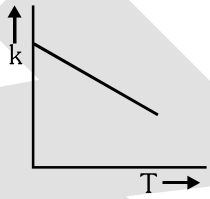
$\dfrac{K_1}{K_0} = 0.5$

14. Plots showing the variation of the rate constant (k) with temperature (T) are given below. The plot that follows Arrhenius equation is – **[JEE 2010]**

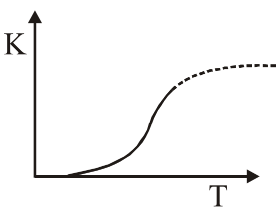
(A) 

(B) 

(C) 

(D) 

Ans. **(A)**

Sol. 

15. The concentration of R in the reaction R $\rightarrow$ P was measured as a function of time and the following data is obtained :

[R] (molar)	1.0	0.75	0.40	0.10
t(min.)	0.0	0.05	0.12	0.18

The order of the reaction is. **[JEE 2010]**

Ans. **Zero**

Sol. By hit and trial method het reaction is zero order.

$K_1 = \dfrac{[A]_0 - [A]_1}{t_1}$

$K_1 = \dfrac{1 - 0.75}{0.05} = 5$

$K_2 = \dfrac{[A]_0 - [A]t_2}{t_2}$

$K_2 = \dfrac{1 - 0.4}{0.12} = 5$

As $K_1 \approx K_1$ so reaction will be zero order reaction.

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EXERCISE # (S-1)

RATE OF REACTION

1. Ammonia and oxygen reacts at higher temperatures as

$$4NH_3(g) + 5O_2(g) \longrightarrow 4NO(g) + 6H_2O(g)$$

In an experiment, the concentration of NO increases by 1.08×10^{-2} mol litre^{-1} in 3 seconds. Calculate.

(i) rate of reaction

(ii) rate of disappearance of ammonia

(iii) rate of formation of water

2. The reaction $2A + B + C \rightarrow D + E$ is found to be first order in A, second order in B and zero order in C.

(i) Give the rate law for the reaction in the form of differential equation.

(ii) What is the effect in rate of increasing concentrations of A, B, and C two times?

3. At 27°C, it was observed during a reaction of hydrogenation that the pressure of hydrogen gas decreases from 2 atmosphere to 1.1 atmosphere in 75 minutes. Calculate the rate of disappearance reaction in M sec^{-1} and in terms of atm min^{-1}. [Use : R = 0.08 atm-L/K-mole]

4. For the elementary reaction $2A + B_2 \longrightarrow 2AB$. Calculate how much the rate of reaction will change if the volume of the vessel is suddenly reduced to one third of its original volume?

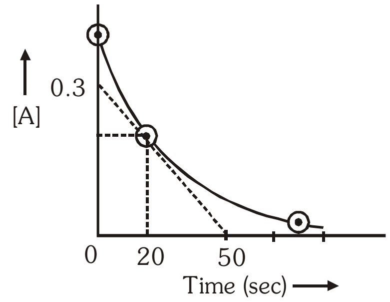
5. For the reaction $3BrO^- \rightarrow BrO_3^- + 2Br^-$ in an alkaline aqueous solution, the value of the second order (in BrO^-) rate constant at 80°C in the rate law for $-\frac{d[BrO^-]}{dt}$ was found to be 0.057 L mol^{-1} s^{-1}. What is the rate constant when the rate law is written for :

(a) $\frac{d[BrO_3^-]}{dt}$, (b) $\frac{d[Br^-]}{dt}$?

6. $xA + yB \rightarrow zC$. If $-\frac{d[A]}{dt} = -\frac{d[B]}{dt} = 1.5\frac{d[C]}{dt}$, then (x + y + z) is. (x, y & z are the lowest integral value.)

7. For the reaction : 3A (g) → 2B(g), the rate of formation of 'B' at 298 K, is represented as $ln\left(\frac{d[B]}{dt}\right) = -0.04 + 2 \times ln[A]$. The order of reaction is-

8. A certain reaction **A → B** follows the given concentration (Molarity) – time graph. Calculate the rate for this reaction at 20 second.



ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Chemical Kinetics\Eng\02_Exercise

9. The reaction $A(g) + 2B(g) \longrightarrow C(g) + D(g)$ is an elementary process. In an experiment, the initial partial pressure of A & B are $P_A = 0.6$ and $P_B = 0.8$ atm. Calculate the ratio of rate of reaction relative to initial rate when P_C becomes 0.2 atm.

ZERO ORDER REACTIONS

10. In the reaction ($A \longrightarrow B$) rate constant is 1.2×10^{-2} M s^{-1} . What is concentration of B after 10 and 20 min., if we start with 10 M of A.

11. From the following data for the zero order reaction : $A \longrightarrow$ products. Calculate the value of k.

Time (min.)	**[A]**
0.0	0.10 M
1.0	0.09 M
2.0	0.08 M

12. A drop of solution (volume 0.10 ml) contains 6×10^{-6} mole of H^+. If the rate constant of disappearance of H^+ is 1×10^7 mole litre^{-1} sec^{-1}, how long would it take for H^+ in drop to disappear?

13. A certain substance A is mixed with an equimolar quantity of substance B. At the end of an hour A is 75% reacted. Calculate the time when A is 10% unreacted. (Given: order of reaction is zero for both A & B.)

14. For a zero order chemical reaction,

$$2NH_3(g) \longrightarrow N_2(g) + 3H_2(g)$$

rate of reaction = 0.1 atm/sec. Initially only NH_3 is present & its pressure = 3 atm. Calculate total pressure at t = 10 sec.

FIRST ORDER REACTIONS

15. A first order reaction is 75% completed in 72 min. How long time will it take for
(i) 50% completion (ii) 87.5% completion

16. A first order reaction is 20% complete in 10 min. Calculate (i) the specific rate constant , (ii) the time taken for the reactions to go to 75% completion. [ln 10 = 2.3, ln2 = 0.7]

17. Show that in case of unimolecular reaction, the time required for 99.9% of the reaction to take place is ten times that required for half of the reaction.

18. A drug is known to be ineffective after it has decomposed 75%. The original concentration of a sample was 500 units/ml. When analyzed 20 months later, the concentration was found to be 250 units/ml. Assuming that decomposition is of I order, what will be the expiry time of the drug?

19. A viral preparation was inactivated in a chemical bath. The inactivation process was found to be first order in virus concentration. At the beginning of the experiment, 2.0 % of the virus was found to be inactivated per minute . Evaluate k for inactivation process. [ln 10 = 2.3, ln 98 = 4.58]

20. The reaction $SO_2Cl_2(g) \longrightarrow SO_2(g) + Cl_2(g)$ is a first order gas reaction with $k = \left(\frac{35}{9}\right) \times 10^{-4}$ sec^{-1} at 320°C. What % of SO_2Cl_2 is decomposed on heating this gas for 90 min. (ln2 = 0.7)

21. The decomposition of a compound A in solution follow first order kinetics. If 10% w/v solution of A is 10% decomposed in 10 minutes at 10°C, then 20% w/v solution of A is .... % decomposed in 20 minutes at 10°C.

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22. Calculate the half-life of the first-order reaction

$$C_2H_4O(g) \longrightarrow CH_4(g) + CO(g)$$

if the initial pressure of $C_2H_4O(g)$ is 80 mm of Hg and the total pressure at the end of 20 minutes is 120 mm of Hg

SECOND ORDER REACTIONS

23. In the II order reaction : $2A \rightarrow A_2$. The rate of formation of A_2 is 10^{-5} M sec^{-1} at 0.01M concentration of A. Calculate the rate constant in the rate of disappearance of 'A'.

24. If $t_{1/2}$ of a second order reaction is 1.0 hr. After what time, the amount of reactant will be 25% of the intial amount?

25. Reaction : $A + B \rightarrow C + D$, follows the rate law : $r = (2 \times 10^{-3} M^{-1} s^{-1})$ [A] [B]. The reaction is started with 1.0 mole each of 'A' and 'B', in constant volume of 5 litre. In what time, the moles of 'A' become 0.25 ?

DETERMINATION OF ORDER OF REACTION & RATE LAW

26. At 800° C, the rate of reaction

$$2\,NO(g) + H_2(g) \longrightarrow N_2(g) + H_2O(g)$$

changes with the concentration of NO(g) and $H_2(g)$ as-

Exp.no.	**[NO] in M**	**[H_2] in M**	$-\frac{1}{2}\frac{d[NO]}{dt}$ **in M sec^{-1}**
(i)	1.5×10^{-3}	4×10^{-3}	4.50×10^{-9}
(ii)	1.5×10^{-3}	2×10^{-3}	2.25×10^{-9}
(iii)	3.0×10^{-3}	2×10^{-3}	9.00×10^{-9}

(a) What is the order of reaction?

(b) What is the rate equation for the reaction?

(c) What is the rate of reaction when $[H_2] = 1.5 \times 10^{-3}$ M and $[NO] = 1.0 \times 10^{-3}$M?

27. The catalytic decomposition of $N_2O(g)$ by gold at 900°C and at an initial pressure of 200 mm is 50% complete in 53 minutes and 73% complete in 100 minutes.

(i) What is the order of the reaction?

(ii) Calculate the velocity constant.

(iii) How much of percentage $N_2O(g)$ will decompose in 100 min. at the same temperature but at initial pressure of 600 mm of Hg ?

28. The pressure of a gas decomposing at the surface of a solid catalyst has been measured at different time and the results are given below

t (sec)	0	100	200	300
Pr. (Pascal)	4×10^3	3.5×10^3	3×10^3	2.5×10^3

Determine the order of reaction and its rate constant.

29. The half life period of decomposition of a compound is 50 minutes. If the initial concentration is halved, the half life period is reduced to 25 minutes. What is the order of reaction?

30. For a chemical reaction $A + B \rightarrow$ products, the order is one with respect to each A and B. The sum of x and y from the following data is.

Rate (mol $l^{-1}s^{-1}$)	**[A](mol l^{-1})**	**[B](mol l^{-1})**
0.10	0.20	0.05
0.40	x	0.05
0.80	0.40	y

ALLEN Session 2019-20

CALCULATION OF RATE CONSTANT USING DIFFERENT PARAMETERS

31. For the Ist order reaction,

$$A \longrightarrow B + C$$

	Time	t	∞
Total pressure of (A + B + C)		P_2	P_3

Find k in term of P_2, P_3 and t

32. For the Ist order reaction $S \longrightarrow G + F$

	Time	t	∞
Rotation of (G + F)		r_t	r_∞

Find k in term of r_t, r_∞ and t

33. The thermal decomposition of dimethyl ether was measured by finding the increase in pressure at 500ºC in the reaction

$$(CH_3)_2O(g) \longrightarrow CH_4(g) + H_2(g) + CO(g)$$

Time (sec.)	200	400	∞
Pressure increase (mm Hg)	540	594	600

The initial pressure of ether was 300 mm Hg. Determine the rate constant of reaction.

34. From the following data, show that decomposition of H_2O_2 in aqueous solution is first order.

Time (in minutes)	0	10	20
Volume (in c.c. of $KMnO_4$)	22.8	11.4	5.7

35. The following data were obtained in experiment of inversion of cane sugar.

Time (minutes)	0	60	120	180	∞
Angle of rotation (degree)	+15	+ 7	+ 3	+1	–1

Show that the reaction is of first order. After what time would you expect a zero reading in polarimeter?

36. The decomposition of N_2O_5 according to the equation $2N_2O_5 (g) \longrightarrow 4 NO_2(g) + O_2(g)$ is a first order reaction. After 60 min. from start of decomposition in a closed rigid vessel, the total pressure developed is found to be 350 mm Hg. On complete decomposition, the total pressure is 500 mm Hg. Calculate the rate constant for disappearance of $N_2O_5(k_{N_2O_5})$ for the reaction.

37. The reaction given below, rate constant for disappearance of A is 8×10^{-3} sec^{-1}. Calculate the time required for the total pressure in a system containing A at an initial pressure of 0.1 atm to rise to 0.145 atm and also find the total pressure after 100 sec. [ln7 = 1.9 , ln 10 = 2.3].

$$2A (g) \longrightarrow 4B(g) + C(g)$$

38. The reaction $A(aq) \longrightarrow B (aq) + C (aq)$ is monitered by measuring optical rotation of reaction mixture at different time interval. The species A, B and C are optically active with specific rotations 20°, 30° and – 40° per mole respectively. Starting with pure A, if the value of optical rotation was found to be 2.5° after 6.93 minutes and optical rotation was –5° after infinite time. Find the rate constant for first order conversion of A into B and C.

ALLEN Session 2019-20

39. For the hydrolysis of ethyl acetate in presence of H^+ as catalyst.

$$CH_3COOC_2H_5(aq.) + H_2O \xrightarrow{H^+} CH_3COOH(aq) + C_2H_5OH(aq.)$$

the experimentally observed rate law is

$$\text{rate} = k\ [\text{ester}]\ [H_2O]\ [H^+].$$

If the value of rate constant is $1.386 \times 10^{-3}\ M^{-2}\ min^{-1}$ and concentration of $[H^+]$ ion is 1.8 M, how many seconds will it take for concentration of ester to become half of initial value ?

40. If 0.01 % of a substance undergoing decomposition is consumed in 1 milli seconds when the concentration is 0.02M and in 0.25 milli seconds when the concentration is 0.04M. The order of reaction is.

PARALLEL , SEQUENTIAL REACTIONS AND REVERSIBLE

41. $A \begin{cases} \xrightarrow{k_1} B \\ \xrightarrow{k_2} C \end{cases}$, $k_1 = x\ hr^{-1}$; $k_1 : k_2 = 1 : 10$. Calculate $\dfrac{[C]}{[A]}$ after one hour from the start of the reaction.

Assuming only A was present in the beginning.

42. How much time would be required for the B to reach maximum concentration for the reaction

$$A \xrightarrow{k_1} B \xrightarrow{k_2} C. \text{ Given } k_1 = \frac{ln2}{4}\ min^{-1},\ k_2 = \frac{ln2}{2}\ min^{-1}$$

43. A gaseous reactant **X** decompose to produce gaseous product B & C in a parallel reaction, both by first order, as follows :

$$X(g) \begin{cases} \xrightarrow{k_1 = 2 \times 10^{-3} min^{-1}} B(g) \\ \xrightarrow[k_2 = 3 \times 10^{-2} min^{-1}]{} C(g) \end{cases}$$

If the decomposition is carried out in a sealed flask, partial pressure of B after very long time was found to be 100 mm Hg. Determine the time when pressure of **X**(g) was 100 mm Hg.
[ln2 = 0.693]

44. The reaction A proceeds in parallel channels $A \begin{cases} \rightarrow B \\ \rightarrow C \end{cases}$. Suppose the half life values for the two branches are 60 minutes and 90 minutes, what is the overall half-life value?

45. For the given sequential reaction

$$A \xrightarrow{k_1} B \xrightarrow{k_2} C$$

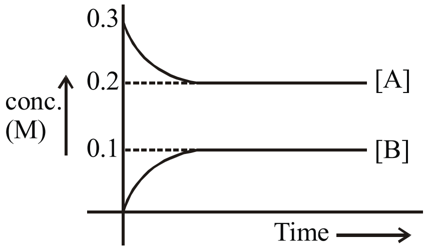
Initial concentration of A is 20M. Calculate the approximate concentration of C after 10 min, if $k_1 = 2 \times 10^8\ min^{-1}$ & $k_2 = 0.0693 min^{-1}$

ALLEN Session 2019-20
node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Chemical Kinetics\Eng\02_Exercise

46. Consider a reversible reaction :

$$A \underset{k_2}{\overset{k_1}{\rightleftharpoons}} B$$

Which is a first order in both the direction ($k_1 = \frac{1.38}{3} \times 10^{-2}\ \text{min}^{-1}$). The variation in concentration is plotted with time as shown below.



Calculate the time (in minute) at which 25% of A would be exhausted. [$\ell n2 = 0.69$]

47. 3 mole mixture of two different substances A & B (mole fraction of A = 2/3) undergoes parallel first order reaction to form product 'C' as follow at 300 K.

$A \xrightarrow{k_1} C \qquad t_{1/2} = 2hr$

$B \xrightarrow{k_2} C \qquad t_{1/2} = 4hr$

The time at which there will be 2 mole of 'C' is

48. For the series of competitive reactions :

$H + HO_2 \xrightarrow{K_1} H_2 + O_2$

$H + HO_2 \xrightarrow{K_2} 2OH$

$H + HO_2 \xrightarrow{K_3} H_2O + O$

It has been found that $K_1 : K_2 : K_3 = 0.60 : 0.30 : 0.10$. The molar ratio of the product, H_2, O_2, OH, H_2O and O, at time t, is -

49. In the parallel reactions : $A \xrightarrow{K_1 = \ln 3\,\text{min}^{-1}} B$ and $A \xrightarrow{K_2 = \ln 3\,\text{min}^{-1}} C$, the time when the concentration of A, B and C becomes equal is -

50. For the sequential reactions :

$A \xrightarrow{K_1 = 0.02\,\text{min}^{-1}} B \xrightarrow{K_2 = 0.02\,\text{min}^{-1}} C$, the initial concentration of 'A' was 0.2M and initially 'B' and 'C' were absent. The time at which the concentration of 'B' becomes maximum and the maximum concentration of 'B' are respectively.

TEMPERATURE DEPENDENCE OF RATE

51. In gaseous reactions important for understanding the upper atmosphere, H_2O and O react bimolecularly to form two OH radicals. ΔH for this reaction is 72 kJ at 500 K and E_a=77 kJ mol^{-1}, then calculate E_a for the bimolecular recombination of 2OH radicals to form H_2O & O at 500 K

52. The specific rate constant for a reaction increases by a factor of 4, if the temperature is changed from 27°C to 47°C. Find the activation energy (in kcal) for the reaction (Use ln 2 = 0.7)

53. Given that the temperature coefficient for the saponification of ethyl acetate by NaOH is 1.75. Calculate activation energy for the saponification of ethyl acetate. Let initial temperature is 298K.
[**Use :** ln (1.75) = 0.56]

ALLEN Session 2019-20

54. The energy of activation for a reaction is at 27ºC 10 kJ/mol. The presence of catalyst lowers the energy of activation by 75%. Find the factor by which rate of reaction increases at 27ºC due to catalyst.
(Take R = 25/3 J/mol-k)

55. At 380°C, the half–life period for the first order decomposition of H_2O_2 is 360 min.The energy of activation of the reaction is 200 kJ mol^{-1}. Calculate the time required for 75% decomposition at 450°C.

Use : $\left[\frac{1}{653}-\frac{1}{723}\right]=1.5\times10^{-4}$, $e^{3.6}$ = 36, R = $\frac{25}{3}$ J / k – mol

56. The rate constant for decomposition of $COCl_2$ (g) according to following reaction $COCl_2(g) \rightarrow CO(g) + Cl_2(g)$; ΔH = 19 kcal/mol, is expressed as

$ln\ k = 15 - \frac{5000}{T}$

Calculate activation energy for given reaction (in Kcal/mol.)

57. At 300 K, 50% of molecule collide with energy greater than or equal to E_a. At what temperature, 25% molecule will have energy greater than or equal to E_a.

58. For a zero order reaction at 200K reaction complete in 5 minutes while at 300K, same reaction completetes in 2.5 minutes. What will be the activation energy in calorie.
(R = 2 Cal/mol-k; *l*n 2 = 0.7)

59. For a first order reaction : A → P, the temperature (T) dependent rate constant (k) was found to follow the equation $\log_{10}k = -(2000)\frac{1}{T} + 6$. Then activation energy equation of reaction will be (ln x = 2.3 × log x)

60. A reaction takes place in three steps. The rate constants are k_1 , k_2 , k_3. The overall rate constant $k=\frac{k_1\sqrt{k_3}}{k_2}$. If energy of activation is 40 , 30 and 20 kJ respectively, the overall energy of activation is :

MECHANISM OF REACTION

61. The reaction $2NO + Br_2 \longrightarrow 2NOBr$, is supposed to follow the following mechanism

(i) $NO + Br_2 \underset{}{\overset{fast}{\rightleftharpoons}} NOBr_2$

(ii) $NOBr_2 + NO \xrightarrow{slow} 2NOBr$

Suggest the rate law expression.

62. For the reaction $2H_2 + 2NO \longrightarrow N_2 + 2H_2O$, the following mechanism has been suggested :

$2NO \rightleftharpoons N_2O_2$ equilibrium constant = K_1 (fast)

$N_2O_2 + H_2 \xrightarrow{k_2} N_2O + H_2O$ (slow)

$N_2O + H_2 \xrightarrow{k_3} N_2 + H_2O$ (fast)

Establish the rate law for given reaction.

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63. For the reaction : $2O_3(g) \longrightarrow 3O_2(g)$

Mechanism :

Step - I $O_3(g) \xrightleftharpoons{\text{fast}} O_2(g) + O(g)$

Step - II $O(g) + O_3(g) \xrightarrow{\text{slow}} 2O_2(g)$

Overall order of reaction based on mechanism is

64. A complex reaction : $2X + Y \rightarrow Z$, takes place in two steps :

$X + Y \xrightarrow{K_1} 2W$

$X + 2W \xrightarrow{K_2} Z$

If $K_1 << K_2$, order of reaction will be -

65. The suggested mechanism for the reaction : $CHCl_3(g) + Cl_2(g) \rightarrow CCl_4(g) + HCl(g)$, is -

$Cl_2 \underset{K_2}{\overset{K_1}{\rightleftharpoons}} 2\dot{C}l(\text{fast})$

$CHCl_3 + \dot{C}l \xrightarrow{K_3} HCl + \dot{C}Cl_3\ (\text{slow})$

$\dot{C}Cl_3 + \dot{C}l \xrightarrow{K_4} CCl_4(\text{fast})$

The experimental rate law consistent with the mehanism.

ALLEN Session 2019-20

EXERCISE # (S-2)

1. The decomposition of a compound P, at temperature T according to the equation

$$2P_{(g)} \longrightarrow 4Q_{(g)} + R_{(g)} + S_{(l)}$$

is the first order reaction. After 30 minutes from the start of decomposition in a closed vessel, the total pressure developed is found to be 317 mm Hg and after a long period of time the total pressure observed to be 617 mm Hg. Calculate the total pressure of the vessel after 75 minutes, if volume of liquid S is supposed to be negligible. Also calculate the time fraction $t_{7/8}$.

Given : Vapour pressure of S (l) at temperature T = 32.5 mm Hg.

[ln = 1.169) = 0.156, ln2 = 0.7, $e^{0.39}$ = 1.5]

2. For the reaction : A + B $\longrightarrow$ Product, rate law is : rate = k $[A]^2$ [B], where k = 5×10^{-5} $(mol/L)^{-2}$ min^{-1} . Determine the **time (in minutes)** in which concentration of 'A' becomes half of its initial concentration, if initial concentration of A and B are 0.2 M and 2×10^3 M respectively.

3. An optically active compound A upon acid catalysed hydrolysis yield two optically active compound B and C by pseudo first order kinetics. The observed rotation of the mixture after 20 min was 5° while after completion of the reaction it was – 20°. If optical rotation per mole of A, B & C are 60°, 40° & – 80°. Calculate half life of the reaction.

4. The reaction

$$cis\text{–}Cr(en)_2(OH)^+_2 \underset{k_2}{\overset{k_1}{\rightleftharpoons}} trans\text{–}Cr(en)_2(OH)^+_2$$

is first order in both directions. At 25°C, the equilibrium constant is 0.1 and the rate constant k_1 is $2 \times 10^{-4} s^{-1}$. In an experiment starting with the pure *cis* form, how long would it take for half the equilibrium amount of the *trans* isomer to be formed ? [ln2 = 0.693]

5. For the two parallel first order reactions A $\xrightarrow{k_1}$ B and A $\xrightarrow{k_2}$ C, show that the activation energy E′ for the disappearance of A is given in terms of activation energies E_1 and E_2 for the two paths by

$$E' = \frac{k_1E_1 + k_2E_2}{k_1 + k_2}$$

6. At room temperature (27°C), orange juice gets spoilt in about 64 hours. In a refrigerator at 7°C, juice can be stored three times as long before it gets spoilt. Estimate

(a) the activation energy of the reaction that causes the spoiling of juice.

(b) How long should it take for juice to get spoilt at 47°C?

[$e^{0.9625}$ = 2.5 , ln3 = 1.1 , R = 2 cal/mol-K]

7. Two reactions (i) A $\rightarrow$ products (ii) B $\rightarrow$ products, follow first order kinetics. The rate of the reaction (i) is doubled when the temperature is raised from 300 K to 310K. The half life for this reaction at 310K is 30 minutes. At 310K temperature and same concentration of reactant in both reaciton B, decomposes twice as fast as A. If the energy of activation for the reaction (ii) is half that of reaction (i), calculate the rate constant of the reaction (ii) at 300K.

ALLEN Session 2019-20

8. The reaction of formation of phosgene from CO and Cl_2 is $CO + Cl_2 \longrightarrow COCl_2$
The proposed mechanism is

(i) $Cl_2 \underset{K-1}{\overset{K_1}{\rightleftharpoons}} 2Cl$ (fast equilibrium)

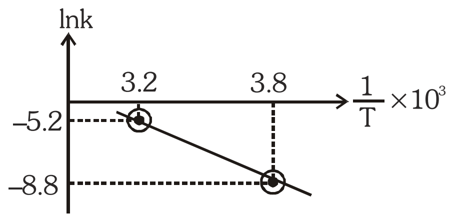
(ii) $Cl + CO \underset{K-2}{\overset{K_2}{\rightleftharpoons}} COCl$ (fast equilibrium)

(iii) $COCl + Cl_2 \xrightarrow{K_3} COCl_2 + Cl$ (slow)

Show that the above mechanism leads to the following rate law $\frac{d[COCl_2]}{dt} = K[CO][Cl_2]^{3/2}$,

where $K = k_3 . \frac{k_2}{k_{-2}}\left(\frac{k_1}{k_{-1}}\right)^{1/2}$.

9. For any elementary reaction, following observation is made :



If the above reaction is carried out in the presence of catalyst at 300K, the reaction proceeds with same rate as without catalyst at 400K. By what amount (in kcal) catalyst has decreased E_a.

10. Milk is pasteurised if it is heated at 67ºC for 4 hours. If $\frac{E_a}{2.303}$ for the process is 23.8 kcal / mol then minimum how much seconds will be required for the the process at 77ºC ? [R = 2cal /K-mol]

11. When CrI_3 reacts with H_2O_2 in presence of NaOH forming Na_2CrO_4, $NaIO_4$ & H_2O then find how many times will be rate of disappearance of NaOH compared to CrI_3.

12. Conversion of A into B and C follows first order kinetics. A, B and C all are optically active, initially A is present in 500 ml solution and 100 minutes after the start of reaction, the resulting solution becomes optically inactive. If angle of rotation of A is +50 degree per molar, B is –20 degree per molar and C is +10 degree per molar, then find the time for completion of $\left(\frac{875}{9}\right)\%$ of the chemical reaction in minutes.

A(aq.) → B(aq.) + C(aq.)

13. A gas phase reaction is,
$A(g) + 2B(g) \rightarrow C(g)$
The reaction was carried out with stoichiometric proportions of A and B and the following data is obtained

Half life (min)	10	160
Initial pressure of A(atm)	40 atm	10 atm

What is overall order of reaction

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Chemical Kinetics\Eng\02_Exercise

14. For the reaction

$A \longrightarrow P$

If ratio of $t_{7/8} : t_{1/4}$ is 7 : 2 what is order of reaction.
$t_{7/8}$ = time in which 7/8 part of reactant is reacted
$t_{1/4}$ = time in which 1/4 part of reactant is reacted

15. A certain reactant A^{n+} is getting converted to $A^{(n+4)+}$ in solution. The rate constant of this reaction is measured by titrating a volume of the solution with a reducing reagent which only reacts with A^{n+} and $A^{(n+4)+}$. In this process, it converts A^{n+} to $A^{(n-2)+}$ and $A^{(n+4)+}$ to $A^{(n-1)+}$.

Time	0	10 min
Volume of reagent consumed	30 ml	45 ml

Calculate the rate constant (in min^{-1}) of the conversion of A^{n+} to $A^{(n+4)+}$ assuming it to be first order reaction.

[n-factor of reagent remain same when it reacts with A^{n+} & $A^{(n+4)+}$]

[Given: *l*n 3/2 = 0.4]

16. Consider two Ist order reactions at 25ºC with same initial concentrations of 1M

I. $A \xrightarrow{t_{1/2} = 40\,\text{min}}$ product (temperature coefficient = 2)

II. $B \xrightarrow{t_{1/2} = 30\,\text{min}}$ product (temperature coefficient = 3)

If both reactions are carried out at 35ºC. Find ratio of concentrations of A & B after one hour.

17. Consider two reactions at 27ºC

(I). $A \longrightarrow B$; (First order kinetics)

(II). $X \longrightarrow Y$; (Zero order kinetics)

If both A & X each having initial concentration 0.1M & same half life period. Then find simplest ratio of rate constants for I & II reactions. [ln2 = 0.7]

18. The rate of first-order reaction is 0.04 mol $litre^{-1}s^{-1}$ at 10 minutes and 0.03 mol $litre^{-1}s^{-1}$ at 22 minutes after initiation. The half life of the reaction (in seconds) is (ln2 = 0.7, ln3 = 1 .1).

19. In order to determine the order of reaction : $A(g) \rightarrow 2B(g)$, vapour density of the system is determined at different stages of reaction, at constant temperature. The reaction is started with pure A(g). From the following data, the order of reaction is -

Time (min)	0	10	20
Vapour density	42	35	30

20. In the reaction : $A \rightarrow P$, the rate is doubled when the concentration of 'A' is quadrupled. If 50% of the reaction occurs in $8\sqrt{2}$ hr, how long (in hours) would it take for the completion of next 50% reaction.

ALLEN Session 2019-20

EXERCISE # (O-1)

SINGLE CORRECT

RATE OF REACTION

1. The rate of a reaction is expressed in different ways as follows :

$$+\frac{1}{2}\frac{d[C]}{dt}=-\frac{1}{3}\frac{d[D]}{dt}=+\frac{1}{4}\frac{d[A]}{dt}=-\frac{d[B]}{dt}$$

The reaction is:

(A) $4A + B \longrightarrow 2C + 3D$
(B) $B + 3D \longrightarrow 4A + 2C$
(C) $A + B \longrightarrow C + D$
(D) $B + D \longrightarrow A + C$

2. For the reaction $2A + 3B \rightarrow 4C$
the rate of reaction may be represented as :–

(A) $r = -2\frac{d(A)}{dt} = -3\frac{d(B)}{dt} = 4\frac{d(C)}{dt}$
(B) $r = -2\frac{d(A)}{dt} = -4\frac{d(B)}{dt} = 3\frac{d(C)}{dt}$
(C) $r = -\frac{1}{2}\frac{d(A)}{dt} = \frac{1}{3}\frac{d(B)}{dt} = \frac{1}{4}\frac{d(C)}{dt}$
(D) $r = -\frac{1}{2}\frac{d(A)}{dt} = -\frac{1}{3}\frac{d(B)}{dt} = \frac{1}{4}\frac{d(C)}{dt}$

3. In a reaction $N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)$, the rate of appearance of NH_3 is 2.5×10^{-4} $molL^{-1}sec^{-1}$. The rate of reaction & rate of disappearance of H_2 will be (In $molL^{-1}sec^{-1}$)

(A) 3.75×10^{-4}, 1.25×10^{-4}
(B) 1.25×10^{-4}, 2.5×10^{-4}
(C) 1.25×10^{-4}, 3.75×10^{-4}
(D) 5.0×10^{-4}, 3.75×10^{-4}

4. For the reaction $4A + B \rightarrow 2C + 2D$, the incorrect statement is :–

(A) The rate of disappearance of B is one fourth the rate of disappearance of A
(B) The rate of appearance of C is half the rate of disappearance of B
(C) The rate of formation of D is half the rate of consumption of A
(D) The rates of formation of C and D are equal

5. Which of the following rate law has an overall order of 0.5 for reaction involving substances x,y and z ?

(A) Rate = k $[C_x]\,[C_y]\,[C_z]$
(B) Rate = k $[C_x]^{0.5}[C_y]^{0.5}[C_z]^{0.5}$
(C) Rate = k $[C_x]^{1.5}\,[C_y]^{-1}[C_z]^{\circ}$
(D) Rate = k $[C_x][C_z]^{\circ} / [C_y]^2$

6. For a reaction $A + B \rightarrow$ products, the rate of the reaction was doubled when the concentration of A was doubled, the rate was again doubled when the conc. of A & B were doubled. The order of the reaction with respect to A & B are:–

(A) 1, 1 (B) 2, 0 (C) 1, 0 (D) 0, 1

7. If the rate of the reaction is equal to the rate constant, the order of the reaction is:-

(A) 0 (B) 1 (C) 2 (D) 3

8. If the first order reaction involves gaseous reactants and gaseous products the units of its rate may be –

(A) atm. (B) atm - sec. (C) atm – sec^{-1} (D) $atm^2\ sec^2$

9. If concentration of reactants is increased by 'x', then the rate constant (k) becomes :-

(A) $\ln\frac{k}{x}$ (B) $\frac{k}{x}$ (C) k + x (D) k

ALLEN Session 2019-20

10. The rate constant of n^{th} order reaction has units

(A) $litre^{1-n}\ mol^{1-n}\ sec^{-1}$ (B) $mol^{n-1}\ litre^{1-n}\ sec^{-1}$
(C) $mol^{1-n}\ litre^{n-1}\ sec^{-1}$ (D) None

11. A reaction is found to have the rate constant x sec^{-1}. By what factor the rate is increased if initial conc. of A is tripled ?

(A) 3 (B) 9 (C) x (D) Remains same

12. Consider following two reactions

$$A \longrightarrow \text{Product} \; ; -\frac{d[A]}{dt} = k_1\,[A]^0$$

$$B \longrightarrow \text{Product} \; ; -\frac{d[B]}{dt} = k_2\,[B]$$

k_1 and k_2 are expressed in terms of molarity (mol L^{-1}) and time (sec) as

(A) sec^{-1}, M sec^{-1} (B) M sec^{-1}, M sec^{-1} (C) sec^{-1}, M^{-1} sec^{-1} (D) M sec^{-1}, sec^{-1}

13. $A(g) \longrightarrow B(g) + 3C(g)$

In a closed container at a given temperature, pressure increases from 100 mm Hg to 160 mm Hg in 10 sec. for reaction.

Then the average rate of reaction in first 10 sec. will be -

(A) 2 mm/sec. (B) 4 mm/sec. (C) 6 mm/sec. (D) 3 mm/sec.

ZERO ORDER REACTION

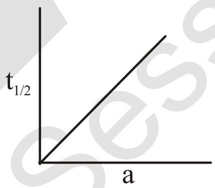
14. H_2 gas is adsorbed on the metal surface like tungsten. This follows .... order reaction -

(A) Third (B) Second (C) Zero (D) First

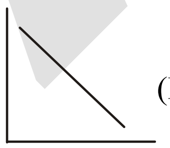
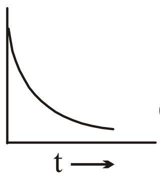
15. The rate constant of a zero order reaction is 0.2 mol $dm^{-3}h^{-1}$. If the concentration of the reactant after 30 minutes is 0.05 mol dm^{-3}. Then its initial concentration would be :

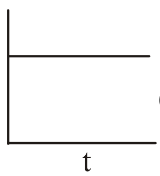
(A) 0.15 mol dm^{-3} (B) 1.05 mol dm^{-3} (C) 0.25 mol dm^{-3} (D) 4.00 mol dm^{-3}

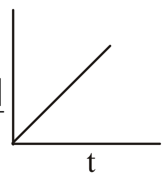
16. Consider the reaction $A \longrightarrow B$, graph between half life ($t_{1/2}$) and initial concentration (a) of the reactant is



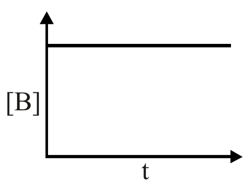
Hence graph between $-\frac{d[A]}{dt}$ and time will be

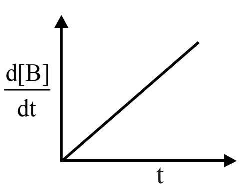
(A) $-\frac{d[A]}{dt}$  (B) $-\frac{d[A]}{dt}$ 

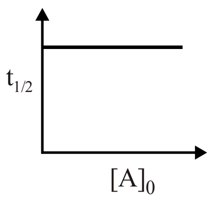
(C) $-\frac{d[A]}{dt}$ 

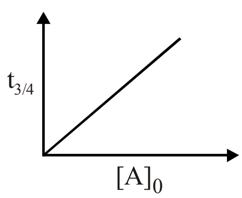
(D) $-\frac{d[A]}{dt}$ 

17. Which graph represents zero order reaction $[A(g) \rightarrow B(g)]$:

(A) 

(B) 

(C) 

(D) 

ALLEN Session 2019-20

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18. K for a zero order reaction is 2×10^{-2} $molL^{-1}$ sec^{-1}. If the concentration of the reactant after 25 sec is 0.5 M, then concentration after 50 sec. must have been.

(A) 0.5 M (B) 0.25 M (C) 0.125 M (D) 0.0 M

FIRST ORDER REACTION

19. The rate constant of a first order reaction is 4×10^{-3} sec^{-1}. At a reactant concentration of 0.02 M, the rate of reaction would be –

(A) 8×10^{-5} M sec^{-1} (B) 4×10^{-3} M sec^{-1}

(C) 2×10^{-1} M sec^{-1} (D) 4×10^{-1} M sec^{-1}

20. In a first order reaction the concentration of the reactant is decreased from 1.0 M to 0.25 M in 20 min. The rate constant of the reaction would be –

(A) 10 min^{-1} (B) 6.931 min^{-1} (C) 0.6931 min^{-1} (D) 0.06931 min^{-1}

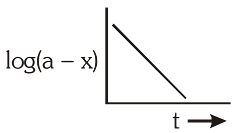
21. A first order reaction has a half life period of 69.3 sec. At 0.10 mol lit^{-1} reactant concentration rate will be –

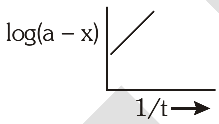
(A) 10^{-4} M sec^{-1} (B) 10^{-3} M sec^{-1} (C) 10^{-1} M sec^{-1} (D) 6.93×10^{-1}Msec^{-1}

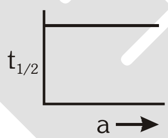
22. What fraction of a reactant (in first order reaction) is left after 40 minute if $t_{½}$ is 20 minute

(A) 1/4 (B) 1/2 (C) 1/8 (D) 1/6

23. Which of the following curve represents a I[st] order reaction :–

(A) 

(B) 

(C) 

(D) 1 & 3 both

24. After how many seconds will the conc. of the reactant in a first order reaction be halved, if the rate constant is 1.155×10^{-3} sec^{-1} :-

(A) 600 (B) 100 (C) 60 (D) 10

25. Correct statement about first order reaction is:-

(A) $t_{completion}$ = finite (B) $t_{1/2} \propto \frac{1}{a}$

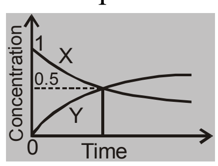
(C) Unit of k is mole lit^{-1} sec^{-1} (D) $t_{1/2} \times k$ = const.

26. For a given reaction of first order, it takes 20 minute for the concentration to drop from 1 M to 0.6 M. The time required for the concentration to drop from 0.6 M to 0.36 M will be :

(A) More than 20 min (B) Less than 20 min

(C) Equal to 20 min (D) Infinity

27. The accompanying figure depicts the change in concentration of species X and Y for the reaction X $\rightarrow$ Y as a function of time. The point of intersection of the two curves represents.



(A) $t_{1/2}$

(B) $t_{3/4}$

(C) $t_{2/3}$

(D) Data insufficient to predict

28. A reaction is of first order. After 100 minutes, 75 g of the reactant A are decomposed when 100g are taken initially. Calculate the time required when 150g of the reactant A are decomposed, the initial weight taken is 200g.

(A) 100 minutes (B) 200 minutes (C) 150 minutes (D) 175 minutes

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29. Consider the reaction :

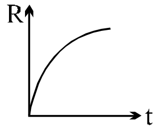
$A \longrightarrow B + C$

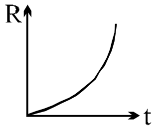
Initial concentration of A is 1 M. 20 minutes time is required for completion of 80 % reaction.

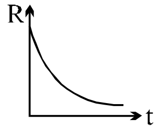
If $\frac{d[B]}{dt} = k[A]$, then half life $(t_{1/2})$ is (Use : ln 5 = 1.6, ln2 = 0.7)

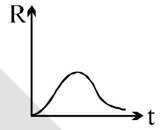
(A) 55.44 min. (B) 50 min (C) 8.75 min (D) 12.5 min

30. If decomposition reaction A (g) $\longrightarrow$ B (g) follows first order kinetics, then the graph of rate of formation (R) of B against time t will be

(A) 

(B) 

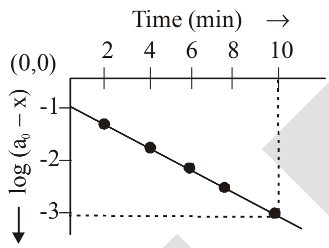
(C) 

(D) 

31. For the first order decomposition of $SO_2Cl_2(g)$,

$SO_2Cl_2(g) \rightarrow SO_2(g) + Cl_2(g)$

a graph of log $(a_0 - x)$ vs t is shown in figure. What is the rate constant (sec^{-1})?



(A) 0.2 (B) 4.6×10^{-1} (C) 7.7×10^{-3} (D) 1.15×10^{-2}

32. The rate constant for a second order reaction is 8×10^{-5} M^{-1} min^{-1}. How long will it take a 1M solution to be reduced to 0.5 M ?

(A) 8.665×10^{3} min (B) 8×10^{-3} min (C) 1.25×10^{4} min (D) 4×10^{-5} min

33. The rate law of the reaction : $A + 2B \rightarrow$ product(P) is given by $\frac{d[P]}{dt} = K[A]^2[B]$. If A is taken in large excess, the order of the reaction will be –

(A) Zero (B) 1 (C) 2 (D) 3

CALCULATION OF ORDER OF REACTION

34. Time required to complete a half fraction of a reaction varies inversely to the concentration of reactant then the order of reaction is –

(A) Zero (B) 1 (C) 2 (D) 3

35. The reaction $L \rightarrow M$ is started with 10 g of L. After 30 and 90 minute, 5 g and 1.25 g of L are left respectively. The order of reaction is –

(A) 0 (B) 2 (C) 1 (D) 3

36. From different sets of data of $t_{1/2}$ at different initial concentrations say 'a' for a given reaction, $[t_{1/2} \propto a]$ is found. The order of reaction is :–

(A) 0 (B) 1 (C) 2 (D) 3

37. At certain temperature, the half life period for the thermal decomposition of a gaseous substance depends on the initial partial pressure of the substance as follows

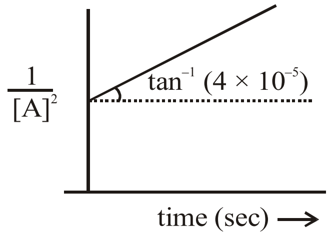
P(mmHg)	500	250
$t_{1/2}$(in min.)	235	950

Find the order of reaction [Given log (23.5) = 1.37 ; log (95) = 1.97; log 2 = 0.30]

(A) 1 (B) 2 (C) 2.5 (D) 3

ALLEN Session 2019-20
node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Chemical Kinetics\Eng\02_Exercise

38. For a reaction A $\longrightarrow$ Product



What is the value of k for the given reaction-

(A) $4 \times 10^{-5}\ M^{-1}s^{-1}$ (B) $\frac{4}{3} \times 10^{-5}\ M^{-2}s^{-1}$ (C) $2 \times 10^{-5}\ M^{-2}s^{-1}$ (D) $\frac{2}{3} \times 10^{-5}\ M^{-2}s^{-1}$

39. Azo isopropane decomposes according to the equation :–

$(CH_3)_2CHN{=}NCH(CH_3)_2(g) \xrightarrow{250-290^0 C} N_2(g) + C_6H_{14}(g)$

It is found to be a first order reaction. If initial pressure is P_o and pressure of the mixture at time t is (P_t) then rate constant K would be :–

(A) $k = \frac{2.303}{t}\log\frac{P_o}{2P_o - P_t}$ (B) $k = \frac{2.303}{t}\log\frac{P_o - P_t}{P_o}$

(C) $k = \frac{2.303}{t}\log\frac{P_o}{P_o - P_t}$ (D) $k = \frac{2.303}{t}\log\frac{2P_o}{2P_o - P_t}$

40. For a 1st order homogeneous gaseous reaction

$A \rightarrow 2B + C$,

if the pressure after time t was P_t and after long time was P_∞, then rate constant (k) in terms of P_t & P_∞ and t is -

(A) $k = \frac{2.303}{t}\log\left(\frac{P_\infty}{P_\infty - P_t}\right)$ (B) $k = \frac{2.303}{t}\log\left(\frac{2P_\infty}{P_\infty - P_t}\right)$

(C) $k = \frac{2.303}{t}\log\left(\frac{2P_\infty}{3(P_\infty - P_t)}\right)$ (D) None of these

41. The first order reaction

$A(g) \rightarrow B(g) + C(g) + D(s)$

taking place at constant pressure and temperature condition. Initially volume of the container containing only A was found to be V_0 and after time 't' it was V_t. Rate constant for the reaction is.

(A) $\frac{1}{t} ln \frac{V_0}{2V_0 - V_t}$ (B) $\frac{1}{t} ln \frac{V_0}{V_0 - V_t}$ (C) $\frac{1}{t} ln \frac{2V_0}{2V_0 - V_t}$ (D) $\frac{1}{t} ln \frac{2V_0}{V_0 + V_t}$

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ALLEN Session 2019-20

42. For the inversion of cane sugar ($C_{12}H_{22}O_{11}$) obeying I order following data were obtained

Time (min.)	0	10	∞
Angle of rotation of solution (degree)	+20	–2.5	–10

What will be rate constant in min^{-1} (**ln 2 = 0.7**)

(A) 0.7 (B) 0.14 (C) 0.21 (D) 0.07

COLLISION THEORY AND ARRHENIUS EQUATION

43. The rate constant for the forward reaction A (g) $\rightleftharpoons$ 2B(g) is $1.5 \times 10^{-3}\ s^{-1}$ at 100 K. If 10^{-5} moles of A and 100 moles of B are present in a 10 litre vessel at equilibrium then rate constant for the backward reaction at this temperature is

(A) 1.50×10^{4} L mol^{-1} s^{-1} (B) 1.5×10^{11} L mol^{-1} s^{-1}

(C) 1.5×10^{10} L mol^{-1} s^{-1} (D) 1.5×10^{-11}L mol^{-1} s^{-1}

44. According to collision theory of reaction –

(A) Every collision between reactant leads to chemical reaction

(B) Rate of reaction is proportional to velocity of molecules

(C) All reactions which occur in gaseous phase are zero order reaction

(D) Rate of reaction is directly proportional to collision frequency.

45. Activation energy of a reaction is –

(A) The energy released during the reaction

(B) The energy evolved when activated complex is formed

(C) Minimum amount of energy needed to overcome the potential barrier of reaction

(D) The energy needed to form one mole of the product

46. The minimum energy for molecules to enter into chemical reaction is called

(A) Kinetic energy (B) Potential energy (C) Threshold energy (D) Activation energy

47. For producing the effective collisions, the colliding molecules must possess:-

(A) A certain minimum amount of energy

(B) Energy equal to or greater than threshold energy

(C) Proper orientation

(D) Threshold energy as well as proper orientation of collision

48. A large increase in the rate of a reaction for a rise in temperature is due to –

(A) Increase in the number of collisions (B) Increase in the number of activated molecules

(C) Lowering of activation energy (D) Shortening of the mean free path

49. Slope of which plot can give the value of activation energy

(A) k versus T (B) $\frac{1}{k}$ versus T (C) Log k versus 1/T (D) C versus T

50. Given that K is the rate constant for any order reaction at temperature T, then the value of $\lim_{T\to\infty} \log k$ ________.

(A) $\frac{A}{2.303}$ (B) A (C) 2.303 A (D) log A

ALLEN Session 2019-20

51. For a certain gaseous reaction a 10° C rise of temperature from 25° C to 35° C doubles the rate of reaction. What is the value of activation energy :–

(A) $\frac{10}{2.303R \times 298 \times 308}$ (B) $\frac{2.303 \times 10}{298 \times 308R}$

(C) $\frac{0.693R \times 10}{290 \times 308}$ (D) $\frac{0.693R \times 298 \times 308}{10}$

52. From the following data, the activation energy (cal/mol) for the reaction $H_2 + I_2 \rightarrow 2HI$, is about

T (in k)	1/T, (in, k^{-1})	$\log_{10}k$
769	1.3×10^{-3}	2.9
667	1.5×10^{-3}	1.1

(A) 4×10^4 (B) 2×10^4 (C) 8×10^4 (D) 3×10^4

53. The rate constant, the activation energy and the Arrhenius parameter (A) of a chemical reaction at 25°C are 3.0×10^{-4} s^{-1}, 104.4 kJ mol^{-1} and $6.0 \times 10^{14} s^{-1}$ respectively. The value of the rate constant at $T \rightarrow \infty$ is

(A) 2.0×10^{18} s^{-1} (B) 6.0×10^{14} s^{-1} (C) infinity (D) 3.6×10^{30} s^{-1}

54. A first order reaction is 50% completed in 20 minutes at 27°C and in 5 min at 47°C. The energy of activation of the reaction is

(A) 100 kJ/mol (B) 55.14 kJ/mol (C) 11.97 kJ/mol (D) 6.65 kJ/mol

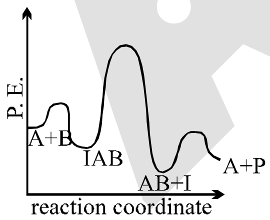
55. For the first order reaction $A \longrightarrow B + C$, carried out at 27 ºC if 3.8×10^{-16} % of the reactant molecules can overcome energy barrier, the E_a (activation energy) of the reaction is [log 3.8 = 0.58, 2.303 × 8.314 × 17.42 = 333.33]

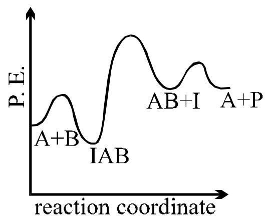
(A) 12 kJ/mole (B) 831.4 kJ/mole (C) 100 kJ/mole (D) 111.11 J/mole

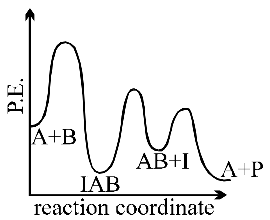
56. The following mechanism has been proposed for the exothermic catalyzed complex reaction.

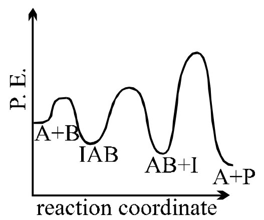
$$A + B \underset{}{\overset{fast}{\rightleftharpoons}} I\,AB \xrightarrow{k_1} AB + I \xrightarrow{k_2} P + A$$

If k_1 is much smaller than k_2. The most suitable qualitative plot of potential energy (P.E.) versus reaction coordinate for the above reaction.

(A) 

(B) 

(C) 

(D) 

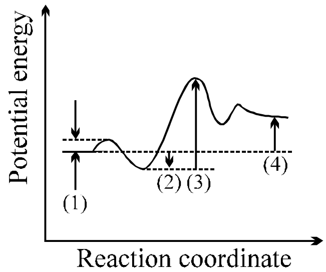
57. Choose the correct set of identifications for the reaction .

Substrate(S) $\xrightarrow{\text{Enzyme(E)}}$ Product (P)

whose mechanism is $E + S \rightleftharpoons ES$

$ES \rightleftharpoons EP$

$EP \rightleftharpoons E + P$



	(1)	(2)	(3)	(4)
(A)	ΔE for $E + S \rightarrow ES$	E_a for $ES \rightarrow EP$	$\Delta E_{overall}$ for $S \rightarrow P$	E_a for $EP \rightarrow E + P$
(B)	E_a for $E + S \rightarrow ES$	ΔE for $E + S \rightarrow ES$	E_a for $ES \rightarrow EP$	$\Delta E_{overall}$ for $S \rightarrow P$
(C)	E_a for $ES \rightarrow EP$	E_a for $EP \rightarrow E + P$	$\Delta E_{overall}$ for $S \rightarrow P$	ΔE for $EP \rightarrow E + P$
(D)	E_a for $E + S \rightarrow ES$	E_a for $ES \rightarrow EP$	E_a for $EP \rightarrow E + P$	$\Delta E_{overall}$ for $S \rightarrow P$
(E)	ΔE for $E + S \rightarrow ES$	$\Delta E_{overall}$ for $S \rightarrow P$	ΔE for $_1$ $EP \rightarrow E + P$	E_a for $EP \rightarrow E + P$

RATE LAW OF MECHANISM OF REACTION

58. Following mechanism has been proposed for a reaction : $2A + B \rightarrow D + E$

Step - I : $A + B \rightarrow C + D$ (slow) **Step - II :** $A + C \rightarrow E$ (fast)

The rate law expression for the reaction is –

(A) $r = k[A]^2 [B]$ (B) $r = k[A] [B]$ (C) $r = k[A]^2$ (D) $r = k[A][C]$

59. The reaction mechanism for the reaction $P \rightarrow R$ is as follows :

$P \underset{k_2}{\overset{k_1}{\rightleftharpoons}} 2Q$ (fast) ; $2Q + P \xrightarrow{K_3} R$ (slow)

the rate law for the main reaction ($P \rightarrow R$) is :

(A) $k_1[P] [Q]$ (B) $k_1k_2[P]$ (C) $\dfrac{k_1k_3[P]^2}{k_2}$ (D) $k_1k_2[a]$

60. The energies of activation for forward and reverse reactions for $A_2 + B_2 \rightleftharpoons 2AB$ are 180 kJ mol^{-1} and 200 kJ mol^{-1} respectively. The presence of a catalyst lowers the activation energy of both (forward and reverse) reactions by 100 kJ mol^{-1}. The enthalpy change of the reaction ($A_2 + B_2 \rightarrow 2AB$) in the presence of catalyst will be (in kJ mol^{-1}) -

(A) 300 (B) 120 (C) 280 (D) –20

ALLEN Session 2019-20
node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Chemical Kinetics\Eng\02_Exercise

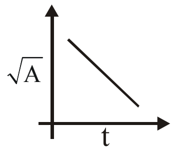
EXERCISE # (O-2)

MORE THAN ONE MAY BE CORRECT

1. For the reaction $A \rightarrow B$, the rate law expression is $-\dfrac{d[A]}{dt} = k\,[A]^{1/2}$. If initial concentration of [A] is $[A]_0$, then

(A) The integrated rate expression is $k = \dfrac{2}{t}(A_0^{1/2} - A^{1/2})$

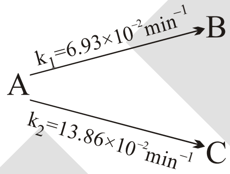
(B) The graph of $\sqrt{A}$ vs t will be



(C) The half life period, $t_{1/2} = \dfrac{K}{2[A]_0^{1/2}}$

(D) The time taken for 75% completion of reaction $t_{3/4} = \dfrac{\sqrt{[A]_0}}{k}$

2. Consider the reaction,



A, B and C all are optically active compound . If optical rotation per unit concentration of A, B and C are 60°, –72°, 42° and initial concentration of A is 2 M then select correct statement(s).

(A) Solution will be optically active and dextrorotatory after very long time

(B) Solution will be optically active and levorotatory after very long time

(C) Half life of reaction is 15 min

(D) After 75% conversion of A into B and C angle of rotation of solution will be 36°.

3. Select **incorrect** statement(s):

(A) Unit of pre-exponential factor (A) for second order reaction is mol L^{-1} s^{-1}.

(B) A zero order reaction must be a complex reaction.

(C) Molecularity is defined only for RDS in a complex reaction.

(D) Rate constant (k) remain unaffected on changing temperature.

4. Which of the following is/are **correct** statement?

(A) Stoichiometry of a reaction tells about the order of the elementary reactions.

(B) For a zero order reaction, rate and the rate constant are identical.

(C) A zero order reaction is controlled by factors other than concentration of reactants.

(D) A zero order reaction is always elementary reaction.

ALLEN Session 2019-20
node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Chemical Kinetics\Eng\02_Exercise

5. For the gas phase reaction : $R-H + X_2 \rightarrow R-X + HX$, following mechanism has been proposed

(i) $X_2 \underset{k_2}{\overset{k_1}{\rightleftharpoons}} 2X^\bullet$ (ii) $X^\bullet + R-H \xrightarrow{k_3} R^\bullet + H-X$ (slowest)

(iii) $R^\bullet + X_2 \xrightarrow{k_4} R-X + X^\bullet$

Based on this, select the correct option(s) :-

(A) Effective rate constant for the formation of RX is $k_3k_4\sqrt{\dfrac{k_1}{k_2}}$ (B) $\dfrac{d[RX]}{dt} \propto [X_2]$

(C) Overall order of the reaction is 3/2 (D) $\dfrac{d[RX]}{dt} \propto [RH]^1$

6. For a first order reaction : $A(g) \rightarrow 2B(g)$

Time(in second)	0	20	40	∞
Total pressure of system (in mm.of Hg)	64	112	124	128

(A) Half life of reaction is 10 sec

(B) Value of rate constant for reaction is $6.93 \times 10^{-3} sec^{-1}$

(C) Total pressure at t = 50 sec will be 126 mm of Hg

(D) Reaction must be a complex reaction

7. Which of the following is **INCORRECT** for first order reaction ?

(A) On introducing catalyst, both rate constant and rate of reaction increases.

(B) On increasing temperature both rate constant & rate of reaction increases.

(C) On decreasing volume both rate constant & rate of gaseous reaction increases.

(D) On increasing concentration of gaseous reactant at constant volume & constant temperature both total pressure and rate of the reaction increases.

8. $2X(g) + Y(g) + 3Z(g) \rightarrow$ Products. The rate equation of above reaction is given by :

Rate = $K[X]^1[Y]^0[Z]^2$. Choose the correct statements

(A) If [z] >> [x] and 75% of X undergoes reaction in 20 sec, then 50% of X will react in 10 sec. (B) Rate of reaction decreases by reducing the concentration of Y to half of the original value

(C) The half life of Z increases by increasing its concentration if [x] >> [z]

(D) On incresing the concentration of X,Y & Z double, rate of reaction becomes 8 times

9. Select the correct statement -

(A) In a mixture of $KMnO_4$ & $H_2C_2O_4$, $KMnO_4$ decolorises faster at higher temperature than lower temperature

(B) A catalyst participate in a chemical reaction by forming temporary bonds with the reactant resulting in an intermediate complex

(C) In collision theory only activation energy determine the criteria for effective collision

(D) Collision theory assumes molecules to be soft spheres & consider their structural aspects.

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Chemical Kinetics\Eng\02_Exercise

10. For the reaction $A + 3B \rightarrow C$, select the correct statement (s)-

(A) $\frac{d[C]}{dt} = \frac{-d[A]}{dt}$ (B) $\frac{3d[C]}{dt} = -\frac{d[B]}{dt}$

(C) Rate law must be $r = k[A][B]^3$ (D) Units for rate of reaction are independent of order of reaction

11. Choose the incorrect statement(s) -

(A) Activation energy of reaction always decreases on decreasing temperature

(B) Order of reaction may change with change in temperature

(C) When slowest step is the first step in a mechanism, then the rate law of overall reaction is the same as the rate law for this step

(D) Rate of photochemical reaction is directly proportional to intensity of absorbed photons.

12. The forward rate constant of a reaction increases by 7% when its temperature is raised from 300K to 301K while its equilibrium constant increases by 3%. Which of the following is/are correct -

(A) Activation energy of forward reaction $(Ea)_F$ is $= 300 \times 301 \times R \times \ln(1.03)$

(B) Standard heat of reaction ($\Delta H°$) is $= 300 \times 301 \times R \times \ln(1.07)$

(C) Activation energy of backward reaction $(Ea)_B$ is $= 300 \times 301 \times R \times \ln\left(\frac{1.07}{1.03}\right)$

(D) Activation energy of backward reaction $(Ea)_B$ is $= 300 \times 301 \times R \times \ln\left(\frac{1.03}{1.07}\right)$

13. Which statement(s) is/are false ?

(A) Every reaction has defined overall order

(B) All rate constant have same dimension.

(C) Every chemical species that appears in the rate law of reaction must be reactant or product in that reaction.

(D) Rate constant is never negative.

ASSERTION & REASONING TYPE QUESTIONS

14. **Statement-1** : A fractional order reaction must be a complex reaction.

Statement-2 : Fractional order of RDS equals to overall order of a complex reaction.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

15. **Statement-1** : The time of completion of reactions of type $A \rightarrow$ product (order <1) may be determined.

Statement-2 : Reactions with order ≥ 1 are either too slow or too fast and hence the time of completion can not be determined.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

16. **Statement-1** : In a reversible endothermic reaction, (E_{act}) of forward reaction is higher than that of backward reaction

Statement-2 : The threshold energy of forward reaction is more than that of backward reaction

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

17. **Statement-1** : A catalyst provides an alternative path to the reaction in which conversion of reactants into products takes place quickly

Statement-2 : The catalyst forms an activated complex of lower potential energy, with the reactants by which more number of molecules are able to cross the barrier per unit of time.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

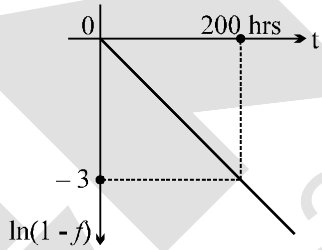
(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

COMPREHENSION

Paragraph for Question Nos. 18 & 19

When metal surface is exposed to air if gets oxidised by first order kinetics if $(1-f)$ is fraction of unoxidised. metal surface thickness and T is total thickness after very long time then it's rate can be represented as.



Rate law : $\frac{df}{dt} = k(1-f)$**, where** $f = x/T$**,**

x = thickness of oxide film at time 't'

& T = thickness of oxide film at t = ∞

A graph of $\ln(1-f)$ vs t is shown in the adjacent figure.

18. The time taken for thickness to grow 50% of 'T' is

(A) 23.1 hrs (B) 46.2 hrs (C) 100 hrs (D) 92.4 hrs

19. The exponential variation of 'f' with t(hrs) is given by

(A) $1-e^{-3t/200}$ (B) $e^{-3t/200}-1$ (C) $e^{-3t/200}$ (D) $e^{3t/200}$

Paragraph for Question Nos. 20 & 21

For a hypothetical elementary reaction : $A \xrightarrow{k_1} 2B$, $A \xrightarrow{k_2} 2C$ where $\frac{k_1}{k_2} = \frac{1}{2}$

Initially only 2 moles of A are present.

20. The total number of moles of A, B & C at the end of 50% reaction are

(A) 2 (B) 3 (C) 4 (D) 5

21. Number of moles of B are

(A) 2 (B) 1.333 (C) 0.667 (D) 0.333

Paragraph for Question Nos. 22 & 23

A reaction is said to be first order if it's rate is proportional to the concentration of reactant. Let us consider a reaction

	A(g)	⟶	B(g)	+ C(g)
At t = 0	a		0	0
At time t	a – x		x	x

The rate of reaction is given by the expression $\frac{dx}{dt} = k(a-x)$ and integrated rate equation for a given reaction is represented as $k = \frac{1}{t} \ln\left(\frac{a}{a-x}\right)$ where a = initial concentration and (a – x) = concentration of A after time t.

ALLEN Session 2019-20
node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Chemical Kinetics\Eng\02_Exercise

22. Thermal decomposition of compound X is a first order reaction. If 75% of X is decomposed in 100 min. How long will it take for 90% of the compound to decompose?

Given : log 2 = 0.30

(A) 190 min (B) 176.66 min (C) 166.66 min (D) 156.66 min

23. Consider a reaction $A(g) \longrightarrow 3B(g) + 2C(g)$ with rate constant $1.386 \times 10^{-2}\ min^{-1}$. Starting with 2 moles of A in 12.5 litre a closed vessel initially, if reaction is allowed to takes place at constant pressure & at 298K then find the concentration of B after 100 min.

(A) 0.04 M (B) 0.36 M (C) 0.09 M (D) None of these

Paragraph for Question Nos. 24 & 25

For the given sequential reaction

$$A \xrightarrow{k_1} B \xrightarrow{k_2} C$$

the concentration of A, B & C at any time 't' is given by

$$[A]_t = [A]_0 e^{-k_1 t}\ ;$$

$$[B]_t = \frac{k_1[A]_0}{(k_2 - k_1)}\left[e^{-k_1 t} - e^{-k_2 t}\right]$$

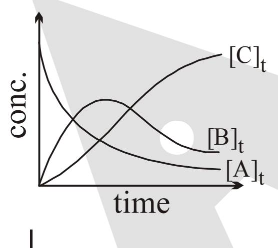
$$[C]_t = [A_0] - ([A]_t + [B]_t)$$

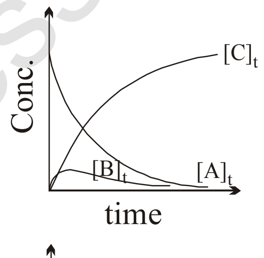
24. The time at which concentration of B is maximum is

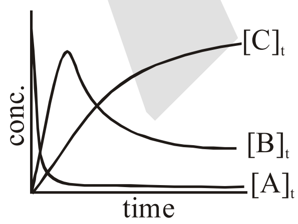
(A) $\frac{k_1}{k_2 - k_1}$ (B) $\frac{1}{k_2 - k_1} \ln \frac{k_1}{k_2}$

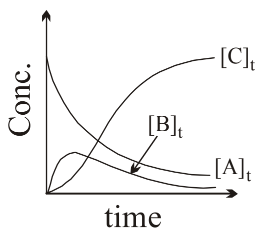
(C) $\frac{1}{k_1 - k_2} \ln \frac{k_1}{k_2}$ (D) $\frac{k_2}{k_2 - k_1}$

25. Select the correct graph if $k_1 = 1000\ s^{-1}$ and $k_2 = 20\ s^{-1}$.

(A) 

(B) 

(C) 

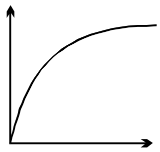
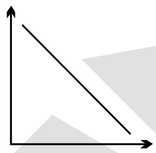
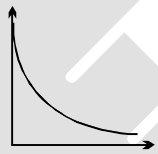
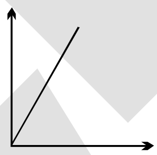
(D) 

Session 2019-20
node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Chemical Kinetics\Eng\02_Exercise

MATCH THE COLUMN

26. For the reaction of type $A(g) \longrightarrow 2B(g)$

Column-I contains four entries and **column-II** contains four entries. Entry of column-I are to be matched with **ONLY ONE ENTRY** of column-II

Column I	**Column II**
(A) $\frac{d[B]}{dt}$ vs $\frac{-d[A]}{dt}$ for first order	(P) 
(B) [A] vs t for first order	(Q) 
(C) [B] vs t for first order	(R) 
(D) [A] vs t for zero order	(S) 

27. **Match the column :**

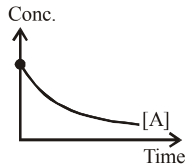
	Column-I		**Column-II**
(A)	Inversion of cane sugar in excess water.	(P)	not 100% complete
(B)	saponification reaction with 1M NaOH	(Q)	pseudo-first order
(C)	decomposition of HI on gold	(R)	zero order
(D)	radioactive decay	(S)	second order

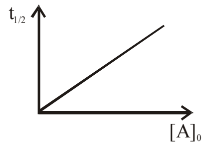
ALLEN Session 2019-20

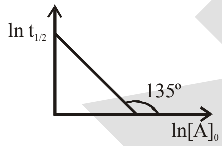
node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Chemical Kinetics\Eng\02_Exercise

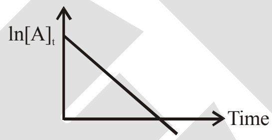
28. For the reaction A + B $\rightarrow$ product, Given : $[A]_0 = [B]_0$

List-I (Observed Rate Law) is - **List-II (Graph)**

(P) $r = k[A]$ (1)


(Q) $r = k[A]^{1/2}[B]^{1/2}$ (2)


(R) $r = k[A]\,[B]$ (3)


(S) $r = k[A]^0\,[B]^0$ (4)


Code:

	P	Q	R	S
(A)	4	1	3	2
(B)	2	3	1	4
(C)	1	2	3	4
(D)	4	3	2	1

❑ Match the Columns for Reaction A $\rightarrow$ P

Column - I	Column - II	Column - III
(I) First Order	(i) Reaction complete in finite time	(P) Rate depends on concentration
(II) Second Order	(ii) Reaction complete in infinite time	(Q) After equal interval of time concentration of reactant left are in G.P.
(III) Third Order	(iii) Half life is independent of concentration of reactant	(R) After equal interval of time concentration of reactant left are in A.P.
(IV) Zero Order	(iv) Half life decreases when concentration of reactant increases	(S) Half life depends on temperature

29. Select only incorrect option :-

(A) (II), (ii), (P) (B) (IV), (iii), (R)

(C) (III), (ii), (P) (D) (I), (ii), (Q)

30. Select only incorrect option :-

(A) (IV), (i), (S) (B) (III), (iv), (P)

(C) (I), (iv), (Q) (D) (II), (iv), (S)

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Chemical Kinetics\Eng\02_Exercise

EXERCISE # (J-MAIN)

1. The rate of a chemical reaction doubles for every 10°C rise of temperature. If the temperature is raised by 50°C, the rate of the reaction increases by about :- **[AIEEE-2011]**

(1) 32 times (2) 64 times (3) 10 times (4) 24 times

2. A reactant (1) forms two products : **[AIEEE-2011]**

$A \xrightarrow{k_1} B$, Activation Energy Ea_1

$A \xrightarrow{k_2} C$, Activation Energy Ea_2

If $Ea_2 = 2\ Ea_1$, then k_1 and k_2 are related as :-

(1) $k_1 = 2k_2e^{Ea_2/RT}$ (2) $k_1 = k_2e^{Ea_1/RT}$ (3) $k_2 = k_1e^{Ea_2/RT}$ (4) $k_1 = [A]\,k_2e^{Ea_1/RT}$

3. For the non-stoichiometric reaction $2A + B \rightarrow C + D$, the following kinetic data were obtained in three separate experiments, all at 298 K. **[J-MAIN 2014]**

Initial Concentration (A)	Initial Concentration (B)	Initial rate of formation of C (mol $L^{-}S^{-}$)
0.1M	0.1M	1.2×10^{-3}
0.1M	0.2M	1.2×10^{-3}
0.2M	0.1M	2.4×10^{-3}

(1) $\frac{dc}{dt}=k[A][B]^2$ (2) $\frac{dc}{dt}=k[A]$ (3) $\frac{dc}{dt}=k[A][B]$ (4) $\frac{dc}{dt}=k[A]^2[B]$

4. Higher order (>3) reactions are rare due to :- **[JEE-MAIN-(Offline)2015]**

(1) shifting of equilibrium towards reactants due to elastic collision

(2) loss of active species on collision

(3) low probability of simultaneous collision of all the reacting species

(4) increase in entropy and activation energy as more molecules are involved.

5. The reaction : $2N_2O_5(g) \rightarrow 4NO_2(g) + O_2(g)$, follows first order kinetics. The pressure of a vessel containing only N_2O_5 was found to increase from 50 mm Hg to 87.5 mm Hg in 30 min. The pressure exerted by the gases after 60 min. will be (Assume temperature remains constant)

(1) 106.25 nm Hg (2) 116.25 nm Hg **[JEE-MAIN-(Online)2015]**

(3) 125 mm Hg (4) 150 mm Hg

6. For the equilibrium, $A(g) \rightleftharpoons B(g)$, ΔH is –40 kJ/mol. If the ratio of the activation energies of the forward (E_f) and reverse (E_b) reactions is $\frac{2}{3}$ then :- **[JEE-MAIN-(Online)2015]**

(1) E_f = 60 kJ/mol; E_b = 100 kJ/mol (2) E_f = 30 kJ/mol; E_b = 70 kJ/mol
(3) E_f = 80 kJ/mol; E_b = 120 kJ/mol (4) E_f = 70kJ/mol; E_b = 30 kJ/mol

Session 2019-20

7. Decomposition of H_2O_2 follows a first order reaction. In fifty minutes the concentration of H_2O_2 decreases from 0.5 to 0.125 M in one such decomposition. When the concentration of H_2O_2 reaches 0.05 M, the rate of formation of O_2 will be :- **[JEE-MAIN-(Offline)2016]**

(1) 1.34×10^{-2} mol min^{-1} (2) 6.93×10^{-2} mol min^{-1}

(3) 6.93×10^{-4} mol min^{-1} (4) 2.66 L min^{-1} at STP

8. The reaction of ozone with oxygen atoms in the presence of chlorine atoms can occur by a two step process shown below : **[JEE-MAIN-(Online)2016]**

$O_3(g) + Cl^{\bullet}(g) \rightarrow O_2(g) + ClO^{\bullet}(g)$ ......(i)

$k_i = 5.2 \times 10^9$ L mol^{-1}s^{-1}

$ClO^{\bullet}(g) + O^{\bullet}(g) \rightarrow O_2(g) + Cl^{\bullet}(g)$

$k_{ii} = 2.6 \times 10^{10}$ L mol^{-1}s^{-1} .....(ii)

The closest rate constant for the overall reaction $O_3(g) + O^{\bullet}(g) \rightarrow 2O_2(g)$ is :

(1) 3.1×10^{10} L mol^{-1}s^{-1} (2) 2.6×10^{10} L mol^{-1}s^{-1}

(3) 5.2×10^9 L mol^{-1}s^{-1} (4) 1.4×10^{20} L mol^{-1}s^{-1}

9. Two reactions R_1 and R_2 have identical pre-exponential factors. Activation energy of R_1 exceeds that of R_2 by 10 kJ mol^{-1}. If k_1 and k_2 are rate constants for reactions R_1 and R_2 respectively at 300 K, then $\ln(k_2/k_1)$ is equal to : **[JEE-MAINS-2017]**

($R = 8.314$ J mol^{-1}K^{-1})

(1) 8 (2) 12 (3) 6 (4) 4

10. The rate of a reaction quadruples when the temperature changes from 300 to 310 K. The activation energy of this reaction is (Assume activation energy and pre-exponential factor are independent of temperature ; $\ln 2 = 0.693$, $R = 8.314$ J mol^{-1}K^{-1}) : **[MAINS-2017(online)]**

(1) 107.2 kJ mol^{-1}K^{-1} (2) 53.6 kJ mol^{-1}K^{-1} (3) 214.4 kJ mol^{-1}K^{-1} (4) 26.8 kJ mol^{-1}K^{-1}

11. The rate of a reaction A doubles on increasing the temperature from 300 to 310 K. By how much, the temperature of reaction B should be increased from 300 K so that rate doubles if activation energy of the reaction B is twice to that of reaction A : **[MAINS-2017(online)]**

(1) 2.45 K (2) 4.92 K (3) 9.84 K (4) 19.67 K

12. At 518° C, the rate of decomposition of a sample of gaseous acetaldehyde, initially at a pressure of 363 Torr, was 1.00 Torr s^{-1} when 5% had reacted and 0.5 Torr s^{-1} when 33% had reacted. The order of the reaction is : **[JEE-MAINS-2018]**

(1) 3 (2) 1 (3) 0 (4) 2

13 For a first order reaction, $A \rightarrow P$, $t_{1/2}$ (half life) is 10 days. The time required for $\frac{1}{4}^{th}$ conversion of A (in days) is :-**(ln 2 = 0.693, ln 3 = 1.1)** **[MAINS-2018(online)]**

(1) 5 (2) 4.1 (3) 3.2 (4) 2.5

14. N_2O_5 decomposes to NO_2 and O_2 and follows first order kinetics. After 50 minutes, the pressure inside the vessel increases from 50 mmHg to 87.5 mmHg. The pressure of the gaseous mixture after 100 minute at constant temperature will be: **[MAINS-2018(online)]**

(1) 116.25 mmHg (2) 175.0 mmHg (3) 106.25 mmHg (4) 136.25 mmHg

ALLEN Session 2019-20

15. The following results were obtained during kinetic studies of the reaction :
$2A + B \rightarrow$ Products

[MAINS-2019(online)]

Experment	**[A] (in mol L^{-1})**	**[B] (in mol L^{-1})**	**Initial Rate of reaction (in mol L^{-1} min^{-1})**
(I)	0.10	0.20	6.93×10^{-3}
(II)	0.10	0.25	6.93×10^{-3}
(III)	0.20	0.30	1.386×10^{-2}

The time (in minutes) required to consume half of A is :

(1) 10 (2) 5 (3) 100 (4) 1

16. For the reaction, $2A + B \rightarrow$ products, when the concentrations of A and B both wrere doubled, the rate of the reaction increased from 0.3 mol $L^{-1}s^{-1}$ to 2.4 mol L^{-1} s^{-1}. When the concentration of A alone is doubled, the rate increased from 0.3 mol $L^{-1}s^{-1}$ to0.6 mol $L^{-1}s^{-1}$**[MAINS-2019(online)]**

Which one of the following statements is correct ?

(1) Order of the reaction with respect to Bis2

(2) Order of the reaction with respect to Ais2

(3) Total order of the reaction is 4

(4) Order of the reaction with respect to B is 1

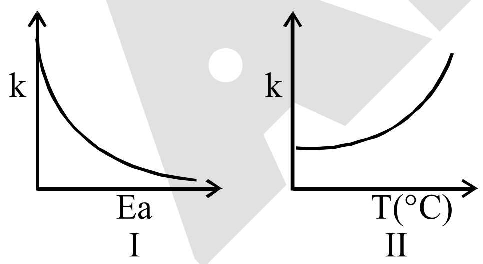
17. For an elementary chemical reaction, **[MAINS-2019(online)]**

$A_2 \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} 2A$, the expression for $\frac{d[A]}{dt}$ is :

(1) $2k_1[A_2]-k_{-1}[A]^2$ (2) $k_1[A_2]-k_{-1}[A]^2$

(3) $2k_1[A_2]-2k_{-1}[A]^2$ (4) $k_1[A_2]+k_{-1}[A]^2$

18. Consider the given plots for a reaction obeying Arrhenius equation (0°C < T < 300°C) : (k and E_a are rate constant and activation energy, respectively) **[MAINS-2019(online)]**



Choose the correct option :

(1) Both I and II are wrong

(2) I is wrong but II is right

(3) Both I and II are correct

(4) I is right but II is wrong

ALLEN Session 2019-20
node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Chemical Kinetics\Eng\02_Exercise

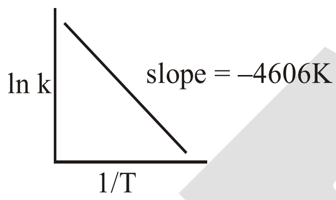
19. The reaction $2X \rightarrow B$ is a zeroth order reaction. If the initial concentration of X is 0.2 M, the half-life is 6 h. When the initial concentration of X is 0.5 M, the time required to reach its final concentration of 0.2 M will be :-

(1) 18.0 h (2) 7.2 h (3) 9.0 h (4) 12.0 h

20. If a reaction follows the Arrhenius equation, the plot lnk vs $\frac{1}{(RT)}$ gives straight line with a gradient (–y) unit. The energy required to activate the reactant is : **[MAINS-2019(online)]**

(1) y unit (2) –y unit (3) yR unit (4) y/R unit

21. For a reaction, consider the plot of ln k versus 1/T given in the figure. If the rate constant of this reaction at 400 K is $10^{-5}\ s^{-1}$, then the rate constant at 500 K is : **[MAINS-2019(online)]**



(1) $2\times10^{-4}\ s^{-1}$ (2) $10^{-4}\ s^{-1}$ (3) $10^{-6}\ s^{-1}$ (4) $4\times10^{-4}\ s^{-1}$

22. Decomposition of X exhibits a rate constant of 0.05 μg/year. How many years are required for the decomposition of 5 μg of X into 2.5 μg ? **[MAINS-2019(online)]**

(1) 50 (2) 25 (3) 20 (4) 40

ALLEN Session 2019-20

EXERCISE # (J-ADVANCED)

1. For the first order reaction, **[JEE 2011]**

$$2N_2O_5(g) \longrightarrow 4NO_2(g) + O_2(g)$$

(A) the concentration of the reactant decreases exponentially with time

(B) the half-life of the reaction decreases with increasing temperature.

(C) the half-life of the reaction depends on the initial concentration of the reactant.

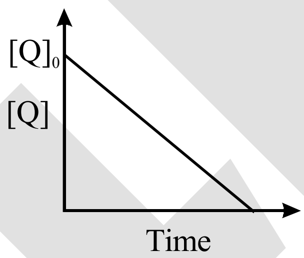
(D) the reaction proceeds to 99.6% completion in eight half-life duration.

2. An organic compound undergoes first-order decomposition . The time taken for its decomposition to 1/8 and 1/10 of its initial concentration are $t_{1/8}$ and $t_{1/10}$ respectively. What is the value of $\frac{[t_{1/8}]}{t_{1/10}} \times 10$?

(take $\log_{10}2 = 0.3$) **[JEE 2012]**

3. In the reaction : **[JEE 2013]**

$$P + Q \longrightarrow R + S$$



the time taken for 75% reaction of P is twice the time taken for 50% reaction of P. The concentration of Q varies with reaction time as shown in the figure. The overall order of the reaction is -

(A) 2 (B) 3 (C) 0 (D) 1

4. For the elementary reaction **M** $\rightarrow$ **N**, the rate of disappearance of **M** increases by a factor of 8 upon doubling the concentration of **M**. The order of the reaction with respect to **M** is

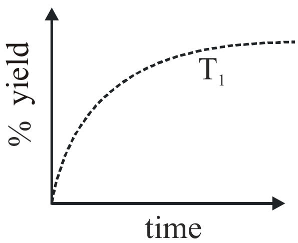
(A) 4 (B) 3 (C) 2 (D) 1 **[JEE 2014]**

5. In dilute aqueous H_2SO_4, the complex diaquodioxalatoferrate(II) is oxidized by MnO_4^-. For this reaction, the ratio of the rate of change of $[H^+]$ to the rate of change of $[MnO_4^-]$ is. **[JEE 2015]**

6. The % yield of ammonia as a function of time in the reaction **[JEE 2015]**

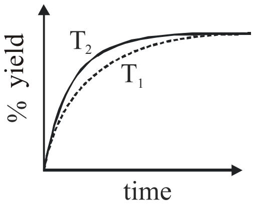
$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g), \Delta H < 0$$

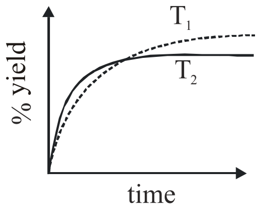
at (P , T_1) is given below -

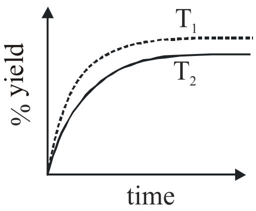


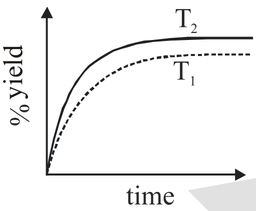
ALLEN Session 2019-20

If this reaction is conducted at (P, T_2), with $T_2 > T_1$, the % yield of ammonia as a function of time is represented by -

(A) 

(B) 

(C) 

(D) 

7. According to the Arrhenius equation, **[JEE 2016]**

(A) A high activation energy usually implies a fast reaction

(B) Rate constant increase with increase in temperature. This is due to a greater number of collisions whose energy exceeds the activation energy

(C) Higher the magnitude of activation energy, stronger is the temperature dependence of the rate constant

(D) The pre-exponential factor is a measure of the rate at which collisions occur, irrespective of their energy.

8. In a bimolecular reaction, the steric factor P was experimentally determined to be 4.5. The correct option(s) among the following is(are) : **[JEE 2017]**

(A) The value of frequency factor predicted by Arrhenius equation is higher than that determined experimentally

(B) The activation energy of the reaction is unaffected by the value of the steric factor

(C) Since P = 4.5, the reaction will not proceed unless an effective catalyst is used.

(D) Experimentally determined value of frequency factor is higher than that predicted by Arrhenius equation.

ALLEN Session 2019-20

ANSWER-KEY

EXERCISE # S-I

1. **Ans.(i)** $r = \frac{1}{4}\frac{d[NO]}{dt} = 9 \times 10^{-4}$ **mol litre^{-1} sec^{-1}, (ii) 3.6×10^{-3} mol litre^{-1}sec^{-1},**
(iii) 5.4×10^{-3} mol litre^{-1}sec^{-1}

$$\text{ROR} = \frac{\text{ROA of NO}}{\text{sto.coeff.of NO}} = \frac{(1.08 \times 10^{-2})/3}{4} = 9 \times 10^{-4}\ \text{M sec}^{-1}$$

2. **Ans. (i)** $\frac{dx}{dt} = k[A][B]^2$**, (ii) rate increases by 8 times**

3. **Ans.** 8.33×10^{-6} **Ms^{-1}, 0.012 atm min^{-1}**

4. **Ans.rate increase by 27 times**

5. **Ans.(a) 0.019 L mol^{-1} s^{-1}, (b) 0.038 L mol^{-1} s^{-1}**

6. **Ans.(8)**
7. **Ans. (2)**

8. **Ans.** 6×10^{-3} **Ms^{-1}**
9. **Ans. 1/6**

10. **Ans. (i) 7.2 M, (ii) 10 M**
11. **Ans. k = 0.01 M min^{-1}**

12. **Ans.** 6×10^{-9} **sec**
13. **Ans. 1.2 hr**

14. **Ans.(5 atm)**
15. **Ans. (i) 36 min. (ii) 108 min.**

16. **Ans. (i) 0.02min^{-1}, (ii) 70 min**
17. **Ans.** $t = 10 \times t_{1/2}$

18. **Ans.40 month**
19. **Ans. 0.02 min^{-1}**

20. **Ans.87.5 %**
21. **Ans. (19)**

22. **Ans.(20 min)**
23. Ans. (2×10^{-1}**M^{-1}s^{-1}**)

24. **Ans.3 hr**
25. **Ans.7500 second**

26. **Ans. (a) Third order, (b)** $r = k[NO]^2[H_2]$**, (c)** 7.5×10^{-10} **M sec^{-1}.**

27. **Ans. (i) first order (ii)** $k = 1.308 \times 10^{-2}$ **min^{-1} (iii) 73%**

28. **Ans. (i) Zero order, (ii) K = 5 Pa/s**
29. **Ans. Zero order**

30. **Ans. (1)**
31. **Ans.** $k = \frac{1}{t} ln \frac{P_3}{2(P_3 - P_2)}$

32. **Ans.** $k = \frac{1}{t} ln \frac{r_\infty}{(r_\infty - r_t)}$
33. **Ans.** 1.15×10^{-2} **sec^{-1}**

34. **Ans. First order**
35. **Ans. 240 min.**

36. **Ans.** $k_1 = 1.55 \times 10^{-2}$ **min^{-1}**
37. **Ans. 50 sec. , 0.1765 atm**

38. **Ans. 0.1 min^{-1}**
39. **Ans. (300 sec.)**

40. **Ans. (3)**
41 **Ans.** $\frac{[C]}{[A]} = \frac{10}{11}(e^{11x} - 1)$

42. **Ans. t = 4 min**
43. **Ans. 86.625 min**

44. **Ans.** $t_{1/2} = 36$ **min.**
45. **Ans. 10 M**

46. **Ans. 100 min.**
47. **Ans.(4 hour)**

48. **Ans. (6 : 6 : 6 : 1 : 1)**
49. **Ans. 30 sec.**

ALLEN Session 2019-20
node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Chemical Kinetics\Eng\02_Exercise

50. Ans. 50 min, $\left(\frac{0.2}{e}\right)M$
51. Ans. 5 kJ mol^{-1}
52. Ans. 13.44 kcal/mole
53. Ans. 10.28 k cal mol^{-1}
54. Ans. (3)
55. Ans. 20 minutes
56. Ans. (10)
57. Ans. (150 K)
58. Ans. (840)
59. Ans. (38.3 kJ mol^{-1})
60. Ans. (20)
61. Ans. $r = K' [NO]^2[Br_2]$
62. Ans. $r = K [NO]^2 [H_2]$, where $K = k_2 \times K_1$
63. Ans. (1)
64. Ans. (2)
65. Ans. ($\text{rate} = K_3 K_1 / K_2[CHCl_3][Cl_2]^{1/2}$)

EXERCISE S-II

1. Ans. 383.2 mm Hg , 403.8 min.
2. Ans. (50)
3. Ans. 20 min
4. Ans. 315 sec.
6. Ans. (a) 9.24 kcal/mole, (b) 25.6 hour
7. Ans. k = 0.0327 min^{-1}
9. Ans. (3)
10. Ans. (1440 sec)
11. Ans. (5)
12. Ans. (200 min)
13. Ans. (3)
14. Ans. (0)
15. Ans. (0.04)
16. Ans. (8)
17. Ans. (14)
18. Ans. (1680)
19. Ans. (0)
20. Ans. (8)

EXERCISE (O-1)

1.	Ans.(B)	2.	Ans.(D)	3.	Ans.(C)	4.	Ans.(B)
5.	Ans.(C)	6.	Ans.(C)	7.	Ans.(A)	8.	Ans.(C)
9.	Ans.(D)	10.	Ans.(C)	11.	Ans.(A)	12.	Ans(D)
13.	Ans(A)	14.	Ans.(C)	15.	Ans.(A)	16.	Ans.(C)
17.	Ans.(D)	18.	Ans.(D)	19.	Ans.(A)	20.	Ans.(D)
21.	Ans.(B)	22.	Ans.(A)	23.	Ans.(D)	24.	Ans.(A)
25.	Ans.(D)	26	Ans.(C)	27.	Ans.(A)	28.	Ans.(A)
29.	Ans.(C)	30.	Ans.(C)	31.	Ans.(C)	32.	Ans.(C)
33.	Ans.(B)	34.	Ans.(C)	35.	Ans.(C)	36.	Ans.(A)
37.	Ans.(D)	38.	Ans (C)	39.	Ans.(A)	40.	Ans.(C)
41.	Ans.(A)	42.	Ans (B)	43.	Ans.(D)	44.	Ans.(D)
45.	Ans.(C)	46.	Ans.(C)	47.	Ans.(D)	48.	Ans.(B)
49.	Ans.(C)	50.	Ans.(D)	51.	Ans.(D)	52.	Ans.(A)
53.	Ans.(B)	54.	Ans.(B)	55.	Ans.(C)	56.	Ans.(A)
57.	Ans.(B)	58.	Ans.(B)	59.	Ans. (C)	60.	Ans.(D)

Session 2019-20

EXERCISE # (O-2)

1.	Ans.(A,B,D)	2	Ans.(A,D)	3	Ans.(A,C, D)	4.	Ans.(A,B,C)
5.	Ans.(C,D)	6.	Ans.(A,C)	7.	Ans. (C)	8.	Ans.(A,D)
9.	Ans.(A, B)	10.	Ans.(A,B,D)	11.	Ans. (A)	12.	Ans.(C)
13.	Ans. (A ,B, C)	14	Ans.(C)	15.	Ans.(C)	16.	Ans.Ans.(C)
17.	Ans.(A)	18.	Ans.(B)	19.	Ans. (A)	20.	Ans.(B)
21.	Ans.(C)	22	Ans.(C)	23.	Ans.(C)	24	Ans.(C)
25.	Ans.(C)						

26. Ans. (A)→(S) ; (B)→(R) ; (C)→(P), (D)→(Q)

27. Ans. (A)→(P,Q) ; (B)→(S,P) ; (C)→(R) ; (D)→(P)

28.	Ans.(A)	29.	Ans.(B)	30.	Ans.(C)

EXERCISE # (J-MAIN)

1.	Ans.(1)	2.	Ans.(2)	3.	Ans.(2)	4.	Ans.(3)
5.	Ans.(1)	6.	Ans.(3)	7.	Ans.(3)	8.	Ans.(3)
9.	Ans.(4)	10.	Ans.(1)	11.	Ans.(2)	12.	Ans.(4)
13.	Ans.(2)	14.	Ans.(3)	15.	Ans.(2)	16.	Ans.(1)
17.	Ans.(3)	18.	Ans.(4)	19.	Ans.(1)	20.	Ans.(1)
21.	Ans.(2)	22.	Ans.(1)				

EXERCISE # (J-ADVANCED)

1.	Ans.(A,B,D)	2.	Ans.(9)	3.	Ans.(D)	4.	Ans.(B)
5.	Ans.(8)	6.	Ans.(B)	7.	Ans.(B,C,D)	8.	Ans.(B,D)

ALLEN Session 2019-20

ELECTROCHEMISTRY

1. INTRODUCTION :

Electrochemistry deals with the study of electrical properties of solutions of electrolytes and the inter-relation of chemical phenomenon and electrical energies.

❖ **Electric Conductors are of two types :**

(i) Metallic conductors (ii) Electrolytic conductors or electrolytes.

(i) Metallic conductors :

The conductors which conduct electric current by movement of free electrons without undergoing any chemical change are known as metallic conductors.

eg. Metals : Cu,Ag,Fe,Al etc., non metals : graphite and various alloys.

(ii) Electrolytic conductors :

Those substances whose aqueous solution conducts the electric current and which are decomposed by the passage of DC current are called electrolytes. In this case, conduction takes place by movement of ions.

Electrolytes also conduct electricity in fused state and undergo decomposition by passage the electric current.

❖ **Strong electrolyte :**

Electrolytes which are completely ionized in aqueous solution or in their molten state, are called ***strong electrolytes***. Their aqueous solutions are strongly conducting.

Ex : All salts, strong acids and strong bases.

❖ **Weak electrolyte :**

Electrolytes which are not completely ionized in aqueous solution are called ***Weak electrolytes.*** Their aqueous solutions are weakly conducting.

Ex: Organic acids CH_3COOH, HCN (Except : Alkyl sulphonic acids, RSO_3H)

Organic base : Amines, Aniline etc.

Note : Ostwald's dilution law is only applicable for weak electrolytes according to which degree of dissociation(α) increases on dilution.

- For weak electrolyte : $\alpha << 1 \Rightarrow$ lesser ions $\Rightarrow$ weakly conducting
- For strong electrolyte : $\alpha = 1$ (always) $\Rightarrow$ more ions $\Rightarrow$ strongly conducting

❖ **Difference between metallic and electrolytic conduction :**

	Metallic conduction	**Electrolytic conduction**
(i)	Flow of electricity takes place due to free electrons without the decomposition of the substance.	Flow of electricity takes place by ions.
(ii)	No transfer of matter takes place.	Transfer of matter takes place.
(iii)	The resistance to the flow of current increases with the increase in temperature and hence the increase in temperature decreases the conduction.	The resistance to the flow of current decreases with the increase in temperature and hence increase in temperature increases the conduction.

ALLEN Session 2019-20

2. ELECTROCHEMICAL CELL :

It is device for inter-converting chemical energy in to electrical energy or vice versa.

Electrochemical cells are of two types

Galvanic cell or Voltaic cell

- A spontaneous chemical reaction generates an electric current | EMF
- Chemical energy converted into electrical energy.

Electrolytic cell.

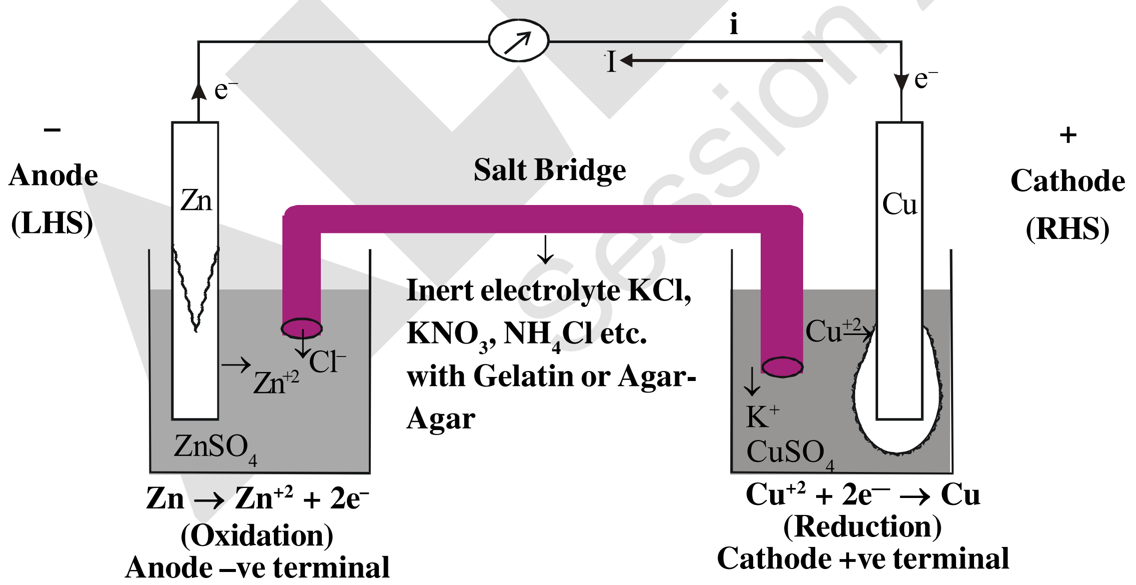
- An electric current drives a non-spontaneous reaction.
- Electrical energy converted into chemical energy.

3. GALVANIC CELL | VOLTAIC CELL

- A cell in which the chemical energy is transformed into electrical energy.
- The chemical reaction occurring in a galvanic cell is a spontaneous redox reaction.
- During the chemical process, the reduction in Gibbs free energy is converted in the form of electrical energy.

$$(DG)_{T,P} = w_{Useful|max.} = -nF\,E_{cell.}$$

Galvanic cell is made up of two half cells i.e., anodic and cathodic. The cell reaction is of redox kind. Oxidation takes place at anode and reduction at cathode. It may be represented by Daniel cell which is a type of Galvanic cell. Zinc rod immersed in $ZnSO_4$ behaves as anode and copper rod immersed in $CuSO_4$ behaves as cathode.



❖ **Oxidation takes place at anode :**

$Zn(s) \longrightarrow Zn^{2+} + 2e^-$ (Loss of electron : Oxidation)

❖ **Reduction takes place at cathode :**

$Cu^{2+} + 2e^- \longrightarrow Cu(s)$ (Gain of electron ; Reduction)

Session 2019-20

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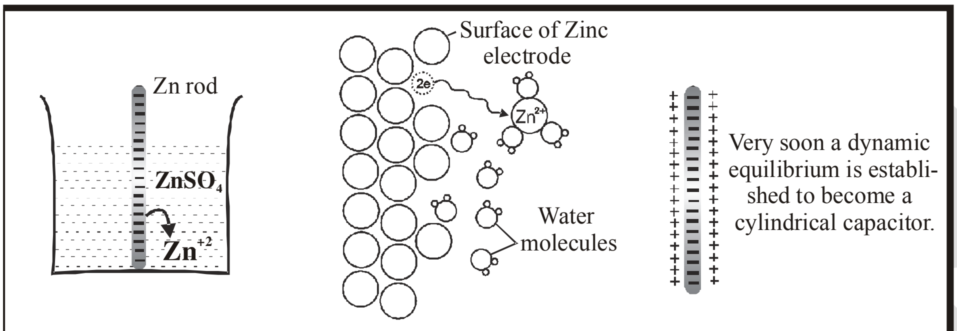
3.1 Construction of Cell :

- It has two half–cells, each having a beaker containing a metal strip that dips in its aqueous solution.
- The metal strips are called ***electrodes*** and are connected by a conducting wire.
- Two solutions are connected by a ***salt bridge.***

3.2 Construction | Working principle of Daniel cell :

I. Anode of Daniel cell : Zn rod is placed in $ZnSO_4$ solution are found to have tendency to go into the solution phase when these are placed in contact with their ions or their salt solutions.



How Zinc atoms become zinc ions at the electrode surface.

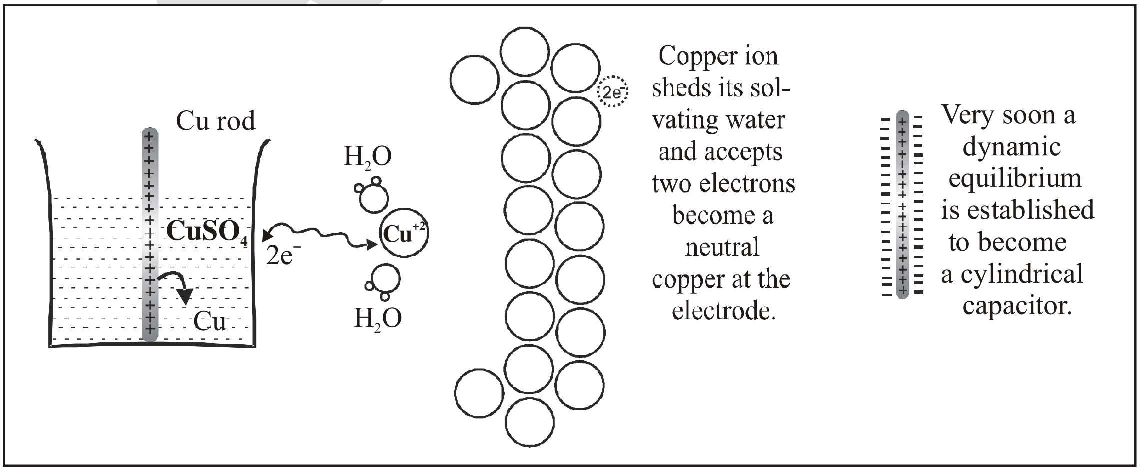
The Zn atom or metal atoms will move in the solution to form Zn^{+2}. After some time following equilibrium will be established.

$$Zn(s) \rightleftharpoons Zn^{2+} + 2e^-$$

There will be accumulation of sufficient negative charge on the rod which will not allow extra zinc ions to move in the solution. i.e. solution will be saturated with Zn^{+2} ions.
The positive charge will be more concentrated near the rod.
The extra positive charge of the solution will be more concentrated around the negatively charged rod. An electrical double layer is developed in the system and hence a potential difference is created between the rod and the solution which is known as electrode potential
This particular electrode is known as anode :

- At anode oxidation will take place. (Release of electron).
- To act as source of electrons.
- It is of negative polarity.
- The electrode oxidation potential is represented by $E_{Zn(s)/Zn^{2+}_{(aq.)}}$ & reduction potential by $E_{Zn^{2+}/Zn}$.

II. Cathode of Daniel cell :



Cu, when placed in contact with their aqueous ions, the ions (Cu^{2+}) from the solution will get deposited on the metal rod.

The following equilibrium will be established :

$$Cu^{2+}(aq) + 2e^{-} \rightleftharpoons Cu(s)$$

So rod will have deficiency of electron (positive charge). Extra negative charge will surround this positively charged rod and form double layer. An electrical double layer is developed in the system and hence a potential difference is created between the rod and the solution which is known as electrode potential. This will be known as cathode.

- At cathode reduction will take place. (Gain of e^{-} will take place)
- Cathode acts as sink of electron.
- Positive polarity will be developed.
- Their electrode reduction potential can be represented by : $E_{Cu^{2+}(aq)|Cu(s)}$.
- **Anode :** $\begin{cases}\text{Is where oxidation occurs}\\ \text{Has a negative sign}\end{cases}$
- **Cathode :** $\begin{cases}\text{Is where reduction occurs}\\ \text{Has a positive sign}\end{cases}$

❖ **Overall process :**

$$Zn(s) + Cu^{2+}(aq) \longrightarrow Cu(s) + Zn^{2+}(aq)$$

In Galvanic cell like Daniel cell ; electrons flow from anode (zinc rod) to the cathode (copper rod) through external circuit ; zinc dissolves as Zn^{2+} ; Cu^{2+} ion in the cathode cell picks up two electron and is deposited at cathode.

Note: The electrode potential will keep on decreasing with time as Zn^{2+} ions increase & Cu^{2+} ions decrease in solution therefore tendency of cell reaction decreases and cell attains equilibrium.

3.3 Functions of Salt Bridge :

- A **salt bridge** is a U-shaped inverted tube that contains a gel permeated with an inert electrolyte.
- It connects the solution of two half cell to complete the circuit.
- It minimise the liquid junction potential, the potential difference between the junction of two liquids)
- It maintains the electrical neutrality of the solution in order to give continuous flow or generation of current " The simultaneous electrical neutrality of the anodic oxidation chamber and cathodic reduction chamber is due to almost same mobility or velocity of K^{+} and NO_3^{-} ions taken into salt bridge.
- If the salt bridge is removed then voltage drops to zero & cell stops working.
- The ions of the inert electrolyte should not react with other ions in the solution and the ions are not oxidised or reduced at the electrodes.
- Generally tube is filled with a paste of agar-agar powder with neutral electrolyte generally not common to anodic|cathodic compartment with porous plugs at each mouth of tube.
- It prevents mechanical mixing of two electrolytic solutions.

ALLEN Session 2019-20

❖ **Liquid Junction Potential :**

The potential difference which arises between two solutions when in contact with each other. Salt bridge removes effects of junction potential by providing appropriate migration of ions.

❖ **Characteristics of electrolyte used in salt bridge :**

1. The electrolyte should be inert.
2. The cations and anions of the electrolyte used should be of ionic mobility.
3. Ions of electrolyte should not react with ions involved in cell reaction.

3.4 Representation of galvanic cell (IUPAC)

The anode is written on the LHS & cathode on the RHS.

we denote salt bridge by two vertical parallel lines (||), if used, in between anode & cathode.

- **Ex.** $Pt(s) \mid H_2(g) \mid H^+(aq.) \parallel H^+(aq.) \mid H_2(g) \mid Pt(s)$
- **Ex.** Daniel cell : $Zn(s) \mid Zn^{2+}(aq) \parallel Cu^{2+}(aq.) \mid Cu(s)$
- **Ex.** For cell reaction : $H_2(g) + Cu^{+2}(aq) \longrightarrow 2H^+(aq) + Cu(s)$

$$Pt \mid H_2(g) \mid H^+(aq) \parallel Cu^{2+}(aq.) \mid Cu(s).$$

3.5 Electrode potential :

When a strip of metal is brought in contact with the solution containing its own ions then the strip of metal gets positively charged or negatively charged and results into a potential being developed between the metallic strip and its solution which is known as electrode potential.

- **At anode :** $M \rightarrow M^{+n} + ne^-$ (Oxidation Potential)
- **At cathode :** $M^{+n} + ne^- \rightarrow M$ (Reduction Potential)

- The value of electrode potential depends upon :

(i) the nature of electrode

(ii) the concentration of solution

(iii) the temperature

3.6 Standard electrode potential (E°) :

If the concentration of ions is unity, temperature is 25°C (or any constant temperature) and pressure is 1 bar (standard conditions), the potential of the electrode is called ***standard electrode potential.***

- The given value of electrode potential is regarded as reduction potential unless it is specifically mentioned that it is an oxidation potential.

3.7 Electromotive force of cell or cell voltage :

The difference in the electrode potentials of the two electrodes of the cell is termed as electro motive force [EMF] or cell voltage which causes current to flow.

$$\mathbf{E_{cell} = E_{red}\,(cathode) - E_{red}\,(anode) = E_{oxi.}(anode) - E_{oxi.}(cathode) = E_{oxi.}(anode) + E_{red}(cathode)}$$

ALLEN Session 2019-20

3.8 Electro chemical series: Standard aqueous electrode potentials at 298 K

Electrode Reduction Reaction	Standard electrode Reduction potential E°, Volts
$Li^+ + e \rightarrow Li$	– 3.05
$K^+ + e^- \rightarrow K$	– 2.93
$Ba^{+2} + 2e^- \rightarrow Ba$	– 2.90
$Ca^{+2} + 2e^- \rightarrow Ca$	– 2.87
$Na^+ + e^- \rightarrow Na$	– 2.71
$Mg^{+2} + 2e^- \rightarrow Mg$	– 2.37
$Al^{+3} + 3e^- \rightarrow Al$	– 1.66
$Mn^{+2} + 2e^- \rightarrow Mn$	– 1.18
$H_2O + e^- \rightarrow \frac{1}{2} H_2 + OH$	– 0.83
$Zn^{+2} + 2e^- \rightarrow Zn$	– 0.76
$Cr^{+3} + 3e^- \rightarrow Cr$	– 0.74
$Fe^{+2} + 2e^- \rightarrow Fe$	– 0.44
$Cd^{+2} + 2e^- \rightarrow Cd$	– 0.40
$Ni^{+2} + 2e^- \rightarrow Ni$	– 0.25
$Sn^{+2} + 2e^- \rightarrow Sn$	– 0.14
$Pb^{+2} + 2e^- \rightarrow Pb$	– 0.13
$2D^+ + e^- \rightarrow D_2$	– 0.01 V
$2H^+ + 2e^- \rightarrow H_2$	0
$AgBr(s) + e^- \rightarrow Ag(s) + Br^-$	+0.09 V
$AgCl + e^- \rightarrow Ag + Cl^-$	0.22 V
$Cu^{+2} + 2e^- \rightarrow Cu$	+ 0.34
$\frac{1}{4} O_2 + \frac{1}{2} H_2O + e^- \rightarrow OH^-$	+0.401 V
$I_2 + 2e^- \rightarrow 2I^-$	+ 0.54
$Q + 2H^+ + 1e^- \rightarrow H_2Q$	+ 0.70
$Hg_2^{+2} + 2e \rightarrow 2Hg$	+ 0.79
$Ag^+ + e^- \rightarrow Ag$	+ 0.80
$Hg^{+2} + 2e^- \rightarrow Hg$	+ 0.85
$Br_2 + 2e^- \rightarrow 2Br^-$	+ 1.08
$\frac{1}{4} O_2 + H^+ + e^- \rightarrow \frac{1}{2} H_2O$	+1.23 V
$Cl_2 + 2e^- \rightarrow 2Cl^-$	+ 1.36
$Pt^{+2} + 2e^- \rightarrow Pt$	+ 1.20
$Au^{+3} + 3e \rightarrow Au$	+ 1.50
$Au^+ + e \rightarrow Au$	+ 1.69
$S_2O_8^{--} + 2e^- \rightarrow 2SO_4^{--}$	+2.0 V
$F_2 + 2e^- \rightarrow 2F^-$	+ 2.87

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• Application of electrochemical series :

(i) Activity of metal : From electrochemical series, the activity of any metal can easily be determined. All the metals which are placed above hydrogen are stronger reducing agents & can easily evolve H_2 gas whereas metals lying below hydrogen are weaker reducing agents cannot lose electrons to H^+ ions & hence can't evolve H_2 gas. For e.g. Na, K, Zn etc. can easily evolve H_2 whereas Cu, Hg, Ag etc. do not have tendency to evolve H_2 gas.

$$Na + H_2SO_4 \rightarrow Na_2SO_4 + H_2\uparrow$$

$$Na + H_2O \rightarrow NaOH + \frac{1}{2}\ H_2\uparrow$$

$$Cu + H_2SO_4 \not\longrightarrow CuSO_4 + H_2\uparrow$$

(ii) Displacement reaction : The active metal can easily displace less active metal from their aq. salt solution for e.g. Zn can replace Cu^{2+} from an aq. solution of $CuSO_4$. But Cu cannot displace Zn^{2+} from solution similarly,

$$2AgNO_3 + Cu \rightarrow Cu(NO_3)_2 + 2Ag,$$

$$CuSO_4 + Ag \not\longrightarrow Ag_2SO_4 + Cu$$

(iii) Feasibility of redox reaction : The feasibility of particular redox reaction can be easily find out from electrochemical series. The metal placed higher or have more reducing property can easily lose electrons to the metal ion present below in series, hence redox reaction become feasible i.e. cell will serve as source of electrical energy. For e.g. $NiSO_4$ solution cannot be placed in Fe vessel because, the redox reaction between them is feasible.

Note: If emf of the cell for redox reaction comes out to be positive , it suggest the redox reaction is spontaneous or feasible. Negative value indicate that redox reaction is not feasible.

–0.25V

$$Ni^{+2} + Fe \longrightarrow Ni(s) + Fe^{+2}$$

+0.44V

–1.36V

$$Cl^- + Cr_2O_7^{--} \not\longrightarrow Cl_2 + Cr^{+3}$$

1.33V

(iv) Oxidising & reducing powers : The metals placed above hydrogen in the electrochemical series are strong reducing agents whereas non-metals placed after hydrogen, are strong oxidising agents.

(v) Displacement of one non-metal from its salt solution by another non-metal : A non metal lower in the series will have more reduction potential and will displace another non-metal with lower reduction potential. e.g. F_2 can displace all halide ion from solution.

$$F_2 + 2KCl \longrightarrow 2KF + Cl_2$$

$$Cl_2 + 2KI \longrightarrow 2KCl + I_2$$

ALLEN Session 2019-20
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EXERCISE # I

1. For the cell reaction $2Ce^{4+} + Co \longrightarrow 2Ce^{3+} + Co^{2+}$

E^o_{cell} is 1.89 V. If $E^o_{Co^{2+}|Co}$ is – 0.28 V, what is the value of $E^o_{Ce^{4+}|Ce^{3+}}$?

2. Determine the standard reduction potential for the half reaction :

$Cl_2 + 2e^- \longrightarrow 2Cl^-$

Given $Pt^{2+} + 2Cl^- \longrightarrow Pt + Cl_2$, $E^o_{Cell} = -0.15$ V

$Pt^{2+} + 2e^- \longrightarrow Pt$ $E° = 1.20$ V

3. Is 1.0 M H^+ solution under H_2SO_4 at 1.0 atm capable of oxidising silver metal in the presence of 1.0 M Ag^+ ion?

$$E^o_{Ag^+|Ag} = 0.80\text{ V},\quad E^o_{H^+|H_2(Pt)} = 0.0\text{ V}$$

4. If $E^o_{Fe^{2+}|Fe} = -0.44$ V, $E^o_{Fe^{3+}|Fe^{2+}} = 0.77$ V. Calculate $E^o_{Fe^{3+}|Fe}$.

5. Which of the following statement is wrong about galvanic cell ?

(A) cathode is positive charged (B) anode is negatively charged

(C) reduction takes place at the anode (D) reduction takes place at the cathode

6. A standard reduction electrode potentials of four metal cations are -

$A^+ = -0.250$ V , $B^+ = -0.140$ V $C^+ = -0.126$ V , $D^+ = -0.402$ V

The metal that displaces A^+ from its aqueous solution is :–

(A) B (B) C (C) D (D) None of the above

7. The standard reduction potentials for two half-cell reactions are given below,

$Cd^{2+}(aq) + 2e^- \rightarrow Cd(s)$, $E° = -0.40V$

$Ag^+(aq) + e^- \rightarrow Ag(s)$, $E° = 0.80V$

The standard free energy change for the reaction

$2Ag^+(aq) + Cd(s) \rightarrow 2Ag(s) + Cd^{2+}(aq)$ is given by :

(A) 115.8 KJ (B) –115.8 KJ (C) –231.6 KJ (D) 231.6KJ

8. The reduction potential values are given below:

$Al^{3+} | Al = -1.67$ volt, $Mg^{2+} | Mg = -2.34$ volt

$Cu^{2+} | Cu = +0.34$ volt, $I_2 | 2I^- = +0.53$ volt

Which one is the best reducing agent ?

(A) Al (B) Mg (C) Cu (D) I_2

9. Standard reduction electrode potentials of three metals A, B and C are respectively + 0.5V, – 3.0V and –1.2 V. The reducing powers of these metals are : **[AIEEE 2003]**

(A) C > B > A (B) A > C > B (C) B > C > A (D) A > B > C

10. The $E^0_{M^{3+}/M^{2+}}$ values for Cr, Mn, Fe and Co are –0.41, + 1.57, +0.77 and + 1.97V respectively. For which one of these metals the change in oxidation state from +2 to +3 is easiest ?

(A) Fe (B) Mn (C) Cr (D) Co **[AIEEE 2004]**

ALLEN Session 2019-20

11. Consider the following E^0 values, **[AIEEE 2004]**

$E^0_{Fe^{3+}/Fe^{2+}} = +0.77V$ $\quad E^0_{Sn^{2+}/Sn} = -0.14V$

Under standard conditions the potential for the reaction,

$Sn(s) + 2Fe^{3+}(aq) \longrightarrow 2Fe^{2+}(aq) + Sn^{2+}(aq)$ is:

(A) 0.91V (B) 1.40V (C) 1.68V (D) 0.63V

12. For a spontaneous reaction the ΔG, equilibrium constant (K) and E^0_{Cell} will be respectively **[AIEEE 2005]**

(A) –ve, < 1, –ve (B) –ve, >1, –ve (C) –ve, >1, +ve (D) +ve, >1, –ve

13. Given: $E°_{Fe^{+3}/Fe} = -0.036V$, $E°_{Fe^{+2}/Fe} = -0.439V$

The value of standard electrode potential for the change.

$Fe^{+3}_{(aq)} + e^- \longrightarrow Fe^{+2}_{(aq)}$ will be :- **[AIEEE 2009]**

(A) 0.770 V (B) –0.27 V (C) –0.072 V (D) 0.385 V

14. The Gibbs energy for the decomposition of Al_2O_3 at 500 °C is as follows **[AIEEE 2010]**

$\frac{2}{3}Al_2O_3 \longrightarrow \frac{4}{3}Al + O_2$, $\Delta_rG = +966$ KJ mol^{-1}

The potential difference needed for electrolytic reduction of Al_2O_3 at 500 °C is at least :-

(A) 5.0 V (B) 4.5 V (C) 3.0 V (D) 2.5 V

15. A standard hydrogen electrode has zero electrode potential because

(A) hydrogen is easier to oxidise (B) electrode potential is assumed to be zero

(C) hydrogen atom has only one electron (D) hydrogen is the lighest element.

16. E° for $F_2 + 2e^- = 2F^-$ is 2.8 V, E° for ½ $F_2 + e^- = F^-$ is ?

(A) 2.8 V (B) 1.4 V (C) –2.8 V (D) –1.4 V

17. If $\Delta G°$ of the cell reaction, $AgCl(s) + ½ H_2(g) \rightarrow Ag(s) + H^+ + Cl^-$ is –21.52 KJ

then $\Delta G°$ of $2AgCl(s) + H_2(g) \rightarrow 2Ag(s) + 2H^+ + 2Cl^-$ is :

(A) –21.52 KJ (B) –10.76 KJ (C) –43.04 KJ (D) 43.04 KJ

18. Given electrode potentials :

$Fe^{3+}_{(aq)} + e^- \longrightarrow Fe^{2+}_{(aq)}$; E° = 0.771 volts $\quad I_{2(g)} + 2e^- \longrightarrow 2I^-_{(aq)}$; E° = 0.536 volts

$E°_{cell}$ for the cell reaction, $2Fe^{3+}_{(aq)} + 2I^-_{(aq)} \longrightarrow 2Fe^{2+}_{(aq)} + I_{2(g)}$ is -

(A) (2 × 0.771 – 0.536) = 1.006 volts (B) (0.771 – 0.5 × 0.536) = 0.503 volts

(C) 0.771 – 0.536 = 0.235 volts (D) 0.536 – 0.771 = –0.235 volts

3.9 NERNST EQUATION :

For a reaction aA + bB = cC + dD

$\Delta G = \Delta G^o + RT\ln Q$

$-nFE = -nFE^o + RT\ln Q$

With the help of Nernst equation, we can calculate the non-standard electrode potential of electrode or EMF of cell.

Nernst equation predicts effects of concentration, pressure or temperature changes on cell EMF.

Nernst equation can be applied on half-cell as well as complete Galvanic cells reaction.

ALLEN Session 2019-20

Nernst equation can be applied on half-cell as well as complete Galvanic cells reaction.

$$\boxed{E_{cell} = E^0 - \frac{RT}{nF}\ln Q = E^o - \frac{2.303RT}{nF}\log Q}$$

Where - E^0 = Standard electrode potential

R = Gas constant

T = Temperature (in K)

F = Faraday (96500 coulomb mol^{-1})

n = No. of e^- gained or lost in balanced equation.

Q = Reaction quotent

$$\frac{2.303\times 8.314\times 298}{96500} = 0.059\,\text{volt} \quad (\text{At } 298\text{ K})$$

Note: (i) For writing Nernst equation, first write balanced cell reaction.

(ii) Nernst equation can be applied on half-cell as well as complete Galvanic cells.

3.10 THERMODYNAMIC TREATMENT OF CELL :

(i) Determination of equilibrium constant : We know, that

$$E = E^\circ - \frac{0.0591}{n}\log Q \qquad ..(i)$$

At equilibrium, the cell potential is zero because cell reactions are balanced, i.e. E = 0 & Q = K_{eq}.

$\therefore$ From E (i), we have

$$0 = E^0 - \frac{0.0591}{n}\log K_{eq} \quad \text{or} \quad K_{eq} = \text{anti}\log\left[\frac{nE^0}{0.0591}\right] = 10^{\frac{nE^o}{0.591}}$$

(ii) Heat of Reaction inside the cell: Let n Faraday charge flows out of a cell of e.m.f. E, then

$$-\Delta G = nFE \qquad (i)$$

Gibbs Helmholtz equation (from thermodynamics) may be given as,

$$\Delta G = \Delta H + T\left[\frac{\partial \Delta G}{\partial T}\right]_p \qquad (ii)$$

From Eqs. (i) and (ii), we have

$$-nFE = \Delta H + T\left[\frac{\partial(-nFE)}{\partial T}\right]_p = \Delta H - nFT\left[\frac{\partial E}{\partial T}\right]_p$$

$$\therefore \quad \Delta H = -nFE + nFT\left[\frac{\partial E}{\partial T}\right]_p$$

ALLEN Session 2019-20

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(iii) Entropy change inside the cell : We know that G = H - TS or $\Delta G = \Delta H - T\Delta S$ ...(i)

where ΔG = Free energy change ; ΔH = Enthalpy change and ΔS = entropy change.

According to Gibbs Helmoholtz equation,

$$\Delta G = \Delta H + T\left[\frac{\partial \Delta G}{\partial T}\right]_p \quad ....(ii)$$

From Eqs. (i) and (ii), we have

$$-T\Delta S = T\left[\frac{\partial \Delta G}{\partial T}\right]_p \quad \text{or} \quad \Delta S = -\left[\frac{\partial \Delta G}{\partial T}\right]_p$$

or $$\Delta S = nF\left[\frac{\partial E}{\partial T}\right]_p$$

where $\left[\frac{\partial E}{\partial T}\right]_p$ is called temperature coefficient of cell.

3.11 DIFFERENT TYPES OF ELECTRODES

I. Metal - Metal ion electrode : Ex. - $M^{+n} | M$

$M^{n+} + ne^- \longrightarrow M(s)$

$$E_{M^{n+}/M} = E^0_{M^{n+}/M} - \frac{0.059}{n}\log\frac{1}{[M^{n+}]}$$

II. Gas - ion Electrode :

Andoe : Pt , H_2 (P atm) | H^+ (cM)

Cathode : H^+(cM) | H_2 (P atm)| Pt

Cathodic raction : $H^+(aq) + e^- \longrightarrow \frac{1}{2}H_2$ (P atm)

$$E = E^\circ - 0.0591 \log \frac{P_{H_2}^{\frac{1}{2}}}{[H^+]} = -0.0591 \text{ pH} \left[\because E^0_{H^+/H_2} = 0 \,\&\, pH_2 = 1\text{bar}\right]$$

III. Oxidation - reduction Electrode (or redox electrode) :

It has same metal (or element) in two different oxidation states in same solution.

Pt | Fe^{2+} , Fe^{3+}

As a reduction electrode

$Fe^{3+} + e^- \longrightarrow Fe^{2+}$

$$E = E^\circ - 0.0591 \log \frac{[Fe^{2+}]}{[Fe^{3+}]}$$

ALLEN Session 2019-20

Also, $Pt \mid Cr_2O_7^{-2}(aq.), Cr^{+3}(aq.), H^+$
$Pt \mid Mn^{+2}(aq.), MnO_4^-(aq.), H^+$

IV. Metal - metal insoluble salt-anion electrode :

In this half cell, a metal coated with its insoluble salt is in contact with a solution containing the anion of the insoluble salt. eg. Silver-Silver Chloride Half Cell:

This half cell is represented as $Cl^-|AgCl|Ag$. The equilibrium reaction that occurs at the electrode is

$AgCl(s) + e^- \rightleftharpoons Ag(s) + Cl^-(aq)$

$$E^0_{Cl^-/AgCl/Ag} = E^0_{Ag^+/Ag} + \frac{0.0591}{1}\log K_{sp}$$

$$E_{Cl^-/AgCl/Ag} = E^0_{Cl^-/AgCl/Ag} - \frac{0.0591}{1}\log\left[Cl^-\right]$$

potential of such cells depends upon the concentration of anions. Such cells can be used as **Reference Electrode**.

3.12 Reference Electrode :

Absolute values of electrode potentials can not be measured. Reference electrodes is an electrode used to measure the electrode potential of other electrodes.

(a) Standard Hydrogen Electrode (SHE) :

It consist of a platinum electrode over which H_2 gas (1 bar pressure) is bubbled and the electrode is immersed in a solution that is 1 M in H^+ at any specified temperature.

$2H^+(1\,M) + 2e^- \longrightarrow H_2(g)(1\,atm)$

The potential of this electrode at all temperature is taken as Zero volt.

[IUPAC convention : $E^\circ_{H^+/H_2} = E^\circ_{H_2/H^+} = 0$]

Calculation of electrode potential :

At Anode

$H_2(g) \rightleftharpoons 2H^+ + 2e^-$

- Oxidation potential O.P. = $E_{H_2(g)/H^+(aq.)}$
- $E^0_{H_2(g)/H^+(aq.)} = SOP = 0$
- $E_{H_2/H^+} \neq 0$

At Cathode

$2H^+ + 2e^- \rightleftharpoons H_2(g)$

- Reduction Potential (R.P.)
  $E_{H^+/H_2(g)} = RP$
- $E^0_{H^+/H_2(g)} = SRP = 0$
- $E_{H^+/H_2} \neq 0$

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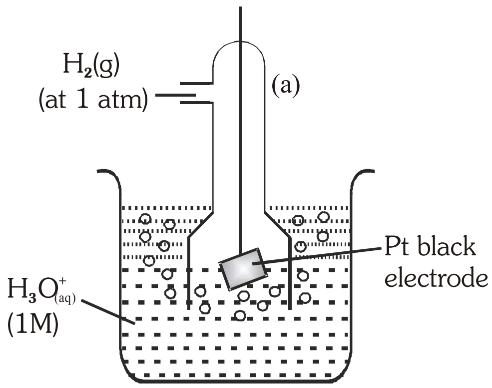


Hydrogen Electrode

- To calculate standard potential of any other electrode a cell is coupled with standard hydrogen electrode (SHE) and it's potential is measured that gives the value of electrode potential of that electrode.

Ex. **Anode** : Zinc electrode

Cathode : SHE

Cell : Zinc electrode | | SHE

$E_{cell} = E_{H^+/H_2(g)} - E^\circ_{Zn2+|Zn}$

$= 0.76$ V (at 298 K experimentally)

So, $E^\circ_{Zn^{2+}|Zn} = -0.76$ V (SRP)

$E^\circ_{Zn|Zn^{2+}(aq)} = 0.76$ V (SOP)

(b) Calomel Electrode :

Cathode : $Cl^-(c\ M)\ |Hg_2Cl_2(s)\ |\ Hg(l)|\ Pt(s)$

It is prepared by a Pt wire in contact with a paste of Hg and Hg_2Cl_2 present in a KCl solution.

reaction $Hg_2Cl_2(s) + 2e^- \longrightarrow 2Hg(l) + 2Cl^-$; $E^0 = +0.27$ V

$$\Rightarrow \boxed{E_{Cl^-/Hg_2Cl_2/Hg} = E^0_{Cl^-/Hg_2Cl_2/Hg} - \frac{0.059}{2}\log[Cl^-]^2}.$$

4. CONCENTRATION CELL

The cells in which electrical current is produced due to transport of a substance from higher to lower concentration. Concentration gradient may arise either in electrode material or in electrolyte. Thus there are two types of concentration cells.

4.1 Electrode concentration cell :

Pt, $H_2(P_1)\ |\ H^+(C)\ |\ H_2(P_2)$, Pt

Here, hydrogen gas is bubbled at two different partial pressures at electrode dipped in the solution of same electrolyte.

Cell process : $1/2H_2(p_1) \rightarrow H^+(c) + e^-$ (Anode process)

$$\frac{H^+(c) + e^- \rightarrow 1/2H_2(p_2)}{1/2H_2(p_1) \rightleftharpoons 1/2H_2(p_2)} \qquad \therefore \qquad E = -\frac{2.303RT}{F}\log\left[\frac{p_2}{p_1}\right]^{1/2}$$

ALLEN Session 2019-20

or $E = \left[\dfrac{2.303RT}{2F}\right]\log\left[\dfrac{p_2}{p_1}\right]$, At 25^0C , $E = \dfrac{0.059}{2F}\log\left[\dfrac{p_1}{p_2}\right]$

For spontanity of such cell reaction, $p_1 > p_2$

4.2 Electrolyte concentration cells:

$Zn(s) \mid ZnSO_4\,(C_1) \parallel ZnSO_4\,(C_2) \mid Zn(s)$

In such cells, concentration gradient arise in electrolyte solutions. Cell process may be given as,

$Zn(s) \rightarrow Zn^{2+}(C_1) + 2e$ (Anodic process)

$$\frac{Zn^{2+}(C_2) + 2e \longrightarrow Zn(s)}{Zn^{2+}(C_2) \rightleftharpoons Zn^{2+}(C_1)}$$ (Cathodic process , Over all process)

$\therefore$ From Nernst equation, we have

$$E = 0 - \frac{2.303RT}{2F}\log\left[\frac{C_1}{C_2}\right] \quad \text{or} \quad E = \frac{2.303RT}{2F}\log\left[\frac{C_2}{C_1}\right]$$

For spontanity of such cell reaction, $C_2 > C_1$

EXERCISE # II

Representation of Cell diagrams, complete and half cell reactions :

19. Write cell reaction of the following cells :

(a) $Ag \mid Ag^+\,(aq) \mid\mid Cu^{2+}\,(aq) \mid Cu$ (b) $Pt \mid Fe^{2+}, Fe^{3+} \mid\mid MnO_4^-, Mn^{2+}, H^+ \mid Pt$

(c) $Pt, Cl_2 \mid Cl^-\,(aq) \mid\mid Ag^+\,(aq) \mid Ag$ (d) $Pt, H_2 \mid H^+\,(aq) \mid\mid Cd^{2+}\,(aq) \mid Cd$

20. Write cell representation for following cells.
(a) $Cd^{2+}\,(aq) + Zn\,(s) \longrightarrow Zn^{2+}\,(aq) + Cd\,(s)$
(b) $2Ag^+\,(aq) + H_2\,(g) \longrightarrow 2H^+\,(aq) + 2Ag\,(s)$
(c) $Cr_2O_7^{2-}\,(a) + 14H^+\,(aq) + 6Fe^{2+}\,(aq) \longrightarrow 6Fe^{3+}\,(aq) + 2Cr^{3+}\,(aq) + 7H_2O\,(l)$

21. Calculate the EMF of a Daniel cell when the concentration of $ZnSO_4$ and $CuSO_4$ are 0.001 M and 0.1M respectively. The standard potential of the cell is 1.1V.

22. Calculate E^0 and E for the cell $Sn \mid Sn^{2+}\,(1M) \parallel Pb^{2+}(10^{-3}M) \mid Pb$, $E^0\,(Sn^{2+}\mid Sn) = -0.14V$, $E^0\,(Pb^{2+}\mid Pb) = -0.13V$. Is cell representation is correct?

23. At what concentration of Cu^{2+} in a solution of $CuSO_4$ will the electrode potential be zero at 25°C? Given : $E^0\,(Cu \mid Cu^{2+}) = -0.34$ V.

24. Calculate the equilibrium constant for the reaction
$Fe^{2+} + Ce^{4+} \rightleftharpoons Fe^{3+} + Ce^{3+}$, [given : $E^0_{Ce^{4+}|Ce^{3+}} = 1.44$ V; $E^0_{Fe^{3+}|Fe^{2+}} = 0.68$ V]

25. Calculate the equilibrium constant for the reaction

$Fe + CuSO_4 \rightleftharpoons FeSO_4 + Cu$ at 25°C.

Given E^0 $(Fe|Fe^{2+}) = 0.44V$, E^0 $(Cu|Cu^{2+}) = -0.337V$.

26. At 25^0C the value of K for the equilibrium $Fe^{3+} + Ag \rightleftharpoons Fe^{2+} + Ag^+$ is 0.531mol|litre. The standard electrode potential for $Ag^+ + e^- \rightleftharpoons Ag$ is 0.799V. What is the standard potential for $Fe^{3+} + e^- \rightleftharpoons Fe^{2+}$?

27. The reduction potential of hydrogen electrode when placed in a buffer solution is found to be – 0.413V. The pH of the buffer is -

28. Calculate the EMF of the following cell

$Zn \mid Zn^{2+} (0.01M) \parallel Zn^{2+} (0.1\ M) \mid Zn$

at 298 K.

29. Calculate pH using the following cell :

$Pt\ (H_2) \mid H^+ (x\ M) \mid\mid H^+ (1\ M) \mid Pt\ (H_2)$ if $E_{cell} = 0.2364$ V.

1 atm 1 atm

30. A standard hydrogen electrode has zero electrode potential because

(A) hydrogen is easier to oxidise

(B) electrode potential is assumed to be zero

(C) hydrogen atom has only one electron

(D) hydrogen is the lighest element.

31. The standard electrode potentials for the reactions

$Ag^+ (a) + e^- \longrightarrow Ag(s)$ $Sn^{2+} (a) + 2e^- \longrightarrow Sn\ (s)$

at 25 °C are 0.80 volt and –0.14 volt, respectively. The standard emf of the cell.

$Sn_{(s)}|Sn^{2+}_{(aq)}(1M)||Ag^+_{(aq)}(1M)|Ag_{(s)}$ is :

(A) 0.66 volt (B) 0.80 volt (C) 1.08 volt (D) 0.94 volt

32. $E°(Ni^{2+}|Ni) = -0.25$ volt, $E°\ (Au^{3+} \mid Au) = 1.50$ volt.

The standard emf of the voltaic cell.

$Ni_{(s)} \mid Ni^{2+}_{(aq)}$ (1.0 M) $\parallel Au^{3+}_{(aq)}$ (1.0 M) $\mid Au_{(s)}$ is :

(A) 1.25 volt (B) –1.75 volt (C) 1.75 volt (D) 4.0 volt

33. E° for $F_2 + 2e^- = 2F^-$ is 2.8 V, E° for $½F_2 + e^- = F^-$ is ?

(A) 2.8 V (B) 1.4 V (C) –2.8 V (D) –1.4 V

34. If $\Delta G°$ of the cell reaction,

$AgCl(s) + ½H_2(g) \rightarrow Ag(s) + H^+ + Cl^-$ is –21.52 KJ

then $\Delta G°$ of $2AgCl(s) + H_2(g) \rightarrow 2Ag(s) + 2H^+ + 2Cl^-$ is :

(A) –21.52 KJ (B) –10.76 KJ (C) –43.04 KJ (D) 43.04 KJ

35. The reduction potential of hydrogen electrode (P_{H_2} = 1 atms; $[H^+]$ = 0.1 M) at 25°C will be -

(A) 0.00 V (B) –0.059 V (C) 0.118 V (D) 0.059 V

36. Which of the following represents the reduction potential of silver wire dipped into 0.1 M $AgNO_3$ solution at 25° C ?

(A) $E°_{red}$ (B) $(E°_{red} + 0.059)$ (C) $(E°_{oxi} - 0.059)$ (D) $(E°_{red} - 0.059)$

ALLEN Session 2019-20

37. For a reaction - $A(s) + 2B^+_{(aq)} \rightarrow A^{2+}_{(aq)} + 2B_{(s)}$ K_C has been found to be 10^{12}. The E°cell is:
(A) 0.354 V (B) 0.708 V (C) 0.0098 V (D) 1.36 V

38. At 25°C the standard emf of cell having reactions involving two electrons change is found to be 0.295V. The equilibrium constant of the reaction is -
(A) 29.5×10^{-2} (B) 10 (C) 10^{10} (D) 29.5×10^{10}

39. For the cell reaction

$$Mg_{(s)} + Zn^{2+}_{(aq)}(1M) \longrightarrow Zn(s) + Mg^{2+}_{(aq)}(1M)$$

The emf has been found to be 1.60 V, E° of the cell is :
(A) –1.60 V (B) 1.60 V (C) 0.0 V (D) 0.16 V

40. The emf of the cell in which the following reaction,
$Zn(s) + Ni^{2+}_{(aq)}\ (a = 0.1) \rightarrow Zn^{2+}_{(aq)}\ (a = 1.0) + Ni(s)$
occurs, is found to be 0.5105 V at 298 K. The standard e.m.f. of the cell is :-
(A) –0.5105 V (B) 0.5400 V (C) 0.4810 V (D) 0.5696 V

41. Given electrode potentials :
$Fe^{3+}_{(aq)} + e^- \longrightarrow Fe^{2+}_{(aq)}$; E° = 0.771 volts $I_{2(g)} + 2e^- \longrightarrow 2I^-_{(aq)}$; E° = 0.536 volts
$E°_{cell}$ for the cell reaction,
$2Fe^{3+}_{(aq)} + 2I^-_{(aq)} \longrightarrow 2Fe^{2+}_{(aq)} + I_{2(g)}$ is -
(A) (2 × 0.771 – 0.536) = 1.006 volts
(B) (0.771 – 0.5 × 0.536) = 0.503 volts
(C) 0.771 – 0.536 = 0.235 volts
(D) 0.536 – 0.771 = –0.235 volts

42. For the redox reaction :
$Zn(s) + Cu^{2+}\ (0.1M) \rightarrow Zn^{2+}\ (1M) + Cu(s)$ taking place in a cell,
$E°_{Cell}$ is 1.10 volt. E_{Cell} for the cell will be $\left(2.303\frac{RT}{F} = 0.0591\right)$ **[AIEEE 2003]**
(A) 1.07 volt (B) 0.82 volt (C) 2.14 volt (D) 1.80 volt

43. For a cell reaction involving a two-electron change, the standard e.m.f. of the cell is found to be 0.295V at 25° C. The equilibrium constant of the reaction at 25°C will be : **[AIEEE 2003]**
(A) 10 (B) 1×10^{10} (C) 1×10^{-10} (D) 29.5×10^{-2}

44. In a cell that utilises the reaction, **[AIEEE 2004]**
$Zn(s) + 2H^+(A) \longrightarrow Zn^{2+}(aq) + H_2(g)$ addition of H_2SO_4 to cathode compartment, will :
(A) increase the E_{cell} and shift equilibrium to the right
(B) lower the E_{cell} and shift equilibrium to the right
(C) lower the E_{cell} and shift equilibrium to the left
(D) increase the E_{cell} and shift equilibrium to the left

ALLEN Session 2019-20

45. The standard e.m.f. of a cell, involving one electron change is found to be 0.591 V at 25°C. The equilibrium constant of the reaction is (F = 96,500C mol^{-1} ; R = 8.314 JK^{-1} mol^{-1}) **[AIEEE 2004]**

(A) 1.0×10^{10} (B) 1.0×10^{5} (C) 1.0×10^{1} (D) 1.0×10^{30}

46. Given : $E^0_{Cr^{3+}/Cr} = -0.72$ V, $E^0_{Fe^{2+}/Fe} = -0.42$V. The potential for the cell **[AIEEE 2008]**

$Cr_{(s)} \mid Cr^{3+}_{(aq)}$ (0.1 M) $\parallel Fe^{2+}_{(aq)}$ (0.01 M)$\mid Fe_{(s)}$ is

(A) 0.26 V (B) 0.339 V (C) – 0.339 V (D) – 0.26 V

47. The cell Zn | $Zn^{+2}_{(aq)}$(1M) || $Cu^{+2}_{(aq)}$(1M) | Cu ($E°_{cell}$ = 1.10 V) was allowed to be completely discharged at 298 K. The relative concentration of Zn^{+2} to Cu^{+2}, $\left\{\frac{[Zn^{+2}]}{[Cu^{+2}]}\right\}$ is : **[AIEEE 2007]**

(A) 9.65×10^4 (B) Antilog (24.08) (C) 37.3 (D) $10^{37.3}$

48. Given the data at 25°C,

$Ag_{(s)} + I^-_{(aq)} \rightarrow AgI_{(s)} + e^-$, E° = 0.152V

$Ag_{(s)} \rightarrow Ag^+_{(aq)} + e^-$, E° = – 0.800 V

What is the value of log K_{sp} for AgI ?

(Where K_{sp}= solubility product)

$\left(2.303\frac{RT}{F} = 0.059V\right)$ **[AIEEE 2006]**

(A) –8.12 (B) +8.612 (C) –37.83 (D) –16.13

5. SOME COMMERCIAL BATTERIES

Any battery or cell that we use as a source of electrical energy is basically an electrochemical cell where oxidising and reducing agents are made to react by using a suitable device. In principle, any redox reaction can be used as the basis of an electrochemical cell, but there are limitations to the use of most reactions as the basis of practical batteries. A battery should be reasonably right and compact and its voltage should not vary appreciably during the use.

There are mainly two types of cells :

(i) primary cells and (ii) secondary cells. In primary cells, the reaction occurs only once and the battery then becomes dead over a period of time and cannot be used again. (For example, dry cell, mercury cell.) Contrary to this, secondary cells can be recharged by passing a current through them so that they can be used again and again.(For example, lead storage battery, nickel- cadmium storage cell.)

ALLEN Session 2019-20

5.1 Primary Batteries

5.1.1 Dry cell or Laclanche cell :

The most familiar type of battery is the dry cell which is a compact of Leclanche cell known after its discoverer Leclanche (fig.) : In this cell, the anode consists of a zinc container and the cathode is a graphite rod surrounded by powdered MnO_2 and carbon. The space between the electrodes is filled with a moist paste of NH_4Cl and $ZnCl_2$. The electrode reactions are complex, but they can be written approximately as follows.

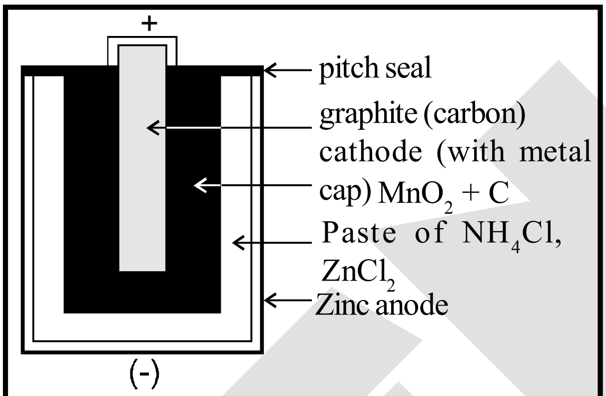


Fig. : A dry cell

- **Anode** $Zn(s) \longrightarrow Zn^{+2} + 2e^-$
- **Cathode** $MnO_2 + NH_4^+ + e^- \longrightarrow MnO(OH) + NH_3$

In the cathode reaction, manganese is reduced from the +4 oxidation state to the +3 state. Ammonia is not liberated as a gas but combines with Zn^{2+} to form $Zn(NH_3)_4^{2+}$ ion. The cell has a potential of nearly 1.5 V.

5.1.2 Mercury cell :

Mercury cell, suitable for low current devices like hearing aids, watches, etc.consists of zinc & mercury amalgam as anode and a paste of HgO and carbon as the cathode. The electrolyte is a paste of KOH and ZnO. The electrode reactions for the cell are given below:

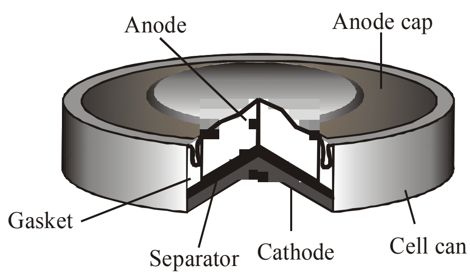
Anode : $Zn(Hg) + 2OH^- \rightarrow ZnO(s) + H_2O + 2e^-$

Cathode : $HgO + 2H_2O + 2e^- \rightarrow Hg(I) + 2OH^-$

The overall reaction is represented by

$$Zn(Hg) + HgO(s) \rightarrow ZnO(s) + Hg(I)$$

The cell potential is approximately 1.35 V and remains constant during its life as the overall reaction does not involve any ion in solution whose concentration can change during its life time.



Commonly used mercury cell. The reducing agent is zinc and the oxidising agent is mercury (II) oxide.

ALLEN Session 2019-20

5.2 Secondary Batteries

A secondary cell after use can be recharged by passing current through it in the opposite direction so that it can be used again. A good secondary cell can undergo a large number of discharging and charging cycles.

5.2.1 Lead storage cell :

The most important secondary cell is the lead storage battery commonly used in automobiles and invertors. It consists of a lead anode and a grid of lead packed with lead dioxide (PbO_2) as cathode. A 38% solution of sulphuric acid is used as an electrolyte.

The cell reactions when the battery is in use are given below:

Anode: $Pb(s) + SO_4^{2-}(aq) \rightarrow PbSO_4(s) + 2e^-$

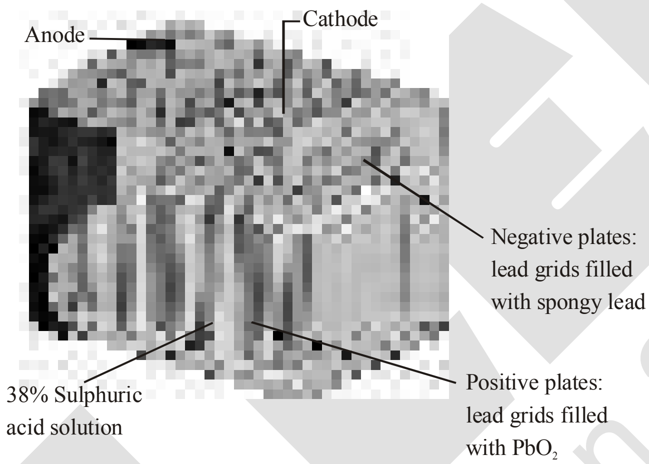
Cathode: $PbO_2(s) + SO_4^{2-}(aq) + 4H^+(aq) + 2e^- \rightarrow PbSO_4(s) + 2H_2O(\ell)$

i.e., overall cell reaction consisting of cathode and anode reactions is:

$$Pb(s) + PbO_2(s) + 2H_2SO_4(aq) \rightarrow 2PbSO_4(s) + 2H_2O(\ell)$$

On charging the battery the reaction is reversed and $PbSO_4(s)$ on anode and cathode is converted into Pb and PbO_2, respectively

.

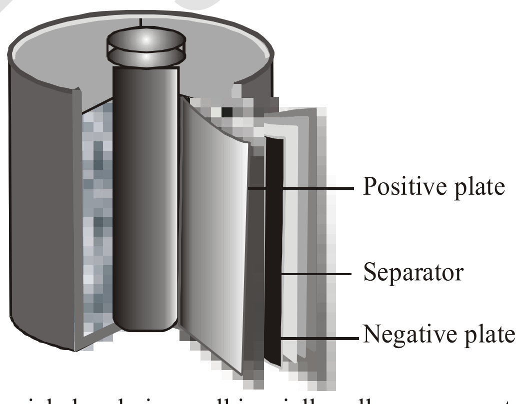


The Lead storage battery

5.2.2 Nickel-cadmium cell :

Another important secondary cell is the nickel-cadmium cell which has longer life than the lead storage cell but more expensive to manufacture. We shall not go into details of working of the cell and the electrode reactions during charging and discharging. The overall reaction during discharge is:

$$Cd(s) + 2Ni(OH)_3(s) \rightarrow CdO(s) + 2Ni(OH)_2(s) + H_2O(\ell)$$



A rechargeable nickel-cadmium cell in a jelly roll arrangement and separated by a layer soaked in moist sodium or potassium hydroxide

Session 2019-20

5.3 FUEL CELLS

Production of electricity by thermal plants is not a very efficient method and is a major source of pollution. In such plants, the chemical energy (heat of combustion) of fossil fuels (coal, gas or oil) is first used for converting water into high pressure steam. This is then used to run a turbine to produce electricity. We know that a galvanic cell directly converts chemical energy into electricity and is highly efficient. It is now possible to make such cells in which reactants are fed continuously to the electrodes and products are removed continuously from the electrolyte compartment. Galvanic cells that are designed to convert the energy of combustion of fuels like hydrogen, methane, methanol, etc. directly into electrical energy are called fuel cells. One of the most successful fuel cells uses the reaction of hydrogen with oxygen to form water . The cell was used for providing electrical power in the Apollo space programme. The water vapours produced during the reaction were condensed and added to the drinking water supply for the astronauts. In the cell, hydrogen and oxygen are bubbled through porous carbon electrodes into concentrated aqueous sodium hydroxide solution. Catalysts like finely divided platinum or palladium metal are incorporated into the electrodes for increasing the rate of electrode reactions. The electrode reactions are given below:

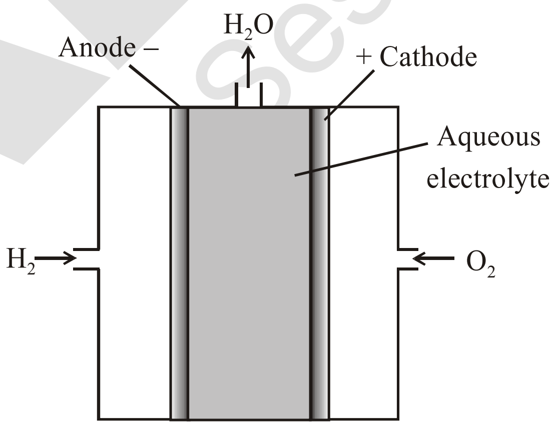
Cathode: $O_2(g) + 2H_2O(I) + 4e^- \rightarrow 4OH^-(aq)$

Anode: $2H_2(g) + 4OH^-(aq) \rightarrow 4H_2O(I) + 4e^-$

Overall reaction being:

$2H_2(g) + O_2(g) \rightarrow 2H_2O(I)$

The cell runs continuously as long as the reactions are supplied Fuel cells produce electricity with an efficiency of about 70% compared to thermal plants whose efficiency is about 40%. There has been tremendous progress in the development of new electrode materials, better catalysts and electrolytes for increasing the efficiency of fuel cells. These have been used in automobiles on an experimental basis. Fuel cells are pollution free and in view of their future importance, a variety of fuel cells have been fabricated and tried.

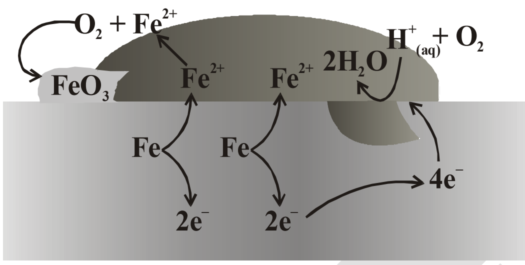


Fuel cell using H_2 and O_2 produces electricity

ALLEN Session 2019-20

6. CORROSION :

Corrosion slowly coats the surfaces of metallic objects with oxides or other salts of the metal. The rusting of iron, tarnishing of silver, development of green coating on copper and bronze are some of the examples of corrosion. It It causes enormous damage to buildings, bridges, ships and to all objects made of metals especially that of iron. We lose crores of rupees every year on account of corrosion. In corrosion, a metal is oxidised by loss of electrons to oxygen and formation of oxides. Corrosion of iron (commonly known as rusting) occurs in presence of water and air. The chemistry of corrosion is quite complex but it may be considered essentially as an electrochemical phenomenon. At a particular spot of an object made of iron, oxidation takes place and that spot behaves as anode and we can write the reaction



Oxidation: $Fe(s) \rightarrow Fe^{2+}(aq) + 2e^-$

Reduction: $O_2(g) + 4H^+(aq) + 4e^- \rightarrow 2H_2O(I)$

Atmospheric

Corrosion of iron in atmosphere

Anode: $2Fe(s) \longrightarrow 2Fe^{2+} + 4e^{-1}$ $\qquad E^\ominus_{(Fe^{2+}/Fe)} = -0.44\ V$

Electrons released at anodic spot move through the metal and go to another spot on the metal and reduce oxygen in presence of H+ (which is believed to be available from H_2CO_3 formed due to dissolution of carbon dioxide from air into water. Hydrogen ion in water may also be available due to dissolution of other acidic oxides from the atmosphere). This spot behaves as cathode with the reaction

Cathode: $O_2(g) + 4H^+(aq) + 4e^- \longrightarrow 2H_2O(I)$ $\qquad E_{H^+|O_2|H_2O} = 1.23\ V$

The overall reaction being:

$2Fe(s) + O_2(g) + 4H^+(aq) \longrightarrow 2Fe^{2+}(aq) + 2H_2O(I)$ $\qquad E^\ominus_{(Cell)} = 1.67\ V$

The ferrous ions are further oxidised by atmospheric oxygen to ferric ions which come out as rust in the form of hydrated ferric oxide ($Fe_2O_3 \cdot xH_2O$) and with further production of hydrogen ions.

Prevention of corrosion is of prime importance. It not only saves money but also helps in preventing accidents such as a bridge collapse or failure of a key component due to corrosion. One of the simplest methods of preventing corrosion is to prevent the surface of the metallic object to come in contact with atmoshphere. This can be done by covering the surface with paint or by some chemicals (e.g. bisphenol). Another simple method is to cover the surface by other metals (Sn, Zn, etc.) that are inert or react to save the object. An electrochemical method is to provide a sacrificial electrode of another metal (like Mg, Zn, etc.). Which corrodes itself but saves the object.

ALLEN Session 2019-20

Ex.1 ***E° of some oxidants are given as :***

$$I_2 + 2e^- \longrightarrow 2I^-, \qquad E^\circ = +0.54\ V$$

$$MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O, \qquad E^\circ = +1.52\ V$$

$$Fe^{3+} + e^- \longrightarrow Fe^{2+}, \qquad E^\circ = +0.77\ V$$

$$Sn^{4+} + 2e^- \longrightarrow Sn^{2+}, \qquad E^\circ = +0.1\ V$$

(a) Select the strongest reductant and oxidant in these.

(b) Select the weakest reductant and oxidant in these.

(c) Select the spontaneous reaction from the changes given below.

(i) $Sn^{4+} + 2Fe^{2+} \longrightarrow Sn^{2+} + 2Fe^{3+}$

(ii) $2Fe^{2+} + I_2 \longrightarrow 2Fe^{3+} + 2I^-$

(iii) $Sn^{4+} + 2I^- \longrightarrow Sn^{2+} + I_2$

(iv) $Sn^{2+} + I_2 \longrightarrow Sn^{4+} + 2I^-$

Sol. (a) More the E°_{OP}, more is the tendency for oxidation. Therefore, since maximum E°_{OP} stands for :

$$Sn^{2+} \longrightarrow Sn^{4+} + 2e^- \qquad E^\circ_{OP} = -0.1V$$

$\therefore$ *Strongest reductant* : Sn^{2+}

and weakest oxidant : Sn^{4+}

(b) More +ve is E°_{RP}, more is the tendency for reduction. Therefore, since maximum E°_{RP} stands for :

$$MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O \quad E^\circ_{RP} = +1.52\ V$$

$\therefore$ *Strongest oxidant* : MnO_4^-

and *weakest reductant* : Mn^{2+}

Note :- *Stronger is oxidant, weaker is its conjugate reducant and vice-versa.*

(c) For (i) $E^\circ_{Cell} = E^\circ_{OP_{Fe^{2+}/Fe^{3+}}} + E^\circ_{RP_{Sn^{4+}/Sn^{2+}}} = -0.77 + 0.1$

Fe^{2+} oxidizes and Sn^{4+} reduces in change.

$\therefore$ $E^\circ_{Cell} = E^\circ_{OP_{Fe^{2+}/Fe^{3+}}} + E^\circ_{RP_{Sn^{4+}/Sn^{2+}}} = -0.77 + 0.1 = -0.67\ V$

E°_{Cell} is negative.

$\therefore$ (i) Is non-spontaneous change.

For (ii) $E^\circ_{Cell} = E^\circ_{OP_{Fe^{2+}/Fe^{3+}}} + E^\circ_{RP_{I_2/I^-}} = -0.77 + 0.54 = -0.23\ V$

$\therefore$ (ii) reaction is non-spontaneous change.

For (iii) $E^\circ_{Cell} = E^\circ_{OP_{I^-/I_2}} + E^\circ_{RP_{Sn^{4+}/Sn^{2+}}} = -0.54 + 0.1 = -0.44\ V$

$\therefore$ (iii) Reaction is non-spontaneous change.

For (iv) $E^\circ_{Cell} = E^\circ_{OP_{Sn^{2+}/Sn^{4+}}} + E^\circ_{RP_{I_2/I^-}} = -0.1 + 0.54 = +0.44\ V$

(iv) Reaction is spontaneous change.

ALLEN Session 2019-20

Ex.1 ***Given the standard electrode potentials ;***

$K^+|K = -2.93$ V, $Ag^+|Ag = 0.80$ V, $Hg^{2+}|Hg = 0.79$ V, $Mg^{2+}|Mg = -2.37$ V, $Cr^{3+}|Cr = -0.74$V.

Arrange these metals in their incresing order of reducing power.

Sol. More is $E°_{RP}$, more is the tendency to get reduced or more is the oxidising power or lesser is reducing power. Ag < Hg < Cr < Mg < K

Ex.2 ***A cell is prepared by dipping a copper rod in 1 M $CuSO_4$ solution and a nickel rod in 1M $NiSO_4$. The standard reduction potentials of copper and nickel electrodes are + 0.34 V and –0.25 V respectively.***

(i) Which electrode will work as anode and which as cathode ?

(ii) What will be the cell reaction ?

(iii) How is the cell represented ?

(iv) Calculate the EMF of the cell.

Sol. (i) The nickel electrode with smaller E° value (–0.25 V) will work as anode while copper electrode with more E° value (+0.34V) will work as cathode.

(ii) The cell reaction may be written as :

At anode : $Ni\,(s) \longrightarrow Ni^{2+}\,(aq.) + 2e^-$

At cathode : $Cu^{2+}\,(aq) + 2e^- \longrightarrow Cu\,(s)$

Cell reaction : $Ni\,(s) + Cu^{2+}\,(aq) \longrightarrow Ni^{2+}\,(aq) + Cu\,(s)$

(iii) The cell may be represented as :

$Ni(s)|Ni^{2+}\,(aq)\,||\,Cu^{2+}\,(aq)|Cu\,(s)$

(iv) EMF of cell $= E^\circ_{cathode} - E^\circ_{anode} = (+0.34) - (-0.25) = 0.59$ V

Ex.4 ***Predict whether the following reaction can occur under standard conditions or not.***

$$Sn^{2+}\,(aq) + Br_2\,(\ell) \longrightarrow Sn^{4+}\,(aq) + 2Br^-(aq)$$

Given : $E°_{Sn^{4+}|Sn^{2+}} = +0.15$; $E°_{Br^2|Br-} = 1.06$V.

Sol. $E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode} = 1.06 - 0.15 = 0.91$V.

Since, E°_{cell} comes out be positive, this means that the reaction can occur.

Ex.5 ***Given that, $Co^{3+} + e^- \longrightarrow Co^{2+}$ $E° = +1.82$V***

$2H_2O \longrightarrow O_2 + 4H^+ + 4e^-$; $E° = -1.23$V.

Explain why Co^{3+} is not stable in aqueous solutions.

Sol. The E°_{cell} can be calculated as follows :

$4 \times [Co^{3+} + e^- \longrightarrow Co^{2+}]$; $E° = +1.82$V

$1 \times [2H_2O \longrightarrow O_2 + 4H^+ + 4e^-]$; $E° = -1.23$ V.

Add : $4Co^{3+} + 2H_2O \longrightarrow 4Co^{2+} + 4H^+ + O_2$; $E° = 1.82 - 1.23 = +0.59$V.

Since, E°_{cell} is positive, the cell reaction is spontaneous. This means that Co^{3+} ions will take part in the reaction. Therefore, Co^{3+} is not stable.

ALLEN Session 2019-20
node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Electrochemistry\Eng\01_Theory.p65

Ex.6 ***The 0.1 M copper sulphate solution in which copper electrode is dipped at 25°C. Calculate the electrode reduction potential of copper electrode.***

[Given : $E^0_{Cu^{2+}|Cu} = 0.34V$ **]**

Sol : $Cu^{+2} + 2e^- \rightarrow Cu$

$$E_{red} = E_{red}^{\ 0} - \frac{0.059}{n}\log Q = E^0_{Cu^{2+}|Cu} - \frac{0.0591}{2}\log\frac{1}{[Cu^{2+}]}$$

$$\text{So } E = 0.34 - \frac{0.0591}{2}\log 10$$

$$= 0.34 - 0.03 = 0.31 \text{ volts}$$

Ex.7 ***Calculate the EMF of the cell : Cr|Cr^{+3} (0.1M) || Fe^{+2} (0.01M)|Fe***
(Given : $E^\circ_{Cr^{+3}|Cr} = -0.75$ V , $E^\circ_{Fe^{+2}|Fe} = -0.45$ V)

Sol. Half cell reactions are :

- **At anode :** $[Cr \rightarrow Cr^{+3} + 3e^-] \times 2$
- **At cathode :** $[Fe^{+2} + 2e^- \rightarrow Fe] \times 3$

Over all reaction : $2Cr(s) + 3Fe^{+2}(aq.) \rightarrow 2Cr^{+3}(aq.) + 3Fe(s.)$

$$E^\circ_{cell} = E^\circ_{Cr|Cr^{+3}} + E^\circ_{Fe^{+3}|Fe} = 0.75 + (-0.45) = 0.30 \text{ V}$$

$$E_{cell} = E^0 - \frac{0.0591}{n}\log Q = 0.30 - \frac{0.0591}{6}\log\frac{\left[Cr^{+3}\right]^2}{\left[Fe^{+2}\right]^3} = 0.30 - \frac{0.0591}{6}\log\frac{[0.1]^2}{[0.01]^3} = 0.26 \text{ V}$$

Ex.8 ***The measured e.m.f. at 25°C for the cell reaction,***
$Zn(s) + Cu^{2+}_{(aq)}(1.0M) \longrightarrow Cu(s) + Zn^{2+}_{(aq)}(0.1\ M)$ is 1.3 volt, Calculate E° for the cell reaction.

Sol. Using Nernst equation (at 298 K), $E_{cell} = E^\circ_{cell} - \frac{0.059}{2}\log\frac{[Zn^{2+}(aq)]}{[Cu^{2+}(aq)]}$

Here, $E_{cell} = 1.3$ V, $[Cu^{2+}(aq)] = 1.0$ M, $[Zn^{2+}(aq)] = 0.1$ M, $E^\circ_{cell} = ?$

Substituting the values , $E^0_{cell} = 1.27$ V

Ex.9 ***The emf of a cell corresponding to the reaction***
$Zn + 2H^+(aq) \longrightarrow Zn^{2+}(0.1M) + H_2(g)$ 1 atm
is 0.28 volt at 25°C. Calculate the pH of the solution at the hydrogen electrode.

$E^\circ_{Zn^{2+}|Zn}$ ***= – 0.76 volt and*** $E^\circ_{H^+|H_2} = 0$

Sol. $E^\circ_{cell} = 0.76 \text{ volt}$

Applying Nernst equation, $E_{cell} = E^\circ_{cell} - \frac{0.0591}{2}\log\frac{[Zn^{2+}]P_{H_2}}{[H^+]^2}$

$0.28 = 0.76 - \frac{0.0591}{2}\log\frac{(0.1)\times 1}{[H^+]^2}$, pH = 8.62

Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Electrochemistry\Eng\01_Theory.p65

Ex.10 The half cell oxidation potential of a half-cell $A^{x+}, A^{(x+n)+}$ | Pt were found to be as follows :

% of reduced form	24.4	48.8
Cell potential |V	0.101	0.115

Determine the value of n.

[take $\frac{2.303RT}{F} = 0.06$, $\log_{10} 24.4 = 1.387$, $\log_{10} 75.6 = 1.878$, $\log_{10} 48.8 = 1.688$, $\log_{10} 51.2 = 1.709$]

Sol. The half-cell reaction is -

$A^{x+} \rightarrow A^{(x+n)+} + ne^-$

Its Nernst equation is -

$$E = E° - 2.303\frac{RT}{nF}\log\frac{[A^{(x+n)+}]}{[A^{x+}]} = E° - \left(\frac{0.06V}{n}\right)\log\left(\frac{\text{oxidized form}}{\text{reduced form}}\right)$$

Substituting the given values, we get

$$0.101\ V = E° - \left(\frac{0.06V}{n}\right)\log\frac{75.6}{24.4} = E° - \left(\frac{0.06V}{n}\right)(0.491)\ ....(i)$$

$$0.115\ V = E° - \left(\frac{0.06V}{n}\right)\log\frac{51.2}{48.8} = E° - \left(\frac{0.06V}{n}\right)(0.021)\ ....(ii)$$

eq. (ii) – (i) , n = 2

Ex.11 ***What is the standard electrode potential for the electrode MnO_4^- | MnO_2 in solution***

$E°_{MnO_4^-|Mn^{2+}}$ ***= 1.51 volt,*** $E°_{MnO_2|Mn^{2+}}$ ***=1.23 volt***

Sol. $MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4\,H_2O$; E° = 1.51 volt

$MnO_2 + 4H^+ + 2e^- \longrightarrow Mn^{2+} + 2\,H_2O$; E° = 1.23 volt

---

$MnO_4^- + 4H^+ + 3e^- \longrightarrow MnO_2 + 2\,H_2O$;E° = ?

$$E° = \frac{5\times1.51 - 2\times1.23}{3} = \frac{7.55 - 2.45}{3} = \frac{5.09}{3} = 1.70\ \text{volt}$$

Ex.12 ***Calculate ΔG° for the reaction : Cu^{2+} (aq) + Fe (s) ⇌ Fe^{2+} (aq) + Cu (s) .***

Given that $E°_{Cu^{2+}|Cu}$ = + 0.34 V, $E°_{Fe^{+2}/Fe}$ ***= – 0.44 V***

Sol. The cell reactions are :

$Fe\ (s) \longrightarrow Fe^{2+}\ (aq) + 2e^-$ .... at (anode)

$Cu^{2+}\ (aq) + 2e^- \longrightarrow Cu\ (s)$ .... at (cathode)

We know that : $\Delta G° = -nF\ E^0_{cell}$; n = 2

$$E°_{cell} = \left[E°_{(Cu^{2+}/Cu)} - E°_{(Fe^{2+}/Fe)}\right] = (+\,0.34\ V) - (-0.44\ V) = +\,0.78\ V \text{ and } F = 96500\ C$$

$$\therefore\ \Delta G° = -\,nF\ E°_{cell} = -(B) \times (96500\ C) \times (+0.78\ V) = -\,150540\ CV = -150540\ J$$

ALLEN Session 2019-20

Ex.13 At 298 K the standard free energy of formation of $H_2O(l)$ is– 256.5 kJ|mol & OH^- is 80 kJ|mol. What will be emf at 298 K of the cell H_2(g, 1 bar) | H^+(1M) | | OH^-(1M) | O_2 (g, 1 bar)

$$H_2 + H_2O + \frac{1}{2}O_2 \longrightarrow 2H^+ + 2OH^-$$

$\Delta G° = -256.5 + 2 \times 80 = -96.5$ kJ

$-\Delta G° = nFE°$

$+ 96.5 \times 1000 = 2 \times 96500 \times E°$

$E° = 0.5$ Volt

Ex.14 ***Calculate the equilibrium constant for the reaction at 298 K.***

$$Zn\ (s) + Cu^{2+}\ (aq) \rightleftharpoons Zn^{2+}(aq) + Cu\ (s)$$

Given , $E^{\circ}_{Zn^{2+}|Zn} = -0.76\ V$ ***and*** $E^{\circ}_{Cu^{2+}|Cu} = +0.34\ V$

Sol. We know that, $\log K_c = \dfrac{nE^0_{cell}}{0.0591}$

$E^{\circ}_{cell} = \left[E^{\circ}_{Cathode} - E^{\circ}_{Anode}\right] = [(+0.34\ V) - (-0.76\ V)] = 1.10\ V, n = 2,$

$\therefore \quad \log K_C = \dfrac{2 \times (1.10\ V)}{(0.0591\ V)} = 37.29$, K_c = Antilog 37.29 = 1.95×10^{37}

Ex.15 ***Calculate the cell e.m.f. and ΔG for the cell reaction at 298 K for the cell.***
Zn (s) | Zn^{2+} (0.0004M) | | Cd^{2+} (0.2M) | Cd(s)

Given, $E^{\circ}_{Zn^{2+}|Zn}$ ***= – 0.763 V ;*** $E^{\circ}_{Cd^{2+}|Cd}$ ***= – 0.403 V at 298 K.***

F = 96500 C mol^{-1}.

Sol. **Step I.** Calculation of cell e.m.f. :

According to Nernst equation,

$$E = E° - \frac{0.0591}{n}\log\frac{[Zn^{2+}(aq)]}{[Cd^{2+}(aq)]}$$

$$E^{\circ}_{cell} = E^{\circ}_{(Cd^{2+}/Cd)} - E^{\circ}_{(Zn^{2+}/Zn)} = (-0.403) - (-0.763) = 0.36\ V$$

$[Zn^+(aq)] = 0.0004$ M, $[Cd^{2+}\ (aq)] = 0.2$ M, n = 2

$$E = (0.36) - \frac{(0.059V)}{2}\log\frac{0.0004}{0.2} = 0.36 - \frac{(0.059\ V)}{2} \times (-2.69990) = 0.36\ V + 0.08 = 0.44\ V$$

Step II. Calculation of ΔG :

$\Delta G = -nFE_{cell}$

$E_{cell} = 0.44$ V, n = 2 mol , F = 96500 c mol^{-1}

$\therefore \quad \Delta G = -(2\ mol) \times (96500\ C\ mol^{-1}) \times (0.44\ V)$

$= -84920\ CV = -84920\ J$

ALLEN Session 2019-20

Ex.16 ***The cell Pt | H_2(g) (1 atm) | H^+ ; pH = x || Normal calomal electrode has EMF of 0.64 volt at 25°C. The standard reduction potential of normal calomal electrode is 0.28 V. What is the pH of solution in anodic compartment. Take $\frac{2.303RT}{F} = 0.06$ at 298 K.***

Sol. $E_{H^+/H_2} = -0.06 \log\frac{1}{[H^+]} = -0.06pH$

$\Rightarrow \quad 0.64 = E_{cathode} - E_{Anode} = 0.28 - (-0.06\ pH)$

$\Rightarrow \quad pH = \frac{0.64 - 0.28}{0.06} = \frac{0.36}{0.06} = 6$

Ex.17 ***Consider a Galvenic cell,***

Zn(s) | Zn^{2+}(0.1 M) || Cu^{2+}(0.1 M) | Cu(s)

by what factor, the electrolyte in anodic half cell should be diluted to increase the emf by 9 milli volt at 298 K.

Ans. $E = E^\circ_{cell} - \frac{0.06}{2}\log\frac{[Zn^{2+}]}{[Cu^{2+}]}$

$E_1 = E^\circ_{cell} - 0.03\ \log\frac{(0.1)}{(0.1)}$

$E_2 = E^\circ_{cell} - 0.03\ \log\frac{(0.1/x)}{(0.1)}$ {x is the factor by which electrolyte is diluted.}

$E_2 - E_1 = 9 \times 10^{-3} = 0 - 0.03\ \log\left(\frac{1}{x}\right)$

$0.009 = 0.03 \log X$

$\frac{9\times10^{-3}}{3\times10^{-2}} = 0.3 = \log X$

$X = 2$

Ex.18 ***A disposable galvanic cell Zn$|Zn^{2+}\|Sn^{2+}|Sn$ is produced using 1.0 mL of 0.5 M $Zn(NO_3)_2$ and 1.0 mL of 0.50 M $Sn(NO_3)_2$. It is needed to power a pace-maker that draws a constant current of 10^{-6} Amp to run it and requires atleast 0.50 V to function. Calculate the value of [Zn^{2+}] when cell reaches 0.5 V at 298 K.***

(Given : E° (Zn^{2+}|Zn) = – 0.76V; E° (Sn^{2+}|Sn) = – 0.14 V).

Ans. **1**

$E = 0.5\ V = 0.62 - \frac{0.059}{2}\log\frac{0.5+x}{0.5-x}$

$\because \quad Zn + \underset{0.5-x}{Sn^{2+}} \rightleftharpoons Sn + \underset{0.5+x}{Zn^{2+}}$

$\Rightarrow \quad \log\frac{0.5+x}{0.5-x} = \frac{0.12\times2}{0.0592} \approx 4$

$\Rightarrow \quad \frac{0.5+x}{0.5-x} = 10^4$

or $\quad x \approx 0.5\ M$

$\therefore \quad [Zn^{2+}] = 1.0\ M$

ALLEN Session 2019-20

Ex.19 ***An alloy of Pb-Ag weighing 54 mg was dissolved in desired amount of HNO_3 & volume was made upto 500ml. An Ag electrode was dipped in solution and then connected to standard hydrogen electrode anode. Then calculate % of Ag in alloy.***

Given : $E_{cell} = 0.5\ V$; $E^{\circ}_{Ag^+|Ag} = 0.8V$ $\quad \dfrac{2.303RT}{F} = 0.06$

Ans. $Ag^+ + 1\ e^- \longrightarrow Ag$

$E_{cell} = 0.5 = 0.8 + \dfrac{0.06}{1} \log [Ag^+]$

$\log [Ag^+] = \dfrac{-0.30}{0.06} = -5$

$[Ag^+] = 10^{-5}$ mol|L

moles of Ag^+ in 500 ml $= \dfrac{10^{-5}}{2}$

Mass of Ag $= \dfrac{10^{-5}}{2} \times 108$

$\% \ Ag = \dfrac{\dfrac{10^{-5}}{2} \times 108}{54 \times 10^{-3}} \times 100 = 1$

Ex.20. ***A solution contains A^+ and B^+ in such a concentration that both deposit simultaneously. If current of 9.65 amp was passed through 100 ml solution for 55 seconds then find the final concentration of A^+ ions if initial concentration of B^+ is 0.1M.***

Given : $A^+ + e^- \longrightarrow A \qquad E^{\circ} = -0.5$ *volt*

$B^+ + e^- \longrightarrow B \qquad E^{\circ} = -0.56$ *volt*

$\dfrac{2.303RT}{F} = 0.06$

Sol. $-0.5 - \dfrac{0.06}{1} \log \dfrac{1}{[A^+]} = -0.56 - \dfrac{0.06}{1} \log \dfrac{1}{[B^+]}$

$0.06 = \dfrac{0.06}{1} \log \dfrac{[B^+]}{[A^+]}$

$\dfrac{[B^+]}{[A^+]} = 10$

$[A^+]_{initial} = 0.01M$

Ex.21 ***While the discharging of a lead storage battery following reaction take place.***

$PbO_2 + Pb + 4H^+ + 2SO_4^{-2} \rightarrow 2PbSO_4 + 2H_2O$; $E^{\circ} = 2.01$

Calculate the energy (in kJ) obtained from a lead storage battery in which 0.014 mol of lead is consumed. Assume a constant concentration of 10.M H_2SO_4 (log2 = 0.3)

Ans. $E = E^{\circ} - \dfrac{0.059}{n} \log \dfrac{1}{[H^+]^4 [SO_4^{-2}]^2} = 2.01 - \dfrac{0.059}{2} \log \dfrac{1}{[20]^4 [10]^2} = 2.22\ V$

Energy = qE

$= 2 \times 0.014 \times 96500 \times 2.22$

$= 6000\ J = 6\ kJ$

ALLEN Session 2019-20

Ex.22 ***Consider the following standard reduction potentials :-***

$Fe^{2+} + 2e^- \rightleftharpoons Fe$; $E^\circ = -0.41\ V$

$Ag^+ + e^- \rightleftharpoons Ag$; $E^\circ = 0.80\ V$

$O_2 + 2H_2O + 4e^- \rightleftharpoons 4OH^-$; $E^\circ = 0.40\ V$

What would happen if a block of silver metal is connected to a buried iron pipe via a wire:-

(A) The silver metal would corrode, a current would be produced in the wire, and O_2 would be reduced on the surface of the iron pipe.

(B) The silver metal would corrode, a current would be produced in the wire, and Fe^{2+} would be reduced on the surface of the iron pipe.

(C) The iron pipe would corrode, a current would be produced in the wire , and Ag^+ would be reduced on the surface of the silver metal.

(D) The iron pipe would corrode, no current would be produced in the wire, and O_2 would be reduced on the surface of the iron pipe.

Sol. **(D)**

7. ELECTROLYSIS :

The process of decomposition of an electrolyte by the passage of electricity is called ***electrolysis*** or electrolytic dissociation. It is carried out in electrolytic cell where electrical energy is converted into chemical energy. For electrolysis to take place two suitable electrodes are immersed in the liquid or solution of an electrolyte containing ions. When an electric potential is applied between the electrodes, the positively charged ions move towards the negative cathode and negatively ions move towards the positive anode, when a cation reaches the cathode, its takes up electron(s) and thus gets its charge neutralised. Thus the gain of electrons (decrease in oxidation number) means reduction takes place at the cathode.

Similarly when an anion it reaches the anode, gives up electron(s) and thus gets discharged. Loss of electrons (Increase in oxidation number) means oxidation takes place at anode.

- The tendency of an electrode to lose electrons is known as the ***oxidation potential***.
- The tendency of an electrode to gain electrons is known as the ***reduction potential***.
- ***Greater oxidation potential means stronger is tendency to get oxidised and act as a reducing agent or reductant.***
- ***Greater reduction potential means stronger is tendency to get reduced and act as an oxidising agent (oxidant)***.

(a) Electrolysis of fused sodium chloride :

When fused sodium chloride is electrolysed, Na^+ ions moves towards the cathode and Cl^- ions moves towards the anode. At cathode Na^+ ions accept electrons to form sodium metal. At anode each Cl^- ion loses an electron to form Cl_2 gas.

- **At anode :** $Cl^- \longrightarrow \frac{1}{2}Cl_2 + e^-$; $E^0_{OP} = -1.36V$
- **At cathode :** $Na^+ + e^- \longrightarrow Na(s)$; $E^0_{RP} = -2.71V$

(b) Electrolysis of aqueous solution of KBr

The solution of KBr contain K^+, Br^- & small amounts of H^+, OH^- (due to small dissociation of water)

- **At anode :** $2Br^-(aq.) \longrightarrow Br_2(g) + 2e^-$
- **At cathode :** $2H_2O(\ell) + 2e^- \longrightarrow H_2(g) + 2OH^-(aq)$
- If more than one types of ions are present at a given electrode, then the one ion is the liberated which requires least energy. The energy required to liberate an ion is provided by the applied potential between electrodes. This potential is called ***discharge or deposition potential***.

Note :

1. In aqueous solution most electropositive metal cations for eg. (s-block & Al^{3+}) will not discharg at cathode instead H_2O is reduced. $\mathbf{2H_2O(\ell) + 2e^- \longrightarrow H_2(g) \text{ - } + 2OH^-(aq)}$

2. In aqueous solution cations of moderately electropositive metals (Mn, Co, Fe, Zn etc.) and least electropositive metals (Cu, Hg, Au, Ag, Pt) get discharged at cathode first.

3. Active vs Inactive electrodes :

- Sometimes the metal electrodes in the cell are active and the metals themselves are components of the half reactions or influence the reaction of electrode.
- For many redox reactions, however, there are no reactants or products capable of serving as electrodes. Inactive electrodes are used, most commonly rods of graphite or platinum, materials that conduct electrons into or out of the cell but cannot take part in the half-reactions.
- Platinum is used $\because$ highly conducting , unreactive.
  highly malleable and ductile.

❖ **Examples of Electrolysis :**

- **Electrolysis of aq. $PbBr_2$** (Using inert (Pt|graphite) electrodes).

Cathode : $Pb^{2+} + 2e^- \rightarrow Pb(s)$ $\quad E^0 = 0.126V$

Anode : $2Br^- \rightarrow Br_2 + 2e^-$ $\quad E^0 = -1.08$ V

$E_{cell} = -0.126 - (0.108) \times 10 = -1.206$ V

$E_{ext} > 1.206$ V

- **Electrolysis of aq. $CuSO_4$** (Using inert (Pt|graphite) electrodes).

Cathode : $Cu^{2+} + 2e^- \rightarrow Cu(s)$ $\quad E^0 = 0.34$ V

$2e + 2H_2O(\ell) \rightarrow H_2(g) + 2OH^-(aq)$ $\quad E^0 = -0.83V$

Anode : $2SO_4^{2-} \rightarrow S_2O_8^{2-} + 2e^-$ $\quad E^0 = -2.05$ V

$2H_2O(\ell) \rightarrow O_2 + 4H^+ + 4e^-$ $\quad E^0 = -1.23$ V

$\therefore$ Cu discharged at cathode and O_2 at anode.

- **Electrolysis of aq. NaCl** (Using inert (Pt|graphite) electrodes).

Cathode : $Na^+ + e^- \rightarrow Na$ $\quad E^0 = -2.71$ V

$2e^- + 2H_2O(\ell) \rightarrow H_2(g) + 2OH^-$ $\quad E^0 = -0.83$ V

Anode : $2Cl^- \rightarrow Cl_2 + 2e^-$ $\quad E^0_{OX} = -1.30$ V

$2H_2O(\ell) \rightarrow O_2 + 4H^+ + 4e^-$ $\quad E^0_{OX} = -1.23$ V

Session 2019-20

Rate of production of Cl_2 is more than rate of production of O_2 gas because of greater activation energy barrier for O_2 production, therefore Cl_2 is released at anode and H_2 at cathode.

Note:(i) As observed from electrode potential values discharge potential for O_2 is less than for Cl_2. According to thermodynamics, oxidation of H_2O to produce O_2 should take place on anode but experimentally (experiment from chemical kinetics) the rate of oxidation of water is found to be very slow. To increase it's rate, the greater potential difference has to be is applied called over voltage or over potential.

Because of this oxidation of Cl^- ions also become feasible and this takes place at anode. Electrode potential values do not take into account such effects as : Activation energy of process or non-uniform ionic concentration in solution.

(ii) Electrode potentials are thermodynamic intensive properties obtained experimentally under ideal & standard conditions. Sometimes in working conditions additional potentials are required for discharging. This difference is termed as overvoltage or overpotential.

❖ **Electrolysis using attackable (reactive) electrodes :**

• **Electrolysis of aq. $CuSO_4$ using Cu electrode.**

Anode (oxidation) : $SO_4^{2-} \rightarrow S_2O_8^{2-} + 2e^-$ $\quad E^0_{OX} = -2.05$ V

$2H_2O(\ell) \rightarrow O_2 + 2H^+ + 4e^-$ $\quad E^0 = -1.23$ V

$Cu(s) \rightarrow Cu^{2+} + 2e^-$ $\quad E^0 = -0.34$ V

Cathode (reduction) : $Cu^{2+} + 2e^- \rightarrow Cu$ $\quad E^0 = +0.34$ V

$2H_2O(\ell) + 2e^- \rightarrow H_2(g) + 2OH^-$ $\quad E^0 = -0.83$ V

$\therefore$ Both oxidation and reduction of copper occurs and density of solution remains constant.

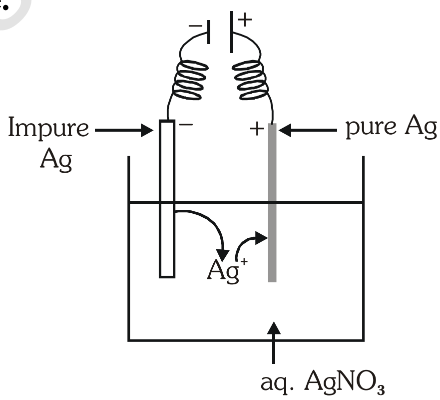
• **Electrolytic Refining**

Electrolysis of $AgNO_3$(aq) using Ag cathode & Ag anode.

Impure metal $\rightarrow$ Anode ;

Pure Metal $\rightarrow$ Cathode ;

Metal salt solution $\rightarrow$ Electrolyte



Anode : $NO_3^- \rightarrow X$ (No reaction)

$2H_2O(\ell) \rightarrow O_2 + 4H^+ + 4e^-$ $\quad E^0 = -1.23$V

$Ag(s) \rightarrow Ag^+(aq) + e^-$ $\quad E^0 = -0.80$V
(impure)

Cathode : $Ag^+ + e^- \rightarrow Ag(s)$ (pure) $\quad E^0 = 0.8$V

$2H_2O(\ell) + 2e^- \rightarrow H_2(g) + 2OH^-$ $\quad E^0 = -0.83$V

$\therefore$ Both oxidation and reduction of Ag occurs and mass transfer of Ag occurs from anode (impure Ag) to cathode (pure Ag). Electrical energy provided by battery is used for mass transfer of Ag from anode to cathode.

ALLEN Session 2019-20
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PRODUCTS OF ELECTROLYSIS OF SOME ELECTROLYTES

S.No.	Electrolyte	Electrode	Product obtained at anode	Product obtained at cathode
(i)	Fused NaCl(molten)	Pt or Graphite	Cl_2	Na
(ii)	Aqueous NaCl(conc.)	Pt or Graphite	Cl_2	H_2
(iii)	dil.NaCl	Pt or Graphite	O_2	H_2
(iv)	Aqueous NaOH	Pt or Graphite	O_2	H_2
(v)	Fused NaOH	Pt or Graphite	O_2	Na
(vi)	Aqueous $CuSO_4$	Pt or Graphite	O_2	Cu
(vii)	Dilute HCl	Pt or Graphite	Cl_2	H_2
(viii)	Dilute H_2SO_4	Pt or Graphite	O_2	H_2
(ix)	Aqueous $AgNO_3$	Pt of Graphite	O_2	Ag
(x)	Aqueous CH_3COONa	Pt of Graphite	$CH_3–CH_3 + CO_2$	H_2

8. FARADAY'S LAWS OF ELECTROLYSIS :

Michael Faraday on basis of experiments deduced two important laws :

(a) **Faraday's first law of electrolysis :** This law states that ***"The amount of a substance deposited or discharged at an electrode is directly proportional to the charge passing through the electrolytic solution".***

If a current of **I** amperes is passed for **t** seconds, (the quantity of charge Q in coulombs). If W gram of substances is deposited by Q coulombs of electricity, then

$$\mathbf{W \propto Q \propto i \times t}$$

$$W = Z \times i \times t = \frac{E}{96500} \times i \times t = \eta \times \frac{E}{F} \times i \times t$$

moles of e^- = $n_e = \dfrac{\eta \times i \times t}{F}$ ***= no. of equivalents of species discharged*** During electrolysis :

Where 1 Faraday (1F) is defined as charge of 1 mole electrons = eN_A = 1 F ≅ 96500 C

Hence faraday (F) is the quantity of charge in coulombs required to deposit one g equivalent of any substance.

E = Equivalent mass of species discharged

η = current efficiency in fraction if current efficiency is not mentioned , by default it is assumed to be 1 (100%).

Z is constant of proportionality and is known as ***electrochemical equivalent***. Its value is different for different species, when Q = 1 coulomb , W = Z, thus electro chemical equivalent may be defined as the weight in grams of an element liberated by the passage of 1 coulomb of electricity.

Electrochemical equivalent of species (Z) = $\dfrac{E}{96500}$ gm | coulomb.

(b) **Faraday's second law :** This law states that the amounts of different substances deposited in different solutions connected in series at electrodes by passage of the same quantity of electricity are proportional to their equivalent masses(E).

W ∝ E (E = equivalent mass)

ALLEN Session 2019-20

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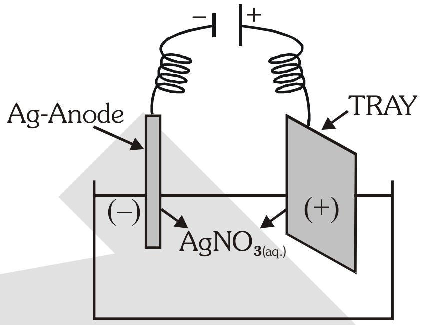
If W_1 and W_2 be the amounts of two different substances deposited at electrodes and E_1 and E_2 be the equivalent weights respectively then -

$$\frac{W_1}{W_2} = \frac{E_1}{E_2}$$

9. APPLICATION OF ELECTROLYSIS



(i) Electroplating : Metal used for plating $\rightarrow$ ANODE
Object to be plated $\rightarrow$ CATHODE

(ii) Electrorefining : Impure metal $\rightarrow$ ANODE (see fig.)
Pure metal $\rightarrow$ CATHODE
Metal salt solution $\rightarrow$ electrolyte

(iii) Electro-metallurgy : (Electrolytic reduction)

Ex. Electrolysis of NaCl (from seawater) Using Hg-cathode.

Anode : $2Cl^- \longrightarrow Cl_{2(g)} + 2e$

Cathode : $Na^+ + e^- \xrightarrow{Hg} Na\text{–}Hg$

EXERCISE # III

49. Calculate the no. of electrons lost or gained during electrolysis of
(a) 3.55 gm of Cl^- ions (b) 1 gm Cu^{2+} ions (c) 2.7 gm of Al^{3+} ions

50. How many faradays of electricity are involved in each of the case
(a) 0.25 mole Al^{3+} is converted to Al.
(b) 27.6 gm of SO_3 is convered to SO_3^{2-}
(c) The Cu^{2+} in 1100 ml of 0.5 M Cu^{2+} is converted to Cu.

51. 0.5 mole of electron is passed through two electrolytic cells in series. One contains silver ions, and the other zinc ions. Assume that only cathode reaction in each cell is the reduction of the ion to the metal. How many gm of each metals will be deposited.

52. If 0.224 litre of H_2 gas is formed at the cathode, how much O_2 gas is formed at the anode under identical conditions?

53. Chromium metal can be plated out from an acidic solution containing CrO_3 according to following equation:
$CrO_3(aq) + 6H^+(aq) + 6e^- \rightarrow Cr(s) + 3H_2O$
Calculate :
(i) How many grams of chromium will be plated out by 24125 coulombs and
(ii) How long will it take to plate out 1.5 gm of chromium by using 12.5 ampere current

54. Which of the substances Na, Hg, S, Pt and graphite can be used as electrodes in electrolytic cells having aqueous solution ?
(A) Hg and Pt (B) Hg, Pt and graphite
(C) Na, S (D) Na, Hg, S

ALLEN Session 2019-20

55. The products formed when an aqueous solution of NaBr is electrolyzed in a cell having inert electrodes are :

(A) Na and Br_2 (B) Na and O_2 (C) H_2, Br_2 and NaOH(D) H_2 and O_2

56. A solution of sodium sulphate in water is electrolysed using inert electrodes. The products at the cathode and anode are respectively.

(A) H_2, O_2 (B) O_2, H_2 (C) O_2, Na (D) none

57. When an aqueous solution of lithium chloride is electrolysed using graphite electrodes

(A) Cl_2 is liberated at the anode.

(B) Li is deposited at the cathode

(C) as the current flows, pH of the solution remains constant

(D) as the current flows, pH of the solution decreases.

58. The amount of an ion discharged during electrolysis is not directly proportional to :

(A) resistance (B) time

(C) current strength (D) electrochemical equivalent of the element

59. Number of electrons involved in the electrodeposition of 63.5 g of Cu from a solution of $CuSO_4$ is : ($N_A = 6 \times 10^{23}$)

(A) 6×10^{23} (B) 3×10^{23} (C) 12×10^{23} (D) 6×10^{22}

60. When one coulomb of electricity is passed through an electrolytic solution the mass deposited on the electrode is equal to :

(A) equivalent weight (B) molecular weight

(C) electrochemical equivalent (D) one gram

61. Electro chemical equivalent of a substance is 0.0006; its e wt. is :

(A) 57.9 (B) 28.95

(C) 115.8 (D) cannot be calculated

62. W g of copper deposited in a copper voltameter when an electric current of 2 ampere is passed for 2 hours. If one ampere of electric current is passed for 4 hours in the same voltameter, copper doposited will be :

(A) W (B) W|2 (C) W|4 (D) 2W

63. When the same electric current is passed through the solution of different electrolytes in series the amounts of elements deposited on the electrodes are in the ratio of their:

(A) atomic number (B) atomic masses (C) specific gravities (D) equivalent masses

64. The amount of electricity that can deposit 108 g. of silver from silver nitrate solution is:

(A) 1 ampere (B) 1 coulomb

(C) 1 Faraday (D) 2 ampere

65. The ratio of weights of hydrogen and magnesium deposited by the same amount of electricity from aqueous H_2SO_4 and fused $MgSO_4$ are :

(A) 1 : 8 (B) 1 : 12 (C) 1 : 16 (D) None of these

66. A current of 9.65 amp. flowing for 10 minute deposits 3.0 g of a metal. The equivalent wt. of the metal is :

(A) 10 (B) 30 (C) 50 (D) 96.5

67. 1 mole of Al is deposited by X coulomb of electricity passing through aluminium nitrate solution. The number of moles of silver deposited by X coulomb of electricity from silver nitrate solution is
(A) 3 (B) 4 (C) 2 (D) 1

68. When an electric current is passed through acid diluted water, 112 ml. of hydrogen gas at STP collects at the cathode in 965 second. The current passed, in ampere is :
(A) 1.0 (B) 0.5 (C) 0.1 (D) 2.0

69. A factory produces 40 kg. of calcium in two hours by electrolysis. How much aluminium can be produced by the same current in two hours :–
(At wt. of Ca = 40, Al = 27)
(A) 22 kg. (B) 18 kg. (C) 9 kg. (D) 27 kg.

70. Calculate the volume of hydrogen at STP obtained by passing a current of 0.536 ampere through acidified water for 30 minutes.
(A) 0.135 litre (B) 0.227 litre (C) 0.057 litre (D) 0.454 litre

71. An electric current is passed through silver voltameter connected to a water voltameter in series. The cathode of the silver voltameter weighed 0.108g more at the end of the electrolysis. The volume of oxygen evolved at STP is :
(A) $56.75 cm^3$ (B) $567.5\ cm^3$ (C) $5.675\ cm^3$ (D) $113.5\ cm^3$

72. When, during electrolysis of a solution of $AgNO_3$ 9650 coulombs of charge pass through the electroplating bath, the mass of silver deposited on the cathode will be : **[AIEEE 2003]**
(A) 21.6g (B) 108g (C) 1.08g (D) 10.8g

73. Aluminium oxide may be electrolysed at 1000°C to furnish aluminium metal (1 Faraday = 96500 Coulombs).The cathode reaction is **[AIEEE 2005]**
$Al^{3+} + 3e^- \longrightarrow Al$
To prepare 5.12 kg of aluminium metal by this method would require,
(A) 5.49×10^4 C of electric charge (B) 5.49×10^1 C of electric charge
(C) 5.49×10^7 C of electric charge (D) 1.83×10^7 C of electric charge

Ex.23 How much charge is present on 1 mole of Cu^{+2} ion in faraday. (1 Faraday = 96500 coulomb)

Ans. 1 mole e^- = 1 mol proton = 1F

2 Faraday

Ex.24 For an element 'X' the process of oxidation is : $X_2O_4^{-2} \longrightarrow$ product
If 965 A current when passed for 100 seconds for 0.1 mol of $X_2O_4^{-2}$, find oxidation state of X in new compound ?

Ans. +3

$X_2O_4^{-2} \longrightarrow 2X^{+n}$

$\because$ oxidation process 3 < n , N-factor = 2 (n – 3)

so $0.1 \times 2(n-3) = \dfrac{i \times t}{96500} = \dfrac{965 \times 100}{96500}$

2(n–3) = 10, n – 3 = 5 , n = 8

ALLEN Session 2019-20

Ex.25 ***How many litres of chlorine at 1atm & 273K will be deposited by 100 amp. current flowing for 5 hours through molten NaCl?***

Sol : $Q = It = 100 \times 5 \times 60 \times 60 = 18 \times 10^5 C$

$$W = ZQ = \frac{E}{96500} \times 18 \times 10^5 = \frac{18E}{96500} \times 10^5 = 662.2 \text{ g}, \left(E_{Cl_2} \frac{71}{2} = 35.5\right)$$

$\because$ Volume of 71 g Cl_2 at 1atm & 273K = 22.4 L

$\therefore$ Volume of 662.2 g Cl_2 at NTP = $\frac{22.4}{71} \times 662.2 = 208.9$ L

Ex.26 ***How much time is required for complete decomposition of two moles of water at anode using 4 amperes current ?***

Sol. $H_2O(l) \longrightarrow H_2(g) + \frac{1}{2}O_2(g)$

For H_2 : $2 \times 2 = \frac{4 \times t}{96500} \Rightarrow t = 96500$ sec,

Ex.27 ***How much charge (in F) must flow through solution during electrolysis of aq. Na_2SO_4 at 0ºC and 1 atm to produce 33.6 L of product gases at 50% current efficiency ?***

Ans. $H_2O \xrightarrow{2F} H_2 + \frac{1}{2}O_2$

V $\quad$ V|2 $\Rightarrow \frac{3}{2}V = 33.6L \Rightarrow V = 22.4$ L

$Q \times \frac{50}{100} = 2F \Rightarrow Q = 4F$

Ex.28 ***Calculate the time required to coat a metal surface of 80 cm^2 with 0.005 mm thick layer of silver (density = 10.5 g cm^{-3}) with the passage of 3A current through silver nitrate solution.***

Sol. $\because$ Volume of layer of silver = $0.005 \times 10^{-1} \times 80 = 0.04$ cm^3

$\therefore$ Mass = Density $\times$ volume = $10.5 \times 0.04 = 0.42$ g

So $w = \frac{E}{96500} \times It \Rightarrow 0.42 = \frac{108}{96500} \times 3 \times t$

$t = \frac{0.42 \times 96500}{108 \times 3} = 125.09$ seconds.

Ex.29 ***A Solution of copper (II) sulphate is electrolysed between copper electrodes by a current of 10.0 amperes passing for one hour. What changes occur at the electrodes and in the solution?***

Sol. According to Faraday's first law of electrolysis :

The reaction at cathode : $Cu^{2+} + 2e^- \longrightarrow Cu$

63.5 $\quad$ 2×96500 $\quad$ C

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The quantity of charge passed = I × t = (10 amp) × (60 × 60s) = 36000 C.

2 × 96500 C of charge deposit copper - 63.5 g

36500 C of charge deposit copper $= \dfrac{(63.5\ g)}{(2\times 96500\ C)}\times(36000\ C) = 11.84\ g$

Thus, 11.84 g of copper will dissolve from the anode and the same amount from the solution will get deposited on the cathode. The concentration of the solution will remain unchanged.

Ex.30 ***An aqueous solution of Na_2SO_4 was electrolysed for 10 min. 82 ml of a gas was produced at anode and collected over water at 27ºC at a total pressure of 580 torr. Determine the current that was used in amp. , Given : Vapour pressure of H_2O at 27ºC = 10 torr, R = 0.082 atm L|mol-K***

Ans **(1.6A)**

At anode reaction will be

$2H_2O \longrightarrow O_2 + 4H^+ + 4e^-$

V_{O_2} collected = 82 ml

By PV = nRT $\quad \dfrac{(580-10)}{760}\times\dfrac{82}{1000} = (n_{O_2})\times 0.0821\times 300$, $n_{O_2} = \dfrac{1}{400}$

By Faraday law $\dfrac{W}{E} = \dfrac{i\times t}{96500}$

$\left(\dfrac{W}{M}\right)\times n = \dfrac{i\times t}{96500}$, $\left(\dfrac{1}{400}\times 4\right) = \dfrac{i\times 10\times 60}{96500}$, i = 1.6 A

Ex.31 ***The same current if passed through solution of silver nitrate and cupric salt connected in series. If the weight of silver deposited is 1.08 g. calculate the weight of copper deposited.***

Sol. According to faradays second law

$\dfrac{W_1}{W_2} = \dfrac{E_1}{E_2} \Rightarrow \dfrac{1.08}{W_2} = \dfrac{108}{31.75} \Rightarrow W_2 = 0.3175\ g$

10 ELECTROLYTIC CONDUCTANCE

10.1 Resistance (R) :

Metallic and electrolytic conductors obey ohm's law according to which the resistance of a conductor is the ratio of the applied potential difference (V) to the current following(I).

$R = \dfrac{V}{I}$ • R is expressed in ohms.

• In the case of solution of electrolytes, the resistance offered by the solution to the flow of current is –

(a) Directly proportional to the distance between the electrodes

$$\mathbf{R} \propto \ell$$

(b) Inversely proportional to the area of cross section of the electrodes

$$R \propto \frac{1}{A}$$

ALLEN Session 2019-20

10.2 Conductance or resistivity (G) :

The conductance of a conductor is equal to reciprocal of resistance.

$G = \frac{1}{R}$ • G is expressed in mho or Ω^{-1} or Siemen(S).

$[1S = 1W^{-1}$ S.I. unit]

10.3 Specific resistance or conductivity (ρ) :

The resistance (R) of a conductor of uniform cross section is directly proportional to its length(ℓ) and inversely proportional to its area of cross section (A).

$R \propto \frac{\ell}{A}$ $\qquad R = \rho \frac{\ell}{A}$

where ρ is a constant and called resistivity or specific resistance.

When $\ell = 1$, $A = 1$, then $\rho = R$ thus the specific resistance may be defined as the resistance of a conductor of unit length and unit area of cross section.

• Unit of $\rho \rightarrow$ ohm.cm

10.4 Specific conductance (κ) :

It is defined as the reciprocal of specific resistance

$$\kappa = \frac{1}{\rho},$$

$$G = \kappa / G^*, \quad G^* = \frac{l}{a} = \text{cell constant}$$

If $\ell = 1$ cm & $A = 1$ cm^2 then $\kappa = G$

Hence conductivity or specific conductance (F) of a solution is defined as the conductance of one centimeter cube of the solution of the electrolyte.

- Cell constant is a fixed quantity for a particular cell and is defined as the distance between two parallel electrodes of a cell divided by the area of cross section of the electrodes.

  κ = G × cell constant
- Unit of $\kappa \rightarrow$ ohm^{-1} cm^{-1}
- SI unit of $\kappa \rightarrow$ S m^{-1}

  1 S m^{-1} = 100 ohm^{-1} cm^{-1}

10.5 Molar conductance (λ_m or $\wedge_m$) : It is defined as the product of specific conductance (κ) and the volume (V in mL) in which contains one mole of the electrolyte.

$\Lambda_m = \kappa \times v$ and $\Lambda_m = \frac{\kappa \times 1000}{M}$ **ohm^{-1} cm^2 mol^{-1}** $\qquad$ **[SI unit : S m^2 mol^{-1}]**

- It can also be defined as conductance of 1 mole electrolyte completely dissolved between two plates separated by unit distance.

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10.6 Equivalent conductance (λ_{eq} or Λ_{eq}) : It is defined as the product of specific conductance (κ) and the volume (V in mL) in which contains one equivalent of the electrolyte.

$\Lambda_m = \kappa \times v$ and $\Lambda_m = \dfrac{\kappa \times 1000}{M}$ **$ohm^{-1}\ cm^2\ eq^{-1}$** **[SI unit : $S\ m^2\ eq^{-1}$]**

- It can also be defined as conductance of 1 equivalent electrolyte completely dissolved between two plates separated by unit distance.

- **Relation between $\Lambda_{eq.}$ and Λ_m :**

$\Lambda_m = \dfrac{\kappa \times 1000}{M}$ and $\Lambda_{eq.} = \dfrac{\kappa \times 1000}{N}$

We know that, ***Normality = Valency Factor × Molarity***

or $N = n \times M \Rightarrow \boxed{\lambda_{eq.} = \dfrac{\lambda_M}{n}}$

\# n = total cationic (or anionic) charge of salt.

Ex. $\Lambda_{eq}[Al_2(SO_4)_3] = \dfrac{\Lambda_m[Al_2(SO_4)_3]}{6}$, $L_{eq\ NaCl} = \dfrac{\Lambda_{m\ NaCl}}{1}$, $L_{eq.CaCl_2} = \dfrac{\Lambda_{m\ CaCl_2}}{2}$

11 EFFECT OF DILUTION ON THE CONDUCTIVITY OF ELECTROLYTES

(i) The degree of ionisation of weak electrolytes increases with the increase of dilution of the solution the conductivity is increases due to increasing the number of ions.

(ii) Effect of dilution on specific conductance :

Specific conductance decreases with the increase of dilution of the solution due to the presence of no. of ions in 1 cm^3 solution decreases conductance also decrease on dilution.

(iii) Effect of dilution on equivalent|molar conductivity :

The equivalent|molar conductivity increases with dilution. For strong electrolyte λ_m or λ_{eq} increases very slowly but for weak electrolytes λ_m & λ_{eq} increase sharply on dilution.

- When the whole of the electrolyte has ionised, further addition of the water brings a small change in the value of equivalent|molar conductance. This stage is called ***infinite dilution***.
- The ratio of equivalent conductivity at any dilution to equivalent conductivity at infinite dilution is called ***conductivity ratio*** or degree of dissociation of solute -

$$\boxed{\alpha = \frac{\lambda_{eq.}}{\lambda_m^{\infty}} = \frac{\lambda_m}{\lambda_m^{\infty}}}$$ [λ_m^{∞} = molar conductance at ∞ dilution.]

❖ **Variation of conductivity and molar conductivity with concentration :**

- Conductance and conductivity always decreases with the decrease in concentration both for weak and strong electrolytes.
- It is because the number of ions per unit volume that carry the current in a solution decreases on dilution decreasing I, C and K.

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ALLEN Session 2019-20

❖ **Strong Electrolytes :**

- For strong electrolytes. Λ increases slowly with dilution and can be represented by the equation

  $\mathbf{\Lambda = \Lambda^\circ - A\, C^{1/2}}$. A = constant

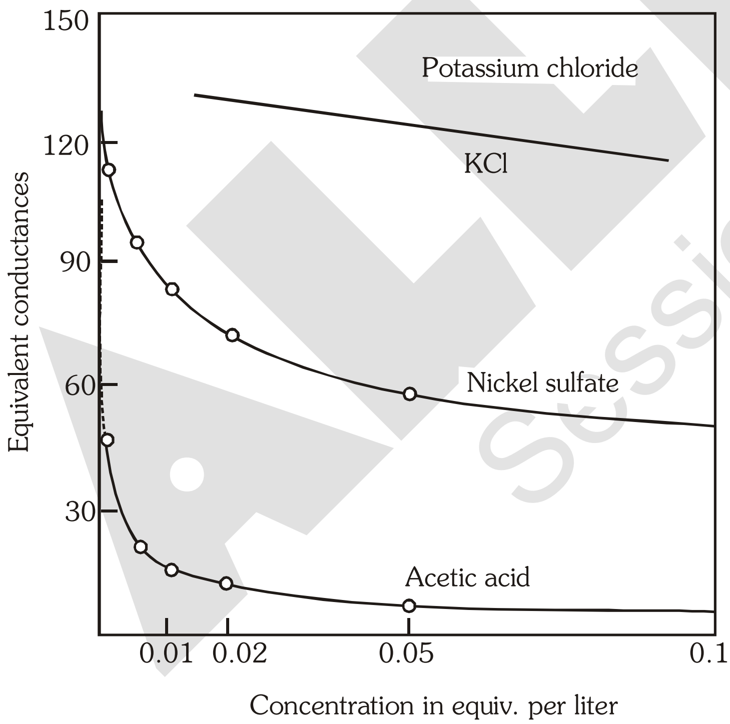
  On dilution interionic separation increases causing free movement & less hindrance. This increases Λ_m & $\Lambda_{eq.}$.

  For strong electrolytes Λ° or Λ^∞ can be calculated graphically from y-intercept.

  It can be seen that if we plot . L_m against $c^{1/2}$, we obtain a straight line with intercept equal to A°_m and slope equal to –A. The value of the constant A for a given solvent and temperature depends on the type of the electrolyte i.e., the charges on the cation and anion produced on the dissociation of the electrolyte in the solution. Thus, NaCl , $CaCl_2$, $MgSO_4$ are known as 1 – 1, 2 – 1 and 2 – 2 electrolytes respectively. All electrolytes of a particular type have the same value for A.

❖ **Weak electrolytes :**

- Weak electrolytes like acetic acid have lower degree of dissociation at higher concentration and hence for such electrolytes, the change in Λ with dilution is due to increase in the number of ions solution containing a given amount of electrolyte.



Conductances of electrolytes of different types

KCl $\rightarrow$ Strong

Ni_2SO_4 $\rightarrow$ Moderate

$CH_3COOH \rightarrow$ Weak electrolyte

Note : (A) Weaker the electrolyte more sharp will be increase of Λ_m or $\Lambda_{eq.}$ on dilution.

(B) same plot is also observed for Λ_m vs. molarity of respective electrolytes.

12. KOHLARAUSCH'S LAW

- "At infinite dilution, when dissociation is complete, each ion makes a definite contribution towards equivalent conductance of the electrolyte irrespective of the nature of the ion with which it is associated and the value of equivalent conductance at infinite dilution for any electrolyte is the sum of contribution of its constituent ions."

i.e., for $A_{n+}B_{n-}$ $\boxed{\begin{array}{l}\Lambda^{\infty}_{eq.} = \Lambda^{\infty}_{eq.}(+) + \Lambda^{\infty}_{eq.}(-)\\ \Lambda^{\infty}_{m} = \nu_{+}\Lambda^{\infty}_{m}(+) + \nu_{-}\Lambda^{\infty}_{m}(-)\end{array}}$

ν_+ = no. of cation in one formula unit of electrolyte.

ν_- = no. of anions in one formula unit of electrolyte.

Note : $\lambda^{\infty} = \lambda^{\circ}$

$$\lambda^0_{eq+} = \frac{\lambda^0_m}{\text{charge on the cation}} \qquad \lambda^0_{eq}\ .Al^{3+} = \frac{\lambda^0_m Al^{3+}}{3}$$

$$\lambda^0_{eq} = \frac{\lambda^0_m}{\text{chargeon the anion}} \qquad \lambda^0_{eq}\text{ , electrolyte} = \frac{\lambda^0_m \text{ electrolyte}}{\text{total +ve charge on cations in electrolyte}}$$

or

total -ve charge on anions in electrolyte

The Independent Migration of Ions. A survey of equivalent conductances at infinite dilution of a number of electrolytes having an ion in common will bring to light certain regularities ;

COMPARISON OF EQUIVALENT CONDUCTANCES AT INFINITE DILUTION

Electrolyte	Λ_0	Electrolyte	Λ_0	Difference
KCl	130.0	NaCl	108.9	21.1
KNO_3	126.3	$NaNO_3$	105.2	21.1
K_2SO_4	133.0	Na_2SO_4	111.9	21.1

Observations of this kind were first made by ***Kohlrausch (1879, 1885)*** by comparing equivalent conductances at high dilutions; described them to the fact that under these conditions every ion makes a definite contribution towards the equivalent conductance of the electrolyte, irrespective of the nature of the other ion with which it is associated in the solution. The value of the equivalent conductance at infinite dilution may thus be regarded as made up of the sum of two independent factors, one characteristic of each ion; this result is known as Kohlrausch's law of independent migration of ions.

The ion conductance is a definite constant for each ion, in a given solvent, its value depending only on the temperature.

It will be seen later that the ion conductances at infinite dilution are related to the speeds with which the ions move under the influence of an applied potential gradient.

ALLEN Session 2019-20

❖ Applications of Kohlarausch's law :

- **Calculate $\Lambda°$ for any electrolyte from the $\Lambda°$ of individual ions.**

An important use of ion conductances is to determine the equivalent conductance at infinite dilution of certain electrolytes which cannot be, or have not been, evaluated from experimental data. For example, with a weak electrolyte the extrapolation to infinite dilution is very uncertain, and with sparingly soluble salts the number of measurements which can be made at appreciably different concentrations is very limited. The value of $\Lambda°$ can, however, so obtained by adding the ion conductances. For example, the equivalent conductance of acetic acid at infinite dilution is the sum of the conductances of the hydrogen and acetate ions; the former is derived from a study of strong acids and the latter from measurements on acetates. It follows, therefore, that at 25°.

$$\Lambda°_{(CH_3CO_2H)} = \lambda^0_{H^+} + \lambda^0_{CH_3CO_2^-} = 349.8 + 40.9 = 390.7 \text{ ohms}^{-1} \text{ cm}^2$$

The same result can be derived in another manner which is often convenient since it avoids the necessity of separating the conductance of an electrolyte into the contributions of its constituent ions. The equivalent conductance of any weak electrolyte MA at infinite dilution it follows, therefore, that

$\Lambda°(MA) = \Lambda°(MCl) + \Lambda°(NaA) - \Lambda°(NaCl)$, [MCl, NaA, NaCl are strong electrolytes]

where $\Lambda°$ (MCl), $\Lambda°$(NaA) and $\Lambda°$(NaCl) are the equivalent conductances at infinite dilution of the chloride of the metal M, i.e., MCI, of the sodium salt of the anion A, i.e., NaA, and of sodium chloride, respectively. Any convenient anion may be used instead of the chloride ion, and similarly the sodium ion may be replaced by another metallic cation or by the hydrogen ion. For example, if M^+ is the hydrogen ion and A^- is the acetate ion, it follows that

$$\begin{aligned}\Lambda°(CH_3COOH) &= \Lambda°(HCl) + \Lambda°(CH_3COONa) - \Lambda°(NaCl)\\ &= 426.16 + 91.0 - 126.45\\ &= 390.71 \text{ ohms}^{-1} \text{ cm}^2 \text{ at } 25.\end{aligned}$$

Similarly :

$$\Lambda_m°[BaSO_4] = \Lambda_m°[BaCl_2] + \Lambda_m°[Na_2SO_4] - 2\Lambda_m°[NaCl]$$

- **Degree of dissociation :**

∴ Degree of dissociation

$$\frac{\text{equivalent conductance at a given concentration}}{\text{equivalent conductance at infinite dilution}}$$

$$\alpha = \frac{\Lambda_m^c}{\Lambda_m^c} = \frac{\Lambda_{eq}^c}{\Lambda_{eq}^c}$$

- **Dissociation constant** of weak electrolyte :

$$K_C = \frac{C\alpha^2}{1-\alpha} \quad \frac{C\left(\frac{\Lambda_m^c}{\Lambda_m^c}\right)^2}{1-\frac{\Lambda_m^c}{\Lambda_m^0}}$$

ALLEN Session 2019-20

- **Solubility(s) and K_{SP} of any sparingly soluble salt.**
  Sparingly soluble salt = Very small solubility
  Solubility = molarity $\cong$ $S \rightarrow 0$
  So, solution can be considered to be of zero conc or infinite dilution.

$$\Lambda_m^s \simeq \Lambda_m^s = \kappa = \frac{1000}{s}$$

$$s = \frac{\kappa \times 1000}{\Lambda_m^0}$$

EXERCISE # IV

74. The resistance of a conductivity cell filled with 0.01N solution of NaCl is 200 ohm at 18°C. Calculate the equivalent conductivity of the solution. The cell constant of the conductivity cell is 0.88 cm^{-1}.
75. The molar conductivity of 0.1 M CH_3COOH solution is 4 S cm^2 $mole^{-1}$. What is the specific conductivity and resistivity of the solution ?
76. The conductivity of pure water in a conductivity cell with electrodes of cross sectional area 4 cm^2 and 2 cm apart is 8×10^{-7} S cm^{-1}.
    (i) What is resistance of conductivity cell ?
    (ii) What current would flow through the cell under an applied potential difference of 1 volt?
77. For 0.01N KCl, the resistivity 800 ohm cm. Calculate the conductivity and equivalent conductance.
78. Equivalent conductance of 0.01 N Na_2SO_4 solution is 120 ohm^{-1} cm^2 eq^{-1}. The equivalent conductance at infinite dilution is 150 ohm^{-1} $cm^2 eq^{-1}$. What is the degree of dissociation in 0.01 N Na_2SO_4 solution?
79. Saturated solution of AgCl at 25°C has specific conductance of 1.12×10^{-6} ohm^{-1} cm^{-1}. The λ_∞ (Ag^+) and $\lambda_\infty(Cl^-)$ are 54 and 58 ohm^{-1} cm^2 | equi. respectively. Calculate the solubility product of AgCl at 25°C.
80. The value of Λ_m^∞ for HCl, NaCl and CH_3CO_2Na are 425, 125 and 100 S cm^2 mol^{-1} respectively. Calculate the value of Λ_m^∞ for acetic acid. If the equivalent conductivity of the given acetic acid is 48 at 25° C, calculate its degree of dissociation.
81. For the strong electroytes NaOH, NaCl and $BaCl_2$ the molar ionic conductivities at infinite dilution are 240×10^{-4}, 125×10^{-4} and 280.0×10^{-4} mho cm^2 mol^{-1} respectively. Calculate the molar conductivity of $Ba(OH)_2$ at infinite dilution.
82. Electrolytic conduction differs from metallic conduction from the fact that in the former
    (A) The resistance increases with increasing temperature
    (B) The resistance decreases with increasing temperature
    (C) The resistance remains constant with increasing temperature
    (D) The resistance is independent of the length of the conductor
83. Which of the following solution of KCl has the lowest value of specific conductance :
    (A) 1 M (B) 0.1 M (C) 0.01 M (D) 0.001 M

ALLEN Session 2019-20

84. Which of the following solutions of KCl has the lowest value of equivalent conductance ?
(A) 1 M (B) 0.1 M (C) .01 M (D) .001 M

85. The molar conductance at infinite dilution of $AgNO_3$, AgCl and NaCl are 115, 120 and 110 respectively. The molar conductance of $NaNO_3$ is :–
(A) 110 (B) 105 (C) 130 (D) 150

86. The equivalent conductivity of 0.1 N CH_3COOH at 25 °C is 80 and at infinite dilution 400. The degree of dissociation of CH_3COOH is :
(A) 1 (B) 0.2 (C) 0.1 (D) 0.5

87. The specific conductance of a 0.01 M solution of KCl is 0.0014 ohm^{-1} cm^{-1} at 25° C. Its equivalent conductance (cm^2 ohm^{-1} $equiv^{-1}$) is :–
(A) 140 (B) 14 (C) 1.4 (D) 0.14

88. The resistance of 0.01 N solution of an electrolyte was found to be 200 ohm at 298 K using a conductivity cell of cell constant 1.5 cm^{-1}. The equivalent conductance of solution is :–
(A) 750 $mhocm^2$ eq^{-1} (B) 75 mho cm^2 eq^{-1}
(C) 750 mho^{-1} cm^2 eq^{-1} (D) 75 $mho^{-1}cm^2$ eq^{-1}

89. The resistance of 0.1 N solution of a acetic acid is 250 ohm. When measured in a cell of cell constant 1.15 cm^{-1}. The equivalent conductance (in ohm^{-1} cm^2 $equiv.^{-1}$) of 0.1 N acetic acid is
(A) 46 (B) 9.2 (C) 18.4 (D) 0.023

90. If 0.01 M solution of an electrolyte has a resistance of 40 ohms in a cell having a cell constant of 0.4 cm^{-1} then its molar conductance in ohm^{-1} cm^2 mol^{-1} is :
(A) 10 (B) 10^2 (C) 10^3 (D) 10^4

91. The conductivity of a saturated solution of $BaSO_4$ is 3.06×10^{-6} ohm^{-1} cm^{-1} and its molar conductance is 1.53 ohm^{-1} cm^2 mol^{-1}. The K_{sp} of $BaSO_4$ will be :
(A) 2×10^{-4} (B) 4×10^{-4} (C) 4×10^{-3} (D) 4×10^{-6}

92. The limiting molar conductivities Λ^0 for NaCl, KBr and KCl are 126,152 and 150Scm^2 mol^{-1} respectively. The Λ^0_m for NaBr is : **[AIEEE 2004]**
(A) 278 S cm^2 mol^{-1} (B) 176 S cm^2 mol^{-1} (C) 128 S cm^2 mol^{-1} (D) 302 S cm^2 mol^{-1}

93.

Electrolyte	Λ^∞ (S cm^2 mol^{-1})
KCl	149.9
KNO_3	145.0
HCl	426.2
NaOAC	91.0
NaCl	126.5

Calculate Λ^∞_{HOAc} using appropriate molar conductances of the electrolytes listed above at infinite dilution in H_2O at 25°C **[AIEEE 2005]**
(A) 390.7 (B) 217.5 (C) 517.2 (D) 552.7

94. The highest electrical conducitivity of the following aqueous solution is of **[AIEEE 2005]**
(A) 0.1 M fluoroacetic acid (B) 0.1 M difluoroacetic acid
(C) 0.1 M acetic acid (D) 0.1 M chloroacetic acid

Session 2019-20

95. The molar conductivities, Λ^0_{NaOAc} and Λ^0_{HCl} at infinite dilution in water at 25°C are 91.0 and 426.2 S cm^2|mol respectively. To calculate Λ^0_{HOAc} the additional value required is :

(A) KCl (B) NaOH (C) NaCl (D) H_2O **[AIEEE 2006]**

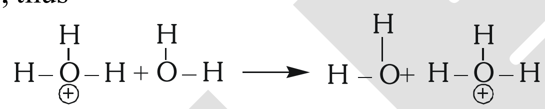
96. Resistance of a conductivity cell filled with a solution of an electrolyte of concentration 0.1M is 100Ω. The conductivity of this solution is 1.29 Sm^{-1}. Resistance of the same cell when filled with 0.02M of the same solution is 520Ω. The molar conductivity of 0.02M solution of the electrolyte will be.

(A) 124×10^{-4} Sm^2 mol^{-1} (B) 1240×10^{-4} Sm^2 mol^{-1} **[AIEEE 2006]**

(C) 1.24×10^{4} Sm^2 mol^{-1} (D) 12.4×10^{-4} Sm^2 mol^{-1}

❖ **Abnormal ion conductances of H^+ and OH^- :**

It is supposed, as already indicated, that the hydrogen ion in water is H_3O^+ with three hydrogen atoms attached to the central oxygen atom. When a potential gradient is applied to an aqueous solution containing hydrogen ions, the latter travel to some extent by the same mechanism as do other ions, but there is in addition another mechanism which permits of a more rapid ionic movement. This second process is believed to involve the transfer of a proton (H^+) from a H_3O^+ ion to an adjacent water molecule; thus



The resulting H_3O^+ ion can now transfer a proton to another water molecule, and in this way the positive charge will be transferred a considerable distance in a short time. The electrical conductance will thus be much greater than that due solely to the normal mechanism.

13. CONDUCTOMETRIC TITRATION :

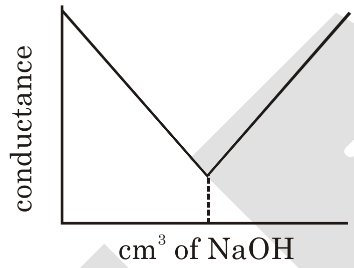
The principle of conductometric titration is based on the fact that during the titration, one of the ions is replaced by the other and invari ably these two ions differ in the ionic conductivity with the result that conductivity of the solution varies during the course of titration. The equivalence point may be located graphically by plotting the change in conductance as a function of the volume of titrant added. In order to reduce the influence of errors in the c onductometric titration to a minimum, the angle between the two branches of the titration curve should be as small as possible (see Fig. 6.2). If the angle is very obtus e, a small error in the conductance data can cause a large deviation. The following app roximate rules will be found

- The smaller the conductivity of the ion which repla ces the reacting ion, the more accurate will be the result. Thus it is preferable C to titrate a silver salt with lithium chloride rather than with HCl. Generally, cations should be titrated with lithium salts and anions with acetates as these ion s have low conductivity
- The larger the conductivity of the anion of the reagent which reacts with the cation to be determined, or vice versa, the more acute is the angle of titration curve.
- The titration of a slightly ionized salt does not g ive good results, since the conductivity increases continuously from the commen cement. Hence, the salt present in the cell should be virtually completely dissociated; for a similar reason; the added reagent should also be as strong electrolyte. The main advantages to the conductometric titration are its applicability to very dilute, and coloured solutions and to system that involver elative incomplete reactions. For example, which neither a potentiometric, nor indica tor method can be used for the neutralization titration of phenol ($K_a \times 10^{-10}$) a conductometric endpoint can be successfully applied.

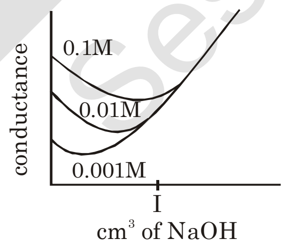
Cation	H_3O^+	NH_4^+	K^+	Na^+	Ag^+	Ca^{2+}	Mg^{2+}
$\lambda^{\infty}_m/(\Omega^{-1}cm^2mol^{-1})$	350.0	73.5	73.5	50.1	62.1	118.0	106.1
Anion	OH^-	Br	Cl^-	NO_3^-	CH_3COO^-	SO_4^{2-}	
$\lambda^{\infty}_m/(\Omega^{-1}cm^2mol^{-1})$	199.2	78.1	76.5	71.4	40.0	159.6	

Some Typical Conductometric Titration Curves are :

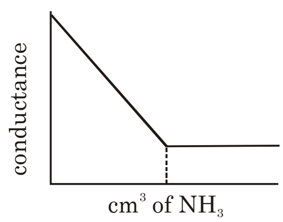
13.1 Before NaOH is added, the conductance is high due to the presence of highly mobile hydrogen ions. When the base is added, the conductance falls due to the replacement of hydrogen ions by the added cation as H^+ ions react with OH - ions to form undissociated water. This decrease in the conducta nce continues till the equivalence point. At the equivalence point, the so lution contains only NaCl. After the equivalence point, the conductance increases due to the large.



13.2 Weak Acid with a Strong Base, e.g. acetic acid with NaOH: Initially the conductance is low due to the feeble ionization of acetic acid. On the addition of base, there is decrease in conductance not only due to the replacement of H^+ by Na^+ but also suppresses the dissociation of acetic acid due to common ion acetate. But very soon, the conductance increases on adding NaOH as NaOH neutralizes the un-dissociated CH_3 COOH to CH_3OONa which is the strong electrolyte. This increase in conductance continues raise up to the equivalence point. The graph near the equivalence point is curved due the hydrolysis of salt CH_3 COONa. Beyond the equivalence point, conductance in creases more rapidly with the addition of NaOH due to the highly conducting OH ions.



13.3 **Strong Acid with a Weak Base, e.g. sulphuric acid with dilute ammonia :** Initially the conductance is high and then it decre ases due to the replacement of H^+. But after the endpoint has been reached the graph becomes almost horizontal, since the excess aqueous ammonia is not appreciably ionised in the presence of ammonium sulphate.



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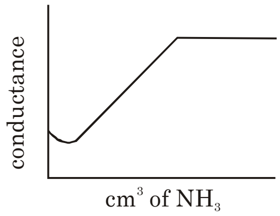
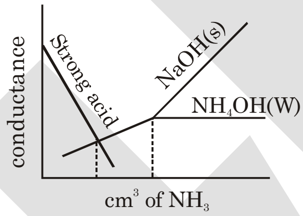
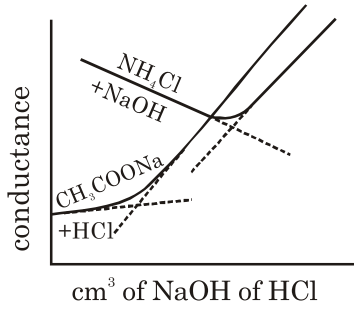
13.4. Weak Acid with a Weak Base : The nature of curve before the equivalence point is similar to the curve obtained by titrating weak acid against strong base. After the equivalence point, conductance virtually remains same as the weak base which is being added is feebly ionized and, therefore, is not much conducting



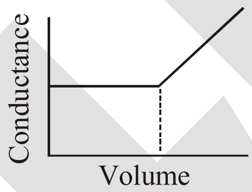
Mixture of a Strong Acid and a Weak Acid vs. a Strong Base or a Weak Base : In this curve there are two break points. The first break point corresponds to the neutralization of strong acid. When the stro ng acid has been completely neutralized only then the weak acid starts neutralizing. The second break point corresponds to the neutralization of weak acid and after that the conductance
increases due to the excess of OH - ions in case of a strong base as the titrant. However, when the titrant is a weak base, it remain s almost constant after the end point similar to



Displacement (or Replacement) Titrations : When a salt of a weak acid is titrated with a strong acid, the anion of the weak acid is replaced by that of the strong acid and weak acid itself is liberated in the undissociated form. Similarly, in the addition of a strong base to the salt of a weak base, the cation of the weak base is replaced by that of the stronger one and the weak base itself is generated in the undissociated form. If for example, M-HCl is added to 0.1 M solution of sodium acetate, the curve shown in Fig.6.7 is obtai ned, the acetate ion is replaced by the chloride ion after the endpoint. The initial increase in conductivity is due to the fact that the conductivity of the chloride ion is slightly geater than that of acetate ion. Until the replace ment is nearly complete, the solution contains enough sodium acetate to suppress the ionization of the liberated acetic acid, so resulting a negligible increase in the conductivity of the solution. However, near the equivalent point, the acetic acid is sufficiently ionized to affect the conductivity and a rounded portion of the curve is obtained. Beyond the equivalence point, when excess of HCl is present (ionization of acetic acid is very much suppressed) therefore, the conductivity arises rapidly. Care must be taken that to titrate a 0.1 M-salt of a weak acid, the dissociation constant should not be more than 5×10^{-4}, for a 0.01 M -salt solution, $K_a < 5 \times 10^{-5}$ and for a 0.001 M-salt solution, $K_a < 5 \times 10^{-6}$, i.e., the ionization constant of the displace acid or base divided by the original concentration of the salt must not exceed above 5×10^{-3} . Fig. 6.6. Also includes the titration of 0.01 M- ammonium chloride solution versus 0.1 M - sodium hydroxide solution. The decrease in conductivity during the displacement is caused by the displacement of ammonium ion of grater conductivity by sodium ion of smaller conductivity.



13.5 Precipitation Titration and Complex Formation Titration : : A reaction may be made the basis of a conductometric precipitation titration provided the reaction product is sparingly soluble or is a stable complex . The solubility of the precipitate (or the dissociation of the complex) should be less than 5%. The addition of ethanol is sometimes recommended to reduce the solubility in the precipitations. An experimental curve is given in F ig. 6.8 (ammonium sulphate in aqueous-ethanol solution with barium acetate). I f the solubility of the precipitate were negligibly small, the conductance at the equivalence point should be given by AB and not the observed AC. The addition of excess of the reagent depresses the solubility of the precipitate and, if the solubility is not too large, the position of the point B can be determine d by continuing the straight portion of the two arms of the curve until they intersect



$AgNO_3$(aq.) vs NaCl

14. IONIC MOBILITY :

It is the speed of ion under unit potential gradient applied through solution.

$$u = \frac{\text{speed of ion (s)}}{\text{Potential gradient}\left(\frac{dV}{dx}\right)} = \frac{cm^2}{\text{volt} - \text{sec.}} = \frac{\Lambda}{ZF}$$

$u/10^{-8}\ m^2\text{-}s^{-1}\text{-}v^{-1}$ in H_2O at 298 K

Li^+	4.01 4.65	Ca^{+2}	6.17	Rb^+	7.92	CH_3COO^-	7.92	CO_3^{--}	7.91	SO_4^{--}	8.29	OH^-	20.64
Na^+	5.19	Ag^+	6.24	H^+	7.92	F^-	7.92	Cl^-	7.96	$[Fe(CN)_6]^{3-}$	7.96		
Cu^{+2}	5.47 5.56	NH_4^+	7.43	H^+	36.23	NO_3^-	7.40	I^-	7.91	$[Fe(CN)_6]^{4-}$	11.4		

Ex.32 The resistance of a 1 N solution of salt is 50 Ω. Calculate the equivalent conductance of the solution, if the two platinum electrodes in solution are 2.1 cm apart and each having an area of 4.2 cm².

Sol : $\kappa = \frac{1}{\rho} = \frac{1}{R}\left(\frac{\ell}{A}\right) = \frac{1}{50} \times \frac{2.1}{4.2} = \frac{1}{100}$ and $\lambda_{eq.} = \frac{\kappa \times 1000}{N} = \frac{1}{100} \times \frac{1000}{1} = 10 = \Omega^{-1}cm^2\text{-}eq^{-1}$

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Ex.33 ***Which of the following have maximum molar conductivity.***

(i) 0.08 M solution and its specific conductivity is $2 \times 10^{-2}\ \Omega^{-1}\ cm^{-1}$.

(ii) 0.1 M solution and its resistivity is $50\ \Omega$ cm.

Sol. (i) $\wedge_m = \dfrac{\kappa \times 1000}{M} = 2 \times 10^{-2} \times \dfrac{1000}{0.08} = 250\ \Omega^{-1}\ cm^2\ mol^{-1}$

(ii) $\wedge_m = \dfrac{\kappa \times 1000}{M}$ $\quad \because\ \kappa = \dfrac{1}{\rho}$ $\quad \therefore\ \Lambda_m = \dfrac{1}{50} \times \dfrac{1000}{0.1} = 200\ \Omega^{-1}\ cm^2\ mol^{-1}$

So, the molar conductivity of 0.08 M solution will be greater than 0.1 M solution.

Ex.34 ***The equivalent conductivity of H_2SO_4 at infinite dilution is 384 $\Omega^{-1}\ cm^2\ eq^{-1}$. If 49 g H_2SO_4 per litre is present in solution and specific resistance is 18.4 Ω–cm then calculate the degree of dissociation.***

Sol : Equivalent of $H_2SO_4 = \dfrac{49}{49} = 1$ N

Specific conductance $= \dfrac{1}{\text{specific resistance}} = \dfrac{1}{18.4}$

$\Rightarrow\ \lambda_{eq.} = \dfrac{1000 \times \kappa}{N} = \dfrac{1000 \times 1}{18.4} = 55$

Degree of dissociation $(\alpha) = \dfrac{\lambda^C_{eq.}}{\lambda^\infty_{eq.}} = \dfrac{55}{384}$

$= 0.14 \Rightarrow \alpha\ \% = 14\%$

Ex.35 Explain following ionic conductance data of 25°C for various fatty acid anions.

Anion	**Formula**	λ^0_- **AT 25ºC (in ohm^{-1}cm^{2}eq^{-1})**
Formate	HCO_2^-	–52
Acetate	$CH_3CO_2^-$	40.9
Propionate	$CH_3CH_2CO_2^-$	35.8
Butyrate	$CH_3(CH_2)_2CO_2^-$	32.6
Valerianate	$CH_3(CH_2)_2CO_2^-$	~29
Caproate	$CH_3(CH_2)_4CO_2^-$	~28

Sol. With increasing chain length bulk increases decreasing ionic mobility and thus equivalent conductance decreases.

$\because$ Charge is identical λ_m also decreases.

Ex.36 ***The resistance of a 0.01 N solution of an electolyte was found to 210 ohm at 298 K using a conductivity cell with a cell constant of 0.88 cm^{-1}. Calculate specific conductance and equivalent conductance of solution.***

Sol. Given, for 0.01 N solution.

R = 210 ohm , $\dfrac{\ell}{a} = 0.88\ cm^{-1}$

Specific conductance,

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ALLEN Session 2019-20

$\therefore \quad \kappa = \frac{1}{R} \times \frac{\ell}{a} \Rightarrow \kappa = \frac{1}{210} \times 0.88 = 4.19 \times 10^{-3}$ mho cm^{-1}

$$\Lambda_{eq} = \frac{\kappa \times 1000}{N} = \frac{4.19 \times 10^{-3} \times 1000}{0.01} = 419 \text{ mho cm}^2 \text{ eq}^{-1}.$$

Ex.37 ***The conductivity of pure water in a conductivity cell with electrodes of cross-sectional area 4 cm^2 placed at a distance 2 cm apart is 8 × 10^{-7} S cm^{-1}. Calculate;***

(a) The resistance of water.

(b) The current that would flow through the cell under the applied potential difference of 1 volt.

Sol. Cell constant = $\frac{\ell}{a} = \frac{2}{4} = \frac{1}{2}$ cm^{-1}

(a) Also, $\kappa = \frac{1}{R} \times \frac{\ell}{a}$

$$R = \frac{1}{\kappa} \times \frac{\ell}{a} = \frac{1}{8 \times 10^{-7}} \times \frac{1}{2} = 6.25 \times 10^5 \text{ ohm}$$

(b) From Ohm's law, $\frac{V}{i} = R$

$$\therefore \quad i = \frac{1}{6.25 \times 10^5} = 1.6 \times 10^{-6} \text{ ampere}$$

Ex.38 Molar conductance of 1 M solution of weak acid HA is 20 ohm^{-1} cm^2 mol^{-1}. Find % dissocaition of HA :

$$\left(\begin{matrix} \overset{\circ}{\Lambda}_m(H^+) = 350 \text{ S cm}^2 \text{ mol}^{-1} \\ \overset{\circ}{\Lambda}_m(A^-) = 50 \text{ S cm}^2 \text{ mol}^{-1} \end{matrix}\right)$$

Ans. $\overset{\circ}{\Lambda}_m(HA) = 350 + 50 = 400$

$$\alpha = \frac{\Lambda_m^C}{\overset{\circ}{\Lambda}_m} \times 100 = \frac{20}{400} \times 100 = 5\ \%$$

Ex.39 ***Conductivity of an aqueous solution of 0.1 M HX (a weak mono-protic acid) is 5 × 10^{-4} Sm^{-1}. Find pK$_a$[HX. Given :*** $\Lambda_m^{\infty}[H^+] = 0.04$ Sm^{2}mol^{-1} ; $\Lambda_m^{\infty}[X^-] = 0.01$ Sm^{2}mol^{-1}

Ans. $HX \rightleftharpoons H^+ + X^-$

$0.1(1-\alpha) \qquad 0.1\alpha \qquad 0.1\alpha$

$$\text{^m} = k \times \frac{1000}{C} \Rightarrow 5 \times 10^6 \times \frac{1000}{0.1} = 0.05\Omega^{-1}\text{cm}^2\text{-mol}^{-1}$$

$$a = \frac{\Lambda_m}{\Lambda_m} = \frac{0.05}{50}\Omega^{-1} - \text{cm}^2\text{mol}^{-1} = 10^{-4}$$

$K_a = C\alpha^2 = 0.1 \times (10^{-4})^2 = 10^{-9}$

$pK_a = 9$

Ex.40 ***Specific conductance of 10^{-4} M n-Butyric acid aqueous solution is 1.9×10^{-9} S m^{-1}. If molar conductance of n-Butyric acid at infinite dilution is 380×10^{-4} S m^2 mol^{-1}, then K_a for n-Butyric acid is :***

Sol. $\Lambda_m = 1000 \times \dfrac{1.9\times10^{-9}}{10^{-4}} = 1.9 \times 10^{-2}$

$\alpha = \dfrac{1.9\times10^{-2}}{380\times10^{-4}} = 0.5$

$K_a = \dfrac{10^{-4}(0.5)^2}{1-0.5} = 5 \times 10^{-5}$ M

Ex.41 ***The specific conductance of a saturated AgCl solution is found to be 2.12×10^{-6} S cm^{-1} and that for water is 6×10^{-8} S cm^{-1}. The solubility of AgCl is :***

($l_{eq.}^{¥}$ = 103 S $equiv^{-1}$ cm^2)

Sol. So.= 241.67 S cm^2 mol^{-1}

$S = \dfrac{(F_{Ag\,sol.} - F_{H_2O})\times1000}{(\Lambda^0_{eq.})_{AgCl}} = 2\times10^{-5}M$

Ex.42 ***The value of μ^∞ for NH_4Cl, NaOH and NaCl are 129.8, 248.1 and 126.4 ohm^{-1} cm^2 mol^{-1} respectively. Calculate μ^∞ for NH_4OH solution.***

Sol. $\mu^\infty_{NH_4OH} = \mu^\infty_{NH_4Cl} + \mu^\infty_{NaOH} - \mu^\infty_{NaCl}$

$= 129.8 + 248.1 - 126.4$

$\mu^\infty_{NH_4OH} = 251.5\ ohm^{-1}\ cm^2\ mol^{-1}$

Ex.43 ***Calculate molar conductance for NH_4OH, given that molar conductances for $Ba(OH)_2$, $BaCl_2$ and NH_4Cl are 523.28, 280.0 and 129.8 ohm^{-1} cm^2 mol^{-1} respectively.***

Sol. $\mu^\infty_{Ba(OH)_2} = \lambda^\infty_{Ba^{2+}} + 2\lambda^\infty_{OH^-} = 523.28$ ....(i)

$\mu^\infty_{BaCl_2} = \lambda^\infty_{Ba^{2+}} + 2\lambda^\infty_{Cl^-} = 280.00$ .... (ii)

$\mu^\infty_{NH_4Cl} = \lambda^\infty_{NH_4^+} + \lambda^\infty_{Cl^-} = 129.80$ .... (iii)

$\mu^\infty_{NH_4OH} = \lambda^\infty_{NH_4^+} + \lambda^\infty_{OH^-} = ?$

$\text{Eq.(iii)} + \dfrac{\text{Eq.(i)}}{2} - \dfrac{\text{Eq.(ii)}}{2}$ will gives

$\lambda^\infty_{NH_4^+} + \lambda^\infty_{OH^-} = \lambda^\infty_{NH_4OH} = \dfrac{502.88}{2} = 251.44\ ohm^{-1}\ cm^2\ mol^{-1}$

ALLEN Session 2019-20

Ex.44 ***The equivalent conductivities of acetic acid at 298 K at the concentrations of 0.1 M and 0.001 M are 5.20 and 49.2 S cm^2 eq.$^{-1}$ respectively. Calculate the degree of dissociation of acetic acid at these concentrations. Given that, $\lambda^\infty(H^+)$ and $\lambda^\infty(CH_3COO^-)$ are 349.8 and 40.9 ohm^{-1} cm^2 mol^{-1} respectively.***

Sol. Degree of dissociation is given by $\alpha = \dfrac{\lambda^c}{\lambda^\infty}$

(i) Evaluation of $\lambda^\infty_{CH_3COOH}$:

$$\lambda^\infty_{CH_3COOH} = \lambda^\infty_{CH_3COO^-} + \lambda^\infty_{H^+}$$

$$= 40.9 + 349.8 = 390.7 \text{ ohm}^{-1} \text{ cm}^2 \text{ eq.}^{-1}$$

(ii) Evaluation of degree of dissociation :

At C = 0.1 M $\alpha = \dfrac{\lambda^c}{\lambda^\infty} = \dfrac{5.20}{390.7} = 0.013$

i.e. 1.3%

At C = 0.001 M $\alpha = \dfrac{\lambda^c}{\lambda^\infty} = \dfrac{49.2}{390.7} = 0.125$

i.e. 12.5%

Ex.45 ***At infinite dilution the molar conductance of Al^{+3} and SO_4^{-2} ion are 189 and 160 Ω^{-1} cm^2 mole^{-1} respectively. Calculate the equivalent and molar conductivity at infinite dilute of $Al_2(SO_4)_3$.***

Sol. $\lambda^\infty_{eq.\left[Al_2(SO_4)_3\right]} = \dfrac{1}{3}\lambda^\infty_{Al^{+3}} + \dfrac{1}{2}\lambda^\infty_{SO_4^{-2}}$

$$= \frac{1}{3} \times 189 + \frac{1}{2} \times 160$$

$$= 143 \ \Omega^{-1} \text{ cm}^2 \text{ eq}^{-1}$$

Molar conductivity= $\lambda_{eq} \times$ V. F. = 143 × 6

$= 858 \ \Omega^{-1} \text{cm}^2 \text{ mol}^{-1}$

ALLEN Session 2019-20

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Ex.46 ***Find Λ_m^∞ (in Ω^{-1} cm^2 mol^{-1}) for strong electrolyte AB_2 in water at 25° from the following data.***

Conc.C(mole/L)	0.25	1
$\wedge_m(W^{-1}cm^2/mol)$	160	150

Sol. $160 = \Lambda_m^\infty - b \times \sqrt{.25}$ ....(A)

$150 = \Lambda_m^\infty - b \times \sqrt{1}$ ....(B)

b = 20 and $\Lambda_m^\infty = 170$

y = 170 – 20x

Intercept = 170

$\Lambda_m^\infty = 170\ \Omega^{-1}\ cm^2\ mol^{-1}$

Ex.47 ***For any sparingly soluble salt $[M(NH_3)_4Br_2]\ H_2PO_2$***

Given : $\lambda^0_{M(NH_3)_4Br_2^+}$ ***= 400 S-m^2–mol^{-1},***

$\lambda^0_{H_2PO_2^-}$ ***= 100 S-m^2–mol^{-1}***

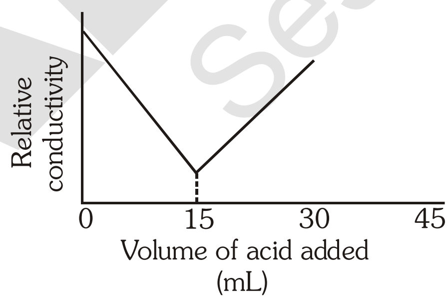
Specific resistance of saturated $[M(NH_3)_4Br_2]H_2PO_2$ solution is 200 Ω-cm.
If solubility product constant of the above salt is 10^{-x}. What will be the value of x.

Ans. $\wedge_m^\infty = \kappa \times \dfrac{1000}{m} \times 10^{-6}$

$500 = \dfrac{1}{200} \times \dfrac{1000}{5} \times 10^{-6}$

S = 10^{-8} mol|L , $K_{sp} = S^2 = 10^{-16}$

Ex.48 ***20 mL of KOH solution was titrated with 0.2 M H_2SO_4 solution in a conductivity cell. The data obtained were plotted to given the graph shown below :***



the concentration of KOH solution was -

(A) 0.3 M ***(B) 0.15 M*** ***(C) 0.12 M*** ***(D) 0.075 M***

Sol. **(A)**

20 × M = 0.2 × 2 × 15 ⇒ M= 0.15

ANSWER KEY

EXERCISE # I

1. Ans. 1.61 V
2. Ans.1.35 V
3. Ans.– 0.80 V, N_0
4. Ans. – 0.0367 V
5. Ans. (C)
6. Ans.(C)
7. Ans. (C)
8. Ans.(B)
9. Ans.(C)
10. Ans.(C)
11. Ans.(A)

12 For a spontaneous reaction the ΔG, equilibrium constant (K) and E^0_{Cell} will be respectively

12 Ans.(C)
13. Ans.(A)
14 Ans.(D)
15. Ans. (B)
16. Ans.(A)
17. Ans.(C)
18. Ans.

EXERCISE # II

19. Ans. (a) $2Ag + Cu^{2+} \longrightarrow 2Ag^+ + Cu$, (b) $MnO_4^- + 5Fe^{2+} + 8H^+ \longrightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O$

(c) $2Cl^- + 2Ag^+ \longrightarrow 2Ag + Cl_2$, (d) $H_2 + Cd^{2+} \longrightarrow Cd + 2H^+$

20. Ans.(a) $Zn \mid Zn^{2+} \mid\mid Cd^{2+} \mid Cd$, (b) $Pt, H_2 \mid H^+ \mid\mid Ag^+ \mid Ag$,

(c) $Pt \mid Fe^{2+}, Fe^{3+} \mid\mid Cr_2O_7^{2-}, H^+, Cr^{3+} \mid Pt$

21. Ans. E =1.159V
22. Ans. $E^0_{cell} = +0.01V$, $E_{cell} = -0.0785V$, correct representation is $Pb|Pb^{2+}(10^{-3}M)||Sn^{2+}(1M)|Sn$
23. Ans. $[Cu^{2+}] = 2.97 \times 10^{-12}M$ for E =0
24. Ans. $K_c = 7.6 \times 10^{12}$
25. Ans. $K_c = 2.18 \times 10^{26}$
26. $E^0 = 0.7826$ V
27. Ans. (7)
28. Ans. 0.0295 V
29. Ans. pH = 4
30. Ans. (B)
31. Ans.(D)
32. Ans.(C)
33. Ans.(A)
34. Ans.(C)
35. Ans.(B)
36. Ans.(D)
37. Ans.(A)
38. Ans.(C)
39. Ans.(B)
40. Ans.(B)
41. Ans.(C)
42. Ans.(A)
43 Ans.(B)
44 Ans.(A)
45. Ans.(A)
46. Ans.(A)
47. Ans.(D)
48. Ans.(D)

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Electrochemistry\Eng\01_Theory.p65

EXERCISE # III

49. Ans. (a) 6.02×10^{22} electrons lost, (b) 1.89×10^{22} electrons gained, (c) (b) 1.80×10^{23} electrons gained
50. Ans. (a) 0.75 F, (b) 0.69 F, (c)1.1 F
51. Ans. (i) 54 gm, (ii) 16.35 gm
52. Ans.0.112 litre
53. Ans. (i) 2.17 gm ; (ii) 1336. 15 sec

54. Ans.(B	55. Ans.(C)	56. Ans.(A)
57. Ans.(A)	58. Ans.(A)	59. Ans.(C)
60. Ans.(C)	61. Ans.(A)	62. Ans.(A)
63. Ans.(D)	64. Ans.(C)	65. Ans.(B)
66. Ans.(C)	67. Ans.(A)	68. Ans.(A)
69. Ans.(B)	70. Ans.(A)	71. Ans.(C)
72. Ans.(D)	73. Ans.(C)	

EXERCISE # IV

74. Ans. 442 S cm^2 equivalent^{-1}
75. Ans. 0.00040 S cm^{-1} ; 2500 ohm cm
76. Ans. (i) 6.25×10^5 ohm, (ii) 1.6×10^{-6} amp
77. Ans. 0.0125 mho g equiv^{-1} m^2, 1.25×10^{-3} mho cm^{-1}
78. Ans. 0.8
79. Ans. 10^{-10} mole^2 |litre^2
80. Ans. (i) 400 S cm^2 mol^{-1} (ii) 12 %
81. Ans. 510×10^{-4} mho cm^2 mol^{-1}

82. Ans.(B)	83. Ans.(D)	84. Ans.(A)
85. Ans.(B)	86. Ans.(B)	87. Ans.(A)
88. Ans.(A)	89. Ans.(A)	90. Ans.(C)
91. Ans.(D)	92 Ans.(C)	93 Ans.(A)
94 Ans.(B))	95 Ans.(C)	96. Ans.(A)

Session 2019-20

SOME PREVIOUS YEAR SOLVED EXAMPLE

1. Find the solubility product of a saturated solution of Ag_2CrO_4 in water at 298 K if the emf of the cell Ag|Ag^+ (satd.Ag_2 CrO_4 soln.) || Ag $^+$(0.1 M) | Ag is 0.164 V at 298K. **[JEE 1998]**

1. Ans.(K_{sp} = 2.287 × 10^{-12})

Sol. Ag|Ag^+ sat. sol^n || Ag^+ (0.1 M)| Ag

$AgClO_4$)

$$0.164 = 0 - \frac{.059}{1}\log\frac{[Ag^+]A}{0.1}$$

$\therefore$ $[Ag^+]_A = (1.66 \times 10^{-4}) \times (.83 \times 10^{-4})$

$= 2.287 \times 10^{-12}$

2. Calculate the equilibrium constant for the reaction, $2Fe^{3+} + 3I^- \rightleftharpoons 2Fe^{2+} + I_3^-$. The standard reduction potentials in acidic conditions are 0.77 and 0.54 V respectively for Fe^{3+} | Fe^{2+} and I_3^- | I^- couples. **[JEE 1998]**

2. Ans. (K_C = 6.26 × 10^7)

Sol. $2Fe^{+3} + 3I^- \rightleftharpoons 2Fe^{2+} + I_3^-$

$E° = 0.77 + (-.54) = 0.23$, $Keq = 10^{\frac{2\times.23}{.059}}$

$Keq = 6.26 \times 10^7$

3. A gas X at 1 atm is bubbled through a solution containing a mixture of 1 M Y and 1 M Z at 25°C. If the reduction potential of Z > Y > X, then **[JEE 1999]**

(A) Y will oxidise X and not Z
(B) Y will oxidise Z and X
(C) Y will oxidise both X and Z
(D) Y will reduce both X and Z.

3. Ans. (A)

4. The following electrochemical cell has been set up

$Pt_{(I)}$ | Fe^{3+}, Fe^{2+}(a =1) || Ce^{4+} , Ce^{3+} (a = 1) | $Pt_{(II)}$

$E^\circ_{Fe^{3+}/Fe^{2+}} = 0.77$ V and $E^\circ_{Ce^{4+}/Ce^{3+}} = 1.61$ V

If an ammetter is connected between the two platinum electrodes. predict the direction of flow of current. Will the current increase or decrease with time? **[JEE 2000]**

4. Ans. (decrease with time)

Sol. $E°_{cell}$ is (+) ve so cell will work and current will flow from cathode to anode. Current will decrease with time

5. Copper sulphate solution (250 mL) was electrolysed using a platinum anode and a copper cathode. A constant current of 2 mA was passed for 16 mintue. It was found that after electrolysis, the absorbance (concentration) of the solution was reduced to 50% of its original value. Calculate the concentration of copper sulphate in the solution to begin with. **[JEE 2000]**

5. Ans.(7.95 × 10^{-5}M)

ALLEN Session 2019-20

Sol. $\left(n_{Cu^{+2}}\right)_{reduced} = \dfrac{2\times10^{-3}\times16\times60}{96500\times2} = \dfrac{.96}{96500}$

$(n_{Cu^{+2}})_{originally\ present} = \dfrac{1.92}{96500}$

$\therefore$ M = $\dfrac{1.92}{96500}\times\dfrac{1000}{250}$ = 7.958 × 10^{-5}

E°_{cell} = – .77 + 1.61 = 0.84

6. For the electrochemical cell, M | M^+ || X^- | X, E° (M^+|M) = 0.44 V and E° (X|X^-) = 0.33V. From this data , one can deduce that **[JEE 2000]**

(A) $M + X \longrightarrow M^+ + X^-$ is the spontaneous reaction

(B) $M^+ + X^- \longrightarrow M + X$ is the spontaneous reaction

(C) E_{cell}= 0.77 V

(D) E_{cell}= – 0.77 V

6. Ans.(B)

Sol. $E^\circ_{cell} = E^0_{MM^+} + E^\circ_{M/M^+} + E^\circ_{X^-/X} = -.44 + -.33 = -0.77$ V

so (B)

7. The standard potential of the following cell is 0.23 V at 15° C & 0.21 V at 35° C

Pt | H_2(g) | HCl (aq) | AgCl(s) | Ag(s)

(i) Write the cell reaction.

(ii) Calculate ΔH^0 ,ΔS^0 for the cell reaction by assuming that these quantities remain unchanged in the range 15°C to 35°C.

(iii) Calculate the solubility of AgCl in water at 25°C. Given standard reduction potential of the Ag^+|Ag couple is 0.80 V at 25°C. **[JEE 2001]**

7. Ans.ΔH^0 = – 49987 Jmol^{-1}, ΔS^0 =– 96.5 J mol^{-1} K^{-1}, s =1.47 ×10^{-5} M

Sol. (i) Anode : $\frac{1}{2}H_2 \longrightarrow H^+ + e^-$

Cathode : $AgCl + e^- \longrightarrow Ag + Cl^-$

$\frac{1}{2}H_2 + AgCl \longrightarrow H^+ + Cl^- + Ag$

(ii) $\dfrac{\partial E}{\partial T} = \dfrac{.21-.23}{308-288} = \dfrac{.02}{10}$ = –2 × 10^{-3}

$\Delta G^\circ = -nFE^\circ$ so $(\Delta G^\circ)_{288K} = -22195$ $(\Delta G^\circ)_{308K} = -20265$ J

now using $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ \Rightarrow \Delta H^\circ = -49987$ J, $\Delta S^\circ = -96.5$ J

(iii) $E^\circ_{25^\circ C} = 0.22V$ $E^\circ_{cell} = E^\circ_{Cl^-/AgCl/Ag} = 0.22$

so –.8 + 0.22 = $\dfrac{0.59}{1}$ log Ksp

$\therefore$ Ksp = 1.47 × $10^{-10} \Rightarrow$ S = 1.21 × 10^{-5}

8. Saturated solution of KNO_3 is used to make salt bridge because

(A) velocity of K^+ is greater than that of NO_3^-

(B) velocity of NO_3^- is greater than that of K^+

(C) velocities of both K^+ and NO_3^- are nearly the same

(D) KNO_3 is highly soluble in water **[JEE 2001]**

8. Ans.(C)

Sol. Fact

9. The correct order of equivalent conductance at infinite dilution of LiCl, NaCl and KCl is

(A) LiCl > NaCl > KCl (B) KCl > NaCl > LiCl **[JEE 2001]**

(C) NaCl > KCl > LiCl (D) LiCl > KCl > NaCl

9. Ans.(B)

Sol. Fact

10. The reaction, **[JEE 2001]**

$3ClO^-(aq) \longrightarrow ClO_3^-(aq) + 2Cl^-(aq)$

is an example of

(A) Oxidation reaction (B) Reduction reaction

(C) Disproportionation reaction (D) Decomposition reaction

10. Ans.(C)

Sol Fact

11. Standard electrode potential data are useful for understanding the suitablilty of an oxidant in a redox titration. Some half cell reactions and their standard potentials are given below: **[JEE 2002]**

$$MnO_4^-(aq) + 8H^+(aq) + 5e^- \longrightarrow Mn^{2+}(aq) + 4H_2O(l);\ E° = 1.51\ V$$

$$Cr_2O_7^{2-}(aq) + 14\ H^+(aq) + 6e^- \longrightarrow 2Cr^{3+}(aq) + 7H_2O(l);\ E° = 1.38\ V$$

$$Fe^{3+}(aq) + e^- \longrightarrow Fe^{2+}(aq);\ E° = 0.77\ V$$

$$Cl_2(g) + 2e^- \longrightarrow 2Cl^-(aq);\ E° = 1.40\ V$$

Identify the only incorrect statement regarding quantitative estimation of aqueous $Fe(NO_3)_2$

(A) MnO_4^- can be used in aqueous HCl

(B) $Cr_2O_7^{2-}$ can be used in aqueous HCl

(C) MnO_4^- can be used in aqueous H_2SO_4

(D) $Cr_2O_7^{2-}$ can be used in aqueous H_2SO_4

11. Ans.(A)

Sol. MnO_4^- will oxidise Cl^- into Cl_2 so MnO_4^- can not be used in aqueous HCl

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Electrochemistry\Eng\01_Theory.p65

12. Two students use same stock solution of $ZnSO_4$ and a solution of $CuSO_4$. The e.m.f of one cell is 0.03 V higher than the other. The conc. of $CuSO_4$ in the cell with higher e.m.f value is 0.5 M. Find out the conc. of $CuSO_4$ in the other cell $\left(\frac{2.303\,RT}{F} = 0.06\right)$. **[JEE 2003]**

12. Ans.(0.05)

Sol. $E_1 = E° - \log \frac{[Zn^{+2}]}{[Cu^{+2}]_1}$

$E_2 = E° - \log \frac{[Zn^{+2}]}{[Cu^{+2}]_2}$

$E_2 - E_1 = -.03 \log \frac{[Cu^{+2}]_1}{[Cu^{+2}]_2} \Rightarrow 0.03 = -\frac{.059}{2} \log \frac{[Cu^{+2}]}{5}$

$\therefore 0.03 = -03 \log \frac{[Cu^{+2}]}{5} \Rightarrow [Cu^{+2}] = .05 \text{ M}$

13. In the electrolytic cell, flow of electrons is from:

(A) Cathode to anode in solution
(B) Cathode to anode through external supply
(C) Cathode to anode through internal supply
(D) Anode to cathode through internal supply. **[JEE 2003]**

13. Ans.(C)

14. Find the equilibrium constant at 298 K for the reaction,

$$Cu^{2+}(aq) + In^{2+}(aq) \rightleftharpoons Cu^{+}(aq) + In^{3+}(aq)$$

Given that $E^{\circ}_{Cu^{2+}|Cu^{+}} = 0.15V$, $E^{\circ}_{In^{3+}|In^{+}} = -0.42V$, $E^{\circ}_{In^{2+}|In^{+}} = -0.40V$ **[JEE 2004]**

14. Ans.($K_C = 10^{10}$)

Sol. $E^{\circ}_{Cell} = E^{\circ}_{In^{+2}/In^{+3}} + E^{\circ}_{Cu^{+2}/Cu^{+}}$

$= .44 + .15 = .59$

$E^{\circ}_{In^{+2}/In^{+3}} = \frac{1\alpha - 0.4 + 2\alpha 0.42}{1} = .44$

$K_{eq} = 10^{\frac{1 \times .59}{.059}} = 10^{10}$

15. $Zn \mid Zn^{2+}$ (a = 0.1M) $\|$ Fe^{2+} (a = 0.01M)|Fe. The emf of the above cell is 0.2905 V. Equilibrium constant for the cell reaction is

(A) $10^{0.32|0.0591}$ (B) $10^{0.32|0.0295}$ (C) $10^{0.26|0.0295}$ (D) $e^{0.32|0.295}$ **[JEE 2004]**

15. Ans.(B)

ALLEN Session 2019-20

Sol. $0.2905 = E° - \frac{.059}{2} \log \frac{.1}{0.01} \Rightarrow E° = .32$

$\therefore K_{sp} = 10^{\frac{2\times.32}{0.0591}} = 10^{\frac{.32}{0.0295}} \Rightarrow B$

16. (a) Calculate ΔG_f^0 of the following reaction

$$Ag^+(aq) + Cl^-(aq) \longrightarrow AgCl(s)$$

Given : ΔG_f^0 (AgCl) =–109 kJ|mole, ΔG_f^0 (Cl^-) = –129 kJ|mole, ΔG_f^0 (Ag^+) = 77 kJ|mole

Represent the above reaction in form of a cell

Calculate E^0 of the cell. Find $\log_{10}K_{SP}$ of AgCl

(b) 6.539×10^{-2} g of metallic Zn (amu = 65.39) was added to 100 ml of saturated solution of AgCl.

Calculate $\log_{10}\frac{[Zn^{2+}]}{[Ag^+]^2}$, given that

$Ag^+ + e^- \longrightarrow Ag \quad E^0 = 0.80$ V ; $Zn^{2+} + 2e^- \longrightarrow Zn \quad E^0 = -0.76$V

Also find how many moles of Ag will be formed? **[JEE 2005]**

16. Ans.(a) E^0 = 0.59 V, $\log_{10}K_{SP}$ = –10 (b) 52.8, 10^{-6} moles

Sol. (a) $\Delta G° = (-109) - [-129 + 77] = -57$

$$E° = \frac{-57\times1000}{1\times96500} = 0.59$$

$$\Delta G° = -2.303RT \log K_{sp} \Rightarrow \log K_{sp} = \frac{-57\times1000}{-2.303\times8.314\times298}$$

$$\log K_{sp} = 9.989 \cong 10$$

(b) $Zn + 2Ag^+ \longrightarrow Zn^{+2} + 2Ag \quad E^\circ_{cell} = 1.56$

$$\therefore \log_{10}\frac{[Zn^{+2}]}{[Ag^+]^2} = \frac{1.56\times2}{.059} = 52.8$$

$$[Ag^+] = \sqrt{Ksp} = \sqrt{10^{-10}} = 10^{-5}$$

$$\therefore n_{Ag^+} = 10^{-5} \times .1 = 10^{-6}$$

17. The half cell reactions for rusting of iron are: **[JEE 2005]**

$2H^+ + \frac{1}{2}O_2 + 2e^- \longrightarrow H_2O$; E^0 = + 1.23 V, $Fe^{2+} + 2e^- \longrightarrow Fe$; E^0 = –0.44 V

ΔG^0 (in kJ) for the reaction is:

(A) –76 (B) –322 (C) –122 (D) –176

17. Ans.(B)

ALLEN Session 2019-20

Sol. $\Delta G° = -nFE°$

$= \frac{-2\times 96500\times 1.67}{1000} - 322$

18. We have taken a saturated solution of AgBr.K_{sp} of AgBr is 12×10^{-14}. If 10^{-7} mole of $AgNO_3$ are added to 1 litre of this solution find conductivity (specific conductance) of this solution in terms of 10^{-7} S m^{-1} **[JEE 2006]**

Given : $\lambda^0_{(Ag^+)} = 6 \times 10^{-3}$ S m² mol⁻¹ ; $\lambda^0_{(Br^-)} = 8 \times 10^{-3}$ S m² mol⁻¹ ; $\lambda^0_{(NO_3^-)} = 7\times 10^{-3}$ S m² mol⁻¹

18. Ans.(55 S m^{-1})

Sol. $AgBr \rightleftharpoons \underset{s}{Ag^+} + \underset{s}{Br^-}$

$AgNO_3 \longrightarrow \underset{10^{-7}}{Ag^+} + \underset{10^{-7}}{NO_3^-}$

$\therefore (K_{sp})_{AgBr} = [Ag^+][Br^-] \Rightarrow 12 \times 10^{-14} = (s + 10^{-7})\cdot(s)$

$s^2 + 10^{-7}s - 12 \times 10^{-14} = 0 \Rightarrow s = 3 \times 10^{-7}$

$\therefore [Ag^+] = 4 \times 10^{-7}$ $[Br] = 3 \times 10^{-7}$ $[NO_3] = 10^{-7}$

now $\Lambda^{\infty}_M = \frac{1000k}{M}$

for $Ag^+ \Rightarrow 6 \times 10^{-3} \times 10^4 = \frac{1000k}{4\times 10^{-7}} \Rightarrow k_{Ag^+} = 24$

$Br^- \Rightarrow 8 \times 10^{-3} \times 10^4 = \frac{1000k}{3\times 10^{-7}} \Rightarrow k_{Br^-} = 24$

$NO_3^- \Rightarrow 7 \times 10^{-3} \times 10^4 = \frac{1000k}{10^{-7}} \Rightarrow R_{NO_3^-} = 7$

Ans. 55

Question No. 19 to 21 (3 questions)

Tollen's reagent is used for the detection of aldehyde when a solution of $AgNO_3$ is added to glucose with NH_4OH then gluconic acid is formed

$Ag^+ + e^- \longrightarrow Ag$; $E^0_{red} = 0.8$ V

$C_6H_{12}O_6 + H_2O \longrightarrow C_6H_{12}O_7$ (Gluconic acid) $+ 2H^+ + 2e^-$; $E^0_{red} = -0.05$ V

$Ag(NH_3)_2^+ + e^- \longrightarrow Ag(s) + 2NH_3$; $E^0 = -0.337$ V

[Use $2.303 \times \frac{RT}{F} = 0.0592$ and $\frac{F}{RT} = 38.92$ at 298 K] **[JEE 2006]**

19. $2Ag^+ + C_6H_{12}O_6 + H_2O \longrightarrow 2Ag(s) + C_6H_{12}O_7 + 2H^+$

Find ln K of this reaction

(A) 66.13 (B) 58.38 (C) 28.30 (D) 46.29

19. Ans.(A)

Sol. $\log K = \frac{2.303\times n\times E°}{.059} = \frac{2.303\times 2\times .85}{.059} = 66.13$

20. When ammonia is added to the solution, pH is raised to 11. Which half-cell reaction is affected by pH and by how much?

(A) E_{oxd} will increase by a factor of 0.65 from E^0_{oxd}

(B) E_{oxd} will decrease by a factor of 0.65 from E^0_{oxd}

(C) E_{red} will increase by a factor of 0.65 from E^0_{red}

(D) E_{red} will decrease by a factor of 0.65 from E^0_{red}

20. Ans.(A)

Sol. Since H^+ is involved in oxidation half reaction so E_{oxd} will be affected and it will increase

$$E_{oxd} = E^\circ_{oxd} - \frac{.059}{2}\log\frac{[C_6H_{12}O_7][H^+]^2}{[C_2H_{12}O_6]}$$

21. Ammonia is always is added in this reaction. Which of the following must be incorrect?

(A) NH_3 combines with Ag^+ to form a complex.

(B) $Ag(NH_3)_2^+$ is a weaker oxidising reagent than Ag^+.

(C) In absence of NH_3 silver salt of gluconic acid is formed.

(D) NH_3 has affected the standard reduction potential of glucose|gluconic acid electrode.

21. Ans.(D)

Sol. E°_{Red} in a constant quantity

Paragraph for Question Nos. 22 to 24 (3 questions)

Chemical reactions involve interaction of atoms and molecules. A large number of atoms|molecules (approximately 6.023×10^{23}) are present in a few grams of any chemical compound varying with their atomic|molecular masses. To handle such large numbers conveniently, the mole concept was introduced. This concept has implications in diverse areas such as analytical chemistry, biochemistry, electrochemistry and radiochemistry. The following example illustrates a typical case, involving chemical| electrochemical reaction, which requires a clear understanding of the mole concept.

A 4.0 molar aqueous solution of NaCl is prepared and 500 mL of this solution is electrolysed. This leads to the evolution of chlorine gas at one of the electrodes **(atomic mass : Na = 23, Hg = 200; 1 Faraday = 96500 coulombs)** **[JEE 2007]**

22. The total number of moles of chlorine gas evolved is

(A) 0.5 (B) 1.0 (C) 2.0 (D) 3.0

22. Ans.(B)

Sol. At anode $\underset{2}{2Cl^-} \longrightarrow \underset{1}{Cl_2} + 2e^-$

23. If the cathode is a Hg electrode, the maximum weight (g) of amalgam formed from this solution is

(A) 200 (B) 225 (C) 400 (D) 446

23. Ans.(D)

Sol. At cathode $\underset{2}{2Na^+} + \underset{2}{2e^-} \longrightarrow 2Na(s)$

$\underset{2}{2Na} + \underset{2}{2Hg} \longrightarrow 2Na\text{–}Hg \quad \therefore W = 446$

Session 2019-20

24. The total charge (coulombs) required for complete electrolysis is
(A) 24125 (B) 48250 (C) 96500 (D) 193000

24. Ans.(D)

Sol. 2 moles on e^- = 2F = 193000 C

Paragraph for Question Nos. 25 & 26 (2 questions)

Redox reactions play a pivoted role in chemistry and biology. The values of standard redox potential (E°) of two half-cell reactions decide which way the reaction is expected to proceed. A simple example is a Daniel cell in which zinc goes into solution and copper gets deposited. Given below are a set of half-cell reactions (acidic medium) along with their E° (V with respect to normal hydrogen electrode) values. Using this data obtain the correct explanations to Questions 14-16.

$I_2 + 2e^- \rightarrow 2I^-$ E° = 0.54
$Cl_2 + 2e^- \rightarrow 2Cl^-$ E° = 1.36
$Mn^{3+} + e^- \rightarrow Mn^{2+}$ E° = 1.50
$Fe^{3+} + e^- \rightarrow Fe^{2+}$ E° = 0.77
$O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$ E° = 1.23 **[JEE 2007]**

25. Among the following, identify the correct statement.
(A) Chloride ion is oxidised by O_2
(B) Fe^{2+} is oxidised by iodine
(C) Iodine ion is oxidised by chlorine
(D) Mn^{2+} is oxidised by chlorine

25. Ans.(C)

Sol. as E° will be positive

26. While Fe^{3+} is stable, Mn^{3+} is not stable in acid solution because
(A) O_2 oxidises Mn^{2+} to Mn^{3+}
(B) O_2 oxidises both Mn^{2+} to Mn^{3+} and Fe^{2+} to Fe^{3+}
(C) Fe^{3+} oxidises H_2O to O_2
(D) Mn^{3+} oxidises H_2O to O_2

26. Ans.(D)

Sol. as E° will be positive

28. For the reaction of NO_3^- ion in an aqueous solution, E° is +0.96 V. Values of E° for some metal ions are given below

$V^{2+}(aq) + 2e^- \rightarrow V$ E° = – 1.19 V
$Fe^{3+}(aq) + 3e^- \rightarrow Fe$ E° = – 0.04 V
$Au^{3+}(aq) + 3e^- \rightarrow Au$ E° = + 1.40 V
$Hg^{2+}(aq) + 2e^- \rightarrow Hg$ E° = + 0.86 V

The pair(s) of metal that is(are) oxidised by NO_3^- in aqueous solution is(are) **[JEE 2009]**
(A) V and Hg
(B) Hg and Fe
(C) Fe and Au
(D) Fe and V

28. Ans.(A,B,D)

Sol. (A,B,D) as E° will be positive

ALLEN Session 2019-20

Paragraph for Questions 29 to 30

The concentration of potassium ions inside a biological cell is at least twenty times higher than the outside. The resulting potential difference across the cell is important in several processes such as transmission of nerve impulses and maintaining the ion balance. A simple model for such a concentration cell involving a metal M is : **[JEE 2010]**

$M(s)|M^+(aq ; 0.05 \text{ molar}) \| M^+(aq ; 1 \text{ molar})|M(s)$

For the above electrolytic cell the magnitude of the cell potential $|E_{cell}| = 70$ mV.

29. For the above cell :-

(A) $E_{cell} < 0$; $\Delta G > 0$ (B) $E_{cell} > 0$; $\Delta G < 0$

(C) $E_{cell} < 0$; $\Delta G^0 > 0$ (D) $E_{cell} > 0$; $\Delta G^0 < 0$

29. Ans.(B)

Sol. $E_1 = -\frac{.059}{1}\log\frac{.05}{1} = (+)\text{ve} \Rightarrow$ so

30. If the 0.05 molar solution of M^+ is replaced by a 0.0025 molar M^+ solution, then the magnitude of the cell potential would be :-

(A) 35 mV (B) 70 mV (C) 140 mV (D) 700 mV

30. Ans.(C)

Sol. $E_2 = -\frac{.059}{1}\log\frac{0.0025}{1}$

$= 2 \times El = 140 \text{ mV} \Rightarrow$ so

ALLEN Session 2019-20

EXERCISE (S-I)

ELECTRODE POTENTIAL CELL EMF.

1. If for the half cell reactions $Cu^{2+} + e^- \longrightarrow Cu^+$ $E° = 0.15\ V$
$Cu^{2+} + 2e^- \longrightarrow Cu$ $E° = 0.34\ V$
Calculate E° of the half cell reaction
$$Cu^+ + e^- \longrightarrow Cu$$
also predict whether Cu^+ undergoes disproportionation or not.

2. For a cell $Mg(s)\ |\ Mg^{2+}(aq)\ ||\ Ag^+\ (aq)\ |\ Ag$,
(i) Calculate the equilibrium constant at 25°C.
(ii) Also find the maximum work per mole Mg that can be obtained by operating the cell in standard condition.
$E^0_{(Mg^{2+}|Mg)} = -2.37V$, $E^0_{(Ag+|Ag)} = 0.8\ V$.

3. The pK_{sp} of AgI is 16.07 . If the E° value for $Ag^+|\ Ag$ is 0.7991 V . Find the E° for the half cell reaction $AgI\ (s) + e^- \longrightarrow Ag + I^-$.

4. A zinc electrode is placed in a 0.1M solution at 25°C. Assuming that the salt (ZnX) is 20% dissociated at this dilutions calculate the electrode reduction potential. $E^0\ (Zn^{2+}|\ Zn) = -0.76V$.

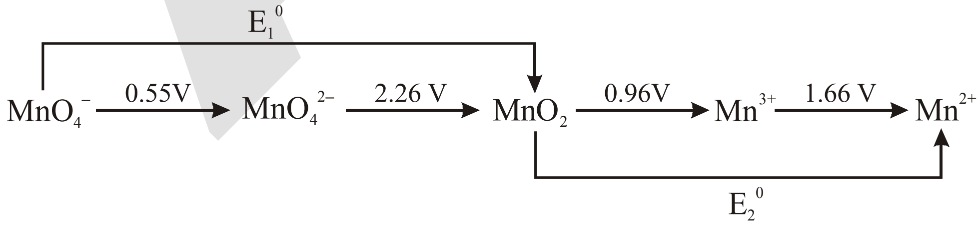
EQUILIBRIUM CONSTANT :

5. The standard reduction potential at 25°C for the reduction of water
$2H_2O + 2e^- \rightleftharpoons H_2 + 2OH^-$ is –0.8277 volt. Calculate the equilibrium constant for the reaction
$2H_2O \rightleftharpoons H_3O^+ + OH^-$ at 25°C.

6. For the reaction, $4Al(s) + 3O_2(g) + 6H_2O + 4\ OH^- \rightleftharpoons 4\ [Al(OH)_4^-]$; $E^\circ_{cell} = 2.73\ V$.
If $\Delta G^\circ_f(OH^-) = -157\ kJ\ mol^{-1}$ and $\Delta G^\circ_f(H_2O) = -237.2\ kJ\ mol^{-1}$, determine $\Delta G^\circ_f\ [Al\ (OH)_4^-]$.

7. For the cell reaction :
$Hg_2Cl_2(s) + 2Ag(s) \rightarrow 2Hg\ (l) + 2AgCl(s)$
temperature coefficient of cell emf is found to be $0.02\ VK^{-1}$. Find $\Delta_r S^o$ for cell reaction in $kJ\ mole^{-1}$

8. From the standard potential in acidic medium as shown in the following latimer diagram , the value of $(E_1^o + E_2^o)$, in volts, is -



CONCENTRATION CELLS :

9. Equinormal Solutions of two weak acids, HA ($pK_a = 3$) and HB ($pK_a = 5$) are each placed in contact with equal pressure of hydrogen electrode at 25°C. When a cell is constructed by interconnecting them through a salt bridge, find the emf of the cell.

ALLEN Session 2019-20
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10. In two vessels each containing 500ml water, 0.5m mol of aniline ($K_b = 10^{-9}$) and 50mmol of HCl are added separately. Two hydrogen electrodes are constructed using these solutions. Calculate the emf of cell made by connecting them appropriately.

ELECTROLYTIC CELL :

11. The electrosynthesis of MnO_2 is carried out from a solution of $MnSO_4$ in H_2SO_4 (aq). If a current of 25.5 ampere is used with a current efficiency of 85%, how long would it take to produce 1 kg of MnO_2?
12. Calculate the quantity of electricity that would be required to reduce 12.3 g of nitrobenzene to aniline, if the current efficiency for the process is 50 percent. If the potential drop across the cell is 3.0 volts, how much energy will be consumed?
13. A constant current of 30 A is passed through an aqueous solution of NaCl for a time of 1.0 hr. How many grams of NaOH are produced? What is volume of Cl_2 gas at 1atm,273K produced (in litre) ?
14. Assume 96500 C as one unit of electricity. If cost of electricity of producing x gm Al is Rs x, what is the cost of electricity of producing x gm Mg?
15. A metal is known to form fluoride MF_2. When 10A of electricity is passed through a molten salt for 330 sec., 1.95g of metal is deposited. Find the atomic weight of M. What will be the quantity of electricity required to deposit the same mass of Cu from $CuSO_4$?
16. After electrolysis of NaCl solution with inert electrodes for a certain period of time. 600 mL of the solution was left. Which was found to be 1N in NaOH. During the same time, 31.75 g of Cu was deposited in the copper voltameter in series with the NaCl electrolytic cell. Calculate the percentage yield of NaOH obtained.
17. A solution of $Ni(NO_3)_2$ is electrolysed between platinum electrodes using a current of 5 ampere for 20 mintue. What mass of Ni is deposited at the cathode?
18. A current of 3.7A is passed for 6hrs. between Ni electrodes in 0.5L of 2M solution of $Ni(NO_3)_2$. What will be the molarity of $Ni(NO_3)_2$ solution at the end of electrolysis?

CONDUCTANCE

19. A solution containing 2.08 g of anhydrous barium chloride is 500 CC of water has a specific conductivity 0.005 $ohm^{-1}cm^{-1}$. What are molar and equivalent conductivities of this solution.

APPLICATION OF KOHLRAUSCH'S LAW

20. Specific conductance of a saturated solution of AgBr is 8.075×10^{-7} $ohm^{-1}cm^{-1}$ at 25^0C. Specific conductance of pure water at 25°C is 0.75×10^{-7} ohm^{-1} cm^{-2}. Λ_m^∞ for KBr , $AgNO_3$ and KNO_3 are 140 , 130 , 110 (S cm^2 mol^{-1}) respectively. Calculate the solubility of AgBr in gm|litre.
21. Hydrofluoric acid is weak acid. At 25°C, the molar conductivity of 0.002M HF is 200 ohm^{-1} cm^2 $mole^{-1}$. If its Λ_m^∞ = 400 ohm^{-1} cm^2 $mole^{-1}$, calculate its degree of dissociation and equilibrium constant at the given concentration.
22. At 25°C, $\lambda_\infty(H^+) = 3.5\times10^{-2}$ S m^2 mol^{-1} and $\lambda_\infty(OH^-) = 2\times10^{-2}$ S m^2mol^{-1}.
Given: Sp. conductnace = 5.5×10^{-6} S m^{-1} for H_2O, determine pH and K_w.

ALLEN Session 2019-20

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EXERCISE (S-II)

1. Calculate the emf of the cell
Pt, H_2(1.0 atm) | CH_3COOH (0.1M) || NH_3(aq, 0.01M) | H_2 (1.0 atm),
Pt $K_a(CH_3COOH) = 1.8 \times 10^{-5}$, $K_b(NH_3) = 1.8 \times 10^{-5}$.

2. The Edison storage cell is represented as Fe(s) | FeO(s) | KOH(aq) | Ni_2O_3 (s) | NiO(s) The half–cell reaction are
$Ni_2O_3(s) + H_2O(i) + 2e^- \rightleftharpoons 2NiO(s) + 2OH^-$, $E^0 = +0.40V$
$FeO(s) + H_2O(l) + 2e^- \rightleftharpoons Fe(s) + 2OH^-$, $E^0 = -0.87V$

(i) What is the cell reaction?

(ii) What is the cell e.m.f.? How does it depend on the concentration of KOH?

(iii) What is the maximum amount of electrical energy that can be obtained from one mole of Ni_2O_3?

3. The standard reduction potential for Cu^{2+} | Cu is 0.34 V. Calculate the reduction potential at pH = 14 for the above couple. K_{sp} of $Cu(OH)_2$ is 1×10^{-19}.

4. The emf of the cell Ag|AgI|KI(0.05M) || $AgNO_3$(0.05M) |Ag is 0.788V. Calculate the solubility product of AgI.

5. Consider the cell Ag|AgBr(s)|Br^-||Cl^- | AgCl(s)| Ag at 25º C . The solubility product constants of AgBr & AgCl are respectively 5×10^{-13} & 1×10^{-10} . For what ratio of the concentrations of Br^- & Cl^- ions would the emf of the cell be zero ?

6. For the galvanic cell : Ag|AgCl(s)| KCl (0.2M) || K Br (0.001 M)| AgBr(s) | Ag,
Calculate the EMF generated and assign correct polarity to each electrode for a spontaneous process after taking into account the cell reaction at 25^0C.

$$[K_{sp(AgCl)} = 2.8 \times 10^{-10}; K_{sp(AgBr)} = 3.3 \times 10^{-13}]$$

7. Given, E° = –0.268 V for the Cl^- | $PbCl_2$ | Pb couple and – 0.126 V for the Pb^{2+} | Pb couple, determine K_{sp} for $PbCl_2$ at 25°C?

8. Calculate the equilibrium constant for the reaction:
$3Sn(s) + 2Cr_2O_7^{2-} + 28H^+ \longrightarrow 3Sn^{4+} + 4Cr^{3+} + 14H_2O$
E^0 for Sn|Sn^{2+} =0.136 V E^0 for Sn^{2+}|Sn^{4+} = – 0.154 V
E^0 for $Cr_2O_7^{2-}$|Cr^{3+} = 1.33 V

9. One of the methods of preparation of per disulphuric acid, $H_2S_2O_8$, involve electrolytic oxidation of H_2SO_4 at anode ($2H_2SO_4 \rightarrow H_2S_2O_8 + 2H^+ + 2e^-$) with oxygen and hydrogen as by–products. In such an electrolysis, 9.722 L of H_2 and 2.35 L of O_2 were generated at STP. What is the weight of $H_2S_2O_8$ formed?

10. A current of 3 amp was passed for 2 hour through a solution of $CuSO_4$,3 g of Cu^{2+} ions were deposited as Cu at cathode. Calculate percentage current efficiency of the process.

11. Dal lake has water 8.2×10^{12} litre approximately. A power reactor produces electricity at the rate of 1.5×10^6coulomb per second at an appropriate voltage.How many years would it take to electrolyse the lake?

ALLEN Session 2019-20
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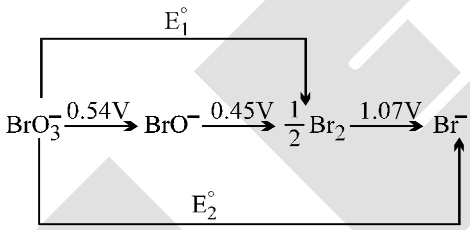
12. The equivalent conductance of 0.10 N solution of $MgCl_2$ is 97.1 mho cm^2 $equi^{-1}$ at 25°C. a cell with electrode that are 1.5 cm^2 in surface area and 0.5 cm apart is filled with 0.1 N $MgCl_2$ solution. How much current will flow when potential difference between the electrodes is 5 volt.

13. When a solution of specific conductance 1.342 ohm^{-1} $metre^{-1}$ was placed in a conductivity cell with parallel electrodes, the resistance was found to be 170.5 ohm. Area of electrodes is 1.86×10^{-4} m^2. Calculate separation of electrodes.

14. The specific conductance at 25°C of a saturated solution of $SrSO_4$ is 1.482×10^{-4} ohm^{-1} cm^{-1} while that of water used is 1.5×10^{-6} mho cm^{-1}. Determine at 25°C the solubility in gm per litre of $SrSO_4$ in water. Molar ionic conductance of Sr^{2+} and SO_4^{2-} ions at infinite dilution are 59.46 and 79.8 ohm^{-1} cm^2 $mole^{-1}$ respectively. [Sr = 87.6 , S = 32 , O = 16]

15. The EMF of the cell M | M^{n+} (0.02M) || H^+ (1M) | $H_2(g)$ (1 atm), Pt at 25°C is 0.81V. Calculate the valency of the metal if the standard oxidation of the metal is 0.76V.

16. From the standard potentials shown in the following diagram, calculate the potentials E_1° and E_2°.



17. Calculate the EMF of the cell,

$$Zn - Hg(c_1M) \mid Zn^{2+}(aq) \mid Hg - Zn(c_2M)$$

at 25°C, if the concentrations of the zinc amalgam are: c_1 = 10g per 100g of mercury and c_2 = 1g per100 g of mercury.

18. How long a current of 2A has to be passed through a solution of $AgNO_3$ to coat a metal surface of 80cm^2 with 5μm thick layer? Density of silver = 10.8g|cm^3.

19. 10g solution of $CuSO_4$ is electrolyzed using 0.01F of electricity. Calculate:
(a)The weight of resulting solution
(b) Equivalents of acid or alkali in the solution.

20. Cadmium amalgam is prepared by electrolysis of a solution of $CdCl_2$ using a mercury cathode. How long should a current of 5A be passed in order to prepare 12% Cd–Hg amalgam on a cathode of 2gm Hg (Cd=112.4)

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Electrochemistry\Eng\02_Ex.p65

EXERCISE (O-I)

GALVANIC CELL

1. The thermodynamic efficiency of cell is given by-

(A) $\frac{\Delta H}{\Delta G}$ (B) $\frac{nFE_{cell}}{\Delta G}$ (C) $-\frac{nFE_{cell}}{\Delta H}$ (D) Zero

2. From the following E° values of half cells,

(i) $A + e \rightarrow A^-$; E° = –0.24 V (ii) $B^- + e \rightarrow B^{2-}$; E° = +1.25 V

(iii) $C^- + 2e \rightarrow C^{3-}$; E° = –1.25 V (iv) $D + 2e \rightarrow D^{2-}$; E° = +0.68 V

What combination of two half cells would result in a cell with the largest potential ?

(A) (ii) and (iii) (B) (ii) and (iv) (C) (i) and (iii) (D) (i) and (iv)

3. Which of the following will increase the voltage of the cell with following cell reaction

$Sn_{(s)} + 2Ag^+_{(aq)} \rightarrow Sn^{+2}_{(aq)} + 2Ag_{(s)}$

(A) Decrease in the concentration of Ag^+ ions

(B) Increase in the concentration of Sn^{+2} ions

(C) Increase in the concentration of Ag^+ ions

(D) (A) & (B) both

4. The standard emf for the cell reaction,

$Zn_{(s)} + Cu^{2+}_{(aq)} \longrightarrow Zn^{2+}_{(aq)} + Cu_{(s)}$ is 1.10 volt at 25 °C. The emf for the cell reaction when 0.1 M Cu^{2+} and 0.1 M Zn^{2+} solution are used at 25°C is :

(A) 1.10 volt (B) 0.110 volt (C) –1.10 volt (D) –0.110 volt

5. What is the potential of the cell containing two hydrogen electrodes as represented below

$Pt \mid H_2(g) \mid H^+_{(aq)}(10^{-8}\ M) \parallel H^+_{(aq)}(0.001\ M) \mid H_2(g)|Pt$

(A) – 0.295 V (B) – 0.0591 V (C) 0.295 V (D) 0.0591 V

6. Consider the cell, $Cu|Cu^{+2}||Ag^+|Ag$. If the concentration of Cu^{+2} and Ag^+ ions becomes ten times the emf of the cell :–

(A) Becomes 10 times (B) Remains same

(C) Increase by 0.0295 V (D) Decrease by 0.0295 V

7. The standard emf of a galvanic cell involving cell reaction with n = 4 is found to be 0.295 V at 25°C. The equilibrium constant of the reaction would be,

(A) 1.0×10^{20} (B) 2.0×10^{11} (C) 4.0×10^{12} (D) 1.0×10^{2}

8. By how much times will potential of half cell $Cu^{+2}|Cu$ change if, the solution is diluted to 100 times at 298 K :–

(A) Increases by 59 mV (B) Decrease by 59 mV

(C) Increases by 29.5 mV (D) Decreases by 29.5 mV

ALLEN Session 2019-20

ELECTROLYTIC CELL

9. When an electric current is passed through a cell containing an electrolyte, positive ions move towards the cathode and negative ions towards the anode. What will happen if the cathode is pulled out of the solution?

(A) The positive ions will start moving towards the anode and negative ions will stop moving.

(B) The negative ions will continue to move towards the anode and the positive ions will stop moving

(C) Both positive and negative ions will move towards the anode.

(D) None of these movements will take place.

10. Electrolysis of a $CuSO_4$ produces :–

(A) An increase in pH (B) A decrease in pH
(C) Either decrease or increase (D) None

11. The same amount of electricity was passed through two separate electrolytic cells containing solutions of nickel nitrate $[Ni(NO_3)_2]$and chromium nitrate $[Cr(NO_3)_3]$ respectively. If 0.3 g of nickel was deposited in the first cell, the amount of chromium deposited is :
(at. wt. of Ni = 59, at. wt. of Cr = 52)

(A) 0.1 g (B) 0.17 g (C) 0.3 g (D) 0.6 g

12. The electric charge for electro deposition of 1 equivalent of a substance is :

(A) one ampere per second (B) 4 faraday
(C) one ampere for one hour (D) charge on one mole of electrons

13. 3.17 g., of a substance was deposited by the flow of 0.1 mole of electrons. The equivalent weight of the substance is :

(A) 3.17 (B) 0.317 (C) 317 (D) 31.7

14. A current of 9.65 amp. passing for 16 min. 40 sec. through a molten tin salt deposits 5.95 g. of tin The oxidation state of the tin in the salt is : (at. wt of Sn = 119)

(A) +4 (B) +3 (C) +2 (D) +1

15. The time required for a current of 3 amp. to decompose electrolytically 18 g of H_2O is:

(A) 18 hour (B) 36 hour (C) 9 hour (D) 18 seconds

16. An ion is reduced to the element when it absorbs 6×10^{20} electrons. The number of gm equivalents of the ion is :

(A) 0.10 (B) 0.01 (C) 0.001 (D) 0.0001

17. How many coulombs of electric charge are required for the oxidation of 1 mole of H_2O to O_2 ?

(A) 9.65×10^4 C (B) 4.825×10^5 C (C) 1.93×10^5 C (D) 1.93×10^4 C

18. The time required to coat ameter surface of 80 cm^2 with 5×10^{-3} cm thick layer of silver (density 1.08 g cm^{-3}) with the passage of 9.65A current through a silver nitrate solution is :

(A) 10 sec. (B) 40 sec. (C) 30 sec. (D) 20 sec.

19. One gm metal M^{+2} was discharged by the passage of 1.2×10^{22} electrons. What is the atomic weight of metal?

(A) 25 (B) 50 (C) 100 (D) 75

ALLEN Session 2019-20

20. One mole of electron passes through each of the solution of $AgNO_3$, $CuSO_4$ and $AlCl_3$ when Ag, Cu and Al are deposited at cathode. The molar ratio of Ag, Cu and Al deposited are

(A) 1 : 1 : 1 (B) 6 : 3 : 2 (C) 6 : 3 : 1 (D) 1 : 3 : 6

21. During electrolysis of an aqueous solution of sodium sulphate, 2.4 L of oxygen at STP was liberated at anode. The volume of hydrogen at STP, liberated at cathode would be

(A) 1.2 L (B) 2.4 L (C) 2.6 L (D) 4.8 L

22. The charge required for the oxidation of one mole Mn_3O_4 into MnO_4^{2-} in presence of alkaline medium is

(A) 5 × 96500 C (B) 96500 C (C) 10 × 96500 C (D) 2 × 96500 C

CONDUCTANCE

23. Equivalent conductances of Ba^{+2} and Cl^- ions are 127 & 76 $ohm^{-1}\,cm^2\,eq^{-1}$ respectively. Equivalent conductance of $BaCl_2$ at infinite dilution is -

(A) 379 (B) 139.5 (C) 203 (D) 330

24. If x is specific resistance of the electrolyte solution and y is the molarity of the solution, then $\wedge_m$ is given by

(A) $\frac{1000x}{y}$ (B) $1000\,\frac{y}{x}$ (C) $\frac{1000}{xy}$ (D) $\frac{xy}{1000}$

ALLEN Session 2019-20

EXERCISE (O-II)

Single correct :

1. Consider the reaction,
$Cl_{2(g)} + 2Br^-_{(aq)} \longrightarrow 2Cl^-_{(aq)} + Br_{2(g)}$
The emf of the cell when $[Cl^-]=[Br^-]=0.01M$ and Cl_2 gas at 1 atm pressure while $Br_2(g)$ at 0.01 atm will be (E° for the above reaction is = 0.29 volt) :
(A) 0.54 volt (B) 0.35 volt (C) 0.24 volt (D) –0.29 volt

2. How much will the reduction potential of a hydrogen electrode change when its solution initially at pH = 0 is neutralised to pH = 7 ?
(A) increase by 0.059V (B) decrease by 0.059V
(C) increase by 0.413V (D) decrease by 0.413V

3. If the pressure of H_2 gas is increased from 1 atm to 100 atm keeping H^+ concentration constant at 1 M, the change in reduction potential of hydrogen half cell at 25°C will be
(A) 0.059 V (B) 0.59 V (C) 0.0295 V (D) 0.118 V

4. A silver wire dipped in 0.1 M HCl solution saturated with AgCl develops oxidation potential of –0.209 V. If $E^\circ_{Ag/Ag^+} = -0.799$ V, the K_{sp} of AgCl in pure water will be
(A) 3×10^{-11} (B) 10^{-11} (C) 4×10^{-11} (D) 3×10^{-11}

5. Salts of A (atomic weight = 7), B (atomic weight =27) and C (atomic weight = 48) were electrolysed under identical conditions using the same quantity of electricity. It was found that when 2.1 g of A was deposited, the weights of B and C deposited were 2.7 and 7.2 g. The valencies of A, B and C respectively are
(A) 3, 1 and 2 (B) 1, 3 and 2 (C) 3, 1 and 3 (D) 2, 3 and 2

6. During electro refining of Cu by electrolysis of an aqueous solution of $CuSO_4$ using copper electrodes, if 2.5 g of Cu is deposited at cathode, then at anode
(A) decrease of more than 2.5 g of mass takes place
(B) 450 ml of O_2 at STP is liberated
(C) 2.5 g of copper is deposited
(D) a decrease of 2.5 g of mass takes place

7. The conductivity of a saturated solution of Ag_3PO_4 is 9×10^{-6} S m^{-1} and its equivalent conductivity is 1.50×10^{-4} S m^2 equivalent^{-1}. The K_{sp} of Ag_3PO_4 is
(A) 4.32×10^{-18} (B) 1.8×10^{-9} (C) 8.64×10^{-13} (D) None of these

8. Equal volumes of 0.015 M CH_3COOH & 0.015 M NaOH are mixed together. What would be molar conductivity of mixture if conductivity of CH_3COONa is 6.3×10^{-4} S cm^{-1}
(A) 8.4 S cm^2 mol^{-1} (B) 84 S cm^2 mol^{-1} (C) 4.2 S cm^2 mol^{-1} (D) 42 S cm^2 mol^{-1}

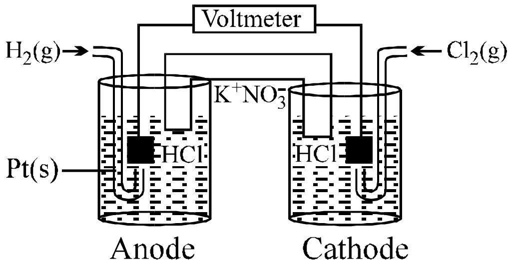
9. For the fuel cell reaction $2H_2(g) + O_2(g) \longrightarrow 2H_2O(l)$; $\Delta_f H^o_{298}(H_2O, l) = -285.5$ kJ|mol

What is ΔS^o_{298} for the given fuel cell reaction?

Given: $O_2(g) + 4H^+(aq) + 4e^- \longrightarrow 2H_2O(l)$ E° = 1.23 V
(A) – 0.322 J|K (B) – 0.635 kJ|K (C) 3.51 kJ|K (D) – 0.322 kJ|K

ALLEN Session 2019-20

10. Consider the following Galvanic cell.



By what value the cell voltage change when concentration of ions in anodic and cathodic compartments both increased by factor of 10 at 298 K

(A) +0.0590 (B) –0.0590 (C) –0.1180 (D) 0

11. The standard reduction potentials at 25°C for the following half reactions are :

Zn^{2+} (aq) + 2e⁻ $\rightleftharpoons$ Zn(s), $E^\circ_{RP} = -0.762V$

Cr^{3+} (aq) + 3e⁻ $\rightleftharpoons$ Cr(s), $E^\circ_{RP} = -0.740V$

$2H^+_{(aq)} + 2e^- \rightleftharpoons H_2(g)$, $E^\circ_{RP} = 0.00$ V

$Fe^{3+}_{(aq)} + 2e^- \rightleftharpoons Fe^{2+}_{(aq)}$, $E^\circ_{RP} = 0.77V$

Which is the strongest reducing agent ?

(A) Zn (B) Cr (C) $H_2(g)$ (D) Fe^{2+} (aq)

12. Using the standard electrode potential values given below, decide which of the statements, I, II, III and IV are correct. Choose the right answer from (A), (B), (C) and (D).

$Fe^{2+}_{(aq)} + 2e^- \rightleftharpoons Fe_{(s)}$; $E^\circ = -0.44$ V

$Cu^{2+}_{(aq)} + 2e^- \rightleftharpoons Cu_{(s)}$; $E^\circ = +0.34$ V

$Ag^{+}_{(aq)} + e^- \rightleftharpoons Ag_{(s)}$; $E^\circ = +0.80$ V

I. Copper can displace iron from $FeSO_4$ solution.

II. Iron can displace copper from $CuSO_4$ solution.

III. Silver can displace copper from $CuSO_4$ solution.

IV. Iron can displace silver from $AgNO_3$ solution.

(A) I and II (B) II and III (C) II and IV (D) I and IV

13. The following facts are available :–

$2X^- + Y_2 \rightarrow 2Y^- + X_2$

$2W^- + Y_2 \rightarrow$ NO reaction

$2Z^- + X_2 \rightarrow 2X^- + Z_2$

Which of the following statements is correct :–

(A) $E^\circ_{W^-/W_2} > E^\circ_{Y^-/Y_2} > E^\circ_{X^-/X_2} > E^\circ_{Z^-/Z_2}$ (B) $E^\circ_{W^-/W_2} < E^\circ_{Y^-/Y_2} < E^\circ_{X^-/X_2} < E^\circ_{Z^-/Z_2}$

(C) $E^\circ_{W^-/W_2} < E^\circ_{Y^-/Y_2} > E^\circ_{X^-/X_2} > E^\circ_{Z^-/Z_2}$ (D) $E^\circ_{W^-/W_2} > E^\circ_{Y^-/Y_2} < E^\circ_{X^-/X_2} < E^\circ_{Z^-/Z_2}$

14. The cost of electricity required to deposit 1 g of Mg is Rs. 5.00. How much would it cost to deposit 9 g of Al (At wt. Al = 27, Mg = 24)

(A) Rs. 10 (B) Rs. 27 (C) Rs. 40 (D) Rs. 60

15. 4.5g of aluminium (at. mass 27 amu) is deposited at cathode from Al^{3+} solution by a certain quantity of electric charge. The volume of hydrogen produced at STP from H^+ ions in solution by the same quantity of electric charge will be –

(A) 45.4L (B) 11.35L (C) 22.7L (D) 5.675 L

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ALLEN Session 2019-20

16. The density of A is 10 g cm^{-3}. The quantity of electricity needed to plate an area 10 cm × 10 cm to a thickness of 10^{-2} cm using ASO_4 solution would be (Atomic mass of A = 193)
(A) 5000 C (B) 10000 C (C) 40000 C (D) 20000 C

17. The resistance of 0.5 M solution of an electrolyte in a cell was found to be 50 Ω. If the electrodes in the cell are 2.2 cm apart and have an area of 4.4 cm^2 then the molar conductivity (in S m^2 mol^{-1}) of the solution is
(A) 0.2 (B) 0.02 (C) 0.002 (D) None of these

18. Equivalent conductance of 0.1 M HA(weak acid) solution is 10 $Scm^2equivalent^{-1}$ and that at infinite dilution is 200 $Scm^2equivalent^{-1}$ Hence pH of HA solution is
(A) 1.3 (B) 1.7 (C) 2.3 (D) 3.7

19. The dissociation constant of n-butyric acid is 1.6×10^{-5} and the molar conductivity at infinite dilution is 380×10^{-4} Sm^2mol^{-1}. The specific conductance of the 0.01 M acid solution is
(A) 1.52×10^{-5} Sm^{-1} (B) 1.52×10^{-2} Sm^{-1}
(C) 1.52×10^{-3} Sm^{-1} (D) None

MULTIPLE CORRECT :

20. During discharging of lead storage battery, which of the following is|are true ?
(A) H_2SO_4 is produced (B) H_2O is consumed
(C) $PbSO_4$ is formed at both electrodes (D) Density of electrolytic solution decreases

21. Which of the following arrangement will produce oxygen at anode during electrolysis ?
(A) Dilute H_2SO_4 solution with Cu electrodes.
(B) Dilute H_2SO_4 solution with inert electrodes.
(C) Fused NaOH with inert electrodes.
(D) Dilute NaCl solution with inert electrodes.

22. If 270.0 g of water is electrolysed during an experiment performed by miss Abhilasha with 75% current efficiency then
(A) 168 L of O_2 (g) will be evolved at anode at 1 atm & 273 K
(B) Total 504 L gases will be produced at 1 atm & 273 K.
(C) 336 L of H_2 (g) will be evolved at anode at 1 atm & 273 K
(D) 45 F electricity will be consumed

23. Pick out the correct statements among the following from inspection of standard reduction potentials (Assume standard state conditions).

$$Cl_2\ (a) + 2e \rightleftharpoons 2Cl^-(a) \qquad E^o_{Cl_2/Cl^-} = +1.36 \text{ volt}$$

$$Br_2\ (a) + 2e \rightleftharpoons 2Br^-(a) \qquad E^o_{Br_2/Br^-} = +1.09 \text{ volt}$$

$$I_2(s) + 2e \rightleftharpoons 2I^-(a) \qquad E^o_{I_2/I^-} = +0.54 \text{ volt}$$

$$S_2O_8^{2-}\ (a) + 2e \rightleftharpoons 2SO_4^{2-}(a) \qquad E^o_{S_2O_8^{2-}/SO_4^{2-}} = +2.00 \text{ volt}$$

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Electrochemistry\Eng\02_Ex.p65

(A) Cl_2 can oxidise SO_4^{2-} from solution

(B) Cl_2 can oxidise Br^- and I^- from aqueous solution

(C) $S_2O_8^{2-}$ can oxidise Cl^-, Br^- and I^- from aqueous solution

(D) $S_2O_8^{2-}$ is added slowly, Br^- can be reduce in presence of Cl^-

24. The EMF of the following cell is 0.22 volt.

Ag(s) | AgCl(s) | KCl (1M) | H^+(1M) | H_2(g) (1atm) ; Pt(s).

Which of the following will decrease the EMF of cell.

(A) increasing pressure of H_2(g) from 1 atm to 2 atm

(B) increasing Cl^- concentration in Anodic compartment

(C) increasing H^+ concentration in cathodic compartment

(D) decreasing KCl concentration in Anodic compartment.

Assertion & Reasoning type questions :

25. Statement -1 : The voltage of mercury cell remains constant for long period of time.

Statement -2 : It is because net cell reaction does not involve active species.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

26. Statement -1 : The SRP of three metallic ions A^+,B^{2+}, C^{3+} are –0.3, –0.5, 0.8 volt respectively, so oxidising power of ions is $C^{3+} > A^+ > B^{2+}$.

Statement -2 : Higher the SRP, higher the oxidising power.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

27. Statement -1 : We can add the electrode potential in order to get electrode potential of net reaction.

Statement -2 : Electrode potential is an intensive property.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

ALLEN Session 2019-20

Table type :

1 TABLE (3Q)

Column-I	Column-II	Column-III
(P) $Zn(s)\vert ZnSO_4(0.1M) \vert\vert Zn(NO_3)$ (0.01M)Zn(s)	(A) Has metal -insoluble salt - anion electrode.	(I) $E_{cell} < 0$
(Q) $Pt, Cl_2(0.1bar)\vert KCl(1M) \vert\vert NaCl(1M)\ Cl_2(1bar), Pt$	(B) Electrolytic concentration cell	(II) $E^0_{cell} = 0$
(R) $Ag(s)\vert AgCl(s)\vert KCl (0.1M)\vert\vert Ag^+(0.1M)\vert Ag(s)$ $Ksp[AgCl] = 10^{-10}$.	(C) Electrode concentration cell	(III) $E_{cell} > 0$
(S) $Pt, H_2(1bar)\vert H_2SO_4 (0.05M)\vert\vert HNO_3(0.1M)\vert H_2$ (1bar), Pt	(D) Has gas-ion electrode	(IV) $E_{cell} = 0$

(1) Use : $\frac{2.303RT}{F} = 0.06$

(2) Assume constant P,T condition of operation.

28. Which option is incorrectly matched ?
(A) P - B - II (B) Q - C - II (C) R - A - I (D) S - D - IV

29. For galvanic cell in option 'Q' on increasing concentration of KCl, cell potential will -
(A) Increase (B) decrease (C) remains constant (D) cannot predict

30. On increasing Ag^+ concentration in anodic compartment in option (R) cell potential will
(A) Remain same (B) increase (C) decrease (D) can't predict

Match the column

31.

Column I	**Column II** (Electrolysis product using inert electrode)
(A) Dilute solution of HCl	(P) O_2 evolved at anode
(B) Dilute solution of NaCl	(Q) H_2 evolved at cathode
(C) Concentrated solution of NaCl	(R) Cl_2 evolved at anode
(D) $AgNO_3$ solution	(S) Ag deposition at cathode

ALLEN Session 2019-20

32. **Column-I** **Column-II**

Cell notation : E_{cell}

(P) $Mn\left|\begin{smallmatrix}Mn^{2+}\\(0.1M)\end{smallmatrix}\right\|\begin{smallmatrix}Cu^{2+}\\(0.01M)\end{smallmatrix}\left|Cu\right.$ (1) 0.33

(Q) $Ag\left|AgBr\right|\begin{smallmatrix}Br^-\\(0.1M)\end{smallmatrix}\left\|\begin{smallmatrix}Ag^+\\(0.01M)\end{smallmatrix}\right|Ag$ (2) 0.6

(R) $Pt, H_2(4atm)\left|\begin{smallmatrix}H^+\\(0.01M)\end{smallmatrix}\right\|\begin{smallmatrix}Cl^-\\(0.2M)\end{smallmatrix}\left|\begin{smallmatrix}Cl_2\\(1atm)\end{smallmatrix}\right|Pt$ (3) 1.49

(S) $Pt,\left|\begin{smallmatrix}Fe^{2+} & Fe^{+3}\\(0.2M) & (0.02M)\end{smallmatrix}\right\|\begin{smallmatrix}MnO_4^- & Mn^{+2} & H^+\\(0.1M), & (0.1M) & ,pH=5\end{smallmatrix}\left|Pt\right.$ (4) 1.54

Given :

$E^0_{Cu^{2+}/Cu} = 0.34V$

$K_{sp}(AgBr) = 10^{-13}$

$E^0_{Mn^{2+}/Mn} = -1.18V$

$\frac{2.303RT}{F} = 0.06$

$E^0_{Ag^+/Ag} = 0.8V$

$E^0_{Fe^{3+}/Fe^{2+}} = 0.77V$

$E^0_{MnO_4^-/Mn^{2+}} = 1.52V$

$E^0_{Cl_2/Cl^-} = 1.36V$

Code:

	P	Q	R	S
(A)	4	3	4	4
(B)	2	3	1	4
(C)	3	2	4	1
(D)	4	3	2	1

ALLEN Session 2019-20

EXERCISE (J-MAINS)

1. Resistance of 0.2 M solution of an electrolyte is 50 Ω. The specific conductance of the solution is 1.3 S m^{-1}. If resistance of the 0.4M solution of the same electrolyte is 260 Ω, its molar conductivity is :- **[AIEEE 2011]**

(1) 6250 S m^2 mol^{-1} (2) 6.25×10^{-4} S m^2 mol^{-1}
(3) 625×10^{-4} S m^2 mol^{-1} (4) 62.5 S m^2 mol^{-1}

2. The reduction potential of hydrogen half-cell will be negative if :- **[AIEEE 2011]**

(1) $p(H_2)$ = 2 atm $[H^+]$ = 1.0 M (2) $p(H_2)$ = 2 atm and $[H^+]$ = 2.0 M
(3) $p(H_2)$ = 1 atm and $[H^+]$ = 2.0 M (4) $p(H_2)$ = 1 atm and $[H^+]$ = 1.0 M

3. The standard reduction potentials for Zn^{2+} | Zn, Ni^{2+} | Ni and Fe^{2+} | Fe are – 0.76, – 0.23 and – 0.44 V respectively. The reaction $X + Y^{+2} \rightarrow X^{2+} + Y$ will be spontaneous when [AIEEE 2012]

(1) X = Zn, Y = Ni (2) X = Ni, Y = Fe (3) X = Ni, Y = Zn (4) X = Fe, Y = Zn

4. Given : **[JEE-MAINS 2013]**

$E^0_{Cr^{3+}/Cr} = -0.74\,V$; $E^0_{MnO_4^-/Mn^{2+}} = 1.51\,V$

$E^0_{Cr_2O_7^{2-}/Cr^{3+}} = 1.33\,V$; $E^0_{Cl/Cl^-} = 1.36\,V$

Based on the data given above, strongest oxidising agent will be :
(1) Cl^- (2) Cr^{3+} (3) Mn^{2+} (4) MnO_4^-

5. The equivalent conductance of NaCl at concentration C and at infinite dilution are λ_C and λ_∞, respectively. The correct relationship between λ_C and λ_∞ is given as : **[JEE-MAINS 2014]**
(where the constant B is postive)

(1) $\lambda_C = \lambda_\infty - (2)\sqrt{C}$ (2) $\lambda_C = \lambda_\infty + (2)\sqrt{C}$
(3) $\lambda_C = \lambda_\infty + (2)\,C$ (4) $\lambda_C = \lambda_\infty - (2)C$

6. Resistance of 0.2 M solution of an electrolyte is 50 Ω. The specific conductance of the solution is 1.4 S m^{-1}. The resistance of 0.5 M solution of the same electrolyte is 280 Ω. The molar conductivity of 0.5 M solution of the electrolyte in S m^2 mol^{-1} is : **[JEE-MAINS 2014]**

(1) 5×10^3 (2) 5×10^2 (3) 5×10^{-4} (4) 5×10^{-3}

7. At 298 K, the standard reduction potentials are 1.51 V for $MnO_4^-|Mn^{2+}$, 1.36 V for $Cl_2|Cl^-$, 1.07 V for $Br_2|Br^-$, and 0.54 V for $I_2|I^-$ .At pH = 3, permanganate is expected to oxidize $\left(\frac{RT}{F} = 0.059\,V\right)$:- **[JEE-MAINS (ONLINE) 2015]**

(1) Cl^- and Br^- (2) Cl^-, Br^- and I^- (3) Br^- and I^- (4) I^- only

8. A variable, opposite external potential (E_{ext}) is applied to the cell
$Zn|Zn^{2+}$ (1 M) || Cu^{2+} (1 M) | Cu, of potential 1.1 V. When E_{ext} < 1.1 V and E_{ext} > 1.1 V, respectively electrons flow from : **[JEE-MAINS (ONLINE) 2015]**
(1) anode to cathode in both cases
(2) anode to cathode and cathode to anode
(3) cathode to anode in both cases
(4) cathode to anode and anode to cathode

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Electrochemistry\Eng\02_Ex.p65

9. Two Faraday of electricity is passed through a solution of $CuSO_4$. The mass of copper deposited at the cathode is : (at. mass of Cu = 63.5 amu) **[JEE-MAINS 2015]**

(1) 2g (2) 127 g (3) 0 g (4) 63.5 g

10. What will occur if a block of copper metal is dropped into a beaker containing a solution of 1M $ZnSO_4$

(1) The copper metal will dissolve and zinc metal will be deposited

(2) No reaction will occur **[JEE-MAINS (ONLINE) 2016]**

(3) The copper metal will dissolve with evolution of oxygen gas

(4) The copper metal will dissolve with evolution of hydrogen gas

11. Oxidation of succinate ion produces ethylene and carbon dioxide gases. On passing 0.2 Faraday electricity through on aqueous solution of potassium succinate, the total volume of gases (at both cathode and anode) at STP (1 atm and 273 K) is : **[JEE-MAINS (ONLINE) 2016]**

(1) 8.96 L (2) 2.24 L (3) 4.48 L (4) 6.72 L

12. Given **[JEE-MAINS - 2017]**

$$E^o_{Cl_2/Cl^-} = 1.36\,V, E^o_{Cr^{3+}/Cr} = -0.74\,V$$

$$E^o_{Cr_2O_7^{2-}/Cr^{3+}} = 1.33\,V, E^o_{MnO_4^-/Mn^{2+}} = 1.51\,V.$$

Among the following, the strongest reducing agent is

(1) Cr (2) Mn^{2+} (3) Cr^{3+} (4) Cl^-

13. What is the standard reduction potential (E°) for $Fe^{3+} \to Fe$? **[JEE-MAINS (ONLINE) 2017]**

Given that :

$Fe^{2+} + 2e^- \to Fe$; $E^o_{Fe^{2+}/Fe} = -0.47$ V

$Fe^{3+} + e^- \to Fe^{2+}$; $E^o_{Fe^{3+}/Fe^{2+}} = +0.77$ V

(1) +0.30 V (2) +0.057 V (3) –0.057 V (4) –0.30 V

14. To find the standard potential of $M^{3+}|M$ electrode, the following cell is constituted: $Pt|M|M^{3+}(0.001\ mol\ L^{-1})|Ag^+(0.01\ mol\ L^{-1})|Ag$ **[JEE-MAINS (ONLINE) 2017]**

The emf of the cell is found to be 0.421 volt at 298 K. The standard potential of half reaction $M^{3+} + 3e^- \to M$ at 298 K will be : (Given $E^{\ominus}_{Ag^+/Ag}$ at 298 K = 0.80 Volt)

(1) +0.30 V (2) +0.057 V (3) –0.057 V (4) –0.30 V

15. How long (approximate) should water be electrolysed by passing through 100 amperes current so that the oxygen released can completely burn 27.66 g of diborane ?**[JEE-MAINS (OFFLINE) 2017]**

(Atomic weight of B = 10.8 u)

(1) 0.8 hours (2) 3.2 hours (3) 1.6 hours (4) 6.4 hours

16 When an electric current is passed through acidified water, 112 mL of hydrogen gas at N.T.P. was collected at the catode in 965 seconds. The current passed, in ampere, is :

[JEE-MAINS (ONLINE) 2018]

(1) 2.0 (2) 1.0 (3) 0.1 (4) 0.5

17. When 9.65 ampere current was passed for 1.0 hour into nitrobenzene in acidic medium, the amount of p-aminophenol produced is :- **[JEE-MAINS (ONLINE) 2018]**

(1) 10.9 g (2) 98.1 g (3) 109.0 g (4) 9.81 g

EXERCISE (J-ADVANCED)

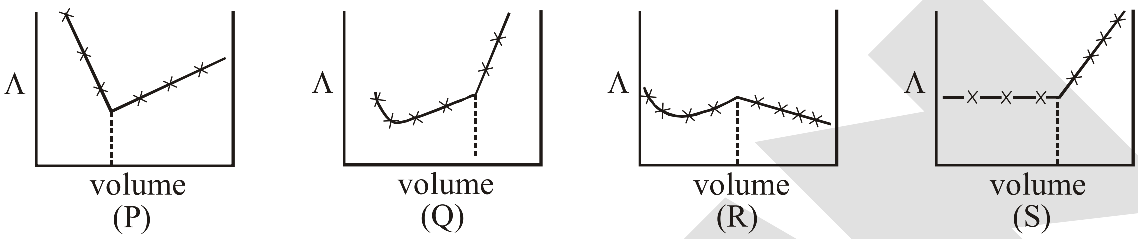
1. Consider the following cell reaction : **[JEE 2011]**

$2Fe_{(s)} + O_{2(g)} + 4H^+_{(aq)} \rightarrow 2Fe^{2+}_{(a)} + 2H_2O(\ell)$ $\quad E° = 1.67\ V$

At$[Fe^{2+}] = 10^{-3}$ M, $P(O_2) = 0.1$ atm and pH = 3, the cell potential at 25°C is -

(A) 1.47 V (B) 1.77 V (C) 1.87 V (D) 1.57 V

2. $AgNO_3$ (a) was added to an aqueous KCl solution gradually and the conductivity of the solution was measured. the plot of conductance (Λ) versus the volume of $AgNO_3$ is - **[JEE 2011]**



(A) (P) (B) (Q) (C) (R) (D) (S)

Paragraph for Question 3 and 4

The electrochemical cell shown below is a concentration cell. **[JEE 2012]**

M | M^{2+} (saturated solution of a sparingly soluble salt, MX_2) || M^{2+} (0.001 mol dm^{-3}) | M

The emf of the cell depends on the difference in concentrations of M^{2+} ions at the two electrodes. The emf of the cell at 298 K is 0.059V.

3. The value of ΔG (kJ mol^{-1}) for the given cell is (take If = 96500 C mol^{-1})

(A) –5.7 (B) 5.7 (C) 11.4 (D) –11.4.

4. The solubility product (K_{sp} ; $mol^3\ dm^{-9}$) of MX_2 at 298 K based on the information available for the given concentration cell is (take 2.303 × R × 298|F = 0.059 V)

(A) 1×10^{-15} (B) 4×10^{-15} (C) 1×10^{-12} (D) 1×10^{-12}

5. The standard reduction potential data at 25ºC is given below **[JEE-Adv. 2013]**

Eº (Fe^{3+}, Fe^{2+}) = +0.77 V ;

Eº (Fe^{2+}, Fe) = –0.44 V ;

Eº (Cu^{2+}, Cu) = +0.34 V ;

Eº (Cu^{+}, Cu) = +0.52 V ;

Eº $(O_2(g) + 4H^+ + 4e^- \rightarrow 2H_2O]$ = +1.23 V ;

Eº $[(O_2(g) + 2H_2O + 4e^- \rightarrow 4OH^-)]$ = +0.40 V ;

Eº (Cr^{3+}, Cr) = –0.74 V ;

Eº (Cr^{2+}, Cr) = –0.91 V ;

Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\Electrochemistry\Eng\02_Ex.p65

Match E° of the redox pair in List-I with the values given in List-II and select the correct answer using the code given below the lists :

	List-I		List-II
(P)	$E^o(Fe^{3+}, Fe)$	(1)	–0.18 V
(Q)	$E^o(4H_2O \rightleftharpoons 4H^+ + 4OH^+)$	(2)	–0.4 V
(R)	$E^o(Cu^{2+} + Cu \rightarrow 2Cu^+)$	(3)	–0.04 V
(S)	$E^o(Cr^{3+}, Cr^{2+})$	(4)	–0.83 V

Codes :

	P	Q	R	S
(A)	4	1	2	3
(B)	2	3	4	1
(C)	1	2	3	4
(D)	3	4	1	2

6. An aqueous solution of X is added slowly to an aqueous solution of Y as shown in List-I. The variation in conductivity of these reactions is given in List-II. Match List-I with List-II and select the correct answer using the code given below the lists : **[JEE-Adv. 2013]**

	List-I		List-II
(P)	$\underset{X}{(C_2H_5)_3N} + \underset{Y}{CH_3COOH}$	(1)	Conductivity decreases and then increases
(Q)	$\underset{X}{KI(0.1M)} + \underset{Y}{AgNO_3(0.01M)}$	(2)	Conductivity decreases and then does not change much
(R)	$\underset{X}{CH_3COOH} + \underset{Y}{KOH}$	(3)	Conductivity increases and then does not change much
(S)	$\underset{X}{NaOH} + \underset{Y}{HI}$	(4)	Conductivity does not change much and then increases

Codes :

	P	Q	R	S
(A)	3	4	2	1
(B)	4	3	2	1
(C)	2	3	4	1
(D)	1	4	3	2

7. In a galvanic cell , the salt bridge - **[JEE-Adv. 2014]**
(A) Does not participate chemically in the cell reaction
(B) Stops the diffusion of ions from one electrode to another
(C) Is necessary for the occurence of the cell reaction
(D) Ensures mixing of the two electrolytic solutions

8. The molar conductivity of a solution of a weak acid HX (0.01 M) is 10 times smaller than the molar conductivity of a solution of a weak acid HY (0.1 M). If $\lambda^0_{X^-} \approx \lambda^0_{Y^-}$, the difference in their pK_a values , $pK_a(HX) - pK_a(HY)$, is (consider degree of ionization of both acids to be <<1).
[JEE-Adv. 2015]

9. All the energy released from the reaction $X \rightarrow Y$, $\Delta_rG^o = -193$ kJ mol^{-1} is used for the oxidizing M^+ and $M^+ \rightarrow M^{3+} + 2e^-$, $E^o = -0.25$ V. **[JEE-Adv. 2015]**

Under standard conditions, the number of moles of M^+ oxidized when one mole of X is converted to Y is - [F = 96500 C mol^{-1}]

ALLEN Session 2019-20

10. For the following electrochemical cell at 298K, **[JEE-Adv. 2016]**

$Pt(s) \mid H_2(g, 1bar) \mid H^+ (aq, 1M) \parallel M^{4+}(a), M^{2+}(a) \mid Pt(s)$

$E_{cell} = 0.092\ V$ when $\frac{[M^{2+}(aq.)]}{[M^{4+}(aq.)]} = 10^x$

Given : $E^0_{M^{4+}/M^{2+}} = 0.151V$; $2.303\ \frac{RT}{F} = 0.059V$

The value of x is -

(A) –2 (B) –1 (C) 1 (D) 2

11. The conductance of a 0.0015 M aqueous solution of a weak monobasic acid was determined by using a conductivity cell consisting of platinized Pt electrodes. The distance between the electrodes is 120 cm with an area of cross section of 1 cm^2. The conductance of this solution was found to be $5 \times 10^{-7}S$. The pH of the solution is 4. The value of limiting molar conductivity (Λ^0_m) of this weak monobasic acid in aqueous solution is $Z \times 10^2 S\ cm^{-1} mol^{-1}$. The value of Z is.

12. For the following cell : **[JEE-Adv. 2017]**

$Zn(s) \mid ZnSO_4\ (aq.) \parallel CuSO_4\ (aq.) \mid Cu(s)$

when the concentration of Zn^{2+} is 10 times the concentration of Cu^{2+} , the expression for ΔG (in J mol^{-1}) is

[F is Faraday constant , R is gas constant, T is temperature , E^o(cell) = 1.1V]

(A) 2.303 RT + 1.1F (B) 2.303 RT – 2.2F

(C) 1.1 F (D) –2.2 F

13. Consider an electrochemical cell: $A(s) \mid A^{n+}\ (aq, 2M) \parallel B^{2n+}\ (aq, 1M) \mid B(s)$. The value of ΔH^θ for the cell reaction is twice that of ΔG^θ at 300 K. If the emf of the cell is zero, the ΔS^θ (in $JK^{-1}\ mol^{-1}$) of the cell reaction per mole of B formed at 300 K is___. **[JEE-Adv. 2018]**

(Given : ln (2) = 0.7, R (universal gas constant) = 8.3 J K^{-1} mol^{-1}. H, S and G are enthalpy, entropy and Gibbs energy, respectively.)

14. For the electrochemical cell,

$Mg(s) \mid Mg^{2+}(aq, 1M) \parallel Cu^{2+}\ (aq, 1M) \mid Cu(s)$

the standard emf of the cell is 2.70 V at 300 K. When the concentration of Mg^{2+} is changed to x M, the cell potential changes to 2.67 V at 300 K. The value of x is____.**[JEE-Adv. 2018]**

(Given, $\frac{F}{R} = 11500\ KV^{-1}$, where F is the Faraday constant and R is the gas constant, ln(10) = 2.30)

ALLEN Session 2019-20

ANSWER KEY

EXERCISE (S-I)

1. Ans. 0.53 V, disproportionation
2. Ans. $K_c = 2.868 \times 10^{107}$, $\Delta G^0 = -611.8$ kJ
3. Ans. $E^0 = -0.14903$V
4. Ans. E = –0.81 V
5. Ans. $K_w \approx 10^{-14}$
6. Ans. -1.30×10^3 kJ mol^{-1}
7. Ans 3.86
8. Ans. (3)
9. Ans. E = 0.059
10. Ans. E = 0.413 V
11. Ans. 1.023×10^5 sec
12. Ans. 115800C, 347.4 kJ
13. Ans. 1.12 mol, 12.535 litre
14. Ans. Rs. 0.75 x
15. Ans. A = 114, Q = 5926.8C
16. Ans. 60 %
17. Ans. 1.825g
18. Ans. 2M
19. Ans. (i) 250 mho $cm^2 mol^{-1}$, (ii) 125 mho cm^2 $equivalent^{-1}$
20. Ans. 9.4×10^{-4} gm|litre
21. Ans. $\alpha = 0.5$, $k = 10 \times 10^{-4}$
22. Ans. (i) 7 (ii) 1×10^{-14}

EXERCISE (S-II)

1. Ans.– 0.46 V
2. Ans.(ii). 1.27 V, (iii) 245.1 kJ
3. Ans. $E^0 = -0.22$ V
4. Ans. $K_{sp} = 1.1 \times 10^{-16}$
5. Ans. $[Br^-] : [Cl^-] = 1 : 200$
6. Ans.–0.037 V
7. Ans. 1.536×10^{-5} M^3
8. Ans. $K = 10^{268}$
9. Ans. 43.456g
10. Ans.42.2 %
11. Ans. 1.9×10^6 year
12. Ans.0.1456 ampere
13. 4.25×10^{-2} metre
14. Ans.0.1934 gm|litre
15. Ans.n = 2
16. Ans.0.52 V, 0.61 V
17. Ans.0.0295 V
18. Ans. t = 193 sec
19. Ans. Final weight = 9.6g, 0.01 Eq of acid
20. Ans. t = 93.65 sec.

EXERCISE (O-I)

1.	Ans. C	2.	Ans.A	3.	Ans.C	4.	Ans.A
5.	Ans.C	6.	Ans.C	7.	Ans.A	8.	Ans.B
9.	Ans.D	10.	Ans.B	11.	Ans.B	12.	Ans.D
13.	Ans.D	14.	Ans.C	15.	Ans.A	16.	Ans.C
17.	Ans.C	18.	Ans.B	19.	Ans.C	20.	Ans.B
21.	Ans.D	22.	Ans.C	23.	Ans.C	24.	Ans.C

ALLEN Session 2019-20

EXERCISE (O-II)

1.	Ans.B	2.	Ans.D	3.	Ans.A	4.	Ans.B
5.	Ans. B	6.	Ans.A	7.	Ans.A	8.	Ans.B
9.	Ans.D	10.	Ans.C	11.	Ans. A	12.	Ans.C
13.	Ans.(B)	14.	Ans.D	15.	Ans.D	16.	Ans.B
17.	Ans.C	18.	Ans.C	19.	Ans.C	20.	Ans. C,D
21.	Ans.B,C,D	22.	Ans.A,B	23.	Ans.B,C	24.	Ans.A,D
25.	Ans.A	26.	Ans.A	27.	Ans.D	28.	Ans.(C)
29.	Ans.(A)	30.	Ans.(C)	31.	Ans. (A) P, Q (B) P, Q (C) Q, R, (D) P,S		
32.	Ans.(C)						

EXERCISE (J-MAINS)

1.	Ans.(2)	2.	Ans.(1)	3.	Ans.(1)	4.	Ans.(4)
5.	Ans. (1)	6.	Ans.(3)	7.	Ans. (3)	8.	Ans. (2)
9.	Ans.(4)	10.	Ans.(2)	11	Ans. (1)	12.	Ans. (1)
13.	Ans.(3)	14.	Ans.(2)	15.	Ans.(2)	16.	Ans.(2)
17.	Ans.(4)						

EXERCISE (J-ADVANCED)

1.	Ans.(D)	2.	Ans.(D)	3.	Ans.(D)	4.	Ans.(B)
5.	Ans.(D)	6.	Ans.(A)	7.	Ans.(A,B)	8.	Ans. (3)
9.	Ans. (4)	10.	Ans.(D)	11.	Ans.(6)	12.	Ans.(B)
13.	Ans.(–11.62)	14.	Ans.(10)				

ALLEN Session 2019-20

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NURTURE COURSE

ELECTRONIC DISPLACEMENT EFFECTS

ELECTRONIC DISPLACEMENT EFFECTS

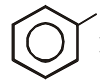
EXERCISE # I

1. Which of the following is false order of – I effect ?

(A) $-F > -Cl > -Br > -I$

(B) $-\overset{\oplus}{N}R_3 > -\overset{\oplus}{N}H_3 > -NO_2$

(C) $-F > -OH > -NH_2$

(D)  $> -C \equiv CH > -H$

2. What is the correct order of inductive effect ?

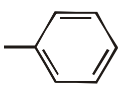
(A) $-O^- > -CH_3 > -CMe_3$

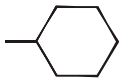
(B) $-CO_2^- > -O^- > CHMe_2$

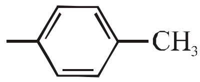
(C) $-O^- > -CH_2Me > -D > -H$

(D) None

3. Which of the following groups have + I effect :

(A) 

(B) 

(C) 

(D) $—CH = CH_2$

4. Which of the following groups have –I effect :

(A) —OH

(B) $-\underset{\underset{O}{\|}}{C}-OH$

(C) $—CH_3$

(D) $—OCH_3$

5. How many of the following groups have + I effect :

(a) —OH
(b) $—O^{\ominus}$
(c) $—NH_2$
(d) $—\overset{\ominus}{N}H$
(e) —COOH
(f) $—COO^{\ominus}$
(g) —Me
(h) —OMe
(i) —F
(j) $—\overset{\oplus}{N}F_3$

6. Which of the following statements is (are) true about resonance.

(a) Resonance is an intramolecular phenomenon.

(b) Resonance involves delocalization of both σ and π electrons.

(c) Resonance involves delocalization of π electrons only.

(d) Resonance decreases potential energy of an acyclic molecule.

(e) Resonance has no effect on the potential energy of a molecule.

(f) Resonance is the only way to increase molecular stability.

(g) Resonance is not the only way to increase molecular stability.

(h) Any resonating molecule is always more stable than any non resonating molecule.

(i) The canonical structure explains all features of a molecule.

(j) The resonance hybrid explains all features of a molecule.

(k) Resonating structures are real and resonance hybrid is imaginary.

(l) Resonance hybrid is real and resonating structures are imaginary.

(m) Resonance hybrid is always more stable than all canonical structures.

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7. Which of the following statement is incorrect ?

(A) Resonating structure are real & have real existence

(B) Equivalent contributing structures make resonance hybrid very stable.

(C) Contributing structures are hypothetical having no real existance

(D) Contributing structures are less stable than the resonance hybrid.

8. Which of the following is most stable.

(A) Conjugated alkadiene ($CH_2 = CH—CH = CH_2$)

(B) Isolated alkadiene ($CH_2 = CH—CH_2—CH = CH_2$)

(C) Cumulated alkadiene ($CH_2 = C = CH_2$)

(D) All are equally stable

9. Arrange the following resonating structure according to their contribution towards resonance hybrid?

(a) $CH_2 = \overset{\oplus}{N} = \overset{\ominus}{N}$ (b) $\overset{\ominus}{C}H_2-N=\overset{\oplus}{N}$ (c) $\overset{\oplus}{C}H_2-\underset{\bullet\bullet}{N}=\overset{\ominus}{N}$ (d) $\overset{\ominus}{C}H_2-\overset{\oplus}{N}\equiv\ddot{N}$

(A) a > d > c > b (B) b > a > c > d (C) a > c > b > d (D) d > a > b > c

10. A canonical structure will be more stable if

(A) it involves cyclic delocalization of (4n + 2) π – electrons than if it involves acyclic delocalization of (4n + 2) π – electrons.

(B) it involves cyclic delocalization (4n) π – electrons than if it involves acyclic delocalization of (4n) π – electrons.

(C) +ve charge is on more electronegative atom than if +ve charge is on less electronegative atom provided atoms are in the same period.

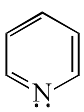
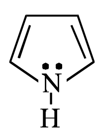
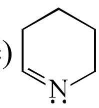
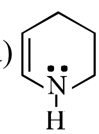
(D) –ve charge is on more electronegative atom than if –ve charge is on less electronegative atom provided atoms are in the same period.

11. Which one of the following pair of structures does not represent the phenomenon of resonance?

(A) $H_2C=CH-\overset{\overset{O}{\|}}{C}-H$; $\overset{+}{C}H_2-CH=\overset{\overset{O^-}{|}}{C}-H$ (B) $CH_2=CH-\overset{+}{C}HCl$; $\overset{+}{C}H_2-CH=CH-Cl$

(C) $(CH_3)_2CH-\overset{\overset{O}{\|}}{C}-O^-$; $(CH_3)_2CH-\overset{\overset{O^-}{|}}{C}=O$ (D) $CH_3-CH_2-\overset{\overset{O}{\|}}{C}-CH_3$; $CH_3-CH=\overset{\overset{O^-}{|}}{C}-CH_3$

12. In which of the following, lone-pair indicated is involved in resonance :

(a)  (b)  (c)  (d) 

(e) $CH_2=CH-CH_2^{\ominus}$ (f) $CH_2=CH-CH=\ddot{N}H$

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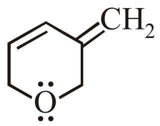
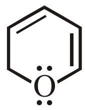
13. In which of the following lone-pair indicated is not involved in resonance :

(a) $CH_2 = CH - \ddot{N}H - CH_3$ (b) $CH_2 = CH - CH = \underset{\cdot\cdot}{\ddot{O}}$

(c) $CH_2 = CH - \underset{\cdot\cdot}{\ddot{O}} - CH = CH_2$ (d) $CH_2 = CH - C \equiv N\!:$

(e)  (f) 

14. Which of the following groups cannot participate in resonance with other suitable group :

(a) $-COOH$ (b) $-COOCH_3$ (c) $-COCl$ (d) $-\overset{\oplus}{N}H_3$

(e) $-\overset{\oplus}{C}H_2$

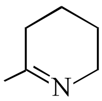
15. Identify electron donating groups in resonance among the following :

(a) $-CONH_2$ (b) $-NO_2$ (c) $-OCOCH_3$ (d) $-COOCH_3$

(e) $-CHO$ (f) $-NHCOCH_3$

16. Identify electron withdrawing groups in resonance among the following :

(a) $-COOH$ (b) $-CONHCH_3$ (c) $-COCl$ (d) $-CN$

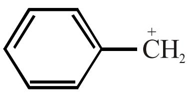
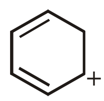
(e) $-O-CH=CH_2$ (f) 

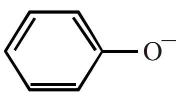
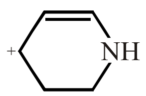
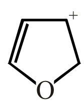
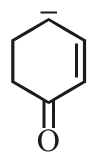
17. Which of the following groups can either donate or withdraw a pair of electrons in resonance depending upon situation :

(a) $-NO_2$ (b) $-NO$ (c) $-CH=CH_2$ (d) $-CHO$

(e) $-NH_2$ (f) $-N=NH$

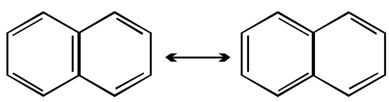
18. Draw the resonance forms to show the delocalization of charges in the following ions

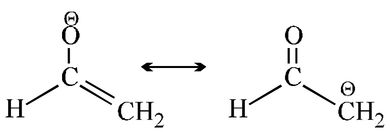
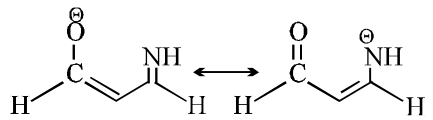
(a) $CH_3-\overset{\overset{O}{\|}}{C}-\overset{\ominus}{C}H_2$ (b) $H-\overset{\overset{O}{\|}}{C}-CH=CH-\overset{\ominus}{C}H_2$ (c)  (d) 

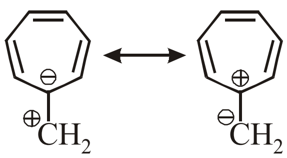
(e)  (f)  (g)  (h) 

(i) $CH_3-CH=CH-CH=CH-\overset{+}{C}H-CH_3$ (j) $CH_3-CH=CH-CH=CH-\overset{+}{C}H_2$

19. Identify less stable canonical structure in each of the following pairs :

(a) $\overset{\oplus}{C}H_2-O-CH_3 \longleftrightarrow CH_2=\overset{\oplus}{O}-CH_3$ (b) 

(c)  (d) 

(e) 

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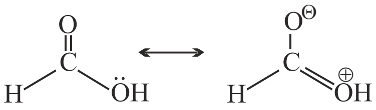
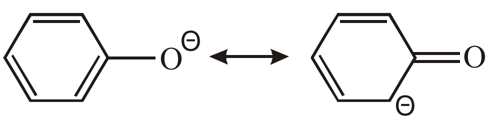
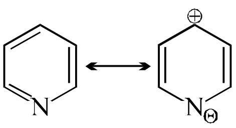
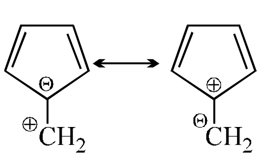
20. Identify more stable canonical structure in each of the following pairs :

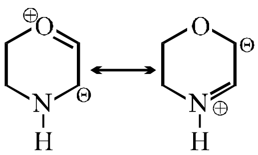
(a) 

(b) 

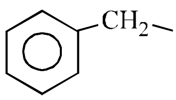
(c) 

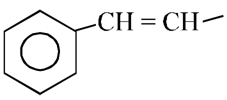
(d) 

(e) $\overset{\oplus}{C}H_2 - CH = CH - \overset{\ominus}{O} \longleftrightarrow CH_2 = CH - CH = O$

(f) 

21. Which of the following group can participate in resonance with other suitable group :

(a) –OH (b) $-CH_2-\overline{C}H_2$ (c) $-CH_2-\overset{\oplus}{C}H_2$ (d) 

(e)  (f) $-BH_2$ (g) $-\overset{\oplus}{P}Ph_3$

22. Consider structural formulas A, B and C:

$H_2\ddot{C}—N\equiv N:$ (A) $H_2C = N = \ddot{N}:$ (B) $H_2C—\ddot{N} = \ddot{N}:$ (C)

(a) Are A, B and C isomers, or are they resonance forms ?
(b) Which structures have a negatively charged carbon?
(c) Which structures have a positively charged carbon?
(d) Which structures have a positively charged nitrogen?
(e) Which structures have a negatively charged nitrogen?
(f) What is the net charge on each structure?
(g) Which is a more stable structure, A or B? Why?
(h) Which is a more stable structure, B or C? Why?

23. In each of the following pairs of resonating structure which resonating structure is more stable :

(a) $\left[CH_3—\ddot{\bar{C}}H–C\equiv N: \longleftrightarrow CH_3—CH=C=\ddot{N}:^{-}\right]$

(b) $\left[CH_3-\overset{O}{\overset{\|}{C}}-\overline{C}H-\overset{O}{\overset{\|}{C}}-CH_3 \longleftrightarrow CH_3-\overset{O^-}{\overset{|}{C}}=CH-\overset{O}{\overset{\|}{C}}-CH_3\right]$

(c) $\left[CH_3—CH_2—\underset{+}{\overset{NH_2}{\overset{|}{C}}}—NH_2 \longleftrightarrow CH_3—CH_2—\overset{NH_2}{\overset{|}{C}}=\overset{+}{N}H_2\right]$

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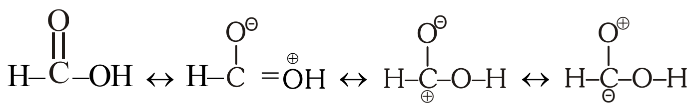
24. Formic acid is considered as a hybrid of the four structures

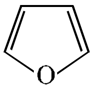
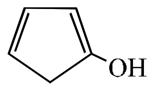


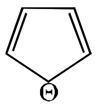
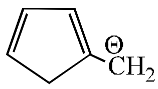
I II III IV

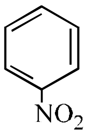
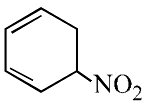
Which of the following order is correct for the stability of four contributing structures.

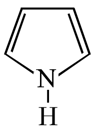
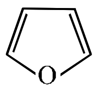
(A) I > II > III > IV (B) I > II > IV > III (C) I > III > II > IV (D) I > IV > III > II

25. In the given pair of compounds select the one in each pair having lesser resonance energy :

(a)  and 

(b)  and 

(c)  and 

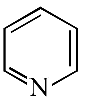
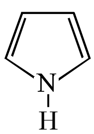
(d)  and 

26. Resonance energy of resonance hybrid of a molecule will be more if :

(a) canonical structures are equivalent than if canonical structures are non-equivalent

(b) molecule is aromatic than if molecule is not aromatic.

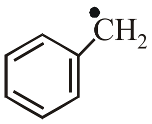
27. In the given pair of compounds select the one in each pair having higher resonance energy :

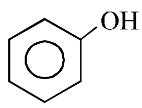
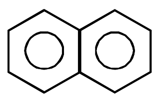
(i)  and 

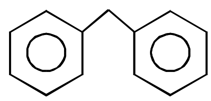
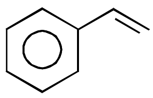
(ii) $CH_2 = CH - O - CH = CH_2$ and $CH_2 = CH - NH - CH = CH_2$

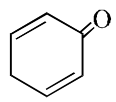
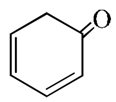
(iii) $CH_2 = CH - \overset{\ominus}{N}H$ and $HN = CH - \overset{\ominus}{N}H$

(iv) $CH_2 = CH - F$ and $CH_2 = CH - Br$

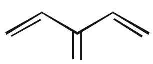
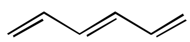
(v)  and $CH_2 = CH - \dot{C}H_2$

(vi)  and 

(vii)  and 

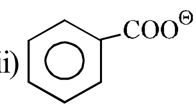
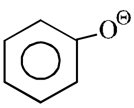
(viii)  and 

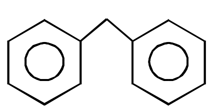
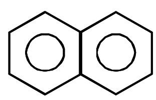
(ix) $CH_2 = CH - OH$ and $CH_2 = CH - CH = CH - OH$

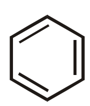
(x)  and 

(xi) CH_3COOH and CH_3COONa

(xii) $CH_2 = CH - \overset{\ominus}{O}$ and $CH_2 = CH - OH$

(xiii)  and 

(xiv)  and 

(xv)  and $CH_2 = CH - CH = CH - CH = CH_2$

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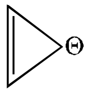
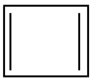
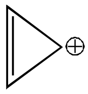
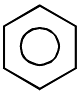
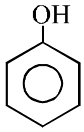
28. In the given pair of compounds select the one in each pair having lesser resonance energy :

(a) CO_3^{2-} and $HCOO^-$

(b)  and $CH_2 = CH - \overset{-}{C}H_2$

(c)  and $CH_2 = CH - CH = CH_2$

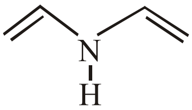
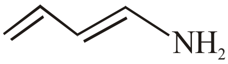
(d)  and $CH_2 = CH - \overset{\oplus}{C}H_2$

(e)  and 

29. In which of the following pairs first one is having more resonance energy than the second one -

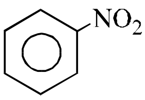
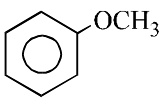
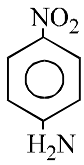
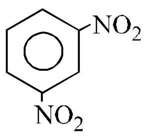
(A)  , 

(B)  , 

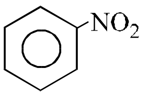
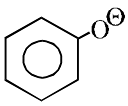
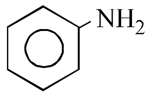
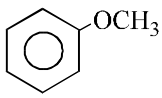
(C)  , 

(D) None of these

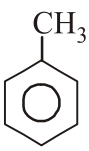
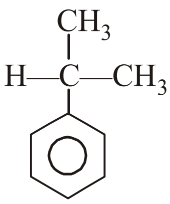
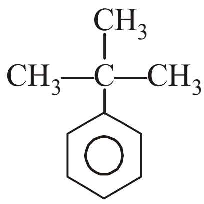
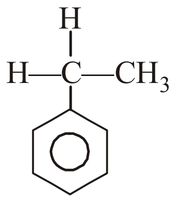
30. In which of the following molecules π – electron density in ring is minimum :

(A) 
(B) 
(C) 
(D) 

31. In which of the following molecules π – electron density in ring is maximum :

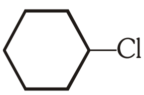
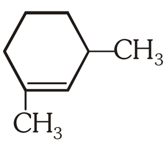
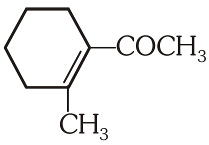
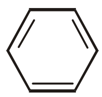
(A) 
(B) 
(C) 
(D) 

32. Arrange following compounds in decreasing order of reactivity of ring towards attack of electron deficient species -

(i) 
(ii) 
(iii) 
(iv) 

(A) i > ii > iii > iv (B) iii > iv > ii > i (C) i > iv > ii > iii (D) i > ii > iv > iii

33. In which of the following molecule all the effect namely inductive, mesomeric & hyperconjugation operate:

(A) 
(B) 
(C) 
(D) 

34. Which one of the following molecules has all the effect, namely inductive, mesomeric and hyperconjugative?

(A) CH_3Cl (B) $CH_3–CH = CH_2$

(C) $CH_3 - CH = CH - \underset{\underset{O}{\|}}{C} - CH_3$ (D) $CH_2 = CH - CH = CH_2$

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35. Select the correct statement.

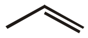
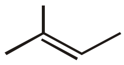
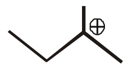
(i) Delocalisation of σ-electron is hyperconjugation.

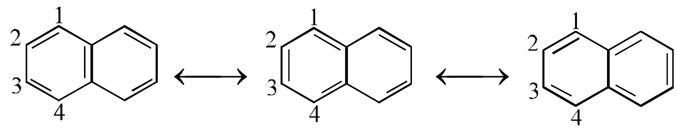
(ii) Delocalisation of π-electron is resonance.

(iii) Permanent partial displacement of σ-electron is inductive effect.

(A) i & iii (B) ii & iii (C) i & ii (D) i, ii, iii

36. Which of the following compound is correctly matched with number of hyperconjugating structures (involving C—H bond) :

(A)  (3) (B)  (9) (C)  (8) (D) $CH_3–C≡C–CH_3$ (5)

37. 

These are three canonical structures of naphthalene. Examine them and find correct statement among the following :

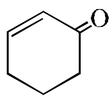
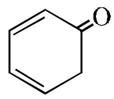
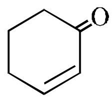
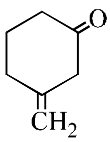
(A) All C – C bonds are of same length

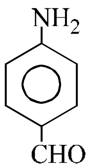
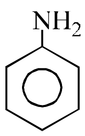
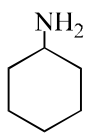
(B) C1 – C2 bond is shorter than C2 – C3 bond.

(C) C1 – C2 bond is longer than C2 – C3 bond

(D) None

38. Which of the following has longest C – O bond :

(A)  (B)  (C)  (D) 

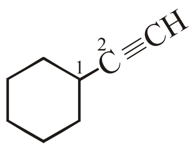
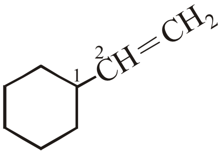
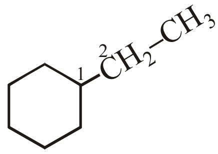
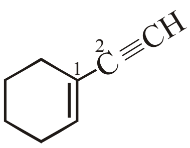
39.    $CH_2 = NH$

I II III IV

Among these compounds, the correct order of C – N bond lengths is :

(A) IV > I > II > III (B) III > I > II > IV (C) III > II > I > IV (D) III > I > IV > II

40. C1 – C2 bond is shortest in

(A)  (B)  (C)  (D) 

41. Which of the following molecule has longest C=C bond length ?

(A) $CH_2=C=CH_2$ (B) $CH_3–CH=CH_2$ (C) $CH_3–\underset{CH_3}{\overset{CH_3}{C}}–CH=CH_2$ (D) $CH_3–\underset{CH_3}{C}=CH_2$

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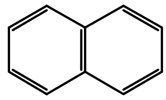
42. Which of the following molecule has shortest C=C bond length ?

(A) $CH_2=C=CH_2$ (B) $CH_3–CH=CH_2$ (C) $CH_3–C(CH_3)_2–CH=CH_2$ (D) $CH_3–C(CH_3)=CH_2$

43. C—C and C=C bond lengths are unequal in :

(A) Benzene (B) 1,3-buta-di-ene (C) 1,3-cyclohexa-di-ene (D) 

44. Among the following molecules, the correct order of C – C bond length is (C_6H_6 is benzene)

(A) $C_2H_6 > C_2H_4 > C_6H_6 > C_2H_2$ (B) $C_2H_6 > C_6H_6 > C_2H_4 > C_2H_2$

(C) $C_2H_4 > C_2H_6 > C_2H_2 > C_6H_6$ (D) $C_2H_6 > C_2H_4 > C_2H_2 > C_6H_6$

45. $CH_3O – CH = CH – NO_2$ I

$CH_2 = CH – NO_2$ II

$CH_2 = CH – Cl$ III

$CH_2 = CH_2$ IV

Which of the following is the correct order of C – C bond lengths among these compounds :

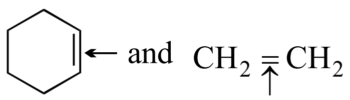
(A) I > II > III > IV (B) IV > III > II > I (C) I > III > II > IV (D) II > III > I > IV

46. Which of the following is (are) the correct order of bond lengths :

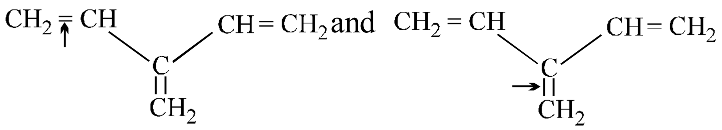
(A) $C – C > C = C > C \equiv C > C \equiv N$ (B) $C = N > C = O > C = C$

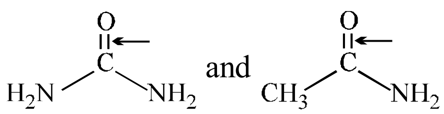
(C) $C = C > C = N > C = O$ (D) $C – C > C = C > C \equiv C > C – H$

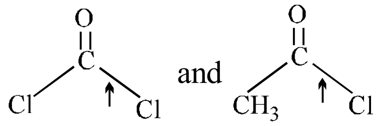
47. In which of the following pairs, indicated bond having less bond dissociation energy :

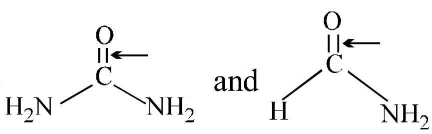
(a) 

(b) $CH_3 – C \equiv CH$ and $HC \equiv CH$

(c) 

(d) 

(e) 

(f) 

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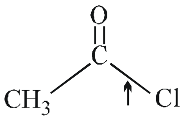
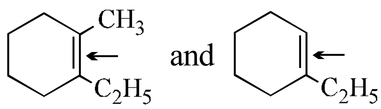
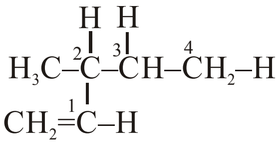
48. In which of the following pairs, indicated bond is of greater strength :

(a) $CH_3-CH_2\underset{\uparrow}{-}Br$ and $CH_3-CH_2\underset{\uparrow}{-}Cl$ (b) $CH_3-CH=CH\underset{\uparrow}{-}Br$ and $CH_3-\underset{\underset{Br}{|}}{CH}-CH_3$

(c) 

and $CH_3-CH_2\underset{\uparrow}{-}Cl$ (d) $CH_2=CH\underset{\uparrow}{-}CH=CH_2$ and $CH_2=CH_2\underset{\uparrow}{-}CH_2-CH_3$

(e) $CH_2=CH\underset{\uparrow}{-}CH=CH_2$ and $CH_2=CH\underset{\uparrow}{-}NO_2$ (f) 

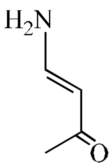
49. 

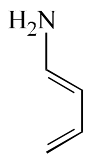
the correct order of bond dissociation energy (provided bond undergoes homolytic cleavage):

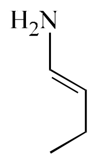
(A) $C^2–H > C^3–H > C^4–H > C^1–H$ (B) $C^2–H > C^3–H > C^1–H > C^4–H$

(C) $C^1–H > C^4–H > C^2–H > C^3–H$ (D) $C^1–H > C^4–H > C^3–H > C^2–H$

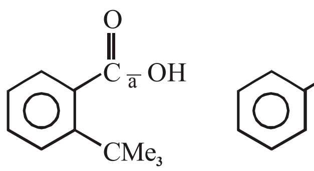
50. Compare the C–N bond-length in the following species:

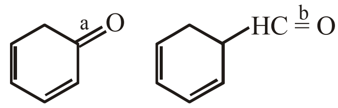
(A) 

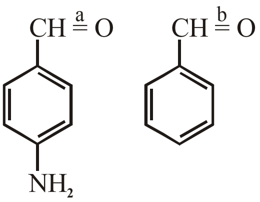
(B) 

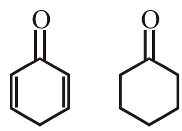
(C) 

51. In which case, C – O bond length is shorter for I[st] compound :

(A) 

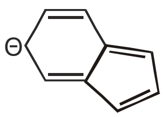
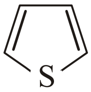
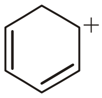
(B) 

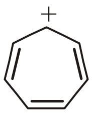
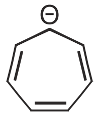
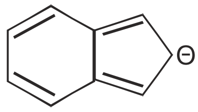
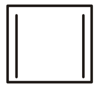
(C) 

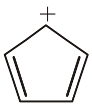
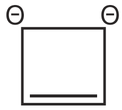
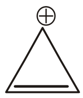
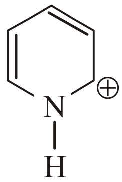
(D) 

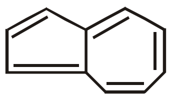
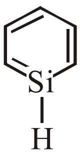
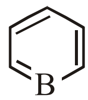
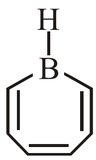
EXERCISE # II

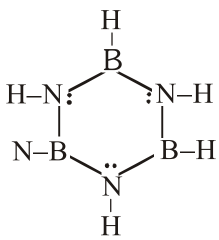
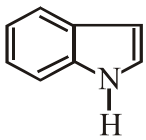
1. In each set of species select the aromatic species.

(i) (a)  (b)  (c) 

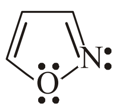
(ii) (a)  (b)  (c)  (d) 

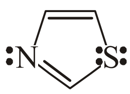
(iii) (a)  (b)  (c)  (d) 

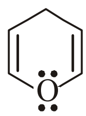
(iv) (a)  (b)  (c)  (d) 

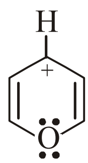
(e)  (f) 

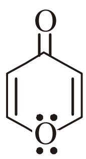
2. Which of the given compound is aromatic, antiaromatic or nonaromatic.

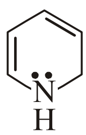
(a) 
isoxazole

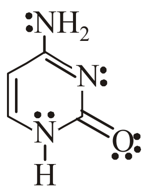
(b) 
1,3-thiazole

(c) 
pyran

(d) 
pyrylium ion

(e) 
γ-pyrone

(f) 
1,2-dihydropyridine

(g) 
cytosine

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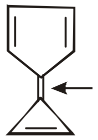
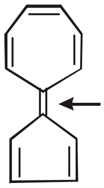
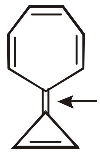
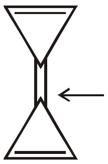
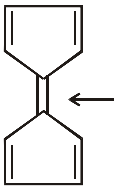
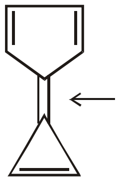
3. (I)  (II)  (III) 

Compare carbon-carbon bond rotation across I, II, III.

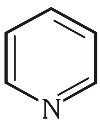
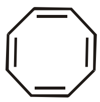
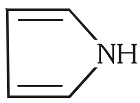
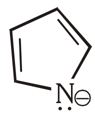
(A) I > II > III (B) I > III > II (C) II > I > III (D) II > III > I

4. Which of the given compunds has minimum rotation energy barrier across indicated carbon-carbon bond.

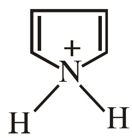
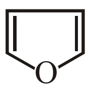
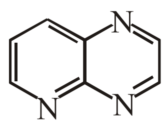
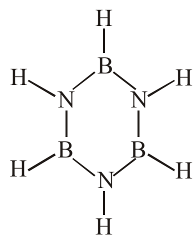
(I)  (II)  (III) 

(A) I (B) II (C) III (D) All are equal

5. Which species is not aromatic ?

(A)  (B)  (C)  (D) 

6. Which of the following are non-aromatic

(A)  (B)  (C)  (D) 

7. Write down the structure of the following molecule and comment on aromaticity ?

(a) $B_3H_3O_3$ (b) $C_3N_3\ (NH_2)_3$ (c) Trimer of isocyanic acid $(HN = C = O)_3$

8. Select the least stable one :

(A) $CH_3-\overset{\oplus}{C}H_2$ (B) $CH_3-CH_2-\overset{\oplus}{C}H_2$

(C) $(H_3C)_2CH-\overset{\oplus}{C}H_2$ (D) $(H_3C)_3C-\overset{\oplus}{C}H_2$

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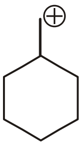
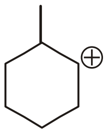
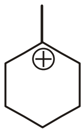
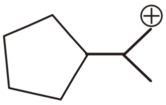
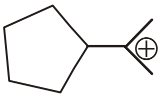
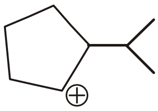
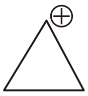
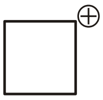
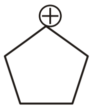
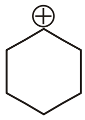
9. Write stability in decreasing order of following intermediates:

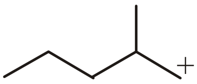
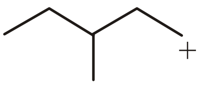
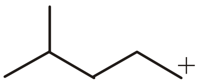
(i) (a) $CH_3-\overset{\oplus}{C}H_2$ (b) $CH_3-\overset{\oplus}{C}H-CH_3$ (c) $CH_3-\overset{\displaystyle CH_3}{\underset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}\oplus$

(ii) (a)  (b)  (c) 

(iii) (a)  (b)  (c) 

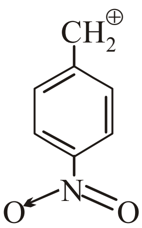
(iv) (a)  (b)  (c)  (d) 

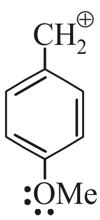
(v) (a) $CF_3-\overset{+}{C}H_2$ (b) $CCl_3-\overset{+}{C}H_2$ (c) $CBr_3-\overset{+}{C}H_2$

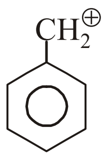
(vi) (a)  (b)  (c) 

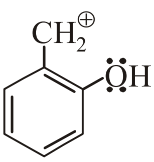
(vii)(a) $HC\equiv\overset{\oplus}{C}$ (b) $CH_2=\overset{\oplus}{C}H$ (c) $CH_3-\overset{\oplus}{C}H_2$

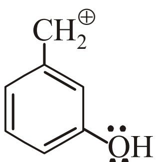
(viii) (a) $C_6H_5^+$ (b) p–$NO_2(C_6H_4)^+$ (c) p–CH_3–$(C_6H_4)^+$ (d) p–Cl–$C_6H_4^+$

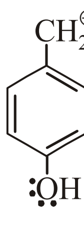
(ix) (a) 

(b) 

(c) 

(x) (a) 

(b) 

(c) 

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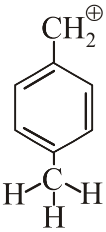
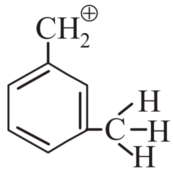
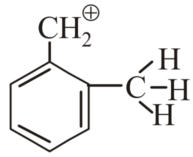
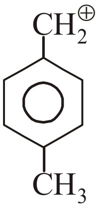
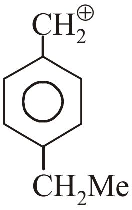
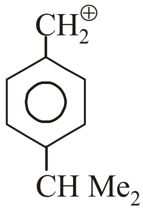
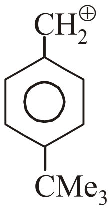
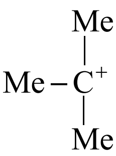
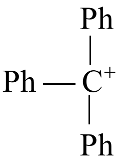
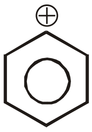
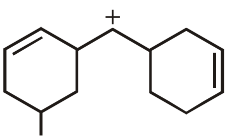
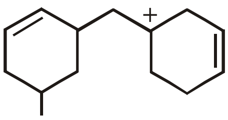
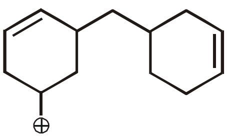
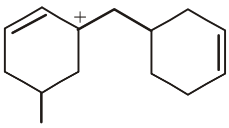
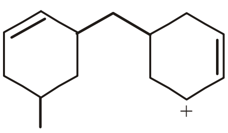
(xi) (a)  (b)  (c) 

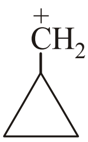
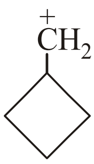
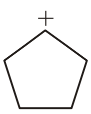
(xii) (a)  (b)  (c)  (d) 

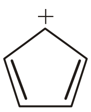
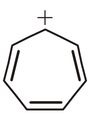
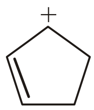
(xiii) (a)  (b)  (c) $Me-\overset{+}{C}H_2$ (d) 

(xiv)(a)  (b) 

(c)  (d) 

(e) 

(xv) (a)  (b)  (c) 

(xvi) (a)  (b)  (c) 

node06\B0AH-AI\Kota\JEE(Advanced)\Nurture\Chem\Sheet\Electronic Displacement\Eng.p65

10. Consider the following statements:

(I) $CH_3O\overset{\oplus}{C}H_2$ is more stable than $CH_3\overset{\oplus}{C}H_2$ (II) $Me_2\overset{\oplus}{C}H$ is more stable than $CH_3CH_2\overset{\oplus}{C}H_2$

(III) $CH_2 = CH - \overset{\oplus}{C}H_2$ is more stable than $CH_3CH_2\overset{\oplus}{C}H_2$ (IV) $CH_2 = \overset{\oplus}{C}H$ is more stable than $CH_3\overset{\oplus}{C}H_2$

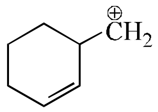
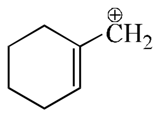
Of these statements:

(A) I and II are correct (B) III and IV are correct

(C) I, II and III are correct (D) II, III and IV are correct

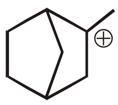
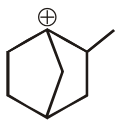
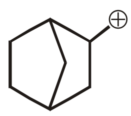
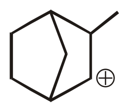
11. In each of the following pairs of ions which ion is more stable:

(a) (I) $C_6H_5-\overset{\oplus}{C}H_2$ and (II) $CH_2=CH-\overset{\oplus}{C}H_2$ (b) (I) $CH_3-\overset{\oplus}{C}H_2$ and (II) $CH_2 = \overset{\oplus}{C}H$

(c) (I)  and (II) 

(d) (I) $CH_3-\underset{\underset{\oplus}{CH_3-C-CH_3}}{\underset{|}{CH}}-CH_3$ and (II) $CH_3-\underset{\underset{\oplus}{CH_3-C-CH_3}}{\underset{|}{N}}-CH_3$

12. Find out correct stability order in the following carbocations-

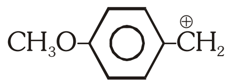
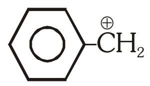
(I)  (II)  (III)  (IV) 

(A) IV > I > III > II (B) IV > III > I > II (C) I > IV > III > II (D) I > III > IV > II

13. Which of the following carbonium ion is most stable ?

(A) Ph_3C^+ (B) $(CH_3)_3\ C^+$ (C) $(CH_3)_2\ CH^+$ (D) $CH_2{=}CH{-}CH_2^+$

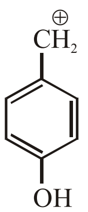
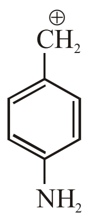
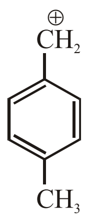
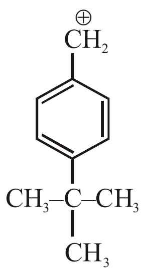
14. Consider the following carbocations

(a)  (b)  (c)  (d) $CH_3-\overset{\oplus}{C}H_2$

The relative stabilities of these carbocations are such that :-

(A) d < b < c < a (B) b < d < c < a

(C) d < b < a < c (D) b < d < a < c

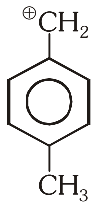
15.    

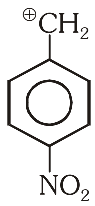
Correct order of carbocation stability is :

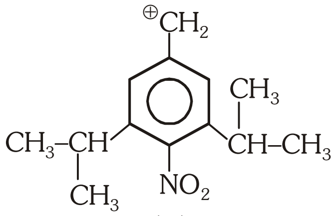
(A) 2 > 1 > 4 > 3 (B) 1 > 2 > 4 > 3 (C) 3 > 4 > 2 > 1 (D) 2 > 1 > 3 > 4

node06\B0AH-AI\Kota\JEE(Advanced)\Nurture\Chem\Sheet\Electronic Displacement\Eng.p65

16. Arrange the following carbocation in the increasing order of stability :

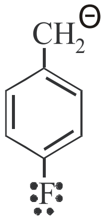
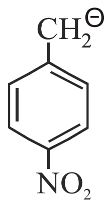

(I)

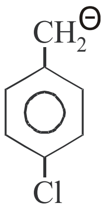
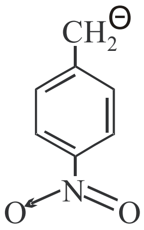
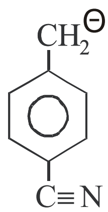
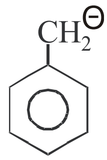

(II)


(III)

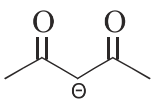
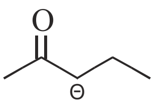
(A) I < II < III (B) II < III < I (C) III < II < I (D) III < I < II

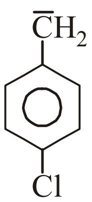
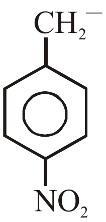
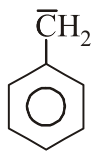
17. Rank the following sets of intermediates in increasing order of their stability.

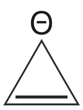
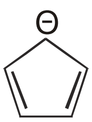
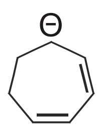
(i) (a)  (b) 

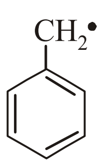
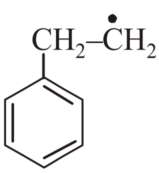
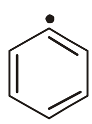
(ii) (a)  (b)  (c)  (d) 

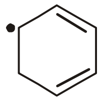
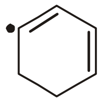
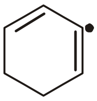
(iii) (a) $\overset{\ominus}{C}H_2-CH=O$ (b) $\overset{\ominus}{C}H_2-CH_3$

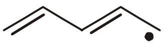
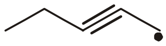
(iv) (a)  (b)  (c) 

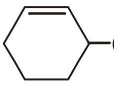
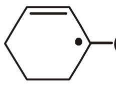
(v) (a)  (b)  (c) 

(vi) (a)  (b)  (c) 

(vii)(a)  (b)  (c) 

(viii) (a)  (b)  (c) 

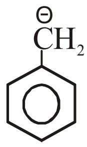
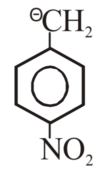
(ix) (a)  (b) 

(x) (a) $CH_2=\dot{C}H$ (b) $CH_2=CH-\dot{C}H_2$ (c)  $-\dot{C}H_2$ (d)  $-CH_3$

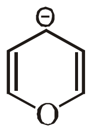
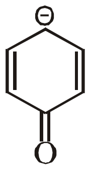
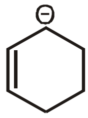
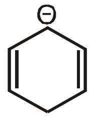
18. Most stable carbanion is :-

(A) $HC\equiv C^{\ominus}$ (B) $C_6H_5^{\ominus}$ (C) $(CH_3)_3C—CH_2^{\ominus}$ (D) $(CH_3)_2C=CH^{\ominus}$

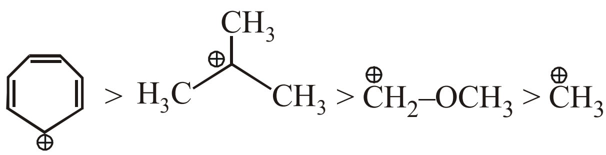
19. Most stable carbanion is :

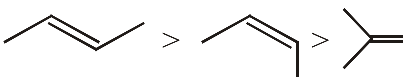
(A) $CH_3^{\ominus}$ (B) $CH_2=CH-\overset{\ominus}{C}H_2$ (C)  (D) 

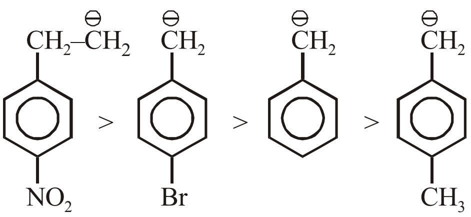
20. Identify the most stable anion.

(A)  (B)  (C)  (D) 

21. Correct order of stability :

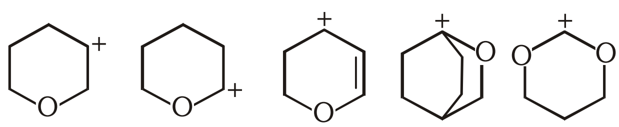
(A) 

(B) 

(C) 

(D) $CH_2=CH-\dot{C}(CH_3)_2 > CH_2=CH-\dot{C}H_2 > CH_3-\dot{C}H_2 > CH_2=\dot{C}H$

22. Rank the following sets of intermediates in increasing order of their stability giving appropriate reasons for your choice.

(a) 

(b) $\bar{C}H_2-\overset{O}{\overset{\|}{C}}-CH_3$, $\bar{C}H_2-\overset{O}{\overset{\|}{C}}-H$, $\bar{C}H_2-\overset{O}{\overset{\|}{C}}-NH_2$, $\bar{C}H_2-\overset{O}{\overset{\|}{C}}-OCH_3$

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23. Select the correct order of stability of carbon free radicals :

I. $Ph-\dot{C}H_2$ II. $Ph-\dot{C}H-CH_2-CH_3$

III. $Ph-\dot{C}H-CH=CH_2$ IV. $Ph-\overset{\bullet}{\underset{\underset{CH_3}{|}}{C}}-CH=CH_2$

(A) IV > III > I > II (B) IV > III > II > I (C) I > II > III > IV (D) I > III > II > IV

24. $\underset{\text{(I)}}{CH_2 = CH - CH = CH - CH_3}$ is more stable than $\underset{\text{(II)}}{CH_3 - CH = C = CH - CH_3}$ because

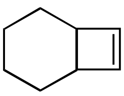
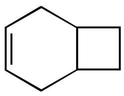
(A) there is resonance in I but not in II (B) there is tautomerism in I but not in II

(C) there is hyperconjugation in I but not in II (D) II has more cononical structures than I.

25. Choose the more stable alkene in each of the following pairs. Explain your reasoning.

(a) 1-Methylcyclohexene or 3-methylcyclohexene

(b) Isopropenylcyclopentane or allylcyclopentane

(c)  or 

26. Match each alkene with the appropriate heat of combustion:

Heats of combustion (kJ/mol) : 5293 ; 4658; 4650; 4638; 4632

(a) 1-Heptene (b) 2,4-Dimethyl-1-pentene

(c) 2,4-Dimethyl-2-pentene (d) 4,4-Dimethyl-2-pentene

(e) 2,4,4-Trimethyl-2-pentene

27. Stability of :

(I) $CH_3-CH=CH-CH_3$ (II) $CH_3-\underset{\underset{CH_3}{|}}{C}=\overset{\overset{CH_3}{|}}{C}-CH_3$

(III) $CH_3-\underset{\underset{CH_3}{|}}{C}=CH_2$ (IV) $CH_3-\underset{\underset{CH_3}{|}}{C}=CH-CH_3$

in the increasing order is :

(A) I < III < IV < II (B) I < II < III < IV

(C) I < IV < III < I (D) II < III < IV < I

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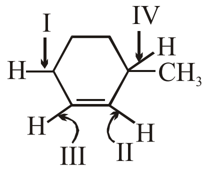
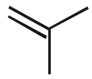
28. Which of the following C-H bonds participate in hyperconjugation ?



(A) I and II (B) I and IV (C) I and III (D) III and IV

29. Rank the following alkenes in decreasing order of heat of combustion values :


(I)

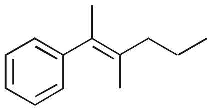
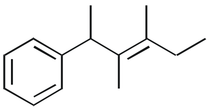
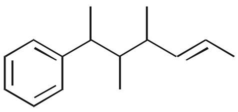
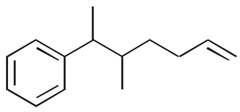

(II)

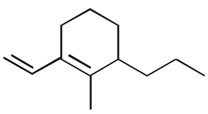
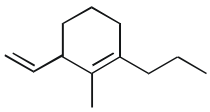
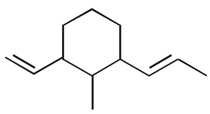
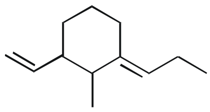

(III)

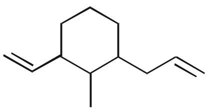

(IV)

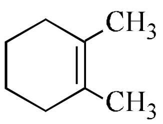
(A) II > III > IV > I
(B) II > IV > III > I
(C) I > III > IV > II
(D) I > IV > III > II

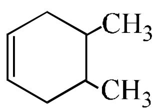
30. Write decreasing order of heat of hydrogenation :

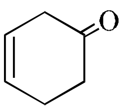
(i) (a)  (b)  (c)  (d) 

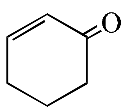
(ii) (a)  (b)  (c)  (d) 

(e) 

(iii) (a) 

(b) 

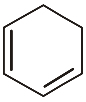
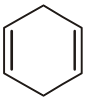
(iv) (a) 

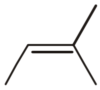
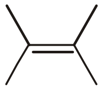
(b) 

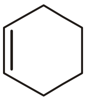
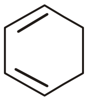
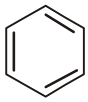
(v) (a) $\begin{matrix} CH_3 & & H \\ & C=C & \\ H & & CH_3 \end{matrix}$ (b) $\begin{matrix} CH_3 & & CH_3 \\ & C=C & \\ H & & H \end{matrix}$

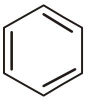
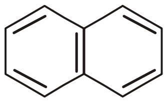
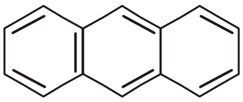
(vi) (a) $CH_2=CH-CH\begin{matrix} CH_3 \\ CH_3 \end{matrix}$ (b) $CH_2=C\begin{matrix} CH_3 \\ CH_2-CH_3 \end{matrix}$

31. Write increasing order of heat of hydrogenation :

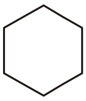
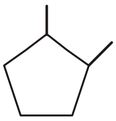
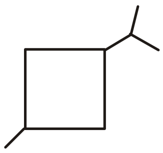
(i) (a)  (b) 

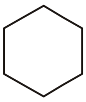
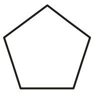
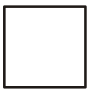
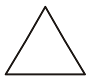
(ii) (a) (b)  (c)  (d)  (e) 

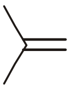
(iii) (a)  (b)  (c) 

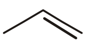
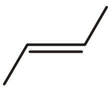
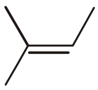
(iv) (a)  (b)  (c)  (HOH per benzene ring)

32. Give decreasing order of heat of combustion (HOC):

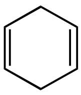
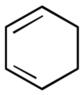
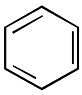
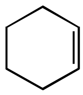
(i) (a)  (b)  (c) 

(ii) (a)  (b)  (c)  (d) 

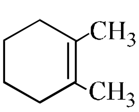
(iii) (a)  (b) 

(iv) (a)  (b)  (c) 

33. Among the following pairs identify the one which gives higher heat of hydrogenation :

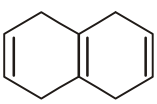
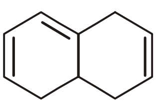
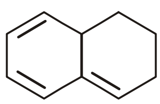
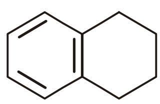
(a)  and  (b)  and 

(c) $CH_3 - CH = CH - CH_3$ and $CH_3 - CH_2 - CH = CH_2$

(d)  and 

34. Arrange the following compounds in order of :

(I) Stability (II) Heat of hydrogenation

(a)  (b)  (c)  (d) 

35. If Heat of hydrogenation of 1-butene is 30 Kcal/mol then heat of hydrogenation of 1,3-butadiene is ?

(A) 30 (B) 60 (C) 57 (D) 25

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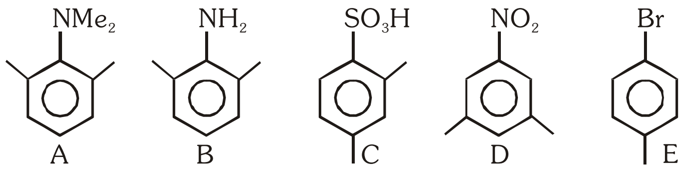
36. 

Steric inhibition of resonance takes place :

(A) In A,B only (B) In A, B, C, E (C) C only (D) In A only

37. Consider the following two structures and choose the correct statements -

(I) (II)

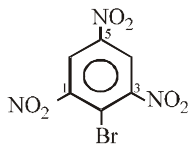
(A) carbon-nitrogen bond length structure I is greater than that in structure II

(B) carbon-nitrogen bond length in structure I is less than in structure II

(C) carbon-nitorogen bond length in both structure is same

(D) It can not be compared

38. Which of the following statements would be true about this compound :



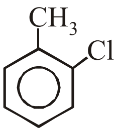
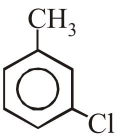
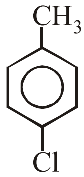
(A) All three C – N bonds are of same length.

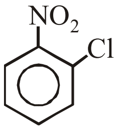
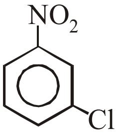
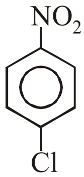
(B) Cl – N and C3 – N bonds are of same length but shorter than C5 – N bond.

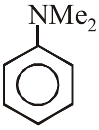
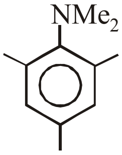
(C) Cl – N and C3 – N bonds are of same length but longer than C5 – N bond.

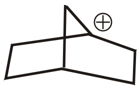
(D) Cl – N and C3 – N bonds are of different length but both are longer than C5 – N bond

39. Arrange given compounds in decreasing order of dipole moment :

(i) (a)  (b)  (c) 

(ii) (a)  (b)  (c) 

(iii) (a)  (b) 

40. Why a cation like  is not possible.

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EXERCISE # III

1. Cyclopentadienyl anion is much more stable than allyl anion because :

(A) Cyclic anion is more stable than acyclic anion

(B) Delocalised anion is more stable than localised anion

(C) Cyclopentadienyl anion is aromatic in nature

(D) None of these

2. Select correct statement regarding given compounds :

CH_3OCH_3 (I) $\qquad$ C_2H_5OH (II)

(A) Boiling point of II is higher than I

(B) Boiling point of II is lower than I

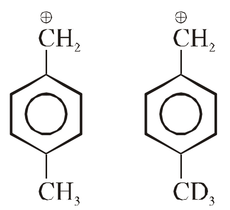
(C) Compound I forms intramolecular H-bonding

(D) Compound II forms intermolecular H-bonding

3. In the compund, $\overset{\text{I}}{CH_3}—\overset{\text{II}}{CH}=\overset{\text{III}}{CH}—\overset{\text{IV}}{C}\equiv N$, the most electronegative carbon is :

(A) I $\qquad$ (B) II $\qquad$ (C) III $\qquad$ (D) IV

4.



(I) $\qquad$ (II)

Carbocation (I) is more stable than carbocation (II), because :

(A) $—CD_3$ has more + I effect than $—CH_3$

(B) $—CH_3$ has more + I effect than $—CD_3$

(C) $—CH_3$ has more + H effect than $—CD_3$

(D) $—CD_3$ has more + H effect than $—CH_3$

5. Select correct statement :

(A) Carbon-oxygen bonds are of equal length in acetate ion

(B) Resonating structures of acetate ion are equivalent

(C) Carbon-oxygen bonds are of unequal length in formate ion

(D) Resonating structures of formate ion are equivalent

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6. Match the column I with column II.

Column-I	**Column-II**
(Group attached with benzene ring)	**(Effect shown by the group)**
(A) $—NO_2$	(P) – R effect
(B) $—O^-$	(Q) + R effect
(C) $—O—CH_3$	(R) + I effect
(D) $—C \equiv N$	(S) – I effect

7.

Column- I	**Column- II**
(Groups attached to phenyl ring)	**(Effect shown)**
(A) $—\ddot{N} = O$	(P) +M
(B) $—CH_3$	(Q) –M
(C) $—\ddot{N}H – C(=O)CH_3$	(R) +H
(D) $– \underset{\underset{O}{\parallel}}{C} – OCH_3$	(S) –I

8. **Match the column :**

Column-I	**Column-II**
(A) Group donate e^- inductively but does not donate / withdraw by resonance	(P) –OH
(B) Group withdraw e^- inductively but does not donate / withdraw by resonance	(Q) $–NO_2$
(C) Group withdraw e^- inductively & donate e^- by resonance	(R) $–CH_2–CH_3$
(D) Group withdraw e^- inductively & withdraw e^- by resonance	(S) $-\overset{+}{N}H_3$
	(T) $–NH_2$

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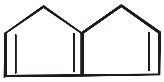
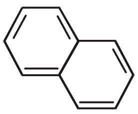
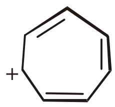
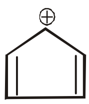
9. Match the column I with column II.

Column-I	**Column-II**
(A) 	(P) Aromatic
(B) 	(Q) Non-aromatic
(C) 	(R) Anti-aromatic
(D) 	(S) Cyclic structure

10. **Statement-I :** $-\overset{O}{\overset{\|}{C}}{}_{a}-OEt$ $\quad -\overset{O}{\overset{\|}{C}}{}_{b}-O-\overset{O}{\overset{\|}{C}}-$ bond length a < b

Because

Statement-II : More is the double bond character less is the bond legnth.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

11. **Statement-I :** $Me_3\overset{+}{C}$ is more stable than $Me_2\overset{+}{C}H$ and $Me_2\overset{+}{C}H$ is more stable than the $Me\overset{+}{C}H_2$.

Because

Statement-II : Greater the number of hyperconjugative structures, more is the stability of carbocation.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

12. **Statement-I :** The potential energy barrier for rotation about C = C bond in 2-butene is much higher than that in ethylene.

Because

Statement-II : Hyperconjugation effect decreases the double bond character.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1

(C) Statement-1 is true, statement-2 is false

(D) Statement-1 is false, statement-2 is true.

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Paragraph for Question 13 to 15

The intramolecular delocalisation of π and non-bonding electrons without any change in the position of atoms is called resonance. Delocalisation may occur in conjugated system involving carbon atom and atom other than carbon. Delocalisation makes system stable. More is the number of resonating structures, more is the stability of the system. A resonating structure is less stable when a higher electronegative atom has positive charge and when identical charges are present on adjacent atoms.

13. The decreasing order of stability of the following resonating structures

$$\underset{(I)}{CH_2=CH-\ddot{\underset{..}{Cl}}:} \longleftrightarrow \underset{(II)}{:\bar{C}H_2-CH=\overset{+}{\underset{..}{Cl}}:} \longleftrightarrow \underset{(III)}{\overset{+}{C}H_2-CH=\ddot{\underset{..}{Cl}}:^{-}}$$ is :

(A) I > II > III (B) II > III > I (C) III > II > I (D) I > III > II

14. If A is $Ph\overset{+}{C}H_2$ and B is $CH_2=CH-\overset{+}{C}H_2$, the greater number of resonating structure is of -

(A) A (B) B (C) both A and B (D) None of these

15. Which of the following pairs represent resonance ?

(A) $CH_2=CHOH$; CH_3CHO (B) $\overset{\ominus}{C}H_2-CHO$; $H_2C=CH-\overset{\ominus}{O}$

(C) $CH_3-\overset{\overset{O}{\|}}{C}-CH_3$; $CH_3-\overset{\overset{OH}{|}}{C}=CH_2$ (D) $CH_3-\overset{\overset{\overset{+}{O}H}{\|}}{C}-CH_3$; $CH_3-\underset{\underset{H}{|}}{\overset{\overset{OH}{|}}{C}}-CH_3$

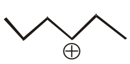
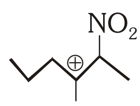
Paragraph for Question 16 to 18

Carbocation is a specie with positively charged carbon atom having six electrons in the valence shell after sharing. Carbocations are formed in the heterolysis of a bond and are planar species. Stability of carbocation is determined by inductive effect, hyperconjugation and resonance effect. Greater the number of contributing structures, more is the stability of a Carbocation. Electron releasing groups (+I effect) increases the stability of a carbocation whereas the electron withdrawing groups (–I effect) have an opposite effect.

16. Which of the following is most stable carbocation ?

(A) $\overset{+}{C}H_3$ (B) $CH_3-\overset{+}{C}H-CH_3$ (C) $CH_3-\overset{+}{C}H_2$ (D) $CH_3-\underset{\underset{CH_3}{|}}{\overset{+}{C}}-CH_3$

17. The most stable carbocation among the following :

(A)  (B)  (C)  (D) 

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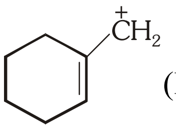
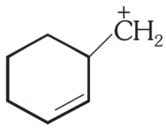
18. In which of the following cases, the carbocation (I) is less stable than the carbocation (II) ?

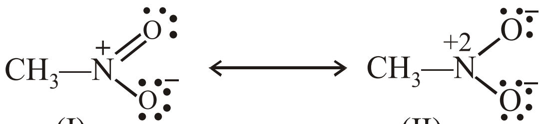
(A) $C_6H_5-\overset{+}{C}H_2$ (I), $CH_2=CH-\overset{+}{C}H_2$ (II)

(B)  (I),  (II)

(C) $CH_2=\overset{+}{C}H$ (I), $CH_3-\overset{+}{C}H_2$ (II)

(D) $H_3C-\overset{+}{C}H_2$ (I), $\underset{\substack{|\\ F}}{CH_2}-\overset{\oplus}{C}H_2$ (II)

19. Examine the structures I and II for nitromethane and choose the statement correctly :



(I) (II)

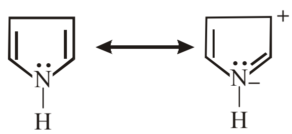
(A) Structure II is unlikely representation because electrons have shifted to oxygen

(B) Structure II is unlikely representation because nitrogen has sextet of electrons

(C) Structure II is acceptable and important

(D) None of these

20. Examine the following two structures for pyrrole and choose the correct statement given below



(I) (II)

(A) II is not an acceptable resonating structure because carbonium ions is less stable than nitride ion

(B) II is not an acceptable resonating structure because there is charge separation

(C) II is not an acceptable resonating structure because nitrogen has ten valance electrons

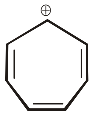
(D) II is an acceptable resonating structure

21. Delocalization of electrons increases molecular stability because :

(A) Potential energy of the molecule decreases (B) Electron-electron repulsion decreases

(C) Both (A) and (B) (D) Electron-electron repulsion increases

22. The most stable and the least stable carbocation among

(I)  (II) $CH_2=CH-\overset{+}{C}H_2$ (III) $C_6H_5-\overset{+}{C}H_2$ (IV) $CH_3-\overset{+}{C}H-CH_3$

are respectively :

(A) II, I (B) III, IV (C) I, II (D) I, IV

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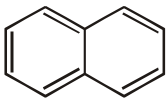
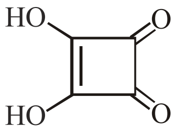
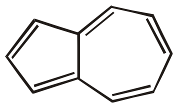
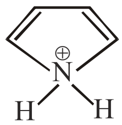
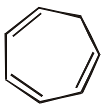
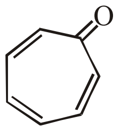
23. Most stable carbocation is formed by the heterolysis of :

(A) $(CH_3)_3CBr$ (B) $(C_6H_5)_3CBr$ (C) $(C_6H_5)_2CHBr$ (D) $C_6H_5CH_2Br$

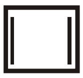
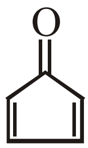
24. Total number of aromatic compounds

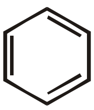
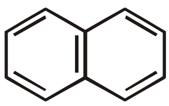
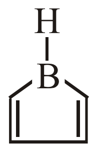
(a) 
(b) 

(c) 
(d) 

(e) 
(f) 

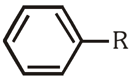
25. Identify total number of compounds which are unstable at room temperature ?

(i) 
(ii) 
(iii) 
(iv) 

(v) 
(vi) 
(vii) 

EXERCISE-IV (J-MAIN)

1. In the following benzyl/allyl system **[AIEEE-2002]**

R–CH=CH_2 or 

(R is alkyl group)

decreasing order of inductive effect is-

(1) $(CH_3)_3C– > (CH_3)_2CH– > CH_3CH_2–$ (2) $CH_3–CH_2– > (CH_3)_2CH– > (CH_3)_3C–$

(3) $(CH_3)_2CH– > CH_3CH_2– > (CH_3)_3CH–$ (4) None of these

2. In the anion $HCOO^–$ the two carbon-oxygen bonds are found to be of equal length. What is the reason for it- **[AIEEE-2003]**

(1) Electronic orbits of carbon atoms are hybridised

(2) The C=O bond is weaker than the C–O bond

(3) The anion $HCOO^–$ has two resonating structure

(4) The anion is obtained by removal of a proton form the acid molecule

3. Which one of the following does not have sp^2 hybridised carbon **[AIEEE-2004]**

(1) Acetamide (2) Acetic acid (3) Acetonitrile (4) Acetone

4. Due to the presence of an unpaired electron, free radicals are - **[AIEEE-2005]**

(1) Chemically inactive (2) Chemically reactive

(3) Cations (4) Anions

5. The increasing order of stability of the following free radicals is **[AIEEE-2006]**

(1) $(C_6H_5)_3\dot{C} < (C_6H_5)_2\dot{C}H < (CH_3)_3\dot{C} < (CH_3)_2\dot{C}H$

(2) $(C_6H_5)_2\dot{C}H < (C_6H_5)_3\dot{C} < (CH_3)_3\dot{C} < (CH_3)_2\dot{C}H$

(3) $(CH_3)_2\dot{C}H < (CH_3)_3\dot{C} < (C_6H_5)_3\dot{C} < (C_6H_5)_2\dot{C}H$

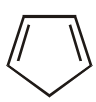
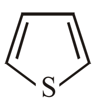
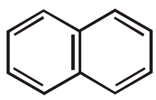
(4) $(CH_3)_2\dot{C}H < (CH_3)_3\dot{C} < (C_6H_5)_2\dot{C}H < (C_6H_5)_3\dot{C}$

6. Arrange the carbanions, $(CH_3)_3\bar{C}$, $\bar{C}Cl_3$, $(CH_3)_2\bar{C}H$, $C_6H_5\bar{C}H_2$, in order of their decreasing stability **[AIEEE-2009]**

(1) $\bar{C}Cl_3 > C_6H_5\bar{C}H_2 > (CH_3)_2\bar{C}H > (CH_3)_3\bar{C}$ (2) $(CH_3)_3\bar{C} > (CH_3)_2\bar{C}H > C_6H_5\bar{C}H_2 > \bar{C}Cl_3$

(3) $C_6H_5\bar{C}H_2 > \bar{C}Cl_3 > (CH_3)_3\bar{C} > (CH_3)_2\bar{C}H$ (4) $(CH_3)_2\bar{C}H > \bar{C}Cl_3 > C_6H_5\bar{C}H_2 > (CH_3)_3\bar{C}$

7. The non aromatic compound among the following is :- **[AIEEE-2011]**

(1)  (2)  (3)  (4) 

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8. ortho-Nitrophenol is less soluble in water than p– and m– Nitrophenols because :-

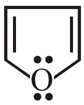
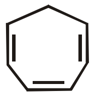
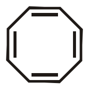
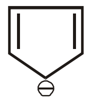
(1) Melting point of o–Nitrophenol is lower than those of m– and p– isomers **[AIEEE-2012]**

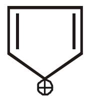
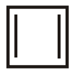
(2) o–Nitrophenol is more volatile in steam than those of m– and p– isomers

(3) o–Nitrophenol shows Intramolecular H–bonding

(4) o–Nitrophenol shows Intermolecular H–bonding

9. Which of the following compounds are antiaromatic :- **[AIEEE-2012(Online)]**

(I) 
(II) 
(III) 
(IV) 

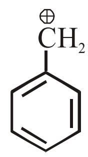
(V) 
(VI) 

(1) (III) and (VI) (2) (II) and (V) (3) (I) and (V) (4) (V) and (VI)

10. Among the following the molecule with the lowest dipole moment is :- **[AIEEE-2012(Online)]**

(1) $CHCl_3$ (2) CH_2Cl_2 (3) CCl_4 (4) CH_3Cl

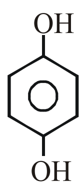
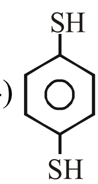
11. The order of stability of the following carbocations **[JEE-MAIN-2013]**

$\underset{\text{I}}{CH_2{=}CH{-}\overset{\oplus}{C}H_2}$; $\underset{\text{II}}{CH_3{-}CH_2{-}\overset{+}{C}H_2}$;  is :-

III

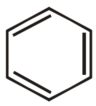
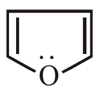
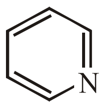
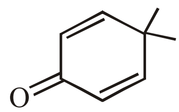
(1) III > II > I (2) II > III > I (3) I > II > III (4) III > I > II

12. For which of the following molecule significant $\mu \neq 0$ **[JEE-MAIN-2014]**

(1) 
(2) 
(3) 
(4) 

(1) Only (3) (2) (3) and (4) (3) Only (1) (4) (1) and (2)

13. Which of the following molecules is least resonance stabilized ? **[JEE-MAIN-2017]**

(1) 
(2) 
(3) 
(4) 

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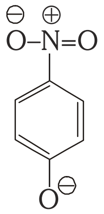
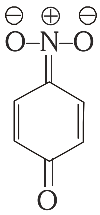
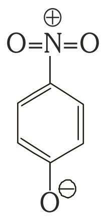
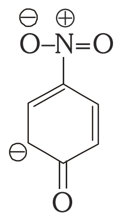
EXERCISE # V (J-ADVANCE)

1. Which one of the following has the smallest heat of hydrogenation per mole of H_2 ? **[IIT-93]**
(A) 1–Butene (B) trans–2–Butene
(C) cis–2–Butene (D) 1, 3–Butadiene

2. Most stable carbonium ion is **[IIT-95]**
(A) p-NO_2—C_6H_4—$\overset{\oplus}{C}H_2$ (B) $C_6H_5\overset{\oplus}{C}H_2$
(C) p-Cl—C_6H_4—$\overset{\oplus}{C}H_2$ (D) p-CH_3O—C_6H_4—$\overset{\oplus}{C}H_2$

3. Arrange the following compounds in order of increasing dipole moment : **[IIT-96]**
toluene (I) m-dichloroobenzene (II)
o-dichlorobenzene (III) p-dichlorobenzene (IV)
(A) I < IV < II < III (B) IV < I < II < III (C) IV < I < III < II (D) IV < II < I < III

4. The most unlikely representation of resonance structure of p-nitrophenoxide ion is - **[IIT-99]**

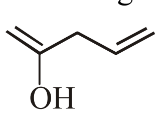
(A)  (B)  (C)  (D) 

5. An aromatic molecule will not **[IIT-99]**
(A) have 4n π electrons (B) have (4n + 2)π electrons
(C) be planar (D) be cyclic

6. **Statement-I :** p-Hydroxybenzoic acid has a lower boiling point that o-hydroxybenzoic acid.
Because
Statement-II : o-Hydroxybenzoic acid has intramolecular hydrogen bonding. **[IIT 2003]**
(A) Statement-I is True, Statement-II is True ; Statement-II is a correct explanation for Statement-I
(B) Statement-I is True, Statement-II is True ; Statement-II is NOT a correct explanation for Statement-I
(C) Statement-I is True, Statement-II is False.
(D) Statement-I is False, Statement-II is True.

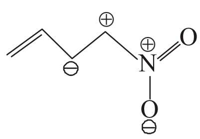
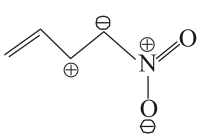
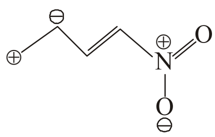
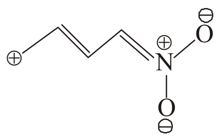
7. Among the following, the molecule with the highest dipole moment is **[IIT-2003]**
(A) CH_3Cl (B) CH_2Cl_2 (C) $CHCl_3$ (D) CCl_4

8. Give resonating structures of following compound. **[IIT 2003]**



9. Which of the following is least stable : **[IIT-2005]**
(A) $CH_3-\overset{\oplus}{O}=CH-\overset{\ominus}{C}H-HC=CH_2$ (B) $CH_3-\overset{\oplus}{O}=CH-CH=HC-\overset{\ominus}{C}H_2$
(C) $CH_3-O-\overset{\oplus}{C}H-\overset{\ominus}{C}H-HC=CH_2$ (D) $CH_3-O-\overset{\ominus}{C}H-\overset{\oplus}{C}H-CH=CH_2$

10. Among the following, the least stable resonance structure is - **[IIT-2007]**

(A)  (B)  (C)  (D) 

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11. The correct stability order for the following species is : **[IIT-2008]**

(I) (II) (III) (IV)

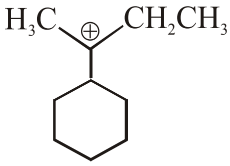
(A) II > IV > I > III (B) I > II > III > IV (C) II > I > IV > III (D) I > III > II > IV

12. The correct stability order of the following resonance structures is **[IIT-2009]**

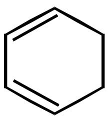
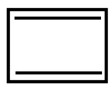
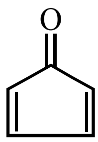
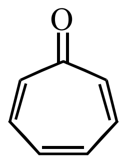
(I) $H_2C=\overset{+}{N}=\overline{N}$ (II) $H_2\overset{+}{C}-N=\overline{N}$ (III) $H_2\overline{C}-\overset{+}{N}\equiv N$ (IV) $H_2\overline{C}-N=\overset{+}{N}$

(A)(I) > (II) > (IV) > (III) (B) (I) > (III) > (II) > (IV)
(C) (II) > (I) > (III) > (IV) (D) (III) > (I) > (IV) > (II)

13. The total number of contributing structures showing hyperconjugation (involving C–H bonds) for the following carbocation is. **[IIT-2011]**



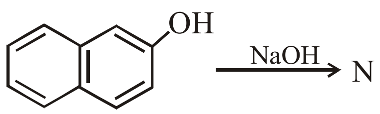
14. Which of the following molecules, in pure from , is (are) **unstable** at room temperature ? **[IIT-2012]**

(A)  (B)  (C)  (D) 

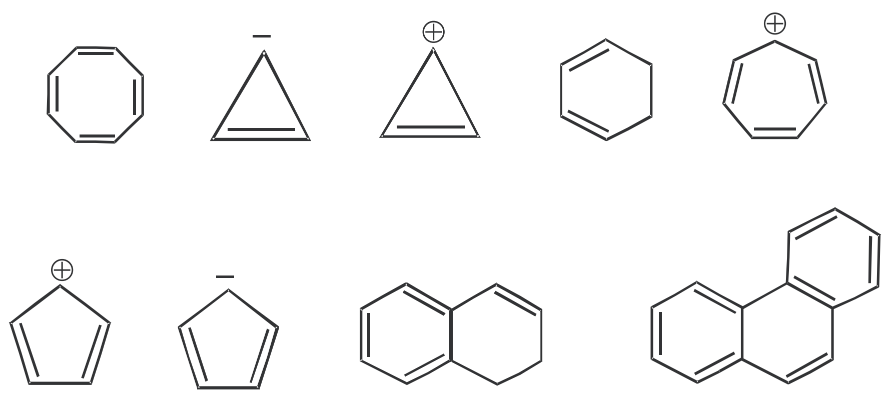
15. The hyperconjugative stbilities of tert-butyl cation and 2-butene, respectively, are due to
(A) $\sigma\rightarrow p$ (empty) and $\sigma\rightarrow\pi$ electron delocalisations **[IIT-2013]**
(B) $\sigma\rightarrow\sigma$ and $\sigma\rightarrow\pi$ electron delocalisations
(C) $\sigma\rightarrow p$ (filled) and $\sigma\rightarrow\pi$ electron delocalisations
(D) p (filled)$\rightarrow\sigma$ and $\sigma\rightarrow\pi$ electron delocalisations

16. The total number of lone-pairs of electrons in melamine is **[IIT-2013]**

17. The number of resonance structures for N is : **[IIT-2015]**



18. Among the following, the number of aromatic compound (s) is- **[JEE - Adv. 2017]**



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ANSWER KEY

EXERCISE # I

1. (D) 2. (C) 3. (B) 4. (A,B,D) 5. 4(b, d, f, g)

6. (a), (c), (d), (g), (j), (ℓ), (m) 7. (A) 8. (A) 9. (A)

10. A,D 11. (D) 12. b, d, e 13. b d e 14. (d) 15. c , f

16. a, b, c, d, f 17. b, c, f 18. 19. a-I , b-II , c-II , d-II , e-I

20. a-I ,b-I, c-I, d-I, e-II, f-II 21. a, e, f, g

22. (a) Resonance forms, (b) A , (c) C , (d) A & B, (e) B & C , (f) 0 , (g) B , (h) B

23. (a) II ; (b) II ; (c) II 24. (A) 25. a-II , b-II , c-II , d-II 26 a & b

27. (i)-I , (ii)-II, (iii)-II, (iv)-I, (v)-I, (vi)-II , (vii)-I, (viii)-II, (ix)-II, (x)-II, (xi)-II ,(xii)-I, (xiii) -I, (xiv)-I, (xv)-I

28. a-II ,b-I, c-I, d-II, e-I 29. (B) 30. (D) 31. (B) 32. (C)

33. (C) 34. (C) 35. (D) 36. (A,B,C)

37. (B) 38. (B) 39. (C) 40. (D) 41. (D) 42. (A)

43. (B,C,D) 44. (B) 45. (A) 46. (A,C,D)

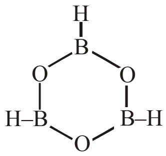
47. a-I ,b-I, c-II, d-I, e-I, f-I 48. a-II ,b-I, c-I, d-I, e-II, f-II 49. (D)

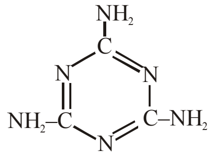
50. C > B > A 51. (A)

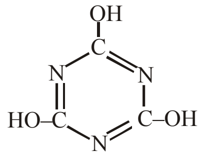
EXERCISE # II

1. (i) a, b ; (ii) a ,c ; (iii) b, c, d ; (iv) a, b, c, d, e, f

2. Aromatic→a, b, d, e, g ; Non-aromatic→c, f 3. (C) 4. (C) 5. (B)

6. (A) 7. (a) 

(b) 

(c) 

8. (D)

9. (i) c > b > a (ii) c > b > a (iii) b > c > a (iv) d > c > b > a (v) c > b > a
(vi) b > c > a (vii) c > b > a (viii) c>a>d>b (ix) b > c > a (x) c > a > b
(xi) c > a > b (xii) a > b > c > d (xiii) b>a>c>d (xiv) d > e > b > a > c
(xv) a > c > b (xvi) b > c > a

10. (C) 11. (a) I ; (b) I ; (c) II ; (d) II ; 12. (C) 13. (A) 14. (A)

15. (D) 16. (B)

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17. (i) a < b (ii) d < a < c < b (iii) b < a (iv) c < b < a (v) c < a < b (vi) a<c< b

(vii) c < b < a (viii) c < b < a (ix) b < a (x) a < c < b < d 18. (A)

19. (D) 20. (B) 21. (D)

22. (a) IV < I < II < III < V (b) III < IV < I < II 23. (B) 24. (A)

25. (a) -I ; (b) - I ; (c) - II 26. (a) 4658 ; (b) 4638 (c) 4632 ; (d) 4650 ; (e) 5293

27. (A) 28. (B) 29. (D)

30. (i) d > c > b > a ; (ii) e > c > d > b > a ; (iii) b > a (iv) a > b (v) b > a ; (vi) a > b

31. (i) a < b ; (ii) e < d < c < b < a ; (iii) a < c < b ; (iv) a > b > c

32. (i) c > b > a ; (ii) a > b > c > d ; (iii) a > b ; (iv) c > b > a

33. a - I ; b - I ; c - II, d - I 34. Stability order : d>c>b> a ; HOH order : a > b > c > d

35. (C) 36. (D) 37. (B) 38. (C)

39. (i) c > b > a (ii) a > b > c (iii) a > b

EXERCISE # III

1. (C) 2. (A,D) 3. (D) 4. (C) 5. (A,B,D)

6. (A)–P,S ; (B)–Q,R ; (C)–Q,S ; (D)–P,S

7. (A) → P, Q, S ; (B) → R ; (C) → P, S ; (D) → Q, S

8. (A)-R ; (B)-S ; (C)-P, T ; (D)-Q 9. (A)–Q,S ; (B)–P,S ; (C)–P,S ; (D)–R,S

10. (D) 11. (A) 12. (D) 13. (A) 14. (A) 15. (B)

16. (D) 17. (C) 18. (C) 19. (B) 20. (C) 21. (C)

22. (D) 23. (B) 24. (4) 25. (4)

EXERCISE # IV (J-MAIN)

1. (1) 2. (3) 3. (3) 4. (2) 5. (4) 6. (1)

7. (1) 8. (3) 9. (4) 10. (3) 11. (4) 12. (2)

13. (4)

EXERCISE # V (J-ADVANCE)

1. (D) 2. (D) 3. (B) 4. (C) 5. (A) 6. (D)

7. (A) 9. (D) 10. (A) 11. (D) 12. (B) 13. (6)

14. (B,C) 15. (A) 16. (6) 17. (9) 18. (5)

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CONTENTS

SALT ANALYSIS

HEATING EFFECT

s-BLOCK ELEMENT

d-BLOCK COMPOUNDS

Session 2019-20

SALT ANALYSIS

Analysis always does not mean breaking of substance into its ultimate constituents. Finding out the nature of substance and identity of its constituents is also analysis and is known as ***qualitative analysis.***

Qualitative analysis of inorganic salts means the identification of cations and anions present in the salt or a mixture of salts. Inorganic salts may be obtained by complete or partial neutralisation of acid with base or vice-versa. In the formation of a salt, the part contributed by the ***acid*** is called ***anion*** and the part contributed by the ***base*** is called ***cation***. For example, in the salts $CuSO_4$ and NaCl, Cu^{2+} and Na^+ ions are cations and SO_4^{2-} and Cl^- ions are anions. Qualitative analysis is carried out on various scales. Amount of substance employed in these is different. In macro analysis, 0.1 to 0.5 g of substance and about 20 mL of solution is used. For semimicro analysis, 0.05 g substance and 1 mL solution is needed while for micro analysis amount required is very small. Qualitative analysis is carried out through the reactions which are easily perceptible to our senses such as sight and smell. Such reactions involve:

(a) Formation of a precipitate

(b) Change in colour

(c) Evolution of gas etc.

Systematic analysis of an inorganic salt involves the following steps:

(i) Preliminary examination of solid salt and its solution.

(ii) Determination of anions by reactions carried out in solution (wet tests) and confirmatory tests.

(iii) Determination of cations by reactions carried out in solution (wet tests) and confirmatory tests.

Although these tests are not conclusive but sometimes they give quite important clues for the presence of certain anions or cations. These tests can be performed within 10^{-15} minutes. These involve noting the general appearance and physical properties, such as colour, smell, solubility etc. of the salt. These are named as ***dry tests.***

Heating of dry salt, blow pipe test, flame tests, borax bead test, sodium carbonate bead test, charcoal cavity test etc. come under dry tests.

Solubility of a salt in water and the pH of aqueous solutions give important information about the nature of ions present in the salt. If a solution of the salt is acidic or basic in nature, this means that it is being hydrolysed in water. If the solution is basic in nature then salt may be some carbonate or sulphide etc. If the solution shows acidic nature then it may be an acid salt or salt of weak base and strong acid. In this case it is best to neutralise the solution with sodium carbonate before testing it for anions.

Gases evolved in the preliminary tests with dil. H_2SO_4/dil. HCl and conc. H_2SO_4 also give good indication about the presence of acid radicals (See Tables 1 and 3). ***Preliminary tests should always be performed before starting the confirmatory tests for the ions.***

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EXPERIMENT 1.1

Aim

To detect one cation and one anion in the given salt from the following ions:

Cations - Pb^{2+}, Cu^{2+}, As^{3+}, Al^{3+}, Fe^{3+}, Mn^{2+}, Ni^{2+}, Zn^{2+}, Co^{2+}, Ca^{2+}, Sr^{2+}, Ba^{2+}, Mg^{2+},NH_4^+

Anions - CO_3^{2-} , S^{2-}, SO_4^{2-}, NO_2^- , NO_3^-, Cl^-, Br^-, I^-, PO_4^{3-}, CH_3COO^-.

(Insoluble salts to be excluded)

Theory

Two basic principles of great use in the analysis are:

(i) the Solubility product

(ii) the Common ion effect.

When ionic product of a salt exceeds its solubility product, precipitation takes place. Ionic product of salt is controlled by making use of common ion effect.

Material Required

• Boiling tube	:	As per need
• Test tubes	:	As per requirement
• Measuring cylinder	:	One
• Test tube stand	:	One
• Test tube holder	:	One
• Delivery tube	:	One
• Corks	:	As per need
• Filter paper	:	As per need
• Reagents	:	As per need

Step - I : Preliminary Test with Dilute Sulphuric Acid $\rightarrow$ In this test the action of dilute sulphuric acid (procedure is given below) on the salt is noted at room temperature and on warming.

Carbonate (CO_3^{2-}), sulphide (S^{2-}), sulphite (SO_3^{2-}), nitrite (NO_2^-) and acetate (CH_3COO^-) react with dilute sulphuric acid to evolve different gases. Study of the characteristics of the gases evolved gives information about the anions. Summary of characteristic properties of gases is given in Table 1.

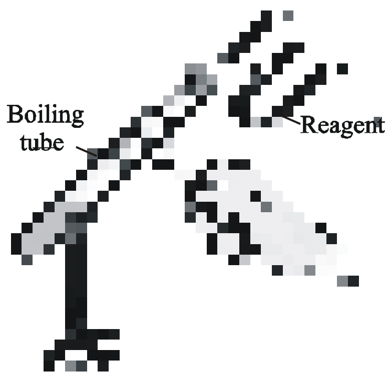

Fig. 7.1 - Testing a gas

Procedure

(a) Take 0.1 g of the salt in a test tube and add 1–2 mL of dilute sulphuric acid. Observe the change, if any, at room temperature. If no gas is evolved, warm the content of the test tube. If gas is evolved test it by using the apparatus shown in Fig.1 and identify the gas evolved (See Table 1).

ALLEN Session 2019-20

Table 1 : Preliminary test with dilute sulphuric acid

Observations	Inference	
	Gas Evolved	Possible Anion
A colourless, odourless gas is evolved with brisk effervescence, which turns lime water milky.	CO_2	Carbonate (CO_3^{2-})
Colourless gas with the smell of rotten eggs is evolved which turns lead acetate paper black.	H_2S	Sulphide (S^{2-})
Colourless gas with a pungent smell, like burning sulphur which turns acidified potassium dichromate solution green.	SO_2	Sulphite (SO_3^{2-})
Brown fumes which turn acidified potassium iodide solution containing starch solution blue.	NO_2	Nitrite (NO_2^-)
Colourless vapours with smell of vinegar. Vapours turn blue litmus red.	CH_3COOH vapours	Acetate, (CH_3COO^-)

Confirmatory tests for CO_3^{2-} , S^{2-}, SO_3^{2-}, NO_2^- and CH_3COO^-

Confirmatory (wet) tests for anions are performed by using water extract when salt is soluble in water and by using sodium carbonate extract when salt is insoluble in water. Confirmation of CO_3^{2-} is done by using aqueous solution of the salt or by using solid salt as such because sodium carbonate extract contains carbonate ions. Water extract is made by dissolving salt in water. Preparation of sodium carbonate extract is given below.

Preparation of sodium carbonate extract

Take 1 g of salt in a porcelain dish or boiling tube. Mix about 3 g of solid sodium carbonate and add 15 mL of distilled water to it. Stir and boil the content for about 10 minutes. Cool, filter and collect the filtrate in a test tube and label it as sodium carbonate extract.

Confirmatory tests for acid radicals, which react with dilute sulphuric acid are given in Table 2.

ALLEN Session 2019-20

Table 2 : Confirmatory tests for CO_3^{2-}, S^{2-}, SO_3^{2-}, NO_3^- , CH_3COO^-

Anion	Confirmatory Test
Carbonate (CO_3^{2-})	Take 0.1 g of salt in a test tube, add dilute sulphuric acid. CO_2 gas is evolved with brisk effervescence which turns lime water milky. On passing the gas for some more time, milkiness disappears.
Sulphide (S^{2-})	Take 1 mL of water extract and make it alkaline by adding ammonium hydroxide or sodium carbonate extract. Add a drop of sodium nitroprusside solution. Purple or violet colouration appears.
Sulphite (SO_3^{2-})	(a) Take 1 mL of water extract or sodium carbonate extract in a test tube and add barium chloride solution. A white precipitate is formed which dissolves in dilute hydrochloric acid and sulphur dioxide gas is also evolved (b) Take the precipitate of step (a) in a test tube and add a few drops of potassium permanganate solution acidified with dil. H_2SO_4. Colour of potassium permanganate solution gets discharged.
Nitrite (NO_2^-)	(a) Take 1 mL of water extract in a test tube. Add a few drops of potassium iodide solution and a few drops of starch solution, acidify with acetic acid. Blue colour appears. (b) Acidify 1 mL of water extract with acetic acid. Add 2-3 drops of sulphanilic acid solution followed by 2-3 drops of 1-naphthylamine reagent. Appearance of red colour indicates the presence of nitrite ion.
Acetate, (CH_3COO^-)	(a) Take 0.1 g of salt in a china dish. Add 1 mL of ethanol and 0.2 mL conc. H_2SO_4 and heat. Fruity odour confirms the presence of acetate ion. (b) Take 0.1 g of salt in a test tube, add 1-2 mL distilled water, shake well filter if necessary. Add 1 to 2 mL neutral ferric chloride solution to the filtrate. Deep red colour appears which disappears on boiling and a brown-red precipitate is formed.

Chemistry of Confirmatory Tests

1. Test for Carbonate ion [CO_3^{2-}]

If there is effervescence with the evolution of a colourless and odourless gas on adding dil. H_2SO_4 to the solid salt, this indicates the presence of carbonate ion.

The gas turns lime water milky due to the formation of $CaCO_3$

$$Na_2CO_3 + H_2SO_4 \longrightarrow Na_2SO_4 + H_2O + CO_2$$

$$Ca(OH)_2 + CO_2 \longrightarrow CaCO_3 + H_2O$$

If CO_2 gas is passed in excess through lime water, the milkiness disappears due to the formation of calcium hydrogen carbonate which is soluble in water.

$$CaCO_3 + CO_2 + H_2O \longrightarrow Ca(HCO_3)_2$$

ALLEN Session 2019-20

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2. Test for Sulphide ion [S^{2-}]

(a) With warm dilute H_2SO_4 a sulphide gives hydrogen sulphide gas which smells like rotten eggs. A piece of filter paper dipped in lead acetate solution turns black on exposure to the gas due to the formation of lead sulphide which is black in colour.

$$Na_2S + H_2SO_4 \longrightarrow Na_2SO_4 + H_2S$$

$$(CH_3COO)_2Pb + H_2S \longrightarrow \underset{\text{Black precipitate}}{\underset{\text{Lead sulphide}}{PbS}} + 2CH_3COOH$$

(b) If the salt is soluble in water, take the solution of salt in water make it alkaline with ammonium hydroxide and add sodium nitroprusside solution. If it is insoluble in water take sodium carbonate extract and add a few drops of sodium nitroprusside solution. Purple or violet colouration due to the formation of complex compound $Na_4[Fe(CN)_5NOS]$ confirms the presence of sulphide ion in the salt.

$$Na_2S + \underset{\text{Sodium nitroprusside}}{Na_2[Fe(CN)_5NO]} \longrightarrow \underset{\text{Complex of Purple colour}}{Na_4[Fe(CN)_5NOS]}$$

3. Test for Sulphite ion [SO_3^{2-}]

(a) On treating sulphite with warm dil. H_2SO_4, SO_2 gas is evolved which is suffocating with the smell of burning sulphur.

$$Na_2SO_3 + H_2SO_4 \longrightarrow Na_2SO_4 + H_2O + SO_2$$

The gas turns potassium dichromate paper acidified with dil. H_2SO_4, green.

$$K_2Cr_2O_7 + H_2SO_4 + 3SO_2 \longrightarrow K_2SO_4 + \underset{\text{sulphate (green)}}{\underset{\text{Chromium}}{Cr_2(SO_4)_3}} + H_2O$$

(b) An aqueous solution or sodium carbonate extract of the salt produces a white precipitate of barium sulphite on addition of barium chloride solution.

$$Na_2SO_3 + BaCl_2 \longrightarrow 2NaCl + \underset{\text{White ppt.}}{BaSO_3}$$

This precipitate gives following tests.

(i) This precipitate on treatment with dilute HCl, dissolves due to decomposition of sulphite by dilute HCl. Evolved SO_2 gas can be tested.

$$BaSO_3 + 2HCl \longrightarrow BaCl_2 + H_2O + SO_2$$

(ii) Precipitate of sulphite decolourises acidified potassium permanganate solution.

$$BaSO_3 + H_2SO_4 \longrightarrow BaSO_4 + H_2O + SO_2$$

$$2KMnO_4 + 3H_2SO_4 \longrightarrow K_2SO_4 + 2MnSO_4 + 3H_2O + 5\,[O]$$

$$SO_2 + H_2O + [O] \longrightarrow H_2SO_4$$

ALLEN Session 2019-20

4. Test for Nitrite ion [NO_2^-]

(a) On treating a solid nitrite with dil. H_2SO_4 and warming , reddish brown fumes of NO_2 gas are evolved. Addition of potassium iodide solution to the salt solution followed by freshly prepared starch solution and acidification with acetic acid produces blue colour. Alternatively, a filter paper moistened with potassium iodide and starch solution and a few drops of acetic acid turns blue on exposure to the gas, due to the interaction of liberated iodine with starch.

(i) $2NaNO_2 + H_2SO_4 \longrightarrow Na_2SO_4 + 2HNO_2$

$3HNO_2 \longrightarrow HNO_3 + 2NO + H_2O$ (disproportionation)

$2NO + O_2 \longrightarrow \underset{\text{Brown gas}}{2NO_2}$

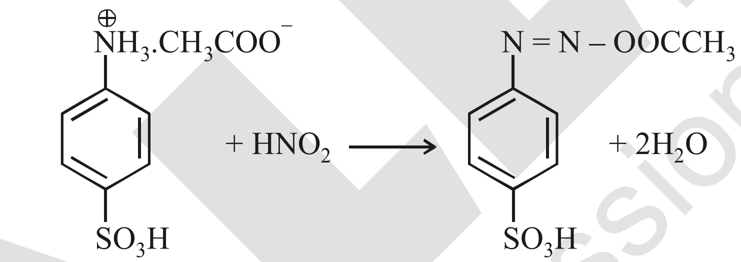
(ii) $NO_2^- + CH_3COOH \longrightarrow HNO_2 + CH_3COO^-$

$2HNO_2 + 2KI + 2CH_3COOH \longrightarrow 2CH_3COOK + 2H_2O + 2NO + I_2$

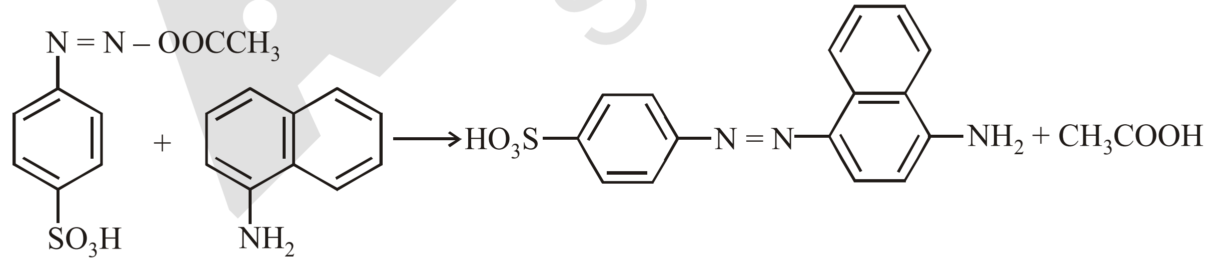
I_2 + Starch $\longrightarrow$ Blue complex

(b) **Sulphanilic acid — 1-naphthylamine reagent test (Griess-llosvay test) :** On adding sulphanilic acid and 1-naphthylamine reagent to the water extract or acidified with acetic acid, sulphanilic acid is diazotised in the reaction by nitrous acid formed. Diazotised acid couples with 1-naphthylamine to form a red azo-dye.

$NO_2^- + CH_3COOH \rightarrow HNO_2 + CH_3COO^-$



(Sulphanilic acid solution)



Diazotised acid 1-Naphthyl amine Red azo-dye

The test solution should be very dilute. In concentrated solutions reaction does not proceed beyond diazotisation.

ALLEN Session 2019-20

5. Test for Acetate ion [CH_3COO^-]

(a) If the salt smells like vinegar on treatment with dil. H_2SO_4, this indicates the presence of acetate ions. Take 0.1 g of salt in a china dish and add 1 mL of ethanol. Then add about 0.2 mL of conc. H_2SO_4 and heat. Fruity odour of ethyl acetate indicates the presence of CH_3COO^- ion.

$$2CH_3COONa + H_2SO_4 \longrightarrow Na_2SO_4 + 2CH_3COOH$$

$$CH_3COOH + C_2H_5OH \longrightarrow \underset{\substack{\text{Ethylacetate} \\ \text{(Fruity odour)}}}{CH_3COOC_2H_5} + H_2O$$

(b) Acetate gives deep red colour on reaction with neutral ferric chloride solution due to the formation of complex ion which decomposes on heating to give Iron (III) dihydroxyacetate as brown red precipitate.

$$6CH_3COO^- + 3Fe^{3+} + 2H_2O \longrightarrow [Fe_3(OH)_2(CH_3COO)_6]^+ + 2H^+$$

$$[Fe_3(OH)_2(CH_3COO)_6]^+ + 4H_2O \longrightarrow \underset{\substack{\text{Iron(III)dihydroxyacetate} \\ \text{(Brown-red precipitate)}}}{3[Fe(OH)_2(CH_3COO)]} + 3CH_3COOH + H^+$$

Step-II : Preliminary Test with Concentrated Sulphuric Acid If no positive result is obtained from dil. H_2SO_4 test, take 0.1 g of salt in a test tube and 3-4 drops of conc. H_2SO_4. Observe the change in the reaction mixture in cold and then warm it. Identify the gas evolved on heating (see Table 3).

Table 3 : Preliminary examination with concentrated sulphuric acid

Observations	Inference	
	Gas/Vapours Evolved	**Possible Anion**
A colourless gas with pungent smell, which gives dense white fumes when a rod dipped in ammonium hydroxide is brought near the mouth of the test tube.	HCl	Chloride, (Cl^-)
Reddish brown gas with a pungent odour is evolved. Intensity of reddish gas increases on heating the reaction mixture after addition of solid MnO_2 to the reaction mixture. Solution also acquires red colour.	Br_2 vapours	Bromide, (Br^-)
Violet vapours, which turn starch paper blue and a layer of violet sublimate is formed on the sides of the tube. Fumes become dense on adding MnO_2 to the reaction mixture.	I_2 vapours	Iodide, (I^-)
Brown fumes evolve which become dense upon heating the reaction mixture after addition of copper turnings and the solution acquires blue colour.	NO_2	Nitrate, (NO_3^-)
Colourless, odourless gas is evolved which turns lime water milky and the gas coming out of lime water burns with a blue flame, if ignited.	CO and CO_2	Oxalate, ($C_2O_4^{2-}$)

Table 4 : Confirmatory tests for Cl^-, Br^-, I^-, NO_3^- and $C_2O_4^{2-}$

Anion	Confirmatory Test
Chloride (Cl^-)	(a) Take 0.1 g of salt in a test tube, add a pinch of manganese dioxide and 3-4 drops of conc. sulphuric acid. Heat the reaction mixture. Greenish yellow chlorine gas is evolved which is detected by its pungent odour and bleaching action. (b) Take 1 mL of sodium carbonate extract in a test tube, acidify it with dil. HNO_3 or take water extract and add silver nitrate solution. A curdy white precipitate is obtained which is soluble in ammonium hydroxide solution. (c) Take 0.1 g salt and a pinch of solid potassium dichromate in a test tube, add conc. H_2SO_4, heat and pass the gas evolved through sodium hydroxide solution. It becomes yellow. Divide the solution into two parts. Acidify one part with acetic acid and add lead acetate solution. A yellow precipitate is formed. Acidify the second part with dilute sulphuric acid and add 1 mL of amyl alcohol followed by 1 mL of 10% hydrogen peroxide. After gentle shaking the organic layer turns blue.
Bromide (Br^-)	(a) Take 0.1 g of salt and a pinch of MnO_2 in a test tube. Add 3-4 drops conc.sulphuric acid and heat. Intense brown fumes are evolved. (b) Neutralise 1 mL of sodium carbonate extract with hydrochloric acid (or take the water extract). Add 1 mL carbon tetrachloride (CCl_4)/ chloroform ($CHCl_3$)/ carbon disulphide. Now add an excess of chlorine water dropwise and shake the test tube. A brown colouration in the organic layer confirms the presence of bromide ion. (c) Acidify 1 mL of sodium carbonate extract with dil. HNO_3 (or take 1 mL water extract) and add silver nitrate solution. A pale yellow precipitate soluble with difficulty in ammonium hydroxide solution is obtained.
Iodide (I^-)	(a) Take 1 mL of salt solution neutralised with HCl and add 1 mL chloroform/carbon tetrachloride/carbon disulphide. Now add an excess of chlorine water drop wise and shake the test tube. A violet colour appears in the organic layer. (b) Take 1 mL of sodium carbonate extract acidify it with dil. HNO_3 (or take water extract). Add, silver nitrate solution. A yellow precipitate insoluble in NH_4OH solution is obtained.

Anion	Confirmatory Test
*Nitrate (NO_3^-)	Take 1 mL of salt solution in water in a test tube. Add 2 mL of conc. H_2SO_4 and mix thoroughly. Cool the mixture under the tap. Add freshly prepared ferrous sulphate along the sides of the test tube without shaking. A dark brown ring is formed at the junction of the two solutions.
Oxalate ($C_2O_4^{2-}$)	(a) Take 1 mL of water extract or sodium carbonate extract acidified with acetic acid and add calcium chloride solution. A white precipitate insoluble in ammonium oxalate and oxalic acid solution but soluble in dilute hydrochloric acid and dilute nitric acid is formed. (b) Take the precipitate from test (a) and dissolve it in dilute H_2SO_4. Add very dilute solution of $KMnO_4$ and warm. Colour of $KMnO_4$ solution is discharged. Pass the gas coming out through lime water. The lime water turns milky.

ALLEN Session 2019-20

Chemistry of Confirmatory Tests

1. Test for Chloride ion [Cl^-]

(a) If on treatment with warm conc. H_2SO_4 the salt gives a colourless gas with pungent smell or if the gas which gives dense white fumes with ammonia solution, then the salt may contain Cl^- ions and the following reaction occurs.

$$NaCl + H_2SO_4 \longrightarrow \underset{\substack{\text{Sodium}\\\text{hydrogen sulphate}}}{NaHSO_4} + \underset{\substack{\text{Hydrogen chloride}\\\text{Colourless gas}}}{HCl}$$

$$HCl + NH_3 \longrightarrow \underset{\substack{\text{Ammonium chloride}\\\text{White fumes}}}{NH_4Cl}$$

(b) If a salt gives effervescence on heating with conc. H_2SO_4 and MnO_2 and a light greenish yellow pungent gas is evolved, this indicates the presence of Cl^- ions.

$$MnO_2 + 2NaCl + 2H_2SO_4 \longrightarrow Na_2SO_4 + MnSO_4 + 2H_2O + Cl_2$$

(c) Salt solution acidified with dilute HNO_3 on addition of silver nitrate solution gives a curdy white precipitate soluble in ammonium hydroxide solution. This indicates the presence of Cl^- ions in the salt.

$$NaCl + AgNO_3 \longrightarrow NaNO_3 + \underset{\substack{\text{Silver chloride}\\\text{(White precipitate)}}}{AgCl}$$

$$AgCl + 2NH_4OH \longrightarrow \underset{\substack{\text{Diammine silver (I)}\\\text{chloride}}}{[Ag(NH_3)_2]Cl} + 2H_2O$$

(d) Mix a little amount of salt and an equal amount of solid potassium dichromate ($K_2Cr_2O_7$) in a test tube and add conc. H_2SO_4 to it. Heat the test tube and pass the evolved gas through sodium hydroxide solution. If a yellow solution is obtained, divide the solution into two parts. Acidify the first part with acetic acid and then add lead acetate solution. Formation of a yellow precipitate of lead chromate confirms the presence of chloride ions in the salt. This test is called ***chromyl chloride test.***

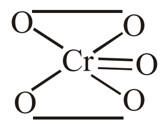
$$4NaCl + K_2Cr_2O_7 + 6H_2SO_4 \longrightarrow 2KHSO_4 + \underset{\substack{\text{(Chromyl}\\\text{chloride)}}}{2CrO_2Cl_2} + 4NaHSO_4 + 3H_2O$$

$$CrO_2Cl_2 + 4NaOH \longrightarrow Na_2CrO_4 + 2NaCl + 2H_2O$$

$$(CH_3COO)_2Pb + \underset{\substack{\text{Sodium}\\\text{chromate}}}{Na_2CrO_4} \longrightarrow \underset{\substack{\text{Lead chromate}\\\text{(Yellow precipitate)}}}{PbCrO_4} + 2CH_3COONa$$

Acidify the second part with dilute sulphuric acid and add small amounts of amyl alcohol and then 1 mL of 10% hydrogen peroxide solution. On gentle shaking organic layer turns blue. CrO_4^{2-} ion formed in the reaction of chromyl chloride with sodium hydroxide reacts with hydrogen peroxide to form chromium pentoxide (CrO_5) (See structure) which dissolves in amyl alcohol to give blue colour.

$$CrO_4^{2-} + 2H^+ + 2H_2O_2 \longrightarrow \underset{\substack{\text{Chromium}\\\text{pentoxide}}}{CrO_5} + 3H_2O$$



Structure of chromium pentoxide

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2. Test for Bromide ion (Br^-)

If on heating the salt with conc. H_2SO_4 reddish brown fumes of bromine are evolved in excess, this indicates the presence of Br^- ions. The fumes get intensified on addition of MnO_2. Bromine vapours turn starch paper yellow.

$$2NaBr + 2H_2SO_4 \longrightarrow Br_2 + SO_2 + Na_2SO_4 + 2H_2O$$
$$2NaBr + MnO_2 + 2H_2SO_4 \longrightarrow Na_2SO_4 + MnSO_4 + 2H_2O + Br_2$$

(a) Add 1 mL of carbon tetrachloride (CCl_4)/chloroform ($CHCl_3$) and excess of freshly prepared chlorine water dropwise to the salt solution in water or sodium carbonate extract neutralised with dilute HCl. Shake the test tube vigorously. The appearance of an orange brown colouration in the organic layer due to the dissolution of bromine in it, confirms the presence of bromide ions.

$$2NaBr + Cl_2 \longrightarrow 2NaCl + Br_2$$

(b) Acidify the sodium carbonate extract of the salt with dil. HNO_3. Add silver nitrate ($AgNO_3$) solution and shake the test tube. A pale yellow precipitate is obtained which dissolves in ammonium hydroxide with difficulty.

$$NaBr + AgNO_3 \longrightarrow NaNO_3 + \underset{\substack{\text{Silver bromide} \\ \text{Pale yellow precipitate}}}{AgBr}$$

3. Test for Iodide ion (I^-)

(a) If on heating the salt with conc. H_2SO_4, deep violet vapours with a pungent smell are evolved. These turns starch paper blue and a violet sublimate is formed on the sides of the test tube, it indicates the presence of I^- ions. Some HI, sulphur dioxide, hydrogen sulphide, and sulphur are also formed due to the following reactions.

$$2NaI + 2H_2SO_4 \longrightarrow Na_2SO_4 + SO_2 + 2H_2O + I_2$$
$$I_2 + \underset{\text{solution}}{\text{Starch}} \longrightarrow \text{Blue colour}$$
$$NaI + H_2SO_4 \longrightarrow NaHSO_4 + HI$$
$$2HI + H_2SO_4 \longrightarrow 2H_2O + I_2 + SO_2$$
$$6NaI + 4H_2SO_4 \longrightarrow 3I_2 + 4H_2O + S + 3Na_2SO_4$$
$$8NaI + 5H_2SO_4 \longrightarrow 4I_2 + H_2S + 4Na_2SO_4 + 4H_2O$$

On adding MnO_2 to the reaction mixture, the violet vapours become dense.

$$2NaI + MnO_2 + 2H_2SO_4 \longrightarrow I_2 + MnSO_4 + Na_2SO_4 + 2H_2O$$

(b) Add 1 mL of $CHCl_3$ or CCl_4 and chlorine water in excess to the salt solution in water or sodium carbonate extract neutralised with dil.HCl and shake the test tube vigorously. Presence of violet colouration in the organic layer confirms the presence of iodide ions.

$$2NaI + Cl_2 \longrightarrow 2NaCl + I_2$$

Iodine dissolves in the organic solvent and the solution becomes violet.

(c) Acidify sodium carbonate extract of the salt with dil.HNO_3 and add $AgNO_3$ solution. Appearance of a yellow precipitate insoluble in excess of NH_4OH confirms the presence of iodide ions.

$$NaI + AgNO_3 \longrightarrow \underset{\substack{\text{silver iodide} \\ \text{(Yellow precipitate)}}}{AgI} + NaNO_3$$

ALLEN Session 2019-20

(a) If on heating the salt with conc. H_2SO_4 light brown fumes are evolved then heat a small quantity of the given salt with few copper turnings or chips and conc. H_2SO_4. Evolution of excess of brown fumes indicates the presence of nitrate ions. The solution turns blue due to the formation of copper sulphate.

$$NaNO_3 + H_2SO_4 \longrightarrow NaHSO_4 + HNO_3$$
$$4HNO_3 \longrightarrow 4NO_2 + O_2 + 2H_2O$$
$$2NaNO_3 + 4H_2SO_4 + 3Cu \longrightarrow \underset{\substack{\text{Copper sulphate} \\ \text{(Blue)}}}{3CuSO_4} + Na_2SO_4 + 4H_2O + 2NO$$
$$2NO + O_2 \longrightarrow \underset{\text{(Brown fumes)}}{2NO_2}$$

(b) Take 1 mL of an aqueous solution of the salt and add 2 mL conc. H_2SO_4 slowly. Mix the solutions thoroughly and cool the test tube under the tap. Now, add freshly prepared ferrous sulphate solution along the sides of the test tube dropwise so that it forms a layer on the top of the liquid already present in the test tube. A dark brown ring is formed at the junction of the two solutions due to the formation of nitroso ferrous sulphate (Fig. 1.2). Alternatively first ferrous sulphate is added and then concentrated sulphuric acid is added.

$$NaNO_3 + H_2SO_4 \longrightarrow NaHSO_4 + HNO_3$$
$$6FeSO_4 + 3H_2SO_4 + 2HNO_3 \longrightarrow 3Fe_2(SO_4)_3 + 4H_2O + 2NO$$
$$FeSO_4 + NO \longrightarrow \underset{\substack{\text{Nitroso ferrous sulphate} \\ \text{(Brown)}}}{[Fe(NO)]SO_4}$$

5. Test for Oxalate ion [$C_2O_4^{2-}$]

If carbon dioxide gas along with carbon monoxide gas is evolved in the preliminary examination with concentrated sulphuric acid, this gives indication about the presence of oxalate ion.

$$Na_2C_2O_4 + \text{Conc.}H_2SO_4 \longrightarrow Na_2SO_4 + H_2O + CO_2 \uparrow + CO \uparrow$$

Oxalate is confirmed by the following tests:

(a) Acidify sodium carbonate extract with acetic acid and add calcium chloride solution. A white precipitate of calcium oxalate, insoluble in ammonium oxalate and oxalic acid solution indicates the presence of oxalate ion.

$$CaCl_2 + Na_2C_2O_4 \longrightarrow \underset{\substack{\text{Calcium oxalate} \\ \text{(White precipitate)}}}{CaC_2O_4} + 2NaCl$$

Session 2019-20

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(b) $KMnO_4$ test

Filter the precipitate from test (a). Add dil. H_2SO_4 to it followed by dilute $KMnO_4$ solution and warm. Pink colour of $KMnO_4$ is discharged:

$$\underset{}{CaC_2O_4} + H_2SO_4 \longrightarrow \underset{\text{Calcium sulphate}}{CaSO_4} + \underset{\text{Oxalic acid}}{H_2C_2O_4}$$

$$2KMnO_4 + 3H_2SO_4 + 5H_2C_2O_4 \longrightarrow 2MnSO_4 + K_2SO_4 + 8H_2O + 10CO_2$$

Pass the gas evolved through lime water. A white precipitate is formed which dissolves on passing the gas for some more time.

Step-III : Test for Sulphate and Phosphate

If no positive test is obtained in Steps-I and II, then tests for the presence of sulphate and phosphate ions are performed. These tests are summarised in Table 5.

Table 5 : Confirmatory tests for Sulphate and Phosphate

Ion	Confirmatory Test
Sulphate (SO_4^{2-})	(a) Take 1 mL water extract of the salt in water or sodium carbonate and after acidifying with dilute hydrochloric acid add $BaCl_2$ solution. White precipitate insoluble in conc. HCl or conc. HNO_3 is obtained. (b) Acidify the aqueous solution or sodium carbonate extract with acetic acid and add lead acetate solution. Appearance of white precipitate confirms the presence of SO_4^{2-} ion.
Phosphate (PO_4^{3-})	(a) Acidify sodium carbonate extract or the solution of the salt in water with conc. HNO_3 and add ammonium molybdate solution and heat to boiling. A canary yellow precipitate is formed.

1. Test of Sulphate ions [SO_4^{2-}]

(a) Aqueous solution or sodium carbonate extract of the salt acidified with acetic acid on addition of barium chloride gives a white precipitate of barium sulphate insoluble in conc. HCl or conc. HNO_3.

$$Na_2SO_4 + BaCl_2 \longrightarrow \underset{\substack{\text{Barium sulphate}\\ \text{(White precipitate)}}}{BaSO_4} + 2NaCl$$

(b) Sulphate ions give white precipitate of lead sulphate when aqueous solution or sodium carbonate extract neutralised with acetic acid is treated with lead acetate solution.

$$Na_2SO_4 + (CH_3COO)_2Pb \longrightarrow \underset{\substack{\text{Lead sulphate}\\ \text{(White precipitate)}}}{PbSO_4} + 2CH_3COONa$$

2. Test for Phosphate ion [PO_4^{3-}]

(a) Add conc. HNO_3 and ammonium molybdate solution to the test solution containing phosphate ions and boil. A yellow colouration in solution or a canary yellow precipitate of ammonium-phosphomolybdate, $(NH_4)_3[P(Mo_3O_{10})_4]$ is formed. Each oxygen of phosphate has been replaced by Mo_3O_{10} group.

$$Na_2HPO_4 + 12(NH_4)_2MoO_4 + 23HNO_3 \longrightarrow \underset{\substack{\text{Canary yellow}\\ \text{precipitate}}}{(NH_4)_3[P(Mo_3O_{10})_4]} + 2NaNO_3 + 21NH_4NO_3 + 12H_2O$$

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ALLEN Session 2019-20

HEATING DEVICES

Heating during the laboratory work can be done with the help of a gas burner, spirit lamp or a kerosene lamp. The gas burner used in the laboratory is usually Bunsen burner . Various parts of Bunsen burner are shown in Fig. The description of these parts is as follows :

(A) Parts of Bunsen Burner

1. The Base

Heavy metallic base is connected to a side tube called gas tube. Gas from the source enters the burner through the gas tube and passes through a small hole called Nipple or Nozzle and enters into the burner tube under increased pressure and can be burnt at the upper end of the burner tube.

2. The Burner Tube

It is a long metallic tube having two holes diametrically opposite to each other near the lower end which form the air vent. The tube can be screwed at the base. The gas coming from the nozzle mixes with the air coming through the air vent and burns at its upper end.

3. The Air Regulator

It is a short metallic cylindrical sleeve with two holes diametrically opposite to each other. When it is fitted to the burner tube, it surrounds the air vent of the burner tube. To control the flow of air through the air vent, size of its hole is adjusted by rotating the sleeve.

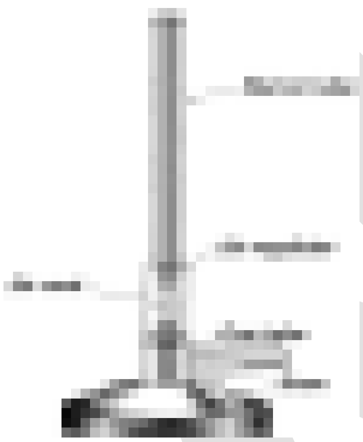

Fig. Bunsen burner

Fig. Parts of Bunsen barner

If the air vent is closed and the gas is ignited, the flame will be large and luminous (smoky and yellow in colour). The light emitted by the flame is due to the radiations given off by the hot carbon particles of partially burnt fuel. The temperature of the flame in this situation is low. If adjustment of sleeve on vent is such that gas mixed with air is fed into the flame, the flame becomes less luminous and finally turns blue. When the flow of air is correctly adjusted, the temperature of the flame becomes quite high. This is called non-luminous flame. Various zones of flame are shown below in Fig.

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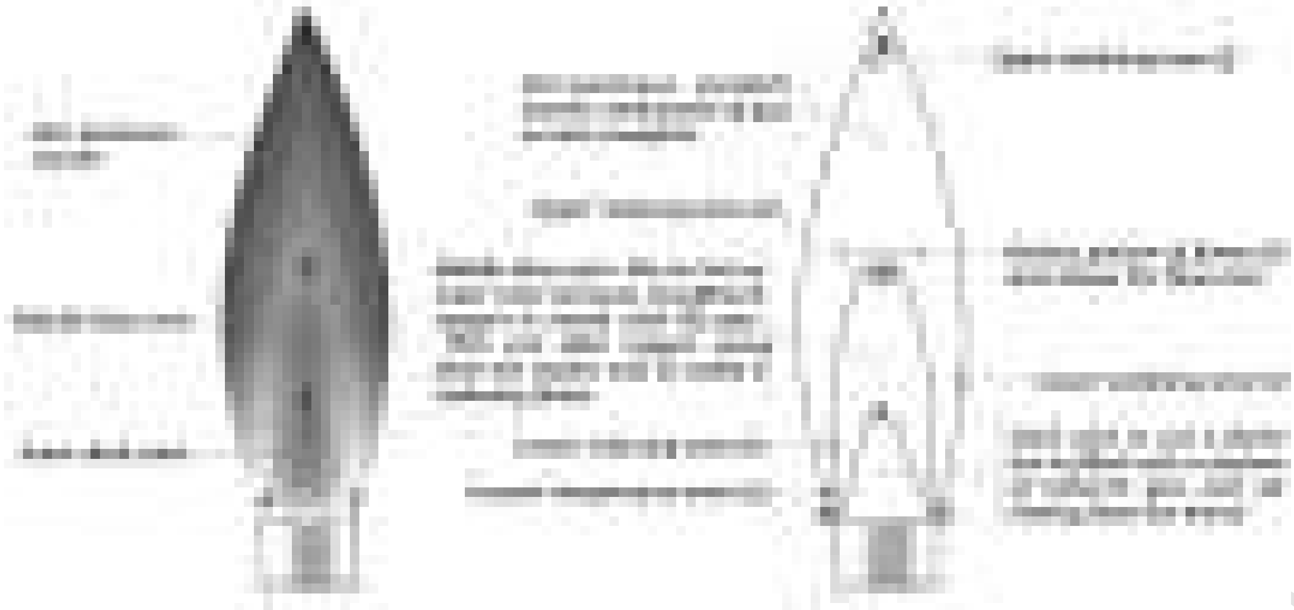
Three distinctly visible parts of the Bunsen flame are described below :



(B) PRINCIPAL PARTS OF BUNSEN FLAME

1. The Inner Dark Cone, A E C

This is innermost dark cone, which is just above the burner tube. It consists of unburnt gases. This zone is the coldest zone of the flame and no combustion takes place here.

2. The Middle Blue Cone, A D C E A

This is middle part of the flame. This becomes luminous when the air vent is slightly closed. Luminosity of this part is due to the presence of unburnt carbon particles produced by decomposition of some gas. These particles get heated up to incandescence and glow but do not burn. Since the combustion is not complete in this part, the temperature is not very high.

3. The Outer Non-luminous Mantle, A B C D A

This is purplish outer cone. It is the hottest part of the flame. It is in direct contact with the atmosphere and combustion is quite complete in this zone.

Bunsen identified six different regions in these three principal parts of the flame:

(i) The upper oxidising zone (f)

Its location is in the non-luminous tip of the flame which is in the air. In comparison to inner portions of the flame large excess of oxygen is present here. The temperature is not as high as in region (c) described below. It may be used for all oxidation processes in which highest temperature of the flame is not required.

(ii) Upper reducing zone (e)

This zone is at the tip of the inner blue cone and is rich in incandescent carbon. It is especially useful for reducing oxide incrustations to the metals.

(iii) Hottest portion of flame (d)

It is the fusion zone. It lies at about one-third of the height of the flame and is approximately equidistant from inside and outside of the mantle i.e. the outermost cone of the flame. Fusibility of the substance can be tested in this region. It can also be employed for testing relative volatility of substances or a mixture of substances.

(iv) Lower oxidising zone (c)

It is located on the outer border of the mantle near the lower part of the flame and may be used for the oxidation of substances dissolved in beads of borax or sodium carbonate etc.

(v) Lower reducing zone (b)

It is situated in the inner edge of the outer mantle near to the blue cone and here reducing gases mix with the oxygen of the air. It is a less powerful reducing zone than (e) and may be employed for the reduction of fused borax and similar beads.

(vi) Lowest temperature zone (a)

Zone (a) of the flame has lowest temperature. It is used for testing volatile substances to determine whether they impart colour to the flame.

(C) STRIKING BACK OF THE BUNSEN BURNER

Striking back is the phenomenon in which flame travels down the burner tube and begins to burn at the nozzle near the base. This happens when vents are fully open.The flow of much air and less gas makes the flame become irregular and it strikes back.

The tube becomes very hot and it may produce burns on touching. This may melt attached rubber tube also. If it happens, put off the burner and cool it under the tap and light it again by keeping the air vent partially opened.

SPIRIT LAMP

If Bunsen burner is not available in the laboratory then spirit lamp can be used for heating. It is a devise in which one end of a wick of cotton thread is dipped in a spirit container and the other end of the wick protrudes out of the nozzle at upper end of the container. Spirit rises upto the upper end of the wick due to the capillary action and can be burnt. The flame is non luminous hence can be used for all heating purposes in the laboratory. To put off the lamp, burning wick is covered with the cover. **Never try to put off the lighted burner by blowing at the flame.**

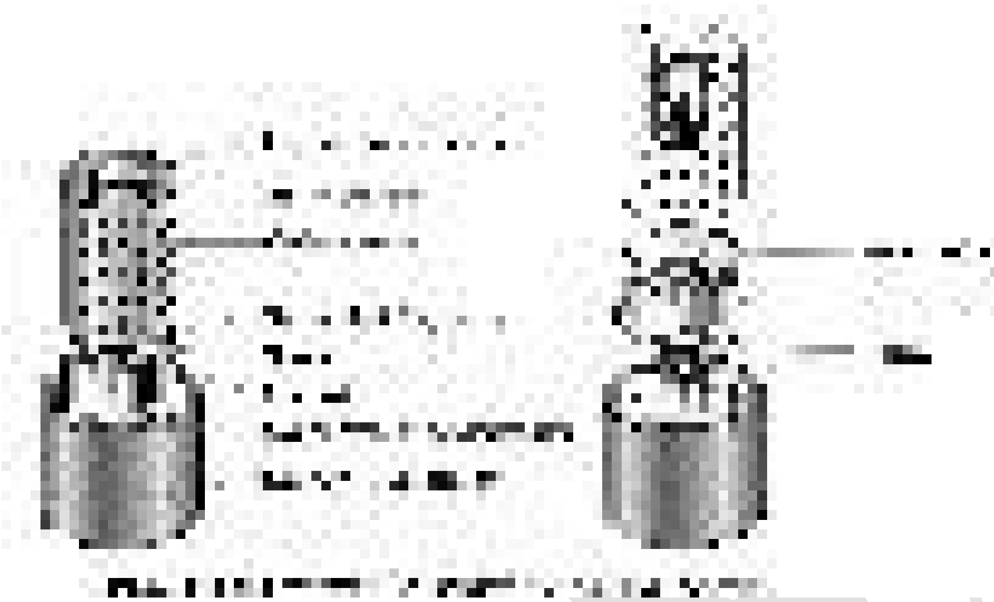
KEROSENE HEATING LAMP

A kerosene lamp has been developed by National Council of Educational Research and Training (NCERT), which is a versatile and cheaper substitute of spirit lamp. It may be used in laboratories as a source of heat whereever spirit and gas burner are not available. Parts of kerosene lamp are shown in Fig.

Working of the Kerosene Lamp

More than half of the container is filled with kerosene. Outer sleeve is removed for lighting the wicks. As the outer sleeve is placed back in position, the flames of four wicks combine to form a big soot-free blue flame.

The lighted heating lamp can be put off only by covering the top of the outer sleeve with a metal or asbestos sheet.



SYSTEMATIC ANALYSIS OF CATIONS

The tests for cations may be carried out according to the following scheme.

Step - I : Preliminary Examination of the Salt for Identification of Cation

1. Colour Test

Observe the colour of the salt carefully, which may provide useful information about the cations. Table 6 gives the characteristic colours of the salts of some cations.

Table 6 Characteristic colours of the some metal ions

Ion	Confirmatory Test
Light green	Fe^{2+}
Yellowis Brown	Fe^{3+}
Blue	Cu^{2+}
Bright green	Ni^{2+}
Blue, Red Violet, Pink	Co^{2+}
Light pink	Mn^{2+}

2. Dry Heating Test

(i) Take about 0.1 g of the dry salt in a clean and dry test tube.

(ii) Heat the above test tube for about one minute and observe the colour of the residue when it is hot and also when it becomes cold. Observation of changes gives indications about the presence of cations, which may not be taken as conclusive evidence (see Table 7).

Session 2019-20

Table 7 : Inferences from the colour of the salt in cold and on heating

Colour when cold	Colour when hot	Inference
Blue	White	Cu^{2+}
Green	Dirty white or yellow	Fe^{2+}
White	Yellow	Zn^{2+}
Pink	Blue	Co^{2+}

3. Flame Test

The chlorides of several metals impart characteristic colour to the flame because they are volatile in non-luminous flame. This test is performed with the help of a platinum wire as follows :

(i) Make a tiny loop at one end of a platinum wire.

(ii) To clean the loop dip it into concentrated hydrochloric acid and hold it in a non-luminous flame (Fig. 1.3).

(iii) Repeat step (ii) until the wire imparts no colour to the flame.

(iv) Put 2-3 drops of concentrated hydrochloric acid on a clean watch glass and make a paste of a small quantity of the salt in it.

(v) Dip the clean loop of the platinum wire in this paste and introduce the loop in the non-luminous (oxidising) flame (Fig. 1.3).

(vi) Observe the colour of the flame first with the naked eye and then through a blue glass and identify the metal ion with the help of Table 8.

Table 8 : Inference from the flame test

Colour fo the flame observed by naked eye	Colour of the flame observed through blue glass	Inference
Green flame with blue centre	Same colour as observed without glass	Cu^{2+}
Crimson red	Purple	Sr^{2+}
Apple green	Bluish green	Ba^{2+}
Brick red	Green	Ca^{2+}

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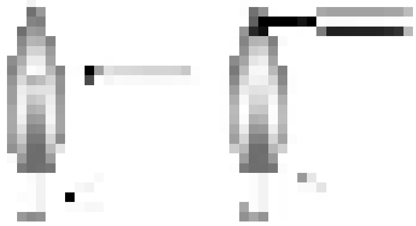
4. Borax Bead Test



This test is employed only for coloured salts because borax reacts with metal salts to form metal borates or metals, which have characteristic colours.

(i) To perform this test make a loop at the end of the platinum wire and heat it in a flame till it is red hot.

(ii) Dip the hot loop into borax powder and heat it again until borax forms a colourless transparent bead on the loop. Before dipping the borax bead in the test salt or mixture, confirm that the bead is transparent and colourless. If it is coloured this means that, the platinum wire is not clean. Then make a fresh bead after cleaning the wire.

(iii) Dip the bead in a small quantity of the dry salt and again hold it in the flame.

(iv) Observe the colour imparted to the bead in the non - luminous flame as well as in the luminous flame while it is hot and when it is cold (Fig. 1.4).

(v) To remove the bead from the platinum wire, heat it to redness and tap the platinum wire with your finger. (Fig.1.5).

On heating, borax loses its water of crystallisation and decomposes to give sodium metaborate and boric anhydride.

$$\underset{\text{Borax}}{Na_2B_4O_7\ .10H_2O} \longrightarrow Na_2B_4O_7 + 10H_2O$$

$$Na_2B_4O_7 \longrightarrow \underset{\text{Sodium metaborate}}{2NaBO_2} + \underset{\text{Boric anhydride}}{B_2O_3}$$

On treatment with metal salt, boric anhydride forms metaborate of the metal which gives different colours in oxidising and reducing flame. For example, in the case of copper sulphate, following reactions occur.

$$CuSO_4 + B_2O_3 \xrightarrow{\text{Non-luminous flame}} \underset{\substack{\text{Cupric metaborate} \\ \text{Blue-green}}}{Cu(BO_2)_2} + SO_3$$

Two reactions may take place in the reducing flame:

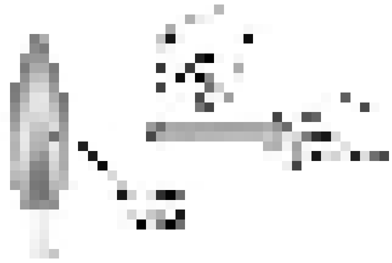
(i) The blue $Cu(BO_2)_2$ is reduced to colourless cuprous metaborate as follows:

$$2Cu(BO_2)_2 + 2NaBO_2 + C \xrightarrow{\text{Luminous flame}} 2CuBO_2 + Na_2B_4O_7 + CO$$

or (ii) Cupric metaborate may be reduced to metallic copper and the bead appears red and opaque.

$$2Cu(BO_2)_2 + 4NaBO_2 + 2C \xrightarrow{\text{Luminous flame}} 2Cu + 2Na_2B_4O_7 + 2CO$$

The preliminary identification of metal ion can be made from Table 9.



ALLEN Session 2019-20

Table 9 : Inference from the borax bead test

Heating in oxidising (non-luminous) flame		Heating in reducing (luminous) flame		Inference
Colour of the salt bead		Colour of the salt bead		
In cold	In hot	In cold	In hot	
Blue	Gren	Red opaque	Colourless	Cu^{2+}
Reddish brown	Violet	Grey	Grey	Ni^{2+}
Light violet	Light violet	Colourless	Colourless	Mn^{2+}
Yellow	Yellowish brown	Green	Green	Fe^{3+}

5. Charcoal Cavity Test

Metallic carbonate when heated in a charcoal cavity decomposes to give corresponding oxide. The oxide appears as a coloured residue in the cavity. Sometimes oxide may be reduced to metal by the carbon of the charcoal cavity.

The test may be performed as follows:

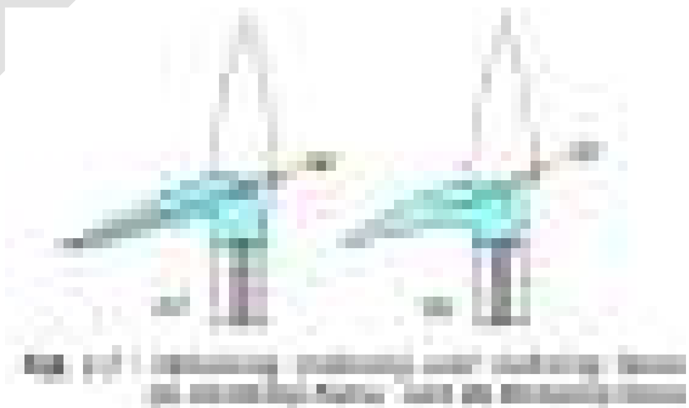
(i) Make a small cavity in a charcoal block with the help of a charcoal borer [Fig.1.6 (a)].

(ii) Fill the cavity with about 0.2 g of the salt and about 0.5 g of anhydrous sodium carbonate.

(iii) Moisten the salt in the cavity with one or two drops of water, otherwise salt/mixture will blow away.

(iv) Use a blowpipe to heat the salt in a luminous (reducing) flame and observe the colour of oxide/ metallic bead formed in the cavity both when hot and cold [Fig. (1.6 b)]. Obtain oxidising and reducing flame as shown in Fig. 1.7 a and b.

(v) Always bore a fresh cavity for testing the new salt.



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When test is performed with $CuSO_4$, the following change occurs.

$$CuSO_4 + Na_2CO_3 \xrightarrow{\text{Heat}} CuCO_3 + Na_2SO_4$$

$$CuCO_3 \xrightarrow{\text{Heat}} CuO + CO_2$$

$$CuO + C \xrightarrow{\text{Heat}} \underset{\text{Red colour}}{Cu} + CO$$

In case of $ZnSO_4$:

$$ZnSO_4 + Na_2CO_3 \xrightarrow{\text{Heat}} ZnCO_3 + Na_2SO_4$$

$$ZnCO_3 \xrightarrow{\text{Heat}} ZnO + CO_2$$

Yellow when hot,
White when cold

The metal ion can be inferred from Table 10.

Table 10 : Inference from the charcoal cavity test

Observations	Inference
Yellow residue when hot and grey metal when cold	Pb^{2+}
White residue with the odour of garlic	As^{3+}
Brown residue	Cd^{2+}
Yellow residue when hot and white when cold	Zn^{2+}

6. Cobalt Nitrate Test

If the residue in the charcoal cavity is white, cobalt nitrate test is performed.

(i) Treat the residue with two or three drops of cobalt nitrate solution.

(ii) Heat it strongly in non-luminous flame with the help of a blow pipe and observe the colour of the residue.

On heating, cobalt nitrate decomposes into cobalt (II) oxide, which gives a characteristic colour with metal oxide present in the cavity.

Thus, with ZnO, Al_2O_3 and MgO, the following reactions occur.

$$2Co(NO_3)_2 \xrightarrow{\text{Heat}} 2CoO + 4NO_2 + O_2$$

$$CoO + ZnO \longrightarrow \underset{\text{Green}}{CoO.ZnO}$$

$$CoO + MgO \longrightarrow \underset{\text{Pink}}{CoO.\ MgO}$$

$$CoO + Al_2O_3 \longrightarrow \underset{\text{Blue}}{CoO.\ Al_2O_3}$$

Session 2019-20

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Step-II : Wet Tests for Identification of Cations

The cations indicated by the preliminary tests given above are confirmed by systematic analysis given below. The first essential step is to prepare a clear and transparent solution of the salt. This is called original solution. It is prepared as follows:

Preparation of Original Solution (O.S.)

To prepare the original solution, following steps are followed one after the other in a systematic order. In case the salt does not dissolve in a particular solvent even on heating, try the next solvent.

The following solvents are tried:

1. Take a little amount of the salt in a clean boiling tube and add a few mL of distilled water and shake it. If the salt does not dissolved, heat the content of the boiling tube till the salt completely dissolves.
2. If the salt is insoluble in water as detailed above, take fresh salt in a clean boiling tube and add a few mL of dil.HCl to it. If the salt is insoluble in cold, heat the boiling tube till the salt is completely dissolved.
3. If the salt does not dissolve either in water or in dilute HCl even on heating, try to dissolve it in a few mL of conc. HCl by heating.
4. If salt does not dissolve in conc. HCl, then dissolve it in dilute nitric acid.
5. If salt does not dissolve even in nitric acid then a mixture of conc. HCl and conc. HNO_3 in the ratio 3 : 1 is tried. This mixture is called aqua regia. A salt not soluble in aqua regia is considered to be an insoluble salt.

Group Analysis

(I) Analysis of Zero group cation (NH_4^+ ion)

(a) Take 0.1 g of salt in a test tube and add 1-2 mL of NaOH solution to it and heat. If there is a smell of ammonia, this indicates the presence of ammonium ions. Bring a glass rod dipped in hydrochloric acid near the mouth of the test tube. White fumes are observed.

(b) Pass the gas through Nessler's reagent. Brown precipitate is obtained.

Chemistry of Confirmatory Tests for NH_4^+ ion

(a) Ammonia gas evolved by the action of sodium hydroxide on ammonium salts reacts with hydrochloric acid to give ammonium chloride, which is visible as dense white fume.

$$(NH_4)_2SO_4 + 2NaOH \longrightarrow Na_2SO_4 + 2NH_3 + 2H_2O$$

$$NH_3 + HCl \longrightarrow NH_4Cl$$

On passing the gas through Nessler's reagent, a brown colouration or a precipitate of basic mercury(II) amido-iodine is formed.

$$2K_2[HgI_4] + NH_3 + 3KOH \longrightarrow \underset{\substack{\text{Basic mercury (II)}\\ \text{amido-iodine}\\ \text{(Brown precipitate)}}}{HgO.Hg(NH_2)I} + 7\,KI + 2H_2O$$

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For the analysis of cations belonging to groups I-VI, the cations are precipitated from the original solution by using the group reagents (see Table 1.11) according to the scheme shown in the flow chart given below: The separation of all the six groups is represented as below :

Flow Chart*

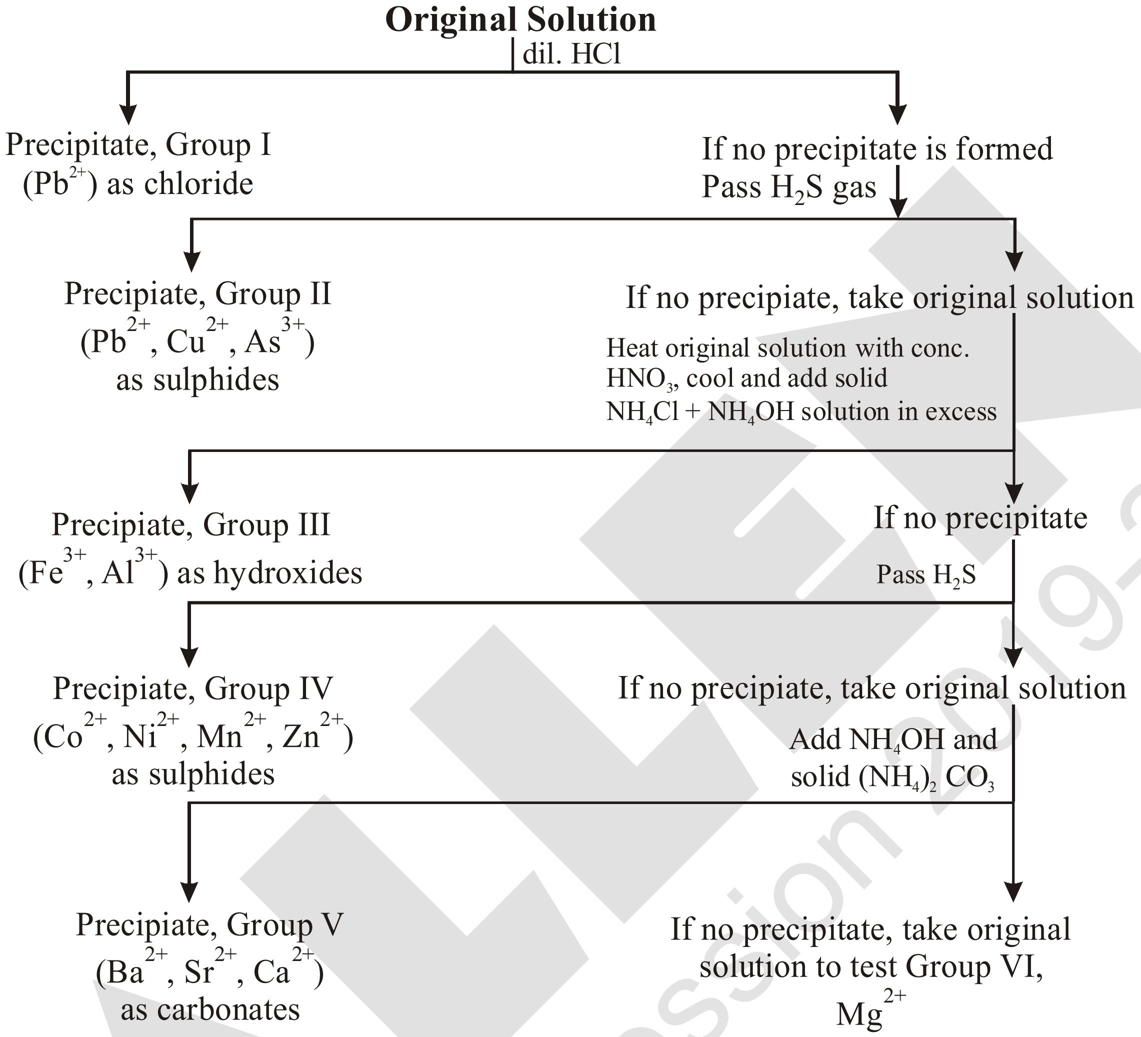

Table 11 : Group reagents for precipitating ions

Group	Cations*	Group Reagent
Group zero	NH_4^+	**None**
Group - I	Pb^{2+}	**Dilute HCl**
Group - II	Pb^{2+}, Cu^{2+}, As^{3+}	**H_2S gas in presence of dil. HCl**
Group - III	Al^{3+}, Fe^{3+}	**NH_4OH in presence of NH_4Cl**
Group - IV	Co^{2+}, Ni^{2+}, Mn^{2+}, Zn^{2+}	**H_2S in presence of NH_4OH**
Group - V	Ba^{2+}, Sr^{2+}, Ca^{2+}	**$(NH_4)_2CO_3$ in presence of NH_4OH**
Group - VI	Mg^{2+}	**None**

ALLEN Session 2019-20

(II) Analysis of Group-I cations

Take a small amount of original solution (if prepared in hot conc. HCl) in a test tube and add cold water to it and cool the test tube under tap water. If a white precipitate appears, this indicates the presence of Pb^{2+} ions in group –I. On the other hand, if the original solution is prepared in water and on addition of dil. HCl, a white precipitate appears, this may also be Pb^{2+}. Confirmatory tests are described below in Table 12.

Table 12 : Confirmatory tests for Group-I cation (Pb^{2+})

Experiment	Observation
Dissolve the precipitate in hot water and divide the hot solution into three parts,	
1. Add potassium idoide solution to the first part.	A yellow precipitate is obtained.
2. To the second part add potassium chromate solution.	A yellow precipitate is obtained which is soluble, in NaOH and insoluble in ammonium acetate solution.
3. To the thrid part of the hot solution add few drops of alcohol and dilute sulphuric acid.	A white precipitate is obtained which is soluble in ammonium acetate solution.

Chemistry of the Confirmatory Tests of Pb^{2+} ions

Lead is precipitated as lead chloride in the first group. The precipitate is soluble in hot water.

1. On adding potassium iodide (KI) solution, a yellow precipitate of lead iodide is obtained which confirms the presence of Pb^{2+} ions.

$$\underset{\text{(Hot solution)}}{PbCl_2} + 2KI \longrightarrow \underset{\text{Yellow precipitate}}{PbI_2} + 2KCl$$

This yellow precipitate (PbI_2) is soluble in boiling water and reappears on cooling as shining crystals.

2. On addition of potassium chromate (K_2CrO_4) solution a yellow precipitate of lead chromate is obtained. This confirms the presence of Pb^{2+} ions.

$$\underset{\text{(Hot solution)}}{PbCl_2} + K_2CrO_4 \longrightarrow \underset{\substack{\text{Lead chromate} \\ \text{(Yellow precipitate)}}}{PbCrO_4} + 2KCl$$

The yellow precipitate ($PbCrO_4$) is soluble in hot NaOH solution.

$$PbCrO_4 + 4NaOH \longrightarrow \underset{\substack{\text{Sodium tetra} \\ \text{hydroxoplumbate (II)}}}{Na_2[Pb(OH)_4]} + Na_2CrO_4$$

3. A white precipitate of lead sulphate ($PbSO_4$) is formed on addition of alcohol followed by dil. H_2SO_4.

$$PbCl_2 + H_2SO_4 \longrightarrow \underset{\substack{\text{Lead sulphate} \\ \text{(White precipitate)}}}{PbSO_4} + 2HCl$$

Lead sulphate is soluble in ammonium acetate solution due to the formation of tetraacetatoplumbate(II) ions. This reaction may be promoted by addition of few drops of acetic acid.

$$PbSO_4 + 4CH_3COONH_4 \longrightarrow \underset{\substack{\text{Ammonium} \\ \text{tetraacetatoplumbate(II)}}}{(NH_4)_2[Pb(CH_3COO)_4]} + (NH_4)_2SO_4$$

ALLEN Session 2019-20

(III) Analysis of Group–II cations

If group-I is absent, add excess of water to the same test tube. Warm the solution and pass H_2S gas for 1-2 minutes (Fig. 1.6). Shake the test tube. If a precipitate appears, this indicates the presence of group-II cations. Pass more H_2S gas through the solution to ensure complete precipitation and separate the precipitate. If the colour of the precipitate is black, itindicates the presence of Cu^{2+} or Pb^{2+} ions.

If it is yellow in colour, then presence of As^{3+} ions is indicated.

Take the precipitate of group-II in a test tube and add excess of yellow ammonium sulphide solution to it. Shake the test tube. If the precipitate is insoluble, group II-A (copper group) is present. If the precipitate is soluble, this indicates the presence of group-II B (arsenic group).

Confirmatory tests for the groups II A and II B are given in Table 13.

Table 13 : Confirmatory tests for the groups II A and II B cations

<table>
<tr><td colspan="2">Black precipitate of Group II A ions Pb^{2+}, Cu^{2+} (insoluble in yellow ammonium sulphide) is formed.</td><td>If a yellow precipitate soluble in yellow ammonium sulphide is formed then As^{3+} ion is present.</td></tr>
<tr><td colspan="2">Boil the precipitate of Group II A with dilute nitric acid and add a few drops of alcohol and dil. H_2SO_4.</td><td rowspan="2">Acidify this solution with dilute HCl. A yellow precipitate is formed. Heat the precipitate with concentrated nitric acid and add ammonium molybdate solution. A canary yellow precipitate is formed.</td></tr>
<tr><td>White precipitate confirms the presence of Pb^{2+} ions. Dissolve the precipitate in ammonium acetate solution. Acidify with acetic acid and divide the solution into two parts.
(i) To the first part add potassium chromate solution, a yellow precipitate is formed.
(ii) To the second part, add potassium iodide solution, a yellow precipitate is formed.</td><td>If no precipitate is formed, add excess of ammonium hydroxide solution. A blue solution is obtained, acidify it with acetic acid and add potassium ferrocyanide solution. A chocolate brown precipitate is formed.</td></tr>
</table>

Session 2019-20
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Group-II A (Copper Group)

Chemistry of confirmatory tests of Group-II A cations

1. Test for Lead ion (Pb^{2+})

Lead sulphide precipitate dissolves in dilute HNO_3. On adding dil. H_2SO_4 and a few drops of alcohol to this solution a white precipitate of lead sulphate appears. This indicates the presence of lead ions.

$$3PbS + 8HNO_3 \longrightarrow 3Pb(NO_3)_2 + 2NO + 4H_2O + 3S$$

$$Pb(NO_3)_2 + H_2SO_4 \longrightarrow PbSO_4 + 2HNO_3$$

The white precipitate dissolves in ammonium acetate solution on boiling. When this solution is acidified with acetic acid and potassium chromate solution is added, a yellow precipitate of $PbCrO_4$ is formed. On adding potassium iodide solution, a yellow precipitate of lead iodide is formed.

$$PbSO_4 + 4CH_3COONH_4 \longrightarrow \underset{\text{Ammonium tetraacetatoplumbate(II)}}{(NH_4)_2[Pb(CH_3COO)_4]} + (NH_4)_2SO_4$$

$$Pb^{2+} + CrO_4^{2-} \longrightarrow \underset{\substack{\text{Lead chromate}\\ \text{(Yellow precipitate)}}}{PbCrO_4} \;;\qquad Pb^{2+} + 2I^- \rightarrow \underset{\substack{\text{Lead iodide}\\ \text{(Yellow precipitate)}}}{PbI_2}$$

2. Test for Copper ion (Cu^{2+})

(a) Copper sulphide dissolves in nitric acid due to the formation of copper nitrate.

$$3CuS + 8HNO_3 \longrightarrow 3Cu(NO_3)_2 + 2NO + 3S + 4H_2O$$

On heating the reaction mixture for long time, sulphur is oxidised to sulphate and copper sulphate is formed and the solution turns blue. A small amount of NH_4OH precipitates basic copper sulphate which is soluble in excess of ammonium hydroxide due to the formation of tetraamminecopper (II) complex.

$$S + 2HNO_3 \longrightarrow H_2SO_4 + 2NO$$

$$2Cu^{2+} + SO_4^{2-} + 2NH_3 + 2H_2O \longrightarrow Cu(OH)_2.CuSO_4 + 2NH_4^+$$

$$Cu(OH)_2.CuSO_4 + 8NH_3 \longrightarrow \underset{\substack{\text{Tetraamminecopper (II)}\\ \text{sulphate (Deep blue)}}}{2[Cu(NH_3)_4]SO_4} + 2OH^- + SO_4^{2-}$$

(b) The blue solution on acidification with acetic acid and then adding potassium ferrocyanide $K_4[Fe(CN)_6]$ solution gives a chocolate colouration due to the formation of copper ferrocyanide

i.e. $Cu_2[Fe(CN)_6]$.

$$[Cu(NH_3)_4]SO_4 + 4CH_3COOH \longrightarrow CuSO_4 + 4CH_3COONH_4$$

$$2CuSO_4 + \underset{\substack{\text{Potassium}\\ \text{hexacyanoferrate (II)}}}{K_4[Fe(CN)_6]} \longrightarrow \underset{\substack{\text{Copper}\\ \text{hexacyanoferrate (II)}\\ \text{(Chocolate brown precipitate)}}}{Cu_2[Fe(CN)_6]} + 2K_2SO_4$$

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Group-II B (Arsenic Group)

If group- II precipitate dissolves in yellow ammonium sulphide and the colour of the solution is yellow, this indicates the presence of As^{3+} ions. Ammonium thioarsenide formed on dissolution of As_2S_3, decomposes with dil. HCl, and a yellow precipitate of arsenic (V) sulphide is formed which dissolves in concentrated nitric acid on heating due to the formation of arsenic acid. On adding ammonium molybdate solution to the reaction mixture and heating, a canary yellow precipitate is formed. This confirms the presence of As^{3+} ions.

$$\underset{\text{Yellow ammonium sulphide}}{As_2S_3 + 3(NH_4)_2S} \longrightarrow 2(NH_4)_3AsS_4 + S$$

$$2(NH_4)_3AsS_4 + 6HCl \rightarrow As_2S_5 + 3H_2S + 6NH_4Cl$$

$$3As_2S_5 + 10HNO_3 + 4H_2O \rightarrow \underset{\text{Arsenic acid}}{6H_3AsO_4} + 10NO + 15S$$

$$\underset{\text{Arsenic acid}}{H_3AsO_4} + \underset{\text{Ammonium molybdate}}{12(NH_4)_2MoO_4} + 21HNO_3 \rightarrow \underset{\substack{\text{Ammonium arsenomolybdate} \\ \text{(yellow precipitate)}}}{(NH_4)_3[As\ (Mo_3O_{10})_4]} + 21NH_4NO_3 + 12H_2O$$

(IV) Analysis of Group–III cations

If group-II is absent, take original solution and add 2-3 drops of conc. HNO_3 to oxidise Fe^{2+} ions to Fe^{3+} ions. Heat the solution for a few minutes. After cooling add a small amount of solid ammonium chloride (NH_4Cl) and an excess of ammonium hydroxide (NH_4OH) solution till it smells of ammonia. Shake the test tube. If a brown or white precipitate is formed, this indicates the presence of group-III cations. Confirmatory tests of group-III cations are summarised in Table 14.

Observe the colour and the nature of the precipitate. A gelatinous white precipitate indicates the presence of aluminium ion (Al^{3+}). If the precipitate is brown in colour, this indicates the presence of ferric ions (Fe^{3+}).

Table 14 : Confirmatory test for Group-III cations

	Brown precipitate Fe^{3+}		**White precipitate Al^{3+}**
	Dissolve the precipitate in dilute HCl and divide the solution into two parts.		Dissolve the white precipitate in dilute HCl and divide into two parts. .
(a)	To the first part add potassium ferrocyanide solution [Potasium hexacyanoferrate (II)]. A blue precipitate/colouration appears.	(a)	To the first part add sodium hydroxide solution and warm. A white gelatinous precipitate soluble in excess of sodium hydroxide solution is obtained
(b)	To the second part add potassium thiocyanate solution. A blood red colouration appears.	(b)	To the second part first add blue litmus solution and then ammonium hydroxide solution drop by drop along the sides of the test tube. A blue floating mass in the colourless solution is obtained.

Session 2019-20

Chemistry of confirmatory tests of Group - III cations

When original solution is heated with concentrated nitric acid, ferrous ion are oxidised to ferric ions.

$$2FeCl_2 + 2HCl + [O] \longrightarrow 2FeCl_3 + H_2O$$

Their group cations are precipitated as their hydroxides, which dissolve in dilute hydrochloric acid due to the formation of corresponding chlorides.

1. Test for Aluminium ions (Al^{3+})

(a) When the solution containing aluminium chloride is treated with sodium hydroxide a white gelationus precipitate of aluminium hydroxide is formed which is soluble in excess of sodium hydroxide solution due to the formation of sodium meta aluminate.

$$AlCl_3 + 3NaOH \longrightarrow Al(OH)_3 + 3NaCl$$

$$\underset{\text{White gelatinous precipitate}}{Al(OH)_3} + NaOH \longrightarrow \underset{\text{Sodium meta aluminate}}{NaAlO_2} + 2H_2O$$

(b) In the second test when blue litmus is added to the solution, a red colouration is obtained due to the acidic nature of the solution. On addition of NH_4OH solution drop by drop, the solution becomes alkaline and aluminium hydroxide is precipitated. Aluminium hydroxide adsorbs blue colour from the solution and forms insoluble adsorption complex named 'lake'. Thus a blue mass floating in the colourless solution is obtained. The test is therefore called **lake test.**

2. Test for ferric ions (Fe^{3+})

Reddish brown precipitate of ferric hydroxide dissolves in hydrochloric acid and ferric chloride is formed.

$$Fe(OH)_3 + 3HCl \longrightarrow FeCl_3 + 3H_2O$$

(a) When the solution containing ferric chloride is treated with potassium ferrocyanide solution a blue precipitate/colouration is obtained. The colour of the precipitate is Prussian blue. It is ferric ferro-cyanide. The reaction takes place as follows:

$$4FeCl_3 + \underset{\text{Potassium ferrocyanide}}{3K_4[Fe(CN)_6]} \longrightarrow \underset{\text{Prussian blue precipitate}}{Fe_4[Fe(CN)_6]_3} + 12KCl$$

If potassium hexacyanoferrate (II) (i.e. potassium ferrocyanide) is added in excess then a product of composition $KFe[Fe(CN)_6]$ is formed. This tends to form a colloidal solution ('soluble Prussian blue') and cannot be filtered.

$$FeCl_3 + K_4[Fe(CN)_6] \longrightarrow \underset{\text{(Soluble prussian blue)}}{KFe[Fe(CN)_6]} + 3KCl$$

(b) To the second part of the solution, add potassium thiocyanate (potassium sulphocyanide) solution. The appearance of a blood red colouration confirms the presence of Fe^{3+} ions.

$$Fe^{3+} + SCN^- \longrightarrow \underset{\text{Blood red colour}}{[Fe(SCN)]^{3+}}$$

(V) Analysis of group-IV cations

If group-III is absent, pass H_2S gas in the solution of group-III for a few minutes. If a precipitate appears (white, black or flesh coloured), this indicates the presence of group-IV cations. Table 15 gives a summary of confirmatory tests of group-IV cations.

ALLEN Session 2019-20

Table 15 : Confirmatory test for Group - IV cations

White precipitate (Zn^{2+})	Flesh coloured precipitate (Mn^{2+})	Black precipitate (Ni^{2+}, Co^{2+})
Dissolve the precipitate in dilute HCl by boiling. Divide the solution into two parts. (a) To the first part add sodium hydroxide solution. A white precipitate soluble in excess of sodium hydroxide solution confirms the presence of Zn^{2+} ions. (b) Neutralise the second part with a mmonium hydroxide solution and add potassium ferrocyanide solution. A bluish white precipitate appears	Dissolve the precipitate in dilute HCl by boiling, then add sodium hydroxide solution in excess. A white precipitate is formed which turns brown on keeping.	Dissolve the precipitate in aqua regia. Heat the solution to dryness and cool. Dissolve the residue in water and divide the solution into two parts. (a) To the first part of the solution add ammonium hydroxide solution till it becomes alkaline. Add a few drops of dimethyl glyoxime and shake the test tube. Formation of a bright red precipitate confirms the presence of Ni^{2+} ions. (b) Neutralise the second part with ammonium hydroxide solution. Acidify it with dilute acetic acid and add solid potassium nitrite. A yellow precipitate confirms the presence of Co^{2+} ions.

Chemistry of confirmatory tests of Group–IV cations

Fourth group cations are precipitated as their sulphides. Observe the colour of the precipitate. A white colour of the precipitate indicates the presence of zinc ions, a flesh colour indicates the presence of manganese ions and a black colour indicates the presence of Ni^{2+} or Co^{2+} ions.

1. Test for Zinc ion (Zn^{2+})

Zinc sulphide dissolves in hydrochloric acid to form zinc chloride.

$$ZnS + 2HCl \longrightarrow ZnCl_2 + H_2S$$

(a) On addition of sodium hydroxide solution it gives a white precipitate of zinc hydroxide, which is soluble in excess of NaOH solution on heating. This confirms the presence of Zn^{2+} ions.

$$ZnCl_2 + 2NaOH \longrightarrow Zn(OH)_2 + 2NaCl$$

$$Zn(OH)_2 + 2NaOH \longrightarrow \underset{\text{Sodium zincate}}{Na_2ZnO_2} + 2H_2O$$

(b) When potassium ferrocyanide $K_4[Fe(CN)_6]$ solution is added to the solution after neutralisation by NH_4OH solution, a white or a bluish white precipitate of zinc ferrocyanide appears.

$$2ZnCl_2 + K_4[Fe(CN)_6] \longrightarrow \underset{\text{Zinc ferrocyanide}}{Zn_2[Fe(CN)_6]} + 4KCl$$

2. Test for Manganese ion (Mn^{2+})

Manganese sulphide precipitate dissolves in dil. HCl on boiling. On addition of NaOH solution in excess, a white precipitate of manganese hydroxide is formed which turns brown due to atmospheric oxidation into hydrated manganese dioxide.

$$MnS + 2HCl \longrightarrow MnCl_2 + H_2S$$

$$MnCl_2 + 2NaOH \longrightarrow \underset{\text{(White precipitate)}}{Mn(OH)_2} + 2NaCl$$

$$Mn(OH)_2 + [O] \longrightarrow \underset{\substack{\text{Hydrated manganese dioxide} \\ \text{(Brown colour)}}}{MnO(OH)_2}$$

ALLEN Session 2019-20

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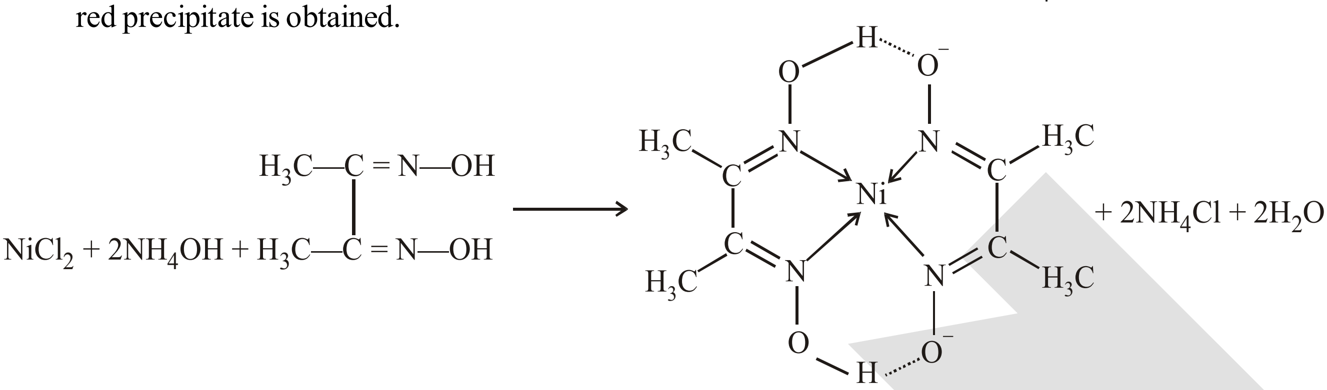
3. Test for Nickel ion (Ni^{2+})

The black precipitate of nickel sulphide dissolves in aqua regia and the reaction takes place as follows:

$$3NiS + 2HNO_3 + 6HCl \longrightarrow 3NiCl_2 + 2NO + 3S + 4H_2O$$

After treatment with aqua regia nickel-chloride is obtained which is soluble in water. When dimethyl glyoxime is added to the aqueous solution of nickel chloride, made alkaline, by adding NH_4OH solution, a brilliant red precipitate is obtained.



Complex of red colour
(Stable form of complex)

4. Test for Cobalt ion (Co^{2+})

Cobalt sulphide dissolves in aqua regia in the same manner as nickel sulphide. When the aqueous solution of the residue obtained after treatment with aqua regia is treated with a strong solution of potassium nitrite after neutralisation with ammonium hydroxide and the solution is acidified with dil. acetic acid, a yellow precipitate of the complex of cobalt named potassium hexanitritocobaltate (III) is formed.

$$CoS + HNO_3 + 3HCl \longrightarrow CoCl_2 + NOCl + S + 2H_2O$$

$$CoCl_2 + 7KNO_2 + 2CH_3COOH \longrightarrow \underset{\substack{\text{Potassium}\\ \text{hexanitritocobaltate(III)}\\ \text{(Yellow precipitate)}}}{K_3[Co(NO_2)_6]} + 2KCl + 2CH_3COOK + NO + H_2O$$

(VI) Analysis of Group–V cations

If group-IV is absent then take original solution and add a small amount of solid NH_4Cl and an excess of NH_4OH solution followed by solid ammonium carbonate $(NH_4)_2CO_3$. If a white precipitate appears, this indicates the presence of group–V cations.

Dissolve the white precipitate by boiling with dilute acetic acid and divide the solution into three parts one each for Ba^{2+}, Sr^{2+} and Ca^{2+} ions. Preserve a small amount of the precipitate for flame test. Summary of confirmatory tests is given in Table 16.

16 : Confirmatory test for Group - V cations

Dissolve the precipitate by boiling with dilute acetic acid and divide the solution into three parts one each for Ba^{2+}, Sr^{2+} and Ca^{2+} ions		
Ba^{2+} ions	**Sr^{2+} ions**	**Ca^{2+} ions**
(a) To the first part add potassium chromate solution. A yellow precipitate appears. **(b) Perform the flame test with the preserved precipitate. A grassy green flame is obtained.**	**(a) If barium is absent, take second part of the solution and add ammonium sulphate solution. Heat and scratch the sides of the test tube with a glass rod and cool. A white precipitate is formed.** **(b) Perform the flame test with the preserved precipitate. A crimson-red flame confirms the presence of Sr^{2+} ions.**	**(a) If both barium and strontium are absent, take the third part of the solution. Add ammonium oxalate solution and shake well. A white precipitate of calcium oxalate is obtained.** **(b) Perform the flame test with the preserved precipitate. A brick red flame, which looks greenish-yellow through blue glass, confirms the presence of Ca^{2+} ions.**

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ALLEN Session 2019-20

Chemistry of Confirmatory Tests of Group–V cations

The Group–V cations are precipitated as their carbonates which dissolve in acetic acid due to the formation of corresponding acetates.

1. Test for Barium ion (Ba^{2+})

(a) Potassium chromate (K_2CrO_4) solution gives a yellow precipitate of barium chromate when the solution of fifth group precipitate in acetic acid is treated with it.

$$BaCO_3 + 2CH_3COOH \longrightarrow (CH_3COO)_2Ba + H_2O + CO_2$$

$$(CH_3COO)_2Ba + K_2CrO_4 \longrightarrow \underset{\substack{\text{Barium chromate} \\ \text{(yellow precipitate)}}}{BaCrO_4} + 2CH_3COOK$$

(b) **Flame test :** Take a platinum wire and dip it in conc. HCl. Heat it strongly until the wire does not impart any colour to the non-luminous flame. Now dip the wire in the paste of the (Group–V) precipitate in conc. HCl. Heat it in the flame. A grassy green colour of the flame confirms the presence of Ba^{2+} ions.

2. Test for Strontium ion (Sr^{2+})

(a) Solution of V group precipitate in acetic acid gives a white precipitate of strontium sulphate with ammonium sulphate solution on heating and scratching the sides of the test tube with a glass rod.

$$SrCO_3 + 2CH_3COOH \longrightarrow (CH_3COO)_2Sr + H_2O + CO_2$$

$$(CH_3COO)_2Sr + (NH_4)_2SO_4 \longrightarrow \underset{\substack{\text{Strontium} \\ \text{sulphate} \\ \text{(White precipitate)}}}{SrSO_4} + 2CH_3COONH_4$$

(b) **Flame test :** Perform the flame test as given in the case of Ba^{2+}. A crimson red flame confirms the presence of Sr^{2+} ions.

3. Test for Calcium ion (Ca^{2+})

(a) Solution of the fifth group precipitate in acetic acid gives a white precipitate with ammonium oxalate solution.

$$CaCO_3 + 2CH_3COOH \longrightarrow (CH_3COO)_2Ca + H_2O + CO_2$$

$$(CH_3COO)_2Ca + \underset{\substack{\text{Ammonium} \\ \text{oxalate}}}{(NH_4)_2C_2O_4} \longrightarrow \underset{\substack{\text{Calcium oxalate} \\ \text{(White precipitate)}}}{(COO)_2Ca} + 2CH_3COONH_4$$

(b) **Flame test :** Perform the flame test as mentioned above. Calcium imparts brick red colour to the flame which looks greenish-yellow through blue glass.

(VII) Analysis of Group–VI cations

If group–V is absent then perform the test for Mg^{2+} ions as given below.

Chemistry of Confirmatory Tests of Group–VI cations

Test for Magnesium ion (Mg^{2+})

(a) If group–V is absent then the solution may contain magnesium carbonate, which is soluble in water in the presence of ammonium salts because the equilibrium is shifted towards the right hand side.

$$NH_4^+ + CO_3^{2-} \longrightarrow NH_3 + HCO_3^-$$

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ALLEN Session 2019-20

The concentration of carbonate ions required to produce a precipitate is not attained. When disodium hydrogenphosphate solution is added and the inner walls of the test tube are scratched with a glass rod, a white crystalline precipitate of magnesium ammonium phosphate is formed which indicates the presence of Mg^{2+} ions.

$$Mg^{2+} + Na_2HPO_4 \longrightarrow \underset{\text{Magnesium ammonium phosphate (White precipitate)}}{Mg(NH_4)PO_4} + NH_4OH + 2Na^+ + H_2O$$

Note down the observations and the inferences of the qualitative analysis in tabular form as given in the specimen record

Precautions

(a) Always use an apron, an eye protector and hand gloves while working in the chemistry laboratory.

(b) Before using any reagent or a chemical, read the label on the bottle carefully. Never use unlabelled reagent.

(c) Do not mix chemicals and reagents unnecessarily. Never taste any chemical.

(d) Be careful in smelling chemicals or vapours.

Always fan the vapours gently towards your nose (Fig. 1.9).

(e) Never add sodium metal to water or throw it in the sink or dustbin.

(f) Always pour acid into water for dilution. Never add water to acid.

(g) Be careful while heating the test tube. The test tube should never point towards yourself or towards your neighbours while heating or adding a reagent. Fig. 1.9 : How to smell a gas

(h) Be careful while dealing with the explosive compounds, inflammable substances, poisonous gases, electric appliances, glass wares, flame and the hot substances.

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Advanced Batch\Chem\Sheet\Salt Analysis & Heating Effect\(I) Salt Analysis\Eng.p65

(i) Keep your working surroundings clean. Never throw papers and glass in the sink. Always use dustbin for this purpose.

(j) Always wash your hands after the completion of the laboratory work.

(k) Always use the reagents in minimum quantity. Use of reagents in excess, not only leads to wastage of chemicals but also causes damage to the environment.

Discussion Questions

(i) What is the difference between a qualitative and a quantitative analysis?

(ii) Can we use glass rod instead of platinum wire for performing the flame test? Explain your answer.

(iii) Why is platinum metal preferred to other metals for the flame test?

(iv) Name the anions detected with the help of dilute H_2SO_4?

(v) Why is dilute H_2SO_4 preferred over dilute HCl while testing anions?

(vi) Name the anions detected by conc. H_2SO_4.

(vii) How is sodium carbonate extract prepared ?

(viii) What is lime water and what happens on passing carbon dioxide gas through it?

(ix) Carbon dioxide gas and sulphur dioxide gas both turn lime water milky. How will you distinguish these two?

(x) How will you test the presence of carbonate ion?

(xi) What is the composition of dark brown ring which is formed at the junction of two layers in the ring test for nitrates?

(xii) Name the radical confirmed by sodium nitroprusside test.

(xiii) What is chromyl chloride test ? How do you justify that CrO_2Cl_2 is acidic in nature?

(xiv) Why do bromides and iodides not give tests similar to chromyl chloride test?

(xv) Describe the layer test for bromide and iodide ions.

(xvi) Why is silver nitrate solution stored in dark coloured bottles?

(xvii) How do you test the presence of sulphide ion?

(xviii) Why does iodine give a blue colour with starch solution?

(xix) What is Nessler's reagent?

(xx) Why is original solution for cations not prepared in conc. HNO_3 or H_2SO_4?

(xxi) Why cannot conc. HCl be used as a group reagent in place of dil. HCl for the precipitation of Ist group cations?

(xxii) How can one prevent the precipitation of Group–IV radicals, with the second group radicals?

(xxiii) Why is it essential to boil off H_2S gas before precipitation of radicals of group–III?

(xxiv) Why is heating with conc. nitric acid done before precipitation of group–III?

(xxv) Can we use ammonium sulphate instead of ammonium chloride in group–III?

(xxvi) Why is NH_4OH added before $(NH_4)_2CO_3$ solution while precipitating group–V cations?

(xxvii) Why do we sometimes get a white precipitate in group–VI even if the salt does not contain Mg^{2+} radical?

(xxviii) What is aqua regia?

(xxix) Name a cation, which is not obtained from a metal.

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Advanced Batch\Chem\Sheet\Salt Analysis & Heating Effect\(I) Salt Analysis\Eng.p65

(xxx) How can you test the presence of ammonium ion?

(xxxi) Why are the group–V radicals tested in the order Ba^{2+}, Sr^{2+} and Ca^{2+} ?

(xxxii) Why does conc. HNO_3 kept in a bottle turn yellow in colour?

(xxxiii) Why should the solution be concentrated before proceeding to group–V?

(xxxiv) Why is the reagent bottle containing sodium hydroxide solution never stoppered?

(xxxv) What do you understand by the term common ion effect?

(xxxvi) Why is zinc sulphide not precipitated in group–II?

SPECIMEN RECORD OF SALT ANALYSIS

Aim

To analyse the given salt for one anion and one cation present in it.

S. No.	Experiment	Observation	Inference
1.	**Noted the colour of the given salt.**	**White**	**Cu^{2+}, Fe^{2+}, Ni^{2+}, Co^{2+}, Mn^{2+} are absent.**
2.	**Noted the smell of the salt.**	**No specific smell.**	**S^{2-}, SO_3^{2-}, CH_3COO^- may be absent.**
3.	**Heated 0.5 g of the salt in a dry test tube and noted the colour of the gas evolved and change in the colour of the residue on heating and cooling.**	**(i) No gas was evolved. (ii) No particular change in colour of the residue is observed when heated and when cooled.**	**(i) CO_3^{2-} may be present, NO_3^-, NO_2^-, Br^- may be absent. (ii) Zn^{2+} may be absent.**
4.	**Prepared a paste of the salt with conc. HCl and performed the flame test.**	**No distinct colour of the flame seen.**	**Ca^{2+}, Sr^{2+}, Ba^{2+} Cu^{2+} may be absent.**
5.	**Borax bead test was not performed as the salt was white in colour.**	—	—
6.	**Treated 0.1 g of salt with 1 mL dil.H_2SO_4 and warmed.**	**No effervescence and evolution of vapours.**	**CO_3^{2-}, SO_3^{2-}, S^{2-}, NO_2^-, CH_3COO^- absent.**
7.	**Heated 0.1 g of salt with 1 mL conc. H_2SO_4.**	**No gas evolved.**	**Cl^-, Br^-, I^-, NO_3^-, $C_2O_4^{2-}$ are absent.**
8.	**Acidified 1mL of aqueous salt solution with conc. HNO_3. Warmed the contents and then added 4-5 drops of ammonium molybdate solution.**	**No yellow precipitate**	**PO_4^{3-} absent.**

Sl. No.	Experiment	Observation	Inference
9.	Acidified water extract of the salt with dil. HCl and then added 2mL of $BaCl_2$ solution.	A white ppt. is obtained which is insoluble in conc. HNO_3 and conc. HCl.	SO_4^{2-} present.
10.	Heated 0.1 g of salt with 2 mL NaOH solution.	Ammonia gas is not evolved.	NH_4^+ absent.
11.	Attempted to prepare original solution of the salt by dissolving 1g of it in 20 mL water.	Clear solution formed	Water soluble salt is present.
12.	To a small part of the above salt solution added 2 mL of dil. HCl.	No white precipitate formed.	Group–I absent.
13.	Passed H_2S gas through one portion of the solution of step 12.	No precipitate formed.	Group–II absent.
14.	Since salt is white, heating with conc. HNO_3 is not required. Added about 0.2 g of solid ammonium chloride and then added excess of ammonium hydroxide to the solution of step 12.	No precipitate formed.	Group–III absent.
15.	Passed H_2S gas through the above solution.	No precipitate formed.	Group–IV absent.
16.	Added excess of ammonium hydroxide solution to the original solution and then added 0.5 g of ammonium carbonate.	No precipitate formed.	Group–V absent.
17.	To the original solution of salt added ammonium hydroxide solution, followed by disodium hydrogen phosphate solution. Heated and scratched the sides of the test tube.	White precipitate	Mg^{2+} confirmed.

Result

the given salt contains

Anion : SO_4^{2-}

Cation : Mg^{2+}

ALLEN Session 2019-20

EXERCISE # I

ANIONS : Class A (Subgroup - I)

1. The colour developed, when sodium sulphide is added to sodium nitroprusside is:
(A) Purple (B) yellow (C) red (D) black

2. When a neutral or slightly alkaline solution of thiosulphate is treated with the $[Ni(en)_3](NO_3)_2$ complex, then
(A) Green precipitate is obtained (B) Brown precipitate is obtained
(C) Violet precipitate is obtained (D) Yellow precipitate is obtained

3. When CH_3COONa heated with solid As_2O_3 then compound X is formed. The smell of compound X is -
(A) Pungent smell (B) Rotten Fish smell (C) Nauseating smell (D) Rotten egg smell

4. NO_2^- ion can be destroyed by -
(A) Sulphamic acid (B) Thiourea (C) Urea (D) All of these

5. Solutions of sodium azide(NaN_3) and iodine (as KI_3) do not react but on addition of a trace of 'X' ion, which acts as a catalyst there is an immediate vigorous evolution of nitrogen. Then 'X' may be:
(A) $S_2O_3^{2-}$ (B) S^{2-} (C) $\overset{\ominus}{S}CN$ (D) All are correct.

6. When $AgNO_3$ react with 'X' ion then initially no visible change occurs due to formation of water soluble complex.Then ion 'X' may be:
(A) SO_3^{2-} (B) $S_2O_3^{2-}$ (C) S^{2-} (D) CO_3^{2-}

7. Match the column

Column-I	**Column-II**
(A) S^{2-}	(P) Produces white ppt. with excess $AgNO_3$
(B) HSO_3^-	(Q) Evolves gas with dil. HCl which turns lime water milky
(C) SO_3^{2-}	(R) Evolves gas with dil.H_2SO_4 which does **not** turn Baryta water milky
(D) $S_2O_3^{2-}$	(S) Produces ppt. with $Pb(OAc)_2$ solution.
	(T) Produces white ppt with $BaCl_2$ solution.

8. Find the number of acidic radical(s) which can form coloured gas when treated with **dil.** H_2SO_4.
CO_3^{2-}, NO_2^-, Br^-, I^-, SO_3^{2-}

Class A (Subgroup - II)

9. Chromyl chloride test is given by -
(A) CH_3Cl (B) AgCl (C) Hg_2Cl_2 (D) NH_4Cl

10. $$BO_3^{3-} + \underset{\text{Conc.}}{H_2SO_4} \xrightarrow{\Delta} \underset{\text{White fumes}}{(P)}$$

$$BO_3^{3-} + \underset{\text{Conc.}}{H_2SO_4} + C_2H_5OH \xrightarrow{\Delta} \underset{\text{Vapours}}{(Q)}$$

P & Q are respectively -
(A) H_3BO_3 , H_3BO_3 (B) $(C_2H_5)_3BO_3$, H_3BO_3
(C) $(C_2H_5)_3BO_3$, $(C_2H_5)_3BO_3$ (D) H_3BO_3, $(C_2H_5)_3BO_3$

11. In layer test of I^- and Br^-. If reddish -brown layer comes first then -
(A) Br^- present (B) I^- absent (C) Both (A) and (B) (D) None of these

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Advanced Batch\Chem\Sheet\Salt Analysis & Heating Effect\(I) Salt Analysis\Eng.p65
ALLEN Session 2019-20

All Anions Of Class A

12. **Statement-1** : When H_2S gas is passed through Na-nitroprusside soluton it gives purple colouration

Statement-2 : H_2S is an weak acid

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

13. When the soda extract containing thiosulphate ion treated with excess of $AgNO_3$ solution followed by boiling, then.

(A) White precipitate is formed (B) Black precipitate is formed

(C) brown precipitate is formed (D) No ppt precipitate is formed

14. "Cacodyl oxide" is formed in the specific test of -

(A) Formate (B) Oxalate (C) Acetate (D) Nitrate

15. An aqueous solution of gas (X) gives the white turbidity on passing H_2S in the solution. Identify (X)

(A) NH_3 (B) SO_2 (C) CO_2 (D) None of these

16. NO_2^- and NO_3^- can be distinguished by which of the following reagent.

(A) dil. H_2SO_4 (B) conc. H_2SO_4

(C) Devarda's alloy + conc.NaOH (D) None of these

17. $[Fe(H_2O)_5NO]^{2+}$ is unstable because -

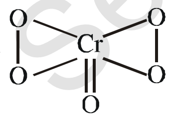
(A) It liberates NO gas on warming

(B) It liberates NO gas on shaking

(C) The charge of central atom is +1 (relatively low enough)

(D) None of these

Class B

18. $Cr_2O_7^{2-} + 4H_2O_2 + 2H^+ \xrightarrow[\text{Solvent}]{\text{Organic}}$ 2  $+ 5H_2O$

In above reaction amyl alcohol is recommended.

Dimethyl ether is not recommended for general use owing to its -

(A) Highly non-flammable character (B) Highly inflammable character

(C) Highly poisonious character (D) None of these

19. If barium sulphate is precipitated in a solution containing potassium permanganate it is coloured pink (violet) by -

(A) Absorption of some of the permanganate (B) Adsorption of some of the permanganate

(C) Both (A) and (B) (D) None of these

ALLEN Session 2019-20

All Anions Of Class A & Class B

20. **List–I (Reaction)** / **List–II (Product)**

(P) $KI + NO_2^- \longrightarrow$ — (1) NH_3

(Q) $NH_4NO_3 \xrightarrow{\Delta}$ — (2) NO

(R) $NO_2^- \xrightarrow{Zn+NaOH}$ — (3) N_2

(S) $(NH_4)_2Cr_2O_7 \xrightarrow{\Delta}$ — (4) N_2O

Code :

	P	Q	R	S		P	Q	R	S
(A)	3	4	1	2	(B)	4	2	1	3
(C)	4	2	3	1	(D)	2	4	1	3

21. **List–I (Reaction)** / **List–II (Product)**

(P) $CO_3^{2-} \xrightarrow{dil.H_2SO_4}$ — (1) H_2S

(Q) $S^{2-} \xrightarrow{dil.H_2SO_4}$ — (2) SO_2

(R) $SO_3^{2-} \xrightarrow{dil.H_2SO_4}$ — (3) CO_2

(S) $S_2O_3^{2-} \xrightarrow{dil.H_2SO_4}$ — (4) $S + SO_2$

Code :

	P	Q	R	S		P	Q	R	S
(A)	3	4	1	2	(B)	2	1	4	3
(C)	3	1	2	4	(D)	2	4	1	3

22. **List–I (Molecule)** / **List–II (Characteristic Odour)**

(P) CO_2 — (1) Rotten egg smell

(Q) SO_2 — (2) Suffocating smell of burning sulphur

(R) H_2S — (3) Vineger like smell

(S) CH_3COOH — (4) Odour less

Code :

	P	Q	R	S		P	Q	R	S
(A)	4	2	1	3	(B)	2	4	1	3
(C)	3	1	2	4	(D)	2	4	1	3

23. **List–I (Acidic radicals)** / **List–II (Test)**

(P) NO_2^- — (1) Green flame test

(Q) BO_3^{3-} — (2) Cacodyl oxide reaction

(R) Br^- — (3) Griess Ilosvay test

(S) CH_3COO^- — (4) Layer test

Code :

	P	Q	R	S		P	Q	R	S
(A)	4	2	1	3	(B)	3	1	4	2
(C)	3	4	2	1	(D)	4	3	2	1

ALLEN Session 2019-20

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CATIONS : DRY TEST

24. Find the number of water of crystallization in microcosmic salt -

(A) 5 (B) 4 (C) 6 (D) 10

25. What is the colour of K^+ through cobalt/double blue glass -

(A) Lilac, (B) Violet (C) Brick red (D) Crimson red

26. What is the colour of $CoO.Al_2O_3$ is -

(A) pink (B) Thenard blue (C) Bluish white (D) None of these

27. The correct formula of Canary yellow ppt and it is the test of ------------ acid radical-

(A) $(NH_4)_2$ $[PMo_{12}O_{40}]$ and phosphate (B) (NH_4) H $[P(Mo_3O_{10})_4]$ and sulphate

(C) $(NH_4)_3$ $[P(Mo_3O_{10})_4]$ and phosphate (D) Na_3 $[P(Mo_3O_{10})_4]$ and phosphate

28. Sodium carbonate bead test generally used for ....compounds.

(A) Mn (B) Cr (C) Zn (D) Cu

WET TEST : GROUP ZERO

29. **Statement-1 :** Test of NH_4^+ can not be done within group analysis

Statement-2 : During group analysis several times NH_4^+ - compound is added at the different steps.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1

(C) Statement-1 is false, statement-2 is true.

(D) Statement-1 is true, statement-2 is false.

GROUP - I

30. Which of the following is not group-I cation though the chlorides of all cations are sparingly soluble in water.

(A) Ag^+ (B) Hg_2^{2+} (C) Cu^+ (D) Pb^{2+}

GROUP - II A

31. Cu^{2+} + KCN (in excess) $\rightarrow$ soluble complex (X). How many statements are correct regarding complex (X) -

(i) the central atom has the co-ordination number of 6

(ii) the central atom has the co-ordination number of 4

(iii) the complex is sq.planar

(iv) the complex is diamagnetic

(v) the complex is paramagnetic

32. $BiCl_3 \xrightarrow{KI}$ black ppt (M) $\xrightarrow{\text{excess KI}}$ soluble complex (N)

Find the number of moles of I^- ions involved for the formation of per mole of (N).

GROUP - II B

33. Sn^{2+} and Sn^{4+} can be distinguished by how many of the following methods -

(i) by passing H_2S in their solution (in acidic medium)

(ii) by addition of NaOH in their solution

(iii) by addition of excess NaOH in their solution

(iv) by addition of dil. HCl in their solution

(v) by addition of $HgCl_2$ solution in their solution

GROUP - III

34. What is the group-III reagent is generally used for group analysis.
(A) $NH_4OH + NH_4NO_3$
(B) $NH_4Cl + (NH_4)_2CO_3$
(C) $NH_4OH + (NH_4)_2SO_4$
(D) $NH_4OH + NH_4Cl$

35. $CrCl_3$ solution + Na_2S solution $\longrightarrow$ ppt(A)
The correct formula and colour of A are
(A) Cr_2S_3, Black
(B) $Cr(OH)_3$, Green
(C) $Na[Cr(OH)_4]$, Green
(D) None of these

GROUP - IV

36. The auxiliary reagent in group-IV reagent is
(A) H_2S (B) dil.HCl (C) NaOH (D) NH_4OH

All Group Cations

37. Which of the following cation gives ppt in two groups during group analysis.
(A) Hg^{2+} (B) Hg_2^{2+} (C) Pb^{2+} (D) Cu^{2+}

38. Which of the following cation produces coloured ppt with Na_2SO_4 solution -
(A) Pb^{2+} solution (B) Ba^{2+} solution (C) Hg^{2+} solution (D) Ca^{2+} solution

39. NH_4^+ and K^+ ions can be distinguished by the use of following reagent
(A) $Na_3[Co(NO_2)_6]$
(B) $Na_2[PtCl_6]$
(C) $HClO_4$ or $NaClO_4$
(D) Boiling with NaOH

40. Which of the following sulphides is yellow in colour?
(A) CuS (B) CdS (C) ZnS (D) CoS

MISCELLANEOUS

41.

List–I (Compound)	List–II (Colour)
(P) HgO	(1) Purple solution
(Q) $BaCO_3$	(2) Yellow ppt
(R) $Na_4[Fe(CN)_5NOS]$	(3) Dark brown
(S) KI_3	(4) White ppt

Code :

	P	Q	R	S
(A)	3	4	1	2
(B)	2	4	1	3
(C)	2	4	3	1
(D)	2	4	3	1

42.

List–I (Basic Radical)	List–II (Group)
(P) Al^{+3}	(1) II group
(Q) Zn^{+2}	(2) V group
(R) Ba^{+2}	(3) IV group
(S) Pb^{+2}	(4) III group

Code :

	P	Q	R	S
(A)	4	2	1	3
(B)	2	4	1	3
(C)	3	1	2	4
(D)	4	3	2	1

43.

List–I (Cations)	List–II (Group reagent)
(P) Co^{+2}	(1) $(NH_4)_2CO_3$ in presence of NH_4Cl
(Q) Fe^{+3}	(2) H_2S gas in acidic medium
(R) Cu^{+2}	(3) H_2S in presence of NH_4OH
(S) Ca^{+2}	(4) NH_4OH in presence of NH_4Cl

Code :

	P	Q	R	S
(A)	4	2	1	3
(B)	3	1	4	2
(C)	1	2	3	4
(D)	3	4	2	1

ALLEN Session 2019-20

EXERCISE # II

ANIONS : Class A (Subgroup - I)

1. **Statement-1:** On passing CO_2 gas through lime water, the solution turns milky.
because
Statement-2: Acid-Base (neutralisation) reaction takes place.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

2. A substance on treatment with dil. H_2SO_4 liberates a colourless gas which produces (i) turbidity with baryta water and (ii) turns acidified dichromate solution green. The reaction indicates the presence of
(A) CO_3^{2-} (B) S^{2-} (C) SO_3^{2-} (D) NO_2^-

3. When $S_2O_3^{2-}$ react with solution of 'X' reagent then reaction is redox followed by precipitation then 'X' is:
(A) $FeCl_3$ solution (B) $AgNO_3$ solution
(C) $CuSO_4$ solution (D) None of these

4. In the test for iodine, when I_2 is treated with sodium thiosulphate, $Na_2S_2O_3$
$Na_2S_2O_3 + I_2 \longrightarrow NaI + ...$
(A) $Na_2S_4O_6$ (B) Na_2SO_4 (C) Na_2S (D) Na_3ISO_4

Paragraph for Q. 5 to Q. 7

$$\begin{tikzcd}
A \arrow[r, "\text{dil HCl}"] \arrow[d, "\text{boil}"', "\text{Sulphur}"] & B_{(g)} \arrow[r, "\text{lime}", "\text{water}"'] & C\ (\text{milky solution}) & \\
D \arrow[r, "\text{dil HCl}"'] & B_{(g)} \arrow[r, "Zn+\text{ dil.}H_2SO_4"] & E_{(g)} \arrow[r, "H_2O_2"'] & F \arrow[d, "\text{boil with (A)}"] \\
 & & G & D \arrow[l, "FeCl_3"']
\end{tikzcd}$$

5. Identify A
(A) CO_3^{2-} (B) SO_3^{2-} (C) $S_2O_3^{2-}$ (D) none of these

6. When A react with $Pb(NO_3)_2$ then compound X is formed. Compound X is oxidized by atmospheric oxygen on boiling, then Y is formed what is the colour of Y
(A) yellow (B) White (C) Black (D) Green

7. When gas E react with sodium nitroprusside in basic medium then compound Z is formed. The colour of compound Z is:
(A) Green (B) purple (C) Reddish brown (D) Black

ALLEN Session 2019-20

Class A (Subgroup - II)

8. When a mixture of solid NaCl, solid $K_2Cr_2O_7$ is heated with conc. H_2SO_4, orange red vapours are obtained. These are of the compound

(A) chromous chloride (B) chromyl chloride

(C) chromic chloride (D) chromic sulphate

9. Which of the following will not give positive chromyl chloride test?

(A) Copper chloride, $CuCl_2$ (B) Mercuric chloride, $HgCl_2$

(C) Zinc chloride, $ZnCl_2$ (D) Aniline chloride, $C_6H_5NH_3Cl$

10. Sodium borate on reaction with conc. H_2SO_4 and C_2H_5OH gives a compound A which burns with a green edged flame. The compound A is

(A) $H_2B_4O_7$ (B) $(C_2H_5)_2B_4O_7$ (C) H_3BO_3 (D) $(C_2H_5)_3BO_3$

11. Nitrate is confirmed by ring test. The brown colour of the ring is due to formation of

(A) ferrous nitrite (B) nitroso ferrous sulphate

(C) ferrous nitrate (D) $FeSO_4NO_2$

12. A salt gives violet vapours when treated with conc. H_2SO_4, it contains

(A) Cl^- (B) I^- (C) Br^- (D) NO_3^-

13. Unknown salt + Al-powder + NaOH (conc.) $\rightarrow$ gas comes out which turns Nesslar's reagent brown. The salt may be -

(A) $NaNO_2$ (B) $NaNO_3$ (C) NH_4Cl (D) NH_4HCO_3

Paragraph for Q. 14 to Q. 17

$$B(g) \xleftarrow[H_2SO_4]{\text{Conc.}} \underset{\text{(Mixture of two ions)}}{(A)} \xrightarrow{\text{(Sulphamic acid)}} \underset{\text{(Only one is present)}}{(C)} \xrightarrow[+\text{ conc. } H_2SO_4]{FeSO_4} (D)\ \text{(Unstable brown coloured compound)}$$

14. Identify mixture of ions (A) -

(A) NO_2^- and Br^- (B) NO_2^- and I^- (C) NO_2^- and NO_3^- (D) None of these

15. What is oxidation state of central atom of (D)

(A) +3 (B) +2 (C) +1 (D) Zero

16. Identify gas B-

(A) Br_2 (B) $Br_2 + NO_2$ (C) NO_2 (D) None of these

17. What is the hybridisation of central atom of D-

(A) d^2sp^3 (B) sp^3d^2 (C) sp^3d (D) sp^3

Session 2019-20
node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Advanced Batch\Chem\Sheet\Salt Analysis & Heating Effect\1) Salt Analysis\Eng.p65

Paragraph for Q. 18 to Q. 20

Unknown salt (M) + conc. H_2SO_4 + $K_2Cr_2O_7 \xrightarrow[\text{Heated}]{\text{Gently}}$ Reddish brown (X) vapour

Reddish brown vapour (X) $\xrightarrow{\text{Passed into NaOH solution}}$ Yellow solution (Y)

Yellow solution (Y) $\xrightarrow[CH_3COOH]{Pb(OAc)_2}$ Yellow ppt. (Z)

18. The salt (M) is/are-
(A) AgCl (B) NH_4Cl + NaBr (C) NaBr (D) $Ca(ClO_4)_2$

19. How many non axial d-orbitals are involved in hybridisation of central atom of compound (X)-
(A) 2 (B) 3 (C) 4 (D) None of these

20. What is the formula of yellow ppt (Z)-
(A) $BaCrO_4$ (B) Na_2CrO_4 (C) Ag_2CrO_4 (D) $PbCrO_4$

CATIONS : DRY TEST

21. In the borax bead test of Co^{2+}, the blue colour of bead is due to the formation of:
(A) B_2O_3 (B) Co_3B_2 (C) $Co(BO_2)_2$ (D) CoO

22. Which of the following leaves no residue on heating?
(A) $Pb(NO_3)_2$ (B) NH_4NO_3 (C) $Cu(NO_3)_2$ (D) $NaNO_3$

23. Which of the following cations is detected by the flame test?
(A) NH_4^+ (B) K^+ (C) Mg^{2+} (D) Al^{3+}

24. Which metal salt gives a violet coloured bead in the borax bead test (oxidising flame, cold)?
(A) Fe^{2+} (B) Ni^{2+} (C) Co^{2+} (D) Mn^{2+}

25. The compound formed in the borax bead test of Cu^{2+} ion in oxidising flame is:
(A) Cu (B) $CuBO_2$ (C) $Cu(BO_2)_2$ (D) None of these

26. In microcosmic salt bead test Co^{2+} produce blue bead due to the formation of -
(A) $Cu(BO_2)_2$ (B) $NaCoPO_4$ (C) $Co_2(PO_4)(BO_2)$ (D) $NaPO_3$

Paragraph for Q. 27 to Q. 30

$$\underset{\text{(Hydrated salt)}}{(A)} \xrightarrow[\text{heated}]{\text{Strongly}} \underbrace{B + C}_{\text{transparent glassy bead}}$$

$$H_3BO_3 \xrightarrow[\text{heated}]{\text{Strongly}} C + D$$

27. Identify C-
(A) $(BN)_X$ (B) $NaPO_3$ (C) B_2O_3 (D) $Mg(NH_4)PO_4$

28. Find the number of water of crystallizations in (A)-
(A) 4 (B) 5 (C) 10 (D) 24

29. How many X–O–X linkages are present in structure of A (X = central atom)-
(A) 4 (B) 3 (C) 5 (D) 2

30. Find the number of tetrahedral and trigonal planar units in structure of A -
(A) 2,1 (B) 2,2 (C) 2,4 (D) 5,2

ALLEN Session 2019-20

WET TEST : GROUP - I

31. Mercurous ion is represented as:

(A) Hg_2^{2+} (B) Hg^{2+} (C) $Hg + Hg^{2+}$ (D) Hg_2^{+}

32. A white sodium salt dissolves readily in water to give a solution which is neutral to litmus. When silver nitrate solution is added to the solution, a white precipitate is obtained which does not dissolve in dil. HNO_3. The anion could be:

(A) CO_3^{2-} (B) Cl^- (C) SO_4^{2-} (D) S^{2-}

33. An aqueous solution of a substance gives a white ppt. on treatment with dil. HCl, which dissolves on heating. When hydrogen sulphide is passed through the hot acidic solution, a black ppt. is obtained. The substance is a

(A) Hg^{2+} salt (B) Cu^{2+} salt (C) Ag^{+} salt (D) Pb^{2+} salt

34. A white ppt obtained in a analysis of a mixture becomes black on treatment with NH_4OH. It may be

(A) $PbCl_2$ (B) AgCl (C) $HgCl_2$ (D) Hg_2Cl_2

GROUP - II

35. When bismuth chloride is poured into a large volume of water the white precipitate produced is

(A) $Bi(OH)_3$ (B) Bi_2O_3 (C) BiOCl (D) Bi_2OCl_3

36. $CuSO_4$ decolourises on addition of excess KCN, the product is

(A) $[Cu(CN)_4]^{2-}$. (B) Cu^{2+} get reduced to form $[Cu(CN)_4]^{3-}$

(C) $Cu(CN)_2$ (D) CuCN

37. When H_2S gas is passed through the HCl containing aqueous solution of $CuCl_2$, $HgCl_2$, $BiCl_3$ and $CoCl_2$, it does not precipitate out:

(A) CuS (B) HgS (C) Bi_2S_3 (D) CoS

38. Which of the following is soluble in yellow ammonium sulphide?

(A) CuS (B) CdS (C) SnS (D) PbS

39. When excess of $SnCl_2$ is added to a solution of $HgCl_2$, a white ppt turning grey is obtained. The grey colour is due to the formation of

(A) Hg_2Cl_2 (B) $SnCl_4$ (C) Sn (D) Hg

40. On passing H_2S gas in II group sometimes the solution turns milky. It indicates the presence of

(A) oxidising agent (B) acidic salt (C) s-block cation (D) reducing agent.

41. Which fo the following yellow coloured sulphide is insouble in yellow ammonium sulphide.

(A) SnS_2 (B) As_2S_5 (C) CdS (D) Bi_2S_3

42. Type of sulphide ppt may be obtained in the group-II ppt during group analysis.

(A) M_2S_3 (B) M_2S (C) MS (D) MS_2

43. The metal ion which is precipitated when H_2S is passed with HCl:

(A) Zn^{2+} (B) Ni^{2+} (C) Cd^{2+} (D) Mn^{2+}

GROUP - III

44. In the precipitation of the iron group in qualitative analysis, ammonium chloride is added before adding ammonium hydroxide to

(A) decrease concentration of OH^- ions. (B) prevent interference by phosphate ions.

(C) increase concentration of Cl^- ions. (D) increase concentration of NH_4^+ ions.

ALLEN Session 2019-20

45. If reddish brown ppt (only) is obtained in group-III during group analysis, then oxidation state of Fe in the original sample may be

(A) +2 (B) +3 (C) +2 and +3 both (D) Neither +2 nor +3

46. If NH_4Cl is not added to the group-III reagent which of the following ppt could be obtained

(A) $Cr(OH)_2$ (B) $Fe(OH)_3$ (C) $Mn(OH)_2$ (D) $Mg(OH)_2$

47. In which of the following cases blue ppt is obtained

(A) $Fe^{2+} + [Fe(CN)_6]^{3-} \longrightarrow$ (B) $Fe^{2+} + [Fe(CN)_6]^{4-} \longrightarrow$

(C) $Fe^{3+} + [Fe(CN)_6]^{4-} \longrightarrow$ (D) $Fe^{3+} + [Fe(CN)_6]^{3-} \xrightarrow{SnCl_2}$

48. What are the following steps are to be done before adding group-III reagent into the group-II filtrate.

(A) Group-II filtrate is to be evapourated to dryness

(B) Group-II filtrate is to be boiled of first

(C) After boiling 2-3 drops of dil.H_2SO_4 is added and boiled again.

(D) After boiling 2-3 drops of HNO_3 is added and boiled again.

49. A pale green crystalline metal salt of M dissolves freely in water. On standing it gives a brown ppt on addition of aqueous NaOH. The metal salt solution also gives a black ppt on bubbling H_2S in basic medium. An aqueous solution of the metal salt decolourizes the pink colour of the permanganate solution. The metal in the metal salt solution is

(A) copper (B) aluminium (C) lead (D) iron

50. Which of the following compound on reaction with NaOH and Na_2O_2 gives yellow colour?

(A) $Cr(OH)_3$ (B) $Zn(OH)_2$ (C) $Al(OH)_3$ (D) None of these

GROUP - IV

51. Colour of nickel chloride solution is

(A) pink (B) black (C) colourless (D) green

52. Dimethyl glyoxime in a suitable solvent was refluxed for 10 minutes with pure pieces of nickel sheet, it will result in

(A) Red ppt (B) Blue ppt. (C) Yellow ppt. (D) No ppt.

53. Which one of the following does not produce metallic sulphide with H_2S?

(A) $ZnCl_2$(Neutral sol^n) (B) $CdCl_{2(aq)}$ (C) $CoCl_{2(aq)}$ (D) $CuCl_{2(aq)}$

54. Which is not dissolved by dil HCl?

(A) ZnS (B) MnS (C) $BaSO_3$ (D) $BaSO_4$

GROUP - V

55. In III group, NH_4Cl is added to decrease concentation of hydroxide ion by NH_4OH. We do not add $(NH_4)_2SO_4$ along with NH_4OH because -

(A) $(NH_4)_2SO_4$ is insoluble in water (B) It precipitate other insoluble sulphates

(C) It is weak electrolyte (D) None of these

GROUP - VI

56. A metal is burnt in air and the ash on moistening smells of ammonia. The metal is

(A) Na (B) Fe (C) Mg (D) Al

57. A metal 'X' on heating in nitrogen gas gives 'Y' . 'Y' on treatment with H_2O gives a colourless gas which when passed through $CuSO_4$ solution gives a blue colour Y is:

(A) $Mg(NO_3)_2$ (B) Mg_3N_2 (C) NH_3 (D) MgO

ALLEN Session 2019-20

MISCELLANEOUS

58. Na_2HPO_4 + Reagent 'M' $\rightarrow$ white ppt. The reagent 'M' is -
(A) $BaCl_2$ solution (B) $AlCl_3$ solution (C) $MnSO_4$ solution (D) $FeCl_3$ solution

59. A white solid is first heated with dil H_2SO_4 and then with conc. H_2SO_4. No action was observed in either case. The solid salt contains
(A) sulphide (B) sulphite (C) thiosulphate (D) sulphate

60. A mixture of chlorides of copper, cadmium, chromium, iron and aluminium was dissolved in water acidified with HCl and hydrogen sulphide gas was passed for sufficient time. It was filtered, boiled and a few drops of nitric acid were added while boiling. To this solution ammonium chloride and sodium hydroxide were added in excess and filtered. The filtrate shall give test for
(A) sodium and iron ion (B) sodium, chromium and aluminium ion
(C) aluminium and iron ion (D) sodium, iron, cadmium and aluminium ion

61. In Nessler's reagent, the ion present is:
(A) HgI^{2-} (B) HgI_4^{2-} (C) Hg^+ (D) Hg^{2+}

62. The cations present in slightly acidic solution are Fe^{3+}, Zn^{2+} and Cu^{2+}. The reagent which when added in excess to this solution would identify and separte Fe^{3+} in one step is:
(A) 2 M HCl (B) 6 M NH_3 (C) 6 M NaOH (D) H_2S gas

63. In the separation of Cu^{2+} and Cd^{2+} in 2^{nd} group qualitative analysis of cation, tetrammine copper (II) sulphate and tetrammine cadmium (II) sulphate react with KCN to form the corresponding cyano complexes. Which one of the following pairs of the complexes and their relative stability enables the separation of Cu^{2+} and Cd^{2+}?
(A) $K_3[Cu(CN)_4]$ more stable and $K_2[Cd(CN)_4]$ less stable.
(B) $K_2[Cu(CN)_4]$ less stable and $K_2[Cd(CN)_4]$ more stable.
(C) $K_2[Cu(CN)_4]$ more stable and $K_2[Cd(CN)_4]$ less stable.
(D) $K_3[Cu(CN)_4]$ less stable and $K_2[Cd(CN)_4]$ more stable.

64. Which one has the minimum solubility product?
(A) AgCl (B) $AlCl_3$ (C) $BaCl_2$ (D) NH_4Cl

65. Which of the following sulphate is insoluble in water?
(A) $CuSO_4$ (B) $CdSO_4$ (C) $PbSO_4$ (D) $Bi_2(SO_4)_3$

66. Which of the following gives blood red colour with KSCN?
(A) Cu^{2+} (B) Fe^{3+} (C) Al^{3+} (D) Zn^{2+}

67. Which one of the following metal sulphide has maximum solubility in water?
(A) HgS, $K_{sp} = 10^{-54}$ (B) CdS, $K_{sp} = 10^{-30}$
(C) FeS, $K_{sp} = 10^{-20}$ (D) ZnS, $K_{sp} = 10^{-22}$

68. Identify the correct order of solubility of Na_2S, CuS and ZnS in aqueous medium is:
(A) CuS > ZnS > Na_2S (B) ZnS > Na_2S > CuS
(C) Na_2S > CuS > ZnS (D) Na_2S > ZnS > CuS

ALLEN Session 2019-20

69. Match the column -

Column-I (Element)	Column-II (Correct characteristics)
(A) Ba	(P) cation in solution produces brick red ppt. with CrO_4^{2-}
(B) Pb	(Q) cation in solution produces yellow ppt .with CrO_4^{2-}
(C) Ag	(R) corresponding salt produces apple green colour in the flame test
(D) Ca	(S) corresponding salt produces brick red colour in the flame test
	(T) cation in solution produces no ppt. with CrO_4^{2-} ion

70.

Column-I Cation in solution	Column-I Correct characteristics when no where excess reagent is used
(A) Ag^+ and Pb^{2+}	(P) can be distinguished by Na_2HPO_4 solution
(B) Zn^{2+} and Mg^{2+}	(Q) can be distinguished by dil.HCl
(C) Pb^{2+} and Hg_2^{2+}	(R) can be distinguished by KI solution
(D) Ag^+ and Fe^{3+}	(S) can not be distinguished by NH_4OH solution

The following column 1, 2, 3 represent the various tests carried out for identification of various group basic radicals, using various reagents and nature of reaction/properties of products observed. Answer the questions that follow

Column-1 - Cations/Basic Radical
Column-2 - Excess Reagent used with cation
Column-3 - Nature of Reaction/Properties of product formed

Column - 1 Cations	Column - 2 Excess Reagent used with cation	Column - 3 Nature of Reaction/ Properties of product formed
(I) Cu^{2+}	(i) KI (< 6M)	(P) Reduction of cation occurs
(II) Fe^{3+}	(ii) $K_4[Fe(CN)_6]$	(Q) Coloured complex formation
(III) Pb^{2+}	(iii) KCN	(R) Precipitation occurs
(IV) Ni^{2+}	(iv) NH_4OH	(S) Diamagnetic & square planar complex formation

71. For a **group-II** basic radical, which is the only **INCORRECT** combination?
(A) (I), (i), (P) (B) (IV), (iii), (S) (C) (III), (iv), (R) (D) (III), (iii), (R)

72. For a **group-IV** basic radical, which is the only **CORRECT** combination?
(A) (I), (iv), (S) (B) (IV), (iii), (P) (C) (II), (iv), (Q) (D) (IV), (iv), (Q)

73. Which combination has a entirely different colour from others?
(A) (IV), (iv), (Q) (B) (I), (iv), (Q) (C) (II), (iii), (Q) (D) (II), (ii), (Q)

74. How many of the following gives green ppt.
(i) $CrCl_3 + NaOH \rightarrow$ (ii) $CrCl_3$ + excess $NaOH \rightarrow$
(iii) $NiCl_2$ + excess $NaOH \rightarrow$ (iv) $NiCl_2$ + excess $NH_4OH \rightarrow$
(v) $Hg_2^{2+} + KI \rightarrow$

75. Find the no. of cation which gives white ppt with $K_4[Fe(CN)_6]$
Sr^{2+} Ca^{2+}, Zn^{2+}, Fe^{3+} , Cu^{2+}

ALLEN Session 2019-20

EXERCISE # JEE MAINS

1. Which products are expected from the disproportionation of hypochlorous acid : **[AIEEE-2002]**
(1) $HClO_3$ and Cl_2O (2) $HClO_2$ and $HClO$ (3) HCl and Cl_2O (4) HCl and $HClO_3$

2. A metal M readily forms its sulphate MSO_4 which is water soluble. It forms oxide MO which becomes inert on heating. It forms insoluble hydroxide which is soluble in NaOH. The metal M is:- **[AIEEE-2002]**
(1) Mg (2) Ba (3) Ca (4) Be

3. Which statement is correct :- **[AIEEE-2003]**
(1) Fe^{3+} ions give deep green precipitate with $K_4[Fe(CN)_6]$
(2) On heating K^+, Ca^{2+} and HCO_3^- ions, we get a precipitate of $K_2[Ca(CO_3)_2]$
(3) Manganese salts give a violet borax bead test in reducing flame
(4) From a mixed precipitate of AgCl and AgI ammonia solution dissolves only AgCl

4. What would happen when a solution of potassium chromate is treated with an excess of dilute nitric acid - **[AIEEE-2003]**
(1) Cr^{3+} and $Cr_2O_7^{2-}$ are formed (2) $Cr_2O_7^{2-}$ and H_2O are formed
(3) $Cr_2O_7^{2-}$ is reduced to +3 state of Cr (4) $Cr_2O_7^{2-}$ is oxidised to +7 state of Cr

5. Ammonia forms the complex ion $[Cu(NH_3)_4]^{2+}$ with copper ions in alkaline solutions but not in acidic solution. What is the reason for it :- **[AIEEE-2003]**
(1) In acidic solutions hydration protects copper ions
(2) In acidic solutions protons coordinate with ammonia molecules forming NH_4^+ ions and NH_3 molecules are not available
(3) In alkaline solutions insoluble $Cu(OH)_2$ is precipitated which is soluble in excess of any alkali
(4) Copper hydroxide is an amphoteric substance

6. Excess of KI reacts with $CuSO_4$ solution and then $Na_2S_2O_3$ solution is added to it. Which of the statements is incorrect for this reaction : **[AIEEE-2004]**
(1) Evolved I_2 is reduced (2) CuI_2 is formed
(3) $Na_2S_2O_3$ is oxidised (4) Cu_2I_2 is formed

7. Calomel on reaction with NH_4OH gives **[AIEEE-2004]**
(1) $HgNH_2Cl$ (2) $NH_2–Hg–Hg–Cl$ (3) Hg_2O (4) HgO

8. One mole of magnesium nitride on reaction with excess of water gives :- **[AIEEE-2004]**
(1) Two mole of HNO_3 (2) Two mole of NH_3
(3) 1 mole of NH_3 (4) 1 mole of HNO_3

9. The products obtained on heating $LiNO_3$ will be :- **[AIEEE-2011]**
(1) $LiNO_2 + O_2$ (2) $Li_2O + NO_2 + O_2$
(3) $Li_3N + O_2$ (4) $Li_2O + NO + O_2$

10. What is the best description of the change that occurs when $Na_2O(s)$ is dissolved in water ? **[AIEEE-2011]**
(1) Oxidation number of sodium decreases
(2) Oxide ion accepts sharing in a pair of electrons
(3) Oxide ion donates a pair of electrons
(4) Oxidation number of oxygen increases

11. Which of the following on thermal-decomposition yields a basic as well as an acidic oxide ? **[AIEEE-2012]**
(1) NH_4NO_3 (2) $NaNO_3$ (3) $KClO_3$ (4) $CaCO_3$

12. The correct statement for the molecule, CsI_3, is: **[JEE(Main)-2014]**
(1) it contains Cs^{3+} and I^- ions (2) it contains Cs^+, I^- and lattice I_2 molecule
(3) it is a covalent molecule (4) it contains Cs^+ and I_3^- ions

13. A metal M on heating in nitrogen gas gives Y. Y on treatment with H_2O gives a colourless gas which when passed through $CuSO_4$ solution gives a blue colour, Y is :- **[JEE(Main)-2012 online_P-4]**

(1) NH_3 (2) MgO (3) Mg_3N_2 (4) $Mg(NO_3)_2$

14. Potassium dichromate when heated with concentrated sulphuric acid and a soluble chloride, gives brown - red vapours of: **[JEE(Main)-2013 online_P-1]**

(1) CrO_3 (2) Cr_2O_3 (3) $CrCl_3$ (4) CrO_2Cl_2

15. Sodium Carbonate cannot be used in place of $(NH_4)_2CO_3$ for the identification of Ca^{2+}, Ba^{2+} and Sr^{2+} ions (in group V) during mixture analysis because : **[JEE(Main)-2013 online_P-1]**

(1) Sodium ions will react with acid radicals (2) Concentration of CO_3^{2-} ions is very low

(3) Mg^{2+} ions will also be precipitated

(4) Na^+ ions will interfere with the detection of Ca^{2+}, Ba^{2+}, Sr^{2+} ions

16. Which of the following statements is incorrect? **[JEE(Main)-2013 online_P-2]**

(1) Fe^{2+} ion also gives blood red colour with SCN^- ion

(2) Cupric ion reacts with excess of ammonia solution to give deep blue colour of $[Cu(NH_3)_4]^{2+}$ ion.

(3) Fe^{3+} ion gives blood red colour with SCN^- ion.

(4) On passing H_2S into Na_2ZnO_2 solution, a white ppt of ZnS is formed.

17. Identify incorrect statement **[JEE(Main)-2013 online_P-3]**

(1) Copper (I) compounds are colourless except where colour results from charge transfer

(2) Copper (I) compounds are diamagnetic

(3) Cu_2S is black (4) Cu_2O is colourless

18. Which one of the following cannot function as an oxidising agent ? **[JEE(Main)-2013 online_P-4]**

(1) $NO_3^-(aq)$ (2) I^- (3) $Cr_2O_7^{2-}$ (4) $S_{(S)}$

19. Which of the following statements about Na_2O_2 is **not** correct ? **[JEE(Main)-2014 online_P-2]**

(1) Na_2O_2 oxidises Cr^{3+} to CrO_4^{2-} in acid medium (2) It is diamagnetic in nature

(3) It is the super oxide of sodium (4) It is a derivative of H_2O_2

20. Consider the following equilibrium **[JEE(Main)-2014 online_P-2]**

$$AgCl\downarrow + 2NH_3 \rightleftharpoons [Ag(NH_3)_2]^+ + Cl^-$$

White precipitate of AgCl appears on adding which of the following?

(1) NH_3 (2) Aqueous NaCl (3) Aqueous NH_4Cl (4) Aqueous HNO_3

21. Consider the reaction **[JEE(Main)-2014 online_P-4]**

$$H_2SO_{3(aq)} + Sn^{4+}_{(aq)} + H_2O_{(l)} \rightarrow Sn^{2+}_{(aq)} + HSO^-_{4(aq)} + 3H^+_{(aq)}$$

Which of the following statements is correct?

(1) H_2SO_3 is the reducing agent because it undergoes oxidation

(2) H_2SO_3 is the reducing agent because it undergoes reduction

(3) Sn^{4+} is the reducing agent because it undergoes oxidation

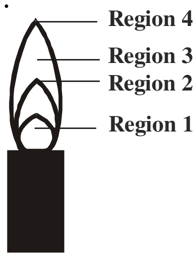
(4) Sn^{4+} is the oxidizing agent because it undergoes oxidation

22. Amongst the following, identify the species with an atom in +6 oxidation state : **[JEE(Main)-2014 online_P-4]**

(1) $[MnO_4]^-$ (2) $[Cr(CN)_6]^{3-}$ (3) Cr_2O_3 (4) CrO_2Cl_2

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Advanced Batch\Chem\Sheet\Salt Analysis & Heating Effect\(I) Salt Analysis\Eng.p65

23. The hottest region of Bunsen flame shown in the figure below is : **[JEE(Main)-2016]**



(1) region 4

(2) region 1

(3) region 2

(4) region 3

24. Sodium extract is heated with concentrated HNO_3 before testing for halogens because :
(1) S^{2-} and CN^{-}, if present are decomposed by conc. HNO_3 and hence do not interfere in the test.
(2) Ag reacts faster with halides in acidic medium **[JEE(Main)-2016 online]**
(3) Ag_2S and AgCN are soluble in acidic medium.
(4) Silver halides are totally insoluble in nitric acid.

25. In the following reactions, ZnO is respectively acting as a/an :
(a) $ZnO + Na_2O \rightarrow Na_2ZnO_2$ (b) $ZnO + CO_2 \rightarrow ZnCO_3$ **[JEE(Main)-2017 off line]**
(1) base and acid (2) base and base (3) acid and acid (4) acid and base

26. The products obtained when chlorine gas reacts with cold and dilute aqueous NaOH are :-
(1) ClO^{-} and ClO_3^{-} (2) ClO_2^{-} and ClO_3^{-} **[JEE(Main)-2017 off line]**
(3) Cl^{-} and ClO^{-} (4) Cl^{-} and ClO_2^{-}

27. Sodium salt of an organic acid 'X' produces effervescence with conc. H_2SO_4. 'X' reacts with the acidified aqueous $CaCl_2$ solution to give a white precipitate which decolourises acidic solution of $KMnO_4$. 'X' is :- **[JEE(Main)-2017 off line]**
(1) C_6H_5COONa (2) HCOONa (3) CH3COONa (4) $Na_2C_2O_4$

28. A solution containing a group-IV cation gives a precipitate on passing H_2S. A solution of this precipitate in dil. HCl produces a white precipitate with NaOH solution and bluish-white precipitate with basic potassium ferrocyanide. The cation is : **[JEE(Main)-2017 on line]**
(1) Mn^{2+} (2) Zn^{2+} (3) Ni^{2+} (4) Co^{2+}

29. Which of the following ions does **not** liberate hydrogen gas on reaction with dilute acids?
(1) Ti^{2+} (2) Cr^{2+} **[JEE(Main)-2017 on line]**
(3) Mn^{2+} (4) V^{2+}

30. When metal 'M' is treated with NaOH, a white gelatinous precipitate 'X' is obtained, which is soluble in excess of NaOH. Compound 'X' when heated strongly gives an oxide which is used in chromatography as an adsorbent. The metal 'M' is **[JEE(Main)-2018 off line]**
(1) Ca (2) Al (3) Fe (4) Zn

31. A white sodium salt dissolves readily in water to give a solution which is neutral to litmus. When silver nitrate solution is added to the aforementioned solution, a white precipitate is obtained which does not dissolve in dil. nitric acid. The anion is : **[JEE(Main)-2018 on line]**
(1) S^{2-} (2) SO_4^{2-} (3) CO_3^{2-} (4) Cl^{-}

32. The incorrect statement is :- **[JEE(Main)-2018 on line]**
(1) Ferric ion gives blood red colour with potassium thiocyanate.
(2) Cu^{2+} and Ni^{2+} ions give black precipitate with H_2S in presence of HCl solution.
(3) Cu^{2+} salts give red coloured borax bead test in reducing flame.
(4) Cu^{2+} ion gives chocolate coloured precipitate with potassium ferrocyanide solution.

33. When XO_2 is fused with an alkali metal hydroxide in presence of an oxidizing agent such as KNO_3; a dark green product is formed which disproportionates in acidic solution to afford a dark purple solution. X is : **[JEE(Main)-2018 on line]**
(1) Ti (2) Cr (3) V (4) Mn

ALLEN Session 2019-20
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EXERCISE # J-ADVANCED

1. Which of the following statement(s) is (are) correct with reference to the ferrous and ferric ions:

(A) Fe^{3+} gives brown colour with potassium ferricyanide **[JEE 1998]**

(B) Fe^{2+} gives blue precipitate with potassium ferricyanide

(C) Fe^{3+} give red colour with potassium thiocyanate

(D) Fe^{2+} gives brown colour with ammonium thiocyanate

2. Which of the following statement(s) is /are correct. When a mixture of NaCl and $K_2Cr_2O_7$ is gently warmed with conc. H_2SO_4? **[JEE 1998]**

(A) A deep red vapours is evolved.

(B) The vapours when passed into NaOH solution gives a yellow solution of Na_2CrO_4

(C) Chlorine gas is evolved

(D) Chromyl chloride is formed.

3. An aqueous solution of a substance gives a white precipitate on treatment with dilute hydrochloric acid, which dissolves on heating. When hydrogen sulphide is passed through the hot acidic solution, a black precipitate is obtained. The substance is a : **[JEE 2000]**

(A) Hg_2^{+} salt (B) Cr^{2+} salt (C) Ag^{+} salt (D) Pb^{2+} salt

4. A gas 'X' is passed through water to form a saturated solution. The aqueous solution on treatment with silver nitrate gives a white precipitate. The saturated aqueous solution also dissolves magnesium ribbon with evolution of a colourless gas 'Y'. Identify 'X' and 'Y': **[JEE 2002(Mains)]**

(A) $X = CO_2$, $Y = Cl_2$ (B) $X = Cl_2$, $Y = CO_2$

(C) $X = Cl_2$, $Y = H_2$ (D) $X = H_2$, $Y = Cl_2$

5. $[X] + H_2SO_4 \rightarrow [Y]$ a colourless gas with irritating smell **[JEE 2003]**

$[Y] + K_2Cr_2O_7 + H_2SO_4 \longrightarrow$ green solution

[X] and [Y] are:

(A) SO_3^{2-}, SO_2 (B) Cl^-, HCl (C) S^{2-}, H_2S (D) CO_3^{2-}, CO_2

6. A sodium salt of an unknown anion when treated with $MgCl_2$ give white precipitate only on boiling. The anion is: **[JEE 2004]**

(A) SO_4^{2-} (B) HCO_3^- (C) CO_3^{2-} (D) NO_3^-

7. $(NH_4)_2Cr_2O_7$ on heating gives a gas which is also given by: **[JEE 2004]**

(A) heating NH_4NO_2 (B) heating NH_4NO_3

(C) $Mg_3N_2 + H_2O$ (D) $NaNO_2 + H_2O_2$

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ALLEN Session 2019-20

8. A metal nitrate reacts with KI to give a black precipitate which on addition of excess of KI convert into orange colour solution. The cation of metal nitrate is: **[JEE 2005]**

(A) Hg^{2+} (B) Bi^{3+} (C) Pb^{2+} (D) Cu^{+}

9. A solution when diluted with H_2O and boiled, it gives a white precipitate. On addition of excess NH_4Cl / NH_4OH, the volume of precipitate decreases leaving behind a white gelatinous precipitate. Identify the precipitate which dissolves in NH_4OH / NH_4Cl. **[JEE 2006]**

(A) $Zn(OH)_2$ (B) $Al(OH)_3$ (C) $Mg(OH)_2$ (D) $Ca(OH)_2$

10. $CuSO_4$ decolourises on addition of excess KCN, the product is: **[JEE 2006]**

(A) $[Cu(CN)_4]^{2-}$ (B) Cu^{2+} get reduced to form $[Cu(CN)_4]^{3-}$

(C) $Cu(CN)_2$ (D) CuCN

11. Consider a titration of potassium dichromate solution with acidified Mohr's salt solution using diphenylamine as indicator. The number of moles of Mohr's salt required per mole of dichromate is: **[JEE 2007]**

(A) 3 (B) 4 (C) 5 (D) 6

12. The species present in solution when CO_2 is dissolved in water are **[JEE 2007]**

(A) $CO_2, H_2CO_3, HCO_3^-, CO_3^{2-}$ (B) H_2CO_3, CO_3^{2-}

(C) CO_3^{2-}, HCO_3^- (D) CO_2, H_2CO_3

13. Sodium fusion extract, obtained from aniline, on treatment with iron (II) sulphate and H_2SO_4 in presence of air gives a Prussian blue precipitate. The blue colour is due to the formation of : **[JEE 2007]**

(A) $Fe_4[Fe(CN)_6]_3$ (B) $Fe_3[Fe(CN)_6]_2$

(C) $Fe_4[Fe(CN)_6]_2$ (D) $Fe_3[Fe(CN)_6]_3$

14. **[JEE 2007]**

Column I	**Column II**
(A) $O_2^{-} \rightarrow O_2 + O_2^{2-}$	(P) Redox reaction
(B) $CrO_4^{2-} + H^+ \rightarrow$	(Q) One of the products has trigonal planar structure
(C) $MnO_4^{-} + NO_2^{-} + H^+ \rightarrow$	(R) Dimeric bridged tetrahedral metal ion
(D) $NO_3^{-} + H_2SO_4 + Fe^{2+} \rightarrow$	(S) Disproportionation

15. A solution of a metal ion when treated with KI gives a red precipitate which dissolves in excess KI to give a colourless solution. Moreover, the solution of metal ion on treatment with a solution of cobalt (II) thiocyanate gives rise to a deep blue crystalline precipitate. The metal ion is **[JEE 2007]**

(A) Pb^{2+} (B) Hg^{2+} (C) Cu^{2+} (D) Co^{2+}

ALLEN Session 2019-20

16. A solution of colourless salt H on boiling with excess NaOH produces a non-flammable gas. The gas evolution ceases after sometime. Upon addition of Zn dust to the same solution, the gas evolution restarts. The colourless salt(s) H is (are) **[JEE 2008]**

(A) NH_4NO_3 (B) NH_4NO_2 (C) NH_4Cl (D) $(NH_4)_2SO_4$

Paragraph for Question Nos. 17 to 19

p-Amino-N, N-dimethylaniline is added to a strongly acidic solution of **X**. The resulting solution is treated with a few drops of aqueous solution of **Y** to yield blue coloration due to the formation of methylene blue. Treatment of the aqueous solution of **Y** with the reagent potassium hexacyanoferrate(II) leads to the formation of an intense blue precipitate. The precipitate dissolves on excess addition of the reagent. Similarly, treatment of the solution of **Y** with the solution of potassium hexacyanoferrate(III) leads to a brown coloration due to the formation of **Z**. **[JEE 2009]**

17. The compound **X** is

(A) $NaNO_3$ (B) NaCl (C) Na_2SO_4 (D) Na_2S

18. The compound **Y** is

(A) $MgCl_2$ (B) $FeCl_2$ (C) $FeCl_3$ (D) $ZnCl_2$

19. The compound Z is

(A) $Mg_2[Fe(CN)_6]$ (B) $Fe[Fe(CN)_6]$ (C) $Fe_4[Fe(CN)_6]_3$ (D) $K_2Zn_3[Fe(CN)_6]_2$

20. Match each of the reactions given in Column I with the corresponding product(s) given in Column II.

[JEE 2009]

Column I	**Column II**
(A) Cu + dil. HNO_3	(P) NO
(B) Cu + conc. HNO_3	(Q) NO_2
(C) Zn + dil. HNO_3	(R) N_2O
(D) Zn + conc. HNO_3	(S) $Cu(NO_3)_2$
	(T) $Zn(NO_3)_2$

21. Passing H_2S gas into a mixture of Mn^{2+}, Ni^{2+}, Cu^{2+} and Hg^{2+} ions in an acidified aqueous solution precipitates **[JEE 2011]**

(A) CuS and HgS (B) MnS and CuS (C) MnS and NiS (D) NiS and HgS

22. Reduction of the metal centre in aqueous permanganate ion involves - **[JEE 2011]**

(A) 3 electrons in neutral medium (B) 5 electrons in neutral medium

(C) 3 electrons in weak alkaline medium (D) 5 electrons in acidic medium

23. The equilibrium **[JEE 2011]**

$$2Cu^{I} \rightleftharpoons Cu^{o} + Cu^{II}$$

in aqueous medium at 25°C shifts towards the left in the presence of

(A) NO_3^- (B) Cl^- (C) SCN^- (D) CN^-

ALLEN Session 2019-20

Paragraph for Questions Nos. 24 to 26

When a metal rod M is dipped into an aqueous colourless concentrated solution of compound N, the solution turns light blue. Addition of aqueous NaCl to the blue solution gives a white precipitate O. Addition of aqueous NH_3 dissolves O and gives are intense blue solution. **[JEE 2011]**

24. The metal rod M is -

(A) Fe (B) Cu (C) Ni (D) Co

25. The compound N is -

(A) $AgNO_3$ (B) $Zn(NO_3)_2$ (C) $Al(NO_3)_3$ (D) $Pb(NO_3)_2$

26. The final solution contains -

(A) $[Pb(NH_3)_4]^{2+}$ and $[CoCl_4]^{2-}$ (B) $[Al(NH_3)_4]^{3+}$ and $[Cu(NH_3)_4]^{2+}$

(C) $[Ag(NH_3)_2]^{+}$ and $[Cu(NH_3)_4]^{2+}$ (D) $[Ag(NH_3)_2]^{+}$ and $[Ni(NH_3)_6]^{2+}$

27. Which of the following hydrogen halides react(s) with $AgNO_3$(aq) to give a precipitate that dissolves in $Na_2S_2O_3$(aq): **[JEE 2012]**

(A) HCl (B) HF (C) HBr (D) HI

28. The reaction of white phosphorus with aqueous NaOH gives phosphine along with another phosphorus containing compound. The reaction type ; the oxidation states of phosphorus in phosphine and the other product are respectively **[JEE 2012]**

(A) redox reaction ; –3 and –5 (B) redox reaction ; +3 and +5

(C) disproportionation reaction ; –3 and +1 (D) disproportionation reaction ; –3 and +3

29. For the given aqueous reactions, which of the statement(s) is (are) true ? **[JEE 2012]**

$$\text{excess KI} + K_3[Fe(CN)_6] \xrightarrow{\text{dilute } H_2SO_4} \text{brownish-yellow solution}$$

$$\text{brownish-yellow solution} \xrightarrow{ZnSO_4} \text{white precipitate} + \underbrace{\text{brownish-yellow filtrate}}$$

$$\text{brownish-yellow filtrate} \xrightarrow{Na_2S_2O_3} \text{colourless solution}$$

(A) The first reaction is a redox reaction.

(B) White precipitate is $Zn_3[Fe(CN)_6]_2$.

(C) Addition of filtrate to starch solution gives blue colour.

(D) White precipitate is soluble in NaOH solution.

30. Upon treatment with ammonical H_2S , the metal ion that precipitates as a sulfide is -

(A) Fe(III) (B) Al(III) (C) Mg(II) (D) Zn (II) **[JEE 2013]**

ALLEN Session 2019-20

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Paragraph for Question 31 and 32

An aqueous solution of a mixture of two inorganic salts, when treated with dilute HCl, gave a precipitate **(P)** and a filtrate **(Q)**. The precipitate **(P)** was found to dissolve in hot water. The filtrate **(Q)** remained unchanged, when treated with H_2S in a dilute mineral acid medium. However, it gave a precipitate **(R)** with H_2S in an ammoniacal medium. The precipitate R gave a coloured solution **(S)**, when treated with H_2O_2 in an aqueous NaOH medium. **[JEE 2013]**

31. The coloured solution **(S)** contains

(A) $Fe_2(SO_4)_3$ (B) $CuSO_4$ (C) $ZnSO_4$ (D) Na_2CrO_4

32. The precipitate **(P)** contains

(A) Pb^{2+} (B) Hg_2^{2+} (C) Ag^+ (D) Hg^{2+}

33. Consider the following list of reagents : **[JEE Adv. 2014]**

Acidified $K_2Cr_2O_7$, alkaline $KMnO_4$, $CuSO_4$, H_2O_2 , Cl_2, O_3, $FeCl_3$, HNO_3 and $Na_2S_2O_3$.

The total number of reagents that can oxidise aqueous iodide to iodine is

34. Among PbS, CuS, HgS, MnS, Ag_2S, NiS, CoS, Bi_2S_3 , and SnS_2 the total number of **BLACK** coloured sulphides is **[JEE Adv. 2014]**

Paragraph for Q.No. 35 to 36

An aqueous solution of metal ion $\mathbf{M_1}$ reacts separately with reagents **Q** and **R** in excess to give tetrahedral and square planar complexes, respectively. An aqueous solution of another metal ion $\mathbf{M_2}$ always forms tetrahedral complexes with these reagents. Aqueous solution of $\mathbf{M_2}$ on reaction with reagent **S** gives white precipitate which dissolves in excess of **S**. The reactions are summarized in the scheme given below. **[JEE Adv. 2014]**

SCHEME:

$$\text{Tetrahedral} \xleftarrow[\text{excess}]{\mathbf{Q}} \mathbf{M_1} \xrightarrow[\text{excess}]{\mathbf{R}} \text{Square planar}$$

$$\text{Tetrahedral} \xleftarrow[\text{excess}]{\mathbf{Q}} \mathbf{M_2} \xrightarrow[\text{excess}]{\mathbf{R}} \text{Tetrahedral}$$

$$\mathbf{M_2} \xrightarrow{\mathbf{S}\text{, stoichiometric amount}} \text{White precipitate} \xrightarrow[\text{excess}]{\mathbf{S}} \text{precipitate dissolves}$$

35. $\mathbf{M_1}$, **Q** and **R**, respectively are

(A) Zn^{2+}, KCN and HCl (B) Ni^{2+}, HCl and KCN

(C) Cd^{2+} , KCN and HCl (D) Co^{2+}, HCl and KCN

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36. Reagent **S** is

(A) $K_4[Fe(CN)_6]$ (B) Na_2HPO_4 (C) K_2CrO_4 (D) KOH

37. Fe^{3+} is reduced to Fe^{2+} by using - **[JEE Adv. 2015]**

(A) H_2O_2 in presence of NaOH (B) Na_2O_2 in water

(C) H_2O_2 in presence of H_2SO_4 (D) Na_2O_2 in presence of H_2SO_4

38. The pair(s) of ions where BOTH the ions are precipitated upon passing H_2S gas in presence of dilute HCl, is(are) **[JEE Adv. 2015]**

(A) Ba^{2+}, Zn^{2+} (B) Bi^{3+}, Fe^{3+} (C) Cu^{2+}, Pb^{2+} (D) Hg^{2+}, Bi^{3+}

39. The reagent(s) that can selectively precipitate S^{2-} from a mixture of S^{2-} and SO_4^{2-} in aqueous soltuion is(are) : **[JEE(Adv.)-2016]**

(A) $CuCl_2$ (B) $BaCl_2$ (C) $Pb(OOCCH_3)_2$ (D) $Na_2[Fe(CN)_5NO]$

40. In the following reaction sequence in aqueous soluiton, the species **X**, **Y** and **Z** respectively, are - **[JEE(Adv.)-2016]**

$$S_2O_3^{2-} \xrightarrow{Ag^+} \underset{\text{clear solution}}{X} \xrightarrow{Ag^+} \underset{\text{white precipitate}}{Y} \xrightarrow{\text{With time}} \underset{\text{black precipitate}}{Z}$$

(A) $[Ag(S_2O_3)_2]^{3-}$, $Ag_2S_2O_3$, Ag_2S (B) $[Ag(S_2O_3)_3]^{5-}$, Ag_2SO_3, Ag_2S

(C) $[Ag(SO_3)_2]^{3-}$, $Ag_2S_2O_3$, Ag (D) $[Ag(SO_3)_3]^{3-}$, Ag_2SO_4, Ag

41. Which of the following combination will produce H_2 gas ? **[JEE(Adv.)-2017]**

(A) Zn metal and NaOH(aq) (B) Au metal and NaCN(aq) in the presence of air

(C) Cu metal and conc. HNO_3 (D) Fe metal and conc. HNO_3

42. Addition of excess aqueous ammonia to a pink coloured aqueous solution of $MCl_2 . 6H_2O$ (X) and NH_4Cl gives an octahedral complex Y in the presence of air. In aqueous solution, complex Y behaves as 1 : 3 electrolyte. The reaction of X with excess HCl at room temperature results in the formation of a blue coloured complex Z. The calculated spin only magnetic moment of X and Z is 3.87 B.M., whereas it is zero for complex Y. **JEE(Adv.)-2017]**

Among the following options, which statements is(are) correct ?

(A) The hybridization of the central metal ion in Y is d^2sp^3

(B) Z is tetrahedral complex

(C) Addition of silver nitrate to Y gives only two equivalents of silver chloride

(D) When X and Z are in equilibrium at 0°C, the colour of the solution is pink

43. The correct option(s) to distinguish nitrate salts of Mn^{2+} and Cu^{2+} taken separately is (are) :-

(A) Mn^{2+} shows the characteristic green colour in the flame test **JEE(Adv.)-2018]**

(B) Only Cu^{2+} shows the formation of precipitate by passing H_2S in acidic medium

(C) Only Mn^{2+} shows the formation of precipitate by passing H_2S in faintly basic medium

(D) Cu^{2+}/Cu has higher reduction potential than Mn^{2+}/Mn (measured under similar conditions)

44. The green colour produced in the borax bead test of a chromium(III) salt is due to-

JEE(Adv.)-2019]

(A) $Cr(BO_2)_3$ (B) CrB (C) $Cr_2(B_4O_7)_3$ (D) Cr_2O_3

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Advanced Batch\Chem\Sheet\Salt Analysis & Heating Effect\(I) Salt Analysis\Eng.p65

ANSWER-KEY

EXERCISE # I

1. (A) 2. (C) 3. (C) 4. (D) 5. (D) 6. (A,B)

7. (A) → R,S; (B) → Q; (C) → P,Q,S,T; (D) → P,Q,S,T 8. (1)

9. (D) 10. (D) 11. (C) 12. (D) 13. (B) 14. (C) 15. (B)

16. (A) 17. (C) 18. (B) 19. (B) 20. (D) 21. (C) 22. (A)

23. (B) 24. (B) 25. (D) 26. (B) 27. (C) 28. (A,B) 29. (A)

30. (C) 31. (2) 32. (4) 33. (2) 34. (D) 35. (B) 36. (D)

37. (C) 38. (C) 39. (C,D) 40. (B) 41. (B) 42. (D) 43. (D)

EXERCISE # II

1. (B) 2. (C) 3. (C) 4. (A) 5. (B) 6. (B) 7. (B)

8. (B) 9. (B) 10. (D) 11. (B) 12. (B) 13. (A,B,C,D) 14. (C)

15. (C) 16. (C) 17. (B) 18. (B) 19. (B) 20. (D) 21. (C)

22. (B) 23. (B) 24. (D) 25. (C) 26. (B) 27. (C) 28. (C)

29. (C) 30. (B) 31. (A) 32. (B) 33. (D) 34. (D) 35. (C)

36. (B) 37. (D) 38. (C) 39. (D) 40. (A) 41. (C) 42. (A,C,D)

43. (C) 44. (A) 45. (A,B,C) 46. (B,C,D) 47. (A,C,D) 48. (B,D) 49. (D)

50. (A) 51. (D) 52. (D) 53. (A,C) 54. (D) 55. (B) 56. (C)

57. (B) 58. (A,B) 59. (D) 60. (B) 61. (B) 62. (B) 63. (A)

64. (A) 65. (C) 66. (B) 67. (C) 68. (D)

69. (A) → Q , R; (B) → Q ; (C) → P ; (D) → S , T 70. (A) → P ; (B) → S ; (C) → R ; (D) → Q, R

71. (B) 72. (D) 73. (C) 74. (3) 75. (2)

EXERCISE # JEE MAINS

1. (4) 2. (4) 3. (4) 4. (2) 5. (2) 6. (2) 7. (1)

8. (2) 9. (2) 10. (3) 11. (4) 12. (4) 13. (3) 14. (4)

15. (3) 16. (1) 17. (4) 18. (2) 19. (3) 20. (4) 21. (1)

22. (4) 23. (3) 24. (1) 25. (4) 26. (3) 27. (4) 28. (2)

29. (3) 30. (2) 31. (4) 32. (2) 33. (4)

EXERCISE # J-ADVANCED

1. (A, B, C) 2. (A, B, D) 3. (D) 4. (C) 5. (A) 6. (B) 7. (A)

8. (B) 9. (A) 10. (B) 11. (D) 12. (A) 13. (A)

14. (A) → P , S ; (B) → R ; (C) → P , Q ; (D) → P 15. (B) 16. (A),(B) 17. (D)

18. (C) 19. (B) 20. (A) → P,S; (B) → Q,S; (C) → R,T; (D) → Q, T 21. (A)

22. (A,C,D) 23. (B,C,D) 24. (B) 25. (A) 26. (C) 27. (A,C,D) 28. (C)

29. (A,C,D) 30. (D) 31. (D) 32. (A) 33. (7) 34. (6) / (7) 35. (B)

36. (D) 37. (A, B) 38. (C,D) 39. (A OR A, C) 40. (A) 41. (A)

42. (A,B,D) 43. (B,D) 44. (A)

ALLEN Session 2019-20

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HEATING EFFECTS

1. HEATING EFFECT OF CARBONATE & BICARBONATE SALTS :

(a) Heating effect of carbonate salts :

Metal carbonate $\xrightarrow{\Delta}$ metal oxide + $CO_2\uparrow$

(i) $MCO_3 \xrightarrow{\Delta} MO + CO_2$ [M = Be, Mg, Ca, Sr, Ba]

(ii) $MgCO_3 \xrightarrow{\Delta} MgO + CO_2$

(iii) $\underbrace{Cu(OH)_2.CuCO_3}_{\text{Basic Cu(II) carbonate}} \xrightarrow{\Delta} \underset{\text{(black)}}{2CuO} + CO_2 + H_2O$

$\qquad \downarrow$ high temperature

$\underset{\text{(red)}}{Cu_2O} + O_2$

(iv) $\underset{\text{white}}{ZnCO_3} \xrightarrow{\Delta} \underset{\substack{\text{Yellow(hot)} \\ \text{white(cold)}}}{ZnO} + CO_2$

(v) $PbCO_3 \xrightarrow{\Delta} \underset{\text{Yellow}}{PbO} + CO_2$

(vi) $Li_2CO_3 \xrightarrow{\Delta} Li_2O + CO_2$

(vii) $\boxed{(NH_4)_2CO_3 \xrightarrow{\Delta} 2NH_3 + H_2O + CO_2 \quad \textbf{Very Important}}$

(viii) $\underset{\text{White lead}}{Pb(OH)_2.2PbCO_3}$

$\xrightarrow{350^\circ C} Pb_3O_4 + CO\uparrow + CO_2\uparrow + H_2O\uparrow$

$\xrightarrow[\text{Bunsen burner temperature}]{700\text{–}800^\circ C} 3PbO + 2CO_2\uparrow + H_2O\uparrow$

(ix) All carbonates **except** (Na, K, Rb, Cs) decompose on heating giving CO_2

(x) Carbonates salts of (Na, K, Rb, Cs) do not decompose on heating, they are melt on high temperature.

(xi) Oxides of heavier metals are less stable so further decompose into metal & oxygen

(xii) $\underset{\substack{\text{white} \\ \text{(yellowish white)}}}{Ag_2CO_3} \xrightarrow{\Delta} \underset{\text{Black}}{2Ag} + CO_2 + \frac{1}{2}O_2$

(xi) $HgCO_3 \xrightarrow{\Delta} Hg + \frac{1}{2}O_2\uparrow + CO_2$

Session 2019-20

(b) Heating effect of bicarbonate :

(i) Metal bicarbonate $\xrightarrow{\Delta}$ metal carbonate + $CO_2\uparrow$

except (Na, K, Rb, Cs)

$\downarrow$

metal oxide + CO_2

(ii) [General reaction $2HCO_3^- \xrightarrow{\Delta} CO_3^{2-} + H_2O + CO_2$]

(iii) All bicarbonates decompose to give carbonates and CO_2. eg.

(iv) $2NaHCO_3 \xrightarrow{\Delta} Na_2CO_3 + CO_2 + H_2O$

(v) $Mg(HCO_3)_2 \xrightarrow{\Delta} MgO + 2CO_2 + H_2O$

2. HEATING EFFECT OF HYDRATED SULPHATE SALTS :

(i) $\underset{\text{(M=Be,Zn,Mg,Ca,Cu,Pb)}}{\text{Metal Sulphate}} \xrightarrow{\Delta} \text{Metal Oxide} + SO_3$

$SO_3 \xrightarrow{T > 800^\circ C} SO_2 + \frac{1}{2}O_2$

(ii) $\underset{\text{Blue vitriol}}{CuSO_4.5H_2O} \xrightarrow[-4H_2O]{100^\circ C} \underset{\text{Bluish White}}{CuSO_4.H_2O} \xrightarrow[-H_2O]{220^\circ C} \underset{\text{White}}{CuSO_4} \xrightarrow{>800^\circ C} CuO + SO_2 + \frac{1}{2}O_2$

$CuO \xrightarrow{1100^\circ C} \frac{1}{2}Cu_2O + \frac{1}{2}O_2$

$CuSO_4 \xrightarrow{750^\circ C} \underset{\text{Black}}{CuO} + SO_3$

(iii) $\boxed{\underset{\text{Green vitriol}}{FeSO_4.7H_2O} \xrightarrow[-7H_2O]{300^\circ C} FeSO_4 \xrightarrow[\text{Strongly}]{\Delta} Fe_2O_3 + SO_2 + SO_3 \text{ (very important)}}$

(iv) $Fe_2(SO_4)_3 \xrightarrow{\Delta} Fe_2O_3 + 3SO_3$

(v) $\underset{\text{gypsum}}{CaSO_4.2H_2O} \xrightarrow{120^\circ C-150^\circ C} (CaSO_4.\frac{1}{2}H_2O) + 1\frac{1}{2}H_2O$

(Plaster of paris) or calcium sulphate hemihydrate

$(CaSO_4.\frac{1}{2}H_2O) \xrightarrow{1\frac{1}{2}H_2O} CaSO_4.2H_2O$ — another crystallographic form of gypsum hard & porous

$(CaSO_4.\frac{1}{2}H_2O) \xrightarrow{\Delta} \underset{\text{Dead burnt}}{CaSO_4} + \frac{1}{2}H_2O$

$CaSO_4 \xrightarrow{1100^\circ C} CaO + SO_3$

(vi) $ZnSO_4.7H_2O \xrightarrow[-H_2O]{70^\circ C} ZnSO_4.6H_2O \xrightarrow[-5H_2O]{70^\circ-200^\circ C} ZnSO_4.H_2O \xrightarrow[-H_2O]{>200^\circ C} ZnSO_4 \xrightarrow{\Delta >800^\circ C} ZnO + SO_2 + \frac{1}{2}O_2$

(vii) $\underset{\text{epsom salt}}{MgSO_4.7H_2O} \xrightarrow[-7H_2O]{\Delta} MgSO_4 \downarrow$ [Same as $ZnSO_4$]

(viii) $2NaHSO_3 \xrightarrow{\Delta} Na_2S_2O_5 + H_2O$

$Na_2S_2O_5 \xrightarrow{\Delta} Na_2SO_3 + SO_2$

$Na_2SO_3 \longrightarrow Na_2S + Na_2SO_4$

(ix) $2NaHSO_4 \xrightarrow{\Delta} Na_2SO_4 + H_2O + SO_3$

(x) $Na_2S_2O_3.5H_2O \xrightarrow{220^\circ C} Na_2S_2O_3 + 5H_2O$

$Na_2S_2O_3 \xrightarrow[\text{(From 4 moles)}]{\Delta} 3Na_2SO_4 + Na_2S_5.$

ALLEN Session 2019-20

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3. HEATING EFFECT OF NITRATE SALTS

(i) Metal nitrate $\xrightarrow{\Delta}$ metal oxide + NO_2 + O_2

(ii) $2M(NO_3)_2 \xrightarrow{\Delta} 2MO + 4NO_2 + O_2$
[M = all bivalent metals eg. Zn^{+2}, Mg^{+2}, Sr^{+2}, Ca^{+2}, Ba^{+2}, Cu^{+2}, Pb^{+2}]

(iii) $2LiNO_3 \xrightarrow{\Delta} Li_2O + 2NO_2 + \frac{1}{2}O_2$

(iv) $\underset{[M=Na,K,Rb,Cs]}{MNO_3} \xrightarrow{\Delta} MNO_2 + \frac{1}{2}O_2$

$\downarrow$ high temperature

$M_2O + N_2 + \frac{3}{2}O_2$

(v) $NaNO_3 \xrightarrow{500°C} NaNO_2 + \frac{1}{2}O_2$

(vi) $NaNO_2 \xrightarrow{800°C} Na_2O + N_2 + \frac{3}{2}O_2$

(vii) $BeCl_2 \xrightarrow{N_2O_4} Be(NO_3)_2.2N_2O_4 \xrightarrow[\text{under vaccum}]{\text{Warm to }50°C} Be(NO_3)_2$

$Be(NO_3)_2 \xrightarrow{700–800°C} BeO + 2NO_2 + 1/2\ O_2$

$Be(NO_3)_2 \xrightarrow{125°C} \underset{\text{basic beryllium nitrate}}{[Be_4O(NO_3)_6]}$

Exception : If formed oxide is of heavier metal then it being less stable and further decomposed in to metal and oxygen.

(viii) $Hg(NO_3)_2 \xrightarrow{\Delta} Hg + 2NO_2 + O_2$

(ix) $2AgNO_3 \xrightarrow{\Delta} 2Ag + 2NO_2 + O_2$

4. HEATING EFFECT OF AMMONIUM SALTS :

If anionic part is oxdising in nature, then N_2 will be the product (some times N_2O).

(i) $\underset{\text{(orange solid)}}{(NH_4)_2Cr_2O_7} \xrightarrow{\Delta} N_2 + \underset{\text{(green)}}{Cr_2O_3} + 4H_2O$ {This reaction is used for making artificial volcano}

(ii) $NH_4NO_2 \xrightarrow{\Delta} N_2 + 2H_2O$

(iii) $NH_4NO_3 \xrightarrow{\Delta} N_2O + 2H_2O$

(iv) $2NH_4ClO_4 \xrightarrow{\Delta} N_2 + Cl_2 + 2O_2 + 4H_2O$

(v) $2NH_4IO_3 \xrightarrow{\Delta} N_2 + I_2 + O_2 + 4H_2O$

If anionic part weakly oxidising or non oxidising in nature then NH_3 will be the product.

(i) $(NH_4)_2HPO_4 \xrightarrow{\Delta} HPO_3 + H_2O + 2NH_3$

(ii) $(NH_4)_2SO_4 \xrightarrow{\Delta} NH_3 + H_2SO_4$

(iii) $2(NH_4)_3PO_4 \xrightarrow{\Delta} 2NH_3 + P_2O_5 + 3H_2O$

(iv) $(NH_4)_2CO_3 \xrightarrow{\Delta} 2NH_3 + H_2O + CO_2$

(v) $NH_4Cl \xrightarrow{\Delta} NH_3\uparrow + HCl\uparrow$

(vi) $(NH_4)_2S \xrightarrow{\Delta} NH_3\uparrow + H_2S\uparrow$

(vii) $NH_4F \xrightarrow{\Delta} NH_3 + HF$

(viii) $(NH_4)_2MoO_4 \xrightarrow{\Delta} NH_3\uparrow + MoO_3 + H_2O$

ALLEN Session 2019-20

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5. HEATING EFFECT OF PHOSPHATE SALTS :

(i) 1° Phosphate salts gives metaphosphate salt on heating.

$$NaH_2PO_4 \xrightarrow{\Delta} H_2O + NaPO_3$$

(ii) 2° Phosphate salts gives pyrophosphate

$$Na_2HPO_4 \xrightarrow{\Delta} H_2O + Na_4P_2O_7$$

(iii) 3° Phosphate salt have no heating effect

$$Na_3PO_4 \xrightarrow{\Delta} \text{No effect}$$

(iv) $\underset{\text{microcosmic salt}}{Na(NH_4)HPO_4.4H_2O} \xrightarrow[-4H_2O]{\Delta} NaNH_4HPO_4 \xrightarrow{\text{High temp.}} NaPO_3 + NH_3 + H_2O$

(v) $2Mg(NH_4)PO_4 \xrightarrow{\Delta} Mg_2P_2O_7 + 2NH_3 + H_2O$

6. HEATING EFFECT OF HALIDES SALTS :

(i) $2FeCl_3 \xrightarrow{\Delta} 2FeCl_2 + Cl_2$

(ii) $AuCl_3 \xrightarrow{\Delta} AuCl + Cl_2$

(iii) $Hg_2Cl_2 \xrightarrow{\Delta} HgCl_2 + Hg$

(iv) $NH_4Cl \xrightarrow{\Delta} NH_3 + HCl$

(v) $Pb(SCN)_4 \xrightarrow{\Delta} Pb(SCN)_2 + (SCN)_2$

(vi) $PbCl_4 \xrightarrow{\Delta} PbCl_2 + Cl_2$

(vii) $PbBr_4 \xrightarrow{\Delta} PbBr_2 + Br_2$ **[PbI_4 does not exists]**

(viii) $\underset{\text{scarlet red}}{HgI_2} \underset{\text{On Rubbing}}{\overset{127°C}{\rightleftarrows}} \underset{\text{yellow}}{HgI_2}$

7. HEATING EFFECT OF HYDRATED CHLORIDE SALTS

(i) $MgCl_2 . 6H_2O \xrightarrow{\Delta} MgO + 2HCl + 5H_2O$

(ii) $2FeCl_3 . 6H_2O \xrightarrow{\Delta} Fe_2O_3 + 6HCl + 9H_2O$

(iii) $2AlCl_3 . 6H_2O \xrightarrow{\Delta} Al_2O_3 + 6HCl + 9H_2O$

(iv) $\underset{\text{Pink}}{CoCl_2.6H_2O} \xrightarrow[-2H_2O]{50°C} \underset{\text{blue}}{CoCl_2.4H_2O} \xrightarrow[-2H_2O]{58°C} \underset{\text{Red violet}}{CoCl_2.2H_2O} \xrightarrow[-2H_2O]{140°C} \underset{\text{Blue}}{CoCl_2}$

Hydrated Co^{2+} salt - Pink   Anhydrous Co^{2+} salt - Blue

ALLEN Session 2019-20
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8. HEATING EFFECT OF OXIDE :

(i) $2Ag_2O \xrightarrow[300°C]{\Delta} 4Ag + O_2$

(ii) $\underset{\text{white}}{ZnO} \underset{\text{Cold}}{\overset{\text{Hot}}{\rightleftharpoons}} \underset{\text{yellow}}{ZnO}$

(iii) $PbO_2 \xrightarrow{\Delta} PbO + \frac{1}{2}O_2$

(iv) $\underset{\text{yellow}}{PbO}$ (Massicot) $\xrightarrow[\text{Cooled \& powdered}]{\text{Fused}} \underset{\text{Litharge (red)}}{PbO}$

(v) $3MnO_2 \xrightarrow{900°C} Mn_3O_4 + O_2$

(vi) $\underset{\text{Red lead}}{Pb_3O_4} \underset{350°C}{\overset{500°C}{\rightleftharpoons}} \underset{\text{Litharge}}{6PbO} + O_2$

(vii) $2CrO_5 \xrightarrow{\Delta} Cr_2O_3 + \frac{7}{2}O_2$

(viii) $\underset{\text{(White)}}{K_2O} \underset{\text{cold}}{\overset{\Delta}{\rightleftharpoons}} \underset{\text{(yellow)}}{K_2O}$

(ix) $I_2O_5 \xrightarrow{\Delta} I_2 + \frac{5}{2}O_2$

(x) $\underset{\text{yellow}}{HgO} \xrightarrow{\Delta} \underset{\text{red}}{HgO} \xrightarrow[\text{strong heating}]{400°C} Hg + \frac{1}{2}O_2$

(xi) $2CrO_3 \xrightarrow{420°C} Cr_2O_3 + \frac{3}{2}O_2$

9. HEATING EFFECT OF PERMANGANATE :

$\underset{\text{dark puple (almost black)}}{2KMnO_4} \xrightarrow{513K} \underset{\text{(green)}}{K_2MnO_4} + \underset{\text{(black)}}{MnO_2} + O_2$

10. HEATING EFFECT OF DICHROMATE & CHROMATE SALTS :

$\underset{\text{orange}}{2K_2Cr_2O_7} \xrightarrow{\Delta} \underset{\text{yellow}}{2K_2CrO_4} + \underset{\text{green}}{Cr_2O_3} + \frac{3}{2}O_2\uparrow$

11. HEATING EFFECT OF ACIDS :

(i) $\underset{\text{conc.}}{2HNO_3} \xrightarrow{\Delta} H_2O + 2NO_2 + \frac{1}{2}O_2$

(ii) $H_2SO_4 \xrightarrow{444°C} H_2O + SO_3$

(iii) $H_2SO_4 \xrightarrow{>800°C} H_2O + SO_2 + \frac{1}{2}O_2$

$$\left.\begin{array}{l}
\text{(iv) } 3H_2SO_3 \longrightarrow 2H_2SO_4 + S\downarrow + H_2O \\
\text{(v) } 3HNO_2 \longrightarrow HNO_3 + 2NO + H_2O \\
\text{(vi) } HClO_3 \longrightarrow HClO_4 + ClO_2 + H_2O \\
\text{(vii) } 3HOCl \longrightarrow 2HCl + HClO_3 \\
\text{(viii) } 4H_3PO_3 \xrightarrow[200°C]{\Delta} 3H_3PO_4 + PH_3 \\
\text{(ix) } 2H_3PO_2 \longrightarrow H_3PO_4 + PH_3 \\
\text{(x) } 2NaH_2PO_2 \longrightarrow Na_2HPO_4 + PH_3
\end{array}\right\}\text{Undergoes dispropor-tionation reaction}$$

(xi) $H_2C_2O_4 \xrightarrow{\Delta} H_2O + CO + CO_2$

(xii) $H_3PO_4 \xrightarrow{220°C} H_4P_2O_7 \xrightarrow{320°C} 4HPO_3 \xrightarrow{>320°C} 2P_2O_5 + 2H_2O$

(xiii) $H_3BO_3 \xrightarrow{100°C} 4HBO_2 \xrightarrow{140°C} H_2B_4O_7 \xrightarrow[\text{hot}]{\text{Red}} 2B_2O_3 + H_2O$

12. HEATING EFFECTS OF ACETATE SALTS

$$\text{Metal acetate} \xrightarrow{\Delta} \underset{(M = \text{LiBe Mg})}{\text{Metal Carbonate}} + CH_3COCH_3$$

$$\text{Metal Carbonate} \xrightarrow{\Delta} \text{Metal oxide} + CO_2$$

(i) $Pb(OAc)_2 \xrightarrow{\Delta} PbO + CO_2 + CH_3COCH_3$

(ii) $Mg(OAc)_2 \xrightarrow{\Delta} MgO + CO_2 + CH_3COCH_3$

(iii) $Be(OAc)_2 \xrightarrow{\Delta} BeO + CO_2 + CH_3COCH_3$

(iv) $Ca(OAc)_2 \xrightarrow{\Delta} CaCO_3 + CH_3COCH_3$

(v) $Ba(OAc)_2 \xrightarrow{\Delta} BaCO_3 + CH_3COCH_3$

(vi) $CH_3CO_2K \xrightarrow{\Delta} K_2CO_3 + CH_3COCH_3$

13. HEATING EFFECTS OF OXALATE SALTS

$$\text{Metal oxalate} \xrightarrow{\Delta} \underset{(M = \text{LiBe Mg})}{\text{Metal Carbonate}} + CO + CO_2$$

$$\text{Metal Carbonate} \xrightarrow{\Delta} \text{Metal oxide} + CO_2$$

(i) $\boxed{7Na_2C_2O_4 \xrightarrow{\Delta} 7Na_2CO_3 + 2CO_2 + 3CO + 2C}$

(ii) $SnC_2O_4 \xrightarrow{\Delta} SnO + CO_2 + CO$

(iii) $FeC_2O_4 \xrightarrow{\Delta} FeO + CO + CO_2$

(iv) $Ag_2C_2O_4 \xrightarrow{\Delta} 2Ag + 2CO_2$

(v) $HgC_2O_4 \xrightarrow{\Delta} Hg + 2CO_2$

14. HEATING EFFECTS OF FORMATE SALTS

(i) $HCO_2Na \xrightarrow{350°C} Na_2C_2O_4 + H_2\uparrow$

(ii) $HCOOAg \xrightarrow{\Delta} HCOOH + 2Ag + \frac{1}{2}\underbrace{O_2 + CO}_{\downarrow \\ CO_2}$

(iii) $(HCOO)_2Hg \longrightarrow HCOOH + Hg + \frac{1}{2}\underbrace{O_2 + CO}_{\downarrow \\ CO_2}$

ALLEN Session 2019-20

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EXERCISE

Single correct

1. Which of the following does not give metal oxide on heating

(A) $NaCO_3$ (B) K_2CO_3 (C) Rb_2CO_3 (D) All of these

2. Which of the following metal bicarbonate will give metal oxide and CO_2 on heating

(A) $NaHCO_3$ (B) $Mg(HCO_3)_2$ (C) $KHCO_3$ (D) Rb_2CO_3

3. Which of the following metal nitrate will give metal and oxygen on heating :

(A) KNO_3 (B) $NaNO_3$ (C) $AgNO_3$ (D) $RbNO_3$

4. Which of the following nitrate will give N_2O on heating :

(A) NH_4NO_3 (B) NH_4NO_2 (C) $NaNO_3$ (D) $AgNO_3$

5. Which of the following ammonium salt will not give acid on heating :

(A) $(NH_4)_2HPO_4$ (B) $(NH_4)_2MoO_4$ (C) $(NH_4)_2SO_4$ (D) NH_4Cl

6. Which of the following halide will not give halogen gas on heating :

(A) $PbCl_4$ (B) $PbBr_4$ (C) Hg_2Cl_2 (D) All of these

7. Select the correct statements

(A) Hydrated Co^{+2} salt is pink (B) Anhydrous Co^{+2} salt is of blue colour

(C) Hybridisation of $CoCl_2.6H_2O$ is sp^3d^2 (D) All of these

8. Which of the following metal sulphate will give SO_2 and SO_3 both gaseous product on heating :

(A) $CuSO_4$ (B) $FeSO_4$ (C) $Fe_2(SO_4)_3$ (D) $CaSO_4$

9. Which of the following compound is called dead burnt plaster :

(A) $CaSO_4.\frac{1}{2}H_2O$ (B) $CaSO_4.2H_2O$ (C) $CaSO_4$ (anhy.) (D) None of these

10. When NaH_2PO_4 is heated then which of the following compound is formed :

(A) $Na_4P_2O_7$ (B) Na_3PO_4 (C) HPO_3 (D) $NaPO_3$

11. When $KMnO_4$ is heated then which of the following compound is formed :

(A) $K_2MnO_4 + MnO_2$ (B) $K_2MnO_4 + MnO$ (C) $MnO_2 + MnO$ (D) No change

12. When CrO_3 is heated then .... + ... are formed :

(A) Cr_2O_3, O_2 (B) CrO_2, O_2 (C) $Cr_2O_7^{-2}$, O_2 (D) None of these

More than one may be correct

13. Which of the following metal carbonate will give of metal and oxyen on heating-

(A) Ag_2CO_3 (B) $HgCO_3$ (C) $(NH_4)_2CO_3$ (D) $PbCO_3$

ALLEN Session 2019-20

14. $\underset{\substack{\text{(Carbonate salt)}\\ \text{(Green colour)}}}{A} \xrightarrow{\Delta} \underset{\text{(Black)}}{B} + CO_2 + H_2O$

$B \xrightarrow[\text{920°C}]{\text{High temperature}} (C) + (D)$

Select the correct statements -

(A) Compound (A) is basic copper carbonate

(B) Compound (B) CuO

(C) Compound (C) is Cu_2O

(D) Compound (D) is paramagnetic in naturue

15. When Ag_2CO_3 is heated then product will be -

(A) Ag_2O (B) Ag (C) O_2 (D) CO_2

16. When compound A (orange red) is heated then green colour oxide of (B) is formed and inert gas (C) is formed then select the correct statements :

(A) Compound (A) is $(NH_4)_2Cr_2O_7$

(B) Compound (B) is used in fire works

(C) Gas C is N_2

(D) Heating effect of (A) is a type of intra molecular redox reaction

17. Which of the following hydrated salts will not become anhydrous on heating :

(A) $MgCl_2.6H_2O$ (B) $FeCl_3.6H_2O$ (C) $AlCl_3.6H_2O$ (D) $CoCl_2.6H_2O$

18. Which of the following metal nitrate produce NO_2 on heating

(A) $Hg(NO_3)_2$ (B) $RbNO_3$ (C) $Pb(NO_3)_2$ (D) $Cu(NO_3)_2$

19. Which of the following oxides turns yellow on heating and becomes white on cooling :

(A) ZnO (B) K_2O (C) PbO (D) Ag_2O

Paragraph for Q. No. 20 to Q. No. 21

$\underset{\text{(Orange solid)}}{(A)} \xrightarrow{\Delta} \underset{\text{(yellow)}}{(B)} + \underset{\text{(green)}}{(C)} + \underset{\text{(Paramagnetic)}}{(D)}$

20. Compound (A) is :

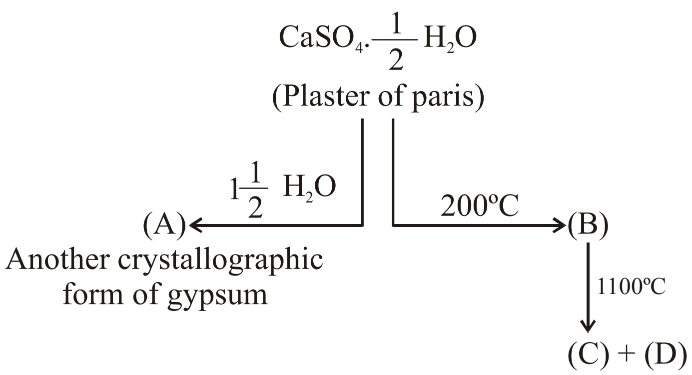
(A) $K_2Cr_2O_7$ (B) K_2CrO_4 (C) Cr_2O_3 (D) O_2

21. Compound (C) is also obtained on heating of :

(A) $(NH_4)_2Cr_2O_7$ (B) NH_4ClO_4 (C) NH_4NO_3 (D) None of these

ALLEN Session 2019-20

Paragarph for Q. No. 22 & 23



22. Compound "A" is :

(A) $CaSO_4.2H_2O$ (B) $2CaSO_4.H_2O$ (C) $CaSO_4.3H_2O$ (D) $CaSO_4.5H_2O$

23. Compound "C" and "D" are respectivelly :

(A) $CaO + CaSO_4$ (B) $CaSO_4 + SO_2$ (C) $CaSO_4 + SO_3$ (D) $CaO + SO_3$

Matrix match

24. Match the column

Column-I	**Column-II**
(A) $Be(NO_3)_2 \xrightarrow{\Delta}$	(P) Gives H_2O
(B) $HNO_2 \xrightarrow{\Delta}$	(Q) Oxyacid is obtained
(C) $NaH_2PO_4 \xrightarrow{\Delta}$	(R) Gives disproportionation reaction
(D) $H_3PO_3 \xrightarrow{\Delta}$	(S) Oxygen gas is evolved

25.

Column-I (Compound)	**Column-II (Products on heating)**
(A) NH_4ClO_4	(P) CO_2 gas is evolved
(B) $Mg(OAc)_2$	(Q) H_2 gas is evolved
(C) HCOONa (above 350ºC)	(R) N_2 gas is evolved
(D) $(NH_4)_2CO_3$	(S) Same gas is evolved which is obtained by heating $(NH_4)_2SO_4$
	(T) Intra molecular redox reaction

ALLEN Session 2019-20

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Integer

26. When calamine is heated then a product (A) is formed then find the total number of following options are correct for compound (A) -

(i) Compound (A) is white in cold conditions

(ii) Compound (A) is yellow in hot conditions

(iii) Compound (A) is called phillosopher's wool

(iv) Compound (A) when combined with CoO, then compound (B) is formed & colour of new compound (B) is green

(v) Compound (B) is called Rinmann's green

27. $NaNO_3 \xrightarrow{500°C} (A) + (B)$

$(A) \xrightarrow{800°C} (C) + (D) + (E)$

Find the number of correct statements

(1) Compound (B) is paramagnetic in nature

(2) Compound (B) when undergoes dimerisation then dimer product is diamagnetic in nature

(3) Bond order of compound (B) is two

(4) D is N_2 gas

(5) Compound B and E are same gas

28. $\underset{\substack{\text{(dark purple}\\\text{almost black)}}}{(A)} \xrightarrow{\Delta} \underset{\text{(green)}}{(B)} + \underset{\text{(black)}}{(C)} + (D)$

Find the number of correct statements

(1) Compound B is K_2MnO_4

(2) Compound C is MnO_2

(3) Compound D is O_2

(4) Compound B is paramagnetic in nature

(5) Compound D has two unpaired electron in bonding molecular orbital

29. Total number of compounds undergoes disproporination redox reaction on heating

MnO_2, $HOCl$, H_3PO_3, HNO_2, CrO_5, $HClO_3$

30. On strong heating of H_3PO_4 and H_3BO_3, sum of oxidation number of P & B in the final product obtained is

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EXERCISE

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	B	C	A	B	C	D	B	C	D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	A	A, B	A,B,C,D	B,C,D	A,B,C,D	A, B, C	A,C,D	A, B	A
Que.	21	22	23	24						
Ans.	A	A	D	(A)$\rightarrow$(S);(B)$\rightarrow$(P,Q,R);(C)$\rightarrow$(P);(D)$\rightarrow$(Q,R)						
Que.	25					26	27	28	29	30
Ans.	(A)$\rightarrow$(R,T);(B)$\rightarrow$(P);(C)$\rightarrow$(P,Q,T);(D)$\rightarrow$(P,S)					5	4	4	4	8

ALLEN Session 2019-20

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s-Block Element

OBJECTIVES

After studying this unit you will be able to :

- Describe the general characteristics of the alkali metals and their compounds;
- Explain the general characteristics of the alkaline earth metals and their compounds;
- Describe the manufacture, properties and uses of industrially important sodium and calcium compounds including Portland cement :
- Appreciate the biological significance of sodium, potassium, magnesium and calcium.

The *s*-block elements of the Periodic Table are those in which the last electron enters the outermost *s*-orbital. As the *s*-orbital can accommodate only two electrons, two groups (1 & 2) belong to the s-block of the Periodic Table. Group 1 of the Periodic Table consists of the elements: lithium, sodium, potassium, rubidium, caesium and francium. They are collectively known as the *alkali metals*. These are so called because they form hydroxides on reaction with water which are strongly alkaline in nature. The elements of Group 2 include beryllium, magnesium, calcium, strontium, barium and radium. These elements with the exception of beryllium are commonly known as the *alkaline earth metals*. These are so called because their oxides and hydroxides are alkaline in nature and these metal oxides are found in the earth's crust*.

Among the alkali metals sodium and potassium are abundant and lithium, rubidium and caesium have much lower abundances. Francium is highly radioactive; its longest-lived isotope 223Fr has a half-life of only 21 minutes. Of the alkaline earth metals calcium and magnesium rank fifth and sixth in abundance respectively in the earth's crust. Strontium and barium have much lower abundances. Beryllium is rare and radium is the rarest of all comprising only 10–10 per cent of igneous rocks .

The general electronic configuration of *s*-block elements is [noble gas]*ns*1 for alkali metals and [noble gas] ns^2 for alkaline earth metals.

* *The thin, rocky outer layer of the Earth is crust. † A type of rock formed from magma (molten rock) that has cooled and hardened.*

Lithium and beryllium, the first elements of Group 1 and Group 2 respectively exhibit some properties which are different from those of the other members of the respective group. In these anomalous properties they resemble the second element of the following group. Thus, lithium shows similarities to magnesium and beryllium to aluminium in many of their properties. This type of diagonal similarity is commonly referred to as *diagonal relationship* in the periodic table. The diagonal relationship is due to the similarity in ionic sizes and /or charge/radius ratio of the elements. Monovalent sodium and potassium ions and divalent magnesium and calcium ions are found in large proportions in biological fluids. These ions perform important biological functions such as maintenance of ion balance and nerve impulse conduction.

ALLEN Session 2019-20

10.1 GROUP 1 ELEMENTS : ALKALI METALS

The alkali metals show regular trends in their physical and chemical properties with the increasing atomic number. The atomic, physical and chemical properties of alkali metals are discussed below.

10.1.1 Electronic Configuration

All the alkali metals have one valence electron, ns^1 outside the noble gas core. The loosely held *s*-electron in the outermost valence shell of these elements makes them the most electropositive metals. They readily lose electron to give monovalent M^+ ions. Hence they are never found in free state in nature.

Element	**Symbol**	**Electronic configuration**
Lithium	Li	$1s^2s^1$
Sodium	Na	$1s^2 2s^2 2p^6 3s^1$
Potassium	K	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$
Rubidium	Rb	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 5s^1$
Caesium	Cs	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 4d^{10} 5s^6 5p^6 6s^1$ or [Xe] $6s^1$
Francium	Fr	[Rn]$7s^1$

10.1.2 Atomic and Ionic Radii

Li < Na < K < Rb < Cs

Increase down the group, because value of n (principal quantum number) increases.

10.1.3 Ionization Enthalpy

Li > Na > K > Rb > Cs.

This is because the effect of increasing size outweighs the increasing nuclear charge, and the outermost electron is very well screened from the nuclear charge.

10.1.4 Hydration Enthalpy

The hydration enthalpies of alkali metal ions decrease with increase in ionic sizes.

$Li^+ > Na^+ > K^+ > Rb^+ > Cs^+$

Li^+ has maximum degree of hydration and for this reason lithium salts are mostly hydrated,

e.g., $LiCl \cdot 2H_2O$

10.1.5 Physical Properties

(i) All the alkali metals are silvery white, soft and light metals.

(ii) Because of the large size, these elements have low density which increases down the group from Li to Cs. However, potassium is lighter than sodium.

Li < **K** < Na < Rb < Cs.

(iii) The melting and boiling points of the alkali metals are low indicating weak metallic bonding due to the presence of only a single valence electron in them.

(iv) The alkali metals and their salts impart characteristic colour to an oxidizing flame. This is because the heat from the flame excites the outermost orbital electron to a higher energy level. When the excited electron comes back to the ground state, there is emission of radiation in the visible region as given below :

Session 2019-20

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Metal	Li	Na	K	Rb	Cs
Colour	Crimson red	Yellow	Violet	Red violet	Blue
λ/nm	670.8	589.2	766.5	780.0	455.5

(v) Alkali metals can therefore, be detected by the respective flame tests and can be determined by flame photometry or atomic absorption spectroscopy.

(vi) These elements when irradiated with light, the light energy absorbed may be sufficient to make an atom lose electron.

Table : Atomic and Physical Properties of the Alkali Metals

Property	**Lithium Li**	**Sodium Na**	**Potassium K**	**Rubidium Rb**	**Caesium Cs**	**Francium Fr**
Atomic number	3	11	19	37	55	87
Atomic mass (g mol^{-1})	6.94	22.99	39.10	85.47	132.91	(223)
Electronic configuration	[He]$2s^1$	[Ne] $3s^1$	[Ar] $4s^1$	[Kr] $5s^1$	[Xe] $6s^1$	[Rn] $7s^1$
Ionization enthalpy/kJ mol^{-1}	520	496	419	403	376	~375
Hydration enthalpy/kJ mol^{-1}	–506	–406	–330	–310	–276	-
Metallic radius/pm	152	186	227	248	265	-
Ionic radius M^+/pm	76	102	138	152	167	(180)
m.p./K	454	371	336	312	302	-
b.p/K	1615	1156	1032	961	944	-
Density / g cm^{-3}	0.53	0.97	0.86	1.53	1.90	-
Standard Potentials E^{-} /V for (M^+/M)	–3.04	–2.714	–2.925	–2.930	–2.927	-
Occurrence in lithosphere	18*	2.27**	1.84**	78-12*	2-6*	~10^{-18}*

**ppm (part per million), ** percentage by weight; † Lithosphere: The Earth's outer layer: its crust and part of the upper mantle.*

This property makes caesium and potassium useful as electrodes in photoelectric cells.

10.1.6 Chemical Properties

The alkali metals are highly reactive due to their large size and low ionization enthalpy. The reactivity of these metals increases down the group.

(i) ***Reactivity towards air :*** The alkali metals tarnish in dry air due to the formation of their oxides which in turn react with moisture to form hydroxides. They burn vigorously in oxygen forming oxides. Lithium forms monoxide, sodium forms peroxide, the other metals form superoxides. The superoxide O_2^- ion is stable only in the presence of large cations such as K, Rb, Cs.

$4Li + O_2 \rightarrow 2Li_2O$ (oxide)

$2Na + O_2 \rightarrow Na_2O_2$ (peroxide)

$M + O_2 \rightarrow MO_2$ (superoxide)

(M = K, Rb, Cs)

Session 2019-20

In all these oxides the oxidation state of the alkali metal is +1. Lithium shows exceptional behaviour in reacting directly with nitrogen of air to form the nitride, Li_3N as well. Because of their high reactivity towards air and water, **alkali metals are normally kept in kerosene oil.**

Problem 10.1

What is the oxidation state of K in KO_2?

Solution

The superoxide species is represented as O_2^-; since the compound is neutral, therefore, the oxidation state of potassium is +1.

(ii) ***Reactivity towards water :*** The alkali metals react with water to form hydroxide and dihydrogen.

$2M + 2H_2O \rightarrow 2M^+ + 2OH^- + H_2$

(M = an alkali metal)

It may be noted that although lithium has most negative $E^\ominus$ value, its reaction with water is less vigorous than that of sodium which has the least negative $E^\ominus$ value among the alkali metals. This behaviour of lithium is attributed to its small size and very high hydration energy. Other metals of the group react explosively with water.

They also react with proton donors such as alcohol, gaseous ammonia and alkynes.

(iii) ***Reactivity towards dihydrogen:*** The alkali metals react with dihydrogen at about 673K (lithium at 1073K) to form hydrides. All the alkali metal hydrides are ionic solids with high melting points.

$2M + H_2 \rightarrow 2M^+H^-$

(iv) ***Reactivity towards halogens :*** The alkali metals readily react vigorously with halogens to form ionic halides, M+X–. However, lithium halides are somewhat covalent. It is because of the high polarisation capability of lithium ion (The distortion of electron cloud of the anion by the cation is called polarisation). The Li+ ion is very small in size and has high tendency to distort electron cloud around the negative halide ion. Since anion with large size can be easily distorted, among halides, lithium iodide is the most covalent in nature.

(v) ***Reducing nature :*** The alkali metals are strong reducing agents, lithium being the most and sodium the least powerful. The standard electrode potential ($E^\ominus$) which measures the reducing power represents the overall change :

$M(s) \rightarrow M(g)$ sublimationenthalpy

$M(g) \rightarrow M^+(g) + e^-$ ionizationenthalpy

$M^+ (g) + H_2O \rightarrow M^+ (aq)$ hydrationenthalpy

With the small size of its ion, lithium has the highest hydration enthalpy which accounts for its high negative $E^\ominus$ value and its high reducing power.

Problem 10.2

The $E^\ominus$ for Cl_2/Cl^- is +1.36, for I_2/I^- is + 0.53, for Ag^+/Ag is +0.79, Na^+/Na is –2.71 and for Li+ /Li is – 3.04. Arrange the following ionic species in decreasing order of reducing strength :

I^-, Ag, Cl^-, Li, Na

ALLEN Session 2019-20

Solution

The order is Li > Na > I– > Ag > Cl–

(vi) ***Solutions in liquid ammonia :*** The alkali metals dissolve in liquid ammonia giving deep blue solutions which are conducting in nature.

$$M+(x+y)NH_3 \rightarrow [M(NH_3)_x]^+ + [e(NH_3)y]^-$$

The blue colour of the solution is due to the ammoniated electron which absorbs energy in the visible region of light and thus imparts blue colour to the solution. The solutions are paramagnetic and on standing slowly liberate hydrogen resulting in the formation of amide.

$M^+_{(am)} + e^- + NH_3(1) \rightarrow MNH_{2(am)} + \frac{1}{2}H_2(g)$ (where 'am' denotes solution in ammonia.) In concentrated solution, the blue colour changes to bronze colour and becomes diamagnetic.

(vii) ***Reaction with H_2***

They react with H_2 forming metal hydride with formula MH which are of ionic nature. Stability of hydride decreases down the group.

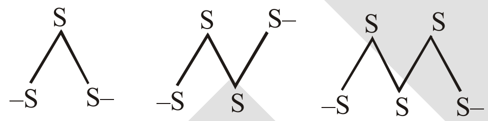
(viii) ***Reaction with N_2***

Only Lithium reacts with N_2 to form ionic lithium nitride Li_3N.

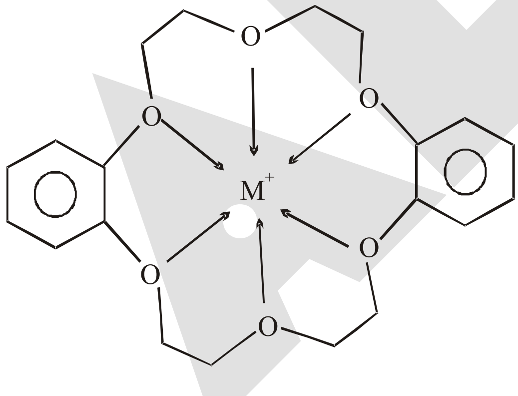
$$3Li + \frac{3}{2}N_2 \rightarrow Li_3N$$

(ix) ***Sulphides***

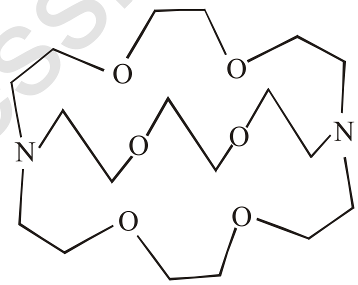
All metals react with S forming sulphides such as Na_2S and Na_2S_n (n = 2, 3, 4, 5 or 6). The polysulphide ions are made from zig-zag chains of sulphur atoms.



(x) ***Crown Ethers and Cryptands :***



Dibenzo-18-Crown-6



Cryptand - 222

$[Na\,(\text{Cryptand } 222)]^+\,Na^-$ [Contains Na^- (sodide ion)]

$[Cs^+(\text{Cryptand - }222)]$ $[(\text{Cyrptand-}222)e^-]$ [electride]

10.1.7 Uses :

Lithium metal is used to make useful alloys, for example with lead to make 'white metal' bearings for motor engines, with aluminium to make aircraft parts, and with magnesium to make armour plates. It is used in thermonuclear reactions.

ALLEN Session 2019-20

Lithium is also used to make electrochemical cells. Sodium is used to make a Na/Pb alloy needed to make $PbEt_4$ and $PbMe_4$. These organolead compounds were earlier used as anti-knock additives to petrol, but nowadays vehicles use lead-free petrol. Liquid sodium metal is used as a coolant in fast breeder nuclear reactors. Potassium has a vital role in biological systems. Potassium chloride is used as a fertilizer. Potassium hydroxide is used in the manufacture of soft soap. It is also used as an excellent absorbent of carbon dioxide. Caesium is used in devising photoelectric cells.

10.2 GENERAL CHARACTERISTICS OF THE COMPOUNDS OF THE ALKALI METALS

All the common compounds of the alkali metals are generally ionic in nature. General characteristics of some of their compounds are discussed here.

10.2.1 Oxides and Hydroxides

On combustion in excess of air, lithium forms mainly the oxide, Li_2O (plus some peroxide Li_2O_2), sodium forms the peroxide, Na_2O_2 (and some oxide Na_2O) whilst potassium, rubidium and caesium form the superoxides, MO_2. Under appropriate conditions pure compounds M_2O, M_2O_2 and MO_2 may be prepared. The increasing stability of the peroxide or superoxide, as the size of the metal ion increases, is due to the stabilisation of large anions by larger cations through lattice energy effects. These oxides are easily hydrolysed by water to form the hydroxides according to the following reactions :

$$M_2O + H_2O \rightarrow 2M^+ + 2OH^-$$

$$M_2O_2 + 2H_2O \rightarrow 2M^+ + 2OH^- + H_2O_2$$

$$2MO_2 + 2H_2O \rightarrow 2M^+ + 2OH^- + H_2O_2 + O_2$$

The oxides and the peroxides are colourless when pure, but the superoxides are yellow or orange in colour. The superoxides are also paramagnetic. Sodium peroxide is widely used as an oxidising agent in inorganic chemistry.

Problem 10.3

Why is KO_2 paramagnetic ?

Solution

The superoxide O_2^- is paramagnetic because of one upaired electron in π^*2p molecular orbital.

The hydroxides which are obtained by the reaction of the oxides with water are all white crystalline solids. The alkali metal hydroxides are the strongest of all bases and dissolve freely in water with evolution of much heat on account of intense hydration.

Sodium Oxide (Na_2O) :

Preparation :

(i) It is obtained by burning sodium at 180°C in a limited supply of air or oxygen and distilling off the excess of sodium in vacuum.

$$2Na + \frac{1}{2}O_2 \xrightarrow{180^\circ} Na_2O$$

(ii) By heating sodium peroxide, nitrate or nitrite with sodium.

$$Na_2O_2 + 2Na \xrightarrow{\Delta} 2Na_2O$$

$$2NaNO_3 + 10Na \xrightarrow{\Delta} 6Na_2O + N_2$$

$$2NaNO_2 + 6Na \xrightarrow{\Delta} 4Na_2O + N_2$$

ALLEN Session 2019-20

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Properties :

(i) It is white amorphous mass.

(ii) It decomposes at 400°C into sodium peroxide and sodium

$$2Na_2O \xrightarrow{400°C} Na_2O_2 + 2Na$$

(iii) It dissolve violently in water, yielding caustic soda.

$$Na_2O + H_2O \longrightarrow 2NaOH$$

Sodium Peroxides (Na_2O_2) :

Preparation: It is formed by heating the metal in excess of air or oxygen at 300°, which is free from moisture and CO_2.

$$2Na + O_2 \longrightarrow Na_2O_2$$

Properties:

(i) It is a pale yellow solid, becoming white in air from the formation of a film of NaOH and Na_2CO_3.

(ii) In cold water (~0°C) produces H_2O_2 but at room temperature produces O_2. In ice-cold mineral acids also produces H_2O_2.

$$Na_2O_2 + 2H_2O \xrightarrow{\sim 0°C} 2NaOH + H_2O_2$$

$$2Na_2O_2 + 2H_2O \xrightarrow{25°C} 4NaOH + O_2$$

$$Na_2O_2 + H_2SO_4 \xrightarrow{\sim 0°C} Na_2SO_4 + H_2O_2$$

(iii) It reacts with CO_2, giving sodium carbonate and oxygen and hence its use for purifying air in a confined space e.g. submarine, ill-ventilated room,

$$2Na_2O_2 + 2CO_2 \longrightarrow 2Na_2CO_3 + O_2$$

(iv) It is an oxidising agent and oxidises charcoal, CO, NH_3, SO_2.

$3Na_2O_2 + 2C \longrightarrow 2Na_2CO_3 + 2Na$ [deposition of metallic Na]

$$CO + Na_2O_2 \longrightarrow Na_2CO_3$$

$$SO_2 + Na_2O_2 \longrightarrow Na_2SO_4$$

$$2NH_3 + 3Na_2O_2 \longrightarrow 6NaOH + N_2$$

(v) It contains peroxide ion $[–O–O–]^{-2}$

Uses:

(i) For preparing H_2O_2, O_2

(ii) Oxygenating the air in submarines

(iii) Oxidising agent in the laboratory.

Oxides of Potassium :

	K_2O,	K_2O_2,	K_2O_3,	KO_2	and	KO_3
Colours :	White	White	Red	Bright Yellow		Orange Solid

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Preparation :

(i) $2KNO_3 + 10K \xrightarrow{\text{heating}} 6K_2O + N_2$

** $\underset{\text{(White)}}{K_2O} \xrightarrow{\text{heating}} \underset{\text{(Yellow)}}{K_2O}$

** $K_2O + H_2O \longrightarrow 2KOH$

(ii) $2K + O_2 \xrightarrow[\text{air at 300°C}]{\text{Controlled}} K_2O_2$ [Props: Similar with Na_2O_2]

(iii) Passage of O_2 through a blue solution of K in liquid NH_3 yields oxides K_2O_2 (white), K_2O_3 (red) and KO_2 (deep yellow) i.e

$$\text{K in liq. } NH_3 \xrightarrow{O_2} \underset{\text{white}}{K_2O_2} \longrightarrow \underset{\text{red}}{K_2O_3} \longrightarrow \underset{\text{yellow}}{KO_2}$$

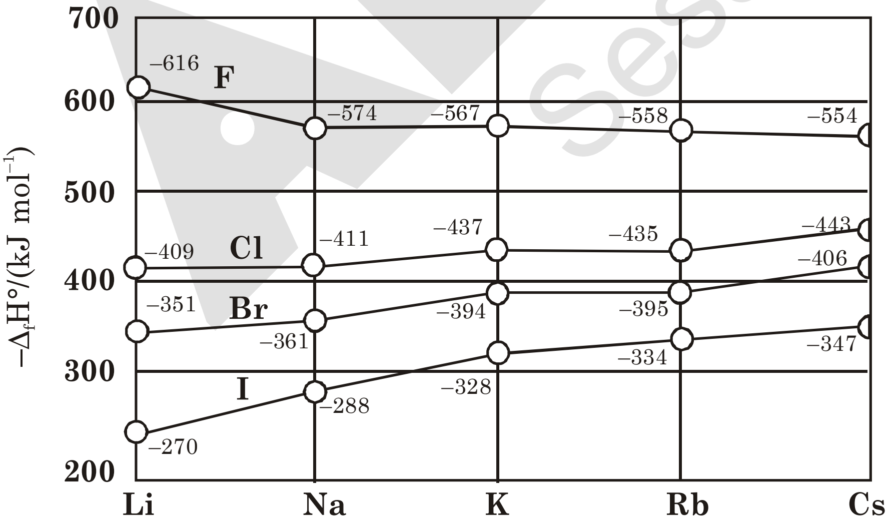
** KO_2 reacts with H_2O and produces H_2O_2 and O_2 both

$$2KO_2 + 2H_2O \xrightarrow{\sim 0°C} 2KOH + H_2O_2 + O_2$$

KO_3 : $\underset{\text{(Dry powdered)}}{KOH} + O_3 \text{ (ozonised oxygen)} \xrightarrow{-10°\text{to}-15°C} \underset{\text{(orange solid)}}{KO_3}$

10.2.2 Halides

The alkali metal halides, MX, (X=F,Cl,Br,I) are all high melting, colourless crystalline solids. They can be prepared by the reaction of the appropriate oxide, hydroxide or carbonate with aqueous hydrohalic acid (HX). All of these halides have high negative enthalpies of formation; the $\Delta_f H^\ominus$ values for fluorides become less negative as we go down the group, whilst the reverse is true for $\Delta_f H^\ominus$ for chlorides, bromides and iodides. For a given metal $\Delta_f H^\ominus$ always becomes less negative from fluoride to iodide.



The standard enthalpies of formation of the halides of Group 1 elements at 298 K

10.2.3 Salts of Oxo-Acids

Oxo-acids are those in which the acidic proton is on a hydroxyl group with an oxo group attached to the same atom e.g., carbonic acid, H_2CO_3 ($OC(OH)_2$; sulphuric acid, H_2SO_4 ($O_2S(OH)_2$). The alkali metals form salts with all the oxo-acids. They are generally soluble in water and thermally stable.

Their carbonates (M_2CO_3) and in most cases the hydrogencarbonates ($MHCO_3$) also are highly stable to heat. As the electropositive character increases down the group, the stability of the carbonates and hydorgencarbonates increases. Lithium carbonate is not so stable to heat; lithium being very small in size polarises a large CO_3^{2-} ion leading to the formation of more stable Li_2O and CO_2. Its hydrogencarbonate does not exist as a solid.

10.3 ANOMALOUS PROPERTIES OF LITHIUM

The anomalous behaviour of lithium is due to the : (i) exceptionally small size of its atom and ion, and (ii) high polarising power (i.e., charge/ radius ratio). As a result, there is increased covalent character of lithium compounds which is responsible for their solubility in organic solvents. Further, lithium shows diagonal relationship to magnesium which has been discussed subsequently.

10.3.1 Points of Difference between Lithium and other Alkali Metals

(i) Lithium is much harder. Its m.p. and b.p. are higher than the other alkali metals.

(ii) Lithium is least reactive but the strongest reducing agent among all the alkali metals. On combustion in air it forms mainly monoxide, Li_2O and the nitride, Li_3N unlike other alkali metals.

(iii) LiCl is deliquescent and crystallises as a hydrate, $LiCl.2H_2O$ whereas other alkali metal chlorides do not form hydrates.

(iv) Lithium hydrogencarbonate is not obtained in the solid form while all other elements form solid hydrogencarbonates.

(v) Lithium unlike other alkali metals forms no ethynide on reaction with ethyne.

(vi) Lithium nitrate when heated gives lithium oxide, Li_2O, whereas other alkali metal nitrates decompose to give the corresponding nitrite.

$4LiNO_3 \rightarrow 2Li_2O + 4NO_2 + O_2$

$2NaNO_3 \rightarrow 2NaNO_2 + O_2$

(vii) LiF and Li_2O are comparatively much less soluble in water than the corresponding compounds of other alkali metals.

10.3.2 Points of Similarities between Lithium and Magnesium

The similarity between lithium and magnesium is particularly striking and arises because of their similar sizes : atomic radii, Li = 152 pm, Mg = 160 pm; ionic radii : Li^+ = 76 pm, Mg^{2+}= 72 pm. The main points of similarity are :

(i) Both lithium and magnesium are harder and lighter than other elements in the respective groups.

(ii) Lithium and magnesium react slowly with water. Their oxides and hydroxides are much less soluble and their hydroxides decompose on heating. Both form a nitride, Li_3N and Mg_3N_2, by direct combination with nitrogen.

(iii) The oxides, Li_2O and MgO do not combine with excess oxygen to give any superoxide.

(iv) The carbonates of lithium and magnesium decompose easily on heating to form the oxides and CO_2. Solid hydrogencarbonates are not formed by lithium and magnesium.

(v) Both LiCl and $MgCl_2$ are soluble in ethanol.

(vi) Both LiCl and $MgCl_2$ are deliquescent and crystallise from aqueous solution as hydrates, $LiCl\cdot2H_2O$ and $MgCl_2\cdot8H_2O$.

ALLEN Session 2019-20

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10.4 SOME IMPORTANT COMPOUNDS OF SODIUM

Industrially important compounds of sodium include sodium carbonate, sodium hydroxide, sodium chloride and sodium bicarbonate. The large scale production of these compounds and their uses are described below :

Sodium Carbonate (Washing Soda), $Na_2CO_3 \cdot 10H_2O$

Sodium carbonate is generally prepared by **Solvay Process**. In this process, advantage is taken of the low solubility of sodium hydrogencarbonate whereby it gets precipitated in the reaction of sodium chloride with ammonium hydrogencarbonate. The latter is prepared by passing CO_2 to a concentrated solution of sodium chloride saturated with ammonia, where ammonium carbonate followed by ammonium hydrogencarbonate are formed.

The equations for the complete process may be written as:

$$2NH_3 + H_2O + CO_2 \rightarrow (NH_4)_2\,CO_3$$

$$(NH_4)_2\,CO_3 + H_2O + CO_2 \rightarrow 2NH_4HCO_3$$

$$NH_4HCO_3 + NaCl \rightarrow NH_4Cl + NaHCO_3$$

Sodium hydrogencarbonate crystal separates. These are heated to give sodium carbonate.

$$2NaHCO_3 \rightarrow Na_2CO_3 + CO_2 + H_2O$$

In this process NH_3 is recovered when the solution containing NH_4Cl is treated with $Ca(OH)_2$. Calcium chloride is obtained as a by-product.

$$2NH_4Cl + Ca(OH)_2 \rightarrow 2NH_3 + CaCl_2 + H_2O$$

It may be mentioned here that Solvay process cannot be extended to the manufacture of potassium carbonate because potassium hydrogencarbonate is too soluble to be precipitated by the addition of ammonium hydrogencarbonate to a saturated solution of potassium chloride.

Properties : Sodium carbonate is a white crystalline solid which exists as a decahydrate, $Na_2CO_3 \cdot 10H_2O$. This is also called washing soda. It is readily soluble in water. On heating, the decahydrate loses its water of crystallisation to form monohydrate. Above 373K, the monohydrate becomes completely anhydrous and changes to a white powder called soda ash.

$$Na_2CO_2.10H_2O \xrightarrow{375K} Na_2CO_3.H_2O + 9H_2O$$

$$Na_2CO_3.H_2O \xrightarrow{>373K} Na_2CO_3 + H_2O$$

Carbonate part of sodium carbonate gets hydrolysed by water to form an alkaline solution.

$$CO_3^{2-} + H_2O \rightarrow HCO_3^{-} + OH^{-}$$

Uses:
- **(i)** It is used in water softening, laundering and cleaning.
- **(ii)** It is used in the manufacture of glass, soap, borax and caustic soda.
- **(iii)** It is used in paper, paints and textile industries.
- **(iv)** It is an important laboratory reagent both in qualitative and quantitative analysis.

Note: K_2CO_3 cannot be prepared by **Solvey process** because $KHCO_3$ is soluble in water and cannot be separated form NH_4Cl.

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Sodium Chloride, NaCl

The most abundant source of sodium chloride is sea water which contains 2.7 to 2.9% by mass of the salt. In tropical countries like India, common salt is generally obtained by evaporation of sea water. Approximately 50 lakh tons of salt are produced annually in India by solar evaporation. Crude sodium chloride, generally obtained by crystallisation of brine solution, contains sodium sulphate, calcium sulphate, calcium chloride and magnesium chloride as impurities. Calcium chloride, $CaCl_2$, and magnesium chloride, $MgCl_2$ are impurities because they are deliquescent (absorb moisture easily from the atmosphere).

To obtain pure sodium chloride, the crude salt is dissolved in minimum amount of water and filtered to remove insoluble impurities. The solution is then saturated with hydrogen chloride gas. Crystals of pure sodium chloride separate out. Calcium and magnesium chloride, being more soluble than sodium chloride, remain in solution.

Sodium chloride melts at 1081K. It has a solubility of 36.0 g in 100 g of water at 273 K. The solubility does not increase appreciably with increase in temperature.

Uses :

(i) It is used as a common salt or table salt for domestic purpose.

(ii) It is used for the preparation of Na_2O_2, NaOH and Na_2CO_3.

(iii) It is used to prepare freezing mixture in laboratory [Ice-common salt mixture is called freezing mixture and temperature goes down to –23°C.]

(iv) For melting ice and snow on road.

Sodium Hydroxide (Caustic Soda), NaOH

Sodium hydroxide is generally prepared commercially by the electrolysis of sodium chloride in Castner-Kellner cell. A brine solution is electrolysed using a mercury cathode and a carbon anode. Sodium metal discharged at the cathode combines with mercury to form sodium amalgam. Chlorine gas is evolved at the anode. **Cathode :** $Na^+ + e^- \xrightarrow{Hg} Na-$ amalgam

$$\textbf{Anode :}\ Cl^- \rightarrow \frac{1}{2}Cl_2 + e^-$$

The amalgam is treated with water to give sodium hydroxide and hydrogen gas.

$$2Na\text{-amalgam} + 2H_2O \rightarrow 2NaOH + 2Hg + H_2$$

Sodium hydroxide is a white, translucent solid. It melts at 591 K. It is readily soluble in water to give a strong alkaline solution. Crystals of sodium hydroxide are deliquescent. The sodium hydroxide solution at the surface reacts with the CO_2 in the atmosphere to form Na_2CO_3.

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Advanced Batch\Chem\Sheet\Salt Analysis & Heating Effect\(III) s-Block\Eng.p65

Uses : It is used in

(i) The manufacture of soap, paper, artificial silk and a number of chemicals,

(ii) In petroleum refining,

(iii) In the purification of bauxite,

(iv) In the textile industries for mercerising cotton fabrics, (v) for the preparation of pure fats and oils, and

(vi) As a laboratory reagent.

***Sodium Hydrogencarbonate (Baking Soda)*, $NaHCO_3$**

Sodium hydrogencarbonate is known as baking soda because it decomposes on heating to generate bubbles of carbon dioxide (leaving holes in cakes or pastries and making them light and fluffy).

Sodium hydrogencarbonate is made by saturating a solution of sodium carbonate with carbon dioxide. The white crystalline powder of sodium hydrogencarbonate, being less soluble, gets separated out.

$$Na_2CO_3 + H_2O + CO_2 \rightarrow 2NaHCO_3$$

Sodium hydrogencarbonate is a mild antiseptic for skin infections. **It is used in fire extinguishers.**

***Potassium carbonate,* K_2CO_3**

It is also called potash or pearl ash. It cannot be made by the use of solvay process as potassium bicarbonate is more soluble than sodium bicarbonate. However, it can be prepared by **Le-Blanc process**. KCl is first converted into K_2SO_4. Potassium sulphate (K_2SO_4) is then heated with $CaCo_3$ and carbon.

$$KCl + H_2SO_4 \longrightarrow KHSO_4 + HCl$$

$$KHSO_4 + KCl \longrightarrow K_2SO_4 + HCl$$

$$K_2SO_4 + CaCO_3 + 2C \longrightarrow K_2CO_3 + CaS + 2CO_2$$

It is a white powder , deliquescent in nature. It is highly soluble in water.

Uses : It is used in the manufacture of hard glass. The mixture of K_2CO_3 and Na_2CO_3 is used a **fusion mixture** in laboratory.

10.5 BIOLOGICAL IMPORTANCE OF SODIUM AND POTASSIUM

A typical 70 kg man contains about 90 g of Na and 170 g of K compared with only 5 g of iron and 0.06 g of copper. Sodium ions are found primarily on the outside of cells, being located in blood plasma and in the interstitial fluid which surrounds the cells. These ions participate in the transmission of nerve signals, in regulating the flow of water across cell membranes and in the transport of sugars and amino acids into cells. Sodium and potassium, although so similar chemically, differ quantitatively in their ability to penetrate cell membranes, in their transport mechanisms and in their efficiency to activate enzymes.

Thus, potassium ions are the most abundant cations within cell fluids, where they activate many enzymes, participate in the oxidation of glucose to produce ATP and, with sodium, are responsible for the transmission of nerve signals.

There is a very considerable variation in the concentration of sodium and potassium ions found on the opposite sides of cell membranes. As a typical example, in blood plasma, sodium is present to the extent of 143 $mmolL^{-1}$, whereas the potassium level is only 5 $mmolL^{-1}$ within the red blood cells. These concentrations change to 10 $mmolL^{-1}$ (Na^+) and 105 $mmolL^{-1}$ (K^+). These ionic gradients demonstrate that a discriminatory mechanism, called the sodium-potassium pump, operates across the cell membranes which consumes more than one-third of the ATP used by a resting animal and about 15 kg per 24 h in a resting human.

10.6 GROUP 2 ELEMENTS : ALKALINE EARTH METALS

The group 2 elements comprise beryllium, magnesium, calcium, strontium, barium and radium. They follow alkali metals in the periodic table. These (except beryllium) are known as alkaline earth metals. The first element beryllium differs from the rest of the members and shows diagonal relationship to aluminium. The atomic and physical properties of the alkaline earth metals are shown in Table.

10.6.1 Electronic Configuration

These elements have two electrons in the *s*-orbital of the valence shell. Their general electronic configuration may be represented as [noble gas] ns^2. Like alkali metals, the compounds of these elements are also predominantly ionic.

Element	Symbol	Electronic configuration
Beryllium	Be	$1s^22s^2$
Magnesium	Mg	$1s^2 2s^22p^63s^2$
Calcium	Ca	$1s^22s^22p^63s^23p^6 4s^2$
Strontium	Sr	$1s^22s^22p^63s^23p^6 3d^{10}4s^24p^65s^2$
Barium	Ba	$1s^22s^22p^63s^23p^63d^{10}4s^24p^6 4d^{10}5s^25p^66s^2$ or [Xe] $6s^2$
Radium	Ra	[Rn]$7s^2$

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10.6.2 Atomic and Ionic Radii

The atomic and ionic radii of the alkaline earth metals are smaller than those of the

Table : Atomic and Physical Properties of the Alkaline Earth Metals

Property	Beryllium Be	Magnesium mg	Calcium Ca	Strontium Sr	Barium Ba	Radium Ra
Atomic number	4	12	20	38	56	88
Atomic mass (g mol^{-1})	9.01	24.31	40.08	87.62	137.33	226.03
Electronic configuration	$[He]2s^2$	$[Ne]\ 3s^2$	$[Ar]\ 4s^2$	$[Kr]\ 5s^2$	$[Xe]\ 6s^2$	$[Rn]\ 7s^2$
Ionization enthalpy (I)/kJ mol^{-1}	899	737	590	549	503	509
Ionization enthyalpy(II) kJ mol^{-1}	1757	1450	1145	1064	965	979
Hydration enthalpy/kJ mol^{-1}	–2494	–1921	–1577	–1443	–1305	-
Metallic radius/pm	111	160	197	215	222	-
Ionic radius M^+/pm	31	72	100	118	135	148
m.p./K	1560	924	1124	1062	1002	973
b.p/K	2745	1363	1767	.1655	2078	(1973)
Density / g cm^{-3}	1.84	1.74	1.55	2.63	3.59	(5.5)
Standard Potentials $E^{\ominus}$ /V for (M^{+2}/M)	–1.97	–2.36	–2.84	–2.89	–2.92	–2.92
Occurrence in lithosphere	2*	2.76**	4.6**	384*	390*	10^{-6}*

**ppm (part per million); ** percentage by weight :*

Corresponding alkali metals in the same periods. This is due to the increased nuclear charge in these elements. Within the group, the atomic and ionic radii increase with increase in atomic number.

10.6.3 Ionization Enthalpies

Ionization Enthalpy

Be > Mg > Ca > Sr > Ba

Down the group IE decreases due to increase in size

Q. IE_1 of AM < IE_1 of AEM

IE_2 of AM > IE_2 of AEM

[where AM = Alkali metal, AEM = Alkaline earth metal]

Reason **:** IE_1 of AEM is large due to increased nuclear charge in AEM as compared to AM but IE_2 of AM is large because second electron in AM is to be removed from cation which has already acquired noble gas configuration.

10.6.4 Hydration Enthalpies

Like alkali metal ions, the hydration enthalpies of alkaline earth metal ions decrease with increase in ionic size down the group.

$Be^{2+} > Mg^{2+} > Ca^{2+} > Sr^{2+} > Ba^{2+}$

ALLEN Session 2019-20
node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Advanced Batch\Chem\Sheet\Salt Analysis & Heating Effect (III) s-Block\Eng.p65

The hydration enthalpies of alkaline earth metal ions are larger than those of alkali metal ions. Thus, compounds of alkaline earth metals are more extensively hydrated than those of alkali metals, e.g., $MgCl_2$ and $CaCl_2$ exist as $MgCl_2.6H_2O$ and $CaCl_2 \cdot 6H_2O$ while NaCl and KCl do not form such hydrates.

10.6.5 Physical Properties

(i) The alkaline earth metals, in general, are silvery white, lustrous and relatively soft but harder than the alkali metals. Beryllium and magnesium appear to be somewhat greyish.

(ii) The melting and boiling points of these metals are higher than the corresponding alkali metals due to smaller sizes. The trend is, however, not systematic.

(iii) Because of the low ionisation enthalpies, they are strongly electropositive in nature. The electropositive character increases down the group from Be to Ba.

(iv) Calcium, strontium and barium impart characteristic brick red, crimson and apple green colours respectively to the flame. In flame the electrons are excited to higher energy levels and when they drop back to the ground state, energy is emitted in the form of visible light. The electrons in beryllium and magnesium are too strongly bound to get excited by flame. Hence, these elements do not impart any colour to the flame. The flame test for Ca, Sr and Ba is helpful in their detection in qualitative analysis and estimation by flame photometry.

(v) The alkaline earth metals like those of alkali metals have high electrical and thermal conductivities which are typical characteristics of metals.

10.6.6 Chemical Properties

The alkaline earth metals are less reactive than the alkali metals. The reactivity of these elements increases on going down the group.

(i) ***Reactivity towards air :*** Beryllium and magnesium are kinetically inert to oxygen and water because of the formation of an oxide film on their surface. However, powdered beryllium burns brilliantly on ignition in air to give BeO and Be_3N_2. Magnesium is more electropositive and burns with dazzling brilliance in air to give MgO and Mg_3N_2. Calcium, strontium and barium are readily attacked by air to form the oxide and nitride.

(ii) ***Reactivity towards water.***

Reaction with H_2O **:** AEM have lesser tendency to react with water as compared to AM. They form hydroxides and liberate H_2 on reaction with H_2O

$$M + 2H_2O \xrightarrow{\Delta} M(OH)_2 + H_2$$

* Be is inert towards water.

* Magnesium react as

$$Mg + 2H_2O \rightarrow Mg(OH)_2 + H_2$$

or $$Mg + H_2O \rightarrow MgO + H_2O$$

MgO forms protective layer, that is why it does not react readily unless layer is removed amalgamating with Hg. Other metals react quite readily (Ca, Sr, Ba).

Note: $Be(OH)_2$ is amphoteric but other hydroxides are basic in nature.

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Advanced Batch\Chem\Sheet\Salt Analysis & Heating Effect\VIII\s-Block\Eng.p65

(iii) ***Reactivity towards the halogens :*** All the alkaline earth metals combine with halogen at elevated temperatures forming their halides.

$M + X_2 \rightarrow MX_2$ (X = F, Cl, Br, l)

Thermal decomposition of $(NH_4)_2BeF_4$ is the best route for the preparation of BeF_2, and $BeCl_2$ is conveniently made from the oxide.

$$BeO + C + Cl_2 \xrightleftharpoons{600-800K} BeCl_2 + CO$$

(iv) ***Reactivity towards hydrogen :*** All the elements except beryllium combine with hydrogen upon heating to form their hydrides, MH_2. BeH_2, however, can be prepared by the reaction of $BeCl_2$ with $LiAlH_4$.

$$2BeCl_2 + LiAlH_4 \rightarrow 2BeH_2 + LiCl + AlCl_3$$

(v) ***Reactivity towards acids :*** AEM react with acids & liberate H_2

$$Mg + 2HCl \rightarrow MgCl_2 + H_2$$

Be is amphoteric as it also react with NaOH, other metals do not react as they are purely basic.

$$Be + 2NaOH \rightarrow Be(OH)_2 \xrightarrow{\text{excess NaOH}} [Be(OH)_4]^{2-}$$

(vi) ***Reducing nature :*** Like alkali metals, the alkaline earth metals are strong reducing agents. This is indicated by large negative values of their reduction potentials. However their reducing power is less than those of their corresponding alkali metals. Beryllium has less negative value compared to other alkaline earth metals. However, its reducing nature is due to large hydration energy associated with the small size of Be^{2+} ion and relatively large value of the atomization enthalpy of the metal.

(vii) ***Solutions in liquid ammonia :*** Like alkali metals, the alkaline earth metals dissolve in liquid ammonia to give deep blue black solutions forming ammoniated ions.

$M + (x + y) NH_3 \rightarrow [M(NH_3)_x]^{2+} + 2[e(NH_3)_y]^-$ (except : Be and Mg)

From these solutions, the ammoniates, $[M(NH_3)_6]^{2+}$ can be recovered.

10.6.7 Uses :

Be : Beryllium is used in the manufacture of alloys. Copper-beryllium alloys are used in the preparation of high strength springs. Metallic beryllium is used for making windows of X-ray tubes.

Mg : Magnesium forms alloys with aluminium, zinc, manganese and tin. Magnesium-aluminium alloys being light in mass are used in air-craft construction. Magnesium (powder and ribbon) is used in flash powders and bulbs, incendiary bombs and signals. A suspension of magnesium hydroxide in water (called *milk of magnesia*) is used as antacid in medicine. Magnesium carbonate is an ingredient of toothpaste.

Ca : Calcium is used in the extraction of metals from oxides which are difficult to reduce with carbon. Calcium and barium metals, owing to their reactivity with oxygen and nitrogen at elevated temperatures, have often been used to remove air from vacuum tubes.

Ra : Radium salts are used in radiotherapy, for example, in the treatment of cancer.

ALLEN Session 2019-20

10.7 GENERAL CHARACTERISTICS OF COMPOUNDS OF THE ALKALINE EARTH METALS

The dipositive oxidation state (M^{2+}) is the predominant valence of Group 2 elements. The alkaline earth metals form compounds which are predominantly ionic but less ionic than the corresponding compounds of alkali metals. This is due to increased nuclear charge and smaller size. The oxides and other compounds of beryllium and magnesium are more covalent than those formed by the heavier and large sized members (Ca, Sr, Ba). The general characteristics of some of the compounds of alkali earth metals are described below.

(i) Oxides and Hydroxides :

The alkaline earth metals burn in oxygen to form the monoxide, MO which, except for BeO, have rock-salt structure. The BeO is essentially covalent in nature. The enthalpies of formation of these oxides are quite high and consequently they are very stable to heat. BeO is amphoteric while oxides of other elements are ionic in nature. All these oxides except BeO are basic in nature and react with water to form sparingly soluble hydroxides.

$$MO + H_2O \rightarrow M(OH)_2$$

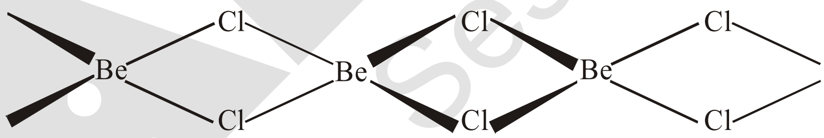
The solubility, thermal stability and the basic character of these hydroxides increase with increasing atomic number from $Mg(OH)_2$ to $Ba(OH)_2$. The alkaline earth metal hydroxides are, however, less basic and less stable than alkali metal hydroxides. Beryllium hydroxide is amphoteric in nature as it reacts with acid and alkali both.

$$Be(OH)_2 + 2OH^- \rightarrow \underset{\text{Beryllate ion}}{[Be(OH)_4]^{2-}}$$

$$Be(OH)_2 + 2HCl \rightarrow 2H_2O + BeCl_2$$

(ii) Halides :

Except for beryllium halides, all other halides of alkaline earth metals are ionic in nature. Beryllium halides are essentially covalent and soluble in organic solvents. Beryllium chloride has a chain structure in the solid state as shown below:



In the vapour phase $BeCl_2$ tends to form a chloro-bridged dimer which dissociates into the linear monomer at high temperatures of the order of 1200 K. The tendency to form halide hydrates gradually decreases (for example, $MgCl_2{\cdot}8H_2O$, $CaCl_2{\cdot}6H_2O$, $SrCl_2{\cdot}6H_2O$ and $BaCl_2{\cdot}2H_2O$) down the group. The dehydration of hydrated chlorides, bromides and iodides of Ca, Sr and Ba can be achieved on heating; however, the corresponding hydrated halides of Be and Mg on heating suffer hydrolysis. The fluorides are relatively less soluble than the chlorides owing to their high lattice energies.

ALLEN Session 2019-20

(iii) ***Salts of Oxoacids :*** The alkaline earth metals also form salts of oxoacids. Some of these are :

Carbonates : Carbonates of alkaline earth metals are insoluble in water and can be precipitated by addition of a sodium or ammonium carbonate solution to a solution of a soluble salt of these metals. The solubility of carbonates in water decreases as the atomic number of the metal ion increases. All the carbonates decompose on heating to give carbon dioxide and the oxide. Beryllium carbonate is unstable and can be kept only in the atmosphere of CO_2. The thermal stability increases with increasing cationic size.

Sulphates : The sulphates of the alkaline earth metals are all white solids and stable to heat. $BeSO_4$, and $MgSO_4$ are readily soluble in water; the solubility decreases from $CaSO_4$ to $BaSO_4$. The greater hydration enthalpies of Be^{2+} and Mg^{2+} ions overcome the lattice enthalpy factor and therefore their sulphates are soluble in water.

Nitrates : The nitrates are made by dissolution of the carbonates in dilute nitric acid. Magnesium nitrate crystallises with six molecules of water, whereas barium nitrate crystallises as the anhydrous salt. This again shows a decreasing tendency to form hydrates with increasing size and decreasing hydration enthalpy. All of them decompose on heating to give the oxide like lithium nitrate.

$$2M\,(NO_3)_2 \rightarrow 2MO + 4NO_2 + O_2$$

(M + Be, Mg, Ca, Sr, Ba)

Problem 10.4

Why does the solubility of alkaline earth metal hydroxides in water increase down the group?

Solution

Among alkaline earth metal hydroxides, the anion being common the cationic radius will influence the lattice enthalpy. Since lattice enthalpy decreases much more than the hydration enthalpy with increasing ionic size, the solubility increases as we go down the group.

Problem 10.5

Why does the solubility of alkaline earth metal carbonates and sulphates in water decrease down the group?

Solution

The size of anions being much larger compared to cations, the lattice enthalpy will remain almost constant within a particular group. Since the hydration enthalpies decrease down the group, solubility will decrease as found for alkaline earth metal carbonates and sulphates.

10.8 ANOMALOUS BEHAVIOUR OF BERYLLIUM

Beryllium, the first member of the Group 2 metals, shows anomalous behaviour as compared to magnesium and rest of the members. Further, it shows diagonal relationship to aluminium which is discussed subsequently.

ALLEN Session 2019-20

(i) Beryllium has exceptionally small atomic and ionic sizes and thus does not compare well with other members of the group. Because of high ionisation enthalpy and small size it forms compounds which are largely covalent and get easily hydrolysed.

(ii) Beryllium does not exhibit coordination number more than four as in its valence shell there are only four orbitals. The remaining members of the group can have a coordination number of six by making use of *d*-orbitals.

(iii) The oxide and hydroxide of beryllium, unlike the hydroxides of other elements in the group, are amphoteric in nature.

10.8.1 Diagonal Relationship between Beryllium and Aluminium

The ionic radius of Be^{2+} is estimated to be 31 pm; the charge/radius ratio is nearly the same as that of the Al^{3+} ion. Hence beryllium resembles aluminium in some ways. Some of the similarities are:

(i) Like aluminium, beryllium is not readily attacked by acids because of the presence of an oxide film on the surface of the metal.

(ii) Beryllium hydroxide dissolves in excess of alkali to give a beryllate ion, $[Be(OH)_4]^{2-}$ just as aluminium hydroxide gives aluminate ion, $[Al(OH)_4]^-$.

(iii) The chlorides of both beryllium and aluminium have Cl^- bridged chloride structure in vapour phase. Both the chlorides are soluble in organic solvents and are strong Lewis acids. They are used as Friedel Craft catalysts.

(iv) Beryllium and aluminium ions have strong tendency to form complexes, BeF_4^{2-}, AlF_6^{3-}

10.9 SOME IMPORTANT COMPOUNDS OF CALCIUM

Important compounds of calcium are calcium oxide, calcium hydroxide, calcium sulphate, calcium carbonate and cement. These are industrially important compounds. The large scale preparation of these compounds and their uses are described below.

***Calcium Oxide or Quick Lime,* CaO**

It is prepared on a commercial scale by heating limestone ($CaCO_3$) in a rotary kiln at 1070-1270 K.

$$CaCO_3 \xrightleftharpoons{heat} CaO + CO_2$$

The carbon dioxide is removed as soon as it is produced to enable the reaction to proceed to completion. Calcium oxide is a white amorphous solid. It has a melting point of 2870 K. On exposure to atmosphere, it absorbs moisture and carbon dioxide.

$$CaO + H_2O \rightarrow Ca(OH)_2$$

$$CaO + CO_2 \rightarrow CaCO_3$$

The addition of limited amount of water breaks the lump of lime. This process is called *slaking of lime*. Quick lime slaked with soda gives solid sodalime. Being a basic oxide, it combines with acidic oxides at high temperature.

$$CaO + SiO_2 \rightarrow CaSiO_3$$

$$6CaO + P_4O_{10} \rightarrow 2Ca_3(PO_4)_2$$

Session 2019-20

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Uses : **(i)** It is an important primary material for manufacturing cement and is the cheapest form of alkali.

(ii) It is used in the manufacture of sodium carbonate from caustic soda.

(iii) It is employed in the purification of sugar and in the manufacture of dye stuffs.

Calcium Hydroxide (Slaked lime), **$Ca(OH)_2$**

Calcium hydroxide is prepared by adding water to quick lime, CaO.

It is a white amorphous powder. It is sparingly soluble in water. The aqueous solution is known as *lime water* and a suspension of slaked lime in water is known as *milk of lime*.

When carbon dioxide is passed through lime water it turns milky due to the formation of calcium carbonate.

$$Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$$

On passing excess of carbon dioxide, the precipitate dissolves to form calcium hydrogencarbonate.

$$CaCO_3 + CO_2 + H_2O \rightarrow Ca\ (HCO_3)_2$$

Milk of lime reacts with chlorine to form hypochlorite, a constituent of bleaching powder.

$$2Ca(OH)_2 + 2Cl_2 \rightarrow CaCl_2 + Ca(OCl)_2 + 2H_2O$$

Uses: **(i)** It is used in the preparation of mortar, a building material.

(ii) It is used in white wash due to its disinfectant nature.

(iii) It is used in glass making, in tanning industry, for the preparation of bleaching powder and for purification of sugar.

Calcium Carbonate, **$CaCO_3$**

Calcium carbonate occurs in nature in several forms like limestone, chalk, marble etc. It can be prepared by passing carbon dioxide through slaked lime or by the addition of sodium carbonate to calcium chloride.

$$Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$$

$$CaCl_2 + Na_2CO_3 \rightarrow CaCO_3 + 2NaCl$$

Excess of carbon dioxide should be avoided since this leads to the formation of water soluble calcium hydrogencarbonate.

Calcium carbonate is a white fluffy powder. It is almost insoluble in water. When heated to 1200 K, it decomposes to evolve carbon dioxide.

$$CaCO_3 \xrightarrow{1200K} CaO + CO_2$$

It reacts with dilute acid to liberate carbon dioxide.

$$CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$$

$$CaCO_3 + H_2SO_4 \rightarrow CaSO_4 + H_2O + CO_2$$

Uses: It is used as a building material in the form of marble and in the manufacture of quick lime. Calcium carbonate along with magnesium carbonate is used as a flux in the extraction of metals such as iron.

ALLEN Session 2019-20

Specially precipitated $CaCO_3$ is extensively used in the manufacture of high quality paper. It is also used as an antacid, mild abrasive in tooth paste, a constituent of chewing gum, and a filler in cosmetics.

***Calcium Sulphate (Plaster of Paris),* $CaSO_4 \cdot \frac{1}{2} H_2O$**

It is a hemihydrate of calcium sulphate. It is obtained when gypsum, $CaSO_4 \cdot 2H_2O$, is heated to 393 K.

$$2(CaSO_4.2H_2O) \rightarrow 2(CaSO_4).H_2O + 3H_2O$$

Above 393 K, no water of crystallisation is left and anhydrous calcium sulphate, $CaSO_4$ is formed. This is known as 'dead burnt plaster'.

It has a remarkable property of setting with water. On mixing with an adequate quantity of water it forms a plastic mass that gets into a hard solid in 5 to 15 minutes.

Uses: The largest use of Plaster of Paris is in the building industry as well as plasters. It is used for immoblising the affected part of organ where there is a bone fracture or sprain. It is also employed in dentistry, in ornamental work and for making casts of statues and busts.

Cement: Cement is an important building material. It was first introduced in England in 1824 by Joseph Aspdin. It is also called **'Portland cement'** because it resembles with the natural limestone quarried in the Isle of Portland, England.

Cement is a product obtained by combining a material rich in lime, CaO with other material such as clay which contains silica, SiO_2 along with the oxides of aluminium, iron and magnesium. The average composition of Portland cement is :

CaO, 50-60%; SiO_2, 20-25%; Al_2O_3, 5-10%; MgO, 2-3%; Fe_2O_3, 1-2% and SO_3, 1-2%. For a good quality cement, the ratio of silica (SiO_2) to alumina (Al_2O_3) should be between 2.5 and 4 and the ratio of lime (CaO) to the total of the oxides of silicon (SiO_2) aluminium (Al_2O_3) and iron (Fe_2O_3) should be as close as possible to 2.

The raw materials for the manufacture of cement are limestone and clay. When clay and lime are strongly heated together they fuse and react to form 'cement clinker'. This clinker is mixed with 2-3% by weight of gypsum ($CaSO_4 \cdot 2H_2O$) to form cement. Thus important ingredients present in Portland cement are dicalcium silicate (Ca_2SiO_4) 26%, tricalcium silicate (Ca_3SiO_5) 51% and tricalcium aluminate ($Ca_3Al_2O_6$) 11%.

Setting of Cement : When mixed with water, the setting of cement takes place to give a hard mass. This is due to the hydration of the molecules of the constituents and their rearrangement. The purpose of adding gypsum is only to slow down the process of setting of the cement so that it gets sufficiently hardened.

Uses: Cement has become a commodity of national necessity for any country next to iron and steel. It is used in concrete and reinforced concrete, in plastering and in the construction of bridges, dams and buildings.

ALLEN Session 2019-20

10.10 BIOLOGICAL IMPORTANCE OF MAGNESIUM AND CALCIUM

An adult body contains about 25 g of Mg and 1200 g of Ca compared with only 5 g of iron and 0.06 g of copper. The daily requirement in the human body has been estimated to be 200 – 300 mg. All enzymes that utilise ATP in phosphate transfer require magnesium as the cofactor. The main pigment for the absorption of light in plants is chlorophyll which contains magnesium. About 99 % of body calcium is present in bones and teeth. It also plays important roles in neuromuscular function, interneuronal transmission, cell membrane integrity and blood coagulation. The calcium concentration in plasma is regulated at about 100 mgL^{-1}. It is maintained by two hormones: calcitonin and parathyroid hormone. Do you know that bone is not an inert and unchanging substance but is continuously being solubilised and redeposited to the extent of 400 mg per day in man? All this calcium passes through the plasma.

Summary : The ***s*-Block** of the periodic table constitutes **Group1** (alkali metals) and **Group 2** (alkaline earth metals). They are so called because their oxides and hydroxides are alkaline in nature. The alkali metals are characterised by one *s*-electron and the alkaline earth metals by two *s*-electrons in the valence shell of their atoms. These are highly reactive metals forming monopositive **(M^+)** and dipositve **(M^{2+})** ions respectively.

There is a regular trend in the physical and chemical properties of the alkali metal with increasing atomic numbers. The **atomic** and **ionic** sizes increase and the **ionization enthalpies** decrease systematically down the group.

Somewhat similar trends are observed among the properties of the alkaline earth metals. The first element in each of these groups, lithium in Group 1 and beryllium in Group 2 shows similarities in properties to the second member of the next group. Such similarities are termed as the **'diagonal relationship'** in the periodic table.

As such these elements are anomalous as far as their group characteristics are concerned.

The alkali metals are silvery white, soft and low melting. They are highly reactive. The compounds of alkali metals are predominantly ionic. Their oxides and hydroxides are soluble in water forming strong alkalies. Important compounds of sodium includes sodium carbonate, sodium chloride, sodium hydroxide and sodium hydrogencarbonate. Sodium hydroxide is manufactured by **Castner-Kellner** process and sodium carbonate by **Solvay** process.

The chemistry of alkaline earth metals is very much like that of the alkali metals. However, some differences arise because of reduced atomic and ionic sizes and increased cationic charges in case of alkaline earth metals. Their oxides and hydroxides are less basic than the alkali metal oxides and hydroxides. Industrially important compounds of calcium include calcium oxide (lime), calcium hydroxide (slaked lime), calcium sulphate **(Plaster of Paris),** calcium carbonate (limestone) and cement. **Portland cement** is an important constructional material.

It is manufactured by heating a pulverised mixture of limestone and clay in a rotary kiln. The clinker thus obtained is mixed with some gypsum (2-3%) to give a fine powder of cement. All these substances find variety of uses in different areas.

Monovalent sodium and potassium ions and divalent magnesium and calcium ions are found in large proportions in **biological fluids**. These ions perform important **biological functions** such as maintenance of ion balance and nerve impulse conduction.

ALLEN Session 2019-20

FEW IMPORTANT POINTS

(i) Magnesium Peroxide (MgO_2) and Calcium Peroxide (CaO_2) are obtained by passing H_2O_2 in a suspension of $Mg(OH)_2$ and $Ca(OH)_2$.

(ii) MgO_2 is used as an antiseptic in tooth paste and as a bleaching agent.

(iii) Preperation of NaOH : Caustication of Na_2CO_3 (Gossage's method):

$$Na_2CO_3 + \underset{\text{(suspension)}}{Ca(OH)_2} \rightarrow 2NaOH + CaCO_3\downarrow$$

Since the K_{sp} ($CaCO_3$) < K_{sp} ($Ca(OH)_2$), the reaction shifts towards right.

(iv) As a reagent KOH is less frequently used but in absorption of CO_2, KOH is preferably used compared to NaOH. Because $KHCO_3$ formed is soluble whereas $NaHCO_3$ is sparingly soluble and may therefore choke the tubes of apparatus used.

(v) Calcium hydroxide is used as a mortar.
[Mortar is a mixture of slaked lime (1 Part) and sand (3 Parts) made into paste with water.]

(vi) NaCl is used to prepare freezing mixture in laboratory [Ice-common salt mixture is called freezing mixture and temperature goes down to –23°C.]

(vii) On heating $MgCl_2{\cdot}6H_2O$ undergoes hydrolysis as follows:

$$MgCl_2{\cdot}6H_2O \xrightarrow{\Delta} Mg(OH)Cl + HCl + 5H_2O$$

$$Mg(OH)Cl \longrightarrow MgO + HCl$$

** Hence, Anh. $MgCl_2$ cannot be prepared by heating this hydrate.

** Because of the formation of HCl , sea water cannot be used in marine boilers which corrodes the iron body.

(viii) Anhydrous $MgCl_2$ can be prepared by heating a double salt like $MgCl_2 . NH_4Cl . 6H_2O$ as follows:

$$MgCl_2 . NH_4Cl . 6H_2O \xrightarrow[\Delta]{-H_2O} MgCl_2 . NH_4Cl \xrightarrow[\Delta]{\text{strong}} MgCl_2 + NH_3 + HCl$$

(ix) Sorel Cement is a mixture of MgO and $MgCl_2$ (paste like) which set to hard mass on standing, this is used in dental filling, flooring etc.

(x) Anh. $CaCl_2$ is used in drying gases and organic compounds but not NH_3 or alcohol due to the formation of $CaCl_2 . 8NH_3$ and $CaCl_2 . 4C_2H_5OH$.

(xi) One interesting feature of the solubility of glauber's salt is; when crystallised at below 32.4°C, then $Na_2SO_4 . 10H_2O$ is obtained but above 32.4°C, Na_2SO_4 (anh.) comes out.

(xii) Leblanc Process (Preparation of Na_2CO_3) :

$$NaCl + H_2SO_4 \text{ (conc.)} \xrightarrow{\text{mild heating}} NaHSO_4 + HCl$$

$$NaCl + NaHSO_4 \xrightarrow[\text{heated}]{\text{Strongly}} \underset{\text{(Salt Cake)}}{Na_2SO_4} + HCl$$

$$Na_2SO_4 + 4C \longrightarrow Na_2S + 4CO\uparrow$$

$$Na_2S + CaCO_3 \longrightarrow Na_2CO_3 + CaS$$

ALLEN Session 2019-20

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EXERCISE # O-1

SELECT THE CORRECT ALTERNATIVE (ONLY ONE CORRECT ANSWER)

Alkali metals

1. Cs^+ ions impart violet colour to Bunsen flame. This is due to the fact that the emitted radiations are of -
(A) high energy (B) lower frequencies
(C) longer wave-lengths (D) zero wave number

2. The reaction of an element A with water produces combustible gas B and an aqueous solution of C. When another substance D reacts with this solution C also produces the same gas B. D also produces the same gas even on reaction with dilute H_2SO_4 at room temperature. Element A imparts golden yellow colour to Bunsen flame. Then, A, B, C and D may be identified as
(A) Na, H_2, NaOH and Zn (B) K, H_2, KOH and Zn
(C) K, H_2, NaOH and Zn (D) Ca, H_2, $CaCOH_2$ and Zn

3. Which of the following carbonate of alkali metals has the least thermal stability?
(A) Li_2CO_3 (B) K_2CO_3 (C) Cs_2CO_3 (D) Na_2CO_3

4. The alkali metals which form normal oxide, peroxide as well as super oxides are
(A) Na, Li (B) K, Li (C) Li, Cs (D) K, Rb

5. The pair of compounds, which cannot exist together in a solution is
(A) $NaHCO_3$ and NaOH (B) Na_2CO_3 and NaOH
(C) $NaHCO_3$ and Na_2CO_3 (D) $NaHCO_3$ and H_2O

6. Solution of sodium metal in liquid ammonia is a strong reducing agent due to presence of
(A) solvated sodium ions (B) solvated hydrogen ions
(C) sodium atoms or sodium hydroxide (D) solvated electrons

7. The order of solubility of lithium halides in non-polar solvents follows the order
(A) LiI > LiBr > LiCl > LiF (B) LiF > LiI > LiBr > LiCl
(C) LiCl > LiF > LiI > LiBr (D) LiBr > LiCl > LiF > LiI

8. The salt which finds uses in qualitative inorganic analysis is
(A) $CuSO_4 \cdot 5H_2O$ or $ZnSO_4 \cdot 5H_2O$ (B) $K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$
(C) $Na(NH_4)HPO_4 \cdot 4H_2O$ (D) $FeSO_4 \cdot (NH_4)_2SO_4 \cdot 6H_2O$

9. Fire extinguishers contain
(A) conc. H_2SO_4 solution (B) H_2SO_4 and $NaHCO_3$ solutions
(C) $NaHCO_3$ solution (D) $CaCO_3$ solution

10. $CsBr_3$ contains
(A) Cs–Br covalent bonds (B) Cs^{3+} and Br^- ions
(C) Cs^+ and Br_3^- ions (D) Cs^{3+} and Br_3^{3-} ions

11. $Na + Al_2O_3 \xrightarrow{\text{High temperature}} X \xrightarrow[\text{water}]{CO_2 \text{ in}} Y$; compound Y is
(A) $NaAlO_2$ (B) $NaHCO_3$ (C) Na_2CO_3 (D) Na_2O_2

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12. aq. NaOH + P_4 (white) $\longrightarrow$ PH_3 + X; compound X is

(A) NaH_2PO_2 (B) $NaHPO_4$ (C) Na_2CO_3 (D) $NaHCO_3$

13. When K_2O is added to water, the solution becomes basic in nature because it contains a significant concentration of

(A) K^+ (B) O^{2-} (C) OH^- (D) O_2^{2-}

14. The order of melting point of chlorides of alkali metals is

(A) LiCl > NaCl > KCl < CsCl (B) LiCl > NaCl > KCl > CsCl

(C) NaCl > KCl > CsCl > LiCl (D) LiCl > NaCl > CsCl > KCl

15. NaOH(Solid) + CO $\xrightarrow{200°C}$ X; product X is

(A) $NaHCO_3$ (B) Na_2CO_3 (C) HCOONa (D) H_2CO_3

16. The aqueous solutions of lithium salts are poor conductor of electricity rather than other alkali metals because of

(A) high ionisation energy

(B) high electronegativity

(C) lower ability of Li^+ ions to polarize water molecules

(D) higher degree of hydration of Li^+ ions

17. In $LiAlH_4$, metal Al is present in

(A) anionic part (B) cationic part

(C) in both anionic and cationic part (D) neither in cationic nor in anionic part

18. Which one of the following fluoride of alkali metals has the highest lattice energy?

(A) LiF (B) CsF (C) NaF (D) KF

19. Crown ethers and cryptands form

(A) complexes with alkali metals

(B) salts of alkali metals

(C) hydroxides of alkali metals used for inorganic quantitative analysis

(D) organic salts of alkali metals

20. The correct order of degree of hydration of M^+ ions of alkali metals is

(A) $Li^+ < K^+ < Na^+ < Rb^+ < Cs^+$ (B) $Li^+ < Na^+ < K^+ < Rb^+ < Cs^+$

(C) $Cs^+ < Rb^+ < K^+ < Na^+ < Li^+$ (D) $Cs^+ < Rb^+ < Na^+ < K^+ < Li^+$

21. The commercial method of preparation of potassium by reduction of molten KCl with metallic sodium at 850°C is based on the fact that

(A) potassium is solid and sodium distils off at 850 °C

(B) potassium being more volatile and distils off thus shifting the reaction forward

(C) sodium is less reactive than potassium at 850 °C with respect to Cl_2

(D) sodium has less affinity to chloride ions in the presence of potassium ion

Alkaline earth metals

22. The 'milk of magnesia' used as an antacid is chemically

(A) $Mg(OH)_2$ (B) MgO (C) $MgCl_2$ (D) MgO + $MgCl_2$

ALLEN Session 2019-20

23. An alkaline earth metal (M) gives a salt with chlorine, which is soluble in water at room temperature. It also forms an insoluble sulphate whose mixture with a sulphide of a transition metal is called 'lithopone' - a white pigment. Metal M is

(A) Ca (B) Mg (C) Ba (D) Sr

24. The hydroxide of IInd A metal, which has the lowest value of solubility product (K_{sp}) at normal temperature (25°C) is

(A) $Ca(OH)_2$ (B) $Mg(OH)_2$ (C) $Sr(OH)_2$ (D) $Be(OH)_2$

25. $$\text{(Yellow ppt) T} \xleftarrow{K_2CrO_4/H^+} X \xrightarrow{\text{dil.HCl}} \text{Y(Yellow ppt)} + Z\uparrow \text{(pungent smelling gas)}$$

If X gives green flame test. Then, X is

(A) $MgSO_4$ (B) BaS_2O_3 (C) $CuSO_4$ (D) PbS_2O_3

26. $Mg_2C_3 + H_2O \longrightarrow X$ (organic compound). Compound X is

(A) C_2H_2 (B) CH_4 (C) propyne (D) ethene

27. The hydration energy of Mg^{2+} is

(A) more than that of Mg^{3+} ion (B) more than that of Na^+ ion

(C) more than that of Al^{3+} ion (D) more than that of Be^{2+} ion

28. The correct order of second ionisation potentials (IP) of Ca, Ba and K is

(A) K > Ca > Ba (B) Ba > Ca > K (C) K > Ba > Ca (D) K = Ba = Ca

29. EDTA is used in the estimation of

(A) Mg^{2+} ions (B) Ca^{2+} ions

(C) both Ca^{2+} and Mg^{2+} ions (D) Mg^{2+} ions but not Ca^{2+} ions

30. The correct order of solubility is

(A) $CaCO_3 < KHCO_3 < NaHCO_3$ (B) $KHCO_3 < CaCO_3 < NaHCO_3$

(C) $NaHCO_3 < CaCO_3 < KHCO_3$ (D) $CaCO_3 < NaHCO_3 < KHCO_3$

31. The complex formation tendency of alkaline earth metals decreases down the group because

(A) atomic size increases

(B) availability of empty d and f-orbitals increases

(C) nuclear charge to volume ratio increases (D) all the above

32. The alkaline earth metals, which do not impart any colour to Bunsen flame are

(A) Be and Mg (B) Mg and Ca (C) Be and Ca (D) Be and Ba

33. $Y \xleftarrow{\Delta, 205°C} CaSO_4 \cdot 2H_2O \xrightarrow{\Delta, 120°C} X$. X and Y are respectively

(A) plaster of paris, dead burnt plaster

(B) dead burnt plaster, plaster of paris

(C) CaO and plaster of paris

(D) plaster of paris, mixture of gases

ALLEN Session 2019-20

34. A metal M readily forms water soluble sulphate, and water insoluble hydroxide $M(OH)_2$. Its oxide MO is amphoteric, hard and having high melting point. The alkaline earth metal M must be

(A) Mg (B) Be (C) Ca (D) Sr

35. $$\text{(White ppt) D} \xleftarrow{Na_2CO_3} \text{A} \xrightarrow[\text{(in acetic acid)}]{K_2CrO_4} \text{B(Yellow ppt)}$$
$$\text{A} \xrightarrow{\text{dil.} H_2SO_4} \text{C(White ppt)}$$

If A is the metallic salt, then the white ppt. of D must be of

(A) stronsium carbonate (B) red lead

(C) barium carbonate (D) calcium carbonate

36. $$\text{(Milky Cloud) C} \xleftarrow{CO_2} \text{A} + Na_2CO_3 \longrightarrow \text{B} + \text{C}$$

The chemical formulae of A and B are

(A) NaOH and $Ca(OH)_2$ (B) $Ca(OH)_2$ and NaOH

(C) NaOH and CaO (D) CaO and $Ca(OH)_2$

37. The correct order of basic-strength of oxides of alkaline earth metals is

(A) BeO > MgO > CaO > SrO

(B) SrO > CaO > MgO > BeO

(C) BeO > CaO > MgO > SrO

(D) SrO > MgO > CaO > BeO

38. $$\text{X} \xrightarrow{N_2,\Delta} \text{Y} \xrightarrow{H_2O} \text{Z(colourless gas)} \xrightarrow[CuSO_4]{} \text{T(blue colour)}$$

Then, substances Y and T are

(A) $Y = Mg_3N_2$ and $T = CuSO_4{\cdot}5H_2O$ (B) $Y = Mg_3N_2$ and $T = CuSO_4{\cdot}4NH_3$

(C) $Y = Mg(NO_3)_2$ and $T = CuO$ (D) $Y = MgO$ and $T = CuSO_4{\cdot}4NH_3$

39. Weakest base among KOH, NaOH, $Ca(OH)_2$ and $Zn(OH)_2$ is

(A) $Ca(OH)_2$ (B) KOH

(C) NaOH (D) $Zn(OH)_2$

40. If X and Y are the second ionisation potentials of alkali and alkaline earth metals of same period, then -

(A) X > Y (B) X < Y (C) X = Y (D) X << Y

41. $\text{X} \xrightarrow{CoCl_2} CaCl_2 + Y\uparrow$; the effective ingredient of X is

(A) OCl^- (B) Cl^- (C) OCl^+ (D) OCl_2^-

42. White heavy precipitate is formed when $BaCl_2$ is added to a clear solution of compound **A**. Precipitate is insoluble in dilute HCl. Then, the compound **A** is

(A) a bicarbonate (B) a carbonate

(C) a sulphate (D) a chloride

ALLEN Session 2019-20

43. $X + C + Cl_2 \xrightarrow[\text{of about 1000 K}]{\text{High temperature}} Y + CO$; $Y + 2H_2O \rightarrow Z + 2HCl$

Compound Y is found in polymeric chain structure and is an electron deficient molecule. Y must be

(A) BeO (B) $BeCl_2$ (C) BeH_2 (D) $AlCl_3$

44. $BeCl_2 + LiAlH_4 \longrightarrow X + LiCl + AlCl_3$

(A) X is LiH (B) X is BeH_2

(C) X is $BeCl_2 \cdot 2H_2O$ (D) None

45. The order of thermal stability of carbonates of IIA group is

(A) $BaCO_3 > SrCO_3 > CaCO_3 > MgCO_3$

(B) $MgCO_3 > CaCO_3 > SrCO_3 > BaCO_3$

(C) $CaCO_3 > SrCO_3 > BaCO_3 > MgCO_3$

(D) $MgCO_3 = CaCO_3 > SrCO_3 = BaCO_3$

46. A pair of substances which gives all the same products on reaction with water is

(A) Mg and MgO (B) Sr and SrO (C) Ca and CaH_2 (D) Be and BeO

47. A metal which is soluble in both water and liquid NH_3 separately -

(A) Cr (B) Mn (C) Ba (D) Al

48. $Be_2C + H_2O \longrightarrow BeO + X$

$CaC_2 + H_2O \longrightarrow Ca(OH)_2 + Y$; then X and Y are respectively

(A) CH_4, CH_4 (B) CH_4, C_2H_6 (C) CH_4, C_2H_2 (D) C_2H_2, CH_4

49. Which of the following groups of elements have chemical properties that are most similar

(A) Na, K, Ca (B) Mg, Sr, Ba (C) Be, Al, Ca (D) Be, Ra, Cs

50. $MgBr_2$ and MgI_2 are soluble in acetone because of

(A) Their ionic nature (B) Their coordinate nature

(C) Their metallic nature (D) Their covalent nature

ALLEN Session 2019-20

EXERCISE # O-2

SELECT THE CORRECT ALTERNATIVES (ONE OR MORE THEN ONE CORRECT ANSWERS)

Alkali metals

1. Nitrogen dioxide can be prepared by heating -

(A) KNO_3 (B) $AgNO_3$ (C) $Pb(NO_3)_2$ (D) $Cu(NO_3)_2$

2. Which of the following compounds are not paramagnetic in nature?

(A) KO_2 (B) K_2O_2 (C) Na_2O_2 (D) RbO_2

3. The golden yellow colour associated with NaCl to Bunsen flame can be explained on the basis of

(A) low ionisation potential of sodium

(B) emission spectrum

(C) photosensitivity of sodium

(D) sublimation of metallic sodium of yellow vapours

4. KO_2 finds use in oxygen cylinders used for space and submarines. The fact(s) related to such use of KO_2 is/are

(A) it produces O_2 (B) it produces O_3

(C) it absorbs CO_2 (D) it absorbs both CO and CO_2

5. The compound(s) which have –O–O– bond(s) is/are

(A) BaO_2 (B) Na_2O_2 (C) CrO_5 (D) Fe_2O_3

6. Highly pure dilute solution of sodium in ammonia

(A) shows blue colouration due to solvated electrons

(B) shows electrical conductivity due to both solvated electrons as well as solvated sodium ions

(C) shows red colouration due to solvated electrons but a bad conductor of electricity

(D) produces hydrogen gas or carbonate

7. Sodium metal is highly reactive and can be stored under

(A) toluene (B) kerosene oil (C) alcohol (D) benzene

Alkaline earth metals

8. The compound(s) of IInd A metals, which are amphoteric in nature is/are

(A) BeO (B) MgO (C) $Be(OH)_2$ (D) $Mg(OH)_2$

9. The correct statement is/are

(A) $BeCl_2$ is a covalent compound (B) $BeCl_2$ is an electron deficient molecule

(C) $BeCl_2$ can form dimer (D) the hybrid state of Be in $BeCl_2$ is sp^2

10. Which of the following substance(s) is/are used in laboratory for drying purposes?

(A) anhydrous P_2O_5 (B) graphite

(C) anhydrous $CaCl_2$ (D) Na_3PO_4

11. Na_2SO_4 is water soluble but $BaSO_4$ is insoluble because

(A) the hydration energy of Na_2SO_4 is higher than that of its lattice energy

(B) the hydration energy of Na_2SO_4 is less than that of its lattice energy

(C) the hydration energy of $BaSO_4$ is less than that of its lattice energy

(D) the hydration energy of $BaSO_4$ is higher than that of its lattice energy

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Advanced Batch\Chem\Sheet\Salt Analysis & Heating Effect\(III) s-Block\Eng.p65

12. Which of the following statements are false?

(A) $BeCl_2$ is a linear molecule in the vapour state but it is polymeric form in the solid state

(B) Calcium hydride is called hydrolith.

(C) Carbides of both Be and Ca react with water to form acetylene

(D) Oxides of both Be and Ca are amphoteric.

13. Which of the following are ionic carbides?

(A) CaC_2 (B) Al_4C_3 (C) SiC (D) Be_2C

14. Which of the following orders are **CORRECT** :

(A) AgCl > AgF : Covalent character order

(B) BaO > BaF_2 : Melting point order

(C) BeF_2 > BaF_2 : Solubility order

(D) $LiNO_3$ < $RbNO_3$: Thermal stability order

15. Which of the following statements are **CORRECT** :

(A) Mg is present in chlorophyll

(B) Alkaline earth metals does not form super oxide

(C) $NaHCO_3$ is known as baking soda

(D) Permanent hardness of water is removed by boiling

16. Which of the following carbides on hydrolysis does not form methane :

(A) Be_2C (B) CaC_2 (C) SrC_2 (D) Mg_2C_3

17. Select the incorrect order for given properties :

(A) Thermal stability : $BaSO_4 > SrSO_4 > CaSO_4$

(B) Solubility : $BaSO_4 > SrSO_4 > CaSO_4$

(C) Thermal stability : $Li_2CO_3 < Na_2CO_3 < K_2CO_3$

(D) Solubility : $Li_2CO_3 > Na_2CO_3 > K_2CO_3$

18. The correct statement(s) is/are

(A) Mg cannot form complexes

(B) Be can form complexes due to a very small atomic size

(C) the first ionisation potential of Be is higher than that of Mg

(D) Mg forms an alkaline hydroxide while Be forms amphoteric oxides

19. Which of the following is are the characteristic of barium?

(A) It produce water soluble sulphide, sulphite and sulphate

(B) It is a silvery white metal

(C) It forms $Ba(NO_3)_2$ which is used in preparation of green fire

(D) It produce blue-black solution in liquid ammonia

ALLEN Session 2019-20

EXERCISE # S-1

NUMERIC GRID TYPE QUESTIONS :

1. Find the number of compounds from the following in which the element in the anionic part is in the minimum oxidation state of it

LiH, Mg_3Bi_2, Al_4C_3, Ca_3P_2, BaO_2

2. How many nitrate groups are present in 1 molecule of Basic beryllium nitrate?

3. Consider the following order :

(1) $CH_4 < CCl_4 < CF_4$: E.N. of central atom C

(2) $Mg^{+2} < K^+ < S^{-2} < Se^{-2}$: Ionic radius

(3) $Be^{+2}_{(aq)} > Mg^{+2}_{(aq)} > Ca^{+2}_{(aq)}$: Ionic mobility

(4) $Be^{+2} > Li^+ > Al^{+3}$: Hydrated size

(5) Be > Li > Cs : Reducing power

(6) $F^{\ominus}_{(aq)} > Cl^{\ominus}_{(aq)} > Br^{\ominus}_{(aq)}$: Electrical conductance in infinite dilute solution

Then calculate value of $|x - y|^2$, where x and y are correct and incorrect orders respectively.

4. Consider the following elements :

Li, Cs, Mg, Pb, Al, N

- x = number of elements which can form MO type of oxides.
- y = the highest oxidation state shown by any one of them.
- z = the number of elements which can form amphoteric oxide(s).

Find the sum of x, y and z.

Fill your answer as sum of digits till you get the single digit answer.

5. Find the number of s-block elements which can produce ammoniated cation and ammoniated electron with liquid ammonia.

Li, Na, K, Rb, Cs, Ca, Sr, Ba

6. How many of the following metal chlorides impart characteristic colour to the oxidising flame.

$LiCl$, $NaCl$, KCl, $BeCl_2$, $MgCl_2$, $CaCl_2$, $SrCl_2$, $BaCl_2$

ALLEN Session 2019-20

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EXERCISE # S-2

COMPREHENSION BASED QUESTIONS

Comprehension # 1

$$A \xrightarrow{\Delta} B\text{ (oxide)} + CO_2$$
$$B + H_2O \longrightarrow C$$
$$C + CO_2 \longrightarrow A\text{ (milky)}$$
$$C + NH_4Cl \xrightarrow{\Delta} D\text{ (gas)}$$
$$D + H_2O + CO_2 \longrightarrow E$$
$$E + NaCl \longrightarrow F$$
$$F \xrightarrow{\Delta} Na_2CO_3 + CO_2 + H_2O$$

1. A is :

(A) $Ca(HCO_3)_2$ (B) $CaCO_3$ (C) CaO (D) Na_2CO_3

2. B and C are :

(A) CaO, $Ca(OH)_2$ (B) $Ca(OH)_2$, $CaCO_3$

(C) $CaCO_3$, $Ca(OH)_2$ (D) $Ca(OH)_2$, CaO

3. D, E and F are :

(A) NH_3, NH_4Cl, NH_4HCO_3 (B) NH_3, NH_4HCO_3, $NaHCO_3$

(C) NH_4HCO_3, Na_2CO_3, $NaHCO_3$ (D) None

Comprehension # 2

Alkali metals readily react with oxyacids forming corresponding salts like M_2CO_3, $MHCO_3$, MNO_3, M_2SO_4 etc. with evolution of hydrogen. They also dissolve in liquid NH_3 but without the evolution of hydrogen. The colour of its dilute solution is blue but when it is heated and concentrated then its colour becomes bronze.

4. Among the nitrate of alkali metals which one can be decomposed to its oxide easily?

(A) $NaNO_3$ (B) KNO_3 (C) $LiNO_3$ (D) $RbNO_3$

5. Among the carbonates of alkali metals which one has highest stability?

(A) Cs_2CO_3 (B) Rb_2CO_3 (C) K_2CO_3 (D) Na_2CO_3

6. Which of the following statement about the sulphate of alkali metal is correct?

(A) Except Li_2SO_4 all sulphate of other alkali metals are soluble in water

(B) All sulphates of alkali metals except lithium sulphate forms alum.

(C) The sulphates of alkali metals cannot be hydrolysed.

(D) All of these

ALLEN Session 2019-20

7. Which of the following statement about solution of alkali metals in liquid ammonia is correct?

(A) The solution have strong oxidizing properties.

(B) Both the dilute solution as well as concentrated solution are paramagnetic in nature

(C) Charge transfer is the responsible for the colour of the solution

(D) None of these

8. Which metal bicarbonates does not exist in solid state?

(i) $LiHCO_3$ (ii) $Ca(HCO_3)_2$ (iii) $Zn(HCO_3)_2$

(iv) $NaHCO_3$ (v) $AgHCO_3$

(A) (i), (ii), (iii), (v) (B) (i), (ii), (iii) (C) (i), (ii), (v) (D) (ii), (iii), (iv)

MATCH THE COLUMN :

9.

Column-I	**Column-II**
(A) Hydrolith	(P) Contain Ca
(B) Nitrolim	(Q) Used as a fertilizer
(C) Dolomite	(R) Used to prepare H_2
(D) Pearl's ash	(S) Contain potassium

10.

Column-I	**Column-II**
(A) Metal sulphate $\xrightarrow{\Delta}$ metal oxide + SO_2 + O_2	(P) Ba
(B) Metal cation + $K_2CrO_4 \longrightarrow$ yellow ppt	(Q) Sr
(C) Metal + $NH_3 \xrightarrow{\text{(liquid)}}$ blue solution	(R) Na
(D) MCl_2 + conc. $H_2SO_4 \longrightarrow$ white ppt.	(S) Mg

MATCH THE CODE :

11.

List–I	**List–II**
(P) CaH_2	(1) Paramagnetic anion
(Q) K_2O_2	(2) Homodiatomic, diamagnetic anion
(R) KO_2	(3) Neutral aqueous solution
(S) NaCl	(4) Gives hydrogen on hydrolysis

Codes :

	P	**Q**	**R**	**S**
(A)	3	2	1	4
(B)	4	2	3	1
(C)	4	3	2	1
(D)	4	2	1	3

ALLEN Session 2019-20

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12. **Column-I** | **Column-II**

Column-I	Column-II
(P) Solvay process used for	(1) NaCl
(Q) Evolve $CO_2\uparrow$ on heating	(2) Na_2O_2
(R) aq. soln. is neutral towards litmus	(3) $NaHCO_3$
(S) Oxone	(4) Na_2CO_3

Codes :

	P	Q	R	S
(A)	3,4	3	1	2
(B)	4,1	1	3	2
(C)	2,3	4	1	3
(D)	2,4	1	3	4

ASSERATION & REASONING :

Questions given below consist of two statements each printed as Assertion (A) and Reason (R); while answering these questions you are required to choose any one of the following four responses:

(A) if both (A) and (R) are true and (R) is the correct explanation of (A)

(B) if both (A) and (R) are true but (R) is not correct explanation of (A)

(C) if (A) is true but (R) is false

(D) if (A) is false and (R) is true

13. **Assertion** : Beryllium does not impart any characteristic colour to the bunsen flame.

Reason : Due to its very high ionization energy, beryllium requires a large amount of energy for exciation of the electrons.

14. **Assertion** : In fused state, calcium chloride cannot be used to dry alcohol or NH_3.

Reason : Anhy. $CaCl_2$ is not a good desiccant.

15. **Assertion** : Diagonal relationship is shown between Be and Al.

Reason : Ionic potential of Be is almost the same as that of Al.

16. **Assertion** : Beryllium halides dissolve in organic solvents.

Reason : Beryllium halides are ionic in character.

17. **Assertion** : $BeCl_2$ fumes in moist air.

Reason : $BeCl_2$ reacts with moisture to form HCl gas.

18. **Assertion** : Calcium carbide on hydrolysis gives methane.

Reason : Calcium carbide contains C_2^{2-} anion.

19. **Assertion** : When CO_2 is passed through lime water, it first turns milky and then the solution becomes clear when the passage of CO_2 is continued.

Reason : The milkiness is due to the formation of insoluble $CaCO_3$ which then changes to soluble $Ca(HCO_3)_2$ when excess of CO_2 is present.

20. **Assertion** : $MgCO_3$ is soluble in water when a current of CO_2 is passed.

Reason : The solubility of $MgCO_3$ is due to the formation of $Mg(HCO_3)_2$.

ALLEN Session 2019-20

MATCHING LIST TYPE 1 × 3 Q. (THREE LIST TYPE Q.)

The following column 1, 2, 3 represent elements of s block and their different oxide formation abilities. Answer the questions that follow

Column-1 - Elements of s-Block

Column-2 - Product formed on reaction with excess oxygen

Column-3 - Characteristics of species form on reaction with excess oxygen

Column - 1 Elements	**Column - 2 Product formed on reaction with excess oxygen**	**Column - 3 Characteristics of species form on reaction with excess oxygen**
(I) Na	(A) Superoxide	(P) Paramagnetic
(II) Ba	(B) Peroxide	(Q) Diamagnetic
(III) K	(C) Monooxide	(R) Bond order = 1.5
(IV) Ca	(D) Dioxide	(S) Bond order = 1

21. Which of the following is an **INCORRECT** match.

(A) (I), (B), (QS) (B) (II), (B), (QS) (C) (II), (A), (PR) (D) (IV), (C), (Q)

22. Which of the following matches will result in species having magnetic moment equal to that of Mn^{+6}

(A) I, (B), QS (B) IV, B, (QS) (C) IV, (A), (PR) (D) III, (A), (PR)

23. On reaction with oxygen, which of the following combination is possible

(A) I, (A, C), (P, Q) (B) I, (B), (Q)

(C) II, (A, B), (P, Q, S) (D) IV, (B, C), Q

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EXERCISE # JEE-MAIN

1. A metal M readily forms its sulphate MSO_4 which is water soluble. It forms oxide MO which becomes inert on heating. It forms insoluble hydroxide which is soluble in NaOH. The metal M is:- **[AIEEE-2002]**

(1) Mg (2) Ba (3) Ca (4) Be

2. KO_2 is used in space and submarines because it **[AIEEE-2002]**

(1) Absorbs CO_2 and increase O_2 concentration

(2) Absorbs moisture

(3) Absorbs CO_2

(4) Produces ozone

3. In curing cement plasters, water is sprinkled from time to time. This helps in :- **[AIEEE-2003]**

(1) Hydrating sand and gravel mixed with cement

(2) Converting sand into silicate

(3) Developing interlocking needle like crystals of hydrated silicates

(4) Keeping it cool

4. The solubilities of carbonates decreases down the magnesium group due to decrease in- **[AIEEE-2003]**

(1) Inter-ionic attraction

(2) Entropy of solution formation

(3) Lattice energy of solids

(4) Hydration energy of cations

5. The substance not likely to contain $CaCO_3$ is :- **[AIEEE-2003]**

(1) Sea shells (2) Dolomite

(3) A marble statue (4) Calcined gypsum

6. One mole of magnesium nitride on reaction with excess of water gives :- **[AIEEE-2004]**

(1) Two mole of HNO_3 (2) Two mole of NH_3

(3) 1 mole of NH_3 (4) 1 mole of HNO_3

7. Berylium and aluminium exhibit many properties which are similar. But the two elements differ in - **[AIEEE-2004]**

(1) Exhibiting maximum covalency in compounds

(2) Forming polymeric hydrides

(3) Forming covalent halides

(4) Exhibiting amphoteric nature in their oxides.

8. The ionic mobility of alkali metal ioins in aqueous solution is maximum for :- **[AIEEE-2006]**

(1) Rb^+ (2) Li^+ (3) Na^+ (4) K^+

9. The products obtained on heating $LiNO_3$ will be :- **[AIEEE-2011]**

(1) $LiNO_2 + O_2$ (2) $Li_2O + NO_2 + O_2$

(3) $Li_3N + O_2$ (4) $Li_2O + NO + O_2$

ALLEN Session 2019-20

10. What is the best description of the change that occurs when $Na_2O(s)$ is dissolved in water ? **[AIEEE-2011]**

(1) Oxidation number of sodium decreases

(2) Oxide ion accepts sharing in a pair of electrons

(3) Oxide ion donates a pair of electrons

(4) Oxidation number of oxygen increases

11. Which of the following on thermal-decomposition yields a basic as well as an acidic oxide ? **[AIEEE-2012]**

(1) NH_4NO_3 (2) $NaNO_3$ (3) $KClO_3$ (4) $CaCO_3$

12. Fire extinguishers contain H_2SO_4 and which one of the following :-**[JEE MAIN-2012, Online]**

(1) $CaCO_3$ (2) $NaHCO_3$ and Na_2CO_3

(3) Na_2CO_3 (4) $NaHCO_3$

13. Based on lattice energy and other considerations, which one of the following alkali metal chloride is expected to have the highest melting point ? **[JEE MAIN-2012, Online]**

(1) RbCl (2) LiCl (3) KCl (4) NaCl

14. Which one of the following will react most vigorously with water ? **[JEE MAIN-2012, Online]**

(1) Li (2) K (3) Rb (4) Na

15. A metal M on heating in nitrogen gas gives Y. Y on treatment with H_2O gives a colourless gas which when passed through $CuSO_4$ solution gives a blue colour, Y is :- **JEE MAIN-2012, Online]**

(1) NH_3 (2) MgO (3) Mg_3N_2 (4) $Mg(NO_3)_2$

16. The correct statement for the molecule, CsI_3, is : **[JEE(Main)-2014]**

(1) it contains Cs^{3+} and I^- ions

(2) it contains Cs^+, I^- and lattice I_2 molecule

(3) it is a covalent molecule

(4) it contains Cs^+ and I_3^- ions

17. Which of the following statements about Na_2O_2 is **not** correct ? **[JEE MAIN-2014, Online]**

(1) Na_2O_2 oxidises Cr^{3+} to CrO_4^{2-} in acid medium

(2) It is diamagnetic in nature

(3) It is the super oxide of sodium

(4) It is a derivative of H_2O_2

18. Amongst LiCl, RbCl, $BeCl_2$ and $MgCl_2$ the compounds with the greatest and the least ionic character, respectively are : **[JEE MAIN-2014, Online]**

(1) RbCl and $MgCl_2$ (2) LiCl and RbCl

(3) $MgCl_2$ and $BeCl_2$ (4) RbCl and $BeCl_2$

19. The correct order of thermal stability of hydroxides is : **JEE(Main)Online-2015]**

(1) $Ba(OH)_2 < Sr(OH)_2 < Ca(OH)_2 < Mg(OH)_2$ (2) $Mg(OH)_2 < Sr(OH)_2 < Ca(OH)_2 < Ba(OH)_2$

(3) $Mg(OH)_2 < Ca(OH)_2 < Sr(OH)_2 < Ba(OH)_2$ (4) $Ba(OH)_2 < Ca(OH)_2 < Sr(OH)_2 < Mg(OH)_2$

20. Which of the alkaline earth metal halides given below is essentially covalent in nature :-

(1) $SrCl_2$ (2) $CaCl_2$ (3) $BeCl_2$ (4) $MgCl_2$

JEE(Main)Online-2015]

21. Which one of the following alkaline earth metal sulphates has its hydration enthalpy greater than its lattice enthalpy ? **[JEE(Main)-2015]**

(1) $BaSO_4$ (2) $SrSO_4$ (3) $CaSO_4$ (4) $BeSO_4$

22. The commercial name for calcium oxide is : **[JEE(Main)-2016]**

(1) Quick lime (2) Milk of lime (3) Limestone (4) Slaked lime

23. The correct order of the solubility of alkaline-earth metal sulphates in water is : **[JEE(Main)-2016]**

(1) Mg < Sr < Ca < Ba (2) Mg < Ca < Sr < Ba

(3) Mg > Ca > Sr > Ba (4) Mg > Sr > Ca > Ba

24. The main oxides formed on combustion of Li, Na and K in excess of air are respectively :

(1) Li_2O, Na_2O_2 and KO_2 (2) Li_2O, Na_2O and KO_2 **[JEE(Main)-2016]**

(3) LiO_2, Na_2O_2 and K_2O (4) Li_2O_2, Na_2O_2 and KO_2

25. Both lithium and magnesium display several similar properties due to the diagonal relationship ; however, the one which is incorrect is : **[JEE(Main)-2017]**

(1) Both form basic carbonates

(2) Both form soluble bicarbonates

(3) Both form nitrides

(4) Nitrates of both Li and Mg yield NO2 and O2 on heating

26. Which of the following ions does **not** liberate hydrogen gas on reaction with dilute acids?

(1) Ti^{2+} (2) Cr^{2+} **[JEE(Main)-2017 on line]**

(3) Mn^{2+} (4) V^{2+}

27. In KO_2, the nature of oxygen species and the oxidation state of oxygen atom are, respectively

[JEE(Main)ONLINE-2018]

(1) Superoxide and –1/2

(2) Oxide and –2

(3) Peroxide and –1/2

(4) Superoxide and –1

ALLEN Session 2019-20

EXERCISE # JEE-ADVANCED

1. The species that do not contain peroxide linkage are - **[JEE 1992]**

(A) PbO_2 (B) H_2O_2 (C) SrO_2 (D) BaO_2

2. Read the following statement and explanation and answer as per the options given below :

Statement-1 : The alkali metals can form ionic hydrides which contain the hydride ion H^-.

Statement-2 : The alkali metals have low electronegativity ; their hydrides conduct electricity when fused and liberate hydrogen at the anode. **[JEE 1994]**

(A) Both 1 and 2 are true and 2 is the correct explanation of 1.

(B) Both 1 and 2 are true but 2 is not the correct explanation of 1.

(C) 1 is true but 2 is false.

(D) 1 is false but 2 is true.

3. The following compounds have been arranged in order of their increasing thermal stabilities. Identify the correct order. **[JEE 1996]**

K_2CO_3(I) $MgCO_3$(II) $CaCO_3$(III) $BeCO_3$(IV)

(A) I < II < III < IV (B) IV < II < III < I

(C) IV < II < I < III (D) II < IV < III < I

4. Property of all the alkaline earth metals that increase with their atomic number is - **[JEE 1997]**

(A) ionisation energy (B) solubility of their hydroxides

(C) solubility of their sulphate (D) electronegativity

5. Highly pure dilute solution of sodium in liquid ammonia - **[JEE 1998]**

(A) shows blue colour (B) exhibits electrical conductivity

(C) produces sodium amide (D) produces hydrogen gas

6. The set representing the correct order of first ionization potential is - **[JEE 2001S]**

(A) K > Na > Li (B) Be > Mg > Ca (C) B > C > N (D) Ge > Si > C

Asseration and Reason

7. This questions contains statement-1 (asseration) and statement-2 (reason) and has 4 choices (a), (b), (c) and (d) out of which only one is correct.

Statement-1 : Alkali metals dissolve in liquid ammonia to give blue solutions. because.

Statement-1 : Alkali metals is liquid ammonia give solvated species of the type $[M(NH_3)_n]^+$ (M = alkali metals). **[JEE 2007]**

(A) Both 1 and 2 are true and 2 is the correct explanation of 1.

(B) Both 1 and 2 are true but 2 is not the correct explanation of 1.

(C) 1 is true but 2 is false.

(D) 1 is false but 2 is true.

8. The compound(s) formed upon combustion of sodium metal in excess air is (are) **[JEE 2009]**

(A) Na_2O_2 (B) Na_2O (C) NaO_2 (D) NaOH

ALLEN Session 2019-20

ANSWER KEY

EXERCISE # O-1

1. (A)	**2.** (A)	**3.** (A)	**4.** (D)
5. (A)	**6.** (D)	**7.** (A)	**8.** (C)
9. (B)	**10.** (C)	**11.** (C)	**12.** (A)
13. (C)	**14.** (C)	**15.** (C)	**16.** (D)
17. (A)	**18.** (A)	**19.** (A)	**20.** (C)
21. (B)	**22.** (A)	**23.** (C)	**24.** (D)
25. (B)	**26.** (C)	**27.** (B)	**28.** (A)
29. (C)	**30.** (D)	**31.** (A)	**32.** (A)
33. (A)	**34.** (B)	**35.** (C)	**36.** (B)
37. (B)	**38.** (B)	**39.** (D)	**40.** (A)
41. (A)	**42.** (C)	**43.** (B)	**44.** (B)
45. (A)	**46.** (C)	**47.** (C)	**48.** (C)
49. (B)	**50.** (D)		

EXERCISE # O-2

1. (B),(C), (D)	**2.** (B),(C)	**3.** (A),(B)	**4.** (A),(C)
5. (A),(B),(C)	**6.** (A), (B)	**7.** (A), (B),(D)	**8.** (A),(C)
9. (A),(B),(C)	**10.** (A),(C)	**11.** (A), (C)	**12.** (C),(D)
13. (A),(B),(D)	**14.** (A),(B),(C),(D)	**15.** (A),(B),(C)	**16.** (B),(C),(D)
17. (B), (D)	**18.** (B),(C),(D)	**19.** (B),(C),(D)	

EXERCISE # S-1

1. (4)	**2. (6)**	**3. (4)**	**4. (10), OMR - (1)**
5. (8)			
6. (6)			

Except Be & Mg other s-block metals impart characteristic colour to oxidising flame.

ALLEN Session 2019-20

EXERCISE # S-2

- **Comprehension Based Questions**

Comprehension # 1

1. (B) **2.** (A) **3.** (B)

Comprehension # 2

4. (C) **5.** (A) **6.** (D) **7.** (D) **8.** (A)

- **Match the column**

9. (A) $\rightarrow$ P,R ; (B) $\rightarrow$ P,Q ; (C) $\rightarrow$ P ; (D) $\rightarrow$ S

10. (A) $\rightarrow$ P,Q,S ; (B) $\rightarrow$ P,Q; (C) $\rightarrow$ P,Q,R ; (D) $\rightarrow$ P,Q

- **Match the code**

11. (D) **12.** (A)

- **Asseration & Reasoning**

13. A	**14. C**	**15. A**	**16. C**
17. A	**18. D**	**19. A**	**20. A**
21. C	**22. D**	**23. B**	

EXERCISE # JEE-MAIN

1. (4)	**2.** (1)	**3.** (3)	**4.** (4)
5. (4)	**6.** (2)	**7.** (1)	**8.** (1)
9. (2)	**10.** (3)	**11.** (4)	**12.** (4)
13. (4)	**14.** (3)	**15.** (3)	**16.** (4)
17. (3)	**18.** (4)	**19.** (3)	**20.** (3)
21. (4)	**22.** (1)	**23.** (3)	**24.** (1)
25. (1)	**26.** (3)	**27.** (1)	

EXERCISE # JEE-ADVANCED

1. A	**2. A**	**3. B**	**4. B**
5. A,B	**6. B**	**7. B**	**8. A,B**

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d-BLOCK COMPOUNDS

TRANSITION ELEMENTS

Definition : They one often called 'transition elements' because their position in the periodic table is between s-block and p-block elements

Typically, the transition elements have incompletely filled d-level. Since Zn group has d^{10} configuration in their ground state as well as in stable oxidation state, they are not considered as transition elements but they ared-block elements.

1st Series										
	Sc	Ti	V	**Cr**	Mn	Fe	Co	Ni	**Cu**	Zn
Z	21	22	23	24	25	26	27	28	29	30
4s	2	2	2	1	2	2	2	2	1	2
3d	1	2	3	5	5	6	7	8	10	10

2nd Series										
	Y	Zr	**Nb**	**Mo**	**Tc**	**Ru**	**Rh**	**Pd**	**Ag**	Cd
Z	39	40	41	42	43	44	45	46	47	48
5s	2	2	1	1	2	1	1	0	1	2
4d	1	2	4	5	5	7	8	10	10	10

3rd Series										
	La	Hf	Ta	W	Re	Os	Ir	**Pt**	**Au**	Hg
Z	57	72	73	74	75	76	77	78	79	80
6s	2	2	2	2	2	2	2	1	1	2
5d	1	2	3	4	5	6	7	9	10	10

4th Series										
	Ac	Rf	Db	Sg	Bh	Hs	Mt	Ds	**Rg**	Uub
Z	89	104	105	106	107	108	109	110	111	112
7s	2	2	2	2	2	2	2	2	1	2
6d	1	2	3	4	5	6	7	8	10	10

General Characteristics **:**

(i) **Metallic character :** They are all metal and good conductor of heat & electricity

(ii) **Electronic configuration :** $(n-1)d^{1-10}ns^{1-2}$

	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
others are as usual				$4s^1$					$4s^1$	
				$3d^5$					$3d^{10}$	

(iii) **M.P.** Cr, Mo, W $\longrightarrow$ Maximum; 6 no. of unpaired e^-s are involved in metallic bonding

Zn, Cd, Hg — lowest m.p. due to no unpaired e^- for metallic bonding

Session 2019-20

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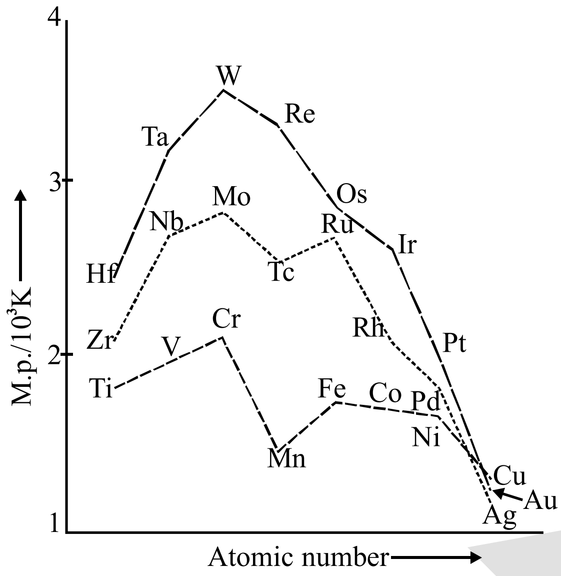
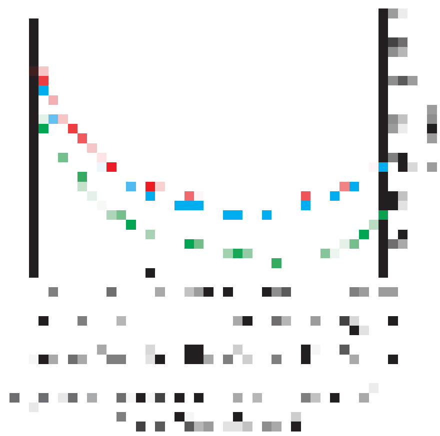


Fig. : Trends in melting points of transition elements

(iv) Variation in atomic radius :

Sc ⟶ Mn Fe Co Ni Cu Zn

decreases   remains same   increases again



(v) Ionisation energy :

1st, 2nd, 3rd IE's are increasing from left to right for 1st Transition series, but not regularly.

For 2nd IE Cr > Fe > Mn and Cu > Zn

For 3rd IE Mn > Cr > Fe and Zn has highest.

(vi) <u>DENSITY</u>

(a) The atomic volume of the transition elements are low, compared with s-block, so their density is comparatively high (D = M/V)

(b) Os (22.57 gm cm^{-3}) and Ir (22.61 gm cm^{-3}) have highest density.

(c) In all the groups (except IIIB) there is normal increase in density from 3d to 4d series, and from 4d to 5d, it increases just double. Due to lanthanide contraction **Ex. :** Ti < Zr << Hf

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(d) **In 3d series**

	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
Density/g cm^{-3}	3.43	4.1	6.07	7.19	7.21	7.8	8.7	8.9	8.9	7.1

(e) In 3d series highest density – Cu lowest density – Sc

(f) Some important orders of density

Fe < Ni < Cu Fe < Cu < Au Fe < Hg < Au

Electronic configurations and some other properties of the first series of transition elements

Element		Sc	Ti	v	Cr	Mn	Fe	Co	Ni	Cu	Zn
Atomic number		21	22	23	24	25	26	27	28	29	30
Electronic configuration											
	M	$3d^1 4s^2$	$3d^2 4s^2$	$3d^3 4s^2$	$3d^5 4s^2$	$3d^5 4s^2$	$3d^6 4s^2$	$3d^7 4s^2$	$3d^{10} 4s^2$	$3d^{10} 4s^1$	$3d^{10} 4s^2$
	M^+	$3d^1 4s^1$	$3d^2 4s^1$	$3d^3 4s^1$	$3d^5$	$3d^5 4s^1$	$3d^6 4s^1$	$3d^7 4s^1$	$3d^8 4s^1$	$3d^{10}$	$3d^{10} 4s^1$
	M^{2+}	$3d^1$	$3d^2$	$3d^3$	$3d^4$	$3d^5$	$3d^6$	$3d^7$	$3d^8$	$3d^9$	$3d^{10}$
	M^{3+}	[Ar]	$3d^1$	$3d^2$	$3d^3$	$3d^4$	$3d^5$	$3d^6$	$3d^7$	–	–
Enthalpy of atomisation, $\Delta_a H^\ominus$/kJ mol^{-1}											
		326	473	515	397	281	416	425	430	339	126
Ionisation Enthalpy, $\Delta_1 H^\ominus$/kJ mol^{-1}											
$\Delta_1 H^\ominus$	I	631	656	650	653	717	762	758	736	745	906
	II	1235	1309	1414	1592	1509	1561	1644	1752	1958	1734
	III	2393	2657	2833	2990	3260	2962	3243	3402	3556	3829
Metallic/Ionic	M	164	147	135	129	137	126	125	125	128	137
radii/pm	M^{2+}	–	–	79	82	82	77	74	70	73	75
	M^{3+}	73	67	64	62	65	65	61	60	–	–
Standard											
electrode	M^{2+}/M	–	–1.63	–1.18	–0.90	–1.18	–0.44	–0.28	–0.25	+0.34	–0.76
potential $E^\ominus$/V	M^{3+}/M^{2+}	–	–0.37	–0.26	–0.41	+1.57	+0.77	+1.97	–	–	–
Density/g cm^{-3}		3.43	4.1	6.07	7.19	7.21	7.8	8.7	8.9	8.9	7.1

VARIABLE OXIDATION STATES POSSIBLE :

(1) The elements which give the greatest number of oxidation states occur in or near the middle of the series. Manganese, for example, exhibits all the oxidation states from +2 to +7.

(2) The lesser number of oxidation states at the extreme ends stems from either too few electrons to lose or share (Sc, Ti) or too many d electrons (hence fewer orbitals available in which to share electrons with others) for higher valence (Cu, Zn).

(3) Thus, early in the series scandium(II) is virtually unknown and titanium (IV) is more stable than Ti(III) or Ti(II).

(4) At the other end, the only oxidation state of zinc is +2 (no d electrons are involved).

(5) The maximum oxidation states of reasonable stability correspond in value to the sum of the s and d electrons upto manganese ($Ti^{IV}O_2$, $V^VO_2^{\ +}$, $Cr^{VI}O_4^{2-}$, $Mn^{VII}O_4^-$) followed by a rather abrupt decrease in stability of higher oxidation states, so that the typical species to follow are $Fe^{II,III}$, $Co^{II,III}$, Ni^{II}, $Cu^{I,II}$, Zn^{II}.

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(6) The variability of oxidation states, a characteristic of transition elements, arises out of incomplete filling of d orbitals in such a way that their oxidation states differ from each other by unity, e.g., V^{II}, V^{III}, V^{IV}, V^{V}.

(7) This is in contrast with the variability of oxidation states of non transition elements where oxidation states normally differ by a unit of two.

(8) An interesting feature in the variability of oxidation states of the d–block elements is noticed among the groups (groups 4 through 10).

(9) In group 6, Mo(VI) and W(VI) are found to be more stable than Cr(VI). Thus Cr(VI) in the form of dichromate in acidic medium is a strong oxidising agent, whereas MoO_3 and WO_3 are not.

(10) Low oxidation states are found when a complex compound has ligands capable of π-acceptor character in addition to the σ-bonding. For example, in $Ni(CO)_4$ and $Fe(CO)_5$, the oxidation state of nickel and iron is zero.

(11) As the oxidation number of a metal increases, ionic character decreases. In the case of Mn, Mn_2O_7 is a covalent green oil. Even CrO_3 and V_2O_5 have low melting points. In these higher oxides, the acidic character is predominant. Thus, Mn_2O_7 gives $HMnO_4$ and CrO_3 gives H_2CrO_4 and $H_2Cr_2O_7$. V_2O_5 is, however, amphoteric though mainly acidic and it gives VO_4^{3-} as well as VO_2^{+} salts. In vanadium there is gradual change from the basic V_2O_3 to less basic V_2O_4 and to amphoteric V_2O_5. V_2O_4 dissolves in acids to give VO^{2+} salts. Similarly, V_2O_5 reacts with alkalies as well as acids to give VO_4^{3-} and VO_2^{+} respectively. The well characterised CrO is basic but Cr_2O_3 is amphoteric.

Oxidation states of the I^st transition series

most common ones are in bold types :

Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
			+1					+1	
	+2	+2	+2	**+2**	**+2**	**+2**	**+2**	**+2**	**+2**
+3	**+3**	**+3**	**+3**	+3	**+3**	**+3**	+3		
	+4	+4	+4	**+4**	+4	+4	+4		
		+5	+5	+5					
			+6	+6	+6				
				+7					

Trends in stability of higher oxidation state :

(1) Table shows the stable halides of the 3*d* series of transition metals. The highest oxidation numbers are achieved in TiX_4 (tetrahalides), VF_5 and CrF_6. The +7 state for Mn is not represented in simple halides but MnO_3F is known, and beyond Mn no metal has a trihalide except FeX_3 and CoF_3.

(2) The ability of fluorine to stabilise the highest oxidation state is due to either higher lattice energy as in the case of CoF_3, or higher bond enthalpy terms for the higher covalent compounds, e.g., VF_5 and CrF_6.

(3) Although V^V is represented only by VF_5, the other halides, however, undergo hydrolysis to give oxohalides, VOX_3.

(4) Another feature of fluorides is their instability in the low oxidation states e.g., VX_2 (X = Cl, Br or I)

Formulas of halides of 3d-metals

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Oxidation Number									
+ 6			CrF_6						
+ 5		VF_5	CrF_5						
+ 4	TiX_4	VX^{I}_4	CrX_4	MnF_4					
+ 3	TiX_3	VX_3	CrX_3	MnF_3	FeX^{I}_3	CoF_3			
+ 2	TiX_2^{III}	VX_2	CrX_2	MnX_2	FeX_2	CoX_2	NiX_2	CuX^{II}_2	ZnX_2
+ 1								CuX^{III}	

Key : $X = F \rightarrow I$; $X^{I} = F \rightarrow Br$; $X^{II} = F \rightarrow Cl$; $X^{III} = Cl \rightarrow I$

and the same applies to CuX. On the other hand, all Cu(II) halides are known except the iodide. In this case, Cu^{2+} oxidises I^- to I_2:

$$2Cu^{2+} + 4I^- \rightarrow Cu_2I_2(s) + I_2$$

(5) However, many copper (I) compounds are unstable in aqueous solution and undergo disproportionation.

$$2Cu^+ \rightarrow Cu^{2+} + Cu$$

(6) The stability of $Cu^{2+}(aq)$ rather than $Cu^+(aq)$ is due to the much more negative $\Delta_{Hyd}H^{\ominus}$ of $Cu^{2+}(aq)$ than Cu^+, which more than compensates for the second ionisation enthalpy of Cu.

(7) The ability of oxygen to stabilise the highest oxidation state is demonstrated in the oxides.

(8) The highest oxidation number in the oxides coincides with the group number and is attained in Sc_2O_3 to Mn_2O_7.

(9) Beyond Group 7, no higher oxides of Fe above Fe_2O_3, are known, although ferrates (VI)$(FeO_4)^{2-}$, are formed in alkaline media but they readily decompose to Fe_2O_3 and O_2.

(10) Besides the oxides, oxocations stabilise V^{v} as VO_2^+, V^{IV} as VO^{2+} and Ti^{IV} as TiO^{2+}

(11) The ability of oxygen to stabilise these high oxidation states exceeds that of fluorine. Thus the highest Mn fluoride is MnF_4 whereas the highest oxide is Mn_2O_7. The ability of oxygen to form multiple bonds to metals explains its superiority.

(12) In the covalent oxide Mn_2O_7, each Mn is tetrahedrally surrounded by O's including a Mn–O–Mn bridge.

(13) The tetrahedral $[MO_4]^n$ ions are known for V^V, Cr^{VI}, Mn^V, Mn^{VI} and Mn^{VII}.

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FORMATION OF COLOURED ION :

<u>Colour</u> : (aquated)

$Ti^{4+} \longrightarrow$ colourless

$V^{4+} \longrightarrow$ blue

$V^{2+} \longrightarrow$ violet

$Cr^{3+} \longrightarrow$ violet

$Mn^{2+} \longrightarrow$ light pink

$Fe^{3+} \longrightarrow$ yellow

$Ni^{2+} \longrightarrow$ green

$Zn^{2+} \longrightarrow$ colourless

$Sc^{3+} \longrightarrow$ colourless

$Ti^{3+} \longrightarrow$ purple

$V^{3+} \longrightarrow$ green

$Cr^{2+} \longrightarrow$ blue

$Mn^{3+} \longrightarrow$ violet

$Fe^{2+} \longrightarrow$ light green

$Co^{2+} \longrightarrow$ pink

$Cu^{2+} \longrightarrow$ blue

CATALYTIC PROPERTIES

(1) The transition metals and their compounds are known for their catalytic activity. This activity is ascribed to their ability to adopt multiple oxidation states and to form complexes. Vanadium(V) oxide (in Contact Process), finely divided iron (in Haber's Process), and nickel (in Catalytic Hydrogenation) are some of the examples.

(2) Catalysts at a solid surface involve the formation of bonds between reactant molecules and atoms of the surface of the catalyst (first row transition metals utilise 3d and 4s electrons for bonding).

(3) This has the effect of increasing the concentration of the reactants at the catalyst surface and also weakening of the bonds in the reacting molecules (the activation energy is lowering).

(4) Also because the transition metal ions can change their oxidation states, they become more effective as catalysts. For example, iron(III) catalyses the reaction between iodide and persulphate ions.

$$2I^- + S_2O_8^{2-} \rightarrow I_2 + 2SO_4^{2-}$$

An explanation of this catalytic action can be given as:

$$2Fe^{3+} + 2I^- \rightarrow 2\,Fe^{2+} + I_2$$

$$2\,Fe^{2+} + S_2O_8^{2-} \rightarrow 2Fe^{3+} + 2SO_4^{2-}$$

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Catalyst		Used
$TiCl_3$	$\longrightarrow$	Used as the Ziegler-Natta catalyst in the production of polythene.
V_2O_5	$\longrightarrow$	Convert SO_2 to SO_3 in the contact process for making H_2SO_4
MnO_2	$\longrightarrow$	Used as a catalyst to decompose $KClO_3$ to give O_2
Fe	$\longrightarrow$	Promoted iron is used in the Haber-Bosch process for making NH_3
$FeCl_3$	$\longrightarrow$	Used in the production of CCl_4 from CS_2 and Cl_2
$PdCl_2$	$\longrightarrow$	Wacker process for converting $C_2H_4 + H_2O + PdCl_2$ to $CH_3CHO + 2HCl + Pd$.
Pd	$\longrightarrow$	Used for hydrogenation (e.g. phenol to cyclohexanone).
Pt/PtO	$\longrightarrow$	Adams catalyst, used for reductions.
Pt	$\longrightarrow$	Formerly used for $SO_2 \longrightarrow SO_3$ in in the contace process for making H_2SO_4
Pt/Rh	$\longrightarrow$	Formerly used in the ostwald process for making HNO_3 to oxidize NH_3 to NO
Cu	$\longrightarrow$	Is used in the direct process for manufacture of $(CH_3)_2SiCl_2$ used to make silicones.
Cu/V	$\longrightarrow$	Oxidation of cyclohexanol/cyclohexanone mixture to adipic acid which is used to make nylon-66
$CuCl_2$	$\longrightarrow$	Deacon process of making Cl_2 from HCl
Ni	$\longrightarrow$	Raney nickel, numerous reduction processes (e.g. manufacture of hexamethylenediamine, productiomn of H_2 from NH_3, reducing anthraquinone to anthraquinol in the production of H_2O_2
$FeSO_4 + H_2O_2$	$\longrightarrow$	Used as Fenton's reagent for oxidizing alcohols to aldehydes.

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Formation of Interstitial Compounds

Interstitial compounds are those which are formed when small atoms like H, C or N are trapped inside the crystal lattices of metals. The principal physical and chemical characteristics of these compounds are as follows:

(i) They have high melting points, higher than those of pure metals.

(ii) They are very hard, some borides approach diamond in hardness.

(iii) They retain metallic conductivity.

(iv) They are chemically inert.

Alloy Formation

An alloy is a blend of metals prepared by mixing the components. Alloys may be homogeneous solid solutions in which the atoms of one metal are distributed randomly among the atoms of the other. Such alloys are formed by atoms with metallic radii that are within about 15 percent of each other. Because of similar radii and other characteristics of transition metals, alloys are readily formed by these metals. The alloys so formed are hard and have often high melting points.

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The best known are ferrous alloys: chromium, vanadium, tungsten, molybdenum and manganese are used for the production of a variety of steels and stainless steel. Alloys of transition metals with non transition metals such as brass (copper-zinc) and bronze (copper-tin), are also of considerable industrial importance.

CHROMATE -DICHROMATE

Preparation : $4FeCr_2O_4 + 8Na_2CO_3 + 7O_2 \xrightarrow[\text{red hot in presence of air}]{1000°-1300°C} \overbrace{8Na_2CrO_4 + 2Fe_2O_3} + 8CO_2\uparrow$

(chromite ore)

$\overbrace{8Na_2CrO_4 + 2Fe_2O_3} \xrightarrow{\text{Dissolve in water and filtered}}$ Residue(Fe_2O_3) + Filtrate(Na_2CrO_4)

[Lime (CaO) added with Na_2CO_3 which keeps the mass porous so that air has access to all parts and prevents fusion]

Then, $\underset{\text{conc.}}{2Na_2CrO_4 + H_2SO_4} \longrightarrow Na_2SO_4\downarrow + Na_2Cr_2O_7 + H_2O$

It's solubility upto 32°C increases and then decreases } Hence, suitable temp. is to be employed to crystallise out Na_2SO_4 first.

Then $Na_2Cr_2O_7$ is crystallised out as $\underset{\text{(red crystal)}}{Na_2Cr_2O_7\cdot 2H_2O}$ on evaporation.

How to get $K_2Cr_2O_7$:

$\underset{\text{hot conc.}}{Na_2Cr_2O_7} + 2KCl \xrightarrow[\text{decomposition}]{\text{double}} K_2Cr_2O_7 + 2NaCl$

NaCl crystallises out first and filtered off. Then $K_2Cr_2O_7$ crystallised out on cooling

The chromates and dichromates are interconvertible in aqueous solution depending upon pH of the solution. The oxidation state of chromium in chromate and dichromate is the same.

$$2\,CrO_4^{2-} + 2H^+ \rightarrow Cr_2O_7^{2-} + H_2O$$

$$Cr_2O_7^{2-} + 2OH^- \rightarrow 2CrO_4^{2-} + H_2O$$

The structures of chromate ion, CrO_4^{2-} and the dichromate ion, $Cr_2O_7^{2-}$ are shown below. The chromate ion is tetrahedral whereas the dichromate ion consists of two tetrahedra sharing one corner with Cr–O–Cr bond angle of 126°. Sodium and potassium dichromates are strong oxidising agents; the sodium salt has a greater solubility in water and is extensively used as an oxidising agent in organic chemistry. Potassium dichromate is used as a primary standard in volumetric analysis. In acidic solution, its oxidising action can be represented as follows:

$$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O \quad (E^\ominus = 1.33V)$$

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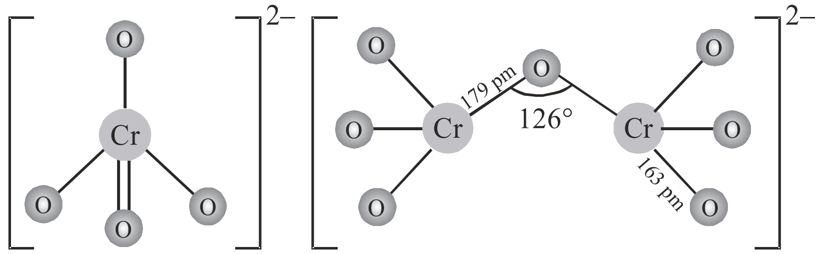

Chromate ion Dichromate ion

Thus, acidified potassium dichromate will oxidise iodides to iodine, sulphides to sulphur, tin(II) to tin(IV) and iron(II) salts to iron(III). The half-reactions are noted below:

$$6\ I^- \rightarrow 3I_2 + 6e^- \ ; \qquad 3\ Sn^{2+} \rightarrow 3Sn^{4+} + 6e^-$$

$$3H_2S \rightarrow 6H^+ + 3S + 6e^- \ ; \qquad 6\ Fe^{2+} \rightarrow 6Fe^{3+} + 6e^-$$

The full ionic equation may be obtained by adding the half-reaction for potassium dichromate to the half-reaction for the reducing agent, for e.g.,

$$Cr_2O_7^{2-} + 14H^+ + 6Fe^{2+} \rightarrow 2Cr^{3+} + 6Fe^{3+} + 7H_2O$$

* Similarities between hexavalent Cr & S-compounds :

(i) SO_3 & $CrO_3 \longrightarrow$ both acidic.

(ii) $S \longrightarrow SO_4^{2-}$, $S_2O_7^{2-}$, $Cr \longrightarrow CrO_4^{2-}$, $Cr_2O_7^{2-}$

(iii) CrO_4^{-2} & SO_4^{2-} are isomorphous

(iv) SO_2Cl_2 & $CrO_2Cl_2 \xrightarrow{OH^-} SO_4^{2-}$ & CrO_4^{2-} respectively.

(v) SO_3Cl^- & $CrO_3Cl^- \xrightarrow{OH^-} SO_4^{2-}$ & CrO_4^{2-}

(vi) CrO_3 & $\beta(SO_3)$ has same structure $-\overset{\overset{O}{\|}}{\underset{\underset{O}{\|}}{Cr}}-O-\overset{\overset{O}{\|}}{\underset{\underset{O}{\|}}{Cr}}-O-\overset{\overset{O}{\|}}{\underset{\underset{O}{\|}}{Cr}}-$

Q. In laboratory $K_2Cr_2O_7$ is used mainly not $Na_2Cr_2O_7$. Why?

Sol. $Na_2Cr_2O_7$ is deliquescent enough and changes its concentration and can not be taken as primary standard solution whereas $K_2Cr_2O_7$ has no water of crystallisation and not deliquescent.

Q. How to standardise $Na_2S_2O_3$ solution in iodometry?

Sol. $K_2Cr_2O_7$ is primary standard $\Rightarrow$ strength is known by weighing the salt in chemical balance and dissolving in measured amount of water.

Then in acidic solution add. KI

$$Cr_2O_7^{2-} + 14H^+ + 6I^- \longrightarrow 2Cr^{3+} + 3I_2 + 7H_2O$$

This I_2 is liberated can be estimated with $S_2O_3^{2-}$.

ALLEN Session 2019-20

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MANGANATE & PERMANGANATE

PREPARATION OF MANGANATE (MnO_4^{2-}) :-

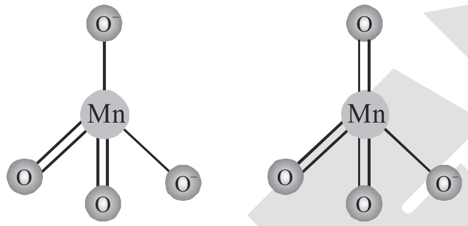
$MnO_2 \xrightarrow[\text{or NaOH, } \Delta \text{ in presence of air}]{KOH} \underset{\text{green melt}}{MnO_4^{2-}}$ [$2MnO_2 + 4KOH + O_2 \longrightarrow 2K_2MnO_4 + 2H_2O$]

$MnO_2 + 2KOH + KNO_3 \longrightarrow K_2MnO_4 + KNO_2 + H_2O$

$3MnO_2 + 6KOH + KClO_3 \longrightarrow 3K_2MnO_4 + KCl + 3H_2O$

green melt $\xrightarrow{\text{Cold Water + little alkali}}$ Green soln. $\xrightarrow{\text{On evaporation}}$

- $\xrightarrow{\text{If KOH used}}$ K_2MnO_4(isomorphrous with K_2SO_4)
- $\xrightarrow{\text{If NaOH used}}$ $Na_2MnO_4.10H_2O$ isomorphrous with $Na_2SO_4.10H_2O$



Tetrahedral manganate (green) ion — Tetrahedral permanganate (purple) ion

In presence of $KClO_3$ & KNO_3 the above reaction is more faster because these two on decomposition provides O_2 easily.

* Manganate is also obtained when $KMnO_4$ is boiled with KOH.

$4KMnO_4 + 4KOH \xrightarrow{\text{boiled}} 4K_2MnO_4 + 2H_2O + O_2$

Properties : The above green solution is quite stable in alkali, but in pure water and in presence of acids, depositing MnO_2 and giving a purple solution of permanganate.

$3K_2MnO_4 + 2H_2O \longrightarrow \underset{\text{purple}}{2KMnO_4} + \underset{\text{drak brown}}{MnO_2 \downarrow} + 4KOH$

Prob. : $E^o_{MnO_4^{2-}/MnO_2} = 2.26$ V ; $E^o_{MnO_4^{2-}/MnO_4^-} = -0.56$ V

Prove that MnO_4^{2-} will disproportionate in acidic medium.

Conversion of MnO_4^{2-} to MnO_4^-

$3K_2MnO_4 + 2H_2SO_4 \longrightarrow 2KMnO_4 + MnO_2 \downarrow + 2K_2SO_4 + 2H_2O$

or $3K_2MnO_4 + 2H_2O + 4CO_2 \longrightarrow 2KMnO_4 + MnO_2 + 4KHCO_3$

But in the above method $\frac{1}{3}$ of Mn is lost as MnO_2 but when oxidised either by Cl_2 or by O_3

$2K_2MnO_4 + Cl_2 \longrightarrow 2KMnO_4 + 2KCl$ [Unwanted MnO_2 does not form]

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OR

$$2K_2MnO_4 + O_3 + H_2O \longrightarrow 2KMnO_4 + 2KOH + O_2$$

OR

$$\underset{\text{Manganate}}{MnO_4^{2-}} \xrightarrow[\text{alkaline solution}]{\text{Electrolytic oxidation in}} \underset{\text{permanganate ion}}{MnO_4^-}$$

Oxidising Prop. of $KMnO_4$: (in acidic medium)

(i) $MnO_4^- + Fe^{+2} + H^+ \longrightarrow Fe^{+3} + Mn^{+2} + H_2O$

(ii) $MnO_4^- + I^- + H^+ \longrightarrow Mn^{+2} + I_2 + H_2O$

(iii) $MnO_4^- + H_2O_2 + H^+ \longrightarrow Mn^{+2} + O_2 + H_2O$

(iv) $MnO_4^- + SO_2 \xrightarrow{H^\oplus} Mn^{+2} + H_2SO_4$

(v) $MnO_4^- + NO_2^- + H^+ \longrightarrow Mn^{+2} + NO_3^- + H_2O$

(vi) $MnO_4^- + H_2C_2O_4 + H^+ \longrightarrow Mn^{+2} + CO_2 + H_2O$

(vii) $MnO_4^- + H_2S \longrightarrow Mn^{2+} + S\downarrow + H_2O$

(viii) $MnO_4^- + S_2O_3^{2-} \longrightarrow Mn^{2+} + S\downarrow + SO_4^{2-}$

(1)* It is not a primary standard since it is difficult to get it in a high degree of purity and free from traces of MnO_2.

(2)* It is slowly reduced to MnO_2 especially in presence of light or acid

$$4MnO_4^- + 4H^+ \longrightarrow 4MnO_2 + 2H_2O + 3O_2$$

Hence it should be kept in dark bottles and standardise just before use.

(ix) $2KMnO_4 + 16HCl \longrightarrow 2KCl + 5Cl_2 + 8H_2O + 2MnCl_2$

Note*: *Permanganate titrations in presence of hydrochloric acid are unsatisfactory since hydrochloric acid is oxidised to chlorine.

Oxidising Prop. of $KMnO_4$ in neutral or faintly alkaline solution.

$$2MnO_4^- + 2OH^- \longrightarrow 2MnO_4^{2-} + H_2O + O.\ \text{Then}\ 2MnO_4^{2-} + 2H_2O \longrightarrow 2MnO_2 + 4OH^- + 2O$$

(i) $2KMnO_4 + H_2O + KI \longrightarrow 2MnO_2 + 2KOH + KIO_3$

(ii) $2KMnO_4 + 3HCO_2K \longrightarrow 2MnO_2 + KHCO_3 + 2K_2CO_3 + H_2O$

(iii) $2KMnO_4 + 3H_2O_2 \longrightarrow 2KOH + 2MnO_2 + 2H_2O + 3O_2$

(iv) Thiosulphate is oxidised almost quantitatively to sulphate:

$$8MnO_4^- + 3S_2O_3^{2-} + H_2O \longrightarrow 8MnO_2 + 6SO_4^{2-} + 2OH^-$$

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Oxidising Prop. in neutral or weakly acidic solution:

(i) $2KMnO_4 + 3MnSO_4 + 2H_2O \xrightarrow{\text{in presence } Zn^{+2} \text{ or } ZnO} 5MnO_2 + K_2SO_4 + 2H_2SO_4$

or $MnO_4^- + Mn^{+2} + 2H_2O \longrightarrow 5MnO_2 + 4H^+$

In absence of Zn^{+2} ions, some of the Mn^{+2} ion may escape, oxidation through the formation of insoluble $Mn^{II}[Mn^{IV}O_3]$ manganous permanganite.

Uses of $KMnO_4$:

Besides its use in analytical chemistry, potassium permanganate is used as a favourite oxidant in preparative organic chemistry. Its uses for the bleaching of wool, cotton, silk and other textile fibres and for the decolourisation of oils are also dependent on its strong oxidising power.

** In laboratory conversion of Mn^{+2} to MnO_4^- is done by :

(i) PbO_2 (ii) $Pb_3O_4 + HNO_3$ (iii) $Pb_2O_3 + HNO_3$ (iv) $NaBiO_3 / H^+$

(v) $(NH_4)_2S_2O_8 / H^+$ (vi) KIO_4 / H^+

Heating effect : $2KMnO_4 \xrightarrow[200°C]{\Delta} \underset{\text{green}}{K_2MnO_4} + \underset{\text{Black}}{MnO_2} + O_2$

$$2K_2MnO_4 \xrightarrow[\text{hot}]{\text{at red}} 2K_2MnO_3 + O_2$$

<u>SILVER AND ITS COMPOUND</u>

(I) Metallic Ag

- $\xrightarrow{\text{aq. regia}}$ Not dissolved
- $\xrightarrow[\text{Moist air}]{\text{Dry or}}$ No effect
- $\xrightarrow{H_2S}$ Black $[4Ag + 2H_2S + O_2 \longrightarrow 2Ag_2S + 2H_2O]$
- $\xrightarrow{H_2SO_4}$ $2Ag + 2H_2SO_4 \longrightarrow Ag_2SO_4 + SO_2 + H_2O$
- $\xrightarrow[\text{hot conc.}]{HNO_3}$ $Ag + 2HNO_3 \longrightarrow AgNO_3 + NO_2 + H_2O$
- $\xrightarrow{HCl}$ Not affected dil. HCl
  $E^o_{Ag/Ag^+} < E^o_{H_2/H^+}$
  Hence in presence of O_2.
  Ag reacts with dil.HCl
  $4Ag + 4HCl + O_2 \longrightarrow 4AgCl\downarrow + 2H_2O$

In the same way in presence of O_2, Ag complexes with NaCN / KCN.

$4Ag + 8KCN + 2H_2O + O_2 \longrightarrow 4K[Ag(CN)_2] + 4KOH$

$AgNO_3$

Prepration : Reaction of Ag with dilute HNO_3 or conc. HNO_3.

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Properties :

(i) It is called as lunar caustic because in contact with skin it produces burning sensation like that of caustic soda with the formation of finely devided silver (black colour)

(ii) Thermal decomposition : $2AgNO_3(s) \rightarrow 2Ag\,(s) + O_2\,(g) + 2NO_2\,(g)$

(iii) Props. of $AgNO_3$

$6AgNO_3 + 3I_2 + 3H_2O \longrightarrow 5AgI + AgIO_3 + 6HNO_3$

(excess)

(iv) $Ag_2SO_4 \xrightarrow{\Delta} 2Ag + SO_2 + O_2$

(v) $\left.\begin{array}{l} A(AgNO_3) \xrightarrow[\text{added}]{B} \text{white ppt appears quickly} \\ B(Na_2S_2O_3) \xrightarrow[\text{added}]{A} \text{It takes time to give white ppt.} \end{array}\right\}$ Explain

(vi) $Ag_2S_2O_3 + H_2O \xrightarrow{\Delta} Ag_2S + H_2SO_4$

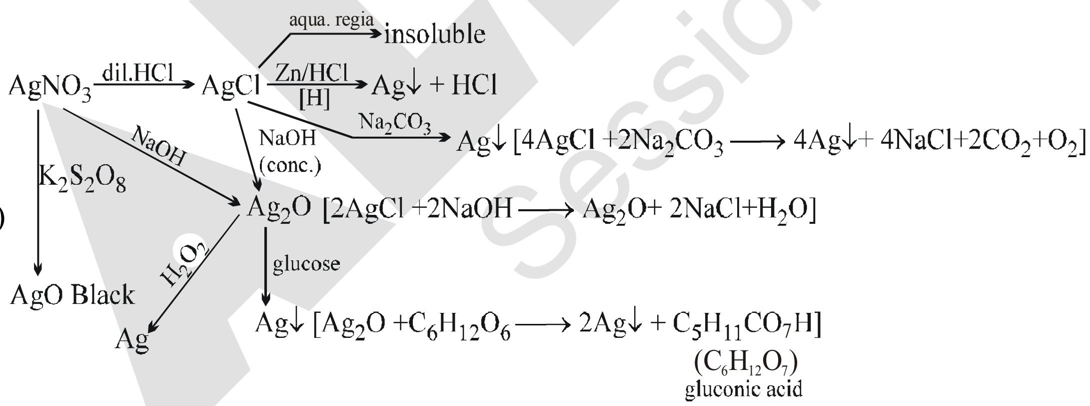
AgCl, AgBr, AgI (but not Ag_2S) are soluble in $Na_2S_2O_3$ forming $[Ag(S_2O_3)_2]^{-3}$ complexes

(vii) $AgBr + AgNO_3 \xrightarrow{KBr} \underset{\text{Pale yellow ppt.}}{AgBr\downarrow} + KNO_3$

Heating effect : $2AgNO_3 \xrightarrow{212°C} 2AgNO_2 + O_2$

$2AgNO_2 \xrightarrow{500°C} 2Ag + 2NO + O_2$

(viii)



$Ag_2O + H_2O_2 \longrightarrow 2Ag + H_2O + O_2$

$K_2S_2O_8 + 2AgNO_3 + 2H_2O \longrightarrow 2AgO + 2KHSO_4 + 2HNO_3$

* AgO supposed to be paramagnetic due to d^9 configuration. But actually it is diamagnetic and exists as $Ag^{I}\,[Ag^{III}O_2]$

* Reaction involved in developer :

$K_2Fe^{II}(C_2O_4)_2 + AgBr \longrightarrow KFe^{III}\,(C_2O_4)_2 + Ag\downarrow + KBr$

ZINC COMPOUNDS

ZnO

It is called as phillospher's wool due to its wooly flock type appearance

Preparation: [1] $2Zn + O_2 \longrightarrow 2ZnO$

[2] Calcination of $ZnCO_3$ or $Zn(NO_3)_2$ or $Zn(OH)_2$

Purest ZnO : $4ZnSO_4 + 4Na_2CO_3 + 3H_2O \longrightarrow \underset{\text{white basic zinc carbonate}}{ZnCO_3{\cdot}3Zn(OH)_2} \downarrow + 4Na_2SO_4 + 3CO_2$

$$ZnCO_3{\cdot}3Zn(OH)_2 \xrightarrow{\Delta} \underset{\text{pure}}{4ZnO} + 3H_2O\uparrow + CO_2\uparrow$$

Properties: 1] $\underset{\text{white}}{ZnO(cold)} \overset{\Delta}{\rightleftharpoons} \underset{\text{yellow}}{ZnO(hot)}$

2] It is insoluble in water

3] It sublimes at 400°C

4] It is amphoteric oxide , react with acid & base both.

$ZnO + 2HCl \longrightarrow ZnCl_2 + H_2O$

$ZnO + H_2SO_4 \longrightarrow ZnSO_4 + H_2O$

$ZnO + 2NaOH \longrightarrow Na_2ZnO_2 + H_2O$

5] $ZnO \longrightarrow Zn$ by H_2 & C

$ZnO + H_2 \xrightarrow{>400°C} Zn + H_2O$

$ZnO + C \longrightarrow Zn + CO$

6] It forms Rinmann's green with $Co(NO_3)_2$

$2Co(NO_3)_2 \longrightarrow 2CoO + 4NO_2 + O_2$

$CoO + ZnO \longrightarrow \underset{\text{Rinmann's green}}{CoZnO_2 \text{ or } CoO{\cdot}ZnO}$

Uses: (1) As white pigment, it is superior than white lead because it does not turn into black

(2) Rinmann's green is used as green pigment

(3) It is used as zinc ointment in medicine

$ZnCl_2$

Preparation : $ZnO + 2HCl \longrightarrow ZnCl_2 + H_2O$

$$\left.\begin{array}{l} ZnCO_3 + 2HCl \longrightarrow ZnCl_2 + H_2O + CO_2 \end{array}\right\} \text{It crystallises as } ZnCl_2{\cdot}2H_2O$$

$Zn(OH)_2 + 2HCl \longrightarrow ZnCl_2 + 2H_2O$

Anh. $ZnCl_2$ cannot be made by heating $ZnCl_2{\cdot}2H_2O$ because

$ZnCl_2{\cdot}2H_2O \xrightarrow{\Delta} Zn(OH)Cl + HCl + H_2O$

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$Zn(OH)Cl \xrightarrow{\Delta} ZnO + HCl$

To get anh. $ZnCl_2$: $Zn + Cl_2 \longrightarrow ZnCl_2$

$Zn + 2HCl(dry) \longrightarrow ZnCl_2 + H_2$

or $Zn + HgCl_2 \longrightarrow ZnCl_2 + Hg$

Properties: (i) It is deliquescent white solid (when anhydrous)

(ii) $ZnCl_2 + H_2S \longrightarrow ZnS$

" $+ NaOH \longrightarrow Zn(OH)_2 \xrightarrow{\text{excess}} Na_2[Zn(OH)_4]$

" $+ NH_4OH \longrightarrow Zn(OH)_2 \xrightarrow{\text{excess}} [Zn(NH_3)_4]^{2+}$

Uses: 1] Used for impregnating timber to prevent destruction by insects

2] As dehydrating agent when anhydrous

3] $ZnO{\cdot}ZnCl_2$ used in dental filling

$ZnSO_4$

Preparation:→

$Zn + \text{dil } H_2SO_4 \longrightarrow ZnSO_4 + H_2$

$ZnO + \text{dil } H_2SO_4 \longrightarrow ZnSO_4 + H_2O$

$ZnCO_3 + \text{dil } H_2SO_4 \longrightarrow ZnSO_4 + H_2O + CO_2$

$$\left.\begin{array}{l} ZnS + 2O_2 \longrightarrow ZnSO_4 \\ ZnS + \frac{3}{2}O_2 \longrightarrow ZnO + SO_2 \end{array}\right\}\text{parallel reaction}$$

$ZnS + 4O_3 \longrightarrow ZnSO_4 + 4O_2$

Props.1] $ZnSO_4{\cdot}7H_2O \xrightarrow{39-70°C} ZnSO_4{\cdot}6H_2O \xrightarrow{>70°C} ZnSO_4{\cdot}H_2O \xrightarrow{>280°C} ZnSO_4$

$ZnSO_4 \xrightarrow{>800°C} \frac{1}{2}O_2 + SO_2 + ZnO$

Uses: 1] in eye lotion

2] Lithophone ($ZnS + BaSO_4$) is used as white pigment

COPPER COMPOUNDS

CuO

Preparation : (i) $CuCO_3.Cu(OH)_2 \xrightarrow{\Delta} 2CuO + H_2O + CO_2$ (Commercial process)

Malachite Green

(native Cu-carbonate)

(ii) $2Cu + O_2 \longrightarrow 2CuO$ & $Cu_2O + \frac{1}{2}O_2 \longrightarrow 2CuO$

(iii) $Cu(OH)_2 \xrightarrow{\Delta} CuO + H_2O$

(iv) $2Cu(NO_3)_2 \xrightarrow{250°C} 2CuO + 4NO_2 + O_2$

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Properties : (i) CuO is insoluble in water

(ii) Readily dissolves in dil. acids

$$CuO + H_2SO_4 \longrightarrow CuSO_4 + H_2O$$
$$HCl \longrightarrow CuCl_2$$
$$HNO_3 \longrightarrow Cu(NO_3)_2$$

(iii) It decomposes when, heated above 1100°C

$$4CuO \longrightarrow 2Cu_2O + O_2$$

(iv) CuO is reduced to Cu by H_2 or C under hot condition

$$CuO + C \longrightarrow Cu + CO\uparrow$$
$$CuO + H_2 \longrightarrow Cu + H_2O\uparrow$$

$CuCl_2$

Preparation:→ $CuO + 2HCl(conc.) \longrightarrow CuCl_2 + H_2O$

$Cu(OH)_2{\cdot}CuCO_3 + 4HCl \longrightarrow 2CuCl_2 + 3H_2O + CO_2$

Properties :→(i) It is crystallised as $CuCl_2{\cdot}2H_2O$ of Emerald green colour

(ii) dil. solution in water is blue in colour due to formation of $[Cu(H_2O)_4]^{2+}$complex.

(iii) When conc. HCl or KCl added to dil. solution of $CuCl_2$ the colour changes into yellow, owing to the formation of $[CuCl_4]^{2-}$

(iv) The conc. aq. solution is green in colour having the two complex ions in equilibrium $2[Cu(H_2O)_4]Cl_2 \rightleftharpoons [Cu(H_2O)_4]^{2+} + [CuCl_4]^{2-} + 4H_2O$

(v) $CuCl_2 \longrightarrow CuCl$ by no. of reagents

(a) $CuCl_2 + \text{Cu-turnings} \xrightarrow{\Delta} 2CuCl$

(b) $2CuCl_2 + H_2SO_3 + H_2O \longrightarrow 2CuCl + 2HCl + 2H_2SO_4$

(c) $2CuCl_2 + Zn/HCl \longrightarrow 2CuCl + ZnCl_2$

(d) $CuCl_2 + SnCl_2 \longrightarrow CuCl + SnCl_4$

** $CuF_2{\cdot}2H_2O \longrightarrow$ light blue

$CuCl_2{\cdot}2H_2O \longrightarrow$ green

$CuBr_2 \longrightarrow$ almost black

{Anh. $CuCl_2$ is dark brown mass obtained by heating $CuCl_2{\cdot}2H_2O$ at 150°C in presence of HCl vap.

CuI_2 does not exist $\quad CuCl_2{\cdot}2H_2O \xrightarrow[\text{HCl gas}]{150°C} CuCl_2 + 2H_2O$

$CuSO_4$

Preparation :→ $CuO + H_2SO_4(dil) \longrightarrow CuSO_4 + H_2O$

$Cu(OH)_2 + H_2SO_4(dil) \longrightarrow CuSO_4 + 2H_2O$

$Cu(OH)_2{\cdot}CuCO_3 + H_2SO_4\ (dil) \rightarrow CuSO_4 + 3H_2O + CO_2$

$Cu + H_2SO_4 + \frac{1}{2}O_2 \longrightarrow CuSO_4 + H_2O$ [Commercial scale]

(Scrap)

$Cu + dil.\ H_2SO_4 \longrightarrow$ no reaction {Cu is a below H in electrochemical series}

ALLEN Session 2019-20

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Properties :→(i) It is crystallised as $CuSO_4 \cdot 5H_2O$

(ii) $$\underset{\text{Blue}}{CuSO_4 \cdot 5H_2O} \underset{\substack{\text{efflorescence}\\ \text{take places}}}{\overset{\substack{\text{On}\\ \text{exposure}}}{\rightleftharpoons}} \underset{\text{Pale blue}}{CuSO_4 \cdot 3H_2O} \overset{100°C}{\rightleftharpoons} \underset{\text{Bluish white}}{CuSO_4 \cdot H_2O}$$

$$CuSO_4 \cdot H_2O \overset{230°C}{\rightleftharpoons} \underset{\text{white}}{CuSO_4(anh.)}$$

$$CuSO_4(anh.) \xrightarrow{800°C} CuO + SO_2 + \frac{1}{2}O_2 \qquad CuSO_4(anh.) \xrightarrow{750°C} CuO + SO_3$$

(iii) Revision with all others reagent

IRON COMPOUNDS

$FeSO_4 \cdot 7H_2O$

Preparation:→(i) Scrap $Fe + \underset{(dil.)}{H_2SO_4} \longrightarrow FeSO_4 + H_2\uparrow$

(ii) From Kipp's waste

$$FeS + H_2SO_4(dil) \longrightarrow FeSO_4 + H_2S\uparrow$$

(iii) $FeS_2 + 2H_2O + \frac{7}{2}O_2 \longrightarrow FeSO_4 + H_2SO_4$

Properties:→ (i) It undergoes aerial oxidation forming basic ferric sulphate

$$4FeSO_4 + H_2O + O_2 \longrightarrow 4Fe(OH)SO_4$$

(ii) $FeSO_4 \cdot 7H_2O \xrightarrow{300°C} \underset{\text{anh. white}}{FeSO_4} \xrightarrow[\text{temp.}]{\text{high}} Fe_2O_3 + SO_2 + SO_3$

(iii) Aq. solution is acidic due to hydrolysis

$$FeSO_4 + 2H_2O \rightleftharpoons \underset{\text{weak base}}{Fe(OH)_2} + H_2SO_4$$

(iv) It is a reducing agent

(a) $Fe^{2+} + MnO_4^- + H^+ \longrightarrow Fe^{3+} + Mn^{2+} + H_2O$

(b) $Fe^{2+} + Cr_2O_7^{2-} + H^+ \longrightarrow Fe^{3+} + Cr^{3+} + H_2O$

(c) $Au^{3+} + Fe^{2+} \longrightarrow Au + Fe^{3+}$

(d) $Fe^{2+} + HgCl_2 \longrightarrow \underset{\text{white ppt.}}{Hg_2Cl_2\downarrow} + Fe^{3+}$

(v) It forms double salt. Example $(NH_4)_2SO_4 \cdot FeSO_4 \cdot 6H_2O$

ALLEN Session 2019-20

FeO(Black)

Preparation : $FeC_2O_4 \xrightarrow[\text{in absence of air}]{\Delta} FeO + CO + CO_2$

Properties : It is stable at high temperature and on cooling slowly disproportionates into Fe_3O_4 and iron

$4FeO \longrightarrow Fe_3O_4 + Fe$

$FeCl_2$

Preparation: $Fe + 2HCl \xrightarrow[\text{a current of HCl}]{\text{heated in}} FeCl_2 + H_2$

OR

$2FeCl_3 + H_2 \xrightarrow{\Delta} 2FeCl_2 + 2HCl$

Properties:$\rightarrow$ (i) It is deliquescent in air like $FeCl_3$

(ii) It is soluble in water, alcohol and ether also because it is sufficiently covalent in nature

(iii) It volatilises at about 1000°C and vapour density indicates the presence of Fe_2Cl_4. Above 1300°C density becomes normal

(iv) It oxidises on heating in air

$12FeCl_2 + 3O_2 \longrightarrow 2Fe_2O_3 + 8FeCl_3$

(v) H_2 evolves on heating in steam

$3FeCl_2 + 4H_2O \longrightarrow Fe_3O_4 + 6HCl + H_2$

(vi) It can exist as different hydrated form

$FeCl_2 \cdot 2H_2O \longrightarrow$ colourless

$FeCl_2 \cdot 4H_2O \longrightarrow$ pale green

$FeCl_2 \cdot 6H_2O \longrightarrow$ green

Ferric Chloride ($FeCl_3$)

This is the most important ferric salt. It is known in anhydrous and hydrated forms. The hydrated form consists of six water molecules, $FeCl_3.6H_2O$.

Preparation :

(i) Anhydrous ferric chloride is obtained by passing dry chlorine gas over heated iron fillings. The vapours are condensed in a bottle attached to the outlet of the tube.

$$2Fe + 3Cl_2 \rightarrow 2FeCl_3$$

(ii) $Fe_2(CO_3)_3 + 6HCl \rightarrow 2FeCl_3 + 3H_2O + 3CO_2$

$Fe(OH)_3 + 3HCl \rightarrow FeCl_3 + 3H_2O$

$Fe_2O_3 + 6HCl \rightarrow 2FeCl_3 + 3H_2O$

The solution on evaporation and cooling deposits yellow crystals of hydrated ferric chloride. $FeCl_3.6H_2O$

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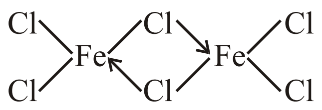
Properties

(i) $Fe_2Cl_6 \xrightleftharpoons{750^\circ C} 2FeCl_3 \xrightleftharpoons{\text{Above } 750^\circ C} 2FeCl_2 + Cl_2$

(ii) 

(iii) It dissolves in water. The solution is acidic in nature due to its hydrolysis as shown below :

$FeCl_3 + 3HOH \rightleftharpoons Fe(OH)_3 + 3HCl$

(iv) $FeCl_3 + 6NH_3 \rightarrow FeCl_3.6NH_3$

(v) Ferric chloride acts as an oxidising agent.

(a) It oxidies stannous chloride to stannic chloride.

$2FeCl_3 + SnCl_2 \rightarrow 2FeCl_2 + SnCl_4$

(b) It oxidises SO_2 to H_2SO_4

$2FeCl_3 + SO_2 + 2H_2O \rightarrow 2FeCl_2 + H_2SO_4 + 2HCl$

(c) It oxidises H_2S to S

$2FeCl_3 + H_2S \rightarrow 2FeCl_2 + 2HCl + S$

(d) It liberates iodine from KI.

$2FeCl_3 + 2KI \rightarrow 2FeCl_2 + 2KCl + I_2$

(e) Nascent hydrogen reduces $FeCl_3$ into $FeCl_2$

$FeCl_3 + [H] \rightarrow FeCl_2 + HCl$

(vi) $FeCl_3 + 3NH_4OH \rightarrow Fe(OH)_3 + 3NH_4Cl$

(vii) $FeCl_3 + NH_4SCN \rightarrow Fe(SCN)Cl_2 + NH_4Cl$

or $FeCl_3 + 3NH_4SCN \rightarrow Fe(SCN)_3 + 3NH_4Cl$

(viii) $4FeCl_3 + 3K_4Fe(CN)_6 \rightarrow \underset{\substack{\text{Prussian blue} \\ \text{(Ferri ferrocyanide)}}}{Fe_4[Fe(CN)_6]_3} + 12KCl$

(ix) On heating hydrated ferric chloride $FeCl_3.6H_2O$, anhydrous ferric chloride is not obtained. It is changed to Fe_2O_3 with evolution of H_2O and HCl.

$2[FeCl_3.6H_2O] \xrightarrow{\text{Heat}} Fe_2O_3 + 6HCl + 9H_2O$

Hydrated ferric chloride may be dehydrated by heating with thionyl chloride.

$FeCl_3.6H_2O + 6SOCl_2 \longrightarrow FeCl_3 + 12HCl + 6SO_2$

ALLEN Session 2019-20

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EXERCISE # O-1

SELECT ONLY ONE IS CORRECT OPTIONS :

General Properties of d-block

1. $N_2(g) + 3H_2(g) \xrightleftharpoons{Fe + Mo} 2NH_3(g)$; Haber's process, Mo is used as

(A) a catalyst (B) a catalytic promoter

(C) an oxidising agent (D) as a catalytic poison

2. An ornamental of gold having 75% of gold, it is of .... carat.

(A) 18 (B) 16 (C) 24 (D) 20

3. Transition elements having more tendency to form complex than representative elements (s and p-block elements) due to -

(A) availability of d-orbitals for bonding

(B) variable oxidation states are not shown by transition elements

(C) all electrons are paired in d-orbitals

(D) *f*-orbitals are available for bonding

4. A compound of mercury used in cosmetics, in Ayurvedic and Yunani medicines and known as Vermilion is -

(A) $HgCl_2$ (B) HgS (C) Hg_2Cl_2 (D) HgI

5. Transition elements are usually characterised by variable oxidation states but Zn does not show this property because of

(A) completion of np-orbitals (B) completion of (n–1)d orbitals

(C) completion of ns-orbitals (D) inert pair effect

6. The d-block element which is a liquid at room temperature, having high specific heat, less reactivity than hydrogen and its chloride (MX_2) is volatile on heating is

(A) Cu (B) Hg (C) Ce (D) Pm

7. Coinage metals show the properties of

(A) typical elements (B) normal elements

(C) inner-transition elements (D) transition element

8. The transition metal used in X-rays tube is

(A) Mo (B) Ta (C) Tc (D) Pm

9. The higher oxidation states of transition elements are found to be in the combination with A and B, which are

(A) F, O (B) O, N (C) O, Cl (D) F, Cl

10. The metals present in insulin and haemoglobin are respectively

(A) Zn, Hg (B) Zn, Fe (C) Co, Fe (D) Mg, Fe

Session 2019-20

11. A metal M which is not affected by strong acids like conc. HNO_3, conc. H_2SO_4 and conc. solution of alkalies like NaOH, KOH forms MCl_3 which finds use for toning in photography. The metal M is

(A) Ag (B) Hg (C) Au (D) Cu

12. Manganese steel is used for making railway tracks because

(A) it is hard with high percentage of Mn

(B) it is soft with high percentage of Mn

(C) it is hard with small concentration of manganese with impurities

(D) it is soft with small concentration of manganese with impurities

13. Transition elements in lower oxidation states act as Lewis acid because

(A) they form complexes (B) they are oxidising agents

(C) they donate electrons (D) they do not show catalytic properties

14. The electrons which take part in order to exhibit variable oxidation states by transition metals are

(A) ns only (B) (n–1)d only

(C) ns and (n–1)d only but not np (D) (n–1)d and np only but not ns

15. Solution of MnO_4^- is purple-coloured due to

(A) d-d-transition (B) charge transfer from O to Mn

(C) due to both d-d-transition and charge transfer (D) none of these

16. An element of 3d-transition series shows two oxidation states x and y, differ by two units then

(A) compounds in oxidation state x are ionic if x > y

(B) compounds in oxidation state x are ionic if x < y

(C) compounds in oxidation state y are covalent if x < y

(D) compounds in oxidation state y are covalent if y < x

Compounds of d-block

17. $(T) \xrightarrow{\text{compd (U)} \ +\text{conc.} H_2SO_4} (V)_{\text{Red gas}} \xrightarrow{NaOH + AgNO_3} (W)_{\text{Red ppt.}} \xrightarrow{NH_3 \text{ soln.}} (X)$

imparts violet colour in the flame test

$(W)_{\text{Red ppt.}} \xrightarrow{\text{dil. HCl}} (Y)_{\text{white ppt.}}$

$(U) \xrightarrow[\Delta]{NaOH} (Z)_{\text{gas}}$ (gives white fumes with HCl)

sublimes on heating

Identify (T) to (Z).

(A) T = $KMnO_4$, U = HCl, V = Cl_2, W = HgI_2, X = $Hg(NH_2)NO_3$, Y = Hg_2Cl_2, Z = N_2

(B) T=$K_2Cr_2O_7$, U=NH_4Cl, V=CrO_2Cl_2, W=Ag_2CrO_4, X=$[Ag(NH_3)_2]^+$, Y=AgCl,Z = NH_3

(C) T = K_2CrO_4, U = KCl, V = CrO_2Cl_2, W = HgI_2, X = Na_2CrO_4, Y = $BaCO_3$, Z = NH_4Cl

(D) T = K_2MnO_4, U = NaCl, V = CrO_3, W = $AgNO_2$, X = $(NH_4)_2CrO_4$, Y = $CaCO_3$, Z = SO_2

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18. The number of moles of acidified $KMnO_4$ required to convert one mole of sulphite ion into sulphate ion is

(A) 2/5 (B) 3/5 (C) 4/5 (D) 1

19. $Cr_2O_7^{2-} \underset{Y}{\overset{X}{\rightleftharpoons}} 2CrO_4^{2-}$, X and Y are respectively

(A) $X = OH^-$, $Y = H^+$ (B) $X = H^+$, $Y = OH^-$

(C) $X = OH^-$, $Y = H_2O_2$ (D) $X = H_2O_2$, $Y = OH^-$

20. CrO_3 dissolves in aqueous NaOH to give

(A) $Cr_2O_7{}^{2-}$ (B) $CrO_4{}^{2-}$ (C) $Cr(OH)_3$ (D) $Cr(OH)_2$

21. During estimation of oxalic acid Vs $KMnO_4$, self indicator is

(A) $KMnO_4$ (B) oxalic acid (C) K_2SO_4 (D) $MnSO_4$

22. Acidified chromic acid + $H_2O_2 \xrightarrow{\text{Org.solvent}} \underset{\text{(blue colour)}}{X + Y}$, X and Y are

(A) CrO_5 and H_2O (B) Cr_2O_3 and H_2O (C) CrO_2 and H_2O (D) CrO and H_2O

23. $\underset{\text{(diatomic covalent molecule)}}{Y} \xleftarrow{KI} CuSO_4 \xrightarrow{\text{dil } H_2SO_4} X(\text{Blue colour})$, X and Y are

(A) $X = I_2$, $Y = [Cu(H_2O)_4]^{2+}$ (B) $X = [Cu(H_2O)_4]^{2+}$, $Y = I_2$

(C) $X = [Cu(H_2O)_4]^{+}$, $Y = I_2$ (D) $X = [Cu(H_2O)_5]^{2+}$, $Y = I_2$

24. $(NH_4)_2Cr_2O_7$ (Ammonium dichromate) is used in fire works. The green coloured powder blown in air is

(A) Cr_2O_3 (B) CrO_2 (C) Cr_2O_4 (D) CrO_3

25. Iron becomes passive by .... due to formation of ....

(A) dil. HCl, Fe_2O_3 (B) 80% conc. HNO_3, Fe_3O_4

(C) conc. H_2SO_4, Fe_3O_4 (D) conc. HCl, Fe_3O_4

26. Bayer's reagent used to detect olifinic double bond is

(A) acidified $KMnO_4$ (B) aqueous $KMnO_4$

(C) 1% alkaline $KMnO_4$ solution (D) $KMnO_4$ in benzene

27. $MnO_4^- + xe^- \xrightarrow[\text{(Alkaline medium)}]{} MnO_4^{2-}$

$\quad + ye^- \text{ (Acidic medium)} \longrightarrow Mn^{2+}$

$\quad + ze^- \text{ (Neutral medium)} \longrightarrow MnO_2$

x, y and z are respectively

(A) 1, 2, 3 (B) 1, 5, 3 (C) 1, 3, 5 (D) 5, 3, 1

28. $Cu + \underset{\text{(hot)}}{\text{conc. } HNO_3} \longrightarrow Cu(NO_3)_2 + X$ (oxide of nitrogen); then X is

(A) N_2O (B) NO_2 (C) NO (D) N_2O_3

29. $CuSO_4$ solution reacts with excess KCN to give

(A) $Cu(CN)_2$ (B) CuCN (C) $K_2[Cu(CN)_2]$ (D) $K_3[Cu(CN)_4]$

ALLEN Session 2019-20

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30. Pick out the incorrect statement:

(A) MnO_2 dissolves in conc. HCl, but does not form Mn^{4+} ions

(B) MnO_2 react with hot concentrated H_2SO_4 liberating oxygen

(C) K_2MnO_4 is formed when MnO_2 in fused KOH is oxidised by air, KNO_3 , PbO_2 or $NaBiO_3$

(D) Decomposition of acidic $KMnO_4$ is not catalysed by sunlight.

31. 1 mole of Fe^{2+} ions are oxidised to Fe^{3+} ions with the help of (in acidic medium)

(A) 1/5 moles of $KMnO_4$ (B) 5/3 moles of $KMnO_4$

(C) 2/5 moles of $KMnO_4$ (D) 5/2 moles of $KMnO_4$

32. To an acidified dichromate solution, a pinch of Na_2O_2 is added and shaken. What is observed:

(A) blue colour (B) Orange colour changing to green

(C) Copious evolution of oxygen (D) Bluish - green precipitate

33. The rusting of iron is formulated as $Fe_2O_3 \cdot xH_2O$ which involves the formation of

(A) Fe_2O_3 (B) $Fe(OH)_3$ (C) $Fe(OH)_2$ (D) $Fe_2O_3 + Fe(OH)_3$

34. Solid $CuSO_4 \cdot 5H_2O$ having covalent, ionic as well as co-ordinate bonds. Copper atom/ion forms .... co-ordinate bonds with water.

(A) 1 (B) 2 (C) 3 (D) 4

35. $\underset{\text{(acidified)}}{KMnO_4} + HCl \longrightarrow H_2O + X(g)$, X is a

(A) red liquid (B) violet gas

(C) greenish yellow gas (D) yellow-brown gas

36. Purple of cassius is:

(A) Pure gold (B) Colliodal solution of gold

(C) Gold (I) hydroxide (D) Gold (III) chloride

37. Amongst the following species, maximum covalent character is exhibited by

(A) $FeCl_2$ (B) $ZnCl_2$ (C) $HgCl_2$ (D) $CdCl_2$

38. Number of moles of $SnCl_2$ required for the reduction of 1 mole of $K_2Cr_2O_7$ into Cr_2O_3 is (in acidic medium)

(A) 3 (B) 2 (C) 1 (D) 1/3

39. Pick out the incorrect statement:

(A) MnO_4^{2-} is quite strongly oxidizing and stable only in very strong alkalies. In dilute alkali, neutral solutions, it disproportionates.

(B) In acidic solutions, MnO_4^- is reduced to Mn^{2+} and thus, $KMnO_4$ is widely used as oxidising agent

(C) $KMnO_4$ does not acts as oxidising agent in alkaline medium

(D) $KMnO_4$ is manufactured by the fusion of pyrolusite ore with KOH in presence of air or KNO_3, followed by electrolytic oxidation in strongly alkaline solution.

40. The aqueous solution of $CuCrO_4$ is green because it contains

(A) green Cu^{2+} ions (B) green CrO_4^{2-} ions

ALLEN Session 2019-20

(C) blue Cu^{2+} ions and green CrO_4^{2-} ions (D) blue Cu^{2+} ions and yellow CrO_4^{2-} ions

41. In nitroprusside ion, the iron exists as Fe^{2+} and NO as NO^+ rather than Fe^{3+} and NO respectively. These forms of ions are established with the help of

(A) magnetic moment in solid state (B) thermal decomposition method

(C) by reaction with KCN (D) by action with K_2SO_4

42. Which of the following reaction is possible at anode?

(A) $2Cr^{3+} + 7H_2O \longrightarrow Cr_2O_7^{2-} + 14\ H^+$ (B) $F_2 \longrightarrow 2\ F^-$

(C) $\frac{1}{2}O_2 + 2H^+ \longrightarrow H_2O$ (D) None of these

43. Colourless solutions of the following four salts are placed separately in four different test tubes and a strip of copper is dipped in each one of these. Which solution will turn blue?

(A) KNO_3 (B) $AgNO_3$ (C) $Zn(NO_3)_2$ (D) $ZnSO_4$

44. When acidified $KMnO_4$ is added to hot oxalic acid solution, the decolourization is slow in the beginning, but becomes very rapid after some time. This is because:

(A) Mn^{2+} acts as autocatalyst (B) CO_2 is formed as the product

(C) Reaction is exothermic (D) MnO_4^- catalyses the reaction

45. Metre scales are made-up of alloy

(A) invar (B) stainless steel (C) elektron (D) magnalium

46. The Ziegler-Natta catalyst used for polymerisation of ethene and styrene is $TiCl_4 + (C_2H_5)_3Al$, the catalysing species (active species) involved in the polymerisation is

(A) $TiCl_4$ (B) $TiCl_3$ (C) $TiCl_2$ (D) TiCl

47. 'Bordeaux mixture' is used as a fungicide. It is a mixture of

(A) $CaSO_4 + Cu(OH)_2$ (B) $CuSO_4 + Ca(OH)_2$

(C) $CuSO_4 + CaO$ (D) $CuO + CaO$

48. Peacock ore is:

(A) FeS_2 (B) $CuFeS_2$ (C) $CuCO_3.Cu(OH)_2$ (D) Cu_5FeS_4

ALLEN Session 2019-20

EXERCISE # O-2

SELECT MORE THAN ONE IS CORRECT OPTIONS

General Properties of d-block

1. Potash alum is a double salt, its aqueous solution shows the characteristics of
(A) Al^{3+} ions (B) K^+ ions (C) SO_4^{2-} ions (D) Al^{3+} ions but not K^+ ions

2. Addition of non-metals like B and C to the interstitial sites of a transition metal results the metal
(A) of more ductability (B) of less ductability
(C) less malleable (D) of more hardness

3. Mercury is a liquid at 0°C because of
(A) very high ionisation energy (B) weak metallic bonds
(C) high heat of hydration (D) high heat of sublimation

4. The correct statement(s) about transition elements is/are
(A) the most stable oxidation state is +3 and its stability decreases across the period
(B) transition elements of 3d-series have almost same atomic sizes from Cr to Cu
(C) the stability of +2 oxidation state increases across the period
(D) some transition elements like Ni, Fe, Cr may show zero oxidation state in some of their compounds

5. The ionisation energies of transition elements are
(A) less than p-block elements (B) more than s-block elements
(C) less than s-block elements (D) more than p-block elements

6. The metal(s) which does/do not form amalgam is/are
(A) Fe (B) Pt (C) Zn (D) Ag

7. Which of the following statements concern with d-block metals?
(A) compounds containing ions of transition elements are usually coloured
(B) Zinc has lowest melting point among 3d-series elements
(C) they show variable oxidation states, which differ by two units only
(D) they easily form complexes

8. The highest oxidation state among transition elements is
(A) + 7 by Mn (B) + 8 by Os (C) + 8 by Ru (D) + 7 by Fe

9. Amphoteric oxide(s) is/are
(A) Al_2O_3 (B) SnO (C) ZnO (D) Fe_2O_3

10. The catalytic activity of transition elements is related to their
(A) variable oxidation states (B) free suface valencies
(C) complex formation ability (D) magnetic moment

11. In the equation: $M + 8CN^- + 2H_2O + O_2 \longrightarrow 4[M(CN)_2]^- + 4OH^-$, metal M is
(A) Ag (B) Au (C) Cu (D) Hg

12. $CuSO_4(aq) + 4NH_3 \longrightarrow X$, then X is
(A) $[Cu(NH_3)_4]^{2+}$ (B) paramagnetic
(C) coloured (D) of a magnetic moment of 1.73 BM

13. Amphoteric oxide(s) of Mn is/are
(A) MnO_2 (B) Mn_3O_4 (C) Mn_2O_7 (D) MnO

14. The lanthanide contraction is responsible for the fact that
(A) Zr and H*f* have same atomic sizes (B) Zr and H*f* have same properties
(C) Zr and H*f* have different atomic sizes (D) Zr and H*f* have different properties

ALLEN Session 2019-20

15. Ion(s) having non zero magnetic moment (spin only) is/are

(A) Sc^{3+} (B) Ti^{3+} (C) Cu^{2+} (D) Zn^{2+}

Compounds of d-block

16. Correct statement(s) is/are

(A) an acidified solution of $K_2Cr_2O_7$ liberates iodine from KI

(B) $K_2Cr_2O_7$ is used as a standard solution for estimation of Fe^{2+} ions

(C) in acidic medium, M = N/6 for $K_2Cr_2O_7$

(D) $(NH_4)_2Cr_2O_7$ on heating decomposes to yield Cr_2O_3 through an endothermic reaction

17. Interstitial compounds are formed by

(A) Co (B) Ni (C) Fe (D) Ca

18. Acidified $KMnO_4$ can be decolourised by

(A) SO_2 (B) H_2O_2 (C) $FeSO_4$ (D) $Fe_2(SO_4)_3$

ALLEN Session 2019-20

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EXERCISE # S-1

NUMERICAL GRID TYPE QUESTIONS :

1. When mixture of NaCl and $K_2Cr_2O_7$ is gently warmed with conc.H_2SO_4 then compound X is formed. What is the oxidation state of central atom of X.

2. Number of ions which gives blue colour in aqueous state :

V^{+4}, Ni^{+2}, Ti^{+3}, Co^{+2}, Fe^{+3}, Cu^{+2}

3. Define the oxidation states of Mn in product of the given reaction

$3K_2MnO_4+2H_2O+4CO_2 \rightarrow 2X+Y+4KHCO_3$

If the oxidation state of Mn in product X and Y are n_1 and n_2 respectively. Then find out the value of (n_1+n_2) -

4. Find number of metal ion which can produce high spin and low spin octahedral complex :

Sc^{+3}, Ti^{+3}, V^{+3}, Cr^{+3}, Mn^{+3}, Fe^{+3}, Co^{+3}, Ni^{+2}

5. How many non-axial d-orbitals are involved in hybridisation of CrO_2Cl_2.

6. Find the number of species from the following which has magnetic moment value of 1.73 B.M.

Fe^{2+}, Cu^{2+}, Ni^{2+}, NO_2, NO_2^-, Ti^{3+}

7. Total no. of moles of Mohr's salt are required for per mole of dichromate ions during volumetric analysis.

8. Find number of reaction(s) in which no redox change takes place -

(I) $BaCl_2 + Na_2CrO_4 \xrightarrow{CH_3COOH}$

(II) $K_2Cr_2O_7 + NaOH \longrightarrow$

(III) $Cr_2O_7^{2-} + NO_3^- \xrightarrow{H^+}$

(IV) $Cr_2O_7^{2-} + C_2H_5OH \xrightarrow{H^+}$

ALLEN Session 2019-20

EXERCISE # S-2

COMPREHENSTION TYPE QUESTIONS :

Comprehension # 01 to 04

Transition metal and their compounds are used as catalysts in industry and in biological system. For example, in the Contact process, vanadium compounds in the +5 state (V_2O_5 or VO_3^-) are used to oxidise SO_2 to SO_3:

$$SO_2 + \frac{1}{2}O_2 \xrightarrow{V_2O_5} SO_3$$

It is thought that the actual oxidation process takes place in two stages. In the first step, V^{5+} in the presence of oxide ions converts SO_2 to SO_3. At the same time, V^{5+} is reduced to V^{4+}.

$$2V^{5+} + O^{2-} + SO_2 \longrightarrow 2V^{4+} + SO_3$$

In the second step, V^{5+} is regenerated from V^{4+} by oxygen :

$$2V^{4+} + \frac{1}{2}O_2 \longrightarrow 2V^{5+} + O^{2-}$$

The overall process is, of course, the sum of these two steps:

$$SO_2 + \frac{1}{2}O_2 \longrightarrow SO_3$$

1. Transition metals and their compounds catalyse reactions because **:**

(A) They have competely filled s-subshell

(B) They have a comparable size due to poor shielding of d-subshell

(C) They introduce an entirely new reaction mechanism with a lower activation energy

(D) They have variable oxidation states differ by two units

2. During the course of the reaction **:**

(A) Catalyst undergoes changes in oxidation state

(B) Catalyst increases the rate constant

(C) Catalyst is regenerated in its orginal form when the reactants form the products

(D) All are correct.

3. Catalytic activity of transition metals depends on **:**

(A) Their ability to exist in different oxidation states

(B) The size of the metal atoms

(C) The number of empty atomic orbitals available

(D) None of these

4. Which of the following ion involved in the above process will show paramagnetism?

(A) V^{5+} (B) V^{4+} (C) O^{2-} (D) VO_3^-

ALLEN Session 2019-20

Comprehension # 05 & 06

(X) is very important laboratory reagent which is prepared by its naturally occuring ore which is called pyrolusite. Pyrolusite when fused with alkali in the presence of O_2, green compound **(Y)** is produced.

(Y) is converted into **(X)** by electrolysis or by using ozone.

5. On small scale **(X)** is prepared by disproportion of **(Y)** in acidic solution. Which of the following is produced by disproportion of **(Y)** in slight alkaline solution.

(A) $KMnO_4$, Mn^{+2} (B) $KMnO_4$, MnO_2

(C) MnO_2, Mn^{+2} (D) K_2MnO_4, Mn^{+2}

6. Select the correct statements :

(A) **(X)** is tetrahedral & diamagnetic

(B) **(Y)** is tetrahedral & paramagnetic

(C) **(X)** produce dimangnese hepta oxide (oily liquid) with conc. H_2SO_4

(D) All are correct

Comprehension # 07 to 09

Due to availability of vacant orbitals of sufficiently low energy, d-block elements form complexes, d-block elements have different properties such as- catalytic, magnetic, alloy formation, interistitial compounds formation. Interistitial compounds are those compounds in which small atoms like carbon and boron fits into interistices of d-block elements crystal. In interistitial compounds, there is no chemical bond formation takes place so , chemical properties remain almost same but physical properties may change.

7. Which of the property of interistitial compounds has the same behaviour as that of the element -

(A) Malleability (B) Ductility

(C) Electrical conductance (D) Hardness

8. Which of the following property gets decreased in interistitial compounds compared to that of the element -

(A) Malleability (B) Metallic lustre

(C) Hardness (D) Density

9. Select correct statement -

(A) Highest oxidation state of 3d-series is +8.

(B) Ni, Cu and Zn are not transition element.

(C) Ziglar natta catalyst contain vanadium.

(D) Aq. solution of Cu^{2+}, Fe^{+3} and Cr^{3+} are blue, yellow and green respectively.

ALLEN Session 2019-20

MATCH THE COLUMN : (Matrix Match)

10. **Column-I (Metals)** | **Column-II**

(A) Zn — (P) Cyanide process involve in the comerical extration

(B) Cu — (Q) Extracted by hydrometallurgical process

(C) Ag — (R) Roasting involve in the comerical extration

(D) Au — (S) Present in Brass

11. **Column-I (catalyst)** | **Column-II (Used)**

(A) $TiCl_4$ — (P) Adams catalyst in reduction

(B) $PdCl_2$ — (Q) In preparation of $(CH_3)_2\ SiCl_2$

(C) Pt/PtO — (R) Used as the ziegler-natta catalyst in polythene production

(D) Cu — (S) Wacker process for converting C_2H_4 to CH_3CHO

SELECT CORRECT CODE :

12. **Column-I** | **Column-II**

(P) Cr_2O_3 — (1) Neutral oxide

(Q) CrO_3 — (2) Amphoteric oxide

(R) Fe_3O_4 — (3) Mix oxide

(S) N_2O — (4) Acidic oxide

Select correct code for matching -

Code :

	P	Q	R	S
(A)	2	4	3	1
(B)	2	3	4	1
(C)	4	2	3	1
(D)	4	3	1	2

13. **Column-I (Metal ion of 3d-series)** | **Column-II (Characterstic)**

(P) Ni^{2+} — (1) produce blue aq. solution

(Q) Cr^{2+} — (2) half filled t_{2g} orbitals in octahedral complex

(R) V^{2+} — (3) diamagnetic ion

(S) Ti^{4+} — (4) calculated $\mu = 2.84$ B.M. (spin only)

Select correct code for matching -

Code :

	P	Q	R	S
(A)	4	2	1	3
(B)	3	1	2	4
(C)	4	1	2	3
(D)	1	2	4	3

ALLEN Session 2019-20

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ASSERATION & REASONING :

Questions given below consist of two statements each printed as Assertion (A) and Reason (R); while answering these questions you are required to choose any one of the following four responses:

(A) if both (A) and (R) are true and (R) is the correct explanation of (A)

(B) if both (A) and (R) are true but (R) is not correct explanation of (A)

(C) if (A) is true but (R) is false

(D) if (A) is false and (R) is true

14. **Assertion** : $KMnO_4$ is purple in colour due to charge transfer.

Reason : In MnO_4^-, there is no electron present in d-orbitals of manganese.

15. **Assertion** : K_2CrO_4 has yellow colour due to charge transfer.

Reason : CrO_4^{2-} ion is tetrahedral in shape.

16. **Assertion** : The highest oxidation state of chromium in its compounds is +6.

Reason : Chromium atom has only six electrons in ns and (n–1) d orbitals.

17. **Assertion** : CrO_3 reacts with HCl to form chromyl chloride gas.

Reason : Chromyl chloride (CrO_2Cl_2) has tetrahedral shape.

18. **Assertion** : Zinc does not show characteristic properties of transition metals.

Reason : In zinc outermost shell is completely filled.

19. **Assertion** : Tungsten has a very high melting point.

Reason : Tungsten is a covalent compound.

20. **Assertion** : Equivalent mass of $KMnO_4$ is equal to one-third of its molecular mass when it acts as an oxidising agent in an alkaline medium.

Reason : Oxidation number of Mn is +7 in $KMnO_4$.

21. **Assertion** : Cu^+ ion is colourless.

Reason : Four water molecules are coordinated to Cu^+ ion in water.

ALLEN Session 2019-20

MATCHING LIST TYPE 1 × 3 Q. (THREE LIST TYPE Q.)

The following column 1, 2, 3 represent the different metals of 3d series and their properties

Answer the questions that follow

Column-1 - Hexa aqua complex of dipositive ion of given metal

Column-2 - Magnetic moment (spin only)

Column-3 - Colour

Column - 1 Hexa aqua complex of dipositive ion of given metal	Column - 2 Magnetic moment (spin only)	Column - 3 Colour
(I) Co	(i) $\sqrt{8}$ B.M.	(P) Green
(II) Ni	(ii) $\sqrt{3}$ B.M.	(Q) Pink
(III) V	(iii) $\sqrt{15}$ B.M.	(R) Colourless
(IV) Zn	(iv) Zero	(S) Violet
		(T) Blue

22. Select **CORRECT** combination for metal which have same magnetism as in $[Ti(H_2O)_6]^{4+}$

(A) (II), (iv), (R) (B) (I), (iv), (R) (C) (IV), (iv), (R) (D) (IV), (iv), (T)

23. Which is **CORRECT** magnetic moment and colour combination of the product when $[Co(H_2O)_6]^{+2}$ is react with conc. HCl solution

(A) (ii), (Q) (B) (ii), (T) (C) (iii), (Q) (D) (iii), (T)

24. Select **CORRECT** combination for hexaaqua complex of metal ion M^{+2} which have half filled t_{2g} configuration (i.e. t_{2g}^3)

(A) (III), (iii), (S) (B) (III), (iii), (P) (C) (III), (ii), (S) (D) (III), (iv), (P)

25. $[M(H_2O)_6]^{+2}$ of a metal in **column-1** have combination (i), (P) in **column-2, 3**. Select **CORRECT** combination for hexaammine complex of same metal ion.

(A) (i), (T) (B) (i), (S) (C) (iv), (R) (D) (iv) (T)

ALLEN Session 2019-20

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EXERCISE # JEE-MAIN

1. What would happen when a solution of potassium chromate is treated with an excess of dilute nitric acid - **[AIEEE-2003]**

(1) Cr^{3+} and $Cr_2O_7^{2-}$ are formed (2) $Cr_2O_7^{2-}$ and H_2O are formed

(3) $Cr_2O_7^{2-}$ is reduced to +3 state of Cr (4) $Cr_2O_7^{2-}$ is oxidised to +7 state of Cr

2. Excess of KI reacts with $CuSO_4$ solution and then $Na_2S_2O_3$ solution is added to it. Which of the statements is incorrect for this reaction : **[AIEEE-2004]**

(1) Evolved I_2 is reduced (2) CuI_2 is formed

(3) $Na_2S_2O_3$ is oxidised (4) Cu_2I_2 is formed

3. Calomel on reaction with NH_4OH gives **[AIEEE-2004]**

(1) $HgNH_2Cl$ (2) NH_2–Hg–Hg–Cl (3) Hg_2O (4) HgO

4. In context with the transition elements, which of the following statements is incorrect? **[AIEEE-2009]**

(1) In the highest oxidation states of the first five transition elements (Sc to Mn), all the 4s and 3d electrons are used for bonding.

(2) Once the d^5 configuration is exceeded, the tendency to involve all the 3d electrons in bonding decreases.

(3) In addition to the normal oxidation states, the zero oxidation state is also shown by these elements in complexes.

(4) In the highest oxidation states, the transition metal show basic character and form cationic complexes.

5. Iron exhibits +2 and +3 oxidation states. Which of the following statements about iron is **incorrect** ? **[AIEEE-2012]**

(1) Ferrous compounds are more easily hydrolysed than the corresponding ferric compounds.

(2) Ferrous oxide is more basic in nature than the ferric oxide.

(3) Ferrous compounds are relatively more ionic than the corresponding ferric compounds.

(4) Ferrous compounds are less volatile than the corresponding ferric compounds.

6. Consider the following reaction : **[JEE MAIN-2013]**

$$xMnO_4^- + yC_2O_4^{2-} + zH^+ \rightarrow xMn^{2+} + 2yCO_2 + \frac{z}{2}H_2O$$

The values of x, y and z in the reaction are respectively :-

(1) 5,2 and 16 (2) 2,5 and 8 (3) 2, 5 and 16 (4) 5,2 and 8

7. Which of the following arrangements does not represent the correct order of the property stated against it ? **[JEE MAIN-2013]**

(1) $V^{2+} < Cr^{2+} < Mn^{2+} < Fe^{2+}$: paramagnetic behaviour

(2) $Ni^{2+} < Co^{2+} < Fe^{2+} < Mn^{2+}$: ionic size

(3) $Co^{3+} < Fe^{3+} < Cr^{3+} < Sc^{3+}$: stability in aqueous solution

(4) Sc < Ti < Cr < Mn : number of oxidation states

8. Potassium dichromate when heated with concentrated sulphuric acid and a soluble chloride, gives brown - red vapours of: **[JEE MAIN-2013, Online]**

(1) CrO_3 (2) Cr_2O_3 (3) $CrCl_3$ (4) CrO_2Cl_2

ALLEN Session 2019-20

9. The element with which of the following outer electron configuration may exhibit the largest number of oxidation states in its compounds : **[JEE MAIN-2013, Online]**

(1) $3d^7 4s^2$ (2) $3d^8 4s^2$ (3) $3d^5 4s^2$ (4) $3d^6 4S^2$

10. When a small amount of $KMnO_4$ is added to concentrated H_2SO_4, a green oily compound is obtained which is highly explosive in nature. Compound may be : **[JEE MAIN-2013, Online]**

(1) Mn_2O_3 (2) $MnSO_4$ (3) Mn_2O_7 (4) MnO_2

11. Which series of reactions correctly represents chemical relations related to iron and its compound?

(1) $Fe \xrightarrow{Cl_2,\ heat} FeCl_3 \xrightarrow{heat,\ air} FeCl_2 \xrightarrow{Zn} Fe$

(2) $Fe \xrightarrow{O_2,\ heat} Fe_3O_4 \xrightarrow{CO,\ 600°C} FeO \xrightarrow{CO,700°C} Fe$

(3) $Fe \xrightarrow{dil\ H_2SO_4} FeSO_4 \xrightarrow{H_2SO_4,\ O_2} Fe_2(SO_4)_3 \xrightarrow{Heat} Fe$

(4) $Fe \xrightarrow{O_2,\ heat} FeO \xrightarrow{dil\ H_2SO_4} FeSO_4 \xrightarrow{Heat} Fe$ **[JEE MAIN-2014]**

12. The equation which is balanced and represents the correct product (s) is : **[JEE MAIN-2014]**

(1) $[Mg(H_2O)_6]^{2+} + (EDTA)^{4-} \xrightarrow{excess\ NaOH} [Mg(EDTA)]^{2+} + 6H_2O$

(2) $CuSO_4 + 4KCN \rightarrow K_2[Cu(CN)_4] + K_2SO_4$

(3) $Li_2O + 2KCl \rightarrow 2LiCl + K_2O$

(4) $[CoCl(NH_3)_5]^+ + 5H^+ \rightarrow Co^{2+} + 5NH_4^+ + Cl^-$

13. Which of the following is **not** formed when H_2S reacts with acidic $K_2Cr_2O_7$ solution ?

(1) K_2SO_4 (2) $Cr_2(SO_4)_3$

(3) S (4) $CrSO_4$ **[JEE MAIN-2014, Online]**

14. Copper becomes green when exposed to moist air for a long period. This is due to :- **[JEE MAIN-2014, Online]**

(1) the formation of a layer of cupric oxide on the surface of copper.

(2) the formation of basic copper sulphate layer on the surface of the metal

(3) the formation of a layer of cupric hydroxide on the surface of copper.

(4) the formation of a layer of basic carbonate of copper on the surface of copper.

15. Which one of the following exhibits the largest number of oxidation states ? **[JEE MAIN-2014, Online]**

(1) Mn(25) (2) V(23) (3) Cr (24) (4) Ti (22)

16. How many electrons are involved in the following redox reaction ? **[JEE MAINS-2014,Online]**

$Cr_2O_7^{2-} + Fe^{2+} + C_2O_4^{2-} \rightarrow Cr^{3+} + Fe^{3+} + CO_2$ (Unbalanced)

(1) 3 (2) 4 (3) 5 (4) 6

17. Amongst the following, identify the species with an atom in +6 oxidation state: **[JEE MAIN-2014, Online]**

(1) $[MnO_4]^-$ (2) $[Cr(CN)_6]^{3-}$ (3) Cr_2O_3 (4) CrO_2Cl_2

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Advanced Batch\Chem\Sheet\Salt Analysis & Heating Effect\VV d-Block\Eng.p65

18. Match the catalysts to the correct processes :- **[JEE MAIN-2015]**

	Catalyst		Process
(A)	$TiCl_3$	(i)	Wacker process
(B)	$PdCl_2$	(ii)	Ziegler-Natta polymerization
(C)	$CuCl_2$	(iii)	Contact process
(D)	V_2O_5	(iv)	Deacon's process

(1) A-ii, B-iii, C-iv, D-i
(2) A-iii, B-i, C-ii, D-iv
(3) A-iii, B-ii, C-iv, D-i
(4) A-ii, B-i, C-iv, D-iii

19. Which of the following statements is false :- **[JEE MAIN-2015, Online]**

(1) $Cr_2O_7^{2-}$ has a Cr – O – Cr bond

(2) CrO_4^{2-} is tetrahedral in shape

(3) $Na_2Cr_2O_7$ is a primary standard in volumetry

(4) $K_2Cr_2O_7$ is less soluble than $Na_2Cr_2O_7$

20. The transition metal ions responsible for colour in ruby and emerald are, respectively : **[Jee Main 2016]**

(1) Cr^{3+} and Cr^{3+} (2) Cr^{3+} and Co^{3+} (3) Co^{3+} and Cr^{3+} (4) Co^{3+} and Co^{3+}

21. Galvanization is applying a coating of :- **[Jee Main 2016]**

(1) Zn (2) Pb (3) Cr (4) Cu

22. Which of the following compounds is metallic and ferromagnetic ? **[Jee Main 2016]**

(1) MnO_2 (2) TiO_2 (3) CrO_2 (4) VO_2

23. Which one of the following species is stable in aqueous solution? **[Jee Main 2016]**

(1) MnO_4^{3-} (2) MnO_4^{2-} (3) Cu^+ (4) Cr^{2+}

24. What will occur if a block of copper metal is dropped into a beaker containing a solution of 1M $ZnSO_4$?

(1) The copper metal will dissolve and zinc metal will be deposited **[Jee Main 2016]**
(2) No reaction will occur
(3) The copper metal will dissolve with evolution of oxygen gas
(4) The copper metal will dissolve with evolution of hydrogen gas

25. Which of the following ions does **not** liberate hydrogen gas on reaction with dilute acids? **[JEE MAIN-2017, Online]**

(1) Ti^{2+} (2) Cr^{2+} (3) Mn^{2+} (4) V^{2+}

26. In the following reactions, ZnO is respectively acting as a/an : **[JEE MAIN-2017]**

(a) $ZnO + Na_2O \rightarrow Na_2ZnO_2$
(b) $ZnO + CO_2 \rightarrow ZnCO_3$

(1) base and acid (2) base and base
(3) acid and acid (4) acid and base

ALLEN session 2019-20

EXERCISE # JEE-ADVANCED

TRUE/FALSE :

1. Cu^+ disproportionates to Cu^{2+} and elemental copper in solution. **[JEE 1991]**

FILL IN THE BLANKS :

2. When Fe(s) is dissolved in aqueous hydrochloric acid in a closed vessel, the work done is .... **[JEE 1997]**

3. Silver jewellery items tarnish slowly in the air due to their reaction with .... **[JEE 1997]**

MULTIPLE CHOICE QUESTIONS WITH ONE CORRECT ANSWER :

4. Which one is solder ? **[JEE 1995]**

(A) Cu and Pb (B) Zn and Cu (C) Pb and Sn (D) Fe and Zn

5. Which compound does not dissolve in hot, dilute HNO_3? **[JEE 1996]**

(A) HgS (B) PbS (C) CuS (D) CdS

6. Ammonium dichromate is used in some fireworks. The green coloured powder blown in the air is - **[JEE 1997]**

(A) CrO_3 (B) Cr_2O_3 (C) Cr (D) $CrO(O_2)$

7. The number of moles of $KMnO_4$ that will be needed to react with one mole of sulphite ion in acidic solution is - **[JEE 1997]**

(A) 2/5 (B) 3/5 (C) 4/5 (D) 1

8. Read the following statement and explanation and answer as per the options given below : **[JEE 1998]**

Assertion : Zn^{2+} is diamagnetic.

Reason : Two electrons are lost from 4s orbital to form Zn^{2+}.

(A) If both assertion and reason are correct, and reason is the correct explanation of the assertion.

(B) If both assertion and reason are correct, but reason is not the correct explanation of the assertion.

(C) If assertion is correct but reason is incorrect.

(D) If assertion is incorrect but reason is correct.

9. In the dichromatic anion, **[JEE 1999]**

(A) 4 Cr – O bonds are equivalent (B) 6 Cr – O bonds are equivalent

(C) all Cr – O bonds are equivalent (D) all Cr – O bonds are non-equivalent

10. Anhydrous ferric chloride is prepared by: **[JEE 2002]**

(A) heating hydrated ferric chloride at a high temperature in a stream of air

(B) heating metallic iron in a stream of dry chlorine gas

(C) reaction of ferric oxide with HCl

(D) reaction of metallic iron with HCl

11. When MnO_2 is fused with KOH, a coloured compound is formed, the product and its colour is:

(A) K_2MnO_4, green (B) $KMnO_4$, purple **[JEE 2003]**

(C) Mn_2O_3, brown (D) Mn_3O_4, black

12. The product of oxidation of I^- with MnO_4^- in alkaline medium is - **[JEE 2004]**

(A) IO_3^- (B) I_2 (C) IO^- (D) IO_4^-

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Advanced Batch\Chem\Sheet\Salt Analysis & Heating Effect\VV) d-block\Eng.p65

13. $(NH_4)_2Cr_2O_7$ on heating liberates a gas. The same gas will be obtained by - **[JEE 2004]**

(A) heating NH_4NO_2 (B) heating NH_4NO_3

(C) treating H_2O_2 with $NaNO_2$ (D) treating Mg_3N_2 with H_2O

14. Which of the following combination will produce H_2 gas ? **[JEE Advance 2017]**

(A) Zn metal and NaOH(aq) (B) Au metal and NaCN(aq) in the presence of air

(C) Cu metal and conc. HNO_3 (D) Fe metal and conc. HNO_3

MULTIPLE CHOICE QUESTIONS WITH ONE OR MORE THAN ONE CORRECT ANSWER :

15. Which of the following alloys contains (s) Cu and Zn ? **[JEE 1993]**

(A) Bronze (B) Brass (C) Gun metal (D) Type metal

16. Addition of high proportions of magnanese makes steel useful in making rails of railroads, becuse manganese. **[JEE 1998]**

(A) gives hardness to steel

(B) helps the formation of oxides of iron

(C) can remove oxygen and sulphur

(D) can show highest oxidation state of +7.

17. The correct statement(s) about Cr^{2+} and Mn^{3+} is (are) **[JEE Advance 2015]**

[Atomic numbers of Cr = 24 and Mn = 25]

(A) Cr^{2+} is a reducing agent

(B) Mn^{3+} is an oxidizing agent

(C) Both Cr^{2+} and Mn^{3+} exhibit d^4 electronic configuration

(D) When Cr^{2+} is used as a reducing agent, the chromium ion attains d^5 electronic configuration

18. Fusion of MnO_2 with KOH in presence of O_2 produces a salt **W**. Alkaline solution of **W** upon eletrolytic oxidation yields another salt **X**. The manganese containing ions present in **W** and **X**, respectively, are **Y** and **Z**. Correct statement(s) is (are) **[JEE Advance 2019]**

(A) **Y** is diamagnetic in nature while **Z** is paramagnetic

(B) Both **Y** and **Z** are coloured and have tetrahedral shape

(C) In both **Y** and **Z** , π-bonding occurs between p-orbitals of oxygen and d-orbitals of manganese.

(D) In aqueous acidic solution, **Y** undergoes disproportionation reaction to give **Z** and MnO_2.

19. Consider the following reactions (unbalanced) **[JEE Advance 2019]**

$Zn + \text{hot conc. } H_2SO_4 \rightarrow G + R + X$

$Zn + \text{conc. } NaOH \rightarrow T + Q$

$G + H_2S + NH_4OH \rightarrow Z \text{ (a precipitate)} + X + Y$

Choose the correct option(s).

(A) The oxidation state of Zn in T is +1

(B) Bond order of Q is 1 in its ground state

(C) Z is dirty white in colour

(D) R is a V-shaped molecule

Session 2019-20

ANSWER KEY

EXERCISE # O-1

1. (B)	2. (A)	3. (A)	4. (B)
5. (B)	6. (B)	7. (D)	8. (A)
9. (A)	10. (B)	11. (C)	12. (A)
13. (A)	14. (C)	15. (B)	16. (B,C)
17. (B)	18. (A)	19. (A)	20. (B)
21. (A)	22. (A)	23. (B)	24. (A)
25. (B)	26. (C)	27. (B)	28. (B)
29. (D)	30. (D)	31. (A)	32. (A,C)
33. (D)	34. (D)	35. (C)	36. (B)
37. (C)	38. (A)	39. (C)	40. (D)
41. (A)	42. (A)	43. (B)	44. (A)
45. (A)	46. (B)	47. (B)	48. (D)

EXERCISE # O-2

1. (A, B, C)	2. (B, C, D)	3. (A, B)	4. (A, C, D)
5. (A, B)	6. (A, B)	7. (A, B, D)	8. (B, C)
9. (A, B, C)	10. (A, B, C)	11. (A, B)	12. (A, B, C, D)
13. (A, B)	14. (A, B)	15. (B, C)	16. (A, B, C)
17. (A, B, C)	18. (A, B, C)		

EXERCISE # S-1

1. **(6),** In this reaction CrO_2Cl_2 is formed in which Cr is +6.
2. **(2),** V^{+4}, Cu^{+2}
3. **(11),** +7, +4
4. **(3),** Mn^{+3}, Fe^{+3}, Co^{+3}
5. CrO_2Cl_2 has d^3S hybridisation and all 3d-orbitals are non-axial which are d_{xy} ; d_{yz} and d_{xz}.
6. **(003)**
7. **(6),** Mohr's salt

   $FeSO_4 .(NH_4)_2SO_4.6H_2O$

   $6\ Fe^{+2} + Cr_2O_7^{2-} + 14\ H^+ \rightarrow 2Cr^{+3} + 7H_2O + 6Fe^{+3}$
8. **3**

(I) $BaCl_2 + Na_2CrO_4 \xrightarrow{CH_3COOH} BaCrO_4 \downarrow + 2NaCl$

(II) $K_2Cr_2O_7 + NaOH \longrightarrow K_2CrO_4$

$CrO_4^{2-} \underset{OH^-}{\overset{H^+}{\rightleftharpoons}} Cr_2O_7^{2-}$

(III) $Cr_2O_7^{2-} + NO_3^- \xrightarrow{H^+}$ no reaction.

(IV) $Cr_2O_7^{2-} + C_2H_5OH \xrightarrow{H^+} CH_3 - COOH + Cr^{+3}$

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Advanced Batch\Chem\Sheet\Salt Analysis & Heating Effect\IV) d-Block\Eng.p65

EXERCISE # S-2

1. (C)	**2. (D)**	**3. (A)**	**4. (B)**
5. (B)	**6. (D)**	**7. (C)**	**8. (A)**
9. (D)	**10. (A)-(R,S) ; (B)-(R,S) ; (C)-(P,Q) ; (D)-(P,Q)**		
11. (A)-(R) ; (B)-(S) ; (C)-(P) ; (D)-(Q)		**12. (A)**	**13. (C)**
14. (B)	**15. (B)**	**16. (A)**	**17.(B)**
18. (C)	**19. (C)**	**20. (B)**	**21. (C)**
22. (C)	**23. (D)**	**24. (A)**	**25. (A)**

EXERCISE # JEE-MAIN

1. (2)	**2. (2)**	**3. (1)**	**4. (4)**	
5. (1)	**6. (3)**	**7. (1)**	**8. (4)**	
9. (3)	**10. (3)**	**11. (2)**	**12. (4)**	
13. (4)	**14. (4)**	**15. (1)**	**16. (4)**	
17. (4)	**18. (4)**	**19. (3)**	**20. (1)**	
21. (1)	**22. (3)**	**23. (2)**	**24. (2)**	**25. (3) 26. (4)**

EXERCISE # JEE-ADVANCED

TRUE/FALSE :

1. True

FILL IN THE BLANKS :

2. Zero **3.** H_2S

MULTIPLE CHOICE QUESTIONS WITH ONE CORRECT ANSWER :

4. (C)	**5.** (A)	**6.** (B)	**7.** (A)
8. (B)	**9.** (B)	**10.** (B)	**11.** (A)
12. (A)	**13.** (A)	**14.** (A)	

MULTIPLE CHOICE QUESTIONS WITH ONE OR MORE THAN ONE CORRECT ANSWER :

15. (B, C) **16.** (A) **17.** (A), (B), (C)

18. Ans.(2,3,4)

$$MnO_2 + 2KOH + \frac{1}{2}O_2 \xrightarrow{\Delta} \underset{(W)}{K_2MnO_4} + H_2O$$

$$\left[(W) = K_2MnO_{4(aq)} \rightleftharpoons 2K^{\oplus}_{(aq)} + \underset{(Y)}{MnO^{2-}_{4(aq)}}\right]$$

$$\underset{}{K_2MnO_4} + H_2O \xrightarrow[\text{Redox}]{\text{Electolytic}} H_2 + KOH + \underset{(X)}{KMnO_4}$$

[anion of X = $\underset{(Z)}{MnO_4^-}$]

$$\left[\because \ \underset{(Y)}{MnO_4^{2-}} \xrightarrow[\text{Oxidation}]{\text{Electrolytic}} \underset{(Z)}{MnO_4^-} + e^-\right]$$

$\because$ In acidic solution; Y undergoes disproportionation reaction

$$\left[3MnO_{4(aq)}^{2-} + 4H^{\oplus} \longrightarrow \underset{(Z)}{2MnO_4^-} + MnO_2 + 2H_2O\right]$$

19. Ans.(2,3,4)

Sol. $Zn + 2H_2SO_4 \text{ (Hot and conc.)} \rightarrow \underset{(G)}{ZnSO_4} + \underset{(R)}{SO_2\uparrow} + \underset{(X)}{2H_2O}$

$Zn + 2NaOH \text{ (conc.)} \rightarrow \underset{(T)}{Na_2ZnO_2} + \underset{(Q)}{H_2\uparrow}$

$ZnSO_4 + H_2S + 2NH_4OH \rightarrow \underset{(Z)}{ZnS\downarrow} + \underset{(X)}{2H_2O} + \underset{(Y)}{(NH_4)_2SO_4}$

ALLEN Session 2019-20

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Important Notes

ALLEN Session 2019-20

p-BLOCK ELEMENT

GROUP 13 ELEMENTS: THE BORON FAMILY :

Boron is a typical non-metal, aluminium is a metal but shows many chemical similarities to boron, and gallium, indium and thallium are almost exclusively metallic in character.

OCCURANCE

Boron : Boron is a fairly rare element, mainly occurs as orthoboric acid, (H_3BO_3), borax, $Na_2B_4O_7 \cdot 10H_2O$, and kernite, $Na_2B_4O_7 \cdot 4H_2O$. There are two isotopic forms of boron ^{10}B (19%) and ^{11}B (81%).

ALUMINIUM :

Aluminium is the most abundant metal and the third most abundant element in the earth's crust (8.3% by mass) after oxygen (45.5%) and Si (27.7%). Bauxite, $Al_2O_3 . 2H_2O$ and cryolite, Na_3AlF_6 are the important minerals of aluminium.

ELECTRONIC CONFIGURATION :

The outer electronic configuration of these elements is ns^2np^1. A close look at the electronic configuration suggests that while boron and aluminium have noble gas core, gallium and indium have noble gas plus 10 *d*-electrons, and thallium has noble gas plus 14 *f*- electrons plus 10 *d*-electron cores. Thus, the electronic structures of these elements are more complex than for the first two groups of elements discussed in unit 10. This difference in electronic structures affects the other properties and consequently the chemistry of all the elements of this group.

ATOMIC RADII :

On moving down the group, for each successive member one extra shell of electrons is added and, therefore, atomic radius is expected to increase. However, a deviation can be seen.

Atomic radii order	Ionic Radii order (+3 OS)
B < Ga < Al < In < Tℓ	B < Al < Ga < In < Tℓ

Atomic radius of Ga is less than that of Al. This can be understood from the variation in the inner core of the electronic configuration. The presence of additional 10 *d*-electrons offer only poor screening effect for the outer electrons from the increased nuclear charge in gallium. Consequently, the atomic radius of gallium (135 pm) is less than that of aluminium (143 pm).

Ionization Enthalpy

The ionisation enthalpy values as expected from the general trends do not decrease smoothly down the group.

Ionization Enthalpies order
B > Tℓ > Ga > Al > In

The decrease from B to Al is associated with increase in size. The observed discontinuity in the ionisation enthalpy values between Al and Ga, and between In and Tl are due to inability of d and f-electrons ,which have low screening effect, to compensate the increase in nuclear charge.

The order of ionisation enthalpies, as expected, is $\Delta_iH_1 < \Delta_iH_2 < \Delta_iH_3$. The sum of the first three ionisation enthalpies for each of the elements is very high. Effect of this will be apparent during study their chemical properties.

Electronegativity

Down the group, electronegativity first decreases from B to Al and then increases marginally. This is because of the discrepancies in atomic size of the elements.

Physical Properties

(i) Boron is non-metallic in nature. It is extremely hard and black coloured solid. It exists in many allotropic forms. Due to very strong crystalline lattice, boron has unusually high melting point.

(ii) Rest of the members are soft metals with low melting point and high electrical conductivity.

(iii) It is worth while to note that gallium with unusually low melting point (303K), could exist in liquid state during summer. Its high boiling point (2676 K) makes it a useful material for measuring high temperatures.

(iv) Density of the elements increases down the group from boron to thallium.

Melting and Boiling points order

M.P. B > Al > Tl > In > Ga

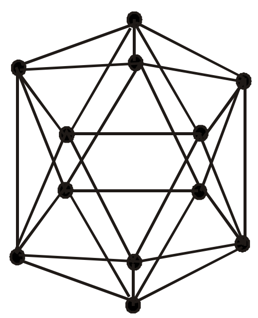
B.P. B > Al > Ga > In > Tl

Electropositive Character

Due to high IE they are less electropositive on moving down the group metallic character increases due to decrease in IE [$\therefore$ B is nonmetals and other elements are metals.]

$$\underbrace{B <}_{\text{Non metal}} \underbrace{Al > Ga > In > Tl}_{\text{Metals}}$$

Note : Boron exists in many allotropic forms. All the allotropes have basic building B_{12} icosahedral units made up of polyhedron having 20 faces and 12 corners.For example one is the simplest form : α - rhombohedral boron.



But Al , In & Tℓ all have close packed metal structure.

Chemical Properties

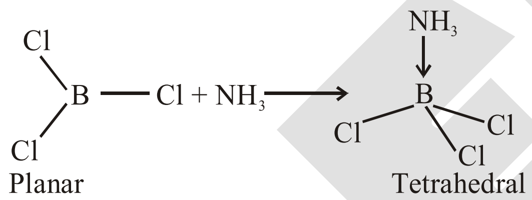
Oxidation state and trends in chemical reactivity

(i) Due to small size of boron, the sum of its first three ionization enthalpies is very high. This prevents it to form +3 ions and forces it to form only covalent compounds. But as we move from B to Al, the sum of the first three ionisation enthalpies of Al considerably decreases, and is therefore able to form Al^{3+} ions. In fact, aluminium is a highly electropositive metal.

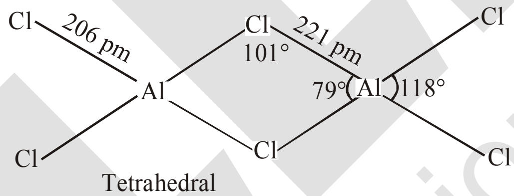
(ii) However, down the group, due to poor shielding effect of intervening d and f orbitals, the increased effective nuclear charge holds ns electrons tightly (responsible for inert pair effect) and thereby, restricting their participation in bonding. As a result of this, only p-orbital electrons may be involved in bonding. In fact in Ga, In and Tℓ, both +1 and +3 oxidation states are observed. The relative stability of +1 oxidation state progressively increases for heavier elements: Al < Ga < In < Tl. In thallium +1 oxidation state is predominant whereas the +3 oxidation state is highly oxidising in character.

(iii) The compounds in +1 oxidation state, as expected from energy considerations, are more ionic than those in +3 oxidation state.

(iv) In trivalent state, the number of electrons around the central atom in a molecule of the compounds of these elements (e.g., boron in BF_3) will be only six. Such **electron deficient** molecules have tendency to accept a pair of electrons to achieve stable electronic configuration and thus, behave as Lewis acids. The tendency to behave as Lewis acid decreases with the increase in the size down the group. BCl_3 easily accepts a lone pair of electrons from ammonia to form $BCl_3.NH_3$.



$AlCl_3$ achieves stability by forming a dimer



(v) In trivalent state most of the compounds being covalent are hydrolysed in water. For example, the trichlorides on hyrolysis in water form tetrahedral $[M(OH)_4]^-$ species; the hybridisation state of element M is sp^3. Aluminium chloride in acidified aqueous solution forms octahedral $[Al(H_2O)_6]^{3+}$ ion. In this complex ion, the 3d orbitals of Al are involved and the hybridisation state of Al is sp^3d^2

REACTIVITY TOWARDS AIR :

(i) Boron is unreactive in crystalline form.

(ii) Aluminium forms a very thin oxide layer on the surface which protects the metal from further attack.

(iii) Amorphous boron and aluminium metal on heating in air form B_2O_3 and Al_2O_3 respectively. With dinitrogen at high temperature they form nitrides.

$$2E(s) + 3O_2(g) \xrightarrow{\Delta} 2E_2O_3(s)$$

$$2E(s) + N_2(g) \xrightarrow{\Delta} 2EN(s)$$

(E = element)

The nature of these oxides varies down the group. Boron trioxide is acidic and reacts with basic (metallic) oxides forming metal borates. Aluminium and gallium oxides are amphoteric and those of indium and thallium are basic in their properties.

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ALLEN Session 2019-20

Reaction with Air at Water

Al should react air to form a very thin oxide film (10^{-4} to 10^{-6} mm thick) on the surface and protects the metal from further attack

$$2\,Al(s) + \frac{3}{2}\,O_2(g) \rightarrow Al_2O_3(s) \qquad \Delta H° = -1670 \text{ kJ/mole (Thermal reaction)}$$

Ga and In are attacked neither by cold water nor hot water unless oxygen is present. $T\ell$ form an oxide on surface.

REACTIVITY TOWARDS ACIDS AND ALKALIES :

Boron does not react with acids and alkalies even at moderate temperature; but aluminium dissolves in mineral acids and aqueous alkalies and thus shows amphoteric character.

Aluminium dissolves in dilute HCl and liberates dihydrogen.

$$2Al(s) + 6HCl(aq) \rightarrow 2Al^{3+}(aq) + 6Cl^{-}(aq) + 3H_2(g)$$

However, concentrated nitric acid renders aluminium passive by forming a protective oxide layer on the surface. Aluminium also reacts with aqueous alkali and liberates dihydrogen.

$$2Al(s) + 2NaOH(aq) + 6H_2O(\ell) \rightarrow 2Na^{+}[Al(OH)_4]^{-}(aq) + 3H_2(g)$$

Sodium tetrahydroxoaluminate(III)

or

$$2NaAlO_2 + 2H_2O$$

Ga, In, $T\ell$ dissolve in dilute acids liberating H_2 Ga is amphoteric like Al and it dissolves in aq. NaOH liberating H_2 and forming gallates.

Reactivity towards halogens :

These elements react with halogens to form trihalides (except $T\ell I_3$).

$$2E(s) + 3X_2(g) \rightarrow 2EX_3(s) \qquad (X = F, Cl, Br, I)$$

IMPORTANT TRENDS AND ANOMALOUS PROPERTIES OF BORON

Certain important trends can be observed in the chemical behaviour of group 13 elements. The tri-chlorides, bromides and iodides of all these elements being covalent in nature are hydrolysed in water. Species like tetrahedral $[M(OH)_4]^{-}$ and octahedral $[M(H_2O)_6]^{3+}$, except in boron, exist in aqueous medium.

The monomeric trihalides, being electron deficient, are strong Lewis acids. Boron trifluoride easily reacts with Lewis bases such as NH_3 to complete octet around boron.

$$F_3B + :NH_3 \longrightarrow (F_3B \leftarrow NH_3)$$

It is due to the absence of *d* orbitals that the maximum covalence of B is 4. Since the *d* orbitals are available with Al and other elements, the maximum covalence can be expected beyond 4. Most of the other metal halides (*e.g.,* $AlCl_3$) are dimerised through halogen bridging (*e.g.,* Al_2Cl_6). The metal species completes its octet by accepting electrons from halogen in these halogen bridged molecules.

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Preparation of Boron :

(i) Preparation of B_2O_3 from Borax or Colemanite

$$Na_2B_4O_7 + HCl/H_2SO_4 \longrightarrow NaX + H_2B_4O_7$$

$$H_2B_4O_7 + 5H_2O \longrightarrow 4H_3BO_3 \xrightarrow{\Delta} B_2O_3 + H_2O$$

(ii) Reduction of B_2O_3

$$B_2O_3 + Na/K/Mg/Al \longrightarrow B + Na_2O/K_2O/MgO/Al_2O_3$$

❑ ***Chemical Properties :***

(i) Burning in air : $4\,B + 3O_2 \longrightarrow 2B_2O_3$

$4Al + 3O_2 \longrightarrow 2Al_2O_3$

(ii) Reaction with water

$B + H_2O$ (Cold & hot) $\longrightarrow$ no reaction

$2B$ (red hot) $+ 3H_2O$ (steam) $\longrightarrow B_2O_3 + H_2$

$(Al + 3H_2O \longrightarrow Al(OH)_3 + \frac{3}{2}H_2)$

(red hot)

(iii) $B + HCl \longrightarrow$ no reaction

$B + H_2SO_4$ (dil) $\longrightarrow$ no reaction

$2B + 3H_2SO_4$ (conc.) $\longrightarrow 2H_3BO_3 + 3SO_2$

$(2Al + 6H_2SO_4 \longrightarrow Al_2(SO_4)_3 + 3SO_2 + 6H_2O)$

$B + 3HNO_3 \longrightarrow H_3BO_3 + 3NO_2$

[$Al + HNO_3(80\%) \longrightarrow Al_2O_3$ (passive layer) and does not react further.]

(iv) $\boxed{2B + 2NaOH + 2H_2O \xrightarrow{\Delta} 2NaBO_2 + 3H_2}$

$2Al + 2NaOH + 2H_2O \longrightarrow 2NaAlO_2 + 3H_2$

(v) $2B + N_2 \longrightarrow 2BN$ $\quad (2Al + N_2 \longrightarrow 2AlN)$

$4B + C \longrightarrow B_4C$ $\quad (4Al + 3C \longrightarrow Al_4C_3)$

(vi) $3Mg + 2B \longrightarrow Mg_3B_2$

SOME IMPORTANT COMPOUNDS OF BORON

Some useful compounds of boron are borax, orthoboric acid and diborane. We will briefly study their chemistry.

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Preparation of Borax :

Borax

$$\underset{\text{Colemanite}}{2CaO\cdot 3B_2O_3} + 2Na_2CO_3 \longrightarrow 2CaCO_3\downarrow + Na_2B_4O_7 + 2NaBO_2$$

$$\xrightarrow[-\,CaCO_3\text{(as residue)}]{\text{Filtered}}\ \underbrace{Na_2B_4O_7 + NaBO_2}_{\textit{in solution}}$$

$$\underbrace{Na_2B_4O_7 + NaBO_2}_{\textit{in solution}} \xrightarrow[\textit{and allowed to crystallise out and filtered}]{\textit{Concentrated}} \underset{\text{in filtrate}}{NaBO_2} + \underset{\text{as residue}}{Na_2B_4O_7\cdot 10H_2O}$$

$$NaBO_2 \xrightarrow{CO_2\text{ passed and crystallise out again}} Na_2B_4O_7\cdot 10H_2O\downarrow$$

$$[4NaBO_2 + CO_2 \longrightarrow Na_2B_4O_7 + Na_2CO_3]$$

Properties :

(i) It is a white crystalline solid of formula $Na_2B_4O_7.10H_2O$. In fact it contains the tetranuclear units $[B_4O_5(OH)_4]^{2-}$ and correct formula; therefore, is $Na_2[B_4O_5(OH)_4].8H_2O$.

(ii) Borax dissolves in water to give an alkaline solution.

$$Na_2B_4O_7 + 7H_2O \rightarrow 2NaOH + \underset{\text{Orthoboric acid}}{4H_3BO_3}$$

(iii) On heating, borax first loses water molecules and swells up. On further heating it turns into a transparent liquid, which solidifies into glass like material known as borax bead.

$$Na_2B_4O_7.10H_2O \xrightarrow{\Delta} Na_2B_4O_7 \xrightarrow{\Delta} \underset{\text{Sodium metaborate}}{2NaBO_2} + \underset{\text{Boric anhydride}}{B_2O_3}$$

The metaborates of many transition metals have characteristic colours and, therefore, borax bead test can be used to identify them in the laboratory. For example, when borax is heated in a Bunsen burner flame with CoO on a loop of platinum wire, a blue coloured $Co(BO_2)_2$ bead is formed.

❑ ***Orthoboric acid :***

Preparation :

(i) It can be prepared by acidifying an aqueous solution of borax.

$$Na_2B_4O_7 + 2HCl + 5H_2O \rightarrow 2NaCl + 4B(OH)_3$$

(ii) It is also formed by the hydrolysis (reaction with water or dilute acid) of most boron compounds (halides, hydrides, etc.)

Property :

(i) Orthoboric acid, H_3BO_3 is a white crystalline solid, with soapy touch.

(ii) It is sparingly soluble in water but highly soluble in hot water.

(iii) H_3BO_3 is soluble in water and behaves as weak monobasic acid. It does not donate protons like most the acids, but rather it accepts OH^- . It therefore is Lewis acid $(B(OH)_3)$

$$B(OH)_3 + 2H_2O \rightleftharpoons H_3O^+ + [B(OH)_4]^-$$

or

H_3BO_3

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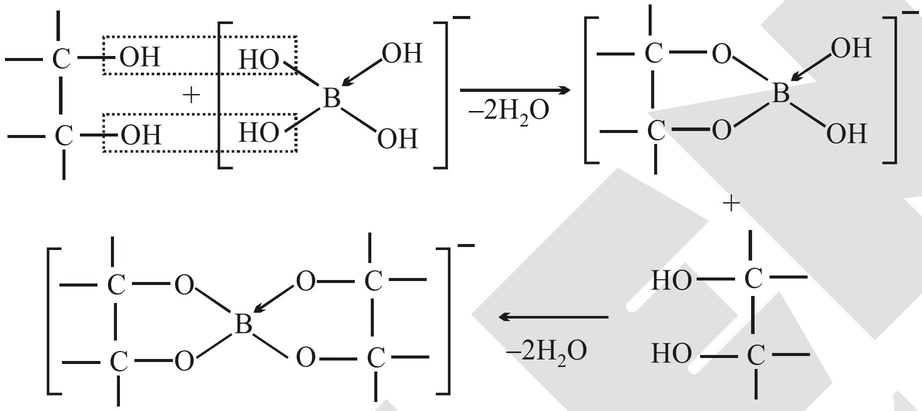
Since $B(OH)_3$ only partially reacts with water to form H_3O^+ and $[B(OH)_4]^-$ it behaves as a weak acid. Thus it cannot be titrated satisfactorily with NaOH as a sharp end point is not obtained. If certain polyhydroxy compounds such as glycerol, mannitol or sugar are added to the titration mixture then $B(OH)_3$ behaves as a strong monobasic acid. and hence can now be titrated with NaOH and end point is diluted using phenolphthalein as indicator.

$B(OH)_3 + NaOH \rightleftharpoons Na[B(OH)_4]$

$NaBO_2 + 2H_2O$

The added compound must be a cis diol to enhance the acidic proprieties in this way the cis-diol forms very stable complexes with $[B(OH)_4]^-$ formed in forward direction above, thus effectively removing it from solution. Hence reaction proceeds in forward direction (Le-Chatelier principle.)



On heating, orthoboric acid above 370K forms metaboric acid, HBO_2 which on further heating yields boric oxide, B_2O_3.

- ***Heating of boric acid :***

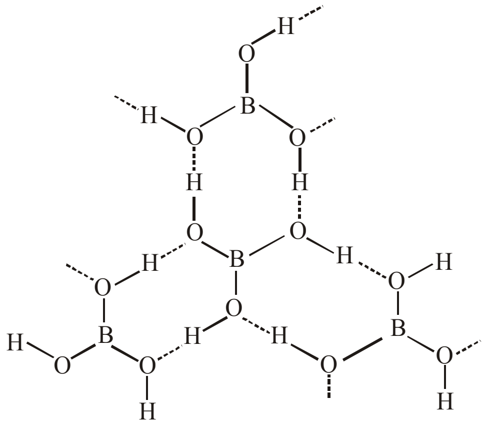
$$H_3BO_3 \xrightarrow{100°C} \underset{\text{Metaboric acid}}{HBO_2} \xrightarrow{140°C} \underset{\text{tetraboric acid}}{H_2B_4O_7} \xrightarrow[\text{red hot}]{\Delta} \underset{\text{Glassy mass}}{B_2O_3}$$

* $H_3BO_3 + H_2O_2 \longrightarrow (H_2O) + (HO)_2B–O–O–H \xrightarrow{NaOH} Na_2\left[(HO)_2 B \begin{matrix} O–O \\ O–O \end{matrix} B(OH)_2\right] \cdot 6H_2O$

Sodium peroxy borate used in washing powder as brightner

STRUCTURE

It has a layer structure in which planar BO_3 units are joined by hydrogen bonds as shown in figure.



Structure of boric acid; the dotted lines represent hydrogen bonds

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Uses of boric acid :

(i) Boric acid is used in manufacturing of optical glasses

(ii) With borax, it is used in the preparation of a buffer solution.

Diborane, B_2H_6

The simplest boron hydride known, is diborane.

Preparation :

(i) It is prepared by treating boron trifluoride with $LiAlH_4$ in diethyl ether.

$$\underset{\text{or } LiBH_4}{3LiAlH_4} + 4BF_3 \longrightarrow 3LiF + \underset{\text{or } 3(BF_3)}{3AlF_3} + 2B_2H_6$$

(ii) **Laboratory** method : For the preparation of diborane involves the oxidation of sodium borohydride with iodine.

$$2NaBH_4 + I_2 \rightarrow B_2H_6 + 2NaI + H_2$$

(iii) **Industrial scale** : By the reaction of BF_3 with sodium hydride.

$$2BF_3 + 6NaH \xrightarrow{450K} B_2H_6 + 6NaF$$

Other reaction of preparation of B_2H_6 :

$$\underset{(10\%)}{Mg_3B_2 + HCl} \longrightarrow B_2H_6 + B_4H_{10} + B_5H_9 \text{ etc.}$$

$$B_4H_{10} \xrightarrow[100°C]{\Delta} B_2H_6 + H_2 + \text{higher borane}$$

$$BCl_3 \text{ (or B Br}_3) + 6H_2 \xrightarrow[\text{discharge at low pressure}]{\text{Electric}} B_2H_6 + 6HCl$$

Properties :

(i) Diborane is a colourless, highly toxic gas with a b.p. of 180 K.

(ii) Diborane catches fire spontaneously upon exposure to air.

(iii) It burns in oxygen releasing an enormous amount of energy.

$$B_2H_6 + 3O_2 \rightarrow B_2O_3 + 3H_2O \ ; \ \Delta_cH^{\ominus} = -1976 \text{ kJ mol}^{-1}$$

Most of the higher boranes are also spontaneously flammable in air.

(iv) Boranes are readily hydrolysed by water to give boric acid.

$$B_2H_6(g) + 6H_2O(\ell) \text{ (Cold is enough)} \rightarrow 2B(OH)_3(aq) + 6H_2(g)$$

$$B_2H_6 + HCl\,(dry) \xrightarrow[AlCl_3]{anh.} B_2H_5Cl + H_2$$

(v) Diborane undergoes cleavage reactions with Lewis bases(L) to give borane adducts, $BH_3.L$

$$B_2H_6 + 2NMe_3 \rightarrow 2BH_3.NMe_3$$

$$B_2H_6 + 2CO \rightarrow 2BH_3.CO$$

Reaction of ammonia with diborane gives initially $B_2H_6.2NH_3$ which is formulated as $[BH_2(NH_3)_2]^+ [BH_4]^-$; further heating gives borazine, $B_3N_3H_6$ known as "inorganic benzene" in view of its ring structure with alternate BH and NH groups.

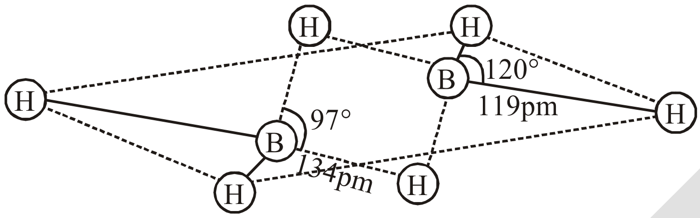
$$3B_2H_6 + 6NH_3 \rightarrow 3[BH_2(NH_3)_2]^+[BH_4]^- \xrightarrow{T=200°C} 2B_3N_3H_6 + 12H_2$$

$$B_3N_3H_6 \xrightarrow{T > 200°C} (BN)_x$$

ALLEN Session 2019-20

Structure & bonding in diborane :

The structure of diborane is shown in figure. The four terminal hydrogen atoms and the two boron atoms lie in one plane. Above and below this plane, there are two bridging hydrogen atoms. The four terminal B–H bonds are regular two centre-two electron bonds while the two bridge (B–H–B) bonds are different and can be described in terms of three centre–two electron bonds shown in figure.



The structure of diborance, B_2H_6

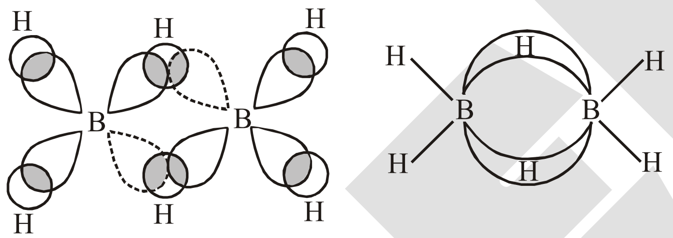


Figure: Bonding in diborane.

Each B atom uses sp^3 hybrid orbitals for bonding. Out of the four sp^3 hybrid orbital on each B atom, one is without an electron shown in broken lines. The terminal B–H bonds are normal 2-centre-2-electron bonds but the two bridge bonds are 3-centre-2-electron bonds. The 3-centre-2-electron bridge bonds are also referred to as banana bonds.

Note: Metal hydrido borates : Boron also forms a series of hydridoborates; the most important one is the tetrahedral $[BH_4]^-$ ion. Tetrahydridoborates of several metals are known. Lithium and sodium tetrahydridoborates, also known as borohydrides, are prepared by the reaction of metal hydrides with B_2H_6 in diethyl ether.

$2MH + B_2H_6 \rightarrow 2\ M^+\ [BH_4]^-$ (M = Li or Na)

Both $LiBH_4$ and $NaBH_4$ are used as reducing agents in organic synthesis. They are useful starting materials for preparing other metal borohydrides.

- ***Al_2O_3 preparation :***

(i) $2Al(OH)_3 \xrightarrow{300°C} Al_2O_3 + 3H_2O$

(ii) $Al_2(SO_4)_3 \xrightarrow{\Delta} Al_2O_3 + 3SO_3$

(iii) $(NH_4)_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O \xrightarrow{\Delta} Al_2O_3 + 2NH_3 + 4SO_3 + 25H_2O$

Uses: (i) In making refractory bricks

(ii) as an abrasive

(iii) To make high alumina cement

- ***$AlCl_3$ preparation :***

(i) $\underset{\text{(over heated)}}{2Al} + \underset{\text{dry}}{6HCl\ (vap.)} \longrightarrow 2AlCl_3 + 3H_2$

(ii) $Al_2O_3 + 3C + 3Cl_2 \xrightarrow[1000°C]{\Delta} 2AlCl_3\ (vap.) + 3CO$

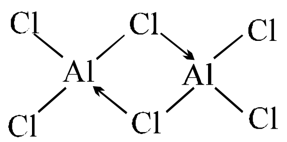
$$2AlCl_3\ (vap.) \xrightarrow{\text{Cooled}} \text{Solid anh. } AlCl_3$$

Properties :

(i) Its anhydrous form is deliquescent and fumes in air.

(ii) It sublimes at 180°C.

(iii) It is covalent and exists in the form of dimer even if in non polar solvents e.g. alcohol, ether, benzene, where it is soluble in fair extent.



Uses: (i) Friedel-Craft reaction

(ii) Dyeing, drug & perfumes etc.

❑ ***Alums :*** $M_2SO_4,\ M'_2(SO_4)_3 \cdot 24\ H_2O$

Properties: Swelling characteristics

where $M = Na^+, K^+, Rb^+, Cs^+, Ag^+, Tl^+, NH_4^+$ (except Li^+)

$M' = Al^{+3}, Cr^{+3}, Fe^{+3}, Mn^{+3}, Co^{+3}$

$K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$	Potash alum
$(NH_4)_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$	Ammonium alum
$K_2SO_4 \cdot Cr_2(SO_4)_3 \cdot 24H_2O$	Chrome alum
$(NH_4)_2SO_4 \cdot Fe_2(SO_4)_3 \cdot 24H_2O$	Ferric alum

Preparation: $Al_2O_3 + 3H_2SO_4 \longrightarrow Al_2(SO_4)_3 + 3H_2O$

$Al_2(SO_4)_3 + K_2SO_4 + \text{aq. sol}^n \longrightarrow \text{crystallise}$

Uses: (i) Act as coagulant

(ii) Purification of water

(iii) Tanning of leather

(iv) Mordant in dying

(v) Antiseptic

USES OF BORON AND ALUMINIUM AND THEIR COMPOUNDS

Boron :

(i) Boron being extremely hard refractory solid of high melting point, low density and very low electrical conductivity, finds many applications.

(ii) Boron fibres are used in making bullet-proof vest and light composite material for aircraft.

(iii) The boron-10 (^{10}B) isotope has high ability to absorb neutrons and, therefore, metal borides are used in nuclear industry as protective shields and control rods.

(iv) The main industrial application of borax and boric acid is in the manufacture of heat resistant glasses(e.g., Pyrex), glass-wool and fibreglass.

(v) Borax is also used as a flux for soldering metals, for heat, scratch and stain resistant glazed coating to earthenwares and as constituent of medicinal soaps.

(vi) An aqueous solution of orthoboric acid is generally used as a mild antiseptic.

ALLEN Session 2019-20

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Aluminium :

(i) Aluminium is a bright silvery-white metal, with high tensile strength.

(ii) It has a high electrical and thermal conductivity.

(iii) On a weight-to-weight basis, the electrical conductivity of aluminium is twice that of copper.

(iv) Aluminium is used extensively in industry and every day life.

(v) It forms alloys with Cu, Mn, Mg, Si and Zn.

(vi) Aluminium and its alloys can be given shapes of pipe, tubes, rods, wires, plates or foils and, therefore, find uses in packing, utensil making, construction, aeroplane and transportation industry.

(vii) The use of aluminium and its compounds for domestic purposes is now reduced considerably because of their toxic nature.

GROUP 14 ELEMENTS :

The carbon family

Carbon (C), silicon (Si), germanium (Ge), tin (Sn) and lead (Pb) are the members of group 14.

- **Occurrence of element**

(i) Carbon : Carbon is the seventeenth most abundant element by mass in the earth's crust. Naturally occurring carbon contains two stable isotopes:^{12}C and ^{13}C. In addition to these, third isotope, ^{14}C is also present. It is a radioactive isotope with halflife 5770 years and used for radiocarbon dating.

(ii) Silicon : Silicon is the second (27.7 % by mass) most abundant element on the earth's crust and is present in nature in the form of silica and silicates. Silicon is a very important component of ceramics, glass and cement.

(iii) Germanium : Germanium exists only in traces.

(iv) Tin : Tin occurs mainly as cassiterite, SnO_2

(v) Lead : Lead as galena, PbS.

Note : Ultrapure form of germanium and silicon are used to make transistors and semiconductor devices.

- **Electronic Configuration**

The valence shell electronic configuration of these elements is ns^2np^2. The inner core of the electronic configuration of elements in this group also differs.

- **Covalent Radius**

There is a considerable increase in covalent radius from C to Si, thereafter from Si to Pb a small increase in radius is observed. This is due to the presence of completely filled d and *f* orbitals in heavier members.

Covalent radii : C < Si < Ge < Sn < Pb

- **Ionization Enthalpy**

The first ionization enthalpy of group 14 members is higher than the corresponding members of group 13. The influence of inner core electrons is visible here also. In general the ionisation enthalpy decreases down the group. Small decrease in $\Delta_i H$ from Si to Ge, Ge to Sn and slight increase in $\Delta_i H$ from Sn to Pb is the consequence of poor shielding effect of intervening *d* and *f* orbitals and increase in size of the atom.

C > Si > Ge > Pb > Sn (IE_1 values)

- **Melting and Boiling Points**

  M.P. : C > Si > Ge > Pb > Sn

  B.P. : Si > Ge > Sn > Pb

- **Electronegativity**

  Due to small size, the elements of this group are slightly more electronegative than group 13 elements. The electronegativity values for elements from Si to Pb are almost the same.

- **Physical Properties**

  All group 14 members are solids. Carbon and silicon are non-metals, germanium is a metalloid, whereas tin and lead are soft metals with low melting points. Melting points and boiling points of group 14 elements are much higher than those of corresponding elements of group 13.

- **Chemical Properties**

  Oxidation states and trends in chemical reactivity

(i) The group 14 elements have four electrons in outermost shell.

(ii) The common oxidation states exhibited by these elements are +4 and +2. Carbon also exhibits negative oxidation states. Since the sum of the first four ionization enthalpies is very high, compounds in +4 oxidation state are generally covalent in nature.

(iii) In heavier members the tendency to show +2 oxidation state increases in the sequence Ge < Sn < Pb. It is due to the inability of ns^2 electrons of valence shell to participate in bonding. The relative stabilities of these two oxidation states vary down the group.

(iv) Carbon and silicon mostly show +4 oxidation state.

(v) Germanium forms stable compounds in +4 state and only few compounds in +2 state.

(vi) Tin forms compounds in both oxidation states (Sn in +2 state is a reducing agent).

(vii) Lead compounds in +2 state are stable and in +4 state are strong oxidising agents.

(viii) In tetravalent state the number of electrons around the central atom in a molecule (*e.g.*, carbon in CCl_4) is eight. Being *electron precise* molecules, they are normally not expected to act as electron acceptor or electron donor species. Although carbon cannot exceed its covalence more than 4, other elements of the group can do so. It is because of the presence of *d* orbital in them. Due to this, their halides undergo hydrolysis and have tendency to form complexes by accepting electron pairs from donor species. For example, the species like, SiF_6^{2-}, $[GeCl_6]^{2-}$, $[Sn(OH)_6]^{2-}$ exist where the hybridisation of the central atom is sp^3d^2.

- ***Reactivity towards oxygen***

  All members when heated in oxygen form oxides. There are mainly two types of oxides, *i.e.,* monoxide and dioxide of formula MO and MO_2 respectively. SiO only exists at high temperature. Oxides in higher oxidation states of elements are generally more acidic than those in lower oxidation states. The dioxides — CO_2, SiO_2 and GeO_2 are acidic, whereas SnO_2 and PbO_2 are amphoteric in nature. Among monoxides, CO is neutral, GeO is distinctly acidic whereas SnO and PbO are amphoteric.

ALLEN Session 2019-20

- **Reactivity towards water**

Carbon, silicon and germanium are not affected by water. Tin decomposes steam to form dioxide and dihydrogen gas.

$Sn + 2H_2O \xrightarrow{\Delta} SnO_2 + 2H_2$

Lead is unaffected by water, probably because of a protective oxide film formation.

- **Reactivity towards halogen**

(i) These elements can form halides of formula MX_2 and MX_4 (where X = F, Cl, Br, I). Except carbon, all other members react directly with halogen under suitable condition to make halides.

(ii) Most of the MX_4 are covalent in nature. The central metal atom in these halides undergoes sp^3 hybridisation and the molecule is tetrahedral in shape. Exceptions are SnF_4 and PbF_4, which are ionic in nature.

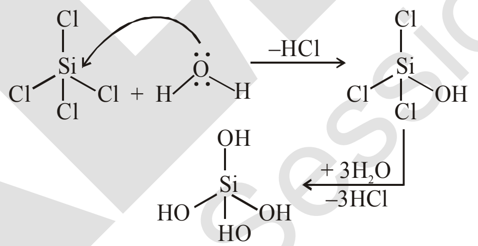
(iii) PbI_4 does not exist because Pb—I bond initially formed during the reaction does not release enough energy to unpair $6s^2$ electrons and excite one of them to higher orbital to have four unpaired electrons around lead atom.

(iv) Heavier members Ge to Pb are able to make halides of formula MX_2.

(v) Stability of dihalides increases down the group. Considering the thermal and chemical stability, GeX_4 is more stable than GeX_2, whereas PbX_2 is more than PbX_4.

(vi) Except CCl_4, other tetrachlorides are easily hydrolysed by water because the central atom can accommodate the lone pair of electrons from oxygen atom of water molecule in *d* orbital.

Hydrolysis can be understood by taking the example of $SiCl_4$. It undergoes hydrolysis by initially accepting lone pair of electrons from water molecule in *d* orbitals of Si, finally leading to the formation of $Si(OH)_4$ as shown below :



Silicic acid

IMPORTANT TRENDS AND ANOMALOUS BEHAVIOUR OF CARBON

Like first member of other groups, carbon also differs from rest of the members of its group. It is due to its smaller size, higher electronegativity, higher ionisation enthalpy unavailability of *d* orbitals.

In carbon, only *s* and *p* orbitals are available for bonding and, therefore, it can accommodate only four pairs of electrons around it. This would limit the maximum covalence to four whereas other members can expand their covalence due to the presence of *d* orbitals.

Carbon also has unique ability to form $p_\pi - p_\pi$ multiple bonds with itself and with other atoms of small size and high electronegativity. Few examples of multiple bonding are: C=C, C≡C, C = O, C = S and C≡N. Heavier elements do not form $p_\pi - p_\pi$ bonds because their atomic orbitals are too large and diffuse to have effective overlapping.

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Catenation Property

Carbon atoms have the tendency to link with one another through covalent bonds to form chains and rings. This property is called **catenation**. This is because C—C bonds are very strong. Down the group the size increases and electronegativity decreases, and, thereby, tendency to show catenation decreases. This can be clearly seen from bond enthalpies values. The order of catenation is C > > Si > Ge ≈ Sn. Lead does not show catenation.

Bond	**Bond enthalpy / kJ mol^{-1}**
C—C	348
Si—Si	297
Ge—Ge	260
Sn—Sn	240

Due to property of catenation and $p_\pi - p_\pi$ bond formation, carbon is able to show allotropic forms.

ALLOTROPES OF CARBON

Carbon exhibits many allotropic forms; both crystalline as well as amorphous. Diamond and graphite are two well-known crystalline forms of carbon. In 1985, third form of carbon known as **fullerenes** was discovered by H.W. Kroto, E.Smalley and R.F.Curl. For this discovery they were awarded the Nobel Prize in 1996.

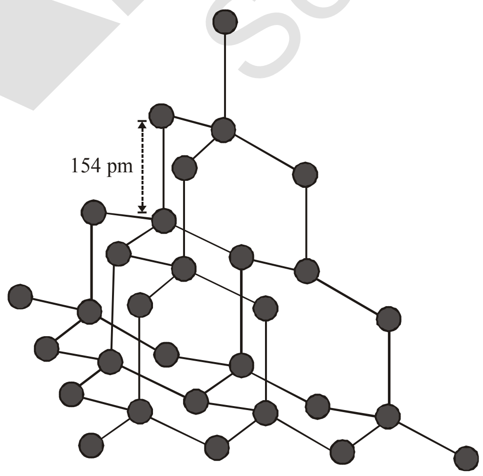
❑ **Diamond**

(i) It has a crystalline lattice.

(ii) In diamond each carbon atom undergoes sp^3 hybridisation and linked to four other carbon atoms by using hybridised orbitals in tetrahedral fashion.

(iii) The C–C bond length is 154 pm.

(iv) The structure extends in space and produces a rigid threedimensional network of carbon atoms. In this



The structure of diamond

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ALLEN Session 2019-20

structure figure directional covalent bonds are present throughout the lattice. It is very difficult to break extended covalent bonding and, therefore, diamond is a hardest substance on the earth.

Use : It is used as an abrasive for sharpening hard tools, in making dyes and in the manufacture of tungsten filaments for electric light bulbs.

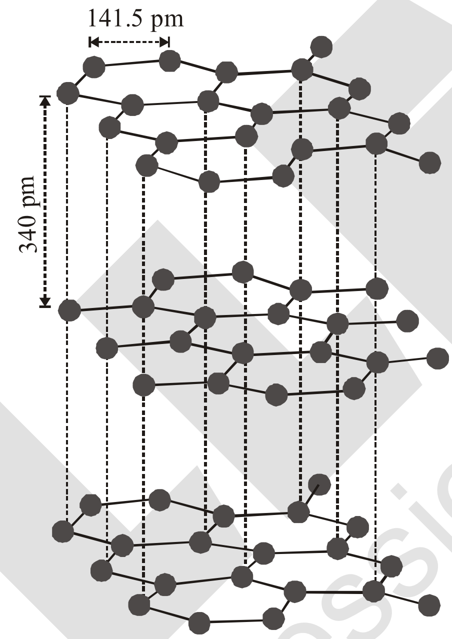
- **Graphite**

(i) Graphite has layered structure figure.

(ii) Layers are held by van der Waals forces and distance between two layers is 340 pm.

(iii) Each layer is composed of planar hexagonal rings of carbon atoms. C—C bond length within the layer is 141.5 pm.

(iv) Each carbon atom in hexagonal ring undergoes sp^2 hybridisation and makes three sigma bonds with three neighbouring carbon atoms. Fourth electron forms a π bond. The electrons are delocalised over the whole sheet. Electrons are mobile and, therefore, graphite conducts electricity along the sheet.



The structure of graphite

(v) Graphite cleaves easily between the layers and, therefore, it is very soft and slippery. For this reason graphite is used as a dry lubricant in machines running at high temperature, where oil cannot be used as a lubricant.

- **Fullerenes**

(i) Fullerenes are made by the heating of graphite in an electric arc in the presence of inert gases such as helium or argon. The sooty material formed by condensation of vapourised C^n small molecules consists of mainly C_{60} with smaller quantity of C_{70} and traces of fullerenes consisting of even number of carbon atoms up to 350 or above. Fullerenes are the only pure form of carbon because they have smooth structure without having 'dangling' bonds. Fullerenes are cage like molecules. C_{60} molecule has a shape like soccer ball and called **Buckminsterfullerene**.

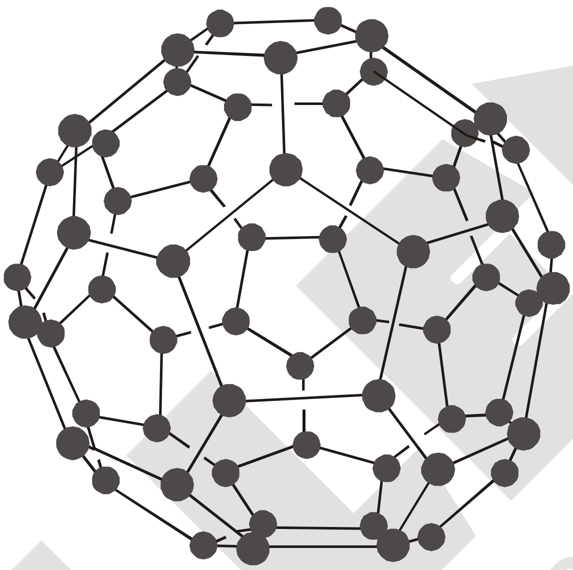
Session 2019-20

(ii) It contains twenty six- membered rings and twelve five membered rings.

(iii) A six membered ring is fused with six or five membered rings but a five membered ring can only fuse with six membered rings.

(iv) All the carbon atoms are equal and they undergo sp^2 hybridisation. Each carbon atom forms three sigma bonds with other three carbon atoms. The remaining electron at each carbon is delocalised in molecular orbitals, which in turn give aromatic character to molecule.

(v) This ball shaped molecule has 60 vertices and each one is occupied by one carbon atom and it also contains both single and double bonds with C–C distances of 143.5 pm and 138.3 pm respectively. Spherical fullerenes are also called *bucky balls* in short.



The structure of C_{60}, Buckminsterfullerene : Note that molecule has the shape of a soccer ball (football).

Note: It is very important to know that graphite is thermodynamically most stable allotrope of carbon and, therefore, $\Delta_f H^\ominus$ of graphite is taken as zero. $\Delta_f H^\ominus$ values of diamond and fullerene, C_{60} are 1.90 and 38.1 kJ mol^{-1}, respectively.

Note: Other forms of elemental carbon like carbon black, coke, and charcoal are all impure forms of graphite or fullerenes. Carbon black is obtained by burning hydrocarbons in a limited supply of air. Charcoal and coke are obtained by heating wood or coal respectively at high temperatures in the absence of air.

❑ **Uses of Carbon**

(i) Graphite fibres embedded in plastic material form high strength, lightweight composites. The composites are used in products such as tennis rackets, fishing rods, aircrafts and canoes.

(ii) Being good conductor, graphite is used for electrodes in batteries and industrial electrolysis.

(iii) Crucibles made from graphite are inert to dilute acids and alkalies.

ALLEN Session 2019-20

(iv) Being highly porous, activated charcoal is used in adsorbing poisonous gases; also used in water filters to remove organic contaminators and in airconditioning system to control odour.

(v) Carbon black is used as black pigment in black ink and as filler in automobile tyres.

(vi) Coke is used as a fuel and largely as a reducing agent in metallurgy.

(vii) Diamond is a precious stone and used in jewellery. It is measured in carats (1 carat = 200 mg).

SOME IMPORTANT COMPOUNDS OF CARBON AND SILICON

- **Types of Carbide**

(i) Ionic and salt like:

Classification on basis of no. of carbon atoms present in hydrocarbon found on their hydrolysis
- (a) C_1 unit
- (b) C_2 unit
- (c) C_3 unit

C_1 unit: Al_4C_3, Be_2C

$Be_2C + 4H_2O \longrightarrow 2Be(OH)_2 + CH_4$

$Al_4C_3 + 12H_2O \longrightarrow 4Al(OH)_3 + 3CH_4$

C_2 unit: CaC_2, BaC_2

$CaC_2 + 2H_2O \longrightarrow Ca(OH)_2 + CH{\equiv}CH$

C_3 unit: Mg_2C_3

$Mg_2C_3 + 4H_2O \longrightarrow 2Mg(OH)_2 + CH_3–C{\equiv}CH$: Propyne

(ii) <u>Covalent carbide</u> : SiC & B_4C

(iii) <u>Interstitial carbide</u> :

(Transition element or inner transitional elements forms this kind of carbide)

Interstitial carbide formation doesn't affect the metallic lusture and electrical conductivity.($\because$ no chemical bond is present, no change in property)

<u>SiC (Carborundum)</u>

- **Preparation**

$SiO_2 + 2C(coke) \xrightarrow[2500°C]{2000\ to} Si + 2CO\uparrow$

$Si + C \xrightarrow[2500°C]{2000\ to} SiC$

Note :

(i) SiC has diamond like or wurtzite structure

(ii) SiC is often dark purple, black or dark green due to traces of Fe and other impurities but pure sample are pale yellow to colourless.

Properties

(i) It is very hard and is used in cutting tools and abrasive powder(polishing material)

(ii) It is very much inert

(iii) It is not being affected by any acid except H_3PO_4

ALLEN Session 2019-20

Carbon Monoxide

Prepration :

(i) Direct oxidation of C in limited supply of oxygen or air yields carbon monoxide.

$$2C(s) + O_2(g) \xrightarrow{\Delta} 2CO(g)$$

(ii) On small scale pure CO is prepared by dehydration of formic acid with concentrated H_2SO_4 at 373 K

$$HCOOH \xrightarrow[\text{conc.}H_2SO_4]{373K} H_2O + CO$$

(iii) On commercial scale it is prepared by the passage of steam over hot coke. The mixture of CO and H_2 thus produced is known as **water gas** or **synthesis gas**.

$$C(s) + H_2O(g) \xrightarrow{473-1273K} CO(g) + H_2(g)$$

When air is used instead of steam, a mixture of CO and N_2 is produced, which is called **producer gas**.

$$2C(s) + O_2(g) + 4N_2(g) \xrightarrow{1273K} 2CO(g) + 4N_2(g) \text{ Producer gas}$$

Water gas and producer gas are very important industrial fuels. Carbon monoxide in water gas or producer gas can undergo further combustion forming carbon dioxide with the liberation of heat.

(iv) By heating potassium ferrocyanide with conc. H_2SO_4 : When potassium ferrocyanide in powdered state is heated with concentrated H_2SO_4, CO is evolved. Dilute H_2SO_4 should never be used because it shall evolve highly poisonous gas HCN.

$$K_4[Fe(CN)_6] + 3H_2SO_4 \rightarrow 2K_2SO_4 + FeSO_4 + 6HCN$$

$$6HCN + 12H_2O \rightarrow \underset{\text{Formic acid}}{6HCOOH} + 6NH_3$$

$$6NH_3 + 3H_2SO_4 \rightarrow 3(NH_4)_2SO_4$$

$$6HCOOH \xrightarrow{H_2SO_4} 6CO + 6H_2O$$

$$K_4[Fe(CN)_6] + 6H_2SO_4 + 6H_2O \rightarrow 2K_2SO_4 + FeSO_4 + 6CO + 3(NH_4)_2SO_4$$

Properties :

(i) Carbon monoxide is a colourless, odourless and almost water insoluble gas.

(ii) It is a powerful reducing agent and reduces almost all metal oxides other than those of the alkali and alkaline earth metals, aluminium and a few transition metals. This property of CO is used in the extraction of many metals from their oxides ores.

$$Fe_2O_3(s) + 3CO(g) \xrightarrow{\Delta} 2Fe(s) + 3CO_2(g)$$

$$ZnO(s) + CO(g) \xrightarrow{\Delta} Zn(s) + CO_2(g)$$

ALLEN Session 2019-20

DETECTION

(a) burns with blue flame

(b) CO is passed through $PdCl_2$ solution giving rise to black ppt.

$$CO + PdCl_2 + H_2O \longrightarrow \underset{}{CO_2}\uparrow + \underset{\text{Black metallic deposition}}{Pd\downarrow} + 2HCl$$

ESTIMATION

$$I_2O_5 + 5CO \longrightarrow I_2 + 5CO_2$$

$$I_2 + S_2O_3^{2-} \longrightarrow 2I^- + S_4O_6^{2-}$$

ABSORBERS

(a) Cu_2Cl_2 : $Cu_2Cl_2 + 2CO + 4H_2O \longrightarrow [CuCl(CO)(H_2O)_2]$

❑ **Bonding in CO mole**

In CO molecule, there are one sigma and two π bonds between carbon and oxygen**.** Because of the presence of a lone pair on carbon, CO molecule acts as a donor and reacts with certain metals when heated to form **metal carbonyls**.

❑ **Poisonous nature of CO**

The highly poisonous nature of CO arises because of its ability to form a **complex with haemoglobin**, which is about 300 times more stable than the oxygen-haemoglobin complex. This prevents haemoglobin in the red blood corpuscles from carrying oxygen round the body and ultimately resulting in death.

❑ **Carbon Dioxide**

Prepration :

(i) It is prepared by complete combustion of carbon and carbon containing fuels in excess of air.

$$C(s) + O_2(g) \xrightarrow{\Delta} CO_2(g)$$

$$CH_4(g) + 2O_2(g) \xrightarrow{\Delta} CO_2(g) + 2H_2O(g)$$

(ii) **Laboratory** by the action of dilute HCl on calcium carbonate.

$$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + CO_2\ (g) + H_2O(\ell)$$

(iii) Commercial scale by heating limestone.

Properties :

(i) It is a colourless and odourless gas.

(ii) Its low solubility in water makes it of immense biochemical and geo-chemical importance.

(iii) With water, it forms carbonic acid, H_2CO_3 which is a weak dibasic acid and dissociates in two steps:

$$H_2CO_3(aq) + H_2O(\ell) \rightleftharpoons HCO_3^-(aq) + H_3O^+(aq)$$

$$HCO_3^-(aq) + H_2O(\ell) \rightleftharpoons CO_3^{2-}(aq) + H_3O^+(aq)$$

H_2CO_3/HCO_3^- buffer system helps to maintain pH of blood between 7.26 to 7.42. Being acidic in nature, it combines with alkalies to form metal carbonates.

ALLEN Session 2019-20

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Use of CO_2

Carbon dioxide, which is normally present to the extent of ~ 0.03 % by volume in the atmosphere, is removed from it by the process known as **photosynthesis**. It is the process by which green plants convert atmospheric CO_2 into carbohydrates such as glucose. The overall chemical change can be expressed as:

$$6CO_2 + 12H_2O \xrightarrow[\text{Chlorphyll}]{h\nu} C_6H_{12}O_6 + 6O_2 + 6H_2O$$

By this process plants make food for themselves as well as for animals and human beings.

Harmful effect of CO_2

Unlike CO, it is not poisonous. But the increase in combustion of fossil fuels and decomposition of limestone for cement manufacture in recent years seem to increase the CO_2 content of the atmosphere. This may lead to increase in **green house effect** and thus, raise the temperature of the atmosphere which might have serious consequences.

(i) Carbon dioxide can be obtained as a solid in the form of **dry ice** by allowing the liquified CO_2 to expand rapidly and dry ice is used as a refrigerant for ice-cream and frozen food.

(ii) Gaseous CO_2 is extensively used to carbonate soft drinks. Being heavy and non-supporter of combustion it is used as fire extinguisher.

(iii) A substantial amount of CO_2 is used to manufacture urea. In CO_2 molecule carbon atom undergoes *sp* hybridisation. Two *sp* hybridised orbitals of carbon atom overlap with two *p* orbitals of oxygen atoms to make two sigma bonds while other two electrons of carbon atom are involved in $p_\pi–p_\pi$ bonding with oxygen atom. This results in its linear shape [with both C–O bonds of equal length (115 pm)] with no dipole moment. The resonance structures are shown below:

$$:\overset{-}{\ddot{\underset{\cdot\cdot}{O}}} - C \equiv \overset{+}{O}: \leftrightarrow :\ddot{O} = C = \ddot{O}: \leftrightarrow :\overset{+}{O} \equiv C - \overset{-}{\ddot{\underset{\cdot\cdot}{O}}}:$$

Resonating structures of carbon dioxide

Note : Carbongene has 95% O_2 and 5% CO_2 and is used as an antidote for poisoning of CO.

- **Teflon** $-(CF_2 - CF_2)_n-$

$$CHCl_3 + HF \xrightarrow[\text{SbFCl}_4 \text{ catalyst}]{\text{heated with}} CF_2HCl \xrightarrow[\text{again } -HCl]{\text{heated}} CF_2{=}CF_2 \xrightarrow[\text{high temp. and press}]{\text{Polymerisation at}} -(CF_2 - CF_2)_n-$$

- **Purpose**

Temperature with standing capacity upto 300°C (1^{st} organic compound withstand this kind of high temperature)

SILICON (Si)

- **Occurrence**

Silicon is the second most abundant (27.2%) element after oxygen (45.5%) in the earth's crust. It does not occur free in nature but in the combined state, it occurs widely in form of silica and silicates. All mineral rocks, clays and soils are built of silicates of magnesium, aluminium, potassium or iron. Aluminium silicate is however the most common constituent of rocks and clays.

Silica is found in the free state in sand, flint and quartz and in the combined state as silicates like

(i) Feldspar – $K_2O.\ Al_2O_3.\ 6SiO_2$

(ii) Kaolinite – $Al_2O_3.\ 2SiO_2.\ 2H_2O$

(iii) Asbestos – CaO. 3MgO. $4SiO_2$

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Preparation

(i) From silica (sand): Elemental silicon is obtained by the reduction of silica (SiO_2) with high purity coke in an electric furnace.

$$SiO_2(s) + 2C(s) \xrightarrow{\text{high temperature}} Si(s) + 2CO(g)$$

(ii) From silicon tetrachloride ($SiCl_4$) or silicon chloroform ($SiHCl_3$) : Silicon of very high purity required for making semiconductors is obtained by reduction of highly purified silicon tetrachloride or silicon chloroform with dihydrogen followed by purification by zone refining.

$$SiCl_4(l) + 2H_2(g) \longrightarrow Si(s) + 4HCl(g)$$

$$SiHCl_3(s) + H_2(g) \longrightarrow Si(s) + 3HCl\ (g)$$

- **Physical Properties :**

(i) Elemental silicon is very hard having diamond like structure.

(ii) It has shining luster with a melting point of 1793 K and boiling point of about 3550 K.

(iii) Silicon exists in three isotopes, i.e. ${}^{28}_{14}Si$, ${}^{29}_{14}Si$ and ${}^{30}_{14}Si$ but ${}^{28}_{14}Si$ is the most common isotope.

- **Chemical Properties :**

Silicon is particularly unreactive at room temperature towards most of the elements except fluorine. Some important chemical reactions of silicon are discussed below.

(i) ***Action of air :*** Silicon reacts with oxygen of air at 1173 K to form silicon dioxide and with nitrogen of air at 1673 K to form silicon nitride,.

$$Si(s) + O_2(g) \xrightarrow{1173K} \underset{\text{Silicon dioxide}}{SiO_2(s)}$$

$$3Si(s) + 2N_2(g) \xrightarrow{1673K} \underset{\text{Silicon nitride}}{Si_3N_4(s)}$$

(ii) ***Action of steam :*** It is slowly attacked by steam when heated to redness liberating dihydrogen gas.

$$Si(s) + 2H_2O(g) \xrightarrow{\text{redness}} SiO_2(s) + 2H_2(g)$$

(iii) ***Reaction with halogens:*** It burns spontaneously in fluorine gas at room temperature to form silicon tetrafluoride (SiF_4).

$$Si(s) + 2F_2(g) \xrightarrow{\text{Room Temperature}} SiF_4(l)$$

However, with other halogens, it combines at high temperatures forming tetrahalides.

(iv) ***Reaction with carbon :*** Silicon combines with carbon at 2500 °C forming silicon carbide (SiC) known as carborundum.

$$Si(s) + C(s) \xrightarrow{2500°C} SiC(s)$$

Carborundum is an extremely hard substance next only to diamond. It is mainly used as an abrasive and as a refractory material.

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Uses :

(i) Silicon is added to steel as such or more usually in form of ferrosilicon (an alloy of Fe and Si) to make it acid-resistant.

(ii) High purity silicon is used as semiconductors in electronic devices such as transistors.

(iii) It is used in the preparation of alloys such as silicon-bronze, magnesium silicon bronze and ferrosilicon.

❑ Compounds of Silicon :

Silane :

$Si_nH_{2n+2} \rightarrow SiH_4$ & Si_2H_6

Only these two are found

Higher molecules are not formed. $\because$ Si can't show catenation property

Hot Mg + Si–vap $\longrightarrow Mg_2Si \xrightarrow{\text{dil.}H_2SO_4} MgSO_4 + SiH_4 + Si_2H_6 + ...$

Silicones

It is an organosilicon polymer

TYPES OF SILICONES :

(i) Linear silicones

$$R_2SiCl_2 + H_2O \xrightarrow[-2HCl]{} R_2Si(OH)_2 \xrightarrow[H_2O]{\Delta} -O-\underset{R}{\overset{R}{|}}{\overset{|}{Si}}-O-\underset{R}{\overset{R}{|}}{\overset{|}{Si}}-O-\underset{R}{\overset{R}{|}}{\overset{|}{Si}}-O-$$

Linear silicone

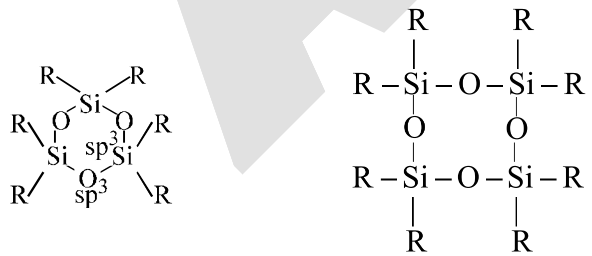
$$R\text{-}Cl \xrightarrow[\text{Silicon}]{\text{Cu powder}} R_2SiCl_2 \xrightarrow{H_2O} R_2Si(OH)_2 \xrightarrow{\text{condensation}} \text{Linear silicone}$$

$$R_2CCl_2 + H_2O \xrightarrow[\substack{-2HCl \\ \text{looses } H_2O \\ \text{readily}}]{} R_2C(OH)_2 \xrightarrow{-H_2O} R-\overset{\overset{O}{\|}}{C}-R$$

(ii) Cyclic silicones

Silicones may have the cyclic structure also having 3, 4, 5 and 6 nos. of silicon atoms within the ring.

Alcohol analogue of silicon is known as silanol



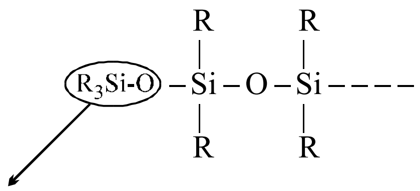
cyclic silicone not planar

(iii) Dimer silicones

$$R_3SiCl \xrightarrow{H_2O} \underset{\text{Silanol}}{R_3SiOH} \xrightarrow[\Delta]{-H_2O} R_3Si\text{-}O\text{-}SiR_3$$

Note

$$R_2SiCl_2 + R_3SiCl \xrightarrow[\text{hydrolysis}]{H_2O} R_2Si(OH)_2 + R_3SiOH \xrightarrow[\text{heating condensation}]{-H_2O}$$

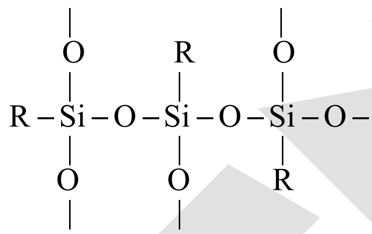


This end of the chain can't be extended hence R_3SiCl is called as chain stopping unit

* Using R_3SiCl in a certain proportion we can control the chain length of the polymer

(iv) Crossed linked silicones

$$RSiCl_3 + H_2O \longrightarrow R\text{-}Si(OH)_3 \xrightarrow{\Delta}$$



cross linked silicone
3 dimensional network

It provides the crosslinking among the chain making the polymer more hard and hence controling the proportion of $RSiCl_3$ we can control the hardness of polymer.

Uses

(1) It can be used as electrical insulator (due to inertness of Si–O–Si bonds)
(2) It is used as water repellant ($\because$ surface is covered) eg. car polish, shoe polish, masonary works in buildings
(3) It is used as antifoaming agent in sewage disposal, beer making and in cooking oil used to prepare potato chips.
(4) As a lubricant in the gear boxes and light weight machinery

SILICA (SiO_2)

Occurrence :

Silica or silicon dioxide occurs in nature in the free state as sand, quartz and flint and in the combined state as silicates like, Feldspar : $K_2O.Al_2O_3.6SiO_2$, Kaolinite : Al_2O_3. $2SiO_2$. $2H_2O$ etc.

Properties :

(i) Pure silica is colourless, but sand is usually coloured yellow or brown due to the presence of ferric oxide as an impurity.
(ii) Silicon dioxide is insoluble in water and all acids except hydrofluoric acid.
$SiO_2(s) + 4HF(l) \longrightarrow SiF_4(l) + 2H_2O(l)$
(iii) It also combines with metallic oxides at high temperature giving silicates e.g.
$SiO_2(s) + CaO(s) \xrightarrow{\Delta} CaSiO_3(s)$
(iv) When silica is heated strongly with metallic salts, silicates are formed and the volatile oxides are driven off as vapours.

$SiO_2(s) + Na_2CO_3(s) \xrightarrow{\text{high temp.}} Na_2SiO_3(s) + CO_2(g)$

$SiO_2(s) + Na_2SO_4(s) \xrightarrow{\text{high temp.}} Na_2SiO_3(s) + SO_3(g)$

$3SiO_2(s) + Ca_3(PO_4)_2(s) \xrightarrow{\text{high temp.}} 3CaSiO_3(s) + P_2O_5(g)$

The first two examples quoted here are important in glass making.

ALLEN Session 2019-20

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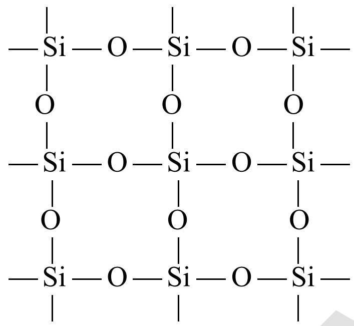
Structures of Silica :

Silica has a three-dimensional network structure. In silica, silicon is sp^3-hybridized and is thus linked to four oxygen atoms and each oxygen atom is linked to two silicon atoms forming a three-dimensional giant molecule as shown in figure. This three-dimensional network structure imparts stability to SiO_2 crystal and hence a large amount of energy is required to break the crystal resulting in high melting point.



Uses :

(i) Sand is used in large quantities to make mortar and cement.

(ii) Being transparent to ultraviolet light, large crystal of quartz are used for making lenses for optical instruments and for controlling the frequency of radio-transmitters.

(iii) Powdered quartz is used for making silica bricks.

(iv) Silica gel ($SiO_2.xH_2O$) is used as a desiccant (for absorbing moisture) and as an adsorbent in chromatography.

Quartz

Quartz is extensively used as a piezoelectric material; it has made possible to develop extremely accurate clocks, modern radio and television broadcasting and mobile radio communications. Silica gel is used as a drying agent and as a support for chromatographic materials and catalysts. Kieselghur, an amorphous form of silica is used in filtration plants.

Silicates

A large number of silicates minerals exist in nature. Some of the examples are feldspar, zeolites, mica and asbestos. Two important man-made silicates are glass and cement.

Zeolites

If aluminium atoms replace few silicon atoms in three-dimensional network of silicon dioxide, overall structure known as aluminosilicate, acquires a negative charge. Cations such as Na^+, K^+ or Ca^{2+} balance the negative charge. Examples are feldspar and zeolites. Zeolites are widely used as a catalyst in petrochemical industries for cracking of hydrocarbons and isomerisation, e.g., ZSM-5 (A type of zeolite) used to convert alcohols directly into gasoline. Hydrated zeolites are used as ion exchangers in softening of "hard" water.

ALLEN Session 2019-20

TIN & ITS COMPOUND

(i) $Sn \xrightarrow[+O_2]{1500^\circ C} SnO_2$ [Burns with a bright flame]

$Sn \xrightarrow{Cl_2, \Delta} SnCl_4$

$Sn \xrightarrow{S, \Delta} SnS_2$

(ii) $Sn + 2H_2O \xrightarrow{\text{room temp.}}$ No reaction

$Sn + 2H_2O \xrightarrow{\text{At high temp.}} SnO_2 + 2H_2$

(iii) Reaction with acid.

$Sn \xrightarrow{\text{dil.HCl}}$ reaction is very slow {Due to nonoxidising nature}

$Sn \xrightarrow{\text{hot.conc.HCl}} Sn + 2HCl \longrightarrow SnCl_2 + H_2\uparrow$

$Sn \xrightarrow{\text{dil. } H_2SO_4}$ dissolve Sn Slowly forming $SnSO_4 + H_2\uparrow$

$Sn \xrightarrow{\text{hot conc. } H_2SO_4} Sn + 4H_2SO_4 \longrightarrow Sn(SO_4)_2 + 2SO_2 + 4H_2O$

$Sn \xrightarrow{\text{cold dil. } HNO_3} 4Sn + 10HNO_3 \longrightarrow 4Sn(NO_3)_2 + NH_4NO_3 + 3H_2O$

$Sn \xrightarrow{\text{hot conc. } HNO_3} 5Sn + 20HNO_3 \longrightarrow \underset{\text{Metastannic acid}}{H_2Sn_5O_{11}.4H_2O} + 20NO_2 + 5H_2O$

$H_2Sn_5O_{11}.4H_2O \xrightarrow{\Delta} SnO_2$

(iv) $Sn + 2NaOH + H_2O \longrightarrow Na_2SnO_3 + 2H_2\uparrow$.

or

KOH [In absence of air K_2SnO_2 forms and in contact with air it readity converts into K_2SnO_3]

Oxides:

SnO (grey)

& SnO_2 (white)

$SnC_2O_4 \xrightarrow[\text{out of contact of air}]{\Delta} SnO \text{ (grey)} + CO + CO_2$

$SnC_2O_4 \xrightarrow{+\frac{1}{2}O_2} SnO_2\text{(White)}$

$Sn + O_2 \xrightarrow[\text{heated}]{\substack{1500^\circ C \\ \text{strongly}}} SnO_2\text{(White)}$

$H_2Sn_5O_{11}.4H_2O \xrightarrow{\Delta} SnO_2\text{(White)}$

$SnCl_2 \xrightarrow{KOH} Sn(OH)_2 \xrightarrow{\Delta \text{ (in absence of air)}} SnO \text{ (grey)}$

Both are amphoteric in nature :

$SnO + H_2SO_4 \longrightarrow SnSO_4 + H_2O$

$SnO + 2HCl \longrightarrow SnCl_2 + H_2O$

$SnO + 2NaOH \text{ or } KOH \xrightarrow{\text{cold}} Na_2SnO_2 \text{ or } K_2SnO_2 + H_2O$

But conc. hot alkali behaves differently.

$2SnO + 2KOH \text{ or } NaOH \longrightarrow K_2SnO_3 \text{ or } Na_2SnO_3 + Sn + H_2O$

* $Bi(OH)_3 + [Sn(OH)_4]^{2-} \longrightarrow \underset{\text{(black)}}{Bi\downarrow} + [Sn(OH)_6]^{2-}$

$SnO_2 + 2H_2SO_4 \xrightarrow{\Delta} Sn(SO_4)_2 + 2H_2O$

(**Soluble only** in hot conc. H_2SO_4)

$SnO_2 + 2NaOH \longrightarrow Na_2SnO_3 + H_2O$

ALLEN Session 2019-20
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$SnCl_2$ & $SnCl_4$:

(1) $Sn + 2HCl \text{ (hot conc.)} \longrightarrow SnCl_2 + H_2\uparrow$

$SnCl_2.2H_2O \xrightarrow{\Delta} Sn(OH)Cl + HCl\uparrow + H_2O\uparrow \Rightarrow$ Hence anh. $SnCl_2$ cannot be obtained.

$\downarrow$

$SnO + HCl\uparrow$

$\{SnCl_4 + 4H_2O \longrightarrow Sn(OH)_4 + 4HCl\uparrow$ fumes comes out$\}$

(2) A piece of Sn is always added to preserve a solution of $SnCl_2$. Explain.

$6SnCl_2 + 2H_2O + O_2 \longrightarrow 2SnCl_4 + 4Sn(OH)Cl\downarrow$ (white ppt)

$SnCl_4 + Sn \longrightarrow 2SnCl_2$

$SnCl_4 + 4H_2O \longrightarrow Sn(OH)_4\downarrow$ (white ppt.) $+ 4HCl$

(3) $SnCl_2 + HCl \longrightarrow HSnCl_3 \xrightarrow{HCl} H_2SnCl_4$

$SnCl_4 + 2HCl \longrightarrow H_2SnCl_6$(Hexachloro stannic (IV) acid)

$SnCl_4 + 2NH_4Cl \longrightarrow (NH_4)_2 SnCl_6$ (colourless crystalline compound known as "pink's salt")

(4) Reducing Properties of $SnCl_2$:

$Sn^{+2} + 2Fe^{+3} \longrightarrow 2Fe^{+2} + Sn^{+4}$

$2Cu^{+2} + Sn^{+2} \longrightarrow 2Cu^{+} + Sn^{+4}$

$Hg^{+2} + Sn^{+2} \longrightarrow Hg\downarrow + Sn^{+4}$

$PhNO_2 + SnCl_2 / HCl \longrightarrow PhNH_2 + Sn^{+4}$

$K_2Cr_2O_7 + SnCl_2 + HCl \longrightarrow Cr^{+3} + Sn^{+4} + KCl + H_2O$

(5) Readily combines with $I_2 \Rightarrow SnCl_2I_2 \Rightarrow$ This reaction is used to estimate tin.

Formation of $SnCl_4$:

(i) $\underset{\text{(molten)}}{Sn} + \underset{\text{(dry)}}{Cl_2 \text{(Excess)}} \longrightarrow SnCl_4$

(ii) $2HgCl_2 + SnCl_2 \longrightarrow 2Hg\downarrow + SnCl_4$

(iii) $Sn + \text{Aq. regia} \longrightarrow SnCl_4 + NO + H_2O$

* $SnCl_4.\ 5H_2O$ is known as butter of tin $\Rightarrow$ used as mordant.

$(NH_4)_2\ SnCl_6$ is known as 'pink's salt' $\Rightarrow$ used in calico printing.

Mosaic gold : SnS_2 yellow crystalline substance :

$Sn + 4NH_4Cl \longrightarrow (NH_4)_2SnCl_4 + 2NH_3 + H_2$

$2(NH_4)_2SnCl_4 + 2S \longrightarrow SnS_2 + 2NH_4Cl + (NH_4)_2SnCl_6$

Note : Mosaic gold used for filling purpose (in joining gold pieces)

* **Distinction of Sn^{+2} / Sn^{+4} :**

(i) H_2S (ii) Hg^{+2} (iii) $Fe^{+3} + K_3[Fe(CN)_6] \xrightarrow{Sn^{+2}}$ Blue ppt.

ALLEN Session 2019-20

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COMPOUNDS OF LEAD

Oxides of lead :

(i) PbO (ii) Pb_3O_4 (Red)

(iii) Pb_2O_3(reddish yellow) (Sesquioxide) (iv) PbO_2 (dark brown)

(1) PbO $\longrightarrow$ Yellow (Massicot) $\xrightarrow[\text{and powdered}]{\text{fused,cooled}}$ Litharge

PbO $\longrightarrow$ Reddish yellow (litharge)

Laboratory Prep^n. :

$Pb(NO_3)_2 \longrightarrow 2PbO + 4NO_2 + O_2$

$PbO_2 \xrightarrow{\text{above } 600°C}$

$Pb_3O_4 \longrightarrow$

$Pb_2O_3 \longrightarrow$

} PbO, hot oxide easily reduced to Pb by H_2 or C.

Preparation of Pb_2O_3 :

$$\underset{\substack{\text{hot sol}^n\text{.}\\ \text{of it in NaOH}}}{2PbO} + \underset{\substack{\text{Limited}\\ \text{amount}}}{NaOCl} \longrightarrow Pb_2O_3 + NaCl$$

$Pb_2O_3 + 2HNO_3 \longrightarrow PbO_2 \downarrow + Pb(NO_3)_2 + H_2O$

This reaction suggests that Pb_2O_3 contains PbO_2.

(2) **Pb_3O_4** : $6PbO + O_2 \underset{500°C}{\overset{350°C}{\rightleftharpoons}} 2Pb_3O_4$ {In the same way, prove that its formula is 2PbO. PbO_2}

$Pb_3O_4 + 4HNO_3$ (cold.conc) or (hot dil.) $\longrightarrow 2Pb(NO_3)_2 + PbO_2\downarrow + 2H_2O$

But $2Pb_3O_4 + 6H_2SO_4 \xrightarrow{\Delta} 6PbSO_4 + 6H_2O + O_2$

$Pb_3O_4 + 8HCl \longrightarrow 3PbCl_2 + 4H_2O + Cl_2$

(3) **PbO_2** : Insoluble in water :

HNO_3, But reacts with HCl and H_2SO_4(hot conc.) but does not react with HNO_3 and soluble in hot NaOH / KOH.

(i) $Pb_3O_4 + 4HNO_3 \longrightarrow 2Pb(NO_3)_2 + PbO_2 + 2H_2O$

(ii) $Pb(OAc)_2 + Ca(OCl)Cl + H_2O \longrightarrow PbO_2$[Brown(dark)] $+ CaCl_2 + 2CH_3CO_2H$

$\downarrow$

Excess bleaching powder is being removed by stirring with HNO_3

Reaction : $PbO_2 + 4HCl \longrightarrow PbCl_2 + Cl_2 + 2H_2O$

$2PbO_2 + 2H_2SO_4 \xrightarrow{\Delta} 2PbSO_4 + 2H_2O + O_2$

$PbO_2 + 2NaOH \longrightarrow Na_2PbO_3 + H_2O$

PbO_2 : Powerful oxidising agent :

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\p-Block Element\Eng\01_Theory.p65

(i) $PbO_2 + SO_2 \longrightarrow PbSO_4$ [spontaneously]

(ii) $PbO_2 + 2HNO_3 + (COOH)_2 \longrightarrow Pb(NO_3)_2 + 2CO_2 + 2H_2O$

(iii) $2Mn(NO_3)_2 + 5PbO_2 + 2H_2SO_4 \longrightarrow 3Pb(NO_3)_2 + 2PbSO_4\downarrow + 2HMnO_4 + 2H_2O$

$PbCl_4$: Exists as $H_2[PbCl_6]$

$PbO_2 + 4HCl \longrightarrow PbCl_4 + 2H_2O$

{ice cold conc. saturated with Cl_2}

$PbCl_4 + 2HCl \longrightarrow H_2PbCl_6$

TetraEthyl lead :

4Na–Pb(alloy 10%–Na.) + $4C_2H_5Cl$(vap.) $\longrightarrow 3Pb + Pb(Et)_4 + 4NaCl$

It is antiknocking agent.

ALLEN Session 2019-20

NITROGEN FAMILY

GROUP-15 ELEMENTS (N, P, As, Sb, Bi)

(i) As we go down the group, there is a shift from non-metallic to metallic through metalloidic character.

(ii) Nitrogen and phosphorus are **non-metals**, arsenic and antimony **metalloids** and bismuth is a **typical metal**.

Occurrence :

Nitrogen : Molecular nitrogen comprises 78% by volume of the atmosphere. It is 33[rd] most abundant element in the earth's crust. In the earth's crust, it occurs as sodium nitrate, $NaNO_3$ (called Chile saltpetre) and potassium nitrate (Indian saltpetre). It is found in the form of proteins and nucleic acid in plants and animals.

Phosphorus :

(i) It is eleventh most abundant element in earth's crust occurs in minerals of the apatite family, $Ca_9(PO_4)_6$. CaX_2 (X = F, Cl or OH) (e.g., fluorapatite $Ca_9\,(PO_4)_6.CaF_2$) and also found as chlorapatite $Ca_9\,(PO_4)_6.CaCl_2$) .

(ii) It is also present in nucleic acid (in DNA and RNA) which are the main components of phosphate rocks.

(iii) Arsenic , antimony and bismuth are found mainly as sulphide minerals.

Electronic Configuration :

The valence shell electronic configuration of these elements is ns^2np^3.

Atomic and Ionic Radii :

Covalent radius : N < P < As < Sb < Bi

Explanation :

Covalent and ionic (in a particular state) radii increase in size down the group. There is a considerable increase in covalent radius from N to P. However, from As to Bi only a small increase in covalent radius is observed. This is due to the presence of completely filled d and/or f orbitals in heavier members.

Ionisation Enthalpy :

N > P > As > Sb > Bi (IE_1 values)

Explanation :

Ionisation enthalpy decreases down the group due to gradual increase in atomic size. Because of the extra stable half-filled p orbitals electronic configuration and smaller size, the ionisation enthalpy of the group 15 elements is much greater than that of group 14 elements in the corresponding periods. The order of successive ionisation enthalpies, as expected is $\Delta_iH_1 < \Delta_iH_2 < \Delta_iH_3$ (See above table).

Electronegativity :

N > P > As > Sb = Bi
(1.9) (1.9)

Session 2019-20

Explanation :

The electronegativity value, in general, decreases down the group with increasing atomic size. However, amongst the heavier elements, the difference is not that much pronounced.

Metallic Character

N < P	As <	Sb < Bi
Non metal	Metalloid	Metals

Physical Properties :

(i) All the elements of this group are polyatomic. Dinitrogen is a diatomic gas while all others are solids.

(ii) Metallic character increases down the group.

(iii) Nitrogen and phosphorus are non-metals, arsenic and antimony metalloids and bismuth is a metal. This is due to decrease in ionisation enthalpy and increase in atomic size.

(iv) The boiling points, in general, increase from top to bottom in the group but the melting point increases upto arsenic and then decreases upto bismuth.

(v) Except nitrogen all the elements show allotropy.

P $\rightarrow$ exists in three allotropic form as white, red and black

As, Sb $\rightarrow$ exist as yellow and grey

Bi $\rightarrow$ exist as α, β, γ, δ allotropic form

Catenation

* The group 15 elements also show catenation property but to much smaller extent than carbon. For example hydrazine (H_2NNH_2) has two N atoms bonded together HN_3 has three N atoms.

HN_3 $\qquad$ $H-\ddot{N}=\overset{+}{N}=\ddot{N}\!:^{\ominus}$

* Among group 15 elements P has the maximum tendency for catenation forming cyclic as well as open chain compounds consisting of many phosphorous atoms.

P_2H_4 has two P atoms bonded together the lesser tendency of elements of group 15 to show catenation in compression to carbon is their low dissociation enthalpies.

C – C	353.3 kJ /mole
N – N	160.8 kJ / mole
P – P	201.6 kJ / mole
As – As	147.4 kJ / mole

Chemical Properties :

Oxidation states and trends in chemical reactivity

(i) The common oxidation states of these elements are –3, +3 and +5.

(ii) The tendency to exhibit –3 oxidation state decreases down the group due to increase in size and metallic character. Bismuth hardly forms any compound in –3 oxidation state.

(iii) The stability of +5 oxidation state decreases down the group. The only well characterised Bi (V) compound is BiF_5.

ALLEN Session 2019-20

(iv) The stability of +5 oxidation state decreases and that of +3 state increases (due to inert pair effect) down the group.

(v) Nitrogen exhibits + 1, + 2, + 4 oxidation states also when it reacts with oxygen. Phosphorus also shows +1 and +4 oxidation states in some oxoacids.

(vi) In the case of nitrogen, all oxidation states from +1 to +4 tend to disproportionate in acid solution. For example,

(i) $3HNO_2 \rightarrow HNO_3 + H_2O + 2NO$

(vii) Similarly, in case of phosphorus nearly all intermediate oxidation states disproportionate into +5 and –3 both in alkali and acid.

(ii) $4H_3PO_3 \xrightarrow{\text{Heat}} 3H_3PO_4 + PH_3$

(viii) +3 oxidation state in case of arsenic, antimony and bismuth becomes increasingly stable with respect to disproportionation.

(ix) Nitrogen is restricted to a maximum covalency of 4 since only four (one s and three p) orbitals are available for bonding.

(x) The heavier elements have vacant d orbitals in the outermost shell which can be used for bonding (covalency) and hence, expand their covalency as in PF_6^-.

Anomalous properties of nitrogen

(i) Nitrogen has unique ability to form p_π - p_π multiple bonds with itself and with other elements having small size and high electronegativity (e.g., C, O).

(ii) Heavier elements of this group do not form p_π-p_π bonds as their atomic orbitals are so large and diffuse that they cannot have effective overlapping.

(iii) Nitrogen exists as a diatomic molecule with a triple bond (one σ and two π) between the two atoms. N_2 bond enthalpy (941.4 kJ mol^{-1}) is very high.

(iv) Phosphorus, arsenic and antimony form single bonds as P–P, As–As and Sb–Sb while bismuth forms metallic bonds in elemental state.

(v) The single N–N bond is weaker than the single P–P bond because of high interelectronic repulsion of the non-bonding electrons, owing to the small bond length. As a result the catenation tendency is weaker in nitrogen.

(vi) Another factor which affects the chemistry of nitrogen is the absence of d orbitals in its valence shell. Besides restricting its covalency to four, nitrogen cannot form d_π–p_π bond as the heavier elements can e.g., $R_3P{=}O$ or $R_3P{=}CH_2$ (R=alkyl group).

(vii) Phosphorus and arsenic can form d_π–d_π bond also with transition metals when their compounds like $P(C_2H_5)_3$ and $As(C_6H_5)_3$ act as ligands.

ALLEN Session 2019-20

(i) Reactivity towards hydrogen:

All the elements of Group 15 form hydrides of the type EH_3 where E = N, P, As, Sb or Bi. Some of the properties of these hydrides are shown in Table. The hydrides show regular gradation in their properties. The stability of hydrides decreases from NH_3 to BiH_3 which can be observed from their bond dissociation enthalpy. Consequently, the reducing character of the hydrides increases. Ammonia is only a mild reducing agent while BiH_3 is the strongest reducing agent amongst all the hydrides. Basicity also decreases in the order $NH_3 > PH_3 > AsH_3 > SbH_3 \geq BiH_3$.

Table : Properties of Hydrides of Group 15 Elements

Property	NH_3	PH_3	AsH_3	SbH_3	BiH_3
Melting point/K	195.2	139.5	156.7	185	—
Boiling point/K	238.5	185.5	210.6	254.6	290
(E–H) distance/pm	101.7	141.9	151.9	170.7	—
HEH angle (°)	107.8	93.6	91.8	91.3	—
$\Delta_f H^\ominus/kJ\ mol^{-1}$	– 46.1	13.4	66.4	145.1	278
$\Delta_{diss} H^\ominus(E-H)/kJ\ mol^{-1}$	389	322	297	255	—

(ii) Reactivity towards oxygen: All these elements form two types of oxides: E_2O_3 and E_2O_5. The oxide in the higher oxidation state of the element is more acidic than that of lower oxidation state. Their acidic character decreases down the group. The oxides of the type E_2O_3 of nitrogen and phosphorus are purely acidic, that of arsenic and antimony amphoteric and those of bismuth predominantly basic.

(iii) Reactivity towards halogens: These elements react to form two series of halides: EX_3 and EX_5. Nitrogen does not form pentahalide due to non-availability of the d orbitals in its valence shell. Pentahalides are more covalent than trihalides. All the trihalides of these elements except those of nitrogen are stable. In case of nitrogen, only NF_3 is known to be stable. Trihalides except BiF_3 are predominantly covalent in nature.

(iv) Reactivity towards metals: All these elements react with metals to form their binary compounds exhibiting –3 oxidation state, such as, Ca_3N_2 (calcium nitride) Ca_3P_2 (calcium phosphide), Na_3As (sodium arsenide), Zn_3Sb_2 (zinc antimonide) and Mg_3Bi_2 (magnesium bismuthide).

DINITROGEN

Preparation :

(a) Commercial preparation :

Dinitrogen is produced **commercially** by the liquefaction and fractional distillation of air. Liquid dinitrogen (b.p. 77.2 K) distils out first leaving behind liquid oxygen (b.p. 90 K).

(b) Laboratory preparation :

(i) Dinitrogen is prepared by treating an aqueous solution of ammonium chloride with sodium nitrite.

$$NH_4Cl(aq) + NaNO_2(aq) \rightarrow N_2(g) + 2H_2O(\ell) + NaCl\ (aq)$$

Small amounts of NO and HNO_3 are also formed in this reaction; these impurities can be removed by passing the gas through aqueous sulphuric acid containing potassium dichromate.

Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\p-Block Element\Eng\01_Theory.p65

(ii) Dinitrogen can also be obtained by the thermal decomposition of ammonium dichromate.

$$(NH_4)_2Cr_2O_7 \xrightarrow{\text{Heat}} N_2 + 4H_2O + Cr_2O_3$$

Note : Very pure nitrogen can be obtained by the thermal decomposition of sodium or barium azide.

$$Ba(N_3)_2 \xrightarrow{\Delta} Ba + 3N_2$$

$$2NaN_3 \xrightarrow{300°C} 2Na + 3N_2$$

(c) Other preparation

$2NH_3 + 3NaOCl \longrightarrow N_2 + 3NaCl + 3H_2O$

$2NH_3 + 3Ca(OCl)_2 \longrightarrow 2N_2 + 3CaCl_2 + 6H_2O$

$8NH_3 + 3Br_2 \longrightarrow N_2 + 6NH_4Br$

$2NO + 2Cu \longrightarrow 2CuO + N_2$

(red,overheated) (Black)

Cl_2 passed into liquid NH_3

$3Cl_2 + 2NH_3 \longrightarrow N_2 + 6HCl$

$6NH_3 + 6HCl \longrightarrow 6NH_4Cl$

$3Cl_2 + 8NH_3 \longrightarrow N_2 + 6NH_4Cl$

In this method conc. of NH_3 should not be lowered down beyond a particular limit.

$3Cl_2 + NH_3 \longrightarrow NCl_3 + 3HCl$

$\downarrow$

(Tremendously explosive)

❑ **Physical properties :**

(i) Dinitrogen is a colourless, odourless, tasteless and non-toxic gas.

(ii) Nitrogen atom has two stable isotopes: ^{14}N and ^{15}N.

(iii) It has a very low solubility in water (23.2 cm^3 per litre of water at 273 K and 1 bar pressure)

(iv) Dinitrogen has low freezing and boiling points.

❑ **Chemical properties**

Reaction with metal : At higher temperatures, it directly combines with some metals to form predominantly ionic nitrides and with non-metals, covalent nitrides. A few typical reactions are:

$$6Li + N_2 \xrightarrow{\text{Heat}} 2Li_3N$$

$$3Mg + N_2 \xrightarrow{\text{Heat}} Mg_3N_2$$

Reaction with metal : It combines with hydrogen at about 773 K in the presence of a catalyst (Haber's Process) to form ammonia:

$$N_2(g) + 3H_2(g) \xrightleftharpoons{773\text{k}} 2NH_3(g)$$

Dinitrogen combines with dioxygen only at very high temperature (at about 2000 K) to form nitric oxide, NO.

$$N_2 + O_2(g) \xrightleftharpoons{\text{Heat}} 2NO(g)$$

ALLEN Session 2019-20

Absorption on calcium carbride

N_2 can be absorbed by calcium carbide at the temperature around 1000°C.

$$CaC_2 + N_2 \xrightarrow{1100^\circ C} \underbrace{CaNCN + C}_{\text{nitrolim}}$$

It is a very good fertiliser.

Cyanamide ion $\quad {}^{2\ominus}N-C\equiv N \;\updownarrow\; {}^{\ominus}N=C=N^{\ominus}$

$$(Ca(NCN) + C) + 3H_2O \longrightarrow CaCO_3 + 2NH_3 + C$$

$\rightarrow$ Slowly decomposes

$\rightarrow NH_2-CO-NH_2$ (Intermidiate formed)

Qus. Why dinitrogen is inert at room temperature ?

Ans. Dinitrogen is inert at room temperature because of the high bond enthalpy of $N\equiv N$ bond. Reactivity, however, increases rapidly with rise in temperature.

TYPES OF NITRIDE :

Salt like or ionic : Li_3N, Na_3N, K_3N, Ca_3N_2, Mg_3N_2, Be_3N_2

Covalent : AlN, BN, Si_3N_4, Ge_3N_4, Sn_3N_4

Interstitial : MN $\underbrace{(M = Sc, Ti, Zr, Hf, La)}_{\text{HCP or FCC}}$

No of metal atom per unit cell is equal to no of octahedral voids per unit cell.

All the octahedral voids are occupied by nitrogen atoms. Hence the fornmula is MN.

HCP : Hexagonal closed packing

FCC : Face centred cubic

Uses :

(i) The main use of dinitrogen is in the manufacture of ammonia and other industrial chemicals containing nitrogen, (e.g., calcium cyanamide).

(ii) It also finds use where an inert atmosphere is required (e.g., in iron and steel industry, inert diluent for reactive chemicals).

(iii) Liquid dinitrogen is used as a refrigerant to preserve biological materials, food items and in cryosurgery.

AMMONIA

Preparation :

(i) Ammonia is present in small quantities in air and soil where it is formed by the decay of nitrogenous organic matter e.g., urea.

$$NH_2CONH_2 + 2H_2O \longrightarrow (NH_4)_2CO_3 \rightleftharpoons 2NH_3 + H_2O + CO_2$$

(ii) Small scale preparation

By the decomposition of ammonium salts when treated with caustic soda or calcium hydroxide.

$$2NH_4Cl + Ca(OH)_2 \rightarrow 2NH_3 + 2H_2O + CaCl_2$$

$$(NH_4)_2SO_4 + 2NaOH \rightarrow 2NH_3 + 2H_2O + Na_2SO_4$$

(iii) Large scale manufacturing (Haber's Process)

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g); \qquad \Delta_f H^\ominus = -46.1 \text{ kJ mol}^{-1}$$

* According to Le Chatelier's principle, high pressure and low temperature would favour the formation of ammonia.

ALLEN Session 2019-20

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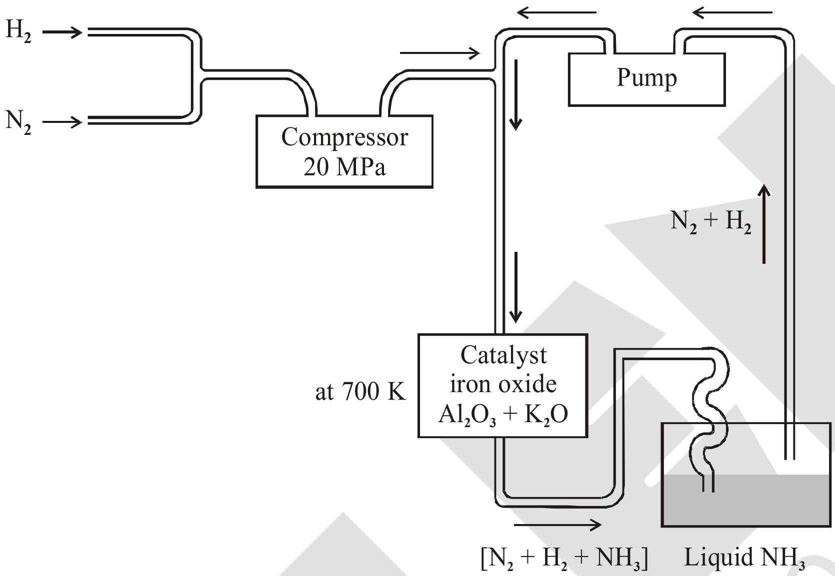
* The optimum conditions for the production of ammonia are a pressure of 200×10^5 Pa (about 200 atm), a temperature of ~ 700 K.

* Use of a catalyst such as iron oxide with small amounts of K_2O and Al_2O_3 to increase the rate of attainment of equilibrium.

* The flow chart for the production of ammonia is shown in figure. Earlier, iron was used as a catalyst with molybdenum as a promoter.



Flow chart for the manufacture of ammonia

Other preparation :

(i) Nitrate or nitrite reduction : $NO_3^- / NO_2^- + Zn$ or $Al + NaOH \longrightarrow NH_3 + [Zn(OH)_4]^{2-}$ or $[Al(OH)_4]^-$

(ii) Metal nitride hydrolysis : $N^{3-} + 3H_2O \longrightarrow NH_3 \uparrow + 3\ OH^-$

Properties :

(i) Ammonia is a colourless gas with a pungent odour.

(ii) Its freezing and boiling points are 198.4 and 239.7 K respectively.

(iii) In the solid and liquid states, it is associated through hydrogen bonds as in the case of water and that accounts for its higher melting and boiling points than expected on the basis of its molecular mass.

(iv) Ammonia gas is highly soluble in water.

(v) **Basic character :**

Its aqueous solution is weakly basic due to the formation of OH^- ions.

$$NH_3(g) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$$

It forms ammonium salts with acids, e.g., NH_4Cl, $(NH_4)_2\ SO_4$, etc.

As a weak base, it precipitates the hydroxides (hydrated oxides in case of some metals) of many metals from their salt solutions.

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\p-Block Element\Eng\01_Theory.p65
Session 2019-20

For example,

$$ZnSO_4(aq) + 2NH_4OH(aq) \longrightarrow \underset{\text{(White ppt)}}{Zn(OH)_2(s)} + (NH_4)_2SO_4(aq)$$

$$FeCl_3(aq) + NH_4OH(aq) \longrightarrow \underset{\text{(brown ppt)}}{Fe_2O_3.xH_2O(s)} + NH_4Cl(aq)$$

In ammonia molecule the presence of a lone pair of electrons on the nitrogen atom of the makes it a Lewis base. It donates the electron pair and forms linkage with metal ions and the formation of such complex compounds finds applications in detection of metal ions such as Cu^{2+}, Ag^+:

$$\underset{\text{(blue)}}{Cu^{2+}(aq)} + 4NH_3(aq) \rightleftharpoons \underset{\text{(deep blue)}}{[Cu(NH_3)_4]^{2+}(aq)}$$

$$\underset{\text{(Colourless)}}{Ag^+(aq)} + Cl^-(aq) \longrightarrow \underset{\text{(white ppt)}}{AgCl(s)}$$

$$\underset{\text{(white ppt)}}{AgCl(s)} + 2NH_3(aq) \longrightarrow \underset{\text{(Colourless)}}{[Ag(NH_3)_2]Cl(aq)}$$

Other reactions

CH_3COOH is strong acid in liq. NH_3 while in water is weak acid.

$$AcOH \rightleftharpoons Ac\overline{O} + H^+$$

$$NH_3 + H^+ \longrightarrow NH_4^+$$

$$H_2O + H^+ \longrightarrow H_3O^+$$

Basicity order $NH_3 > H_2O$

more solvation of H^+ in NH_3.

Hydrolysis and Ammonolysis occurs is a same way.

$$SiCl_4 + 4H_2O \longrightarrow 4HCl + Si(OH)_4 \xrightarrow{\Delta} SiO_2 + 2H_2O$$

$$SiCl_4 + 8NH_3 \longrightarrow 4NH_4Cl + Si(NH_2)_4 \xrightarrow{\Delta} Si_3N_4 + NH_3\uparrow$$

Rate of hydrolysis and Ammonolysis will be affected by the presence of HCl vapour & NH_4Cl vapour respectively.

Uses :

(i) Ammonia is used to produce various nitrogenous fertilisers (ammonium nitrate, urea, ammonium phosphate and ammonium sulphate).

(ii) In the manufacture of some inorganic nitrogen compounds, the most important one being nitric acid.

(iii) Liquid ammonia is also used as a refrigerant.

BONDING IN AMMONIA :

The ammonia molecule is trigonal pyramidal with the nitrogen atom at the apex. It has three bond pairs and one lone pair of electrons.

ALLEN Session 2019-20

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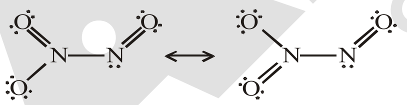
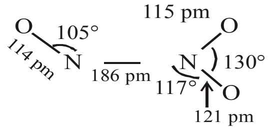
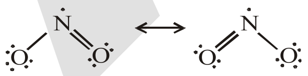
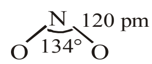
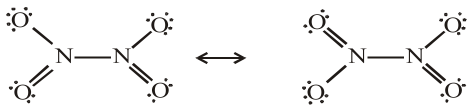
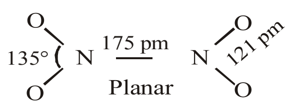
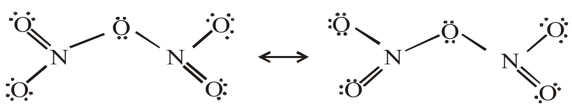
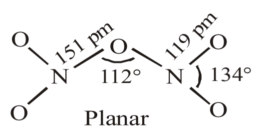
OXIDES OF NITROGEN :

Nitrogen forms a number of oxides in different oxidation states. The names, formulas, preparation and physical appearance of these oxides are given in Table.

Oxides of Nitrogen

Name	Formula	Oxidation state of nitrogen	Common methods of preparation	Physical appearance and chemical nature
Dinitrogen oxide [Nitrogen oxide]	N_2O	+1	$NH_4NO_3 \xrightarrow{\text{Heat}} N_2O + 2H_2O$	Colourless gas, neutral
Nitrogen monoxide [Nitrogen (II) oxide]	NO	+2	$2NaNO_2 + 2FeSO_4 + 3H_2SO_4 \rightarrow Fe_2(SO_4)_3 + 2NaHSO_4 + 2H_2O + 2NO$	Colourless gas, neutral
Dinitrogen trioxide [Nitrogen (III) oxide]	N_2O_3	+3	$2NO + N_2O_4 \xrightarrow{-30°C} 2N_2O_3$	Pale Blue solid (MP = –100.1°C), acidic, Intense blue liquid (–30ºC)
Nitrogen dioxide [Nitrogen (IV) oxide]	NO_2	+4	$2Pb(NO_3)_2 \xrightarrow{673K} 4NO_2 + 2PbO + O_2$	brown gas, acidic
Dinitrogen tetroxide [Nitrogen (IV) oxide]	N_2O_4	+4	$2NO_2 \underset{\text{Heat}}{\overset{\text{Cool}}{\rightleftharpoons}} N_2O_4$	Colourless solid/ liquid, acidic
Dinitrogen pentaoxide [Nitrogen(V) oxide]	N_2O_5	+5	$4HNO_3 + P_4O_{10} \rightarrow 4HPO_3 + 2N_2O_5$	colourless solid, acidic

Structure of Oxides of Nitrogen

Formula	Resonance structures	Bond Parameters
N_2O	$\ddot{N}=N=\ddot{O} \leftrightarrow :N\equiv N-\ddot{O}:$	N—N—O 113 pm 119 pm Linear
NO	$:N=\ddot{O} \leftrightarrow :\dot{N}=\ddot{O}:$	N—O 115 pm
N_2O_3		
No_2		 Angular
N_2O_4		
N_2O_5		 Planar

Session 2019-20

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Preparation:

1. **N_2O**

(i) $NH_4NO_3 \longrightarrow N_2O + H_2O$

(ii) $(NH_4)_2SO_4 + NaNO_3 \longrightarrow NH_4NO_3 + Na_2SO_4$

$$NH_4NO_3 \downarrow N_2O + 2H_2O$$

(iii) $\underset{\text{(dil.\& cold)}}{Zn + HNO_3} \longrightarrow Zn(NO_3)_2 + N_2O + H_2O$

2. **NO**

(i) $Cu + \underset{\text{hot}}{HNO_3\ (1:1)} \longrightarrow Cu(NO_3)_2 + NO + H_2O$

(ii) $KNO_3 + FeSO_4 + H_2SO_4 \longrightarrow Fe_2(SO_4)_3 + K_2SO_4 + NO + H_2O$

$FeSO_4 + NO \longrightarrow FeSO_4 \cdot NO \xrightarrow{\Delta} FeSO_4 + NO\uparrow$

(iii) Oswald process–Restricted oxidation of NH_3.

Industrial process.

$4NH_3 + 5O_2 \xrightarrow[750°C,\ Pt-Cat]{6\ atm} 4NO + 6H_2O$

3. **N_2O_3**

(i) $HNO_3 + As_2O_3 \longrightarrow H_3AsO_4 + N_2O_3$

(ii) $Cu + HNO_3(6M) \longrightarrow Cu(NO_3)_2 + \underbrace{(NO + NO_2)}_{\downarrow\ \text{Cool}(-30°C)\ \text{Blue liq}\ (N_2O_3)}$

4. **NO_2**

(i) $M(NO_3)_2 \xrightarrow{\Delta} MO + 2NO_2 + \frac{1}{2}O_2$

M = Pb, Cu, Ba, Ca

(ii) $(Cu, Pb, Ag) + \underset{\text{(hot \& conc.)}}{HNO_3} \longrightarrow \text{M-nitrate} + NO_2 + H_2O$

5. **N_2O_5**

(i) $2HNO_3 + P_2O_5 \longrightarrow 2HPO_3 + N_2O_5$

(ii) $4AgNO_3 + 2Cl_2\text{(dry gas)} \longrightarrow 4AgCl + 2N_2O_5 + O_2$

ALLEN Session 2019-20

Properties:

(I) Decoposition Behaviour

(i) $N_2O \xrightarrow{500°C-900°C} 2N_2 + O_2$

(ii) $2NO \xrightarrow{800°C} N_2 + O_2$

(iii) $N_2O_3 \xrightarrow{\text{Room temp.}} NO_2 + NO$

(Blue liq.) at (–30°C)

(iv) $2NO_2 \xrightarrow{620°C} 2NO + O_2$

$$\underset{\substack{\text{(white solid)} \\ \text{at } (-11°C)}}{N_2O_4} \xrightarrow{-11°C} \underset{\text{Brown gas}}{2NO_2}$$

(v) $$\underset{\substack{\text{colourless} \\ \text{solid}}}{N_2O_5} \xrightarrow{30°C} \underset{\substack{\text{yellow} \\ \text{liq.}}}{N_2O_5} \xrightarrow{40°C} 2NO_2 + \frac{1}{2}O_2$$

(II) Reaction with H_2O & NaOH

		H_2O	NaOH
(i)	N_2O :	Fairly soluble in water and produces neutral solution	-------
(ii)	NO :	Sparingly soluble in water and produces neutral sol^n.	-------
(iii)	N_2O_3:	$2HNO_2$ Hence it is known as anhydride of HNO_2	$NaNO_2$
(iv)	NO_2:	$HNO_2 + HNO_3$ called as mixed anhydride	$NaNO_2 + NaNO_3$
(v)	N_2O_5:	$2HNO_3$ called as anhydride of HNO_3	$NaNO_3$

Other properties:

N_2O : $2N_2O \longrightarrow 2N_2 + O_2$ $\left\{\begin{array}{l}\text{mixture contains} \\ 33\%\, O_2 \text{ compared} \\ \text{to } 20\% \text{ in air}\end{array}\right.$

Hence, it is a better supporter for combustion.

$S + N_2O \longrightarrow SO_2 + N_2$

$P + N_2O \longrightarrow P_2O_5 + N_2$

$Mg + N_2O \longrightarrow MgO + N_2$

$Na + N_2O \longrightarrow Na_2O + N_2$

$Cu + N_2O \longrightarrow CuO + N_2$

$H_2 + N_2O \longrightarrow H_2O + N_2$

ALLEN Session 2019-20

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NO : (i) It burns : $NO + \frac{1}{2}O_2 \longrightarrow NO_2$

(ii) It supports combustion also for molten sulphur and hot phosphorous.

$S + 2NO \longrightarrow SO_2 + N_2$

$2P + 5NO \longrightarrow P_2O_5 + \frac{5}{2}N_2$

(iii) It is being absorbed by $FeSO_4$ solution.

(iv) It is having reducing property.

$KMnO_4 + NO + H_2SO_4 \longrightarrow K_2SO_4 + MnSO_4 + HNO_3 + H_2O$

$HOCl + NO + H_2O \longrightarrow HNO_3 + HCl$

(v) NO shows oxidising property also.

$SO_2 + 2NO + H_2O \longrightarrow H_2SO_4 + N_2O$

$H_2S + 2NO \longrightarrow H_2O + S\downarrow + N_2O$

$3SnCl_2 + 2NO + 6HCl \longrightarrow 3SnCl_4 + 2NH_2OH$

(Used for NH_2OH preparation)

(vi) NO combines with X_2 (X_2= $Cl_2Br_2F_2$) to produce NO X

$2NO + X_2 \longrightarrow 2NOX$

N_2O_3 : No more properties.

NO_2 : (1) It is having oxidising property.

$S + NO_2 \longrightarrow SO_2 + NO$

$P + NO_2 \longrightarrow P_2O_5 + NO$

$C + NO_2 \longrightarrow CO_2 + NO$

$SO_2 + NO_2 + H_2O \longrightarrow H_2SO_4 + NO$

$H_2S + NO_2 \longrightarrow H_2O + S\downarrow + NO$

$CO + NO_2 \longrightarrow CO_2 + NO$

NO not formed : $2KI + 2NO_2 \longrightarrow I_2 + 2KNO_2$

(2) Reducing property of NO_2.

$KMnO_4 + NO_2 + H_2SO_4 \longrightarrow K_2SO_4 + MnSO_4 + HNO_3 + H_2O$

$$O_3^0 + 2\overset{+4}{N}O_2 \longrightarrow \underbrace{O_2^0}_{\textit{not the reduction product of } O_3} + \overset{+5}{N_2}O_5$$

N_2O_5 :

$I_2 + 5N_2O_5 \longrightarrow I_2O_5 + 10NO_2$

I_2O_5 is used for the estimation of CO

$I_2O_5 + 5CO \longrightarrow I_2 + 5CO_2$

$I_2 + 2S_2O_3^{2-} \longrightarrow 2I^- + S_4O_6^{2-}$

$N_2O_5 + NaCl \longrightarrow NaNO_3 + NO_2Cl$

This proves that N_2O_5 is consisting of ion pair of NO_2^+ & NO_3^-

ALLEN Session 2019-20

OXOACIDS OF NITROGEN

$H_2N_2O_2$ (hyponitrous acid), HNO_2 (nitrous acid) and HNO_3 (nitric acid). Amongst them HNO_3 is the most important.

NITROUS ACID (HNO_2)

Preparation

(a) M-nitrite $\xrightarrow[\text{HCl or } H_2SO_4]{\text{dil.acid}}$ HNO_2

(b) $N_2O_3 + H_2O \longrightarrow 2HNO_2$

Properties

(a) **Oxidising property** : Because of its easy oxidation to liberate nascent oxygen, it acts as a strong oxidant $2HNO_2 \longrightarrow H_2O + 2NO + (O)$

$2KI + 2HNO_2 + 2HCl \longrightarrow 2KCl + 2H_2O + 2NO + I_2$

$SnCl_2 + 2HNO_2 + 2HCl \longrightarrow SnCl_4 + 2NO + 2H_2O$

$SO_2 + 2HNO_2 \longrightarrow H_2SO_4 + 2NO$

$H_2S + 2HNO_2 \longrightarrow 2H_2O + S\downarrow + 2NO$

$2FeSO_4 + 2HNO_2 + H_2SO_4 \longrightarrow Fe_2(SO_4)_3 + 2NO + 2H_2O$

$Na_3AsO_3 + 2HNO_2 \longrightarrow Na_3AsO_4 + 2NO + H_2O$

(b) **Reducing property** : Nitrous acid also acts as a reducing agent as it can be oxidised into nitric acid.

$HNO_2 + (O) \longrightarrow HNO_3$

$2KMnO_4 + 5HNO_2 + 3H_2SO_4 \longrightarrow K_2SO_4 + 2MnSO_4 + 5HNO_3 + 3H_2O$

$K_2Cr_2O_7 + 3HNO_2 + 4H_2SO_4 \longrightarrow K_2SO_4 + Cr_2(SO_4)_3 + 3HNO_3 + 4H_2O$

$H_2O_2 + HNO_2 \longrightarrow H_2O + HNO_3$

(c) **Reaction with NH_3 / $-NH_2$ compounds :**

$2HNO_2 + \underset{\text{Urea}}{NH_2CONH_2} \longrightarrow 2N_2 + CO_2 + 3H_2O$

$HNO_2 + NH_3 \longrightarrow NH_4NO_2 \longrightarrow N_2 + 2H_2O$

$HNO_2 + C_2H_5NH_2 \longrightarrow C_2H_5OH + N_2 + H_2O$

$HNO_2 + C_6H_5\text{-}NH_2\cdot HCl \xrightarrow{<5^\circ C} \underset{\text{Benzene diazonium chloride}}{C_6H_5N = NCl} + 2H_2O$

$HNO_2 + \text{thiourea} \longrightarrow N_2\uparrow + H^+ + SCN^- + 2H_2O$

$HNO_2 + \text{sulphamic acid} \longrightarrow N_2\uparrow + 2H^+ + SO_4^{2-} + H_2O$

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NITRIC ACID

It was named aqua fortis (means strong water) by alchemists.

Preparation :

Laboratory Method : By heating KNO_3 or $NaNO_3$ and concentrated H_2SO_4 in a glass retort.

$$KNO_3 / NaNO_3 + H_2SO_4 \rightarrow KHSO_4 / NaHSO_4 + HNO_3$$

Large scale preparation (Ostwald's process) :

(i) This method is based upon catalytic oxidation of NH_3 by atmospheric oxygen.

$$4NH_3(g) + \underset{\text{(From air)}}{5O_2(g)} \xrightarrow[500\text{K}, 9\text{bar}]{\text{Pt/Rh}-\text{gauge catalyst}} 4NO(g) + 6H_2O(g)$$

(ii) Nitric oxide thus formed combines with oxygen giving NO_2.

$$2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$$

(iii) Nitrogen dioxide so formed, dissolves in water to give HNO_3.

$$3NO_2(g) + H_2O(\ell) \longrightarrow 2HNO_3(aq) + NO(g)$$

NO thus formed is recycled and the aqueous HNO_3 can be concentrated by distillation upto ~ 68% by mass. Further concentration to 98% can be achieved by dehydration with concentrated H_2SO_4.

Birkeland Eyde Process or arc process

step 1 $\quad N_2 + O_2 \xrightarrow[\text{Electric Arc}]{3000^0\text{C}} 2NO - \text{heat}$

step 2 $\quad NO + O_2 \longrightarrow NO_2$

step 3 $\quad NO_2 + H_2O \longrightarrow HNO_2 + HNO_3$

step 4 $\quad HNO_2 \longrightarrow HNO_3 + NO + H_2O$

Properties

Physical properties

It has extremely corrosive action on the skin and causes painful sores.

(i) It is a colourless liquid (f.p. 231.4 K and b.p. 355.6 K).

(ii) Laboratory grade nitric acid contains ~ 68% of the HNO_3 by mass and has a specific gravity of 1.504.

(iii) Nitric acid usually acquires yellow or brown colour due to its decomposition by sunlight into NO_2.

$$4HNO_3 \xrightarrow{\text{Sunlight}} 4NO_2 + 2H_2O + O_2$$

The yellow or brown colour of the acid can be removed by warming it to 60-80°C and bubbling dry air through it.

Chemical properties

Acidic character in aqueous solution, nitric acid behaves as a strong acid giving hydronium and nitrate ions.

$HNO_3(aq) + H_2O(l) \rightarrow H_3O^+(aq) + NO_3^-$ (aq)

Oxidising nature: Nitric acid acts as a strong oxidising agent as it decomposes to give nascent oxygen easily.

$$2HNO_3 \longrightarrow H_2O + 2NO_2 + O$$

or

$$2HNO_3 \longrightarrow H_2O + 2NO + 3O$$

(i) Oxidation of non-metals : The nascent oxygen oxidises various non-metals to their corresponding oxyacids of highest oxidation state.

(1) Sulphur is oxidised to sulphuric acid

$$\underset{\text{conc.and hot}}{S + 6HNO_3} \longrightarrow H_2SO_4 + 6NO_2 + 2H_2O$$

(2) Carbon is oxidised to carbonic acid

$C + 4HNO_3 \rightarrow H_2CO_3 + 4NO_2 + 2H_2O$

(3) Phosphorus is oxidised to orthophosphoric acid.

$2P + 10HNO_3 \rightarrow 2H_3PO_4 + 10NO_2 + 2H_2O$

conc. and hot

(4) Iodine is oxidised to iodic acid

$I_2 + 10HNO_3 \rightarrow 2HIO_3 + 10NO_2 + 4H_2O$

conc. and hot

(ii) Oxidation of metalloids

Metalloids like non-metals also form oxyacids of highest oxidation state.

(1) Arsenic is oxidised to arsenic acid

$2As + 10HNO_3 \rightarrow 2H_3AsO_4 + 10NO_2 + 2H_2O$

or $As + 5HNO_3 \rightarrow H_3AsO_4 + 5NO_2 + H_2O$

conc. and hot

(2) Antimony is oxidised to antimonic acid

$Sb + 5HNO_3 \rightarrow H_3SbO_4 + 5NO_2 + H_2O$

conc. and hot

(3) Tin is oxidised to meta-stannic acid.

$Sn + 2HNO_3 \rightarrow H_2SnO_3 + 4NO_2 + H_2O$

(iii) Oxidation of Compounds:

(1) Sulphur dioxide is oxidised to sulphuric acid

$SO_2 + 2HNO_3 \rightarrow H_2SO_4 + 2NO_2$

(2) Hydrogen sulphiode is oxidised to sulphur

$H_2S + 2HNO_3 \rightarrow 2NO_2 + 2H_2O + S$

(3) Ferrous sulphate is oxidised to ferric sulphate in presence of H_2SO_4

$6FeSO_4 + 3H_2SO_4 + 2HNO_3 \rightarrow 3Fe_2(SO_4)_3 + 2NO + 4H_2O$

(4) Iodine is liberated from KI.

$6KI + 8HNO_3 \rightarrow 6KNO_3 + 2NO + 3I_2 + 4H_2O$

(5) HBr, HI are oxidised to Br_2 and I_2, respectively.

$2HBr + 2HNO_3 \rightarrow Br_2 + 2NO_2 + 2H_2O$

Similarly, $2HI + 2HNO_3 \rightarrow I_2 + 2NO_2 + 2H_2O$

(6) Ferrous sulphide is oxidised to ferric sulphate

$FeS + HNO_3 \rightarrow Fe_2(SO_4)_3 + 8NO_2 + 4H_2O$

(7) Stannous chloride is oxidised to stannic chloride is presence of HCl.

$$2HNO_3 + 14H \rightarrow \underset{\text{Hydroxylamine}}{NH_2OH} + NH_3 + 5H_2O$$

$NH_3 + HNO_3 \rightarrow NH_4NO_3$

$7SnCl_2 + 14HCl + 3HNO_3 \rightarrow 7SnCl_4 + NH_2OH + NH_4NO_3 + 5H_2O$

Session 2019-20

(8) Oxidation of organic compounds .

Sawdust catches fire when nitric acid is poured on it.

Turpentine oil bursts into flames when treated with fuming nitric acid.

Toluene is oxidised to benzoic acid with dil. HNO_3.

Cane sugar is oxidised to oxalic acid.

$$C_{12}H_{22}O_{11} + 36HNO_3 \rightarrow 6(COOH)_2 + 36NO_2 + 23H_2O$$

(ii) Reaction with metal concentrated nitric acid is a strong oxidising agent and attacks most metals except noble metals such as gold and platinum. Au & Pt dissolve in aqua regia a mixture of 25% conc. HNO_3 & 75% conc. HCl. The products of oxidation depend upon the concentration of the acid, temperature and the nature of the material undergoing oxidation.

Ex.

$$3Cu + 8\ HNO_3(\text{dilute}) \rightarrow 3Cu(NO_3)_2 + 2NO + 4H_2O$$

$$Cu + 4HNO_3(\text{conc.}) \rightarrow Cu(NO_3)_2 + 2NO_2 + 2H_2O$$

$$4Zn + 10HNO_3(\text{dilute}) \rightarrow 4\ Zn(NO_3)_2 + 5H_2O + N_2O$$

$$Zn + 4HNO_3(\text{conc.}) \rightarrow Zn(NO_3)_2 + 2H_2O + 2NO_2$$

Some metals (e.g., Cr, Al) do not dissolve in concentrated nitric acid because of the formation of a passive film of oxide on the surface.

Armstrong postulated that primary action of nitric acid is to produce hydrogen in the nascent form. Before this hydrogen is allowed to escape, it reduces the nitric acid into number of products like NO_2, NO, N_2O, N_2 or NH_3 according to the following reactions:

$$\text{Metal} + HNO_3 \longrightarrow \text{Nitrate} + H$$

$$2HNO_3 + 2H \longrightarrow 2NO + 2H_2O$$

$$2HNO_3 + 6H \longrightarrow 2NO + 4H_2O$$

$$2HNO_3 + 10H \longrightarrow N_2 + 6H_2O$$

$$2HNO_3 + 16\ H \longrightarrow 2NH_3 + 6H_2O$$

The progress of the reaction is controlled by a number of factors :

(a) the nature of the metal,

(b) the concentration of the acid,

(c) the temperature of the reaction,

(d) the presence of other impurities.

Concentration of nitric acid	Metal	Main Products
Very dilute HNO_3 (6%)	Mg, Mn	H_2 + Metal nitrate
	Fe, Zn, Sn	NH_4NO_3 + metal nitrate + H_2O
	Pb, Cu, Ag, Hg	NO + metal nitrate + H_2O

Session 2019-20

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Dilute HNO_3 (20%)	Fe, Zn	N_2O + metal nitrate + H_2O
	Sn	$NH_4NO_3 + Sn(NO_3)_2$
Conc. HNO_3(70%)	Zn, Fe, Pb, Cu, Ag	NO_2 + metal nitrate + H_2O
	Sn	$NO_2 + H_2SnO_3$ Metastannic acid

Action on Proteins :
Nitric acid attacks proteins forming a yellow nitro compound called xanthoprotein. It, therefore, stains skin and renders wool yellow. This property is utilized for the test of proteins.

Uses : The major use of nitric acid is in the manufacture of ammonium nitrate for fertilisers and other nitrates for use in explosives and pyrotechnics. It is also used for the preparation of nitroglycerin, trinitrotoluene and other organic nitro compounds. Other major uses are in the **pickling of stainless steel**, etching of metals and as an oxidiser in rocket fuels.

Pickling is a metal surface treatment used to remove impurities, such as stains, inorganic contaminants, rust or scale from ferrous metals, copper, precious metals and aluminium alloys. A solution called pickle liquor, which contains strong acids, is used to remove surface impurities.

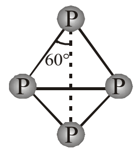
ALLOTROPIC FORMS OF PHOSPHORUS

Phosphorus is found in many allotropic forms, the important ones being white, red and black. White phosphorus is a translucent white waxy solid. It is poisonous, insoluble in water but soluble in carbon disulphide and glows in dark (chemiluminescence). It dissolves in boiling NaOH solution in an inert atmosphere giving PH_3. $P_4 + 3NaOH + 3H_2O \longrightarrow PH_3 + \underset{\text{(Sodium hypophosphite)}}{3NaH_2PO_2}$

White phosphorus is less stable and therefore, more reactive than the other solid phases under normal conditions because of angular strain in the P_4 molecule where the angles are only 60°. It readily catches fire in air to give dense white fumes of P_4O_{10}.

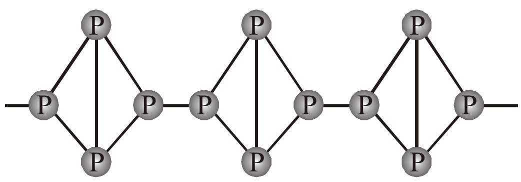
$$P_4 + 5O_2 \rightarrow P_4O_{10}$$

It consists of discrete tetrahedral P_4 molecule as shown in Fig.



White Phosphorus

Red phosphorus is obtained by heating white phosphorus at 573K in an inert atmosphere for several days. When red phosphorus is heated under high pressure, a series of phases of black phosphorus is formed. Red phosphorus possesses iron grey lustre. It is odourless, nonpoisonous and insoluble in water as well as in carbon disulphide. Chemically, red phosphorus is much less reactive than white phosphorus. It does not glow in the dark.
It is polymeric, consisting of chains of P_4 tetrahedra linked together in the manner as shown in Fig.



Red Phosphorus

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Black phosphorus has two forms α-black phosphorus and β-black phosphorus. α-Black phosphorus is formed when red phosphorus is heated in a sealed tube at 803K. It can be sublimed in air and has opaque monoclinic or rhombohedral crystals. It does not oxidise in air. β-Black phosphorus is prepared by heating white phosphorus at 473 K under high pressure. It does not burn in air upto 673 K.

Note : If white phosphorus is heated to about 250ºC, or a lower temperature in the presence of sunlight, then red phosphorus is formed. Heating white phosphorus under high pressure results in a highly polymerized form of P called black phosphorus. This is thermodynamically the most stable allotrope.

Comparison between White and Red Phosphorus

Property	White phosphorus	Red phosphorus
Physical state	Soft waxy solid.	Brittle powder.
Colour	White when pure. Attains yellow colour on standing.	Red.
Odour	Garlic	Odourless.
Solubility in water	Insoluble.	insoluble
Solubility in CS_2	Soluble.	Insoluble.
Physiological action	Poisonous.	Non-poisonous.
Chemical activity	Very active.	Less active.
Stability	Unstable.	Stable.
Phosphorescence	Glows in dark	Does not glow in dark.
Molecular formula	P_4	Complex polymer.

Prepration of white 'P'

(i) $\underbrace{\text{Bone ash or Apatite rock}}_{\text{both have same formula}} \Big\{ \ Ca_3(PO_4)_2 + 3SiO_2 \xrightarrow{1200°C} 3CaSiO_3 + P_2O_5$

$$\underset{}{2P_2O_5} + \underset{\text{(Coke)}}{10C} \xrightarrow{1500°C} \underset{\text{white 'P'}}{P_4} + 10CO\uparrow$$

(ii) $Ca_3(PO_4)_2 + 3H_2SO_4 \text{ (conc.)} \longrightarrow 3CaSO_4 + 2H_3PO_4$

$$H_3PO_4 \xrightarrow[-H_2O]{320°C} \underset{\text{meta phosphoric acid}}{HPO_3}$$

$$\underset{\text{Coke}}{12C} + 4HPO_3 \xrightarrow{1000°C} 2H_2\uparrow + 12CO\uparrow + \underset{\text{white 'P'}}{P_4}$$

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Reactions of 'P'

* $P + H_2SO_4$ (hot & conc.) $\longrightarrow H_3PO_4 + SO_2 + H_2O$

  $P + KIO_3 + H_2SO_4 \longrightarrow H_3PO_4 + I_2 + K_2SO_4$

* Reaction with hot metal —

  $3Na + P \longrightarrow Na_3P$

  $3Mg + 2P \longrightarrow Mg_3P_2$

  $3Ca + 2P \longrightarrow Ca_3P_2$

  $2Cu + 2P \longrightarrow Cu_3P_2$

  $Al + P \longrightarrow AlP$

PHOSPHINE

Preparation

(i) Phosphine is prepared by the reaction of calcium phosphide with water or dilute HCl.

$Ca_3P_2 + 6H_2O \rightarrow 3Ca(OH)_2 + 2PH_3$

$Ca_3P_2 + 6HCl \rightarrow 3CaCl_2 + 2PH_3$

(ii) Laboratory prepration it is prepared by heating white phosphorus with concentrated NaOH solution in an inert atmosphere of CO_2.

$$P_4 + 3NaOH + 3H_2O \longrightarrow PH_3 + \underset{\text{(sodium hypophosphite)}}{3NaH_2PO_2}$$

Pure PH_3 is non inflammable but becomes inflammable owingto the presence of P_2H_4 or P_4 vapours. To purify it from the impurities, it is absorbed in HI to form phosphonium iodide (PH_4I) which on treating with KOH gives off phosphine.

Other preparation

$$\left.\begin{array}{l} \underset{(PH_3+HI)}{PH_4I} + KOH \longrightarrow KI + PH_3 + H_2O \\ 2AlP + 3H_2SO_4 \longrightarrow Al_2(SO_4)_3 + PH_3\uparrow \end{array}\right\} \text{Purest } PH_3$$

$$4H_3PO_3 \xrightarrow{\Delta} PH_3 + 3H_3PO_4$$

$$2H_3PO_2 \xrightarrow{\Delta} PH_3 + 3H_3PO_4$$

Phyiscal Properties :

(i) It is a colourless gas with rotten fish smell and is highly poisonous.

(ii) It explodes in contact with traces of oxidising agents like HNO_3, Cl_2 and Br_2 vapours.

(iii) It is slightly soluble in water but soluble in CS_2. The solution of PH_3 in water decomposes in presence of light giving red phosphorus and H_2.

Chemical Properties :

(i) It absorbed in copper sulphate or mercuric chloride solution, the corresponding phosphides are obtained.

$$3CuSO_4 + 2PH_3 \rightarrow Cu_3P_2 + 3H_2SO_4$$

$$3HgCl_2 + 2PH_3 \rightarrow Hg_3P_2 + 6HCl$$

Phosphine is weakly basic and like ammonia, gives phosphonium compounds with acids e.g.,

$$PH_3 + HBr \rightarrow PH_4Br$$

(ii) $PH_3 + O_2 \xrightarrow{150^\circ} P_2O_5 + H_2O$

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(iii) $PH_3 + 3Cl_2 \longrightarrow PCl_3 + 3HCl$

(iv) $PH_3 + 4N_2O \xrightarrow[\text{sparking}]{\text{electrical}} H_3PO_4 + 4N_2$

(v) $PH_3 + 6AgNO_3 \longrightarrow [Ag_3P \cdot 3AgNO_3\downarrow] + 3HNO_3$

yellow ppt.

$Ag_3P \cdot 3AgNO_3 + 3H_2O \xrightarrow{\text{on standing}} 6Ag\downarrow + 3HNO_3 + H_3PO_3$

Black ppt.

(vi) $PH_3 + 4HCHO + HCl \longrightarrow [P(CH_2OH)_4]^+Cl^-$

white/colourless solid which is used for making fire-proof cotton fabrics

Note :

Like NH_3, PH_3 also can form addition product.

$\Downarrow$ $\searrow$

$CaCl_2 \cdot 8NH_3$ $Cu_2Cl_2 \cdot 2PH_3$, $AlCl_3 \cdot 2PH_3$, $SnCl_4 \cdot 2PH_3$

PH_3 can be absorbed by Ca(OCl)Cl.

$PH_3 + 3Ca(OCl)Cl + 3H_2O \longrightarrow PCl_3 + 3HCl + 3Ca(OH)_2$

$2NH_3 + 3Ca(OCl)Cl \longrightarrow N_2 + 3CaCl_2 + 3H_2O$

Uses :

(i) The spontaneous combustion of phosphine is technically used in Holme's signals. Containers containing calcium carbide and calcium phosphide are pierced and thrown in the sea when the gases evolved burn and serve as a signal.

(ii) It is also used in smoke screens.

PHOSPHORUS HALIDES

Phosphorus forms two types of halides, PX_3 (X = F, Cl, Br, I) and PX_5 (X = F, Cl, Br).

PHOSPHORUS TRICHLORIDE

Preparation

(i) By passing dry chlorine over heated white phosphorus.

$$P_4 + 6Cl_2 \rightarrow 4PCl_3$$

(ii) By the action of thionyl chloride with white phosphorus.

$$P_4 + 8SOCl_2 \rightarrow 4PCl_3 + 4SO_2 + 2S_2Cl_2$$

Properties

(i) It is a colourless oily liquid

(ii) Hydrolyses in the presence of moisture.

$$PCl_3 + 3H_2O \rightarrow H_3PO_3 + 3HCl$$

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(iii) It reacts with organic compounds containing –OH group such as CH_3COOH, C_2H_5OH.

$$3CH_3COOH + PCl_3 \rightarrow 3CH_3COCl + H_3PO_3$$
$$3C_2H_5OH + PCl_3 \rightarrow 3C_2H_5Cl + H_3PO_3$$

PHOSPHORUS PENTACHLORIDE

Preparation

(i) By the reaction of white phosphorus with excess of dry chlorine.

$$P_4 + 10Cl_2 \rightarrow 4PCl_5$$

(ii) By the action of SO_2Cl_2 on phosphorus.

$$P_4 + 10SO_2Cl_2 \rightarrow 4PCl_5 + 10SO_2$$

Properties :

(i) PCl_5 is a yellowish white powder

(ii) It hydroslysis in moist air to $POCl_3$ and finally gets converted to phosphoric acid.

$$PCl_5 + H_2O \rightarrow POCl_3 + 2HCl$$
$$POCl_3 + 3H_2O \rightarrow H_3PO_4 + 3HCl$$

(iii) When heated, it sublimes but decomposes on stronger heating.

$$PCl_5 \xrightarrow{\text{Heat}} PCl_3 + Cl_2$$

(iv) It reacts with organic compounds containing –OH group converting them to chloro derivatives.

$$C_2H_5OH + PCl_5 \rightarrow C_2H_5Cl + POCl_3 + HCl$$
$$CH_3COOH + PCl_5 \rightarrow CH_3COCl + POCl_3 + HCl$$

(v) Finely divided metals on heating with PCl_5 give corresponding chlorides.

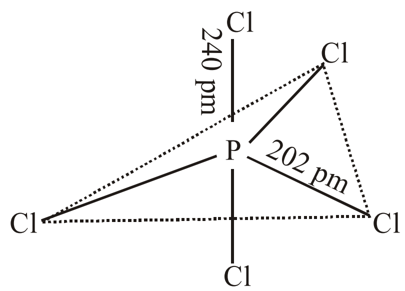
$$2Ag + PCl_5 \rightarrow 2AgCl + PCl_3$$
$$Sn + 2PCl_5 \rightarrow SnCl_4 + 2PCl_3$$

Uses :

It is used in the synthesis of some organic compounds, e.g., C_2H_5Cl, CH_3COCl.

Note :

In gaseous and liquid phases, it has a trigonal bipyramidal structure as shown. The three equitorial P–Cl bonds are equivalent, while the two axial bonds are longer than equitorial bonds. This is due to the fact that the axial bond pairs suffer more repulsion as compared to equitorial bond pairs.



Inthe solid state it exists as an ionic solid, $[PCl_4]^+[PCl_6]^-$ in which the cation, $[PCl_4]^+$ is tetrahedral and the anion, $[PCl_6]^-$ is octahedral.

OXIDES OF PHOSPHORUS

It forms three important oxides which exist in dimeric forms.

PHOSPHORUS TRIOXIDE (P_4O_6)

Preparation

Phosphorus trioxides is formed when phosphorus is burnt in a limited supply of air and inert atmosphere.

$P_4 + 3O_2$ (limited) $\longrightarrow P_4O_6$

Properties

(a) **Heating in air :** On heating in air, it forms phosphorus pentoxide.

$P_4O_6 + 2O_2 \longrightarrow P_4O_{10}$

Phosphorus (V) oxide

(b) **Action of water :** It dissolves in cold water to give phosphorus acid.

$P_4O_6 + 6H_2O$ (cold) $\longrightarrow 4H_3PO_3$

Phosphorus acid

It is, therefore, considered as anhydride of phosphorus acid.

Note: With hot water, it gives phosphoric acid and inflammable phosphine.

Structure

(a) Each atom of phosphorus in P_4O_6 is present at the corner of a tetrahedron

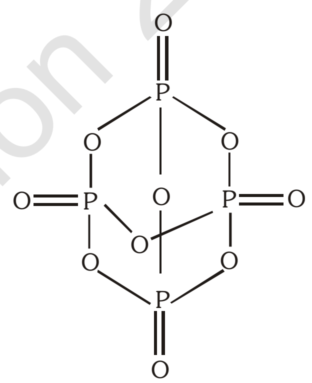
(b) Each phosphorus atom is covalently bonded to three oxygen atoms and each oxygen atom is bonded to two phosphorus atoms.

(c) It is clear from the structure that the six oxygen atoms lie along the edges of the tetrahedron of P atoms.

PHOSPHORUS (V) OXIDE (P_4O_{10})

Preparation : It is prepared by heating white phosphorus in excess of air.

$P_4 + 5O_2$ (excess) $\xrightarrow{\Delta} P_4O_{10}$



Properties

(a) It is snowy white solid.

(b) **Action with water :** It readily dissolves in cold water forming metaphosphoric acid.

$P_4O_{10} + 2H_2O$ (Cold) $\longrightarrow 4HPO_3$

Metaphosphoric acid.

With hot water it gives phosphoric acid.

$P_4O_{10} + 6H_2O$ (Hot) $\longrightarrow 4H_3PO_4$

Phosphoric acid

(c) **Dehydrating nature :** Phosphorus pentoxide has strong affinity for water and, therefore, acts as a powerful dehydrating agent. It extracts water from many inorganic and organic compounds.

(d) P_4O_{10} is a very strong dehydrating agent and extracts water from many compounds including sulphuric acid and nitric acid.

$2HClO_4 \xrightarrow[-H_2O]{P_4O_{10}} Cl_2O_7$

Chlorine (VII) oxide

$2CH_3CONH_2 \xrightarrow[-H_2O]{P_4O_{10}} CH_3CN$

Acetamide — Methyl cyanide

Structure

(a) Its structure is similar to that of P_4O_6.

(b) In addition, each phosphorus atom forms a double bond with oxygen atom as shown in figure.

OXOACIDS OF PHOSPHORUS :

The important oxoacids of phosphorus with their formulae, methods of preparation and the presence of some characteristic bonds in their structures are given in a table.

Oxoacids of Phosphorus

Name	Formula	Oxidation state of Phosphorus	Characteristic bonds and their number	Preparation
Hypophosphorus (Phosphinic)	H_3PO_2	+ 1	One P — OH Two P — H One P = O	white P_4 + alkali
Orthophosphorous (Phosphonic)	H_3PO_3	+ 3	Two P — OH One P — H One P = O	$P_2O_3 + H_2O$
Pyrophosphorous	$H_4P_2O_5$	+ 3	Two P — OH Two P — H Two P = O	$PCl_3 + H_3PO_3$
Hypophosphoric	$H_4P_2O_6$	+ 4	Four P — OH Two P = O One P — P	red P_4 + alkali
Orthophosphoric	H_3PO_4	+ 5	Three P — OH One P = O	$P_4O_{10} + H_2O$
Pyrophosphoric	$H_4P_2O_7$	+ 5	Four P — OH Two P = O One P — O — P	heat phosphoric acid
Metaphosphoric*	$(HPO_3)_n$	+ 5	Three P — OH Three P = O Three P — O — P	phosphorous acid + Br_2, heat in a sealed tube

STRUCTURE OF OXOACID :

In oxoacids phosphorus is tetrahedrally surrounded by other atoms. All these acids contain at least one P=O bond and one P–OH bond. The oxoacids in which phosphorus has lower oxidation state (less than +5) contain, in addition to P=O and P–OH bonds, either P–P (e.g., in $H_4P_2O_6$) or P–H (e.g., in H_3PO_2) bonds but not both. H_3PO_3 and H_3PO_2 are diabasic and monobasic respectively.

Note : -

(i) These acids in +3 oxidation state of phosphorus tend to disproportionate to higher and lower oxidation states. For example, phophorous acid on heating disproportionates to give orthophosphoric acid (or phosphoric acid) and phosphine.

$$4H_3PO_3 \rightarrow 3H_3PO_4 + PH_3$$

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(ii) The acids which contain P–H bond have strong reducing properties. Thus, hypophosphorous acid is a good reducing agent as it contains two P–H bonds and reduces, $AgNO_3$ to metallic silver.

$$4AgNO_3 + 2H_2O + H_3PO_2 \rightarrow 4Ag + 4HNO_3 + H_3PO_4$$

Heating Effect :

$$H_3PO_2 \xrightarrow{140°C} PH_3 + H_3PO_4$$

$$\underset{\text{orthophosphoric acid}}{H_3PO_4} \xrightarrow[220°C]{\text{gentle heat}} \underset{\text{Pyrophosphoric acid}}{H_4P_2O_7} \xrightarrow[320°C]{\text{strong heat}} \underset{\text{metaphosphoric acid}}{(HPO_3)_n}$$

Graham salt

Graham's salt is the best known of these long chain polyphosphates, and is formed by quenching molten $NaPO_3$. Graham's salt is soluble in water. These solutions give precipitates with metal ions such as Pb^{2+} and Ag^+ but not with Ca^{2+} and Mg^{2+}. Graham's salt is sold commercially under the trade name Calgon. In industry it is incorrectly called sodium hexametaphosphate crystallizing. It is widely used for softening water.

$$NaH_2PO_4 \xrightarrow{> 240°C} \underset{\text{Sodium trimetaphosphate}}{(NaPO_3)_3} \xrightarrow{625°C} NaPO_3(\text{liquid melt}) \xrightarrow[\text{cooling}]{\text{rapid}} \underset{\text{(Graham's salt) (glass)}}{(NaPO_3)_n}$$

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OXYGEN FAMILY

GROUP 16 ELEMENTS (O, S, Se, Te, Po)

This is sometimes known as group of chalcogens.

- **Occurrence**

Oxygen is the most abundant of all the elements on earth crust. Oxygen forms about 46.6% by mass of earth's crust. Dry air contains 20.946% oxygen by volume. However, the abundance of sulphur in the earth's crust is only 0.03-0.1%. Combined sulphur exists primarily as sulphates such as gypsum $CaSO_4.2H_2O$, epsom salt $MgSO_4.7H_2O$, baryte $BaSO_4$ and sulphides such as galena PbS, zinc blende ZnS, copper pyrites $CuFeS_2$. Traces of sulphur occur as hydrogen sulphide in volcanoes. Organic materials such as eggs, proteins, garlic, onion, mustard, hair and wool contain sulphur.

Selenium and tellurium are also found as metal selenides and tellurides in sulphide ores. Polonium occurs in nature as a decay product of thorium and uranium minerals.

- **Electronic Configuration**

ns^2np^4 is the general valence shell electronic configuration.

- **Atomic and Ionic Radii :** Covalent radius : O < S < Se < Te
- **Ionisation Enthalpy :** O > S > Se > Te > Po (IE_1 values)
- **Most Negative Electron Gain Enthalpy:** S > Se > Te > Po > O
- **Electronegativity :** O > S > Se > Te
- **Metallic Character :** O < S < Se < Te < Po
- **Melting and Boiling points :**

M.P. : Te > Po > Se > S > O

B.P. : Te > Po > Se > S > O

Elemental State

Oxygen exist as diatomic molecular gas in this case there is $p\pi - p\pi$ overlap thus two O atoms form double bond O = O. The intermolecular forces in O_2 are weak VB forces. $\therefore$ O_2 exist as gas . On the other hand, other elements of family do not form stable $p\pi - p\pi$ bonds and do not exist as M_2 molecules. Other atoms are linked by single bonds and form poly atomic complex molecules for eg. $S - S_8$, $Se - Se_8$

Allotropy

All element exhibit allotropy for e.g.

Oxygen – O_2 and O_3

Liquid O_2 - pale blue

Solid O_2 - blue

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Sulphur -

The main allotropic forms are

(i) Rhombic sulphur (α sulphur)

(ii) Monoclinic (β sulphur)

(iii) Plastic sulphur (δ sulphur)

Catenation

In this group only S has a strong tendency for catenation. Oxygen has this tendency to a limited extent.

H_2O_2 H – O – O – H (Poly oxides)

H_2S_2 H – S – S – H (Polysulphides or polysulphones)

H_2S_3 H – S – S – S – H

H_2S_4 H – S – S – S – S – H

❑ **Physical Properties**

(i) Oxygen and sulphur are non-metals, selenium and tellurium metalloids, whereas polonium is a metal.

(ii) Polonium is radioactive and is short lived (Half-life 13.8 days).

(iii) All these elements exhibit allotropy.

(iv) The melting and boiling points increase with an increase in atomic number down the group. The large difference between the melting and boiling points of oxygen and sulphur may be explained on the basis of their atomicity; oxygen exists as diatomic molecule (O_2) whereas sulphur exists as polyatomic molecule (S_8).

❑ **Chemical Properties**

Oxidation states and trends in chemical reactivity :

(i) The stability of -2 oxidation state decreases down the group. Polonium hardly shows –2 oxidation state.

(ii) Electronegativity of oxygen is very high, it shows only negative oxidation state as –2 except

Oxygen shows oxidation states of +2 and +1 in oxygen fluorides OF_2 and O_2F_2 respectively.

(iii) Elements of the group exhibit+ 2, + 4, + 6 oxidation states but + 4 and + 6 are more common.

(iv) Sulphur, selenium and tellurium usually show + 4 oxidation state in their compounds with oxygen and + 6 with fluorine.

(v) The stability of + 6 oxidation state decreases down the group and stability of + 4 oxidation state increases (inert pair effect).

(vi) Bonding in +4 and +6 oxidation states is primarily covalent.

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Anomalous behaviour of oxygen

The anomalous behaviour of oxygen, like other members of p-block present in second period is due to its small size and high electronegativity. One typical example of effects of small size and high electronegativity is the presence of strong hydrogen bonding in H_2O which is not found in H_2S. The absence of d orbitals in oxygen limits its covalency to four and in practice, rarely exceeds two. On the other hand, in case of other elements of the group, the valence shells can be expanded and covalency exceeds four.

(I) Reactivity with hydrogen:

(i) All the elements of Group 16 form hydrides of the type H_2E (E = O, S, Se, Te, Po).

(ii) Their acidic character increases from H_2O to H_2Te. The increase in acidic character can be explained in terms of decrease in bond enthalpy for the dissociation of H–E bond down the group. Owing to the decrease in enthalpy for the dissociation of H–E bond down the group, the thermal stability of hydrides also decreases from H_2O to H_2Po.

(iii) All the hydrides except water possess reducing property and this character increases from H_2S to H_2Te.

Properties of Hydrides of Group 16 Elements

Property	H_2O	H_2S	H_2Se	H_2Te
m.p./K	273	188	208	222
b.p./K	373	213	232	269
H—E distance /pm	96	134	146	169
HEH angle (°)	104	92	91	90
$\Delta_f H$/kJ mol^{-1}	– 286	– 20	73	100
$\Delta_{diss}H$ (H – E)kJ mol^{-1}	463	347	276	238
Dissociation constant[a]	1.8×10^{-16}	1.3×10^{-7}	1.3×10^{-4}	2.3×10^{-3}

(II) Reactivity with oxygen:

(i) All these elements form oxides of the EO_2 and EO_3 types where E = S, Se, Te or Po.

(ii) Ozone (O_3) and sulphur dioxide (SO_2) are gases while selenium dioxide (SeO_2) is solid.

(iii) Reducing property of dioxide decreases from SO_2 to TeO_2; SO_2 is reducing while TeO_2 is an oxidising agent.

(iv) Besides EO_2 type, sulphur, selenium and tellurium also form EO_3 type oxides (SO_3, SeO_3, TeO_3). Both types of oxides are acidic in nature.

(III) Reactivity towards the halogens:

(i) Elements of Group 16 form a large number of halides of the type, EX_6, EX_4 and EX_2 where E is an element of the group and X is a halogen.

(ii) The stability of the halides decreases in the order $F^- > Cl^- > Br^- > I^-$.

(iii) Amongst hexahalides, hexafluorides are the only stable halides. All hexafluorides are gaseous in nature. They have octahedral structure. Sulphur hexafluoride, SF_6 is exceptionally stable for steric reasons.

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Amongst tetrafluorides, SF_4 is a gas, SeF_4 a liquid and TeF_4 a solid. These fluorides have sp^3d hybridisation and thus, have trigonal bipyramidal structures in which one of the equatorial positions is occupied by a lone pair of electrons. This geometry is also regarded as see-saw geometry. All elements except oxygen form dichlorides and dibromides (because they form oxides). These dihalides are formed by sp^3 hybridisation and thus, have tetrahedral structure. The well known monohalides are dimeric in nature. Examples are S_2F_2, S_2Cl_2, S_2Br_2, Se_2Cl_2 and Se_2Br_2. These dimeric halides undergo disproportionation as given below : $2Se_2Cl_2 \rightarrow SeCl_4 + 3Se$

DIOXYGEN

(a) Laboratory method

(i) By heating oxygen containing salts such as chlorates, nitrates and permanganates.

$$2KClO_3 \xrightarrow[MnO_2]{Heat} 2KCl + 3O_2$$

(ii) By the thermal decomposition of the oxides of metals low in the electrochemical series and higher oxides of some metals.

$2Ag_2O(s) \rightarrow 4Ag(s) + O_2(g)$; $2Pb_3O_4(s) \rightarrow 6PbO(s) + O_2(g)$

$2HgO(s) \rightarrow 2Hg(\ell) + O_2(g)$; $2PbO_2(s) \rightarrow 2PbO(s) + O_2(g)$

(iii) Hydrogen peroxide is readily decomposed into water and dioxygen by catalysts such as finely divided metals and manganese dioxide.

$$2H_2O_2(aq) \rightarrow 2H_2O(\ell) + O_2(g)$$

(b) Large scale preparation : It can be prepared from water or air. Electrolysis of water leads to the release of hydrogen at the cathode and oxygen at the anode.

(c) Industrially method : Dioxygen is obtained from air by first removing carbon dioxide and water vapour and then, the remaining gases are liquefied and fractionally distilled to give dinitrogen and dioxygen.

Properties

(i) Dioxygen is a colourless and odourless gas.

(ii) Its solubility in water is to the extent of 3.08 cm^3 in 100 cm^3 water at 293 K which is just sufficient for the vital support of marine and aquatic life.

(iii) It liquefies at 90 K and freezes at 55 K.

(iv) Oxygen atom has three stable isotopes: ^{16}O, ^{17}O and ^{18}O. Molecular oxygen, O_2 is unique in being paramagnetic inspite of having even number of electrons.

(v) Dioxygen directly reacts with nearly all metals and non-metals except some metals (e.g., Au, Pt) and some noble gases. Its combination with other elements is often strongly exothermic which helps in sustaining the reaction. However, to initiate the reaction, some external heating is required as bond dissociation enthalpy of oxgyen-oxygen double bond is high (493.4 kJ mol^{-1}). Some of the reactions of dioxygen with metals, non-metals and other compounds are as follows :

$$2Ca + O_2 \rightarrow 2CaO$$
$$4Al + 3O_2 \rightarrow 2Al_2O_3$$
$$P_4 + 5O_2 \rightarrow P_4O_{10}$$
$$C + O_2 \rightarrow CO_2$$
$$2ZnS + 3O_2 \rightarrow 2ZnO + 2SO_2$$
$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$$

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Some compounds are catalytically oxidised. For example,

$$2SO_2 + O_2 \xrightarrow{V_2O_5} 2SO_3$$

$$4HCl + O_2 \xrightarrow{CuCl_2} 2Cl_2 + 2H_2O$$

Uses: (i) It's importance in normal respiration and combustion processes, oxygen is used in oxyacetylene welding, in the manufacture of many metals, particularly steel.

(ii) Oxygen cylinders are widely used in hospitals, high altitude flying and in mountaineering.

(iii) The combustion of fuels, e.g., hydrazines in liquid oxygen, provides tremendous thrust in rockets.

SIMPLE OXIDES

A binary compound of oxygen with another element is called oxide. In many cases one element forms two or more oxides. The oxides vary widely in their nature and properties. Oxides can be simple (e.g., MgO, Al_2O_3) or mixed (Pb_3O_4, Fe_3O_4).

Types of simple oxide : Simple oxides can be classified on the basis of their acidic, basic or amphoteric character.

Acidic oxide : An oxide that combines with water to give an acid is termed acidic oxide (e.g., SO_2, Cl_2O_7, CO_2, N_2O_5). For example, SO_2 combines with water to give H_2SO_3, an acid.

$$SO_2 + H_2O \rightarrow H_2SO_3$$

As a general rule, only non-metal oxides are acidic but oxides of some metals in high oxidation state also have acidic character (e.g., Mn_2O_7, CrO_3, V_2O_5).

Basic oxide : The oxides which give a base with water are known as basic oxides (e.g., Na_2O, CaO, BaO). In general, metallic oxides are basic. For example, CaO combines with water to give $Ca(OH)_2$, a base.

$$CaO + H_2O \rightarrow Ca(OH)_2$$

Amphoteric oxide :

Some metallic oxides exhibit a dual behaviour. They show characteristics of both acidic as well as basic oxides. Such oxides are known as amphoteric oxides. They react with acids as well as alkalies. For example, Al_2O_3 reacts with acids as well as alkalies.

$Al_2O_3(s) + 6HCl(aq) + 9H_2O(\ell) \rightarrow 2[Al(H_2O)_6]^{3+}(aq) + 6Cl^-(aq)$

$Al_2O_3(s) + 6NaOH(aq) + 3H_2O(\ell) \rightarrow 2Na_3[Al(OH)_6]\ (aq)$

Neutral oxide:

There are some oxides which are neither acidic nor basic. Such oxides are known as neutral oxides. Examples of neutral oxides are CO, NO and N_2O.

OZONE

(i) Ozone is an allotropic form of oxygen and is diamagnetic.

(ii) It is too reactive to remain for long in the atmosphere at sea level. At a height of about 20 kilometres, it is formed from atmospheric oxygen in the presence of sunlight. This ozone layer protects the earth's surface from an excessive concentration of ultraviolet (UV) radiations.

Threats to ozone layer

(i) Experiments have shown that nitrogen oxides (particularly nitric oxide) combine very rapidly with ozone and there is, thus, the possibility that nitrogen oxides emitted from the exhaust systems of supersonic jet aeroplanes might be slowly depleting the concentration of the ozone layer in the upper atmosphere.

$$NO(g) + O_3(g) \rightarrow NO_2(g) + O_2(g)$$

(ii) Another threat to this ozone layer is probably posed by the use of freons which are used in aerosol sprays and as refrigerants.

ALLEN Session 2019-20

Preparation

When a slow dry stream of oxygen is passed through a silent electrical discharge, conversion of oxygen to ozone (10%) occurs. The product is known as ozonised oxygen.

$3O_2 \rightarrow 2O_3$ ΔH^{-} (298 K) = +142 kJ mol^{-1}

Since the formation of ozone from oxygen is an **endothermic** process, it is necessary to use a silent electrical discharge in its preparation to prevent its decomposition. If concentration of ozone greater than 10 percent is required, a battery of ozonisers can be used, and pure ozone (b.p. –112.4°C) can be condensed in a vessel surrounded by liquid oxygen.

Ques. Ozone is thermodynamically unstable with respect to oxygen. Explain ?

Sol. Because its decomposition into oxygen results in the liberation of heat (ΔH is negative) and an increase in entropy (ΔS is positive). These two effects reinforce each other, resulting in large negative Gibbs energy change (ΔG) for its conversion into oxygen.

Note :

$$\left.\begin{array}{l} 2F_2 + 2H_2O \longrightarrow 4HF + O_2 \\ F_2 + 3H_2O \longrightarrow 6HF + O_3 \end{array}\right] \text{Ozonised oxygen}$$

] is separated by passing into spiral tube cooled by liq. air. Ozone condenses at –112.4°C.

[b.p. of O_2 –183°C ; b.p. of liq. air is –190°C]

Properties

(i) Pure ozone is a pale blue gas, dark blue liquid and violet-black solid.

(ii) Ozone has a characteristic fishy smell and in small concentrations it is harmless.

Toxic effect :

(a) Toxic enough (more toxic than KCN). It's intense blue colour is due to the absorption of red light.

(b) However, if the concentration rises above about 100 parts per million, breathing becomes uncomfortable resulting in headache and nausea.

Oxidizing properties

It is one of best oxidising agent, in acid solution, its standard, reduction potential value is 2.07 V.

$O_3 + 2H^+ + 2e \longrightarrow O_2 + H_2O$ $E° = +2.07$ V

It is next to F_2. [above 2.07 V, only F_2, F_2O are there]

It is not really surprising, therefore, high concentrations of ozone can be dangerously explosive. Due to the ease with which it liberates atoms of nascent oxygen ($O_3 \rightarrow O_2 + O$), it acts as a powerful oxidising agent. For example, it oxidises lead sulphide to lead sulphate and iodide ions to iodine.

$$PbS(s) + 4O_3(g) \rightarrow PbSO_4(s) + 4O_2(g)$$

$$2I^-(aq) + H_2O(\ell) + O_3(g) \rightarrow 2OH^-(aq) + I_2(s) + O_2(g)$$

(i) Metal Sulphides to Sulphates.

$MS + 4O_3 \longrightarrow MSO_4 + 4O_2$ [M = Pb, Cu, Zn, Cd]

(ii) $2HX + O_3 \longrightarrow X_2 + H_2O + O_2$ [X = Cl, Br, I]

(iii) $NaNO_2 + O_3 \longrightarrow NaNO_3 + O_2$

$Na_2SO_3 + O_3 \longrightarrow Na_2SO_4 + O_2$

$Na_2AsO_3 + O_3 \longrightarrow Na_3AsO_4 + O_2$

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(iv) Moist S, P, As + $O_3 \Rightarrow$

$S + H_2O + 3O_3 \longrightarrow H_2SO_4 + 3O_2$

$2P + 3H_2O + 5O_3 \longrightarrow 2H_3PO_4 + 5O_2$

$2As + 3H_2O + 5O_3 \longrightarrow 2H_3AsO_4 + 5O_2$

(v) Moist $I_2 \longrightarrow HIO_3$ whereas dry iodine $\longrightarrow I_4O_9$ (yellow)

$I_2 + 5O_3 + H_2O \longrightarrow 2HIO_3 + 5O_2$

$2I_2 + 9O_3 \longrightarrow I_4O_9 + 9O_2$ (I_4O_9 exists as I^{3+} and $3IO_3^-$)

(vi) $2K_2MnO_4 + O_3 + H_2O \longrightarrow 2KMnO_4 + 2KOH + O_2$

$2K_4[Fe(CN)_6] + O_3 + H_2O \longrightarrow 2K_3[Fe(CN)_6] + 2KOH + O_2$

$2FeSO_4 + O_3 + H_2SO_4 \longrightarrow Fe_2(SO_4)_3 + O_2 + H_2O$

(vii)(a) 2KI (acidified) + O_3 + 2HCl $\longrightarrow I_2 + 2KCl + H_2O + O_2$

(b) 2KI (neutral) + $O_3 + H_2O \longrightarrow \underbrace{I_2 + 2KOH}_{KI+KOI} + O_2$

$KI + KOI + 2HCl \longrightarrow 2KCl + I_2 + H_2O$

$\}$ O_3 is estimated by this reaction

$I_2 + 2S_2O_3^{2-} \longrightarrow S_4O_6^{2-} + 2I$

(c) alk. $\left\{ \begin{array}{l} KI + 3O_3 \longrightarrow KIO_3 + 3O_2 \\ KI + 4O_3 \longrightarrow KIO_4 + 4O_2 \end{array} \right\}$

(viii) Hg loses its fluidity (tailing of Hg)

$2Hg + O_3 \longrightarrow Hg_2O + O_2$

similarly $2Ag + O_3 \longrightarrow \underset{\text{Brown}}{Ag_2O} + O_2$

(ix) $BaO_2 + O_3 \rightarrow BaO + 2O_2$

$H_2O_2 + O_3 \longrightarrow H_2O + 2O_2$

$Na_2O_2 + O_3 + H_2O \longrightarrow 2NaOH + 2O_2$

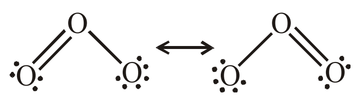
(x) $2KOH + 5O_3 \longrightarrow 2KO_3 + 5O_2 + H_2O$

In all above reaction O_3 gives up O_2 but some reactions are there which consumes all O-atom.

(i) $3SO_2 + O_3 \longrightarrow 3SO_3$

(ii) $3SnCl_2 + 6HCl + O_3 \longrightarrow 3SnCl_4 + 3H_2O$

Bonding in ozone : In ozone the two oxygen-oxygen bond lengths in the ozone molecule are identical (128 pm) and the molecule is angular as expected with a bond angle of about 117°. It is a resonance hybrid of two main forms given below:



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Absorbent : (i) Turpentine oil (ii) Oil of cinnamon

Quantitative method for the estimating on Ozone : Ozone reacts with an excess of potassium iodide solution buffered with a borate buffer (pH 9.2), iodine is liberated which can be titrated against a standard solution of sodium thiosulphate.

Uses : (i) Sterilising water

(ii) Detection of position of the double bond in the unsaturated compound.

(iii) It is used as a germicide, disinfectant and for sterilising water.

(iv) It is also used for bleaching oils, ivory, flour, starch, etc.

(v) It acts as an oxidising agent in the manufacture of potassium permanganate.

❑ HYDROGEN PEROXIDE (H_2O_2)

Hydrogen peroxide is an important chemical used in pollution control treatment of domestic and industrial effluents.

Preparation

It can be prepared by the following methods.

(i) $Na_2O_2 + H_2O$ (ice cold water) $\longrightarrow 2NaOH + H_2O_2$

(ii) Acidifying barium peroxide and removing excess water by evaporation under reduced pressure gives hydrogen peroxide.

$BaO_2.8H_2O(s) + H_2SO_4(aq) \rightarrow BaSO_4(s) + H_2O_2(aq) + 8H_2O(\ell)$

Instead of H_2SO_4, H_3PO_4 is added now-a-days because H_2SO_4 catalyses the decomposition of H_2O_2 whereas H_3PO_4 favours to restore it.

$3BaO_2+2H_3PO_4 \rightarrow Ba_3(PO_4)_2+3H_2O_2$

and, $Ba_3(PO_4)_2+3H_2SO_4 \rightarrow 3BaSO_4+ 2H_3PO_4$ (reused again)

(iii) Peroxodisulphate, obtained by electrolytic oxidation of acidified sulphate solutions at high current density, on hydrolysis yields hydrogen peroxide.

$2H_2SO_4 \rightleftharpoons 2H^+ + 2HSO_4^-$

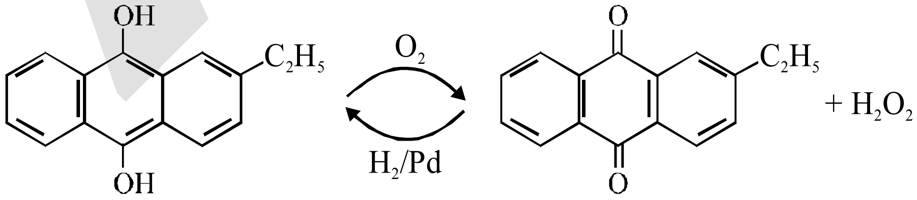
$2HSO_4^- \longrightarrow H_2S_2O_8 + 2e.$ [At anode] [At cathode $2H^+ + 2e \rightarrow H_2$]

$H_2S_2O_8 + 2H_2O \longrightarrow 2H_2SO_4 + H_2O_2$

This method is now used for the laboratory preparation of D_2O_2.

$K_2S_2O_8(s) + 2D_2O(\ell) \longrightarrow 2KDSO_4(aq) + D_2O_2(\ell)$

(iii) Industrially it is prepared by the autooxidation of 2-alklylanthraquinols.



2-ethylanthraquinol 2-ethylanthraquinone

In this case 1% H_2O_2 is formed. It is extracted with water and concentrated to ~30% (by mass) by distillation under reduced pressure. It can be further concentrated to ~85% by careful distillation under low pressure. The remaining water can be frozen out to obtain pure H_2O_2.

ALLEN Session 2019-20

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- **Physical Properties**

In the pure state H_2O_2 is an almost colourless (very pale blue) liquid. Its important physical properties are given in Table. H_2O_2 is miscible with water in all proportions and forms a hydrate $H_2O_2.H_2O$ (mp 221K). A 30% solution of H_2O_2 is marketed as '100 volume' hydrogen peroxide. It means that one millilitre of 30% H_2O_2 solution will give 100 mL of oxygen at STP. Commercially marketed sample is 10 V, which means that the sample contains 3% H_2O_2.

30% (w / v) or "100 V" H_2O_2 solution is called **per hydrol**.

Problem

Calculate the strength of 10 volume solution of hydrogen peroxide.

Solution

10 volume solution of H_2O_2 means that 1L of this H_2O_2 solution will give 10 L of oxygen at STP

$2H_2O_2(\ell) \rightarrow O_2(g) + H_2O(\ell)$

On the basis of above equation 22.7 L of O_2 is produced from 68 g H_2O_2 at STP 10 L of O_2 at STP is produced from

$\frac{68 \times 10}{22.7}$ g = 29.9 g   30 g H_2O_2

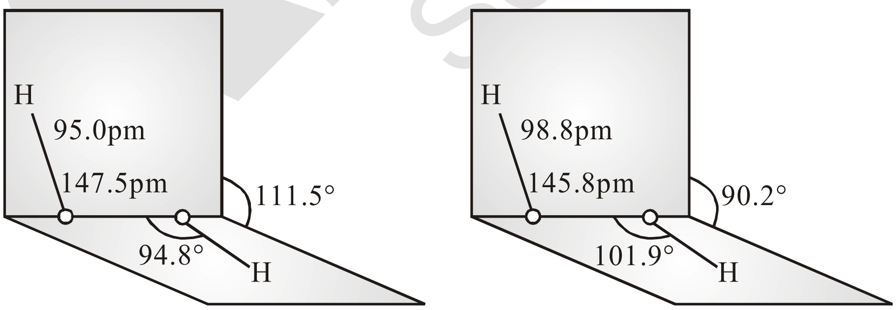
Therefore, strength of H_2O_2 in 10 volume H_2O_2 solution = 30 g/L **=** 3% H_2O_2 solution

Physical Properties of Hydrogen Peroxide

Melting point/K	272.4	Density (liquid at 298K)/g cm^{-3}	1.44
Boiling point (exrapolated)/K	423	Viscosity (290 K)/centipoise	1.25
Vapour pressure (298K) mmHg	1.9	Dielectric constant (298K)/C^2/N m^2	70.7
Density (solid at 268.5K)/g cm^{-3}	1.64	Electrical conductivity (298K)/Ω^{-1} cm^{-1}	5.1×10^{-8}

Structure

Hydrogen peroxide has a non-planar structure. The molecular dimensions in the gas phase and solid phase are shown in Fig.



(a) Gas phase   (a) Solid phase

***Fig.** (a) H_2O_2 structure in gas phase, dihedral angle is 111.5°. (b) H_2O_2 structure in solid phase at 110K, dihedral angle is 90.2°.*

Chemical Properties:

(i) Acidic nature : $H_2O_2 + 2NaOH \longrightarrow Na_2O_2 + H_2O$

$H_2O_2 + Ba(OH)_2 \longrightarrow BaO_2 + 2H_2O$

$H_2O_2 + Na_2CO_3 \longrightarrow Na_2O_2 + CO_2 + H_2O$

(ii) It is oxidant as well as reductant.

$H_2O_2 + 2H^+ + 2e \rightarrow 2H_2O$ [reaction in acidic medium]

$H_2O_2 + 2e \rightarrow 2OH^-$ [rxn^n in alkali medium]

Oxidising Properties:

(i) $PbS + 4H_2O_2 \longrightarrow PbSO_4 + 4H_2O$ (Used in washing of oil painting)

(ii) $NaNO_2 + H_2O_2 \longrightarrow NaNO_3 + H_2O$

$Na_2SO_3 + H_2O_2 \longrightarrow Na_2SO_4 + H_2O$

$Na_3AsO_3 + H_2O_2 \longrightarrow Na_3AsO_4 + H_2O$

$2KI + H_2O_2 \longrightarrow 2KOH + I_2$

$H_2S + H_2O_2 \longrightarrow S\downarrow + 2H_2O$

$\begin{bmatrix} X_2 + H_2O_2 \longrightarrow 2HX + O_2 \quad X = Cl, Br. \\ \text{S.R.P order of } Cl_2 > Br_2 > H_2O_2 > I_2 \end{bmatrix}$

$H_2SO_4 + 2FeSO_4 + H_2O_2 \longrightarrow Fe_2(SO_4)_3 + 2H_2O$

$2\,K_4[Fe(CN)_6] + H_2O_2 + H_2SO_4 \longrightarrow 2K_3[Fe(CN)_6] + K_2SO_4 + 2H_2O$

$2[Cr(OH)_4]^- + 3H_2O_2 + 2OH^- \longrightarrow 2CrO_4^{2-} + 8H_2O$

$CrO_4^{2-} + 2H^+ + 2H_2O_2 \rightarrow CrO_5\text{ (Blue)}\downarrow + 3H_2O$

$4CrO_5 + 12H^+ \longrightarrow 4Cr^{+3} + 7O_2 + 6H_2O$

$Mn^{+2} + OH^- + H_2O_2 \longrightarrow MnO_2 + 2H_2O \Rightarrow$ This reaction can be utilised to detect NH_3

Reducing properties:

(a) $Ag_2O + H_2O_2 \longrightarrow 2Ag + H_2O + O_2$

(b) $O_3 + H_2O_2 \longrightarrow H_2O + 2O_2$

(c) $MnO_2 + H_2O_2 + H_2SO_4 \rightarrow MnSO_4 + 2H_2O + O_2$

(d) $PbO_2 + H_2O_2 \rightarrow PbO + H_2O + O_2$

(e) $Pb_3O_4 + 4HNO_3 \longrightarrow 2Pb(NO_3)_2 + PbO_2 + 2H_2O$

$PbO_2 + H_2O_2 \longrightarrow PbO + H_2O + O_2$

$PbO + 2HNO_3 \longrightarrow Pb(NO_3)_2 + H_2O$

$Pb_3O_4 + H_2O_2 + 6HNO_3 \longrightarrow 3Pb(NO_3)_2 + 4H_2O + O_2$

(f) $X_2 + H_2O_2 \longrightarrow 2HX + O_2$ [X = Cl, Br]

$2KMnO_4 + 3H_2O_2 \rightarrow 2KOH + 2MnO_2 + 2H_2O + 3O_2$

$2MnO_4^- + 2OH^- \longrightarrow 2MnO_4^{2-} + H_2O + O$

ALLEN Session 2019-20

$2MnO_4^{2-} + 2H_2O \longrightarrow 2MnO_2 + 4OH^- + 2O$

$2MnO_4^- + H_2O \longrightarrow 2MnO_2 + 2OH^- + 3O$

(g) $2KMnO_4 + 5H_2O_2 + 3H_2SO_4 \longrightarrow 2MnSO_4 + K_2SO_4 + 5O_2 + 8H_2O$

(h) $2[Fe(CN)_6]^{3-} + 2OH^- + H_2O_2 \longrightarrow 2[Fe(CN)_6]^{4-} + 2H_2O + O_2$

(i) $NaOCl + H_2O_2 \longrightarrow NaCl + H_2O + O_2$

(j) $NaIO_4 + H_2O_2 \longrightarrow NaIO_3 + H_2O + O_2$

It acts as an oxidising as well as reducing agent in both acidic and alkaline media. Simple reactions are described below.

(i) Oxidising action in acidic medium

(a) $2Fe^{2+}(aq) + 2H^+(aq) + H_2O_2(aq) \rightarrow 2Fe^{3+}(aq) + 2H_2O(\ell)$

(b) $PbS(s) + 4H_2O_2(aq) \rightarrow PbSO_4(s) + 4H_2O(\ell)$

(ii) Reducing action in acidic medium

(a) $2MnO_4^- + 6H^+ + 5H_2O_2 \rightarrow 2Mn^{2+} + 8H_2O + 5O_2$

(b) $HOCl + H_2O_2 \rightarrow H_3O^+ + Cl^- + O_2$

(c) $CrO_4^{2-} + 2H^+ + 2H_2O_2 \rightarrow CrO_5$ (Blue) $+ 3H_2O$

$4CrO_5 + 12H^+ \longrightarrow 4Cr^{+3} + 7O_2 + 6H_2O$

(iii) Oxidising action in basic medium

(a) $2Fe^{2+} + H_2O_2 \rightarrow 2Fe^{3+} + 2OH^-$

(b) $Mn^{2+} + H_2O_2 \rightarrow Mn^{4+} + 2OH^-$

(iv) Reducing action in basic medium

(a) $I_2 + H_2O_2 + 2OH^- \rightarrow 2I^- + 2H_2O + O_2$

(b) $2MnO_4^- + 3H_2O_2 \rightarrow 2MnO_2 + 3O_2 + 2H_2O + 2OH^-$

- **Storage**

H_2O_2 decomposes slowly on exposure to light.

$2H_2O_2(\ell) \rightarrow 2H_2O(\ell) + O_2(g)$

In the presence of metal surfaces or traces of alkali (present in glass containers), the above reaction is catalysed. It is, therefore, stored in wax-lined glass or plastic vessels in dark. Acetanalide or Glycerol or Urea can be added as a stabiliser. It is kept away from dust because dust can induce explosive decomposition of the compound.

Uses

Its wide scale use has led to tremendous increase in the industrial production of H_2O_2.

Some of the uses are listed below :

(i) In daily life it is used as a hair bleach and as a mild disinfectant. As an antiseptic it is sold in the market as perhydrol.

ALLEN Session 2019-20

(ii) It is used to manufacture chemicals like sodium perborate and per-carbonate, which are used in high quality detergents.

(iii) It is used in the synthesis of hydroquinone, tartaric acid and certain food products and pharmaceuticals (cephalosporin) etc.

(iv) It is employed in the industries as a bleaching agent for textiles, paper pulp, leather, oils, fats, etc.

(v) As a rocket propellant:

$NH_2.NH_2 + 2H_2O_2 \longrightarrow N_2 + 4H_2O$ [highly exothermic and large increase in volume]

(vi) In detection of Cr^{+3}, Ti^{+4} etc.

$$Ti(SO_4)_2 + H_2O_2 + 2H_2O \longrightarrow \underset{\substack{\text{Yellow or orange}\\ \text{Pertitanic acid}}}{H_2TiO_4} + 2H_2SO_4$$

(vii) Nowadays it is also used in Environmental (Green) Chemistry. For example, in pollution control treatment of domestic and industrial effluents, oxidation of cyanides, restoration of aerobic conditions to sewage wastes, etc.

ALLOTROPIC FORMS OF SULPHUR

Sulphur forms numerous allotropes of which the yellow rhombic (α-sulphur) and monoclinic (β -sulphur) forms are the most important. The stable form at room temperature is rhombic sulphur, which transforms to monoclinic sulphur when heated above 369 K.

$$\alpha - \text{sulphur} \underset{<369K}{\overset{>369K}{\rightleftharpoons}} \beta - \text{sulphur}$$

At 369 K both the forms are stable. This temperature is called transition temperature.

- **Rhombic sulphur (α-sulphur)**

(i) This allotrope is yellow in colour, m.p. 385.8 K and specific gravity 2.06.

(ii) Rhombic sulphur crystals are formed on evaporating the solution of roll sulphur in CS_2.

(iii) It is insoluble in water but dissolves to some extent in benzene, alcohol and ether.

(iv) It is readily soluble in CS_2.

- **Monoclinic sulphur (β-sulphur)**

(i) Its m.p. is 393 K and specific gravity 1.98.

(ii) It is soluble in CS_2.

(iii) This form of sulphur is prepared by melting rhombic sulphur in a dish and cooling, till crust is formed. Two holes are made in the crust and the remaining liquid poured out. On removing the crust, colourless needle shaped crystals of β-sulphur are formed. It is stable above 369 K and transforms into α-sulphur below it. Conversely, α-sulphur is stable below 369 K and transforms into β-sulphur above this.

Structure of α and β sulphur

Both rhombic and monoclinic sulphur have S_8 molecules. These S_8 molecules are packed to give different crystal structures. The S_8 ring in both the forms is puckered and has a crown shape.

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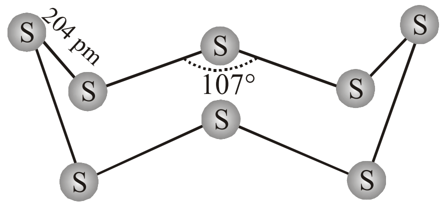
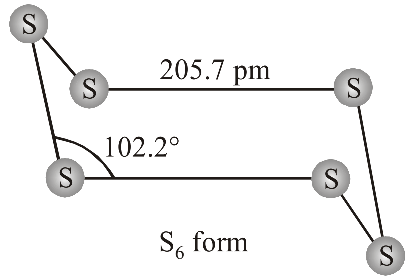


S_8 ring in rhombic sulphur



S_6 form

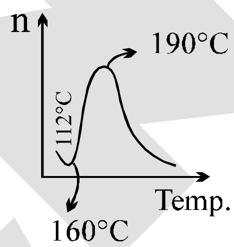
Several other modifications of sulphur containing 6-20 sulphur atoms per ring have been synthesised in the last two decades. In cyclo-S_6, the ring adopts the chair form. At elevated temperatures (~1000 K), S_2 is the dominant species and is paramagnetic like O_2.

Note: Viscosity of 'S' with temperature :

m.p. of 'S' $\longrightarrow$ 112.8°C.

> 112.8°C to 160°C $\Rightarrow$ slow decreases due to S_8 rings slip and roll over one another easily.

> 160°C, increases sharply due to breaking of S_8 rings into chains and polymerses into large size chain.



❑ **Amorphous forms are**

(i) Plastic sulphur (ii) Milk of sulphur (iii) Colloidal sulphur

SULPHUR DIOXIDE

Preparation

Sulphur dioxide is formed together with a little (6-8%) sulphur trioxide when sulphur is burnt in air or oxygen:

$$S(s) + O_2(g) \rightarrow SO_2\ (g)$$

laboratory method by treating a sulphite with dilute sulphuric acid.

$$SO_3^{2-}(aq) + 2H^+(aq) \rightarrow H_2O(\ell) + SO_2(g)$$

other preparation :

$$Cu + 2H_2SO_4\ (conc.) \longrightarrow CuSO_4 + 2H_2O + SO_2$$

$$Hg + H_2SO_4 \longrightarrow HgSO_4 + H_2O + SO_2$$

$$2Ag + H_2SO_4 \longrightarrow Ag_2SO_4 + H_2O + SO_2$$

$$S + 2H_2SO_4 \longrightarrow 3SO_2 + 2H_2O$$

$$(Charcoal)C + 2H_2SO_4 \longrightarrow CO_2 + 2SO_2 + 2H_2O$$

$$NaHSO_3 + H_2SO_4 \longrightarrow NaHSO_4 + H_2O + SO_2$$

ALLEN Session 2019-20

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Industrial method, by-product of the roasting of sulphide ores.

$$4FeS_2(s) + 11O_2(g) \rightarrow 2Fe_2O_3(s) + 8SO_2(g)$$

The gas after drying is liquefied under pressure and stored in steel cylinders.

Properties

(i) Sulphur dioxide is a colourless gas with pungent smell.

(ii) It is highly soluble in water.

(iii) It liquefies at room temperature under a pressure of two atmospheres and boils at 263 K.

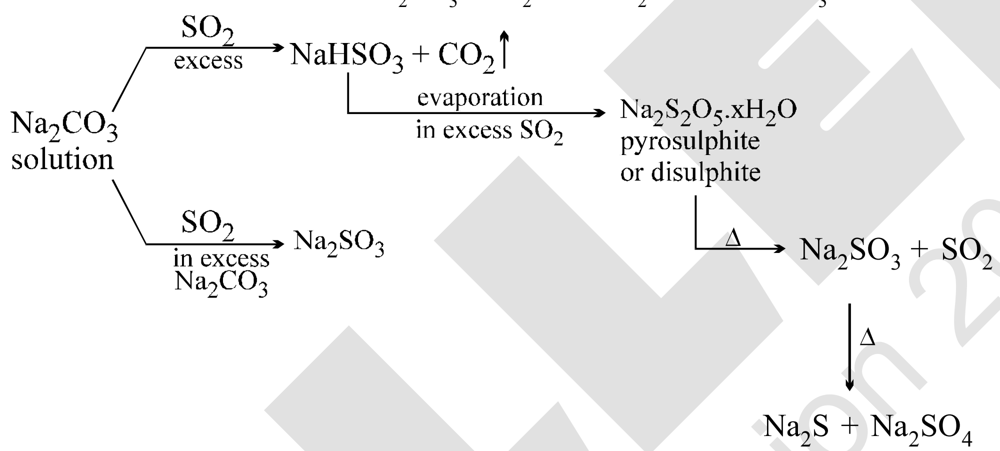
(iv) **Acidic character:** sulphur dioxide, when passed through water, forms a solution of sulphurous acid.

$$SO_2(g) + H_2O(\ell) \rightleftharpoons H_2SO_3(aq)$$

It reacts readily with sodium hydroxide solution, forming sodium sulphite, which then reacts with more sulphur dioxide to form sodium hydrogen sulphite.

$$2NaOH + SO_2 \rightarrow Na_2SO_3 + H_2O$$

$$Na_2SO_3 + H_2O + SO_2 \rightarrow 2NaHSO_3$$



In its reaction with water and alkalies, the behaviour of sulphur dioxide is very similar to that of carbon dioxide. Sulphur dioxide reacts with chlorine in the presence of charcoal (which acts as a catalyst) to give sulphuryl chloride, SO_2Cl_2. It is oxidised to sulphur trioxide by oxygen in the presence of vanadium(V) oxide catalyst.

$$SO_2(g) + Cl_2(g) \xrightarrow[\text{of charcoal}]{\text{in presence}} SO_2Cl_2(\ell)$$

$$2SO_2(g) + O_2(g) \xrightarrow{V_2O_5} 2SO_3(g)$$

Reducing properties

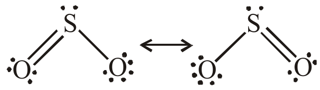
When moist, sulphur dioxide behaves as a reducing agent. For example, it converts iron(III) ions to iron(II) ions and decolourises acidified potassium permanganate(VII) solution; the latter reaction is a convenient test for the gas.

$2Fe^{3+} + SO_2 + 2H_2O \rightarrow 2Fe^{2+} + SO_4^{2-} + 4H^+$

$5SO_2 + 2MnO_4^- + 2H_2O \rightarrow 5SO_4^{2-} + 4H^+ + 2Mn^{2+}$

ALLEN Session 2019-20

Bonding in SO_2 : The molecule of SO_2 is angular. It is a resonance hybrid of the two canonical forms:



Uses:

(i) It is used refining petroleum and sugar

(ii) It is used in bleaching wool and silk

(iii) It is used as an anti-chlor, disinfectant and preservative. Sulphuric acid, sodium hydrogen sulphite and calcium hydrogen sulphite (industrial chemicals) are manufactured from sulphur dioxide. Liquid SO_2 is used as a solvent to dissolve a number of organic and inorganic chemicals.

<u>HYDROGEN SULPHIDE (H_2S) SULPHURATED HYDROGEN</u>

Preparation

By the action of dil. HCl or H_2SO_4 on iron pyrites.

$$FeS + H_2SO_4 \longrightarrow FeSO_4 + H_2S\uparrow$$

Note: **Drying agent for this gas : fused $CaCl_2$, Al_2O_3 (dehydrated) P_2O_5 etc. But not H_2SO_4, because $H_2SO_4 + H_2S \longrightarrow 2H_2O + SO_2 + S$**

Properties

It is a colourless gas having an offensive smell of rotten eggs.

(a) It burn in air with blue flame

$$2H_2S + O_2 \longrightarrow 2H_2O + S$$

If the air supply is in excess

$$2H_2S + 3O_2 \longrightarrow 2H_2O + 2SO_2$$

(b) It is a mild acid.

$$H_2S \longrightarrow H^+ + SH^- \quad , \qquad SH^- \longrightarrow H^+ + S^{-2}$$

(c) It act as a reducing agent. It reduces halogen into corresponding hydroacid.

$$H_2S + X_2 \longrightarrow 2HX + S$$

Tests of H_2S

(a) Unpleasant odour resembling that of rotten eggs.

(b) It turns lead acetate into paper black

$$(CH_3COO)_2Pb + H_2S \longrightarrow PbS\downarrow(\text{Black}) + 2\ CH_3COOH$$

(c) It gives a violet colouration with a alkaline solution of sodium nitroprusside.

Structure of H_2S

(a) Similar to structure of water molecule i.e. V– shaped structure with bond length (H–S) 1.35A° and bond angle (H–S–H) is 92.5°

Uses

(a) It is mainly employed in salt analysis for the detection of cation.

(b) Reducing agent for H_2SO_4, $KMnO_4$, $K_2Cr_2O_7$, O_3, H_2O_2, $FeCl_3$

ALLEN Session 2019-20

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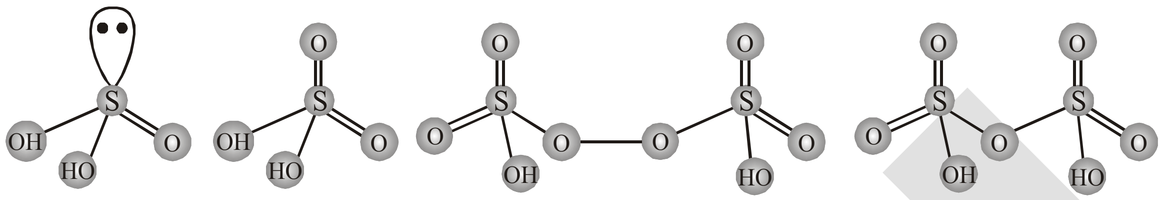
OXOACIDS OF SULPHUR

Sulphur forms a number of oxoacids such as H_2SO_3, $H_2S_2O_3$, $H_2S_2O_4$, $H_2S_2O_5$, $H_2S_xO_6$ (x = 2 to 5), H_2SO_4, $H_2S_2O_7$, H_2SO_5, $H_2S_2O_8$. Some of these acids are unstable and cannot be isolated. They are known in aqueous solution or in the form of their salts. Structures of some important oxoacids are shown in Fig.



Sulphurous acid (H_2SO_3) Sulphuric acid (H_2SO_4) Peroxodisulphuric acid ($H_2S_2O_8$) Pyrosulphuric acid (Oleum) ($H_2S_2O_7$)

Structures of some important oxoacids of sulphur

SULPHURIC ACID

Sulphuric acid is one of the most important industrial chemicals worldwide.

Industrial Manufacturing (Contact process)

Steps involved :

(i) Burning of sulphur or sulphide ores in air to generate SO_2.

(ii) Conversion of SO_2 to SO_3 by the reaction with oxygen in the presence of a catalyst (V_2O_5):

The SO_2 produced is purified by removing dust and other impurities such as arsenic compounds. The key step in the manufacture of H_2SO_4 is the catalytic oxidation of SO_2 with O_2 to give SO_3 in the presence of V_2O_5 (catalyst).

$$2SO_2(g) + O_2(g) \xrightarrow{V_2O_5} 2SO_3(g) \qquad \Delta_r H^{\ominus} = -196.6 \text{ kJ mol}^{-1}$$

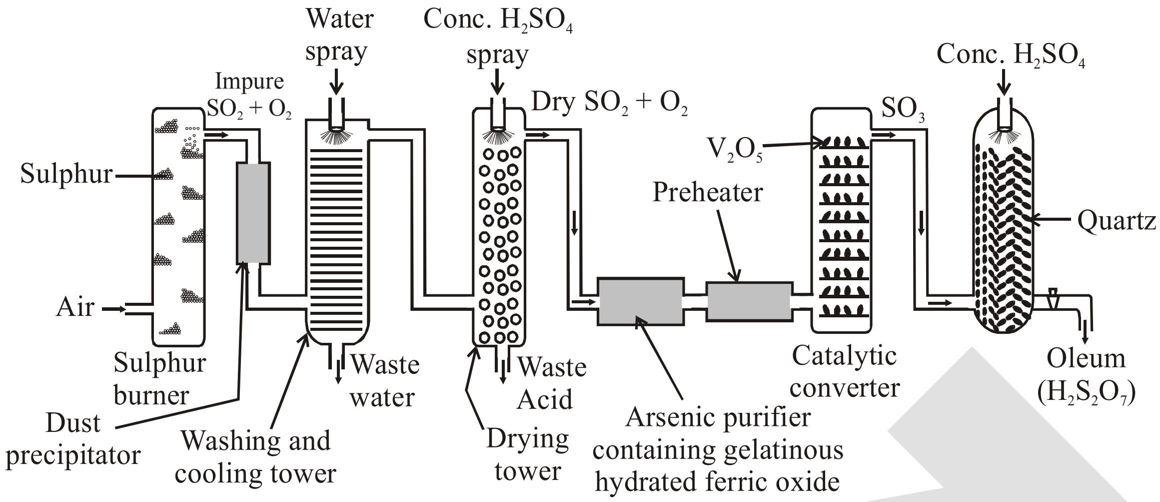
The reaction is exothermic, reversible and the forward reaction leads to a decrease in volume. Therefore, low temperature and high pressure are the favourable conditions for maximum yield. But the temperature should not be very low otherwise rate of reaction will become slow. In practice, the plant is operated at a pressure of 2 bar and a temperature of 720 K.

(iii) Absorption of SO_3 in H_2SO_4 to give Oleum ($H_2S_2O_7$) :

The SO_3 gas from the catalytic converter is absorbed in concentrated H_2SO_4 to produce oleum. Dilution of oleum with water gives H_2SO_4 of the desired concentration. In the industry two steps are carried out simultaneously to make the process a continuous one and also to reduce the cost.

$$SO_3 + H_2SO_4 \rightarrow \underset{\text{(Oleum)}}{H_2S_2O_7}$$

Session 2019-20



Flow diagram for the manufacture of sulphuric acid

The sulphuric acid obtained by Contact process is 96-98% pure.

P_2O_5 is stronger dehydrating agent than H_2SO_4 : $H_2SO_4 + P_2O_5 \longrightarrow 2HPO_3 + SO_3$

Properties

(i) Sulphuric acid is a colourless, dense, oily liquid with a specific gravity of 1.84 at 298 K.

(ii) The acid freezes at 283 K and boils at 611 K.

(iii) It dissolves in water with the evolution of a large quantity of heat. Hence, care must be taken while preparing sulphuric acid solution from concentrated sulphuric acid. The concentrated acid must be added slowly into water with constant stirring.

Chemical properties

The chemical reactions of sulphuric acid are as a result of the following characteristics:

(a) low volatility
(b) strong acidic character
(c) strong affinity for water and
(d) ability to act as an oxidising agent.

Acidic character :

In aqueous solution, sulphuric acid ionises in two steps.

$$H_2SO_4(aq) + H_2O(\ell) \rightarrow H_3O^+(aq) + HSO_4^-(aq):\ K_{a_1} = \text{Very large } (K_{a1} > 10)$$

$$HSO_4^-(aq) + H_2O(\ell) \rightarrow H_3O^+(aq) + SO_4^{2-}(aq);\ K_{a_2} = 1.2 \times 10^{-2}$$

The larger value of Ka_1 (Ka_1 >10) means that H_2SO_4 is largely dissociated into H^+ and HSO_4^-. Greater the value of dissociation constant (K_a), the stronger is the acid.

The acid forms two series of salts: normal sulphates (such as sodium sulphate and copper sulphate) and acid sulphates (e.g., sodium hydrogen sulphate). Sulphuric acid, because of its low volatility can be used to manufacture more volatile acids from their corresponding salts.

$$2MX + H_2SO_4 \rightarrow 2HX + M_2SO_4 \ (X = F, Cl, NO_3)$$
$$(M = \text{Metal})$$

ALLEN Session 2019-20

Dehydrating Property :

Concentrated sulphuric acid is a strong dehydrating agent. Many wet gases can be dried by passing them through sulphuric acid, provided the gases do not react with the acid. Sulphuric acid removes water from organic compounds; it is evident by its charring action on carbohydrates.

$$C_{12}H_{22}O_{11} \xrightarrow[\text{Conc.}]{H_2SO_4} 12C + 11H_2O$$

$$C_6H_{12}O_6 \xrightarrow[-6H_2O]{H_2SO_4} 6C$$

$$(COOH)_2 \xrightarrow[-H_2O]{H_2SO_4} CO + CO_2$$

$$HCO_2H \xrightarrow[-H_2O]{} CO$$

Oxidizing Nature :

Hot concentrated sulphuric acid is a moderately strong oxidising agent. In this respect, it is intermediate between phosphoric and nitric acids. Both metals and non-metals are oxidised by concentrated sulphuric acid, which is reduced to SO_2.

$Cu + 2H_2SO_4(\text{conc.}) \rightarrow CuSO_4 + SO_2 + 2H_2O$

$HBr / HI + H_2SO_4(\text{conc.}) \rightarrow Br_2/I_2 + SO_2 + H_2O$

$3S + 2H_2SO_4(\text{conc.}) \rightarrow 3SO_2 + 2H_2O$

$C + 2H_2SO_4(\text{conc.}) \rightarrow CO_2 + 2SO_2 + 2H_2O$

$2P + 5H_2SO_4\ (\text{conc.}) \rightarrow H_3PO_4 + 5SO_2$

H_2SO_4 & SO_3 :

$$\overbrace{SO_2 + Cl_2}^{\text{Both gas}} \longrightarrow SO_2Cl_2$$

$$H_2SO_4 + 2PCl_5 \longrightarrow \underset{\substack{\downarrow \\ \text{good chlorinating agent}}}{SO_2Cl_2} + 2POCl_3 + 2HCl$$

Uses: Sulphuric acid is a very important industrial chemical. A nation's industrial strength can be judged by the quantity of sulphuric acid it produces and consumes. It is needed for the manufacture of hundreds of other compounds and also in many industrial processes. The bulk of sulphuric acid produced is used in the manufacture of fertilisers (e.g., ammonium sulphate, superphosphate). Other uses are in:

(i) petroleum refining

(ii) manufacture of pigments, paints and dyestuff intermediates

(iii) detergent industry

(iv) metallurgical applications (e.g., cleansing metals before enameling, electroplating and galvanising

(v) storage batteries

(vi) in the manufacture of nitrocellulose products and

(vii) as a laboratory reagent.

Session 2019-20

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SODIUM THIOSULPHATE

Preparation:

(i) Na_2SO_3 solution + S (powder) $\xrightarrow{\text{boiling}}$ $Na_2S_2O_3$ $\xrightarrow{\text{evaporation}}$ $Na_2S_2O_3.5H_2O$, monoclinic crystal

Na_2CO_3 excess $\xrightarrow{SO_2 \text{ passed}}$ $\begin{cases} Na_2CO_3 + 2SO_2 + H_2O \longrightarrow 2NaHSO_3 + CO_2 \\ 2NaHSO_3 + Na_2CO_3 \longrightarrow Na_2SO_3 + H_2O + CO_2 \end{cases}$

(ii) $\underset{\text{Salt cake}}{Na_2SO_4} + \underset{\text{Coke}}{4C} \xrightarrow{\text{roasting}} Na_2S + 4CO$

$\downarrow$ SO_2 passed into it

$Na_2S_2O_3$ [$3SO_2 + 2Na_2S \longrightarrow 2Na_2S_2O_3 + S$]

(iii) $2Na_2S + Na_2CO_3 + 4SO_2 \longrightarrow 3Na_2S_2O_3 + CO_2$

(iv) $6NaOH + 4S \longrightarrow Na_2S_2O_3 + 2Na_2S + 3H_2O$

$3Ca(OH)_2 + 12\ S \longrightarrow CaS_2O_3 + 2CaS_5 + 3H_2O$

(v) $Na_2SO_3 + Na_2S + I_2 \longrightarrow Na_2S_2O_3 + 2NaI$

(vi) $2Na_2S + 2O_2 + H_2O \longrightarrow Na_2S_2O_3 + 2NaOH$

[Na_2S is readily oxidised in air giving rise to $Na_2S_2O_3$]

Properties: (i) $4Na_2S_2O_3 \xrightarrow{\Delta} Na_2S_5 + 3Na_2SO_4$

(ii) $Na_2S_2O_3 + 2H^+ \longrightarrow H_2S_2O_3 \xrightarrow{\Delta} H_2O + SO_2 + S\downarrow$ (White turbidity)

Reaction: (i) $Na_2S_2O_3 + I_2 \longrightarrow S_4O_6^{2-} + 2I^-$

$+\ Cl_2 - \text{water} \longrightarrow SO_4^{2-} + S + 2HCl$

$+\ Br_2 - \text{water} \longrightarrow SO_4^{2-} + S + 2HBr$

$+\ 4OI^- + 2OH^- \longrightarrow 2SO_4^{2-} + 4I^- + H_2O$

$+\ 4Cl_2 + 5H_2O \longrightarrow \underbrace{Na_2SO_4 + H_2SO_4}_{2NaHSO_4} + 8HCl$

HALOGEN FAMILY

GROUP 17 ELEMENTS (F, Cl, Br, I, At)

These are collectively known as the halogens (Greek halo means salt and genes means born i.e., salt producers). Astatine is a radioactive element.

❑ **Occurrence**

(i) Fluorine and chlorine are fairly abundant while bromine and iodine less so.

(ii) Fluorine is present mainly as insoluble fluorides (fluorspar CaF_2, cryolite Na_3AlF_6 and fluoroapatite $3Ca_3(PO_4)_2.CaF_2$) and small quantities are present in soil, river water plants and bones and teeth of animals.

(iii) Sea water contains chlorides, bromides and iodides of sodium, potassium, magnesium and calcium, but is mainly sodium chloride solution (2.5% by mass).

(iv) The deposits of dried up seas contain these compounds, e.g., sodium chloride and carnallite, $KCl.MgCl_2.6H_2O$. Certain forms of marine life contain iodine in their systems; various seaweeds, for example, contain upto 0.5% of iodine and Chile saltpetre contains upto 0.2% of sodium iodate.

Electronic Configuration

The electronic configuration of outermost shell 17^{th} group element is (ns^2np^5).

❑ **Atomic and ionic radii :** F < Cl < Br < I

❑ **Ionisation Enthalpy :** F > Cl > Br > I

❑ **Most Negative Electron Gain Enthalpy:** Cl > F > Br > I

It is due to small size of fluorine atom. As a result, there are strong interelectronic repulsions in the relatively small 2p orbitals of fluorine and thus, the incoming electron does not experience much attraction.

❑ **Electronegativity :** F > Cl > Br > I

❑ **Physical Properties**

(i) Their melting and boiling points steadily increase with atomic number.

(ii) All halogens are coloured. For example, F_2 is a yellow gas, Cl_2 greenish yellow gas, Br_2 red liquid and I_2 violet coloured solid. Reason : Decrease in HOMO-LUMO gap.

(iii) Fluorine and chlorine react with water. Bromine and iodine are only sparingly soluble in water but are soluble in various organic solvents such as chloroform, carbon tetrachloride, carbon disulphide and hydrocarbons to give coloured solutions.

(iv) **Bond energy order ;** $Cl_2 > Br_2 > F_2 > I_2$

A reason for this anomaly is the relatively large electron-electron repulsion among the lone pairs in F_2 molecule where they are much closer to each other than in case of Cl_2.

ALLEN Session 2019-20

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❑ Chemical Properties

Oxidation states :

(i) All the halogens exhibit –1 oxidation state. However, chlorine, bromine and iodine exhibit + 1, + 3, + 5 and + 7 oxidation states also as explained below:

	n*s*	n*p*	n*d*	
Halogen atom in ground state (other than fluorine)	[↑↓]	[↑↓] [↑↓] [↑]	[ ] [ ] [ ] [ ] [ ]	1 unpaired electron accounts for – 1 or + 1 oxidation states
First excited state	[↑↓]	[↑↓] [↑] [↑]	[↑] [ ] [ ] [ ] [ ]	3 unpaired electron accounts for + 3 oxidation states
Second excited state	[↑↓]	[↑] [↑] [↑]	[↑] [↑] [ ] [ ] [ ]	5 unpaired electron accounts for + 5 oxidation states
Third excited state	[↑]	[↑] [↑] [↑]	[↑] [↑] [↑] [ ] [ ]	7 unpaired electron accounts for + 7 oxidation states

(ii) The higher oxidation states of chlorine, bromine and iodine are realised mainly when the halogens are in combination with the small and highly electronegative fluorine and oxygen atoms. e.g., in interhalogens, oxides and oxoacids.

(iii) The oxidation states of +4 and +6 occur in the oxides and oxoacids of chlorine and bromine.

(iv) The fluorine atom has no d orbitals in its valence shell and therefore cannot expand its octet. Being the most electronegative, it exhibits only –1 oxidation state.

Chemical reactivity

(i) All the halogens are highly reactive.

(ii) They react with metals and non-metals to form halides and the reactivity of the halogens decreases down the group. i.e. the order is $F_2 > Cl_2 > Br_2 > I_2$

(iii) The ready acceptance of an electron is the reason for the strong oxidising nature of halogens. F_2 is the strongest oxidising halogen and it oxidises other halide ions in solution or even in the solid phase. In general, a halogen oxidises halide ions of higher atomic number.

$$F_2 + 2X^- \rightarrow 2F^- + X_2 \ (X = Cl, Br \text{ or } I)$$

$$Cl_2 + 2X^- \rightarrow 2Cl^- + X_2 \ (X = Br \text{ or } I)$$

$$Br_2 + 2I^- \rightarrow 2Br^- + I_2$$

The decreasing oxidising ability of the halogens in aqueous solution down the group is evident from their standard electrode potentials, which are dependent on the parameters as follows :

$$\frac{1}{2}X_2(g) \xrightarrow{1/2\,\Delta_{diss}H^\ominus} X(g) \xrightarrow{\Delta_{eg}H^\ominus} X^-(g) \xrightarrow{\Delta_{hyd}H^\ominus} X^-(aq)$$

(1) **Reactivity towards water**

The relative oxidising power of halogens can further be illustrated by their reactions with water. Fluorine oxidises water to oxygen whereas chlorine and bromine react with water to form corresponding hydrohalic and hypohalous acids. The reaction of iodine with water is nonspontaneous. In fact, I^- can be oxidised by oxygen in acidic medium; just the reverse of the reaction is observed with fluorine.

ALLEN Session 2019-20

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$$2F_2(g) + H_2O(l) \rightarrow 4H^+(aq) + 4F^-(aq) + O_2(g)$$

$$X_2(g) + H_2O(l) \rightarrow HX(aq) + HOX(aq)$$

(where X = Cl or Br)

$$4I^-(aq) + 4H^+(aq) + O_2(g) \rightarrow 2I_2(s) + 2H_2O(l)$$

Note : Anomalous behaviour of fluorine

(i) Fluorine is anomalous in many properties. For example, ionisation enthalpy, electronegativity, and electrode potentials are all higher for fluorine than expected from the trends set by other halogens. Also, ionic and covalent radii, m.p. and b.p., enthalpy of bond dissociation and electron gain enthalpy are quite lower than expected.

(ii) Most of the reactions of fluorine are exothermic (due to the small and strong bond formed by it with other elements).

(iii) It forms only one oxoacid while other halogens form a number of oxoacids.

(iv) Hydrogen fluoride is a liquid (b.p. 293 K) due to strong hydrogen bonding. Other hydrogen halides are gases.

Reason for the anomalous behaviour of fluorine

The anomalous behaviour of fluorine is due to its small size, highest electronegativity, low F-F bond dissociation enthalpy, and non availability of d orbitals in valence shell.

Properties of Hydrogen Halides

Property	HF	HCl	HBr	HI
Melting point/K	190	159	185	222
Boiling point/K	293	189	206	238
Bond length (H – X)/pm	91.7	127.4	141.4	160.9
$\Delta_{diss}H^{\ominus}$/kJ mol^{-1}	57.4	432	363	295
pK_a	3.2	– 7.0	– 9.5	– 10.0

(2) **Reactivity towards hydrogen :** They all react with hydrogen to give hydrogen halides but affinity for hydrogen decreases from fluorine to iodine. Hydrogen halides dissolve in water to form hydrohalic acids. Some of the properties of hydrogen halides are :

(i) The acidic strength order : HF < HCl < HBr < HI

(ii) The stability order of these halides : H–F > H–Cl > H–Br > H–I.

(3) **Reactivity towards oxygen :**

(i) Halogens form many oxides with oxygen but most of them are unstable. Fluorine forms two oxides OF_2 and O_2F_2. However, only OF_2 is thermally stable at 298 K. These oxides are essentially oxygen fluorides because of the higher electronegativity of fluorine than oxygen. Both are strong fluorinating agents. O_2F_2 oxidises plutonium to PuF_6 and the reaction is used in removing plutonium as PuF_6 from spent nuclear fuel.

Session 2019-20

(ii) Chlorine, bromine and iodine form oxides in which the oxidation states of these halogens range from +1 to +7.

(iii) A combination of kinetic and thermodynamic factors lead to the generally decreasing order of stability of oxides formed by halogens, I > Cl > Br.

(iv) The higher oxides of halogens tend to be more stable than the lower ones.

(v) Chlorine oxides, Cl_2O, ClO_2, Cl_2O_6 and Cl_2O_7 are highly reactive oxidising agents and tend to explode. ClO_2 is used as a bleaching agent for paper pulp and textiles and in water treatment.

(vi) The bromine oxides, Br_2O, BrO_2 , BrO_3 are the least stable halogen oxides (middle row anomally) and exist only at low temperatures. They are very powerful oxidising agents.

(vii) The iodine oxides, I_2O_4 , I_2O_5, I_2O_7 are insoluble solids and decompose on heating. I_2O_5 is a very good oxidising agent and is used in the estimation of carbon monoxide.

(4) **Reactivity towards metals :** Halogens react with metals to form metal halides. For example, bromine reacts with magnesium to give magnesium bromide.

$$Mg(s) + Br_2(l) \rightarrow MgBr_2(s)$$

The ionic character of the halides decreases in the order MF > MCl > MBr > MI where M is monovalent metal. If a metal exhibits more than one oxidation state, the halides in higher oxidation state will be more covalent than the one in lower oxidation state. For example, $SnCl_4$, $PbCl_4$, $SbCl_5$ and UF_6 are more covalent than $SnCl_2$, $PbCl_2$, $SbCl_3$ and UF_4 respectively.

(5) **Reactivity of halogens towards other halogens :** Halogens combine amongst themselves to form a number of compounds known as interhalogens of the types XX' , XX_3', XX_5' and XX_7' where X is a larger size halogen and X′ is smaller size halogen.

ALLEN Session 2019-20

FLUORINE

Method of Preparation :

Moissan process : [By electrolysis of KHF_2 (which is obtained from CaF_2)]

$CaF_2 + H_2SO_4 \longrightarrow CaSO_4 + 2HF$

$HF + KF \longrightarrow KHF_2$

$\left\{\text{KF decreases the m.p. of the mix. depending upon the composition, due to formation of } KHF_2.\right\}$

$KHF_2 \xrightarrow{\text{Electrolysis}} H_2 \text{ (at cathode)} + F_2 \text{ (at anode)}$

BROMINE

$2NaBr + Cl_2 \rightarrow 2NaCl + Br_2$

IODINE

From Sea-weed process : Sea weed $\xrightarrow{\Delta}$ Ash $\longrightarrow$ Iodide dissolved in boiling water $\xrightarrow{H_2O_2\ [O]}$ I_2 $\longrightarrow$ Solvent extraction $\longrightarrow$ Evaporation $\longrightarrow$ Pure I_2

CHLORINE

Chlorine was discovered in 1774 by Scheele by the action of HCl on MnO_2.

In 1810 Davy established its elementary nature and suggested the name chlorine on account of its colour (Greek, chloros = greenish yellow).

Preparation

(i) By electrolysis of aq. NaCl :

$$2NaCl + 2H_2O \longrightarrow 2NaOH + \underbrace{H_2}_{\text{at cathode}} + \underset{\text{(anode)}}{Cl_2}$$

$$\underset{\text{(Molten)}}{2NaCl} \xrightarrow{\text{Electrolysis}} \underset{\text{(cathode)}}{2Na} + \underset{\text{(anode)}}{Cl_2}$$

(ii) By heating manganese dioxide with concentrated hydrochloric acid.

$$MnO_2 + 4HCl \rightarrow MnCl_2 + Cl_2 + 2H_2O$$

However, a mixture of common salt and concentrated H_2SO_4 is used in place of HCl.

$$4NaCl + MnO_2 + 4H_2SO_4 \rightarrow MnCl_2 + 4NaHSO_4 + 2H_2O + Cl_2$$

(iii) By the action of HCl on potassium permanganate.

$$2KMnO_4 + 16HCl \rightarrow 2KCl + 2MnCl_2 + 8H_2O + 5Cl_2$$

❑ **Manufacture of chlorine**

(i) **Deacon's process :** By oxidation of hydrogen chloride gas by atmospheric oxygen in the presence of $CuCl_2$ (catalyst) at 723 K.

$$4HCl + O_2 \xrightarrow{CuCl_2} 2Cl_2 + 2H_2O$$

(ii) **Electrolytic process :** Chlorine is obtained by the electrolysis of brine (concentrated NaCl solution). Chlorine is liberated at anode. It is also obtained as a by–product in many chemical industries.

Properties

(i) It is a greenish yellow gas with pungent and suffocating odour.

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(ii) It is about 2-5 times heavier than air.

(iii) It can be liquefied easily into greenish yellow liquid which boils at 239 K.

(iv) It is soluble in water. Chlorine reacts with a number of metals and non-metals to form chlorides.

$2Al + 3Cl_2 \rightarrow 2AlCl_3$; $P_4 + 6Cl_2 \rightarrow 4PCl_3$

$2Na + Cl_2 \rightarrow 2NaCl$; $S_8 + 4Cl_2 \rightarrow 4S_2Cl_2$

$2Fe + 3Cl_2 \rightarrow 2FeCl_3$;

Reaction with hydrogen

It has great affinity for hydrogen. It reacts with compounds containing hydrogen to form HCl.

$H_2 + Cl_2 \rightarrow 2HCl$

$H_2S + Cl_2 \rightarrow 2HCl + S$

$C_{10}H_{16} + 8Cl_2 \rightarrow 16HCl + 10C$

Reaction with ammonia

With excess ammonia, chlorine gives nitrogen and ammonium chloride whereas with excess chlorine, nitrogen trichloride (explosive) is formed.

$\underset{\text{(excess)}}{8NH_3} + 3Cl_2 \rightarrow 6NH_4Cl + N_2$; $NH_3 + \underset{\text{(excess)}}{3Cl_2} \rightarrow NCl_3 + 3HCl$

Reaction with alkalies

With cold and dilute alkalies chlorine produces a mixture of chloride and hypochlorite but with hot and concentrated alkalies it gives chloride and chlorate.

$2NaOH + Cl_2 \rightarrow NaCl + NaOCl + H_2O$
(cold and dilute)

$6\ NaOH + 3Cl_2 \rightarrow 5NaCl + NaClO_3 + 3H_2O$
(hot and conc.)

but on acidification the disproportionated product gives back the same element.

$X^- + OX^- + 2H^+ \longrightarrow X_2 + H_2O$

$5X^- + XO_3^- + 6H^+ \longrightarrow 3X_2 + 3H_2O$

[X = Cl, Br, I]

Reaction with slaked lime

With dry slaked lime it gives bleaching powder.

$2Ca(OH)_2 + 2Cl_2 \rightarrow Ca(OCl)_2 + CaCl_2 + 2H_2O$

The composition of bleaching powder is $\mathbf{Ca(OCl)_2.CaCl_2.Ca(OH)_2.2H_2O}$.

Reaction with hydrocabon

Chlorine reacts with hydrocarbons and gives substitution products with saturated hydrocarbons and addition products with unsaturated hydrocarbons. For example,

$$\underset{\text{Methane}}{CH_4} + Cl_2 \xrightarrow{UV} \underset{\text{Methyl Chloride}}{CH_3Cl} + HCl$$

$$\underset{\text{Ethene}}{C_2H_4} + Cl_2 \xrightarrow{\text{Room temperature}} \underset{\text{1,2-dichloroethane}}{C_2H_4Cl_2}$$

Note :

Chlorine water on standing loses its yellow colour due to the formation of HCl and HOCl. Hypochlorous acid (HOCl) so formed, gives nascent oxygen which is responsible for oxidising and bleaching properties of chlorine.

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\p-Block Element\Eng\01_Theory.p65

Session 2019-20

(i) It oxidises ferrous to ferric and sulphite to sulphate. Chlorine oxidises sulphur dioxide to sulphur trioxide and iodine to iodate. In the presence of water they form sulphuric acid and iodic acid respectively.

$2FeSO_4 + H_2SO_4 + Cl_2 \rightarrow Fe_2(SO_4)_3 + 2HCl$

$Na_2SO_3 + Cl_2 + H_2O \rightarrow Na_2SO_4 + 2HCl$

$SO_2 + 2H_2O + Cl_2 \rightarrow H_2SO_4 + 2HCl$

$I_2 + 6H_2O + 5Cl_2 \rightarrow 2HIO_3 + 10HCl$

(ii) It is a powerful bleaching agent; bleaching action is due to oxidation.

$$Cl_2 + H_2O \rightarrow 2HCl + O$$

Coloured substance + O → Colourless substance

Uses: It is used (i) for bleaching woodpulp (required for the manufacture of paper and rayon), bleaching cotton and textiles, (ii) in the extraction of gold and platinum (iii) in the manufacture of dyes, drugs and organic compounds such as CCl_4, $CHCl_3$, DDT, refrigerants, etc. (iv) in sterilising drinking water and (v) preparation of poisonous gases such as phosgene ($COCl_2$), tear gas (CCl_3NO_2), mustard gas ($ClCH_2CH_2SCH_2CH_2Cl$). (vi) It bleaches vegetable or organic matter in the presence of moisture. Bleaching effect of chlorine is permanent.

Bleaching Powder : $Ca\begin{matrix} \diagup Cl \\ \diagdown OCl \end{matrix}$

Preparation.: $Cl_2(g) + \underset{\text{Slaked lime}}{Ca(OH)_2} \xrightarrow{40°C} Ca(OCl)Cl + H_2O$

(a) On long standing it undergoes

(i) auto oxiation $6Ca(OCl)Cl \longrightarrow Ca(ClO_3)_2 + 5CaCl_2$

(ii) $2Ca(OCl)Cl \xrightarrow[\text{Cat.}]{CoCl_2} 2CaCl_2 + O_2\uparrow$

(iii) $Ca(OCl)Cl + H_2O \longrightarrow Ca(OH)_2 + Cl_2$

Oxidising Properties:

$CaOCl_2 + H_2S \longrightarrow S + CaCl_2 + H_2O$

$CaOCl_2 + 2FeSO_4 + H_2SO_4 \longrightarrow Fe_2(SO_4)_3 + CaCl_2 + H_2O$

$CaOCl_2 + KNO_2 \longrightarrow CaCl_2 + KNO_3$

$3CaOCl_2 + 2NH_3 \longrightarrow 3CaCl_2 + 3H_2O + N_2$

$CaOCl_2 + 2KI + 2HCl \longrightarrow CaCl_2 + 2KCl + H_2O + I_2$

$CaOCl_2 + 2KI + 2AcOH \longrightarrow CaCl_2 + 2KOAc + H_2O + I_2$

$CaOCl_2 + Na_3AsO_3 \longrightarrow Na_3AsO_4 + CaCl_2$

Reaction with acid:

$CaOCl_2 + 2HCl \longrightarrow CaCl_2 + H_2O + Cl_2$; $Ca(OCl)Cl + H_2SO_4 \longrightarrow CaSO_4 + H_2O + Cl_2$

$Ca(OCl)Cl + CO_2 \longrightarrow CaCO_3 + Cl_2$

Note: ClO_2 does not dimerise because odd electron undergoes delocalisation (in its own vacant 3d-orbital)

Cl_2O_4 ($Cl.ClO_4$) is not the dimer of ClO_2. Actually it is Cl-perchlorate.

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\p-Block Element\Eng\01_Theory.p65

HYDROGEN CHLORIDE

(i) Glauber prepared this acid in 1648 by heating common salt with concentrated sulphuric acid.

(ii) Davy in 1810 showed that it is a compound of hydrogen and chlorine.

Preparation

Laboratory method : it is prepared by heating sodium chloride with concentrated sulphuric acid.

$$NaCl + H_2SO_4 \xrightarrow{420K} NaHSO_4 + HCl$$

$$NaHSO_4 + NaCl \xrightarrow{823K} Na_2SO_4 + HCl$$

HCl gas can be dried by passing through concentrated sulphuric acid.

Properties

(i) It is a colourless and pungent smelling gas. Due to strong affinity for water, conc. HCl pulls moisture of air towards self. The moisture forms droplets of water and hence, cloudy white fumes appear.

(ii) It is easily liquefied to a colourless liquid (b.p.189 K) and freezes to a white crystalline solid (f.p. 159 K).

(iii) It is extremely soluble in water

(iv) **Acidic character :** It ionises as follows

$$HCl(g) + H_2O(l) \rightarrow H_3O^+(aq) + Cl^-(aq)\ K_a = 10^7$$

Its aqueous solution is called hydrochloric acid. High value of dissociation constant (K_a) indicates that it is a strong acid in water. It reacts with NH_3 and gives white fumes of NH_4Cl.

$$NH_3 + HCl \rightarrow NH_4Cl$$

Note : Aqua regia

When three parts of concentrated HCl and one part of concentrated HNO_3 are mixed, aqua regia is formed which is used for dissolving noble metals, e.g., gold, platinum.

$$Au + 4H^+ + NO_3^- + 4Cl^- \rightarrow AuCl_4^- + NO + 2H_2O$$

$$3Pt + 16H^+ + 4NO_3^- + 18Cl^- \rightarrow 3PtCl_6^{2-} + 4NO + 8H_2O$$

Reaction with salts

Hydrochloric acid decomposes salts of weaker acids, e.g., carbonates, hydrogencarbonates, sulphites, etc.

$$Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2$$

$$NaHCO_3 + HCl \rightarrow NaCl + H_2O + CO_2$$

$$Na_2SO_3 + 2HCl \rightarrow 2NaCl + H_2O + SO_2$$

Uses:

(i) It is used in the manufacture of chlorine, NH_4Cl and glucose (from corn starch)

(ii) It is used for extracting glue from bones and purifying bone black

(iii) It is used in medicine and as a laboratory reagent.

OXOACIDS OF HALOGENS

(i) Due to high electronegativity and small size, fluorine forms only one oxoacid, HOF known as fluoric (I) acid or hypofluorous acid.

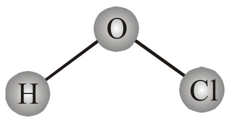
(ii) The other halogens form several oxoacids.

(iii) Most of them cannot be isolated in pure state. They are stable only in aqueous solutions or in the form of their salts.

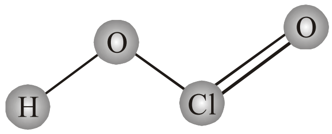
ALLEN Session 2019-20

Oxoacids of Halogens

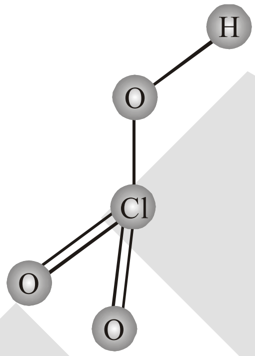
Halic (I) acid (Hypohalous acid)	**HOF** (Hypofluorous acid)	**HOCl** (Hypochlorous acid)	**HOBr** (Hypobromous acid)	**HOI** (Hypoiodous acid)
Halic (III) acid (Halous acid)	—	**HOClO** (Chlorous acid)	—	—
Halic (V) acid (Halic acid)	—	**$HOClO_2$** (Chloric acid)	**$HOBrO_2$** (Bromic acid)	**$HOIO_2$** (Iodic acid)
Halic (VII) acid (Perhalic acid)	—	**$HOClO_3$** (Perchloric acid)	**$HOBrO_3$** (Perbromic acid)	**$HOIO_3$** (Periodic acid)



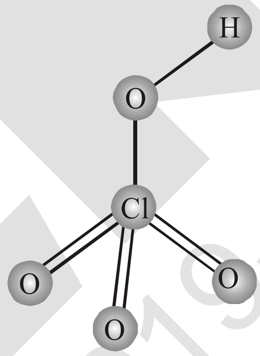
Hypochlorous acid



Chlorous acid



Chloric acid



Perchloric acid

The structures of oxoacids of chlorine

ALLEN Session 2019-20

INTERHALOGEN COMPOUNDS

When two different halogens react with each other, interhalogen compounds are formed. They can be assigned general compositions as XX′ , XX_3', XX_5' and XX_7' where X is halogen of larger size and X′ of smaller size and X is more electropositive than X′. As the ratio between radii of X and X′ increases, the number of atoms per molecule also increases. Thus, iodine (VII) fluoride should have maximum number of atoms as the ratio of radii between I and F should be maximum. That is why its formula is IF_7 (having maximum number of atoms).

$5IF \longrightarrow IF_5 + 2I_2$ [The overall system gains B.E. by 250 kJ/mol]

There are never more than two halogens in a molecule.

Bonds are essentially covalent and b.p. increases as the E.N. difference increases.

AX_5 & AX_7 type formed by large atoms like Br & I to accommodate more atoms around it.

The interhalogens are generally more reactive than the halogens (except F_2) due to weaker A–X bonds compared toX–X bond.

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Preparation

The interhalogen compounds can be prepared by the direct combination or by the action of halogen on lower interhalogen compounds. The product formed depends upon some specific conditions, For e.g.,

$$\underset{\text{(equal volume)}}{Cl_2 + F_2} \xrightarrow{473K} 2ClF; \qquad \underset{\text{(excess)}}{I_2 + 3Cl_2} \longrightarrow 2ICl_3$$

$$Cl_2 + \underset{\text{(excess)}}{3F_2} \xrightarrow{573K} 2ClF_3; \qquad \underset{\text{(diluted with water)}}{Br_2 + 3F_2} \longrightarrow 2BrF_3$$

$$\underset{\text{(equimolar)}}{I_2 + Cl_2} \longrightarrow 2ICl; \qquad \underset{\text{(excess)}}{Br_2 + 5F_2} \longrightarrow 2BrF_5$$

Properties

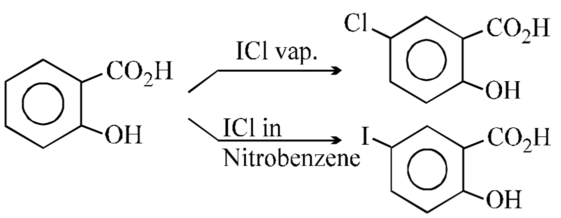
Some properties of interhalogen compounds are given in Table below

Some Properties of Interhalogen Compounds

Type	Formula	Physical stae and colour	Structure
XX'_1	ClF	colourless gas	—
	BrF	pale brown gas	—
	IF[a]	Detected spectroscopically	—
	BrCl[b]	gas	—
	ICl	ruby red solid (α-form)	—
		brown red solid (β-form)	—
	IBr	black solid	—
XX'_3	ClF_3	colourless gas	Bent T-shaped
	BrF_3	yellow green liquid	Bent T-shaped
	IF_3	yellow powder	Bent T-shaped
	ICl_3[c]	orange solid	Bent T-shaped
Xx'_5	IF_5	colourless gas but solid below 77 K	square pyramidal
	BrF_5	colourless liquid	square pyramidal
	ClF_5	colourless liquid	square pyramidal
Xx'_7	IF_7	colourless gas	pentagonal bipyramidal

[a]Very unstable; [b]The pure solid is known at room temperature; [c]Dimerises as Cl–bridged dimer (I_2Cl_6)

One pecularity with ICl :



is chlorination (I)

is Iodination (II)

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Session 2019-20

(i) These are all covalent molecules and are diamagnetic in nature.
(ii) They are volatile solids or liquids at 298 K except ClF which is a gas.
(iii) Their physical properties are intermediate between those of constituent halogens except that their m.p. and b.p. are a little higher than expected.
(iv) Their chemical reactions can be compared with the individual halogens. In general, interhalogen compounds are more reactive than halogens (except fluorine). This is because X–X′ bond in interhalogens is weaker than X–X bond in halogens except F–F bond.
(v) All these undergo hydrolysis giving halide ion derived from the smaller halogen and a hypohalite (when XX′), halite (when XX'_3), halate (when XX'_5) and perhalate (when XX'_7) anion derived from the larger halogen.

$$XX' + H_2O \rightarrow HX' + HOX$$

Their molecular structures are very interesting which can be explained on the basis of VSEPR theory. The XX_3 compounds have the bent 'T' shape, XX_5 compounds square pyramidal and IF_7 has pentagonal bipyramidal structures.

Uses: (i) These compounds can be used as non aqueous solvents.
(ii) Interhalogen compounds are very useful fluorinating agents. ClF_3 and BrF_3 are used for the production of UF_6 in the enrichment of ^{235}U.

$$U(s) + 3ClF_3(l) \rightarrow UF_6(g) + 3ClF(g)$$

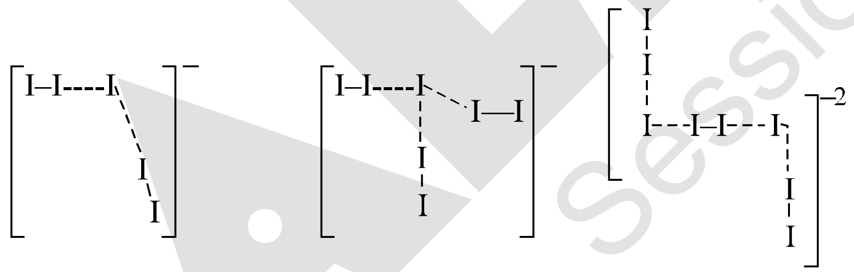
POLYHALIDES

(i) $KI + I_2 \longrightarrow KI_3$
(ii) $ICl + KCl \longrightarrow K^+ [ICl_2]^-$
(iii) $ICl_3 + KCl \longrightarrow K^+[ICl_4]^-$
(iv) $IF_5 + CsF \longrightarrow Cs^+[IF_6]^-$
(v) $ICl + KBr \longrightarrow K^+[BrICl]^-$

$Rb[ICl_2] \xrightarrow{\Delta} RbCl + ICl$ [not $RbI + Cl_2$]

Here the products on heating depends on the lattice energy of the product halide. The lattice energy of alkali halide with smaller halogen is highest since the interatomic distance is least.

Structure of I_5^-, I_7^-, I_8^{-2}



in $\left[N(CH_3)_4\right]^+ I_7^-$ in Cs_2I_8

I_3^-, Br_3^-, Cl_3^-, F_3^- are known Cl_3^- compounds are very less.

Stability order : $I_3^- > Br_3^- > Cl_3^- > F_3^-$ depends upon the donating ability of X^-.

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\p-Block Element\Eng\01_Theory.p65

PSEUDO HALOGENS

There are univalent ion consisting of two or more atoms of which at least one is N, that have properties similar to those of the halide ions. E.g.

(i) Na-salts are soluble in water but Ag-salts are insoluble in water.

(ii) H-compounds are acids like HX. (iii) Some anions can be oxidised to give molecules X_2.

Anions :	**Acids**	**Dimer**
CN^-	HCN	$(CN)_2$
SCN^-	HSCN(thiocyanic acid)	$(SCN)_2$
$SeCN^-$		$(SeCN)_2$
OCN^-	HOCN (cyanic acid)	
NCN^{2-}(Bivalent)	H_2NCN (cyanamide)	
ONC^-	HONC (Fulminic acid)	
N_3^-	HN_3 (Hydrazoic acid)	

$CN^{\ominus}$ shows maximum similarites with Cl^-, Br^-, I^-

(i) forms HCN (ii) forms$(CN)_2$ (iii) AgCN, $Pb(CN)_2$, are insoluble

(iv) Interpseudo halogen compounds ClCN, BrCN, ICN can be formed

(v) AgCN is insoluble in H_2O but soluble in NH_3

(vi) forms large no.of complexes.e.g. $[Cu(CN)_4]^{3-}$ & $[CuCl_4]^{-3}$

$[Co(CN)_6]^{-3}$ & $[CoCl_6]^{-3}$

NOBLE GASES FAMILY
GROUP 18 ELEMENTS (He, Ne, Ar, Kr, Xe, Rn)

Occurrence

(i) All the noble gases except radon occur in the atmosphere.

Relative abundance : Ar is highest (Ne, Kr, He, Rn)

(ii) Their atmospheric abundance in dry air is ~ 1% by volume of which argon is the major constituent.

(iii) Helium and sometimes neon are found in minerals of radioactive origin e.g., pitchblende, monazite, cleveite.

(iv) The main commercial source of helium is natural gas.

(v) Xenon and radon are the rarest elements of the group.

(vi) Radon is obtained as a decay product of ^{226}Ra.

$$^{226}_{88}Ra \rightarrow {}^{222}_{86}Rn + {}^{4}_{2}He$$

(vii) He liquid can exist in two forms. I-form when changes to II-form at λ-point temperature many physical properties change abruptly.

e.g.

(i) Sp. heat changes by a factor of 10

(ii) Thermal conductivity increases by 10^6 and it becomes 800 times faster than Cu

(iii) It shows zero resistance

(iv) It can flow up the sides of the vessel

ALLEN Session 2019-20

Qus. Why 18 group element are termed as noble gas ?

Ans. All these are gases and chemically unreactive. They form very few compounds. Because of this they are termed noble gases.

- **Electronic Configuration**

General electronic configuration of 18 group element is ns^2np^6 except helium which has $1s^2$. Many of the properties of noble gases including their inactive nature are ascribed to their closed shell structures.

- **Ionisation Enthalpy**

Ionsiation energy decreases down the group with increase in atomic size.

He > Ne > Ar > Kr > Xe > Rn (I.E. order)

Due to stable electronic configuration these gases exhibit very high ionisation enthalpy.

- **Atomic Radii**

Atomic radii increase down the group with increase in atomic number.

He < Ne < Ar < Kr < Xe < Rn (atomic radius order)

- **Electron Gain Enthalpy**

They have large positive values of electron gain enthalpy due to stable electronic configurations, and there for have no tendency to accept the electron

- **Melting point and boiling point**

He < Ne < Ar < Kr < Xe < Rn (Melting point order)

↓

(–269°C)

B.P. order : He < Ne < Ar < Kr < Xe < Rn (Boiling point order)

- **Density order :**

He < Ne < Ar < Kr < Xe < Rn (Density order)

Physical properties :

(i) All the noble gases are monoatomic.

(ii) They are colourless, odourless and tasteless.

(iii) They are sparingly soluble in water.

(iv) They have very low melting and boiling points because the only type of interatomic interaction in these elements is weak dispersion forces.

(v) Helium has the lowest boiling point (4.2 K) of any known substance.

(vi) It has an unusual property of diffusing through most commonly used laboratory materials such as rubber, glass or plastics.

ALLEN Session 2019-20

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Chemical Properties

In general, noble gases are least reactive. Their inertness to chemical reactivity is attributed to the following reasons:

(i) The noble gases except helium ($1s^2$) have completely filled ns^2np^6 electronic configuration in their valence shell.

(ii) They have high ionisation enthalpy and more positive electron gain enthalpy.

Note : The reactivity of noble gases has been investigated occasionally, In March 1962, Neil Bartlett, then at the University of British Columbia, observed the reaction of a noble gas. First, he prepared a red compound which is formulated as $O_2^+ PtF_6^-$. He, then realised that the first ionisation enthalpy of molecular oxygen (1175 $kJmol^{-1}$) was almost identical with that of xenon (1170 kJ mol^{-1}). He made efforts to prepare same type of compound with Xe and was successful in preparing another red colour compound $Xe^+PtF_6^-$ by mixing PtF_6 and xenon. After this discovery, a number of xenon compounds mainly with most electronegative elements like fluorine and oxygen, have been synthesised. The compounds of krypton are fewer. Only the difluoride (KrF_2) has been studied in detail. Compounds of radon have not been isolated but only identified (e.g., RnF_2) by radiotracer technique. No true compounds of Ar, Ne or He are yet known.

FLUORIDES OF XENON

Preparation

$$\underset{\text{(xenon in excess)}}{Xe(g) + F_2(g)} \xrightarrow{673K, 1\ bar} XeF_2(s)$$

$$\underset{(1:5\ ratio)}{Xe(g) + 2F_2(g)} \xrightarrow{873K,\ 7bar} XeF_4(s)$$

$$\underset{(1:20\ ratio)}{Xe(g) + 3F_2(g)} \xrightarrow{573K,\ 60-70 bar} XeF_6(s)$$

$$XeF_4 + O_2F_2 \xrightarrow{143K} XeF_6 + O_2$$

Physical properties

XeF_2, XeF_4 and XeF_6 are colourless crystalline solids and sublime readily at 298 K.

Chemical properties

(i) Hydrolysis : They are readily hydrolysed even by traces of water. For example, XeF_2 is hydrolysed to give Xe, HF and O_2.

$$2XeF_2(s) + 2H_2O(l) \rightarrow 2Xe\ (g) + 4\ HF(aq) + O_2(g)$$

Hydrolysis of XeF_4 and XeF_6 with water gives XeO_3.

$$6XeF_4 + 12H_2O \rightarrow 4Xe + 2XeO_3 + 24HF + 3O_2$$

$$XeF_6 + 3H_2O \rightarrow XeO_3 + 6\ HF$$

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\p-Block Element\Eng\01_Theory.p65

Partial hydrolysis of XeF_6 gives oxyfluorides, $XeOF_4$ and XeO_2F_2.

$XeF_6 + H_2O \rightarrow XeOF_4 + 2HF$

$XeF_6 + 2H_2O \rightarrow XeO_2F_2 + 4HF$

Note : **Hydrolysis in alkaline medium**

$2XeF_2 + 4OH^- \longrightarrow 2Xe + 4F^- + 2H_2O + O_2$

$XeF_6 + 7OH^- \longrightarrow \underset{\text{Xenate ion}}{HXeO_4^-} + 3H_2O + 6F^-$

$2HXeO_4^- + 2OH^- \longrightarrow XeO_6^{4-}\downarrow + Xe + 2H_2O + O_2$

(ii) As fluorinating agents : They are powerful fluorinating agents.

$2SF_4 + XeF_4 \longrightarrow 2SF_6 + Xe$; $\quad Pt + XeF_4 \longrightarrow PtF_4 + Xe$

$NO + XeF_2 \longrightarrow NOF + Xe$

(iii) As fluoride donor

$XeF_2 + MF_5 \longrightarrow [XeF]^+ [MF_6]^-$ (M = As, Sb, P)

$XeF_6 + BF_3 \longrightarrow [XeF_5]^+ [BF_4^-]$

$XeF_6 + HF \longrightarrow [XeF_5]^+ [HF_2]^-$

(iv) As Fluoride acceptor

$XeF_6 + MF \longrightarrow M^+ [XeF_7]^-$ (M = Na, K, Rb, Cs)

$2Cs^+[XeF_7]^- \xrightarrow[50°C]{\Delta} XeF_6 + Cs_2[XeF_8]$

$XeF_4 + MF \longrightarrow M^+ + XeF_5^-$

(alkali metals fluoride)

(v) Reaction with SiO_2

SiO_2 also converts XeF_6 into $XeOF_4$

$2XeF_6 + SiO_2 \longrightarrow SiF_4\uparrow + 2XeOF_4$

$XeO_3 + 2XeF_6 \longrightarrow 3XeOF_4$

Similarly, $XeO_3 + XeOF_4 \longrightarrow 2XeO_2F_2$

ALLEN Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\p-Block Element\Eng\01_Theory.p65

(vi) Oxidizing properties

H_2 reduces Xe – fluorides to Xe

$XeF_2 + H_2 \longrightarrow Xe + 2HF$ and so on

Xe - fluorides oxidise Cl^- to Cl_2 and I^- to I_2

$XeF_2 + 2HCl \longrightarrow 2HF + Xe + Cl_2$

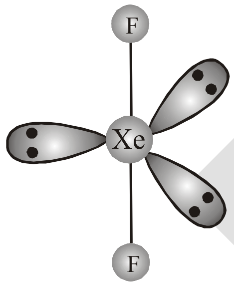
$XeF_4 + 4KI \longrightarrow 4KF + Xe + 2I_2$

Noble gas hydrate (clathrate compound) : Ar, Kr, Xe can form clathrate compounds but He, Ne cannot due to their smaller size.

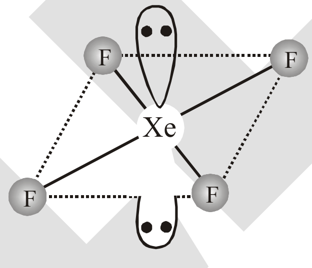
eg. $Xe \cdot 6H_2O$ ⎫ formed only when
$Ar \cdot 6H_2O$ ⎬ water freezes at high
$Kr \cdot 6H_2O$ ⎭ pressure together with noble gas

(a) Structure and bonding

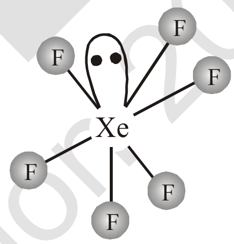
XeF_2 and XeF_4 have linear and square planar structures respectively. XeF_6 has seven electron pairs (6 bonding pairs and one lone pair) and would, thus, have a distorted octahedral structure as found experimentally in the gas phase. Xenon fluorides react with fluoride ion acceptors to form cationic species and fluoride ion donors to form fluoroanions.



(a) Linear



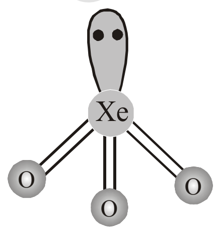
(b) Square plannar



(c) Distorted octahedral

(b) Xenon-oxygen compounds

XeO_3 is a colourless, white hygroscopic explosive solid and has a pyramidal molecular structure.



Pyramidal

$XeOF_4$ is a colourless volatile liquid and has a square pyramidal molecular structure.

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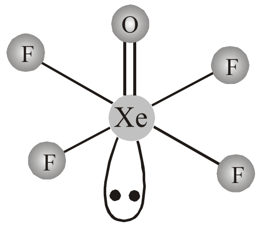

Square pyramidal

Uses of helium :

(i) He is a non-inflammable and light gas. Hence, it is used in filling balloons for meteorological observations.

(ii) It is also used in gas-cooled nuclear reactors.

(iii) It is used in cryoscopy to obtain the very low temperature required for superconductor and laser (b.p. 4.2 K) finds use as cryogenic agent for carrying out various experiments at low temperatures.

(iv) It is used to produce and sustain powerful superconducting magnets which form an essential part of modern NMR spectrometers and Magnetic Resonance Imaging (MRI) systems for clinical diagnosis.

(v) It is used as a diluent for oxygen in modern diving apparatus because of its very low solubility in blood. He is used in preference to N_2 to dil. O_2 in the gas cylinders used by divers. This is because N_2 is quite soluble in blood, so a sudden change in pressure causes degassing and gives bubbles of N_2 in the blood. This causes the painful condition called bends. He is slightly soluble so the risk of bends is reduced.

USES OF NEON :

(i) Ne is used in discharge tubes and fluorescent bulbs for advertisement display purposes.

(ii) Neon bulbs are used in botanical gardens and in green houses.

USES OF ARGON :

(i) Argon is used mainly to provide an inert atmosphere in high temperature metallurgical processes (arc welding of metals or alloys) and for filling electric bulbs.

(ii) It is also used in the laboratory for handling substances that are air-sensitive.

USES OF XENON AND KRYPTON :

There are no significant uses of Xenon and Krypton. They are used in light bulbs designed for special purposes.

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SOLVED EXAMPLE

1. Though nitrogen exhibits +5 oxidation state, it does not form pentahalide. Give reason.

Sol. Nitrogen with n = 2, has s and p orbitals only. It does not have d orbitals to expand its covalency beyond four. That is why it does not form pentahalide.

2. PH_3 has lower boiling point than NH_3. Why?

Sol. Unlike NH_3, PH_3 molecules are not associated through hydrogen bonding in liquid state. That is why the boiling point of PH_3 is lower than NH_3.

3. Write the reaction of thermal decomposition of sodium azide.

Sol. Thermal decomposition of sodium azide gives dinitrogen gas.

$2NaN_3 \rightarrow 2Na + 3N_2$

4. Why does NH_3 act as a Lewis base ?

Sol. Nitrogen atom in NH_3 has one lone pair of electrons which is available for donation. Therefore, it acts as a Lewis base.

5. NH_3 can't be dried by H_2SO_4, P_2O_5 and anh.$CaCl_2$

Sol. because : $2NH_3 + H_2SO_4 \longrightarrow (NH_4)_2\ SO_4$

$H_2O + NH_3 + P_2O_5 \longrightarrow (NH_4)_3\ PO_4$

$CaCl_2 + 8NH_3 \longrightarrow \underset{\text{forms adduct}}{CaCl_2 \cdot 8NH_3}$

Quick lime is used for this purpose

$$\left.\begin{array}{r} CaO + H_2O \longrightarrow Ca(OH)_2\ (\text{base}) \\ NH_3\ (\text{base}) \end{array}\right\} \textit{Hence no interaction}$$

6. Why does NO_2 dimerise ?

Sol. NO_2 contains odd number of valence electrons. It behaves as a typical odd molecule. On dimerisation, it is converted to stable N_2O_4 molecule with even number of electrons.

7. In what way can it be proved that PH_3 is basic in nature?

Sol. PH_3 reacts with acids like HI to form PH_4I which shows that it is basic in nature.

$$PH_3 + HI \rightarrow PH_4I$$

Due to lone pair on phosphorus atom, PH_3 is acting as a Lewis base in the above reaction.

8. Why does PCl_3 fume in moisture ?

Sol. PCl_3 hydrolyses in the presence of moisture giving fumes of HCl.

$$PCl_3 + 3H_2O \rightarrow H_3PO_3 + 3HCl$$

9. Are all the five bonds in PCl_5 molecule equivalent? Justify your answer.

Sol. PCl_5 has a trigonal bipyramidal structure and the three equitorial P-Cl bonds are equivalent, while the two axial bonds are different and longer than equitorial bonds.

10. How do you account for the reducing behaviour of H_3PO_2 on the basis of its structure?

Sol. In H_3PO_2, two H atoms are bonded directly to P atom which imparts reducing character to the acid.

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11. Elements of Group 16 generally show lower value of first ionisation enthalpy compared to the corresponding periods of group 15. Why?

Sol Due to extra stable half-filled p orbitals electronic configurations of Group 15 elements, larger amount of energy is required to remove electrons compared to Group 16 elements.

12. H_2S is less acidic than H_2Te. Why?

Sol. Due to the decrease in bond (E–H) dissociation enthalpy down the group, acidic character increases.

13. Which form of sulphur shows paramagnetic behaviour ?

Sol. In vapour state sulphur partly exists as S_2 molecule which has two unpaired electrons in the antibonding π^* orbitals like O_2 and, hence, exhibits paramagnetism.

14. What happens when

(i) Concentrated H_2SO_4 is added to calcium fluoride

(ii) SO_3 is passed through water?

Sol. (i) It forms hydrogen fluoride

$$CaF_2 + H_2SO_4 \rightarrow CaSO_4 + 2HF$$

(ii) It dissolves SO_3 to give H_2SO_4.

$$SO_3 + H_2O \rightarrow H_2SO_4$$

15. Halogens have maximum negative electron gain enthalpy in the respective periods of the periodic table. Why?

Sol. Halogens have the smallest size in their respective periods and therefore high effective nuclear charge. As a consequence, they readily accept one electron to acquire noble gas electronic configuration.

16. Although electron gain enthalpy of fluorine is less negative as compared to chlorine, fluorine is a stronger oxidising agent than chlorine. Why?

Sol. It is due to

(i) low enthalpy of dissociation of F-F bond

(ii) high hydration enthalpy of F^-

17. Fluorine exhibits only –1 oxidation state whereas other halogens exhibit + 1, + 3, + 5 and + 7 oxidation states also. Explain.

Sol. Fluorine is the most electronegative element and cannot exhibit any positive oxidation state. Other halogens have d orbitals and therefore, can expand their octets and show + 1, + 3, + 5 and + 7 oxidation states also.

18. Write the balanced chemical equation for the reaction of Cl_2 with hot and concentrated NaOH. Is this reaction a disproportionation reaction? Justify.

Sol. $3Cl_2 + 6NaOH \rightarrow 5NaCl + NaClO_3 + 3H_2O$

Yes, chlorine from zero oxidation state is changed to –1 and +5 oxidation states.

19. CaF_2 used in HF prep^n. must be free from SiO_2. Explain.

Ans. $CaF_2 + H_2SO_4 \longrightarrow CaSO_4 + HF$

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If SiO_2 present as impurity

$4HF + SiO_2 \longrightarrow SiF_4 + 2H_2O$ } Hence presence of one molecule SiO_2

$SiF_4 + 2HF \longrightarrow H_2[SiF_6]$ } consumes 6 molecule of HF

HF can not be stored in glass vessel due to same reason.

20. When HCl reacts with finely powdered iron, it forms ferrous chloride and not ferric chloride. Why?

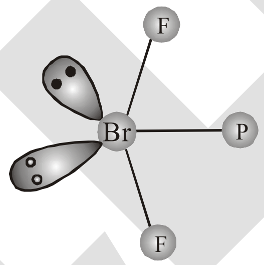
Sol. Its reaction with iron produces H_2.

$Fe + 2HCl \rightarrow FeCl_2 + H_2$

Liberation of hydrogen prevents the formation of ferric chloride.

21. Discuss the molecular shape of BrF_3 on the basis of VSEPR theory.

Sol. The central atom Br has seven electrons in the valence shell. Three of these will form electronpair bonds with three fluorine atoms leaving behind four electrons. Thus, there are three bond pairs and two lone pairs. According to VSEPR theory, these will occupy the corners of a trigonal bipyramid. The two lone pairs will occupy the equatorial positions to minimise lone pair-lone pair and the bond pair-lone pair repulsions which are greater than the bond pair-bond pair repulsions. In addition, the axial fluorine atoms will be bent towards the equitorial fluorine in order to minimise the lone-pair-lone pair repulsions. The shape would be that of a slightly bent 'T'.



22. Why are the elements of Group 18 known as noble gases ?

Sol. The elements present in Group 18 have their valence shell orbitals completely filled and, therefore, react with a few elements only under certain conditions. Therefore, they are now known as noble gases.

23. Noble gases have very low boiling points. Why?

Sol. Noble gases being monoatomic have no interatomic forces except weak dispersion forces and therefore, they are liquefied at very low temperatures. Hence, they have low boiling points.

24. Does the hydrolysis of XeF_6 lead to a redox reaction?

Sol. No, the products of hydrolysis are $XeOF_4$ and XeO_2F_2 where the oxidation states of all the elements remain the same as it was in the reacting state.

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25. Standard electrode potential values, $E^{\ominus}$ for Al^{3+}/Al is –1.66 V and that of Tl^{3+}/Tl is +1.26 V. Predict about the formation of M^{3+} ion in solution and compare the electropositive character of the two metals.

Sol. Standard electrode potential values for two half cell reactions suggest that aluminium has high tendency to make $Al^{3+}(aq)$ ions, whereas Tl^{3+} is not only unstable in solution but is a powerful oxidising agent also. Thus Tl^{+} is more stable in solution than Tl^{3+}. Aluminium being able to form +3 ions easily, is more electropositive than thallium.

26. White fumes appear around the bottle of anhydrous aluminium chloride. Give reason.

Sol. Anhydrous aluminium chloride is partially hydrolysed with atmospheric moisture to liberate HCl gas. Moist HCl appears white in colour.

27. Boron is unable to form BF_6^{3-} ion. Explain.

Sol. Due to non-availability of d orbitals, boron is unable to expand its octet. Therefore, the maximum covalence of boron cannot exceed 4.

28. Why is boric acid considered as a weak acid ?

Sol. Because it is not able to release H^+ ions on its own. It receives OH^- ions from water molecule to complete its octet and in turn releases H^+ ions.

29. Select the member(s) of group 14 that (i) forms the most acidic dioxide, (ii) is commonly found in +2 oxidation state, (iii) used as semiconductor.

Sol. (i) carbon (ii) lead (iii) silicon and germanium

30. $[SiF_6]^{2-}$ is known whereas $[SiCl_6]^{2-}$ not. Give possible reasons.

Sol. The main reasons are :

(i) six large chloride ions cannot be accommodated around Si^{4+} due to limitation of its size.

(ii) interaction between lone pair of chloride ion and Si^{4+} is not very strong.

31. Diamond is covalent, yet it has high melting point. Why ?

Sol. Diamond has a three-dimensional network involving strong C—C bonds, which are very difficult to break and, in turn has high melting point.

32. SiH_4 is more reactive than CH_4. Explain

Reasons

(i) $Si^{\delta+}-H^{\delta-}$ in $C^{\delta-}-H^{\delta+}$

C - more electronegative than H

Si less electronegative than H

So bond polarity is reversed when Nu^- attacks, it faces repulsion in C but not in Si

(ii) Silicon is having vacant d orbital which is not in case of carbon

(iii) Silicon is larger in size compared to C. By which the incoming Nu^- doesn't face any steric hindrance to attack at Si whereas CH_4 is tightly held from all sides.

ALLEN Session 2019-20

EXERCISE # I

Only one option is correct :

1. PH_3(Phosphine) when passed in aqueous solution of $CuSO_4$ it produce -
(A) Blue precipitate of $Cu(OH)_2$
(B) dark blue solution of $[Cu(PH_3)_4]SO_4$
(C) Black precipitate of Cu_3P_2
(D) Colorless solution of $[Cu(H_2O)_4]^+$

2. $H_3PO_2 \xrightarrow{\Delta} (X) + PH_3$; is
(A) Dehydration reaction
(B) Oxidation reaction
(C) Disproportionation reaction
(D) Dephosphorelation reaction

3. Which of the following species is not a pseudohalide?
(A) CNO^- (B) $RCOO^-$ (C) OCN^- (D) N_3^-

4. An orange solid (X) on heating, gives a colourless gas (Y) and a only green residue (Z). Gas (Y) on treatment with Mg, produces a white solid substance ....
(A) Mg_3N_2 (B) MgO (C) Mg_2O_3 (D) $MgCl_2$

5. Conc. HNO_3 is yellow coloured liquid due to
(A) dissolution of NO in conc. HNO_3
(B) dissolution of NO_2 in conc. HNO_3
(C) dissolution of N_2O in conc. HNO_3
(D) dissolution of N_2O_3 in conc. HNO_3

6. A gas at low temperature does not react with the most of compounds. It is almost inert and is used to create inert atmosphere in bulbs. The combustion of this gas is exceptionally an endothermic reaction. Based on the given information, we can conclude that the gas is
(A) oxygen
(B) nitrogen
(C) carbon mono-oxide
(D) hydrogen

7. When chlorine gas is passed through an aqueous solution of a potassium halide in the presence of chloroform, a voilet colouration is obtained. On passing more of chlorine water, the voilet colour is disappeared and solution becomes colourless. This test confirms the presence of .... in aqueous solution.
(A) chlorine (B) fluorine (C) bromine (D) iodine

8. $H_3PO_2 \xrightarrow{140°C} A \xrightarrow{220°C} B \xrightarrow{320°C} C$
Compound (C) is
(A) H_2PO_3 (B) H_3PO_3 (C) $(HPO_3)_n$ (D) $H_4P_2O_7$

9. An explosive compound (A) reacts with water to produce NH_4OH and HOCl. Then, the compound (A), is
(A) TNG (B) NCl_3 (C) PCl_3 (D) HNO_3

10. An inorganic compound (A) made of two most occuring elements into the earth crust, having a polymeric tetrahedral network structure. With carbon, compound (A) produces a poisonous gas (B) which is the most stable diatomic molecule. Compounds (A) and (B) will be
(A) SiO_2, CO_2 (B) SiO_2, CO (C) SiC,CO (D) SiO_2, N_2

ALLEN Session 2019-20

11. A sulphate of a metal (A) on heating evolves two gases (B) and (C) and an oxide (D). Gas (B) turns $K_2Cr_2O_7$ paper green while gas (C) forms a trimer in which there is no S–S bond. Compound (D) with HCl, forms a Lewis acid (E) which exists as a dimer. Compounds (A), (B), (C), (D) and (E) are respectively

(A) $FeSO_4$, SO_2, SO_3, Fe_2O_3, $FeCl_3$ (B) $Al_2(SO_4)_3$, SO_2, SO_3, Al_2O_3, $FeCl_3$

(C) FeS, SO_2, SO_3, $FeSO_4$, $FeCl_3$ (D) FeS, SO_2, SO_3 , $Fe_2(PO_4)_3$, $FeCl_2$

12. A tetra-atomic molecule (A) on reaction with nitrogen(I)oxide, produces two substances (B) and (C). (B) is a dehydrating agent while substance (C) is a diatomic gas which shows almost inert behaviour. The substances (A) and (B) and (C) respectively will be

(A) P_4, P_4O_{10}, N_2 (B) P_4, N_2O_5, N_2 (C) P_4, P_2O_3, Ar (D) P_4, P_2O_3, H_2

13. First compound of inert gases was prepared by scientist Neil Bartlett in 1962. This compound is

(A) $XePtF_6$ (B) XeO_3 (C) XeF_6 (D) $XeOF_4$

14. Carbongene has X% of CO_2 and is used as an antidote for poisoning of Y . Then, X and Y are

(A) X = 95% and Y = lead poisoning (B) X = 5% and Y = CO poisoning

(C) X = 30% and Y = CO_2 poisoning (D) X = 45% and Y = CO poisoning

15. The correct order of acidic strength of oxides of nitrogen is

(A) $NO < NO_2 < N_2O < N_2O_3 < N_2O_5$ (B) $N_2O < NO < N_2O_3 < N_2O_4 < N_2O_5$

(C) $NO < N_2O < N_2O_3 < N_2O_5 < N_2O_4$ (D) $NO < N_2O < N_2O_5 < N_2O_3 < N_2O_4$

16. $H_3BO_3 \xrightarrow{T_1} X \xrightarrow{T_2} Y \xrightarrow{\text{red hot}} B_2O_3$

if $T_1 < T_2$ then X and Y respectively are

(A) X = Metaboric acid and Y = Tetraboric acid

(B) X = Tetraboric acid and Y = Metaboric acid

(C) X = Borax and Y = Metaboric acid

(D) X = Tetraboric acid and Y = Borax

17. When conc. H_2SO_4 was treated with $K_4[Fe(CN)_6]$, CO gas was evolved. By mistake, somebody used dilute H_2SO_4 instead of conc. H_2SO_4 then the gas evolved was

(A) CO (B) HCN (C) N_2 (D) CO_2

18. An inorganic white crystalline compound (A) has a rock salt structure. (A) on reaction with conc. H_2SO_4 and MnO_2, evolves a pungent smelling, greenish-yellow gas (B). Compound (A) gives white ppt. of (C) with $AgNO_3$ solution. Compounds (A), (B) and (C) will be respectively

(A) NaCl, Cl_2, AgCl (B) NaBr, Br_2, NaBr

(C) NaCl, Cl_2, Ag_2SO_4 (D) Na_2CO_3 , CO_2, Ag_2CO_3

19. $RCl \xrightarrow[\text{Si}]{\text{Cu-powder}} R_2SiCl_2 \xrightarrow{H_2O} R_2Si(OH)_2 \xrightarrow{\text{condensation}} A$

Compound (A) is

(A) a linear silicone (B) a chlorosilane (C) a linear silane (D) a network silane

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20. When oxalic acid reacts with conc. H_2SO_4 , two gases produced are of neutral and acidic in nature respectively. Potassium hydroxide absorbs one of the two gases. The product formed during this absorption and the gas which gets absorbed are respectively

(A) K_2CO_3 and CO_2 (B) $KHCO_3$ and CO_2

(C) K_2CO_3 and CO (D) $KHCO_3$ and CO

21. Conc. H_2SO_4 cannot be used to prepare HBr from NaBr because it

(A) reacts slowly with NaBr (B) oxidises HBr

(C) reduces HBr (D) disproportionates HBr

22. Ammonia can be dried by

(A) conc. H_2SO_4 (B) P_4O_{10} (C) CaO (D) anhydrous $CaCl_2$

23. When chlorine reacts with a gas X, an explosive inorganic compound Y is formed. Then X and Y will be

(A) X = O_2 and Y = NCl_3 (B) X = NH_3 and Y=NCl_3

(C) X = O_2 and Y = NH_4Cl (D) X = NH_3 and Y = NH_4Cl

24. $HNO_3 + P_4O_{10} \longrightarrow HPO_3 + A$; the product A is

(A) N_2O (B) N_2O_3 (C) NO_2 (D) N_2O_5

25. Which of the following is the correct order of acidic strength?

(A) $Cl_2O_7 > SO_3 > P_4O_{10}$ (B) $CO_2 > N_2O_5 > SO_3$

(C) $Na_2O > MgO > Al_2O_3$ (D) $K_2O > CaO > MgO$

26. $Ca + C_2 \longrightarrow CaC_2 \xrightarrow{N_2} A$

Compound (A) is used as a/an

(A) fertilizer (B) dehydrating agent (C) oxidising agent (D) reducing agent

27. A gas which exists in three allotropic forms α, β and γ is

(A) SO_2 (B) SO_3 (C) CO_2 (D) NH_3

28. A red coloured mixed oxide (X) on treatment with conc. HNO_3 gives a compound (Y). (Y) with HCl, produces a chloride compound (Z) which can also be produced by treating (X) with conc. HCl. Compounds (X) , (Y), and (Z) will be

(A) Mn_3O_4, MnO_2, $MnCl_2$ (B) Pb_3O_4, PbO_2, $PbCl_2$

(C) Fe_3O_4, Fe_2O_3, $FeCl_2$ (D) Fe_3O_4, Fe_2O_3, $FeCl_3$

29. One mole of calcium phosphide on reaction with excess of water gives

(A) one mole of phosphine (B) two moles of phosphoric acid

(C) two moles of phosphine (D) one mole of phosphorus penta-oxide

30. $NaH_2PO_4 \xrightarrow{> 240°C} \underset{\text{Sodium trimetaphosphate}}{(NaPO_3)_3} \xrightarrow{625°C} NaPO_3\text{(liquid melt)} \xrightarrow[\text{cooling}]{\text{rapid}} D\text{(glass)}$

Compound (D) is known as

(A) Microcosmic salt (B) Graham's salt (C) Fischer's salt (D) Switzer's Salt

ALLEN Session 2019-20

31. Three allotropes (A), (B) and (C) of phosphorous in the following change are respectively

$$A \xrightarrow[1200\ \text{atm}]{473\ \text{K}} B$$
$$A \xrightarrow[CO_2\text{–atm}]{573\ \text{K}} C$$

(A) white, β-black, red (B) β-black, white, red

(C) red, β-black, white (D) red, violet, β-black

32. When an inorganic compound reacts with SO_2 in aqueous medium, produces (A). (A) on reaction with Na_2CO_3, gives compound (B) which with sulphur, gives a substance (C) used in photography. Compound (C) is

(A) Na_2S (B) $Na_2S_2O_7$ (C) Na_2SO_4 (D) $Na_2S_2O_3$

33. $B(OH)_3 + NaOH \rightleftharpoons NaBO_2 + Na[B(OH)_4] + H_2O$

How can this reaction is made to proceed in forward direction?

(A) addition of cis 1,2 diol (B) addititon of borax

(C) addition of trans 1,2 diol (D) addition of Na_2HPO_4

34. Which is the compound responsible for the flickering light called **will-o-the-wisp**, some times seen in the Marsh.

(A) PH_3 (B) P_2H_4 (C) H_2S (D) $PH_3 + H_2S$

35. The gun powder is consisting of'______'+ sulphur + Charcoal what is the missing substance for gun powder

(A) $LiNO_3$ (B) NH_4NO_2 (C) KNO_3 (D) (A) and (B) mixture

36. An aqueous solution of borax is

(A) Neutral (B) Amphoteric (C) Basic (D) Acidic

37. Boric acid is polymeric due to

(A) Its acidic nature

(B) The presence of hydrogen bonds

(C) Its monobasic nature

(D) Its geometry

38. The type of hybridisation of boron in diborane is

(A) sp (B) sp^2 (C) sp^3 (D) dsp^2

39. Thermodynamically the most stable form of carbon is

(A) Diamond (B) Graphite (C) Fullerenes (D) Coal

40. Elements of group 14

(A) Exhibit oxidation state of +4 only (B) Exhibit oxidation state of +2 and +4 only

(C) Form M^{2-} and M^{4+} ions (D) Form M^{2+} and M^{4+} ions

ALLEN Session 2019-20

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41. $A + Br_2 \rightarrow N_2 + (B)$

$$(B) \xrightarrow{NaOH} (A)$$

if A is a basic gas then identified (A) and (B)

(A) NH_3, NH_4Br (B) NH_3, N_2O (C) NH_3, N_2O_5 (D) None of these

Question No. 42 to 47 (6 questions)

Questions given below consist of two statements each printed as Assertion (A) and Reason (R); while answering these questions you are required to choose any one of the following four responses:

(A) if both (A) and (R) are true and (R) is the correct explanation of (A)

(B) if both (A) and (R) are true but (R) is not correct explanation of (A)

(C) if (A) is true but (R) is false

(D) if (A) is false and (R) is true

42. **Assertion :** Borax bead test is applicable only to coloured salt.

Reason : In borax bead test, coloured salts are decomposed to give coloured metal meta borates.

43. **Assertion :** Aluminium and zinc metal evolve H_2 gas from NaOH solution

Reason : Several non-metals such as P, S, Cl, etc. convert a hydride instead of producing H_2 gas from NaOH.

44. **Assertion :** Conc. H_2SO_4 can not be used to prepare pure HBr from NaBr

Reason : It reacts slowly with NaBr.

45. **Assertion :** Oxygen is more electronegative than sulphur, yet H_2S is acidic, while H_2O is neutral.

Reason : H–S bond is weaker than O–H bond.

46. **Assertion :** Chlorine gas disproportionates in hot & conc. NaOH solution.

Reason : NaCl and NaOCl are formed in the above reaction.

47. **Assertion :** Liquid IF_5 conducts electricity.

Reason : Liquid IF_5 self ionizes as, $2IF_5 \rightleftharpoons IF_4^+ + IF_6^-$

ALLEN Session 2019-20

EXERCISE # II

***One or more than one option may be correct* :**

1. When a compound X reacts with ozone in aqueous medium, a compound Y is produced. Ozone also reacts with Y and produces compound Z. Z acts as an oxidising agent, then X, Y and Z will be

(A) X = HI, Y = I_2 and Z = HIO_3

(B) X = KI, Y = I_2 and Z = HIO_3

(C) X = KI, Y = I_2 and Z = HIO_4

(D) X = HI, Y = I_2 and Z = HIO_4

2. Which of the following statements is/are correct regarding B_2H_6?

(A) banana bonds are longer but stronger than normal B–H bonds

(B) B_2H_6 is also known as 3c–2e compound

(C) the hybrid state of B in B_2H_6 is sp^3 while that of sp^2 in BH_3

(D) it cannot be prepared by reacting BF_3 with $LiBH_4$ in the presence of dry ether

3. Which of the following statements is/are correct regarding inter-halogen compounds of AB_x type?

(A) x may be 1,3,5 and 7

(B) A is a more electronegative halogen than B

(C) FBr_3 cannot exit

(D) The interhalogens are generally more reactive than the halogens (except F_2) due to weaker A–X bonds compared to X–X bond.

4. When an inorganic compound (X) having 3c-2e as well as 2c-2e bonds reacts with ammonia gas at a certain temperature, gives a compound (Y) iso-structural with benzene. Compound (X) with ammonia at a high temperature, produces a slippery substance (Z). Then

(A) (X) is B_2H_6

(B) (Z) is known as inorganic graphite

(C) (Z) having structure similar to graphite

(D) (Z) having structure similar to (X)

5. Boric acid

(A) exists in polymeric form due to inter-molecular hydrogen bonding.

(B) is used in manufacturing of optical glasses.

(C) is a tri-basic acid

(D) with borax, it is used in the preparation of a buffer solution.

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6. The correct statement(s) related to allotropes of carbon is/are

(A) graphite is the thermodynamically most stable allotrope of carbon and having a two dimensional sheet like structure of hexagonal rings of carbon (sp^2)

(B) diamond is the hardest allotrope of carbon and having a three dimensional network structure of C(sp^3)

(C) fullerene (C_{60}) is recently discovered non-crystalline allotrope of carbon having a football-like structure.

(D) Vander Waal's force of attraction acts between the layers of graphite 6.14 Å away from each other

7. $Al_2(SO_4)_3 + NH_4OH \longrightarrow X$, then

(A) X is a white coloured compound

(B) X is insoluble in excess of NH_4OH

(C) X is soluble in NaOH

(D) X cannot be used as an antacid

8. The species that undergo(es) disproportionation in an alkaline medium is/are

(A) Cl_2

(B) MnO_4^{2-}

(C) P_4

(D) ClO_4^-

9. Select correct statement(s):

(A) Borax is used as a buffer

(B) 1 M borax solution reacts with equal volumes of 2 M HCl solution

(C) Titration of borax can be made using methyl orange as the indicator

(D) Coloured bead obtained in borax-bead test contains metaborate

10. Which of the following is / are correct for group 14 elements?

(A) The stability of dihalides are in the order $CX_2 < SiX_2 < GeX_2 < SnX_2 < PbX_2$

(B) The ability to form $p\pi–p\pi$ multiple bonds among themselves increases down the group

(C) The tendency for catenation decreases down the group

(D) They all form oxides with the formula MO_2.

11. Zeolite is used in which of the following cases :

(A) Conversion of alcohols into gasoline

(B) Cracking of hydrocarbon

(C) Isomerisation of hydrocarbons

(D) Softening of hard water

12. Which of the following oxides are mixed oxide ?

(A) PbO_2

(B) SnO_2

(C) Pb_2O_3

(D) Pb_3O_4

13. Which of the following oxide(s) gives brown ppt on reaction with conc. HNO_3 :

(A) PbO

(B) SnO

(C) Pb_2O_3

(D) Pb_3O_4

14. Which of the following reaction produces PH_3 :

(A) $Ca_3P_2 + H_2O \rightarrow$

(B) $P_4 + NaOH \rightarrow$

(C) $PH_4I + KOH \rightarrow$

(D) $H_3PO_2 \xrightarrow{\Delta}$

15. Which of the following halides is least stable and has doubtful existence ?

(A) CCl_4

(B) GeI_4

(C) SnI_4

(D) PbI_4

ALLEN Session 2019-20

16. $Ca_2B_6O_{11} + Na_2CO_3 \xrightarrow{\Delta} [X] + CaCO_3 + NaBO_2$ (Unbalanced equation)

Correct statement for [X]

(A) Structure of anion of crystalline (X) has one boron atom sp^3 hybridised and other three boron atoms sp^2 hybridised

(B) (X) with NaOH(aq.) gives a compound which on reaction with H_2O_2 in alkaline medium yields a compound used as brightner in soaps

(C) Hydrolysis of (X) with HCl or H_2SO_4 yields a compound which on reaction with HF gives fluoroboric acid

(D) [X] on heating with cobalt salt in oxidising flame gives blue coloured bead

17. $(A) + 2C \xrightarrow[2500°C]{2000°C \text{ to}} (B) + 2CO\uparrow$

$(B) + \text{Carbon} \xrightarrow[2500°C]{2000°C \text{ to}} (C)$

If A is an example of 3-d silicate then select the correct statements about (C)

(A) Central atom of C is sp^3 hybridised

(B) (C) is non planar and all atoms are sp^3 hybridised

(C) (C) has diamond like structure, and it is coloured when impurity is present but pale yellow to colourless solid at room temperature

(D) (C) is silicon carbide (SiC) and it is not being affected by any acid except H_3PO_4

ALLEN Session 2019-20

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EXERCISE # III

Paragraph for Question No. 1 & 2

$$\underset{\text{(tetrameric element)}}{A} + \text{alkaline solution} \rightarrow B + C$$

$$A \xrightarrow{\text{limited } Cl_2} (D)$$

$$C \xrightarrow{\text{limited } Cl_2} (D)$$

$$(F) \xleftarrow{3H_2O} (D) \xrightarrow{CH_3COOH} E + F$$

1. When D react with C_2H_5OH then product will be

(A) $C_2H_5Cl,\ H_3PO_4$ (B) $C_2H_5Cl,\ H_3PO_3$

(C) $CH_3COCl,\ H_3PO_3$ (D) Only H_3PO_3

2. B can be absorbed by :

(A) Ca(OCl) Cl (B) H_2S (C) Both (D) None

Paragraph for Question No. 3 to 6

$$HCOOH \xrightarrow[\text{Conc.} H_2SO_4]{373K} H_2O + (X)$$

$$C(s) + H_2O \xrightarrow{423-1273K} (X) + H_2(g)$$

3. Select the correct statement about (X)

(A) (X) is a colourless, odourless and almost water insoluble gas

(B) X is highly poisonous and burns with blue flame

(C) When (X) gas is passed through $PdCl_2$ solution giving rise to black ppt

(D) All of these

4. Mixture of (X) gas + H_2 is called

(A) Water gas or synthesis gas (B) Producer gas

(C) Methane gas (D) None of these

5. In second reaction when air is used instead of steam a mixture of (X) gas and N_2 is produced which is called

(A) Water gas (B) Synthesis gas (C) Producer gas (D) Carbon dioxide gas

6. Select the correct statement about (X)

(A) (X) gas is estimated by I_2O_5 (B) Cu_2Cl_2 is absorber of (X) gas

(C) (X) gas is the purifying agent for Ni (D) All of these

ALLEN Session 2019-20

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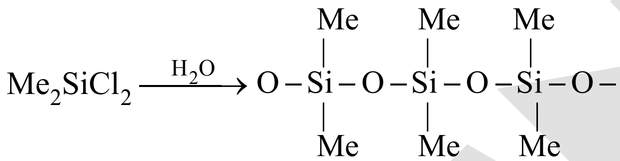
Paragraph for Question No. 7 & 8

Read the following write-ups and answer the questions at the end of it.

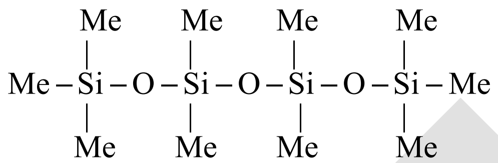
Silicons are synthetic polymers containing repreated R_2SiO units. Since, the empirical formula is that of a ketone (R_2CO), the name silicone has been given to these materials. Silicones can be made into oils, rubbery elastomers and resins. They find a variety of applications because of their chemical inertness, water repelling nature, heat-resistance and good electrical insulating property.

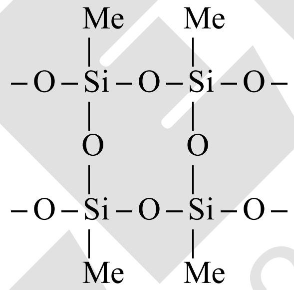
Commercial silicon polymers are usually methyl derivatives and to a lesser extent phenyl derivatives and are synthesised by the hydrolysis of

R_2SiCl_2[R=methyl (Me) or phenyl (ϕ)]



7. If we mix Me_3SiCl with Me_2SiCl_2, we get silicones of the type:

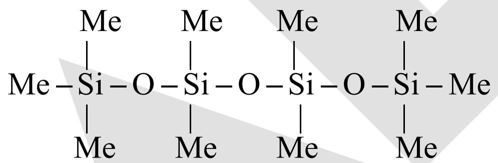
(A) 

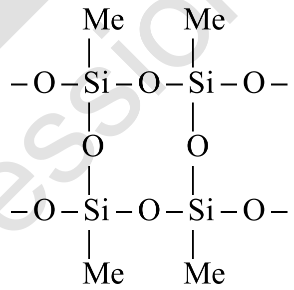
(B) 

(C) both of the above

(D) none of the above

8. If we start with $MeSiCl_3$ as the starting material, silicones formed is:

(A) 

(B) 

(C) Both of the above

(D) None of the above

Paragraph for Question No. 9 & 10

CO gas is absorbed by aqueous suspension of cuprous chloride forming the complex like $[CuCl(CO)(H_2O)_2]$.

9. Comment on the shape of the above complex.

(A) Tetrahedral (B) TBP (C) Square planar (D) Cannot be predicted

10. Choose the correct statement regarding the above molecule

(A) Cl-atom is separated by equal angle from both of the water molecule

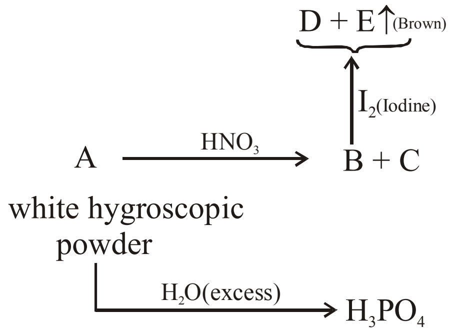
(B) Magnetic moment of the above complex is 1.73 B.M.

(C) There are two stereo isomer for the above complex.

(D) Both (A) and (C)

ALLEN Session 2019-20

Paragraph for Question No. 11 to 12



11. Which of the following compound is not having +5 oxidation state on its central atom

(A) B (B) D (C) C (D) E

12. "A" on reacting with Acetamide yields a compound "F". The compound "F" contains

(A) 4 σ bonds, 2 π bonds

(B) 5 σ bonds, 1 π bonds

(C) 5 σ bonds, 2 π bonds

(D) 2 σ bonds, 2 π bonds

13. Match List-I with List-II

List-I (Chemical reaction)	**List-II (Name of process)**
(I) $4NH_3 + 5O_2 \xrightarrow{800°C/Pt} 4NO + 6H_2O$	(a) Contact process
(II) $4HCl + O_2 \xrightarrow{230°C/CuCl_2} 2Cl_2 + 2H_2O$	(b) Ostwald's process
(III) $2SO_2 + O_2 \xrightarrow{450-500°/V_2O_5} 2SO_3$	(c) Deacon's process
(IV) $2N_2 + 3H_2 \xrightarrow{Fe+Mo} 2NH_3$	(d) Haber's proces

(A) I-a, II-b, III-d, IV-c (B) I-b, II-c, III-a, IV-d

(C) I-a, II-d, III-c, IV-b (D) I-a, II-c, III-b, IV-d

14.

Column-I	**Column-II**
(P) Dry ice	(1) Used as antidote for CO-poisoning
(Q) Carbongene	(2) Used as nonstick coating
(R) Carborundum	(3) Used as refrigerant
(S) Teflon	(4) Used as abrassive

Code :

	P	Q	R	S		P	Q	R	S
(A)	4	1	3	2	(B)	4	2	1	3
(C)	3	1	4	2	(D)	1	4	3	2

ALLEN Session 2019-20

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15. **Column-I** — **Compound** | **Column-II** — **Correct statement for compounds given**

(P) $SnCl_2$ — (1) Used in printing technology

(Q) Butter of tin — (2) Used for gilding purpose (in joining gold pieces)

(R) Mosaic gold — (3) Reducing agent

(S) Pink's salt — (4) Mordant

Code :

	P	Q	R	S
(A)	4	3	2	1
(B)	3	4	2	1
(C)	2	1	3	4
(D)	1	3	4	2

16. **Column-I (Metal)** | **Column-II (Correct statements)**

(P) Fe — (1) Produces NO with 20% HNO_3

(Q) Cu — (2) Produces NH_4NO_3 with 6% HNO_3

(R) Pb — (3) Produces NO_2 with 70% HNO_3

(S) Sn — (4) Produces NH_4NO_3 with 20% HNO_3

Code :

	P	Q	R	S
(A)	2, 1	1	3, 4	2, 3
(B)	2, 3	1, 3	1, 3	2, 3, 4
(C)	1, 3	1, 2	3, 4	2
(D)	1, 4	2, 3	1, 3	1, 4

17. **Column-I (Reactions)** | **Column-II (Correct statements)**

(P) $XeF_2 + PF_5 \rightarrow$ — (1) Fluoride of Xe acts as fluoride acceptor

(Q) $XeF_4 + Pt \rightarrow$ — (2) Fluoride of Xe undergoes disproportion

(R) $XeF_4 + H_2O \rightarrow$ — (3) Fluoride of Xe acts as fluorinating agent

(S) $XeF_6 + CsF \rightarrow$ — (4) Fluoride of Xe act as fluoride donor

Code :

	P	Q	R	S
(A)	4	2	3	1
(B)	3	2	1	4
(C)	4	3	2	1
(D)	3	4	2	1

ALLEN Session 2019-20

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18. **Column-I (Substances)** **Column-II (Can be prepared by)**

Column-I (Substances)	Column-II (Can be prepared by)
(P) O_3	(1) Acidification of BaO_2 with H_3PO_4
(Q) Bleaching powder	(2) Birkeland Eyde process
(R) H_2O_2	(3) Dry O_2 is passed through a silent electrical discharge
(S) HNO_3	(4) Cl_2 gas is passed through slaked lime

Code :

	P	Q	R	S		P	Q	R	S
(A)	3	4	1	2	(B)	1	3	4	2
(C)	2	1	3	4	(D)	4	1	2	3

MATCHING LIST TYPE 1 × 3 Q. (19 to 21)

For the following molecules in column-I, match the correct order of properties with column-II & column-III according to the questions asked.

Column - 1 Molecules	Column - 2 Properties	Column - 3 Correct order
(I) NH_3, PH_3, AsH_3, SbH_3	(i) Bond angle	(p) Increasing order
(II) H_2O, H_2S, H_2Se, H_2Te	(ii) Reducing character	(q)Decreasing order
(III) HF, HCl, HBr, HI	(iii) Intermolecular force	(r) All equal
(IV) CH_4, SiH_4, GeH_4, SnH_4	(iv) Boiling Point	(s) Irregular order

19. Which is the only **CORRECT** combination?

(A) (I), (iv), (p) (B) (II), (i), (p) (C) (III), (iv), (q) (D) (IV), (iii), (p)

20. Which is the only **INCORRECT** combination?

(A) (I), (ii), (p) (B) (IV), (i), (r) (C) (III), (iii), (q) (D) (II), (iv), (s)

21. In which combination, Drago's Rule plays an insignificant role in prediction of orders?

(A) (I), (i), (q) (B) (II), (iv), (s) (C) (IV), (i), (r) (D) (III), (ii), (p)

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MATCHING LIST TYPE 1 × 3 Q. (22 to 24)

Molecule	Hybridization of central atom(s)	Properties related to structure
(P) $(HBNH)_3$	(I) sp^2	(i) Cyclic structure
(Q) C_3O_2	(II) sp^3	(ii) Has $(X-\ddot{\underset{\cdot\cdot}{O}}-X)$ linkage (X = central atom)
(R) $P_4O_{12}^{4-}$	(III) sp^3d	(iii) Planar
(S) N_2O_4	(IV) sp	(iv) Has (X – X) linkage

22. Identify **CORRECT** match

(A) P, II, iii (B) P, I, i (C) P, IV, iii (D) Q, I, i

23. Identify **CORRECT** match

(A) Q, IV, iii (B) Q, IV, i (C) Q, I, ii (D) R, I, ii

24. Identify **CORRECT** match

(A) R, II, iv (B) R, II, ii (C) S, III, ii (D) S, IV, i

ALLEN Session 2019-20

EXERCISE # JEE-MAINS

1. Which products are expected from the disproportionation of hypochlorous acid : [AIEEE-2002]
(1) $HClO_3$ and Cl_2O (2) $HClO_2$ and HClO (3) HCl and Cl_2O (4) HCl and $HClO_3$

2. Identify the **INCORRECT** statement among the following : [AIEEE-2002]
(1) Ozone reacts with SO_2 to give SO_3
(2) Silicon reacts with NaOH(aq.) in the presence of air to give Na_2SiO_3 and H_2O
(3) Cl_2 reacts with excess of NH_3 to give N_2 and HCl
(4) Br_2 reacts with hot and strong NaOH solution to give NaBr, $NaBrO_4$ and H_2O

3. Aluminium is industrially prepared by: [AIEEE-2002]
(1) Fused cryolite (2) Bauxite ore (3) Alunite (4) Borax

4. For making good quality mirrors, plates of float glass are used. These are obtained by floating molten glass over a liquid metal which does not solidify before glass. The metal used can be : [AIEEE-2003]
(1) Sodium (2) Magnesium (3) Mercury (4) Tin

5. What may be expected when phosphine gas is mixed with chlorine gas: [AIEEE-2003]
(1) PCl_5 and HCl are formed and mixture cools down
(2) $PH_3.Cl_2$ is formed with warming up
(3) The mixture only cools down
(4) PCl_3 and HCl are formed and the mixture warms up

6. Graphite is a soft solid lubricant extremely difficult to melt. The reason for this anomalous behaviour is that graphite : [AIEEE-2003]
(1) Has molecules of variable molecular masses like polymers
(2) Has carbon atoms arranged in large plated of rings of strongly bonded carbon atoms with weak interplate bonds
(3) Is a non crystalline substance
(4) Is an allotropic form of diamond

7. Concentrated hydrochloric acid when kept in open air sometimes produces a cloud of white fumes. This is due to : [AIEEE-2003]
(1) Strong affinity of HCl gas for moisture in air results in forming of droplets of liquid solution which appears like a cloudy smoke
(2) Due to strong affinity for water, conc. HCl pulls moisture of air towards self. The moisture forms droplets of water and hence the cloud
(3) Conc. HCl emits strongly smelling HCl gas all the time
(4) Oxygen in air reacts with emitted HCl gas to form a cloud of Cl_2 gas

ALLEN Session 2019-20

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8. Aluminium chloride exists as dimer, Al_2Cl_6 in solid state as well as in solution of non-polar solvents such as benzene. When dissolved in water, it gives- **[AIEEE-2004]**

(1) $Al^{3+} + 3Cl^-$ (2) $[Al(H_2O)_6]^{3+} + 3Cl^-$

(3) $[Al(OH)_6]^{3-} + 3HCl$ (4) $Al_2O_3 + 6HCl$

9. The soldiers of Napolean army while at Alps during freezing winter suffered a serious problem as regards to the tin buttons of their uniforms. White Metallic tin buttons get converted to grey powder. This transformation is related to:- **[AIEEE-2004]**

(1) An interaction with water vapour contained in humid air

(2) A change in crystalline structure of tin

(3) A change in the partial pressure of O_2 in air

(4) An interaction with N_2 of air at low temperature

10. Which one of the following statements regarding helium is incorrect **[AIEEE-2004]**

(1) It is used to produce and sustain powerful superconducting magnets

(2) It is used as a cryogenic agent for carrying out experiments at low temperatures

(3) It is used to fill gas balloons instead of hydrogen because it is lighter than hydrogen and non-inflammable

(4) It is used in gas-cooled nuclear reactors

11. The number of hydrogen atoms attached to phosphorus atom in hypophosphorous acid is : **[AIEEE-2005]**

(1) Zero (2) Two (3) One (4) Three

12. Heating an aqueous solution of aluminium chloride to dryness will give :- **[AIEEE-2005]**

(1) $AlCl_3$ (2) Al_2Cl_6 (3) Al_2O_3 (4) $Al(OH)Cl_2$

13. Which one of the following is the correct statement **[AIEEE-2005]**

(1) Boric acid is a protonic acid

(2) Beryllium exhibits coordination number of six

(3) Chlorides of both beryllium and aluminium have bridged chloride structures in solid phase

(4) B_2H_6, $2NH_3$ is known as "inorganic benzene"

14. In silicon dioxide : **[AIEEE-2005]**

(1) Each silicon atom is surrounded by four oxygen atoms and each oxygen atom is bonded to two silicon atoms

(2) Each silicon atom is surrounded by two oxygen atoms and each oxygen atom is bonded to two silicon atoms

(3) Silicon atom is bonded to two oxygen atoms

(4) There are double bonds between silicon and oxygen atoms

ALLEN Session 2019-20

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15. Regular use of which of the following fertilizer increases the acidity of soil : **[AIEEE-2007]**
(1) Potassium nitrate (2) Urea
(3) Superphosphate of lime (4) Ammonium sulphate

16. The stability of dihalides of Si, Ge, Sn and Pb increases steadily in the sequence: **[AIEEE-2007]**
(1) $GeX_2 << SiX_2 << SnX_2 << PbX_2$ (2) $SiX_2 << GeX_2 << PbX_2 << SnX_2$
(3) $SiX_2 << GeX_2 << SnX_2 << PbX_2$ (4) $PbX_2 << SnX_2 << GeX_2 << SiX_2$

17. Among the following substituted silanes the one which will give rise to cross linked silicone polymer on hydrolysis is **[AIEEE-2008]**
(1) R_4Si (2) $RSiCl_3$ (3) R_2SiCl_2 (4) R_3SiCl

18. Which one of the following reactions of Xenon compounds is not feasible ? **[AIEEE-2009]**
(1) $2XeF_2 + 2H_2O \rightarrow 2Xe + 4HF + O_2$ (2) $XeF_6 + RbF \rightarrow Rb[XeF_7]$
(3) $XeO_3 + 6HF \rightarrow XeF_6 + 3H_2O$ (4) $3XeF_4 + 6H_2O \rightarrow 2Xe + XeO_3 + 12HF + 1.5O_2$

19. Which of the following statement is wrong ? **[AIEEE-2011]**
(1) Single N–N bond is weaker than the single P–P bond
(2) N_2O_4 has two resonance structures
(3) The stability of hydrides increases from NH_3 to BiH_3 in group 15 of the periodic table
(4) Nitrogen cannot form $d\pi$-$p\pi$ bond

20. Which of the following statements regarding sulphur is incorrect ? **[AIEEE-2011]**
(1) At 600°C the gas mainly consists of S_2 molceules
(2) The oxidation state of sulphur is never less than +4 in its compounds
(3) S_2 molecule is paramagnetic
(4) The vapour at 200°C consists mostly of S_8 rings

21. Boron cannot form which one of the following anions ? **[AIEEE-2011]**
(1) $B(OH)_4^-$ (2) BO_2^- (3) BF_6^{3-} (4) BH_4^-

22. In view of the signs of $\Delta_rG°$ for the following reactions
$PbO_2 + Pb \rightarrow 2\ PbO,\ \Delta_rG° < 0$
$SnO_2 + Sn \rightarrow 2\ SnO,\ \Delta_rG° > 0,$
Which oxidation states are more characteristic for lead and tin ? **[AIEEE-2011]**
(1) For lead + 4, for tin + 2 (2) For lead + 2, for tin + 2
(3) For lead + 4, for tin + 4 (4) For lead + 2, for tin + 4

23. The number of S–S bonds in SO_3, $S_2O_3^{2-}$, $S_2O_6^{2-}$ and $S_2O_8^{2-}$ respectively are :- **[Jee Main(Online)-2012]**
(1) 1, 0, 1, 0 (2) 0, 1, 1, 0 (3) 1, 0, 0, 1 (4) 0, 1, 0, 1

24. Which one of the following depletes ozone layer ? **[Jee Main(Online)-2012]**
(1) NO and freons (2) SO_2 (3) CO (4) CO_2

ALLEN Session 2019-20

25. In which of the following arrangements, the sequence is not strictly according to the property written against it ? **[Jee Main(Online)-2012]**

(1) $CO_2 < SiO_2 < SnO_2 < PbO_2$: increasing oxidising power

(2) B < C < O < N : increasing first ionisation enthalpy

(3) $NH_3 < PH_3 < AsH_3 < SbH_3$: increasing basic strength

(4) HF < HCl < HBr < HI : increasing acid strength

26. The formation of molecular complex $BF_3 - NH_3$ results in a change in hybridisation of boron :- **[JEE(Main) Online–2012]**

(1) from sp^3 to sp^3d (2) from sp^2 to dsp^2

(3) from sp^3 to sp^2 (4) from sp^2 to sp^3

27. The catenation tendency of C, Si and Ge is in the order Ge < Si < C. The bond energies (in kJ mol^{-1}) of C — C, Si —Si and Ge—Ge bonds are respectively : **[JEE(Main) Online–2013]**

(1) 348, 260, 297 (2) 348, 297, 260 (3) 297, 348, 260 (4) 260, 297, 348

28. The gas evolved on heating CaF_2 and SiO_2 with concentrated H_2SO_4, on hydrolysis gives a white gelatinous precipitate. The precipitate is: **[Jee Main(Online)-2014]**

(1) silica gel (2) silicic acid

(3) hydrofluosilicic acid (4) calciumfluorosilicate

29. Which of the following series correctly represents relations between the elements from X to Y? **[Jee Main(Online)-2014]**

	$X \rightarrow Y$
(1) $_{18}Ar \rightarrow {}_{54}Xe$	Noble character increases
(2) $_{3}Li \rightarrow {}_{19}K$	Ionization enthalpy increases
(3) $_{6}C \rightarrow {}_{32}Ge$	Atomic radii increases
(4) $_{9}F \rightarrow {}_{35}Br$	Electron gain enthalpy with negative sign increases

30. Which of the following statements about the depletion of ozone layer is correct?

[Jee Main(Online)-2014]

(1) The problem of ozone depletion is more serious at poles because ice crystals in the clouds over poles act as catalyst for photochemical reactions involving the decomposition of ozone by $Cl^{\bullet}$ and $ClO^{\bullet}$ radicals

(2) The problem of ozone depletion is less serious at poles because NO_2 solidifies and is not available for consuming $ClO^{\bullet}$ radicals

(3) Oxides of nitrogen also do not react with ozone in stratosphere

(4) Freons, chlorofluorocarbons, are inert chemically, they do not react with ozone in stratosphere

31. Which of the following xenon-OXO compounds may not be obtained by hydrolysis of xenon fluorides ? **[Jee Main(Online)-2014]**

(1) XeO_2F_2 (2) XeO_3 (3) XeO_4 (4) $XeOF_4$

ALLEN Session 2019-20

32. Hydrogen peroxide acts both as an oxidising and as a reducing agent depending upon the nature of the reacting species. In which of the following cases H_2O_2 acts as a reducing agent in acid medium ? **[Jee Main(Online)-2014]**

(1) MnO_4^- (2) SO_3^{2-} (3) KI (4) $Cr_2O_7^{2-}$

33. Consider the reaction **[Jee Main(Online)-2014]**

$$H_2SO_{3(aq)} + Sn^{4+}_{(aq)} + H_2O_{(l)} \rightarrow Sn^{2+}_{(aq)} + HSO^-_{4(aq)} + 3H^+_{(aq)}$$

Which of the following statements is correct?

(1) H_2SO_3 is the reducing agent because it undergoes oxidation

(2) H_2SO_3 is the reducing agent because it undergoes reduction

(3) Sn^{4+} is the reducing agent because it undergoes oxidation

(4) Sn^{4+} is the oxidizing agent because it undergoes oxidation

34. In the following sets of reactants which two sets best exhibit the amphoteric character of Al_2O_3. xH_2O ? **[JEE(Main) Online–2014]**

Set-1 : $Al_2O_3.xH_2O(s)$ and OH^- (aq)

Set-2 : $Al_2O_3.xH_2O(s)$ and H_2O (ℓ)

Set-3 : $Al_2O_3.xH_2O(s)$ and H^+ (aq)

Set-4 : $Al_2O_3.xH_2O(s)$ and NH_3 (aq)

(1) 1 and 2 (2) 2 and 4 (3) 1 and 3 (4) 3 and 4

35. Which of the following compounds has a P-P bond :- **[Jee Main(Online)-2015]**

(1) $H_4P_2O_5$ (2) $(HPO_3)_3$ (3) $H_4P_2O_7$ (4) $H_4P_2O_6$

36. Chlorine water on standing loses its colour and forms :- **[Jee Main(Online)-2015]**

(1) HCl and $HClO_2$ (2) HCl only

(3) HOCl and $HOCl_2$ (4) HCl and HOCl

37. Which among the following is the most reactive ? **[Jee Main-2015]**

(1) I_2 (2) ICl (3) Cl_2 (4) Br_2

38. Which one has the highest boiling point ? **[Jee Main-2015]**

(1) Kr (2) Xe (3) He (4) Ne

39. From the following statements regarding H_2O_2, choose the incorrect statement : **[Jee Main-2015]**

(1) It has to be stored in plastic or wax lined glass bottles in dark

(2) It has to be kept away from dust

(3) It can act only as an oxidizing agent

(4) It decomposes on exposure to light

ALLEN Session 2019-20

40. The reaction of zinc with dilute and concentrated nitric acid, respectively produces : **[JEE (Main) 2016]**

(1) NO_2 and N_2O (2) N_2O and NO_2 (3) NO_2 and NO (4) NO and N_2O

41. The non-metal that does not exhibit positive oxidation state is : **[JEE (Main) 2016]**

(1) Oxygen (2) Fluorine (3) Iodine (4) Chlorine

42. Which intermolecular force is most responsible in allowing xenon gas to liquefy? **[JEE (Main) Online 2016]**

(1) Ionic (2) Instantaneous dipole- induced dipole

(3) Dipole - dipole (4) Ion - dipole

43. The following statements concern elements in the periodic table. Which of the following is true ?

(1) The group 13 elements are all metals. **[JEE (Main) Online 2016]**

(2) For group 15 elements, the stability of +5 oxidation state increases down the group.

(3) All the elements in Group 17 are gases.

(4) Elements of group 16 have lower ionization enthalpy values compared to those of group 15 in the corresponding periods.

44. ***Assertion :*** Among the carbon allotropes, diamond is an insulator, whereas, graphite is a good conductor of electricity. **[JEE (Main) Online 2016]**

Reason : Hybridization of carbon in diamond and graphite are sp^3 and sp^2, respectively.

(1) Assertion is incorrect statement, but the reason is correct.

(2)Both assertion and reason are correct, and the reason is the correct explanation for the assertion.

(3) Both assertion and reason are incorrect.

(4) Both assertion and reson are correct, but the reason is not the correct explanation for the assertion.

45. Identify the incorrect statement : **[JEE (Main) Online 2016]**

(1) S_8 ring has a crown shape.

(2) The S–S–S bond angles in the S_8 and S_6 rings are the same

(3) S_2 is paramagnetic like oxygen

(4) Rhombic and monoclinic sulphur have S_8 molecules.

46. Which of the following reactions is an example of a redox reaction ? **[JEE (Main) 2017]**

(1) $XeF_4 + O_2F_2 \rightarrow XeF_6 + O_2$

(2) $XeF_2 + PF_5 \rightarrow [XeF]^+PF_6^-$

(3) $XeF_6 + H_2O \rightarrow XeOF_4 + 2HF$

(4) $XeF_6 + 2H_2O \rightarrow XeO_2F_2 + 4HF$

47. The products obtained when chlorine gas reacts with cold and dilute aqueous NaOH are :-

(1) ClO^- and ClO_3^- (2) ClO_2^- and ClO_3^- **[JEE (Main) 2017]**

(3) Cl^- and ClO^- (4) Cl^- and ClO_2^-

Session 2019-20

node06\B0AI-B0\Kota\JEE(Advanced)\Enthusiast\Chem\Sheet\p-Block Element\Eng\02_Ex.p65

48. In which of the following reaction, hydrogen peroxide acts as an oxidizing agent ?

(1) $I_2 + H_2O_2 + 2OH^- \rightarrow 2I^- + 2H_2O + O_2$ **[JEE (Main) ONLINE 2017]**

(2) $HOCl + H_2O_2 \rightarrow H_3O^+ + Cl^- + O_2$

(3) $PbS + 4H_2O_2 \rightarrow PbSO_4 + 4H_2O$

(4) $2MnO_4^- + 3H_2O_2 \rightarrow 2MnO_2 + 3O_2 + 2H_2O + 2OH^-$

49. XeF_6 on partial hydrolysis with water produces a compound 'X'. The same compound 'X' is formed when XeF_6 reacts with silica. The compound 'X' is:- **[JEE (Main) ONLINE 2017]**

(1) XeO_3 (2) XeF_4 (3) $XeOF_4$ (4) XeF_2

50. The compound that does not produce nitrogen gas by the thermal decomposition is

[JEE (Main) OFFLINE 2018]

(1) $(NH_4)_2Cr_2O_7$ (2) NH_4NO_2 (3) $(NH_4)_2SO_4$ (4) $Ba(N_3)_2$

51. For per gram of reactant, the maximum quantity of N_2 gas is produced in which of the following thermal decomposition reactions ? **[JEE (Main) ONLINE 2018]**

(Given : Atomic wt. – Cr = 52u, Ba = 137u)

(1) $2NH_4NO_3(s) \rightarrow 2N_2(g) + 4H_2O(g) + O_2(g)$

(2) $Ba(N_3)_2(s) \rightarrow Ba(s) + 3N_2(g)$

(3) $(NH_4)_2Cr_2O_7(s) \rightarrow N_2(g) + 4H_2O(g)$

(4) $2NH_3(g) \rightarrow N_2(g) + 3H_2(g)$

52. Lithium aluminium hydride reacts with silicon tetrachloride to form :- **[JEE (Main) ONLINE 2018]**

(1) LiCl, $AlCl_3$ and SiH_4 (2) LiCl, AlH_3 and SiH_4

(3) LiH, $AlCl_3$ and $SiCl_2$ (4) LiH, AlH_3 and SiH_4

53. Xenon hexafluoride on partial hydrolysis produces compounds 'X' and 'Y' Compounds 'X' and 'Y' and the oxidation state of Xe are respectively : **[JEE (Main) ONLINE 2018]**

(1) $XeO_2F_2(+6)$ and $XeO_2(+4)$ (2) $XeOF_4(+6)$ and $XeO_2F_2(+6)$

(3) $XeOF_4(+6)$ and $XeO_3(+6)$ (4) $XeO_2(+4)$ and $XeO_3(+6)$

54. Which of the following is a lewis acid ? **[JEE (Main) ONLINE 2018]**

(1) NaH (2) NF_3 (3) PH_3 (4) $B(CH_3)_3$

ALLEN Session 2019-20
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EXERCISE # J-ADVANCED

(IIT JEE ASKED QUESTIONS)

Fill in the blanks

1. Give reason : **[IIT- 2000]**

Why elemental nitrogen exists as a diatomic molecule whereas elemental phosphorus is a tetra atomic molecule.

2. Give an example of oxidation of one halide by another halogen. Explain the feasibility of the reaction. **[IIT- 2000]**

3. Compounds X on reduction with $LiAlH_4$ gives a hydride Y containing 21.72% hydrogen alongwith other products. The compound Y reacts with air explosively resulting in boron trioxide. Identify X and Y. Give balanced reactions involved in the formation of Y and its reaction with air Draw the structure of Y. **[IIT- 2001]**

4. Starting from $SiCl_4$, prepare the following in steps not exceeding the number given in parenthesis (reactions only) **[IIT- 2001]**

(i) Silicon (1)

(ii) Linear silicon containing methyl group only (4)

(iii) Na_2SiO_3 (3)

5. Write the balanced chemical equation for developing photographic films. **[IIT- 2001]**

6. Identify (X) in the following synthetic scheme and write their structures. **[IIT- 2001]**

$Ba\overset{*}{C}O_3 + H_2SO_4 \longrightarrow$ X (gas) (C denotes C^{14})

7. Write the balanced equations for the reactions of the following compounds with water **[IIT-2002]**

(i) Al_4Cl_3 (ii) CaNCN (iii) BF_3 (iv) NCl_3 (v) XeF_3

8. Write the balanced equations for the reactions of the following compounds with water: **[IIT- 2002]**

(i) Al_4C_3 (ii) CaNCN (iii) BF_3 (iv) NCl_3 (v) XeF_4

9. Identify the following:

$$\underset{(aq)}{Na_2CO_3} \xrightarrow{SO_2} A \xrightarrow{Na_2CO_3} B \xrightarrow[\Delta]{\text{elemental S}} C \xrightarrow{I_2} D$$

[IIT- 2003]

Also mention the oxidation state of S in all the compounds.

10. Arrange the following oxides in the increasing order of Bronsted basicity. **[IIT- 2004]**

Cl_2O_7, BaO,SO_3, CO_2, B_2O_3

11. The number of P—O—P bonds in cyclic tetrametaphosphoric acid is – **[IIT-2000]**

(A) Zero (B) Two (C) Three (D) Four

12. The correct order of acidic strength is – **[IIT- 2000]**

(A) $Cl_2O_7 > SO_2 > P_4O_{10}$ (B) $CO_2 > N_2O_5 > SO_3$

(C) $Na_2O > MgO > Al_2O_3$ (D) $K_2O > CaO > MgO$

ALLEN Session 2019-20
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13. Amongst H_2O, H_2S, H_2Se and H_2Te, the one with the highest boiling point is – **[IIT- 2000]**

(A) H_2O because of hydrogen bonding (B) H_2Te because of higher molecular weight

(C) H_2S because of hydrogen bonding (D) H_2Se because of lower molecular weight.

14. Ammonia can be dried by – **[IIT- 2000]**

(A) Conc. H_2SO_4 (B) P_4O_{10} (C) CaO (D) Anhydrous $CaCl_2$

15. Which of the following are hydrolysed – **[REE 2000]**

(A) NCl_3 (B) BCl_3 (C) CCl_4 (D) $SiCl_4$

16. The set with correct order of acidity is – **[IIT- 2001]**

(A) $HClO < HClO_2 < HClO_3 < HClO_4$

(B) $HClO_4 < HClO_3 < HClO_2 < HClO$

(C) $HClO < HClO_4 < HClO_3 < HClO_2$

(D) $HClO_4 < HClO_2 < HClO_3 < HClO$

17. The reaction, $3ClO^-(aq) \longrightarrow ClO_3^-(aq) + 2Cl^-(aq)$ is an example of – **[IIT- 2001]**

(A) Oxidation reaction (B) reduction reaction

(C) Disproportionation reaction (D) Decomposition reaction

18. The number of S–S bonds in sulphur trioxide trimer, (S_3O_9) is – **[IIT- 2001]**

(A) Three (B) Two (C) One (D) Zero

19. **Statement-I :** Between $SiCl_4$ and CCl_4, only $SiCl_4$ reacts with water **[IIT- 2001]**

Because :

Statement-II : $SiCl_4$ is ionic and CCl_4 is covalent

(A) If both assertion and reason are correct and reason is the corect explanation of the assertion

(B) If both assertion and reason are correct, but reason is not the correct explanation of the assertion

(C) If assertion is correct, but reason is incorrect

(D) If assertion is incorrect, but reason is correct.

20. Polyphosphates are used as water softening agents because they – **[IIT- 2002]**

(A) Form soluble complexes with anionic species

(B) Precipitate anionic species

(C) Form soluble complexes with cationic species

(D) Precipitate cationic species

21. Identify the correct order of solubility of Na_2S, CuS, and ZnS in aqueous medium – **[IIT- 2002]**

(A) $CuS > ZnS > Na_2S$ (B) $ZnS > Na_2S > CuS$

(C) $Na_2S > CuS > ZnS$ (D) $Na_2S > ZnS > CuS$

22. Identify, the correct order of acidic strength of CO_2, CuO, CaO, H_2O – **[IIT- 2002]**

(A) $CaO < CuO < H_2O < CO_2$ (B) $H_2O < CuO < CaO < CO_2$

(C) $CaO < H_2O < CuO < CO_2$ (D) $H_2O < CO_2 < CaO < CuO$

ALLEN Session 2019-20

23. H_3BO_3 is – **[IIT- 2002, 3]**

(A) Monobasic acid and weak Lewis acid
(B) Monobasic and weak Bronsted acid
(C) Monobasic and strong Lewis acid
(D) Tribasic and weak Bronsted acid

24. When I^- is oxidised by MnO_4^- in alkaline medium, I^- converts into – **[IIT- 2003]**

(A) IO_3^- (B) I_2 (C) IO_4^- (D) IO^-

25. **[IIT- 2003]**

	Column-I (Change)		**Column-II (Given change is done by)**
(A)	$Bi^{3+} \longrightarrow (BiO)^+$	(p)	Heat
(B)	$[AlO_2]^- \longrightarrow Al(OH)_3$	(q)	Hydrolysis
(C)	$SiO_4^{4-} \longrightarrow Si_2O_7^{6-}$	(r)	Acidification
(D)	$(B_4O_7^{2-}) \longrightarrow [B(OH)_3]$	(s)	Dilution by water

26. $(Me)_2\ SiCl_2$ on hydrolysis will produce – **[IIT- 2003]**

(A) $(Me)_2\ Si(OH)_2$
(B) $(Me)_2\ Si = O$
(C) $[—(Me)_2\ Si—O—]_n$
(D) $Me_2\ SiCl(OH)$

27. Which is the most thermodynamically stable allotropic form of phosphorus? **[IIT- 2004]**

(A) Red (B) White (C) Black (D) Yellow

28. When PbO_2 reacts with conc. HNO_3 the gas evolved may be : **[2005]**

(A) NO_2 (B) O_2 (C) N_2 (D) N_2O

29. Which of the following is not oxidised by O_3? **[IIT- 2005]**

(A) KI (B) $FeSO_4$ (C) $KMnO_4$ (D) K_2MnO_4

30. Which blue-liquid is obtained on reacting equimolar amounts of two gases at – 30ºC ? **[IIT- 2005]**

(A) N_2O (B) N_2O_3 (C) N_2O_4 (D) N_2O_5

31. $B(OH)_3 + NaOH \rightleftharpoons NaBO_2 + Na[B(OH)_4] + H_2O$ how can this reaction is made to proceed in forward direction ? **[IIT- 2006]**

(A) Addition of cis 1, 2 diol
(B) Addition of borax
(C) Addition of trans 1, 2 diol
(D) Addition of Na_2HPO_4

32. Among the following, the paramagnetic compound is – **[IIT- 2007]**

(A) Na_2O_2 (B) O_3 (C) N_2O (D) KO_2

33. **Statement-I** : Boron always forms covalent bond **[2007]**

Because :

Statement-II : The small size of B^{3+} favours formation of covalent bond.

(A) Statement-I is True, Statement-II is True, Statement-II is a correct explanation for Statement-I
(B) Statement-I is True, Statement-II is Ture, Statement-II is not a correct explanation for Statement-II
(C) Statement-I is True, Statement-II is False
(D) Statement-I is False, Statement-II is True

ALLEN Session 2019-20

34. **Statement-I** : In water, orthoboric acid behaves as a weak monobasic acid. **[2007]**

Statement-II : In water, orthoboric acid acts as a proton donor.

(A) Statement-I is True, Statement-II is True, Statement-II is a correct explanation for Statement-I

(B) Statement-I is True, Statement-II is True, Statement-II is not a correct explanation for Statement-II

(C) Statement-I is True, Statement-II is False

(D) Statement-I is False, Statement-II is True

Comprehension # 1 (Q. 35 to 37)

The noble gases have closed-shell electronic configuration and are monoatomic gases under normal conditions. The low boiling point of the lighter noble gases are due to weak dispersion forces between the atoms and the absence of other interatomic interactions. The direct reaction of xenon with fluorine leads to a series of compounds with oxidation number + 2, + 4 and + 6. XeF_4 reacts violently with water to give XeO_3. The compounds of xenon exhibit rich stereochemistry and their geometries can be deduced considering the total number of electron pairs in the valence shell. **[IIT- 2007]**

35. Argon is used in arc welding because of its –

(A) Low reactivity with metal
(B) Ability to lower the melting point of metal
(C) Flammability
(D) High calorific value

36. The structure of XeO_3 is –

(A) Linear (B) Planar (C) Pyramidal (D) T-shaped

37. XeF_4 and XeF_6 are expected to be –

(A) Oxidising agent (B) Reducing agent (C) Unreactive (D) Strongly basic

Comprehension # 2 (Q.38 to 40)

There are some deposits of nitrates and phosphates in earth's crust. Nitrates are more soluble in water. Nitrates are difficult to reduce under the laboratory conditions but microbes do it easily. Ammonia forms large number of complexes with transition metal ions. Hybridization easily explains the ease of sigma donation capability of NH_3 and PH_3. Phosphine is a flammable gas and is prepared from white phosphorous. **IIT- 2008]**

38. Among the following, the correct statement is :-

(A) Phosphates have no biological significance in humans

(B) Between nitrates and phosphates, phosphates are less abundant in earth's crust

(C) Between nitrates and phosphates, nitrates are less abundant in earth's crust

(D) Oxidation of nitrates is possible in soil

39. Among the following, the correct statement is :-

(A) Between NH_3 and PH_3, NH_3 is a better electron donor because the lone pair of electrons occupies spherical 's' orbital and is less directional

(B) Between NH_3 and PH_3, PH_3 is a better electron donor because the lone pair of electrons occupies sp^3 orbital and is more directional

(C) Between NH_3 and PH_3, NH_3 is a better electron donor because the lone pair of electrons occupies sp^3 orbital and is more directional

(D) Between NH_3 and PH_3. PH_3 is a better electron donor because the lone pair of electrons occupies spherical 's' orbital and is less directional.

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40. White phosphorus on reaction with NaOH gives PH_3 as one of the products. This is a :-

(A) dimerization reaction (B) disproportionation reaction

(C) condensation reaction (D) precipitation reaction

41. The reaction of P_4 with X leads selectively to P_4O_6. The X is **[JEE 2009]**

(A) Dry O_2 (B) A mixture of O_2 and N_2

(C) Moist O_2 (D) O_2 in the presence of aqueous NaOH

42. The reaction of white phosphorus with aqueous NaOH gives phosphine along with another phosphorus containing compound. The reaction type ; the oxidation states of phosphorus in phosphine and the other product are respectively **[JEE 2012]**

(A) redox reaction ; -3 and –5 (B) redox reaction ; +3 and +5

(C) disproportionation reaction ; -3 and +1 (D) disproportionation reaction ; -3 and +3

43. Bleaching powder contains a salt of an oxoacid as one of its components. The anhydride of that oxoacid is :

(A) Cl_2O (B) Cl_2O_7 (C) ClO_2 (D) Cl_2O_6 **[JEE 2012]**

44. With respect to graphite and diamond, which of the statement(s) given below is (are) correct ?

(A) Graphite is harder than diamond. **[JEE 2012]**

(B) Graphite has higher electrical conductivity than diamond.

(C) Graphite has higher thermal conductivity than diamond.

(D) Graphite has higher C–C bond order than diamond.

45. Concentrated nitric acid, upon long standing, turns yellow-brown due to the formation of -

(A) NO (B) NO_2 (C) N_2O (D) N_2O_4 **[JEE 2013]**

46. The correct statement(s) about O_3 is(are) **[JEE 2013]**

(A) O–O bond lengths are equal (B) Thermal decomposition of O_3 is endothermic

(C) O_3 is diamagnetic in nature (D) O_3 has a bent structure

Comprehension # 3 (Q. 47 and 48)

The reaction of Cl_2 gas with cold dilute and hot concentrated NaOH in water give sodium salt of two (different) oxoacids of chlorine P and Q respectively. The Cl_2 gas reacts with SO_2 gas , in presence of charcoal to give a product R. R reacts with white phosphorous to give a compound S. On hydrolysis, S gives as oxoacid of phosphorous T.

47. R, S and T , respectively are - **[JEE 2013]**

(A) SO_2Cl_2, PCl_5 and H_3PO_4 (B) SO_2Cl_2, PCl_3 and H_3PO_3

(C) $SOCl_2$, PCl_3 and H_3PO_2 (D) $SOCl_2$, PCl_3 and H_3PO_3

48. P and Q, respectively, are the sodium salts of -

(A) Hypochlorus and chloric acid (B) Hypochlorus and chlorus acid

(C) Chloric and perchloric acids (D) Chloric and hypochlorus acids

ALLEN Session 2019-20

49. The unbalanced chemical reactions given in List-I show missing reagent or condition (?) which are provided in List-II. Match List-I with List-II and select the correct answer using the code given below the lists : **[JEE 2013]**

	List-I		**List-II**
(P)	$PbO_2 + H_2SO_4 \xrightarrow{?} PbSO_4 + O_2$ + other product	(1)	NO
(Q)	$Na_2S_2O_3 + H_2O \xrightarrow{?} NaHSO_4$ + other product	(2)	I_2
(R)	$N_2H_4 \xrightarrow{?} \overset{\circ}{N}_2$ + other product	(3)	Warm
(S)	$XeF_2 \xrightarrow{?} Xe$ + other product	(4)	Cl_2

Codes :

	P	Q	R	S		P	Q	R	S
(A)	4	2	3	1	(B)	3	2	1	4
(C)	1	4	2	3	(D)	3	4	2	1

50. Under ambient conditions, the total number of gases released as products in the final step of the reaction scheme shown below is **[JEE Adv. 2014]**

$$XeF_6 \xrightarrow[\text{Hydrolysis}]{\text{Complete}} P + \text{other product}$$

$$P \xrightarrow{OH^- / H_2O} Q$$

$$Q \xrightarrow{\text{slow disproportionation in } OH^- / H_2O} \text{products}$$

(A) 0 (B) 1 (C) 2 (D) 3

51. The product formed in the reaction of $SOCl_2$ with white phosphorous is **[JEE Adv. 2014]**

(A) PCl_3 (B) SO_2Cl_2 (C) SCl_2 (D) $POCl_3$

52. The **CORRECT** statements(s) for orthoboric acid is / are - **[JEE Adv. 2014]**

(A) It behaves as a weak acid in water due to self ionization

(B) Acidity of its aqueous solution increases upon addition of ethylene glycol

(C) It has a three dimensional structure due to hydrogen bonding.

(D) It is a weak electrolyte in water

53. The correct statement(s) regarding, (i) HClO, (ii) $HClO_2$, (iii) $HClO_3$ and (iv) $HClO_4$, is(are)

(A) The number of Cl=O bonds in (ii) and (iii) together is two **[JEE Adv. 2015]**

(B) The number of lone pairs of electrons on Cl in (ii) and (iii) together is three

(C) The hybridization of Cl in (iv) is sp^3

(D) Amongst (i) to (iv), the strongest acid is (i)

ALLEN Session 2019-20

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54. When O_2 is adsorbed on a metallic surface, electron transfer occurs from the metal to O_2. The **TRUE**, statement (s) regarding this adsorption is (are) **[JEE Adv. 2015]**

(A) O_2 is physisorbed (B) heat is released

(C) occupancy of π^*_{2p} of O_2 is increased (D) bond length of O_2 is increased

55. Under hydrolytic conditions, the compounds used for preparation of linear polymer and for chain termination, respectively, are **[JEE (Adv.) 2015]**

(A) CH_3SiCl_3 and $Si(CH_3)_4$ (B) $(CH_3)_2SiCl_2$ and $(CH_3)_3SiCl$

(C) $(CH_3)_2SiCl_2$ and CH_3SiCl_3 (D) $SiCl_4$ and $(CH_3)_3SiCl$

56. Three moles of B_2H_6 are completely reacted with methanol. The number of moles of boron containing product formed is - **[JEE (Adv.) 2015]**

57. The increasing order of atomic radii of the following group 13 elements is : **[JEE Adv. 2016]**

(A) Al < Ga < In < Tl (B) Ga < Al < In < Tl

(C) Al < In < Ga < Tl (D) Al < Ga < Tl < In

58. The crystalline form of borax has **[JEE Adv. 2016]**

(A) Tetranuclear $[B_4O_5(OH)_4]^{2-}$ unit

(B) All boron atoms in the same plane

(C) Equal number of sp^2 and sp^3 hybridized boron atoms

(D) One terminal hydroxide per boron atom

59. The nitrogen containing compound produced in the reaction of HNO_3 with P_4O_{10}

(A) can also be prepared by reaction of P_4 and HNO_3 **[JEE Adv. 2016]**

(B) is diamagnetic

(C) contains one N-N bond

(D) reacts with Na metal producing a brown gas

60. The correct statements(s) about the oxoacids, $HClO_4$ and HClO, is (are) - **[JEE Adv. 2017]**

(A) $HClO_4$ is more acidic than HClO because of the resonance stabilization of its anion

(B) $HClO_4$ is formed in the reaction between Cl_2 and H_2O

(C) The central atom in Both $HClO_4$ and HClO is sp^3 hybridized

(D) The conjugate base of $HClO_4$ is weaker base than H_2O

61. The colour of the X_2 molecules of group 17 elements changes gradually from yellow to violet down the group. This is due to - **[JEE Adv. 2017]**

(A) the physical state of X_2 at room temperature changes from gas to solid down the group

(B) decrease in HOMO-LUMO gap down the group

(C) decrease in π^*-σ^* gap down the group

(D) decrease in ionization energy down the group

ALLEN Session 2019-20

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62. The order of the oxidation state of the phosphorus atom in H_3PO_2, H_3PO_4, H_3PO_3 and $H_4P_2O_6$ is **[JEE Adv. 2017]**

(A) $H_3PO_4 > H_4P_2O_6 > H_3PO_3 > H_3PO_2$

(B) $H_3PO_3 > H_3PO_2 > H_3PO_4 > H_4P_2O_6$

(C) $H_3PO_2 > H_3PO_3 > H_4P_2O_6 > H_3PO_4$

(D) $H_3PO_4 > H_3PO_2 > H_3PO_3 > H_4P_2O_6$

63. The option(s) with only amphoteric oxides is (are): **[JEE Adv. 2017]**

(A) Cr_2O_3, CrO, SnO, PbO

(B) NO, B_2O_3, PbO, SnO_2

(C) Cr_2O_3, BeO, SnO, SnO_2

(D) ZnO, Al_2O_3, PbO, PbO_2

64. Among the following, the correct statement(s) is are **[JEE Adv. 2017]**

(A) $Al(CH_3)_3$ has the three-centre two-electron bonds in its dimeric structure

(B) $AlCl_3$ has the three-centre two-electron bonds in its dimeric structure

(C) BH_3 has the three-centre two-electron bonds in its dimeric structure

(D) The Lewis acidity of BCl_3 is greater than that of $AlCl_3$

Paragraph for Q.65 & 66

Upon heating $KClO_3$ in the presence of catalytic amount of MnO_2, a gas **W** is formed. Excess amount of **W** reacts with white phosphorus to give **X.** The reaction of **X** with pure HNO_3 gives **Y** and **Z**. **[JEE Adv. 2017]**

65. **W** and **X** are, respectively

(A) O_3 and P_4O_6 (B) O_2 and P_4O_{10} (C) O_3 and P_4O_{10} (D) O_2 and P_4O_6

66. **Y** and **Z** are , respectively

(A) N_2O_4 and H_3PO_3 (B) N_2O_4 and HPO_3

(C) N_2O_5 and HPO_3 (D) N_2O_3 and H_3PO_4

67. The compound(s) which generate(s) N_2 gas upon thermal decomposition below 300°C is (are) **(IIT JEE 2018)**

(A) NH_4NO_3 (B) $(NH_4)_2Cr_2O_7$ (C) $Ba(N_3)_2$ (D) Mg_3N_2

INTEGER

68. The total number of compounds having at least one bridging oxo group among the molecules given below is______. **[JEE Adv. 2018]**

N_2O_3, N_2O_5, P_4O_6, P_4O_7, $H_4P_2O_5$, $H_5P_3O_{10}$, $H_2S_2O_3$, $H_2S_2O_5$

ALLEN Session 2019-20

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69. A tin chloride **Q** undergoes the following reactions (not balanced) **[JEE Adv. 2019]**

$\mathbf{Q} + Cl^- \rightarrow \mathbf{X}$

$\mathbf{Q} + Me_3N \rightarrow \mathbf{Y}$

$\mathbf{Q} + CuCl_2 \rightarrow \mathbf{Z} + CuCl$

X is a monoanion having pyramidal geometry. Both **Y** and **Z** are neutral compounds. Choose the correct option(s).

(1) The central atoms in **X** is sp^3 hybridized

(2) The oxidation state of the central atom in **Z** is +2

(3) The central atom in **Z** has one lone pair of electrons

(4) There is a coordinate bond in **Y**

70. At 143 K. the reaction of XeF_4 with O_2F_2 produces a xenon compound **Y**. The total number of lone pair(s) of electrons present on the whole molecule of **Y** is ______ **[JEE Adv. 2019]**

71. The amount of water produced (in g) in the oxidation of 1 mole of rhombic sulphur by conc.HNO_3 to a compound with the highest oxidation state of sulphur is ___ **[JEE Adv. 2019]**

(Given data : Molar mass of water = 18 g mol^{-1})

72. Each of the following options contains a set of four molecules. Identify the option(s) where all four molecules possess permanent dipole moment at room temperature. **[JEE Adv. 2019]**

(1) $BeCl_2$, CO_2, BCl_3, $CHCl_3$

(2) SO_2, C_6H_5Cl, H_2Se, BrF_5

(3) BF_3, O_3, SF_6, XeF_6

(4) NO_2, NH_3, $POCl_3$, CH_3Cl

73. Among B_2H_6, $B_3N_3H_6$, N_2O, N_2O_4, $H_2S_2O_3$ and $H_2S_2O_8$, the total number of molecules containing covalent bond between two atoms of the same kind is ——— **[JEE Adv. 2019]**

ALLEN Session 2019-20

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